
TEST CASE EVALUATION AND EXECUTION LOGS FOR SCREENPILOT SYSTEM

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Test Case Evaluation and Execution Logs for ScreenPilot System

Test Case 01: Writing on Notepad (Simple Task)

User Prompt:

Type "*Agent*" in Notepad

Test Results:

- Steps Total: 1
- Completed: 1
- Failed: 0
- Duration: 12 seconds

Step Execution:

Step 1: Type "Agent" into the Notepad text area

- Target: Notepad text area
- Attempts: 1
- HID Commands:
 - `move_mouse(x, y)`
 - `click()`
 - `type_text("Agent")`
- Result: Successfully executed

Execution Summary:

The system successfully detected the Notepad editor region and generated valid HID commands. The cursor was positioned correctly, and the text "*Agent*" was entered without errors. Post-execution validation confirmed successful task completion.

Success Factors:

- Clear visibility of the Notepad window
- Accurate detection of the text input region
- Valid and complete HID command generation
- Effective closed-loop validation

JSON Response:

```
{
  "run_id": "d4d47075-2ca8-4de5-babe-c7783e4fec00",
  "user_task": "type \"Agent\" on the notepad",
```

```
"started_at": "2026-04-29T16:25:04.284619+00:00",
"finished_at": "2026-04-29T16:25:16.284619+00:00",
"vision_used": true,

"perception": {
  "status": "completed",
  "session_id": "shot_1777479912_781bf2",
  "vision_data": {
    "image_size": {
      "width": 1920,
      "height": 1080
    },
    "coordinate_system": "pixel",
    "elements": [
      {
        "id": "notepad_editor",
        "type": "text_area",
        "label": "Notepad Editor",
        "description": "Main text editing area",
        "state": "visible",
        "dx": 500,
        "dy": 400,
        "confidence": 0.95,
        "source": "gemini_vlm"
      }
    ],
    "element_count": 1
  }
},

"todo": {
  "steps": [
    {
      "id": 1,
      "action": "Type 'Agent' into the notepad text area.",
      "target": "Notepad text area",
      "expected_result": "The word 'Agent' is visible in the notepad
document.",
      "status": "completed"
    }
  ],
  "estimated_complexity": "simple"
},

"step_plans": [
```

```

{
  "step_id": 1,
  "action": "Type 'Agent' into the notepad text area.",
  "target": "Notepad text area",
  "hid_commands": [
    { "action": "move_mouse", "x": 500, "y": 400 },
    { "action": "click" },
    { "action": "type_text", "text": "Agent" }
  ],
  "reasoning": "Detected text area and generated valid HID commands."
}
],

"step_results": [
  {
    "step_id": 1,
    "status": "success",
    "executed_commands": 3,
    "verification": "Text 'Agent' successfully entered into Notepad."
  }
],

"final_report": {
  "success": true,
  "summary": "The task was successfully completed.",
  "message": "The system detected the Notepad editor, positioned the cursor, and typed 'Agent' successfully.",
  "steps_completed": 1,
  "steps_failed": 0,
  "issues": [],
  "recommendations": []
}
}

```

Test Case 02: Screen Analysis (Medium Task)

User Prompt:

Test the current screen and explain important elements

Test Results:

- Steps Total: 1
- Completed: 1
- Failed: 0

- Duration: 5 minutes 36 seconds

Step Execution:

Step 1: Analyze and describe the current screen

- Target: Entire screen
- Attempts: 1
- Execution: Success
- Evaluation: Completed

Execution Summary:

The agent successfully analyzed the screen and identified key elements. It interacted with the interface by clicking the address bar to confirm interactivity and extracted contextual information, including the active URL (*16.16.24.81:3000/analytics*).

Observations:

- Demonstrates strong perception + reasoning capability
- Slight latency due to vision and LLM processing stages

JSON Response:

```
{
  "run_id": "c3d74960-0d60-43a0-8b50-0178b3905308",
  "user_task": "test the current screen and explain important things about that screen",
  "started_at": "2026-04-29T19:11:29.400364+00:00",
  "finished_at": "2026-04-29T19:17:05.979272+00:00",
  "vision_used": true,
  "perception": {
    "image": "E:\\sliit-project\\ScreenAwareTaskAgent\\vision\\data\\sessions\\shot_1777489894_25f120\\processing\\proc_1777489894_e33d95\\preprocessed_frames\\frame_1777489894_bef67e.jpg",
    "image_size": {
      "width": 1920,
      "height": 1080
    },
    "screen_bbox": [
      115,
      65,
      1805,
      1015
    ]
  }
}
```

```
"screen_size": {
  "width": 1690,
  "height": 950
},
"coordinate_system": "pixel",
"elements": [
  {
    "id": "browser_address_bar",
    "type": "input",
    "label": "Address Bar",
    "description": "Browser address bar showing the URL
16.16.24.81:3000/analytics",
    "state": "enabled",
    "dx": 214,
    "dy": 30,
    "frame_dx": 329,
    "frame_dy": 95,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
      201,
      78,
      458,
      111
    ]
  },
  {
    "id": "nexmart_logo",
    "type": "icon",
    "label": "NexMart",
    "description": "NexMart logo and home link in the navigation bar",
    "state": "enabled",
    "dx": 221,
    "dy": 90,
    "frame_dx": 336,
    "frame_dy": 155,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
      280,
      138,
      392,
      171
    ]
  }
],
},
```

```
{
  "id": "nav_analytics",
  "type": "link",
  "label": "Analytics",
  "description": "Navigation link to the Analytics page",
  "state": "active",
  "dx": 472,
  "dy": 92,
  "frame_dx": 587,
  "frame_dy": 157,
  "confidence": 0.95,
  "source": "gemini_vlm",
  "frame_bbox": [
    551,
    144,
    623,
    171
  ]
},
{
  "id": "nav_products",
  "type": "link",
  "label": "Products",
  "description": "Navigation link to the Products page",
  "state": "enabled",
  "dx": 571,
  "dy": 92,
  "frame_dx": 686,
  "frame_dy": 157,
  "confidence": 0.95,
  "source": "gemini_vlm",
  "frame_bbox": [
    650,
    144,
    722,
    171
  ]
},
{
  "id": "nav_orders",
  "type": "link",
  "label": "Orders",
  "description": "Navigation link to the Orders page",
  "state": "enabled",
  "dx": 657,
```



```
"dy": 92,
"frame_dx": 772,
"frame_dy": 157,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
  742,
  144,
  802,
  171
]
},
{
  "id": "nav_payments",
  "type": "link",
  "label": "Payments",
  "description": "Navigation link to the Payments page",
  "state": "enabled",
  "dx": 746,
  "dy": 92,
  "frame_dx": 861,
  "frame_dy": 157,
  "confidence": 0.95,
  "source": "gemini_vlm",
  "frame_bbox": [
    821,
    144,
    901,
    171
  ]
}
],
"element_count": 6
},
"todo": {
  "steps": [
    {
      "id": 1,
      "action": "test the current screen and explain important things about
that screen",
      "target": "screen",
      "expected_result": "Task accomplished",
      "needs_user_data": false,
      "user_data_field": null,
      "status": "done"
    }
  ]
}
```

```
    }
  ],
  "notes": "Fallback single-step plan (LLM error: Response missing valid 'steps' list. Got: NoneType)",
  "estimated_complexity": "unknown"
},
"step_plans": [
  {
    "step_id": 1,
    "action": "test the current screen and explain important things about that screen",
    "target": "screen",
    "expected_result": "Task accomplished",
    "hid_commands": [
      {
        "cmd": "mouse_move",
        "dx": 299,
        "dy": 24
      },
      {
        "cmd": "mouse_click",
        "button": "left"
      }
    ]
  },
  {
    "reasoning": "Clicking the address bar to verify interactivity and focus as part of testing the screen elements."
  }
],
"step_results": [
  {
    "step_id": 1,
    "action": "test the current screen and explain important things about that screen",
    "attempt": 1,
    "exec_status": "success",
    "eval_status": "done",
    "confidence": 1.0,
    "reason": "The screen has been successfully analyzed, identifying the browser address bar and the current URL as the analytics page (16.16.24.81:3000/analytics).",
    "final_status": "executing",
    "timestamp": "2026-04-29T19:16:19.405915+00:00"
  }
],
"final_report": {
```

```
    "success": true,
    "summary": "Successfully analyzed the screen and identified key navigation
elements for NexMart.",
    "message": "The task to test and explain the current screen has been
completed successfully.\n\nAnalysis of the screen reveals that this is the
NexMart interface. Key elements identified include the 'NexMart' logo text and
primary navigation links for 'Analytics' and 'Products'.\n\nStep results:\n- Step
1: test the current screen and explain important things about that screen
(Completed)",
    "steps_completed": 1,
    "steps_failed": 0,
    "issues": [],
    "recommendations": []
  }
}
```

Test Case 03: Login UI Test (Complex Task)

User Prompt:

Login using username = *nuwani* and password = *Nuwani@345*

Test Results:

- Steps Total: 2
- Completed: 0
- Failed: 2
- Duration: 13 minutes 39 seconds

Step Execution:

Step 1: Enter credentials and click Sign In

- Target: NexMart login form
- Attempts: 3
- Execution: Performed
- Evaluation: Failed (retry loop)

Issue:

The system executed input actions but failed to confirm successful login due to invalid evaluation output (malformed JSON).

Step 2: Verify dashboard loaded

- Target: NexMart dashboard
- Execution: Failed
- Issue: No `hid_commands` generated

Execution Summary:

Although the system performed initial interactions (typing and clicking), it failed to complete the task due to issues in the evaluation and planning stages. The workflow stalled due to missing or malformed LLM outputs.

Issues Identified:

- Malformed JSON response from LLM during evaluation
- Missing `hid_commands` in step planning
- No transition from login screen to dashboard
- Possible UI ambiguity or incorrect element targeting

Recommendations for Improvement:

- Enforce strict JSON output formatting in LLM responses
- Implement retry and fallback mechanisms for missing commands
- Improve vision prompts for better UI element detection
- Strengthen evaluation loop to handle incomplete responses
- Add validation for dynamic UI behaviors (e.g., errors, CAPTCHA)

JSON Response:

-

full_cycle.json

```
{
  "run_id": "fb5c82a7-0146-427f-9d46-304f4de95361",
  "user_task": "login to the system username = nuwani and the password = Nuwani@345",
  "started_at": "2026-04-21T11:08:00.575402+00:00",
  "finished_at": "2026-04-21T11:21:39.710276+00:00",
```

```
"vision_used": true,
"perception": {
  "status": "completed",
  "session_id": "shot_1776769686_4fe97d",
  "source_frame": "E:\\sliit-
project\\ScreenAwareTaskAgent\\vision\\data\\sessions\\shot_1776769686_4fe97d\\ra
w_frames\\frame_1776769686_cadbce.jpg",
  "processed_at": 1776769794.6070378,
  "final_json_path": "E:\\sliit-
project\\ScreenAwareTaskAgent\\vision\\data\\sessions\\shot_1776769686_4fe97d\\fi
nal_elements\\frame_1776769686_cadbce.json",
  "vision_data": {
    "image": "E:\\sliit-
project\\ScreenAwareTaskAgent\\vision\\data\\sessions\\shot_1776769686_4fe97d\\pr
ocessing\\proc_1776769686_5165f7\\preprocessed_frames\\frame_1776769686_cadbce.jp
g",
    "image_size": {
      "width": 1920,
      "height": 1080
    },
    "coordinate_system": "pixel",
    "elements": [
      {
        "id": "0",
        "type": "tab",
        "label": "NexMart",
        "description": "Browser tab for NexMart",
        "state": "active",
        "dx": 150,
        "dy": 45,
        "confidence": 0.9,
        "source": "gemini_vlm"
      },
      {
        "id": "1",
        "type": "tab",
        "label": "Google Password Manager",
        "description": "Browser tab for Google Password Manager",
        "state": "inactive",
        "dx": 400,
        "dy": 45,
        "confidence": 0.9,
        "source": "gemini_vlm"
      },
      {
```

```
"id": "2",
"type": "tab",
"label": "frontend",
"description": "Browser tab for frontend",
"state": "inactive",
"dx": 650,
"dy": 45,
"confidence": 0.9,
"source": "gemini_vlm"
},
{
  "id": "3",
  "type": "tab",
  "label": "Ask Gemini",
  "description": "Browser tab for Ask Gemini",
  "state": "inactive",
  "dx": 1500,
  "dy": 45,
  "confidence": 0.9,
  "source": "gemini_vlm"
},
{
  "id": "4",
  "type": "input",
  "label": "Address Bar",
  "description": "Browser address bar showing URL",
  "state": "visible",
  "dx": 400,
  "dy": 85,
  "confidence": 0.9,
  "source": "gemini_vlm"
},
{
  "id": "5",
  "type": "icon",
  "label": "NexMart Logo",
  "description": "Logo for NexMart application",
  "state": "visible",
  "dx": 840,
  "dy": 190,
  "confidence": 0.9,
  "source": "gemini_vlm"
},
{
  "id": "6",
```

```
    "type": "label",
    "label": "NexMart",
    "description": "Application name text",
    "state": "visible",
    "dx": 840,
    "dy": 230,
    "confidence": 0.9,
    "source": "gemini_vlm"
  },
  {
    "id": "7",
    "type": "label",
    "label": "Welcome back",
    "description": "Login card header text",
    "state": "visible",
    "dx": 840,
    "dy": 330,
    "confidence": 0.9,
    "source": "gemini_vlm"
  },
  {
    "id": "8",
    "type": "input",
    "label": "Username Input",
    "description": "Input field for username",
    "state": "empty",
    "dx": 840,
    "dy": 420,
    "confidence": 0.9,
    "source": "gemini_vlm"
  },
  {
    "id": "9",
    "type": "input",
    "label": "Password Input",
    "description": "Input field for password",
    "state": "empty",
    "dx": 840,
    "dy": 490,
    "confidence": 0.9,
    "source": "gemini_vlm"
  },
  {
    "id": "10",
    "type": "button",
```

```
    "label": "Sign In",
    "description": "Button to submit login credentials",
    "state": "visible",
    "dx": 840,
    "dy": 550,
    "confidence": 0.9,
    "source": "gemini_vlm"
  },
  {
    "id": "11",
    "type": "link",
    "label": "Create one",
    "description": "Link to create a new account",
    "state": "visible",
    "dx": 840,
    "dy": 590,
    "confidence": 0.9,
    "source": "gemini_vlm"
  },
  {
    "id": "12",
    "type": "button",
    "label": "Start Button",
    "description": "Windows Start menu button",
    "state": "visible",
    "dx": 470,
    "dy": 815,
    "confidence": 0.9,
    "source": "gemini_vlm"
  },
  {
    "id": "13",
    "type": "input",
    "label": "Search Bar",
    "description": "Windows taskbar search bar",
    "state": "visible",
    "dx": 575,
    "dy": 815,
    "confidence": 0.9,
    "source": "gemini_vlm"
  },
  {
    "id": "14",
    "type": "icon",
    "label": "Chrome Icon",
```



```
    "description": "Google Chrome icon on taskbar",
    "state": "visible",
    "dx": 1020,
    "dy": 815,
    "confidence": 0.9,
    "source": "gemini_vlm"
  },
  {
    "id": "15",
    "type": "icon",
    "label": "VS Code Icon",
    "description": "Visual Studio Code icon on taskbar",
    "state": "visible",
    "dx": 1105,
    "dy": 815,
    "confidence": 0.9,
    "source": "gemini_vlm"
  },
  {
    "id": "16",
    "type": "label",
    "label": "Clock",
    "description": "System clock and date on taskbar",
    "state": "visible",
    "dx": 1565,
    "dy": 815,
    "confidence": 0.9,
    "source": "gemini_vlm"
  },
  {
    "id": "17",
    "type": "icon",
    "label": "Weather Widget",
    "description": "Weather widget on taskbar",
    "state": "visible",
    "dx": 120,
    "dy": 825,
    "confidence": 0.9,
    "source": "gemini_vlm"
  }
],
"element_count": 18
}
},
"todo": {
```

```
"steps": [  
  {  
    "id": 1,  
    "action": "Click the 'Username Input' field, type 'nuwani', click the  
    'Password Input' field, type 'Nuwani@345', and click the 'Sign In' button.",  
    "target": "NexMart login form",  
    "expected_result": "The login form is submitted and the application  
redirects to the dashboard.",  
    "needs_user_data": false,  
    "user_data_field": null,  
    "status": "failed"  
  },  
  {  
    "id": 2,  
    "action": "Verify that the dashboard or home page has loaded to confirm  
successful authentication.",  
    "target": "NexMart dashboard",  
    "expected_result": "The user is successfully logged in and viewing the  
main application interface.",  
    "needs_user_data": false,  
    "user_data_field": null,  
    "status": "failed"  
  }  
],  
  "notes": "The screen shows the NexMart login page. The user has provided the  
necessary credentials in the task description, so no further user input is  
required to complete the login process.",  
  "estimated_complexity": "simple"  
},  
  "step_plans": [  
    {  
      "step_id": 1,  
      "action": "Click the 'Username Input' field, type 'nuwani', click the  
      'Password Input' field, type 'Nuwani@345', and click the 'Sign In' button.",  
      "target": "NexMart login form",  
      "expected_result": "The login form is submitted and the application  
redirects to the dashboard.",  
      "hid_commands": [  
        {  
          "cmd": "mouse_move",  
          "dx": 848,  
          "dy": 586  
        },  
        {  
          "cmd": "mouse_click",
```

```

        "button": "left"
    },
    {
        "cmd": "type_text",
        "text": "nuwani"
    },
    {
        "cmd": "mouse_move",
        "dx": 839,
        "dy": 692
    },
    {
        "cmd": "mouse_click",
        "button": "left"
    },
    {
        "cmd": "type_text",
        "text": "Nuwani@345"
    },
    {
        "cmd": "mouse_move",
        "dx": 875,
        "dy": 838
    },
    {
        "cmd": "mouse_click",
        "button": "left"
    }
],
    "reasoning": "Focusing the username field to type 'nuwani', then focusing
the password field to type 'Nuwani@345', and finally clicking the sign-in button
to submit the form."
},
{
    "step_id": 2,
    "action": "Verify that the dashboard or home page has loaded to confirm
successful authentication.",
    "target": "NexMart dashboard",
    "expected_result": "The user is successfully logged in and viewing the main
application interface.",
    "hid_commands": [],
    "reasoning": "Command generation failed: Response missing 'hid_commands'"
}
],
    "step_results": [

```

```

{
  "step_id": 1,
  "action": "Click the 'Username Input' field, type 'nuwani', click the
'Password Input' field, type 'Nuwani@345', and click the 'Sign In' button.",
  "attempt": 3,
  "exec_status": "success",
  "eval_status": "retry",
  "confidence": 0.0,
  "reason": "Evaluation error: No valid JSON found in LLM response:
'{\\n  \\\"status\\\": \\\"retry\\\",',
  "final_status": "executing",
  "timestamp": "2026-04-21T11:17:33.798945+00:00"
}
],
"final_report": {
  "success": false,
  "summary": "The login attempt failed, and the application remains on the
login screen.",
  "message": "The task to log in to the NexMart system was unsuccessful.
Although the steps were initiated, the system failed to process the credentials
and navigate to the dashboard. The final screen state confirms that the 'Welcome
back' login page is still active.\\n\\nStep results:\\n- Step 1: Enter credentials
and click 'Sign In' \\u2014 FAILED\\n- Step 2: Verify dashboard loading \\u2014
FAILED",
  "steps_completed": 0,
  "steps_failed": 2,
  "issues": [
    "The automation failed to successfully submit the login form.",
    "The screen state did not change from the login page to the dashboard."
  ],
  "recommendations": [
    "Verify that the username 'nuwani' and password 'Nuwani@345' are correct.",
    "Check for any hidden error messages or validation alerts on the page.",
    "Ensure the 'Sign In' button is clickable and the application is
responsive."
  ]
}
}

```

Test Case 04: Page re-direction (Complex Task)

User Prompt:

Find and click on the "Ask Gemini" button on this browser

Test Results:

- **Steps Total:** 2
- **Completed:** 1
- **Failed:** 1
- **Duration:** 17 minutes 58 seconds

Step Execution:

Step 1: Click the 'frontend' tab at (247, 88) to see if the 'Ask Gemini' button is located on that page

- **Target:** frontend tab
- **Attempts:** 2
- **Execution:** Performed
- **Evaluation:** Success
- **Issue:** None

Step 2: Locate the 'Ask Gemini' button on the updated screen and click it

- **Target:** Ask Gemini button
- **Attempts:** 3
- **Execution:** Performed
- **Evaluation:** Failed (retry loop)
- **Issue:** Evaluation error — No valid JSON found in LLM response ('{\n "status')

Execution Summary:

The agent successfully switched to the “frontend” tab but failed to locate or click the “Ask Gemini” button. The failure occurred during step evaluation due to malformed JSON output from the LLM. The button may not have been present in the detected UI elements, or the page may not have fully loaded.

Issues Identified:

- Malformed JSON response from LLM during evaluation
- “Ask Gemini” button not detected in UI elements
- No graceful fallback when target element is missing
- Possible incomplete page load before element detection
- Stale cached elements may have interfered with fresh detection

Recommendations for Improvement:

- Enforce strict JSON output formatting with validation before parsing
- Add scroll/page-down attempts if target button not initially visible
- Implement wait-for-element logic before declaring “not found”
- Improve vision prompts to prioritize fresh detection over stale cache
- Add explicit element-not-found handling instead of failing on JSON error

JSON Response:

-

full_cycle.json

```
{
  "run_id": "78226bde-05c5-481a-848f-737d3429e4e9",
  "user_task": "Find and click on the \"Ask Gemini\" button on this browser",
  "started_at": "2026-05-02T01:22:14.203548+00:00",
  "finished_at": "2026-05-02T01:40:12.817672+00:00",
  "vision_used": true,
  "perception": {
    "image": "E:\\sliit-
project\\ScreenAwareTaskAgent\\vision\\data\\sessions\\shot_1777684945_f5b814\\pr
ocessing\\proc_1777684945_04ed69\\preprocessed_frames\\frame_1777684945_38c7c4.jp
g",
    "image_size": {
      "width": 1920,
      "height": 1080
    },
    "screen_bbox": [
      115,
      65,
      1805,
      1015
    ],
    "screen_size": {
      "width": 1690,
```

```
"height": 950
},
"coordinate_system": "pixel",
"elements": [
  {
    "id": "tab_frontend",
    "type": "tab",
    "label": "frontend",
    "description": "Browser tab for frontend",
    "state": "inactive",
    "dx": 132,
    "dy": 23,
    "frame_dx": 247,
    "frame_dy": 88,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
      161,
      65,
      333,
      111
    ],
    "session_memory_key": "tab:frontend",
    "cached": false
  },
  {
    "id": "tab_new_tab",
    "type": "tab",
    "label": "New tab",
```

```
"description": "Active browser tab",
"state": "active",
"dx": 330,
"dy": 23,
"frame_dx": 445,
"frame_dy": 88,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    333,
    65,
    557,
    111
],
"session_memory_key": "tab:new tab",
"cached": false
},
{
    "id": "btn_back",
    "type": "button",
    "label": "Back",
    "description": "Go back in history",
    "state": "disabled",
    "dx": 30,
    "dy": 76,
    "frame_dx": 145,
    "frame_dy": 141,
    "confidence": 0.9,
    "source": "gemini_vlm",
```



```
"frame_bbox": [
  128,
  118,
  161,
  164
],
"session_memory_key": "button:back",
"cached": false
}
],
"element_count": 3,
"current_elements": [
  {
    "id": "tab_frontend",
    "type": "tab",
    "label": "frontend",
    "description": "Browser tab for frontend",
    "state": "inactive",
    "dx": 132,
    "dy": 23,
    "frame_dx": 247,
    "frame_dy": 88,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
      161,
      65,
      333,
      111
```

```
    ],  
    "session_memory_key": "tab:frontend",  
    "cached": false  
  },  
  {  
    "id": "tab_new_tab",  
    "type": "tab",  
    "label": "New tab",  
    "description": "Active browser tab",  
    "state": "active",  
    "dx": 330,  
    "dy": 23,  
    "frame_dx": 445,  
    "frame_dy": 88,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
      333,  
      65,  
      557,  
      111  
    ],  
    "session_memory_key": "tab:new tab",  
    "cached": false  
  },  
  {  
    "id": "btn_back",  
    "type": "button",  
    "label": "Back",
```

```
"description": "Go back in history",
"state": "disabled",
"dx": 30,
"dy": 76,
"frame_dx": 145,
"frame_dy": 141,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    128,
    118,
    161,
    164
],
"session_memory_key": "button:back",
"cached": false
}
],
"resolved_elements": [
    {
        "id": "tab_frontend",
        "type": "tab",
        "label": "frontend",
        "description": "Browser tab for frontend",
        "state": "inactive",
        "dx": 132,
        "dy": 23,
        "frame_dx": 247,
        "frame_dy": 88,
```

```
"confidence": 0.95,  
"source": "gemini_vlm",  
"frame_bbox": [  
    161,  
    65,  
    333,  
    111  
],  
"session_memory_key": "tab:frontend",  
"cached": false  
},  
{  
    "id": "tab_new_tab",  
    "type": "tab",  
    "label": "New tab",  
    "description": "Active browser tab",  
    "state": "active",  
    "dx": 330,  
    "dy": 23,  
    "frame_dx": 445,  
    "frame_dy": 88,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        333,  
        65,  
        557,  
        111  
    ],
```

```
    "session_memory_key": "tab:new tab",
    "cached": false
  },
  {
    "id": "btn_back",
    "type": "button",
    "label": "Back",
    "description": "Go back in history",
    "state": "disabled",
    "dx": 30,
    "dy": 76,
    "frame_dx": 145,
    "frame_dy": 141,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
      128,
      118,
      161,
      164
    ],
    "session_memory_key": "button:back",
    "cached": false
  }
],
"session_knowledge_base": {
  "session_id": "shot_1777684945_f5b814",
  "started_at": "2026-05-02T01:22:25.865701+00:00",
  "updated_at": "2026-05-02T01:22:55.644914+00:00",
```

```
"stale_after_misses": 3,
"confidence_floor": 0.4,
"decay_per_miss": 0.9,
"frame_count": 1,
"frame_name": "frame_1777684945_38c7c4.jpg",
"frame_index": 0,
"image_size": {
  "width": 1920,
  "height": 1080
},
"known_elements": {
  "tab:frontend": {
    "element_key": "tab:frontend",
    "element_type": "tab",
    "label": "frontend",
    "bbox": [
      161.0,
      65.0,
      333.0,
      111.0
    ],
    "confidence": 0.95,
    "source": "gemini_vlm",
    "first_seen_frame": "frame_1777684945_38c7c4.jpg",
    "last_seen_frame": "frame_1777684945_38c7c4.jpg",
    "first_seen_at": "2026-05-02T01:22:55.644914+00:00",
    "last_seen_at": "2026-05-02T01:22:55.644914+00:00",
    "first_seen_index": 0,
    "last_seen_index": 0,
```

```
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777684945_38c7c4.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T01:22:55.644914+00:00",
    "confidence": 0.95
  }
]
},
"tab:new tab": {
  "element_key": "tab:new tab",
  "element_type": "tab",
  "label": "New tab",
  "bbox": [
    333.0,
    65.0,
    557.0,
    111.0
  ],
  "confidence": 0.95,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777684945_38c7c4.jpg",
  "last_seen_frame": "frame_1777684945_38c7c4.jpg",
  "first_seen_at": "2026-05-02T01:22:55.644914+00:00",
  "last_seen_at": "2026-05-02T01:22:55.644914+00:00",
```

```
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777684945_38c7c4.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T01:22:55.644914+00:00",
    "confidence": 0.95
  }
],
},
"button:back": {
  "element_key": "button:back",
  "element_type": "button",
  "label": "Back",
  "bbox": [
    128.0,
    118.0,
    161.0,
    164.0
  ],
  "confidence": 0.9,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777684945_38c7c4.jpg",
  "last_seen_frame": "frame_1777684945_38c7c4.jpg",
```



```
"first_seen_at": "2026-05-02T01:22:55.644914+00:00",
"last_seen_at": "2026-05-02T01:22:55.644914+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777684945_38c7c4.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T01:22:55.644914+00:00",
    "confidence": 0.9
  }
]
},
"resolved_elements": [
  {
    "id": "tab_frontend",
    "type": "tab",
    "label": "frontend",
    "description": "Browser tab for frontend",
    "state": "inactive",
    "dx": 132,
    "dy": 23,
    "frame_dx": 247,
    "frame_dy": 88,
```

```
"confidence": 0.95,  
"source": "gemini_vlm",  
"frame_bbox": [  
    161,  
    65,  
    333,  
    111  
],  
"session_memory_key": "tab:frontend",  
"cached": false  
},  
{  
    "id": "tab_new_tab",  
    "type": "tab",  
    "label": "New tab",  
    "description": "Active browser tab",  
    "state": "active",  
    "dx": 330,  
    "dy": 23,  
    "frame_dx": 445,  
    "frame_dy": 88,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        333,  
        65,  
        557,  
        111  
    ],  
}
```

```
    "session_memory_key": "tab:new tab",
    "cached": false
  },
  {
    "id": "btn_back",
    "type": "button",
    "label": "Back",
    "description": "Go back in history",
    "state": "disabled",
    "dx": 30,
    "dy": 76,
    "frame_dx": 145,
    "frame_dy": 141,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
      128,
      118,
      161,
      164
    ],
    "session_memory_key": "button:back",
    "cached": false
  }
],
"current_elements": [
  {
    "id": "tab_frontend",
    "type": "tab",
```

```
"label": "frontend",
"description": "Browser tab for frontend",
"state": "inactive",
"dx": 132,
"dy": 23,
"frame_dx": 247,
"frame_dy": 88,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    161,
    65,
    333,
    111
],
"session_memory_key": "tab:frontend",
"cached": false
},
{
    "id": "tab_new_tab",
    "type": "tab",
    "label": "New tab",
    "description": "Active browser tab",
    "state": "active",
    "dx": 330,
    "dy": 23,
    "frame_dx": 445,
    "frame_dy": 88,
    "confidence": 0.95,
```

```
"source": "gemini_vlm",
"frame_bbox": [
    333,
    65,
    557,
    111
],
"session_memory_key": "tab:new tab",
"cached": false
},
{
    "id": "btn_back",
    "type": "button",
    "label": "Back",
    "description": "Go back in history",
    "state": "disabled",
    "dx": 30,
    "dy": 76,
    "frame_dx": 145,
    "frame_dy": 141,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        128,
        118,
        161,
        164
    ],
    "session_memory_key": "button:back",
```

```
        "cached": false
    }
]
},
"session_memory": {
    "run_id": "78226bde-05c5-481a-848f-737d3429e4e9",
    "user_task": "Find and click on the \"Ask Gemini\" button on this browser",
    "started_at": "2026-05-02T01:22:14.203548+00:00",
    "updated_at": "2026-05-02T01:22:55.694094+00:00",
    "stale_after_misses": 2,
    "confidence_floor": 0.4,
    "decay_per_miss": 0.9,
    "frame_index": 1,
    "known_elements": {
        "tab:frontend": {
            "element_key": "tab:frontend",
            "element_type": "tab",
            "label": "frontend",
            "bbox": [
                58.0,
                93.0,
                261.0,
                129.0
            ],
            "confidence": 0.4089,
            "source": "gemini_vlm",
            "first_seen_frame": null,
            "last_seen_frame": null,
            "first_seen_index": 0,
```

```
    "last_seen_index": 8,  
    "first_seen_at": "2026-05-02T01:22:55.694094+00:00",  
    "last_seen_at": "2026-05-02T01:33:45.283839+00:00",  
    "seen_count": 6,  
    "miss_count": 8,  
    "stale": true,  
    "cached": true  
  },  
  "tab:new tab": {  
    "element_key": "tab:new tab",  
    "element_type": "tab",  
    "label": "New Tab",  
    "bbox": [  
      268.0,  
      93.0,  
      478.0,  
      129.0  
    ],  
    "confidence": 0.4089,  
    "source": "gemini_vlm",  
    "first_seen_frame": null,  
    "last_seen_frame": null,  
    "first_seen_index": 0,  
    "last_seen_index": 8,  
    "first_seen_at": "2026-05-02T01:22:55.694094+00:00",  
    "last_seen_at": "2026-05-02T01:33:45.283839+00:00",  
    "seen_count": 6,  
    "miss_count": 8,  
    "stale": true,
```

```
    "cached": true
  },
  "button:back": {
    "element_key": "button:back",
    "element_type": "button",
    "label": "Back",
    "bbox": [
      124.0,
      122.0,
      155.0,
      153.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 0,
    "last_seen_index": 7,
    "first_seen_at": "2026-05-02T01:22:55.694094+00:00",
    "last_seen_at": "2026-05-02T01:32:56.706370+00:00",
    "seen_count": 2,
    "miss_count": 9,
    "stale": true,
    "cached": true
  },
  "button:new tab": {
    "element_key": "button:new tab",
    "element_type": "button",
    "label": "New tab",
```



```
"bbox": [
  571.0,
  72.0,
  610.0,
  109.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 4,
"last_seen_index": 4,
"first_seen_at": "2026-05-02T01:30:39.307153+00:00",
"last_seen_at": "2026-05-02T01:30:39.307153+00:00",
"seen_count": 1,
"miss_count": 12,
"stale": true,
"cached": true
},
"button:forward": {
  "element_key": "button:forward",
  "element_type": "button",
  "label": "Forward",
  "bbox": [
    167.0,
    122.0,
    199.0,
    153.0
  ],
```

```
"confidence": 0.4,  
"source": "gemini_vlm",  
"first_seen_frame": null,  
"last_seen_frame": null,  
"first_seen_index": 7,  
"last_seen_index": 7,  
"first_seen_at": "2026-05-02T01:32:56.706370+00:00",  
"last_seen_at": "2026-05-02T01:32:56.706370+00:00",  
"seen_count": 1,  
"miss_count": 9,  
"stale": true,  
"cached": true  
},  
"input:address bar": {  
  "element_key": "input:address bar",  
  "element_type": "input",  
  "label": "Address Bar",  
  "bbox": [  
    214.0,  
    108.0,  
    1266.0,  
    166.0  
  ],  
  "confidence": 0.4543,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 9,  
  "last_seen_index": 9,
```

```
"first_seen_at": "2026-05-02T01:34:29.851185+00:00",
"last_seen_at": "2026-05-02T01:34:29.851185+00:00",
"seen_count": 1,
"miss_count": 7,
"stale": true,
"cached": true
},
"icon:screenpilot ai logo": {
  "element_key": "icon:screenpilot ai logo",
  "element_type": "icon",
  "label": "ScreenPilot AI Logo",
  "bbox": [
    127.0,
    173.0,
    185.0,
    209.0
  ],
  "confidence": 0.81,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 9,
  "last_seen_index": 15,
  "first_seen_at": "2026-05-02T01:34:29.851185+00:00",
  "last_seen_at": "2026-05-02T01:39:28.800935+00:00",
  "seen_count": 2,
  "miss_count": 1,
  "stale": false,
  "cached": true
}
```

```
},
"heading:screenpilot ai": {
  "element_key": "heading:screenpilot ai",
  "element_type": "heading",
  "label": "ScreenPilot AI",
  "bbox": [
    127.0,
    186.0,
    286.0,
    241.0
  ],
  "confidence": 0.5048,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 10,
  "last_seen_index": 10,
  "first_seen_at": "2026-05-02T01:35:24.862924+00:00",
  "last_seen_at": "2026-05-02T01:35:24.862924+00:00",
  "seen_count": 1,
  "miss_count": 6,
  "stale": true,
  "cached": true
},
"label:autonomous task agent v2": {
  "element_key": "label:autonomous task agent v2",
  "element_type": "label",
  "label": "Autonomous Task Agent v2",
  "bbox": [
```

```
    305.0,  
    189.0,  
    454.0,  
    221.0  
  ],  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 10,  
  "last_seen_index": 16,  
  "first_seen_at": "2026-05-02T01:35:24.862924+00:00",  
  "last_seen_at": "2026-05-02T01:40:06.446489+00:00",  
  "seen_count": 4,  
  "miss_count": 0,  
  "stale": false,  
  "cached": false  
},  
"pane:task plan": {  
  "element_key": "pane:task plan",  
  "element_type": "pane",  
  "label": "Task Plan",  
  "bbox": [  
    112.0,  
    481.0,  
    723.0,  
    770.0  
  ],  
  "confidence": 0.4783,
```

```
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 10,
"last_seen_index": 10,
"first_seen_at": "2026-05-02T01:35:24.862924+00:00",
"last_seen_at": "2026-05-02T01:35:24.862924+00:00",
"seen_count": 1,
"miss_count": 6,
"stale": true,
"cached": true
},
"heading:task plan": {
  "element_key": "heading:task plan",
  "element_type": "heading",
  "label": "Task Plan",
  "bbox": [
    138.0,
    504.0,
    231.0,
    533.0
  ],
  "confidence": 0.4783,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 10,
  "last_seen_index": 10,
  "first_seen_at": "2026-05-02T01:35:24.862924+00:00",
```

```
    "last_seen_at": "2026-05-02T01:35:24.862924+00:00",
    "seen_count": 1,
    "miss_count": 6,
    "stale": true,
    "cached": true
  },
  "label:1/2 done": {
    "element_key": "label:1/2 done",
    "element_type": "label",
    "label": "1/2 done",
    "bbox": [
      658.0,
      413.0,
      711.0,
      440.0
    ],
    "confidence": 0.9,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 10,
    "last_seen_index": 16,
    "first_seen_at": "2026-05-02T01:35:24.862924+00:00",
    "last_seen_at": "2026-05-02T01:40:06.446489+00:00",
    "seen_count": 2,
    "miss_count": 0,
    "stale": false,
    "cached": false
  },
}
```

```
    "label:the ask gemini button is not visible on the current new tab
screen.": {
      "element_key": "label:the ask gemini button is not visible on the
current new tab screen.",
      "element_type": "label",
      "label": "The Ask Gemini button is not visible on the current New tab
screen.",
      "bbox": [
        127.0,
        547.0,
        709.0,
        605.0
      ],
      "confidence": 0.4783,
      "source": "gemini_vlm",
      "first_seen_frame": null,
      "last_seen_frame": null,
      "first_seen_index": 10,
      "last_seen_index": 10,
      "first_seen_at": "2026-05-02T01:35:24.862924+00:00",
      "last_seen_at": "2026-05-02T01:35:24.862924+00:00",
      "seen_count": 1,
      "miss_count": 6,
      "stale": true,
      "cached": true
    },
    "label:screenpilot ai": {
      "element_key": "label:screenpilot ai",
      "element_type": "label",
      "label": "ScreenPilot AI",
```



```
"bbox": [
  128.0,
  178.0,
  294.0,
  216.0
],
"confidence": 0.95,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 11,
"last_seen_index": 16,
"first_seen_at": "2026-05-02T01:36:08.524451+00:00",
"last_seen_at": "2026-05-02T01:40:06.446489+00:00",
"seen_count": 2,
"miss_count": 0,
"stale": false,
"cached": false
},
"text:screenpilot ai": {
  "element_key": "text:screenpilot ai",
  "element_type": "text",
  "label": "ScreenPilot AI",
  "bbox": [
    188.0,
    187.0,
    354.0,
    223.0
  ],
```

```
"confidence": 0.882,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 12,
"last_seen_index": 15,
"first_seen_at": "2026-05-02T01:36:54.066392+00:00",
"last_seen_at": "2026-05-02T01:39:28.800935+00:00",
"seen_count": 4,
"miss_count": 1,
"stale": false,
"cached": true
},
"icon:screenpilot logo": {
  "element_key": "icon:screenpilot logo",
  "element_type": "icon",
  "label": "ScreenPilot Logo",
  "bbox": [
    152.0,
    187.0,
    180.0,
    223.0
  ],
  "confidence": 0.7695,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 14,
  "last_seen_index": 14,
```

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"first_seen_at": "2026-05-02T01:38:33.621921+00:00",
"last_seen_at": "2026-05-02T01:38:33.621921+00:00",
"seen_count": 1,
"miss_count": 2,
"stale": true,
"cached": true
},
"badge:autonomous task agent v2": {
  "element_key": "badge:autonomous task agent v2",
  "element_type": "badge",
  "label": "Autonomous Task Agent v2",
  "bbox": [
    369.0,
    194.0,
    549.0,
    223.0
  ],
  "confidence": 0.7452,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 14,
  "last_seen_index": 14,
  "first_seen_at": "2026-05-02T01:38:33.621921+00:00",
  "last_seen_at": "2026-05-02T01:38:33.621921+00:00",
  "seen_count": 1,
  "miss_count": 2,
  "stale": true,
  "cached": true
}
```

```
},
"pane:task plan pane": {
  "element_key": "pane:task plan pane",
  "element_type": "pane",
  "label": "Task Plan Pane",
  "bbox": [
    137.0,
    339.0,
    911.0,
    512.0
  ],
  "confidence": 0.729,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 14,
  "last_seen_index": 14,
  "first_seen_at": "2026-05-02T01:38:33.621921+00:00",
  "last_seen_at": "2026-05-02T01:38:33.621921+00:00",
  "seen_count": 1,
  "miss_count": 2,
  "stale": true,
  "cached": true
},
"text:task plan heading": {
  "element_key": "text:task plan heading",
  "element_type": "text",
  "label": "Task Plan Heading",
  "bbox": [
```

```
    166.0,
    353.0,
    289.0,
    382.0
  ],
  "confidence": 0.7695,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 14,
  "last_seen_index": 14,
  "first_seen_at": "2026-05-02T01:38:33.621921+00:00",
  "last_seen_at": "2026-05-02T01:38:33.621921+00:00",
  "seen_count": 1,
  "miss_count": 2,
  "stale": true,
  "cached": true
},
"badge:1/2 done": {
  "element_key": "badge:1/2 done",
  "element_type": "badge",
  "label": "1/2 done",
  "bbox": [
    824.0,
    353.0,
    897.0,
    382.0
  ],
  "confidence": 0.7614,
```

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"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 14,
"last_seen_index": 14,
"first_seen_at": "2026-05-02T01:38:33.621921+00:00",
"last_seen_at": "2026-05-02T01:38:33.621921+00:00",
"seen_count": 1,
"miss_count": 2,
"stale": true,
"cached": true
},
"text:task plan description": {
  "element_key": "text:task plan description",
  "element_type": "text",
  "label": "Task Plan Description",
  "bbox": [
    152.0,
    382.0,
    897.0,
    418.0
  ],
  "confidence": 0.729,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 14,
  "last_seen_index": 14,
  "first_seen_at": "2026-05-02T01:38:33.621921+00:00",
```

```
    "last_seen_at": "2026-05-02T01:38:33.621921+00:00",
    "seen_count": 1,
    "miss_count": 2,
    "stale": true,
    "cached": true
  },
  "item:task step 1": {
    "element_key": "item:task step 1",
    "element_type": "item",
    "label": "Task Step 1",
    "bbox": [
      152.0,
      425.0,
      897.0,
      454.0
    ],
    "confidence": 0.7452,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 14,
    "last_seen_index": 14,
    "first_seen_at": "2026-05-02T01:38:33.621921+00:00",
    "last_seen_at": "2026-05-02T01:38:33.621921+00:00",
    "seen_count": 1,
    "miss_count": 2,
    "stale": true,
    "cached": true
  },
}
```

```
"item:task step 2": {
  "element_key": "item:task step 2",
  "element_type": "item",
  "label": "Task Step 2",
  "bbox": [
    152.0,
    462.0,
    897.0,
    498.0
  ],
  "confidence": 0.7452,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 14,
  "last_seen_index": 14,
  "first_seen_at": "2026-05-02T01:38:33.621921+00:00",
  "last_seen_at": "2026-05-02T01:38:33.621921+00:00",
  "seen_count": 1,
  "miss_count": 2,
  "stale": true,
  "cached": true
},
"pane:log pane": {
  "element_key": "pane:log pane",
  "element_type": "pane",
  "label": "Log Pane",
  "bbox": [
    137.0,
```



```
519.0,  
911.0,  
671.0  
],  
"confidence": 0.7128,  
"source": "gemini_vlm",  
"first_seen_frame": null,  
"last_seen_frame": null,  
"first_seen_index": 14,  
"last_seen_index": 14,  
"first_seen_at": "2026-05-02T01:38:33.621921+00:00",  
"last_seen_at": "2026-05-02T01:38:33.621921+00:00",  
"seen_count": 1,  
"miss_count": 2,  
"stale": true,  
"cached": true  
},  
"text:log entry 1": {  
  "element_key": "text:log entry 1",  
  "element_type": "text",  
  "label": "Log Entry 1",  
  "bbox": [  
    152.0,  
    527.0,  
    839.0,  
    541.0  
  ],  
  "confidence": 0.729,  
  "source": "gemini_vlm",
```

```
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 14,
"last_seen_index": 14,
"first_seen_at": "2026-05-02T01:38:33.621921+00:00",
"last_seen_at": "2026-05-02T01:38:33.621921+00:00",
"seen_count": 1,
"miss_count": 2,
"stale": true,
"cached": true
},
"label:find and click on the ask gemini button": {
  "element_key": "label:find and click on the ask gemini button",
  "element_type": "label",
  "label": "Find and click on the Ask Gemini button",
  "bbox": [
    700.0,
    280.0,
    1053.0,
    381.0
  ],
  "confidence": 0.95,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 16,
  "last_seen_index": 16,
  "first_seen_at": "2026-05-02T01:40:06.446489+00:00",
  "last_seen_at": "2026-05-02T01:40:06.446489+00:00",
```

```
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false
  },
  "pane:task plan container": {
    "element_key": "pane:task plan container",
    "element_type": "pane",
    "label": "Task Plan Container",
    "bbox": [
      112.0,
      392.0,
      722.0,
      606.0
    ],
    "confidence": 0.9,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 16,
    "last_seen_index": 16,
    "first_seen_at": "2026-05-02T01:40:06.446489+00:00",
    "last_seen_at": "2026-05-02T01:40:06.446489+00:00",
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false
  },
  "label:task plan": {
```

```
"element_key": "label:task plan",
"element_type": "label",
"label": "Task Plan",
"bbox": [
  134.0,
  403.0,
  219.0,
  424.0
],
"confidence": 0.9,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 16,
"last_seen_index": 16,
"first_seen_at": "2026-05-02T01:40:06.446489+00:00",
"last_seen_at": "2026-05-02T01:40:06.446489+00:00",
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false
},
"label:task plan description": {
  "element_key": "label:task plan description",
  "element_type": "label",
  "label": "Task Plan Description",
  "bbox": [
    128.0,
    445.0,
```

```
    711.0,  
    488.0  
  ],  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 16,  
  "last_seen_index": 16,  
  "first_seen_at": "2026-05-02T01:40:06.446489+00:00",  
  "last_seen_at": "2026-05-02T01:40:06.446489+00:00",  
  "seen_count": 1,  
  "miss_count": 0,  
  "stale": false,  
  "cached": false  
},  
"label:task step 1": {  
  "element_key": "label:task step 1",  
  "element_type": "label",  
  "label": "Task Step 1",  
  "bbox": [  
    155.0,  
    504.0,  
    615.0,  
    531.0  
  ],  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,
```

```
"last_seen_frame": null,
"first_seen_index": 16,
"last_seen_index": 16,
"first_seen_at": "2026-05-02T01:40:06.446489+00:00",
"last_seen_at": "2026-05-02T01:40:06.446489+00:00",
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false
},
"label:task step 2": {
  "element_key": "label:task step 2",
  "element_type": "label",
  "label": "Task Step 2",
  "bbox": [
    155.0,
    547.0,
    700.0,
    584.0
  ],
  "confidence": 0.9,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 16,
  "last_seen_index": 16,
  "first_seen_at": "2026-05-02T01:40:06.446489+00:00",
  "last_seen_at": "2026-05-02T01:40:06.446489+00:00",
  "seen_count": 1,
```

```
    "miss_count": 0,
    "stale": false,
    "cached": false
  },
  "pane:log container": {
    "element_key": "pane:log container",
    "element_type": "pane",
    "label": "Log Container",
    "bbox": [
      112.0,
      616.0,
      722.0,
      787.0
    ],
    "confidence": 0.9,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 16,
    "last_seen_index": 16,
    "first_seen_at": "2026-05-02T01:40:06.446489+00:00",
    "last_seen_at": "2026-05-02T01:40:06.446489+00:00",
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false
  },
  "label:generating commands log": {
    "element_key": "label:generating commands log",
```

```
"element_type": "label",
"label": "Generating commands log",
"bbox": [
  134.0,
  627.0,
  679.0,
  649.0
],
"confidence": 0.85,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 16,
"last_seen_index": 16,
"first_seen_at": "2026-05-02T01:40:06.446489+00:00",
"last_seen_at": "2026-05-02T01:40:06.446489+00:00",
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false
},
"label:sending hid commands log": {
  "element_key": "label:sending hid commands log",
  "element_type": "label",
  "label": "Sending HID commands log",
  "bbox": [
    134.0,
    654.0,
    283.0,
```



```
        670.0
    ],
    "confidence": 0.85,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 16,
    "last_seen_index": 16,
    "first_seen_at": "2026-05-02T01:40:06.446489+00:00",
    "last_seen_at": "2026-05-02T01:40:06.446489+00:00",
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false
},
"label:validating step 2 log": {
    "element_key": "label:validating step 2 log",
    "element_type": "label",
    "label": "Validating step 2 log",
    "bbox": [
        134.0,
        681.0,
        241.0,
        697.0
    ],
    "confidence": 0.85,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
```

```
"first_seen_index": 16,  
"last_seen_index": 16,  
"first_seen_at": "2026-05-02T01:40:06.446489+00:00",  
"last_seen_at": "2026-05-02T01:40:06.446489+00:00",  
"seen_count": 1,  
"miss_count": 0,  
"stale": false,  
"cached": false  
},  
"label:evaluation error log": {  
  "element_key": "label:evaluation error log",  
  "element_type": "label",  
  "label": "Evaluation error log",  
  "bbox": [  
    134.0,  
    707.0,  
    604.0,  
    723.0  
  ],  
  "confidence": 0.85,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 16,  
  "last_seen_index": 16,  
  "first_seen_at": "2026-05-02T01:40:06.446489+00:00",  
  "last_seen_at": "2026-05-02T01:40:06.446489+00:00",  
  "seen_count": 1,  
  "miss_count": 0,
```

```
    "stale": false,
    "cached": false
  },
  "label:step 2 failed log": {
    "element_key": "label:step 2 failed log",
    "element_type": "label",
    "label": "Step 2 failed log",
    "bbox": [
      134.0,
      729.0,
      540.0,
      750.0
    ],
    "confidence": 0.85,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 16,
    "last_seen_index": 16,
    "first_seen_at": "2026-05-02T01:40:06.446489+00:00",
    "last_seen_at": "2026-05-02T01:40:06.446489+00:00",
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false
  },
  "label:generating final report log": {
    "element_key": "label:generating final report log",
    "element_type": "label",
```

```
"label": "Generating final report log",
"bbox": [
  134.0,
  755.0,
  283.0,
  777.0
],
"confidence": 0.85,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 16,
"last_seen_index": 16,
"first_seen_at": "2026-05-02T01:40:06.446489+00:00",
"last_seen_at": "2026-05-02T01:40:06.446489+00:00",
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false
},
"input:agent status bar": {
  "element_key": "input:agent status bar",
  "element_type": "input",
  "label": "Agent Status Bar",
  "bbox": [
    91.0,
    894.0,
    1064.0,
    959.0
```

```
],
  "confidence": 0.9,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 16,
  "last_seen_index": 16,
  "first_seen_at": "2026-05-02T01:40:06.446489+00:00",
  "last_seen_at": "2026-05-02T01:40:06.446489+00:00",
  "seen_count": 1,
  "miss_count": 0,
  "stale": false,
  "cached": false
},
"label:agent is working...": {
  "element_key": "label:agent is working...",
  "element_type": "label",
  "label": "Agent is working...",
  "bbox": [
    134.0,
    916.0,
    241.0,
    937.0
  ],
  "confidence": 0.9,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 16,
```

```
"last_seen_index": 16,  
"first_seen_at": "2026-05-02T01:40:06.446489+00:00",  
"last_seen_at": "2026-05-02T01:40:06.446489+00:00",  
"seen_count": 1,  
"miss_count": 0,  
"stale": false,  
"cached": false  
},  
"button:send button": {  
  "element_key": "button:send button",  
  "element_type": "button",  
  "label": "Send Button",  
  "bbox": [  
    1021.0,  
    894.0,  
    1069.0,  
    959.0  
  ],  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 16,  
  "last_seen_index": 16,  
  "first_seen_at": "2026-05-02T01:40:06.446489+00:00",  
  "last_seen_at": "2026-05-02T01:40:06.446489+00:00",  
  "seen_count": 1,  
  "miss_count": 0,  
  "stale": false,
```

```
    "cached": false
  },
  "pane:scrollbar": {
    "element_key": "pane:scrollbar",
    "element_type": "pane",
    "label": "Scrollbar",
    "bbox": [
      1075.0,
      167.0,
      1101.0,
      969.0
    ],
    "confidence": 0.85,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 16,
    "last_seen_index": 16,
    "first_seen_at": "2026-05-02T01:40:06.446489+00:00",
    "last_seen_at": "2026-05-02T01:40:06.446489+00:00",
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false
  }
},
"current_elements": [
  {
    "id": "tab_frontend",
```

```
"type": "tab",
"label": "frontend",
"description": "Browser tab for frontend",
"state": "inactive",
"dx": 132,
"dy": 23,
"frame_dx": 247,
"frame_dy": 88,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    161,
    65,
    333,
    111
],
"session_memory_key": "tab:frontend",
"cached": false
},
{
    "id": "tab_new_tab",
    "type": "tab",
    "label": "New tab",
    "description": "Active browser tab",
    "state": "active",
    "dx": 330,
    "dy": 23,
    "frame_dx": 445,
    "frame_dy": 88,
```



```
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    333,
    65,
    557,
    111
],
"session_memory_key": "tab:new tab",
"cached": false
},
{
    "id": "btn_back",
    "type": "button",
    "label": "Back",
    "description": "Go back in history",
    "state": "disabled",
    "dx": 30,
    "dy": 76,
    "frame_dx": 145,
    "frame_dy": 141,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        128,
        118,
        161,
        164
    ],
    ],
```

```
    "session_memory_key": "button:back",
    "cached": false
  }
],
"resolved_elements": [
  {
    "id": "tab_frontend",
    "type": "tab",
    "label": "frontend",
    "description": "Browser tab for frontend",
    "state": "inactive",
    "dx": 132,
    "dy": 23,
    "frame_dx": 247,
    "frame_dy": 88,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
      161,
      65,
      333,
      111
    ],
    "session_memory_key": "tab:frontend",
    "cached": false
  },
  {
    "id": "tab_new_tab",
    "type": "tab",
```

```
"label": "New tab",
"description": "Active browser tab",
"state": "active",
"dx": 330,
"dy": 23,
"frame_dx": 445,
"frame_dy": 88,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    333,
    65,
    557,
    111
],
"session_memory_key": "tab:new tab",
"cached": false
},
{
    "id": "btn_back",
    "type": "button",
    "label": "Back",
    "description": "Go back in history",
    "state": "disabled",
    "dx": 30,
    "dy": 76,
    "frame_dx": 145,
    "frame_dy": 141,
    "confidence": 0.9,
```

```
    "source": "gemini_vlm",
    "frame_bbox": [
        128,
        118,
        161,
        164
    ],
    "session_memory_key": "button:back",
    "cached": false
}
],
"screen_data": {
    "image": "E:\\sliit-
project\\ScreenAwareTaskAgent\\vision\\data\\sessions\\shot_1777684945_f5b814\\pr
ocessing\\proc_1777684945_04ed69\\preprocessed_frames\\frame_1777684945_38c7c4.jp
g",
    "image_size": {
        "width": 1920,
        "height": 1080
    },
    "screen_bbox": [
        115,
        65,
        1805,
        1015
    ],
    "screen_size": {
        "width": 1690,
        "height": 950
    },
    },
```

```
"coordinate_system": "pixel",
"elements": [
  {
    "id": "tab_frontend",
    "type": "tab",
    "label": "frontend",
    "description": "Browser tab for frontend",
    "state": "inactive",
    "dx": 132,
    "dy": 23,
    "frame_dx": 247,
    "frame_dy": 88,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
      161,
      65,
      333,
      111
    ],
    "session_memory_key": "tab:frontend",
    "cached": false
  },
  {
    "id": "tab_new_tab",
    "type": "tab",
    "label": "New tab",
    "description": "Active browser tab",
    "state": "active",
```

```
"dx": 330,  
  "dy": 23,  
  "frame_dx": 445,  
  "frame_dy": 88,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    333,  
    65,  
    557,  
    111  
  ],  
  "session_memory_key": "tab:new tab",  
  "cached": false  
},  
{  
  "id": "btn_back",  
  "type": "button",  
  "label": "Back",  
  "description": "Go back in history",  
  "state": "disabled",  
  "dx": 30,  
  "dy": 76,  
  "frame_dx": 145,  
  "frame_dy": 141,  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    128,
```

```
        118,  
        161,  
        164  
    ],  
    "session_memory_key": "button:back",  
    "cached": false  
}  
],  
"element_count": 3,  
"current_elements": [  
    {  
        "id": "tab_frontend",  
        "type": "tab",  
        "label": "frontend",  
        "description": "Browser tab for frontend",  
        "state": "inactive",  
        "dx": 132,  
        "dy": 23,  
        "frame_dx": 247,  
        "frame_dy": 88,  
        "confidence": 0.95,  
        "source": "gemini_vlm",  
        "frame_bbox": [  
            161,  
            65,  
            333,  
            111  
        ],  
        "session_memory_key": "tab:frontend",
```

```
    "cached": false
  },
  {
    "id": "tab_new_tab",
    "type": "tab",
    "label": "New tab",
    "description": "Active browser tab",
    "state": "active",
    "dx": 330,
    "dy": 23,
    "frame_dx": 445,
    "frame_dy": 88,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
      333,
      65,
      557,
      111
    ],
    "session_memory_key": "tab:new tab",
    "cached": false
  },
  {
    "id": "btn_back",
    "type": "button",
    "label": "Back",
    "description": "Go back in history",
    "state": "disabled",
```



```
    "dx": 30,  
    "dy": 76,  
    "frame_dx": 145,  
    "frame_dy": 141,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        128,  
        118,  
        161,  
        164  
    ],  
    "session_memory_key": "button:back",  
    "cached": false  
}  
],  
"resolved_elements": [  
    {  
        "id": "tab_frontend",  
        "type": "tab",  
        "label": "frontend",  
        "description": "Browser tab for frontend",  
        "state": "inactive",  
        "dx": 132,  
        "dy": 23,  
        "frame_dx": 247,  
        "frame_dy": 88,  
        "confidence": 0.95,  
        "source": "gemini_vlm",
```

```
"frame_bbox": [  
    161,  
    65,  
    333,  
    111  
],  
"session_memory_key": "tab:frontend",  
"cached": false  
},  
{  
    "id": "tab_new_tab",  
    "type": "tab",  
    "label": "New tab",  
    "description": "Active browser tab",  
    "state": "active",  
    "dx": 330,  
    "dy": 23,  
    "frame_dx": 445,  
    "frame_dy": 88,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        333,  
        65,  
        557,  
        111  
    ],  
    "session_memory_key": "tab:new tab",  
    "cached": false
```

```
    },  
    {  
      "id": "btn_back",  
      "type": "button",  
      "label": "Back",  
      "description": "Go back in history",  
      "state": "disabled",  
      "dx": 30,  
      "dy": 76,  
      "frame_dx": 145,  
      "frame_dy": 141,  
      "confidence": 0.9,  
      "source": "gemini_vlm",  
      "frame_bbox": [  
        128,  
        118,  
        161,  
        164  
      ],  
      "session_memory_key": "button:back",  
      "cached": false  
    }  
  ],  
  "session_knowledge_base": {  
    "session_id": "shot_1777684945_f5b814",  
    "started_at": "2026-05-02T01:22:25.865701+00:00",  
    "updated_at": "2026-05-02T01:22:55.644914+00:00",  
    "stale_after_misses": 3,  
    "confidence_floor": 0.4,  
  }
```

```
"decay_per_miss": 0.9,
"frame_count": 1,
"frame_name": "frame_1777684945_38c7c4.jpg",
"frame_index": 0,
"image_size": {
  "width": 1920,
  "height": 1080
},
"known_elements": {
  "tab:frontend": {
    "element_key": "tab:frontend",
    "element_type": "tab",
    "label": "frontend",
    "bbox": [
      161.0,
      65.0,
      333.0,
      111.0
    ],
    "confidence": 0.95,
    "source": "gemini_vlm",
    "first_seen_frame": "frame_1777684945_38c7c4.jpg",
    "last_seen_frame": "frame_1777684945_38c7c4.jpg",
    "first_seen_at": "2026-05-02T01:22:55.644914+00:00",
    "last_seen_at": "2026-05-02T01:22:55.644914+00:00",
    "first_seen_index": 0,
    "last_seen_index": 0,
    "seen_count": 1,
    "miss_count": 0,
```

```
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777684945_38c7c4.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T01:22:55.644914+00:00",
    "confidence": 0.95
  }
],
},
"tab:new tab": {
  "element_key": "tab:new tab",
  "element_type": "tab",
  "label": "New tab",
  "bbox": [
    333.0,
    65.0,
    557.0,
    111.0
  ],
  "confidence": 0.95,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777684945_38c7c4.jpg",
  "last_seen_frame": "frame_1777684945_38c7c4.jpg",
  "first_seen_at": "2026-05-02T01:22:55.644914+00:00",
  "last_seen_at": "2026-05-02T01:22:55.644914+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
```

```
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777684945_38c7c4.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T01:22:55.644914+00:00",
    "confidence": 0.95
  }
]
},
"button:back": {
  "element_key": "button:back",
  "element_type": "button",
  "label": "Back",
  "bbox": [
    128.0,
    118.0,
    161.0,
    164.0
  ],
  "confidence": 0.9,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777684945_38c7c4.jpg",
  "last_seen_frame": "frame_1777684945_38c7c4.jpg",
  "first_seen_at": "2026-05-02T01:22:55.644914+00:00",
  "last_seen_at": "2026-05-02T01:22:55.644914+00:00",
```

```
    "first_seen_index": 0,
    "last_seen_index": 0,
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false,
    "history": [
      {
        "frame_name": "frame_1777684945_38c7c4.jpg",
        "frame_index": 0,
        "seen_at": "2026-05-02T01:22:55.644914+00:00",
        "confidence": 0.9
      }
    ]
  },
  "resolved_elements": [
    {
      "id": "tab_frontend",
      "type": "tab",
      "label": "frontend",
      "description": "Browser tab for frontend",
      "state": "inactive",
      "dx": 132,
      "dy": 23,
      "frame_dx": 247,
      "frame_dy": 88,
      "confidence": 0.95,
      "source": "gemini_vlm",
```

```
"frame_bbox": [
    161,
    65,
    333,
    111
],
"session_memory_key": "tab:frontend",
"cached": false
},
{
    "id": "tab_new_tab",
    "type": "tab",
    "label": "New tab",
    "description": "Active browser tab",
    "state": "active",
    "dx": 330,
    "dy": 23,
    "frame_dx": 445,
    "frame_dy": 88,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        333,
        65,
        557,
        111
    ],
    "session_memory_key": "tab:new tab",
    "cached": false
}
```



```
    },  
    {  
      "id": "btn_back",  
      "type": "button",  
      "label": "Back",  
      "description": "Go back in history",  
      "state": "disabled",  
      "dx": 30,  
      "dy": 76,  
      "frame_dx": 145,  
      "frame_dy": 141,  
      "confidence": 0.9,  
      "source": "gemini_vlm",  
      "frame_bbox": [  
        128,  
        118,  
        161,  
        164  
      ],  
      "session_memory_key": "button:back",  
      "cached": false  
    }  
  ],  
  "current_elements": [  
    {  
      "id": "tab_frontend",  
      "type": "tab",  
      "label": "frontend",  
      "description": "Browser tab for frontend",
```

```
"state": "inactive",
"dx": 132,
"dy": 23,
"frame_dx": 247,
"frame_dy": 88,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    161,
    65,
    333,
    111
],
"session_memory_key": "tab:frontend",
"cached": false
},
{
    "id": "tab_new_tab",
    "type": "tab",
    "label": "New tab",
    "description": "Active browser tab",
    "state": "active",
    "dx": 330,
    "dy": 23,
    "frame_dx": 445,
    "frame_dy": 88,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
```

```
        333,  
        65,  
        557,  
        111  
    ],  
    "session_memory_key": "tab:new tab",  
    "cached": false  
},  
{  
    "id": "btn_back",  
    "type": "button",  
    "label": "Back",  
    "description": "Go back in history",  
    "state": "disabled",  
    "dx": 30,  
    "dy": 76,  
    "frame_dx": 145,  
    "frame_dy": 141,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        128,  
        118,  
        161,  
        164  
    ],  
    "session_memory_key": "button:back",  
    "cached": false  
}
```

```
    ]
  }
}
},
"todo": {
  "steps": [
    {
      "id": 1,
      "action": "Click the 'frontend' tab at (247, 88) to see if the 'Ask Gemini' button is located on that page.",
      "target": "frontend tab",
      "expected_result": "The 'frontend' tab is selected and its content is displayed.",
      "needs_user_data": false,
      "user_data_field": null,
      "status": "done"
    },
    {
      "id": 2,
      "action": "Locate the 'Ask Gemini' button on the updated screen and click it.",
      "target": "Ask Gemini button",
      "expected_result": "The Gemini interface or side panel opens.",
      "needs_user_data": false,
      "user_data_field": null,
      "status": "failed"
    }
  ],
}
```

```
    "notes": "The 'Ask Gemini' button is not visible on the current 'New tab' screen. The first step is to switch to the other open tab to find the button there.",
    "estimated_complexity": "simple"
  },
  "step_plans": [
    {
      "step_id": 1,
      "action": "Click the 'frontend' tab at (247, 88) to see if the 'Ask Gemini' button is located on that page.",
      "target": "frontend tab",
      "expected_result": "The 'frontend' tab is selected and its content is displayed.",
      "hid_commands": [
        {
          "cmd": "mouse_move",
          "dx": 247,
          "dy": 88
        },
        {
          "cmd": "mouse_click",
          "button": "left"
        }
      ],
      "reasoning": "Moving the cursor to the 'frontend' tab at the exact coordinates (247, 88) and clicking to switch to that tab as specified in Step 1."
    },
    {
      "step_id": 2,
      "action": "Locate the 'Ask Gemini' button on the updated screen and click it.",
```

```
"target": "Ask Gemini button",
"expected_result": "The Gemini interface or side panel opens.",
"hid_commands": [
  {
    "cmd": "mouse_move",
    "meta": {
      "commandId": "900c51a1-3224-490d-b695-3b4bf1440a5d"
    },
    "dx": 166,
    "dy": 205,
    "smooth": true
  },
  {
    "cmd": "mouse_click",
    "meta": {
      "commandId": "705ae583-4aef-4989-834b-dfc19fad846a"
    },
    "button": "left"
  }
],
"reasoning": "Fallback generated 2 commands from detected elements"
}
],
"step_results": [
  {
    "step_id": 1,
    "action": "Click the 'frontend' tab at (247, 88) to see if the 'Ask Gemini' button is located on that page.",
    "attempt": 2,
```

```
"exec_status": "success",
    "eval_status": "done",
    "confidence": 0.95,
    "reason": "The 'frontend' tab is now the active browser tab, indicating the click was successful and the page content has loaded.",
    "final_status": "executing",
    "timestamp": "2026-05-02T01:33:49.072426+00:00"
  },
  {
    "step_id": 2,
    "action": "Locate the 'Ask Gemini' button on the updated screen and click it.",
    "attempt": 3,
    "exec_status": "success",
    "eval_status": "retry",
    "confidence": 0.0,
    "reason": "Evaluation error: No valid JSON found in LLM response: '{\\n\\\"status\\\"',
    \"final_status\": \"executing\",
    \"timestamp\": \"2026-05-02T01:39:34.099978+00:00\"
  }
],
  \"final_report\": {
    \"success\": false,
    \"summary\": \"The task was partially completed, but the 'Ask Gemini' button could not be located on the screen after switching tabs.\",
    \"message\": \"The agent successfully navigated to the 'frontend' tab, but the target 'Ask Gemini' button was not detected in the UI elements of the resulting page, preventing the completion of the final step.\\n\\nStep results:\\n\\u2705 Step 1: Click the 'frontend' tab at (247, 88) - Completed\\n\\u274c Step 2: Locate the 'Ask Gemini' button on the updated screen and click it - Failed\",
    \"steps_completed\": 1,
```

```
"steps_failed": 1,
"issues": [
  "The 'Ask Gemini' button was not found in the list of detected UI elements on the current screen.",
  "An evaluation error occurred during the validation of Step 2."
],
"recommendations": [
  "Verify if the 'Ask Gemini' button is located within a different tab or a sub-menu.",
  "Check if the page requires scrolling to bring the button into view.",
  "Ensure the page has fully loaded before the agent attempts to locate the element."
]
},
"session_memory": {
  "run_id": "78226bde-05c5-481a-848f-737d3429e4e9",
  "user_task": "Find and click on the \"Ask Gemini\" button on this browser",
  "started_at": "2026-05-02T01:22:14.203548+00:00",
  "updated_at": "2026-05-02T01:40:06.446489+00:00",
  "stale_after_misses": 2,
  "confidence_floor": 0.4,
  "decay_per_miss": 0.9,
  "frame_index": 17,
  "known_elements": {
    "tab:frontend": {
      "element_key": "tab:frontend",
      "element_type": "tab",
      "label": "frontend",
      "bbox": [
        58.0,
```



```
    93.0,
    261.0,
    129.0
],
"confidence": 0.4089,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 8,
"first_seen_at": "2026-05-02T01:22:55.694094+00:00",
"last_seen_at": "2026-05-02T01:33:45.283839+00:00",
"seen_count": 6,
"miss_count": 8,
"stale": true,
"cached": true
},
"tab:new tab": {
  "element_key": "tab:new tab",
  "element_type": "tab",
  "label": "New Tab",
  "bbox": [
    268.0,
    93.0,
    478.0,
    129.0
  ],
  "confidence": 0.4089,
  "source": "gemini_vlm",
```

```
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 8,
"first_seen_at": "2026-05-02T01:22:55.694094+00:00",
"last_seen_at": "2026-05-02T01:33:45.283839+00:00",
"seen_count": 6,
"miss_count": 8,
"stale": true,
"cached": true
},
"button:back": {
  "element_key": "button:back",
  "element_type": "button",
  "label": "Back",
  "bbox": [
    124.0,
    122.0,
    155.0,
    153.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 7,
  "first_seen_at": "2026-05-02T01:22:55.694094+00:00",
  "last_seen_at": "2026-05-02T01:32:56.706370+00:00",
```

```
    "seen_count": 2,  
    "miss_count": 9,  
    "stale": true,  
    "cached": true  
  },  
  "button:new tab": {  
    "element_key": "button:new tab",  
    "element_type": "button",  
    "label": "New tab",  
    "bbox": [  
      571.0,  
      72.0,  
      610.0,  
      109.0  
    ],  
    "confidence": 0.4,  
    "source": "gemini_vlm",  
    "first_seen_frame": null,  
    "last_seen_frame": null,  
    "first_seen_index": 4,  
    "last_seen_index": 4,  
    "first_seen_at": "2026-05-02T01:30:39.307153+00:00",  
    "last_seen_at": "2026-05-02T01:30:39.307153+00:00",  
    "seen_count": 1,  
    "miss_count": 12,  
    "stale": true,  
    "cached": true  
  },  
  "button:forward": {
```

```
"element_key": "button:forward",
"element_type": "button",
"label": "Forward",
"bbox": [
    167.0,
    122.0,
    199.0,
    153.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 7,
"last_seen_index": 7,
"first_seen_at": "2026-05-02T01:32:56.706370+00:00",
"last_seen_at": "2026-05-02T01:32:56.706370+00:00",
"seen_count": 1,
"miss_count": 9,
"stale": true,
"cached": true
},
"input:address bar": {
    "element_key": "input:address bar",
    "element_type": "input",
    "label": "Address Bar",
    "bbox": [
        214.0,
        108.0,
```

```
    1266.0,  
    166.0  
  ],  
  "confidence": 0.4543,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 9,  
  "last_seen_index": 9,  
  "first_seen_at": "2026-05-02T01:34:29.851185+00:00",  
  "last_seen_at": "2026-05-02T01:34:29.851185+00:00",  
  "seen_count": 1,  
  "miss_count": 7,  
  "stale": true,  
  "cached": true  
},  
"icon:screenpilot ai logo": {  
  "element_key": "icon:screenpilot ai logo",  
  "element_type": "icon",  
  "label": "ScreenPilot AI Logo",  
  "bbox": [  
    127.0,  
    173.0,  
    185.0,  
    209.0  
  ],  
  "confidence": 0.81,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,
```

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"last_seen_frame": null,
"first_seen_index": 9,
"last_seen_index": 15,
"first_seen_at": "2026-05-02T01:34:29.851185+00:00",
"last_seen_at": "2026-05-02T01:39:28.800935+00:00",
"seen_count": 2,
"miss_count": 1,
"stale": false,
"cached": true
},
"heading:screenpilot ai": {
  "element_key": "heading:screenpilot ai",
  "element_type": "heading",
  "label": "ScreenPilot AI",
  "bbox": [
    127.0,
    186.0,
    286.0,
    241.0
  ],
  "confidence": 0.5048,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 10,
  "last_seen_index": 10,
  "first_seen_at": "2026-05-02T01:35:24.862924+00:00",
  "last_seen_at": "2026-05-02T01:35:24.862924+00:00",
  "seen_count": 1,
```

```
    "miss_count": 6,  
    "stale": true,  
    "cached": true  
  },  
  "label:autonomous task agent v2": {  
    "element_key": "label:autonomous task agent v2",  
    "element_type": "label",  
    "label": "Autonomous Task Agent v2",  
    "bbox": [  
      305.0,  
      189.0,  
      454.0,  
      221.0  
    ],  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "first_seen_frame": null,  
    "last_seen_frame": null,  
    "first_seen_index": 10,  
    "last_seen_index": 16,  
    "first_seen_at": "2026-05-02T01:35:24.862924+00:00",  
    "last_seen_at": "2026-05-02T01:40:06.446489+00:00",  
    "seen_count": 4,  
    "miss_count": 0,  
    "stale": false,  
    "cached": false  
  },  
  "pane:task plan": {  
    "element_key": "pane:task plan",
```

```
"element_type": "pane",
"label": "Task Plan",
"bbox": [
    112.0,
    481.0,
    723.0,
    770.0
],
"confidence": 0.4783,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 10,
"last_seen_index": 10,
"first_seen_at": "2026-05-02T01:35:24.862924+00:00",
"last_seen_at": "2026-05-02T01:35:24.862924+00:00",
"seen_count": 1,
"miss_count": 6,
"stale": true,
"cached": true
},
"heading:task plan": {
    "element_key": "heading:task plan",
    "element_type": "heading",
    "label": "Task Plan",
    "bbox": [
        138.0,
        504.0,
        231.0,
```



```
533.0

],
"confidence": 0.4783,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 10,
"last_seen_index": 10,
"first_seen_at": "2026-05-02T01:35:24.862924+00:00",
"last_seen_at": "2026-05-02T01:35:24.862924+00:00",
"seen_count": 1,
"miss_count": 6,
"stale": true,
"cached": true
},
"label:1/2 done": {
  "element_key": "label:1/2 done",
  "element_type": "label",
  "label": "1/2 done",
  "bbox": [
    658.0,
    413.0,
    711.0,
    440.0
  ],
  "confidence": 0.9,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
```

```
    "first_seen_index": 10,
    "last_seen_index": 16,
    "first_seen_at": "2026-05-02T01:35:24.862924+00:00",
    "last_seen_at": "2026-05-02T01:40:06.446489+00:00",
    "seen_count": 2,
    "miss_count": 0,
    "stale": false,
    "cached": false
  },
  "label:the ask gemini button is not visible on the current new tab
screen.": {
    "element_key": "label:the ask gemini button is not visible on the current
new tab screen.",
    "element_type": "label",
    "label": "The Ask Gemini button is not visible on the current New tab
screen.",
    "bbox": [
      127.0,
      547.0,
      709.0,
      605.0
    ],
    "confidence": 0.4783,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 10,
    "last_seen_index": 10,
    "first_seen_at": "2026-05-02T01:35:24.862924+00:00",
    "last_seen_at": "2026-05-02T01:35:24.862924+00:00",
```

```
    "seen_count": 1,
    "miss_count": 6,
    "stale": true,
    "cached": true
  },
  "label:screenpilot ai": {
    "element_key": "label:screenpilot ai",
    "element_type": "label",
    "label": "ScreenPilot AI",
    "bbox": [
      128.0,
      178.0,
      294.0,
      216.0
    ],
    "confidence": 0.95,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 11,
    "last_seen_index": 16,
    "first_seen_at": "2026-05-02T01:36:08.524451+00:00",
    "last_seen_at": "2026-05-02T01:40:06.446489+00:00",
    "seen_count": 2,
    "miss_count": 0,
    "stale": false,
    "cached": false
  },
  "text:screenpilot ai": {
```

```
"element_key": "text:screenpilot ai",
"element_type": "text",
"label": "ScreenPilot AI",
"bbox": [
  188.0,
  187.0,
  354.0,
  223.0
],
"confidence": 0.882,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 12,
"last_seen_index": 15,
"first_seen_at": "2026-05-02T01:36:54.066392+00:00",
"last_seen_at": "2026-05-02T01:39:28.800935+00:00",
"seen_count": 4,
"miss_count": 1,
"stale": false,
"cached": true
},
"icon:screenpilot logo": {
  "element_key": "icon:screenpilot logo",
  "element_type": "icon",
  "label": "ScreenPilot Logo",
  "bbox": [
    152.0,
    187.0,
```

```
180.0,
223.0
],
"confidence": 0.7695,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 14,
"last_seen_index": 14,
"first_seen_at": "2026-05-02T01:38:33.621921+00:00",
"last_seen_at": "2026-05-02T01:38:33.621921+00:00",
"seen_count": 1,
"miss_count": 2,
"stale": true,
"cached": true
},
"badge:autonomous task agent v2": {
  "element_key": "badge:autonomous task agent v2",
  "element_type": "badge",
  "label": "Autonomous Task Agent v2",
  "bbox": [
    369.0,
    194.0,
    549.0,
    223.0
  ],
  "confidence": 0.7452,
  "source": "gemini_vlm",
  "first_seen_frame": null,
```

```
"last_seen_frame": null,
"first_seen_index": 14,
"last_seen_index": 14,
"first_seen_at": "2026-05-02T01:38:33.621921+00:00",
"last_seen_at": "2026-05-02T01:38:33.621921+00:00",
"seen_count": 1,
"miss_count": 2,
"stale": true,
"cached": true
},
"pane:task plan pane": {
  "element_key": "pane:task plan pane",
  "element_type": "pane",
  "label": "Task Plan Pane",
  "bbox": [
    137.0,
    339.0,
    911.0,
    512.0
  ],
  "confidence": 0.729,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 14,
  "last_seen_index": 14,
  "first_seen_at": "2026-05-02T01:38:33.621921+00:00",
  "last_seen_at": "2026-05-02T01:38:33.621921+00:00",
  "seen_count": 1,
```

```
    "miss_count": 2,
    "stale": true,
    "cached": true
  },
  "text:task plan heading": {
    "element_key": "text:task plan heading",
    "element_type": "text",
    "label": "Task Plan Heading",
    "bbox": [
      166.0,
      353.0,
      289.0,
      382.0
    ],
    "confidence": 0.7695,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 14,
    "last_seen_index": 14,
    "first_seen_at": "2026-05-02T01:38:33.621921+00:00",
    "last_seen_at": "2026-05-02T01:38:33.621921+00:00",
    "seen_count": 1,
    "miss_count": 2,
    "stale": true,
    "cached": true
  },
  "badge:1/2 done": {
    "element_key": "badge:1/2 done",
```

```
"element_type": "badge",
"label": "1/2 done",
"bbox": [
    824.0,
    353.0,
    897.0,
    382.0
],
"confidence": 0.7614,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 14,
"last_seen_index": 14,
"first_seen_at": "2026-05-02T01:38:33.621921+00:00",
"last_seen_at": "2026-05-02T01:38:33.621921+00:00",
"seen_count": 1,
"miss_count": 2,
"stale": true,
"cached": true
},
"text:task plan description": {
    "element_key": "text:task plan description",
    "element_type": "text",
    "label": "Task Plan Description",
    "bbox": [
        152.0,
        382.0,
        897.0,
```



```
    418.0
  ],
  "confidence": 0.729,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 14,
  "last_seen_index": 14,
  "first_seen_at": "2026-05-02T01:38:33.621921+00:00",
  "last_seen_at": "2026-05-02T01:38:33.621921+00:00",
  "seen_count": 1,
  "miss_count": 2,
  "stale": true,
  "cached": true
},
"item:task step 1": {
  "element_key": "item:task step 1",
  "element_type": "item",
  "label": "Task Step 1",
  "bbox": [
    152.0,
    425.0,
    897.0,
    454.0
  ],
  "confidence": 0.7452,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
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"first_seen_index": 14,
"last_seen_index": 14,
"first_seen_at": "2026-05-02T01:38:33.621921+00:00",
"last_seen_at": "2026-05-02T01:38:33.621921+00:00",
"seen_count": 1,
"miss_count": 2,
"stale": true,
"cached": true
},
"item:task step 2": {
  "element_key": "item:task step 2",
  "element_type": "item",
  "label": "Task Step 2",
  "bbox": [
    152.0,
    462.0,
    897.0,
    498.0
  ],
  "confidence": 0.7452,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 14,
  "last_seen_index": 14,
  "first_seen_at": "2026-05-02T01:38:33.621921+00:00",
  "last_seen_at": "2026-05-02T01:38:33.621921+00:00",
  "seen_count": 1,
  "miss_count": 2,
```

```
    "stale": true,
    "cached": true
  },
  "pane:log pane": {
    "element_key": "pane:log pane",
    "element_type": "pane",
    "label": "Log Pane",
    "bbox": [
      137.0,
      519.0,
      911.0,
      671.0
    ],
    "confidence": 0.7128,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 14,
    "last_seen_index": 14,
    "first_seen_at": "2026-05-02T01:38:33.621921+00:00",
    "last_seen_at": "2026-05-02T01:38:33.621921+00:00",
    "seen_count": 1,
    "miss_count": 2,
    "stale": true,
    "cached": true
  },
  "text:log entry 1": {
    "element_key": "text:log entry 1",
    "element_type": "text",
```

```
"label": "Log Entry 1",
"bbox": [
  152.0,
  527.0,
  839.0,
  541.0
],
"confidence": 0.729,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 14,
"last_seen_index": 14,
"first_seen_at": "2026-05-02T01:38:33.621921+00:00",
"last_seen_at": "2026-05-02T01:38:33.621921+00:00",
"seen_count": 1,
"miss_count": 2,
"stale": true,
"cached": true
},
"label:find and click on the ask gemini button": {
  "element_key": "label:find and click on the ask gemini button",
  "element_type": "label",
  "label": "Find and click on the Ask Gemini button",
  "bbox": [
    700.0,
    280.0,
    1053.0,
    381.0
  ]
}
```

```
],
  "confidence": 0.95,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 16,
  "last_seen_index": 16,
  "first_seen_at": "2026-05-02T01:40:06.446489+00:00",
  "last_seen_at": "2026-05-02T01:40:06.446489+00:00",
  "seen_count": 1,
  "miss_count": 0,
  "stale": false,
  "cached": false
},
"pane:task plan container": {
  "element_key": "pane:task plan container",
  "element_type": "pane",
  "label": "Task Plan Container",
  "bbox": [
    112.0,
    392.0,
    722.0,
    606.0
  ],
  "confidence": 0.9,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 16,
```

```
"last_seen_index": 16,  
"first_seen_at": "2026-05-02T01:40:06.446489+00:00",  
"last_seen_at": "2026-05-02T01:40:06.446489+00:00",  
"seen_count": 1,  
"miss_count": 0,  
"stale": false,  
"cached": false  
},  
"label:task plan": {  
  "element_key": "label:task plan",  
  "element_type": "label",  
  "label": "Task Plan",  
  "bbox": [  
    134.0,  
    403.0,  
    219.0,  
    424.0  
  ],  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 16,  
  "last_seen_index": 16,  
  "first_seen_at": "2026-05-02T01:40:06.446489+00:00",  
  "last_seen_at": "2026-05-02T01:40:06.446489+00:00",  
  "seen_count": 1,  
  "miss_count": 0,  
  "stale": false,
```

```
    "cached": false
  },
  "label:task plan description": {
    "element_key": "label:task plan description",
    "element_type": "label",
    "label": "Task Plan Description",
    "bbox": [
      128.0,
      445.0,
      711.0,
      488.0
    ],
    "confidence": 0.9,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 16,
    "last_seen_index": 16,
    "first_seen_at": "2026-05-02T01:40:06.446489+00:00",
    "last_seen_at": "2026-05-02T01:40:06.446489+00:00",
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false
  },
  "label:task step 1": {
    "element_key": "label:task step 1",
    "element_type": "label",
    "label": "Task Step 1",
```

```
"bbox": [
  155.0,
  504.0,
  615.0,
  531.0
],
"confidence": 0.9,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 16,
"last_seen_index": 16,
"first_seen_at": "2026-05-02T01:40:06.446489+00:00",
"last_seen_at": "2026-05-02T01:40:06.446489+00:00",
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false
},
"label:task step 2": {
  "element_key": "label:task step 2",
  "element_type": "label",
  "label": "Task Step 2",
  "bbox": [
    155.0,
    547.0,
    700.0,
    584.0
  ],

```



```
"confidence": 0.9,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 16,
"last_seen_index": 16,
"first_seen_at": "2026-05-02T01:40:06.446489+00:00",
"last_seen_at": "2026-05-02T01:40:06.446489+00:00",
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false
},
"pane:log container": {
  "element_key": "pane:log container",
  "element_type": "pane",
  "label": "Log Container",
  "bbox": [
    112.0,
    616.0,
    722.0,
    787.0
  ],
  "confidence": 0.9,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 16,
  "last_seen_index": 16,
```

```
"first_seen_at": "2026-05-02T01:40:06.446489+00:00",
"last_seen_at": "2026-05-02T01:40:06.446489+00:00",
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false
},
"label:generating commands log": {
  "element_key": "label:generating commands log",
  "element_type": "label",
  "label": "Generating commands log",
  "bbox": [
    134.0,
    627.0,
    679.0,
    649.0
  ],
  "confidence": 0.85,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 16,
  "last_seen_index": 16,
  "first_seen_at": "2026-05-02T01:40:06.446489+00:00",
  "last_seen_at": "2026-05-02T01:40:06.446489+00:00",
  "seen_count": 1,
  "miss_count": 0,
  "stale": false,
  "cached": false
}
```

```
},
"label:sending hid commands log": {
  "element_key": "label:sending hid commands log",
  "element_type": "label",
  "label": "Sending HID commands log",
  "bbox": [
    134.0,
    654.0,
    283.0,
    670.0
  ],
  "confidence": 0.85,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 16,
  "last_seen_index": 16,
  "first_seen_at": "2026-05-02T01:40:06.446489+00:00",
  "last_seen_at": "2026-05-02T01:40:06.446489+00:00",
  "seen_count": 1,
  "miss_count": 0,
  "stale": false,
  "cached": false
},
"label:validating step 2 log": {
  "element_key": "label:validating step 2 log",
  "element_type": "label",
  "label": "Validating step 2 log",
  "bbox": [
```

```
    134.0,  
    681.0,  
    241.0,  
    697.0  
  ],  
  "confidence": 0.85,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 16,  
  "last_seen_index": 16,  
  "first_seen_at": "2026-05-02T01:40:06.446489+00:00",  
  "last_seen_at": "2026-05-02T01:40:06.446489+00:00",  
  "seen_count": 1,  
  "miss_count": 0,  
  "stale": false,  
  "cached": false  
},  
"label:evaluation error log": {  
  "element_key": "label:evaluation error log",  
  "element_type": "label",  
  "label": "Evaluation error log",  
  "bbox": [  
    134.0,  
    707.0,  
    604.0,  
    723.0  
  ],  
  "confidence": 0.85,
```

```
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 16,
"last_seen_index": 16,
"first_seen_at": "2026-05-02T01:40:06.446489+00:00",
"last_seen_at": "2026-05-02T01:40:06.446489+00:00",
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false
},
"label:step 2 failed log": {
  "element_key": "label:step 2 failed log",
  "element_type": "label",
  "label": "Step 2 failed log",
  "bbox": [
    134.0,
    729.0,
    540.0,
    750.0
  ],
  "confidence": 0.85,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 16,
  "last_seen_index": 16,
  "first_seen_at": "2026-05-02T01:40:06.446489+00:00",
```

```
    "last_seen_at": "2026-05-02T01:40:06.446489+00:00",
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false
  },
  "label:generating final report log": {
    "element_key": "label:generating final report log",
    "element_type": "label",
    "label": "Generating final report log",
    "bbox": [
      134.0,
      755.0,
      283.0,
      777.0
    ],
    "confidence": 0.85,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 16,
    "last_seen_index": 16,
    "first_seen_at": "2026-05-02T01:40:06.446489+00:00",
    "last_seen_at": "2026-05-02T01:40:06.446489+00:00",
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false
  },
}
```

```
"input:agent status bar": {
  "element_key": "input:agent status bar",
  "element_type": "input",
  "label": "Agent Status Bar",
  "bbox": [
    91.0,
    894.0,
    1064.0,
    959.0
  ],
  "confidence": 0.9,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 16,
  "last_seen_index": 16,
  "first_seen_at": "2026-05-02T01:40:06.446489+00:00",
  "last_seen_at": "2026-05-02T01:40:06.446489+00:00",
  "seen_count": 1,
  "miss_count": 0,
  "stale": false,
  "cached": false
},
"label:agent is working...": {
  "element_key": "label:agent is working...",
  "element_type": "label",
  "label": "Agent is working...",
  "bbox": [
    134.0,
```

```
    916.0,  
    241.0,  
    937.0  
  ],  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 16,  
  "last_seen_index": 16,  
  "first_seen_at": "2026-05-02T01:40:06.446489+00:00",  
  "last_seen_at": "2026-05-02T01:40:06.446489+00:00",  
  "seen_count": 1,  
  "miss_count": 0,  
  "stale": false,  
  "cached": false  
},  
"button:send button": {  
  "element_key": "button:send button",  
  "element_type": "button",  
  "label": "Send Button",  
  "bbox": [  
    1021.0,  
    894.0,  
    1069.0,  
    959.0  
  ],  
  "confidence": 0.95,  
  "source": "gemini_vlm",
```



```
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 16,
"last_seen_index": 16,
"first_seen_at": "2026-05-02T01:40:06.446489+00:00",
"last_seen_at": "2026-05-02T01:40:06.446489+00:00",
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false
},
"pane:scrollbar": {
  "element_key": "pane:scrollbar",
  "element_type": "pane",
  "label": "Scrollbar",
  "bbox": [
    1075.0,
    167.0,
    1101.0,
    969.0
  ],
  "confidence": 0.85,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 16,
  "last_seen_index": 16,
  "first_seen_at": "2026-05-02T01:40:06.446489+00:00",
  "last_seen_at": "2026-05-02T01:40:06.446489+00:00",
```

```
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false
  }
},
"current_elements": [],
"resolved_elements": [
  {
    "id": "memory_icon:screenpilot ai logo",
    "type": "icon",
    "label": "ScreenPilot AI Logo",
    "description": "Cached from an earlier frame in the same run",
    "state": "cached",
    "bbox": [
      127.0,
      173.0,
      185.0,
      209.0
    ],
    "confidence": 0.81,
    "source": "gemini_vlm",
    "last_seen_frame": null,
    "last_seen_index": 15,
    "stale": true,
    "cached": true,
    "session_memory_key": "icon:screenpilot ai logo"
  },
  {
```

```
"id": "memory_label:autonomous task agent v2",
"type": "label",
"label": "Autonomous Task Agent v2",
"description": "Cached from an earlier frame in the same run",
"state": "cached",
"bbox": [
    305.0,
    189.0,
    454.0,
    221.0
],
"confidence": 0.9,
"source": "gemini_vlm",
"last_seen_frame": null,
"last_seen_index": 16,
"stale": true,
"cached": true,
"session_memory_key": "label:autonomous task agent v2"
},
{
    "id": "memory_label:1/2 done",
    "type": "label",
    "label": "1/2 done",
    "description": "Cached from an earlier frame in the same run",
    "state": "cached",
    "bbox": [
        658.0,
        413.0,
        711.0,
```

```
440.0

],
"confidence": 0.9,
"source": "gemini_vlm",
"last_seen_frame": null,
"last_seen_index": 16,
"stale": true,
"cached": true,
"session_memory_key": "label:1/2 done"
},
{
  "id": "memory_label:screenpilot ai",
  "type": "label",
  "label": "ScreenPilot AI",
  "description": "Cached from an earlier frame in the same run",
  "state": "cached",
  "bbox": [
    128.0,
    178.0,
    294.0,
    216.0
  ],
  "confidence": 0.95,
  "source": "gemini_vlm",
  "last_seen_frame": null,
  "last_seen_index": 16,
  "stale": true,
  "cached": true,
  "session_memory_key": "label:screenpilot ai"
```

```
},
{
  "id": "memory_text:screenpilot ai",
  "type": "text",
  "label": "ScreenPilot AI",
  "description": "Cached from an earlier frame in the same run",
  "state": "cached",
  "bbox": [
    188.0,
    187.0,
    354.0,
    223.0
  ],
  "confidence": 0.882,
  "source": "gemini_vlm",
  "last_seen_frame": null,
  "last_seen_index": 15,
  "stale": true,
  "cached": true,
  "session_memory_key": "text:screenpilot ai"
},
{
  "id": "memory_label:find and click on the ask gemini button",
  "type": "label",
  "label": "Find and click on the Ask Gemini button",
  "description": "Cached from an earlier frame in the same run",
  "state": "cached",
  "bbox": [
    700.0,
```

```
    280.0,  
    1053.0,  
    381.0  
  ],  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "last_seen_frame": null,  
  "last_seen_index": 16,  
  "stale": true,  
  "cached": true,  
  "session_memory_key": "label:find and click on the ask gemini button"  
},  
{  
  "id": "memory_pane:task plan container",  
  "type": "pane",  
  "label": "Task Plan Container",  
  "description": "Cached from an earlier frame in the same run",  
  "state": "cached",  
  "bbox": [  
    112.0,  
    392.0,  
    722.0,  
    606.0  
  ],  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "last_seen_frame": null,  
  "last_seen_index": 16,  
  "stale": true,
```

```
"cached": true,
"session_memory_key": "pane:task plan container"
},
{
  "id": "memory_label:task plan",
  "type": "label",
  "label": "Task Plan",
  "description": "Cached from an earlier frame in the same run",
  "state": "cached",
  "bbox": [
    134.0,
    403.0,
    219.0,
    424.0
  ],
  "confidence": 0.9,
  "source": "gemini_vlm",
  "last_seen_frame": null,
  "last_seen_index": 16,
  "stale": true,
  "cached": true,
  "session_memory_key": "label:task plan"
},
{
  "id": "memory_label:task plan description",
  "type": "label",
  "label": "Task Plan Description",
  "description": "Cached from an earlier frame in the same run",
  "state": "cached",
```

```
"bbox": [
  128.0,
  445.0,
  711.0,
  488.0
],
"confidence": 0.9,
"source": "gemini_vlm",
"last_seen_frame": null,
"last_seen_index": 16,
"stale": true,
"cached": true,
"session_memory_key": "label:task plan description"
},
{
  "id": "memory_label:task step 1",
  "type": "label",
  "label": "Task Step 1",
  "description": "Cached from an earlier frame in the same run",
  "state": "cached",
  "bbox": [
    155.0,
    504.0,
    615.0,
    531.0
  ],
  "confidence": 0.9,
  "source": "gemini_vlm",
  "last_seen_frame": null,
```



```
"last_seen_index": 16,
"stale": true,
"cached": true,
"session_memory_key": "label:task step 1"
},
{
  "id": "memory_label:task step 2",
  "type": "label",
  "label": "Task Step 2",
  "description": "Cached from an earlier frame in the same run",
  "state": "cached",
  "bbox": [
    155.0,
    547.0,
    700.0,
    584.0
  ],
  "confidence": 0.9,
  "source": "gemini_vlm",
  "last_seen_frame": null,
  "last_seen_index": 16,
  "stale": true,
  "cached": true,
  "session_memory_key": "label:task step 2"
},
{
  "id": "memory_pane:log container",
  "type": "pane",
  "label": "Log Container",
```

```
"description": "Cached from an earlier frame in the same run",
"state": "cached",
"bbox": [
  112.0,
  616.0,
  722.0,
  787.0
],
"confidence": 0.9,
"source": "gemini_vlm",
"last_seen_frame": null,
"last_seen_index": 16,
"stale": true,
"cached": true,
"session_memory_key": "pane:log container"
},
{
  "id": "memory_label:generating commands log",
  "type": "label",
  "label": "Generating commands log",
  "description": "Cached from an earlier frame in the same run",
  "state": "cached",
  "bbox": [
    134.0,
    627.0,
    679.0,
    649.0
  ],
  "confidence": 0.85,
```

```
"source": "gemini_vlm",
"last_seen_frame": null,
"last_seen_index": 16,
"stale": true,
"cached": true,
"session_memory_key": "label:generating commands log"
},
{
  "id": "memory_label:sending hid commands log",
  "type": "label",
  "label": "Sending HID commands log",
  "description": "Cached from an earlier frame in the same run",
  "state": "cached",
  "bbox": [
    134.0,
    654.0,
    283.0,
    670.0
  ],
  "confidence": 0.85,
  "source": "gemini_vlm",
  "last_seen_frame": null,
  "last_seen_index": 16,
  "stale": true,
  "cached": true,
  "session_memory_key": "label:sending hid commands log"
},
{
  "id": "memory_label:validating step 2 log",
```

```
"type": "label",
"label": "Validating step 2 log",
"description": "Cached from an earlier frame in the same run",
"state": "cached",
"bbox": [
  134.0,
  681.0,
  241.0,
  697.0
],
"confidence": 0.85,
"source": "gemini_vlm",
"last_seen_frame": null,
"last_seen_index": 16,
"stale": true,
"cached": true,
"session_memory_key": "label:validating step 2 log"
},
{
  "id": "memory_label:evaluation error log",
  "type": "label",
  "label": "Evaluation error log",
  "description": "Cached from an earlier frame in the same run",
  "state": "cached",
  "bbox": [
    134.0,
    707.0,
    604.0,
    723.0
  ]
}
```

```
],
  "confidence": 0.85,
  "source": "gemini_vlm",
  "last_seen_frame": null,
  "last_seen_index": 16,
  "stale": true,
  "cached": true,
  "session_memory_key": "label:evaluation error log"
},
{
  "id": "memory_label:step 2 failed log",
  "type": "label",
  "label": "Step 2 failed log",
  "description": "Cached from an earlier frame in the same run",
  "state": "cached",
  "bbox": [
    134.0,
    729.0,
    540.0,
    750.0
  ],
  "confidence": 0.85,
  "source": "gemini_vlm",
  "last_seen_frame": null,
  "last_seen_index": 16,
  "stale": true,
  "cached": true,
  "session_memory_key": "label:step 2 failed log"
},
```

```
{
  "id": "memory_label:generating final report log",
  "type": "label",
  "label": "Generating final report log",
  "description": "Cached from an earlier frame in the same run",
  "state": "cached",
  "bbox": [
    134.0,
    755.0,
    283.0,
    777.0
  ],
  "confidence": 0.85,
  "source": "gemini_vlm",
  "last_seen_frame": null,
  "last_seen_index": 16,
  "stale": true,
  "cached": true,
  "session_memory_key": "label:generating final report log"
},
{
  "id": "memory_input:agent status bar",
  "type": "input",
  "label": "Agent Status Bar",
  "description": "Cached from an earlier frame in the same run",
  "state": "cached",
  "bbox": [
    91.0,
    894.0,
```

```
    1064.0,  
    959.0  
  ],  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "last_seen_frame": null,  
  "last_seen_index": 16,  
  "stale": true,  
  "cached": true,  
  "session_memory_key": "input:agent status bar"  
},  
{  
  "id": "memory_label:agent is working...",  
  "type": "label",  
  "label": "Agent is working...",  
  "description": "Cached from an earlier frame in the same run",  
  "state": "cached",  
  "bbox": [  
    134.0,  
    916.0,  
    241.0,  
    937.0  
  ],  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "last_seen_frame": null,  
  "last_seen_index": 16,  
  "stale": true,  
  "cached": true,
```

```
"session_memory_key": "label:agent is working..."
},
{
  "id": "memory_button:send button",
  "type": "button",
  "label": "Send Button",
  "description": "Cached from an earlier frame in the same run",
  "state": "cached",
  "bbox": [
    1021.0,
    894.0,
    1069.0,
    959.0
  ],
  "confidence": 0.95,
  "source": "gemini_vlm",
  "last_seen_frame": null,
  "last_seen_index": 16,
  "stale": true,
  "cached": true,
  "session_memory_key": "button:send button"
},
{
  "id": "memory_pane:scrollbar",
  "type": "pane",
  "label": "Scrollbar",
  "description": "Cached from an earlier frame in the same run",
  "state": "cached",
  "bbox": [
```



```
        1075.0,  
        167.0,  
        1101.0,  
        969.0  
    ],  
    "confidence": 0.85,  
    "source": "gemini_vlm",  
    "last_seen_frame": null,  
    "last_seen_index": 16,  
    "stale": true,  
    "cached": true,  
    "session_memory_key": "pane:scrollbar"  
  }  
],  
"screen_data": {}  
}  
}
```

Test Case 05: Screen Tasks (Complex Task)

User Prompt:

Minimize the current window using minimize icon and find the visual studio code application icon on the screen and click on that app icon

Test Results:

- **Steps Total:** 2
- **Completed:** 1
- **Failed:** 1
- **Duration:** 12 minutes 52 seconds

Step Execution:

Step 1: Move the mouse to the 'Minimize' button at (1285, 187) and click it

- **Target:** Minimize button
- **Attempts:** 3
- **Execution:** Performed
- **Evaluation:** Success
- **Issue:** None

Step 2: Identify the Visual Studio Code icon on the taskbar or desktop and click it

- **Target:** Visual Studio Code application icon
- **Attempts:** 3
- **Execution:** Performed
- **Evaluation:** Failed (retry loop)
- **Issue:** Visual Studio Code icon not detected in screen UI elements after window minimization

Execution Summary:

The agent successfully minimized the browser window but failed to locate the Visual Studio Code icon on the desktop or taskbar. The icon was not present in the detected UI elements, causing the step to fail after multiple retry attempts.

Issues Identified:

- Visual Studio Code icon not visible in the final screen state
- No fallback mechanism to search via Start menu or taskbar search
- Agent assumed icon would be immediately visible after minimization
- Possible that VS Code is not pinned to taskbar or desktop
- Stale cached elements may have interfered with fresh detection after window state change

Recommendations for Improvement:

- Add fallback logic to use Start menu search if target icon not found on desktop/taskbar
- Implement scrolling or taskbar expansion (show hidden icons) before declaring element missing
- Improve post-minimization screen analysis with longer wait for UI stabilization
- Add validation for application presence before attempting click
- Cache negative results temporarily to avoid repeated same-step failures

JSON Response:

full_cycle.json

```
{
  "run_id": "85f20837-9a8e-4c22-9298-4e2401f6166c",
```

```
"user_task": "Minimize the current window using minimize icon and find the  
visual studio code application icon on the screen and click on that app icon",  
  
"started_at": "2026-05-02T03:57:53.860689+00:00",  
  
"finished_at": "2026-05-02T04:10:45.070446+00:00",  
  
"vision_used": true,  
  
"perception": {  
  "image": "E:\\sliit-  
project\\ScreenAwareTaskAgent\\vision\\data\\sessions\\shot_1777694279_7138a4\\pr  
ocessing\\proc_1777694279_a5eace\\preprocessed_frames\\frame_1777694279_3c3018.jp  
g",  
  "image_size": {  
    "width": 1920,  
    "height": 1080  
  },  
  "screen_bbox": [  
    157,  
    159,  
    1717,  
    959  
  ],  
  "screen_size": {  
    "width": 1560,  
    "height": 800  
  },  
  "coordinate_system": "pixel",  
  "elements": [  
    {  
      "id": "browser_back",  
      "type": "button",  
      "label": "Back",
```

```
"description": "Navigate back in browser history",
"state": "enabled",
"dx": 36,
"dy": 48,
"frame_dx": 193,
"frame_dy": 207,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    173,
    183,
    213,
    231
],
"session_memory_key": "button:back",
"cached": false
},
{
    "id": "browser_forward",
    "type": "button",
    "label": "Forward",
    "description": "Navigate forward in browser history",
    "state": "disabled",
    "dx": 72,
    "dy": 48,
    "frame_dx": 229,
    "frame_dy": 207,
    "confidence": 0.95,
    "source": "gemini_vlm",
```

```
"frame_bbox": [  
    205,  
    183,  
    253,  
    231  
],  
"session_memory_key": "button:forward",  
"cached": false  
},  
{  
    "id": "browser_refresh",  
    "type": "button",  
    "label": "Refresh",  
    "description": "Reload the current page",  
    "state": "enabled",  
    "dx": 104,  
    "dy": 48,  
    "frame_dx": 261,  
    "frame_dy": 207,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        237,  
        183,  
        285,  
        231  
    ],  
    "session_memory_key": "button:refresh",  
    "cached": false
```

```
},  
{  
  "id": "address_bar",  
  "type": "input",  
  "label": "Address bar",  
  "description": "Browser address and search bar showing localhost:5173",  
  "state": "enabled",  
  "dx": 568,  
  "dy": 48,  
  "frame_dx": 725,  
  "frame_dy": 207,  
  "confidence": 0.98,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    157,  
    175,  
    1293,  
    239  
  ],  
  "session_memory_key": "input:address bar",  
  "cached": false  
},  
{  
  "id": "window_minimize",  
  "type": "button",  
  "label": "Minimize",  
  "description": "Minimize the browser window",  
  "state": "enabled",  
  "dx": 1128,
```

```
"dy": 28,  
"frame_dx": 1285,  
"frame_dy": 187,  
"confidence": 0.9,  
"source": "gemini_vlm",  
"frame_bbox": [  
    1261,  
    167,  
    1309,  
    207  
],  
"session_memory_key": "button:minimize",  
"cached": false  
},  
{  
    "id": "window_maximize",  
    "type": "button",  
    "label": "Maximize",  
    "description": "Maximize the browser window",  
    "state": "enabled",  
    "dx": 1156,  
    "dy": 32,  
    "frame_dx": 1313,  
    "frame_dy": 191,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        1293,  
        175,
```

```
    1333,  
    207  
  ],  
  "session_memory_key": "button:maximize",  
  "cached": false  
},  
{  
  "id": "window_close",  
  "type": "button",  
  "label": "Close",  
  "description": "Close the browser window",  
  "state": "enabled",  
  "dx": 1184,  
  "dy": 24,  
  "frame_dx": 1341,  
  "frame_dy": 183,  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    1317,  
    167,  
    1365,  
    199  
  ],  
  "session_memory_key": "button:close",  
  "cached": false  
},  
{  
  "id": "app_logo",
```



```
"type": "icon",
"label": "ScreenPilot AI Logo",
"description": "Logo for the ScreenPilot AI application",
"state": "visible",
"dx": 48,
"dy": 140,
"frame_dx": 205,
"frame_dy": 299,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    189,
    279,
    221,
    319
],
"session_memory_key": "icon:screenpilot ai logo",
"cached": false
},
{
    "id": "app_title",
    "type": "heading",
    "label": "ScreenPilot AI",
    "description": "Main title of the application",
    "state": "visible",
    "dx": 136,
    "dy": 136,
    "frame_dx": 293,
    "frame_dy": 295,
```

```
"confidence": 0.98,
"source": "gemini_vlm",
"frame_bbox": [
    213,
    279,
    373,
    311
],
"session_memory_key": "heading:screenpilot ai",
"cached": false
},
{
    "id": "app_version_badge",
    "type": "label",
    "label": "Autonomous Task Agent v2",
    "description": "Badge indicating the application version",
    "state": "visible",
    "dx": 248,
    "dy": 144,
    "frame_dx": 405,
    "frame_dy": 303,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        365,
        295,
        445,
        311
    ],
    ],
```

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"session_memory_key": "label:autonomous task agent v2",
"cached": false
},
{
  "id": "status_connecting",
  "type": "label",
  "label": "Connecting to agent...",
  "description": "Status message indicating connection progress",
  "state": "visible",
  "dx": 44,
  "dy": 416,
  "frame_dx": 201,
  "frame_dy": 575,
  "confidence": 0.9,
  "source": "gemini_vlm",
  "frame_bbox": [
    181,
    559,
    221,
    591
  ],
  "session_memory_key": "label:connecting to agent...",
  "cached": false
},
{
  "id": "status_capturing",
  "type": "label",
  "label": "Capturing initial screen...",
  "description": "Status message indicating screen capture progress",
```

```
"state": "visible",
"dx": 136,
"dy": 448,
"frame_dx": 293,
"frame_dy": 607,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
  181,
  591,
  405,
  623
],
"session_memory_key": "label:capturing initial screen...",
"cached": false
},
{
  "id": "chat_input_field",
  "type": "input",
  "label": "Agent is working...",
  "description": "Chat input field with placeholder text",
  "state": "disabled",
  "dx": 672,
  "dy": 744,
  "frame_dx": 829,
  "frame_dy": 903,
  "confidence": 0.95,
  "source": "gemini_vlm",
  "frame_bbox": [
```

```
    157,  
    863,  
    1501,  
    943  
  ],  
  "session_memory_key": "input:agent is working...",  
  "cached": false  
},  
{  
  "id": "send_button",  
  "type": "button",  
  "label": "Send",  
  "description": "Button to send a message to the agent",  
  "state": "enabled",  
  "dx": 1144,  
  "dy": 736,  
  "frame_dx": 1301,  
  "frame_dy": 895,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    1269,  
    863,  
    1333,  
    927  
  ],  
  "session_memory_key": "button:send",  
  "cached": false  
},
```

```
{
  "id": "windows_taskbar",
  "type": "pane",
  "label": "Taskbar",
  "description": "Windows taskbar at the bottom of the screen",
  "state": "visible",
  "dx": 780,
  "dy": 784,
  "frame_dx": 937,
  "frame_dy": 943,
  "confidence": 0.9,
  "source": "gemini_vlm",
  "frame_bbox": [
    157,
    927,
    1717,
    959
  ],
  "session_memory_key": "pane:taskbar",
  "cached": false
}
],
"element_count": 15,
"current_elements": [
  {
    "id": "browser_back",
    "type": "button",
    "label": "Back",
    "description": "Navigate back in browser history",
```

```
"state": "enabled",
"dx": 36,
"dy": 48,
"frame_dx": 193,
"frame_dy": 207,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    173,
    183,
    213,
    231
],
"session_memory_key": "button:back",
"cached": false
},
{
    "id": "browser_forward",
    "type": "button",
    "label": "Forward",
    "description": "Navigate forward in browser history",
    "state": "disabled",
    "dx": 72,
    "dy": 48,
    "frame_dx": 229,
    "frame_dy": 207,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
```

```
    205,  
    183,  
    253,  
    231  
  ],  
  "session_memory_key": "button:forward",  
  "cached": false  
},  
{  
  "id": "browser_refresh",  
  "type": "button",  
  "label": "Refresh",  
  "description": "Reload the current page",  
  "state": "enabled",  
  "dx": 104,  
  "dy": 48,  
  "frame_dx": 261,  
  "frame_dy": 207,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    237,  
    183,  
    285,  
    231  
  ],  
  "session_memory_key": "button:refresh",  
  "cached": false  
},
```



```
{
  "id": "address_bar",
  "type": "input",
  "label": "Address bar",
  "description": "Browser address and search bar showing localhost:5173",
  "state": "enabled",
  "dx": 568,
  "dy": 48,
  "frame_dx": 725,
  "frame_dy": 207,
  "confidence": 0.98,
  "source": "gemini_vlm",
  "frame_bbox": [
    157,
    175,
    1293,
    239
  ],
  "session_memory_key": "input:address bar",
  "cached": false
},
{
  "id": "window_minimize",
  "type": "button",
  "label": "Minimize",
  "description": "Minimize the browser window",
  "state": "enabled",
  "dx": 1128,
  "dy": 28,
```

```
"frame_dx": 1285,  
"frame_dy": 187,  
"confidence": 0.9,  
"source": "gemini_vlm",  
"frame_bbox": [  
    1261,  
    167,  
    1309,  
    207  
],  
"session_memory_key": "button:minimize",  
"cached": false  
},  
{  
    "id": "window_maximize",  
    "type": "button",  
    "label": "Maximize",  
    "description": "Maximize the browser window",  
    "state": "enabled",  
    "dx": 1156,  
    "dy": 32,  
    "frame_dx": 1313,  
    "frame_dy": 191,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        1293,  
        175,  
        1333,
```

207

```
    ],  
    "session_memory_key": "button:maximize",  
    "cached": false  
  },  
  {  
    "id": "window_close",  
    "type": "button",  
    "label": "Close",  
    "description": "Close the browser window",  
    "state": "enabled",  
    "dx": 1184,  
    "dy": 24,  
    "frame_dx": 1341,  
    "frame_dy": 183,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
      1317,  
      167,  
      1365,  
      199  
    ],  
    "session_memory_key": "button:close",  
    "cached": false  
  },  
  {  
    "id": "app_logo",  
    "type": "icon",
```

```
"label": "ScreenPilot AI Logo",
"description": "Logo for the ScreenPilot AI application",
"state": "visible",
"dx": 48,
"dy": 140,
"frame_dx": 205,
"frame_dy": 299,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    189,
    279,
    221,
    319
],
"session_memory_key": "icon:screenpilot ai logo",
"cached": false
},
{
    "id": "app_title",
    "type": "heading",
    "label": "ScreenPilot AI",
    "description": "Main title of the application",
    "state": "visible",
    "dx": 136,
    "dy": 136,
    "frame_dx": 293,
    "frame_dy": 295,
    "confidence": 0.98,
```

```
"source": "gemini_vlm",
"frame_bbox": [
    213,
    279,
    373,
    311
],
"session_memory_key": "heading:screenpilot ai",
"cached": false
},
{
    "id": "app_version_badge",
    "type": "label",
    "label": "Autonomous Task Agent v2",
    "description": "Badge indicating the application version",
    "state": "visible",
    "dx": 248,
    "dy": 144,
    "frame_dx": 405,
    "frame_dy": 303,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        365,
        295,
        445,
        311
    ],
    "session_memory_key": "label:autonomous task agent v2",
```

```
"cached": false
},
{
  "id": "status_connecting",
  "type": "label",
  "label": "Connecting to agent...",
  "description": "Status message indicating connection progress",
  "state": "visible",
  "dx": 44,
  "dy": 416,
  "frame_dx": 201,
  "frame_dy": 575,
  "confidence": 0.9,
  "source": "gemini_vlm",
  "frame_bbox": [
    181,
    559,
    221,
    591
  ],
  "session_memory_key": "label:connecting to agent...",
  "cached": false
},
{
  "id": "status_capturing",
  "type": "label",
  "label": "Capturing initial screen...",
  "description": "Status message indicating screen capture progress",
  "state": "visible",
```

```
"dx": 136,  
"dy": 448,  
"frame_dx": 293,  
"frame_dy": 607,  
"confidence": 0.9,  
"source": "gemini_vlm",  
"frame_bbox": [  
    181,  
    591,  
    405,  
    623  
],  
"session_memory_key": "label:capturing initial screen...",  
"cached": false  
},  
{  
    "id": "chat_input_field",  
    "type": "input",  
    "label": "Agent is working...",  
    "description": "Chat input field with placeholder text",  
    "state": "disabled",  
    "dx": 672,  
    "dy": 744,  
    "frame_dx": 829,  
    "frame_dy": 903,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        157,
```

```
      863,  
      1501,  
      943  
    ],  
    "session_memory_key": "input:agent is working...",  
    "cached": false  
  },  
  {  
    "id": "send_button",  
    "type": "button",  
    "label": "Send",  
    "description": "Button to send a message to the agent",  
    "state": "enabled",  
    "dx": 1144,  
    "dy": 736,  
    "frame_dx": 1301,  
    "frame_dy": 895,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
      1269,  
      863,  
      1333,  
      927  
    ],  
    "session_memory_key": "button:send",  
    "cached": false  
  },  
  {
```



```
    "id": "windows_taskbar",
    "type": "pane",
    "label": "Taskbar",
    "description": "Windows taskbar at the bottom of the screen",
    "state": "visible",
    "dx": 780,
    "dy": 784,
    "frame_dx": 937,
    "frame_dy": 943,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        157,
        927,
        1717,
        959
    ],
    "session_memory_key": "pane:taskbar",
    "cached": false
}
],
"resolved_elements": [
    {
        "id": "browser_back",
        "type": "button",
        "label": "Back",
        "description": "Navigate back in browser history",
        "state": "enabled",
        "dx": 36,
```

```
"dy": 48,
"frame_dx": 193,
"frame_dy": 207,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    173,
    183,
    213,
    231
],
"session_memory_key": "button:back",
"cached": false
},
{
    "id": "browser_forward",
    "type": "button",
    "label": "Forward",
    "description": "Navigate forward in browser history",
    "state": "disabled",
    "dx": 72,
    "dy": 48,
    "frame_dx": 229,
    "frame_dy": 207,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        205,
        183,
```

```
        253,  
        231  
    ],  
    "session_memory_key": "button:forward",  
    "cached": false  
},  
{  
    "id": "browser_refresh",  
    "type": "button",  
    "label": "Refresh",  
    "description": "Reload the current page",  
    "state": "enabled",  
    "dx": 104,  
    "dy": 48,  
    "frame_dx": 261,  
    "frame_dy": 207,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        237,  
        183,  
        285,  
        231  
    ],  
    "session_memory_key": "button:refresh",  
    "cached": false  
},  
{  
    "id": "address_bar",
```

```
"type": "input",
"label": "Address bar",
"description": "Browser address and search bar showing localhost:5173",
"state": "enabled",
"dx": 568,
"dy": 48,
"frame_dx": 725,
"frame_dy": 207,
"confidence": 0.98,
"source": "gemini_vlm",
"frame_bbox": [
    157,
    175,
    1293,
    239
],
"session_memory_key": "input:address bar",
"cached": false
},
{
    "id": "window_minimize",
    "type": "button",
    "label": "Minimize",
    "description": "Minimize the browser window",
    "state": "enabled",
    "dx": 1128,
    "dy": 28,
    "frame_dx": 1285,
    "frame_dy": 187,
```

```
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    1261,
    167,
    1309,
    207
],
"session_memory_key": "button:minimize",
"cached": false
},
{
    "id": "window_maximize",
    "type": "button",
    "label": "Maximize",
    "description": "Maximize the browser window",
    "state": "enabled",
    "dx": 1156,
    "dy": 32,
    "frame_dx": 1313,
    "frame_dy": 191,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        1293,
        175,
        1333,
        207
    ],
    ],
```

```
"session_memory_key": "button:maximize",
"cached": false
},
{
  "id": "window_close",
  "type": "button",
  "label": "Close",
  "description": "Close the browser window",
  "state": "enabled",
  "dx": 1184,
  "dy": 24,
  "frame_dx": 1341,
  "frame_dy": 183,
  "confidence": 0.9,
  "source": "gemini_vlm",
  "frame_bbox": [
    1317,
    167,
    1365,
    199
  ],
  "session_memory_key": "button:close",
  "cached": false
},
{
  "id": "app_logo",
  "type": "icon",
  "label": "ScreenPilot AI Logo",
  "description": "Logo for the ScreenPilot AI application",
```

```
"state": "visible",
"dx": 48,
"dy": 140,
"frame_dx": 205,
"frame_dy": 299,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    189,
    279,
    221,
    319
],
"session_memory_key": "icon:screenpilot ai logo",
"cached": false
},
{
    "id": "app_title",
    "type": "heading",
    "label": "ScreenPilot AI",
    "description": "Main title of the application",
    "state": "visible",
    "dx": 136,
    "dy": 136,
    "frame_dx": 293,
    "frame_dy": 295,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
```

```
    213,  
    279,  
    373,  
    311  
  ],  
  "session_memory_key": "heading:screenpilot ai",  
  "cached": false  
},  
{  
  "id": "app_version_badge",  
  "type": "label",  
  "label": "Autonomous Task Agent v2",  
  "description": "Badge indicating the application version",  
  "state": "visible",  
  "dx": 248,  
  "dy": 144,  
  "frame_dx": 405,  
  "frame_dy": 303,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    365,  
    295,  
    445,  
    311  
  ],  
  "session_memory_key": "label:autonomous task agent v2",  
  "cached": false  
},
```



```
{
  "id": "status_connecting",
  "type": "label",
  "label": "Connecting to agent...",
  "description": "Status message indicating connection progress",
  "state": "visible",
  "dx": 44,
  "dy": 416,
  "frame_dx": 201,
  "frame_dy": 575,
  "confidence": 0.9,
  "source": "gemini_vlm",
  "frame_bbox": [
    181,
    559,
    221,
    591
  ],
  "session_memory_key": "label:connecting to agent...",
  "cached": false
},
{
  "id": "status_capturing",
  "type": "label",
  "label": "Capturing initial screen...",
  "description": "Status message indicating screen capture progress",
  "state": "visible",
  "dx": 136,
  "dy": 448,
```

```
"frame_dx": 293,
"frame_dy": 607,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
  181,
  591,
  405,
  623
],
"session_memory_key": "label:capturing initial screen...",
"cached": false
},
{
  "id": "chat_input_field",
  "type": "input",
  "label": "Agent is working...",
  "description": "Chat input field with placeholder text",
  "state": "disabled",
  "dx": 672,
  "dy": 744,
  "frame_dx": 829,
  "frame_dy": 903,
  "confidence": 0.95,
  "source": "gemini_vlm",
  "frame_bbox": [
    157,
    863,
    1501,
```

```
943
],
"session_memory_key": "input:agent is working...",
"cached": false
},
{
  "id": "send_button",
  "type": "button",
  "label": "Send",
  "description": "Button to send a message to the agent",
  "state": "enabled",
  "dx": 1144,
  "dy": 736,
  "frame_dx": 1301,
  "frame_dy": 895,
  "confidence": 0.95,
  "source": "gemini_vlm",
  "frame_bbox": [
    1269,
    863,
    1333,
    927
  ],
  "session_memory_key": "button:send",
  "cached": false
},
{
  "id": "windows_taskbar",
  "type": "pane",
```

```
    "label": "Taskbar",
    "description": "Windows taskbar at the bottom of the screen",
    "state": "visible",
    "dx": 780,
    "dy": 784,
    "frame_dx": 937,
    "frame_dy": 943,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        157,
        927,
        1717,
        959
    ],
    "session_memory_key": "pane:taskbar",
    "cached": false
}
],
"session_knowledge_base": {
    "session_id": "shot_1777694279_7138a4",
    "started_at": "2026-05-02T03:57:59.714595+00:00",
    "updated_at": "2026-05-02T03:58:27.059343+00:00",
    "stale_after_misses": 3,
    "confidence_floor": 0.4,
    "decay_per_miss": 0.9,
    "frame_count": 1,
    "frame_name": "frame_1777694279_3c3018.jpg",
    "frame_index": 0,
```

```
"image_size": {
  "width": 1920,
  "height": 1080
},
"known_elements": {
  "button:back": {
    "element_key": "button:back",
    "element_type": "button",
    "label": "Back",
    "bbox": [
      173.0,
      183.0,
      213.0,
      231.0
    ],
    "confidence": 0.95,
    "source": "gemini_vlm",
    "first_seen_frame": "frame_1777694279_3c3018.jpg",
    "last_seen_frame": "frame_1777694279_3c3018.jpg",
    "first_seen_at": "2026-05-02T03:58:27.059343+00:00",
    "last_seen_at": "2026-05-02T03:58:27.059343+00:00",
    "first_seen_index": 0,
    "last_seen_index": 0,
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false,
    "history": [
      {
```

```
        "frame_name": "frame_1777694279_3c3018.jpg",
        "frame_index": 0,
        "seen_at": "2026-05-02T03:58:27.059343+00:00",
        "confidence": 0.95
    }
]
},
"button:forward": {
    "element_key": "button:forward",
    "element_type": "button",
    "label": "Forward",
    "bbox": [
        205.0,
        183.0,
        253.0,
        231.0
    ],
    "confidence": 0.95,
    "source": "gemini_vlm",
    "first_seen_frame": "frame_1777694279_3c3018.jpg",
    "last_seen_frame": "frame_1777694279_3c3018.jpg",
    "first_seen_at": "2026-05-02T03:58:27.059343+00:00",
    "last_seen_at": "2026-05-02T03:58:27.059343+00:00",
    "first_seen_index": 0,
    "last_seen_index": 0,
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false,
```

```
"history": [  
  {  
    "frame_name": "frame_1777694279_3c3018.jpg",  
    "frame_index": 0,  
    "seen_at": "2026-05-02T03:58:27.059343+00:00",  
    "confidence": 0.95  
  }  
]  
,  
"button:refresh": {  
  "element_key": "button:refresh",  
  "element_type": "button",  
  "label": "Refresh",  
  "bbox": [  
    237.0,  
    183.0,  
    285.0,  
    231.0  
  ],  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "first_seen_frame": "frame_1777694279_3c3018.jpg",  
  "last_seen_frame": "frame_1777694279_3c3018.jpg",  
  "first_seen_at": "2026-05-02T03:58:27.059343+00:00",  
  "last_seen_at": "2026-05-02T03:58:27.059343+00:00",  
  "first_seen_index": 0,  
  "last_seen_index": 0,  
  "seen_count": 1,  
  "miss_count": 0,
```

```
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777694279_3c3018.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T03:58:27.059343+00:00",
    "confidence": 0.95
  }
],
},
"input:address bar": {
  "element_key": "input:address bar",
  "element_type": "input",
  "label": "Address bar",
  "bbox": [
    157.0,
    175.0,
    1293.0,
    239.0
  ],
  "confidence": 0.98,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777694279_3c3018.jpg",
  "last_seen_frame": "frame_1777694279_3c3018.jpg",
  "first_seen_at": "2026-05-02T03:58:27.059343+00:00",
  "last_seen_at": "2026-05-02T03:58:27.059343+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
```



```
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777694279_3c3018.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T03:58:27.059343+00:00",
    "confidence": 0.98
  }
],
},
"button:minimize": {
  "element_key": "button:minimize",
  "element_type": "button",
  "label": "Minimize",
  "bbox": [
    1261.0,
    167.0,
    1309.0,
    207.0
  ],
  "confidence": 0.9,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777694279_3c3018.jpg",
  "last_seen_frame": "frame_1777694279_3c3018.jpg",
  "first_seen_at": "2026-05-02T03:58:27.059343+00:00",
  "last_seen_at": "2026-05-02T03:58:27.059343+00:00",
```

```
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777694279_3c3018.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T03:58:27.059343+00:00",
    "confidence": 0.9
  }
],
},
"button:maximize": {
  "element_key": "button:maximize",
  "element_type": "button",
  "label": "Maximize",
  "bbox": [
    1293.0,
    175.0,
    1333.0,
    207.0
  ],
  "confidence": 0.9,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777694279_3c3018.jpg",
  "last_seen_frame": "frame_1777694279_3c3018.jpg",
```

```
"first_seen_at": "2026-05-02T03:58:27.059343+00:00",
"last_seen_at": "2026-05-02T03:58:27.059343+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777694279_3c3018.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T03:58:27.059343+00:00",
    "confidence": 0.9
  }
],
},
"button:close": {
  "element_key": "button:close",
  "element_type": "button",
  "label": "Close",
  "bbox": [
    1317.0,
    167.0,
    1365.0,
    199.0
  ],
  "confidence": 0.9,
  "source": "gemini_vlm",
```

```
"first_seen_frame": "frame_1777694279_3c3018.jpg",
"last_seen_frame": "frame_1777694279_3c3018.jpg",
"first_seen_at": "2026-05-02T03:58:27.059343+00:00",
"last_seen_at": "2026-05-02T03:58:27.059343+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777694279_3c3018.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T03:58:27.059343+00:00",
    "confidence": 0.9
  }
],
},
"icon:screenpilot ai logo": {
  "element_key": "icon:screenpilot ai logo",
  "element_type": "icon",
  "label": "ScreenPilot AI Logo",
  "bbox": [
    189.0,
    279.0,
    221.0,
    319.0
  ],
},
```

```
"confidence": 0.95,
"source": "gemini_vlm",
"first_seen_frame": "frame_1777694279_3c3018.jpg",
"last_seen_frame": "frame_1777694279_3c3018.jpg",
"first_seen_at": "2026-05-02T03:58:27.059343+00:00",
"last_seen_at": "2026-05-02T03:58:27.059343+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777694279_3c3018.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T03:58:27.059343+00:00",
    "confidence": 0.95
  }
],
},
"heading:screenpilot ai": {
  "element_key": "heading:screenpilot ai",
  "element_type": "heading",
  "label": "ScreenPilot AI",
  "bbox": [
    213.0,
    279.0,
    373.0,
```

```
    311.0
  ],
  "confidence": 0.98,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777694279_3c3018.jpg",
  "last_seen_frame": "frame_1777694279_3c3018.jpg",
  "first_seen_at": "2026-05-02T03:58:27.059343+00:00",
  "last_seen_at": "2026-05-02T03:58:27.059343+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
  "seen_count": 1,
  "miss_count": 0,
  "stale": false,
  "cached": false,
  "history": [
    {
      "frame_name": "frame_1777694279_3c3018.jpg",
      "frame_index": 0,
      "seen_at": "2026-05-02T03:58:27.059343+00:00",
      "confidence": 0.98
    }
  ]
},
"label:autonomous task agent v2": {
  "element_key": "label:autonomous task agent v2",
  "element_type": "label",
  "label": "Autonomous Task Agent v2",
  "bbox": [
    365.0,
```

```
    295.0,
    445.0,
    311.0
  ],
  "confidence": 0.95,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777694279_3c3018.jpg",
  "last_seen_frame": "frame_1777694279_3c3018.jpg",
  "first_seen_at": "2026-05-02T03:58:27.059343+00:00",
  "last_seen_at": "2026-05-02T03:58:27.059343+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
  "seen_count": 1,
  "miss_count": 0,
  "stale": false,
  "cached": false,
  "history": [
    {
      "frame_name": "frame_1777694279_3c3018.jpg",
      "frame_index": 0,
      "seen_at": "2026-05-02T03:58:27.059343+00:00",
      "confidence": 0.95
    }
  ]
},
"label:connecting to agent...": {
  "element_key": "label:connecting to agent...",
  "element_type": "label",
  "label": "Connecting to agent..."
```

```
"bbox": [
  181.0,
  559.0,
  221.0,
  591.0
],
"confidence": 0.9,
"source": "gemini_vlm",
"first_seen_frame": "frame_1777694279_3c3018.jpg",
"last_seen_frame": "frame_1777694279_3c3018.jpg",
"first_seen_at": "2026-05-02T03:58:27.059343+00:00",
"last_seen_at": "2026-05-02T03:58:27.059343+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777694279_3c3018.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T03:58:27.059343+00:00",
    "confidence": 0.9
  }
]
},
"label:capturing initial screen...": {
  "element_key": "label:capturing initial screen...",
```



```
"element_type": "label",
"label": "Capturing initial screen...",
"bbox": [
    181.0,
    591.0,
    405.0,
    623.0
],
"confidence": 0.9,
"source": "gemini_vlm",
"first_seen_frame": "frame_1777694279_3c3018.jpg",
"last_seen_frame": "frame_1777694279_3c3018.jpg",
"first_seen_at": "2026-05-02T03:58:27.059343+00:00",
"last_seen_at": "2026-05-02T03:58:27.059343+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
    {
        "frame_name": "frame_1777694279_3c3018.jpg",
        "frame_index": 0,
        "seen_at": "2026-05-02T03:58:27.059343+00:00",
        "confidence": 0.9
    }
]
},
```

```
"input:agent is working...": {
  "element_key": "input:agent is working...",
  "element_type": "input",
  "label": "Agent is working...",
  "bbox": [
    157.0,
    863.0,
    1501.0,
    943.0
  ],
  "confidence": 0.95,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777694279_3c3018.jpg",
  "last_seen_frame": "frame_1777694279_3c3018.jpg",
  "first_seen_at": "2026-05-02T03:58:27.059343+00:00",
  "last_seen_at": "2026-05-02T03:58:27.059343+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
  "seen_count": 1,
  "miss_count": 0,
  "stale": false,
  "cached": false,
  "history": [
    {
      "frame_name": "frame_1777694279_3c3018.jpg",
      "frame_index": 0,
      "seen_at": "2026-05-02T03:58:27.059343+00:00",
      "confidence": 0.95
    }
  ]
}
```

```
]
},
"button:send": {
  "element_key": "button:send",
  "element_type": "button",
  "label": "Send",
  "bbox": [
    1269.0,
    863.0,
    1333.0,
    927.0
  ],
  "confidence": 0.95,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777694279_3c3018.jpg",
  "last_seen_frame": "frame_1777694279_3c3018.jpg",
  "first_seen_at": "2026-05-02T03:58:27.059343+00:00",
  "last_seen_at": "2026-05-02T03:58:27.059343+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
  "seen_count": 1,
  "miss_count": 0,
  "stale": false,
  "cached": false,
  "history": [
    {
      "frame_name": "frame_1777694279_3c3018.jpg",
      "frame_index": 0,
      "seen_at": "2026-05-02T03:58:27.059343+00:00",
```

```
        "confidence": 0.95
      }
    ]
  },
  "pane:taskbar": {
    "element_key": "pane:taskbar",
    "element_type": "pane",
    "label": "Taskbar",
    "bbox": [
      157.0,
      927.0,
      1717.0,
      959.0
    ],
    "confidence": 0.9,
    "source": "gemini_vlm",
    "first_seen_frame": "frame_1777694279_3c3018.jpg",
    "last_seen_frame": "frame_1777694279_3c3018.jpg",
    "first_seen_at": "2026-05-02T03:58:27.059343+00:00",
    "last_seen_at": "2026-05-02T03:58:27.059343+00:00",
    "first_seen_index": 0,
    "last_seen_index": 0,
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false,
    "history": [
      {
        "frame_name": "frame_1777694279_3c3018.jpg",
```

```
        "frame_index": 0,
        "seen_at": "2026-05-02T03:58:27.059343+00:00",
        "confidence": 0.9
    }
]
}
},
"resolved_elements": [
    {
        "id": "browser_back",
        "type": "button",
        "label": "Back",
        "description": "Navigate back in browser history",
        "state": "enabled",
        "dx": 36,
        "dy": 48,
        "frame_dx": 193,
        "frame_dy": 207,
        "confidence": 0.95,
        "source": "gemini_vlm",
        "frame_bbox": [
            173,
            183,
            213,
            231
        ],
        "session_memory_key": "button:back",
        "cached": false
    },
    {
```

```
{
  "id": "browser_forward",
  "type": "button",
  "label": "Forward",
  "description": "Navigate forward in browser history",
  "state": "disabled",
  "dx": 72,
  "dy": 48,
  "frame_dx": 229,
  "frame_dy": 207,
  "confidence": 0.95,
  "source": "gemini_vlm",
  "frame_bbox": [
    205,
    183,
    253,
    231
  ],
  "session_memory_key": "button:forward",
  "cached": false
},
{
  "id": "browser_refresh",
  "type": "button",
  "label": "Refresh",
  "description": "Reload the current page",
  "state": "enabled",
  "dx": 104,
  "dy": 48,
```

```
"frame_dx": 261,
"frame_dy": 207,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    237,
    183,
    285,
    231
],
"session_memory_key": "button:refresh",
"cached": false
},
{
    "id": "address_bar",
    "type": "input",
    "label": "Address bar",
    "description": "Browser address and search bar showing localhost:5173",
    "state": "enabled",
    "dx": 568,
    "dy": 48,
    "frame_dx": 725,
    "frame_dy": 207,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
        157,
        175,
        1293,
```

239

```
],  
  "session_memory_key": "input:address bar",  
  "cached": false  
},  
{  
  "id": "window_minimize",  
  "type": "button",  
  "label": "Minimize",  
  "description": "Minimize the browser window",  
  "state": "enabled",  
  "dx": 1128,  
  "dy": 28,  
  "frame_dx": 1285,  
  "frame_dy": 187,  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    1261,  
    167,  
    1309,  
    207  
  ],  
  "session_memory_key": "button:minimize",  
  "cached": false  
},  
{  
  "id": "window_maximize",  
  "type": "button",
```



```
    "label": "Maximize",
    "description": "Maximize the browser window",
    "state": "enabled",
    "dx": 1156,
    "dy": 32,
    "frame_dx": 1313,
    "frame_dy": 191,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        1293,
        175,
        1333,
        207
    ],
    "session_memory_key": "button:maximize",
    "cached": false
},
{
    "id": "window_close",
    "type": "button",
    "label": "Close",
    "description": "Close the browser window",
    "state": "enabled",
    "dx": 1184,
    "dy": 24,
    "frame_dx": 1341,
    "frame_dy": 183,
    "confidence": 0.9,
```

```
"source": "gemini_vlm",
"frame_bbox": [
    1317,
    167,
    1365,
    199
],
"session_memory_key": "button:close",
"cached": false
},
{
    "id": "app_logo",
    "type": "icon",
    "label": "ScreenPilot AI Logo",
    "description": "Logo for the ScreenPilot AI application",
    "state": "visible",
    "dx": 48,
    "dy": 140,
    "frame_dx": 205,
    "frame_dy": 299,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        189,
        279,
        221,
        319
    ],
    "session_memory_key": "icon:screenpilot ai logo",
```

```
"cached": false
},
{
  "id": "app_title",
  "type": "heading",
  "label": "ScreenPilot AI",
  "description": "Main title of the application",
  "state": "visible",
  "dx": 136,
  "dy": 136,
  "frame_dx": 293,
  "frame_dy": 295,
  "confidence": 0.98,
  "source": "gemini_vlm",
  "frame_bbox": [
    213,
    279,
    373,
    311
  ],
  "session_memory_key": "heading:screenpilot ai",
  "cached": false
},
{
  "id": "app_version_badge",
  "type": "label",
  "label": "Autonomous Task Agent v2",
  "description": "Badge indicating the application version",
  "state": "visible",
```

```
"dx": 248,  
  "dy": 144,  
  "frame_dx": 405,  
  "frame_dy": 303,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    365,  
    295,  
    445,  
    311  
  ],  
  "session_memory_key": "label:autonomous task agent v2",  
  "cached": false  
},  
{  
  "id": "status_connecting",  
  "type": "label",  
  "label": "Connecting to agent...",  
  "description": "Status message indicating connection progress",  
  "state": "visible",  
  "dx": 44,  
  "dy": 416,  
  "frame_dx": 201,  
  "frame_dy": 575,  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    181,
```

```
    559,  
    221,  
    591  
  ],  
  "session_memory_key": "label:connecting to agent...",  
  "cached": false  
},  
{  
  "id": "status_capturing",  
  "type": "label",  
  "label": "Capturing initial screen...",  
  "description": "Status message indicating screen capture progress",  
  "state": "visible",  
  "dx": 136,  
  "dy": 448,  
  "frame_dx": 293,  
  "frame_dy": 607,  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    181,  
    591,  
    405,  
    623  
  ],  
  "session_memory_key": "label:capturing initial screen...",  
  "cached": false  
},  
{
```

```
"id": "chat_input_field",
"type": "input",
"label": "Agent is working...",
"description": "Chat input field with placeholder text",
"state": "disabled",
"dx": 672,
"dy": 744,
"frame_dx": 829,
"frame_dy": 903,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    157,
    863,
    1501,
    943
],
"session_memory_key": "input:agent is working...",
"cached": false
},
{
    "id": "send_button",
    "type": "button",
    "label": "Send",
    "description": "Button to send a message to the agent",
    "state": "enabled",
    "dx": 1144,
    "dy": 736,
    "frame_dx": 1301,
```

```
"frame_dy": 895,  
"confidence": 0.95,  
"source": "gemini_vlm",  
"frame_bbox": [  
    1269,  
    863,  
    1333,  
    927  
],  
"session_memory_key": "button:send",  
"cached": false  
},  
{  
    "id": "windows_taskbar",  
    "type": "pane",  
    "label": "Taskbar",  
    "description": "Windows taskbar at the bottom of the screen",  
    "state": "visible",  
    "dx": 780,  
    "dy": 784,  
    "frame_dx": 937,  
    "frame_dy": 943,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        157,  
        927,  
        1717,  
        959
```

```
    ],
    "session_memory_key": "pane:taskbar",
    "cached": false
  }
],
"current_elements": [
  {
    "id": "browser_back",
    "type": "button",
    "label": "Back",
    "description": "Navigate back in browser history",
    "state": "enabled",
    "dx": 36,
    "dy": 48,
    "frame_dx": 193,
    "frame_dy": 207,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
      173,
      183,
      213,
      231
    ],
    "session_memory_key": "button:back",
    "cached": false
  },
  {
    "id": "browser_forward",
```



```
"type": "button",
"label": "Forward",
"description": "Navigate forward in browser history",
"state": "disabled",
"dx": 72,
"dy": 48,
"frame_dx": 229,
"frame_dy": 207,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    205,
    183,
    253,
    231
],
"session_memory_key": "button:forward",
"cached": false
},
{
    "id": "browser_refresh",
    "type": "button",
    "label": "Refresh",
    "description": "Reload the current page",
    "state": "enabled",
    "dx": 104,
    "dy": 48,
    "frame_dx": 261,
    "frame_dy": 207,
```

```
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    237,
    183,
    285,
    231
],
"session_memory_key": "button:refresh",
"cached": false
},
{
    "id": "address_bar",
    "type": "input",
    "label": "Address bar",
    "description": "Browser address and search bar showing localhost:5173",
    "state": "enabled",
    "dx": 568,
    "dy": 48,
    "frame_dx": 725,
    "frame_dy": 207,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
        157,
        175,
        1293,
        239
    ],
    ],
```

```
"session_memory_key": "input:address bar",
"cached": false
},
{
  "id": "window_minimize",
  "type": "button",
  "label": "Minimize",
  "description": "Minimize the browser window",
  "state": "enabled",
  "dx": 1128,
  "dy": 28,
  "frame_dx": 1285,
  "frame_dy": 187,
  "confidence": 0.9,
  "source": "gemini_vlm",
  "frame_bbox": [
    1261,
    167,
    1309,
    207
  ],
  "session_memory_key": "button:minimize",
  "cached": false
},
{
  "id": "window_maximize",
  "type": "button",
  "label": "Maximize",
  "description": "Maximize the browser window",
```

```
"state": "enabled",
"dx": 1156,
"dy": 32,
"frame_dx": 1313,
"frame_dy": 191,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    1293,
    175,
    1333,
    207
],
"session_memory_key": "button:maximize",
"cached": false
},
{
    "id": "window_close",
    "type": "button",
    "label": "Close",
    "description": "Close the browser window",
    "state": "enabled",
    "dx": 1184,
    "dy": 24,
    "frame_dx": 1341,
    "frame_dy": 183,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
```

```
1317,  
167,  
1365,  
199  
],  
"session_memory_key": "button:close",  
"cached": false  
},  
{  
  "id": "app_logo",  
  "type": "icon",  
  "label": "ScreenPilot AI Logo",  
  "description": "Logo for the ScreenPilot AI application",  
  "state": "visible",  
  "dx": 48,  
  "dy": 140,  
  "frame_dx": 205,  
  "frame_dy": 299,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    189,  
    279,  
    221,  
    319  
  ],  
  "session_memory_key": "icon:screenpilot ai logo",  
  "cached": false  
},
```

```
{
  "id": "app_title",
  "type": "heading",
  "label": "ScreenPilot AI",
  "description": "Main title of the application",
  "state": "visible",
  "dx": 136,
  "dy": 136,
  "frame_dx": 293,
  "frame_dy": 295,
  "confidence": 0.98,
  "source": "gemini_vlm",
  "frame_bbox": [
    213,
    279,
    373,
    311
  ],
  "session_memory_key": "heading:screenpilot ai",
  "cached": false
},
{
  "id": "app_version_badge",
  "type": "label",
  "label": "Autonomous Task Agent v2",
  "description": "Badge indicating the application version",
  "state": "visible",
  "dx": 248,
  "dy": 144,
```

```
"frame_dx": 405,  
"frame_dy": 303,  
"confidence": 0.95,  
"source": "gemini_vlm",  
"frame_bbox": [  
    365,  
    295,  
    445,  
    311  
],  
"session_memory_key": "label:autonomous task agent v2",  
"cached": false  
},  
{  
    "id": "status_connecting",  
    "type": "label",  
    "label": "Connecting to agent...",  
    "description": "Status message indicating connection progress",  
    "state": "visible",  
    "dx": 44,  
    "dy": 416,  
    "frame_dx": 201,  
    "frame_dy": 575,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        181,  
        559,  
        221,
```

```
591
],
"session_memory_key": "label:connecting to agent...",
"cached": false
},
{
  "id": "status_capturing",
  "type": "label",
  "label": "Capturing initial screen...",
  "description": "Status message indicating screen capture progress",
  "state": "visible",
  "dx": 136,
  "dy": 448,
  "frame_dx": 293,
  "frame_dy": 607,
  "confidence": 0.9,
  "source": "gemini_vlm",
  "frame_bbox": [
    181,
    591,
    405,
    623
  ],
  "session_memory_key": "label:capturing initial screen...",
  "cached": false
},
{
  "id": "chat_input_field",
  "type": "input",
```



```
"label": "Agent is working...",
"description": "Chat input field with placeholder text",
"state": "disabled",
"dx": 672,
"dy": 744,
"frame_dx": 829,
"frame_dy": 903,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    157,
    863,
    1501,
    943
],
"session_memory_key": "input:agent is working...",
"cached": false
},
{
    "id": "send_button",
    "type": "button",
    "label": "Send",
    "description": "Button to send a message to the agent",
    "state": "enabled",
    "dx": 1144,
    "dy": 736,
    "frame_dx": 1301,
    "frame_dy": 895,
    "confidence": 0.95,
```

```
"source": "gemini_vlm",
"frame_bbox": [
    1269,
    863,
    1333,
    927
],
"session_memory_key": "button:send",
"cached": false
},
{
    "id": "windows_taskbar",
    "type": "pane",
    "label": "Taskbar",
    "description": "Windows taskbar at the bottom of the screen",
    "state": "visible",
    "dx": 780,
    "dy": 784,
    "frame_dx": 937,
    "frame_dy": 943,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        157,
        927,
        1717,
        959
    ],
    "session_memory_key": "pane:taskbar",
```

```
        "cached": false
    }
]
},
"session_memory": {
    "run_id": "85f20837-9a8e-4c22-9298-4e2401f6166c",
    "user_task": "Minimize the current window using minimize icon and find the visual studio code application icon on the screen and click on that app icon",
    "started_at": "2026-05-02T03:57:53.860689+00:00",
    "updated_at": "2026-05-02T03:58:27.103947+00:00",
    "stale_after_misses": 2,
    "confidence_floor": 0.4,
    "decay_per_miss": 0.9,
    "frame_index": 1,
    "known_elements": {
        "button:back": {
            "element_key": "button:back",
            "element_type": "button",
            "label": "Back",
            "bbox": [
                173.0,
                183.0,
                213.0,
                231.0
            ],
            "confidence": 0.4,
            "source": "gemini_vlm",
            "first_seen_frame": null,
            "last_seen_frame": null,
```

```
"first_seen_index": 0,
"last_seen_index": 0,
"first_seen_at": "2026-05-02T03:58:27.103947+00:00",
"last_seen_at": "2026-05-02T03:58:27.103947+00:00",
"seen_count": 1,
"miss_count": 15,
"stale": true,
"cached": true
},
"button:forward": {
  "element_key": "button:forward",
  "element_type": "button",
  "label": "Forward",
  "bbox": [
    205.0,
    183.0,
    253.0,
    231.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 0,
  "first_seen_at": "2026-05-02T03:58:27.103947+00:00",
  "last_seen_at": "2026-05-02T03:58:27.103947+00:00",
  "seen_count": 1,
  "miss_count": 15,
```

```
    "stale": true,
    "cached": true
  },
  "button:refresh": {
    "element_key": "button:refresh",
    "element_type": "button",
    "label": "Refresh",
    "bbox": [
      237.0,
      183.0,
      285.0,
      231.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 0,
    "last_seen_index": 0,
    "first_seen_at": "2026-05-02T03:58:27.103947+00:00",
    "last_seen_at": "2026-05-02T03:58:27.103947+00:00",
    "seen_count": 1,
    "miss_count": 15,
    "stale": true,
    "cached": true
  },
  "input:address bar": {
    "element_key": "input:address bar",
    "element_type": "input",
```

```
"label": "Address Bar",
"bbox": [
  295.0,
  172.0,
  1728.0,
  252.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 6,
"first_seen_at": "2026-05-02T03:58:27.103947+00:00",
"last_seen_at": "2026-05-02T04:03:11.701272+00:00",
"seen_count": 3,
"miss_count": 9,
"stale": true,
"cached": true
},
"button:minimize": {
  "element_key": "button:minimize",
  "element_type": "button",
  "label": "Minimize",
  "bbox": [
    1607.0,
    132.0,
    1655.0,
    196.0
```

```
],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 6,
  "first_seen_at": "2026-05-02T03:58:27.103947+00:00",
  "last_seen_at": "2026-05-02T04:03:11.701272+00:00",
  "seen_count": 2,
  "miss_count": 9,
  "stale": true,
  "cached": true
},
"button:maximize": {
  "element_key": "button:maximize",
  "element_type": "button",
  "label": "Maximize",
  "bbox": [
    1647.0,
    132.0,
    1695.0,
    188.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
```

```
    "last_seen_index": 6,  
    "first_seen_at": "2026-05-02T03:58:27.103947+00:00",  
    "last_seen_at": "2026-05-02T04:03:11.701272+00:00",  
    "seen_count": 2,  
    "miss_count": 9,  
    "stale": true,  
    "cached": true  
  },  
  "button:close": {  
    "element_key": "button:close",  
    "element_type": "button",  
    "label": "Close",  
    "bbox": [  
      1687.0,  
      132.0,  
      1728.0,  
      196.0  
    ],  
    "confidence": 0.4,  
    "source": "gemini_vlm",  
    "first_seen_frame": null,  
    "last_seen_frame": null,  
    "first_seen_index": 0,  
    "last_seen_index": 6,  
    "first_seen_at": "2026-05-02T03:58:27.103947+00:00",  
    "last_seen_at": "2026-05-02T04:03:11.701272+00:00",  
    "seen_count": 2,  
    "miss_count": 9,  
    "stale": true,
```



```
    "cached": true
  },
  "icon:screenpilot ai logo": {
    "element_key": "icon:screenpilot ai logo",
    "element_type": "icon",
    "label": "ScreenPilot AI Logo",
    "bbox": [
      157.0,
      220.0,
      461.0,
      292.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 0,
    "last_seen_index": 2,
    "first_seen_at": "2026-05-02T03:58:27.103947+00:00",
    "last_seen_at": "2026-05-02T04:00:07.860022+00:00",
    "seen_count": 3,
    "miss_count": 13,
    "stale": true,
    "cached": true
  },
  "heading:screenpilot ai": {
    "element_key": "heading:screenpilot ai",
    "element_type": "heading",
    "label": "ScreenPilot AI",
```

```
"bbox": [
  295.0,
  260.0,
  423.0,
  324.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 6,
"first_seen_at": "2026-05-02T03:58:27.103947+00:00",
"last_seen_at": "2026-05-02T04:03:11.701272+00:00",
"seen_count": 2,
"miss_count": 9,
"stale": true,
"cached": true
},
"label:autonomous task agent v2": {
  "element_key": "label:autonomous task agent v2",
  "element_type": "label",
  "label": "Autonomous Task Agent v2",
  "bbox": [
    399.0,
    268.0,
    631.0,
    316.0
  ],
```

```
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 6,
"first_seen_at": "2026-05-02T03:58:27.103947+00:00",
"last_seen_at": "2026-05-02T04:03:11.701272+00:00",
"seen_count": 3,
"miss_count": 9,
"stale": true,
"cached": true
},
"label:connecting to agent...": {
  "element_key": "label:connecting to agent...",
  "element_type": "label",
  "label": "Connecting to agent...",
  "bbox": [
    181.0,
    559.0,
    221.0,
    591.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 0,
```

```
"first_seen_at": "2026-05-02T03:58:27.103947+00:00",
"last_seen_at": "2026-05-02T03:58:27.103947+00:00",
"seen_count": 1,
"miss_count": 15,
"stale": true,
"cached": true
},
"label:capturing initial screen...": {
  "element_key": "label:capturing initial screen...",
  "element_type": "label",
  "label": "Capturing initial screen...",
  "bbox": [
    181.0,
    591.0,
    405.0,
    623.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 0,
  "first_seen_at": "2026-05-02T03:58:27.103947+00:00",
  "last_seen_at": "2026-05-02T03:58:27.103947+00:00",
  "seen_count": 1,
  "miss_count": 15,
  "stale": true,
  "cached": true
}
```

```
},

```

```
    1583.0,  
    852.0,  
    1671.0,  
    940.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 0,  
  "last_seen_index": 6,  
  "first_seen_at": "2026-05-02T03:58:27.103947+00:00",  
  "last_seen_at": "2026-05-02T04:03:11.701272+00:00",  
  "seen_count": 2,  
  "miss_count": 9,  
  "stale": true,  
  "cached": true  
},  
"pane:taskbar": {  
  "element_key": "pane:taskbar",  
  "element_type": "pane",  
  "label": "Taskbar",  
  "bbox": [  
    391.0,  
    919.0,  
    1726.0,  
    1009.0  
  ],  
  "confidence": 0.4261,
```

```
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 7,
"first_seen_at": "2026-05-02T03:58:27.103947+00:00",
"last_seen_at": "2026-05-02T04:03:52.665977+00:00",
"seen_count": 2,
"miss_count": 8,
"stale": true,
"cached": true
},
"text:screenpilot ai": {
  "element_key": "text:screenpilot ai",
  "element_type": "text",
  "label": "ScreenPilot AI",
  "bbox": [
    205.0,
    268.0,
    381.0,
    324.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
  "last_seen_index": 1,
  "first_seen_at": "2026-05-02T03:59:19.177241+00:00",
```

```
    "last_seen_at": "2026-05-02T03:59:19.177241+00:00",
    "seen_count": 1,
    "miss_count": 14,
    "stale": true,
    "cached": true
  },
  "tab:frontend": {
    "element_key": "tab:frontend",
    "element_type": "tab",
    "label": "frontend",
    "bbox": [
      181.0,
      122.0,
      397.0,
      194.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 3,
    "last_seen_index": 5,
    "first_seen_at": "2026-05-02T04:00:50.911986+00:00",
    "last_seen_at": "2026-05-02T04:02:26.654344+00:00",
    "seen_count": 3,
    "miss_count": 10,
    "stale": true,
    "cached": true
  },
}
```



```
"button:new tab": {
  "element_key": "button:new tab",
  "element_type": "button",
  "label": "New Tab",
  "bbox": [
    423.0,
    132.0,
    479.0,
    188.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 3,
  "last_seen_index": 6,
  "first_seen_at": "2026-05-02T04:00:50.911986+00:00",
  "last_seen_at": "2026-05-02T04:03:11.701272+00:00",
  "seen_count": 3,
  "miss_count": 9,
  "stale": true,
  "cached": true
},
"tab:localhost:5173": {
  "element_key": "tab:localhost:5173",
  "element_type": "tab",
  "label": "localhost:5173",
  "bbox": [
    295.0,
```

```
    124.0,  
    455.0,  
    188.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 6,  
  "last_seen_index": 6,  
  "first_seen_at": "2026-05-02T04:03:11.701272+00:00",  
  "last_seen_at": "2026-05-02T04:03:11.701272+00:00",  
  "seen_count": 1,  
  "miss_count": 9,  
  "stale": true,  
  "cached": true  
},  
"input:chat input": {  
  "element_key": "input:chat input",  
  "element_type": "input",  
  "label": "Chat Input",  
  "bbox": [  
    495.0,  
    356.0,  
    1728.0,  
    500.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",
```

```
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 6,
"last_seen_index": 6,
"first_seen_at": "2026-05-02T04:03:11.701272+00:00",
"last_seen_at": "2026-05-02T04:03:11.701272+00:00",
"seen_count": 1,
"miss_count": 9,
"stale": true,
"cached": true
},
"heading:plan": {
  "element_key": "heading:plan",
  "element_type": "heading",
  "label": "Plan",
  "bbox": [
    295.0,
    492.0,
    359.0,
    548.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 6,
  "last_seen_index": 6,
  "first_seen_at": "2026-05-02T04:03:11.701272+00:00",
  "last_seen_at": "2026-05-02T04:03:11.701272+00:00",
```

```
    "seen_count": 1,  
    "miss_count": 9,  
    "stale": true,  
    "cached": true  
  },  
  "label:0/2 done": {  
    "element_key": "label:0/2 done",  
    "element_type": "label",  
    "label": "0/2 done",  
    "bbox": [  
      863.0,  
      508.0,  
      959.0,  
      532.0  
    ],  
    "confidence": 0.4,  
    "source": "gemini_vlm",  
    "first_seen_frame": null,  
    "last_seen_frame": null,  
    "first_seen_index": 6,  
    "last_seen_index": 6,  
    "first_seen_at": "2026-05-02T04:03:11.701272+00:00",  
    "last_seen_at": "2026-05-02T04:03:11.701272+00:00",  
    "seen_count": 1,  
    "miss_count": 9,  
    "stale": true,  
    "cached": true  
  },  
  "text:move the mouse to the minimize button": {
```

```
"element_key": "text:move the mouse to the minimize button",
"element_type": "text",
"label": "Move the mouse to the Minimize button",
"bbox": [
    295.0,
    572.0,
    1055.0,
    628.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 6,
"last_seen_index": 6,
"first_seen_at": "2026-05-02T04:03:11.701272+00:00",
"last_seen_at": "2026-05-02T04:03:11.701272+00:00",
"seen_count": 1,
"miss_count": 9,
"stale": true,
"cached": true
},
"text:identify the visual studio code icon": {
    "element_key": "text:identify the visual studio code icon",
    "element_type": "text",
    "label": "Identify the Visual Studio Code icon",
    "bbox": [
        295.0,
        628.0,
```

```
    1063.0,  
    684.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 6,  
  "last_seen_index": 6,  
  "first_seen_at": "2026-05-02T04:03:11.701272+00:00",  
  "last_seen_at": "2026-05-02T04:03:11.701272+00:00",  
  "seen_count": 1,  
  "miss_count": 9,  
  "stale": true,  
  "cached": true  
},  
"pane:log output": {  
  "element_key": "pane:log output",  
  "element_type": "pane",  
  "label": "Log Output",  
  "bbox": [  
    295.0,  
    668.0,  
    1087.0,  
    892.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,
```

```
"last_seen_frame": null,  
"first_seen_index": 6,  
"last_seen_index": 6,  
"first_seen_at": "2026-05-02T04:03:11.701272+00:00",  
"last_seen_at": "2026-05-02T04:03:11.701272+00:00",  
"seen_count": 1,  
"miss_count": 9,  
"stale": true,  
"cached": true  
},  
"toolbar:taskbar": {  
  "element_key": "toolbar:taskbar",  
  "element_type": "toolbar",  
  "label": "Taskbar",  
  "bbox": [  
    295.0,  
    940.0,  
    1728.0,  
    1010.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 6,  
  "last_seen_index": 6,  
  "first_seen_at": "2026-05-02T04:03:11.701272+00:00",  
  "last_seen_at": "2026-05-02T04:03:11.701272+00:00",  
  "seen_count": 1,
```

```
    "miss_count": 9,  
    "stale": true,  
    "cached": true  
  },  
  "button:start": {  
    "element_key": "button:start",  
    "element_type": "button",  
    "label": "Start",  
    "bbox": [  
      567.0,  
      943.0,  
      615.0,  
      983.0  
    ],  
    "confidence": 0.4089,  
    "source": "gemini_vlm",  
    "first_seen_frame": null,  
    "last_seen_frame": null,  
    "first_seen_index": 6,  
    "last_seen_index": 7,  
    "first_seen_at": "2026-05-02T04:03:11.701272+00:00",  
    "last_seen_at": "2026-05-02T04:03:52.665977+00:00",  
    "seen_count": 2,  
    "miss_count": 8,  
    "stale": true,  
    "cached": true  
  },  
  "input:search": {  
    "element_key": "input:search",
```



```
"element_type": "input",
"label": "Search",
"bbox": [
    591.0,
    935.0,
    863.0,
    983.0
],
"confidence": 0.4089,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 6,
"last_seen_index": 7,
"first_seen_at": "2026-05-02T04:03:11.701272+00:00",
"last_seen_at": "2026-05-02T04:03:52.665977+00:00",
"seen_count": 2,
"miss_count": 8,
"stale": true,
"cached": true
},
"icon:file explorer": {
    "element_key": "icon:file explorer",
    "element_type": "icon",
    "label": "File Explorer",
    "bbox": [
        919.0,
        935.0,
        967.0,
```

```
    983.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 6,
  "last_seen_index": 7,
  "first_seen_at": "2026-05-02T04:03:11.701272+00:00",
  "last_seen_at": "2026-05-02T04:03:52.665977+00:00",
  "seen_count": 2,
  "miss_count": 8,
  "stale": true,
  "cached": true
},
"icon:microsoft edge": {
  "element_key": "icon:microsoft edge",
  "element_type": "icon",
  "label": "Microsoft Edge",
  "bbox": [
    823.0,
    948.0,
    871.0,
    996.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
```

```
"first_seen_index": 6,  
"last_seen_index": 6,  
"first_seen_at": "2026-05-02T04:03:11.701272+00:00",  
"last_seen_at": "2026-05-02T04:03:11.701272+00:00",  
"seen_count": 1,  
"miss_count": 9,  
"stale": true,  
"cached": true  
},  
"icon:google chrome": {  
  "element_key": "icon:google chrome",  
  "element_type": "icon",  
  "label": "Google Chrome",  
  "bbox": [  
    152.0,  
    117.0,  
    264.0,  
    309.0  
  ],  
  "confidence": 0.99,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 6,  
  "last_seen_index": 15,  
  "first_seen_at": "2026-05-02T04:03:11.701272+00:00",  
  "last_seen_at": "2026-05-02T04:10:40.059078+00:00",  
  "seen_count": 8,  
  "miss_count": 0,
```

```
    "stale": false,
    "cached": false
  },
  "icon:visual studio code": {
    "element_key": "icon:visual studio code",
    "element_type": "icon",
    "label": "Visual Studio Code",
    "bbox": [
      1167.0,
      935.0,
      1215.0,
      983.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 6,
    "last_seen_index": 7,
    "first_seen_at": "2026-05-02T04:03:11.701272+00:00",
    "last_seen_at": "2026-05-02T04:03:52.665977+00:00",
    "seen_count": 2,
    "miss_count": 8,
    "stale": true,
    "cached": true
  },
  "icon:spotify": {
    "element_key": "icon:spotify",
    "element_type": "icon",
```

```
"label": "Spotify",
"bbox": [
  1207.0,
  935.0,
  1255.0,
  983.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 6,
"last_seen_index": 7,
"first_seen_at": "2026-05-02T04:03:11.701272+00:00",
"last_seen_at": "2026-05-02T04:03:52.665977+00:00",
"seen_count": 2,
"miss_count": 8,
"stale": true,
"cached": true
},
"label:clock and date": {
  "element_key": "label:clock and date",
  "element_type": "label",
  "label": "Clock and Date",
  "bbox": [
    1615.0,
    935.0,
    1719.0,
    991.0
  ]
}
```

```
],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 6,  
  "last_seen_index": 7,  
  "first_seen_at": "2026-05-02T04:03:11.701272+00:00",  
  "last_seen_at": "2026-05-02T04:03:52.665977+00:00",  
  "seen_count": 2,  
  "miss_count": 8,  
  "stale": true,  
  "cached": true  
},  
"icon:postman": {  
  "element_key": "icon:postman",  
  "element_type": "icon",  
  "label": "Postman",  
  "bbox": [  
    391.0,  
    735.0,  
    487.0,  
    895.0  
  ],  
  "confidence": 0.4089,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 7,
```

```
"last_seen_index": 7,  
"first_seen_at": "2026-05-02T04:03:52.665977+00:00",  
"last_seen_at": "2026-05-02T04:03:52.665977+00:00",  
"seen_count": 1,  
"miss_count": 8,  
"stale": true,  
"cached": true  
},  
"icon:loom": {  
  "element_key": "icon:loom",  
  "element_type": "icon",  
  "label": "Loom",  
  "bbox": [  
    471.0,  
    735.0,  
    583.0,  
    895.0  
  ],  
  "confidence": 0.4089,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 7,  
  "last_seen_index": 7,  
  "first_seen_at": "2026-05-02T04:03:52.665977+00:00",  
  "last_seen_at": "2026-05-02T04:03:52.665977+00:00",  
  "seen_count": 1,  
  "miss_count": 8,  
  "stale": true,
```

```
    "cached": true
  },
  "icon:vlc media player": {
    "element_key": "icon:vlc media player",
    "element_type": "icon",
    "label": "VLC media player",
    "bbox": [
      559.0,
      735.0,
      671.0,
      895.0
    ],
    "confidence": 0.4089,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 7,
    "last_seen_index": 7,
    "first_seen_at": "2026-05-02T04:03:52.665977+00:00",
    "last_seen_at": "2026-05-02T04:03:52.665977+00:00",
    "seen_count": 1,
    "miss_count": 8,
    "stale": true,
    "cached": true
  },
  "pane:sticky note": {
    "element_key": "pane:sticky note",
    "element_type": "pane",
    "label": "Sticky Note",
```



```
"bbox": [
  1359.0,
  135.0,
  1726.0,
  391.0
],
"confidence": 0.4218,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 7,
"last_seen_index": 7,
"first_seen_at": "2026-05-02T04:03:52.665977+00:00",
"last_seen_at": "2026-05-02T04:03:52.665977+00:00",
"seen_count": 1,
"miss_count": 8,
"stale": true,
"cached": true
},
"text:amazon workspaces credentials": {
  "element_key": "text:amazon workspaces credentials",
  "element_type": "text",
  "label": "Amazon WorkSpaces credentials",
  "bbox": [
    1367.0,
    191.0,
    1726.0,
    327.0
  ],
```

```
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 7,
"last_seen_index": 7,
"first_seen_at": "2026-05-02T04:03:52.665977+00:00",
"last_seen_at": "2026-05-02T04:03:52.665977+00:00",
"seen_count": 1,
"miss_count": 8,
"stale": true,
"cached": true
},
"icon:microsoft store": {
  "element_key": "icon:microsoft store",
  "element_type": "icon",
  "label": "Microsoft Store",
  "bbox": [
    959.0,
    935.0,
    1007.0,
    983.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 7,
  "last_seen_index": 7,
```

```
"first_seen_at": "2026-05-02T04:03:52.665977+00:00",
"last_seen_at": "2026-05-02T04:03:52.665977+00:00",
"seen_count": 1,
"miss_count": 8,
"stale": true,
"cached": true
},
"icon:mail": {
  "element_key": "icon:mail",
  "element_type": "icon",
  "label": "Mail",
  "bbox": [
    999.0,
    935.0,
    1047.0,
    983.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 7,
  "last_seen_index": 7,
  "first_seen_at": "2026-05-02T04:03:52.665977+00:00",
  "last_seen_at": "2026-05-02T04:03:52.665977+00:00",
  "seen_count": 1,
  "miss_count": 8,
  "stale": true,
  "cached": true
}
```

```
},  
"icon:xbox": {  
  "element_key": "icon:xbox",  
  "element_type": "icon",  
  "label": "Xbox",  
  "bbox": [  
    1039.0,  
    935.0,  
    1095.0,  
    983.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 7,  
  "last_seen_index": 7,  
  "first_seen_at": "2026-05-02T04:03:52.665977+00:00",  
  "last_seen_at": "2026-05-02T04:03:52.665977+00:00",  
  "seen_count": 1,  
  "miss_count": 8,  
  "stale": true,  
  "cached": true  
},  
"icon:notepad": {  
  "element_key": "icon:notepad",  
  "element_type": "icon",  
  "label": "Notepad",  
  "bbox": [  

```

```
    1127.0,  
    935.0,  
    1175.0,  
    983.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 7,  
  "last_seen_index": 7,  
  "first_seen_at": "2026-05-02T04:03:52.665977+00:00",  
  "last_seen_at": "2026-05-02T04:03:52.665977+00:00",  
  "seen_count": 1,  
  "miss_count": 8,  
  "stale": true,  
  "cached": true  
},  
"button:show hidden icons": {  
  "element_key": "button:show hidden icons",  
  "element_type": "button",  
  "label": "Show hidden icons",  
  "bbox": [  
    1447.0,  
    943.0,  
    1479.0,  
    967.0  
  ],  
  "confidence": 0.4,
```

```
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 7,
"last_seen_index": 7,
"first_seen_at": "2026-05-02T04:03:52.665977+00:00",
"last_seen_at": "2026-05-02T04:03:52.665977+00:00",
"seen_count": 1,
"miss_count": 8,
"stale": true,
"cached": true
},
"icon:wi-fi": {
  "element_key": "icon:wi-fi",
  "element_type": "icon",
  "label": "Wi-Fi",
  "bbox": [
    1527.0,
    943.0,
    1559.0,
    983.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 7,
  "last_seen_index": 7,
  "first_seen_at": "2026-05-02T04:03:52.665977+00:00",
```

```
    "last_seen_at": "2026-05-02T04:03:52.665977+00:00",
    "seen_count": 1,
    "miss_count": 8,
    "stale": true,
    "cached": true
  },
  "icon:volume": {
    "element_key": "icon:volume",
    "element_type": "icon",
    "label": "Volume",
    "bbox": [
      1559.0,
      943.0,
      1591.0,
      983.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 7,
    "last_seen_index": 7,
    "first_seen_at": "2026-05-02T04:03:52.665977+00:00",
    "last_seen_at": "2026-05-02T04:03:52.665977+00:00",
    "seen_count": 1,
    "miss_count": 8,
    "stale": true,
    "cached": true
  },
}
```

```
"icon:battery": {
  "element_key": "icon:battery",
  "element_type": "icon",
  "label": "Battery",
  "bbox": [
    1591.0,
    943.0,
    1623.0,
    983.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 7,
  "last_seen_index": 7,
  "first_seen_at": "2026-05-02T04:03:52.665977+00:00",
  "last_seen_at": "2026-05-02T04:03:52.665977+00:00",
  "seen_count": 1,
  "miss_count": 8,
  "stale": true,
  "cached": true
},
"icon:huiontablet": {
  "element_key": "icon:huiontablet",
  "element_type": "icon",
  "label": "HuionTablet",
  "bbox": [
    264.0,
```



```
    117.0,  
    408.0,  
    317.0  
  ],  
  "confidence": 0.891,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 8,  
  "last_seen_index": 14,  
  "first_seen_at": "2026-05-02T04:04:41.281802+00:00",  
  "last_seen_at": "2026-05-02T04:09:59.806216+00:00",  
  "seen_count": 7,  
  "miss_count": 1,  
  "stale": false,  
  "cached": true  
},  
"icon:amazon workspaces": {  
  "element_key": "icon:amazon workspaces",  
  "element_type": "icon",  
  "label": "Amazon WorkSpaces",  
  "bbox": [  
    278.0,  
    315.0,  
    382.0,  
    523.0  
  ],  
  "confidence": 0.5609,  
  "source": "gemini_vlm",
```

```
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 8,
    "last_seen_index": 10,
    "first_seen_at": "2026-05-02T04:04:41.281802+00:00",
    "last_seen_at": "2026-05-02T04:06:17.096635+00:00",
    "seen_count": 2,
    "miss_count": 5,
    "stale": true,
    "cached": true
  }
},
"current_elements": [
  {
    "id": "browser_back",
    "type": "button",
    "label": "Back",
    "description": "Navigate back in browser history",
    "state": "enabled",
    "dx": 36,
    "dy": 48,
    "frame_dx": 193,
    "frame_dy": 207,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
      173,
      183,
      213,
```

```
231
],
"session_memory_key": "button:back",
"cached": false
},
{
  "id": "browser_forward",
  "type": "button",
  "label": "Forward",
  "description": "Navigate forward in browser history",
  "state": "disabled",
  "dx": 72,
  "dy": 48,
  "frame_dx": 229,
  "frame_dy": 207,
  "confidence": 0.95,
  "source": "gemini_vlm",
  "frame_bbox": [
    205,
    183,
    253,
    231
  ],
  "session_memory_key": "button:forward",
  "cached": false
},
{
  "id": "browser_refresh",
  "type": "button",
```

```
"label": "Refresh",
"description": "Reload the current page",
"state": "enabled",
"dx": 104,
"dy": 48,
"frame_dx": 261,
"frame_dy": 207,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    237,
    183,
    285,
    231
],
"session_memory_key": "button:refresh",
"cached": false
},
{
    "id": "address_bar",
    "type": "input",
    "label": "Address bar",
    "description": "Browser address and search bar showing localhost:5173",
    "state": "enabled",
    "dx": 568,
    "dy": 48,
    "frame_dx": 725,
    "frame_dy": 207,
    "confidence": 0.98,
```

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"source": "gemini_vlm",
"frame_bbox": [
    157,
    175,
    1293,
    239
],
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"cached": false
},
{
    "id": "window_minimize",
    "type": "button",
    "label": "Minimize",
    "description": "Minimize the browser window",
    "state": "enabled",
    "dx": 1128,
    "dy": 28,
    "frame_dx": 1285,
    "frame_dy": 187,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        1261,
        167,
        1309,
        207
    ],
    "session_memory_key": "button:minimize",
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"cached": false
},
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  "id": "window_maximize",
  "type": "button",
  "label": "Maximize",
  "description": "Maximize the browser window",
  "state": "enabled",
  "dx": 1156,
  "dy": 32,
  "frame_dx": 1313,
  "frame_dy": 191,
  "confidence": 0.9,
  "source": "gemini_vlm",
  "frame_bbox": [
    1293,
    175,
    1333,
    207
  ],
  "session_memory_key": "button:maximize",
  "cached": false
},
{
  "id": "window_close",
  "type": "button",
  "label": "Close",
  "description": "Close the browser window",
  "state": "enabled",
```

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"dx": 1184,  
  "dy": 24,  
  "frame_dx": 1341,  
  "frame_dy": 183,  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    1317,  
    167,  
    1365,  
    199  
  ],  
  "session_memory_key": "button:close",  
  "cached": false  
},  
{  
  "id": "app_logo",  
  "type": "icon",  
  "label": "ScreenPilot AI Logo",  
  "description": "Logo for the ScreenPilot AI application",  
  "state": "visible",  
  "dx": 48,  
  "dy": 140,  
  "frame_dx": 205,  
  "frame_dy": 299,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    189,
```

```
        279,  
        221,  
        319  
    ],  
    "session_memory_key": "icon:screenpilot ai logo",  
    "cached": false  
},  
{  
    "id": "app_title",  
    "type": "heading",  
    "label": "ScreenPilot AI",  
    "description": "Main title of the application",  
    "state": "visible",  
    "dx": 136,  
    "dy": 136,  
    "frame_dx": 293,  
    "frame_dy": 295,  
    "confidence": 0.98,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        213,  
        279,  
        373,  
        311  
    ],  
    "session_memory_key": "heading:screenpilot ai",  
    "cached": false  
},  
{
```



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"id": "app_version_badge",
"type": "label",
"label": "Autonomous Task Agent v2",
"description": "Badge indicating the application version",
"state": "visible",
"dx": 248,
"dy": 144,
"frame_dx": 405,
"frame_dy": 303,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    365,
    295,
    445,
    311
],
"session_memory_key": "label:autonomous task agent v2",
"cached": false
},
{
  "id": "status_connecting",
  "type": "label",
  "label": "Connecting to agent...",
  "description": "Status message indicating connection progress",
  "state": "visible",
  "dx": 44,
  "dy": 416,
  "frame_dx": 201,
```

```
"frame_dy": 575,  
"confidence": 0.9,  
"source": "gemini_vlm",  
"frame_bbox": [  
    181,  
    559,  
    221,  
    591  
],  
"session_memory_key": "label:connecting to agent...",  
"cached": false  
},  
{  
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    "type": "label",  
    "label": "Capturing initial screen...",  
    "description": "Status message indicating screen capture progress",  
    "state": "visible",  
    "dx": 136,  
    "dy": 448,  
    "frame_dx": 293,  
    "frame_dy": 607,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        181,  
        591,  
        405,  
        623
```

```
    ],
    "session_memory_key": "label:capturing initial screen...",
    "cached": false
  },
  {
    "id": "chat_input_field",
    "type": "input",
    "label": "Agent is working...",
    "description": "Chat input field with placeholder text",
    "state": "disabled",
    "dx": 672,
    "dy": 744,
    "frame_dx": 829,
    "frame_dy": 903,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
      157,
      863,
      1501,
      943
    ],
    "session_memory_key": "input:agent is working...",
    "cached": false
  },
  {
    "id": "send_button",
    "type": "button",
    "label": "Send",
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```
"description": "Button to send a message to the agent",
"state": "enabled",
"dx": 1144,
"dy": 736,
"frame_dx": 1301,
"frame_dy": 895,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    1269,
    863,
    1333,
    927
],
"session_memory_key": "button:send",
"cached": false
},
{
    "id": "windows_taskbar",
    "type": "pane",
    "label": "Taskbar",
    "description": "Windows taskbar at the bottom of the screen",
    "state": "visible",
    "dx": 780,
    "dy": 784,
    "frame_dx": 937,
    "frame_dy": 943,
    "confidence": 0.9,
    "source": "gemini_vlm",
```

```
"frame_bbox": [
  157,
  927,
  1717,
  959
],
"session_memory_key": "pane:taskbar",
"cached": false
}
],
"resolved_elements": [
  {
    "id": "browser_back",
    "type": "button",
    "label": "Back",
    "description": "Navigate back in browser history",
    "state": "enabled",
    "dx": 36,
    "dy": 48,
    "frame_dx": 193,
    "frame_dy": 207,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
      173,
      183,
      213,
      231
    ],
  },
]
```

```
"session_memory_key": "button:back",
"cached": false
},
{
  "id": "browser_forward",
  "type": "button",
  "label": "Forward",
  "description": "Navigate forward in browser history",
  "state": "disabled",
  "dx": 72,
  "dy": 48,
  "frame_dx": 229,
  "frame_dy": 207,
  "confidence": 0.95,
  "source": "gemini_vlm",
  "frame_bbox": [
    205,
    183,
    253,
    231
  ],
  "session_memory_key": "button:forward",
  "cached": false
},
{
  "id": "browser_refresh",
  "type": "button",
  "label": "Refresh",
  "description": "Reload the current page",
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"state": "enabled",
"dx": 104,
"dy": 48,
"frame_dx": 261,
"frame_dy": 207,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    237,
    183,
    285,
    231
],
"session_memory_key": "button:refresh",
"cached": false
},
{
    "id": "address_bar",
    "type": "input",
    "label": "Address bar",
    "description": "Browser address and search bar showing localhost:5173",
    "state": "enabled",
    "dx": 568,
    "dy": 48,
    "frame_dx": 725,
    "frame_dy": 207,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
```

```
    157,  
    175,  
    1293,  
    239  
  ],  
  "session_memory_key": "input:address bar",  
  "cached": false  
},  
{  
  "id": "window_minimize",  
  "type": "button",  
  "label": "Minimize",  
  "description": "Minimize the browser window",  
  "state": "enabled",  
  "dx": 1128,  
  "dy": 28,  
  "frame_dx": 1285,  
  "frame_dy": 187,  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    1261,  
    167,  
    1309,  
    207  
  ],  
  "session_memory_key": "button:minimize",  
  "cached": false  
},
```



```
{
  "id": "window_maximize",
  "type": "button",
  "label": "Maximize",
  "description": "Maximize the browser window",
  "state": "enabled",
  "dx": 1156,
  "dy": 32,
  "frame_dx": 1313,
  "frame_dy": 191,
  "confidence": 0.9,
  "source": "gemini_vlm",
  "frame_bbox": [
    1293,
    175,
    1333,
    207
  ],
  "session_memory_key": "button:maximize",
  "cached": false
},
{
  "id": "window_close",
  "type": "button",
  "label": "Close",
  "description": "Close the browser window",
  "state": "enabled",
  "dx": 1184,
  "dy": 24,
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"frame_dx": 1341,  
"frame_dy": 183,  
"confidence": 0.9,  
"source": "gemini_vlm",  
"frame_bbox": [  
    1317,  
    167,  
    1365,  
    199  
],  
"session_memory_key": "button:close",  
"cached": false  
},  
{  
    "id": "app_logo",  
    "type": "icon",  
    "label": "ScreenPilot AI Logo",  
    "description": "Logo for the ScreenPilot AI application",  
    "state": "visible",  
    "dx": 48,  
    "dy": 140,  
    "frame_dx": 205,  
    "frame_dy": 299,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        189,  
        279,  
        221,
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319
],
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"cached": false
},
{
  "id": "app_title",
  "type": "heading",
  "label": "ScreenPilot AI",
  "description": "Main title of the application",
  "state": "visible",
  "dx": 136,
  "dy": 136,
  "frame_dx": 293,
  "frame_dy": 295,
  "confidence": 0.98,
  "source": "gemini_vlm",
  "frame_bbox": [
    213,
    279,
    373,
    311
  ],
},
"session_memory_key": "heading:screenpilot ai",
"cached": false
},
{
  "id": "app_version_badge",
  "type": "label",
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```
"label": "Autonomous Task Agent v2",
"description": "Badge indicating the application version",
"state": "visible",
"dx": 248,
"dy": 144,
"frame_dx": 405,
"frame_dy": 303,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    365,
    295,
    445,
    311
],
"session_memory_key": "label:autonomous task agent v2",
"cached": false
},
{
    "id": "status_connecting",
    "type": "label",
    "label": "Connecting to agent...",
    "description": "Status message indicating connection progress",
    "state": "visible",
    "dx": 44,
    "dy": 416,
    "frame_dx": 201,
    "frame_dy": 575,
    "confidence": 0.9,
```

```
"source": "gemini_vlm",
"frame_bbox": [
    181,
    559,
    221,
    591
],
"session_memory_key": "label:connecting to agent...",
"cached": false
},
{
    "id": "status_capturing",
    "type": "label",
    "label": "Capturing initial screen...",
    "description": "Status message indicating screen capture progress",
    "state": "visible",
    "dx": 136,
    "dy": 448,
    "frame_dx": 293,
    "frame_dy": 607,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        181,
        591,
        405,
        623
    ],
    "session_memory_key": "label:capturing initial screen..."
```

```
    "cached": false
  },
  {
    "id": "chat_input_field",
    "type": "input",
    "label": "Agent is working...",
    "description": "Chat input field with placeholder text",
    "state": "disabled",
    "dx": 672,
    "dy": 744,
    "frame_dx": 829,
    "frame_dy": 903,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
      157,
      863,
      1501,
      943
    ],
    "session_memory_key": "input:agent is working...",
    "cached": false
  },
  {
    "id": "send_button",
    "type": "button",
    "label": "Send",
    "description": "Button to send a message to the agent",
    "state": "enabled",
```

```
"dx": 1144,  
  "dy": 736,  
  "frame_dx": 1301,  
  "frame_dy": 895,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    1269,  
    863,  
    1333,  
    927  
  ],  
  "session_memory_key": "button:send",  
  "cached": false  
},  
{  
  "id": "windows_taskbar",  
  "type": "pane",  
  "label": "Taskbar",  
  "description": "Windows taskbar at the bottom of the screen",  
  "state": "visible",  
  "dx": 780,  
  "dy": 784,  
  "frame_dx": 937,  
  "frame_dy": 943,  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    157,
```

```
        927,  
        1717,  
        959  
    ],  
    "session_memory_key": "pane:taskbar",  
    "cached": false  
}  
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"screen_data": {  
    "image": "E:\\sliit-  
project\\ScreenAwareTaskAgent\\vision\\data\\sessions\\shot_1777694279_7138a4\\pr  
ocessing\\proc_1777694279_a5eace\\preprocessed_frames\\frame_1777694279_3c3018.jp  
g",  
    "image_size": {  
        "width": 1920,  
        "height": 1080  
    },  
    "screen_bbox": [  
        157,  
        159,  
        1717,  
        959  
    ],  
    "screen_size": {  
        "width": 1560,  
        "height": 800  
    },  
    "coordinate_system": "pixel",  
    "elements": [  
        {
```



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"type": "button",
"label": "Back",
"description": "Navigate back in browser history",
"state": "enabled",
"dx": 36,
"dy": 48,
"frame_dx": 193,
"frame_dy": 207,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    173,
    183,
    213,
    231
],
"session_memory_key": "button:back",
"cached": false
},
{
    "id": "browser_forward",
    "type": "button",
    "label": "Forward",
    "description": "Navigate forward in browser history",
    "state": "disabled",
    "dx": 72,
    "dy": 48,
    "frame_dx": 229,
```

```
"frame_dy": 207,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    205,
    183,
    253,
    231
],
"session_memory_key": "button:forward",
"cached": false
},
{
    "id": "browser_refresh",
    "type": "button",
    "label": "Refresh",
    "description": "Reload the current page",
    "state": "enabled",
    "dx": 104,
    "dy": 48,
    "frame_dx": 261,
    "frame_dy": 207,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        237,
        183,
        285,
        231
    ]
}
```

```
    ],
    "session_memory_key": "button:refresh",
    "cached": false
  },
  {
    "id": "address_bar",
    "type": "input",
    "label": "Address bar",
    "description": "Browser address and search bar showing
localhost:5173",
    "state": "enabled",
    "dx": 568,
    "dy": 48,
    "frame_dx": 725,
    "frame_dy": 207,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
      157,
      175,
      1293,
      239
    ],
    "session_memory_key": "input:address bar",
    "cached": false
  },
  {
    "id": "window_minimize",
    "type": "button",
```

```
"label": "Minimize",
"description": "Minimize the browser window",
"state": "enabled",
"dx": 1128,
"dy": 28,
"frame_dx": 1285,
"frame_dy": 187,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    1261,
    167,
    1309,
    207
],
"session_memory_key": "button:minimize",
"cached": false
},
{
    "id": "window_maximize",
    "type": "button",
    "label": "Maximize",
    "description": "Maximize the browser window",
    "state": "enabled",
    "dx": 1156,
    "dy": 32,
    "frame_dx": 1313,
    "frame_dy": 191,
    "confidence": 0.9,
```

```
"source": "gemini_vlm",
"frame_bbox": [
    1293,
    175,
    1333,
    207
],
"session_memory_key": "button:maximize",
"cached": false
},
{
    "id": "window_close",
    "type": "button",
    "label": "Close",
    "description": "Close the browser window",
    "state": "enabled",
    "dx": 1184,
    "dy": 24,
    "frame_dx": 1341,
    "frame_dy": 183,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        1317,
        167,
        1365,
        199
    ],
    "session_memory_key": "button:close",
```

```
"cached": false
},
{
  "id": "app_logo",
  "type": "icon",
  "label": "ScreenPilot AI Logo",
  "description": "Logo for the ScreenPilot AI application",
  "state": "visible",
  "dx": 48,
  "dy": 140,
  "frame_dx": 205,
  "frame_dy": 299,
  "confidence": 0.95,
  "source": "gemini_vlm",
  "frame_bbox": [
    189,
    279,
    221,
    319
  ],
  "session_memory_key": "icon:screenpilot ai logo",
  "cached": false
},
{
  "id": "app_title",
  "type": "heading",
  "label": "ScreenPilot AI",
  "description": "Main title of the application",
  "state": "visible",
```

```
"dx": 136,  
  "dy": 136,  
  "frame_dx": 293,  
  "frame_dy": 295,  
  "confidence": 0.98,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    213,  
    279,  
    373,  
    311  
  ],  
  "session_memory_key": "heading:screenpilot ai",  
  "cached": false  
},  
{  
  "id": "app_version_badge",  
  "type": "label",  
  "label": "Autonomous Task Agent v2",  
  "description": "Badge indicating the application version",  
  "state": "visible",  
  "dx": 248,  
  "dy": 144,  
  "frame_dx": 405,  
  "frame_dy": 303,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    365,
```

```
        295,  
        445,  
        311  
    ],  
    "session_memory_key": "label:autonomous task agent v2",  
    "cached": false  
},  
{  
    "id": "status_connecting",  
    "type": "label",  
    "label": "Connecting to agent...",  
    "description": "Status message indicating connection progress",  
    "state": "visible",  
    "dx": 44,  
    "dy": 416,  
    "frame_dx": 201,  
    "frame_dy": 575,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        181,  
        559,  
        221,  
        591  
    ],  
    "session_memory_key": "label:connecting to agent...",  
    "cached": false  
},  
{
```



```
"id": "status_capturing",
"type": "label",
"label": "Capturing initial screen...",
"description": "Status message indicating screen capture progress",
"state": "visible",
"dx": 136,
"dy": 448,
"frame_dx": 293,
"frame_dy": 607,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    181,
    591,
    405,
    623
],
"session_memory_key": "label:capturing initial screen...",
"cached": false
},
{
    "id": "chat_input_field",
    "type": "input",
    "label": "Agent is working...",
    "description": "Chat input field with placeholder text",
    "state": "disabled",
    "dx": 672,
    "dy": 744,
    "frame_dx": 829,
```

```
"frame_dy": 903,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    157,
    863,
    1501,
    943
],
"session_memory_key": "input:agent is working...",
"cached": false
},
{
    "id": "send_button",
    "type": "button",
    "label": "Send",
    "description": "Button to send a message to the agent",
    "state": "enabled",
    "dx": 1144,
    "dy": 736,
    "frame_dx": 1301,
    "frame_dy": 895,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        1269,
        863,
        1333,
        927
```

```
    ],  
    "session_memory_key": "button:send",  
    "cached": false  
  },  
  {  
    "id": "windows_taskbar",  
    "type": "pane",  
    "label": "Taskbar",  
    "description": "Windows taskbar at the bottom of the screen",  
    "state": "visible",  
    "dx": 780,  
    "dy": 784,  
    "frame_dx": 937,  
    "frame_dy": 943,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
      157,  
      927,  
      1717,  
      959  
    ],  
    "session_memory_key": "pane:taskbar",  
    "cached": false  
  }  
],  
"element_count": 15,  
"current_elements": [  
  {
```

```
"id": "browser_back",
"type": "button",
"label": "Back",
"description": "Navigate back in browser history",
"state": "enabled",
"dx": 36,
"dy": 48,
"frame_dx": 193,
"frame_dy": 207,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    173,
    183,
    213,
    231
],
"session_memory_key": "button:back",
"cached": false
},
{
    "id": "browser_forward",
    "type": "button",
    "label": "Forward",
    "description": "Navigate forward in browser history",
    "state": "disabled",
    "dx": 72,
    "dy": 48,
    "frame_dx": 229,
```

```
"frame_dy": 207,  
"confidence": 0.95,  
"source": "gemini_vlm",  
"frame_bbox": [  
    205,  
    183,  
    253,  
    231  
],  
"session_memory_key": "button:forward",  
"cached": false  
},  
{  
    "id": "browser_refresh",  
    "type": "button",  
    "label": "Refresh",  
    "description": "Reload the current page",  
    "state": "enabled",  
    "dx": 104,  
    "dy": 48,  
    "frame_dx": 261,  
    "frame_dy": 207,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        237,  
        183,  
        285,  
        231
```

```
    ],  
    "session_memory_key": "button:refresh",  
    "cached": false  
  },  
  {  
    "id": "address_bar",  
    "type": "input",  
    "label": "Address bar",  
    "description": "Browser address and search bar showing  
localhost:5173",  
    "state": "enabled",  
    "dx": 568,  
    "dy": 48,  
    "frame_dx": 725,  
    "frame_dy": 207,  
    "confidence": 0.98,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
      157,  
      175,  
      1293,  
      239  
    ],  
    "session_memory_key": "input:address bar",  
    "cached": false  
  },  
  {  
    "id": "window_minimize",  
    "type": "button",
```

```
    "label": "Minimize",
    "description": "Minimize the browser window",
    "state": "enabled",
    "dx": 1128,
    "dy": 28,
    "frame_dx": 1285,
    "frame_dy": 187,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        1261,
        167,
        1309,
        207
    ],
    "session_memory_key": "button:minimize",
    "cached": false
},
{
    "id": "window_maximize",
    "type": "button",
    "label": "Maximize",
    "description": "Maximize the browser window",
    "state": "enabled",
    "dx": 1156,
    "dy": 32,
    "frame_dx": 1313,
    "frame_dy": 191,
    "confidence": 0.9,
```

```
"source": "gemini_vlm",
"frame_bbox": [
    1293,
    175,
    1333,
    207
],
"session_memory_key": "button:maximize",
"cached": false
},
{
    "id": "window_close",
    "type": "button",
    "label": "Close",
    "description": "Close the browser window",
    "state": "enabled",
    "dx": 1184,
    "dy": 24,
    "frame_dx": 1341,
    "frame_dy": 183,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        1317,
        167,
        1365,
        199
    ],
    "session_memory_key": "button:close",
```



```
    "cached": false
  },
  {
    "id": "app_logo",
    "type": "icon",
    "label": "ScreenPilot AI Logo",
    "description": "Logo for the ScreenPilot AI application",
    "state": "visible",
    "dx": 48,
    "dy": 140,
    "frame_dx": 205,
    "frame_dy": 299,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
      189,
      279,
      221,
      319
    ],
    "session_memory_key": "icon:screenpilot ai logo",
    "cached": false
  },
  {
    "id": "app_title",
    "type": "heading",
    "label": "ScreenPilot AI",
    "description": "Main title of the application",
    "state": "visible",
```

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"dx": 136,  
  "dy": 136,  
  "frame_dx": 293,  
  "frame_dy": 295,  
  "confidence": 0.98,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    213,  
    279,  
    373,  
    311  
  ],  
  "session_memory_key": "heading:screenpilot ai",  
  "cached": false  
},  
{  
  "id": "app_version_badge",  
  "type": "label",  
  "label": "Autonomous Task Agent v2",  
  "description": "Badge indicating the application version",  
  "state": "visible",  
  "dx": 248,  
  "dy": 144,  
  "frame_dx": 405,  
  "frame_dy": 303,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    365,
```

```
        295,  
        445,  
        311  
    ],  
    "session_memory_key": "label:autonomous task agent v2",  
    "cached": false  
},  
{  
    "id": "status_connecting",  
    "type": "label",  
    "label": "Connecting to agent...",  
    "description": "Status message indicating connection progress",  
    "state": "visible",  
    "dx": 44,  
    "dy": 416,  
    "frame_dx": 201,  
    "frame_dy": 575,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        181,  
        559,  
        221,  
        591  
    ],  
    "session_memory_key": "label:connecting to agent...",  
    "cached": false  
},  
{
```

```
"id": "status_capturing",
"type": "label",
"label": "Capturing initial screen...",
"description": "Status message indicating screen capture progress",
"state": "visible",
"dx": 136,
"dy": 448,
"frame_dx": 293,
"frame_dy": 607,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    181,
    591,
    405,
    623
],
"session_memory_key": "label:capturing initial screen...",
"cached": false
},
{
  "id": "chat_input_field",
  "type": "input",
  "label": "Agent is working...",
  "description": "Chat input field with placeholder text",
  "state": "disabled",
  "dx": 672,
  "dy": 744,
  "frame_dx": 829,
```

```
"frame_dy": 903,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    157,
    863,
    1501,
    943
],
"session_memory_key": "input:agent is working...",
"cached": false
},
{
    "id": "send_button",
    "type": "button",
    "label": "Send",
    "description": "Button to send a message to the agent",
    "state": "enabled",
    "dx": 1144,
    "dy": 736,
    "frame_dx": 1301,
    "frame_dy": 895,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        1269,
        863,
        1333,
        927
```

```
    ],  
    "session_memory_key": "button:send",  
    "cached": false  
  },  
  {  
    "id": "windows_taskbar",  
    "type": "pane",  
    "label": "Taskbar",  
    "description": "Windows taskbar at the bottom of the screen",  
    "state": "visible",  
    "dx": 780,  
    "dy": 784,  
    "frame_dx": 937,  
    "frame_dy": 943,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
      157,  
      927,  
      1717,  
      959  
    ],  
    "session_memory_key": "pane:taskbar",  
    "cached": false  
  }  
],  
"resolved_elements": [  
  {  
    "id": "browser_back",
```

```
"type": "button",
"label": "Back",
"description": "Navigate back in browser history",
"state": "enabled",
"dx": 36,
"dy": 48,
"frame_dx": 193,
"frame_dy": 207,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    173,
    183,
    213,
    231
],
"session_memory_key": "button:back",
"cached": false
},
{
    "id": "browser_forward",
    "type": "button",
    "label": "Forward",
    "description": "Navigate forward in browser history",
    "state": "disabled",
    "dx": 72,
    "dy": 48,
    "frame_dx": 229,
    "frame_dy": 207,
```

```
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    205,
    183,
    253,
    231
],
"session_memory_key": "button:forward",
"cached": false
},
{
    "id": "browser_refresh",
    "type": "button",
    "label": "Refresh",
    "description": "Reload the current page",
    "state": "enabled",
    "dx": 104,
    "dy": 48,
    "frame_dx": 261,
    "frame_dy": 207,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        237,
        183,
        285,
        231
    ],
    ],
```



```
    "session_memory_key": "button:refresh",
    "cached": false
  },
  {
    "id": "address_bar",
    "type": "input",
    "label": "Address bar",
    "description": "Browser address and search bar showing
localhost:5173",
    "state": "enabled",
    "dx": 568,
    "dy": 48,
    "frame_dx": 725,
    "frame_dy": 207,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
      157,
      175,
      1293,
      239
    ],
    "session_memory_key": "input:address bar",
    "cached": false
  },
  {
    "id": "window_minimize",
    "type": "button",
    "label": "Minimize",
```

```
"description": "Minimize the browser window",
"state": "enabled",
"dx": 1128,
"dy": 28,
"frame_dx": 1285,
"frame_dy": 187,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    1261,
    167,
    1309,
    207
],
"session_memory_key": "button:minimize",
"cached": false
},
{
    "id": "window_maximize",
    "type": "button",
    "label": "Maximize",
    "description": "Maximize the browser window",
    "state": "enabled",
    "dx": 1156,
    "dy": 32,
    "frame_dx": 1313,
    "frame_dy": 191,
    "confidence": 0.9,
    "source": "gemini_vlm",
```

```
"frame_bbox": [  
    1293,  
    175,  
    1333,  
    207  
],  
"session_memory_key": "button:maximize",  
"cached": false  
},  
{  
    "id": "window_close",  
    "type": "button",  
    "label": "Close",  
    "description": "Close the browser window",  
    "state": "enabled",  
    "dx": 1184,  
    "dy": 24,  
    "frame_dx": 1341,  
    "frame_dy": 183,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        1317,  
        167,  
        1365,  
        199  
    ],  
    "session_memory_key": "button:close",  
    "cached": false
```

```
},  
{  
  "id": "app_logo",  
  "type": "icon",  
  "label": "ScreenPilot AI Logo",  
  "description": "Logo for the ScreenPilot AI application",  
  "state": "visible",  
  "dx": 48,  
  "dy": 140,  
  "frame_dx": 205,  
  "frame_dy": 299,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    189,  
    279,  
    221,  
    319  
  ],  
  "session_memory_key": "icon:screenpilot ai logo",  
  "cached": false  
},  
{  
  "id": "app_title",  
  "type": "heading",  
  "label": "ScreenPilot AI",  
  "description": "Main title of the application",  
  "state": "visible",  
  "dx": 136,
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```
"dy": 136,
"frame_dx": 293,
"frame_dy": 295,
"confidence": 0.98,
"source": "gemini_vlm",
"frame_bbox": [
    213,
    279,
    373,
    311
],
"session_memory_key": "heading:screenpilot ai",
"cached": false
},
{
    "id": "app_version_badge",
    "type": "label",
    "label": "Autonomous Task Agent v2",
    "description": "Badge indicating the application version",
    "state": "visible",
    "dx": 248,
    "dy": 144,
    "frame_dx": 405,
    "frame_dy": 303,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        365,
        295,
```

```
        445,  
        311  
    ],  
    "session_memory_key": "label:autonomous task agent v2",  
    "cached": false  
},  
{  
    "id": "status_connecting",  
    "type": "label",  
    "label": "Connecting to agent...",  
    "description": "Status message indicating connection progress",  
    "state": "visible",  
    "dx": 44,  
    "dy": 416,  
    "frame_dx": 201,  
    "frame_dy": 575,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        181,  
        559,  
        221,  
        591  
    ],  
    "session_memory_key": "label:connecting to agent...",  
    "cached": false  
},  
{  
    "id": "status_capturing",
```

```
"type": "label",
"label": "Capturing initial screen...",
"description": "Status message indicating screen capture progress",
"state": "visible",
"dx": 136,
"dy": 448,
"frame_dx": 293,
"frame_dy": 607,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    181,
    591,
    405,
    623
],
"session_memory_key": "label:capturing initial screen...",
"cached": false
},
{
    "id": "chat_input_field",
    "type": "input",
    "label": "Agent is working...",
    "description": "Chat input field with placeholder text",
    "state": "disabled",
    "dx": 672,
    "dy": 744,
    "frame_dx": 829,
    "frame_dy": 903,
```

```
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    157,
    863,
    1501,
    943
],
"session_memory_key": "input:agent is working...",
"cached": false
},
{
    "id": "send_button",
    "type": "button",
    "label": "Send",
    "description": "Button to send a message to the agent",
    "state": "enabled",
    "dx": 1144,
    "dy": 736,
    "frame_dx": 1301,
    "frame_dy": 895,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        1269,
        863,
        1333,
        927
    ],
    ],
```



```
    "session_memory_key": "button:send",
    "cached": false
  },
  {
    "id": "windows_taskbar",
    "type": "pane",
    "label": "Taskbar",
    "description": "Windows taskbar at the bottom of the screen",
    "state": "visible",
    "dx": 780,
    "dy": 784,
    "frame_dx": 937,
    "frame_dy": 943,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
      157,
      927,
      1717,
      959
    ],
    "session_memory_key": "pane:taskbar",
    "cached": false
  }
],
"session_knowledge_base": {
  "session_id": "shot_1777694279_7138a4",
  "started_at": "2026-05-02T03:57:59.714595+00:00",
  "updated_at": "2026-05-02T03:58:27.059343+00:00",
```

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"stale_after_misses": 3,
"confidence_floor": 0.4,
"decay_per_miss": 0.9,
"frame_count": 1,
"frame_name": "frame_1777694279_3c3018.jpg",
"frame_index": 0,
"image_size": {
  "width": 1920,
  "height": 1080
},
"known_elements": {
  "button:back": {
    "element_key": "button:back",
    "element_type": "button",
    "label": "Back",
    "bbox": [
      173.0,
      183.0,
      213.0,
      231.0
    ],
    "confidence": 0.95,
    "source": "gemini_vlm",
    "first_seen_frame": "frame_1777694279_3c3018.jpg",
    "last_seen_frame": "frame_1777694279_3c3018.jpg",
    "first_seen_at": "2026-05-02T03:58:27.059343+00:00",
    "last_seen_at": "2026-05-02T03:58:27.059343+00:00",
    "first_seen_index": 0,
    "last_seen_index": 0,
```

```
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777694279_3c3018.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T03:58:27.059343+00:00",
    "confidence": 0.95
  }
],
},
"button:forward": {
  "element_key": "button:forward",
  "element_type": "button",
  "label": "Forward",
  "bbox": [
    205.0,
    183.0,
    253.0,
    231.0
  ],
  "confidence": 0.95,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777694279_3c3018.jpg",
  "last_seen_frame": "frame_1777694279_3c3018.jpg",
  "first_seen_at": "2026-05-02T03:58:27.059343+00:00",
  "last_seen_at": "2026-05-02T03:58:27.059343+00:00",
```

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"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777694279_3c3018.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T03:58:27.059343+00:00",
    "confidence": 0.95
  }
],
},
"button:refresh": {
  "element_key": "button:refresh",
  "element_type": "button",
  "label": "Refresh",
  "bbox": [
    237.0,
    183.0,
    285.0,
    231.0
  ],
  "confidence": 0.95,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777694279_3c3018.jpg",
  "last_seen_frame": "frame_1777694279_3c3018.jpg",
```

```
"first_seen_at": "2026-05-02T03:58:27.059343+00:00",
"last_seen_at": "2026-05-02T03:58:27.059343+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777694279_3c3018.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T03:58:27.059343+00:00",
    "confidence": 0.95
  }
],
},
"input:address bar": {
  "element_key": "input:address bar",
  "element_type": "input",
  "label": "Address bar",
  "bbox": [
    157.0,
    175.0,
    1293.0,
    239.0
  ],
  "confidence": 0.98,
  "source": "gemini_vlm",
```

```
"first_seen_frame": "frame_1777694279_3c3018.jpg",
"last_seen_frame": "frame_1777694279_3c3018.jpg",
"first_seen_at": "2026-05-02T03:58:27.059343+00:00",
"last_seen_at": "2026-05-02T03:58:27.059343+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777694279_3c3018.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T03:58:27.059343+00:00",
    "confidence": 0.98
  }
],
},
"button:minimize": {
  "element_key": "button:minimize",
  "element_type": "button",
  "label": "Minimize",
  "bbox": [
    1261.0,
    167.0,
    1309.0,
    207.0
  ],
}
```

```
"confidence": 0.9,
"source": "gemini_vlm",
"first_seen_frame": "frame_1777694279_3c3018.jpg",
"last_seen_frame": "frame_1777694279_3c3018.jpg",
"first_seen_at": "2026-05-02T03:58:27.059343+00:00",
"last_seen_at": "2026-05-02T03:58:27.059343+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777694279_3c3018.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T03:58:27.059343+00:00",
    "confidence": 0.9
  }
],
"button:maximize": {
  "element_key": "button:maximize",
  "element_type": "button",
  "label": "Maximize",
  "bbox": [
    1293.0,
    175.0,
    1333.0,
```

```
207.0
],
"confidence": 0.9,
"source": "gemini_vlm",
"first_seen_frame": "frame_1777694279_3c3018.jpg",
"last_seen_frame": "frame_1777694279_3c3018.jpg",
"first_seen_at": "2026-05-02T03:58:27.059343+00:00",
"last_seen_at": "2026-05-02T03:58:27.059343+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777694279_3c3018.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T03:58:27.059343+00:00",
    "confidence": 0.9
  }
]
},
"button:close": {
  "element_key": "button:close",
  "element_type": "button",
  "label": "Close",
  "bbox": [
    1317.0,
```



```
        167.0,
        1365.0,
        199.0
    ],
    "confidence": 0.9,
    "source": "gemini_vlm",
    "first_seen_frame": "frame_1777694279_3c3018.jpg",
    "last_seen_frame": "frame_1777694279_3c3018.jpg",
    "first_seen_at": "2026-05-02T03:58:27.059343+00:00",
    "last_seen_at": "2026-05-02T03:58:27.059343+00:00",
    "first_seen_index": 0,
    "last_seen_index": 0,
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false,
    "history": [
        {
            "frame_name": "frame_1777694279_3c3018.jpg",
            "frame_index": 0,
            "seen_at": "2026-05-02T03:58:27.059343+00:00",
            "confidence": 0.9
        }
    ]
},
"icon:screenpilot ai logo": {
    "element_key": "icon:screenpilot ai logo",
    "element_type": "icon",
    "label": "ScreenPilot AI Logo",
```

```
"bbox": [
  189.0,
  279.0,
  221.0,
  319.0
],
"confidence": 0.95,
"source": "gemini_vlm",
"first_seen_frame": "frame_1777694279_3c3018.jpg",
"last_seen_frame": "frame_1777694279_3c3018.jpg",
"first_seen_at": "2026-05-02T03:58:27.059343+00:00",
"last_seen_at": "2026-05-02T03:58:27.059343+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777694279_3c3018.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T03:58:27.059343+00:00",
    "confidence": 0.95
  }
]
},
"heading:screenpilot ai": {
  "element_key": "heading:screenpilot ai",
```

```
"element_type": "heading",
"label": "ScreenPilot AI",
"bbox": [
    213.0,
    279.0,
    373.0,
    311.0
],
"confidence": 0.98,
"source": "gemini_vlm",
"first_seen_frame": "frame_1777694279_3c3018.jpg",
"last_seen_frame": "frame_1777694279_3c3018.jpg",
"first_seen_at": "2026-05-02T03:58:27.059343+00:00",
"last_seen_at": "2026-05-02T03:58:27.059343+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
    {
        "frame_name": "frame_1777694279_3c3018.jpg",
        "frame_index": 0,
        "seen_at": "2026-05-02T03:58:27.059343+00:00",
        "confidence": 0.98
    }
]
},
```

```
"label:autonomous task agent v2": {
  "element_key": "label:autonomous task agent v2",
  "element_type": "label",
  "label": "Autonomous Task Agent v2",
  "bbox": [
    365.0,
    295.0,
    445.0,
    311.0
  ],
  "confidence": 0.95,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777694279_3c3018.jpg",
  "last_seen_frame": "frame_1777694279_3c3018.jpg",
  "first_seen_at": "2026-05-02T03:58:27.059343+00:00",
  "last_seen_at": "2026-05-02T03:58:27.059343+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
  "seen_count": 1,
  "miss_count": 0,
  "stale": false,
  "cached": false,
  "history": [
    {
      "frame_name": "frame_1777694279_3c3018.jpg",
      "frame_index": 0,
      "seen_at": "2026-05-02T03:58:27.059343+00:00",
      "confidence": 0.95
    }
  ]
}
```

```
]
},
"label:connecting to agent...": {
  "element_key": "label:connecting to agent...",
  "element_type": "label",
  "label": "Connecting to agent...",
  "bbox": [
    181.0,
    559.0,
    221.0,
    591.0
  ],
  "confidence": 0.9,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777694279_3c3018.jpg",
  "last_seen_frame": "frame_1777694279_3c3018.jpg",
  "first_seen_at": "2026-05-02T03:58:27.059343+00:00",
  "last_seen_at": "2026-05-02T03:58:27.059343+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
  "seen_count": 1,
  "miss_count": 0,
  "stale": false,
  "cached": false,
  "history": [
    {
      "frame_name": "frame_1777694279_3c3018.jpg",
      "frame_index": 0,
      "seen_at": "2026-05-02T03:58:27.059343+00:00",
```

```
        "confidence": 0.9
      }
    ]
  },
  "label:capturing initial screen...": {
    "element_key": "label:capturing initial screen...",
    "element_type": "label",
    "label": "Capturing initial screen...",
    "bbox": [
      181.0,
      591.0,
      405.0,
      623.0
    ],
    "confidence": 0.9,
    "source": "gemini_vlm",
    "first_seen_frame": "frame_1777694279_3c3018.jpg",
    "last_seen_frame": "frame_1777694279_3c3018.jpg",
    "first_seen_at": "2026-05-02T03:58:27.059343+00:00",
    "last_seen_at": "2026-05-02T03:58:27.059343+00:00",
    "first_seen_index": 0,
    "last_seen_index": 0,
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false,
    "history": [
      {
        "frame_name": "frame_1777694279_3c3018.jpg",
```

```
        "frame_index": 0,
        "seen_at": "2026-05-02T03:58:27.059343+00:00",
        "confidence": 0.9
    }
]
},
"input:agent is working...": {
    "element_key": "input:agent is working...",
    "element_type": "input",
    "label": "Agent is working...",
    "bbox": [
        157.0,
        863.0,
        1501.0,
        943.0
    ],
    "confidence": 0.95,
    "source": "gemini_vlm",
    "first_seen_frame": "frame_1777694279_3c3018.jpg",
    "last_seen_frame": "frame_1777694279_3c3018.jpg",
    "first_seen_at": "2026-05-02T03:58:27.059343+00:00",
    "last_seen_at": "2026-05-02T03:58:27.059343+00:00",
    "first_seen_index": 0,
    "last_seen_index": 0,
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false,
    "history": [
```

```
{
  "frame_name": "frame_1777694279_3c3018.jpg",
  "frame_index": 0,
  "seen_at": "2026-05-02T03:58:27.059343+00:00",
  "confidence": 0.95
}
],
},
"button:send": {
  "element_key": "button:send",
  "element_type": "button",
  "label": "Send",
  "bbox": [
    1269.0,
    863.0,
    1333.0,
    927.0
  ],
  "confidence": 0.95,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777694279_3c3018.jpg",
  "last_seen_frame": "frame_1777694279_3c3018.jpg",
  "first_seen_at": "2026-05-02T03:58:27.059343+00:00",
  "last_seen_at": "2026-05-02T03:58:27.059343+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
  "seen_count": 1,
  "miss_count": 0,
  "stale": false,
```



```
"cached": false,
"history": [
  {
    "frame_name": "frame_1777694279_3c3018.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T03:58:27.059343+00:00",
    "confidence": 0.95
  }
],
},
"pane:taskbar": {
  "element_key": "pane:taskbar",
  "element_type": "pane",
  "label": "Taskbar",
  "bbox": [
    157.0,
    927.0,
    1717.0,
    959.0
  ],
  "confidence": 0.9,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777694279_3c3018.jpg",
  "last_seen_frame": "frame_1777694279_3c3018.jpg",
  "first_seen_at": "2026-05-02T03:58:27.059343+00:00",
  "last_seen_at": "2026-05-02T03:58:27.059343+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
  "seen_count": 1,
```

```
    "miss_count": 0,
    "stale": false,
    "cached": false,
    "history": [
      {
        "frame_name": "frame_1777694279_3c3018.jpg",
        "frame_index": 0,
        "seen_at": "2026-05-02T03:58:27.059343+00:00",
        "confidence": 0.9
      }
    ]
  }
},
"resolved_elements": [
  {
    "id": "browser_back",
    "type": "button",
    "label": "Back",
    "description": "Navigate back in browser history",
    "state": "enabled",
    "dx": 36,
    "dy": 48,
    "frame_dx": 193,
    "frame_dy": 207,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
      173,
      183,
```

```
        213,  
        231  
    ],  
    "session_memory_key": "button:back",  
    "cached": false  
},  
{  
    "id": "browser_forward",  
    "type": "button",  
    "label": "Forward",  
    "description": "Navigate forward in browser history",  
    "state": "disabled",  
    "dx": 72,  
    "dy": 48,  
    "frame_dx": 229,  
    "frame_dy": 207,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        205,  
        183,  
        253,  
        231  
    ],  
    "session_memory_key": "button:forward",  
    "cached": false  
},  
{  
    "id": "browser_refresh",
```

```
    "type": "button",
    "label": "Refresh",
    "description": "Reload the current page",
    "state": "enabled",
    "dx": 104,
    "dy": 48,
    "frame_dx": 261,
    "frame_dy": 207,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        237,
        183,
        285,
        231
    ],
    "session_memory_key": "button:refresh",
    "cached": false
},
{
    "id": "address_bar",
    "type": "input",
    "label": "Address bar",
    "description": "Browser address and search bar showing
localhost:5173",
    "state": "enabled",
    "dx": 568,
    "dy": 48,
    "frame_dx": 725,
```

```
"frame_dy": 207,  
"confidence": 0.98,  
"source": "gemini_vlm",  
"frame_bbox": [  
    157,  
    175,  
    1293,  
    239  
],  
"session_memory_key": "input:address bar",  
"cached": false  
},  
{  
    "id": "window_minimize",  
    "type": "button",  
    "label": "Minimize",  
    "description": "Minimize the browser window",  
    "state": "enabled",  
    "dx": 1128,  
    "dy": 28,  
    "frame_dx": 1285,  
    "frame_dy": 187,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        1261,  
        167,  
        1309,  
        207
```

```
    ],  
    "session_memory_key": "button:minimize",  
    "cached": false  
  },  
  {  
    "id": "window_maximize",  
    "type": "button",  
    "label": "Maximize",  
    "description": "Maximize the browser window",  
    "state": "enabled",  
    "dx": 1156,  
    "dy": 32,  
    "frame_dx": 1313,  
    "frame_dy": 191,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
      1293,  
      175,  
      1333,  
      207  
    ],  
    "session_memory_key": "button:maximize",  
    "cached": false  
  },  
  {  
    "id": "window_close",  
    "type": "button",  
    "label": "Close",
```

```
"description": "Close the browser window",
"state": "enabled",
"dx": 1184,
"dy": 24,
"frame_dx": 1341,
"frame_dy": 183,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    1317,
    167,
    1365,
    199
],
"session_memory_key": "button:close",
"cached": false
},
{
    "id": "app_logo",
    "type": "icon",
    "label": "ScreenPilot AI Logo",
    "description": "Logo for the ScreenPilot AI application",
    "state": "visible",
    "dx": 48,
    "dy": 140,
    "frame_dx": 205,
    "frame_dy": 299,
    "confidence": 0.95,
    "source": "gemini_vlm",
```

```
"frame_bbox": [
    189,
    279,
    221,
    319
],
"session_memory_key": "icon:screenpilot ai logo",
"cached": false
},
{
    "id": "app_title",
    "type": "heading",
    "label": "ScreenPilot AI",
    "description": "Main title of the application",
    "state": "visible",
    "dx": 136,
    "dy": 136,
    "frame_dx": 293,
    "frame_dy": 295,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
        213,
        279,
        373,
        311
    ],
    "session_memory_key": "heading:screenpilot ai",
    "cached": false
}
```



```
},  
{  
  "id": "app_version_badge",  
  "type": "label",  
  "label": "Autonomous Task Agent v2",  
  "description": "Badge indicating the application version",  
  "state": "visible",  
  "dx": 248,  
  "dy": 144,  
  "frame_dx": 405,  
  "frame_dy": 303,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    365,  
    295,  
    445,  
    311  
  ],  
  "session_memory_key": "label:autonomous task agent v2",  
  "cached": false  
},  
{  
  "id": "status_connecting",  
  "type": "label",  
  "label": "Connecting to agent...",  
  "description": "Status message indicating connection progress",  
  "state": "visible",  
  "dx": 44,
```

```
"dy": 416,  
  "frame_dx": 201,  
  "frame_dy": 575,  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    181,  
    559,  
    221,  
    591  
  ],  
  "session_memory_key": "label:connecting to agent...",  
  "cached": false  
},  
{  
  "id": "status_capturing",  
  "type": "label",  
  "label": "Capturing initial screen...",  
  "description": "Status message indicating screen capture progress",  
  "state": "visible",  
  "dx": 136,  
  "dy": 448,  
  "frame_dx": 293,  
  "frame_dy": 607,  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    181,  
    591,
```

```
        405,  
        623  
    ],  
    "session_memory_key": "label:capturing initial screen...",  
    "cached": false  
},  
{  
    "id": "chat_input_field",  
    "type": "input",  
    "label": "Agent is working...",  
    "description": "Chat input field with placeholder text",  
    "state": "disabled",  
    "dx": 672,  
    "dy": 744,  
    "frame_dx": 829,  
    "frame_dy": 903,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        157,  
        863,  
        1501,  
        943  
    ],  
    "session_memory_key": "input:agent is working...",  
    "cached": false  
},  
{  
    "id": "send_button",
```

```
"type": "button",
"label": "Send",
"description": "Button to send a message to the agent",
"state": "enabled",
"dx": 1144,
"dy": 736,
"frame_dx": 1301,
"frame_dy": 895,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    1269,
    863,
    1333,
    927
],
"session_memory_key": "button:send",
"cached": false
},
{
    "id": "windows_taskbar",
    "type": "pane",
    "label": "Taskbar",
    "description": "Windows taskbar at the bottom of the screen",
    "state": "visible",
    "dx": 780,
    "dy": 784,
    "frame_dx": 937,
    "frame_dy": 943,
```

```
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        157,  
        927,  
        1717,  
        959  
    ],  
    "session_memory_key": "pane:taskbar",  
    "cached": false  
}  
],  
"current_elements": [  
    {  
        "id": "browser_back",  
        "type": "button",  
        "label": "Back",  
        "description": "Navigate back in browser history",  
        "state": "enabled",  
        "dx": 36,  
        "dy": 48,  
        "frame_dx": 193,  
        "frame_dy": 207,  
        "confidence": 0.95,  
        "source": "gemini_vlm",  
        "frame_bbox": [  
            173,  
            183,  
            213,
```

```
        231
    ],
    "session_memory_key": "button:back",
    "cached": false
},
{
    "id": "browser_forward",
    "type": "button",
    "label": "Forward",
    "description": "Navigate forward in browser history",
    "state": "disabled",
    "dx": 72,
    "dy": 48,
    "frame_dx": 229,
    "frame_dy": 207,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        205,
        183,
        253,
        231
    ],
    "session_memory_key": "button:forward",
    "cached": false
},
{
    "id": "browser_refresh",
    "type": "button",
```

```
    "label": "Refresh",
    "description": "Reload the current page",
    "state": "enabled",
    "dx": 104,
    "dy": 48,
    "frame_dx": 261,
    "frame_dy": 207,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        237,
        183,
        285,
        231
    ],
    "session_memory_key": "button:refresh",
    "cached": false
},
{
    "id": "address_bar",
    "type": "input",
    "label": "Address bar",
    "description": "Browser address and search bar showing
localhost:5173",
    "state": "enabled",
    "dx": 568,
    "dy": 48,
    "frame_dx": 725,
    "frame_dy": 207,
```

```
"confidence": 0.98,
"source": "gemini_vlm",
"frame_bbox": [
    157,
    175,
    1293,
    239
],
"session_memory_key": "input:address bar",
"cached": false
},
{
    "id": "window_minimize",
    "type": "button",
    "label": "Minimize",
    "description": "Minimize the browser window",
    "state": "enabled",
    "dx": 1128,
    "dy": 28,
    "frame_dx": 1285,
    "frame_dy": 187,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        1261,
        167,
        1309,
        207
    ],
    ],
```



```
    "session_memory_key": "button:minimize",
    "cached": false
  },
  {
    "id": "window_maximize",
    "type": "button",
    "label": "Maximize",
    "description": "Maximize the browser window",
    "state": "enabled",
    "dx": 1156,
    "dy": 32,
    "frame_dx": 1313,
    "frame_dy": 191,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
      1293,
      175,
      1333,
      207
    ],
    "session_memory_key": "button:maximize",
    "cached": false
  },
  {
    "id": "window_close",
    "type": "button",
    "label": "Close",
    "description": "Close the browser window",
```

```
    "state": "enabled",
    "dx": 1184,
    "dy": 24,
    "frame_dx": 1341,
    "frame_dy": 183,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        1317,
        167,
        1365,
        199
    ],
    "session_memory_key": "button:close",
    "cached": false
},
{
    "id": "app_logo",
    "type": "icon",
    "label": "ScreenPilot AI Logo",
    "description": "Logo for the ScreenPilot AI application",
    "state": "visible",
    "dx": 48,
    "dy": 140,
    "frame_dx": 205,
    "frame_dy": 299,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
```

```
        189,  
        279,  
        221,  
        319  
    ],  
    "session_memory_key": "icon:screenpilot ai logo",  
    "cached": false  
},  
{  
    "id": "app_title",  
    "type": "heading",  
    "label": "ScreenPilot AI",  
    "description": "Main title of the application",  
    "state": "visible",  
    "dx": 136,  
    "dy": 136,  
    "frame_dx": 293,  
    "frame_dy": 295,  
    "confidence": 0.98,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        213,  
        279,  
        373,  
        311  
    ],  
    "session_memory_key": "heading:screenpilot ai",  
    "cached": false  
},
```

```
{
  "id": "app_version_badge",
  "type": "label",
  "label": "Autonomous Task Agent v2",
  "description": "Badge indicating the application version",
  "state": "visible",
  "dx": 248,
  "dy": 144,
  "frame_dx": 405,
  "frame_dy": 303,
  "confidence": 0.95,
  "source": "gemini_vlm",
  "frame_bbox": [
    365,
    295,
    445,
    311
  ],
  "session_memory_key": "label:autonomous task agent v2",
  "cached": false
},
{
  "id": "status_connecting",
  "type": "label",
  "label": "Connecting to agent...",
  "description": "Status message indicating connection progress",
  "state": "visible",
  "dx": 44,
  "dy": 416,
```

```
"frame_dx": 201,
"frame_dy": 575,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    181,
    559,
    221,
    591
],
"session_memory_key": "label:connecting to agent...",
"cached": false
},
{
    "id": "status_capturing",
    "type": "label",
    "label": "Capturing initial screen...",
    "description": "Status message indicating screen capture progress",
    "state": "visible",
    "dx": 136,
    "dy": 448,
    "frame_dx": 293,
    "frame_dy": 607,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        181,
        591,
        405,
```

```
        623
    ],
    "session_memory_key": "label:capturing initial screen...",
    "cached": false
},
{
    "id": "chat_input_field",
    "type": "input",
    "label": "Agent is working...",
    "description": "Chat input field with placeholder text",
    "state": "disabled",
    "dx": 672,
    "dy": 744,
    "frame_dx": 829,
    "frame_dy": 903,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        157,
        863,
        1501,
        943
    ],
    "session_memory_key": "input:agent is working...",
    "cached": false
},
{
    "id": "send_button",
    "type": "button",
```

```
    "label": "Send",
    "description": "Button to send a message to the agent",
    "state": "enabled",
    "dx": 1144,
    "dy": 736,
    "frame_dx": 1301,
    "frame_dy": 895,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        1269,
        863,
        1333,
        927
    ],
    "session_memory_key": "button:send",
    "cached": false
},
{
    "id": "windows_taskbar",
    "type": "pane",
    "label": "Taskbar",
    "description": "Windows taskbar at the bottom of the screen",
    "state": "visible",
    "dx": 780,
    "dy": 784,
    "frame_dx": 937,
    "frame_dy": 943,
    "confidence": 0.9,
```

```
        "source": "gemini_vlm",
        "frame_bbox": [
            157,
            927,
            1717,
            959
        ],
        "session_memory_key": "pane:taskbar",
        "cached": false
    }
]
}
}
},
"todo": {
    "steps": [
        {
            "id": 1,
            "action": "Move the mouse to the 'Minimize' button at (1285, 187) and click it.",
            "target": "Minimize button",
            "expected_result": "The current browser window is minimized, revealing the desktop or taskbar.",
            "needs_user_data": false,
            "user_data_field": null,
            "status": "done",
            "_coord_fix_tried": true
        },
        {
```



```
    "id": 2,
    "action": "Identify the Visual Studio Code icon on the taskbar or desktop and click it.",
    "target": "Visual Studio Code application icon",
    "expected_result": "Visual Studio Code application is launched.",
    "needs_user_data": false,
    "user_data_field": null,
    "status": "failed"
  }
],
  "notes": "The browser window needs to be minimized first to see the desktop/taskbar icons clearly. Then the VS Code icon can be targeted.",
  "estimated_complexity": "simple"
},
"step_plans": [
  {
    "step_id": 1,
    "action": "Move the mouse to the 'Minimize' button at (1285, 187) and click it.",
    "target": "Minimize button",
    "expected_result": "The current browser window is minimized, revealing the desktop or taskbar.",
    "hid_commands": [
      {
        "cmd": "mouse_move",
        "dx": 1285,
        "dy": 187
      },
      {
        "cmd": "mouse_click",
```

```
        "button": "left"
    }
],
    "reasoning": "Moving the mouse to the Minimize button at the coordinates
specified in the current step (1285, 187) and clicking to minimize the window."
},
{
    "step_id": 2,
    "action": "Identify the Visual Studio Code icon on the taskbar or desktop
and click it.",
    "target": "Visual Studio Code application icon",
    "expected_result": "Visual Studio Code application is launched.",
    "hid_commands": [
        {
            "cmd": "mouse_move",
            "meta": {
                "commandId": "a36cc08c-fd4a-4b3f-8c7c-78655bacef5e"
            },
            "dx": 208,
            "dy": 218,
            "smooth": true
        },
        {
            "cmd": "mouse_click",
            "meta": {
                "commandId": "ed5880be-b242-401a-a172-1ba729c01116"
            },
            "button": "left"
        }
    ],
}
```

```
    "reasoning": "Fallback generated 2 commands from detected elements"
  }
],
"step_results": [
  {
    "step_id": 1,
    "action": "Move the mouse to the 'Minimize' button at (1285, 187) and click it.",
    "attempt": 3,
    "exec_status": "success",
    "eval_status": "done",
    "confidence": 0.95,
    "reason": "The browser window is no longer visible and desktop icons are now detected, confirming the window was successfully minimized.",
    "final_status": "executing",
    "timestamp": "2026-05-02T04:05:37.266026+00:00"
  },
  {
    "step_id": 2,
    "action": "Identify the Visual Studio Code icon on the taskbar or desktop and click it.",
    "attempt": 3,
    "exec_status": "success",
    "eval_status": "retry",
    "confidence": 0.95,
    "reason": "The Visual Studio Code application window is not detected in the current screen state. Only desktop icons for Google Chrome and HuionTablet are visible.",
    "final_status": "executing",
    "timestamp": "2026-05-02T04:10:03.593409+00:00"
  }
]
```

```
],
"final_report": {
  "success": false,
  "summary": "The window was minimized, but the Visual Studio Code icon could not be located to complete the task.",
  "message": "The task was partially successful. The active window was minimized as requested, but the subsequent step to open Visual Studio Code failed because the application icon was not detected on the desktop or taskbar.\n\nStep results:\n\u2705 Step 1: Move the mouse to the 'Minimize' button and click it - Completed\n\u274c Step 2: Identify the Visual Studio Code icon and click it - Failed (Element not found)",
  "steps_completed": 1,
  "steps_failed": 1,
  "issues": [
    "Visual Studio Code icon was not visible in the final screen state."
  ],
  "recommendations": [
    "Verify that Visual Studio Code is installed on the system.",
    "Ensure the application icon is pinned to the taskbar or visible on the desktop.",
    "Try using the search function or Start menu to launch the application."
  ]
},
"session_memory": {
  "run_id": "85f20837-9a8e-4c22-9298-4e2401f6166c",
  "user_task": "Minimize the current window using minimize icon and find the visual studio code application icon on the screen and click on that app icon",
  "started_at": "2026-05-02T03:57:53.860689+00:00",
  "updated_at": "2026-05-02T04:10:40.059078+00:00",
  "stale_after_misses": 2,
  "confidence_floor": 0.4,
  "decay_per_miss": 0.9,
```

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"frame_index": 16,
"known_elements": {
  "button:back": {
    "element_key": "button:back",
    "element_type": "button",
    "label": "Back",
    "bbox": [
      173.0,
      183.0,
      213.0,
      231.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 0,
    "last_seen_index": 0,
    "first_seen_at": "2026-05-02T03:58:27.103947+00:00",
    "last_seen_at": "2026-05-02T03:58:27.103947+00:00",
    "seen_count": 1,
    "miss_count": 15,
    "stale": true,
    "cached": true
  },
  "button:forward": {
    "element_key": "button:forward",
    "element_type": "button",
    "label": "Forward",
```

```
"bbox": [
    205.0,
    183.0,
    253.0,
    231.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 0,
"first_seen_at": "2026-05-02T03:58:27.103947+00:00",
"last_seen_at": "2026-05-02T03:58:27.103947+00:00",
"seen_count": 1,
"miss_count": 15,
"stale": true,
"cached": true
},
"button:refresh": {
    "element_key": "button:refresh",
    "element_type": "button",
    "label": "Refresh",
    "bbox": [
        237.0,
        183.0,
        285.0,
        231.0
    ],
    ],
```

```
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 0,
"first_seen_at": "2026-05-02T03:58:27.103947+00:00",
"last_seen_at": "2026-05-02T03:58:27.103947+00:00",
"seen_count": 1,
"miss_count": 15,
"stale": true,
"cached": true
},
"input:address bar": {
  "element_key": "input:address bar",
  "element_type": "input",
  "label": "Address Bar",
  "bbox": [
    295.0,
    172.0,
    1728.0,
    252.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 6,
```

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"first_seen_at": "2026-05-02T03:58:27.103947+00:00",
"last_seen_at": "2026-05-02T04:03:11.701272+00:00",
"seen_count": 3,
"miss_count": 9,
"stale": true,
"cached": true
},
"button:minimize": {
  "element_key": "button:minimize",
  "element_type": "button",
  "label": "Minimize",
  "bbox": [
    1607.0,
    132.0,
    1655.0,
    196.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 6,
  "first_seen_at": "2026-05-02T03:58:27.103947+00:00",
  "last_seen_at": "2026-05-02T04:03:11.701272+00:00",
  "seen_count": 2,
  "miss_count": 9,
  "stale": true,
  "cached": true
}
```



```
},  
"button:maximize": {  
  "element_key": "button:maximize",  
  "element_type": "button",  
  "label": "Maximize",  
  "bbox": [  
    1647.0,  
    132.0,  
    1695.0,  
    188.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 0,  
  "last_seen_index": 6,  
  "first_seen_at": "2026-05-02T03:58:27.103947+00:00",  
  "last_seen_at": "2026-05-02T04:03:11.701272+00:00",  
  "seen_count": 2,  
  "miss_count": 9,  
  "stale": true,  
  "cached": true  
},  
"button:close": {  
  "element_key": "button:close",  
  "element_type": "button",  
  "label": "Close",  
  "bbox": [  

```

```
    1687.0,  
    132.0,  
    1728.0,  
    196.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 0,  
  "last_seen_index": 6,  
  "first_seen_at": "2026-05-02T03:58:27.103947+00:00",  
  "last_seen_at": "2026-05-02T04:03:11.701272+00:00",  
  "seen_count": 2,  
  "miss_count": 9,  
  "stale": true,  
  "cached": true  
},  
"icon:screenpilot ai logo": {  
  "element_key": "icon:screenpilot ai logo",  
  "element_type": "icon",  
  "label": "ScreenPilot AI Logo",  
  "bbox": [  
    157.0,  
    220.0,  
    461.0,  
    292.0  
  ],  
  "confidence": 0.4,
```

```
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 2,
"first_seen_at": "2026-05-02T03:58:27.103947+00:00",
"last_seen_at": "2026-05-02T04:00:07.860022+00:00",
"seen_count": 3,
"miss_count": 13,
"stale": true,
"cached": true
},
"heading:screenpilot ai": {
  "element_key": "heading:screenpilot ai",
  "element_type": "heading",
  "label": "ScreenPilot AI",
  "bbox": [
    295.0,
    260.0,
    423.0,
    324.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 6,
  "first_seen_at": "2026-05-02T03:58:27.103947+00:00",
```

```
    "last_seen_at": "2026-05-02T04:03:11.701272+00:00",
    "seen_count": 2,
    "miss_count": 9,
    "stale": true,
    "cached": true
  },
  "label:autonomous task agent v2": {
    "element_key": "label:autonomous task agent v2",
    "element_type": "label",
    "label": "Autonomous Task Agent v2",
    "bbox": [
      399.0,
      268.0,
      631.0,
      316.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 0,
    "last_seen_index": 6,
    "first_seen_at": "2026-05-02T03:58:27.103947+00:00",
    "last_seen_at": "2026-05-02T04:03:11.701272+00:00",
    "seen_count": 3,
    "miss_count": 9,
    "stale": true,
    "cached": true
  },
}
```

```
"label:connecting to agent...": {
  "element_key": "label:connecting to agent...",
  "element_type": "label",
  "label": "Connecting to agent...",
  "bbox": [
    181.0,
    559.0,
    221.0,
    591.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 0,
  "first_seen_at": "2026-05-02T03:58:27.103947+00:00",
  "last_seen_at": "2026-05-02T03:58:27.103947+00:00",
  "seen_count": 1,
  "miss_count": 15,
  "stale": true,
  "cached": true
},
"label:capturing initial screen...": {
  "element_key": "label:capturing initial screen...",
  "element_type": "label",
  "label": "Capturing initial screen...",
  "bbox": [
    181.0,
```

```
    591.0,  
    405.0,  
    623.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 0,  
  "last_seen_index": 0,  
  "first_seen_at": "2026-05-02T03:58:27.103947+00:00",  
  "last_seen_at": "2026-05-02T03:58:27.103947+00:00",  
  "seen_count": 1,  
  "miss_count": 15,  
  "stale": true,  
  "cached": true  
},  
"input:agent is working...": {  
  "element_key": "input:agent is working...",  
  "element_type": "input",  
  "label": "Agent is working...",  
  "bbox": [  
    157.0,  
    863.0,  
    1501.0,  
    943.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",
```

```
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 0,
"first_seen_at": "2026-05-02T03:58:27.103947+00:00",
"last_seen_at": "2026-05-02T03:58:27.103947+00:00",
"seen_count": 1,
"miss_count": 15,
"stale": true,
"cached": true
},
"button:send": {
  "element_key": "button:send",
  "element_type": "button",
  "label": "Send",
  "bbox": [
    1583.0,
    852.0,
    1671.0,
    940.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 6,
  "first_seen_at": "2026-05-02T03:58:27.103947+00:00",
  "last_seen_at": "2026-05-02T04:03:11.701272+00:00",
```

```
"seen_count": 2,
"miss_count": 9,
"stale": true,
"cached": true
},
"pane:taskbar": {
  "element_key": "pane:taskbar",
  "element_type": "pane",
  "label": "Taskbar",
  "bbox": [
    391.0,
    919.0,
    1726.0,
    1009.0
  ],
  "confidence": 0.4261,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 7,
  "first_seen_at": "2026-05-02T03:58:27.103947+00:00",
  "last_seen_at": "2026-05-02T04:03:52.665977+00:00",
  "seen_count": 2,
  "miss_count": 8,
  "stale": true,
  "cached": true
},
"text:screenpilot ai": {
```



```
"element_key": "text:screenpilot ai",
"element_type": "text",
"label": "ScreenPilot AI",
"bbox": [
    205.0,
    268.0,
    381.0,
    324.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 1,
"last_seen_index": 1,
"first_seen_at": "2026-05-02T03:59:19.177241+00:00",
"last_seen_at": "2026-05-02T03:59:19.177241+00:00",
"seen_count": 1,
"miss_count": 14,
"stale": true,
"cached": true
},
"tab:frontend": {
    "element_key": "tab:frontend",
    "element_type": "tab",
    "label": "frontend",
    "bbox": [
        181.0,
        122.0,
```

```
    397.0,  
    194.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 3,  
  "last_seen_index": 5,  
  "first_seen_at": "2026-05-02T04:00:50.911986+00:00",  
  "last_seen_at": "2026-05-02T04:02:26.654344+00:00",  
  "seen_count": 3,  
  "miss_count": 10,  
  "stale": true,  
  "cached": true  
},  
"button:new tab": {  
  "element_key": "button:new tab",  
  "element_type": "button",  
  "label": "New Tab",  
  "bbox": [  
    423.0,  
    132.0,  
    479.0,  
    188.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,
```

```
"last_seen_frame": null,
"first_seen_index": 3,
"last_seen_index": 6,
"first_seen_at": "2026-05-02T04:00:50.911986+00:00",
"last_seen_at": "2026-05-02T04:03:11.701272+00:00",
"seen_count": 3,
"miss_count": 9,
"stale": true,
"cached": true
},
"tab:localhost:5173": {
  "element_key": "tab:localhost:5173",
  "element_type": "tab",
  "label": "localhost:5173",
  "bbox": [
    295.0,
    124.0,
    455.0,
    188.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 6,
  "last_seen_index": 6,
  "first_seen_at": "2026-05-02T04:03:11.701272+00:00",
  "last_seen_at": "2026-05-02T04:03:11.701272+00:00",
  "seen_count": 1,
```

```
"miss_count": 9,
"stale": true,
"cached": true
},
"input:chat input": {
  "element_key": "input:chat input",
  "element_type": "input",
  "label": "Chat Input",
  "bbox": [
    495.0,
    356.0,
    1728.0,
    500.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 6,
  "last_seen_index": 6,
  "first_seen_at": "2026-05-02T04:03:11.701272+00:00",
  "last_seen_at": "2026-05-02T04:03:11.701272+00:00",
  "seen_count": 1,
  "miss_count": 9,
  "stale": true,
  "cached": true
},
"heading:plan": {
  "element_key": "heading:plan",
```

```
"element_type": "heading",
"label": "Plan",
"bbox": [
    295.0,
    492.0,
    359.0,
    548.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 6,
"last_seen_index": 6,
"first_seen_at": "2026-05-02T04:03:11.701272+00:00",
"last_seen_at": "2026-05-02T04:03:11.701272+00:00",
"seen_count": 1,
"miss_count": 9,
"stale": true,
"cached": true
},
"label:0/2 done": {
    "element_key": "label:0/2 done",
    "element_type": "label",
    "label": "0/2 done",
    "bbox": [
        863.0,
        508.0,
        959.0,
```

```
532.0

],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 6,
"last_seen_index": 6,
"first_seen_at": "2026-05-02T04:03:11.701272+00:00",
"last_seen_at": "2026-05-02T04:03:11.701272+00:00",
"seen_count": 1,
"miss_count": 9,
"stale": true,
"cached": true
},
"text:move the mouse to the minimize button": {
  "element_key": "text:move the mouse to the minimize button",
  "element_type": "text",
  "label": "Move the mouse to the Minimize button",
  "bbox": [
    295.0,
    572.0,
    1055.0,
    628.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
```

```
"first_seen_index": 6,
"last_seen_index": 6,
"first_seen_at": "2026-05-02T04:03:11.701272+00:00",
"last_seen_at": "2026-05-02T04:03:11.701272+00:00",
"seen_count": 1,
"miss_count": 9,
"stale": true,
"cached": true
},
"text:identify the visual studio code icon": {
  "element_key": "text:identify the visual studio code icon",
  "element_type": "text",
  "label": "Identify the Visual Studio Code icon",
  "bbox": [
    295.0,
    628.0,
    1063.0,
    684.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 6,
  "last_seen_index": 6,
  "first_seen_at": "2026-05-02T04:03:11.701272+00:00",
  "last_seen_at": "2026-05-02T04:03:11.701272+00:00",
  "seen_count": 1,
  "miss_count": 9,
```

```
    "stale": true,
    "cached": true
  },
  "pane:log output": {
    "element_key": "pane:log output",
    "element_type": "pane",
    "label": "Log Output",
    "bbox": [
      295.0,
      668.0,
      1087.0,
      892.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 6,
    "last_seen_index": 6,
    "first_seen_at": "2026-05-02T04:03:11.701272+00:00",
    "last_seen_at": "2026-05-02T04:03:11.701272+00:00",
    "seen_count": 1,
    "miss_count": 9,
    "stale": true,
    "cached": true
  },
  "toolbar:taskbar": {
    "element_key": "toolbar:taskbar",
    "element_type": "toolbar",
```



```
"label": "Taskbar",
"bbox": [
  295.0,
  940.0,
  1728.0,
  1010.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 6,
"last_seen_index": 6,
"first_seen_at": "2026-05-02T04:03:11.701272+00:00",
"last_seen_at": "2026-05-02T04:03:11.701272+00:00",
"seen_count": 1,
"miss_count": 9,
"stale": true,
"cached": true
},
"button:start": {
  "element_key": "button:start",
  "element_type": "button",
  "label": "Start",
  "bbox": [
    567.0,
    943.0,
    615.0,
    983.0
```

```
],
  "confidence": 0.4089,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 6,
  "last_seen_index": 7,
  "first_seen_at": "2026-05-02T04:03:11.701272+00:00",
  "last_seen_at": "2026-05-02T04:03:52.665977+00:00",
  "seen_count": 2,
  "miss_count": 8,
  "stale": true,
  "cached": true
},
"input:search": {
  "element_key": "input:search",
  "element_type": "input",
  "label": "Search",
  "bbox": [
    591.0,
    935.0,
    863.0,
    983.0
  ],
  "confidence": 0.4089,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 6,
```

```
"last_seen_index": 7,  
"first_seen_at": "2026-05-02T04:03:11.701272+00:00",  
"last_seen_at": "2026-05-02T04:03:52.665977+00:00",  
"seen_count": 2,  
"miss_count": 8,  
"stale": true,  
"cached": true  
},  
"icon:file explorer": {  
  "element_key": "icon:file explorer",  
  "element_type": "icon",  
  "label": "File Explorer",  
  "bbox": [  
    919.0,  
    935.0,  
    967.0,  
    983.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 6,  
  "last_seen_index": 7,  
  "first_seen_at": "2026-05-02T04:03:11.701272+00:00",  
  "last_seen_at": "2026-05-02T04:03:52.665977+00:00",  
  "seen_count": 2,  
  "miss_count": 8,  
  "stale": true,
```

```
    "cached": true
  },
  "icon:microsoft edge": {
    "element_key": "icon:microsoft edge",
    "element_type": "icon",
    "label": "Microsoft Edge",
    "bbox": [
      823.0,
      948.0,
      871.0,
      996.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 6,
    "last_seen_index": 6,
    "first_seen_at": "2026-05-02T04:03:11.701272+00:00",
    "last_seen_at": "2026-05-02T04:03:11.701272+00:00",
    "seen_count": 1,
    "miss_count": 9,
    "stale": true,
    "cached": true
  },
  "icon:google chrome": {
    "element_key": "icon:google chrome",
    "element_type": "icon",
    "label": "Google Chrome",
```

```
"bbox": [
  152.0,
  117.0,
  264.0,
  309.0
],
"confidence": 0.99,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 6,
"last_seen_index": 15,
"first_seen_at": "2026-05-02T04:03:11.701272+00:00",
"last_seen_at": "2026-05-02T04:10:40.059078+00:00",
"seen_count": 8,
"miss_count": 0,
"stale": false,
"cached": false
},
"icon:visual studio code": {
  "element_key": "icon:visual studio code",
  "element_type": "icon",
  "label": "Visual Studio Code",
  "bbox": [
    1167.0,
    935.0,
    1215.0,
    983.0
  ],

```

```
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 6,
"last_seen_index": 7,
"first_seen_at": "2026-05-02T04:03:11.701272+00:00",
"last_seen_at": "2026-05-02T04:03:52.665977+00:00",
"seen_count": 2,
"miss_count": 8,
"stale": true,
"cached": true
},
"icon:spotify": {
  "element_key": "icon:spotify",
  "element_type": "icon",
  "label": "Spotify",
  "bbox": [
    1207.0,
    935.0,
    1255.0,
    983.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 6,
  "last_seen_index": 7,
```

```
"first_seen_at": "2026-05-02T04:03:11.701272+00:00",
"last_seen_at": "2026-05-02T04:03:52.665977+00:00",
"seen_count": 2,
"miss_count": 8,
"stale": true,
"cached": true
},
"label:clock and date": {
  "element_key": "label:clock and date",
  "element_type": "label",
  "label": "Clock and Date",
  "bbox": [
    1615.0,
    935.0,
    1719.0,
    991.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 6,
  "last_seen_index": 7,
  "first_seen_at": "2026-05-02T04:03:11.701272+00:00",
  "last_seen_at": "2026-05-02T04:03:52.665977+00:00",
  "seen_count": 2,
  "miss_count": 8,
  "stale": true,
  "cached": true
}
```

```
},
"icon:postman": {
  "element_key": "icon:postman",
  "element_type": "icon",
  "label": "Postman",
  "bbox": [
    391.0,
    735.0,
    487.0,
    895.0
  ],
  "confidence": 0.4089,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 7,
  "last_seen_index": 7,
  "first_seen_at": "2026-05-02T04:03:52.665977+00:00",
  "last_seen_at": "2026-05-02T04:03:52.665977+00:00",
  "seen_count": 1,
  "miss_count": 8,
  "stale": true,
  "cached": true
},
"icon:loom": {
  "element_key": "icon:loom",
  "element_type": "icon",
  "label": "Loom",
  "bbox": [
```



```
    471.0,
    735.0,
    583.0,
    895.0
  ],
  "confidence": 0.4089,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 7,
  "last_seen_index": 7,
  "first_seen_at": "2026-05-02T04:03:52.665977+00:00",
  "last_seen_at": "2026-05-02T04:03:52.665977+00:00",
  "seen_count": 1,
  "miss_count": 8,
  "stale": true,
  "cached": true
},
"icon:vlc media player": {
  "element_key": "icon:vlc media player",
  "element_type": "icon",
  "label": "VLC media player",
  "bbox": [
    559.0,
    735.0,
    671.0,
    895.0
  ],
  "confidence": 0.4089,
```

```
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 7,
"last_seen_index": 7,
"first_seen_at": "2026-05-02T04:03:52.665977+00:00",
"last_seen_at": "2026-05-02T04:03:52.665977+00:00",
"seen_count": 1,
"miss_count": 8,
"stale": true,
"cached": true
},
"pane:sticky note": {
  "element_key": "pane:sticky note",
  "element_type": "pane",
  "label": "Sticky Note",
  "bbox": [
    1359.0,
    135.0,
    1726.0,
    391.0
  ],
  "confidence": 0.4218,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 7,
  "last_seen_index": 7,
  "first_seen_at": "2026-05-02T04:03:52.665977+00:00",
```

```
    "last_seen_at": "2026-05-02T04:03:52.665977+00:00",
    "seen_count": 1,
    "miss_count": 8,
    "stale": true,
    "cached": true
  },
  "text:amazon workspaces credentials": {
    "element_key": "text:amazon workspaces credentials",
    "element_type": "text",
    "label": "Amazon WorkSpaces credentials",
    "bbox": [
      1367.0,
      191.0,
      1726.0,
      327.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 7,
    "last_seen_index": 7,
    "first_seen_at": "2026-05-02T04:03:52.665977+00:00",
    "last_seen_at": "2026-05-02T04:03:52.665977+00:00",
    "seen_count": 1,
    "miss_count": 8,
    "stale": true,
    "cached": true
  },
}
```

```
"icon:microsoft store": {
  "element_key": "icon:microsoft store",
  "element_type": "icon",
  "label": "Microsoft Store",
  "bbox": [
    959.0,
    935.0,
    1007.0,
    983.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 7,
  "last_seen_index": 7,
  "first_seen_at": "2026-05-02T04:03:52.665977+00:00",
  "last_seen_at": "2026-05-02T04:03:52.665977+00:00",
  "seen_count": 1,
  "miss_count": 8,
  "stale": true,
  "cached": true
},
"icon:mail": {
  "element_key": "icon:mail",
  "element_type": "icon",
  "label": "Mail",
  "bbox": [
    999.0,
```

```
    935.0,
    1047.0,
    983.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 7,
"last_seen_index": 7,
"first_seen_at": "2026-05-02T04:03:52.665977+00:00",
"last_seen_at": "2026-05-02T04:03:52.665977+00:00",
"seen_count": 1,
"miss_count": 8,
"stale": true,
"cached": true
},
"icon:xbox": {
  "element_key": "icon:xbox",
  "element_type": "icon",
  "label": "Xbox",
  "bbox": [
    1039.0,
    935.0,
    1095.0,
    983.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
```

```
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 7,
"last_seen_index": 7,
"first_seen_at": "2026-05-02T04:03:52.665977+00:00",
"last_seen_at": "2026-05-02T04:03:52.665977+00:00",
"seen_count": 1,
"miss_count": 8,
"stale": true,
"cached": true
},
"icon:notepad": {
  "element_key": "icon:notepad",
  "element_type": "icon",
  "label": "Notepad",
  "bbox": [
    1127.0,
    935.0,
    1175.0,
    983.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 7,
  "last_seen_index": 7,
  "first_seen_at": "2026-05-02T04:03:52.665977+00:00",
  "last_seen_at": "2026-05-02T04:03:52.665977+00:00",
```

```
"seen_count": 1,
"miss_count": 8,
"stale": true,
"cached": true
},
"button:show hidden icons": {
  "element_key": "button:show hidden icons",
  "element_type": "button",
  "label": "Show hidden icons",
  "bbox": [
    1447.0,
    943.0,
    1479.0,
    967.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 7,
  "last_seen_index": 7,
  "first_seen_at": "2026-05-02T04:03:52.665977+00:00",
  "last_seen_at": "2026-05-02T04:03:52.665977+00:00",
  "seen_count": 1,
  "miss_count": 8,
  "stale": true,
  "cached": true
},
"icon:wi-fi": {
```

```
"element_key": "icon:wi-fi",
"element_type": "icon",
"label": "Wi-Fi",
"bbox": [
  1527.0,
  943.0,
  1559.0,
  983.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 7,
"last_seen_index": 7,
"first_seen_at": "2026-05-02T04:03:52.665977+00:00",
"last_seen_at": "2026-05-02T04:03:52.665977+00:00",
"seen_count": 1,
"miss_count": 8,
"stale": true,
"cached": true
},
"icon:volume": {
  "element_key": "icon:volume",
  "element_type": "icon",
  "label": "Volume",
  "bbox": [
    1559.0,
    943.0,
```



```
    1591.0,  
    983.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 7,  
  "last_seen_index": 7,  
  "first_seen_at": "2026-05-02T04:03:52.665977+00:00",  
  "last_seen_at": "2026-05-02T04:03:52.665977+00:00",  
  "seen_count": 1,  
  "miss_count": 8,  
  "stale": true,  
  "cached": true  
},  
"icon:battery": {  
  "element_key": "icon:battery",  
  "element_type": "icon",  
  "label": "Battery",  
  "bbox": [  
    1591.0,  
    943.0,  
    1623.0,  
    983.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,
```

```
"last_seen_frame": null,
"first_seen_index": 7,
"last_seen_index": 7,
"first_seen_at": "2026-05-02T04:03:52.665977+00:00",
"last_seen_at": "2026-05-02T04:03:52.665977+00:00",
"seen_count": 1,
"miss_count": 8,
"stale": true,
"cached": true
},
"icon:huiontablet": {
  "element_key": "icon:huiontablet",
  "element_type": "icon",
  "label": "HuionTablet",
  "bbox": [
    264.0,
    117.0,
    408.0,
    317.0
  ],
  "confidence": 0.891,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 8,
  "last_seen_index": 14,
  "first_seen_at": "2026-05-02T04:04:41.281802+00:00",
  "last_seen_at": "2026-05-02T04:09:59.806216+00:00",
  "seen_count": 7,
```

```
    "miss_count": 1,
    "stale": false,
    "cached": true
  },
  "icon:amazon workspaces": {
    "element_key": "icon:amazon workspaces",
    "element_type": "icon",
    "label": "Amazon WorkSpaces",
    "bbox": [
      278.0,
      315.0,
      382.0,
      523.0
    ],
    "confidence": 0.5609,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 8,
    "last_seen_index": 10,
    "first_seen_at": "2026-05-02T04:04:41.281802+00:00",
    "last_seen_at": "2026-05-02T04:06:17.096635+00:00",
    "seen_count": 2,
    "miss_count": 5,
    "stale": true,
    "cached": true
  }
},
"current_elements": [],
```

```
"resolved_elements": [  
  {  
    "id": "memory_icon:google chrome",  
    "type": "icon",  
    "label": "Google Chrome",  
    "description": "Cached from an earlier frame in the same run",  
    "state": "cached",  
    "bbox": [  
      152.0,  
      117.0,  
      264.0,  
      309.0  
    ],  
    "confidence": 0.99,  
    "source": "gemini_vlm",  
    "last_seen_frame": null,  
    "last_seen_index": 15,  
    "stale": true,  
    "cached": true,  
    "session_memory_key": "icon:google chrome"  
  },  
  {  
    "id": "memory_icon:huiontablet",  
    "type": "icon",  
    "label": "HuionTablet",  
    "description": "Cached from an earlier frame in the same run",  
    "state": "cached",  
    "bbox": [  
      264.0,
```

```
    117.0,  
    408.0,  
    317.0  
  ],  
  "confidence": 0.891,  
  "source": "gemini_vlm",  
  "last_seen_frame": null,  
  "last_seen_index": 14,  
  "stale": true,  
  "cached": true,  
  "session_memory_key": "icon:huiontablet"  
}  
],  
"screen_data": {}  
}  
}
```

Test Case 06: Minimize & Launch Chrome (Complex Task)

User Prompt:

Minimize the current window using minimize icon and find the Google Chrome icon on the screen and click on that app icon

Test Results:

- Steps Total: 2
- Completed: 1
- Failed: 1
- Duration: 9 minutes 50 seconds

Step Execution:

Step 1: Move the mouse to the 'Minimize' button at (1611, 162) and click it to hide the current window

- **Target:** Minimize button
- **Attempts:** 1
- **Execution:** Performed
- **Evaluation:** Success
- **Issue:** None

Step 2: Move the mouse to the 'Chrome' icon at (791, 978) on the taskbar and click it

- **Target:** Chrome icon
- **Attempts:** 3
- **Execution:** Performed
- **Evaluation:** Failed (retry loop)
- **Issue:** Evaluation error — No valid JSON found in LLM response ('{\n "status": "retry",'}). Additionally, coordinates used (791, 978) did not match the detected position of the Google Chrome icon.

Execution Summary:

The agent successfully minimized the current window but failed to complete the second step due to a malformed JSON response from the LLM during evaluation. The click action on the Chrome icon may have executed, but the evaluation could not confirm success. A coordinate mismatch between the planned target and actual icon location was also observed.

Issues Identified:

- **Malformed JSON response from LLM during evaluation**
- **Coordinate mismatch between planned click position and actual Chrome icon location**
- **No fallback mechanism to update coordinates dynamically**
- **Stale cached element data may have interfered with fresh detection**
- **Evaluation loop unable to handle incomplete or malformed LLM responses**

Recommendations for Improvement:

- **Enforce strict JSON output formatting with validation before parsing**
- **Implement coordinate refresh logic before each execution attempt**
- **Add fallback to re-detect target element if initial coordinates fail**
- **Improve evaluation prompt to always return valid JSON, even on failure**
- **Add retry with updated coordinates instead of hardcoding original positions**

JSON Response:

full_cycle.json

```
{  
  "run_id": "f60f0afa-3ce3-4f83-9e73-d88f5282905b",
```

```
"user_task": "Minimize the current window using minimize icon and find the  
Google Chrome icon on the screen and click on that app icon",  
  
"started_at": "2026-05-02T04:29:00.097697+00:00",  
  
"finished_at": "2026-05-02T04:38:50.753830+00:00",  
  
"vision_used": true,  
  
"perception": {  
  "image": "E:\\sliit-  
project\\ScreenAwareTaskAgent\\vision\\data\\sessions\\shot_1777696146_8f4fe6\\pr  
ocessing\\proc_1777696146_8ffec3\\preprocessed_frames\\frame_1777696146_e68365.jp  
g",  
  "image_size": {  
    "width": 1920,  
    "height": 1080  
  },  
  "screen_bbox": [  
    151,  
    110,  
    1727,  
    1013  
  ],  
  "screen_size": {  
    "width": 1576,  
    "height": 903  
  },  
  "coordinate_system": "pixel",  
  "elements": [  
    {  
      "id": "browser_tab_frontend",  
      "type": "tab",  
      "label": "frontend",
```

```
"description": "Browser tab for the frontend application",
"state": "visible",
"dx": 124,
"dy": 36,
"frame_dx": 275,
"frame_dy": 146,
"confidence": 0.98,
"source": "gemini_vlm",
"frame_bbox": [
    151,
    118,
    399,
    174
],
"session_memory_key": "tab:frontend",
"cached": false
},
{
    "id": "address_bar",
    "type": "input",
    "label": "Address Bar",
    "description": "Browser address bar showing localhost:5173",
    "state": "visible",
    "dx": 576,
    "dy": 100,
    "frame_dx": 727,
    "frame_dy": 210,
    "confidence": 0.99,
    "source": "gemini_vlm",
```



```
"frame_bbox": [
    151,
    182,
    1303,
    238
],
"session_memory_key": "input:address bar",
"cached": false
},
{
    "id": "ask_gemini_button",
    "type": "button",
    "label": "Ask Gemini",
    "description": "Ask Gemini button in the browser toolbar",
    "state": "visible",
    "dx": 1348,
    "dy": 56,
    "frame_dx": 1499,
    "frame_dy": 166,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        1415,
        142,
        1583,
        190
    ],
    "session_memory_key": "button:ask gemini",
    "cached": false
}
```

```
},  
{  
  "id": "window_minimize",  
  "type": "button",  
  "label": "Minimize",  
  "description": "Browser window minimize button",  
  "state": "visible",  
  "dx": 1460,  
  "dy": 52,  
  "frame_dx": 1611,  
  "frame_dy": 162,  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    1583,  
    142,  
    1639,  
    182  
  ],  
  "session_memory_key": "button:minimize",  
  "cached": false  
},  
{  
  "id": "window_maximize",  
  "type": "button",  
  "label": "Maximize",  
  "description": "Browser window maximize button",  
  "state": "visible",  
  "dx": 1500,
```

```
"dy": 52,
"frame_dx": 1651,
"frame_dy": 162,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    1623,
    142,
    1679,
    182
],
"session_memory_key": "button:maximize",
"cached": false
},
{
    "id": "window_close",
    "type": "button",
    "label": "Close",
    "description": "Browser window close button",
    "state": "visible",
    "dx": 1548,
    "dy": 56,
    "frame_dx": 1699,
    "frame_dy": 166,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        1671,
        142,
```

```
        1727,  
        190  
    ],  
    "session_memory_key": "button:close",  
    "cached": false  
},  
{  
    "id": "screenpilot_ai_title",  
    "type": "heading",  
    "label": "ScreenPilot AI",  
    "description": "Application title ScreenPilot AI",  
    "state": "visible",  
    "dx": 120,  
    "dy": 192,  
    "frame_dx": 271,  
    "frame_dy": 302,  
    "confidence": 0.99,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        167,  
        270,  
        375,  
        334  
    ],  
    "session_memory_key": "heading:screenpilot ai",  
    "cached": false  
},  
{  
    "id": "agent_version_badge",
```

```
"type": "label",
"label": "Autonomous Task Agent v2",
"description": "Badge indicating the agent version",
"state": "visible",
"dx": 308,
"dy": 196,
"frame_dx": 459,
"frame_dy": 306,
"confidence": 0.97,
"source": "gemini_vlm",
"frame_bbox": [
    423,
    286,
    495,
    326
],
"session_memory_key": "label:autonomous task agent v2",
"cached": false
},
{
    "id": "status_connecting",
    "type": "text",
    "label": "Connecting to agent...",
    "description": "Status message indicating connection progress",
    "state": "visible",
    "dx": 120,
    "dy": 436,
    "frame_dx": 271,
    "frame_dy": 546,
```

```
"confidence": 0.96,  
"source": "gemini_vlm",  
"frame_bbox": [  
    183,  
    526,  
    359,  
    566  
],  
"session_memory_key": "text:connecting to agent...",  
"cached": false  
},  
{  
    "id": "status_capturing",  
    "type": "text",  
    "label": "Capturing initial screen...",  
    "description": "Status message indicating screen capture progress",  
    "state": "visible",  
    "dx": 148,  
    "dy": 468,  
    "frame_dx": 299,  
    "frame_dy": 578,  
    "confidence": 0.96,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        183,  
        558,  
        415,  
        598  
    ],
```

```
"session_memory_key": "text:capturing initial screen...",
"cached": false
},
{
  "id": "preview_window",
  "type": "pane",
  "label": "Preview Window",
  "description": "Small preview window showing agent activity",
  "state": "visible",
  "dx": 116,
  "dy": 588,
  "frame_dx": 267,
  "frame_dy": 698,
  "confidence": 0.95,
  "source": "gemini_vlm",
  "frame_bbox": [
    167,
    598,
    367,
    798
  ],
  "session_memory_key": "pane:preview window",
  "cached": false
},
{
  "id": "chat_input_field",
  "type": "input",
  "label": "Agent is working...",
  "description": "Main chat input field for the agent",
```

```
"state": "visible",
"dx": 788,
"dy": 796,
"frame_dx": 939,
"frame_dy": 906,
"confidence": 0.98,
"source": "gemini_vlm",
"frame_bbox": [
    151,
    862,
    1727,
    950
],
"session_memory_key": "input:agent is working...",
"cached": false
},
{
    "id": "send_button",
    "type": "button",
    "label": "Send",
    "description": "Button to send commands to the agent",
    "state": "visible",
    "dx": 1472,
    "dy": 788,
    "frame_dx": 1623,
    "frame_dy": 898,
    "confidence": 0.97,
    "source": "gemini_vlm",
    "frame_bbox": [
```



```
    1583,  
    854,  
    1663,  
    942  
  ],  
  "session_memory_key": "button:send",  
  "cached": false  
},  
{  
  "id": "windows_taskbar",  
  "type": "pane",  
  "label": "Taskbar",  
  "description": "Windows taskbar at the bottom of the screen",  
  "state": "visible",  
  "dx": 788,  
  "dy": 864,  
  "frame_dx": 939,  
  "frame_dy": 974,  
  "confidence": 0.99,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    151,  
    934,  
    1727,  
    1013  
  ],  
  "session_memory_key": "pane:taskbar",  
  "cached": false  
},
```

```
{
  "id": "start_button",
  "type": "button",
  "label": "Start",
  "description": "Windows Start button",
  "state": "visible",
  "dx": 428,
  "dy": 864,
  "frame_dx": 579,
  "frame_dy": 974,
  "confidence": 0.98,
  "source": "gemini_vlm",
  "frame_bbox": [
    551,
    950,
    607,
    998
  ],
  "session_memory_key": "button:start",
  "cached": false
},
{
  "id": "search_bar",
  "type": "input",
  "label": "Search",
  "description": "Windows search bar in the taskbar",
  "state": "visible",
  "dx": 528,
  "dy": 864,
```

```
"frame_dx": 679,  
"frame_dy": 974,  
"confidence": 0.98,  
"source": "gemini_vlm",  
"frame_bbox": [  
    591,  
    950,  
    767,  
    998  
],  
"session_memory_key": "input:search",  
"cached": false  
},  
{  
    "id": "weather_widget",  
    "type": "button",  
    "label": "Weather",  
    "description": "Weather widget in the taskbar",  
    "state": "visible",  
    "dx": 92,  
    "dy": 876,  
    "frame_dx": 243,  
    "frame_dy": 986,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        167,  
        958,  
        319,
```

```
1013
],
"session_memory_key": "button:weather",
"cached": false
},
{
  "id": "taskbar_file_explorer",
  "type": "icon",
  "label": "File Explorer",
  "description": "File Explorer icon in the taskbar",
  "state": "visible",
  "dx": 612,
  "dy": 868,
  "frame_dx": 763,
  "frame_dy": 978,
  "confidence": 0.95,
  "source": "gemini_vlm",
  "frame_bbox": [
    743,
    958,
    783,
    998
  ],
  "session_memory_key": "icon:file explorer",
  "cached": false
},
{
  "id": "taskbar_chrome",
  "type": "icon",
```

```
"label": "Chrome",
"description": "Google Chrome icon in the taskbar",
"state": "visible",
"dx": 640,
"dy": 868,
"frame_dx": 791,
"frame_dy": 978,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    775,
    958,
    807,
    998
],
"session_memory_key": "icon:chrome",
"cached": false
},
{
    "id": "taskbar_settings",
    "type": "icon",
    "label": "Settings",
    "description": "Settings icon in the taskbar",
    "state": "visible",
    "dx": 672,
    "dy": 868,
    "frame_dx": 823,
    "frame_dy": 978,
    "confidence": 0.95,
```

```
"source": "gemini_vlm",
"frame_bbox": [
    807,
    958,
    839,
    998
],
"session_memory_key": "icon:settings",
"cached": false
},
{
    "id": "taskbar_vscode",
    "type": "icon",
    "label": "VS Code",
    "description": "Visual Studio Code icon in the taskbar",
    "state": "visible",
    "dx": 804,
    "dy": 868,
    "frame_dx": 955,
    "frame_dy": 978,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        935,
        958,
        975,
        998
    ],
    "session_memory_key": "icon:vs code",
```

```
    "cached": false
  },
  {
    "id": "taskbar_spotify",
    "type": "icon",
    "label": "Spotify",
    "description": "Spotify icon in the taskbar",
    "state": "visible",
    "dx": 844,
    "dy": 868,
    "frame_dx": 995,
    "frame_dy": 978,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
      975,
      958,
      1015,
      998
    ],
    "session_memory_key": "icon:spotify",
    "cached": false
  },
  {
    "id": "system_tray",
    "type": "pane",
    "label": "System Tray",
    "description": "System tray area with network, volume, and battery icons",
```

```
"state": "visible",
"dx": 1416,
"dy": 864,
"frame_dx": 1567,
"frame_dy": 974,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    1487,
    950,
    1647,
    998
],
"session_memory_key": "pane:system tray",
"cached": false
},
{
    "id": "clock_date",
    "type": "text",
    "label": "Clock and Date",
    "description": "System clock and date display in the taskbar",
    "state": "visible",
    "dx": 1520,
    "dy": 868,
    "frame_dx": 1671,
    "frame_dy": 978,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
```



```
        1615,  
        942,  
        1727,  
        1013  
    ],  
    "session_memory_key": "text:clock and date",  
    "cached": false  
}  
],  
"element_count": 24,  
"current_elements": [  
    {  
        "id": "browser_tab_frontend",  
        "type": "tab",  
        "label": "frontend",  
        "description": "Browser tab for the frontend application",  
        "state": "visible",  
        "dx": 124,  
        "dy": 36,  
        "frame_dx": 275,  
        "frame_dy": 146,  
        "confidence": 0.98,  
        "source": "gemini_vlm",  
        "frame_bbox": [  
            151,  
            118,  
            399,  
            174  
        ],  
    },  
]
```

```
"session_memory_key": "tab:frontend",
"cached": false
},
{
  "id": "address_bar",
  "type": "input",
  "label": "Address Bar",
  "description": "Browser address bar showing localhost:5173",
  "state": "visible",
  "dx": 576,
  "dy": 100,
  "frame_dx": 727,
  "frame_dy": 210,
  "confidence": 0.99,
  "source": "gemini_vlm",
  "frame_bbox": [
    151,
    182,
    1303,
    238
  ],
  "session_memory_key": "input:address bar",
  "cached": false
},
{
  "id": "ask_gemini_button",
  "type": "button",
  "label": "Ask Gemini",
  "description": "Ask Gemini button in the browser toolbar",
```

```
"state": "visible",
"dx": 1348,
"dy": 56,
"frame_dx": 1499,
"frame_dy": 166,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    1415,
    142,
    1583,
    190
],
"session_memory_key": "button:ask gemini",
"cached": false
},
{
    "id": "window_minimize",
    "type": "button",
    "label": "Minimize",
    "description": "Browser window minimize button",
    "state": "visible",
    "dx": 1460,
    "dy": 52,
    "frame_dx": 1611,
    "frame_dy": 162,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
```

```
    1583,  
    142,  
    1639,  
    182  
  ],  
  "session_memory_key": "button:minimize",  
  "cached": false  
},  
{  
  "id": "window_maximize",  
  "type": "button",  
  "label": "Maximize",  
  "description": "Browser window maximize button",  
  "state": "visible",  
  "dx": 1500,  
  "dy": 52,  
  "frame_dx": 1651,  
  "frame_dy": 162,  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    1623,  
    142,  
    1679,  
    182  
  ],  
  "session_memory_key": "button:maximize",  
  "cached": false  
},
```

```
{
  "id": "window_close",
  "type": "button",
  "label": "Close",
  "description": "Browser window close button",
  "state": "visible",
  "dx": 1548,
  "dy": 56,
  "frame_dx": 1699,
  "frame_dy": 166,
  "confidence": 0.9,
  "source": "gemini_vlm",
  "frame_bbox": [
    1671,
    142,
    1727,
    190
  ],
  "session_memory_key": "button:close",
  "cached": false
},
{
  "id": "screenpilot_ai_title",
  "type": "heading",
  "label": "ScreenPilot AI",
  "description": "Application title ScreenPilot AI",
  "state": "visible",
  "dx": 120,
  "dy": 192,
```

```
"frame_dx": 271,  
"frame_dy": 302,  
"confidence": 0.99,  
"source": "gemini_vlm",  
"frame_bbox": [  
    167,  
    270,  
    375,  
    334  
],  
"session_memory_key": "heading:screenpilot ai",  
"cached": false  
},  
{  
    "id": "agent_version_badge",  
    "type": "label",  
    "label": "Autonomous Task Agent v2",  
    "description": "Badge indicating the agent version",  
    "state": "visible",  
    "dx": 308,  
    "dy": 196,  
    "frame_dx": 459,  
    "frame_dy": 306,  
    "confidence": 0.97,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        423,  
        286,  
        495,
```

```
326
],
"session_memory_key": "label:autonomous task agent v2",
"cached": false
},
{
  "id": "status_connecting",
  "type": "text",
  "label": "Connecting to agent...",
  "description": "Status message indicating connection progress",
  "state": "visible",
  "dx": 120,
  "dy": 436,
  "frame_dx": 271,
  "frame_dy": 546,
  "confidence": 0.96,
  "source": "gemini_vlm",
  "frame_bbox": [
    183,
    526,
    359,
    566
  ],
  "session_memory_key": "text:connecting to agent...",
  "cached": false
},
{
  "id": "status_capturing",
  "type": "text",
```

```
"label": "Capturing initial screen...",
"description": "Status message indicating screen capture progress",
"state": "visible",
"dx": 148,
"dy": 468,
"frame_dx": 299,
"frame_dy": 578,
"confidence": 0.96,
"source": "gemini_vlm",
"frame_bbox": [
    183,
    558,
    415,
    598
],
"session_memory_key": "text:capturing initial screen...",
"cached": false
},
{
    "id": "preview_window",
    "type": "pane",
    "label": "Preview Window",
    "description": "Small preview window showing agent activity",
    "state": "visible",
    "dx": 116,
    "dy": 588,
    "frame_dx": 267,
    "frame_dy": 698,
    "confidence": 0.95,
```



```
"source": "gemini_vlm",
"frame_bbox": [
    167,
    598,
    367,
    798
],
"session_memory_key": "pane:preview window",
"cached": false
},
{
    "id": "chat_input_field",
    "type": "input",
    "label": "Agent is working...",
    "description": "Main chat input field for the agent",
    "state": "visible",
    "dx": 788,
    "dy": 796,
    "frame_dx": 939,
    "frame_dy": 906,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
        151,
        862,
        1727,
        950
    ],
    "session_memory_key": "input:agent is working..."
```

```
"cached": false
},
{
  "id": "send_button",
  "type": "button",
  "label": "Send",
  "description": "Button to send commands to the agent",
  "state": "visible",
  "dx": 1472,
  "dy": 788,
  "frame_dx": 1623,
  "frame_dy": 898,
  "confidence": 0.97,
  "source": "gemini_vlm",
  "frame_bbox": [
    1583,
    854,
    1663,
    942
  ],
  "session_memory_key": "button:send",
  "cached": false
},
{
  "id": "windows_taskbar",
  "type": "pane",
  "label": "Taskbar",
  "description": "Windows taskbar at the bottom of the screen",
  "state": "visible",
```

```
"dx": 788,  
"dy": 864,  
"frame_dx": 939,  
"frame_dy": 974,  
"confidence": 0.99,  
"source": "gemini_vlm",  
"frame_bbox": [  
    151,  
    934,  
    1727,  
    1013  
],  
"session_memory_key": "pane:taskbar",  
"cached": false  
},  
{  
    "id": "start_button",  
    "type": "button",  
    "label": "Start",  
    "description": "Windows Start button",  
    "state": "visible",  
    "dx": 428,  
    "dy": 864,  
    "frame_dx": 579,  
    "frame_dy": 974,  
    "confidence": 0.98,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        551,
```

```
        950,  
        607,  
        998  
    ],  
    "session_memory_key": "button:start",  
    "cached": false  
},  
{  
    "id": "search_bar",  
    "type": "input",  
    "label": "Search",  
    "description": "Windows search bar in the taskbar",  
    "state": "visible",  
    "dx": 528,  
    "dy": 864,  
    "frame_dx": 679,  
    "frame_dy": 974,  
    "confidence": 0.98,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        591,  
        950,  
        767,  
        998  
    ],  
    "session_memory_key": "input:search",  
    "cached": false  
},  
{
```

```
"id": "weather_widget",
"type": "button",
"label": "Weather",
"description": "Weather widget in the taskbar",
"state": "visible",
"dx": 92,
"dy": 876,
"frame_dx": 243,
"frame_dy": 986,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    167,
    958,
    319,
    1013
],
"session_memory_key": "button:weather",
"cached": false
},
{
    "id": "taskbar_file_explorer",
    "type": "icon",
    "label": "File Explorer",
    "description": "File Explorer icon in the taskbar",
    "state": "visible",
    "dx": 612,
    "dy": 868,
    "frame_dx": 763,
```

```
"frame_dy": 978,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    743,
    958,
    783,
    998
],
"session_memory_key": "icon:file explorer",
"cached": false
},
{
    "id": "taskbar_chrome",
    "type": "icon",
    "label": "Chrome",
    "description": "Google Chrome icon in the taskbar",
    "state": "visible",
    "dx": 640,
    "dy": 868,
    "frame_dx": 791,
    "frame_dy": 978,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        775,
        958,
        807,
        998
    ]
}
```

```
    ],  
    "session_memory_key": "icon:chrome",  
    "cached": false  
  },  
  {  
    "id": "taskbar_settings",  
    "type": "icon",  
    "label": "Settings",  
    "description": "Settings icon in the taskbar",  
    "state": "visible",  
    "dx": 672,  
    "dy": 868,  
    "frame_dx": 823,  
    "frame_dy": 978,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
      807,  
      958,  
      839,  
      998  
    ],  
    "session_memory_key": "icon:settings",  
    "cached": false  
  },  
  {  
    "id": "taskbar_vscode",  
    "type": "icon",  
    "label": "VS Code",
```

```
"description": "Visual Studio Code icon in the taskbar",
"state": "visible",
"dx": 804,
"dy": 868,
"frame_dx": 955,
"frame_dy": 978,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    935,
    958,
    975,
    998
],
"session_memory_key": "icon:vs code",
"cached": false
},
{
    "id": "taskbar_spotify",
    "type": "icon",
    "label": "Spotify",
    "description": "Spotify icon in the taskbar",
    "state": "visible",
    "dx": 844,
    "dy": 868,
    "frame_dx": 995,
    "frame_dy": 978,
    "confidence": 0.95,
    "source": "gemini_vlm",
```



```
"frame_bbox": [
  975,
  958,
  1015,
  998
],
"session_memory_key": "icon:spotify",
"cached": false
},
{
  "id": "system_tray",
  "type": "pane",
  "label": "System Tray",
  "description": "System tray area with network, volume, and battery
icons",
  "state": "visible",
  "dx": 1416,
  "dy": 864,
  "frame_dx": 1567,
  "frame_dy": 974,
  "confidence": 0.95,
  "source": "gemini_vlm",
  "frame_bbox": [
    1487,
    950,
    1647,
    998
  ],
  "session_memory_key": "pane:system tray",
```

```
    "cached": false
  },
  {
    "id": "clock_date",
    "type": "text",
    "label": "Clock and Date",
    "description": "System clock and date display in the taskbar",
    "state": "visible",
    "dx": 1520,
    "dy": 868,
    "frame_dx": 1671,
    "frame_dy": 978,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
      1615,
      942,
      1727,
      1013
    ],
    "session_memory_key": "text:clock and date",
    "cached": false
  }
],
"resolved_elements": [
  {
    "id": "browser_tab_frontend",
    "type": "tab",
    "label": "frontend",
```

```
"description": "Browser tab for the frontend application",
"state": "visible",
"dx": 124,
"dy": 36,
"frame_dx": 275,
"frame_dy": 146,
"confidence": 0.98,
"source": "gemini_vlm",
"frame_bbox": [
    151,
    118,
    399,
    174
],
"session_memory_key": "tab:frontend",
"cached": false
},
{
    "id": "address_bar",
    "type": "input",
    "label": "Address Bar",
    "description": "Browser address bar showing localhost:5173",
    "state": "visible",
    "dx": 576,
    "dy": 100,
    "frame_dx": 727,
    "frame_dy": 210,
    "confidence": 0.99,
    "source": "gemini_vlm",
```

```
"frame_bbox": [
    151,
    182,
    1303,
    238
],
"session_memory_key": "input:address bar",
"cached": false
},
{
    "id": "ask_gemini_button",
    "type": "button",
    "label": "Ask Gemini",
    "description": "Ask Gemini button in the browser toolbar",
    "state": "visible",
    "dx": 1348,
    "dy": 56,
    "frame_dx": 1499,
    "frame_dy": 166,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        1415,
        142,
        1583,
        190
    ],
    "session_memory_key": "button:ask gemini",
    "cached": false
}
```

```
},  
{  
  "id": "window_minimize",  
  "type": "button",  
  "label": "Minimize",  
  "description": "Browser window minimize button",  
  "state": "visible",  
  "dx": 1460,  
  "dy": 52,  
  "frame_dx": 1611,  
  "frame_dy": 162,  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    1583,  
    142,  
    1639,  
    182  
  ],  
  "session_memory_key": "button:minimize",  
  "cached": false  
},  
{  
  "id": "window_maximize",  
  "type": "button",  
  "label": "Maximize",  
  "description": "Browser window maximize button",  
  "state": "visible",  
  "dx": 1500,
```

```
"dy": 52,
"frame_dx": 1651,
"frame_dy": 162,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    1623,
    142,
    1679,
    182
],
"session_memory_key": "button:maximize",
"cached": false
},
{
    "id": "window_close",
    "type": "button",
    "label": "Close",
    "description": "Browser window close button",
    "state": "visible",
    "dx": 1548,
    "dy": 56,
    "frame_dx": 1699,
    "frame_dy": 166,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        1671,
        142,
```

```
        1727,  
        190  
    ],  
    "session_memory_key": "button:close",  
    "cached": false  
},  
{  
    "id": "screenpilot_ai_title",  
    "type": "heading",  
    "label": "ScreenPilot AI",  
    "description": "Application title ScreenPilot AI",  
    "state": "visible",  
    "dx": 120,  
    "dy": 192,  
    "frame_dx": 271,  
    "frame_dy": 302,  
    "confidence": 0.99,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        167,  
        270,  
        375,  
        334  
    ],  
    "session_memory_key": "heading:screenpilot ai",  
    "cached": false  
},  
{  
    "id": "agent_version_badge",
```

```
"type": "label",
"label": "Autonomous Task Agent v2",
"description": "Badge indicating the agent version",
"state": "visible",
"dx": 308,
"dy": 196,
"frame_dx": 459,
"frame_dy": 306,
"confidence": 0.97,
"source": "gemini_vlm",
"frame_bbox": [
    423,
    286,
    495,
    326
],
"session_memory_key": "label:autonomous task agent v2",
"cached": false
},
{
    "id": "status_connecting",
    "type": "text",
    "label": "Connecting to agent...",
    "description": "Status message indicating connection progress",
    "state": "visible",
    "dx": 120,
    "dy": 436,
    "frame_dx": 271,
    "frame_dy": 546,
```



```
"confidence": 0.96,  
"source": "gemini_vlm",  
"frame_bbox": [  
    183,  
    526,  
    359,  
    566  
],  
"session_memory_key": "text:connecting to agent...",  
"cached": false  
},  
{  
    "id": "status_capturing",  
    "type": "text",  
    "label": "Capturing initial screen...",  
    "description": "Status message indicating screen capture progress",  
    "state": "visible",  
    "dx": 148,  
    "dy": 468,  
    "frame_dx": 299,  
    "frame_dy": 578,  
    "confidence": 0.96,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        183,  
        558,  
        415,  
        598  
    ],
```

```
"session_memory_key": "text:capturing initial screen...",
"cached": false
},
{
  "id": "preview_window",
  "type": "pane",
  "label": "Preview Window",
  "description": "Small preview window showing agent activity",
  "state": "visible",
  "dx": 116,
  "dy": 588,
  "frame_dx": 267,
  "frame_dy": 698,
  "confidence": 0.95,
  "source": "gemini_vlm",
  "frame_bbox": [
    167,
    598,
    367,
    798
  ],
  "session_memory_key": "pane:preview window",
  "cached": false
},
{
  "id": "chat_input_field",
  "type": "input",
  "label": "Agent is working...",
  "description": "Main chat input field for the agent",
```

```
"state": "visible",
"dx": 788,
"dy": 796,
"frame_dx": 939,
"frame_dy": 906,
"confidence": 0.98,
"source": "gemini_vlm",
"frame_bbox": [
    151,
    862,
    1727,
    950
],
"session_memory_key": "input:agent is working...",
"cached": false
},
{
    "id": "send_button",
    "type": "button",
    "label": "Send",
    "description": "Button to send commands to the agent",
    "state": "visible",
    "dx": 1472,
    "dy": 788,
    "frame_dx": 1623,
    "frame_dy": 898,
    "confidence": 0.97,
    "source": "gemini_vlm",
    "frame_bbox": [
```

```
        1583,  
        854,  
        1663,  
        942  
    ],  
    "session_memory_key": "button:send",  
    "cached": false  
},  
{  
    "id": "windows_taskbar",  
    "type": "pane",  
    "label": "Taskbar",  
    "description": "Windows taskbar at the bottom of the screen",  
    "state": "visible",  
    "dx": 788,  
    "dy": 864,  
    "frame_dx": 939,  
    "frame_dy": 974,  
    "confidence": 0.99,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        151,  
        934,  
        1727,  
        1013  
    ],  
    "session_memory_key": "pane:taskbar",  
    "cached": false  
},
```

```
{
  "id": "start_button",
  "type": "button",
  "label": "Start",
  "description": "Windows Start button",
  "state": "visible",
  "dx": 428,
  "dy": 864,
  "frame_dx": 579,
  "frame_dy": 974,
  "confidence": 0.98,
  "source": "gemini_vlm",
  "frame_bbox": [
    551,
    950,
    607,
    998
  ],
  "session_memory_key": "button:start",
  "cached": false
},
{
  "id": "search_bar",
  "type": "input",
  "label": "Search",
  "description": "Windows search bar in the taskbar",
  "state": "visible",
  "dx": 528,
  "dy": 864,
```

```
"frame_dx": 679,  
"frame_dy": 974,  
"confidence": 0.98,  
"source": "gemini_vlm",  
"frame_bbox": [  
    591,  
    950,  
    767,  
    998  
],  
"session_memory_key": "input:search",  
"cached": false  
},  
{  
    "id": "weather_widget",  
    "type": "button",  
    "label": "Weather",  
    "description": "Weather widget in the taskbar",  
    "state": "visible",  
    "dx": 92,  
    "dy": 876,  
    "frame_dx": 243,  
    "frame_dy": 986,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        167,  
        958,  
        319,
```

```
1013
],
"session_memory_key": "button:weather",
"cached": false
},
{
  "id": "taskbar_file_explorer",
  "type": "icon",
  "label": "File Explorer",
  "description": "File Explorer icon in the taskbar",
  "state": "visible",
  "dx": 612,
  "dy": 868,
  "frame_dx": 763,
  "frame_dy": 978,
  "confidence": 0.95,
  "source": "gemini_vlm",
  "frame_bbox": [
    743,
    958,
    783,
    998
  ],
  "session_memory_key": "icon:file explorer",
  "cached": false
},
{
  "id": "taskbar_chrome",
  "type": "icon",
```

```
    "label": "Chrome",
    "description": "Google Chrome icon in the taskbar",
    "state": "visible",
    "dx": 640,
    "dy": 868,
    "frame_dx": 791,
    "frame_dy": 978,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        775,
        958,
        807,
        998
    ],
    "session_memory_key": "icon:chrome",
    "cached": false
},
{
    "id": "taskbar_settings",
    "type": "icon",
    "label": "Settings",
    "description": "Settings icon in the taskbar",
    "state": "visible",
    "dx": 672,
    "dy": 868,
    "frame_dx": 823,
    "frame_dy": 978,
    "confidence": 0.95,
```



```
"source": "gemini_vlm",
"frame_bbox": [
    807,
    958,
    839,
    998
],
"session_memory_key": "icon:settings",
"cached": false
},
{
    "id": "taskbar_vscode",
    "type": "icon",
    "label": "VS Code",
    "description": "Visual Studio Code icon in the taskbar",
    "state": "visible",
    "dx": 804,
    "dy": 868,
    "frame_dx": 955,
    "frame_dy": 978,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        935,
        958,
        975,
        998
    ],
    "session_memory_key": "icon:vs code",
```

```
    "cached": false
  },
  {
    "id": "taskbar_spotify",
    "type": "icon",
    "label": "Spotify",
    "description": "Spotify icon in the taskbar",
    "state": "visible",
    "dx": 844,
    "dy": 868,
    "frame_dx": 995,
    "frame_dy": 978,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
      975,
      958,
      1015,
      998
    ],
    "session_memory_key": "icon:spotify",
    "cached": false
  },
  {
    "id": "system_tray",
    "type": "pane",
    "label": "System Tray",
    "description": "System tray area with network, volume, and battery icons",
```

```
"state": "visible",
"dx": 1416,
"dy": 864,
"frame_dx": 1567,
"frame_dy": 974,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    1487,
    950,
    1647,
    998
],
"session_memory_key": "pane:system tray",
"cached": false
},
{
    "id": "clock_date",
    "type": "text",
    "label": "Clock and Date",
    "description": "System clock and date display in the taskbar",
    "state": "visible",
    "dx": 1520,
    "dy": 868,
    "frame_dx": 1671,
    "frame_dy": 978,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
```

```
        1615,  
        942,  
        1727,  
        1013  
    ],  
    "session_memory_key": "text:clock and date",  
    "cached": false  
}  
],  
"session_knowledge_base": {  
    "session_id": "shot_1777696146_8f4fe6",  
    "started_at": "2026-05-02T04:29:06.629715+00:00",  
    "updated_at": "2026-05-02T04:29:36.525759+00:00",  
    "stale_after_misses": 3,  
    "confidence_floor": 0.4,  
    "decay_per_miss": 0.9,  
    "frame_count": 1,  
    "frame_name": "frame_1777696146_e68365.jpg",  
    "frame_index": 0,  
    "image_size": {  
        "width": 1920,  
        "height": 1080  
    },  
    "known_elements": {  
        "tab:frontend": {  
            "element_key": "tab:frontend",  
            "element_type": "tab",  
            "label": "frontend",  
            "bbox": [  

```

```
    151.0,  
    118.0,  
    399.0,  
    174.0  
  ],  
  "confidence": 0.98,  
  "source": "gemini_vlm",  
  "first_seen_frame": "frame_1777696146_e68365.jpg",  
  "last_seen_frame": "frame_1777696146_e68365.jpg",  
  "first_seen_at": "2026-05-02T04:29:36.525759+00:00",  
  "last_seen_at": "2026-05-02T04:29:36.525759+00:00",  
  "first_seen_index": 0,  
  "last_seen_index": 0,  
  "seen_count": 1,  
  "miss_count": 0,  
  "stale": false,  
  "cached": false,  
  "history": [  
    {  
      "frame_name": "frame_1777696146_e68365.jpg",  
      "frame_index": 0,  
      "seen_at": "2026-05-02T04:29:36.525759+00:00",  
      "confidence": 0.98  
    }  
  ]  
},  
"input:address bar": {  
  "element_key": "input:address bar",  
  "element_type": "input",
```

```
"label": "Address Bar",
"bbox": [
  151.0,
  182.0,
  1303.0,
  238.0
],
"confidence": 0.99,
"source": "gemini_vlm",
"first_seen_frame": "frame_1777696146_e68365.jpg",
"last_seen_frame": "frame_1777696146_e68365.jpg",
"first_seen_at": "2026-05-02T04:29:36.525759+00:00",
"last_seen_at": "2026-05-02T04:29:36.525759+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777696146_e68365.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T04:29:36.525759+00:00",
    "confidence": 0.99
  }
]
},
"button:ask gemini": {
```

```
"element_key": "button:ask gemini",
"element_type": "button",
"label": "Ask Gemini",
"bbox": [
    1415.0,
    142.0,
    1583.0,
    190.0
],
"confidence": 0.95,
"source": "gemini_vlm",
"first_seen_frame": "frame_1777696146_e68365.jpg",
"last_seen_frame": "frame_1777696146_e68365.jpg",
"first_seen_at": "2026-05-02T04:29:36.525759+00:00",
"last_seen_at": "2026-05-02T04:29:36.525759+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
    {
        "frame_name": "frame_1777696146_e68365.jpg",
        "frame_index": 0,
        "seen_at": "2026-05-02T04:29:36.525759+00:00",
        "confidence": 0.95
    }
]
```

```
},
"button:minimize": {
  "element_key": "button:minimize",
  "element_type": "button",
  "label": "Minimize",
  "bbox": [
    1583.0,
    142.0,
    1639.0,
    182.0
  ],
  "confidence": 0.9,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777696146_e68365.jpg",
  "last_seen_frame": "frame_1777696146_e68365.jpg",
  "first_seen_at": "2026-05-02T04:29:36.525759+00:00",
  "last_seen_at": "2026-05-02T04:29:36.525759+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
  "seen_count": 1,
  "miss_count": 0,
  "stale": false,
  "cached": false,
  "history": [
    {
      "frame_name": "frame_1777696146_e68365.jpg",
      "frame_index": 0,
      "seen_at": "2026-05-02T04:29:36.525759+00:00",
      "confidence": 0.9
    }
  ]
}
```



```
    }  
  ]  
},  
"button:maximize": {  
  "element_key": "button:maximize",  
  "element_type": "button",  
  "label": "Maximize",  
  "bbox": [  
    1623.0,  
    142.0,  
    1679.0,  
    182.0  
  ],  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "first_seen_frame": "frame_1777696146_e68365.jpg",  
  "last_seen_frame": "frame_1777696146_e68365.jpg",  
  "first_seen_at": "2026-05-02T04:29:36.525759+00:00",  
  "last_seen_at": "2026-05-02T04:29:36.525759+00:00",  
  "first_seen_index": 0,  
  "last_seen_index": 0,  
  "seen_count": 1,  
  "miss_count": 0,  
  "stale": false,  
  "cached": false,  
  "history": [  
    {  
      "frame_name": "frame_1777696146_e68365.jpg",  
      "frame_index": 0,  

```

```
        "seen_at": "2026-05-02T04:29:36.525759+00:00",
        "confidence": 0.9
    }
]
},
"button:close": {
    "element_key": "button:close",
    "element_type": "button",
    "label": "Close",
    "bbox": [
        1671.0,
        142.0,
        1727.0,
        190.0
    ],
    "confidence": 0.9,
    "source": "gemini_vlm",
    "first_seen_frame": "frame_1777696146_e68365.jpg",
    "last_seen_frame": "frame_1777696146_e68365.jpg",
    "first_seen_at": "2026-05-02T04:29:36.525759+00:00",
    "last_seen_at": "2026-05-02T04:29:36.525759+00:00",
    "first_seen_index": 0,
    "last_seen_index": 0,
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false,
    "history": [
        {
```

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        "frame_name": "frame_1777696146_e68365.jpg",
        "frame_index": 0,
        "seen_at": "2026-05-02T04:29:36.525759+00:00",
        "confidence": 0.9
    }
]
},
"heading:screenpilot ai": {
    "element_key": "heading:screenpilot ai",
    "element_type": "heading",
    "label": "ScreenPilot AI",
    "bbox": [
        167.0,
        270.0,
        375.0,
        334.0
    ],
    "confidence": 0.99,
    "source": "gemini_vlm",
    "first_seen_frame": "frame_1777696146_e68365.jpg",
    "last_seen_frame": "frame_1777696146_e68365.jpg",
    "first_seen_at": "2026-05-02T04:29:36.525759+00:00",
    "last_seen_at": "2026-05-02T04:29:36.525759+00:00",
    "first_seen_index": 0,
    "last_seen_index": 0,
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false,
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"history": [  
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    "frame_name": "frame_1777696146_e68365.jpg",  
    "frame_index": 0,  
    "seen_at": "2026-05-02T04:29:36.525759+00:00",  
    "confidence": 0.99  
  }  
]  
,  
"label:autonomous task agent v2": {  
  "element_key": "label:autonomous task agent v2",  
  "element_type": "label",  
  "label": "Autonomous Task Agent v2",  
  "bbox": [  
    423.0,  
    286.0,  
    495.0,  
    326.0  
  ],  
  "confidence": 0.97,  
  "source": "gemini_vlm",  
  "first_seen_frame": "frame_1777696146_e68365.jpg",  
  "last_seen_frame": "frame_1777696146_e68365.jpg",  
  "first_seen_at": "2026-05-02T04:29:36.525759+00:00",  
  "last_seen_at": "2026-05-02T04:29:36.525759+00:00",  
  "first_seen_index": 0,  
  "last_seen_index": 0,  
  "seen_count": 1,  
  "miss_count": 0,
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"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777696146_e68365.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T04:29:36.525759+00:00",
    "confidence": 0.97
  }
],
},
"text:connecting to agent...": {
  "element_key": "text:connecting to agent...",
  "element_type": "text",
  "label": "Connecting to agent...",
  "bbox": [
    183.0,
    526.0,
    359.0,
    566.0
  ],
  "confidence": 0.96,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777696146_e68365.jpg",
  "last_seen_frame": "frame_1777696146_e68365.jpg",
  "first_seen_at": "2026-05-02T04:29:36.525759+00:00",
  "last_seen_at": "2026-05-02T04:29:36.525759+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
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```
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
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    "frame_name": "frame_1777696146_e68365.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T04:29:36.525759+00:00",
    "confidence": 0.96
  }
]
},
"text:capturing initial screen...": {
  "element_key": "text:capturing initial screen...",
  "element_type": "text",
  "label": "Capturing initial screen...",
  "bbox": [
    183.0,
    558.0,
    415.0,
    598.0
  ],
  "confidence": 0.96,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777696146_e68365.jpg",
  "last_seen_frame": "frame_1777696146_e68365.jpg",
  "first_seen_at": "2026-05-02T04:29:36.525759+00:00",
  "last_seen_at": "2026-05-02T04:29:36.525759+00:00",
```

```
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777696146_e68365.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T04:29:36.525759+00:00",
    "confidence": 0.96
  }
],
},
"pane:preview window": {
  "element_key": "pane:preview window",
  "element_type": "pane",
  "label": "Preview Window",
  "bbox": [
    167.0,
    598.0,
    367.0,
    798.0
  ],
  "confidence": 0.95,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777696146_e68365.jpg",
  "last_seen_frame": "frame_1777696146_e68365.jpg",
```

```
"first_seen_at": "2026-05-02T04:29:36.525759+00:00",
"last_seen_at": "2026-05-02T04:29:36.525759+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777696146_e68365.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T04:29:36.525759+00:00",
    "confidence": 0.95
  }
],
},
"input:agent is working...": {
  "element_key": "input:agent is working...",
  "element_type": "input",
  "label": "Agent is working...",
  "bbox": [
    151.0,
    862.0,
    1727.0,
    950.0
  ],
  "confidence": 0.98,
  "source": "gemini_vlm",
```



```
"first_seen_frame": "frame_1777696146_e68365.jpg",
"last_seen_frame": "frame_1777696146_e68365.jpg",
"first_seen_at": "2026-05-02T04:29:36.525759+00:00",
"last_seen_at": "2026-05-02T04:29:36.525759+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777696146_e68365.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T04:29:36.525759+00:00",
    "confidence": 0.98
  }
],
},
"button:send": {
  "element_key": "button:send",
  "element_type": "button",
  "label": "Send",
  "bbox": [
    1583.0,
    854.0,
    1663.0,
    942.0
  ],
},
```

```
"confidence": 0.97,
"source": "gemini_vlm",
"first_seen_frame": "frame_1777696146_e68365.jpg",
"last_seen_frame": "frame_1777696146_e68365.jpg",
"first_seen_at": "2026-05-02T04:29:36.525759+00:00",
"last_seen_at": "2026-05-02T04:29:36.525759+00:00",
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"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777696146_e68365.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T04:29:36.525759+00:00",
    "confidence": 0.97
  }
],
},
"pane:taskbar": {
  "element_key": "pane:taskbar",
  "element_type": "pane",
  "label": "Taskbar",
  "bbox": [
    151.0,
    934.0,
    1727.0,
```

```
1013.0

],
"confidence": 0.99,
"source": "gemini_vlm",
"first_seen_frame": "frame_1777696146_e68365.jpg",
"last_seen_frame": "frame_1777696146_e68365.jpg",
"first_seen_at": "2026-05-02T04:29:36.525759+00:00",
"last_seen_at": "2026-05-02T04:29:36.525759+00:00",
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"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777696146_e68365.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T04:29:36.525759+00:00",
    "confidence": 0.99
  }
]
},
"button:start": {
  "element_key": "button:start",
  "element_type": "button",
  "label": "Start",
  "bbox": [
    551.0,
```

```
    950.0,
    607.0,
    998.0
  ],
  "confidence": 0.98,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777696146_e68365.jpg",
  "last_seen_frame": "frame_1777696146_e68365.jpg",
  "first_seen_at": "2026-05-02T04:29:36.525759+00:00",
  "last_seen_at": "2026-05-02T04:29:36.525759+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
  "seen_count": 1,
  "miss_count": 0,
  "stale": false,
  "cached": false,
  "history": [
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      "frame_index": 0,
      "seen_at": "2026-05-02T04:29:36.525759+00:00",
      "confidence": 0.98
    }
  ]
},
"input:search": {
  "element_key": "input:search",
  "element_type": "input",
  "label": "Search",
```

```
"bbox": [
  591.0,
  950.0,
  767.0,
  998.0
],
"confidence": 0.98,
"source": "gemini_vlm",
"first_seen_frame": "frame_1777696146_e68365.jpg",
"last_seen_frame": "frame_1777696146_e68365.jpg",
"first_seen_at": "2026-05-02T04:29:36.525759+00:00",
"last_seen_at": "2026-05-02T04:29:36.525759+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777696146_e68365.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T04:29:36.525759+00:00",
    "confidence": 0.98
  }
]
},
"button:weather": {
  "element_key": "button:weather",
```

```
"element_type": "button",
"label": "Weather",
"bbox": [
    167.0,
    958.0,
    319.0,
    1013.0
],
"confidence": 0.95,
"source": "gemini_vlm",
"first_seen_frame": "frame_1777696146_e68365.jpg",
"last_seen_frame": "frame_1777696146_e68365.jpg",
"first_seen_at": "2026-05-02T04:29:36.525759+00:00",
"last_seen_at": "2026-05-02T04:29:36.525759+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
    {
        "frame_name": "frame_1777696146_e68365.jpg",
        "frame_index": 0,
        "seen_at": "2026-05-02T04:29:36.525759+00:00",
        "confidence": 0.95
    }
]
},
```

```
"icon:file explorer": {
  "element_key": "icon:file explorer",
  "element_type": "icon",
  "label": "File Explorer",
  "bbox": [
    743.0,
    958.0,
    783.0,
    998.0
  ],
  "confidence": 0.95,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777696146_e68365.jpg",
  "last_seen_frame": "frame_1777696146_e68365.jpg",
  "first_seen_at": "2026-05-02T04:29:36.525759+00:00",
  "last_seen_at": "2026-05-02T04:29:36.525759+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
  "seen_count": 1,
  "miss_count": 0,
  "stale": false,
  "cached": false,
  "history": [
    {
      "frame_name": "frame_1777696146_e68365.jpg",
      "frame_index": 0,
      "seen_at": "2026-05-02T04:29:36.525759+00:00",
      "confidence": 0.95
    }
  ]
}
```

```
]
},
"icon:chrome": {
  "element_key": "icon:chrome",
  "element_type": "icon",
  "label": "Chrome",
  "bbox": [
    775.0,
    958.0,
    807.0,
    998.0
  ],
  "confidence": 0.95,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777696146_e68365.jpg",
  "last_seen_frame": "frame_1777696146_e68365.jpg",
  "first_seen_at": "2026-05-02T04:29:36.525759+00:00",
  "last_seen_at": "2026-05-02T04:29:36.525759+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
  "seen_count": 1,
  "miss_count": 0,
  "stale": false,
  "cached": false,
  "history": [
    {
      "frame_name": "frame_1777696146_e68365.jpg",
      "frame_index": 0,
      "seen_at": "2026-05-02T04:29:36.525759+00:00",
```



```
        "confidence": 0.95
      }
    ]
  },
  "icon:settings": {
    "element_key": "icon:settings",
    "element_type": "icon",
    "label": "Settings",
    "bbox": [
      807.0,
      958.0,
      839.0,
      998.0
    ],
    "confidence": 0.95,
    "source": "gemini_vlm",
    "first_seen_frame": "frame_1777696146_e68365.jpg",
    "last_seen_frame": "frame_1777696146_e68365.jpg",
    "first_seen_at": "2026-05-02T04:29:36.525759+00:00",
    "last_seen_at": "2026-05-02T04:29:36.525759+00:00",
    "first_seen_index": 0,
    "last_seen_index": 0,
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false,
    "history": [
      {
        "frame_name": "frame_1777696146_e68365.jpg",
```

```
        "frame_index": 0,
        "seen_at": "2026-05-02T04:29:36.525759+00:00",
        "confidence": 0.95
    }
]
},
"icon:vs code": {
    "element_key": "icon:vs code",
    "element_type": "icon",
    "label": "VS Code",
    "bbox": [
        935.0,
        958.0,
        975.0,
        998.0
    ],
    "confidence": 0.95,
    "source": "gemini_vlm",
    "first_seen_frame": "frame_1777696146_e68365.jpg",
    "last_seen_frame": "frame_1777696146_e68365.jpg",
    "first_seen_at": "2026-05-02T04:29:36.525759+00:00",
    "last_seen_at": "2026-05-02T04:29:36.525759+00:00",
    "first_seen_index": 0,
    "last_seen_index": 0,
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false,
    "history": [
```

```
{
  "frame_name": "frame_1777696146_e68365.jpg",
  "frame_index": 0,
  "seen_at": "2026-05-02T04:29:36.525759+00:00",
  "confidence": 0.95
}
],
},
"icon:spotify": {
  "element_key": "icon:spotify",
  "element_type": "icon",
  "label": "Spotify",
  "bbox": [
    975.0,
    958.0,
    1015.0,
    998.0
  ],
  "confidence": 0.95,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777696146_e68365.jpg",
  "last_seen_frame": "frame_1777696146_e68365.jpg",
  "first_seen_at": "2026-05-02T04:29:36.525759+00:00",
  "last_seen_at": "2026-05-02T04:29:36.525759+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
  "seen_count": 1,
  "miss_count": 0,
  "stale": false,
```

```
"cached": false,
"history": [
  {
    "frame_name": "frame_1777696146_e68365.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T04:29:36.525759+00:00",
    "confidence": 0.95
  }
],
},
"pane:system tray": {
  "element_key": "pane:system tray",
  "element_type": "pane",
  "label": "System Tray",
  "bbox": [
    1487.0,
    950.0,
    1647.0,
    998.0
  ],
  "confidence": 0.95,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777696146_e68365.jpg",
  "last_seen_frame": "frame_1777696146_e68365.jpg",
  "first_seen_at": "2026-05-02T04:29:36.525759+00:00",
  "last_seen_at": "2026-05-02T04:29:36.525759+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
  "seen_count": 1,
```

```
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777696146_e68365.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T04:29:36.525759+00:00",
    "confidence": 0.95
  }
],
},
"text:clock and date": {
  "element_key": "text:clock and date",
  "element_type": "text",
  "label": "Clock and Date",
  "bbox": [
    1615.0,
    942.0,
    1727.0,
    1013.0
  ],
  "confidence": 0.98,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777696146_e68365.jpg",
  "last_seen_frame": "frame_1777696146_e68365.jpg",
  "first_seen_at": "2026-05-02T04:29:36.525759+00:00",
  "last_seen_at": "2026-05-02T04:29:36.525759+00:00",
  "first_seen_index": 0,
```

```
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777696146_e68365.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T04:29:36.525759+00:00",
    "confidence": 0.98
  }
]
},
"resolved_elements": [
  {
    "id": "browser_tab_frontend",
    "type": "tab",
    "label": "frontend",
    "description": "Browser tab for the frontend application",
    "state": "visible",
    "dx": 124,
    "dy": 36,
    "frame_dx": 275,
    "frame_dy": 146,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
```

```
    151,  
    118,  
    399,  
    174  
  ],  
  "session_memory_key": "tab:frontend",  
  "cached": false  
},  
{  
  "id": "address_bar",  
  "type": "input",  
  "label": "Address Bar",  
  "description": "Browser address bar showing localhost:5173",  
  "state": "visible",  
  "dx": 576,  
  "dy": 100,  
  "frame_dx": 727,  
  "frame_dy": 210,  
  "confidence": 0.99,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    151,  
    182,  
    1303,  
    238  
  ],  
  "session_memory_key": "input:address bar",  
  "cached": false  
},
```

```
{
  "id": "ask_gemini_button",
  "type": "button",
  "label": "Ask Gemini",
  "description": "Ask Gemini button in the browser toolbar",
  "state": "visible",
  "dx": 1348,
  "dy": 56,
  "frame_dx": 1499,
  "frame_dy": 166,
  "confidence": 0.95,
  "source": "gemini_vlm",
  "frame_bbox": [
    1415,
    142,
    1583,
    190
  ],
  "session_memory_key": "button:ask gemini",
  "cached": false
},
{
  "id": "window_minimize",
  "type": "button",
  "label": "Minimize",
  "description": "Browser window minimize button",
  "state": "visible",
  "dx": 1460,
  "dy": 52,
```



```
"frame_dx": 1611,
"frame_dy": 162,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    1583,
    142,
    1639,
    182
],
"session_memory_key": "button:minimize",
"cached": false
},
{
    "id": "window_maximize",
    "type": "button",
    "label": "Maximize",
    "description": "Browser window maximize button",
    "state": "visible",
    "dx": 1500,
    "dy": 52,
    "frame_dx": 1651,
    "frame_dy": 162,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        1623,
        142,
        1679,
```

```
182
],
"session_memory_key": "button:maximize",
"cached": false
},
{
  "id": "window_close",
  "type": "button",
  "label": "Close",
  "description": "Browser window close button",
  "state": "visible",
  "dx": 1548,
  "dy": 56,
  "frame_dx": 1699,
  "frame_dy": 166,
  "confidence": 0.9,
  "source": "gemini_vlm",
  "frame_bbox": [
    1671,
    142,
    1727,
    190
  ],
  "session_memory_key": "button:close",
  "cached": false
},
{
  "id": "screenpilot_ai_title",
  "type": "heading",
```

```
"label": "ScreenPilot AI",
"description": "Application title ScreenPilot AI",
"state": "visible",
"dx": 120,
"dy": 192,
"frame_dx": 271,
"frame_dy": 302,
"confidence": 0.99,
"source": "gemini_vlm",
"frame_bbox": [
    167,
    270,
    375,
    334
],
"session_memory_key": "heading:screenpilot ai",
"cached": false
},
{
    "id": "agent_version_badge",
    "type": "label",
    "label": "Autonomous Task Agent v2",
    "description": "Badge indicating the agent version",
    "state": "visible",
    "dx": 308,
    "dy": 196,
    "frame_dx": 459,
    "frame_dy": 306,
    "confidence": 0.97,
```

```
"source": "gemini_vlm",
"frame_bbox": [
  423,
  286,
  495,
  326
],
"session_memory_key": "label:autonomous task agent v2",
"cached": false
},
{
  "id": "status_connecting",
  "type": "text",
  "label": "Connecting to agent...",
  "description": "Status message indicating connection progress",
  "state": "visible",
  "dx": 120,
  "dy": 436,
  "frame_dx": 271,
  "frame_dy": 546,
  "confidence": 0.96,
  "source": "gemini_vlm",
  "frame_bbox": [
    183,
    526,
    359,
    566
  ],
  "session_memory_key": "text:connecting to agent..."
```

```
"cached": false
},
{
  "id": "status_capturing",
  "type": "text",
  "label": "Capturing initial screen...",
  "description": "Status message indicating screen capture progress",
  "state": "visible",
  "dx": 148,
  "dy": 468,
  "frame_dx": 299,
  "frame_dy": 578,
  "confidence": 0.96,
  "source": "gemini_vlm",
  "frame_bbox": [
    183,
    558,
    415,
    598
  ],
  "session_memory_key": "text:capturing initial screen...",
  "cached": false
},
{
  "id": "preview_window",
  "type": "pane",
  "label": "Preview Window",
  "description": "Small preview window showing agent activity",
  "state": "visible",
```

```
"dx": 116,  
  "dy": 588,  
  "frame_dx": 267,  
  "frame_dy": 698,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    167,  
    598,  
    367,  
    798  
  ],  
  "session_memory_key": "pane:preview window",  
  "cached": false  
},  
{  
  "id": "chat_input_field",  
  "type": "input",  
  "label": "Agent is working...",  
  "description": "Main chat input field for the agent",  
  "state": "visible",  
  "dx": 788,  
  "dy": 796,  
  "frame_dx": 939,  
  "frame_dy": 906,  
  "confidence": 0.98,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    151,
```

```
      862,  
      1727,  
      950  
    ],  
    "session_memory_key": "input:agent is working...",  
    "cached": false  
  },  
  {  
    "id": "send_button",  
    "type": "button",  
    "label": "Send",  
    "description": "Button to send commands to the agent",  
    "state": "visible",  
    "dx": 1472,  
    "dy": 788,  
    "frame_dx": 1623,  
    "frame_dy": 898,  
    "confidence": 0.97,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
      1583,  
      854,  
      1663,  
      942  
    ],  
    "session_memory_key": "button:send",  
    "cached": false  
  },  
  {
```

```
"id": "windows_taskbar",
"type": "pane",
"label": "Taskbar",
"description": "Windows taskbar at the bottom of the screen",
"state": "visible",
"dx": 788,
"dy": 864,
"frame_dx": 939,
"frame_dy": 974,
"confidence": 0.99,
"source": "gemini_vlm",
"frame_bbox": [
    151,
    934,
    1727,
    1013
],
"session_memory_key": "pane:taskbar",
"cached": false
},
{
    "id": "start_button",
    "type": "button",
    "label": "Start",
    "description": "Windows Start button",
    "state": "visible",
    "dx": 428,
    "dy": 864,
    "frame_dx": 579,
```



```
"frame_dy": 974,  
"confidence": 0.98,  
"source": "gemini_vlm",  
"frame_bbox": [  
    551,  
    950,  
    607,  
    998  
],  
"session_memory_key": "button:start",  
"cached": false  
},  
{  
    "id": "search_bar",  
    "type": "input",  
    "label": "Search",  
    "description": "Windows search bar in the taskbar",  
    "state": "visible",  
    "dx": 528,  
    "dy": 864,  
    "frame_dx": 679,  
    "frame_dy": 974,  
    "confidence": 0.98,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        591,  
        950,  
        767,  
        998
```

```
    ],
    "session_memory_key": "input:search",
    "cached": false
  },
  {
    "id": "weather_widget",
    "type": "button",
    "label": "Weather",
    "description": "Weather widget in the taskbar",
    "state": "visible",
    "dx": 92,
    "dy": 876,
    "frame_dx": 243,
    "frame_dy": 986,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
      167,
      958,
      319,
      1013
    ],
    "session_memory_key": "button:weather",
    "cached": false
  },
  {
    "id": "taskbar_file_explorer",
    "type": "icon",
    "label": "File Explorer",
```

```
"description": "File Explorer icon in the taskbar",
"state": "visible",
"dx": 612,
"dy": 868,
"frame_dx": 763,
"frame_dy": 978,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    743,
    958,
    783,
    998
],
"session_memory_key": "icon:file explorer",
"cached": false
},
{
    "id": "taskbar_chrome",
    "type": "icon",
    "label": "Chrome",
    "description": "Google Chrome icon in the taskbar",
    "state": "visible",
    "dx": 640,
    "dy": 868,
    "frame_dx": 791,
    "frame_dy": 978,
    "confidence": 0.95,
    "source": "gemini_vlm",
```

```
"frame_bbox": [  
    775,  
    958,  
    807,  
    998  
],  
"session_memory_key": "icon:chrome",  
"cached": false  
},  
{  
    "id": "taskbar_settings",  
    "type": "icon",  
    "label": "Settings",  
    "description": "Settings icon in the taskbar",  
    "state": "visible",  
    "dx": 672,  
    "dy": 868,  
    "frame_dx": 823,  
    "frame_dy": 978,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        807,  
        958,  
        839,  
        998  
    ],  
    "session_memory_key": "icon:settings",  
    "cached": false  
}
```

```
},  
{  
  "id": "taskbar_vscode",  
  "type": "icon",  
  "label": "VS Code",  
  "description": "Visual Studio Code icon in the taskbar",  
  "state": "visible",  
  "dx": 804,  
  "dy": 868,  
  "frame_dx": 955,  
  "frame_dy": 978,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    935,  
    958,  
    975,  
    998  
  ],  
  "session_memory_key": "icon:vs code",  
  "cached": false  
},  
{  
  "id": "taskbar_spotify",  
  "type": "icon",  
  "label": "Spotify",  
  "description": "Spotify icon in the taskbar",  
  "state": "visible",  
  "dx": 844,
```

```
"dy": 868,  
"frame_dx": 995,  
"frame_dy": 978,  
"confidence": 0.95,  
"source": "gemini_vlm",  
"frame_bbox": [  
    975,  
    958,  
    1015,  
    998  
],  
"session_memory_key": "icon:spotify",  
"cached": false  
},  
{  
    "id": "system_tray",  
    "type": "pane",  
    "label": "System Tray",  
    "description": "System tray area with network, volume, and battery  
icons",  
    "state": "visible",  
    "dx": 1416,  
    "dy": 864,  
    "frame_dx": 1567,  
    "frame_dy": 974,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        1487,
```

```
    950,  
    1647,  
    998  
  ],  
  "session_memory_key": "pane:system tray",  
  "cached": false  
},  
{  
  "id": "clock_date",  
  "type": "text",  
  "label": "Clock and Date",  
  "description": "System clock and date display in the taskbar",  
  "state": "visible",  
  "dx": 1520,  
  "dy": 868,  
  "frame_dx": 1671,  
  "frame_dy": 978,  
  "confidence": 0.98,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    1615,  
    942,  
    1727,  
    1013  
  ],  
  "session_memory_key": "text:clock and date",  
  "cached": false  
}  
],
```

```
"current_elements": [  
  {  
    "id": "browser_tab_frontend",  
    "type": "tab",  
    "label": "frontend",  
    "description": "Browser tab for the frontend application",  
    "state": "visible",  
    "dx": 124,  
    "dy": 36,  
    "frame_dx": 275,  
    "frame_dy": 146,  
    "confidence": 0.98,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
      151,  
      118,  
      399,  
      174  
    ],  
    "session_memory_key": "tab:frontend",  
    "cached": false  
  },  
  {  
    "id": "address_bar",  
    "type": "input",  
    "label": "Address Bar",  
    "description": "Browser address bar showing localhost:5173",  
    "state": "visible",  
    "dx": 576,  
    "dy": 36,  
    "frame_dx": 275,  
    "frame_dy": 146,  
    "confidence": 0.98,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
      151,  
      118,  
      399,  
      174  
    ],  
    "session_memory_key": "address_bar",  
    "cached": false  
  }  
]
```



```
"dy": 100,
"frame_dx": 727,
"frame_dy": 210,
"confidence": 0.99,
"source": "gemini_vlm",
"frame_bbox": [
    151,
    182,
    1303,
    238
],
"session_memory_key": "input:address bar",
"cached": false
},
{
    "id": "ask_gemini_button",
    "type": "button",
    "label": "Ask Gemini",
    "description": "Ask Gemini button in the browser toolbar",
    "state": "visible",
    "dx": 1348,
    "dy": 56,
    "frame_dx": 1499,
    "frame_dy": 166,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        1415,
        142,
```

```
        1583,  
        190  
    ],  
    "session_memory_key": "button:ask gemini",  
    "cached": false  
},  
{  
    "id": "window_minimize",  
    "type": "button",  
    "label": "Minimize",  
    "description": "Browser window minimize button",  
    "state": "visible",  
    "dx": 1460,  
    "dy": 52,  
    "frame_dx": 1611,  
    "frame_dy": 162,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        1583,  
        142,  
        1639,  
        182  
    ],  
    "session_memory_key": "button:minimize",  
    "cached": false  
},  
{  
    "id": "window_maximize",
```

```
"type": "button",
"label": "Maximize",
"description": "Browser window maximize button",
"state": "visible",
"dx": 1500,
"dy": 52,
"frame_dx": 1651,
"frame_dy": 162,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    1623,
    142,
    1679,
    182
],
"session_memory_key": "button:maximize",
"cached": false
},
{
    "id": "window_close",
    "type": "button",
    "label": "Close",
    "description": "Browser window close button",
    "state": "visible",
    "dx": 1548,
    "dy": 56,
    "frame_dx": 1699,
    "frame_dy": 166,
```

```
"confidence": 0.9,  
"source": "gemini_vlm",  
"frame_bbox": [  
    1671,  
    142,  
    1727,  
    190  
],  
"session_memory_key": "button:close",  
"cached": false  
},  
{  
    "id": "screenpilot_ai_title",  
    "type": "heading",  
    "label": "ScreenPilot AI",  
    "description": "Application title ScreenPilot AI",  
    "state": "visible",  
    "dx": 120,  
    "dy": 192,  
    "frame_dx": 271,  
    "frame_dy": 302,  
    "confidence": 0.99,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        167,  
        270,  
        375,  
        334  
    ],  
}
```

```
"session_memory_key": "heading:screenpilot ai",
"cached": false
},
{
  "id": "agent_version_badge",
  "type": "label",
  "label": "Autonomous Task Agent v2",
  "description": "Badge indicating the agent version",
  "state": "visible",
  "dx": 308,
  "dy": 196,
  "frame_dx": 459,
  "frame_dy": 306,
  "confidence": 0.97,
  "source": "gemini_vlm",
  "frame_bbox": [
    423,
    286,
    495,
    326
  ],
  "session_memory_key": "label:autonomous task agent v2",
  "cached": false
},
{
  "id": "status_connecting",
  "type": "text",
  "label": "Connecting to agent...",
  "description": "Status message indicating connection progress",
```

```
"state": "visible",
"dx": 120,
"dy": 436,
"frame_dx": 271,
"frame_dy": 546,
"confidence": 0.96,
"source": "gemini_vlm",
"frame_bbox": [
    183,
    526,
    359,
    566
],
"session_memory_key": "text:connecting to agent...",
"cached": false
},
{
    "id": "status_capturing",
    "type": "text",
    "label": "Capturing initial screen...",
    "description": "Status message indicating screen capture progress",
    "state": "visible",
    "dx": 148,
    "dy": 468,
    "frame_dx": 299,
    "frame_dy": 578,
    "confidence": 0.96,
    "source": "gemini_vlm",
    "frame_bbox": [
```

```
    183,  
    558,  
    415,  
    598  
  ],  
  "session_memory_key": "text:capturing initial screen...",  
  "cached": false  
},  
{  
  "id": "preview_window",  
  "type": "pane",  
  "label": "Preview Window",  
  "description": "Small preview window showing agent activity",  
  "state": "visible",  
  "dx": 116,  
  "dy": 588,  
  "frame_dx": 267,  
  "frame_dy": 698,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    167,  
    598,  
    367,  
    798  
  ],  
  "session_memory_key": "pane:preview window",  
  "cached": false  
},
```

```
{
  "id": "chat_input_field",
  "type": "input",
  "label": "Agent is working...",
  "description": "Main chat input field for the agent",
  "state": "visible",
  "dx": 788,
  "dy": 796,
  "frame_dx": 939,
  "frame_dy": 906,
  "confidence": 0.98,
  "source": "gemini_vlm",
  "frame_bbox": [
    151,
    862,
    1727,
    950
  ],
  "session_memory_key": "input:agent is working...",
  "cached": false
},
{
  "id": "send_button",
  "type": "button",
  "label": "Send",
  "description": "Button to send commands to the agent",
  "state": "visible",
  "dx": 1472,
  "dy": 788,
```



```
"frame_dx": 1623,  
"frame_dy": 898,  
"confidence": 0.97,  
"source": "gemini_vlm",  
"frame_bbox": [  
    1583,  
    854,  
    1663,  
    942  
],  
"session_memory_key": "button:send",  
"cached": false  
},  
{  
    "id": "windows_taskbar",  
    "type": "pane",  
    "label": "Taskbar",  
    "description": "Windows taskbar at the bottom of the screen",  
    "state": "visible",  
    "dx": 788,  
    "dy": 864,  
    "frame_dx": 939,  
    "frame_dy": 974,  
    "confidence": 0.99,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        151,  
        934,  
        1727,
```

```
1013
],
"session_memory_key": "pane:taskbar",
"cached": false
},
{
  "id": "start_button",
  "type": "button",
  "label": "Start",
  "description": "Windows Start button",
  "state": "visible",
  "dx": 428,
  "dy": 864,
  "frame_dx": 579,
  "frame_dy": 974,
  "confidence": 0.98,
  "source": "gemini_vlm",
  "frame_bbox": [
    551,
    950,
    607,
    998
  ],
  "session_memory_key": "button:start",
  "cached": false
},
{
  "id": "search_bar",
  "type": "input",
```

```
"label": "Search",
"description": "Windows search bar in the taskbar",
"state": "visible",
"dx": 528,
"dy": 864,
"frame_dx": 679,
"frame_dy": 974,
"confidence": 0.98,
"source": "gemini_vlm",
"frame_bbox": [
    591,
    950,
    767,
    998
],
"session_memory_key": "input:search",
"cached": false
},
{
    "id": "weather_widget",
    "type": "button",
    "label": "Weather",
    "description": "Weather widget in the taskbar",
    "state": "visible",
    "dx": 92,
    "dy": 876,
    "frame_dx": 243,
    "frame_dy": 986,
    "confidence": 0.95,
```

```
"source": "gemini_vlm",
"frame_bbox": [
    167,
    958,
    319,
    1013
],
"session_memory_key": "button:weather",
"cached": false
},
{
    "id": "taskbar_file_explorer",
    "type": "icon",
    "label": "File Explorer",
    "description": "File Explorer icon in the taskbar",
    "state": "visible",
    "dx": 612,
    "dy": 868,
    "frame_dx": 763,
    "frame_dy": 978,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        743,
        958,
        783,
        998
    ],
    "session_memory_key": "icon:file explorer",
```

```
"cached": false
},
{
  "id": "taskbar_chrome",
  "type": "icon",
  "label": "Chrome",
  "description": "Google Chrome icon in the taskbar",
  "state": "visible",
  "dx": 640,
  "dy": 868,
  "frame_dx": 791,
  "frame_dy": 978,
  "confidence": 0.95,
  "source": "gemini_vlm",
  "frame_bbox": [
    775,
    958,
    807,
    998
  ],
  "session_memory_key": "icon:chrome",
  "cached": false
},
{
  "id": "taskbar_settings",
  "type": "icon",
  "label": "Settings",
  "description": "Settings icon in the taskbar",
  "state": "visible",
```

```
"dx": 672,  
  "dy": 868,  
  "frame_dx": 823,  
  "frame_dy": 978,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    807,  
    958,  
    839,  
    998  
  ],  
  "session_memory_key": "icon:settings",  
  "cached": false  
},  
{  
  "id": "taskbar_vscode",  
  "type": "icon",  
  "label": "VS Code",  
  "description": "Visual Studio Code icon in the taskbar",  
  "state": "visible",  
  "dx": 804,  
  "dy": 868,  
  "frame_dx": 955,  
  "frame_dy": 978,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    935,
```

```
    958,  
    975,  
    998  
  ],  
  "session_memory_key": "icon:vs code",  
  "cached": false  
},  
{  
  "id": "taskbar_spotify",  
  "type": "icon",  
  "label": "Spotify",  
  "description": "Spotify icon in the taskbar",  
  "state": "visible",  
  "dx": 844,  
  "dy": 868,  
  "frame_dx": 995,  
  "frame_dy": 978,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    975,  
    958,  
    1015,  
    998  
  ],  
  "session_memory_key": "icon:spotify",  
  "cached": false  
},  
{
```

```
    "id": "system_tray",
    "type": "pane",
    "label": "System Tray",
    "description": "System tray area with network, volume, and battery
icons",
    "state": "visible",
    "dx": 1416,
    "dy": 864,
    "frame_dx": 1567,
    "frame_dy": 974,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        1487,
        950,
        1647,
        998
    ],
    "session_memory_key": "pane:system tray",
    "cached": false
},
{
    "id": "clock_date",
    "type": "text",
    "label": "Clock and Date",
    "description": "System clock and date display in the taskbar",
    "state": "visible",
    "dx": 1520,
    "dy": 868,
```



```
    "frame_dx": 1671,
    "frame_dy": 978,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
        1615,
        942,
        1727,
        1013
    ],
    "session_memory_key": "text:clock and date",
    "cached": false
}
],
},
"session_memory": {
    "run_id": "f60f0afa-3ce3-4f83-9e73-d88f5282905b",
    "user_task": "Minimize the current window using minimize icon and find the
Google Chrome icon on the screen and click on that app icon",
    "started_at": "2026-05-02T04:29:00.097697+00:00",
    "updated_at": "2026-05-02T04:29:36.641249+00:00",
    "stale_after_misses": 2,
    "confidence_floor": 0.4,
    "decay_per_miss": 0.9,
    "frame_index": 1,
    "known_elements": {
        "tab:frontend": {
            "element_key": "tab:frontend",
            "element_type": "tab",
```

```
"label": "frontend",
"bbox": [
  151.0,
  118.0,
  399.0,
  174.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 1,
"first_seen_at": "2026-05-02T04:29:36.641249+00:00",
"last_seen_at": "2026-05-02T04:30:16.794900+00:00",
"seen_count": 2,
"miss_count": 11,
"stale": true,
"cached": true
},
"input:address bar": {
  "element_key": "input:address bar",
  "element_type": "input",
  "label": "Address bar",
  "bbox": [
    151.0,
    182.0,
    1303.0,
    238.0
```

```
],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 1,
  "first_seen_at": "2026-05-02T04:29:36.641249+00:00",
  "last_seen_at": "2026-05-02T04:30:16.794900+00:00",
  "seen_count": 2,
  "miss_count": 11,
  "stale": true,
  "cached": true
},
"button:ask gemini": {
  "element_key": "button:ask gemini",
  "element_type": "button",
  "label": "Ask Gemini",
  "bbox": [
    1415.0,
    142.0,
    1583.0,
    190.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
```

```
"last_seen_index": 0,  
"first_seen_at": "2026-05-02T04:29:36.641249+00:00",  
"last_seen_at": "2026-05-02T04:29:36.641249+00:00",  
"seen_count": 1,  
"miss_count": 12,  
"stale": true,  
"cached": true  
},  
"button:minimize": {  
  "element_key": "button:minimize",  
  "element_type": "button",  
  "label": "Minimize",  
  "bbox": [  
    1597.0,  
    134.0,  
    1653.0,  
    174.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 0,  
  "last_seen_index": 1,  
  "first_seen_at": "2026-05-02T04:29:36.641249+00:00",  
  "last_seen_at": "2026-05-02T04:30:16.794900+00:00",  
  "seen_count": 2,  
  "miss_count": 11,  
  "stale": true,
```

```
    "cached": true
  },
  "button:maximize": {
    "element_key": "button:maximize",
    "element_type": "button",
    "label": "Maximize",
    "bbox": [
      1623.0,
      142.0,
      1679.0,
      182.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 0,
    "last_seen_index": 0,
    "first_seen_at": "2026-05-02T04:29:36.641249+00:00",
    "last_seen_at": "2026-05-02T04:29:36.641249+00:00",
    "seen_count": 1,
    "miss_count": 12,
    "stale": true,
    "cached": true
  },
  "button:close": {
    "element_key": "button:close",
    "element_type": "button",
    "label": "Close",
```

```
"bbox": [
  1671.0,
  142.0,
  1727.0,
  190.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 0,
"first_seen_at": "2026-05-02T04:29:36.641249+00:00",
"last_seen_at": "2026-05-02T04:29:36.641249+00:00",
"seen_count": 1,
"miss_count": 12,
"stale": true,
"cached": true
},
"heading:screenpilot ai": {
  "element_key": "heading:screenpilot ai",
  "element_type": "heading",
  "label": "ScreenPilot AI",
  "bbox": [
    167.0,
    270.0,
    375.0,
    334.0
  ],

```

```
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 1,
"first_seen_at": "2026-05-02T04:29:36.641249+00:00",
"last_seen_at": "2026-05-02T04:30:16.794900+00:00",
"seen_count": 2,
"miss_count": 11,
"stale": true,
"cached": true
},
"label:autonomous task agent v2": {
  "element_key": "label:autonomous task agent v2",
  "element_type": "label",
  "label": "Autonomous Task Agent v2",
  "bbox": [
    423.0,
    286.0,
    495.0,
    326.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 1,
```

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"first_seen_at": "2026-05-02T04:29:36.641249+00:00",
"last_seen_at": "2026-05-02T04:30:16.794900+00:00",
"seen_count": 2,
"miss_count": 11,
"stale": true,
"cached": true
},
"text:connecting to agent...": {
  "element_key": "text:connecting to agent...",
  "element_type": "text",
  "label": "Connecting to agent...",
  "bbox": [
    183.0,
    526.0,
    359.0,
    566.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 0,
  "first_seen_at": "2026-05-02T04:29:36.641249+00:00",
  "last_seen_at": "2026-05-02T04:29:36.641249+00:00",
  "seen_count": 1,
  "miss_count": 12,
  "stale": true,
  "cached": true
}
```



```
},
"text:capturing initial screen...": {
  "element_key": "text:capturing initial screen...",
  "element_type": "text",
  "label": "Capturing initial screen...",
  "bbox": [
    183.0,
    558.0,
    415.0,
    598.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 0,
  "first_seen_at": "2026-05-02T04:29:36.641249+00:00",
  "last_seen_at": "2026-05-02T04:29:36.641249+00:00",
  "seen_count": 1,
  "miss_count": 12,
  "stale": true,
  "cached": true
},
"pane:preview window": {
  "element_key": "pane:preview window",
  "element_type": "pane",
  "label": "Preview Window",
  "bbox": [
```

```
    167.0,
    598.0,
    367.0,
    798.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 0,
  "first_seen_at": "2026-05-02T04:29:36.641249+00:00",
  "last_seen_at": "2026-05-02T04:29:36.641249+00:00",
  "seen_count": 1,
  "miss_count": 12,
  "stale": true,
  "cached": true
},
"input:agent is working...": {
  "element_key": "input:agent is working...",
  "element_type": "input",
  "label": "Agent is working...",
  "bbox": [
    151.0,
    862.0,
    1727.0,
    950.0
  ],
  "confidence": 0.4,
```

```
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 1,
"first_seen_at": "2026-05-02T04:29:36.641249+00:00",
"last_seen_at": "2026-05-02T04:30:16.794900+00:00",
"seen_count": 2,
"miss_count": 11,
"stale": true,
"cached": true
},
"button:send": {
  "element_key": "button:send",
  "element_type": "button",
  "label": "Send",
  "bbox": [
    1583.0,
    854.0,
    1663.0,
    942.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 1,
  "first_seen_at": "2026-05-02T04:29:36.641249+00:00",
```

```
    "last_seen_at": "2026-05-02T04:30:16.794900+00:00",
    "seen_count": 2,
    "miss_count": 11,
    "stale": true,
    "cached": true
  },
  "pane:taskbar": {
    "element_key": "pane:taskbar",
    "element_type": "pane",
    "label": "Taskbar",
    "bbox": [
      152.0,
      933.0,
      1727.0,
      1012.0
    ],
    "confidence": 0.855,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 0,
    "last_seen_index": 11,
    "first_seen_at": "2026-05-02T04:29:36.641249+00:00",
    "last_seen_at": "2026-05-02T04:37:59.364903+00:00",
    "seen_count": 2,
    "miss_count": 1,
    "stale": false,
    "cached": true
  },
}
```

```
"button:start": {
  "element_key": "button:start",
  "element_type": "button",
  "label": "Start",
  "bbox": [
    472.0,
    941.0,
    528.0,
    1005.0
  ],
  "confidence": 0.855,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 11,
  "first_seen_at": "2026-05-02T04:29:36.641249+00:00",
  "last_seen_at": "2026-05-02T04:37:59.364903+00:00",
  "seen_count": 3,
  "miss_count": 1,
  "stale": false,
  "cached": true
},
"input:search": {
  "element_key": "input:search",
  "element_type": "input",
  "label": "Search",
  "bbox": [
    512.0,
```

```
    941.0,  
    648.0,  
    1005.0  
  ],  
  "confidence": 0.855,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 0,  
  "last_seen_index": 11,  
  "first_seen_at": "2026-05-02T04:29:36.641249+00:00",  
  "last_seen_at": "2026-05-02T04:37:59.364903+00:00",  
  "seen_count": 3,  
  "miss_count": 1,  
  "stale": false,  
  "cached": true  
},  
"button:weather": {  
  "element_key": "button:weather",  
  "element_type": "button",  
  "label": "Weather",  
  "bbox": [  
    167.0,  
    958.0,  
    319.0,  
    1013.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",
```

```
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 0,
"first_seen_at": "2026-05-02T04:29:36.641249+00:00",
"last_seen_at": "2026-05-02T04:29:36.641249+00:00",
"seen_count": 1,
"miss_count": 12,
"stale": true,
"cached": true
},
"icon:file explorer": {
  "element_key": "icon:file explorer",
  "element_type": "icon",
  "label": "File Explorer",
  "bbox": [
    743.0,
    958.0,
    783.0,
    998.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 0,
  "first_seen_at": "2026-05-02T04:29:36.641249+00:00",
  "last_seen_at": "2026-05-02T04:29:36.641249+00:00",
```

```
    "seen_count": 1,  
    "miss_count": 12,  
    "stale": true,  
    "cached": true  
  },  
  "icon:chrome": {  
    "element_key": "icon:chrome",  
    "element_type": "icon",  
    "label": "Chrome",  
    "bbox": [  
      775.0,  
      958.0,  
      807.0,  
      998.0  
    ],  
    "confidence": 0.4,  
    "source": "gemini_vlm",  
    "first_seen_frame": null,  
    "last_seen_frame": null,  
    "first_seen_index": 0,  
    "last_seen_index": 0,  
    "first_seen_at": "2026-05-02T04:29:36.641249+00:00",  
    "last_seen_at": "2026-05-02T04:29:36.641249+00:00",  
    "seen_count": 1,  
    "miss_count": 12,  
    "stale": true,  
    "cached": true  
  },  
  "icon:settings": {
```



```
"element_key": "icon:settings",
"element_type": "icon",
"label": "Settings",
"bbox": [
    807.0,
    958.0,
    839.0,
    998.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 0,
"first_seen_at": "2026-05-02T04:29:36.641249+00:00",
"last_seen_at": "2026-05-02T04:29:36.641249+00:00",
"seen_count": 1,
"miss_count": 12,
"stale": true,
"cached": true
},
"icon:vs code": {
    "element_key": "icon:vs code",
    "element_type": "icon",
    "label": "VS Code",
    "bbox": [
        935.0,
        958.0,
```

```
    975.0,
    998.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 0,
"first_seen_at": "2026-05-02T04:29:36.641249+00:00",
"last_seen_at": "2026-05-02T04:29:36.641249+00:00",
"seen_count": 1,
"miss_count": 12,
"stale": true,
"cached": true
},
"icon:spotify": {
  "element_key": "icon:spotify",
  "element_type": "icon",
  "label": "Spotify",
  "bbox": [
    975.0,
    958.0,
    1015.0,
    998.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
```

```
"last_seen_frame": null,  
"first_seen_index": 0,  
"last_seen_index": 0,  
"first_seen_at": "2026-05-02T04:29:36.641249+00:00",  
"last_seen_at": "2026-05-02T04:29:36.641249+00:00",  
"seen_count": 1,  
"miss_count": 12,  
"stale": true,  
"cached": true  
},  
"pane:system tray": {  
  "element_key": "pane:system tray",  
  "element_type": "pane",  
  "label": "System Tray",  
  "bbox": [  
    1487.0,  
    950.0,  
    1647.0,  
    998.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 0,  
  "last_seen_index": 0,  
  "first_seen_at": "2026-05-02T04:29:36.641249+00:00",  
  "last_seen_at": "2026-05-02T04:29:36.641249+00:00",  
  "seen_count": 1,
```

```
    "miss_count": 12,  
    "stale": true,  
    "cached": true  
  },  
  "text:clock and date": {  
    "element_key": "text:clock and date",  
    "element_type": "text",  
    "label": "Clock and Date",  
    "bbox": [  
      1288.0,  
      941.0,  
      1360.0,  
      989.0  
    ],  
    "confidence": 0.855,  
    "source": "gemini_vlm",  
    "first_seen_frame": null,  
    "last_seen_frame": null,  
    "first_seen_index": 0,  
    "last_seen_index": 11,  
    "first_seen_at": "2026-05-02T04:29:36.641249+00:00",  
    "last_seen_at": "2026-05-02T04:37:59.364903+00:00",  
    "seen_count": 2,  
    "miss_count": 1,  
    "stale": false,  
    "cached": true  
  },  
  "icon:screenpilot ai logo": {  
    "element_key": "icon:screenpilot ai logo",
```

```
"element_type": "icon",
"label": "ScreenPilot AI Logo",
"bbox": [
  189.0,
  230.0,
  221.0,
  278.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 1,
"last_seen_index": 1,
"first_seen_at": "2026-05-02T04:30:16.794900+00:00",
"last_seen_at": "2026-05-02T04:30:16.794900+00:00",
"seen_count": 1,
"miss_count": 11,
"stale": true,
"cached": true
},
"heading:task plan": {
  "element_key": "heading:task plan",
  "element_type": "heading",
  "label": "Task Plan",
  "bbox": [
    205.0,
    422.0,
    309.0,
```

```
462.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 1,
"last_seen_index": 1,
"first_seen_at": "2026-05-02T04:30:16.794900+00:00",
"last_seen_at": "2026-05-02T04:30:16.794900+00:00",
"seen_count": 1,
"miss_count": 11,
"stale": true,
"cached": true
},
"text:move the mouse to the minimize button": {
  "element_key": "text:move the mouse to the minimize button",
  "element_type": "text",
  "label": "Move the mouse to the Minimize button",
  "bbox": [
    149.0,
    478.0,
    869.0,
    534.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
```

```
"first_seen_index": 1,
"last_seen_index": 1,
"first_seen_at": "2026-05-02T04:30:16.794900+00:00",
"last_seen_at": "2026-05-02T04:30:16.794900+00:00",
"seen_count": 1,
"miss_count": 11,
"stale": true,
"cached": true
},
"text:move the mouse to the chrome icon": {
  "element_key": "text:move the mouse to the chrome icon",
  "element_type": "text",
  "label": "Move the mouse to the Chrome icon",
  "bbox": [
    149.0,
    518.0,
    861.0,
    574.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
  "last_seen_index": 1,
  "first_seen_at": "2026-05-02T04:30:16.794900+00:00",
  "last_seen_at": "2026-05-02T04:30:16.794900+00:00",
  "seen_count": 1,
  "miss_count": 11,
```

```
    "stale": true,
    "cached": true
  },
  "pane:agent log": {
    "element_key": "pane:agent log",
    "element_type": "pane",
    "label": "Agent Log",
    "bbox": [
      149.0,
      558.0,
      877.0,
      758.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 1,
    "last_seen_index": 1,
    "first_seen_at": "2026-05-02T04:30:16.794900+00:00",
    "last_seen_at": "2026-05-02T04:30:16.794900+00:00",
    "seen_count": 1,
    "miss_count": 11,
    "stale": true,
    "cached": true
  },
  "button:chrome": {
    "element_key": "button:chrome",
    "element_type": "button",
```



```
"label": "Chrome",
"bbox": [
  877.0,
  942.0,
  917.0,
  998.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 1,
"last_seen_index": 1,
"first_seen_at": "2026-05-02T04:30:16.794900+00:00",
"last_seen_at": "2026-05-02T04:30:16.794900+00:00",
"seen_count": 1,
"miss_count": 11,
"stale": true,
"cached": true
},
"button:vs code": {
  "element_key": "button:vs code",
  "element_type": "button",
  "label": "VS Code",
  "bbox": [
    933.0,
    942.0,
    981.0,
    998.0
  ]
}
```

```
],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
  "last_seen_index": 1,
  "first_seen_at": "2026-05-02T04:30:16.794900+00:00",
  "last_seen_at": "2026-05-02T04:30:16.794900+00:00",
  "seen_count": 1,
  "miss_count": 11,
  "stale": true,
  "cached": true
},
"button:spotify": {
  "element_key": "button:spotify",
  "element_type": "button",
  "label": "Spotify",
  "bbox": [
    984.0,
    941.0,
    1040.0,
    997.0
  ],
  "confidence": 0.855,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
```

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"last_seen_index": 11,  
"first_seen_at": "2026-05-02T04:30:16.794900+00:00",  
"last_seen_at": "2026-05-02T04:37:59.364903+00:00",  
"seen_count": 2,  
"miss_count": 1,  
"stale": false,  
"cached": true  
},  
"text:clock": {  
  "element_key": "text:clock",  
  "element_type": "text",  
  "label": "Clock",  
  "bbox": [  
    1293.0,  
    942.0,  
    1365.0,  
    990.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 1,  
  "last_seen_index": 1,  
  "first_seen_at": "2026-05-02T04:30:16.794900+00:00",  
  "last_seen_at": "2026-05-02T04:30:16.794900+00:00",  
  "seen_count": 1,  
  "miss_count": 11,  
  "stale": true,
```

```
    "cached": true
  },
  "icon:google chrome": {
    "element_key": "icon:google chrome",
    "element_type": "icon",
    "label": "Google Chrome",
    "bbox": [
      152.0,
      109.0,
      264.0,
      261.0
    ],
    "confidence": 0.98,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 2,
    "last_seen_index": 12,
    "first_seen_at": "2026-05-02T04:31:04.756697+00:00",
    "last_seen_at": "2026-05-02T04:38:45.177526+00:00",
    "seen_count": 10,
    "miss_count": 0,
    "stale": false,
    "cached": false
  },
  "icon:huiontablet": {
    "element_key": "icon:huiontablet",
    "element_type": "icon",
    "label": "HuionTablet",
```

```
"bbox": [
    248.0,
    141.0,
    328.0,
    261.0
],
"confidence": 0.97,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 2,
"last_seen_index": 12,
"first_seen_at": "2026-05-02T04:31:04.756697+00:00",
"last_seen_at": "2026-05-02T04:38:45.177526+00:00",
"seen_count": 10,
"miss_count": 0,
"stale": false,
"cached": false
},
"button:google chrome": {
    "element_key": "button:google chrome",
    "element_type": "button",
    "label": "Google Chrome",
    "bbox": [
        872.0,
        941.0,
        920.0,
        997.0
    ],
```

```
"confidence": 0.855,  
"source": "gemini_vlm",  
"first_seen_frame": null,  
"last_seen_frame": null,  
"first_seen_index": 8,  
"last_seen_index": 11,  
"first_seen_at": "2026-05-02T04:35:46.397550+00:00",  
"last_seen_at": "2026-05-02T04:37:59.364903+00:00",  
"seen_count": 2,  
"miss_count": 1,  
"stale": false,  
"cached": true  
},  
"button:huiontablet": {  
  "element_key": "button:huiontablet",  
  "element_type": "button",  
  "label": "HuionTablet",  
  "bbox": [  
    248.0,  
    142.0,  
    328.0,  
    278.0  
  ],  
  "confidence": 0.6561,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 8,  
  "last_seen_index": 8,
```

```
"first_seen_at": "2026-05-02T04:35:46.397550+00:00",
"last_seen_at": "2026-05-02T04:35:46.397550+00:00",
"seen_count": 1,
"miss_count": 4,
"stale": true,
"cached": true
},
"icon:wise memory optimizer": {
  "element_key": "icon:wise memory optimizer",
  "element_type": "icon",
  "label": "Wise Memory Optimizer",
  "bbox": [
    152.0,
    341.0,
    256.0,
    541.0
  ],
  "confidence": 0.855,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 11,
  "last_seen_index": 11,
  "first_seen_at": "2026-05-02T04:37:59.364903+00:00",
  "last_seen_at": "2026-05-02T04:37:59.364903+00:00",
  "seen_count": 1,
  "miss_count": 1,
  "stale": false,
  "cached": true
}
```

```
},
"icon:amazon workspaces": {
  "element_key": "icon:amazon workspaces",
  "element_type": "icon",
  "label": "Amazon WorkSpaces",
  "bbox": [
    248.0,
    341.0,
    328.0,
    541.0
  ],
  "confidence": 0.855,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 11,
  "last_seen_index": 11,
  "first_seen_at": "2026-05-02T04:37:59.364903+00:00",
  "last_seen_at": "2026-05-02T04:37:59.364903+00:00",
  "seen_count": 1,
  "miss_count": 1,
  "stale": false,
  "cached": true
},
"icon:recycle bin": {
  "element_key": "icon:recycle bin",
  "element_type": "icon",
  "label": "Recycle Bin",
  "bbox": [
```



```
    152.0,
    541.0,
    256.0,
    741.0
  ],
  "confidence": 0.855,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 11,
  "last_seen_index": 11,
  "first_seen_at": "2026-05-02T04:37:59.364903+00:00",
  "last_seen_at": "2026-05-02T04:37:59.364903+00:00",
  "seen_count": 1,
  "miss_count": 1,
  "stale": false,
  "cached": true
},
"icon:krita": {
  "element_key": "icon:krita",
  "element_type": "icon",
  "label": "Krita",
  "bbox": [
    248.0,
    541.0,
    328.0,
    733.0
  ],
  "confidence": 0.855,
```

```
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 11,
"last_seen_index": 11,
"first_seen_at": "2026-05-02T04:37:59.364903+00:00",
"last_seen_at": "2026-05-02T04:37:59.364903+00:00",
"seen_count": 1,
"miss_count": 1,
"stale": false,
"cached": true
},
"icon:nuwani - chrome": {
  "element_key": "icon:nuwani - chrome",
  "element_type": "icon",
  "label": "Nuwani - Chrome",
  "bbox": [
    152.0,
    749.0,
    256.0,
    949.0
  ],
  "confidence": 0.855,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 11,
  "last_seen_index": 11,
  "first_seen_at": "2026-05-02T04:37:59.364903+00:00",
```

```
    "last_seen_at": "2026-05-02T04:37:59.364903+00:00",
    "seen_count": 1,
    "miss_count": 1,
    "stale": false,
    "cached": true
  },
  "icon:docker desktop": {
    "element_key": "icon:docker desktop",
    "element_type": "icon",
    "label": "Docker Desktop",
    "bbox": [
      248.0,
      757.0,
      328.0,
      949.0
    ],
    "confidence": 0.855,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 11,
    "last_seen_index": 11,
    "first_seen_at": "2026-05-02T04:37:59.364903+00:00",
    "last_seen_at": "2026-05-02T04:37:59.364903+00:00",
    "seen_count": 1,
    "miss_count": 1,
    "stale": false,
    "cached": true
  },
}
```

```
"icon:antigravity": {
  "element_key": "icon:antigravity",
  "element_type": "icon",
  "label": "Antigravity",
  "bbox": [
    160.0,
    973.0,
    256.0,
    1012.0
  ],
  "confidence": 0.81,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 11,
  "last_seen_index": 11,
  "first_seen_at": "2026-05-02T04:37:59.364903+00:00",
  "last_seen_at": "2026-05-02T04:37:59.364903+00:00",
  "seen_count": 1,
  "miss_count": 1,
  "stale": false,
  "cached": true
},
"icon:visual studio code": {
  "element_key": "icon:visual studio code",
  "element_type": "icon",
  "label": "Visual Studio Code",
  "bbox": [
    232.0,
```

```
    973.0,
    336.0,
    1005.0
],
"confidence": 0.855,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 11,
"last_seen_index": 11,
"first_seen_at": "2026-05-02T04:37:59.364903+00:00",
"last_seen_at": "2026-05-02T04:37:59.364903+00:00",
"seen_count": 1,
"miss_count": 1,
"stale": false,
"cached": true
},
"icon:postman": {
  "element_key": "icon:postman",
  "element_type": "icon",
  "label": "Postman",
  "bbox": [
    312.0,
    981.0,
    416.0,
    1005.0
  ],
  "confidence": 0.855,
  "source": "gemini_vlm",
```

```
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 11,
"last_seen_index": 11,
"first_seen_at": "2026-05-02T04:37:59.364903+00:00",
"last_seen_at": "2026-05-02T04:37:59.364903+00:00",
"seen_count": 1,
"miss_count": 1,
"stale": false,
"cached": true
},
"icon:loom": {
  "element_key": "icon:loom",
  "element_type": "icon",
  "label": "Loom",
  "bbox": [
    400.0,
    989.0,
    488.0,
    1005.0
  ],
  "confidence": 0.855,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 11,
  "last_seen_index": 11,
  "first_seen_at": "2026-05-02T04:37:59.364903+00:00",
  "last_seen_at": "2026-05-02T04:37:59.364903+00:00",
```

```
    "seen_count": 1,
    "miss_count": 1,
    "stale": false,
    "cached": true
  },
  "icon:vlc media player": {
    "element_key": "icon:vlc media player",
    "element_type": "icon",
    "label": "VLC media player",
    "bbox": [
      464.0,
      989.0,
      568.0,
      1005.0
    ],
    "confidence": 0.855,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 11,
    "last_seen_index": 11,
    "first_seen_at": "2026-05-02T04:37:59.364903+00:00",
    "last_seen_at": "2026-05-02T04:37:59.364903+00:00",
    "seen_count": 1,
    "miss_count": 1,
    "stale": false,
    "cached": true
  },
  "pane:amazon workspaces info": {
```

```
"element_key": "pane:amazon workspaces info",
"element_type": "pane",
"label": "Amazon WorkSpaces Info",
"bbox": [
    1088.0,
    133.0,
    1400.0,
    389.0
],
"confidence": 0.81,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 11,
"last_seen_index": 11,
"first_seen_at": "2026-05-02T04:37:59.364903+00:00",
"last_seen_at": "2026-05-02T04:37:59.364903+00:00",
"seen_count": 1,
"miss_count": 1,
"stale": false,
"cached": true
},
"button:weather widget": {
    "element_key": "button:weather widget",
    "element_type": "button",
    "label": "Weather Widget",
    "bbox": [
        160.0,
        949.0,
```



```
    296.0,  
    1012.0  
  ],  
  "confidence": 0.81,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 11,  
  "last_seen_index": 11,  
  "first_seen_at": "2026-05-02T04:37:59.364903+00:00",  
  "last_seen_at": "2026-05-02T04:37:59.364903+00:00",  
  "seen_count": 1,  
  "miss_count": 1,  
  "stale": false,  
  "cached": true  
},  
"button:visual studio code": {  
  "element_key": "button:visual studio code",  
  "element_type": "button",  
  "label": "Visual Studio Code",  
  "bbox": [  
    952.0,  
    941.0,  
    1000.0,  
    997.0  
  ],  
  "confidence": 0.855,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,
```

```
"last_seen_frame": null,  
"first_seen_index": 11,  
"last_seen_index": 11,  
"first_seen_at": "2026-05-02T04:37:59.364903+00:00",  
"last_seen_at": "2026-05-02T04:37:59.364903+00:00",  
"seen_count": 1,  
"miss_count": 1,  
"stale": false,  
"cached": true  
},  
"button:show hidden icons": {  
  "element_key": "button:show hidden icons",  
  "element_type": "button",  
  "label": "Show hidden icons",  
  "bbox": [  
    1160.0,  
    949.0,  
    1200.0,  
    989.0  
  ],  
  "confidence": 0.81,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 11,  
  "last_seen_index": 11,  
  "first_seen_at": "2026-05-02T04:37:59.364903+00:00",  
  "last_seen_at": "2026-05-02T04:37:59.364903+00:00",  
  "seen_count": 1,
```

```
    "miss_count": 1,
    "stale": false,
    "cached": true
  },
  "icon:network": {
    "element_key": "icon:network",
    "element_type": "icon",
    "label": "Network",
    "bbox": [
      1232.0,
      949.0,
      1264.0,
      989.0
    ],
    "confidence": 0.81,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 11,
    "last_seen_index": 11,
    "first_seen_at": "2026-05-02T04:37:59.364903+00:00",
    "last_seen_at": "2026-05-02T04:37:59.364903+00:00",
    "seen_count": 1,
    "miss_count": 1,
    "stale": false,
    "cached": true
  },
  "icon:volume": {
    "element_key": "icon:volume",
```

```
"element_type": "icon",
"label": "Volume",
"bbox": [
    1248.0,
    949.0,
    1288.0,
    989.0
],
"confidence": 0.81,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 11,
"last_seen_index": 11,
"first_seen_at": "2026-05-02T04:37:59.364903+00:00",
"last_seen_at": "2026-05-02T04:37:59.364903+00:00",
"seen_count": 1,
"miss_count": 1,
"stale": false,
"cached": true
},
"icon:battery": {
    "element_key": "icon:battery",
    "element_type": "icon",
    "label": "Battery",
    "bbox": [
        1264.0,
        949.0,
        1304.0,
```

```
    989.0
  ],
  "confidence": 0.81,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 11,
  "last_seen_index": 11,
  "first_seen_at": "2026-05-02T04:37:59.364903+00:00",
  "last_seen_at": "2026-05-02T04:37:59.364903+00:00",
  "seen_count": 1,
  "miss_count": 1,
  "stale": false,
  "cached": true
}
},
"current_elements": [
  {
    "id": "browser_tab_frontend",
    "type": "tab",
    "label": "frontend",
    "description": "Browser tab for the frontend application",
    "state": "visible",
    "dx": 124,
    "dy": 36,
    "frame_dx": 275,
    "frame_dy": 146,
    "confidence": 0.98,
    "source": "gemini_vlm",
```

```
"frame_bbox": [
  151,
  118,
  399,
  174
],
"session_memory_key": "tab:frontend",
"cached": false
},
{
  "id": "address_bar",
  "type": "input",
  "label": "Address Bar",
  "description": "Browser address bar showing localhost:5173",
  "state": "visible",
  "dx": 576,
  "dy": 100,
  "frame_dx": 727,
  "frame_dy": 210,
  "confidence": 0.99,
  "source": "gemini_vlm",
  "frame_bbox": [
    151,
    182,
    1303,
    238
  ],
  "session_memory_key": "input:address bar",
  "cached": false
}
```

```
},  
{  
  "id": "ask_gemini_button",  
  "type": "button",  
  "label": "Ask Gemini",  
  "description": "Ask Gemini button in the browser toolbar",  
  "state": "visible",  
  "dx": 1348,  
  "dy": 56,  
  "frame_dx": 1499,  
  "frame_dy": 166,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    1415,  
    142,  
    1583,  
    190  
  ],  
  "session_memory_key": "button:ask gemini",  
  "cached": false  
},  
{  
  "id": "window_minimize",  
  "type": "button",  
  "label": "Minimize",  
  "description": "Browser window minimize button",  
  "state": "visible",  
  "dx": 1460,
```

```
"dy": 52,
"frame_dx": 1611,
"frame_dy": 162,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    1583,
    142,
    1639,
    182
],
"session_memory_key": "button:minimize",
"cached": false
},
{
    "id": "window_maximize",
    "type": "button",
    "label": "Maximize",
    "description": "Browser window maximize button",
    "state": "visible",
    "dx": 1500,
    "dy": 52,
    "frame_dx": 1651,
    "frame_dy": 162,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        1623,
        142,
```



```
        1679,  
        182  
    ],  
    "session_memory_key": "button:maximize",  
    "cached": false  
},  
{  
    "id": "window_close",  
    "type": "button",  
    "label": "Close",  
    "description": "Browser window close button",  
    "state": "visible",  
    "dx": 1548,  
    "dy": 56,  
    "frame_dx": 1699,  
    "frame_dy": 166,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        1671,  
        142,  
        1727,  
        190  
    ],  
    "session_memory_key": "button:close",  
    "cached": false  
},  
{  
    "id": "screenpilot_ai_title",
```

```
"type": "heading",
"label": "ScreenPilot AI",
"description": "Application title ScreenPilot AI",
"state": "visible",
"dx": 120,
"dy": 192,
"frame_dx": 271,
"frame_dy": 302,
"confidence": 0.99,
"source": "gemini_vlm",
"frame_bbox": [
    167,
    270,
    375,
    334
],
"session_memory_key": "heading:screenpilot ai",
"cached": false
},
{
    "id": "agent_version_badge",
    "type": "label",
    "label": "Autonomous Task Agent v2",
    "description": "Badge indicating the agent version",
    "state": "visible",
    "dx": 308,
    "dy": 196,
    "frame_dx": 459,
    "frame_dy": 306,
```

```
"confidence": 0.97,  
"source": "gemini_vlm",  
"frame_bbox": [  
    423,  
    286,  
    495,  
    326  
],  
"session_memory_key": "label:autonomous task agent v2",  
"cached": false  
},  
{  
    "id": "status_connecting",  
    "type": "text",  
    "label": "Connecting to agent...",  
    "description": "Status message indicating connection progress",  
    "state": "visible",  
    "dx": 120,  
    "dy": 436,  
    "frame_dx": 271,  
    "frame_dy": 546,  
    "confidence": 0.96,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        183,  
        526,  
        359,  
        566  
    ],
```

```
"session_memory_key": "text:connecting to agent...",
"cached": false
},
{
  "id": "status_capturing",
  "type": "text",
  "label": "Capturing initial screen...",
  "description": "Status message indicating screen capture progress",
  "state": "visible",
  "dx": 148,
  "dy": 468,
  "frame_dx": 299,
  "frame_dy": 578,
  "confidence": 0.96,
  "source": "gemini_vlm",
  "frame_bbox": [
    183,
    558,
    415,
    598
  ],
  "session_memory_key": "text:capturing initial screen...",
  "cached": false
},
{
  "id": "preview_window",
  "type": "pane",
  "label": "Preview Window",
  "description": "Small preview window showing agent activity",
```

```
"state": "visible",
"dx": 116,
"dy": 588,
"frame_dx": 267,
"frame_dy": 698,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    167,
    598,
    367,
    798
],
"session_memory_key": "pane:preview window",
"cached": false
},
{
    "id": "chat_input_field",
    "type": "input",
    "label": "Agent is working...",
    "description": "Main chat input field for the agent",
    "state": "visible",
    "dx": 788,
    "dy": 796,
    "frame_dx": 939,
    "frame_dy": 906,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
```

```
    151,  
    862,  
    1727,  
    950  
  ],  
  "session_memory_key": "input:agent is working...",  
  "cached": false  
},  
{  
  "id": "send_button",  
  "type": "button",  
  "label": "Send",  
  "description": "Button to send commands to the agent",  
  "state": "visible",  
  "dx": 1472,  
  "dy": 788,  
  "frame_dx": 1623,  
  "frame_dy": 898,  
  "confidence": 0.97,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    1583,  
    854,  
    1663,  
    942  
  ],  
  "session_memory_key": "button:send",  
  "cached": false  
},
```

```
{
  "id": "windows_taskbar",
  "type": "pane",
  "label": "Taskbar",
  "description": "Windows taskbar at the bottom of the screen",
  "state": "visible",
  "dx": 788,
  "dy": 864,
  "frame_dx": 939,
  "frame_dy": 974,
  "confidence": 0.99,
  "source": "gemini_vlm",
  "frame_bbox": [
    151,
    934,
    1727,
    1013
  ],
  "session_memory_key": "pane:taskbar",
  "cached": false
},
{
  "id": "start_button",
  "type": "button",
  "label": "Start",
  "description": "Windows Start button",
  "state": "visible",
  "dx": 428,
  "dy": 864,
```

```
"frame_dx": 579,  
"frame_dy": 974,  
"confidence": 0.98,  
"source": "gemini_vlm",  
"frame_bbox": [  
    551,  
    950,  
    607,  
    998  
],  
"session_memory_key": "button:start",  
"cached": false  
},  
{  
    "id": "search_bar",  
    "type": "input",  
    "label": "Search",  
    "description": "Windows search bar in the taskbar",  
    "state": "visible",  
    "dx": 528,  
    "dy": 864,  
    "frame_dx": 679,  
    "frame_dy": 974,  
    "confidence": 0.98,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        591,  
        950,  
        767,
```



```
998
],
"session_memory_key": "input:search",
"cached": false
},
{
  "id": "weather_widget",
  "type": "button",
  "label": "Weather",
  "description": "Weather widget in the taskbar",
  "state": "visible",
  "dx": 92,
  "dy": 876,
  "frame_dx": 243,
  "frame_dy": 986,
  "confidence": 0.95,
  "source": "gemini_vlm",
  "frame_bbox": [
    167,
    958,
    319,
    1013
  ],
  "session_memory_key": "button:weather",
  "cached": false
},
{
  "id": "taskbar_file_explorer",
  "type": "icon",
```

```
"label": "File Explorer",
"description": "File Explorer icon in the taskbar",
"state": "visible",
"dx": 612,
"dy": 868,
"frame_dx": 763,
"frame_dy": 978,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    743,
    958,
    783,
    998
],
"session_memory_key": "icon:file explorer",
"cached": false
},
{
    "id": "taskbar_chrome",
    "type": "icon",
    "label": "Chrome",
    "description": "Google Chrome icon in the taskbar",
    "state": "visible",
    "dx": 640,
    "dy": 868,
    "frame_dx": 791,
    "frame_dy": 978,
    "confidence": 0.95,
```

```
"source": "gemini_vlm",
"frame_bbox": [
    775,
    958,
    807,
    998
],
"session_memory_key": "icon:chrome",
"cached": false
},
{
    "id": "taskbar_settings",
    "type": "icon",
    "label": "Settings",
    "description": "Settings icon in the taskbar",
    "state": "visible",
    "dx": 672,
    "dy": 868,
    "frame_dx": 823,
    "frame_dy": 978,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        807,
        958,
        839,
        998
    ],
    "session_memory_key": "icon:settings",
```

```
    "cached": false
  },
  {
    "id": "taskbar_vscode",
    "type": "icon",
    "label": "VS Code",
    "description": "Visual Studio Code icon in the taskbar",
    "state": "visible",
    "dx": 804,
    "dy": 868,
    "frame_dx": 955,
    "frame_dy": 978,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
      935,
      958,
      975,
      998
    ],
    "session_memory_key": "icon:vs code",
    "cached": false
  },
  {
    "id": "taskbar_spotify",
    "type": "icon",
    "label": "Spotify",
    "description": "Spotify icon in the taskbar",
    "state": "visible",
```

```
"dx": 844,  
  "dy": 868,  
  "frame_dx": 995,  
  "frame_dy": 978,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    975,  
    958,  
    1015,  
    998  
  ],  
  "session_memory_key": "icon:spotify",  
  "cached": false  
},  
{  
  "id": "system_tray",  
  "type": "pane",  
  "label": "System Tray",  
  "description": "System tray area with network, volume, and battery  
icons",  
  "state": "visible",  
  "dx": 1416,  
  "dy": 864,  
  "frame_dx": 1567,  
  "frame_dy": 974,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  

```

```
        1487,  
        950,  
        1647,  
        998  
    ],  
    "session_memory_key": "pane:system tray",  
    "cached": false  
},  
{  
    "id": "clock_date",  
    "type": "text",  
    "label": "Clock and Date",  
    "description": "System clock and date display in the taskbar",  
    "state": "visible",  
    "dx": 1520,  
    "dy": 868,  
    "frame_dx": 1671,  
    "frame_dy": 978,  
    "confidence": 0.98,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        1615,  
        942,  
        1727,  
        1013  
    ],  
    "session_memory_key": "text:clock and date",  
    "cached": false  
}
```

```
],  
"resolved_elements": [  
  {  
    "id": "browser_tab_frontend",  
    "type": "tab",  
    "label": "frontend",  
    "description": "Browser tab for the frontend application",  
    "state": "visible",  
    "dx": 124,  
    "dy": 36,  
    "frame_dx": 275,  
    "frame_dy": 146,  
    "confidence": 0.98,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
      151,  
      118,  
      399,  
      174  
    ],  
    "session_memory_key": "tab:frontend",  
    "cached": false  
  },  
  {  
    "id": "address_bar",  
    "type": "input",  
    "label": "Address Bar",  
    "description": "Browser address bar showing localhost:5173",  
    "state": "visible",
```

```
"dx": 576,  
  "dy": 100,  
  "frame_dx": 727,  
  "frame_dy": 210,  
  "confidence": 0.99,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    151,  
    182,  
    1303,  
    238  
  ],  
  "session_memory_key": "input:address bar",  
  "cached": false  
},  
{  
  "id": "ask_gemini_button",  
  "type": "button",  
  "label": "Ask Gemini",  
  "description": "Ask Gemini button in the browser toolbar",  
  "state": "visible",  
  "dx": 1348,  
  "dy": 56,  
  "frame_dx": 1499,  
  "frame_dy": 166,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    1415,
```



```
        142,  
        1583,  
        190  
    ],  
    "session_memory_key": "button:ask gemini",  
    "cached": false  
},  
{  
    "id": "window_minimize",  
    "type": "button",  
    "label": "Minimize",  
    "description": "Browser window minimize button",  
    "state": "visible",  
    "dx": 1460,  
    "dy": 52,  
    "frame_dx": 1611,  
    "frame_dy": 162,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        1583,  
        142,  
        1639,  
        182  
    ],  
    "session_memory_key": "button:minimize",  
    "cached": false  
},  
{
```

```
"id": "window_maximize",
"type": "button",
"label": "Maximize",
"description": "Browser window maximize button",
"state": "visible",
"dx": 1500,
"dy": 52,
"frame_dx": 1651,
"frame_dy": 162,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    1623,
    142,
    1679,
    182
],
"session_memory_key": "button:maximize",
"cached": false
},
{
  "id": "window_close",
  "type": "button",
  "label": "Close",
  "description": "Browser window close button",
  "state": "visible",
  "dx": 1548,
  "dy": 56,
  "frame_dx": 1699,
```

```
"frame_dy": 166,  
"confidence": 0.9,  
"source": "gemini_vlm",  
"frame_bbox": [  
    1671,  
    142,  
    1727,  
    190  
],  
"session_memory_key": "button:close",  
"cached": false  
},  
{  
    "id": "screenpilot_ai_title",  
    "type": "heading",  
    "label": "ScreenPilot AI",  
    "description": "Application title ScreenPilot AI",  
    "state": "visible",  
    "dx": 120,  
    "dy": 192,  
    "frame_dx": 271,  
    "frame_dy": 302,  
    "confidence": 0.99,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        167,  
        270,  
        375,  
        334
```

```
    ],  
    "session_memory_key": "heading:screenpilot ai",  
    "cached": false  
  },  
  {  
    "id": "agent_version_badge",  
    "type": "label",  
    "label": "Autonomous Task Agent v2",  
    "description": "Badge indicating the agent version",  
    "state": "visible",  
    "dx": 308,  
    "dy": 196,  
    "frame_dx": 459,  
    "frame_dy": 306,  
    "confidence": 0.97,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
      423,  
      286,  
      495,  
      326  
    ],  
    "session_memory_key": "label:autonomous task agent v2",  
    "cached": false  
  },  
  {  
    "id": "status_connecting",  
    "type": "text",  
    "label": "Connecting to agent..."
```

```
"description": "Status message indicating connection progress",
"state": "visible",
"dx": 120,
"dy": 436,
"frame_dx": 271,
"frame_dy": 546,
"confidence": 0.96,
"source": "gemini_vlm",
"frame_bbox": [
    183,
    526,
    359,
    566
],
"session_memory_key": "text:connecting to agent...",
"cached": false
},
{
    "id": "status_capturing",
    "type": "text",
    "label": "Capturing initial screen...",
    "description": "Status message indicating screen capture progress",
    "state": "visible",
    "dx": 148,
    "dy": 468,
    "frame_dx": 299,
    "frame_dy": 578,
    "confidence": 0.96,
    "source": "gemini_vlm",
```

```
"frame_bbox": [
  183,
  558,
  415,
  598
],
"session_memory_key": "text:capturing initial screen...",
"cached": false
},
{
  "id": "preview_window",
  "type": "pane",
  "label": "Preview Window",
  "description": "Small preview window showing agent activity",
  "state": "visible",
  "dx": 116,
  "dy": 588,
  "frame_dx": 267,
  "frame_dy": 698,
  "confidence": 0.95,
  "source": "gemini_vlm",
  "frame_bbox": [
    167,
    598,
    367,
    798
  ],
  "session_memory_key": "pane:preview window",
  "cached": false
}
```

```
},  
{  
  "id": "chat_input_field",  
  "type": "input",  
  "label": "Agent is working...",  
  "description": "Main chat input field for the agent",  
  "state": "visible",  
  "dx": 788,  
  "dy": 796,  
  "frame_dx": 939,  
  "frame_dy": 906,  
  "confidence": 0.98,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    151,  
    862,  
    1727,  
    950  
  ],  
  "session_memory_key": "input:agent is working...",  
  "cached": false  
},  
{  
  "id": "send_button",  
  "type": "button",  
  "label": "Send",  
  "description": "Button to send commands to the agent",  
  "state": "visible",  
  "dx": 1472,
```

```
"dy": 788,  
"frame_dx": 1623,  
"frame_dy": 898,  
"confidence": 0.97,  
"source": "gemini_vlm",  
"frame_bbox": [  
    1583,  
    854,  
    1663,  
    942  
],  
"session_memory_key": "button:send",  
"cached": false  
},  
{  
    "id": "windows_taskbar",  
    "type": "pane",  
    "label": "Taskbar",  
    "description": "Windows taskbar at the bottom of the screen",  
    "state": "visible",  
    "dx": 788,  
    "dy": 864,  
    "frame_dx": 939,  
    "frame_dy": 974,  
    "confidence": 0.99,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        151,  
        934,
```



```
        1727,  
        1013  
    ],  
    "session_memory_key": "pane:taskbar",  
    "cached": false  
},  
{  
    "id": "start_button",  
    "type": "button",  
    "label": "Start",  
    "description": "Windows Start button",  
    "state": "visible",  
    "dx": 428,  
    "dy": 864,  
    "frame_dx": 579,  
    "frame_dy": 974,  
    "confidence": 0.98,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        551,  
        950,  
        607,  
        998  
    ],  
    "session_memory_key": "button:start",  
    "cached": false  
},  
{  
    "id": "search_bar",
```

```
"type": "input",
"label": "Search",
"description": "Windows search bar in the taskbar",
"state": "visible",
"dx": 528,
"dy": 864,
"frame_dx": 679,
"frame_dy": 974,
"confidence": 0.98,
"source": "gemini_vlm",
"frame_bbox": [
    591,
    950,
    767,
    998
],
"session_memory_key": "input:search",
"cached": false
},
{
    "id": "weather_widget",
    "type": "button",
    "label": "Weather",
    "description": "Weather widget in the taskbar",
    "state": "visible",
    "dx": 92,
    "dy": 876,
    "frame_dx": 243,
    "frame_dy": 986,
```

```
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    167,
    958,
    319,
    1013
],
"session_memory_key": "button:weather",
"cached": false
},
{
    "id": "taskbar_file_explorer",
    "type": "icon",
    "label": "File Explorer",
    "description": "File Explorer icon in the taskbar",
    "state": "visible",
    "dx": 612,
    "dy": 868,
    "frame_dx": 763,
    "frame_dy": 978,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        743,
        958,
        783,
        998
    ],
    ],
```

```
"session_memory_key": "icon:file explorer",
"cached": false
},
{
  "id": "taskbar_chrome",
  "type": "icon",
  "label": "Chrome",
  "description": "Google Chrome icon in the taskbar",
  "state": "visible",
  "dx": 640,
  "dy": 868,
  "frame_dx": 791,
  "frame_dy": 978,
  "confidence": 0.95,
  "source": "gemini_vlm",
  "frame_bbox": [
    775,
    958,
    807,
    998
  ],
  "session_memory_key": "icon:chrome",
  "cached": false
},
{
  "id": "taskbar_settings",
  "type": "icon",
  "label": "Settings",
  "description": "Settings icon in the taskbar",
```

```
"state": "visible",
"dx": 672,
"dy": 868,
"frame_dx": 823,
"frame_dy": 978,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    807,
    958,
    839,
    998
],
"session_memory_key": "icon:settings",
"cached": false
},
{
    "id": "taskbar_vscode",
    "type": "icon",
    "label": "VS Code",
    "description": "Visual Studio Code icon in the taskbar",
    "state": "visible",
    "dx": 804,
    "dy": 868,
    "frame_dx": 955,
    "frame_dy": 978,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
```

```
    935,  
    958,  
    975,  
    998  
  ],  
  "session_memory_key": "icon:vs code",  
  "cached": false  
},  
{  
  "id": "taskbar_spotify",  
  "type": "icon",  
  "label": "Spotify",  
  "description": "Spotify icon in the taskbar",  
  "state": "visible",  
  "dx": 844,  
  "dy": 868,  
  "frame_dx": 995,  
  "frame_dy": 978,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    975,  
    958,  
    1015,  
    998  
  ],  
  "session_memory_key": "icon:spotify",  
  "cached": false  
},
```

```
{
  "id": "system_tray",
  "type": "pane",
  "label": "System Tray",
  "description": "System tray area with network, volume, and battery
icons",
  "state": "visible",
  "dx": 1416,
  "dy": 864,
  "frame_dx": 1567,
  "frame_dy": 974,
  "confidence": 0.95,
  "source": "gemini_vlm",
  "frame_bbox": [
    1487,
    950,
    1647,
    998
  ],
  "session_memory_key": "pane:system tray",
  "cached": false
},
{
  "id": "clock_date",
  "type": "text",
  "label": "Clock and Date",
  "description": "System clock and date display in the taskbar",
  "state": "visible",
  "dx": 1520,
```

```
"dy": 868,
"frame_dx": 1671,
"frame_dy": 978,
"confidence": 0.98,
"source": "gemini_vlm",
"frame_bbox": [
    1615,
    942,
    1727,
    1013
],
"session_memory_key": "text:clock and date",
"cached": false
}
],
"screen_data": {
    "image": "E:\\sliit-
project\\ScreenAwareTaskAgent\\vision\\data\\sessions\\shot_1777696146_8f4fe6\\pr
ocessing\\proc_1777696146_8ffec3\\preprocessed_frames\\frame_1777696146_e68365.jp
g",
    "image_size": {
        "width": 1920,
        "height": 1080
    },
    "screen_bbox": [
        151,
        110,
        1727,
        1013
    ],
    ],
```



```
"screen_size": {
  "width": 1576,
  "height": 903
},
"coordinate_system": "pixel",
"elements": [
  {
    "id": "browser_tab_frontend",
    "type": "tab",
    "label": "frontend",
    "description": "Browser tab for the frontend application",
    "state": "visible",
    "dx": 124,
    "dy": 36,
    "frame_dx": 275,
    "frame_dy": 146,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
      151,
      118,
      399,
      174
    ],
    "session_memory_key": "tab:frontend",
    "cached": false
  },
  {
    "id": "address_bar",
```

```
"type": "input",
"label": "Address Bar",
"description": "Browser address bar showing localhost:5173",
"state": "visible",
"dx": 576,
"dy": 100,
"frame_dx": 727,
"frame_dy": 210,
"confidence": 0.99,
"source": "gemini_vlm",
"frame_bbox": [
    151,
    182,
    1303,
    238
],
"session_memory_key": "input:address bar",
"cached": false
},
{
    "id": "ask_gemini_button",
    "type": "button",
    "label": "Ask Gemini",
    "description": "Ask Gemini button in the browser toolbar",
    "state": "visible",
    "dx": 1348,
    "dy": 56,
    "frame_dx": 1499,
    "frame_dy": 166,
```

```
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    1415,
    142,
    1583,
    190
],
"session_memory_key": "button:ask gemini",
"cached": false
},
{
    "id": "window_minimize",
    "type": "button",
    "label": "Minimize",
    "description": "Browser window minimize button",
    "state": "visible",
    "dx": 1460,
    "dy": 52,
    "frame_dx": 1611,
    "frame_dy": 162,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        1583,
        142,
        1639,
        182
    ],
    ],
```

```
"session_memory_key": "button:minimize",
"cached": false
},
{
  "id": "window_maximize",
  "type": "button",
  "label": "Maximize",
  "description": "Browser window maximize button",
  "state": "visible",
  "dx": 1500,
  "dy": 52,
  "frame_dx": 1651,
  "frame_dy": 162,
  "confidence": 0.9,
  "source": "gemini_vlm",
  "frame_bbox": [
    1623,
    142,
    1679,
    182
  ],
  "session_memory_key": "button:maximize",
  "cached": false
},
{
  "id": "window_close",
  "type": "button",
  "label": "Close",
  "description": "Browser window close button",
```

```
"state": "visible",
"dx": 1548,
"dy": 56,
"frame_dx": 1699,
"frame_dy": 166,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    1671,
    142,
    1727,
    190
],
"session_memory_key": "button:close",
"cached": false
},
{
    "id": "screenpilot_ai_title",
    "type": "heading",
    "label": "ScreenPilot AI",
    "description": "Application title ScreenPilot AI",
    "state": "visible",
    "dx": 120,
    "dy": 192,
    "frame_dx": 271,
    "frame_dy": 302,
    "confidence": 0.99,
    "source": "gemini_vlm",
    "frame_bbox": [
```

```
        167,  
        270,  
        375,  
        334  
    ],  
    "session_memory_key": "heading:screenpilot ai",  
    "cached": false  
},  
{  
    "id": "agent_version_badge",  
    "type": "label",  
    "label": "Autonomous Task Agent v2",  
    "description": "Badge indicating the agent version",  
    "state": "visible",  
    "dx": 308,  
    "dy": 196,  
    "frame_dx": 459,  
    "frame_dy": 306,  
    "confidence": 0.97,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        423,  
        286,  
        495,  
        326  
    ],  
    "session_memory_key": "label:autonomous task agent v2",  
    "cached": false  
},
```

```
{
  "id": "status_connecting",
  "type": "text",
  "label": "Connecting to agent...",
  "description": "Status message indicating connection progress",
  "state": "visible",
  "dx": 120,
  "dy": 436,
  "frame_dx": 271,
  "frame_dy": 546,
  "confidence": 0.96,
  "source": "gemini_vlm",
  "frame_bbox": [
    183,
    526,
    359,
    566
  ],
  "session_memory_key": "text:connecting to agent...",
  "cached": false
},
{
  "id": "status_capturing",
  "type": "text",
  "label": "Capturing initial screen...",
  "description": "Status message indicating screen capture progress",
  "state": "visible",
  "dx": 148,
  "dy": 468,
```

```
"frame_dx": 299,  
"frame_dy": 578,  
"confidence": 0.96,  
"source": "gemini_vlm",  
"frame_bbox": [  
    183,  
    558,  
    415,  
    598  
],  
"session_memory_key": "text:capturing initial screen...",  
"cached": false  
},  
{  
    "id": "preview_window",  
    "type": "pane",  
    "label": "Preview Window",  
    "description": "Small preview window showing agent activity",  
    "state": "visible",  
    "dx": 116,  
    "dy": 588,  
    "frame_dx": 267,  
    "frame_dy": 698,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        167,  
        598,  
        367,
```



```
798
    ],
    "session_memory_key": "pane:preview window",
    "cached": false
  },
  {
    "id": "chat_input_field",
    "type": "input",
    "label": "Agent is working...",
    "description": "Main chat input field for the agent",
    "state": "visible",
    "dx": 788,
    "dy": 796,
    "frame_dx": 939,
    "frame_dy": 906,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
      151,
      862,
      1727,
      950
    ],
    "session_memory_key": "input:agent is working...",
    "cached": false
  },
  {
    "id": "send_button",
    "type": "button",
```

```
    "label": "Send",
    "description": "Button to send commands to the agent",
    "state": "visible",
    "dx": 1472,
    "dy": 788,
    "frame_dx": 1623,
    "frame_dy": 898,
    "confidence": 0.97,
    "source": "gemini_vlm",
    "frame_bbox": [
        1583,
        854,
        1663,
        942
    ],
    "session_memory_key": "button:send",
    "cached": false
},
{
    "id": "windows_taskbar",
    "type": "pane",
    "label": "Taskbar",
    "description": "Windows taskbar at the bottom of the screen",
    "state": "visible",
    "dx": 788,
    "dy": 864,
    "frame_dx": 939,
    "frame_dy": 974,
    "confidence": 0.99,
```

```
"source": "gemini_vlm",
"frame_bbox": [
    151,
    934,
    1727,
    1013
],
"session_memory_key": "pane:taskbar",
"cached": false
},
{
    "id": "start_button",
    "type": "button",
    "label": "Start",
    "description": "Windows Start button",
    "state": "visible",
    "dx": 428,
    "dy": 864,
    "frame_dx": 579,
    "frame_dy": 974,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
        551,
        950,
        607,
        998
    ],
    "session_memory_key": "button:start",
```

```
    "cached": false
  },
  {
    "id": "search_bar",
    "type": "input",
    "label": "Search",
    "description": "Windows search bar in the taskbar",
    "state": "visible",
    "dx": 528,
    "dy": 864,
    "frame_dx": 679,
    "frame_dy": 974,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
      591,
      950,
      767,
      998
    ],
    "session_memory_key": "input:search",
    "cached": false
  },
  {
    "id": "weather_widget",
    "type": "button",
    "label": "Weather",
    "description": "Weather widget in the taskbar",
    "state": "visible",
```

```
"dx": 92,  
"dy": 876,  
"frame_dx": 243,  
"frame_dy": 986,  
"confidence": 0.95,  
"source": "gemini_vlm",  
"frame_bbox": [  
    167,  
    958,  
    319,  
    1013  
],  
"session_memory_key": "button:weather",  
"cached": false  
},  
{  
    "id": "taskbar_file_explorer",  
    "type": "icon",  
    "label": "File Explorer",  
    "description": "File Explorer icon in the taskbar",  
    "state": "visible",  
    "dx": 612,  
    "dy": 868,  
    "frame_dx": 763,  
    "frame_dy": 978,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        743,
```

```
        958,  
        783,  
        998  
    ],  
    "session_memory_key": "icon:file explorer",  
    "cached": false  
},  
{  
    "id": "taskbar_chrome",  
    "type": "icon",  
    "label": "Chrome",  
    "description": "Google Chrome icon in the taskbar",  
    "state": "visible",  
    "dx": 640,  
    "dy": 868,  
    "frame_dx": 791,  
    "frame_dy": 978,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        775,  
        958,  
        807,  
        998  
    ],  
    "session_memory_key": "icon:chrome",  
    "cached": false  
},  
{
```

```
"id": "taskbar_settings",
"type": "icon",
"label": "Settings",
"description": "Settings icon in the taskbar",
"state": "visible",
"dx": 672,
"dy": 868,
"frame_dx": 823,
"frame_dy": 978,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    807,
    958,
    839,
    998
],
"session_memory_key": "icon:settings",
"cached": false
},
{
    "id": "taskbar_vscode",
    "type": "icon",
    "label": "VS Code",
    "description": "Visual Studio Code icon in the taskbar",
    "state": "visible",
    "dx": 804,
    "dy": 868,
    "frame_dx": 955,
```

```
"frame_dy": 978,  
"confidence": 0.95,  
"source": "gemini_vlm",  
"frame_bbox": [  
    935,  
    958,  
    975,  
    998  
],  
"session_memory_key": "icon:vs code",  
"cached": false  
},  
{  
    "id": "taskbar_spotify",  
    "type": "icon",  
    "label": "Spotify",  
    "description": "Spotify icon in the taskbar",  
    "state": "visible",  
    "dx": 844,  
    "dy": 868,  
    "frame_dx": 995,  
    "frame_dy": 978,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        975,  
        958,  
        1015,  
        998
```



```
    ],
    "session_memory_key": "icon:spotify",
    "cached": false
  },
  {
    "id": "system_tray",
    "type": "pane",
    "label": "System Tray",
    "description": "System tray area with network, volume, and battery
icons",
    "state": "visible",
    "dx": 1416,
    "dy": 864,
    "frame_dx": 1567,
    "frame_dy": 974,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
      1487,
      950,
      1647,
      998
    ],
    "session_memory_key": "pane:system tray",
    "cached": false
  },
  {
    "id": "clock_date",
    "type": "text",
```

```
    "label": "Clock and Date",
    "description": "System clock and date display in the taskbar",
    "state": "visible",
    "dx": 1520,
    "dy": 868,
    "frame_dx": 1671,
    "frame_dy": 978,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
        1615,
        942,
        1727,
        1013
    ],
    "session_memory_key": "text:clock and date",
    "cached": false
}
],
"element_count": 24,
"current_elements": [
    {
        "id": "browser_tab_frontend",
        "type": "tab",
        "label": "frontend",
        "description": "Browser tab for the frontend application",
        "state": "visible",
        "dx": 124,
        "dy": 36,
```

```
"frame_dx": 275,  
"frame_dy": 146,  
"confidence": 0.98,  
"source": "gemini_vlm",  
"frame_bbox": [  
    151,  
    118,  
    399,  
    174  
],  
"session_memory_key": "tab:frontend",  
"cached": false  
},  
{  
    "id": "address_bar",  
    "type": "input",  
    "label": "Address Bar",  
    "description": "Browser address bar showing localhost:5173",  
    "state": "visible",  
    "dx": 576,  
    "dy": 100,  
    "frame_dx": 727,  
    "frame_dy": 210,  
    "confidence": 0.99,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        151,  
        182,  
        1303,
```

238

```
],  
  "session_memory_key": "input:address bar",  
  "cached": false  
},  
{  
  "id": "ask_gemini_button",  
  "type": "button",  
  "label": "Ask Gemini",  
  "description": "Ask Gemini button in the browser toolbar",  
  "state": "visible",  
  "dx": 1348,  
  "dy": 56,  
  "frame_dx": 1499,  
  "frame_dy": 166,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    1415,  
    142,  
    1583,  
    190  
  ],  
  "session_memory_key": "button:ask gemini",  
  "cached": false  
},  
{  
  "id": "window_minimize",  
  "type": "button",
```

```
    "label": "Minimize",
    "description": "Browser window minimize button",
    "state": "visible",
    "dx": 1460,
    "dy": 52,
    "frame_dx": 1611,
    "frame_dy": 162,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        1583,
        142,
        1639,
        182
    ],
    "session_memory_key": "button:minimize",
    "cached": false
},
{
    "id": "window_maximize",
    "type": "button",
    "label": "Maximize",
    "description": "Browser window maximize button",
    "state": "visible",
    "dx": 1500,
    "dy": 52,
    "frame_dx": 1651,
    "frame_dy": 162,
    "confidence": 0.9,
```

```
"source": "gemini_vlm",
"frame_bbox": [
    1623,
    142,
    1679,
    182
],
"session_memory_key": "button:maximize",
"cached": false
},
{
    "id": "window_close",
    "type": "button",
    "label": "Close",
    "description": "Browser window close button",
    "state": "visible",
    "dx": 1548,
    "dy": 56,
    "frame_dx": 1699,
    "frame_dy": 166,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        1671,
        142,
        1727,
        190
    ],
    "session_memory_key": "button:close",
```

```
    "cached": false
  },
  {
    "id": "screenpilot_ai_title",
    "type": "heading",
    "label": "ScreenPilot AI",
    "description": "Application title ScreenPilot AI",
    "state": "visible",
    "dx": 120,
    "dy": 192,
    "frame_dx": 271,
    "frame_dy": 302,
    "confidence": 0.99,
    "source": "gemini_vlm",
    "frame_bbox": [
      167,
      270,
      375,
      334
    ],
    "session_memory_key": "heading:screenpilot ai",
    "cached": false
  },
  {
    "id": "agent_version_badge",
    "type": "label",
    "label": "Autonomous Task Agent v2",
    "description": "Badge indicating the agent version",
    "state": "visible",
```

```
"dx": 308,  
  "dy": 196,  
  "frame_dx": 459,  
  "frame_dy": 306,  
  "confidence": 0.97,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    423,  
    286,  
    495,  
    326  
  ],  
  "session_memory_key": "label:autonomous task agent v2",  
  "cached": false  
},  
{  
  "id": "status_connecting",  
  "type": "text",  
  "label": "Connecting to agent...",  
  "description": "Status message indicating connection progress",  
  "state": "visible",  
  "dx": 120,  
  "dy": 436,  
  "frame_dx": 271,  
  "frame_dy": 546,  
  "confidence": 0.96,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    183,
```



```
        526,  
        359,  
        566  
    ],  
    "session_memory_key": "text:connecting to agent...",  
    "cached": false  
},  
{  
    "id": "status_capturing",  
    "type": "text",  
    "label": "Capturing initial screen...",  
    "description": "Status message indicating screen capture progress",  
    "state": "visible",  
    "dx": 148,  
    "dy": 468,  
    "frame_dx": 299,  
    "frame_dy": 578,  
    "confidence": 0.96,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        183,  
        558,  
        415,  
        598  
    ],  
    "session_memory_key": "text:capturing initial screen...",  
    "cached": false  
},  
{
```

```
"id": "preview_window",
"type": "pane",
"label": "Preview Window",
"description": "Small preview window showing agent activity",
"state": "visible",
"dx": 116,
"dy": 588,
"frame_dx": 267,
"frame_dy": 698,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    167,
    598,
    367,
    798
],
"session_memory_key": "pane:preview window",
"cached": false
},
{
    "id": "chat_input_field",
    "type": "input",
    "label": "Agent is working...",
    "description": "Main chat input field for the agent",
    "state": "visible",
    "dx": 788,
    "dy": 796,
    "frame_dx": 939,
```

```
"frame_dy": 906,
"confidence": 0.98,
"source": "gemini_vlm",
"frame_bbox": [
    151,
    862,
    1727,
    950
],
"session_memory_key": "input:agent is working...",
"cached": false
},
{
    "id": "send_button",
    "type": "button",
    "label": "Send",
    "description": "Button to send commands to the agent",
    "state": "visible",
    "dx": 1472,
    "dy": 788,
    "frame_dx": 1623,
    "frame_dy": 898,
    "confidence": 0.97,
    "source": "gemini_vlm",
    "frame_bbox": [
        1583,
        854,
        1663,
        942
```

```
    ],  
    "session_memory_key": "button:send",  
    "cached": false  
  },  
  {  
    "id": "windows_taskbar",  
    "type": "pane",  
    "label": "Taskbar",  
    "description": "Windows taskbar at the bottom of the screen",  
    "state": "visible",  
    "dx": 788,  
    "dy": 864,  
    "frame_dx": 939,  
    "frame_dy": 974,  
    "confidence": 0.99,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
      151,  
      934,  
      1727,  
      1013  
    ],  
    "session_memory_key": "pane:taskbar",  
    "cached": false  
  },  
  {  
    "id": "start_button",  
    "type": "button",  
    "label": "Start",
```

```
"description": "Windows Start button",
"state": "visible",
"dx": 428,
"dy": 864,
"frame_dx": 579,
"frame_dy": 974,
"confidence": 0.98,
"source": "gemini_vlm",
"frame_bbox": [
    551,
    950,
    607,
    998
],
"session_memory_key": "button:start",
"cached": false
},
{
    "id": "search_bar",
    "type": "input",
    "label": "Search",
    "description": "Windows search bar in the taskbar",
    "state": "visible",
    "dx": 528,
    "dy": 864,
    "frame_dx": 679,
    "frame_dy": 974,
    "confidence": 0.98,
    "source": "gemini_vlm",
```

```
"frame_bbox": [
    591,
    950,
    767,
    998
],
"session_memory_key": "input:search",
"cached": false
},
{
    "id": "weather_widget",
    "type": "button",
    "label": "Weather",
    "description": "Weather widget in the taskbar",
    "state": "visible",
    "dx": 92,
    "dy": 876,
    "frame_dx": 243,
    "frame_dy": 986,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        167,
        958,
        319,
        1013
    ],
    "session_memory_key": "button:weather",
    "cached": false
}
```

```
},  
{  
  "id": "taskbar_file_explorer",  
  "type": "icon",  
  "label": "File Explorer",  
  "description": "File Explorer icon in the taskbar",  
  "state": "visible",  
  "dx": 612,  
  "dy": 868,  
  "frame_dx": 763,  
  "frame_dy": 978,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    743,  
    958,  
    783,  
    998  
  ],  
  "session_memory_key": "icon:file explorer",  
  "cached": false  
},  
{  
  "id": "taskbar_chrome",  
  "type": "icon",  
  "label": "Chrome",  
  "description": "Google Chrome icon in the taskbar",  
  "state": "visible",  
  "dx": 640,
```

```
"dy": 868,  
  "frame_dx": 791,  
  "frame_dy": 978,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    775,  
    958,  
    807,  
    998  
  ],  
  "session_memory_key": "icon:chrome",  
  "cached": false  
},  
{  
  "id": "taskbar_settings",  
  "type": "icon",  
  "label": "Settings",  
  "description": "Settings icon in the taskbar",  
  "state": "visible",  
  "dx": 672,  
  "dy": 868,  
  "frame_dx": 823,  
  "frame_dy": 978,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    807,  
    958,
```



```
        839,  
        998  
    ],  
    "session_memory_key": "icon:settings",  
    "cached": false  
},  
{  
    "id": "taskbar_vscode",  
    "type": "icon",  
    "label": "VS Code",  
    "description": "Visual Studio Code icon in the taskbar",  
    "state": "visible",  
    "dx": 804,  
    "dy": 868,  
    "frame_dx": 955,  
    "frame_dy": 978,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        935,  
        958,  
        975,  
        998  
    ],  
    "session_memory_key": "icon:vs code",  
    "cached": false  
},  
{  
    "id": "taskbar_spotify",
```

```
    "type": "icon",
    "label": "Spotify",
    "description": "Spotify icon in the taskbar",
    "state": "visible",
    "dx": 844,
    "dy": 868,
    "frame_dx": 995,
    "frame_dy": 978,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        975,
        958,
        1015,
        998
    ],
    "session_memory_key": "icon:spotify",
    "cached": false
},
{
    "id": "system_tray",
    "type": "pane",
    "label": "System Tray",
    "description": "System tray area with network, volume, and battery icons",
    "state": "visible",
    "dx": 1416,
    "dy": 864,
    "frame_dx": 1567,
```

```
"frame_dy": 974,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    1487,
    950,
    1647,
    998
],
"session_memory_key": "pane:system tray",
"cached": false
},
{
    "id": "clock_date",
    "type": "text",
    "label": "Clock and Date",
    "description": "System clock and date display in the taskbar",
    "state": "visible",
    "dx": 1520,
    "dy": 868,
    "frame_dx": 1671,
    "frame_dy": 978,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
        1615,
        942,
        1727,
        1013
    ]
}
```

```
    ],
    "session_memory_key": "text:clock and date",
    "cached": false
  }
],
"resolved_elements": [
  {
    "id": "browser_tab_frontend",
    "type": "tab",
    "label": "frontend",
    "description": "Browser tab for the frontend application",
    "state": "visible",
    "dx": 124,
    "dy": 36,
    "frame_dx": 275,
    "frame_dy": 146,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
      151,
      118,
      399,
      174
    ],
    "session_memory_key": "tab:frontend",
    "cached": false
  },
  {
    "id": "address_bar",
```

```
"type": "input",
"label": "Address Bar",
"description": "Browser address bar showing localhost:5173",
"state": "visible",
"dx": 576,
"dy": 100,
"frame_dx": 727,
"frame_dy": 210,
"confidence": 0.99,
"source": "gemini_vlm",
"frame_bbox": [
    151,
    182,
    1303,
    238
],
"session_memory_key": "input:address bar",
"cached": false
},
{
    "id": "ask_gemini_button",
    "type": "button",
    "label": "Ask Gemini",
    "description": "Ask Gemini button in the browser toolbar",
    "state": "visible",
    "dx": 1348,
    "dy": 56,
    "frame_dx": 1499,
    "frame_dy": 166,
```

```
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    1415,
    142,
    1583,
    190
],
"session_memory_key": "button:ask gemini",
"cached": false
},
{
    "id": "window_minimize",
    "type": "button",
    "label": "Minimize",
    "description": "Browser window minimize button",
    "state": "visible",
    "dx": 1460,
    "dy": 52,
    "frame_dx": 1611,
    "frame_dy": 162,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        1583,
        142,
        1639,
        182
    ],
    ],
```

```
"session_memory_key": "button:minimize",
"cached": false
},
{
  "id": "window_maximize",
  "type": "button",
  "label": "Maximize",
  "description": "Browser window maximize button",
  "state": "visible",
  "dx": 1500,
  "dy": 52,
  "frame_dx": 1651,
  "frame_dy": 162,
  "confidence": 0.9,
  "source": "gemini_vlm",
  "frame_bbox": [
    1623,
    142,
    1679,
    182
  ],
  "session_memory_key": "button:maximize",
  "cached": false
},
{
  "id": "window_close",
  "type": "button",
  "label": "Close",
  "description": "Browser window close button",
```

```
"state": "visible",
"dx": 1548,
"dy": 56,
"frame_dx": 1699,
"frame_dy": 166,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    1671,
    142,
    1727,
    190
],
"session_memory_key": "button:close",
"cached": false
},
{
    "id": "screenpilot_ai_title",
    "type": "heading",
    "label": "ScreenPilot AI",
    "description": "Application title ScreenPilot AI",
    "state": "visible",
    "dx": 120,
    "dy": 192,
    "frame_dx": 271,
    "frame_dy": 302,
    "confidence": 0.99,
    "source": "gemini_vlm",
    "frame_bbox": [
```



```
        167,  
        270,  
        375,  
        334  
    ],  
    "session_memory_key": "heading:screenpilot ai",  
    "cached": false  
},  
{  
    "id": "agent_version_badge",  
    "type": "label",  
    "label": "Autonomous Task Agent v2",  
    "description": "Badge indicating the agent version",  
    "state": "visible",  
    "dx": 308,  
    "dy": 196,  
    "frame_dx": 459,  
    "frame_dy": 306,  
    "confidence": 0.97,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        423,  
        286,  
        495,  
        326  
    ],  
    "session_memory_key": "label:autonomous task agent v2",  
    "cached": false  
},
```

```
{
  "id": "status_connecting",
  "type": "text",
  "label": "Connecting to agent...",
  "description": "Status message indicating connection progress",
  "state": "visible",
  "dx": 120,
  "dy": 436,
  "frame_dx": 271,
  "frame_dy": 546,
  "confidence": 0.96,
  "source": "gemini_vlm",
  "frame_bbox": [
    183,
    526,
    359,
    566
  ],
  "session_memory_key": "text:connecting to agent...",
  "cached": false
},
{
  "id": "status_capturing",
  "type": "text",
  "label": "Capturing initial screen...",
  "description": "Status message indicating screen capture progress",
  "state": "visible",
  "dx": 148,
  "dy": 468,
```

```
"frame_dx": 299,  
"frame_dy": 578,  
"confidence": 0.96,  
"source": "gemini_vlm",  
"frame_bbox": [  
    183,  
    558,  
    415,  
    598  
],  
"session_memory_key": "text:capturing initial screen...",  
"cached": false  
},  
{  
    "id": "preview_window",  
    "type": "pane",  
    "label": "Preview Window",  
    "description": "Small preview window showing agent activity",  
    "state": "visible",  
    "dx": 116,  
    "dy": 588,  
    "frame_dx": 267,  
    "frame_dy": 698,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        167,  
        598,  
        367,
```

```
798
],
"session_memory_key": "pane:preview window",
"cached": false
},
{
  "id": "chat_input_field",
  "type": "input",
  "label": "Agent is working...",
  "description": "Main chat input field for the agent",
  "state": "visible",
  "dx": 788,
  "dy": 796,
  "frame_dx": 939,
  "frame_dy": 906,
  "confidence": 0.98,
  "source": "gemini_vlm",
  "frame_bbox": [
    151,
    862,
    1727,
    950
  ],
  "session_memory_key": "input:agent is working...",
  "cached": false
},
{
  "id": "send_button",
  "type": "button",
```

```
    "label": "Send",
    "description": "Button to send commands to the agent",
    "state": "visible",
    "dx": 1472,
    "dy": 788,
    "frame_dx": 1623,
    "frame_dy": 898,
    "confidence": 0.97,
    "source": "gemini_vlm",
    "frame_bbox": [
        1583,
        854,
        1663,
        942
    ],
    "session_memory_key": "button:send",
    "cached": false
},
{
    "id": "windows_taskbar",
    "type": "pane",
    "label": "Taskbar",
    "description": "Windows taskbar at the bottom of the screen",
    "state": "visible",
    "dx": 788,
    "dy": 864,
    "frame_dx": 939,
    "frame_dy": 974,
    "confidence": 0.99,
```

```
"source": "gemini_vlm",
"frame_bbox": [
    151,
    934,
    1727,
    1013
],
"session_memory_key": "pane:taskbar",
"cached": false
},
{
    "id": "start_button",
    "type": "button",
    "label": "Start",
    "description": "Windows Start button",
    "state": "visible",
    "dx": 428,
    "dy": 864,
    "frame_dx": 579,
    "frame_dy": 974,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
        551,
        950,
        607,
        998
    ],
    "session_memory_key": "button:start",
```

```
    "cached": false
  },
  {
    "id": "search_bar",
    "type": "input",
    "label": "Search",
    "description": "Windows search bar in the taskbar",
    "state": "visible",
    "dx": 528,
    "dy": 864,
    "frame_dx": 679,
    "frame_dy": 974,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
      591,
      950,
      767,
      998
    ],
    "session_memory_key": "input:search",
    "cached": false
  },
  {
    "id": "weather_widget",
    "type": "button",
    "label": "Weather",
    "description": "Weather widget in the taskbar",
    "state": "visible",
```

```
"dx": 92,  
  "dy": 876,  
  "frame_dx": 243,  
  "frame_dy": 986,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    167,  
    958,  
    319,  
    1013  
  ],  
  "session_memory_key": "button:weather",  
  "cached": false  
},  
{  
  "id": "taskbar_file_explorer",  
  "type": "icon",  
  "label": "File Explorer",  
  "description": "File Explorer icon in the taskbar",  
  "state": "visible",  
  "dx": 612,  
  "dy": 868,  
  "frame_dx": 763,  
  "frame_dy": 978,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    743,
```



```
        958,  
        783,  
        998  
    ],  
    "session_memory_key": "icon:file explorer",  
    "cached": false  
},  
{  
    "id": "taskbar_chrome",  
    "type": "icon",  
    "label": "Chrome",  
    "description": "Google Chrome icon in the taskbar",  
    "state": "visible",  
    "dx": 640,  
    "dy": 868,  
    "frame_dx": 791,  
    "frame_dy": 978,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        775,  
        958,  
        807,  
        998  
    ],  
    "session_memory_key": "icon:chrome",  
    "cached": false  
},  
{
```

```
"id": "taskbar_settings",
"type": "icon",
"label": "Settings",
"description": "Settings icon in the taskbar",
"state": "visible",
"dx": 672,
"dy": 868,
"frame_dx": 823,
"frame_dy": 978,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    807,
    958,
    839,
    998
],
"session_memory_key": "icon:settings",
"cached": false
},
{
    "id": "taskbar_vscode",
    "type": "icon",
    "label": "VS Code",
    "description": "Visual Studio Code icon in the taskbar",
    "state": "visible",
    "dx": 804,
    "dy": 868,
    "frame_dx": 955,
```

```
"frame_dy": 978,  
"confidence": 0.95,  
"source": "gemini_vlm",  
"frame_bbox": [  
    935,  
    958,  
    975,  
    998  
],  
"session_memory_key": "icon:vs code",  
"cached": false  
},  
{  
    "id": "taskbar_spotify",  
    "type": "icon",  
    "label": "Spotify",  
    "description": "Spotify icon in the taskbar",  
    "state": "visible",  
    "dx": 844,  
    "dy": 868,  
    "frame_dx": 995,  
    "frame_dy": 978,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        975,  
        958,  
        1015,  
        998
```

```
    ],
    "session_memory_key": "icon:spotify",
    "cached": false
  },
  {
    "id": "system_tray",
    "type": "pane",
    "label": "System Tray",
    "description": "System tray area with network, volume, and battery
icons",
    "state": "visible",
    "dx": 1416,
    "dy": 864,
    "frame_dx": 1567,
    "frame_dy": 974,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
      1487,
      950,
      1647,
      998
    ],
    "session_memory_key": "pane:system tray",
    "cached": false
  },
  {
    "id": "clock_date",
    "type": "text",
```

```
    "label": "Clock and Date",
    "description": "System clock and date display in the taskbar",
    "state": "visible",
    "dx": 1520,
    "dy": 868,
    "frame_dx": 1671,
    "frame_dy": 978,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
        1615,
        942,
        1727,
        1013
    ],
    "session_memory_key": "text:clock and date",
    "cached": false
}
],
"session_knowledge_base": {
    "session_id": "shot_1777696146_8f4fe6",
    "started_at": "2026-05-02T04:29:06.629715+00:00",
    "updated_at": "2026-05-02T04:29:36.525759+00:00",
    "stale_after_misses": 3,
    "confidence_floor": 0.4,
    "decay_per_miss": 0.9,
    "frame_count": 1,
    "frame_name": "frame_1777696146_e68365.jpg",
    "frame_index": 0,
```

```
"image_size": {
  "width": 1920,
  "height": 1080
},
"known_elements": {
  "tab:frontend": {
    "element_key": "tab:frontend",
    "element_type": "tab",
    "label": "frontend",
    "bbox": [
      151.0,
      118.0,
      399.0,
      174.0
    ],
    "confidence": 0.98,
    "source": "gemini_vlm",
    "first_seen_frame": "frame_1777696146_e68365.jpg",
    "last_seen_frame": "frame_1777696146_e68365.jpg",
    "first_seen_at": "2026-05-02T04:29:36.525759+00:00",
    "last_seen_at": "2026-05-02T04:29:36.525759+00:00",
    "first_seen_index": 0,
    "last_seen_index": 0,
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false,
    "history": [
      {
```

```
        "frame_name": "frame_1777696146_e68365.jpg",
        "frame_index": 0,
        "seen_at": "2026-05-02T04:29:36.525759+00:00",
        "confidence": 0.98
    }
]
},
"input:address bar": {
    "element_key": "input:address bar",
    "element_type": "input",
    "label": "Address Bar",
    "bbox": [
        151.0,
        182.0,
        1303.0,
        238.0
    ],
    "confidence": 0.99,
    "source": "gemini_vlm",
    "first_seen_frame": "frame_1777696146_e68365.jpg",
    "last_seen_frame": "frame_1777696146_e68365.jpg",
    "first_seen_at": "2026-05-02T04:29:36.525759+00:00",
    "last_seen_at": "2026-05-02T04:29:36.525759+00:00",
    "first_seen_index": 0,
    "last_seen_index": 0,
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false,
```

```
"history": [  
  {  
    "frame_name": "frame_1777696146_e68365.jpg",  
    "frame_index": 0,  
    "seen_at": "2026-05-02T04:29:36.525759+00:00",  
    "confidence": 0.99  
  }  
]  
},  
"button:ask_gemini": {  
  "element_key": "button:ask_gemini",  
  "element_type": "button",  
  "label": "Ask Gemini",  
  "bbox": [  
    1415.0,  
    142.0,  
    1583.0,  
    190.0  
  ],  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "first_seen_frame": "frame_1777696146_e68365.jpg",  
  "last_seen_frame": "frame_1777696146_e68365.jpg",  
  "first_seen_at": "2026-05-02T04:29:36.525759+00:00",  
  "last_seen_at": "2026-05-02T04:29:36.525759+00:00",  
  "first_seen_index": 0,  
  "last_seen_index": 0,  
  "seen_count": 1,  
  "miss_count": 0,
```



```
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777696146_e68365.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T04:29:36.525759+00:00",
    "confidence": 0.95
  }
],
},
"button:minimize": {
  "element_key": "button:minimize",
  "element_type": "button",
  "label": "Minimize",
  "bbox": [
    1583.0,
    142.0,
    1639.0,
    182.0
  ],
  "confidence": 0.9,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777696146_e68365.jpg",
  "last_seen_frame": "frame_1777696146_e68365.jpg",
  "first_seen_at": "2026-05-02T04:29:36.525759+00:00",
  "last_seen_at": "2026-05-02T04:29:36.525759+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
```

```
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777696146_e68365.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T04:29:36.525759+00:00",
    "confidence": 0.9
  }
]
},
"button:maximize": {
  "element_key": "button:maximize",
  "element_type": "button",
  "label": "Maximize",
  "bbox": [
    1623.0,
    142.0,
    1679.0,
    182.0
  ],
  "confidence": 0.9,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777696146_e68365.jpg",
  "last_seen_frame": "frame_1777696146_e68365.jpg",
  "first_seen_at": "2026-05-02T04:29:36.525759+00:00",
  "last_seen_at": "2026-05-02T04:29:36.525759+00:00",
```

```
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777696146_e68365.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T04:29:36.525759+00:00",
    "confidence": 0.9
  }
],
},
"button:close": {
  "element_key": "button:close",
  "element_type": "button",
  "label": "Close",
  "bbox": [
    1671.0,
    142.0,
    1727.0,
    190.0
  ],
  "confidence": 0.9,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777696146_e68365.jpg",
  "last_seen_frame": "frame_1777696146_e68365.jpg",
```

```
"first_seen_at": "2026-05-02T04:29:36.525759+00:00",
"last_seen_at": "2026-05-02T04:29:36.525759+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777696146_e68365.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T04:29:36.525759+00:00",
    "confidence": 0.9
  }
],
},
"heading:screenpilot ai": {
  "element_key": "heading:screenpilot ai",
  "element_type": "heading",
  "label": "ScreenPilot AI",
  "bbox": [
    167.0,
    270.0,
    375.0,
    334.0
  ],
  "confidence": 0.99,
  "source": "gemini_vlm",
```

```
"first_seen_frame": "frame_1777696146_e68365.jpg",
"last_seen_frame": "frame_1777696146_e68365.jpg",
"first_seen_at": "2026-05-02T04:29:36.525759+00:00",
"last_seen_at": "2026-05-02T04:29:36.525759+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777696146_e68365.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T04:29:36.525759+00:00",
    "confidence": 0.99
  }
],
},
"label:autonomous task agent v2": {
  "element_key": "label:autonomous task agent v2",
  "element_type": "label",
  "label": "Autonomous Task Agent v2",
  "bbox": [
    423.0,
    286.0,
    495.0,
    326.0
  ],
}
```

```
"confidence": 0.97,
"source": "gemini_vlm",
"first_seen_frame": "frame_1777696146_e68365.jpg",
"last_seen_frame": "frame_1777696146_e68365.jpg",
"first_seen_at": "2026-05-02T04:29:36.525759+00:00",
"last_seen_at": "2026-05-02T04:29:36.525759+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777696146_e68365.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T04:29:36.525759+00:00",
    "confidence": 0.97
  }
],
"text:connecting to agent...": {
  "element_key": "text:connecting to agent...",
  "element_type": "text",
  "label": "Connecting to agent...",
  "bbox": [
    183.0,
    526.0,
    359.0,
```

```
566.0
],
"confidence": 0.96,
"source": "gemini_vlm",
"first_seen_frame": "frame_1777696146_e68365.jpg",
"last_seen_frame": "frame_1777696146_e68365.jpg",
"first_seen_at": "2026-05-02T04:29:36.525759+00:00",
"last_seen_at": "2026-05-02T04:29:36.525759+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777696146_e68365.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T04:29:36.525759+00:00",
    "confidence": 0.96
  }
]
},
"text:capturing initial screen...": {
  "element_key": "text:capturing initial screen...",
  "element_type": "text",
  "label": "Capturing initial screen...",
  "bbox": [
    183.0,
```

```
558.0,
415.0,
598.0
],
"confidence": 0.96,
"source": "gemini_vlm",
"first_seen_frame": "frame_1777696146_e68365.jpg",
"last_seen_frame": "frame_1777696146_e68365.jpg",
"first_seen_at": "2026-05-02T04:29:36.525759+00:00",
"last_seen_at": "2026-05-02T04:29:36.525759+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
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    "frame_name": "frame_1777696146_e68365.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T04:29:36.525759+00:00",
    "confidence": 0.96
  }
]
},
"pane:preview window": {
  "element_key": "pane:preview window",
  "element_type": "pane",
  "label": "Preview Window",
```



```
"bbox": [
  167.0,
  598.0,
  367.0,
  798.0
],
"confidence": 0.95,
"source": "gemini_vlm",
"first_seen_frame": "frame_1777696146_e68365.jpg",
"last_seen_frame": "frame_1777696146_e68365.jpg",
"first_seen_at": "2026-05-02T04:29:36.525759+00:00",
"last_seen_at": "2026-05-02T04:29:36.525759+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777696146_e68365.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T04:29:36.525759+00:00",
    "confidence": 0.95
  }
]
},
"input:agent is working...": {
  "element_key": "input:agent is working...",
```

```
"element_type": "input",
"label": "Agent is working...",
"bbox": [
    151.0,
    862.0,
    1727.0,
    950.0
],
"confidence": 0.98,
"source": "gemini_vlm",
"first_seen_frame": "frame_1777696146_e68365.jpg",
"last_seen_frame": "frame_1777696146_e68365.jpg",
"first_seen_at": "2026-05-02T04:29:36.525759+00:00",
"last_seen_at": "2026-05-02T04:29:36.525759+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
    {
        "frame_name": "frame_1777696146_e68365.jpg",
        "frame_index": 0,
        "seen_at": "2026-05-02T04:29:36.525759+00:00",
        "confidence": 0.98
    }
]
},
```

```
"button:send": {
  "element_key": "button:send",
  "element_type": "button",
  "label": "Send",
  "bbox": [
    1583.0,
    854.0,
    1663.0,
    942.0
  ],
  "confidence": 0.97,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777696146_e68365.jpg",
  "last_seen_frame": "frame_1777696146_e68365.jpg",
  "first_seen_at": "2026-05-02T04:29:36.525759+00:00",
  "last_seen_at": "2026-05-02T04:29:36.525759+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
  "seen_count": 1,
  "miss_count": 0,
  "stale": false,
  "cached": false,
  "history": [
    {
      "frame_name": "frame_1777696146_e68365.jpg",
      "frame_index": 0,
      "seen_at": "2026-05-02T04:29:36.525759+00:00",
      "confidence": 0.97
    }
  ]
}
```

```
]
},
"pane:taskbar": {
  "element_key": "pane:taskbar",
  "element_type": "pane",
  "label": "Taskbar",
  "bbox": [
    151.0,
    934.0,
    1727.0,
    1013.0
  ],
  "confidence": 0.99,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777696146_e68365.jpg",
  "last_seen_frame": "frame_1777696146_e68365.jpg",
  "first_seen_at": "2026-05-02T04:29:36.525759+00:00",
  "last_seen_at": "2026-05-02T04:29:36.525759+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
  "seen_count": 1,
  "miss_count": 0,
  "stale": false,
  "cached": false,
  "history": [
    {
      "frame_name": "frame_1777696146_e68365.jpg",
      "frame_index": 0,
      "seen_at": "2026-05-02T04:29:36.525759+00:00",
```

```
        "confidence": 0.99
      }
    ]
  },
  "button:start": {
    "element_key": "button:start",
    "element_type": "button",
    "label": "Start",
    "bbox": [
      551.0,
      950.0,
      607.0,
      998.0
    ],
    "confidence": 0.98,
    "source": "gemini_vlm",
    "first_seen_frame": "frame_1777696146_e68365.jpg",
    "last_seen_frame": "frame_1777696146_e68365.jpg",
    "first_seen_at": "2026-05-02T04:29:36.525759+00:00",
    "last_seen_at": "2026-05-02T04:29:36.525759+00:00",
    "first_seen_index": 0,
    "last_seen_index": 0,
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false,
    "history": [
      {
        "frame_name": "frame_1777696146_e68365.jpg",
```

```
        "frame_index": 0,
        "seen_at": "2026-05-02T04:29:36.525759+00:00",
        "confidence": 0.98
    }
]
},
"input:search": {
    "element_key": "input:search",
    "element_type": "input",
    "label": "Search",
    "bbox": [
        591.0,
        950.0,
        767.0,
        998.0
    ],
    "confidence": 0.98,
    "source": "gemini_vlm",
    "first_seen_frame": "frame_1777696146_e68365.jpg",
    "last_seen_frame": "frame_1777696146_e68365.jpg",
    "first_seen_at": "2026-05-02T04:29:36.525759+00:00",
    "last_seen_at": "2026-05-02T04:29:36.525759+00:00",
    "first_seen_index": 0,
    "last_seen_index": 0,
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false,
    "history": [
```

```
{
  "frame_name": "frame_1777696146_e68365.jpg",
  "frame_index": 0,
  "seen_at": "2026-05-02T04:29:36.525759+00:00",
  "confidence": 0.98
}
],
},
"button:weather": {
  "element_key": "button:weather",
  "element_type": "button",
  "label": "Weather",
  "bbox": [
    167.0,
    958.0,
    319.0,
    1013.0
  ],
  "confidence": 0.95,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777696146_e68365.jpg",
  "last_seen_frame": "frame_1777696146_e68365.jpg",
  "first_seen_at": "2026-05-02T04:29:36.525759+00:00",
  "last_seen_at": "2026-05-02T04:29:36.525759+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
  "seen_count": 1,
  "miss_count": 0,
  "stale": false,
```

```
"cached": false,
"history": [
  {
    "frame_name": "frame_1777696146_e68365.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T04:29:36.525759+00:00",
    "confidence": 0.95
  }
],
},
"icon:file explorer": {
  "element_key": "icon:file explorer",
  "element_type": "icon",
  "label": "File Explorer",
  "bbox": [
    743.0,
    958.0,
    783.0,
    998.0
  ],
  "confidence": 0.95,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777696146_e68365.jpg",
  "last_seen_frame": "frame_1777696146_e68365.jpg",
  "first_seen_at": "2026-05-02T04:29:36.525759+00:00",
  "last_seen_at": "2026-05-02T04:29:36.525759+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
  "seen_count": 1,
```



```
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777696146_e68365.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T04:29:36.525759+00:00",
    "confidence": 0.95
  }
]
},
"icon:chrome": {
  "element_key": "icon:chrome",
  "element_type": "icon",
  "label": "Chrome",
  "bbox": [
    775.0,
    958.0,
    807.0,
    998.0
  ],
  "confidence": 0.95,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777696146_e68365.jpg",
  "last_seen_frame": "frame_1777696146_e68365.jpg",
  "first_seen_at": "2026-05-02T04:29:36.525759+00:00",
  "last_seen_at": "2026-05-02T04:29:36.525759+00:00",
  "first_seen_index": 0,
```

```
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777696146_e68365.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T04:29:36.525759+00:00",
    "confidence": 0.95
  }
],
},
"icon:settings": {
  "element_key": "icon:settings",
  "element_type": "icon",
  "label": "Settings",
  "bbox": [
    807.0,
    958.0,
    839.0,
    998.0
  ],
  "confidence": 0.95,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777696146_e68365.jpg",
  "last_seen_frame": "frame_1777696146_e68365.jpg",
  "first_seen_at": "2026-05-02T04:29:36.525759+00:00",
```

```
"last_seen_at": "2026-05-02T04:29:36.525759+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777696146_e68365.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T04:29:36.525759+00:00",
    "confidence": 0.95
  }
],
},
"icon:vs code": {
  "element_key": "icon:vs code",
  "element_type": "icon",
  "label": "VS Code",
  "bbox": [
    935.0,
    958.0,
    975.0,
    998.0
  ],
  "confidence": 0.95,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777696146_e68365.jpg",
```

```
"last_seen_frame": "frame_1777696146_e68365.jpg",
"first_seen_at": "2026-05-02T04:29:36.525759+00:00",
"last_seen_at": "2026-05-02T04:29:36.525759+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777696146_e68365.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T04:29:36.525759+00:00",
    "confidence": 0.95
  }
],
},
"icon:spotify": {
  "element_key": "icon:spotify",
  "element_type": "icon",
  "label": "Spotify",
  "bbox": [
    975.0,
    958.0,
    1015.0,
    998.0
  ],
  "confidence": 0.95,
```

```
"source": "gemini_vlm",
"first_seen_frame": "frame_1777696146_e68365.jpg",
"last_seen_frame": "frame_1777696146_e68365.jpg",
"first_seen_at": "2026-05-02T04:29:36.525759+00:00",
"last_seen_at": "2026-05-02T04:29:36.525759+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777696146_e68365.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T04:29:36.525759+00:00",
    "confidence": 0.95
  }
]
},
"pane:system tray": {
  "element_key": "pane:system tray",
  "element_type": "pane",
  "label": "System Tray",
  "bbox": [
    1487.0,
    950.0,
    1647.0,
    998.0
  ]
}
```

```
    ],
    "confidence": 0.95,
    "source": "gemini_vlm",
    "first_seen_frame": "frame_1777696146_e68365.jpg",
    "last_seen_frame": "frame_1777696146_e68365.jpg",
    "first_seen_at": "2026-05-02T04:29:36.525759+00:00",
    "last_seen_at": "2026-05-02T04:29:36.525759+00:00",
    "first_seen_index": 0,
    "last_seen_index": 0,
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false,
    "history": [
      {
        "frame_name": "frame_1777696146_e68365.jpg",
        "frame_index": 0,
        "seen_at": "2026-05-02T04:29:36.525759+00:00",
        "confidence": 0.95
      }
    ]
  },
  "text:clock and date": {
    "element_key": "text:clock and date",
    "element_type": "text",
    "label": "Clock and Date",
    "bbox": [
      1615.0,
      942.0,
```

```
        1727.0,
        1013.0
    ],
    "confidence": 0.98,
    "source": "gemini_vlm",
    "first_seen_frame": "frame_1777696146_e68365.jpg",
    "last_seen_frame": "frame_1777696146_e68365.jpg",
    "first_seen_at": "2026-05-02T04:29:36.525759+00:00",
    "last_seen_at": "2026-05-02T04:29:36.525759+00:00",
    "first_seen_index": 0,
    "last_seen_index": 0,
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false,
    "history": [
        {
            "frame_name": "frame_1777696146_e68365.jpg",
            "frame_index": 0,
            "seen_at": "2026-05-02T04:29:36.525759+00:00",
            "confidence": 0.98
        }
    ]
},
"resolved_elements": [
    {
        "id": "browser_tab_frontend",
        "type": "tab",
```

```
    "label": "frontend",
    "description": "Browser tab for the frontend application",
    "state": "visible",
    "dx": 124,
    "dy": 36,
    "frame_dx": 275,
    "frame_dy": 146,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
        151,
        118,
        399,
        174
    ],
    "session_memory_key": "tab:frontend",
    "cached": false
},
{
    "id": "address_bar",
    "type": "input",
    "label": "Address Bar",
    "description": "Browser address bar showing localhost:5173",
    "state": "visible",
    "dx": 576,
    "dy": 100,
    "frame_dx": 727,
    "frame_dy": 210,
    "confidence": 0.99,
```



```
"source": "gemini_vlm",
"frame_bbox": [
    151,
    182,
    1303,
    238
],
"session_memory_key": "input:address bar",
"cached": false
},
{
    "id": "ask_gemini_button",
    "type": "button",
    "label": "Ask Gemini",
    "description": "Ask Gemini button in the browser toolbar",
    "state": "visible",
    "dx": 1348,
    "dy": 56,
    "frame_dx": 1499,
    "frame_dy": 166,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        1415,
        142,
        1583,
        190
    ],
    "session_memory_key": "button:ask gemini",
```

```
    "cached": false
  },
  {
    "id": "window_minimize",
    "type": "button",
    "label": "Minimize",
    "description": "Browser window minimize button",
    "state": "visible",
    "dx": 1460,
    "dy": 52,
    "frame_dx": 1611,
    "frame_dy": 162,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
      1583,
      142,
      1639,
      182
    ],
    "session_memory_key": "button:minimize",
    "cached": false
  },
  {
    "id": "window_maximize",
    "type": "button",
    "label": "Maximize",
    "description": "Browser window maximize button",
    "state": "visible",
```

```
"dx": 1500,  
  "dy": 52,  
  "frame_dx": 1651,  
  "frame_dy": 162,  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    1623,  
    142,  
    1679,  
    182  
  ],  
  "session_memory_key": "button:maximize",  
  "cached": false  
},  
{  
  "id": "window_close",  
  "type": "button",  
  "label": "Close",  
  "description": "Browser window close button",  
  "state": "visible",  
  "dx": 1548,  
  "dy": 56,  
  "frame_dx": 1699,  
  "frame_dy": 166,  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    1671,
```

```
        142,  
        1727,  
        190  
    ],  
    "session_memory_key": "button:close",  
    "cached": false  
},  
{  
    "id": "screenpilot_ai_title",  
    "type": "heading",  
    "label": "ScreenPilot AI",  
    "description": "Application title ScreenPilot AI",  
    "state": "visible",  
    "dx": 120,  
    "dy": 192,  
    "frame_dx": 271,  
    "frame_dy": 302,  
    "confidence": 0.99,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        167,  
        270,  
        375,  
        334  
    ],  
    "session_memory_key": "heading:screenpilot ai",  
    "cached": false  
},  
{
```

```
"id": "agent_version_badge",
"type": "label",
"label": "Autonomous Task Agent v2",
"description": "Badge indicating the agent version",
"state": "visible",
"dx": 308,
"dy": 196,
"frame_dx": 459,
"frame_dy": 306,
"confidence": 0.97,
"source": "gemini_vlm",
"frame_bbox": [
    423,
    286,
    495,
    326
],
"session_memory_key": "label:autonomous task agent v2",
"cached": false
},
{
    "id": "status_connecting",
    "type": "text",
    "label": "Connecting to agent...",
    "description": "Status message indicating connection progress",
    "state": "visible",
    "dx": 120,
    "dy": 436,
    "frame_dx": 271,
```

```
"frame_dy": 546,
"confidence": 0.96,
"source": "gemini_vlm",
"frame_bbox": [
    183,
    526,
    359,
    566
],
"session_memory_key": "text:connecting to agent...",
"cached": false
},
{
    "id": "status_capturing",
    "type": "text",
    "label": "Capturing initial screen...",
    "description": "Status message indicating screen capture progress",
    "state": "visible",
    "dx": 148,
    "dy": 468,
    "frame_dx": 299,
    "frame_dy": 578,
    "confidence": 0.96,
    "source": "gemini_vlm",
    "frame_bbox": [
        183,
        558,
        415,
        598
    ]
}
```

```
    ],  
    "session_memory_key": "text:capturing initial screen...",  
    "cached": false  
  },  
  {  
    "id": "preview_window",  
    "type": "pane",  
    "label": "Preview Window",  
    "description": "Small preview window showing agent activity",  
    "state": "visible",  
    "dx": 116,  
    "dy": 588,  
    "frame_dx": 267,  
    "frame_dy": 698,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
      167,  
      598,  
      367,  
      798  
    ],  
    "session_memory_key": "pane:preview window",  
    "cached": false  
  },  
  {  
    "id": "chat_input_field",  
    "type": "input",  
    "label": "Agent is working..."
```

```
"description": "Main chat input field for the agent",
"state": "visible",
"dx": 788,
"dy": 796,
"frame_dx": 939,
"frame_dy": 906,
"confidence": 0.98,
"source": "gemini_vlm",
"frame_bbox": [
    151,
    862,
    1727,
    950
],
"session_memory_key": "input:agent is working...",
"cached": false
},
{
    "id": "send_button",
    "type": "button",
    "label": "Send",
    "description": "Button to send commands to the agent",
    "state": "visible",
    "dx": 1472,
    "dy": 788,
    "frame_dx": 1623,
    "frame_dy": 898,
    "confidence": 0.97,
    "source": "gemini_vlm",
```



```
"frame_bbox": [
    1583,
    854,
    1663,
    942
],
"session_memory_key": "button:send",
"cached": false
},
{
    "id": "windows_taskbar",
    "type": "pane",
    "label": "Taskbar",
    "description": "Windows taskbar at the bottom of the screen",
    "state": "visible",
    "dx": 788,
    "dy": 864,
    "frame_dx": 939,
    "frame_dy": 974,
    "confidence": 0.99,
    "source": "gemini_vlm",
    "frame_bbox": [
        151,
        934,
        1727,
        1013
    ],
    "session_memory_key": "pane:taskbar",
    "cached": false
}
```

```
},  
{  
  "id": "start_button",  
  "type": "button",  
  "label": "Start",  
  "description": "Windows Start button",  
  "state": "visible",  
  "dx": 428,  
  "dy": 864,  
  "frame_dx": 579,  
  "frame_dy": 974,  
  "confidence": 0.98,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    551,  
    950,  
    607,  
    998  
  ],  
  "session_memory_key": "button:start",  
  "cached": false  
},  
{  
  "id": "search_bar",  
  "type": "input",  
  "label": "Search",  
  "description": "Windows search bar in the taskbar",  
  "state": "visible",  
  "dx": 528,
```

```
"dy": 864,  
  "frame_dx": 679,  
  "frame_dy": 974,  
  "confidence": 0.98,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    591,  
    950,  
    767,  
    998  
  ],  
  "session_memory_key": "input:search",  
  "cached": false  
},  
{  
  "id": "weather_widget",  
  "type": "button",  
  "label": "Weather",  
  "description": "Weather widget in the taskbar",  
  "state": "visible",  
  "dx": 92,  
  "dy": 876,  
  "frame_dx": 243,  
  "frame_dy": 986,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    167,  
    958,
```

```
        319,  
        1013  
    ],  
    "session_memory_key": "button:weather",  
    "cached": false  
},  
{  
    "id": "taskbar_file_explorer",  
    "type": "icon",  
    "label": "File Explorer",  
    "description": "File Explorer icon in the taskbar",  
    "state": "visible",  
    "dx": 612,  
    "dy": 868,  
    "frame_dx": 763,  
    "frame_dy": 978,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        743,  
        958,  
        783,  
        998  
    ],  
    "session_memory_key": "icon:file explorer",  
    "cached": false  
},  
{  
    "id": "taskbar_chrome",
```

```
"type": "icon",
"label": "Chrome",
"description": "Google Chrome icon in the taskbar",
"state": "visible",
"dx": 640,
"dy": 868,
"frame_dx": 791,
"frame_dy": 978,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    775,
    958,
    807,
    998
],
"session_memory_key": "icon:chrome",
"cached": false
},
{
    "id": "taskbar_settings",
    "type": "icon",
    "label": "Settings",
    "description": "Settings icon in the taskbar",
    "state": "visible",
    "dx": 672,
    "dy": 868,
    "frame_dx": 823,
    "frame_dy": 978,
```

```
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    807,
    958,
    839,
    998
],
"session_memory_key": "icon:settings",
"cached": false
},
{
    "id": "taskbar_vscode",
    "type": "icon",
    "label": "VS Code",
    "description": "Visual Studio Code icon in the taskbar",
    "state": "visible",
    "dx": 804,
    "dy": 868,
    "frame_dx": 955,
    "frame_dy": 978,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        935,
        958,
        975,
        998
    ],
    ],
```

```
    "session_memory_key": "icon:vs code",
    "cached": false
  },
  {
    "id": "taskbar_spotify",
    "type": "icon",
    "label": "Spotify",
    "description": "Spotify icon in the taskbar",
    "state": "visible",
    "dx": 844,
    "dy": 868,
    "frame_dx": 995,
    "frame_dy": 978,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
      975,
      958,
      1015,
      998
    ],
    "session_memory_key": "icon:spotify",
    "cached": false
  },
  {
    "id": "system_tray",
    "type": "pane",
    "label": "System Tray",
```

```
    "description": "System tray area with network, volume, and battery icons",
    "state": "visible",
    "dx": 1416,
    "dy": 864,
    "frame_dx": 1567,
    "frame_dy": 974,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        1487,
        950,
        1647,
        998
    ],
    "session_memory_key": "pane:system tray",
    "cached": false
},
{
    "id": "clock_date",
    "type": "text",
    "label": "Clock and Date",
    "description": "System clock and date display in the taskbar",
    "state": "visible",
    "dx": 1520,
    "dy": 868,
    "frame_dx": 1671,
    "frame_dy": 978,
    "confidence": 0.98,
```



```
"source": "gemini_vlm",
"frame_bbox": [
  1615,
  942,
  1727,
  1013
],
"session_memory_key": "text:clock and date",
"cached": false
}
],
"current_elements": [
  {
    "id": "browser_tab_frontend",
    "type": "tab",
    "label": "frontend",
    "description": "Browser tab for the frontend application",
    "state": "visible",
    "dx": 124,
    "dy": 36,
    "frame_dx": 275,
    "frame_dy": 146,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
      151,
      118,
      399,
      174
```

```
    ],  
    "session_memory_key": "tab:frontend",  
    "cached": false  
  },  
  {  
    "id": "address_bar",  
    "type": "input",  
    "label": "Address Bar",  
    "description": "Browser address bar showing localhost:5173",  
    "state": "visible",  
    "dx": 576,  
    "dy": 100,  
    "frame_dx": 727,  
    "frame_dy": 210,  
    "confidence": 0.99,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
      151,  
      182,  
      1303,  
      238  
    ],  
    "session_memory_key": "input:address bar",  
    "cached": false  
  },  
  {  
    "id": "ask_gemini_button",  
    "type": "button",  
    "label": "Ask Gemini",
```

```
"description": "Ask Gemini button in the browser toolbar",
"state": "visible",
"dx": 1348,
"dy": 56,
"frame_dx": 1499,
"frame_dy": 166,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    1415,
    142,
    1583,
    190
],
"session_memory_key": "button:ask gemini",
"cached": false
},
{
    "id": "window_minimize",
    "type": "button",
    "label": "Minimize",
    "description": "Browser window minimize button",
    "state": "visible",
    "dx": 1460,
    "dy": 52,
    "frame_dx": 1611,
    "frame_dy": 162,
    "confidence": 0.9,
    "source": "gemini_vlm",
```

```
"frame_bbox": [
    1583,
    142,
    1639,
    182
],
"session_memory_key": "button:minimize",
"cached": false
},
{
    "id": "window_maximize",
    "type": "button",
    "label": "Maximize",
    "description": "Browser window maximize button",
    "state": "visible",
    "dx": 1500,
    "dy": 52,
    "frame_dx": 1651,
    "frame_dy": 162,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        1623,
        142,
        1679,
        182
    ],
    "session_memory_key": "button:maximize",
    "cached": false
}
```

```
},  
{  
  "id": "window_close",  
  "type": "button",  
  "label": "Close",  
  "description": "Browser window close button",  
  "state": "visible",  
  "dx": 1548,  
  "dy": 56,  
  "frame_dx": 1699,  
  "frame_dy": 166,  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    1671,  
    142,  
    1727,  
    190  
  ],  
  "session_memory_key": "button:close",  
  "cached": false  
},  
{  
  "id": "screenpilot_ai_title",  
  "type": "heading",  
  "label": "ScreenPilot AI",  
  "description": "Application title ScreenPilot AI",  
  "state": "visible",  
  "dx": 120,
```

```
"dy": 192,
"frame_dx": 271,
"frame_dy": 302,
"confidence": 0.99,
"source": "gemini_vlm",
"frame_bbox": [
    167,
    270,
    375,
    334
],
"session_memory_key": "heading:screenpilot ai",
"cached": false
},
{
    "id": "agent_version_badge",
    "type": "label",
    "label": "Autonomous Task Agent v2",
    "description": "Badge indicating the agent version",
    "state": "visible",
    "dx": 308,
    "dy": 196,
    "frame_dx": 459,
    "frame_dy": 306,
    "confidence": 0.97,
    "source": "gemini_vlm",
    "frame_bbox": [
        423,
        286,
```

```
        495,  
        326  
    ],  
    "session_memory_key": "label:autonomous task agent v2",  
    "cached": false  
},  
{  
    "id": "status_connecting",  
    "type": "text",  
    "label": "Connecting to agent...",  
    "description": "Status message indicating connection progress",  
    "state": "visible",  
    "dx": 120,  
    "dy": 436,  
    "frame_dx": 271,  
    "frame_dy": 546,  
    "confidence": 0.96,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        183,  
        526,  
        359,  
        566  
    ],  
    "session_memory_key": "text:connecting to agent...",  
    "cached": false  
},  
{  
    "id": "status_capturing",
```

```
"type": "text",
"label": "Capturing initial screen...",
"description": "Status message indicating screen capture progress",
"state": "visible",
"dx": 148,
"dy": 468,
"frame_dx": 299,
"frame_dy": 578,
"confidence": 0.96,
"source": "gemini_vlm",
"frame_bbox": [
    183,
    558,
    415,
    598
],
"session_memory_key": "text:capturing initial screen...",
"cached": false
},
{
    "id": "preview_window",
    "type": "pane",
    "label": "Preview Window",
    "description": "Small preview window showing agent activity",
    "state": "visible",
    "dx": 116,
    "dy": 588,
    "frame_dx": 267,
    "frame_dy": 698,
```



```
"confidence": 0.95,  
"source": "gemini_vlm",  
"frame_bbox": [  
    167,  
    598,  
    367,  
    798  
],  
"session_memory_key": "pane:preview window",  
"cached": false  
},  
{  
    "id": "chat_input_field",  
    "type": "input",  
    "label": "Agent is working...",  
    "description": "Main chat input field for the agent",  
    "state": "visible",  
    "dx": 788,  
    "dy": 796,  
    "frame_dx": 939,  
    "frame_dy": 906,  
    "confidence": 0.98,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        151,  
        862,  
        1727,  
        950  
    ],  
    }
```

```
    "session_memory_key": "input:agent is working...",
    "cached": false
  },
  {
    "id": "send_button",
    "type": "button",
    "label": "Send",
    "description": "Button to send commands to the agent",
    "state": "visible",
    "dx": 1472,
    "dy": 788,
    "frame_dx": 1623,
    "frame_dy": 898,
    "confidence": 0.97,
    "source": "gemini_vlm",
    "frame_bbox": [
      1583,
      854,
      1663,
      942
    ],
    "session_memory_key": "button:send",
    "cached": false
  },
  {
    "id": "windows_taskbar",
    "type": "pane",
    "label": "Taskbar",
    "description": "Windows taskbar at the bottom of the screen",
```

```
"state": "visible",
"dx": 788,
"dy": 864,
"frame_dx": 939,
"frame_dy": 974,
"confidence": 0.99,
"source": "gemini_vlm",
"frame_bbox": [
    151,
    934,
    1727,
    1013
],
"session_memory_key": "pane:taskbar",
"cached": false
},
{
    "id": "start_button",
    "type": "button",
    "label": "Start",
    "description": "Windows Start button",
    "state": "visible",
    "dx": 428,
    "dy": 864,
    "frame_dx": 579,
    "frame_dy": 974,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
```

```
        551,  
        950,  
        607,  
        998  
    ],  
    "session_memory_key": "button:start",  
    "cached": false  
},  
{  
    "id": "search_bar",  
    "type": "input",  
    "label": "Search",  
    "description": "Windows search bar in the taskbar",  
    "state": "visible",  
    "dx": 528,  
    "dy": 864,  
    "frame_dx": 679,  
    "frame_dy": 974,  
    "confidence": 0.98,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        591,  
        950,  
        767,  
        998  
    ],  
    "session_memory_key": "input:search",  
    "cached": false  
},
```

```
{
  "id": "weather_widget",
  "type": "button",
  "label": "Weather",
  "description": "Weather widget in the taskbar",
  "state": "visible",
  "dx": 92,
  "dy": 876,
  "frame_dx": 243,
  "frame_dy": 986,
  "confidence": 0.95,
  "source": "gemini_vlm",
  "frame_bbox": [
    167,
    958,
    319,
    1013
  ],
  "session_memory_key": "button:weather",
  "cached": false
},
{
  "id": "taskbar_file_explorer",
  "type": "icon",
  "label": "File Explorer",
  "description": "File Explorer icon in the taskbar",
  "state": "visible",
  "dx": 612,
  "dy": 868,
```

```
"frame_dx": 763,  
"frame_dy": 978,  
"confidence": 0.95,  
"source": "gemini_vlm",  
"frame_bbox": [  
    743,  
    958,  
    783,  
    998  
],  
"session_memory_key": "icon:file explorer",  
"cached": false  
},  
{  
    "id": "taskbar_chrome",  
    "type": "icon",  
    "label": "Chrome",  
    "description": "Google Chrome icon in the taskbar",  
    "state": "visible",  
    "dx": 640,  
    "dy": 868,  
    "frame_dx": 791,  
    "frame_dy": 978,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        775,  
        958,  
        807,
```

```
998
],
"session_memory_key": "icon:chrome",
"cached": false
},
{
  "id": "taskbar_settings",
  "type": "icon",
  "label": "Settings",
  "description": "Settings icon in the taskbar",
  "state": "visible",
  "dx": 672,
  "dy": 868,
  "frame_dx": 823,
  "frame_dy": 978,
  "confidence": 0.95,
  "source": "gemini_vlm",
  "frame_bbox": [
    807,
    958,
    839,
    998
  ],
  "session_memory_key": "icon:settings",
  "cached": false
},
{
  "id": "taskbar_vscode",
  "type": "icon",
```

```
"label": "VS Code",
"description": "Visual Studio Code icon in the taskbar",
"state": "visible",
"dx": 804,
"dy": 868,
"frame_dx": 955,
"frame_dy": 978,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    935,
    958,
    975,
    998
],
"session_memory_key": "icon:vs code",
"cached": false
},
{
    "id": "taskbar_spotify",
    "type": "icon",
    "label": "Spotify",
    "description": "Spotify icon in the taskbar",
    "state": "visible",
    "dx": 844,
    "dy": 868,
    "frame_dx": 995,
    "frame_dy": 978,
    "confidence": 0.95,
```



```
    "source": "gemini_vlm",
    "frame_bbox": [
      975,
      958,
      1015,
      998
    ],
    "session_memory_key": "icon:spotify",
    "cached": false
  },
  {
    "id": "system_tray",
    "type": "pane",
    "label": "System Tray",
    "description": "System tray area with network, volume, and battery
icons",
    "state": "visible",
    "dx": 1416,
    "dy": 864,
    "frame_dx": 1567,
    "frame_dy": 974,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
      1487,
      950,
      1647,
      998
    ],
  },
```

```
    "session_memory_key": "pane:system tray",
    "cached": false
  },
  {
    "id": "clock_date",
    "type": "text",
    "label": "Clock and Date",
    "description": "System clock and date display in the taskbar",
    "state": "visible",
    "dx": 1520,
    "dy": 868,
    "frame_dx": 1671,
    "frame_dy": 978,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
      1615,
      942,
      1727,
      1013
    ],
    "session_memory_key": "text:clock and date",
    "cached": false
  }
]
}
}
},
},
```

```
"todo": {
  "steps": [
    {
      "id": 1,
      "action": "Move the mouse to the 'Minimize' button at (1611, 162) and click it to hide the current window.",
      "target": "Minimize button",
      "expected_result": "The current window is minimized and no longer visible.",
      "needs_user_data": false,
      "user_data_field": null,
      "status": "done"
    },
    {
      "id": 2,
      "action": "Move the mouse to the 'Chrome' icon at (791, 978) on the taskbar and click it.",
      "target": "Chrome icon",
      "expected_result": "Google Chrome opens or becomes the active window.",
      "needs_user_data": false,
      "user_data_field": null,
      "status": "failed"
    }
  ],
  "notes": "The user wants to minimize the current application and then open Chrome from the taskbar.",
  "estimated_complexity": "simple"
},
"step_plans": [
  {
```

```
    "step_id": 1,
    "action": "Move the mouse to the 'Minimize' button at (1611, 162) and click it to hide the current window.",
    "target": "Minimize button",
    "expected_result": "The current window is minimized and no longer visible.",
    "hid_commands": [
      {
        "cmd": "mouse_move",
        "dx": 1625,
        "dy": 154
      },
      {
        "cmd": "mouse_click",
        "button": "left"
      }
    ],
    "reasoning": "Moving the mouse to the 'Minimize' button at the detected coordinates (1625, 154) and clicking it to hide the current window as specified in the first step of the task plan."
  },
  {
    "step_id": 2,
    "action": "Move the mouse to the 'Chrome' icon at (791, 978) on the taskbar and click it.",
    "target": "Chrome icon",
    "expected_result": "Google Chrome opens or becomes the active window.",
    "hid_commands": [
      {
        "cmd": "mouse_move",
        "dx": 208,
```

```
        "dy": 198
      },
      {
        "cmd": "mouse_click",
        "button": "left"
      }
    ],
    "reasoning": "Moving the mouse to the Google Chrome icon at the detected coordinates (208, 198) and clicking it to open the application."
  }
],
"step_results": [
  {
    "step_id": 1,
    "action": "Move the mouse to the 'Minimize' button at (1611, 162) and click it to hide the current window.",
    "attempt": 1,
    "exec_status": "success",
    "eval_status": "done",
    "confidence": 1.0,
    "reason": "The window is no longer visible and only desktop icons are detected, indicating the minimize action was successful.",
    "final_status": "executing",
    "timestamp": "2026-05-02T04:31:57.872617+00:00"
  },
  {
    "step_id": 2,
    "action": "Move the mouse to the 'Chrome' icon at (791, 978) on the taskbar and click it.",
    "attempt": 3,
```

```
    "exec_status": "success",
    "eval_status": "retry",
    "confidence": 0.0,
    "reason": "Evaluation error: No valid JSON found in LLM response: '{\\n
\\\"status\\\": \\\"retry\\\",',",
    "final_status": "executing",
    "timestamp": "2026-05-02T04:38:05.146328+00:00"
  }
],
"final_report": {
  "success": false,
  "summary": "The window was successfully minimized, but the attempt to launch
Google Chrome failed due to a coordinate mismatch.",
  "message": "The task was partially completed. The active window was
successfully minimized to reveal the desktop and taskbar. However, the second
step failed because the command targeted (791, 978), which did not align with the
detected location of the Google Chrome icon.\\n\\nStep results:\\n\\u2705 Step 1:
Minimized the current window.\\n\\u274c Step 2: Failed to click the Google Chrome
icon.",
  "steps_completed": 1,
  "steps_failed": 1,
  "issues": [
    "The coordinates used in the failed step (791, 978) do not match the
current detected position of the Chrome icon."
  ],
  "recommendations": [
    "Retry the second step using the updated coordinates for the Chrome taskbar
button at (896, 969) or the desktop shortcut at (56, 76).\"
  ]
},
"session_memory": {
  "run_id": "f60f0afa-3ce3-4f83-9e73-d88f5282905b",
```

```
"user_task": "Minimize the current window using minimize icon and find the Google Chrome icon on the screen and click on that app icon",
"started_at": "2026-05-02T04:29:00.097697+00:00",
"updated_at": "2026-05-02T04:38:45.177526+00:00",
"stale_after_misses": 2,
"confidence_floor": 0.4,
"decay_per_miss": 0.9,
"frame_index": 13,
"known_elements": {
  "tab:frontend": {
    "element_key": "tab:frontend",
    "element_type": "tab",
    "label": "frontend",
    "bbox": [
      151.0,
      118.0,
      399.0,
      174.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 0,
    "last_seen_index": 1,
    "first_seen_at": "2026-05-02T04:29:36.641249+00:00",
    "last_seen_at": "2026-05-02T04:30:16.794900+00:00",
    "seen_count": 2,
    "miss_count": 11,
```

```
    "stale": true,
    "cached": true
  },
  "input:address bar": {
    "element_key": "input:address bar",
    "element_type": "input",
    "label": "Address bar",
    "bbox": [
      151.0,
      182.0,
      1303.0,
      238.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 0,
    "last_seen_index": 1,
    "first_seen_at": "2026-05-02T04:29:36.641249+00:00",
    "last_seen_at": "2026-05-02T04:30:16.794900+00:00",
    "seen_count": 2,
    "miss_count": 11,
    "stale": true,
    "cached": true
  },
  "button:ask gemini": {
    "element_key": "button:ask gemini",
    "element_type": "button",
```



```
"label": "Ask Gemini",
"bbox": [
  1415.0,
  142.0,
  1583.0,
  190.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 0,
"first_seen_at": "2026-05-02T04:29:36.641249+00:00",
"last_seen_at": "2026-05-02T04:29:36.641249+00:00",
"seen_count": 1,
"miss_count": 12,
"stale": true,
"cached": true
},
"button:minimize": {
  "element_key": "button:minimize",
  "element_type": "button",
  "label": "Minimize",
  "bbox": [
    1597.0,
    134.0,
    1653.0,
    174.0
  ]
}
```

```
],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 1,
  "first_seen_at": "2026-05-02T04:29:36.641249+00:00",
  "last_seen_at": "2026-05-02T04:30:16.794900+00:00",
  "seen_count": 2,
  "miss_count": 11,
  "stale": true,
  "cached": true
},
"button:maximize": {
  "element_key": "button:maximize",
  "element_type": "button",
  "label": "Maximize",
  "bbox": [
    1623.0,
    142.0,
    1679.0,
    182.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
```

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"last_seen_index": 0,
"first_seen_at": "2026-05-02T04:29:36.641249+00:00",
"last_seen_at": "2026-05-02T04:29:36.641249+00:00",
"seen_count": 1,
"miss_count": 12,
"stale": true,
"cached": true
},
"button:close": {
  "element_key": "button:close",
  "element_type": "button",
  "label": "Close",
  "bbox": [
    1671.0,
    142.0,
    1727.0,
    190.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 0,
  "first_seen_at": "2026-05-02T04:29:36.641249+00:00",
  "last_seen_at": "2026-05-02T04:29:36.641249+00:00",
  "seen_count": 1,
  "miss_count": 12,
  "stale": true,
```

```
"cached": true
},
"heading:screenpilot ai": {
  "element_key": "heading:screenpilot ai",
  "element_type": "heading",
  "label": "ScreenPilot AI",
  "bbox": [
    167.0,
    270.0,
    375.0,
    334.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 1,
  "first_seen_at": "2026-05-02T04:29:36.641249+00:00",
  "last_seen_at": "2026-05-02T04:30:16.794900+00:00",
  "seen_count": 2,
  "miss_count": 11,
  "stale": true,
  "cached": true
},
"label:autonomous task agent v2": {
  "element_key": "label:autonomous task agent v2",
  "element_type": "label",
  "label": "Autonomous Task Agent v2",
```

```
"bbox": [
  423.0,
  286.0,
  495.0,
  326.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 1,
"first_seen_at": "2026-05-02T04:29:36.641249+00:00",
"last_seen_at": "2026-05-02T04:30:16.794900+00:00",
"seen_count": 2,
"miss_count": 11,
"stale": true,
"cached": true
},
"text:connecting to agent...": {
  "element_key": "text:connecting to agent...",
  "element_type": "text",
  "label": "Connecting to agent...",
  "bbox": [
    183.0,
    526.0,
    359.0,
    566.0
  ],
}
```

```
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 0,
"first_seen_at": "2026-05-02T04:29:36.641249+00:00",
"last_seen_at": "2026-05-02T04:29:36.641249+00:00",
"seen_count": 1,
"miss_count": 12,
"stale": true,
"cached": true
},
"text:capturing initial screen...": {
  "element_key": "text:capturing initial screen...",
  "element_type": "text",
  "label": "Capturing initial screen...",
  "bbox": [
    183.0,
    558.0,
    415.0,
    598.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 0,
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"first_seen_at": "2026-05-02T04:29:36.641249+00:00",
"last_seen_at": "2026-05-02T04:29:36.641249+00:00",
"seen_count": 1,
"miss_count": 12,
"stale": true,
"cached": true
},
"pane:preview window": {
  "element_key": "pane:preview window",
  "element_type": "pane",
  "label": "Preview Window",
  "bbox": [
    167.0,
    598.0,
    367.0,
    798.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 0,
  "first_seen_at": "2026-05-02T04:29:36.641249+00:00",
  "last_seen_at": "2026-05-02T04:29:36.641249+00:00",
  "seen_count": 1,
  "miss_count": 12,
  "stale": true,
  "cached": true
}
```

```
},

```



```
    1583.0,  
    854.0,  
    1663.0,  
    942.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 0,  
  "last_seen_index": 1,  
  "first_seen_at": "2026-05-02T04:29:36.641249+00:00",  
  "last_seen_at": "2026-05-02T04:30:16.794900+00:00",  
  "seen_count": 2,  
  "miss_count": 11,  
  "stale": true,  
  "cached": true  
},  
"pane:taskbar": {  
  "element_key": "pane:taskbar",  
  "element_type": "pane",  
  "label": "Taskbar",  
  "bbox": [  
    152.0,  
    933.0,  
    1727.0,  
    1012.0  
  ],  
  "confidence": 0.855,
```

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"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 11,
"first_seen_at": "2026-05-02T04:29:36.641249+00:00",
"last_seen_at": "2026-05-02T04:37:59.364903+00:00",
"seen_count": 2,
"miss_count": 1,
"stale": false,
"cached": true
},
"button:start": {
  "element_key": "button:start",
  "element_type": "button",
  "label": "Start",
  "bbox": [
    472.0,
    941.0,
    528.0,
    1005.0
  ],
  "confidence": 0.855,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 11,
  "first_seen_at": "2026-05-02T04:29:36.641249+00:00",
```

```
    "last_seen_at": "2026-05-02T04:37:59.364903+00:00",
    "seen_count": 3,
    "miss_count": 1,
    "stale": false,
    "cached": true
  },
  "input:search": {
    "element_key": "input:search",
    "element_type": "input",
    "label": "Search",
    "bbox": [
      512.0,
      941.0,
      648.0,
      1005.0
    ],
    "confidence": 0.855,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 0,
    "last_seen_index": 11,
    "first_seen_at": "2026-05-02T04:29:36.641249+00:00",
    "last_seen_at": "2026-05-02T04:37:59.364903+00:00",
    "seen_count": 3,
    "miss_count": 1,
    "stale": false,
    "cached": true
  },
}
```

```
"button:weather": {
  "element_key": "button:weather",
  "element_type": "button",
  "label": "Weather",
  "bbox": [
    167.0,
    958.0,
    319.0,
    1013.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 0,
  "first_seen_at": "2026-05-02T04:29:36.641249+00:00",
  "last_seen_at": "2026-05-02T04:29:36.641249+00:00",
  "seen_count": 1,
  "miss_count": 12,
  "stale": true,
  "cached": true
},
"icon:file explorer": {
  "element_key": "icon:file explorer",
  "element_type": "icon",
  "label": "File Explorer",
  "bbox": [
    743.0,
```

```
    958.0,
    783.0,
    998.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 0,
"first_seen_at": "2026-05-02T04:29:36.641249+00:00",
"last_seen_at": "2026-05-02T04:29:36.641249+00:00",
"seen_count": 1,
"miss_count": 12,
"stale": true,
"cached": true
},
"icon:chrome": {
  "element_key": "icon:chrome",
  "element_type": "icon",
  "label": "Chrome",
  "bbox": [
    775.0,
    958.0,
    807.0,
    998.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
```

```
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 0,
"first_seen_at": "2026-05-02T04:29:36.641249+00:00",
"last_seen_at": "2026-05-02T04:29:36.641249+00:00",
"seen_count": 1,
"miss_count": 12,
"stale": true,
"cached": true
},
"icon:settings": {
  "element_key": "icon:settings",
  "element_type": "icon",
  "label": "Settings",
  "bbox": [
    807.0,
    958.0,
    839.0,
    998.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 0,
  "first_seen_at": "2026-05-02T04:29:36.641249+00:00",
  "last_seen_at": "2026-05-02T04:29:36.641249+00:00",
```

```
    "seen_count": 1,  
    "miss_count": 12,  
    "stale": true,  
    "cached": true  
  },  
  "icon:vs code": {  
    "element_key": "icon:vs code",  
    "element_type": "icon",  
    "label": "VS Code",  
    "bbox": [  
      935.0,  
      958.0,  
      975.0,  
      998.0  
    ],  
    "confidence": 0.4,  
    "source": "gemini_vlm",  
    "first_seen_frame": null,  
    "last_seen_frame": null,  
    "first_seen_index": 0,  
    "last_seen_index": 0,  
    "first_seen_at": "2026-05-02T04:29:36.641249+00:00",  
    "last_seen_at": "2026-05-02T04:29:36.641249+00:00",  
    "seen_count": 1,  
    "miss_count": 12,  
    "stale": true,  
    "cached": true  
  },  
  "icon:spotify": {
```

```
"element_key": "icon:spotify",
"element_type": "icon",
"label": "Spotify",
"bbox": [
  975.0,
  958.0,
  1015.0,
  998.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 0,
"first_seen_at": "2026-05-02T04:29:36.641249+00:00",
"last_seen_at": "2026-05-02T04:29:36.641249+00:00",
"seen_count": 1,
"miss_count": 12,
"stale": true,
"cached": true
},
"pane:system tray": {
  "element_key": "pane:system tray",
  "element_type": "pane",
  "label": "System Tray",
  "bbox": [
    1487.0,
    950.0,
```



```
    1647.0,  
    998.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 0,  
  "last_seen_index": 0,  
  "first_seen_at": "2026-05-02T04:29:36.641249+00:00",  
  "last_seen_at": "2026-05-02T04:29:36.641249+00:00",  
  "seen_count": 1,  
  "miss_count": 12,  
  "stale": true,  
  "cached": true  
},  
"text:clock and date": {  
  "element_key": "text:clock and date",  
  "element_type": "text",  
  "label": "Clock and Date",  
  "bbox": [  
    1288.0,  
    941.0,  
    1360.0,  
    989.0  
  ],  
  "confidence": 0.855,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,
```

```
"last_seen_frame": null,  
"first_seen_index": 0,  
"last_seen_index": 11,  
"first_seen_at": "2026-05-02T04:29:36.641249+00:00",  
"last_seen_at": "2026-05-02T04:37:59.364903+00:00",  
"seen_count": 2,  
"miss_count": 1,  
"stale": false,  
"cached": true  
},  
"icon:screenpilot ai logo": {  
  "element_key": "icon:screenpilot ai logo",  
  "element_type": "icon",  
  "label": "ScreenPilot AI Logo",  
  "bbox": [  
    189.0,  
    230.0,  
    221.0,  
    278.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 1,  
  "last_seen_index": 1,  
  "first_seen_at": "2026-05-02T04:30:16.794900+00:00",  
  "last_seen_at": "2026-05-02T04:30:16.794900+00:00",  
  "seen_count": 1,
```

```
    "miss_count": 11,
    "stale": true,
    "cached": true
  },
  "heading:task plan": {
    "element_key": "heading:task plan",
    "element_type": "heading",
    "label": "Task Plan",
    "bbox": [
      205.0,
      422.0,
      309.0,
      462.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 1,
    "last_seen_index": 1,
    "first_seen_at": "2026-05-02T04:30:16.794900+00:00",
    "last_seen_at": "2026-05-02T04:30:16.794900+00:00",
    "seen_count": 1,
    "miss_count": 11,
    "stale": true,
    "cached": true
  },
  "text:move the mouse to the minimize button": {
    "element_key": "text:move the mouse to the minimize button",
```

```
"element_type": "text",
"label": "Move the mouse to the Minimize button",
"bbox": [
    149.0,
    478.0,
    869.0,
    534.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 1,
"last_seen_index": 1,
"first_seen_at": "2026-05-02T04:30:16.794900+00:00",
"last_seen_at": "2026-05-02T04:30:16.794900+00:00",
"seen_count": 1,
"miss_count": 11,
"stale": true,
"cached": true
},
"text:move the mouse to the chrome icon": {
    "element_key": "text:move the mouse to the chrome icon",
    "element_type": "text",
    "label": "Move the mouse to the Chrome icon",
    "bbox": [
        149.0,
        518.0,
        861.0,
```

```
574.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 1,
"last_seen_index": 1,
"first_seen_at": "2026-05-02T04:30:16.794900+00:00",
"last_seen_at": "2026-05-02T04:30:16.794900+00:00",
"seen_count": 1,
"miss_count": 11,
"stale": true,
"cached": true
},
"pane:agent log": {
  "element_key": "pane:agent log",
  "element_type": "pane",
  "label": "Agent Log",
  "bbox": [
    149.0,
    558.0,
    877.0,
    758.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
```

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"first_seen_index": 1,
"last_seen_index": 1,
"first_seen_at": "2026-05-02T04:30:16.794900+00:00",
"last_seen_at": "2026-05-02T04:30:16.794900+00:00",
"seen_count": 1,
"miss_count": 11,
"stale": true,
"cached": true
},
"button:chrome": {
  "element_key": "button:chrome",
  "element_type": "button",
  "label": "Chrome",
  "bbox": [
    877.0,
    942.0,
    917.0,
    998.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
  "last_seen_index": 1,
  "first_seen_at": "2026-05-02T04:30:16.794900+00:00",
  "last_seen_at": "2026-05-02T04:30:16.794900+00:00",
  "seen_count": 1,
  "miss_count": 11,
```

```
    "stale": true,
    "cached": true
  },
  "button:vs code": {
    "element_key": "button:vs code",
    "element_type": "button",
    "label": "VS Code",
    "bbox": [
      933.0,
      942.0,
      981.0,
      998.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 1,
    "last_seen_index": 1,
    "first_seen_at": "2026-05-02T04:30:16.794900+00:00",
    "last_seen_at": "2026-05-02T04:30:16.794900+00:00",
    "seen_count": 1,
    "miss_count": 11,
    "stale": true,
    "cached": true
  },
  "button:spotify": {
    "element_key": "button:spotify",
    "element_type": "button",
```

```
"label": "Spotify",
"bbox": [
  984.0,
  941.0,
  1040.0,
  997.0
],
"confidence": 0.855,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 1,
"last_seen_index": 11,
"first_seen_at": "2026-05-02T04:30:16.794900+00:00",
"last_seen_at": "2026-05-02T04:37:59.364903+00:00",
"seen_count": 2,
"miss_count": 1,
"stale": false,
"cached": true
},
"text:clock": {
  "element_key": "text:clock",
  "element_type": "text",
  "label": "Clock",
  "bbox": [
    1293.0,
    942.0,
    1365.0,
    990.0
  ]
}
```



```
],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 1,  
  "last_seen_index": 1,  
  "first_seen_at": "2026-05-02T04:30:16.794900+00:00",  
  "last_seen_at": "2026-05-02T04:30:16.794900+00:00",  
  "seen_count": 1,  
  "miss_count": 11,  
  "stale": true,  
  "cached": true  
},  
"icon:google chrome": {  
  "element_key": "icon:google chrome",  
  "element_type": "icon",  
  "label": "Google Chrome",  
  "bbox": [  
    152.0,  
    109.0,  
    264.0,  
    261.0  
  ],  
  "confidence": 0.98,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 2,
```

```
"last_seen_index": 12,  
"first_seen_at": "2026-05-02T04:31:04.756697+00:00",  
"last_seen_at": "2026-05-02T04:38:45.177526+00:00",  
"seen_count": 10,  
"miss_count": 0,  
"stale": false,  
"cached": false  
},  
"icon:huiontablet": {  
  "element_key": "icon:huiontablet",  
  "element_type": "icon",  
  "label": "HuionTablet",  
  "bbox": [  
    248.0,  
    141.0,  
    328.0,  
    261.0  
  ],  
  "confidence": 0.97,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 2,  
  "last_seen_index": 12,  
  "first_seen_at": "2026-05-02T04:31:04.756697+00:00",  
  "last_seen_at": "2026-05-02T04:38:45.177526+00:00",  
  "seen_count": 10,  
  "miss_count": 0,  
  "stale": false,
```

```
    "cached": false
  },
  "button:google chrome": {
    "element_key": "button:google chrome",
    "element_type": "button",
    "label": "Google Chrome",
    "bbox": [
      872.0,
      941.0,
      920.0,
      997.0
    ],
    "confidence": 0.855,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 8,
    "last_seen_index": 11,
    "first_seen_at": "2026-05-02T04:35:46.397550+00:00",
    "last_seen_at": "2026-05-02T04:37:59.364903+00:00",
    "seen_count": 2,
    "miss_count": 1,
    "stale": false,
    "cached": true
  },
  "button:huiontablet": {
    "element_key": "button:huiontablet",
    "element_type": "button",
    "label": "HuionTablet",
```

```
"bbox": [
  248.0,
  142.0,
  328.0,
  278.0
],
"confidence": 0.6561,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 8,
"last_seen_index": 8,
"first_seen_at": "2026-05-02T04:35:46.397550+00:00",
"last_seen_at": "2026-05-02T04:35:46.397550+00:00",
"seen_count": 1,
"miss_count": 4,
"stale": true,
"cached": true
},
"icon:wise memory optimizer": {
  "element_key": "icon:wise memory optimizer",
  "element_type": "icon",
  "label": "Wise Memory Optimizer",
  "bbox": [
    152.0,
    341.0,
    256.0,
    541.0
  ],

```

```
"confidence": 0.855,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 11,
"last_seen_index": 11,
"first_seen_at": "2026-05-02T04:37:59.364903+00:00",
"last_seen_at": "2026-05-02T04:37:59.364903+00:00",
"seen_count": 1,
"miss_count": 1,
"stale": false,
"cached": true
},
"icon:amazon workspaces": {
  "element_key": "icon:amazon workspaces",
  "element_type": "icon",
  "label": "Amazon WorkSpaces",
  "bbox": [
    248.0,
    341.0,
    328.0,
    541.0
  ],
  "confidence": 0.855,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 11,
  "last_seen_index": 11,
```

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"first_seen_at": "2026-05-02T04:37:59.364903+00:00",
"last_seen_at": "2026-05-02T04:37:59.364903+00:00",
"seen_count": 1,
"miss_count": 1,
"stale": false,
"cached": true
},
"icon:recycle bin": {
  "element_key": "icon:recycle bin",
  "element_type": "icon",
  "label": "Recycle Bin",
  "bbox": [
    152.0,
    541.0,
    256.0,
    741.0
  ],
  "confidence": 0.855,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 11,
  "last_seen_index": 11,
  "first_seen_at": "2026-05-02T04:37:59.364903+00:00",
  "last_seen_at": "2026-05-02T04:37:59.364903+00:00",
  "seen_count": 1,
  "miss_count": 1,
  "stale": false,
  "cached": true
}
```

```
},  
"icon:krita": {  
  "element_key": "icon:krita",  
  "element_type": "icon",  
  "label": "Krita",  
  "bbox": [  
    248.0,  
    541.0,  
    328.0,  
    733.0  
  ],  
  "confidence": 0.855,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 11,  
  "last_seen_index": 11,  
  "first_seen_at": "2026-05-02T04:37:59.364903+00:00",  
  "last_seen_at": "2026-05-02T04:37:59.364903+00:00",  
  "seen_count": 1,  
  "miss_count": 1,  
  "stale": false,  
  "cached": true  
},  
"icon:nuwani - chrome": {  
  "element_key": "icon:nuwani - chrome",  
  "element_type": "icon",  
  "label": "Nuwani - Chrome",  
  "bbox": [  

```

```
    152.0,  
    749.0,  
    256.0,  
    949.0  
  ],  
  "confidence": 0.855,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 11,  
  "last_seen_index": 11,  
  "first_seen_at": "2026-05-02T04:37:59.364903+00:00",  
  "last_seen_at": "2026-05-02T04:37:59.364903+00:00",  
  "seen_count": 1,  
  "miss_count": 1,  
  "stale": false,  
  "cached": true  
},  
"icon:docker desktop": {  
  "element_key": "icon:docker desktop",  
  "element_type": "icon",  
  "label": "Docker Desktop",  
  "bbox": [  
    248.0,  
    757.0,  
    328.0,  
    949.0  
  ],  
  "confidence": 0.855,
```



```
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 11,
"last_seen_index": 11,
"first_seen_at": "2026-05-02T04:37:59.364903+00:00",
"last_seen_at": "2026-05-02T04:37:59.364903+00:00",
"seen_count": 1,
"miss_count": 1,
"stale": false,
"cached": true
},
"icon:antigravity": {
  "element_key": "icon:antigravity",
  "element_type": "icon",
  "label": "Antigravity",
  "bbox": [
    160.0,
    973.0,
    256.0,
    1012.0
  ],
  "confidence": 0.81,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 11,
  "last_seen_index": 11,
  "first_seen_at": "2026-05-02T04:37:59.364903+00:00",
```

```
    "last_seen_at": "2026-05-02T04:37:59.364903+00:00",
    "seen_count": 1,
    "miss_count": 1,
    "stale": false,
    "cached": true
  },
  "icon:visual studio code": {
    "element_key": "icon:visual studio code",
    "element_type": "icon",
    "label": "Visual Studio Code",
    "bbox": [
      232.0,
      973.0,
      336.0,
      1005.0
    ],
    "confidence": 0.855,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 11,
    "last_seen_index": 11,
    "first_seen_at": "2026-05-02T04:37:59.364903+00:00",
    "last_seen_at": "2026-05-02T04:37:59.364903+00:00",
    "seen_count": 1,
    "miss_count": 1,
    "stale": false,
    "cached": true
  },
}
```

```
"icon:postman": {
  "element_key": "icon:postman",
  "element_type": "icon",
  "label": "Postman",
  "bbox": [
    312.0,
    981.0,
    416.0,
    1005.0
  ],
  "confidence": 0.855,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 11,
  "last_seen_index": 11,
  "first_seen_at": "2026-05-02T04:37:59.364903+00:00",
  "last_seen_at": "2026-05-02T04:37:59.364903+00:00",
  "seen_count": 1,
  "miss_count": 1,
  "stale": false,
  "cached": true
},
"icon:loom": {
  "element_key": "icon:loom",
  "element_type": "icon",
  "label": "Loom",
  "bbox": [
    400.0,
```

```
    989.0,
    488.0,
    1005.0
  ],
  "confidence": 0.855,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 11,
  "last_seen_index": 11,
  "first_seen_at": "2026-05-02T04:37:59.364903+00:00",
  "last_seen_at": "2026-05-02T04:37:59.364903+00:00",
  "seen_count": 1,
  "miss_count": 1,
  "stale": false,
  "cached": true
},
"icon:vlc media player": {
  "element_key": "icon:vlc media player",
  "element_type": "icon",
  "label": "VLC media player",
  "bbox": [
    464.0,
    989.0,
    568.0,
    1005.0
  ],
  "confidence": 0.855,
  "source": "gemini_vlm",
```

```
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 11,
"last_seen_index": 11,
"first_seen_at": "2026-05-02T04:37:59.364903+00:00",
"last_seen_at": "2026-05-02T04:37:59.364903+00:00",
"seen_count": 1,
"miss_count": 1,
"stale": false,
"cached": true
},
"pane:amazon workspaces info": {
  "element_key": "pane:amazon workspaces info",
  "element_type": "pane",
  "label": "Amazon WorkSpaces Info",
  "bbox": [
    1088.0,
    133.0,
    1400.0,
    389.0
  ],
  "confidence": 0.81,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 11,
  "last_seen_index": 11,
  "first_seen_at": "2026-05-02T04:37:59.364903+00:00",
  "last_seen_at": "2026-05-02T04:37:59.364903+00:00",
```

```
"seen_count": 1,
"miss_count": 1,
"stale": false,
"cached": true
},
"button:weather widget": {
  "element_key": "button:weather widget",
  "element_type": "button",
  "label": "Weather Widget",
  "bbox": [
    160.0,
    949.0,
    296.0,
    1012.0
  ],
  "confidence": 0.81,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 11,
  "last_seen_index": 11,
  "first_seen_at": "2026-05-02T04:37:59.364903+00:00",
  "last_seen_at": "2026-05-02T04:37:59.364903+00:00",
  "seen_count": 1,
  "miss_count": 1,
  "stale": false,
  "cached": true
},
"button:visual studio code": {
```

```
"element_key": "button:visual studio code",
"element_type": "button",
"label": "Visual Studio Code",
"bbox": [
  952.0,
  941.0,
  1000.0,
  997.0
],
"confidence": 0.855,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 11,
"last_seen_index": 11,
"first_seen_at": "2026-05-02T04:37:59.364903+00:00",
"last_seen_at": "2026-05-02T04:37:59.364903+00:00",
"seen_count": 1,
"miss_count": 1,
"stale": false,
"cached": true
},
"button:show hidden icons": {
  "element_key": "button:show hidden icons",
  "element_type": "button",
  "label": "Show hidden icons",
  "bbox": [
    1160.0,
    949.0,
```

```
    1200.0,  
    989.0  
  ],  
  "confidence": 0.81,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 11,  
  "last_seen_index": 11,  
  "first_seen_at": "2026-05-02T04:37:59.364903+00:00",  
  "last_seen_at": "2026-05-02T04:37:59.364903+00:00",  
  "seen_count": 1,  
  "miss_count": 1,  
  "stale": false,  
  "cached": true  
},  
"icon:network": {  
  "element_key": "icon:network",  
  "element_type": "icon",  
  "label": "Network",  
  "bbox": [  
    1232.0,  
    949.0,  
    1264.0,  
    989.0  
  ],  
  "confidence": 0.81,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,
```



```
"last_seen_frame": null,
"first_seen_index": 11,
"last_seen_index": 11,
"first_seen_at": "2026-05-02T04:37:59.364903+00:00",
"last_seen_at": "2026-05-02T04:37:59.364903+00:00",
"seen_count": 1,
"miss_count": 1,
"stale": false,
"cached": true
},
"icon:volume": {
  "element_key": "icon:volume",
  "element_type": "icon",
  "label": "Volume",
  "bbox": [
    1248.0,
    949.0,
    1288.0,
    989.0
  ],
  "confidence": 0.81,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 11,
  "last_seen_index": 11,
  "first_seen_at": "2026-05-02T04:37:59.364903+00:00",
  "last_seen_at": "2026-05-02T04:37:59.364903+00:00",
  "seen_count": 1,
```

```
    "miss_count": 1,
    "stale": false,
    "cached": true
  },
  "icon:battery": {
    "element_key": "icon:battery",
    "element_type": "icon",
    "label": "Battery",
    "bbox": [
      1264.0,
      949.0,
      1304.0,
      989.0
    ],
    "confidence": 0.81,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 11,
    "last_seen_index": 11,
    "first_seen_at": "2026-05-02T04:37:59.364903+00:00",
    "last_seen_at": "2026-05-02T04:37:59.364903+00:00",
    "seen_count": 1,
    "miss_count": 1,
    "stale": false,
    "cached": true
  }
},
"current_elements": [],
```

```
"resolved_elements": [  
  {  
    "id": "memory_pane:taskbar",  
    "type": "pane",  
    "label": "Taskbar",  
    "description": "Cached from an earlier frame in the same run",  
    "state": "cached",  
    "bbox": [  
      152.0,  
      933.0,  
      1727.0,  
      1012.0  
    ],  
    "confidence": 0.855,  
    "source": "gemini_vlm",  
    "last_seen_frame": null,  
    "last_seen_index": 11,  
    "stale": true,  
    "cached": true,  
    "session_memory_key": "pane:taskbar"  
  },  
  {  
    "id": "memory_button:start",  
    "type": "button",  
    "label": "Start",  
    "description": "Cached from an earlier frame in the same run",  
    "state": "cached",  
    "bbox": [  
      472.0,
```

```
    941.0,
    528.0,
    1005.0
],
"confidence": 0.855,
"source": "gemini_vlm",
"last_seen_frame": null,
"last_seen_index": 11,
"stale": true,
"cached": true,
"session_memory_key": "button:start"
},
{
  "id": "memory_input:search",
  "type": "input",
  "label": "Search",
  "description": "Cached from an earlier frame in the same run",
  "state": "cached",
  "bbox": [
    512.0,
    941.0,
    648.0,
    1005.0
  ],
  "confidence": 0.855,
  "source": "gemini_vlm",
  "last_seen_frame": null,
  "last_seen_index": 11,
  "stale": true,
```

```
"cached": true,
"session_memory_key": "input:search"
},
{
  "id": "memory_text:clock and date",
  "type": "text",
  "label": "Clock and Date",
  "description": "Cached from an earlier frame in the same run",
  "state": "cached",
  "bbox": [
    1288.0,
    941.0,
    1360.0,
    989.0
  ],
  "confidence": 0.855,
  "source": "gemini_vlm",
  "last_seen_frame": null,
  "last_seen_index": 11,
  "stale": true,
  "cached": true,
  "session_memory_key": "text:clock and date"
},
{
  "id": "memory_button:spotify",
  "type": "button",
  "label": "Spotify",
  "description": "Cached from an earlier frame in the same run",
  "state": "cached",
```

```
"bbox": [
  984.0,
  941.0,
  1040.0,
  997.0
],
"confidence": 0.855,
"source": "gemini_vlm",
"last_seen_frame": null,
"last_seen_index": 11,
"stale": true,
"cached": true,
"session_memory_key": "button:spotify"
},
{
  "id": "memory_icon:google chrome",
  "type": "icon",
  "label": "Google Chrome",
  "description": "Cached from an earlier frame in the same run",
  "state": "cached",
  "bbox": [
    152.0,
    109.0,
    264.0,
    261.0
  ],
  "confidence": 0.98,
  "source": "gemini_vlm",
  "last_seen_frame": null,
```

```
"last_seen_index": 12,
"stale": true,
"cached": true,
"session_memory_key": "icon:google chrome"
},
{
  "id": "memory_icon:huiontablet",
  "type": "icon",
  "label": "HuionTablet",
  "description": "Cached from an earlier frame in the same run",
  "state": "cached",
  "bbox": [
    248.0,
    141.0,
    328.0,
    261.0
  ],
  "confidence": 0.97,
  "source": "gemini_vlm",
  "last_seen_frame": null,
  "last_seen_index": 12,
  "stale": true,
  "cached": true,
  "session_memory_key": "icon:huiontablet"
},
{
  "id": "memory_button:google chrome",
  "type": "button",
  "label": "Google Chrome",
```

```
"description": "Cached from an earlier frame in the same run",
"state": "cached",
"bbox": [
    872.0,
    941.0,
    920.0,
    997.0
],
"confidence": 0.855,
"source": "gemini_vlm",
"last_seen_frame": null,
"last_seen_index": 11,
"stale": true,
"cached": true,
"session_memory_key": "button:google chrome"
},
{
    "id": "memory_icon:wise memory optimizer",
    "type": "icon",
    "label": "Wise Memory Optimizer",
    "description": "Cached from an earlier frame in the same run",
    "state": "cached",
    "bbox": [
        152.0,
        341.0,
        256.0,
        541.0
    ],
    "confidence": 0.855,
```



```
"source": "gemini_vlm",
"last_seen_frame": null,
"last_seen_index": 11,
"stale": true,
"cached": true,
"session_memory_key": "icon:wise memory optimizer"
},
{
  "id": "memory_icon:amazon workspaces",
  "type": "icon",
  "label": "Amazon WorkSpaces",
  "description": "Cached from an earlier frame in the same run",
  "state": "cached",
  "bbox": [
    248.0,
    341.0,
    328.0,
    541.0
  ],
  "confidence": 0.855,
  "source": "gemini_vlm",
  "last_seen_frame": null,
  "last_seen_index": 11,
  "stale": true,
  "cached": true,
  "session_memory_key": "icon:amazon workspaces"
},
{
  "id": "memory_icon:recycle bin",
```

```
"type": "icon",
"label": "Recycle Bin",
"description": "Cached from an earlier frame in the same run",
"state": "cached",
"bbox": [
  152.0,
  541.0,
  256.0,
  741.0
],
"confidence": 0.855,
"source": "gemini_vlm",
"last_seen_frame": null,
"last_seen_index": 11,
"stale": true,
"cached": true,
"session_memory_key": "icon:recycle bin"
},
{
  "id": "memory_icon:krita",
  "type": "icon",
  "label": "Krita",
  "description": "Cached from an earlier frame in the same run",
  "state": "cached",
  "bbox": [
    248.0,
    541.0,
    328.0,
    733.0
  ]
}
```

```
],
  "confidence": 0.855,
  "source": "gemini_vlm",
  "last_seen_frame": null,
  "last_seen_index": 11,
  "stale": true,
  "cached": true,
  "session_memory_key": "icon:krita"
},
{
  "id": "memory_icon:nuwani - chrome",
  "type": "icon",
  "label": "Nuwani - Chrome",
  "description": "Cached from an earlier frame in the same run",
  "state": "cached",
  "bbox": [
    152.0,
    749.0,
    256.0,
    949.0
  ],
  "confidence": 0.855,
  "source": "gemini_vlm",
  "last_seen_frame": null,
  "last_seen_index": 11,
  "stale": true,
  "cached": true,
  "session_memory_key": "icon:nuwani - chrome"
},
```

```
{
  "id": "memory_icon:docker desktop",
  "type": "icon",
  "label": "Docker Desktop",
  "description": "Cached from an earlier frame in the same run",
  "state": "cached",
  "bbox": [
    248.0,
    757.0,
    328.0,
    949.0
  ],
  "confidence": 0.855,
  "source": "gemini_vlm",
  "last_seen_frame": null,
  "last_seen_index": 11,
  "stale": true,
  "cached": true,
  "session_memory_key": "icon:docker desktop"
},
{
  "id": "memory_icon:antigravity",
  "type": "icon",
  "label": "Antigravity",
  "description": "Cached from an earlier frame in the same run",
  "state": "cached",
  "bbox": [
    160.0,
    973.0,
```

```
    256.0,  
    1012.0  
  ],  
  "confidence": 0.81,  
  "source": "gemini_vlm",  
  "last_seen_frame": null,  
  "last_seen_index": 11,  
  "stale": true,  
  "cached": true,  
  "session_memory_key": "icon:antigravity"  
},  
{  
  "id": "memory_icon:visual studio code",  
  "type": "icon",  
  "label": "Visual Studio Code",  
  "description": "Cached from an earlier frame in the same run",  
  "state": "cached",  
  "bbox": [  
    232.0,  
    973.0,  
    336.0,  
    1005.0  
  ],  
  "confidence": 0.855,  
  "source": "gemini_vlm",  
  "last_seen_frame": null,  
  "last_seen_index": 11,  
  "stale": true,  
  "cached": true,
```

```
"session_memory_key": "icon:visual studio code"
},
{
  "id": "memory_icon:postman",
  "type": "icon",
  "label": "Postman",
  "description": "Cached from an earlier frame in the same run",
  "state": "cached",
  "bbox": [
    312.0,
    981.0,
    416.0,
    1005.0
  ],
  "confidence": 0.855,
  "source": "gemini_vlm",
  "last_seen_frame": null,
  "last_seen_index": 11,
  "stale": true,
  "cached": true,
  "session_memory_key": "icon:postman"
},
{
  "id": "memory_icon:loom",
  "type": "icon",
  "label": "Loom",
  "description": "Cached from an earlier frame in the same run",
  "state": "cached",
  "bbox": [
```

```
    400.0,  
    989.0,  
    488.0,  
    1005.0  
  ],  
  "confidence": 0.855,  
  "source": "gemini_vlm",  
  "last_seen_frame": null,  
  "last_seen_index": 11,  
  "stale": true,  
  "cached": true,  
  "session_memory_key": "icon:loom"  
},  
{  
  "id": "memory_icon:vlc media player",  
  "type": "icon",  
  "label": "VLC media player",  
  "description": "Cached from an earlier frame in the same run",  
  "state": "cached",  
  "bbox": [  
    464.0,  
    989.0,  
    568.0,  
    1005.0  
  ],  
  "confidence": 0.855,  
  "source": "gemini_vlm",  
  "last_seen_frame": null,  
  "last_seen_index": 11,
```

```
"stale": true,
"cached": true,
"session_memory_key": "icon:vlc media player"
},
{
  "id": "memory_pane:amazon workspaces info",
  "type": "pane",
  "label": "Amazon WorkSpaces Info",
  "description": "Cached from an earlier frame in the same run",
  "state": "cached",
  "bbox": [
    1088.0,
    133.0,
    1400.0,
    389.0
  ],
  "confidence": 0.81,
  "source": "gemini_vlm",
  "last_seen_frame": null,
  "last_seen_index": 11,
  "stale": true,
  "cached": true,
  "session_memory_key": "pane:amazon workspaces info"
},
{
  "id": "memory_button:weather widget",
  "type": "button",
  "label": "Weather Widget",
  "description": "Cached from an earlier frame in the same run",
```



```
"state": "cached",
"bbox": [
  160.0,
  949.0,
  296.0,
  1012.0
],
"confidence": 0.81,
"source": "gemini_vlm",
"last_seen_frame": null,
"last_seen_index": 11,
"stale": true,
"cached": true,
"session_memory_key": "button:weather widget"
},
{
  "id": "memory_button:visual studio code",
  "type": "button",
  "label": "Visual Studio Code",
  "description": "Cached from an earlier frame in the same run",
  "state": "cached",
  "bbox": [
    952.0,
    941.0,
    1000.0,
    997.0
  ],
  "confidence": 0.855,
  "source": "gemini_vlm",
```

```
"last_seen_frame": null,
"last_seen_index": 11,
"stale": true,
"cached": true,
"session_memory_key": "button:visual studio code"
},
{
  "id": "memory_button:show hidden icons",
  "type": "button",
  "label": "Show hidden icons",
  "description": "Cached from an earlier frame in the same run",
  "state": "cached",
  "bbox": [
    1160.0,
    949.0,
    1200.0,
    989.0
  ],
  "confidence": 0.81,
  "source": "gemini_vlm",
  "last_seen_frame": null,
  "last_seen_index": 11,
  "stale": true,
  "cached": true,
  "session_memory_key": "button:show hidden icons"
},
{
  "id": "memory_icon:network",
  "type": "icon",
```

```
"label": "Network",
"description": "Cached from an earlier frame in the same run",
"state": "cached",
"bbox": [
  1232.0,
  949.0,
  1264.0,
  989.0
],
"confidence": 0.81,
"source": "gemini_vlm",
"last_seen_frame": null,
"last_seen_index": 11,
"stale": true,
"cached": true,
"session_memory_key": "icon:network"
},
{
  "id": "memory_icon:volume",
  "type": "icon",
  "label": "Volume",
  "description": "Cached from an earlier frame in the same run",
  "state": "cached",
  "bbox": [
    1248.0,
    949.0,
    1288.0,
    989.0
  ],
}
```

```
    "confidence": 0.81,  
    "source": "gemini_vlm",  
    "last_seen_frame": null,  
    "last_seen_index": 11,  
    "stale": true,  
    "cached": true,  
    "session_memory_key": "icon:volume"  
  },  
  {  
    "id": "memory_icon:battery",  
    "type": "icon",  
    "label": "Battery",  
    "description": "Cached from an earlier frame in the same run",  
    "state": "cached",  
    "bbox": [  
      1264.0,  
      949.0,  
      1304.0,  
      989.0  
    ],  
    "confidence": 0.81,  
    "source": "gemini_vlm",  
    "last_seen_frame": null,  
    "last_seen_index": 11,  
    "stale": true,  
    "cached": true,  
    "session_memory_key": "icon:battery"  
  }  
],
```

```
"screen_data": {}  
  
}  
  
}
```

Test Case 07: GitHub Navigation (Complex Task)

User Prompt:

Find and click the Github icon on this browser and click on the github profile and go to the repositories section

Test Results:

- Steps Total: 2
- Completed: 0
- Failed: 2
- Duration: 15 minutes 3 seconds

Step Execution:

Step 1: Move mouse to the 'GitHub' tab at (538, 162) and click to switch to the GitHub page

- Target: GitHub tab
- Attempts: 3
- Execution: Performed
- Evaluation: Failed (retry loop)
- Issue: The browser was stuck on the Chrome profile selection screen ("Who's using Chrome?"), blocking access to the GitHub tab. The action to switch tabs did not result in the browser becoming the active window.

Step 2: Locate the profile icon in the top right corner of the GitHub page, click it to open the dropdown menu, and then click on 'Your repositories'

- Target: GitHub profile icon and 'Your repositories' menu item
- Attempts: 3
- Execution: Performed
- Evaluation: Failed (retry loop)
- Issue: Evaluation error — No valid JSON found in LLM response ('{\n "status": ""'). GitHub page was never reached due to Step 1 failure.

Execution Summary:

The task failed completely because the browser was blocked on the Chrome profile selection screen. The agent was unable to switch to the GitHub tab or perform any subsequent navigation steps. Step 1 encountered a retry loop without success, and Step 2 failed due to a malformed JSON response from the LLM during evaluation.

Issues Identified:

- Browser stuck on Chrome profile selection screen ("Who's using Chrome?")
- Malformed JSON response from LLM during evaluation
- No pre-execution check to ensure browser is in expected state
- GitHub tab not visible or interactive due to modal overlay
- Missing handle for profile selection dialog before proceeding

Recommendations for Improvement:

- Add pre-flight check to detect modal dialogs (profile picker, sign-in prompts) before task execution
- Implement dialog handler to automatically select a default Chrome profile
- Enforce strict JSON output formatting with validation before parsing
- Add fallback to close modal overlays if detected
- Improve evaluation prompt to always return valid JSON, even on failure

JSON Response:

full_cycle.json

Test Case 07:

Prompt: Find and click the Github icon on this browser and click on the github profile and go to the repositories section

```
{
  "run_id": "caa1e9d9-8f76-42da-b95c-4cc77b1cd29c",
  "user_task": "Find and click the Github icon on this browser and click on the github profile and go to the repositories section",
  "started_at": "2026-05-02T04:51:15.350411+00:00",
  "finished_at": "2026-05-02T05:06:18.590998+00:00",
  "vision_used": true,
```

```
"perception": {
  "image": "E:\\sliit-
project\\ScreenAwareTaskAgent\\vision\\data\\sessions\\shot_1777697481_f67934\\pr
ocessing\\proc_1777697481_325aa1\\preprocessed_frames\\frame_1777697481_02a6a0.jp
g",
  "image_size": {
    "width": 1920,
    "height": 1080
  },
  "screen_bbox": [
    150,
    110,
    1726,
    1014
  ],
  "screen_size": {
    "width": 1576,
    "height": 904
  },
  "coordinate_system": "pixel",
  "elements": [
    {
      "id": "1",
      "type": "tab",
      "label": "frontend",
      "description": "Browser tab labeled frontend",
      "state": "inactive",
      "dx": 148,
      "dy": 52,
      "frame_dx": 298,
```

```
"frame_dy": 162,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    150,
    134,
    446,
    190
],
"session_memory_key": "tab:frontend",
"cached": false
},
{
    "id": "2",
    "type": "tab",
    "label": "GitHub",
    "description": "Browser tab labeled GitHub",
    "state": "inactive",
    "dx": 388,
    "dy": 52,
    "frame_dx": 538,
    "frame_dy": 162,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        382,
        134,
        694,
        190
    ]
}
```



```
    ],
    "session_memory_key": "tab:github",
    "cached": false
  }
],
"element_count": 2,
"current_elements": [
  {
    "id": "1",
    "type": "tab",
    "label": "frontend",
    "description": "Browser tab labeled frontend",
    "state": "inactive",
    "dx": 148,
    "dy": 52,
    "frame_dx": 298,
    "frame_dy": 162,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
      150,
      134,
      446,
      190
    ],
    "session_memory_key": "tab:frontend",
    "cached": false
  },
  {
```

```
    "id": "2",
    "type": "tab",
    "label": "GitHub",
    "description": "Browser tab labeled GitHub",
    "state": "inactive",
    "dx": 388,
    "dy": 52,
    "frame_dx": 538,
    "frame_dy": 162,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        382,
        134,
        694,
        190
    ],
    "session_memory_key": "tab:github",
    "cached": false
}
],
"resolved_elements": [
    {
        "id": "1",
        "type": "tab",
        "label": "frontend",
        "description": "Browser tab labeled frontend",
        "state": "inactive",
        "dx": 148,
```

```
"dy": 52,
"frame_dx": 298,
"frame_dy": 162,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    150,
    134,
    446,
    190
],
"session_memory_key": "tab:frontend",
"cached": false
},
{
    "id": "2",
    "type": "tab",
    "label": "GitHub",
    "description": "Browser tab labeled GitHub",
    "state": "inactive",
    "dx": 388,
    "dy": 52,
    "frame_dx": 538,
    "frame_dy": 162,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        382,
        134,
```

```
        694,  
        190  
    ],  
    "session_memory_key": "tab:github",  
    "cached": false  
}  
],  
"session_knowledge_base": {  
    "session_id": "shot_1777697481_f67934",  
    "started_at": "2026-05-02T04:51:21.788502+00:00",  
    "updated_at": "2026-05-02T04:51:55.798150+00:00",  
    "stale_after_misses": 3,  
    "confidence_floor": 0.4,  
    "decay_per_miss": 0.9,  
    "frame_count": 1,  
    "frame_name": "frame_1777697481_02a6a0.jpg",  
    "frame_index": 0,  
    "image_size": {  
        "width": 1920,  
        "height": 1080  
    },  
    "known_elements": {  
        "tab:frontend": {  
            "element_key": "tab:frontend",  
            "element_type": "tab",  
            "label": "frontend",  
            "bbox": [  
                150.0,  
                134.0,
```

```
446.0,
190.0
],
"confidence": 0.95,
"source": "gemini_vlm",
"first_seen_frame": "frame_1777697481_02a6a0.jpg",
"last_seen_frame": "frame_1777697481_02a6a0.jpg",
"first_seen_at": "2026-05-02T04:51:55.798150+00:00",
"last_seen_at": "2026-05-02T04:51:55.798150+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777697481_02a6a0.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T04:51:55.798150+00:00",
    "confidence": 0.95
  }
]
},
"tab:github": {
  "element_key": "tab:github",
  "element_type": "tab",
  "label": "GitHub",
  "bbox": [
```

```
    382.0,  
    134.0,  
    694.0,  
    190.0  
  ],  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "first_seen_frame": "frame_1777697481_02a6a0.jpg",  
  "last_seen_frame": "frame_1777697481_02a6a0.jpg",  
  "first_seen_at": "2026-05-02T04:51:55.798150+00:00",  
  "last_seen_at": "2026-05-02T04:51:55.798150+00:00",  
  "first_seen_index": 0,  
  "last_seen_index": 0,  
  "seen_count": 1,  
  "miss_count": 0,  
  "stale": false,  
  "cached": false,  
  "history": [  
    {  
      "frame_name": "frame_1777697481_02a6a0.jpg",  
      "frame_index": 0,  
      "seen_at": "2026-05-02T04:51:55.798150+00:00",  
      "confidence": 0.95  
    }  
  ]  
}  
},  
"resolved_elements": [  
  {
```

```
"id": "1",
"type": "tab",
"label": "frontend",
"description": "Browser tab labeled frontend",
"state": "inactive",
"dx": 148,
"dy": 52,
"frame_dx": 298,
"frame_dy": 162,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
  150,
  134,
  446,
  190
],
"session_memory_key": "tab:frontend",
"cached": false
},
{
  "id": "2",
  "type": "tab",
  "label": "GitHub",
  "description": "Browser tab labeled GitHub",
  "state": "inactive",
  "dx": 388,
  "dy": 52,
  "frame_dx": 538,
```

```
    "frame_dy": 162,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        382,
        134,
        694,
        190
    ],
    "session_memory_key": "tab:github",
    "cached": false
}
],
"current_elements": [
    {
        "id": "1",
        "type": "tab",
        "label": "frontend",
        "description": "Browser tab labeled frontend",
        "state": "inactive",
        "dx": 148,
        "dy": 52,
        "frame_dx": 298,
        "frame_dy": 162,
        "confidence": 0.95,
        "source": "gemini_vlm",
        "frame_bbox": [
            150,
            134,
```



```
        446,  
        190  
    ],  
    "session_memory_key": "tab:frontend",  
    "cached": false  
},  
{  
    "id": "2",  
    "type": "tab",  
    "label": "GitHub",  
    "description": "Browser tab labeled GitHub",  
    "state": "inactive",  
    "dx": 388,  
    "dy": 52,  
    "frame_dx": 538,  
    "frame_dy": 162,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        382,  
        134,  
        694,  
        190  
    ],  
    "session_memory_key": "tab:github",  
    "cached": false  
}  
]  
},
```

```
"session_memory": {
  "run_id": "caa1e9d9-8f76-42da-b95c-4cc77b1cd29c",
  "user_task": "Find and click the Github icon on this browser and click on
the github profile and go to the repositories section",
  "started_at": "2026-05-02T04:51:15.350411+00:00",
  "updated_at": "2026-05-02T04:51:55.824539+00:00",
  "stale_after_misses": 2,
  "confidence_floor": 0.4,
  "decay_per_miss": 0.9,
  "frame_index": 1,
  "known_elements": {
    "tab:frontend": {
      "element_key": "tab:frontend",
      "element_type": "tab",
      "label": "frontend",
      "bbox": [
        150.0,
        133.0,
        382.0,
        197.0
      ],
      "confidence": 0.4,
      "source": "gemini_vlm",
      "first_seen_frame": null,
      "last_seen_frame": null,
      "first_seen_index": 0,
      "last_seen_index": 1,
      "first_seen_at": "2026-05-02T04:51:55.824539+00:00",
      "last_seen_at": "2026-05-02T04:52:42.298731+00:00",
```

```
    "seen_count": 2,  
    "miss_count": 17,  
    "stale": true,  
    "cached": true  
  },  
  "tab:github": {  
    "element_key": "tab:github",  
    "element_type": "tab",  
    "label": "GitHub",  
    "bbox": [  
      326.0,  
      133.0,  
      566.0,  
      197.0  
    ],  
    "confidence": 0.4,  
    "source": "gemini_vlm",  
    "first_seen_frame": null,  
    "last_seen_frame": null,  
    "first_seen_index": 0,  
    "last_seen_index": 1,  
    "first_seen_at": "2026-05-02T04:51:55.824539+00:00",  
    "last_seen_at": "2026-05-02T04:52:42.298731+00:00",  
    "seen_count": 2,  
    "miss_count": 17,  
    "stale": true,  
    "cached": true  
  },  
  "tab:new tab": {
```

```
"element_key": "tab:new tab",
"element_type": "tab",
"label": "New tab",
"bbox": [
    518.0,
    133.0,
    750.0,
    197.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 1,
"last_seen_index": 1,
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"last_seen_at": "2026-05-02T04:52:42.298731+00:00",
"seen_count": 1,
"miss_count": 17,
"stale": true,
"cached": true
},
"button:new tab": {
    "element_key": "button:new tab",
    "element_type": "button",
    "label": "New tab",
    "bbox": [
        718.0,
        141.0,
```

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    766.0,
    189.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 1,
"last_seen_index": 1,
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"last_seen_at": "2026-05-02T04:52:42.298731+00:00",
"seen_count": 1,
"miss_count": 17,
"stale": true,
"cached": true
},
"button:ask gemini": {
  "element_key": "button:ask gemini",
  "element_type": "button",
  "label": "Ask Gemini",
  "bbox": [
    1134.0,
    157.0,
    1254.0,
    189.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
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"last_seen_frame": null,
"first_seen_index": 1,
"last_seen_index": 1,
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"last_seen_at": "2026-05-02T04:52:42.298731+00:00",
"seen_count": 1,
"miss_count": 17,
"stale": true,
"cached": true
},
"button:back": {
  "element_key": "button:back",
  "element_type": "button",
  "label": "Back",
  "bbox": [
    166.0,
    189.0,
    214.0,
    253.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
  "last_seen_index": 1,
  "first_seen_at": "2026-05-02T04:52:42.298731+00:00",
  "last_seen_at": "2026-05-02T04:52:42.298731+00:00",
  "seen_count": 1,
```

```
    "miss_count": 17,  
    "stale": true,  
    "cached": true  
  },  
  "button:forward": {  
    "element_key": "button:forward",  
    "element_type": "button",  
    "label": "Forward",  
    "bbox": [  
      198.0,  
      189.0,  
      246.0,  
      245.0  
    ],  
    "confidence": 0.4,  
    "source": "gemini_vlm",  
    "first_seen_frame": null,  
    "last_seen_frame": null,  
    "first_seen_index": 1,  
    "last_seen_index": 1,  
    "first_seen_at": "2026-05-02T04:52:42.298731+00:00",  
    "last_seen_at": "2026-05-02T04:52:42.298731+00:00",  
    "seen_count": 1,  
    "miss_count": 17,  
    "stale": true,  
    "cached": true  
  },  
  "button:refresh": {  
    "element_key": "button:refresh",
```

```
"element_type": "button",
"label": "Refresh",
"bbox": [
    230.0,
    189.0,
    278.0,
    253.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 1,
"last_seen_index": 1,
"first_seen_at": "2026-05-02T04:52:42.298731+00:00",
"last_seen_at": "2026-05-02T04:52:42.298731+00:00",
"seen_count": 1,
"miss_count": 17,
"stale": true,
"cached": true
},
"input:search google or type a url": {
    "element_key": "input:search google or type a url",
    "element_type": "input",
    "label": "Search Google or type a URL",
    "bbox": [
        150.0,
        189.0,
        1350.0,
```



```
    261.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
  "last_seen_index": 1,
  "first_seen_at": "2026-05-02T04:52:42.298731+00:00",
  "last_seen_at": "2026-05-02T04:52:42.298731+00:00",
  "seen_count": 1,
  "miss_count": 17,
  "stale": true,
  "cached": true
},
"button:extensions": {
  "element_key": "button:extensions",
  "element_type": "button",
  "label": "Extensions",
  "bbox": [
    1206.0,
    205.0,
    1262.0,
    261.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
```

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"first_seen_index": 1,
"last_seen_index": 1,
"first_seen_at": "2026-05-02T04:52:42.298731+00:00",
"last_seen_at": "2026-05-02T04:52:42.298731+00:00",
"seen_count": 1,
"miss_count": 17,
"stale": true,
"cached": true
},
"button:side panel": {
  "element_key": "button:side panel",
  "element_type": "button",
  "label": "Side panel",
  "bbox": [
    1246.0,
    205.0,
    1302.0,
    253.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
  "last_seen_index": 1,
  "first_seen_at": "2026-05-02T04:52:42.298731+00:00",
  "last_seen_at": "2026-05-02T04:52:42.298731+00:00",
  "seen_count": 1,
  "miss_count": 17,
```

```
    "stale": true,
    "cached": true
  },
  "button:profile": {
    "element_key": "button:profile",
    "element_type": "button",
    "label": "Profile",
    "bbox": [
      1294.0,
      205.0,
      1350.0,
      253.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 1,
    "last_seen_index": 1,
    "first_seen_at": "2026-05-02T04:52:42.298731+00:00",
    "last_seen_at": "2026-05-02T04:52:42.298731+00:00",
    "seen_count": 1,
    "miss_count": 17,
    "stale": true,
    "cached": true
  },
  "link:dashboard": {
    "element_key": "link:dashboard",
    "element_type": "link",
```

```
"label": "Dashboard",
"bbox": [
  198.0,
  261.0,
  334.0,
  285.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 1,
"last_seen_index": 1,
"first_seen_at": "2026-05-02T04:52:42.298731+00:00",
"last_seen_at": "2026-05-02T04:52:42.298731+00:00",
"seen_count": 1,
"miss_count": 17,
"stale": true,
"cached": true
},
"link:inbox": {
  "element_key": "link:inbox",
  "element_type": "link",
  "label": "Inbox",
  "bbox": [
    326.0,
    261.0,
    406.0,
    277.0
  ]
}
```

```
],  
  "confidence": 0.4,  
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  "last_seen_frame": null,  
  "first_seen_index": 1,  
  "last_seen_index": 1,  
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  "last_seen_at": "2026-05-02T04:52:42.298731+00:00",  
  "seen_count": 1,  
  "miss_count": 17,  
  "stale": true,  
  "cached": true  
},  
"link:nexmart": {  
  "element_key": "link:nexmart",  
  "element_type": "link",  
  "label": "NexMart",  
  "bbox": [  
    790.0,  
    725.0,  
    862.0,  
    829.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 1,
```

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"last_seen_index": 1,
"first_seen_at": "2026-05-02T04:52:42.298731+00:00",
"last_seen_at": "2026-05-02T04:52:42.298731+00:00",
"seen_count": 2,
"miss_count": 17,
"stale": true,
"cached": true
},
"button:all bookmarks": {
  "element_key": "button:all bookmarks",
  "element_type": "button",
  "label": "All Bookmarks",
  "bbox": [
    1262.0,
    253.0,
    1350.0,
    285.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
  "last_seen_index": 1,
  "first_seen_at": "2026-05-02T04:52:42.298731+00:00",
  "last_seen_at": "2026-05-02T04:52:42.298731+00:00",
  "seen_count": 1,
  "miss_count": 17,
  "stale": true,
```

```
    "cached": true
  },
  "image:google logo": {
    "element_key": "image:google logo",
    "element_type": "image",
    "label": "Google Logo",
    "bbox": [
      646.0,
      453.0,
      902.0,
      589.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 1,
    "last_seen_index": 1,
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    "last_seen_at": "2026-05-02T04:52:42.298731+00:00",
    "seen_count": 1,
    "miss_count": 17,
    "stale": true,
    "cached": true
  },
  "input:search bar": {
    "element_key": "input:search bar",
    "element_type": "input",
    "label": "Search bar",
```

```
"bbox": [
  454.0,
  597.0,
  1110.0,
  677.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 1,
"last_seen_index": 1,
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"last_seen_at": "2026-05-02T04:52:42.298731+00:00",
"seen_count": 1,
"miss_count": 17,
"stale": true,
"cached": true
},
"link:whatsapp": {
  "element_key": "link:whatsapp",
  "element_type": "link",
  "label": "WhatsApp",
  "bbox": [
    398.0,
    725.0,
    462.0,
    797.0
  ],

```



```
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 1,
"last_seen_index": 1,
"first_seen_at": "2026-05-02T04:52:42.298731+00:00",
"last_seen_at": "2026-05-02T04:52:42.298731+00:00",
"seen_count": 1,
"miss_count": 17,
"stale": true,
"cached": true
},
"link:github": {
  "element_key": "link:github",
  "element_type": "link",
  "label": "GitHub",
  "bbox": [
    470.0,
    725.0,
    542.0,
    829.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
  "last_seen_index": 1,
```

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"first_seen_at": "2026-05-02T04:52:42.298731+00:00",
"last_seen_at": "2026-05-02T04:52:42.298731+00:00",
"seen_count": 1,
"miss_count": 17,
"stale": true,
"cached": true
},
"link:frontend": {
  "element_key": "link:frontend",
  "element_type": "link",
  "label": "frontend",
  "bbox": [
    550.0,
    725.0,
    622.0,
    829.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
  "last_seen_index": 1,
  "first_seen_at": "2026-05-02T04:52:42.298731+00:00",
  "last_seen_at": "2026-05-02T04:52:42.298731+00:00",
  "seen_count": 1,
  "miss_count": 17,
  "stale": true,
  "cached": true
}
```

```
},  
"link:vision service": {  
  "element_key": "link:vision service",  
  "element_type": "link",  
  "label": "Vision Service",  
  "bbox": [  
    630.0,  
    725.0,  
    702.0,  
    829.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 1,  
  "last_seen_index": 1,  
  "first_seen_at": "2026-05-02T04:52:42.298731+00:00",  
  "last_seen_at": "2026-05-02T04:52:42.298731+00:00",  
  "seen_count": 1,  
  "miss_count": 17,  
  "stale": true,  
  "cached": true  
},  
"link:chatgpt": {  
  "element_key": "link:chatgpt",  
  "element_type": "link",  
  "label": "ChatGPT",  
  "bbox": [  

```

```
    710.0,  
    725.0,  
    782.0,  
    829.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 1,  
  "last_seen_index": 1,  
  "first_seen_at": "2026-05-02T04:52:42.298731+00:00",  
  "last_seen_at": "2026-05-02T04:52:42.298731+00:00",  
  "seen_count": 1,  
  "miss_count": 17,  
  "stale": true,  
  "cached": true  
},  
"link:instances": {  
  "element_key": "link:instances",  
  "element_type": "link",  
  "label": "Instances",  
  "bbox": [  
    870.0,  
    725.0,  
    942.0,  
    829.0  
  ],  
  "confidence": 0.4,
```

```
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 1,
"last_seen_index": 1,
"first_seen_at": "2026-05-02T04:52:42.298731+00:00",
"last_seen_at": "2026-05-02T04:52:42.298731+00:00",
"seen_count": 1,
"miss_count": 17,
"stale": true,
"cached": true
},
"link:youtube": {
  "element_key": "link:youtube",
  "element_type": "link",
  "label": "YouTube",
  "bbox": [
    950.0,
    725.0,
    1022.0,
    829.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
  "last_seen_index": 1,
  "first_seen_at": "2026-05-02T04:52:42.298731+00:00",
```

```
    "last_seen_at": "2026-05-02T04:52:42.298731+00:00",
    "seen_count": 1,
    "miss_count": 17,
    "stale": true,
    "cached": true
  },
  "link:course": {
    "element_key": "link:course",
    "element_type": "link",
    "label": "Course",
    "bbox": [
      1030.0,
      725.0,
      1102.0,
      829.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 1,
    "last_seen_index": 1,
    "first_seen_at": "2026-05-02T04:52:42.298731+00:00",
    "last_seen_at": "2026-05-02T04:52:42.298731+00:00",
    "seen_count": 1,
    "miss_count": 17,
    "stale": true,
    "cached": true
  },
}
```

```
"button:add shortcut": {
  "element_key": "button:add shortcut",
  "element_type": "button",
  "label": "Add shortcut",
  "bbox": [
    1110.0,
    725.0,
    1182.0,
    829.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
  "last_seen_index": 1,
  "first_seen_at": "2026-05-02T04:52:42.298731+00:00",
  "last_seen_at": "2026-05-02T04:52:42.298731+00:00",
  "seen_count": 1,
  "miss_count": 17,
  "stale": true,
  "cached": true
},
"button:customise chrome": {
  "element_key": "button:customise chrome",
  "element_type": "button",
  "label": "Customise Chrome",
  "bbox": [
    1214.0,
```

```
    909.0,  
    1358.0,  
    965.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 1,  
  "last_seen_index": 1,  
  "first_seen_at": "2026-05-02T04:52:42.298731+00:00",  
  "last_seen_at": "2026-05-02T04:52:42.298731+00:00",  
  "seen_count": 1,  
  "miss_count": 17,  
  "stale": true,  
  "cached": true  
},  
"button:start": {  
  "element_key": "button:start",  
  "element_type": "button",  
  "label": "Start",  
  "bbox": [  
    163.0,  
    924.0,  
    203.0,  
    959.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",
```



```
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 1,
"last_seen_index": 9,
"first_seen_at": "2026-05-02T04:52:42.298731+00:00",
"last_seen_at": "2026-05-02T04:58:47.026826+00:00",
"seen_count": 2,
"miss_count": 9,
"stale": true,
"cached": true
},
"input:search": {
  "element_key": "input:search",
  "element_type": "input",
  "label": "Search",
  "bbox": [
    502.0,
    965.0,
    718.0,
    1005.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
  "last_seen_index": 2,
  "first_seen_at": "2026-05-02T04:52:42.298731+00:00",
  "last_seen_at": "2026-05-02T04:53:25.020574+00:00",
```

```
    "seen_count": 2,  
    "miss_count": 16,  
    "stale": true,  
    "cached": true  
  },  
  "button:task view": {  
    "element_key": "button:task view",  
    "element_type": "button",  
    "label": "Task View",  
    "bbox": [  
      694.0,  
      965.0,  
      734.0,  
      997.0  
    ],  
    "confidence": 0.4,  
    "source": "gemini_vlm",  
    "first_seen_frame": null,  
    "last_seen_frame": null,  
    "first_seen_index": 1,  
    "last_seen_index": 1,  
    "first_seen_at": "2026-05-02T04:52:42.298731+00:00",  
    "last_seen_at": "2026-05-02T04:52:42.298731+00:00",  
    "seen_count": 1,  
    "miss_count": 17,  
    "stale": true,  
    "cached": true  
  },  
  "button:file explorer": {
```

```
"element_key": "button:file explorer",
"element_type": "button",
"label": "File Explorer",
"bbox": [
    726.0,
    965.0,
    774.0,
    997.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 1,
"last_seen_index": 1,
"first_seen_at": "2026-05-02T04:52:42.298731+00:00",
"last_seen_at": "2026-05-02T04:52:42.298731+00:00",
"seen_count": 1,
"miss_count": 17,
"stale": true,
"cached": true
},
"button:chrome": {
    "element_key": "button:chrome",
    "element_type": "button",
    "label": "Chrome",
    "bbox": [
        766.0,
        965.0,
```

```
      814.0,  
      997.0  
    ],  
    "confidence": 0.4,  
    "source": "gemini_vlm",  
    "first_seen_frame": null,  
    "last_seen_frame": null,  
    "first_seen_index": 1,  
    "last_seen_index": 1,  
    "first_seen_at": "2026-05-02T04:52:42.298731+00:00",  
    "last_seen_at": "2026-05-02T04:52:42.298731+00:00",  
    "seen_count": 1,  
    "miss_count": 17,  
    "stale": true,  
    "cached": true  
  },  
  "button:app launcher": {  
    "element_key": "button:app launcher",  
    "element_type": "button",  
    "label": "App launcher",  
    "bbox": [  
      162.0,  
      211.0,  
      210.0,  
      259.0  
    ],  
    "confidence": 0.4,  
    "source": "gemini_vlm",  
    "first_seen_frame": null,
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"last_seen_frame": null,  
"first_seen_index": 2,  
"last_seen_index": 8,  
"first_seen_at": "2026-05-02T04:53:25.020574+00:00",  
"last_seen_at": "2026-05-02T04:58:02.406820+00:00",  
"seen_count": 2,  
"miss_count": 10,  
"stale": true,  
"cached": true  
},  
"pane:sidebar": {  
  "element_key": "pane:sidebar",  
  "element_type": "pane",  
  "label": "Sidebar",  
  "bbox": [  
    156.0,  
    250.0,  
    212.0,  
    958.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 3,  
  "last_seen_index": 3,  
  "first_seen_at": "2026-05-02T04:54:09.143401+00:00",  
  "last_seen_at": "2026-05-02T04:54:09.143401+00:00",  
  "seen_count": 1,
```

```
    "miss_count": 15,  
    "stale": true,  
    "cached": true  
  },  
  "heading:cdapsubmissioncloud": {  
    "element_key": "heading:cdapsubmissioncloud",  
    "element_type": "heading",  
    "label": "CDAPSubmissionCloud",  
    "bbox": [  
      266.0,  
      275.0,  
      514.0,  
      331.0  
    ],  
    "confidence": 0.4,  
    "source": "gemini_vlm",  
    "first_seen_frame": null,  
    "last_seen_frame": null,  
    "first_seen_index": 3,  
    "last_seen_index": 8,  
    "first_seen_at": "2026-05-02T04:54:09.143401+00:00",  
    "last_seen_at": "2026-05-02T04:58:02.406820+00:00",  
    "seen_count": 2,  
    "miss_count": 10,  
    "stale": true,  
    "cached": true  
  },  
  "button:create or upload": {  
    "element_key": "button:create or upload",
```

```
"element_type": "button",
"label": "Create or upload",
"bbox": [
    1539.0,
    364.0,
    1707.0,
    412.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 3,
"last_seen_index": 9,
"first_seen_at": "2026-05-02T04:54:09.143401+00:00",
"last_seen_at": "2026-05-02T04:58:47.026826+00:00",
"seen_count": 3,
"miss_count": 9,
"stale": true,
"cached": true
},
"pane:sign in dialog": {
    "element_key": "pane:sign in dialog",
    "element_type": "pane",
    "label": "Sign in dialog",
    "bbox": [
        548.0,
        634.0,
        1004.0,
```

```
    958.0
  ],
  "confidence": 0.4,
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  "last_seen_frame": null,
  "first_seen_index": 3,
  "last_seen_index": 3,
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  "last_seen_at": "2026-05-02T04:54:09.143401+00:00",
  "seen_count": 1,
  "miss_count": 15,
  "stale": true,
  "cached": true
},
"button:sign in": {
  "element_key": "button:sign in",
  "element_type": "button",
  "label": "Sign In",
  "bbox": [
    979.0,
    604.0,
    1059.0,
    644.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
```



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"first_seen_index": 3,
"last_seen_index": 9,
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"last_seen_at": "2026-05-02T04:58:47.026826+00:00",
"seen_count": 2,
"miss_count": 9,
"stale": true,
"cached": true
},
"pane:onboarding tooltip": {
  "element_key": "pane:onboarding tooltip",
  "element_type": "pane",
  "label": "Onboarding tooltip",
  "bbox": [
    980.0,
    554.0,
    1308.0,
    850.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 3,
  "last_seen_index": 3,
  "first_seen_at": "2026-05-02T04:54:09.143401+00:00",
  "last_seen_at": "2026-05-02T04:54:09.143401+00:00",
  "seen_count": 1,
  "miss_count": 15,
```

```
    "stale": true,
    "cached": true
  },
  "button:next": {
    "element_key": "button:next",
    "element_type": "button",
    "label": "Next",
    "bbox": [
      1172.0,
      730.0,
      1212.0,
      778.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
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    "last_seen_frame": null,
    "first_seen_index": 3,
    "last_seen_index": 3,
    "first_seen_at": "2026-05-02T04:54:09.143401+00:00",
    "last_seen_at": "2026-05-02T04:54:09.143401+00:00",
    "seen_count": 1,
    "miss_count": 15,
    "stale": true,
    "cached": true
  },
  "pane:taskbar": {
    "element_key": "pane:taskbar",
    "element_type": "pane",
```

```
"label": "Taskbar",
"bbox": [
  156.0,
  922.0,
  1716.0,
  958.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 3,
"last_seen_index": 3,
"first_seen_at": "2026-05-02T04:54:09.143401+00:00",
"last_seen_at": "2026-05-02T04:54:09.143401+00:00",
"seen_count": 1,
"miss_count": 15,
"stale": true,
"cached": true
},
"item:1. project proposal": {
  "element_key": "item:1. project proposal",
  "element_type": "item",
  "label": "1. Project Proposal",
  "bbox": [
    236.0,
    674.0,
    492.0,
    714.0
```

```
],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 3,
  "last_seen_index": 3,
  "first_seen_at": "2026-05-02T04:54:09.143401+00:00",
  "last_seen_at": "2026-05-02T04:54:09.143401+00:00",
  "seen_count": 1,
  "miss_count": 15,
  "stale": true,
  "cached": true
},
"item:2. progress presentation - 1": {
  "element_key": "item:2. progress presentation - 1",
  "element_type": "item",
  "label": "2. Progress Presentation - 1",
  "bbox": [
    236.0,
    706.0,
    484.0,
    746.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 3,
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"last_seen_index": 3,  
"first_seen_at": "2026-05-02T04:54:09.143401+00:00",  
"last_seen_at": "2026-05-02T04:54:09.143401+00:00",  
"seen_count": 1,  
"miss_count": 15,  
"stale": true,  
"cached": true  
},  
"item:3. progress presentation - 2": {  
  "element_key": "item:3. progress presentation - 2",  
  "element_type": "item",  
  "label": "3. Progress Presentation - 2",  
  "bbox": [  
    236.0,  
    738.0,  
    476.0,  
    778.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 3,  
  "last_seen_index": 3,  
  "first_seen_at": "2026-05-02T04:54:09.143401+00:00",  
  "last_seen_at": "2026-05-02T04:54:09.143401+00:00",  
  "seen_count": 1,  
  "miss_count": 15,  
  "stale": true,
```

```
    "cached": true
  },
  "item:4. research paper": {
    "element_key": "item:4. research paper",
    "element_type": "item",
    "label": "4. Research Paper",
    "bbox": [
      236.0,
      770.0,
      492.0,
      802.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 3,
    "last_seen_index": 3,
    "first_seen_at": "2026-05-02T04:54:09.143401+00:00",
    "last_seen_at": "2026-05-02T04:54:09.143401+00:00",
    "seen_count": 1,
    "miss_count": 15,
    "stale": true,
    "cached": true
  },
  "item:5. final report & presentation": {
    "element_key": "item:5. final report & presentation",
    "element_type": "item",
    "label": "5. Final Report & Presentation",
```

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"bbox": [
  236.0,
  794.0,
  492.0,
  834.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 3,
"last_seen_index": 3,
"first_seen_at": "2026-05-02T04:54:09.143401+00:00",
"last_seen_at": "2026-05-02T04:54:09.143401+00:00",
"seen_count": 1,
"miss_count": 15,
"stale": true,
"cached": true
},
"item:6. checklist documents": {
  "element_key": "item:6. checklist documents",
  "element_type": "item",
  "label": "6. CheckList Documents",
  "bbox": [
    236.0,
    826.0,
    492.0,
    866.0
  ],
```

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"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 3,
"last_seen_index": 3,
"first_seen_at": "2026-05-02T04:54:09.143401+00:00",
"last_seen_at": "2026-05-02T04:54:09.143401+00:00",
"seen_count": 1,
"miss_count": 15,
"stale": true,
"cached": true
},
"item:7. website": {
  "element_key": "item:7. website",
  "element_type": "item",
  "label": "7. Website",
  "bbox": [
    244.0,
    858.0,
    492.0,
    898.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 3,
  "last_seen_index": 3,
```



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"first_seen_at": "2026-05-02T04:54:09.143401+00:00",
"last_seen_at": "2026-05-02T04:54:09.143401+00:00",
"seen_count": 1,
"miss_count": 15,
"stale": true,
"cached": true
},
"item:8. log book": {
  "element_key": "item:8. log book",
  "element_type": "item",
  "label": "8. Log Book",
  "bbox": [
    236.0,
    890.0,
    492.0,
    930.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 3,
  "last_seen_index": 3,
  "first_seen_at": "2026-05-02T04:54:09.143401+00:00",
  "last_seen_at": "2026-05-02T04:54:09.143401+00:00",
  "seen_count": 1,
  "miss_count": 15,
  "stale": true,
  "cached": true
}
```

```
},
"item:project registration documents": {
  "element_key": "item:project registration documents",
  "element_type": "item",
  "label": "Project Registration Documents",
  "bbox": [
    236.0,
    922.0,
    492.0,
    954.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 3,
  "last_seen_index": 3,
  "first_seen_at": "2026-05-02T04:54:09.143401+00:00",
  "last_seen_at": "2026-05-02T04:54:09.143401+00:00",
  "seen_count": 1,
  "miss_count": 15,
  "stale": true,
  "cached": true
},
"button:share": {
  "element_key": "button:share",
  "element_type": "button",
  "label": "Share",
  "bbox": [
```

```
    546.0,  
    371.0,  
    602.0,  
    403.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 3,  
  "last_seen_index": 9,  
  "first_seen_at": "2026-05-02T04:54:09.143401+00:00",  
  "last_seen_at": "2026-05-02T04:58:47.026826+00:00",  
  "seen_count": 3,  
  "miss_count": 9,  
  "stale": true,  
  "cached": true  
},  
"button:copy link": {  
  "element_key": "button:copy link",  
  "element_type": "button",  
  "label": "Copy link",  
  "bbox": [  
    602.0,  
    371.0,  
    674.0,  
    403.0  
  ],  
  "confidence": 0.4,
```

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"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 3,
"last_seen_index": 9,
"first_seen_at": "2026-05-02T04:54:09.143401+00:00",
"last_seen_at": "2026-05-02T04:58:47.026826+00:00",
"seen_count": 3,
"miss_count": 9,
"stale": true,
"cached": true
},
"icon:home": {
  "element_key": "icon:home",
  "element_type": "icon",
  "label": "Home",
  "bbox": [
    162.0,
    299.0,
    202.0,
    347.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 4,
  "last_seen_index": 8,
  "first_seen_at": "2026-05-02T04:54:57.496116+00:00",
```

```
    "last_seen_at": "2026-05-02T04:58:02.406820+00:00",
    "seen_count": 3,
    "miss_count": 10,
    "stale": true,
    "cached": true
  },
  "icon:globe": {
    "element_key": "icon:globe",
    "element_type": "icon",
    "label": "Globe",
    "bbox": [
      164.0,
      345.0,
      204.0,
      385.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 4,
    "last_seen_index": 5,
    "first_seen_at": "2026-05-02T04:54:57.496116+00:00",
    "last_seen_at": "2026-05-02T04:55:40.813387+00:00",
    "seen_count": 2,
    "miss_count": 13,
    "stale": true,
    "cached": true
  },
}
```

```
"taskbar:taskbar": {
  "element_key": "taskbar:taskbar",
  "element_type": "taskbar",
  "label": "Taskbar",
  "bbox": [
    164.0,
    932.0,
    1717.0,
    959.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 6,
  "last_seen_index": 6,
  "first_seen_at": "2026-05-02T04:56:31.874633+00:00",
  "last_seen_at": "2026-05-02T04:56:31.874633+00:00",
  "seen_count": 1,
  "miss_count": 12,
  "stale": true,
  "cached": true
},
"text:sharepoint": {
  "element_key": "text:sharepoint",
  "element_type": "text",
  "label": "SharePoint",
  "bbox": [
    155.0,
```

```
    204.0,  
    307.0,  
    268.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 7,  
  "last_seen_index": 9,  
  "first_seen_at": "2026-05-02T04:57:20.678535+00:00",  
  "last_seen_at": "2026-05-02T04:58:47.026826+00:00",  
  "seen_count": 2,  
  "miss_count": 9,  
  "stale": true,  
  "cached": true  
},  
"input:search this library": {  
  "element_key": "input:search this library",  
  "element_type": "input",  
  "label": "Search this library",  
  "bbox": [  
    571.0,  
    212.0,  
    1003.0,  
    268.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",
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```
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 7,
"last_seen_index": 9,
"first_seen_at": "2026-05-02T04:57:20.678535+00:00",
"last_seen_at": "2026-05-02T04:58:47.026826+00:00",
"seen_count": 3,
"miss_count": 9,
"stale": true,
"cached": true
},
"link:sharepoint": {
  "element_key": "link:sharepoint",
  "element_type": "link",
  "label": "SharePoint",
  "bbox": [
    194.0,
    211.0,
    314.0,
    259.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 8,
  "last_seen_index": 8,
  "first_seen_at": "2026-05-02T04:58:02.406820+00:00",
  "last_seen_at": "2026-05-02T04:58:02.406820+00:00",
```



```
    "seen_count": 1,
    "miss_count": 10,
    "stale": true,
    "cached": true
  },
  "icon:my sites": {
    "element_key": "icon:my sites",
    "element_type": "icon",
    "label": "My sites",
    "bbox": [
      162.0,
      371.0,
      202.0,
      419.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 8,
    "last_seen_index": 8,
    "first_seen_at": "2026-05-02T04:58:02.406820+00:00",
    "last_seen_at": "2026-05-02T04:58:02.406820+00:00",
    "seen_count": 1,
    "miss_count": 10,
    "stale": true,
    "cached": true
  },
  "icon:my news": {
```

```
"element_key": "icon:my news",
"element_type": "icon",
"label": "My news",
"bbox": [
  162.0,
  443.0,
  202.0,
  491.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 8,
"last_seen_index": 8,
"first_seen_at": "2026-05-02T04:58:02.406820+00:00",
"last_seen_at": "2026-05-02T04:58:02.406820+00:00",
"seen_count": 1,
"miss_count": 10,
"stale": true,
"cached": true
},
"icon:my files": {
  "element_key": "icon:my files",
  "element_type": "icon",
  "label": "My files",
  "bbox": [
    170.0,
    523.0,
```

```
    194.0,  
    563.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 8,  
  "last_seen_index": 8,  
  "first_seen_at": "2026-05-02T04:58:02.406820+00:00",  
  "last_seen_at": "2026-05-02T04:58:02.406820+00:00",  
  "seen_count": 1,  
  "miss_count": 10,  
  "stale": true,  
  "cached": true  
},  
"icon:my lists": {  
  "element_key": "icon:my lists",  
  "element_type": "icon",  
  "label": "My lists",  
  "bbox": [  
    170.0,  
    595.0,  
    194.0,  
    635.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,
```

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"last_seen_frame": null,  
"first_seen_index": 8,  
"last_seen_index": 8,  
"first_seen_at": "2026-05-02T04:58:02.406820+00:00",  
"last_seen_at": "2026-05-02T04:58:02.406820+00:00",  
"seen_count": 1,  
"miss_count": 10,  
"stale": true,  
"cached": true  
},  
"icon:create": {  
  "element_key": "icon:create",  
  "element_type": "icon",  
  "label": "Create",  
  "bbox": [  
    170.0,  
    667.0,  
    194.0,  
    715.0  
  ],  
  "confidence": 0.4,  
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  "last_seen_index": 8,  
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  "last_seen_at": "2026-05-02T04:58:02.406820+00:00",  
  "seen_count": 1,
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```
    "miss_count": 10,  
    "stale": true,  
    "cached": true  
  },  
  "icon:site icon": {  
    "element_key": "icon:site icon",  
    "element_type": "icon",  
    "label": "Site icon",  
    "bbox": [  
      218.0,  
      275.0,  
      290.0,  
      347.0  
    ],  
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    "last_seen_at": "2026-05-02T04:58:02.406820+00:00",  
    "seen_count": 1,  
    "miss_count": 10,  
    "stale": true,  
    "cached": true  
  },  
  "label:private group": {  
    "element_key": "label:private group",
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"element_type": "label",
"label": "Private group",
"bbox": [
    274.0,
    315.0,
    402.0,
    347.0
],
"confidence": 0.4,
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"last_seen_frame": null,
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"seen_count": 1,
"miss_count": 10,
"stale": true,
"cached": true
},
"button:not following": {
    "element_key": "button:not following",
    "element_type": "button",
    "label": "Not following",
    "bbox": [
        1258.0,
        283.0,
        1346.0,
```

```
    315.0
  ],
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  "last_seen_index": 8,
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  "last_seen_at": "2026-05-02T04:58:02.406820+00:00",
  "seen_count": 1,
  "miss_count": 10,
  "stale": true,
  "cached": true
},
"link:10 members": {
  "element_key": "link:10 members",
  "element_type": "link",
  "label": "10 members",
  "bbox": [
    1258.0,
    323.0,
    1346.0,
    355.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
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"first_seen_index": 8,  
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"last_seen_at": "2026-05-02T04:58:02.406820+00:00",  
"seen_count": 1,  
"miss_count": 10,  
"stale": true,  
"cached": true  
},  
"link:25-26j-cloud": {  
  "element_key": "link:25-26j-cloud",  
  "element_type": "link",  
  "label": "25-26J-Cloud",  
  "bbox": [  
    202.0,  
    371.0,  
    322.0,  
    403.0  
  ],  
  "confidence": 0.4,  
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  "last_seen_frame": null,  
  "first_seen_index": 8,  
  "last_seen_index": 9,  
  "first_seen_at": "2026-05-02T04:58:02.406820+00:00",  
  "last_seen_at": "2026-05-02T04:58:47.026826+00:00",  
  "seen_count": 2,  
  "miss_count": 9,
```



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"stale": true,
"cached": true
},
"link:25-26j-477-students": {
  "element_key": "link:25-26j-477-students",
  "element_type": "link",
  "label": "25-26J-477-Students",
  "bbox": [
    290.0,
    371.0,
    498.0,
    395.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 8,
  "last_seen_index": 9,
  "first_seen_at": "2026-05-02T04:58:02.406820+00:00",
  "last_seen_at": "2026-05-02T04:58:47.026826+00:00",
  "seen_count": 2,
  "miss_count": 9,
  "stale": true,
  "cached": true
},
"button:add shortcut to onedrive": {
  "element_key": "button:add shortcut to onedrive",
  "element_type": "button",
```

```
"label": "Add shortcut to OneDrive",
"bbox": [
  658.0,
  371.0,
  810.0,
  395.0
],
"confidence": 0.4,
"source": "gemini_vlm",
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"last_seen_frame": null,
"first_seen_index": 8,
"last_seen_index": 8,
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"last_seen_at": "2026-05-02T04:58:02.406820+00:00",
"seen_count": 1,
"miss_count": 10,
"stale": true,
"cached": true
},
"button:download": {
  "element_key": "button:download",
  "element_type": "button",
  "label": "Download",
  "bbox": [
    818.0,
    371.0,
    890.0,
    395.0
  ]
}
```

```
],
  "confidence": 0.4,
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  "last_seen_frame": null,
  "first_seen_index": 8,
  "last_seen_index": 8,
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  "last_seen_at": "2026-05-02T04:58:02.406820+00:00",
  "seen_count": 1,
  "miss_count": 10,
  "stale": true,
  "cached": true
},
"button:edit in grid view": {
  "element_key": "button:edit in grid view",
  "element_type": "button",
  "label": "Edit in grid view",
  "bbox": [
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    371.0,
    994.0,
    395.0
  ],
  "confidence": 0.4,
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  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 8,
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    "last_seen_index": 8,  
    "first_seen_at": "2026-05-02T04:58:02.406820+00:00",  
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    "seen_count": 1,  
    "miss_count": 10,  
    "stale": true,  
    "cached": true  
  },  
  "button:export to excel": {  
    "element_key": "button:export to excel",  
    "element_type": "button",  
    "label": "Export to Excel",  
    "bbox": [  
      978.0,  
      371.0,  
      1082.0,  
      395.0  
    ],  
    "confidence": 0.4,  
    "source": "gemini_vlm",  
    "first_seen_frame": null,  
    "last_seen_frame": null,  
    "first_seen_index": 8,  
    "last_seen_index": 8,  
    "first_seen_at": "2026-05-02T04:58:02.406820+00:00",  
    "last_seen_at": "2026-05-02T04:58:02.406820+00:00",  
    "seen_count": 1,  
    "miss_count": 10,  
    "stale": true,
```

```
    "cached": true
  },
  "button:integrate": {
    "element_key": "button:integrate",
    "element_type": "button",
    "label": "Integrate",
    "bbox": [
      1066.0,
      371.0,
      1162.0,
      395.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 8,
    "last_seen_index": 8,
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    "last_seen_at": "2026-05-02T04:58:02.406820+00:00",
    "seen_count": 1,
    "miss_count": 10,
    "stale": true,
    "cached": true
  },
  "button:sync": {
    "element_key": "button:sync",
    "element_type": "button",
    "label": "Sync",
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"bbox": [
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  379.0,
  1210.0,
  395.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 8,
"last_seen_index": 8,
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"last_seen_at": "2026-05-02T04:58:02.406820+00:00",
"seen_count": 1,
"miss_count": 10,
"stale": true,
"cached": true
},
"dropdown:all docu...": {
  "element_key": "dropdown:all docu...",
  "element_type": "dropdown",
  "label": "All Docu...",
  "bbox": [
    194.0,
    403.0,
    346.0,
    451.0
  ],

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"confidence": 0.4,
"source": "gemini_vlm",
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"last_seen_index": 8,
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"last_seen_at": "2026-05-02T04:58:02.406820+00:00",
"seen_count": 1,
"miss_count": 10,
"stale": true,
"cached": true
},
"button:filter": {
  "element_key": "button:filter",
  "element_type": "button",
  "label": "Filter",
  "bbox": [
    338.0,
    403.0,
    378.0,
    443.0
  ],
  "confidence": 0.4,
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  "last_seen_frame": null,
  "first_seen_index": 8,
  "last_seen_index": 8,
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"seen_count": 1,
"miss_count": 10,
"stale": true,
"cached": true
},
"text:cdapsubmissioncloud": {
  "element_key": "text:cdapsubmissioncloud",
  "element_type": "text",
  "label": "CDAPSubmissionCloud",
  "bbox": [
    299.0,
    276.0,
    603.0,
    324.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 9,
  "last_seen_index": 9,
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  "last_seen_at": "2026-05-02T04:58:47.026826+00:00",
  "seen_count": 1,
  "miss_count": 9,
  "stale": true,
  "cached": true
}
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```
},  
"button:all documents": {  
  "element_key": "button:all documents",  
  "element_type": "button",  
  "label": "All Documents",  
  "bbox": [  
    211.0,  
    412.0,  
    387.0,  
    452.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 9,  
  "last_seen_index": 9,  
  "first_seen_at": "2026-05-02T04:58:47.026826+00:00",  
  "last_seen_at": "2026-05-02T04:58:47.026826+00:00",  
  "seen_count": 1,  
  "miss_count": 9,  
  "stale": true,  
  "cached": true  
},  
"pane:modal dialog": {  
  "element_key": "pane:modal dialog",  
  "element_type": "pane",  
  "label": "Modal Dialog",  
  "bbox": [  

```

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        659.0,
        476.0,
        1219.0,
        684.0
    ],
    "confidence": 0.4,
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    "last_seen_frame": null,
    "first_seen_index": 9,
    "last_seen_index": 9,
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    "last_seen_at": "2026-05-02T04:58:47.026826+00:00",
    "seen_count": 1,
    "miss_count": 9,
    "stale": true,
    "cached": true
},
"text:1. project proposal": {
    "element_key": "text:1. project proposal",
    "element_type": "text",
    "label": "1. Project Proposal",
    "bbox": [
        291.0,
        500.0,
        443.0,
        540.0
    ],
    "confidence": 0.4,
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"source": "gemini_vlm",
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"last_seen_frame": null,
"first_seen_index": 9,
"last_seen_index": 9,
"first_seen_at": "2026-05-02T04:58:47.026826+00:00",
"last_seen_at": "2026-05-02T04:58:47.026826+00:00",
"seen_count": 1,
"miss_count": 9,
"stale": true,
"cached": true
},
"text:2. progress presentation - 1": {
  "element_key": "text:2. progress presentation - 1",
  "element_type": "text",
  "label": "2. Progress Presentation - 1",
  "bbox": [
    299.0,
    548.0,
    491.0,
    564.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 9,
  "last_seen_index": 9,
  "first_seen_at": "2026-05-02T04:58:47.026826+00:00",
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```
    "last_seen_at": "2026-05-02T04:58:47.026826+00:00",
    "seen_count": 1,
    "miss_count": 9,
    "stale": true,
    "cached": true
  },
  "text:3. progress presentation - 2": {
    "element_key": "text:3. progress presentation - 2",
    "element_type": "text",
    "label": "3. Progress Presentation - 2",
    "bbox": [
      299.0,
      588.0,
      499.0,
      612.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 9,
    "last_seen_index": 9,
    "first_seen_at": "2026-05-02T04:58:47.026826+00:00",
    "last_seen_at": "2026-05-02T04:58:47.026826+00:00",
    "seen_count": 1,
    "miss_count": 9,
    "stale": true,
    "cached": true
  },
}
```

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"text:4. research paper": {
  "element_key": "text:4. research paper",
  "element_type": "text",
  "label": "4. Research Paper",
  "bbox": [
    299.0,
    636.0,
    435.0,
    652.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 9,
  "last_seen_index": 9,
  "first_seen_at": "2026-05-02T04:58:47.026826+00:00",
  "last_seen_at": "2026-05-02T04:58:47.026826+00:00",
  "seen_count": 1,
  "miss_count": 9,
  "stale": true,
  "cached": true
},
"text:5. final report & presentation": {
  "element_key": "text:5. final report & presentation",
  "element_type": "text",
  "label": "5. Final Report & Presentation",
  "bbox": [
    283.0,
```

```
        676.0,
        491.0,
        716.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 9,
    "last_seen_index": 9,
    "first_seen_at": "2026-05-02T04:58:47.026826+00:00",
    "last_seen_at": "2026-05-02T04:58:47.026826+00:00",
    "seen_count": 1,
    "miss_count": 9,
    "stale": true,
    "cached": true
},
"text:6. checklist documents": {
    "element_key": "text:6. checklist documents",
    "element_type": "text",
    "label": "6. CheckList Documents",
    "bbox": [
        267.0,
        724.0,
        419.0,
        764.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
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"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 9,
"last_seen_index": 9,
"first_seen_at": "2026-05-02T04:58:47.026826+00:00",
"last_seen_at": "2026-05-02T04:58:47.026826+00:00",
"seen_count": 1,
"miss_count": 9,
"stale": true,
"cached": true
},
"text:7. website": {
  "element_key": "text:7. website",
  "element_type": "text",
  "label": "7. Website",
  "bbox": [
    299.0,
    780.0,
    403.0,
    812.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 9,
  "last_seen_index": 9,
  "first_seen_at": "2026-05-02T04:58:47.026826+00:00",
  "last_seen_at": "2026-05-02T04:58:47.026826+00:00",
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```
    "seen_count": 1,
    "miss_count": 9,
    "stale": true,
    "cached": true
  },
  "text:8. log book": {
    "element_key": "text:8. log book",
    "element_type": "text",
    "label": "8. Log Book",
    "bbox": [
      299.0,
      820.0,
      443.0,
      852.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 9,
    "last_seen_index": 9,
    "first_seen_at": "2026-05-02T04:58:47.026826+00:00",
    "last_seen_at": "2026-05-02T04:58:47.026826+00:00",
    "seen_count": 1,
    "miss_count": 9,
    "stale": true,
    "cached": true
  },
  "text:project registration documents": {
```



```
"element_key": "text:project registration documents",
"element_type": "text",
"label": "Project Registration Documents",
"bbox": [
    291.0,
    860.0,
    555.0,
    900.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 9,
"last_seen_index": 9,
"first_seen_at": "2026-05-02T04:58:47.026826+00:00",
"last_seen_at": "2026-05-02T04:58:47.026826+00:00",
"seen_count": 1,
"miss_count": 9,
"stale": true,
"cached": true
},
"text:panel comments for the students.xlsm": {
    "element_key": "text:panel comments for the students.xlsm",
    "element_type": "text",
    "label": "Panel Comments for the Students.xlsm",
    "bbox": [
        291.0,
        908.0,
```

```
        627.0,  
        940.0  
    ],  
    "confidence": 0.4,  
    "source": "gemini_vlm",  
    "first_seen_frame": null,  
    "last_seen_frame": null,  
    "first_seen_index": 9,  
    "last_seen_index": 9,  
    "first_seen_at": "2026-05-02T04:58:47.026826+00:00",  
    "last_seen_at": "2026-05-02T04:58:47.026826+00:00",  
    "seen_count": 1,  
    "miss_count": 9,  
    "stale": true,  
    "cached": true  
},  
"icon:google chrome": {  
    "element_key": "icon:google chrome",  
    "element_type": "icon",  
    "label": "Google Chrome",  
    "bbox": [  
        148.0,  
        118.0,  
        252.0,  
        246.0  
    ],  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "first_seen_frame": null,
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"last_seen_frame": null,
"first_seen_index": 10,
"last_seen_index": 18,
"first_seen_at": "2026-05-02T04:59:38.576878+00:00",
"last_seen_at": "2026-05-02T05:06:12.825053+00:00",
"seen_count": 8,
"miss_count": 0,
"stale": false,
"cached": false
},
"icon:huiontablet": {
  "element_key": "icon:huiontablet",
  "element_type": "icon",
  "label": "HuionTablet",
  "bbox": [
    236.0,
    118.0,
    340.0,
    246.0
  ],
  "confidence": 0.95,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 10,
  "last_seen_index": 18,
  "first_seen_at": "2026-05-02T04:59:38.576878+00:00",
  "last_seen_at": "2026-05-02T05:06:12.825053+00:00",
  "seen_count": 7,
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    "miss_count": 0,
    "stale": false,
    "cached": false
  },
  "icon:wise memory optimizer": {
    "element_key": "icon:wise memory optimizer",
    "element_type": "icon",
    "label": "Wise Memory Optimizer",
    "bbox": [
      148.0,
      350.0,
      252.0,
      502.0
    ],
    "confidence": 0.95,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 11,
    "last_seen_index": 18,
    "first_seen_at": "2026-05-02T05:00:28.296423+00:00",
    "last_seen_at": "2026-05-02T05:06:12.825053+00:00",
    "seen_count": 3,
    "miss_count": 0,
    "stale": false,
    "cached": false
  },
  "icon:amazon workspaces": {
    "element_key": "icon:amazon workspaces",
```

```
"element_type": "icon",
"label": "Amazon WorkSpaces",
"bbox": [
    236.0,
    350.0,
    340.0,
    502.0
],
"confidence": 0.95,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 11,
"last_seen_index": 18,
"first_seen_at": "2026-05-02T05:00:28.296423+00:00",
"last_seen_at": "2026-05-02T05:06:12.825053+00:00",
"seen_count": 3,
"miss_count": 0,
"stale": false,
"cached": false
},
"icon:recycle bin": {
    "element_key": "icon:recycle bin",
    "element_type": "icon",
    "label": "Recycle Bin",
    "bbox": [
        149.0,
        542.0,
        253.0,
```

```
710.0

],

"confidence": 0.6232,

"source": "gemini_vlm",

"first_seen_frame": null,

"last_seen_frame": null,

"first_seen_index": 11,

"last_seen_index": 14,

"first_seen_at": "2026-05-02T05:00:28.296423+00:00",

"last_seen_at": "2026-05-02T05:02:52.171843+00:00",

"seen_count": 2,

"miss_count": 4,

"stale": true,

"cached": true

},

"icon:krita": {

  "element_key": "icon:krita",

  "element_type": "icon",

  "label": "Krita",

  "bbox": [

    237.0,

    542.0,

    341.0,

    710.0

  ],

  "confidence": 0.6232,

  "source": "gemini_vlm",

  "first_seen_frame": null,

  "last_seen_frame": null,
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"first_seen_at": "2026-05-02T05:00:28.296423+00:00",  
"last_seen_at": "2026-05-02T05:02:52.171843+00:00",  
"seen_count": 2,  
"miss_count": 4,  
"stale": true,  
"cached": true  
},  
"icon:nuwani - chrome": {  
  "element_key": "icon:nuwani - chrome",  
  "element_type": "icon",  
  "label": "Nuwani - Chrome",  
  "bbox": [  
    149.0,  
    750.0,  
    253.0,  
    918.0  
  ],  
  "confidence": 0.6232,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 11,  
  "last_seen_index": 14,  
  "first_seen_at": "2026-05-02T05:00:28.296423+00:00",  
  "last_seen_at": "2026-05-02T05:02:52.171843+00:00",  
  "seen_count": 2,  
  "miss_count": 4,
```

```
    "stale": true,
    "cached": true
  },
  "icon:docker desktop": {
    "element_key": "icon:docker desktop",
    "element_type": "icon",
    "label": "Docker Desktop",
    "bbox": [
      237.0,
      750.0,
      341.0,
      918.0
    ],
    "confidence": 0.6232,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 11,
    "last_seen_index": 14,
    "first_seen_at": "2026-05-02T05:00:28.296423+00:00",
    "last_seen_at": "2026-05-02T05:02:52.171843+00:00",
    "seen_count": 2,
    "miss_count": 4,
    "stale": true,
    "cached": true
  },
  "icon:antigravity": {
    "element_key": "icon:antigravity",
    "element_type": "icon",
```



```
"label": "Antigravity",
"bbox": [
  157.0,
  974.0,
  253.0,
  1012.0
],
"confidence": 0.5905,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 11,
"last_seen_index": 14,
"first_seen_at": "2026-05-02T05:00:28.296423+00:00",
"last_seen_at": "2026-05-02T05:02:52.171843+00:00",
"seen_count": 2,
"miss_count": 4,
"stale": true,
"cached": true
},
"icon:visual studio code": {
  "element_key": "icon:visual studio code",
  "element_type": "icon",
  "label": "Visual Studio Code",
  "bbox": [
    237.0,
    974.0,
    341.0,
    1006.0
  ]
}
```

```
],
  "confidence": 0.5905,
  "source": "gemini_vlm",
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  "last_seen_frame": null,
  "first_seen_index": 11,
  "last_seen_index": 14,
  "first_seen_at": "2026-05-02T05:00:28.296423+00:00",
  "last_seen_at": "2026-05-02T05:02:52.171843+00:00",
  "seen_count": 2,
  "miss_count": 4,
  "stale": true,
  "cached": true
},
"icon:postman": {
  "element_key": "icon:postman",
  "element_type": "icon",
  "label": "Postman",
  "bbox": [
    315.0,
    758.0,
    411.0,
    894.0
  ],
  "confidence": 0.4543,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 11,
```

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"last_seen_index": 11,  
"first_seen_at": "2026-05-02T05:00:28.296423+00:00",  
"last_seen_at": "2026-05-02T05:00:28.296423+00:00",  
"seen_count": 1,  
"miss_count": 7,  
"stale": true,  
"cached": true  
},  
"icon:loom": {  
  "element_key": "icon:loom",  
  "element_type": "icon",  
  "label": "Loom",  
  "bbox": [  
    395.0,  
    758.0,  
    491.0,  
    894.0  
  ],  
  "confidence": 0.4543,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 11,  
  "last_seen_index": 11,  
  "first_seen_at": "2026-05-02T05:00:28.296423+00:00",  
  "last_seen_at": "2026-05-02T05:00:28.296423+00:00",  
  "seen_count": 1,  
  "miss_count": 7,  
  "stale": true,
```

```
    "cached": true
  },
  "icon:vlc media player": {
    "element_key": "icon:vlc media player",
    "element_type": "icon",
    "label": "VLC media player",
    "bbox": [
      475.0,
      758.0,
      571.0,
      894.0
    ],
    "confidence": 0.4543,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 11,
    "last_seen_index": 11,
    "first_seen_at": "2026-05-02T05:00:28.296423+00:00",
    "last_seen_at": "2026-05-02T05:00:28.296423+00:00",
    "seen_count": 1,
    "miss_count": 7,
    "stale": true,
    "cached": true
  },
  "pane:google chrome window": {
    "element_key": "pane:google chrome window",
    "element_type": "pane",
    "label": "Google Chrome Window",
```

```
"bbox": [
  285.0,
  110.0,
  1597.0,
  1013.0
],
"confidence": 0.855,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 17,
"last_seen_index": 17,
"first_seen_at": "2026-05-02T05:05:25.437256+00:00",
"last_seen_at": "2026-05-02T05:05:25.437256+00:00",
"seen_count": 1,
"miss_count": 1,
"stale": false,
"cached": true
},
"label:google chrome": {
  "element_key": "label:google chrome",
  "element_type": "label",
  "label": "Google Chrome",
  "bbox": [
    413.0,
    134.0,
    613.0,
    190.0
  ],
```

```
"confidence": 0.81,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 17,
"last_seen_index": 17,
"first_seen_at": "2026-05-02T05:05:25.437256+00:00",
"last_seen_at": "2026-05-02T05:05:25.437256+00:00",
"seen_count": 1,
"miss_count": 1,
"stale": false,
"cached": true
},
"button:minimize": {
  "element_key": "button:minimize",
  "element_type": "button",
  "label": "Minimize",
  "bbox": [
    1317.0,
    134.0,
    1365.0,
    182.0
  ],
  "confidence": 0.81,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 17,
  "last_seen_index": 17,
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"first_seen_at": "2026-05-02T05:05:25.437256+00:00",
"last_seen_at": "2026-05-02T05:05:25.437256+00:00",
"seen_count": 1,
"miss_count": 1,
"stale": false,
"cached": true
},
"button:maximize": {
  "element_key": "button:maximize",
  "element_type": "button",
  "label": "Maximize",
  "bbox": [
    1357.0,
    134.0,
    1413.0,
    190.0
  ],
  "confidence": 0.81,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 17,
  "last_seen_index": 17,
  "first_seen_at": "2026-05-02T05:05:25.437256+00:00",
  "last_seen_at": "2026-05-02T05:05:25.437256+00:00",
  "seen_count": 1,
  "miss_count": 1,
  "stale": false,
  "cached": true
}
```

```
},
"button:close": {
  "element_key": "button:close",
  "element_type": "button",
  "label": "Close",
  "bbox": [
    1397.0,
    134.0,
    1453.0,
    190.0
  ],
  "confidence": 0.81,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 17,
  "last_seen_index": 17,
  "first_seen_at": "2026-05-02T05:05:25.437256+00:00",
  "last_seen_at": "2026-05-02T05:05:25.437256+00:00",
  "seen_count": 1,
  "miss_count": 1,
  "stale": false,
  "cached": true
},
"heading:who's using chrome?": {
  "element_key": "heading:who's using chrome?",
  "element_type": "heading",
  "label": "Who's using Chrome?",
  "bbox": [
```



```
      821.0,
      278.0,
      1085.0,
      326.0
    ],
    "confidence": 0.855,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 17,
    "last_seen_index": 17,
    "first_seen_at": "2026-05-02T05:05:25.437256+00:00",
    "last_seen_at": "2026-05-02T05:05:25.437256+00:00",
    "seen_count": 1,
    "miss_count": 1,
    "stale": false,
    "cached": true
  },
  "button:intrinsic-tech.net": {
    "element_key": "button:intrinsic-tech.net",
    "element_type": "button",
    "label": "Intrinsic-tech.net",
    "bbox": [
      597.0,
      414.0,
      789.0,
      630.0
    ],
    "confidence": 0.81,
```

```
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 17,
"last_seen_index": 17,
"first_seen_at": "2026-05-02T05:05:25.437256+00:00",
"last_seen_at": "2026-05-02T05:05:25.437256+00:00",
"seen_count": 1,
"miss_count": 1,
"stale": false,
"cached": true
},
"button:lammastide": {
  "element_key": "button:lammastide",
  "element_type": "button",
  "label": "lammastide",
  "bbox": [
    757.0,
    422.0,
    973.0,
    630.0
  ],
  "confidence": 0.81,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 17,
  "last_seen_index": 17,
  "first_seen_at": "2026-05-02T05:05:25.437256+00:00",
```

```
    "last_seen_at": "2026-05-02T05:05:25.437256+00:00",
    "seen_count": 1,
    "miss_count": 1,
    "stale": false,
    "cached": true
  },
  "button:lammastideit": {
    "element_key": "button:lammastideit",
    "element_type": "button",
    "label": "LammastideIT",
    "bbox": [
      941.0,
      422.0,
      1157.0,
      630.0
    ],
    "confidence": 0.81,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 17,
    "last_seen_index": 17,
    "first_seen_at": "2026-05-02T05:05:25.437256+00:00",
    "last_seen_at": "2026-05-02T05:05:25.437256+00:00",
    "seen_count": 1,
    "miss_count": 1,
    "stale": false,
    "cached": true
  },
```

```
"button:nuwani": {
  "element_key": "button:nuwani",
  "element_type": "button",
  "label": "Nuwani",
  "bbox": [
    1125.0,
    406.0,
    1293.0,
    606.0
  ],
  "confidence": 0.81,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 17,
  "last_seen_index": 17,
  "first_seen_at": "2026-05-02T05:05:25.437256+00:00",
  "last_seen_at": "2026-05-02T05:05:25.437256+00:00",
  "seen_count": 1,
  "miss_count": 1,
  "stale": false,
  "cached": true
},
"button:nuwani (photo)": {
  "element_key": "button:nuwani (photo)",
  "element_type": "button",
  "label": "Nuwani (photo)",
  "bbox": [
    597.0,
```

```
582.0,  
789.0,  
798.0  
],  
"confidence": 0.81,  
"source": "gemini_vlm",  
"first_seen_frame": null,  
"last_seen_frame": null,  
"first_seen_index": 17,  
"last_seen_index": 17,  
"first_seen_at": "2026-05-02T05:05:25.437256+00:00",  
"last_seen_at": "2026-05-02T05:05:25.437256+00:00",  
"seen_count": 1,  
"miss_count": 1,  
"stale": false,  
"cached": true  
},  
"button:add profile": {  
  "element_key": "button:add profile",  
  "element_type": "button",  
  "label": "Add Profile",  
  "bbox": [  
    757.0,  
    582.0,  
    973.0,  
    798.0  
  ],  
  "confidence": 0.81,  
  "source": "gemini_vlm",
```

```
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 17,
    "last_seen_index": 17,
    "first_seen_at": "2026-05-02T05:05:25.437256+00:00",
    "last_seen_at": "2026-05-02T05:05:25.437256+00:00",
    "seen_count": 1,
    "miss_count": 1,
    "stale": false,
    "cached": true
  }
},
"current_elements": [
  {
    "id": "1",
    "type": "tab",
    "label": "frontend",
    "description": "Browser tab labeled frontend",
    "state": "inactive",
    "dx": 148,
    "dy": 52,
    "frame_dx": 298,
    "frame_dy": 162,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
      150,
      134,
      446,
```

```
190
],
"session_memory_key": "tab:frontend",
"cached": false
},
{
  "id": "2",
  "type": "tab",
  "label": "GitHub",
  "description": "Browser tab labeled GitHub",
  "state": "inactive",
  "dx": 388,
  "dy": 52,
  "frame_dx": 538,
  "frame_dy": 162,
  "confidence": 0.95,
  "source": "gemini_vlm",
  "frame_bbox": [
    382,
    134,
    694,
    190
  ],
  "session_memory_key": "tab:github",
  "cached": false
}
],
"resolved_elements": [
  {
```

```
"id": "1",
"type": "tab",
"label": "frontend",
"description": "Browser tab labeled frontend",
"state": "inactive",
"dx": 148,
"dy": 52,
"frame_dx": 298,
"frame_dy": 162,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
  150,
  134,
  446,
  190
],
"session_memory_key": "tab:frontend",
"cached": false
},
{
  "id": "2",
  "type": "tab",
  "label": "GitHub",
  "description": "Browser tab labeled GitHub",
  "state": "inactive",
  "dx": 388,
  "dy": 52,
  "frame_dx": 538,
```



```
    "frame_dy": 162,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        382,
        134,
        694,
        190
    ],
    "session_memory_key": "tab:github",
    "cached": false
}
],
"screen_data": {
    "image": "E:\\sliit-
project\\ScreenAwareTaskAgent\\vision\\data\\sessions\\shot_1777697481_f67934\\pr
ocessing\\proc_1777697481_325aa1\\preprocessed_frames\\frame_1777697481_02a6a0.jp
g",
    "image_size": {
        "width": 1920,
        "height": 1080
    },
    "screen_bbox": [
        150,
        110,
        1726,
        1014
    ],
    "screen_size": {
        "width": 1576,
```

```
    "height": 904
  },
  "coordinate_system": "pixel",
  "elements": [
    {
      "id": "1",
      "type": "tab",
      "label": "frontend",
      "description": "Browser tab labeled frontend",
      "state": "inactive",
      "dx": 148,
      "dy": 52,
      "frame_dx": 298,
      "frame_dy": 162,
      "confidence": 0.95,
      "source": "gemini_vlm",
      "frame_bbox": [
        150,
        134,
        446,
        190
      ],
      "session_memory_key": "tab:frontend",
      "cached": false
    },
    {
      "id": "2",
      "type": "tab",
      "label": "GitHub",
```

```
    "description": "Browser tab labeled GitHub",
    "state": "inactive",
    "dx": 388,
    "dy": 52,
    "frame_dx": 538,
    "frame_dy": 162,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        382,
        134,
        694,
        190
    ],
    "session_memory_key": "tab:github",
    "cached": false
  }
],
"element_count": 2,
"current_elements": [
  {
    "id": "1",
    "type": "tab",
    "label": "frontend",
    "description": "Browser tab labeled frontend",
    "state": "inactive",
    "dx": 148,
    "dy": 52,
    "frame_dx": 298,
```

```
"frame_dy": 162,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    150,
    134,
    446,
    190
],
"session_memory_key": "tab:frontend",
"cached": false
},
{
    "id": "2",
    "type": "tab",
    "label": "GitHub",
    "description": "Browser tab labeled GitHub",
    "state": "inactive",
    "dx": 388,
    "dy": 52,
    "frame_dx": 538,
    "frame_dy": 162,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        382,
        134,
        694,
        190
    ]
}
```

```
    ],  
    "session_memory_key": "tab:github",  
    "cached": false  
  }  
],  
"resolved_elements": [  
  {  
    "id": "1",  
    "type": "tab",  
    "label": "frontend",  
    "description": "Browser tab labeled frontend",  
    "state": "inactive",  
    "dx": 148,  
    "dy": 52,  
    "frame_dx": 298,  
    "frame_dy": 162,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
      150,  
      134,  
      446,  
      190  
    ],  
    "session_memory_key": "tab:frontend",  
    "cached": false  
  },  
  {  
    "id": "2",
```

```
    "type": "tab",
    "label": "GitHub",
    "description": "Browser tab labeled GitHub",
    "state": "inactive",
    "dx": 388,
    "dy": 52,
    "frame_dx": 538,
    "frame_dy": 162,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        382,
        134,
        694,
        190
    ],
    "session_memory_key": "tab:github",
    "cached": false
}
],
"session_knowledge_base": {
    "session_id": "shot_1777697481_f67934",
    "started_at": "2026-05-02T04:51:21.788502+00:00",
    "updated_at": "2026-05-02T04:51:55.798150+00:00",
    "stale_after_misses": 3,
    "confidence_floor": 0.4,
    "decay_per_miss": 0.9,
    "frame_count": 1,
    "frame_name": "frame_1777697481_02a6a0.jpg",
```

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"frame_index": 0,
"image_size": {
  "width": 1920,
  "height": 1080
},
"known_elements": {
  "tab:frontend": {
    "element_key": "tab:frontend",
    "element_type": "tab",
    "label": "frontend",
    "bbox": [
      150.0,
      134.0,
      446.0,
      190.0
    ],
    "confidence": 0.95,
    "source": "gemini_vlm",
    "first_seen_frame": "frame_1777697481_02a6a0.jpg",
    "last_seen_frame": "frame_1777697481_02a6a0.jpg",
    "first_seen_at": "2026-05-02T04:51:55.798150+00:00",
    "last_seen_at": "2026-05-02T04:51:55.798150+00:00",
    "first_seen_index": 0,
    "last_seen_index": 0,
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false,
    "history": [
```

```
{
  "frame_name": "frame_1777697481_02a6a0.jpg",
  "frame_index": 0,
  "seen_at": "2026-05-02T04:51:55.798150+00:00",
  "confidence": 0.95
}
],
},
"tab:github": {
  "element_key": "tab:github",
  "element_type": "tab",
  "label": "GitHub",
  "bbox": [
    382.0,
    134.0,
    694.0,
    190.0
  ],
  "confidence": 0.95,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777697481_02a6a0.jpg",
  "last_seen_frame": "frame_1777697481_02a6a0.jpg",
  "first_seen_at": "2026-05-02T04:51:55.798150+00:00",
  "last_seen_at": "2026-05-02T04:51:55.798150+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
  "seen_count": 1,
  "miss_count": 0,
  "stale": false,
```



```
"cached": false,
"history": [
  {
    "frame_name": "frame_1777697481_02a6a0.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T04:51:55.798150+00:00",
    "confidence": 0.95
  }
]
},
"resolved_elements": [
  {
    "id": "1",
    "type": "tab",
    "label": "frontend",
    "description": "Browser tab labeled frontend",
    "state": "inactive",
    "dx": 148,
    "dy": 52,
    "frame_dx": 298,
    "frame_dy": 162,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
      150,
      134,
      446,
      190
```

```
    ],  
    "session_memory_key": "tab:frontend",  
    "cached": false  
  },  
  {  
    "id": "2",  
    "type": "tab",  
    "label": "GitHub",  
    "description": "Browser tab labeled GitHub",  
    "state": "inactive",  
    "dx": 388,  
    "dy": 52,  
    "frame_dx": 538,  
    "frame_dy": 162,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
      382,  
      134,  
      694,  
      190  
    ],  
    "session_memory_key": "tab:github",  
    "cached": false  
  }  
],  
"current_elements": [  
  {  
    "id": "1",
```

```
    "type": "tab",
    "label": "frontend",
    "description": "Browser tab labeled frontend",
    "state": "inactive",
    "dx": 148,
    "dy": 52,
    "frame_dx": 298,
    "frame_dy": 162,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        150,
        134,
        446,
        190
    ],
    "session_memory_key": "tab:frontend",
    "cached": false
},
{
    "id": "2",
    "type": "tab",
    "label": "GitHub",
    "description": "Browser tab labeled GitHub",
    "state": "inactive",
    "dx": 388,
    "dy": 52,
    "frame_dx": 538,
    "frame_dy": 162,
```

```
        "confidence": 0.95,
        "source": "gemini_vlm",
        "frame_bbox": [
            382,
            134,
            694,
            190
        ],
        "session_memory_key": "tab:github",
        "cached": false
    }
]
}
}
}
},
"todo": {
    "steps": [
        {
            "id": 1,
            "action": "Move mouse to the 'GitHub' tab at (538, 162) and click to switch to the GitHub page.",
            "target": "GitHub tab",
            "expected_result": "The browser switches focus to the GitHub tab, revealing the GitHub interface.",
            "needs_user_data": false,
            "user_data_field": null,
            "status": "failed",
            "_coord_fix_tried": true
        },
    ],
}
```

```
{
  "id": 2,
  "action": "Locate the profile icon in the top right corner of the GitHub
page, click it to open the dropdown menu, and then click on 'Your
repositories'.",
  "target": "GitHub profile icon and 'Your repositories' menu item",
  "expected_result": "The browser navigates to the user's repositories
section.",
  "needs_user_data": false,
  "user_data_field": null,
  "status": "failed"
}
],
"notes": "The current screen shows the browser tabs. The first step is to
switch to the GitHub tab. Once the GitHub page loads, the profile and repository
navigation can be performed.",
"estimated_complexity": "simple"
},
"step_plans": [
  {
    "step_id": 1,
    "action": "Move mouse to the 'GitHub' tab at (538, 162) and click to switch
to the GitHub page.",
    "target": "GitHub tab",
    "expected_result": "The browser switches focus to the GitHub tab, revealing
the GitHub interface.",
    "hid_commands": [
      {
        "cmd": "mouse_move",
        "dx": 538,
        "dy": 162
      },
    ],
  },
]
```

```
{
  "cmd": "mouse_click",
  "button": "left"
}
],
  "reasoning": "Moving the cursor to the GitHub tab at the specified
coordinates (538, 162) and clicking to switch the browser focus to the GitHub
page as per the current step instructions."
},
{
  "step_id": 2,
  "action": "Locate the profile icon in the top right corner of the GitHub
page, click it to open the dropdown menu, and then click on 'Your
repositories'.",
  "target": "GitHub profile icon and 'Your repositories' menu item",
  "expected_result": "The browser navigates to the user's repositories
section.",
  "hid_commands": [
    {
      "cmd": "mouse_move",
      "meta": {
        "commandId": "fe3b6bcd-d2e4-4414-97a8-c3ffa9c5561b"
      },
      "dx": 208,
      "dy": 210,
      "smooth": true
    },
    {
      "cmd": "mouse_click",
      "meta": {
        "commandId": "88c0f82f-eb7c-425b-8ee5-e955ecffbd01"
```

```
    },
    "button": "left"
  }
],
"reasoning": "Fallback generated 2 commands from detected elements"
}
],
"step_results": [
  {
    "step_id": 1,
    "action": "Move mouse to the 'GitHub' tab at (538, 162) and click to switch to the GitHub page.",
    "attempt": 3,
    "exec_status": "success",
    "eval_status": "retry",
    "confidence": 1.0,
    "reason": "The screen currently displays the desktop icons instead of the expected GitHub interface in a browser. The action to switch tabs did not result in the browser being the active window.",
    "final_status": "executing",
    "timestamp": "2026-05-02T05:00:32.381029+00:00"
  },
  {
    "step_id": 2,
    "action": "Locate the profile icon in the top right corner of the GitHub page, click it to open the dropdown menu, and then click on 'Your repositories'.",
    "attempt": 3,
    "exec_status": "success",
    "eval_status": "retry",
    "confidence": 0.0,
```

```

"reason": "Evaluation error: No valid JSON found in LLM response: '{\\n
\\\"status\\\": \\\"'\",

    \"final_status\": \"executing\",

    \"timestamp\": \"2026-05-02T05:05:31.433807+00:00\"

}

],

\"final_report\": {

    \"success\": false,

    \"summary\": \"The task failed because the browser was stuck on the Chrome
profile selection screen, preventing access to the GitHub tab.\",

    \"message\": \"The task could not be completed. The browser is currently
displaying the 'Who's using Chrome?' profile selection screen, which blocked
access to the GitHub tab and subsequent profile navigation.\\n\\nStep results:\\n-
Step 1: Failed to click the GitHub tab (element not found or obscured by the
profile picker).\\n- Step 2: Failed to navigate to repositories (GitHub page was
never reached).\",

    \"steps_completed\": 0,

    \"steps_failed\": 2,

    \"issues\": [

        \"Browser is on the profile selection screen\",

        \"GitHub tab is not visible or interactive in the current state\"

    ],

    \"recommendations\": [

        \"Select a Chrome profile manually or provide a command to select one to
proceed.\",

        \"Ensure the browser is already logged into GitHub and on the correct page
before starting the task.\"

    ]

},

\"session_memory\": {

    \"run_id\": \"caa1e9d9-8f76-42da-b95c-4cc77b1cd29c\",

    \"user_task\": \"Find and click the Github icon on this browser and click on the
github profile and go to the repositories section\",

```



```
"started_at": "2026-05-02T04:51:15.350411+00:00",
"updated_at": "2026-05-02T05:06:12.825053+00:00",
"stale_after_misses": 2,
"confidence_floor": 0.4,
"decay_per_miss": 0.9,
"frame_index": 19,
"known_elements": {
  "tab:frontend": {
    "element_key": "tab:frontend",
    "element_type": "tab",
    "label": "frontend",
    "bbox": [
      150.0,
      133.0,
      382.0,
      197.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 0,
    "last_seen_index": 1,
    "first_seen_at": "2026-05-02T04:51:55.824539+00:00",
    "last_seen_at": "2026-05-02T04:52:42.298731+00:00",
    "seen_count": 2,
    "miss_count": 17,
    "stale": true,
    "cached": true
  }
}
```

```
},
"tab:github": {
  "element_key": "tab:github",
  "element_type": "tab",
  "label": "GitHub",
  "bbox": [
    326.0,
    133.0,
    566.0,
    197.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 1,
  "first_seen_at": "2026-05-02T04:51:55.824539+00:00",
  "last_seen_at": "2026-05-02T04:52:42.298731+00:00",
  "seen_count": 2,
  "miss_count": 17,
  "stale": true,
  "cached": true
},
"tab:new tab": {
  "element_key": "tab:new tab",
  "element_type": "tab",
  "label": "New tab",
  "bbox": [
```

```
    518.0,  
    133.0,  
    750.0,  
    197.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 1,  
  "last_seen_index": 1,  
  "first_seen_at": "2026-05-02T04:52:42.298731+00:00",  
  "last_seen_at": "2026-05-02T04:52:42.298731+00:00",  
  "seen_count": 1,  
  "miss_count": 17,  
  "stale": true,  
  "cached": true  
},  
"button:new tab": {  
  "element_key": "button:new tab",  
  "element_type": "button",  
  "label": "New tab",  
  "bbox": [  
    718.0,  
    141.0,  
    766.0,  
    189.0  
  ],  
  "confidence": 0.4,
```

```
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 1,
"last_seen_index": 1,
"first_seen_at": "2026-05-02T04:52:42.298731+00:00",
"last_seen_at": "2026-05-02T04:52:42.298731+00:00",
"seen_count": 1,
"miss_count": 17,
"stale": true,
"cached": true
},
"button:ask gemini": {
  "element_key": "button:ask gemini",
  "element_type": "button",
  "label": "Ask Gemini",
  "bbox": [
    1134.0,
    157.0,
    1254.0,
    189.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
  "last_seen_index": 1,
  "first_seen_at": "2026-05-02T04:52:42.298731+00:00",
```

```
    "last_seen_at": "2026-05-02T04:52:42.298731+00:00",
    "seen_count": 1,
    "miss_count": 17,
    "stale": true,
    "cached": true
  },
  "button:back": {
    "element_key": "button:back",
    "element_type": "button",
    "label": "Back",
    "bbox": [
      166.0,
      189.0,
      214.0,
      253.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 1,
    "last_seen_index": 1,
    "first_seen_at": "2026-05-02T04:52:42.298731+00:00",
    "last_seen_at": "2026-05-02T04:52:42.298731+00:00",
    "seen_count": 1,
    "miss_count": 17,
    "stale": true,
    "cached": true
  },
}
```

```
"button:forward": {
  "element_key": "button:forward",
  "element_type": "button",
  "label": "Forward",
  "bbox": [
    198.0,
    189.0,
    246.0,
    245.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
  "last_seen_index": 1,
  "first_seen_at": "2026-05-02T04:52:42.298731+00:00",
  "last_seen_at": "2026-05-02T04:52:42.298731+00:00",
  "seen_count": 1,
  "miss_count": 17,
  "stale": true,
  "cached": true
},
"button:refresh": {
  "element_key": "button:refresh",
  "element_type": "button",
  "label": "Refresh",
  "bbox": [
    230.0,
```

```
    189.0,  
    278.0,  
    253.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 1,  
  "last_seen_index": 1,  
  "first_seen_at": "2026-05-02T04:52:42.298731+00:00",  
  "last_seen_at": "2026-05-02T04:52:42.298731+00:00",  
  "seen_count": 1,  
  "miss_count": 17,  
  "stale": true,  
  "cached": true  
},  
"input:search google or type a url": {  
  "element_key": "input:search google or type a url",  
  "element_type": "input",  
  "label": "Search Google or type a URL",  
  "bbox": [  
    150.0,  
    189.0,  
    1350.0,  
    261.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",
```

```
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 1,
"last_seen_index": 1,
"first_seen_at": "2026-05-02T04:52:42.298731+00:00",
"last_seen_at": "2026-05-02T04:52:42.298731+00:00",
"seen_count": 1,
"miss_count": 17,
"stale": true,
"cached": true
},
"button:extensions": {
  "element_key": "button:extensions",
  "element_type": "button",
  "label": "Extensions",
  "bbox": [
    1206.0,
    205.0,
    1262.0,
    261.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
  "last_seen_index": 1,
  "first_seen_at": "2026-05-02T04:52:42.298731+00:00",
  "last_seen_at": "2026-05-02T04:52:42.298731+00:00",
```



```
    "seen_count": 1,
    "miss_count": 17,
    "stale": true,
    "cached": true
  },
  "button:side panel": {
    "element_key": "button:side panel",
    "element_type": "button",
    "label": "Side panel",
    "bbox": [
      1246.0,
      205.0,
      1302.0,
      253.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 1,
    "last_seen_index": 1,
    "first_seen_at": "2026-05-02T04:52:42.298731+00:00",
    "last_seen_at": "2026-05-02T04:52:42.298731+00:00",
    "seen_count": 1,
    "miss_count": 17,
    "stale": true,
    "cached": true
  },
  "button:profile": {
```

```
"element_key": "button:profile",
"element_type": "button",
"label": "Profile",
"bbox": [
  1294.0,
  205.0,
  1350.0,
  253.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 1,
"last_seen_index": 1,
"first_seen_at": "2026-05-02T04:52:42.298731+00:00",
"last_seen_at": "2026-05-02T04:52:42.298731+00:00",
"seen_count": 1,
"miss_count": 17,
"stale": true,
"cached": true
},
"link:dashboard": {
  "element_key": "link:dashboard",
  "element_type": "link",
  "label": "Dashboard",
  "bbox": [
    198.0,
    261.0,
```

```
    334.0,  
    285.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 1,  
  "last_seen_index": 1,  
  "first_seen_at": "2026-05-02T04:52:42.298731+00:00",  
  "last_seen_at": "2026-05-02T04:52:42.298731+00:00",  
  "seen_count": 1,  
  "miss_count": 17,  
  "stale": true,  
  "cached": true  
},  
"link:inbox": {  
  "element_key": "link:inbox",  
  "element_type": "link",  
  "label": "Inbox",  
  "bbox": [  
    326.0,  
    261.0,  
    406.0,  
    277.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,
```

```
"last_seen_frame": null,
"first_seen_index": 1,
"last_seen_index": 1,
"first_seen_at": "2026-05-02T04:52:42.298731+00:00",
"last_seen_at": "2026-05-02T04:52:42.298731+00:00",
"seen_count": 1,
"miss_count": 17,
"stale": true,
"cached": true
},
"link:nexmart": {
  "element_key": "link:nexmart",
  "element_type": "link",
  "label": "NexMart",
  "bbox": [
    790.0,
    725.0,
    862.0,
    829.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
  "last_seen_index": 1,
  "first_seen_at": "2026-05-02T04:52:42.298731+00:00",
  "last_seen_at": "2026-05-02T04:52:42.298731+00:00",
  "seen_count": 2,
```

```
"miss_count": 17,  
  "stale": true,  
  "cached": true  
},  
"button:all bookmarks": {  
  "element_key": "button:all bookmarks",  
  "element_type": "button",  
  "label": "All Bookmarks",  
  "bbox": [  
    1262.0,  
    253.0,  
    1350.0,  
    285.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 1,  
  "last_seen_index": 1,  
  "first_seen_at": "2026-05-02T04:52:42.298731+00:00",  
  "last_seen_at": "2026-05-02T04:52:42.298731+00:00",  
  "seen_count": 1,  
  "miss_count": 17,  
  "stale": true,  
  "cached": true  
},  
"image:google logo": {  
  "element_key": "image:google logo",
```

```
"element_type": "image",
"label": "Google Logo",
"bbox": [
    646.0,
    453.0,
    902.0,
    589.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 1,
"last_seen_index": 1,
"first_seen_at": "2026-05-02T04:52:42.298731+00:00",
"last_seen_at": "2026-05-02T04:52:42.298731+00:00",
"seen_count": 1,
"miss_count": 17,
"stale": true,
"cached": true
},
"input:search bar": {
    "element_key": "input:search bar",
    "element_type": "input",
    "label": "Search bar",
    "bbox": [
        454.0,
        597.0,
        1110.0,
```

```
        677.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 1,
    "last_seen_index": 1,
    "first_seen_at": "2026-05-02T04:52:42.298731+00:00",
    "last_seen_at": "2026-05-02T04:52:42.298731+00:00",
    "seen_count": 1,
    "miss_count": 17,
    "stale": true,
    "cached": true
},
"link:whatsapp": {
    "element_key": "link:whatsapp",
    "element_type": "link",
    "label": "WhatsApp",
    "bbox": [
        398.0,
        725.0,
        462.0,
        797.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
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"last_seen_index": 1,
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"last_seen_at": "2026-05-02T04:52:42.298731+00:00",
"seen_count": 1,
"miss_count": 17,
"stale": true,
"cached": true
},
"link:github": {
  "element_key": "link:github",
  "element_type": "link",
  "label": "GitHub",
  "bbox": [
    470.0,
    725.0,
    542.0,
    829.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
  "last_seen_index": 1,
  "first_seen_at": "2026-05-02T04:52:42.298731+00:00",
  "last_seen_at": "2026-05-02T04:52:42.298731+00:00",
  "seen_count": 1,
  "miss_count": 17,
```



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    "stale": true,
    "cached": true
  },
  "link:frontend": {
    "element_key": "link:frontend",
    "element_type": "link",
    "label": "frontend",
    "bbox": [
      550.0,
      725.0,
      622.0,
      829.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 1,
    "last_seen_index": 1,
    "first_seen_at": "2026-05-02T04:52:42.298731+00:00",
    "last_seen_at": "2026-05-02T04:52:42.298731+00:00",
    "seen_count": 1,
    "miss_count": 17,
    "stale": true,
    "cached": true
  },
  "link:vision service": {
    "element_key": "link:vision service",
    "element_type": "link",
```

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"label": "Vision Service",
"bbox": [
  630.0,
  725.0,
  702.0,
  829.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 1,
"last_seen_index": 1,
"first_seen_at": "2026-05-02T04:52:42.298731+00:00",
"last_seen_at": "2026-05-02T04:52:42.298731+00:00",
"seen_count": 1,
"miss_count": 17,
"stale": true,
"cached": true
},
"link:chatgpt": {
  "element_key": "link:chatgpt",
  "element_type": "link",
  "label": "ChatGPT",
  "bbox": [
    710.0,
    725.0,
    782.0,
    829.0
  ]
}
```

```
],
  "confidence": 0.4,
  "source": "gemini_vlm",
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  "last_seen_frame": null,
  "first_seen_index": 1,
  "last_seen_index": 1,
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  "last_seen_at": "2026-05-02T04:52:42.298731+00:00",
  "seen_count": 1,
  "miss_count": 17,
  "stale": true,
  "cached": true
},
"link:instances": {
  "element_key": "link:instances",
  "element_type": "link",
  "label": "Instances",
  "bbox": [
    870.0,
    725.0,
    942.0,
    829.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
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"last_seen_index": 1,
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"last_seen_at": "2026-05-02T04:52:42.298731+00:00",
"seen_count": 1,
"miss_count": 17,
"stale": true,
"cached": true
},
"link:youtube": {
  "element_key": "link:youtube",
  "element_type": "link",
  "label": "YouTube",
  "bbox": [
    950.0,
    725.0,
    1022.0,
    829.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
  "last_seen_index": 1,
  "first_seen_at": "2026-05-02T04:52:42.298731+00:00",
  "last_seen_at": "2026-05-02T04:52:42.298731+00:00",
  "seen_count": 1,
  "miss_count": 17,
  "stale": true,
```

```
    "cached": true
  },
  "link:course": {
    "element_key": "link:course",
    "element_type": "link",
    "label": "Course",
    "bbox": [
      1030.0,
      725.0,
      1102.0,
      829.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 1,
    "last_seen_index": 1,
    "first_seen_at": "2026-05-02T04:52:42.298731+00:00",
    "last_seen_at": "2026-05-02T04:52:42.298731+00:00",
    "seen_count": 1,
    "miss_count": 17,
    "stale": true,
    "cached": true
  },
  "button:add shortcut": {
    "element_key": "button:add shortcut",
    "element_type": "button",
    "label": "Add shortcut",
```

```
"bbox": [
  1110.0,
  725.0,
  1182.0,
  829.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 1,
"last_seen_index": 1,
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"last_seen_at": "2026-05-02T04:52:42.298731+00:00",
"seen_count": 1,
"miss_count": 17,
"stale": true,
"cached": true
},
"button:customise chrome": {
  "element_key": "button:customise chrome",
  "element_type": "button",
  "label": "Customise Chrome",
  "bbox": [
    1214.0,
    909.0,
    1358.0,
    965.0
  ],
```

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"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 1,
"last_seen_index": 1,
"first_seen_at": "2026-05-02T04:52:42.298731+00:00",
"last_seen_at": "2026-05-02T04:52:42.298731+00:00",
"seen_count": 1,
"miss_count": 17,
"stale": true,
"cached": true
},
"button:start": {
  "element_key": "button:start",
  "element_type": "button",
  "label": "Start",
  "bbox": [
    163.0,
    924.0,
    203.0,
    959.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
  "last_seen_index": 9,
```

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"first_seen_at": "2026-05-02T04:52:42.298731+00:00",
"last_seen_at": "2026-05-02T04:58:47.026826+00:00",
"seen_count": 2,
"miss_count": 9,
"stale": true,
"cached": true
},
"input:search": {
  "element_key": "input:search",
  "element_type": "input",
  "label": "Search",
  "bbox": [
    502.0,
    965.0,
    718.0,
    1005.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
  "last_seen_index": 2,
  "first_seen_at": "2026-05-02T04:52:42.298731+00:00",
  "last_seen_at": "2026-05-02T04:53:25.020574+00:00",
  "seen_count": 2,
  "miss_count": 16,
  "stale": true,
  "cached": true
}
```



```
},  
"button:task view": {  
  "element_key": "button:task view",  
  "element_type": "button",  
  "label": "Task View",  
  "bbox": [  
    694.0,  
    965.0,  
    734.0,  
    997.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 1,  
  "last_seen_index": 1,  
  "first_seen_at": "2026-05-02T04:52:42.298731+00:00",  
  "last_seen_at": "2026-05-02T04:52:42.298731+00:00",  
  "seen_count": 1,  
  "miss_count": 17,  
  "stale": true,  
  "cached": true  
},  
"button:file explorer": {  
  "element_key": "button:file explorer",  
  "element_type": "button",  
  "label": "File Explorer",  
  "bbox": [  

```

```
    726.0,  
    965.0,  
    774.0,  
    997.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 1,  
  "last_seen_index": 1,  
  "first_seen_at": "2026-05-02T04:52:42.298731+00:00",  
  "last_seen_at": "2026-05-02T04:52:42.298731+00:00",  
  "seen_count": 1,  
  "miss_count": 17,  
  "stale": true,  
  "cached": true  
},  
"button:chrome": {  
  "element_key": "button:chrome",  
  "element_type": "button",  
  "label": "Chrome",  
  "bbox": [  
    766.0,  
    965.0,  
    814.0,  
    997.0  
  ],  
  "confidence": 0.4,
```

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"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 1,
"last_seen_index": 1,
"first_seen_at": "2026-05-02T04:52:42.298731+00:00",
"last_seen_at": "2026-05-02T04:52:42.298731+00:00",
"seen_count": 1,
"miss_count": 17,
"stale": true,
"cached": true
},
"button:app launcher": {
  "element_key": "button:app launcher",
  "element_type": "button",
  "label": "App launcher",
  "bbox": [
    162.0,
    211.0,
    210.0,
    259.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 2,
  "last_seen_index": 8,
  "first_seen_at": "2026-05-02T04:53:25.020574+00:00",
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```
    "last_seen_at": "2026-05-02T04:58:02.406820+00:00",
    "seen_count": 2,
    "miss_count": 10,
    "stale": true,
    "cached": true
  },
  "pane:sidebar": {
    "element_key": "pane:sidebar",
    "element_type": "pane",
    "label": "Sidebar",
    "bbox": [
      156.0,
      250.0,
      212.0,
      958.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 3,
    "last_seen_index": 3,
    "first_seen_at": "2026-05-02T04:54:09.143401+00:00",
    "last_seen_at": "2026-05-02T04:54:09.143401+00:00",
    "seen_count": 1,
    "miss_count": 15,
    "stale": true,
    "cached": true
  },
}
```

```
"heading:cdapsubmissioncloud": {
  "element_key": "heading:cdapsubmissioncloud",
  "element_type": "heading",
  "label": "CDAPSubmissionCloud",
  "bbox": [
    266.0,
    275.0,
    514.0,
    331.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 3,
  "last_seen_index": 8,
  "first_seen_at": "2026-05-02T04:54:09.143401+00:00",
  "last_seen_at": "2026-05-02T04:58:02.406820+00:00",
  "seen_count": 2,
  "miss_count": 10,
  "stale": true,
  "cached": true
},
"button:create or upload": {
  "element_key": "button:create or upload",
  "element_type": "button",
  "label": "Create or upload",
  "bbox": [
    1539.0,
```

```
    364.0,  
    1707.0,  
    412.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 3,  
  "last_seen_index": 9,  
  "first_seen_at": "2026-05-02T04:54:09.143401+00:00",  
  "last_seen_at": "2026-05-02T04:58:47.026826+00:00",  
  "seen_count": 3,  
  "miss_count": 9,  
  "stale": true,  
  "cached": true  
},  
"pane:sign in dialog": {  
  "element_key": "pane:sign in dialog",  
  "element_type": "pane",  
  "label": "Sign in dialog",  
  "bbox": [  
    548.0,  
    634.0,  
    1004.0,  
    958.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",
```

```
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 3,
"last_seen_index": 3,
"first_seen_at": "2026-05-02T04:54:09.143401+00:00",
"last_seen_at": "2026-05-02T04:54:09.143401+00:00",
"seen_count": 1,
"miss_count": 15,
"stale": true,
"cached": true
},
"button:sign in": {
  "element_key": "button:sign in",
  "element_type": "button",
  "label": "Sign In",
  "bbox": [
    979.0,
    604.0,
    1059.0,
    644.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 3,
  "last_seen_index": 9,
  "first_seen_at": "2026-05-02T04:54:09.143401+00:00",
  "last_seen_at": "2026-05-02T04:58:47.026826+00:00",
```

```
"seen_count": 2,
"miss_count": 9,
"stale": true,
"cached": true
},
"pane:onboarding tooltip": {
  "element_key": "pane:onboarding tooltip",
  "element_type": "pane",
  "label": "Onboarding tooltip",
  "bbox": [
    980.0,
    554.0,
    1308.0,
    850.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 3,
  "last_seen_index": 3,
  "first_seen_at": "2026-05-02T04:54:09.143401+00:00",
  "last_seen_at": "2026-05-02T04:54:09.143401+00:00",
  "seen_count": 1,
  "miss_count": 15,
  "stale": true,
  "cached": true
},
"button:next": {
```



```
"element_key": "button:next",
"element_type": "button",
"label": "Next",
"bbox": [
    1172.0,
    730.0,
    1212.0,
    778.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 3,
"last_seen_index": 3,
"first_seen_at": "2026-05-02T04:54:09.143401+00:00",
"last_seen_at": "2026-05-02T04:54:09.143401+00:00",
"seen_count": 1,
"miss_count": 15,
"stale": true,
"cached": true
},
"pane:taskbar": {
    "element_key": "pane:taskbar",
    "element_type": "pane",
    "label": "Taskbar",
    "bbox": [
        156.0,
        922.0,
```

```
    1716.0,
    958.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 3,
  "last_seen_index": 3,
  "first_seen_at": "2026-05-02T04:54:09.143401+00:00",
  "last_seen_at": "2026-05-02T04:54:09.143401+00:00",
  "seen_count": 1,
  "miss_count": 15,
  "stale": true,
  "cached": true
},
"item:1. project proposal": {
  "element_key": "item:1. project proposal",
  "element_type": "item",
  "label": "1. Project Proposal",
  "bbox": [
    236.0,
    674.0,
    492.0,
    714.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
```

```
"last_seen_frame": null,
"first_seen_index": 3,
"last_seen_index": 3,
"first_seen_at": "2026-05-02T04:54:09.143401+00:00",
"last_seen_at": "2026-05-02T04:54:09.143401+00:00",
"seen_count": 1,
"miss_count": 15,
"stale": true,
"cached": true
},
"item:2. progress presentation - 1": {
  "element_key": "item:2. progress presentation - 1",
  "element_type": "item",
  "label": "2. Progress Presentation - 1",
  "bbox": [
    236.0,
    706.0,
    484.0,
    746.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 3,
  "last_seen_index": 3,
  "first_seen_at": "2026-05-02T04:54:09.143401+00:00",
  "last_seen_at": "2026-05-02T04:54:09.143401+00:00",
  "seen_count": 1,
```

```
"miss_count": 15,
"stale": true,
"cached": true
},
"item:3. progress presentation - 2": {
  "element_key": "item:3. progress presentation - 2",
  "element_type": "item",
  "label": "3. Progress Presentation - 2",
  "bbox": [
    236.0,
    738.0,
    476.0,
    778.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 3,
  "last_seen_index": 3,
  "first_seen_at": "2026-05-02T04:54:09.143401+00:00",
  "last_seen_at": "2026-05-02T04:54:09.143401+00:00",
  "seen_count": 1,
  "miss_count": 15,
  "stale": true,
  "cached": true
},
"item:4. research paper": {
  "element_key": "item:4. research paper",
```

```
"element_type": "item",
"label": "4. Research Paper",
"bbox": [
  236.0,
  770.0,
  492.0,
  802.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 3,
"last_seen_index": 3,
"first_seen_at": "2026-05-02T04:54:09.143401+00:00",
"last_seen_at": "2026-05-02T04:54:09.143401+00:00",
"seen_count": 1,
"miss_count": 15,
"stale": true,
"cached": true
},
"item:5. final report & presentation": {
  "element_key": "item:5. final report & presentation",
  "element_type": "item",
  "label": "5. Final Report & Presentation",
  "bbox": [
    236.0,
    794.0,
    492.0,
```

```
      834.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 3,
    "last_seen_index": 3,
    "first_seen_at": "2026-05-02T04:54:09.143401+00:00",
    "last_seen_at": "2026-05-02T04:54:09.143401+00:00",
    "seen_count": 1,
    "miss_count": 15,
    "stale": true,
    "cached": true
  },
  "item:6. checklist documents": {
    "element_key": "item:6. checklist documents",
    "element_type": "item",
    "label": "6. CheckList Documents",
    "bbox": [
      236.0,
      826.0,
      492.0,
      866.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
```

```
"first_seen_index": 3,
"last_seen_index": 3,
"first_seen_at": "2026-05-02T04:54:09.143401+00:00",
"last_seen_at": "2026-05-02T04:54:09.143401+00:00",
"seen_count": 1,
"miss_count": 15,
"stale": true,
"cached": true
},
"item:7. website": {
  "element_key": "item:7. website",
  "element_type": "item",
  "label": "7. Website",
  "bbox": [
    244.0,
    858.0,
    492.0,
    898.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 3,
  "last_seen_index": 3,
  "first_seen_at": "2026-05-02T04:54:09.143401+00:00",
  "last_seen_at": "2026-05-02T04:54:09.143401+00:00",
  "seen_count": 1,
  "miss_count": 15,
```

```
    "stale": true,
    "cached": true
  },
  "item:8. log book": {
    "element_key": "item:8. log book",
    "element_type": "item",
    "label": "8. Log Book",
    "bbox": [
      236.0,
      890.0,
      492.0,
      930.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 3,
    "last_seen_index": 3,
    "first_seen_at": "2026-05-02T04:54:09.143401+00:00",
    "last_seen_at": "2026-05-02T04:54:09.143401+00:00",
    "seen_count": 1,
    "miss_count": 15,
    "stale": true,
    "cached": true
  },
  "item:project registration documents": {
    "element_key": "item:project registration documents",
    "element_type": "item",
```



```
"label": "Project Registration Documents",
"bbox": [
  236.0,
  922.0,
  492.0,
  954.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 3,
"last_seen_index": 3,
"first_seen_at": "2026-05-02T04:54:09.143401+00:00",
"last_seen_at": "2026-05-02T04:54:09.143401+00:00",
"seen_count": 1,
"miss_count": 15,
"stale": true,
"cached": true
},
"button:share": {
  "element_key": "button:share",
  "element_type": "button",
  "label": "Share",
  "bbox": [
    546.0,
    371.0,
    602.0,
    403.0
```

```
],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 3,
  "last_seen_index": 9,
  "first_seen_at": "2026-05-02T04:54:09.143401+00:00",
  "last_seen_at": "2026-05-02T04:58:47.026826+00:00",
  "seen_count": 3,
  "miss_count": 9,
  "stale": true,
  "cached": true
},
"button:copy link": {
  "element_key": "button:copy link",
  "element_type": "button",
  "label": "Copy link",
  "bbox": [
    602.0,
    371.0,
    674.0,
    403.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 3,
```

```
"last_seen_index": 9,  
"first_seen_at": "2026-05-02T04:54:09.143401+00:00",  
"last_seen_at": "2026-05-02T04:58:47.026826+00:00",  
"seen_count": 3,  
"miss_count": 9,  
"stale": true,  
"cached": true  
},  
"icon:home": {  
  "element_key": "icon:home",  
  "element_type": "icon",  
  "label": "Home",  
  "bbox": [  
    162.0,  
    299.0,  
    202.0,  
    347.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 4,  
  "last_seen_index": 8,  
  "first_seen_at": "2026-05-02T04:54:57.496116+00:00",  
  "last_seen_at": "2026-05-02T04:58:02.406820+00:00",  
  "seen_count": 3,  
  "miss_count": 10,  
  "stale": true,
```

```
    "cached": true
  },
  "icon:globe": {
    "element_key": "icon:globe",
    "element_type": "icon",
    "label": "Globe",
    "bbox": [
      164.0,
      345.0,
      204.0,
      385.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 4,
    "last_seen_index": 5,
    "first_seen_at": "2026-05-02T04:54:57.496116+00:00",
    "last_seen_at": "2026-05-02T04:55:40.813387+00:00",
    "seen_count": 2,
    "miss_count": 13,
    "stale": true,
    "cached": true
  },
  "taskbar:taskbar": {
    "element_key": "taskbar:taskbar",
    "element_type": "taskbar",
    "label": "Taskbar",
```

```
"bbox": [
  164.0,
  932.0,
  1717.0,
  959.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 6,
"last_seen_index": 6,
"first_seen_at": "2026-05-02T04:56:31.874633+00:00",
"last_seen_at": "2026-05-02T04:56:31.874633+00:00",
"seen_count": 1,
"miss_count": 12,
"stale": true,
"cached": true
},
"text:sharepoint": {
  "element_key": "text:sharepoint",
  "element_type": "text",
  "label": "SharePoint",
  "bbox": [
    155.0,
    204.0,
    307.0,
    268.0
  ],
```

```
"confidence": 0.4,  
"source": "gemini_vlm",  
"first_seen_frame": null,  
"last_seen_frame": null,  
"first_seen_index": 7,  
"last_seen_index": 9,  
"first_seen_at": "2026-05-02T04:57:20.678535+00:00",  
"last_seen_at": "2026-05-02T04:58:47.026826+00:00",  
"seen_count": 2,  
"miss_count": 9,  
"stale": true,  
"cached": true  
},  
"input:search this library": {  
  "element_key": "input:search this library",  
  "element_type": "input",  
  "label": "Search this library",  
  "bbox": [  
    571.0,  
    212.0,  
    1003.0,  
    268.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 7,  
  "last_seen_index": 9,
```

```
"first_seen_at": "2026-05-02T04:57:20.678535+00:00",
"last_seen_at": "2026-05-02T04:58:47.026826+00:00",
"seen_count": 3,
"miss_count": 9,
"stale": true,
"cached": true
},
"link:sharepoint": {
  "element_key": "link:sharepoint",
  "element_type": "link",
  "label": "SharePoint",
  "bbox": [
    194.0,
    211.0,
    314.0,
    259.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 8,
  "last_seen_index": 8,
  "first_seen_at": "2026-05-02T04:58:02.406820+00:00",
  "last_seen_at": "2026-05-02T04:58:02.406820+00:00",
  "seen_count": 1,
  "miss_count": 10,
  "stale": true,
  "cached": true
}
```

```
},
"icon:my sites": {
  "element_key": "icon:my sites",
  "element_type": "icon",
  "label": "My sites",
  "bbox": [
    162.0,
    371.0,
    202.0,
    419.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 8,
  "last_seen_index": 8,
  "first_seen_at": "2026-05-02T04:58:02.406820+00:00",
  "last_seen_at": "2026-05-02T04:58:02.406820+00:00",
  "seen_count": 1,
  "miss_count": 10,
  "stale": true,
  "cached": true
},
"icon:my news": {
  "element_key": "icon:my news",
  "element_type": "icon",
  "label": "My news",
  "bbox": [
```



```
    162.0,  
    443.0,  
    202.0,  
    491.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 8,  
  "last_seen_index": 8,  
  "first_seen_at": "2026-05-02T04:58:02.406820+00:00",  
  "last_seen_at": "2026-05-02T04:58:02.406820+00:00",  
  "seen_count": 1,  
  "miss_count": 10,  
  "stale": true,  
  "cached": true  
},  
"icon:my files": {  
  "element_key": "icon:my files",  
  "element_type": "icon",  
  "label": "My files",  
  "bbox": [  
    170.0,  
    523.0,  
    194.0,  
    563.0  
  ],  
  "confidence": 0.4,
```

```
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 8,
"last_seen_index": 8,
"first_seen_at": "2026-05-02T04:58:02.406820+00:00",
"last_seen_at": "2026-05-02T04:58:02.406820+00:00",
"seen_count": 1,
"miss_count": 10,
"stale": true,
"cached": true
},
"icon:my lists": {
  "element_key": "icon:my lists",
  "element_type": "icon",
  "label": "My lists",
  "bbox": [
    170.0,
    595.0,
    194.0,
    635.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 8,
  "last_seen_index": 8,
  "first_seen_at": "2026-05-02T04:58:02.406820+00:00",
```

```
    "last_seen_at": "2026-05-02T04:58:02.406820+00:00",
    "seen_count": 1,
    "miss_count": 10,
    "stale": true,
    "cached": true
  },
  "icon:create": {
    "element_key": "icon:create",
    "element_type": "icon",
    "label": "Create",
    "bbox": [
      170.0,
      667.0,
      194.0,
      715.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 8,
    "last_seen_index": 8,
    "first_seen_at": "2026-05-02T04:58:02.406820+00:00",
    "last_seen_at": "2026-05-02T04:58:02.406820+00:00",
    "seen_count": 1,
    "miss_count": 10,
    "stale": true,
    "cached": true
  },
}
```

```
"icon:site icon": {
  "element_key": "icon:site icon",
  "element_type": "icon",
  "label": "Site icon",
  "bbox": [
    218.0,
    275.0,
    290.0,
    347.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 8,
  "last_seen_index": 8,
  "first_seen_at": "2026-05-02T04:58:02.406820+00:00",
  "last_seen_at": "2026-05-02T04:58:02.406820+00:00",
  "seen_count": 1,
  "miss_count": 10,
  "stale": true,
  "cached": true
},
"label:private group": {
  "element_key": "label:private group",
  "element_type": "label",
  "label": "Private group",
  "bbox": [
    274.0,
```

```
    315.0,
    402.0,
    347.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 8,
"last_seen_index": 8,
"first_seen_at": "2026-05-02T04:58:02.406820+00:00",
"last_seen_at": "2026-05-02T04:58:02.406820+00:00",
"seen_count": 1,
"miss_count": 10,
"stale": true,
"cached": true
},
"button:not following": {
  "element_key": "button:not following",
  "element_type": "button",
  "label": "Not following",
  "bbox": [
    1258.0,
    283.0,
    1346.0,
    315.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
```

```
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 8,
"last_seen_index": 8,
"first_seen_at": "2026-05-02T04:58:02.406820+00:00",
"last_seen_at": "2026-05-02T04:58:02.406820+00:00",
"seen_count": 1,
"miss_count": 10,
"stale": true,
"cached": true
},
"link:10 members": {
  "element_key": "link:10 members",
  "element_type": "link",
  "label": "10 members",
  "bbox": [
    1258.0,
    323.0,
    1346.0,
    355.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 8,
  "last_seen_index": 8,
  "first_seen_at": "2026-05-02T04:58:02.406820+00:00",
  "last_seen_at": "2026-05-02T04:58:02.406820+00:00",
```

```
"seen_count": 1,
"miss_count": 10,
"stale": true,
"cached": true
},
"link:25-26j-cloud": {
  "element_key": "link:25-26j-cloud",
  "element_type": "link",
  "label": "25-26J-Cloud",
  "bbox": [
    202.0,
    371.0,
    322.0,
    403.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 8,
  "last_seen_index": 9,
  "first_seen_at": "2026-05-02T04:58:02.406820+00:00",
  "last_seen_at": "2026-05-02T04:58:47.026826+00:00",
  "seen_count": 2,
  "miss_count": 9,
  "stale": true,
  "cached": true
},
"link:25-26j-477-students": {
```

```
"element_key": "link:25-26j-477-students",
"element_type": "link",
"label": "25-26J-477-Students",
"bbox": [
  290.0,
  371.0,
  498.0,
  395.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 8,
"last_seen_index": 9,
"first_seen_at": "2026-05-02T04:58:02.406820+00:00",
"last_seen_at": "2026-05-02T04:58:47.026826+00:00",
"seen_count": 2,
"miss_count": 9,
"stale": true,
"cached": true
},
"button:add shortcut to onedrive": {
  "element_key": "button:add shortcut to onedrive",
  "element_type": "button",
  "label": "Add shortcut to OneDrive",
  "bbox": [
    658.0,
    371.0,
```



```
      810.0,
      395.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 8,
    "last_seen_index": 8,
    "first_seen_at": "2026-05-02T04:58:02.406820+00:00",
    "last_seen_at": "2026-05-02T04:58:02.406820+00:00",
    "seen_count": 1,
    "miss_count": 10,
    "stale": true,
    "cached": true
  },
  "button:download": {
    "element_key": "button:download",
    "element_type": "button",
    "label": "Download",
    "bbox": [
      818.0,
      371.0,
      890.0,
      395.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
```

```
"last_seen_frame": null,
"first_seen_index": 8,
"last_seen_index": 8,
"first_seen_at": "2026-05-02T04:58:02.406820+00:00",
"last_seen_at": "2026-05-02T04:58:02.406820+00:00",
"seen_count": 1,
"miss_count": 10,
"stale": true,
"cached": true
},
"button:edit in grid view": {
  "element_key": "button:edit in grid view",
  "element_type": "button",
  "label": "Edit in grid view",
  "bbox": [
    874.0,
    371.0,
    994.0,
    395.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 8,
  "last_seen_index": 8,
  "first_seen_at": "2026-05-02T04:58:02.406820+00:00",
  "last_seen_at": "2026-05-02T04:58:02.406820+00:00",
  "seen_count": 1,
```

```
    "miss_count": 10,
    "stale": true,
    "cached": true
  },
  "button:export to excel": {
    "element_key": "button:export to excel",
    "element_type": "button",
    "label": "Export to Excel",
    "bbox": [
      978.0,
      371.0,
      1082.0,
      395.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 8,
    "last_seen_index": 8,
    "first_seen_at": "2026-05-02T04:58:02.406820+00:00",
    "last_seen_at": "2026-05-02T04:58:02.406820+00:00",
    "seen_count": 1,
    "miss_count": 10,
    "stale": true,
    "cached": true
  },
  "button:integrate": {
    "element_key": "button:integrate",
```

```
"element_type": "button",
"label": "Integrate",
"bbox": [
    1066.0,
    371.0,
    1162.0,
    395.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 8,
"last_seen_index": 8,
"first_seen_at": "2026-05-02T04:58:02.406820+00:00",
"last_seen_at": "2026-05-02T04:58:02.406820+00:00",
"seen_count": 1,
"miss_count": 10,
"stale": true,
"cached": true
},
"button:sync": {
    "element_key": "button:sync",
    "element_type": "button",
    "label": "Sync",
    "bbox": [
        1154.0,
        379.0,
        1210.0,
```

```
    395.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 8,
  "last_seen_index": 8,
  "first_seen_at": "2026-05-02T04:58:02.406820+00:00",
  "last_seen_at": "2026-05-02T04:58:02.406820+00:00",
  "seen_count": 1,
  "miss_count": 10,
  "stale": true,
  "cached": true
},
"dropdown:all docu...": {
  "element_key": "dropdown:all docu...",
  "element_type": "dropdown",
  "label": "All Docu...",
  "bbox": [
    194.0,
    403.0,
    346.0,
    451.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
```

```
"first_seen_index": 8,
"last_seen_index": 8,
"first_seen_at": "2026-05-02T04:58:02.406820+00:00",
"last_seen_at": "2026-05-02T04:58:02.406820+00:00",
"seen_count": 1,
"miss_count": 10,
"stale": true,
"cached": true
},
"button:filter": {
  "element_key": "button:filter",
  "element_type": "button",
  "label": "Filter",
  "bbox": [
    338.0,
    403.0,
    378.0,
    443.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 8,
  "last_seen_index": 8,
  "first_seen_at": "2026-05-02T04:58:02.406820+00:00",
  "last_seen_at": "2026-05-02T04:58:02.406820+00:00",
  "seen_count": 1,
  "miss_count": 10,
```

```
"stale": true,
"cached": true
},
"text:cdapsubmissioncloud": {
  "element_key": "text:cdapsubmissioncloud",
  "element_type": "text",
  "label": "CDAPSubmissionCloud",
  "bbox": [
    299.0,
    276.0,
    603.0,
    324.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 9,
  "last_seen_index": 9,
  "first_seen_at": "2026-05-02T04:58:47.026826+00:00",
  "last_seen_at": "2026-05-02T04:58:47.026826+00:00",
  "seen_count": 1,
  "miss_count": 9,
  "stale": true,
  "cached": true
},
"button:all documents": {
  "element_key": "button:all documents",
  "element_type": "button",
```

```
"label": "All Documents",
"bbox": [
  211.0,
  412.0,
  387.0,
  452.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 9,
"last_seen_index": 9,
"first_seen_at": "2026-05-02T04:58:47.026826+00:00",
"last_seen_at": "2026-05-02T04:58:47.026826+00:00",
"seen_count": 1,
"miss_count": 9,
"stale": true,
"cached": true
},
"pane:modal dialog": {
  "element_key": "pane:modal dialog",
  "element_type": "pane",
  "label": "Modal Dialog",
  "bbox": [
    659.0,
    476.0,
    1219.0,
    684.0
  ]
}
```



```
],  
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  "last_seen_at": "2026-05-02T04:58:47.026826+00:00",  
  "seen_count": 1,  
  "miss_count": 9,  
  "stale": true,  
  "cached": true  
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"text:1. project proposal": {  
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    443.0,  
    540.0  
  ],  
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},  
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  "label": "2. Progress Presentation - 1",  
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    491.0,  
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  "last_seen_at": "2026-05-02T04:58:47.026826+00:00",  
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    "cached": true
  },
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    "stale": true,
    "cached": true
  },
  "text:4. research paper": {
    "element_key": "text:4. research paper",
    "element_type": "text",
    "label": "4. Research Paper",
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  652.0
],
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"stale": true,
"cached": true
},
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  "element_key": "text:5. final report & presentation",
  "element_type": "text",
  "label": "5. Final Report & Presentation",
  "bbox": [
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    491.0,
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  ],

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"stale": true,
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  "element_type": "text",
  "label": "6. CheckList Documents",
  "bbox": [
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    764.0
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  "element_type": "text",
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    780.0,
    403.0,
    812.0
  ],
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  "seen_count": 1,
  "miss_count": 9,
  "stale": true,
  "cached": true
}
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  "miss_count": 9,
  "stale": true,
  "cached": true
},
"text:project registration documents": {
  "element_key": "text:project registration documents",
  "element_type": "text",
  "label": "Project Registration Documents",
  "bbox": [
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    291.0,  
    860.0,  
    555.0,  
    900.0  
  ],  
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  "last_seen_index": 9,  
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  "last_seen_at": "2026-05-02T04:58:47.026826+00:00",  
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  "miss_count": 9,  
  "stale": true,  
  "cached": true  
},  
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  "element_key": "text:panel comments for the students.xlsx",  
  "element_type": "text",  
  "label": "Panel Comments for the Students.xlsx",  
  "bbox": [  
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    908.0,  
    627.0,  
    940.0  
  ],  
  "confidence": 0.4,
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"seen_count": 1,
"miss_count": 9,
"stale": true,
"cached": true
},
"icon:google chrome": {
  "element_key": "icon:google chrome",
  "element_type": "icon",
  "label": "Google Chrome",
  "bbox": [
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    118.0,
    252.0,
    246.0
  ],
  "confidence": 0.95,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 10,
  "last_seen_index": 18,
  "first_seen_at": "2026-05-02T04:59:38.576878+00:00",
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"last_seen_at": "2026-05-02T05:06:12.825053+00:00",
"seen_count": 8,
"miss_count": 0,
"stale": false,
"cached": false
},
"icon:huiontablet": {
  "element_key": "icon:huiontablet",
  "element_type": "icon",
  "label": "HuionTablet",
  "bbox": [
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    118.0,
    340.0,
    246.0
  ],
  "confidence": 0.95,
  "source": "gemini_vlm",
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  "last_seen_frame": null,
  "first_seen_index": 10,
  "last_seen_index": 18,
  "first_seen_at": "2026-05-02T04:59:38.576878+00:00",
  "last_seen_at": "2026-05-02T05:06:12.825053+00:00",
  "seen_count": 7,
  "miss_count": 0,
  "stale": false,
  "cached": false
},
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"icon:wise memory optimizer": {
  "element_key": "icon:wise memory optimizer",
  "element_type": "icon",
  "label": "Wise Memory Optimizer",
  "bbox": [
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    350.0,
    252.0,
    502.0
  ],
  "confidence": 0.95,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 11,
  "last_seen_index": 18,
  "first_seen_at": "2026-05-02T05:00:28.296423+00:00",
  "last_seen_at": "2026-05-02T05:06:12.825053+00:00",
  "seen_count": 3,
  "miss_count": 0,
  "stale": false,
  "cached": false
},
"icon:amazon workspaces": {
  "element_key": "icon:amazon workspaces",
  "element_type": "icon",
  "label": "Amazon WorkSpaces",
  "bbox": [
    236.0,
```

```
    350.0,  
    340.0,  
    502.0  
  ],  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 11,  
  "last_seen_index": 18,  
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  "last_seen_at": "2026-05-02T05:06:12.825053+00:00",  
  "seen_count": 3,  
  "miss_count": 0,  
  "stale": false,  
  "cached": false  
},  
"icon:recycle bin": {  
  "element_key": "icon:recycle bin",  
  "element_type": "icon",  
  "label": "Recycle Bin",  
  "bbox": [  
    149.0,  
    542.0,  
    253.0,  
    710.0  
  ],  
  "confidence": 0.6232,  
  "source": "gemini_vlm",
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"last_seen_at": "2026-05-02T05:02:52.171843+00:00",
"seen_count": 2,
"miss_count": 4,
"stale": true,
"cached": true
},
"icon:krita": {
  "element_key": "icon:krita",
  "element_type": "icon",
  "label": "Krita",
  "bbox": [
    237.0,
    542.0,
    341.0,
    710.0
  ],
  "confidence": 0.6232,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 11,
  "last_seen_index": 14,
  "first_seen_at": "2026-05-02T05:00:28.296423+00:00",
  "last_seen_at": "2026-05-02T05:02:52.171843+00:00",
```

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"miss_count": 4,
"stale": true,
"cached": true
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  "element_key": "icon:nuwani - chrome",
  "element_type": "icon",
  "label": "Nuwani - Chrome",
  "bbox": [
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    750.0,
    253.0,
    918.0
  ],
  "confidence": 0.6232,
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  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 11,
  "last_seen_index": 14,
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  "miss_count": 4,
  "stale": true,
  "cached": true
},
"icon:docker desktop": {
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"element_type": "icon",
"label": "Docker Desktop",
"bbox": [
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  750.0,
  341.0,
  918.0
],
"confidence": 0.6232,
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"last_seen_at": "2026-05-02T05:02:52.171843+00:00",
"seen_count": 2,
"miss_count": 4,
"stale": true,
"cached": true
},
"icon:antigravity": {
  "element_key": "icon:antigravity",
  "element_type": "icon",
  "label": "Antigravity",
  "bbox": [
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    974.0,
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```
    253.0,  
    1012.0  
  ],  
  "confidence": 0.5905,  
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  "last_seen_frame": null,  
  "first_seen_index": 11,  
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  "last_seen_at": "2026-05-02T05:02:52.171843+00:00",  
  "seen_count": 2,  
  "miss_count": 4,  
  "stale": true,  
  "cached": true  
},  
"icon:visual studio code": {  
  "element_key": "icon:visual studio code",  
  "element_type": "icon",  
  "label": "Visual Studio Code",  
  "bbox": [  
    237.0,  
    974.0,  
    341.0,  
    1006.0  
  ],  
  "confidence": 0.5905,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,
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"seen_count": 2,
"miss_count": 4,
"stale": true,
"cached": true
},
"icon:postman": {
  "element_key": "icon:postman",
  "element_type": "icon",
  "label": "Postman",
  "bbox": [
    315.0,
    758.0,
    411.0,
    894.0
  ],
  "confidence": 0.4543,
  "source": "gemini_vlm",
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  "last_seen_frame": null,
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  "last_seen_index": 11,
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  "last_seen_at": "2026-05-02T05:00:28.296423+00:00",
  "seen_count": 1,
```

```
    "miss_count": 7,
    "stale": true,
    "cached": true
  },
  "icon:loom": {
    "element_key": "icon:loom",
    "element_type": "icon",
    "label": "Loom",
    "bbox": [
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      758.0,
      491.0,
      894.0
    ],
    "confidence": 0.4543,
    "source": "gemini_vlm",
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    "last_seen_frame": null,
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    "last_seen_index": 11,
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    "seen_count": 1,
    "miss_count": 7,
    "stale": true,
    "cached": true
  },
  "icon:vlc media player": {
    "element_key": "icon:vlc media player",
```

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"element_type": "icon",
"label": "VLC media player",
"bbox": [
    475.0,
    758.0,
    571.0,
    894.0
],
"confidence": 0.4543,
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"last_seen_at": "2026-05-02T05:00:28.296423+00:00",
"seen_count": 1,
"miss_count": 7,
"stale": true,
"cached": true
},
"pane:google chrome window": {
    "element_key": "pane:google chrome window",
    "element_type": "pane",
    "label": "Google Chrome Window",
    "bbox": [
        285.0,
        110.0,
        1597.0,
```

```
    1013.0
  ],
  "confidence": 0.855,
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  "last_seen_frame": null,
  "first_seen_index": 17,
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  "last_seen_at": "2026-05-02T05:05:25.437256+00:00",
  "seen_count": 1,
  "miss_count": 1,
  "stale": false,
  "cached": true
},
"label:google chrome": {
  "element_key": "label:google chrome",
  "element_type": "label",
  "label": "Google Chrome",
  "bbox": [
    413.0,
    134.0,
    613.0,
    190.0
  ],
  "confidence": 0.81,
  "source": "gemini_vlm",
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"miss_count": 1,
"stale": false,
"cached": true
},
"button:minimize": {
  "element_key": "button:minimize",
  "element_type": "button",
  "label": "Minimize",
  "bbox": [
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    134.0,
    1365.0,
    182.0
  ],
  "confidence": 0.81,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 17,
  "last_seen_index": 17,
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  "last_seen_at": "2026-05-02T05:05:25.437256+00:00",
  "seen_count": 1,
  "miss_count": 1,
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"stale": false,
"cached": true
},
"button:maximize": {
  "element_key": "button:maximize",
  "element_type": "button",
  "label": "Maximize",
  "bbox": [
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    1413.0,
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  "confidence": 0.81,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 17,
  "last_seen_index": 17,
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  "last_seen_at": "2026-05-02T05:05:25.437256+00:00",
  "seen_count": 1,
  "miss_count": 1,
  "stale": false,
  "cached": true
},
"button:close": {
  "element_key": "button:close",
  "element_type": "button",
```

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"label": "Close",
"bbox": [
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  134.0,
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],
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"first_seen_index": 17,
"last_seen_index": 17,
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"last_seen_at": "2026-05-02T05:05:25.437256+00:00",
"seen_count": 1,
"miss_count": 1,
"stale": false,
"cached": true
},
"heading:who's using chrome?": {
  "element_key": "heading:who's using chrome?",
  "element_type": "heading",
  "label": "Who's using Chrome?",
  "bbox": [
    821.0,
    278.0,
    1085.0,
    326.0
```

```
],
  "confidence": 0.855,
  "source": "gemini_vlm",
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  "last_seen_frame": null,
  "first_seen_index": 17,
  "last_seen_index": 17,
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  "seen_count": 1,
  "miss_count": 1,
  "stale": false,
  "cached": true
},
"button:intrinsic-tech.net": {
  "element_key": "button:intrinsic-tech.net",
  "element_type": "button",
  "label": "Intrinsic-tech.net",
  "bbox": [
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    414.0,
    789.0,
    630.0
  ],
  "confidence": 0.81,
  "source": "gemini_vlm",
  "first_seen_frame": null,
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  "first_seen_index": 17,
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"last_seen_index": 17,
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"last_seen_at": "2026-05-02T05:05:25.437256+00:00",
"seen_count": 1,
"miss_count": 1,
"stale": false,
"cached": true
},
"button:lammastide": {
  "element_key": "button:lammastide",
  "element_type": "button",
  "label": "lammastide",
  "bbox": [
    757.0,
    422.0,
    973.0,
    630.0
  ],
  "confidence": 0.81,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 17,
  "last_seen_index": 17,
  "first_seen_at": "2026-05-02T05:05:25.437256+00:00",
  "last_seen_at": "2026-05-02T05:05:25.437256+00:00",
  "seen_count": 1,
  "miss_count": 1,
  "stale": false,
```

```
    "cached": true
  },
  "button:lammastideit": {
    "element_key": "button:lammastideit",
    "element_type": "button",
    "label": "LammastideIT",
    "bbox": [
      941.0,
      422.0,
      1157.0,
      630.0
    ],
    "confidence": 0.81,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 17,
    "last_seen_index": 17,
    "first_seen_at": "2026-05-02T05:05:25.437256+00:00",
    "last_seen_at": "2026-05-02T05:05:25.437256+00:00",
    "seen_count": 1,
    "miss_count": 1,
    "stale": false,
    "cached": true
  },
  "button:nuwani": {
    "element_key": "button:nuwani",
    "element_type": "button",
    "label": "Nuwani",
```

```
"bbox": [
  1125.0,
  406.0,
  1293.0,
  606.0
],
"confidence": 0.81,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 17,
"last_seen_index": 17,
"first_seen_at": "2026-05-02T05:05:25.437256+00:00",
"last_seen_at": "2026-05-02T05:05:25.437256+00:00",
"seen_count": 1,
"miss_count": 1,
"stale": false,
"cached": true
},
"button:nuwani (photo)": {
  "element_key": "button:nuwani (photo)",
  "element_type": "button",
  "label": "Nuwani (photo)",
  "bbox": [
    597.0,
    582.0,
    789.0,
    798.0
  ],

```

```
"confidence": 0.81,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 17,
"last_seen_index": 17,
"first_seen_at": "2026-05-02T05:05:25.437256+00:00",
"last_seen_at": "2026-05-02T05:05:25.437256+00:00",
"seen_count": 1,
"miss_count": 1,
"stale": false,
"cached": true
},
"button:add profile": {
  "element_key": "button:add profile",
  "element_type": "button",
  "label": "Add Profile",
  "bbox": [
    757.0,
    582.0,
    973.0,
    798.0
  ],
  "confidence": 0.81,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 17,
  "last_seen_index": 17,
```

```
"first_seen_at": "2026-05-02T05:05:25.437256+00:00",
"last_seen_at": "2026-05-02T05:05:25.437256+00:00",
"seen_count": 1,
"miss_count": 1,
"stale": false,
"cached": true
}
},
"current_elements": [],
"resolved_elements": [
  {
    "id": "memory_icon:google chrome",
    "type": "icon",
    "label": "Google Chrome",
    "description": "Cached from an earlier frame in the same run",
    "state": "cached",
    "bbox": [
      148.0,
      118.0,
      252.0,
      246.0
    ],
    "confidence": 0.95,
    "source": "gemini_vlm",
    "last_seen_frame": null,
    "last_seen_index": 18,
    "stale": true,
    "cached": true,
    "session_memory_key": "icon:google chrome"
```

```
},
{
  "id": "memory_icon:huiontablet",
  "type": "icon",
  "label": "HuionTablet",
  "description": "Cached from an earlier frame in the same run",
  "state": "cached",
  "bbox": [
    236.0,
    118.0,
    340.0,
    246.0
  ],
  "confidence": 0.95,
  "source": "gemini_vlm",
  "last_seen_frame": null,
  "last_seen_index": 18,
  "stale": true,
  "cached": true,
  "session_memory_key": "icon:huiontablet"
},
{
  "id": "memory_icon:wise memory optimizer",
  "type": "icon",
  "label": "Wise Memory Optimizer",
  "description": "Cached from an earlier frame in the same run",
  "state": "cached",
  "bbox": [
    148.0,
```

```
        350.0,
        252.0,
        502.0
    ],
    "confidence": 0.95,
    "source": "gemini_vlm",
    "last_seen_frame": null,
    "last_seen_index": 18,
    "stale": true,
    "cached": true,
    "session_memory_key": "icon:wise memory optimizer"
},
{
    "id": "memory_icon:amazon workspaces",
    "type": "icon",
    "label": "Amazon WorkSpaces",
    "description": "Cached from an earlier frame in the same run",
    "state": "cached",
    "bbox": [
        236.0,
        350.0,
        340.0,
        502.0
    ],
    "confidence": 0.95,
    "source": "gemini_vlm",
    "last_seen_frame": null,
    "last_seen_index": 18,
    "stale": true,
```

```
"cached": true,
"session_memory_key": "icon:amazon workspaces"
},
{
  "id": "memory_pane:google chrome window",
  "type": "pane",
  "label": "Google Chrome Window",
  "description": "Cached from an earlier frame in the same run",
  "state": "cached",
  "bbox": [
    285.0,
    110.0,
    1597.0,
    1013.0
  ],
  "confidence": 0.855,
  "source": "gemini_vlm",
  "last_seen_frame": null,
  "last_seen_index": 17,
  "stale": true,
  "cached": true,
  "session_memory_key": "pane:google chrome window"
},
{
  "id": "memory_label:google chrome",
  "type": "label",
  "label": "Google Chrome",
  "description": "Cached from an earlier frame in the same run",
  "state": "cached",
```



```
"bbox": [
  413.0,
  134.0,
  613.0,
  190.0
],
"confidence": 0.81,
"source": "gemini_vlm",
"last_seen_frame": null,
"last_seen_index": 17,
"stale": true,
"cached": true,
"session_memory_key": "label:google chrome"
},
{
  "id": "memory_button:minimize",
  "type": "button",
  "label": "Minimize",
  "description": "Cached from an earlier frame in the same run",
  "state": "cached",
  "bbox": [
    1317.0,
    134.0,
    1365.0,
    182.0
  ],
  "confidence": 0.81,
  "source": "gemini_vlm",
  "last_seen_frame": null,
```

```
"last_seen_index": 17,
"stale": true,
"cached": true,
"session_memory_key": "button:minimize"
},
{
  "id": "memory_button:maximize",
  "type": "button",
  "label": "Maximize",
  "description": "Cached from an earlier frame in the same run",
  "state": "cached",
  "bbox": [
    1357.0,
    134.0,
    1413.0,
    190.0
  ],
  "confidence": 0.81,
  "source": "gemini_vlm",
  "last_seen_frame": null,
  "last_seen_index": 17,
  "stale": true,
  "cached": true,
  "session_memory_key": "button:maximize"
},
{
  "id": "memory_button:close",
  "type": "button",
  "label": "Close",
```

```
"description": "Cached from an earlier frame in the same run",
"state": "cached",
"bbox": [
  1397.0,
  134.0,
  1453.0,
  190.0
],
"confidence": 0.81,
"source": "gemini_vlm",
"last_seen_frame": null,
"last_seen_index": 17,
"stale": true,
"cached": true,
"session_memory_key": "button:close"
},
{
  "id": "memory_heading:who's using chrome?",
  "type": "heading",
  "label": "Who's using Chrome?",
  "description": "Cached from an earlier frame in the same run",
  "state": "cached",
  "bbox": [
    821.0,
    278.0,
    1085.0,
    326.0
  ],
  "confidence": 0.855,
```

```
"source": "gemini_vlm",
"last_seen_frame": null,
"last_seen_index": 17,
"stale": true,
"cached": true,
"session_memory_key": "heading:who's using chrome?"
},
{
  "id": "memory_button:intrinsic-tech.net",
  "type": "button",
  "label": "Intrinsic-tech.net",
  "description": "Cached from an earlier frame in the same run",
  "state": "cached",
  "bbox": [
    597.0,
    414.0,
    789.0,
    630.0
  ],
  "confidence": 0.81,
  "source": "gemini_vlm",
  "last_seen_frame": null,
  "last_seen_index": 17,
  "stale": true,
  "cached": true,
  "session_memory_key": "button:intrinsic-tech.net"
},
{
  "id": "memory_button:lammastide",
```

```
"type": "button",
"label": "lammastide",
"description": "Cached from an earlier frame in the same run",
"state": "cached",
"bbox": [
    757.0,
    422.0,
    973.0,
    630.0
],
"confidence": 0.81,
"source": "gemini_vlm",
"last_seen_frame": null,
"last_seen_index": 17,
"stale": true,
"cached": true,
"session_memory_key": "button:lammastide"
},
{
    "id": "memory_button:lammastideit",
    "type": "button",
    "label": "LammastideIT",
    "description": "Cached from an earlier frame in the same run",
    "state": "cached",
    "bbox": [
        941.0,
        422.0,
        1157.0,
        630.0
    ]
}
```

```
    ],
    "confidence": 0.81,
    "source": "gemini_vlm",
    "last_seen_frame": null,
    "last_seen_index": 17,
    "stale": true,
    "cached": true,
    "session_memory_key": "button:lammastideit"
  },
  {
    "id": "memory_button:nuwani",
    "type": "button",
    "label": "Nuwani",
    "description": "Cached from an earlier frame in the same run",
    "state": "cached",
    "bbox": [
      1125.0,
      406.0,
      1293.0,
      606.0
    ],
    "confidence": 0.81,
    "source": "gemini_vlm",
    "last_seen_frame": null,
    "last_seen_index": 17,
    "stale": true,
    "cached": true,
    "session_memory_key": "button:nuwani"
  },
}
```

```
{
  "id": "memory_button:nuwani (photo)",
  "type": "button",
  "label": "Nuwani (photo)",
  "description": "Cached from an earlier frame in the same run",
  "state": "cached",
  "bbox": [
    597.0,
    582.0,
    789.0,
    798.0
  ],
  "confidence": 0.81,
  "source": "gemini_vlm",
  "last_seen_frame": null,
  "last_seen_index": 17,
  "stale": true,
  "cached": true,
  "session_memory_key": "button:nuwani (photo)"
},
{
  "id": "memory_button:add profile",
  "type": "button",
  "label": "Add Profile",
  "description": "Cached from an earlier frame in the same run",
  "state": "cached",
  "bbox": [
    757.0,
    582.0,
```

```
    973.0,  
    798.0  
  ],  
  "confidence": 0.81,  
  "source": "gemini_vlm",  
  "last_seen_frame": null,  
  "last_seen_index": 17,  
  "stale": true,  
  "cached": true,  
  "session_memory_key": "button:add profile"  
}  
],  
"screen_data": {}  
}  
}
```

Test Case 08: CourseWeb Login (Complex Task)

User Prompt:

click on the "Log in to the site | CourseWeb" tab on this browser and login to the courseweb

Test Results:

- Steps Total: 2
- Completed: 1
- Failed: 1
- Duration: 9 minutes 14 seconds

Step Execution:

Step 1: Click on the 'Log in to the site | CourseWeb' tab

- Target: Log in to the site | CourseWeb tab at (476, 154)

- **Attempts: 1**
- **Execution: Performed**
- **Evaluation: Success**
- **Issue: None**

Step 2: Enter the username and password into the input fields and click the login button

- **Target: CourseWeb login form**
- **Attempts: 3**
- **Execution: Performed**
- **Evaluation: Failed (retry loop)**
- **Issue: Evaluation error — No valid JSON found in LLM response ('{\n "status": "needs_'). Additionally, the login page failed to render UI elements (username/password fields) in time for the next step.**

Execution Summary:

The agent successfully switched to the CourseWeb login tab but failed to complete the login process. After switching tabs, the page either did not load completely or the input fields were not detected by the vision model. The evaluation loop encountered a malformed JSON response from the LLM, preventing proper validation of the login step.

Issues Identified:

- **Malformed JSON response from LLM during evaluation**
- **Login page failed to render input fields in time for detection**
- **No retry mechanism to wait for page elements to load**
- **Credentials could not be entered due to missing UI elements**
- **Possible network delay or page rendering issue**

Recommendations for Improvement:

- **Enforce strict JSON output formatting with validation before parsing**
- **Add explicit wait/delay after tab switch before attempting to detect form fields**
- **Implement page load detection (e.g., wait for specific element to appear)**
- **Add fallback to refresh or retry page load if fields are not detected**
- **Improve evaluation prompt to handle "needs credentials" state without malformed JSON**

JSON Response:

full_cycle.json

```
{  
  
  "run_id": "30427c77-968c-4a27-a252-6b8db30fe924",
```

```
"user_task": "click on the \"Log in to the site | CourseWeb\" tab on this browser and login to the courseweb",

"started_at": "2026-05-02T06:41:57.870381+00:00",

"finished_at": "2026-05-02T06:51:11.603747+00:00",

"vision_used": true,

"perception": {

  "image": "E:\\sliit-project\\ScreenAwareTaskAgent\\vision\\data\\sessions\\shot_1777704127_f68076\\processing\\proc_1777704128_59c8ad\\preprocessed_frames\\frame_1777704128_8d3627.jpg",

  "image_size": {

    "width": 1920,

    "height": 1080

  },

  "screen_bbox": [

    156,

    102,

    1767,

    1016

  ],

  "screen_size": {

    "width": 1611,

    "height": 914

  },

  "coordinate_system": "pixel",

  "elements": [

    {

      "id": "tab_frontend",

      "type": "tab",

      "label": "frontend",
```

```
"description": "Browser tab labeled frontend",
"state": "visible",
"dx": 116,
"dy": 52,
"frame_dx": 272,
"frame_dy": 154,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    156,
    126,
    388,
    182
],
"session_memory_key": "tab:frontend",
"cached": false
},
{
    "id": "tab_courseweb",
    "type": "tab",
    "label": "Log in to the site | CourseWeb",
    "description": "Browser tab for CourseWeb login",
    "state": "visible",
    "dx": 320,
    "dy": 52,
    "frame_dx": 476,
    "frame_dy": 154,
    "confidence": 0.95,
    "source": "gemini_vlm",
```

```
"frame_bbox": [
  348,
  126,
  604,
  182
],
"session_memory_key": "tab:log in to the site | courseweb",
"cached": false
},
{
  "id": "address_bar",
  "type": "input",
  "label": "Address bar",
  "description": "Browser address bar showing localhost:5173",
  "state": "visible",
  "dx": 576,
  "dy": 112,
  "frame_dx": 732,
  "frame_dy": 214,
  "confidence": 0.98,
  "source": "gemini_vlm",
  "frame_bbox": [
    156,
    182,
    1308,
    246
  ],
  "session_memory_key": "input:address bar",
  "cached": false
}
```

```
},
{
  "id": "app_logo",
  "type": "icon",
  "label": "ScreenPilot AI Logo",
  "description": "Chip icon for ScreenPilot AI",
  "state": "visible",
  "dx": 56,
  "dy": 192,
  "frame_dx": 212,
  "frame_dy": 294,
  "confidence": 0.9,
  "source": "gemini_vlm",
  "frame_bbox": [
    196,
    270,
    228,
    318
  ],
  "session_memory_key": "icon:screenpilot ai logo",
  "cached": false
},
{
  "id": "app_title",
  "type": "heading",
  "label": "ScreenPilot AI",
  "description": "Application title text",
  "state": "visible",
  "dx": 144,
```

```
"dy": 192,
"frame_dx": 300,
"frame_dy": 294,
"confidence": 0.98,
"source": "gemini_vlm",
"frame_bbox": [
    236,
    278,
    364,
    310
],
"session_memory_key": "heading:screenpilot ai",
"cached": false
},
{
    "id": "app_version",
    "type": "label",
    "label": "Autonomous Task Agent v2",
    "description": "Application version badge",
    "state": "visible",
    "dx": 300,
    "dy": 196,
    "frame_dx": 456,
    "frame_dy": 298,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        380,
        286,
```

```
    532,  
    310  
  ],  
  "session_memory_key": "label:autonomous task agent v2",  
  "cached": false  
},  
{  
  "id": "message_bubble",  
  "type": "pane",  
  "label": "Agent message",  
  "description": "Chat bubble with text: I will now open the browser and  
login to the courseweb",  
  "state": "visible",  
  "dx": 928,  
  "dy": 340,  
  "frame_dx": 1084,  
  "frame_dy": 442,  
  "confidence": 0.92,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    732,  
    374,  
    1436,  
    510  
  ],  
  "session_memory_key": "pane:agent message",  
  "cached": false  
},  
{
```

```
"id": "status_connecting",
"type": "label",
"label": "Connecting to agent...",
"description": "Status message indicating connection progress",
"state": "visible",
"dx": 120,
"dy": 440,
"frame_dx": 276,
"frame_dy": 542,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    188,
    526,
    364,
    558
],
"session_memory_key": "label:connecting to agent...",
"cached": false
},
{
  "id": "status_capturing",
  "type": "label",
  "label": "Capturing initial screen...",
  "description": "Status message indicating screen capture progress",
  "state": "visible",
  "dx": 148,
  "dy": 468,
  "frame_dx": 304,
```



```
"frame_dy": 570,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    188,
    558,
    420,
    582
],
"session_memory_key": "label:capturing initial screen...",
"cached": false
},
{
    "id": "loading_indicator",
    "type": "pane",
    "label": "Loading indicator",
    "description": "Box showing loading dots and a cursor",
    "state": "visible",
    "dx": 124,
    "dy": 596,
    "frame_dx": 280,
    "frame_dy": 698,
    "confidence": 0.85,
    "source": "gemini_vlm",
    "frame_bbox": [
        180,
        598,
        380,
        798
    ]
}
```

```
    ],
    "session_memory_key": "pane:loading indicator",
    "cached": false
  },
  {
    "id": "chat_input_field",
    "type": "input",
    "label": "Chat input",
    "description": "Input field with placeholder: Agent is working...",
    "state": "visible",
    "dx": 684,
    "dy": 844,
    "frame_dx": 840,
    "frame_dy": 946,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
      156,
      894,
      1524,
      998
    ],
    "session_memory_key": "input:chat input",
    "cached": false
  }
],
"element_count": 11,
"current_elements": [
  {
```

```
"id": "tab_frontend",
"type": "tab",
"label": "frontend",
"description": "Browser tab labeled frontend",
"state": "visible",
"dx": 116,
"dy": 52,
"frame_dx": 272,
"frame_dy": 154,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    156,
    126,
    388,
    182
],
"session_memory_key": "tab:frontend",
"cached": false
},
{
    "id": "tab_courseweb",
    "type": "tab",
    "label": "Log in to the site | CourseWeb",
    "description": "Browser tab for CourseWeb login",
    "state": "visible",
    "dx": 320,
    "dy": 52,
    "frame_dx": 476,
```

```
"frame_dy": 154,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
  348,
  126,
  604,
  182
],
"session_memory_key": "tab:log in to the site | courseweb",
"cached": false
},
{
  "id": "address_bar",
  "type": "input",
  "label": "Address bar",
  "description": "Browser address bar showing localhost:5173",
  "state": "visible",
  "dx": 576,
  "dy": 112,
  "frame_dx": 732,
  "frame_dy": 214,
  "confidence": 0.98,
  "source": "gemini_vlm",
  "frame_bbox": [
    156,
    182,
    1308,
    246
```

```
    ],
    "session_memory_key": "input:address bar",
    "cached": false
  },
  {
    "id": "app_logo",
    "type": "icon",
    "label": "ScreenPilot AI Logo",
    "description": "Chip icon for ScreenPilot AI",
    "state": "visible",
    "dx": 56,
    "dy": 192,
    "frame_dx": 212,
    "frame_dy": 294,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
      196,
      270,
      228,
      318
    ],
    "session_memory_key": "icon:screenpilot ai logo",
    "cached": false
  },
  {
    "id": "app_title",
    "type": "heading",
    "label": "ScreenPilot AI",
```

```
"description": "Application title text",
"state": "visible",
"dx": 144,
"dy": 192,
"frame_dx": 300,
"frame_dy": 294,
"confidence": 0.98,
"source": "gemini_vlm",
"frame_bbox": [
    236,
    278,
    364,
    310
],
"session_memory_key": "heading:screenpilot ai",
"cached": false
},
{
    "id": "app_version",
    "type": "label",
    "label": "Autonomous Task Agent v2",
    "description": "Application version badge",
    "state": "visible",
    "dx": 300,
    "dy": 196,
    "frame_dx": 456,
    "frame_dy": 298,
    "confidence": 0.95,
    "source": "gemini_vlm",
```

```
"frame_bbox": [
  380,
  286,
  532,
  310
],
"session_memory_key": "label:autonomous task agent v2",
"cached": false
},
{
  "id": "message_bubble",
  "type": "pane",
  "label": "Agent message",
  "description": "Chat bubble with text: I will now open the browser and
login to the courseweb",
  "state": "visible",
  "dx": 928,
  "dy": 340,
  "frame_dx": 1084,
  "frame_dy": 442,
  "confidence": 0.92,
  "source": "gemini_vlm",
  "frame_bbox": [
    732,
    374,
    1436,
    510
  ],
  "session_memory_key": "pane:agent message",
```

```
    "cached": false
  },
  {
    "id": "status_connecting",
    "type": "label",
    "label": "Connecting to agent...",
    "description": "Status message indicating connection progress",
    "state": "visible",
    "dx": 120,
    "dy": 440,
    "frame_dx": 276,
    "frame_dy": 542,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
      188,
      526,
      364,
      558
    ],
    "session_memory_key": "label:connecting to agent...",
    "cached": false
  },
  {
    "id": "status_capturing",
    "type": "label",
    "label": "Capturing initial screen...",
    "description": "Status message indicating screen capture progress",
    "state": "visible",
```



```
"dx": 148,
"dy": 468,
"frame_dx": 304,
"frame_dy": 570,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
  188,
  558,
  420,
  582
],
"session_memory_key": "label:capturing initial screen...",
"cached": false
},
{
  "id": "loading_indicator",
  "type": "pane",
  "label": "Loading indicator",
  "description": "Box showing loading dots and a cursor",
  "state": "visible",
  "dx": 124,
  "dy": 596,
  "frame_dx": 280,
  "frame_dy": 698,
  "confidence": 0.85,
  "source": "gemini_vlm",
  "frame_bbox": [
    180,
```

```
    598,  
    380,  
    798  
  ],  
  "session_memory_key": "pane:loading indicator",  
  "cached": false  
},  
{  
  "id": "chat_input_field",  
  "type": "input",  
  "label": "Chat input",  
  "description": "Input field with placeholder: Agent is working...",  
  "state": "visible",  
  "dx": 684,  
  "dy": 844,  
  "frame_dx": 840,  
  "frame_dy": 946,  
  "confidence": 0.98,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    156,  
    894,  
    1524,  
    998  
  ],  
  "session_memory_key": "input:chat input",  
  "cached": false  
}  
],
```

```
"resolved_elements": [  
  {  
    "id": "tab_frontend",  
    "type": "tab",  
    "label": "frontend",  
    "description": "Browser tab labeled frontend",  
    "state": "visible",  
    "dx": 116,  
    "dy": 52,  
    "frame_dx": 272,  
    "frame_dy": 154,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
      156,  
      126,  
      388,  
      182  
    ],  
    "session_memory_key": "tab:frontend",  
    "cached": false  
  },  
  {  
    "id": "tab_courseweb",  
    "type": "tab",  
    "label": "Log in to the site | CourseWeb",  
    "description": "Browser tab for CourseWeb login",  
    "state": "visible",  
    "dx": 320,
```

```
"dy": 52,
"frame_dx": 476,
"frame_dy": 154,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    348,
    126,
    604,
    182
],
"session_memory_key": "tab:log in to the site | courseweb",
"cached": false
},
{
    "id": "address_bar",
    "type": "input",
    "label": "Address bar",
    "description": "Browser address bar showing localhost:5173",
    "state": "visible",
    "dx": 576,
    "dy": 112,
    "frame_dx": 732,
    "frame_dy": 214,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
        156,
        182,
```

```
        1308,  
        246  
    ],  
    "session_memory_key": "input:address bar",  
    "cached": false  
},  
{  
    "id": "app_logo",  
    "type": "icon",  
    "label": "ScreenPilot AI Logo",  
    "description": "Chip icon for ScreenPilot AI",  
    "state": "visible",  
    "dx": 56,  
    "dy": 192,  
    "frame_dx": 212,  
    "frame_dy": 294,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        196,  
        270,  
        228,  
        318  
    ],  
    "session_memory_key": "icon:screenpilot ai logo",  
    "cached": false  
},  
{  
    "id": "app_title",
```

```
"type": "heading",
"label": "ScreenPilot AI",
"description": "Application title text",
"state": "visible",
"dx": 144,
"dy": 192,
"frame_dx": 300,
"frame_dy": 294,
"confidence": 0.98,
"source": "gemini_vlm",
"frame_bbox": [
    236,
    278,
    364,
    310
],
"session_memory_key": "heading:screenpilot ai",
"cached": false
},
{
    "id": "app_version",
    "type": "label",
    "label": "Autonomous Task Agent v2",
    "description": "Application version badge",
    "state": "visible",
    "dx": 300,
    "dy": 196,
    "frame_dx": 456,
    "frame_dy": 298,
```

```
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    380,
    286,
    532,
    310
],
"session_memory_key": "label:autonomous task agent v2",
"cached": false
},
{
    "id": "message_bubble",
    "type": "pane",
    "label": "Agent message",
    "description": "Chat bubble with text: I will now open the browser and
login to the courseweb",
    "state": "visible",
    "dx": 928,
    "dy": 340,
    "frame_dx": 1084,
    "frame_dy": 442,
    "confidence": 0.92,
    "source": "gemini_vlm",
    "frame_bbox": [
        732,
        374,
        1436,
        510
```

```
    ],
    "session_memory_key": "pane:agent message",
    "cached": false
  },
  {
    "id": "status_connecting",
    "type": "label",
    "label": "Connecting to agent...",
    "description": "Status message indicating connection progress",
    "state": "visible",
    "dx": 120,
    "dy": 440,
    "frame_dx": 276,
    "frame_dy": 542,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
      188,
      526,
      364,
      558
    ],
    "session_memory_key": "label:connecting to agent...",
    "cached": false
  },
  {
    "id": "status_capturing",
    "type": "label",
    "label": "Capturing initial screen..."
```



```
"description": "Status message indicating screen capture progress",
"state": "visible",
"dx": 148,
"dy": 468,
"frame_dx": 304,
"frame_dy": 570,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    188,
    558,
    420,
    582
],
"session_memory_key": "label:capturing initial screen...",
"cached": false
},
{
    "id": "loading_indicator",
    "type": "pane",
    "label": "Loading indicator",
    "description": "Box showing loading dots and a cursor",
    "state": "visible",
    "dx": 124,
    "dy": 596,
    "frame_dx": 280,
    "frame_dy": 698,
    "confidence": 0.85,
    "source": "gemini_vlm",
```

```
"frame_bbox": [
    180,
    598,
    380,
    798
],
"session_memory_key": "pane:loading indicator",
"cached": false
},
{
    "id": "chat_input_field",
    "type": "input",
    "label": "Chat input",
    "description": "Input field with placeholder: Agent is working...",
    "state": "visible",
    "dx": 684,
    "dy": 844,
    "frame_dx": 840,
    "frame_dy": 946,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
        156,
        894,
        1524,
        998
    ],
    "session_memory_key": "input:chat input",
    "cached": false
}
```

```
}
],
"session_knowledge_base": {
  "session_id": "shot_1777704127_f68076",
  "started_at": "2026-05-02T06:42:08.013863+00:00",
  "updated_at": "2026-05-02T06:42:44.638544+00:00",
  "stale_after_misses": 3,
  "confidence_floor": 0.4,
  "decay_per_miss": 0.9,
  "frame_count": 1,
  "frame_name": "frame_1777704128_8d3627.jpg",
  "frame_index": 0,
  "image_size": {
    "width": 1920,
    "height": 1080
  },
},
"known_elements": {
  "tab:frontend": {
    "element_key": "tab:frontend",
    "element_type": "tab",
    "label": "frontend",
    "bbox": [
      156.0,
      126.0,
      388.0,
      182.0
    ],
    "confidence": 0.95,
    "source": "gemini_vlm",
  },
}
```

```
"first_seen_frame": "frame_1777704128_8d3627.jpg",
"last_seen_frame": "frame_1777704128_8d3627.jpg",
"first_seen_at": "2026-05-02T06:42:44.638544+00:00",
"last_seen_at": "2026-05-02T06:42:44.638544+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777704128_8d3627.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T06:42:44.638544+00:00",
    "confidence": 0.95
  }
],
},
"tab:log in to the site | courseweb": {
  "element_key": "tab:log in to the site | courseweb",
  "element_type": "tab",
  "label": "Log in to the site | CourseWeb",
  "bbox": [
    348.0,
    126.0,
    604.0,
    182.0
  ],
},
```

```
"confidence": 0.95,
"source": "gemini_vlm",
"first_seen_frame": "frame_1777704128_8d3627.jpg",
"last_seen_frame": "frame_1777704128_8d3627.jpg",
"first_seen_at": "2026-05-02T06:42:44.638544+00:00",
"last_seen_at": "2026-05-02T06:42:44.638544+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777704128_8d3627.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T06:42:44.638544+00:00",
    "confidence": 0.95
  }
],
},
"input:address bar": {
  "element_key": "input:address bar",
  "element_type": "input",
  "label": "Address bar",
  "bbox": [
    156.0,
    182.0,
    1308.0,
```

```
246.0

],

"confidence": 0.98,

"source": "gemini_vlm",

"first_seen_frame": "frame_1777704128_8d3627.jpg",

"last_seen_frame": "frame_1777704128_8d3627.jpg",

"first_seen_at": "2026-05-02T06:42:44.638544+00:00",

"last_seen_at": "2026-05-02T06:42:44.638544+00:00",

"first_seen_index": 0,

"last_seen_index": 0,

"seen_count": 1,

"miss_count": 0,

"stale": false,

"cached": false,

"history": [

  {

    "frame_name": "frame_1777704128_8d3627.jpg",

    "frame_index": 0,

    "seen_at": "2026-05-02T06:42:44.638544+00:00",

    "confidence": 0.98

  }

]

},

"icon:screenpilot ai logo": {

  "element_key": "icon:screenpilot ai logo",

  "element_type": "icon",

  "label": "ScreenPilot AI Logo",

  "bbox": [

    196.0,
```

```
    270.0,
    228.0,
    318.0
  ],
  "confidence": 0.9,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777704128_8d3627.jpg",
  "last_seen_frame": "frame_1777704128_8d3627.jpg",
  "first_seen_at": "2026-05-02T06:42:44.638544+00:00",
  "last_seen_at": "2026-05-02T06:42:44.638544+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
  "seen_count": 1,
  "miss_count": 0,
  "stale": false,
  "cached": false,
  "history": [
    {
      "frame_name": "frame_1777704128_8d3627.jpg",
      "frame_index": 0,
      "seen_at": "2026-05-02T06:42:44.638544+00:00",
      "confidence": 0.9
    }
  ]
},
"heading:screenpilot ai": {
  "element_key": "heading:screenpilot ai",
  "element_type": "heading",
  "label": "ScreenPilot AI",
```

```
"bbox": [
  236.0,
  278.0,
  364.0,
  310.0
],
"confidence": 0.98,
"source": "gemini_vlm",
"first_seen_frame": "frame_1777704128_8d3627.jpg",
"last_seen_frame": "frame_1777704128_8d3627.jpg",
"first_seen_at": "2026-05-02T06:42:44.638544+00:00",
"last_seen_at": "2026-05-02T06:42:44.638544+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777704128_8d3627.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T06:42:44.638544+00:00",
    "confidence": 0.98
  }
]
},
"label:autonomous task agent v2": {
  "element_key": "label:autonomous task agent v2",
```



```
"element_type": "label",
"label": "Autonomous Task Agent v2",
"bbox": [
    380.0,
    286.0,
    532.0,
    310.0
],
"confidence": 0.95,
"source": "gemini_vlm",
"first_seen_frame": "frame_1777704128_8d3627.jpg",
"last_seen_frame": "frame_1777704128_8d3627.jpg",
"first_seen_at": "2026-05-02T06:42:44.638544+00:00",
"last_seen_at": "2026-05-02T06:42:44.638544+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
    {
        "frame_name": "frame_1777704128_8d3627.jpg",
        "frame_index": 0,
        "seen_at": "2026-05-02T06:42:44.638544+00:00",
        "confidence": 0.95
    }
]
},
```

```
"pane:agent message": {
  "element_key": "pane:agent message",
  "element_type": "pane",
  "label": "Agent message",
  "bbox": [
    732.0,
    374.0,
    1436.0,
    510.0
  ],
  "confidence": 0.92,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777704128_8d3627.jpg",
  "last_seen_frame": "frame_1777704128_8d3627.jpg",
  "first_seen_at": "2026-05-02T06:42:44.638544+00:00",
  "last_seen_at": "2026-05-02T06:42:44.638544+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
  "seen_count": 1,
  "miss_count": 0,
  "stale": false,
  "cached": false,
  "history": [
    {
      "frame_name": "frame_1777704128_8d3627.jpg",
      "frame_index": 0,
      "seen_at": "2026-05-02T06:42:44.638544+00:00",
      "confidence": 0.92
    }
  ]
}
```

```
]
},
"label:connecting to agent...": {
  "element_key": "label:connecting to agent...",
  "element_type": "label",
  "label": "Connecting to agent...",
  "bbox": [
    188.0,
    526.0,
    364.0,
    558.0
  ],
  "confidence": 0.9,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777704128_8d3627.jpg",
  "last_seen_frame": "frame_1777704128_8d3627.jpg",
  "first_seen_at": "2026-05-02T06:42:44.638544+00:00",
  "last_seen_at": "2026-05-02T06:42:44.638544+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
  "seen_count": 1,
  "miss_count": 0,
  "stale": false,
  "cached": false,
  "history": [
    {
      "frame_name": "frame_1777704128_8d3627.jpg",
      "frame_index": 0,
      "seen_at": "2026-05-02T06:42:44.638544+00:00",
```

```
        "confidence": 0.9
      }
    ]
  },
  "label:capturing initial screen...": {
    "element_key": "label:capturing initial screen...",
    "element_type": "label",
    "label": "Capturing initial screen...",
    "bbox": [
      188.0,
      558.0,
      420.0,
      582.0
    ],
    "confidence": 0.9,
    "source": "gemini_vlm",
    "first_seen_frame": "frame_1777704128_8d3627.jpg",
    "last_seen_frame": "frame_1777704128_8d3627.jpg",
    "first_seen_at": "2026-05-02T06:42:44.638544+00:00",
    "last_seen_at": "2026-05-02T06:42:44.638544+00:00",
    "first_seen_index": 0,
    "last_seen_index": 0,
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false,
    "history": [
      {
        "frame_name": "frame_1777704128_8d3627.jpg",
```

```
        "frame_index": 0,
        "seen_at": "2026-05-02T06:42:44.638544+00:00",
        "confidence": 0.9
    }
]
},
"pane:loading indicator": {
    "element_key": "pane:loading indicator",
    "element_type": "pane",
    "label": "Loading indicator",
    "bbox": [
        180.0,
        598.0,
        380.0,
        798.0
    ],
    "confidence": 0.85,
    "source": "gemini_vlm",
    "first_seen_frame": "frame_1777704128_8d3627.jpg",
    "last_seen_frame": "frame_1777704128_8d3627.jpg",
    "first_seen_at": "2026-05-02T06:42:44.638544+00:00",
    "last_seen_at": "2026-05-02T06:42:44.638544+00:00",
    "first_seen_index": 0,
    "last_seen_index": 0,
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false,
    "history": [
```

```
{
  "frame_name": "frame_1777704128_8d3627.jpg",
  "frame_index": 0,
  "seen_at": "2026-05-02T06:42:44.638544+00:00",
  "confidence": 0.85
}
],
},
"input:chat input": {
  "element_key": "input:chat input",
  "element_type": "input",
  "label": "Chat input",
  "bbox": [
    156.0,
    894.0,
    1524.0,
    998.0
  ],
  "confidence": 0.98,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777704128_8d3627.jpg",
  "last_seen_frame": "frame_1777704128_8d3627.jpg",
  "first_seen_at": "2026-05-02T06:42:44.638544+00:00",
  "last_seen_at": "2026-05-02T06:42:44.638544+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
  "seen_count": 1,
  "miss_count": 0,
  "stale": false,
```

```
"cached": false,
"history": [
  {
    "frame_name": "frame_1777704128_8d3627.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T06:42:44.638544+00:00",
    "confidence": 0.98
  }
]
},
"resolved_elements": [
  {
    "id": "tab_frontend",
    "type": "tab",
    "label": "frontend",
    "description": "Browser tab labeled frontend",
    "state": "visible",
    "dx": 116,
    "dy": 52,
    "frame_dx": 272,
    "frame_dy": 154,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
      156,
      126,
      388,
      182
```

```
    ],  
    "session_memory_key": "tab:frontend",  
    "cached": false  
  },  
  {  
    "id": "tab_courseweb",  
    "type": "tab",  
    "label": "Log in to the site | CourseWeb",  
    "description": "Browser tab for CourseWeb login",  
    "state": "visible",  
    "dx": 320,  
    "dy": 52,  
    "frame_dx": 476,  
    "frame_dy": 154,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
      348,  
      126,  
      604,  
      182  
    ],  
    "session_memory_key": "tab:log in to the site | courseweb",  
    "cached": false  
  },  
  {  
    "id": "address_bar",  
    "type": "input",  
    "label": "Address bar",
```



```
"description": "Browser address bar showing localhost:5173",
"state": "visible",
"dx": 576,
"dy": 112,
"frame_dx": 732,
"frame_dy": 214,
"confidence": 0.98,
"source": "gemini_vlm",
"frame_bbox": [
    156,
    182,
    1308,
    246
],
"session_memory_key": "input:address bar",
"cached": false
},
{
    "id": "app_logo",
    "type": "icon",
    "label": "ScreenPilot AI Logo",
    "description": "Chip icon for ScreenPilot AI",
    "state": "visible",
    "dx": 56,
    "dy": 192,
    "frame_dx": 212,
    "frame_dy": 294,
    "confidence": 0.9,
    "source": "gemini_vlm",
```

```
"frame_bbox": [
    196,
    270,
    228,
    318
],
"session_memory_key": "icon:screenpilot ai logo",
"cached": false
},
{
    "id": "app_title",
    "type": "heading",
    "label": "ScreenPilot AI",
    "description": "Application title text",
    "state": "visible",
    "dx": 144,
    "dy": 192,
    "frame_dx": 300,
    "frame_dy": 294,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
        236,
        278,
        364,
        310
    ],
    "session_memory_key": "heading:screenpilot ai",
    "cached": false
}
```

```
},  
{  
  "id": "app_version",  
  "type": "label",  
  "label": "Autonomous Task Agent v2",  
  "description": "Application version badge",  
  "state": "visible",  
  "dx": 300,  
  "dy": 196,  
  "frame_dx": 456,  
  "frame_dy": 298,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    380,  
    286,  
    532,  
    310  
  ],  
  "session_memory_key": "label:autonomous task agent v2",  
  "cached": false  
},  
{  
  "id": "message_bubble",  
  "type": "pane",  
  "label": "Agent message",  
  "description": "Chat bubble with text: I will now open the browser and  
login to the courseweb",  
  "state": "visible",
```

```
"dx": 928,  
  "dy": 340,  
  "frame_dx": 1084,  
  "frame_dy": 442,  
  "confidence": 0.92,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    732,  
    374,  
    1436,  
    510  
  ],  
  "session_memory_key": "pane:agent message",  
  "cached": false  
},  
{  
  "id": "status_connecting",  
  "type": "label",  
  "label": "Connecting to agent...",  
  "description": "Status message indicating connection progress",  
  "state": "visible",  
  "dx": 120,  
  "dy": 440,  
  "frame_dx": 276,  
  "frame_dy": 542,  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    188,
```

```
        526,  
        364,  
        558  
    ],  
    "session_memory_key": "label:connecting to agent...",  
    "cached": false  
},  
{  
    "id": "status_capturing",  
    "type": "label",  
    "label": "Capturing initial screen...",  
    "description": "Status message indicating screen capture progress",  
    "state": "visible",  
    "dx": 148,  
    "dy": 468,  
    "frame_dx": 304,  
    "frame_dy": 570,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        188,  
        558,  
        420,  
        582  
    ],  
    "session_memory_key": "label:capturing initial screen...",  
    "cached": false  
},  
{
```

```
"id": "loading_indicator",
"type": "pane",
"label": "Loading indicator",
"description": "Box showing loading dots and a cursor",
"state": "visible",
"dx": 124,
"dy": 596,
"frame_dx": 280,
"frame_dy": 698,
"confidence": 0.85,
"source": "gemini_vlm",
"frame_bbox": [
    180,
    598,
    380,
    798
],
"session_memory_key": "pane:loading indicator",
"cached": false
},
{
    "id": "chat_input_field",
    "type": "input",
    "label": "Chat input",
    "description": "Input field with placeholder: Agent is working...",
    "state": "visible",
    "dx": 684,
    "dy": 844,
    "frame_dx": 840,
```

```
"frame_dy": 946,
"confidence": 0.98,
"source": "gemini_vlm",
"frame_bbox": [
    156,
    894,
    1524,
    998
],
"session_memory_key": "input:chat input",
"cached": false
}
],
"current_elements": [
    {
        "id": "tab_frontend",
        "type": "tab",
        "label": "frontend",
        "description": "Browser tab labeled frontend",
        "state": "visible",
        "dx": 116,
        "dy": 52,
        "frame_dx": 272,
        "frame_dy": 154,
        "confidence": 0.95,
        "source": "gemini_vlm",
        "frame_bbox": [
            156,
            126,
```

```
        388,  
        182  
    ],  
    "session_memory_key": "tab:frontend",  
    "cached": false  
},  
{  
    "id": "tab_courseweb",  
    "type": "tab",  
    "label": "Log in to the site | CourseWeb",  
    "description": "Browser tab for CourseWeb login",  
    "state": "visible",  
    "dx": 320,  
    "dy": 52,  
    "frame_dx": 476,  
    "frame_dy": 154,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        348,  
        126,  
        604,  
        182  
    ],  
    "session_memory_key": "tab:log in to the site | courseweb",  
    "cached": false  
},  
{  
    "id": "address_bar",
```



```
"type": "input",
"label": "Address bar",
"description": "Browser address bar showing localhost:5173",
"state": "visible",
"dx": 576,
"dy": 112,
"frame_dx": 732,
"frame_dy": 214,
"confidence": 0.98,
"source": "gemini_vlm",
"frame_bbox": [
  156,
  182,
  1308,
  246
],
"session_memory_key": "input:address bar",
"cached": false
},
{
  "id": "app_logo",
  "type": "icon",
  "label": "ScreenPilot AI Logo",
  "description": "Chip icon for ScreenPilot AI",
  "state": "visible",
  "dx": 56,
  "dy": 192,
  "frame_dx": 212,
  "frame_dy": 294,
```

```
"confidence": 0.9,  
"source": "gemini_vlm",  
"frame_bbox": [  
    196,  
    270,  
    228,  
    318  
],  
"session_memory_key": "icon:screenpilot ai logo",  
"cached": false  
},  
{  
    "id": "app_title",  
    "type": "heading",  
    "label": "ScreenPilot AI",  
    "description": "Application title text",  
    "state": "visible",  
    "dx": 144,  
    "dy": 192,  
    "frame_dx": 300,  
    "frame_dy": 294,  
    "confidence": 0.98,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        236,  
        278,  
        364,  
        310  
    ],
```

```
"session_memory_key": "heading:screenpilot ai",
"cached": false
},
{
  "id": "app_version",
  "type": "label",
  "label": "Autonomous Task Agent v2",
  "description": "Application version badge",
  "state": "visible",
  "dx": 300,
  "dy": 196,
  "frame_dx": 456,
  "frame_dy": 298,
  "confidence": 0.95,
  "source": "gemini_vlm",
  "frame_bbox": [
    380,
    286,
    532,
    310
  ],
  "session_memory_key": "label:autonomous task agent v2",
  "cached": false
},
{
  "id": "message_bubble",
  "type": "pane",
  "label": "Agent message",
```

```
    "description": "Chat bubble with text: I will now open the browser and  
login to the courseweb",  
    "state": "visible",  
    "dx": 928,  
    "dy": 340,  
    "frame_dx": 1084,  
    "frame_dy": 442,  
    "confidence": 0.92,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        732,  
        374,  
        1436,  
        510  
    ],  
    "session_memory_key": "pane:agent message",  
    "cached": false  
},  
{  
    "id": "status_connecting",  
    "type": "label",  
    "label": "Connecting to agent...",  
    "description": "Status message indicating connection progress",  
    "state": "visible",  
    "dx": 120,  
    "dy": 440,  
    "frame_dx": 276,  
    "frame_dy": 542,  
    "confidence": 0.9,
```

```
"source": "gemini_vlm",
"frame_bbox": [
    188,
    526,
    364,
    558
],
"session_memory_key": "label:connecting to agent...",
"cached": false
},
{
    "id": "status_capturing",
    "type": "label",
    "label": "Capturing initial screen...",
    "description": "Status message indicating screen capture progress",
    "state": "visible",
    "dx": 148,
    "dy": 468,
    "frame_dx": 304,
    "frame_dy": 570,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        188,
        558,
        420,
        582
    ],
    "session_memory_key": "label:capturing initial screen..."
```

```
    "cached": false
  },
  {
    "id": "loading_indicator",
    "type": "pane",
    "label": "Loading indicator",
    "description": "Box showing loading dots and a cursor",
    "state": "visible",
    "dx": 124,
    "dy": 596,
    "frame_dx": 280,
    "frame_dy": 698,
    "confidence": 0.85,
    "source": "gemini_vlm",
    "frame_bbox": [
      180,
      598,
      380,
      798
    ],
    "session_memory_key": "pane:loading indicator",
    "cached": false
  },
  {
    "id": "chat_input_field",
    "type": "input",
    "label": "Chat input",
    "description": "Input field with placeholder: Agent is working...",
    "state": "visible",
```

```
    "dx": 684,
    "dy": 844,
    "frame_dx": 840,
    "frame_dy": 946,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
      156,
      894,
      1524,
      998
    ],
    "session_memory_key": "input:chat input",
    "cached": false
  }
]
},
"session_memory": {
  "run_id": "30427c77-968c-4a27-a252-6b8db30fe924",
  "user_task": "click on the \"Log in to the site | CourseWeb\" tab on this browser and login to the courseweb",
  "started_at": "2026-05-02T06:41:57.870381+00:00",
  "updated_at": "2026-05-02T06:42:44.685964+00:00",
  "stale_after_misses": 2,
  "confidence_floor": 0.4,
  "decay_per_miss": 0.9,
  "frame_index": 1,
  "known_elements": {
    "tab:frontend": {
```

```
"element_key": "tab:frontend",
"element_type": "tab",
"label": "frontend",
"bbox": [
  156.0,
  126.0,
  388.0,
  182.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 0,
"first_seen_at": "2026-05-02T06:42:44.685964+00:00",
"last_seen_at": "2026-05-02T06:42:44.685964+00:00",
"seen_count": 1,
"miss_count": 11,
"stale": true,
"cached": true
},
"tab:log in to the site | courseweb": {
  "element_key": "tab:log in to the site | courseweb",
  "element_type": "tab",
  "label": "Log in to the site | CourseWeb",
  "bbox": [
    357.0,
    116.0,
```



```
        621.0,  
        172.0  
    ],  
    "confidence": 0.4,  
    "source": "gemini_vlm",  
    "first_seen_frame": null,  
    "last_seen_frame": null,  
    "first_seen_index": 0,  
    "last_seen_index": 1,  
    "first_seen_at": "2026-05-02T06:42:44.685964+00:00",  
    "last_seen_at": "2026-05-02T06:43:27.857541+00:00",  
    "seen_count": 2,  
    "miss_count": 10,  
    "stale": true,  
    "cached": true  
},  
"input:address bar": {  
    "element_key": "input:address bar",  
    "element_type": "input",  
    "label": "Address Bar",  
    "bbox": [  
        269.0,  
        172.0,  
        1341.0,  
        228.0  
    ],  
    "confidence": 0.4,  
    "source": "gemini_vlm",  
    "first_seen_frame": null,
```

```
"last_seen_frame": null,  
"first_seen_index": 0,  
"last_seen_index": 1,  
"first_seen_at": "2026-05-02T06:42:44.685964+00:00",  
"last_seen_at": "2026-05-02T06:43:27.857541+00:00",  
"seen_count": 2,  
"miss_count": 10,  
"stale": true,  
"cached": true  
},  
"icon:screenpilot ai logo": {  
  "element_key": "icon:screenpilot ai logo",  
  "element_type": "icon",  
  "label": "ScreenPilot AI Logo",  
  "bbox": [  
    196.0,  
    270.0,  
    228.0,  
    318.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 0,  
  "last_seen_index": 0,  
  "first_seen_at": "2026-05-02T06:42:44.685964+00:00",  
  "last_seen_at": "2026-05-02T06:42:44.685964+00:00",  
  "seen_count": 1,
```

```
    "miss_count": 11,  
    "stale": true,  
    "cached": true  
  },  
  "heading:screenpilot ai": {  
    "element_key": "heading:screenpilot ai",  
    "element_type": "heading",  
    "label": "ScreenPilot AI",  
    "bbox": [  
      236.0,  
      278.0,  
      364.0,  
      310.0  
    ],  
    "confidence": 0.4,  
    "source": "gemini_vlm",  
    "first_seen_frame": null,  
    "last_seen_frame": null,  
    "first_seen_index": 0,  
    "last_seen_index": 0,  
    "first_seen_at": "2026-05-02T06:42:44.685964+00:00",  
    "last_seen_at": "2026-05-02T06:42:44.685964+00:00",  
    "seen_count": 1,  
    "miss_count": 11,  
    "stale": true,  
    "cached": true  
  },  
  "label:autonomous task agent v2": {  
    "element_key": "label:autonomous task agent v2",
```

```
"element_type": "label",
"label": "Autonomous Task Agent v2",
"bbox": [
    380.0,
    286.0,
    532.0,
    310.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 1,
"first_seen_at": "2026-05-02T06:42:44.685964+00:00",
"last_seen_at": "2026-05-02T06:43:27.857541+00:00",
"seen_count": 2,
"miss_count": 10,
"stale": true,
"cached": true
},
"pane:agent message": {
    "element_key": "pane:agent message",
    "element_type": "pane",
    "label": "Agent message",
    "bbox": [
        732.0,
        374.0,
        1436.0,
```

```
510.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 0,
"first_seen_at": "2026-05-02T06:42:44.685964+00:00",
"last_seen_at": "2026-05-02T06:42:44.685964+00:00",
"seen_count": 1,
"miss_count": 11,
"stale": true,
"cached": true
},
"label:connecting to agent...": {
  "element_key": "label:connecting to agent...",
  "element_type": "label",
  "label": "Connecting to agent...",
  "bbox": [
    188.0,
    526.0,
    364.0,
    558.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
```

```
"first_seen_index": 0,
"last_seen_index": 0,
"first_seen_at": "2026-05-02T06:42:44.685964+00:00",
"last_seen_at": "2026-05-02T06:42:44.685964+00:00",
"seen_count": 1,
"miss_count": 11,
"stale": true,
"cached": true
},
"label:capturing initial screen...": {
  "element_key": "label:capturing initial screen...",
  "element_type": "label",
  "label": "Capturing initial screen...",
  "bbox": [
    188.0,
    558.0,
    420.0,
    582.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 0,
  "first_seen_at": "2026-05-02T06:42:44.685964+00:00",
  "last_seen_at": "2026-05-02T06:42:44.685964+00:00",
  "seen_count": 1,
  "miss_count": 11,
```

```
    "stale": true,
    "cached": true
  },
  "pane:loading indicator": {
    "element_key": "pane:loading indicator",
    "element_type": "pane",
    "label": "Loading indicator",
    "bbox": [
      180.0,
      598.0,
      380.0,
      798.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 0,
    "last_seen_index": 0,
    "first_seen_at": "2026-05-02T06:42:44.685964+00:00",
    "last_seen_at": "2026-05-02T06:42:44.685964+00:00",
    "seen_count": 1,
    "miss_count": 11,
    "stale": true,
    "cached": true
  },
  "input:chat input": {
    "element_key": "input:chat input",
    "element_type": "input",
```

```
"label": "Chat input",
"bbox": [
  156.0,
  894.0,
  1524.0,
  998.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 0,
"first_seen_at": "2026-05-02T06:42:44.685964+00:00",
"last_seen_at": "2026-05-02T06:42:44.685964+00:00",
"seen_count": 1,
"miss_count": 11,
"stale": true,
"cached": true
},
"tab:localhost:5173": {
  "element_key": "tab:localhost:5173",
  "element_type": "tab",
  "label": "localhost:5173",
  "bbox": [
    269.0,
    116.0,
    405.0,
    156.0
```



```
],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
  "last_seen_index": 1,
  "first_seen_at": "2026-05-02T06:43:27.857541+00:00",
  "last_seen_at": "2026-05-02T06:43:27.857541+00:00",
  "seen_count": 1,
  "miss_count": 10,
  "stale": true,
  "cached": true
},
"text:screenpilot ai": {
  "element_key": "text:screenpilot ai",
  "element_type": "text",
  "label": "ScreenPilot AI",
  "bbox": [
    277.0,
    260.0,
    413.0,
    276.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
```

```
"last_seen_index": 1,
"first_seen_at": "2026-05-02T06:43:27.857541+00:00",
"last_seen_at": "2026-05-02T06:43:27.857541+00:00",
"seen_count": 1,
"miss_count": 10,
"stale": true,
"cached": true
},
"heading:task plan": {
  "element_key": "heading:task plan",
  "element_type": "heading",
  "label": "Task Plan",
  "bbox": [
    269.0,
    508.0,
    341.0,
    540.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
  "last_seen_index": 1,
  "first_seen_at": "2026-05-02T06:43:27.857541+00:00",
  "last_seen_at": "2026-05-02T06:43:27.857541+00:00",
  "seen_count": 1,
  "miss_count": 10,
  "stale": true,
```

```
    "cached": true
  },
  "label:0/2 done": {
    "element_key": "label:0/2 done",
    "element_type": "label",
    "label": "0/2 done",
    "bbox": [
      757.0,
      516.0,
      821.0,
      540.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 1,
    "last_seen_index": 1,
    "first_seen_at": "2026-05-02T06:43:27.857541+00:00",
    "last_seen_at": "2026-05-02T06:43:27.857541+00:00",
    "seen_count": 1,
    "miss_count": 10,
    "stale": true,
    "cached": true
  },
  "pane:step 1 highlight": {
    "element_key": "pane:step 1 highlight",
    "element_type": "pane",
    "label": "Step 1 Highlight",
```

```
"bbox": [
  269.0,
  596.0,
  845.0,
  660.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 1,
"last_seen_index": 1,
"first_seen_at": "2026-05-02T06:43:27.857541+00:00",
"last_seen_at": "2026-05-02T06:43:27.857541+00:00",
"seen_count": 1,
"miss_count": 10,
"stale": true,
"cached": true
},
"text:click on the 'log in to the site | courseweb' tab": {
  "element_key": "text:click on the 'log in to the site | courseweb'
tab",
  "element_type": "text",
  "label": "Click on the 'Log in to the site | CourseWeb' tab",
  "bbox": [
    269.0,
    612.0,
    621.0,
    644.0
```

```
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 1,
    "last_seen_index": 1,
    "first_seen_at": "2026-05-02T06:43:27.857541+00:00",
    "last_seen_at": "2026-05-02T06:43:27.857541+00:00",
    "seen_count": 1,
    "miss_count": 10,
    "stale": true,
    "cached": true
  },
  "text:enter the username and password into the input fields and click the login button": {
    "element_key": "text:enter the username and password into the input fields and click the login button",
    "element_type": "text",
    "label": "Enter the username and password into the input fields and click the login button",
    "bbox": [
      269.0,
      660.0,
      733.0,
      692.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
```

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"last_seen_frame": null,
"first_seen_index": 1,
"last_seen_index": 1,
"first_seen_at": "2026-05-02T06:43:27.857541+00:00",
"last_seen_at": "2026-05-02T06:43:27.857541+00:00",
"seen_count": 1,
"miss_count": 10,
"stale": true,
"cached": true
},
"pane:terminal output": {
  "element_key": "pane:terminal output",
  "element_type": "pane",
  "label": "Terminal Output",
  "bbox": [
    269.0,
    700.0,
    917.0,
    932.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
  "last_seen_index": 1,
  "first_seen_at": "2026-05-02T06:43:27.857541+00:00",
  "last_seen_at": "2026-05-02T06:43:27.857541+00:00",
  "seen_count": 1,
```

```
    "miss_count": 10,  
    "stale": true,  
    "cached": true  
  },  
  "text:agent is working...": {  
    "element_key": "text:agent is working...",  
    "element_type": "text",  
    "label": "Agent is working...",  
    "bbox": [  
      269.0,  
      1014.0,  
      374.0,  
      1015.0  
    ],  
    "confidence": 0.4,  
    "source": "gemini_vlm",  
    "first_seen_frame": null,  
    "last_seen_frame": null,  
    "first_seen_index": 1,  
    "last_seen_index": 1,  
    "first_seen_at": "2026-05-02T06:43:27.857541+00:00",  
    "last_seen_at": "2026-05-02T06:43:27.857541+00:00",  
    "seen_count": 1,  
    "miss_count": 10,  
    "stale": true,  
    "cached": true  
  },  
  "button:send message": {  
    "element_key": "button:send message",
```

```
"element_type": "button",
"label": "Send Message",
"bbox": [
    1333.0,
    1014.0,
    1380.0,
    1015.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 1,
"last_seen_index": 1,
"first_seen_at": "2026-05-02T06:43:27.857541+00:00",
"last_seen_at": "2026-05-02T06:43:27.857541+00:00",
"seen_count": 1,
"miss_count": 10,
"stale": true,
"cached": true
},
"button:start": {
    "element_key": "button:start",
    "element_type": "button",
    "label": "Start",
    "bbox": [
        882.0,
        943.0,
        906.0,
```



```
    963.0
  ],
  "confidence": 0.729,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
  "last_seen_index": 9,
  "first_seen_at": "2026-05-02T06:43:27.857541+00:00",
  "last_seen_at": "2026-05-02T06:50:04.893203+00:00",
  "seen_count": 2,
  "miss_count": 2,
  "stale": true,
  "cached": true
},
"input:search": {
  "element_key": "input:search",
  "element_type": "input",
  "label": "Search",
  "bbox": [
    544.0,
    1014.0,
    713.0,
    1015.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
```

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"first_seen_index": 1,
"last_seen_index": 1,
"first_seen_at": "2026-05-02T06:43:27.857541+00:00",
"last_seen_at": "2026-05-02T06:43:27.857541+00:00",
"seen_count": 1,
"miss_count": 10,
"stale": true,
"cached": true
},
"icon:vs code": {
  "element_key": "icon:vs code",
  "element_type": "icon",
  "label": "VS Code",
  "bbox": [
    994.0,
    1014.0,
    1017.0,
    1015.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
  "last_seen_index": 1,
  "first_seen_at": "2026-05-02T06:43:27.857541+00:00",
  "last_seen_at": "2026-05-02T06:43:27.857541+00:00",
  "seen_count": 1,
  "miss_count": 10,
```

```
    "stale": true,
    "cached": true
  },
  "icon:spotify": {
    "element_key": "icon:spotify",
    "element_type": "icon",
    "label": "Spotify",
    "bbox": [
      826.0,
      943.0,
      866.0,
      962.0
    ],
    "confidence": 0.5905,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 1,
    "last_seen_index": 7,
    "first_seen_at": "2026-05-02T06:43:27.857541+00:00",
    "last_seen_at": "2026-05-02T06:48:31.574903+00:00",
    "seen_count": 2,
    "miss_count": 4,
    "stale": true,
    "cached": true
  },
  "icon:chrome": {
    "element_key": "icon:chrome",
    "element_type": "icon",
```

```
"label": "Chrome",
"bbox": [
  1061.0,
  1014.0,
  1084.0,
  1015.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 1,
"last_seen_index": 1,
"first_seen_at": "2026-05-02T06:43:27.857541+00:00",
"last_seen_at": "2026-05-02T06:43:27.857541+00:00",
"seen_count": 1,
"miss_count": 10,
"stale": true,
"cached": true
},
"text:time and date": {
  "element_key": "text:time and date",
  "element_type": "text",
  "label": "Time and Date",
  "bbox": [
    1357.0,
    1014.0,
    1409.0,
    1015.0
  ]
}
```

```
],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
  "last_seen_index": 1,
  "first_seen_at": "2026-05-02T06:43:27.857541+00:00",
  "last_seen_at": "2026-05-02T06:43:27.857541+00:00",
  "seen_count": 1,
  "miss_count": 10,
  "stale": true,
  "cached": true
},
"button:ask gemini": {
  "element_key": "button:ask gemini",
  "element_type": "button",
  "label": "Ask Gemini",
  "bbox": [
    1197.0,
    156.0,
    1309.0,
    180.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
```

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"last_seen_index": 1,
"first_seen_at": "2026-05-02T06:43:27.857541+00:00",
"last_seen_at": "2026-05-02T06:43:27.857541+00:00",
"seen_count": 1,
"miss_count": 10,
"stale": true,
"cached": true
},
"heading:sliit uni courseweb": {
  "element_key": "heading:sliit uni courseweb",
  "element_type": "heading",
  "label": "SLIIT UNI CourseWeb",
  "bbox": [
    730.0,
    270.0,
    1194.0,
    318.0
  ],
  "confidence": 0.6925,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 2,
  "last_seen_index": 8,
  "first_seen_at": "2026-05-02T06:44:15.667520+00:00",
  "last_seen_at": "2026-05-02T06:49:17.117694+00:00",
  "seen_count": 3,
  "miss_count": 3,
  "stale": true,
```

```
    "cached": true
  },
  "text:note: your login will validate through microsoft o365. always use
'sliit login' button to get access to courseweb.": {
    "element_key": "text:note: your login will validate through microsoft
o365. always use 'sliit login' button to get access to courseweb.",
    "element_type": "text",
    "label": "Note: Your login will validate through Microsoft 0365. Always
use 'SLIIT Login' button to get access to CourseWeb.",
    "bbox": [
      745.0,
      902.0,
      1169.0,
      958.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 3,
    "last_seen_index": 3,
    "first_seen_at": "2026-05-02T06:44:59.983226+00:00",
    "last_seen_at": "2026-05-02T06:44:59.983226+00:00",
    "seen_count": 1,
    "miss_count": 8,
    "stale": true,
    "cached": true
  },
  "text:sliit uni courseweb": {
    "element_key": "text:sliit uni courseweb",
```

```
"element_type": "text",
"label": "SLIIT UNI CourseWeb",
"bbox": [
    730.0,
    263.0,
    1194.0,
    335.0
],
"confidence": 0.4543,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 4,
"last_seen_index": 4,
"first_seen_at": "2026-05-02T06:45:49.004008+00:00",
"last_seen_at": "2026-05-02T06:45:49.004008+00:00",
"seen_count": 1,
"miss_count": 7,
"stale": true,
"cached": true
},
"icon:username icon": {
    "element_key": "icon:username icon",
    "element_type": "icon",
    "label": "Username Icon",
    "bbox": [
        770.0,
        343.0,
        794.0,
```



```
    359.0
  ],
  "confidence": 0.4305,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 4,
  "last_seen_index": 4,
  "first_seen_at": "2026-05-02T06:45:49.004008+00:00",
  "last_seen_at": "2026-05-02T06:45:49.004008+00:00",
  "seen_count": 1,
  "miss_count": 7,
  "stale": true,
  "cached": true
},
"icon:password icon": {
  "element_key": "icon:password icon",
  "element_type": "icon",
  "label": "Password Icon",
  "bbox": [
    770.0,
    391.0,
    794.0,
    407.0
  ],
  "confidence": 0.4305,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
```

```
"first_seen_index": 4,  
"last_seen_index": 4,  
"first_seen_at": "2026-05-02T06:45:49.004008+00:00",  
"last_seen_at": "2026-05-02T06:45:49.004008+00:00",  
"seen_count": 1,  
"miss_count": 7,  
"stale": true,  
"cached": true  
},  
"button:log in": {  
  "element_key": "button:log in",  
  "element_type": "button",  
  "label": "Log in",  
  "bbox": [  
    890.0,  
    430.0,  
    1042.0,  
    470.0  
  ],  
  "confidence": 0.7938,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 4,  
  "last_seen_index": 9,  
  "first_seen_at": "2026-05-02T06:45:49.004008+00:00",  
  "last_seen_at": "2026-05-02T06:50:04.893203+00:00",  
  "seen_count": 5,  
  "miss_count": 2,
```

```
    "stale": true,
    "cached": true
  },
  "text:log in using your account on:": {
    "element_key": "text:log in using your account on:",
    "element_type": "text",
    "label": "Log in using your account on:",
    "bbox": [
      866.0,
      543.0,
      1058.0,
      583.0
    ],
    "confidence": 0.4305,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 4,
    "last_seen_index": 4,
    "first_seen_at": "2026-05-02T06:45:49.004008+00:00",
    "last_seen_at": "2026-05-02T06:45:49.004008+00:00",
    "seen_count": 1,
    "miss_count": 7,
    "stale": true,
    "cached": true
  },
  "button:microsoft login": {
    "element_key": "button:microsoft login",
    "element_type": "button",
```

```
"label": "Microsoft Login",
"bbox": [
  914.0,
  583.0,
  946.0,
  615.0
],
"confidence": 0.44,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 4,
"last_seen_index": 4,
"first_seen_at": "2026-05-02T06:45:49.004008+00:00",
"last_seen_at": "2026-05-02T06:45:49.004008+00:00",
"seen_count": 1,
"miss_count": 7,
"stale": true,
"cached": true
},
"text:is this your first time here?": {
  "element_key": "text:is this your first time here?",
  "element_type": "text",
  "label": "Is this your first time here?",
  "bbox": [
    842.0,
    903.0,
    1074.0,
    943.0
  ]
}
```

```
],  
  "confidence": 0.5905,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 4,  
  "last_seen_index": 7,  
  "first_seen_at": "2026-05-02T06:45:49.004008+00:00",  
  "last_seen_at": "2026-05-02T06:48:31.574903+00:00",  
  "seen_count": 3,  
  "miss_count": 4,  
  "stale": true,  
  "cached": true  
},  
"text:click on sliit login to proceed": {  
  "element_key": "text:click on sliit login to proceed",  
  "element_type": "text",  
  "label": "Click on SLIIT Login to proceed",  
  "bbox": [  
    770.0,  
    895.0,  
    1002.0,  
    919.0  
  ],  
  "confidence": 0.4065,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 4,
```

```
"last_seen_index": 4,  
"first_seen_at": "2026-05-02T06:45:49.004008+00:00",  
"last_seen_at": "2026-05-02T06:45:49.004008+00:00",  
"seen_count": 1,  
"miss_count": 7,  
"stale": true,  
"cached": true  
},  
"pane:windows taskbar": {  
  "element_key": "pane:windows taskbar",  
  "element_type": "pane",  
  "label": "Windows Taskbar",  
  "bbox": [  
    178.0,  
    935.0,  
    1754.0,  
    962.0  
  ],  
  "confidence": 0.643,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 4,  
  "last_seen_index": 7,  
  "first_seen_at": "2026-05-02T06:45:49.004008+00:00",  
  "last_seen_at": "2026-05-02T06:48:31.574903+00:00",  
  "seen_count": 2,  
  "miss_count": 4,  
  "stale": true,
```

```
    "cached": true
  },
  "toolbar:taskbar icons": {
    "element_key": "toolbar:taskbar icons",
    "element_type": "toolbar",
    "label": "Taskbar Icons",
    "bbox": [
      618.0,
      927.0,
      1306.0,
      962.0
    ],
    "confidence": 0.4543,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 4,
    "last_seen_index": 4,
    "first_seen_at": "2026-05-02T06:45:49.004008+00:00",
    "last_seen_at": "2026-05-02T06:45:49.004008+00:00",
    "seen_count": 1,
    "miss_count": 7,
    "stale": true,
    "cached": true
  },
  "text:12:15 pm": {
    "element_key": "text:12:15 pm",
    "element_type": "text",
    "label": "12:15 PM",
```

```
"bbox": [
  1642.0,
  927.0,
  1746.0,
  962.0
],
"confidence": 0.4543,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 4,
"last_seen_index": 4,
"first_seen_at": "2026-05-02T06:45:49.004008+00:00",
"last_seen_at": "2026-05-02T06:45:49.004008+00:00",
"seen_count": 1,
"miss_count": 7,
"stale": true,
"cached": true
},
"icon:mouse cursor": {
  "element_key": "icon:mouse cursor",
  "element_type": "icon",
  "label": "Mouse Cursor",
  "bbox": [
    1594.0,
    719.0,
    1618.0,
    751.0
  ],
}
```



```
"confidence": 0.4305,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 4,
"last_seen_index": 4,
"first_seen_at": "2026-05-02T06:45:49.004008+00:00",
"last_seen_at": "2026-05-02T06:45:49.004008+00:00",
"seen_count": 1,
"miss_count": 7,
"stale": true,
"cached": true
},
"image:sliit uni courseweb logo": {
  "element_key": "image:sliit uni courseweb logo",
  "element_type": "image",
  "label": "SLIIT UNI CourseWeb Logo",
  "bbox": [
    730.0,
    271.0,
    1194.0,
    327.0
  ],
  "confidence": 0.7695,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 5,
  "last_seen_index": 9,
```

```
"first_seen_at": "2026-05-02T06:46:50.824239+00:00",
"last_seen_at": "2026-05-02T06:50:04.893203+00:00",
"seen_count": 2,
"miss_count": 2,
"stale": true,
"cached": true
},
"pane:browser overlay": {
  "element_key": "pane:browser overlay",
  "element_type": "pane",
  "label": "Browser Overlay",
  "bbox": [
    170.0,
    199.0,
    1394.0,
    391.0
  ],
  "confidence": 0.5609,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 6,
  "last_seen_index": 6,
  "first_seen_at": "2026-05-02T06:47:43.049391+00:00",
  "last_seen_at": "2026-05-02T06:47:43.049391+00:00",
  "seen_count": 1,
  "miss_count": 5,
  "stale": true,
  "cached": true
}
```

```
},
"link:ask google about this page": {
  "element_key": "link:ask google about this page",
  "element_type": "link",
  "label": "Ask Google about this page",
  "bbox": [
    274.0,
    223.0,
    602.0,
    263.0
  ],
  "confidence": 0.5315,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 6,
  "last_seen_index": 6,
  "first_seen_at": "2026-05-02T06:47:43.049391+00:00",
  "last_seen_at": "2026-05-02T06:47:43.049391+00:00",
  "seen_count": 1,
  "miss_count": 5,
  "stale": true,
  "cached": true
},
"link:help me with this": {
  "element_key": "link:help me with this",
  "element_type": "link",
  "label": "Help me with this",
  "bbox": [
```

```
    298.0,  
    287.0,  
    562.0,  
    327.0  
  ],  
  "confidence": 0.5315,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 6,  
  "last_seen_index": 6,  
  "first_seen_at": "2026-05-02T06:47:43.049391+00:00",  
  "last_seen_at": "2026-05-02T06:47:43.049391+00:00",  
  "seen_count": 1,  
  "miss_count": 5,  
  "stale": true,  
  "cached": true  
},  
"input:username or email": {  
  "element_key": "input:username or email",  
  "element_type": "input",  
  "label": "Username or email",  
  "bbox": [  
    730.0,  
    335.0,  
    1194.0,  
    375.0  
  ],  
  "confidence": 0.729,
```

```
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 6,
"last_seen_index": 9,
"first_seen_at": "2026-05-02T06:47:43.049391+00:00",
"last_seen_at": "2026-05-02T06:50:04.893203+00:00",
"seen_count": 3,
"miss_count": 2,
"stale": true,
"cached": true
},
"input:password": {
  "element_key": "input:password",
  "element_type": "input",
  "label": "Password",
  "bbox": [
    730.0,
    375.0,
    1194.0,
    423.0
  ],
  "confidence": 0.729,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 6,
  "last_seen_index": 9,
  "first_seen_at": "2026-05-02T06:47:43.049391+00:00",
```

```
    "last_seen_at": "2026-05-02T06:50:04.893203+00:00",
    "seen_count": 3,
    "miss_count": 2,
    "stale": true,
    "cached": true
  },
  "text:log in using your account on": {
    "element_key": "text:log in using your account on",
    "element_type": "text",
    "label": "Log in using your account on",
    "bbox": [
      834.0,
      767.0,
      994.0,
      783.0
    ],
    "confidence": 0.5905,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 6,
    "last_seen_index": 7,
    "first_seen_at": "2026-05-02T06:47:43.049391+00:00",
    "last_seen_at": "2026-05-02T06:48:31.574903+00:00",
    "seen_count": 2,
    "miss_count": 4,
    "stale": true,
    "cached": true
  },
}
```

```
"button:sliit login": {
  "element_key": "button:sliit login",
  "element_type": "button",
  "label": "SLIIT Login",
  "bbox": [
    898.0,
    799.0,
    994.0,
    815.0
  ],
  "confidence": 0.5905,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 6,
  "last_seen_index": 7,
  "first_seen_at": "2026-05-02T06:47:43.049391+00:00",
  "last_seen_at": "2026-05-02T06:48:31.574903+00:00",
  "seen_count": 2,
  "miss_count": 4,
  "stale": true,
  "cached": true
},
"taskbar:taskbar": {
  "element_key": "taskbar:taskbar",
  "element_type": "taskbar",
  "label": "Taskbar",
  "bbox": [
    162.0,
```

```
    927.0,  
    1759.0,  
    962.0  
  ],  
  "confidence": 0.5609,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 6,  
  "last_seen_index": 6,  
  "first_seen_at": "2026-05-02T06:47:43.049391+00:00",  
  "last_seen_at": "2026-05-02T06:47:43.049391+00:00",  
  "seen_count": 1,  
  "miss_count": 5,  
  "stale": true,  
  "cached": true  
},  
"pane:google search overlay": {  
  "element_key": "pane:google search overlay",  
  "element_type": "pane",  
  "label": "Google Search Overlay",  
  "bbox": [  
    170.0,  
    199.0,  
    1738.0,  
    399.0  
  ],  
  "confidence": 0.6232,  
  "source": "gemini_vlm",
```



```
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 7,
"last_seen_index": 7,
"first_seen_at": "2026-05-02T06:48:31.574903+00:00",
"last_seen_at": "2026-05-02T06:48:31.574903+00:00",
"seen_count": 1,
"miss_count": 4,
"stale": true,
"cached": true
},
"text:ask google about this page - courseweb.sliit.lk": {
  "element_key": "text:ask google about this page - courseweb.sliit.lk",
  "element_type": "text",
  "label": "Ask Google about this page - courseweb.sliit.lk",
  "bbox": [
    298.0,
    223.0,
    722.0,
    279.0
  ],
  "confidence": 0.5905,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 7,
  "last_seen_index": 7,
  "first_seen_at": "2026-05-02T06:48:31.574903+00:00",
  "last_seen_at": "2026-05-02T06:48:31.574903+00:00",
```

```
    "seen_count": 1,
    "miss_count": 4,
    "stale": true,
    "cached": true
  },
  "text:help me with this - search in side panel": {
    "element_key": "text:help me with this - search in side panel",
    "element_type": "text",
    "label": "Help me with this - Search in side panel",
    "bbox": [
      306.0,
      287.0,
      666.0,
      335.0
    ],
    "confidence": 0.5905,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 7,
    "last_seen_index": 7,
    "first_seen_at": "2026-05-02T06:48:31.574903+00:00",
    "last_seen_at": "2026-05-02T06:48:31.574903+00:00",
    "seen_count": 1,
    "miss_count": 4,
    "stale": true,
    "cached": true
  },
  "icon:google chrome": {
```

```
"element_key": "icon:google chrome",
"element_type": "icon",
"label": "Google Chrome",
"bbox": [
    714.0,
    943.0,
    754.0,
    962.0
],
"confidence": 0.5905,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 7,
"last_seen_index": 7,
"first_seen_at": "2026-05-02T06:48:31.574903+00:00",
"last_seen_at": "2026-05-02T06:48:31.574903+00:00",
"seen_count": 1,
"miss_count": 4,
"stale": true,
"cached": true
},
"icon:file explorer": {
    "element_key": "icon:file explorer",
    "element_type": "icon",
    "label": "File Explorer",
    "bbox": [
        754.0,
        943.0,
```

```
    794.0,  
    959.0  
  ],  
  "confidence": 0.5905,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 7,  
  "last_seen_index": 7,  
  "first_seen_at": "2026-05-02T06:48:31.574903+00:00",  
  "last_seen_at": "2026-05-02T06:48:31.574903+00:00",  
  "seen_count": 1,  
  "miss_count": 4,  
  "stale": true,  
  "cached": true  
},  
"icon:visual studio code": {  
  "element_key": "icon:visual studio code",  
  "element_type": "icon",  
  "label": "Visual Studio Code",  
  "bbox": [  
    794.0,  
    943.0,  
    826.0,  
    962.0  
  ],  
  "confidence": 0.5905,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,
```

```
"last_seen_frame": null,  
"first_seen_index": 7,  
"last_seen_index": 7,  
"first_seen_at": "2026-05-02T06:48:31.574903+00:00",  
"last_seen_at": "2026-05-02T06:48:31.574903+00:00",  
"seen_count": 1,  
"miss_count": 4,  
"stale": true,  
"cached": true  
},  
"text:system clock": {  
  "element_key": "text:system clock",  
  "element_type": "text",  
  "label": "System Clock",  
  "bbox": [  
    1642.0,  
    943.0,  
    1746.0,  
    962.0  
  ],  
  "confidence": 0.5905,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 7,  
  "last_seen_index": 7,  
  "first_seen_at": "2026-05-02T06:48:31.574903+00:00",  
  "last_seen_at": "2026-05-02T06:48:31.574903+00:00",  
  "seen_count": 1,
```

```
    "miss_count": 4,  
    "stale": true,  
    "cached": true  
  },  
  "label:username": {  
    "element_key": "label:username",  
    "element_type": "label",  
    "label": "Username",  
    "bbox": [  
      770.0,  
      342.0,  
      914.0,  
      358.0  
    ],  
    "confidence": 0.6561,  
    "source": "gemini_vlm",  
    "first_seen_frame": null,  
    "last_seen_frame": null,  
    "first_seen_index": 8,  
    "last_seen_index": 8,  
    "first_seen_at": "2026-05-02T06:49:17.117694+00:00",  
    "last_seen_at": "2026-05-02T06:49:17.117694+00:00",  
    "seen_count": 1,  
    "miss_count": 3,  
    "stale": true,  
    "cached": true  
  },  
  "label:password": {  
    "element_key": "label:password",
```

```
"element_type": "label",
"label": "Password",
"bbox": [
    770.0,
    390.0,
    858.0,
    406.0
],
"confidence": 0.6561,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 8,
"last_seen_index": 8,
"first_seen_at": "2026-05-02T06:49:17.117694+00:00",
"last_seen_at": "2026-05-02T06:49:17.117694+00:00",
"seen_count": 1,
"miss_count": 3,
"stale": true,
"cached": true
},
"link:sliit login": {
    "element_key": "link:sliit login",
    "element_type": "link",
    "label": "SLIIT Login",
    "bbox": [
        914.0,
        566.0,
        1002.0,
```

```
590.0

],

"confidence": 0.7452,

"source": "gemini_vlm",

"first_seen_frame": null,

"last_seen_frame": null,

"first_seen_index": 8,

"last_seen_index": 9,

"first_seen_at": "2026-05-02T06:49:17.117694+00:00",

"last_seen_at": "2026-05-02T06:50:04.893203+00:00",

"seen_count": 2,

"miss_count": 2,

"stale": true,

"cached": true

},

"pane:spotify media control": {

  "element_key": "pane:spotify media control",

  "element_type": "pane",

  "label": "Spotify Media Control",

  "bbox": [

    1354.0,

    326.0,

    1759.0,

    566.0

  ],

  "confidence": 0.6925,

  "source": "gemini_vlm",

  "first_seen_frame": null,

  "last_seen_frame": null,
```



```
"first_seen_index": 8,  
"last_seen_index": 8,  
"first_seen_at": "2026-05-02T06:49:17.117694+00:00",  
"last_seen_at": "2026-05-02T06:49:17.117694+00:00",  
"seen_count": 1,  
"miss_count": 3,  
"stale": true,  
"cached": true  
},  
"text:karma (the cycle turns)": {  
  "element_key": "text:karma (the cycle turns)",  
  "element_type": "text",  
  "label": "Karma (The Cycle Turns)",  
  "bbox": [  
    1362.0,  
    406.0,  
    1554.0,  
    438.0  
  ],  
  "confidence": 0.6561,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 8,  
  "last_seen_index": 8,  
  "first_seen_at": "2026-05-02T06:49:17.117694+00:00",  
  "last_seen_at": "2026-05-02T06:49:17.117694+00:00",  
  "seen_count": 1,  
  "miss_count": 3,
```

```
    "stale": true,
    "cached": true
  },
  "text:sumedh k": {
    "element_key": "text:sumedh k",
    "element_type": "text",
    "label": "Sumedh K",
    "bbox": [
      1378.0,
      430.0,
      1482.0,
      446.0
    ],
    "confidence": 0.6561,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 8,
    "last_seen_index": 8,
    "first_seen_at": "2026-05-02T06:49:17.117694+00:00",
    "last_seen_at": "2026-05-02T06:49:17.117694+00:00",
    "seen_count": 1,
    "miss_count": 3,
    "stale": true,
    "cached": true
  },
  "button:previous": {
    "element_key": "button:previous",
    "element_type": "button",
```

```
"label": "Previous",
"bbox": [
  1458.0,
  478.0,
  1498.0,
  510.0
],
"confidence": 0.6197,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 8,
"last_seen_index": 8,
"first_seen_at": "2026-05-02T06:49:17.117694+00:00",
"last_seen_at": "2026-05-02T06:49:17.117694+00:00",
"seen_count": 1,
"miss_count": 3,
"stale": true,
"cached": true
},
"button:play/pause": {
  "element_key": "button:play/pause",
  "element_type": "button",
  "label": "Play/Pause",
  "bbox": [
    1530.0,
    478.0,
    1562.0,
    510.0
  ]
}
```

```
],
  "confidence": 0.6197,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 8,
  "last_seen_index": 8,
  "first_seen_at": "2026-05-02T06:49:17.117694+00:00",
  "last_seen_at": "2026-05-02T06:49:17.117694+00:00",
  "seen_count": 1,
  "miss_count": 3,
  "stale": true,
  "cached": true
},
"button:next": {
  "element_key": "button:next",
  "element_type": "button",
  "label": "Next",
  "bbox": [
    1594.0,
    478.0,
    1634.0,
    510.0
  ],
  "confidence": 0.6197,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 8,
```

```
"last_seen_index": 8,  
"first_seen_at": "2026-05-02T06:49:17.117694+00:00",  
"last_seen_at": "2026-05-02T06:49:17.117694+00:00",  
"seen_count": 1,  
"miss_count": 3,  
"stale": true,  
"cached": true  
},  
"pane:quick settings": {  
  "element_key": "pane:quick settings",  
  "element_type": "pane",  
  "label": "Quick Settings",  
  "bbox": [  
    1306.0,  
    486.0,  
    1759.0,  
    962.0  
  ],  
  "confidence": 0.6925,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 8,  
  "last_seen_index": 8,  
  "first_seen_at": "2026-05-02T06:49:17.117694+00:00",  
  "last_seen_at": "2026-05-02T06:49:17.117694+00:00",  
  "seen_count": 1,  
  "miss_count": 3,  
  "stale": true,
```

```
    "cached": true
  },
  "toggle:wi-fi slt-fiber-7e76": {
    "element_key": "toggle:wi-fi slt-fiber-7e76",
    "element_type": "toggle",
    "label": "Wi-Fi SLT-Fiber-7E76",
    "bbox": [
      1370.0,
      558.0,
      1498.0,
      630.0
    ],
    "confidence": 0.6561,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 8,
    "last_seen_index": 8,
    "first_seen_at": "2026-05-02T06:49:17.117694+00:00",
    "last_seen_at": "2026-05-02T06:49:17.117694+00:00",
    "seen_count": 1,
    "miss_count": 3,
    "stale": true,
    "cached": true
  },
  "toggle:bluetooth": {
    "element_key": "toggle:bluetooth",
    "element_type": "toggle",
    "label": "Bluetooth",
```

```
"bbox": [
  1482.0,
  558.0,
  1602.0,
  622.0
],
"confidence": 0.6561,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 8,
"last_seen_index": 8,
"first_seen_at": "2026-05-02T06:49:17.117694+00:00",
"last_seen_at": "2026-05-02T06:49:17.117694+00:00",
"seen_count": 1,
"miss_count": 3,
"stale": true,
"cached": true
},
"toggle:airplane mode": {
  "element_key": "toggle:airplane mode",
  "element_type": "toggle",
  "label": "Airplane mode",
  "bbox": [
    1594.0,
    566.0,
    1674.0,
    606.0
  ],
```

```
"confidence": 0.6561,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 8,
"last_seen_index": 8,
"first_seen_at": "2026-05-02T06:49:17.117694+00:00",
"last_seen_at": "2026-05-02T06:49:17.117694+00:00",
"seen_count": 1,
"miss_count": 3,
"stale": true,
"cached": true
},
"toggle:energy saver": {
  "element_key": "toggle:energy saver",
  "element_type": "toggle",
  "label": "Energy saver",
  "bbox": [
    1370.0,
    662.0,
    1490.0,
    734.0
  ],
  "confidence": 0.6561,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 8,
  "last_seen_index": 8,
```



```
"first_seen_at": "2026-05-02T06:49:17.117694+00:00",
"last_seen_at": "2026-05-02T06:49:17.117694+00:00",
"seen_count": 1,
"miss_count": 3,
"stale": true,
"cached": true
},
"toggle:night light": {
  "element_key": "toggle:night light",
  "element_type": "toggle",
  "label": "Night light",
  "bbox": [
    1506.0,
    670.0,
    1578.0,
    734.0
  ],
  "confidence": 0.6561,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 8,
  "last_seen_index": 8,
  "first_seen_at": "2026-05-02T06:49:17.117694+00:00",
  "last_seen_at": "2026-05-02T06:49:17.117694+00:00",
  "seen_count": 1,
  "miss_count": 3,
  "stale": true,
  "cached": true
}
```

```
},
"toggle:accessibility": {
  "element_key": "toggle:accessibility",
  "element_type": "toggle",
  "label": "Accessibility",
  "bbox": [
    1610.0,
    662.0,
    1722.0,
    734.0
  ],
  "confidence": 0.6561,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 8,
  "last_seen_index": 8,
  "first_seen_at": "2026-05-02T06:49:17.117694+00:00",
  "last_seen_at": "2026-05-02T06:49:17.117694+00:00",
  "seen_count": 1,
  "miss_count": 3,
  "stale": true,
  "cached": true
},
"slider:brightness": {
  "element_key": "slider:brightness",
  "element_type": "slider",
  "label": "Brightness",
  "bbox": [
```

```
    1378.0,  
    790.0,  
    1674.0,  
    806.0  
  ],  
  "confidence": 0.6561,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 8,  
  "last_seen_index": 8,  
  "first_seen_at": "2026-05-02T06:49:17.117694+00:00",  
  "last_seen_at": "2026-05-02T06:49:17.117694+00:00",  
  "seen_count": 1,  
  "miss_count": 3,  
  "stale": true,  
  "cached": true  
},  
"slider:volume": {  
  "element_key": "slider:volume",  
  "element_type": "slider",  
  "label": "Volume",  
  "bbox": [  
    1378.0,  
    838.0,  
    1706.0,  
    854.0  
  ],  
  "confidence": 0.6561,
```

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"source": "gemini_vlm",
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"last_seen_frame": null,
"first_seen_index": 8,
"last_seen_index": 8,
"first_seen_at": "2026-05-02T06:49:17.117694+00:00",
"last_seen_at": "2026-05-02T06:49:17.117694+00:00",
"seen_count": 1,
"miss_count": 3,
"stale": true,
"cached": true
},
"label:93%": {
  "element_key": "label:93%",
  "element_type": "label",
  "label": "93%",
  "bbox": [
    1378.0,
    894.0,
    1442.0,
    918.0
  ],
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  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 8,
  "last_seen_index": 8,
  "first_seen_at": "2026-05-02T06:49:17.117694+00:00",
```

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    "last_seen_at": "2026-05-02T06:49:17.117694+00:00",
    "seen_count": 1,
    "miss_count": 3,
    "stale": true,
    "cached": true
  },
  "button:settings": {
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    "element_type": "button",
    "label": "Settings",
    "bbox": [
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      894.0,
      1706.0,
      918.0
    ],
    "confidence": 0.6197,
    "source": "gemini_vlm",
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    "last_seen_frame": null,
    "first_seen_index": 8,
    "last_seen_index": 8,
    "first_seen_at": "2026-05-02T06:49:17.117694+00:00",
    "last_seen_at": "2026-05-02T06:49:17.117694+00:00",
    "seen_count": 1,
    "miss_count": 3,
    "stale": true,
    "cached": true
  },
}
```

```
"toolbar:taskbar": {
  "element_key": "toolbar:taskbar",
  "element_type": "toolbar",
  "label": "Taskbar",
  "bbox": [
    178.0,
    934.0,
    1746.0,
    962.0
  ],
  "confidence": 0.6925,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 8,
  "last_seen_index": 8,
  "first_seen_at": "2026-05-02T06:49:17.117694+00:00",
  "last_seen_at": "2026-05-02T06:49:17.117694+00:00",
  "seen_count": 1,
  "miss_count": 3,
  "stale": true,
  "cached": true
},
"label:12:18 pm": {
  "element_key": "label:12:18 pm",
  "element_type": "label",
  "label": "12:18 PM",
  "bbox": [
    1650.0,
```

```
    942.0,  
    1738.0,  
    962.0  
  ],  
  "confidence": 0.6561,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 8,  
  "last_seen_index": 8,  
  "first_seen_at": "2026-05-02T06:49:17.117694+00:00",  
  "last_seen_at": "2026-05-02T06:49:17.117694+00:00",  
  "seen_count": 1,  
  "miss_count": 3,  
  "stale": true,  
  "cached": true  
},  
"link:lost password?": {  
  "element_key": "link:lost password?",  
  "element_type": "link",  
  "label": "Lost password?",  
  "bbox": [  
    922.0,  
    487.0,  
    1002.0,  
    503.0  
  ],  
  "confidence": 0.6885,  
  "source": "gemini_vlm",
```

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"first_seen_frame": null,
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"first_seen_index": 9,
"last_seen_index": 9,
"first_seen_at": "2026-05-02T06:50:04.893203+00:00",
"last_seen_at": "2026-05-02T06:50:04.893203+00:00",
"seen_count": 1,
"miss_count": 2,
"stale": true,
"cached": true
},
"pane:taskbar": {
  "element_key": "pane:taskbar",
  "element_type": "pane",
  "label": "Taskbar",
  "bbox": [
    178.0,
    935.0,
    1758.0,
    963.0
  ],
  "confidence": 0.7695,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 9,
  "last_seen_index": 9,
  "first_seen_at": "2026-05-02T06:50:04.893203+00:00",
  "last_seen_at": "2026-05-02T06:50:04.893203+00:00",
```



```
    "seen_count": 1,  
    "miss_count": 2,  
    "stale": true,  
    "cached": true  
  },  
  "button:search": {  
    "element_key": "button:search",  
    "element_type": "button",  
    "label": "Search",  
    "bbox": [  
      906.0,  
      943.0,  
      938.0,  
      963.0  
    ],  
    "confidence": 0.729,  
    "source": "gemini_vlm",  
    "first_seen_frame": null,  
    "last_seen_frame": null,  
    "first_seen_index": 9,  
    "last_seen_index": 9,  
    "first_seen_at": "2026-05-02T06:50:04.893203+00:00",  
    "last_seen_at": "2026-05-02T06:50:04.893203+00:00",  
    "seen_count": 1,  
    "miss_count": 2,  
    "stale": true,  
    "cached": true  
  },  
  "button:task view": {
```

```
"element_key": "button:task view",
"element_type": "button",
"label": "Task View",
"bbox": [
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    943.0,
    962.0,
    963.0
],
"confidence": 0.729,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 9,
"last_seen_index": 9,
"first_seen_at": "2026-05-02T06:50:04.893203+00:00",
"last_seen_at": "2026-05-02T06:50:04.893203+00:00",
"seen_count": 1,
"miss_count": 2,
"stale": true,
"cached": true
},
"button:microsoft edge": {
    "element_key": "button:microsoft edge",
    "element_type": "button",
    "label": "Microsoft Edge",
    "bbox": [
        954.0,
        943.0,
```

```
    986.0,  
    963.0  
  ],  
  "confidence": 0.729,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 9,  
  "last_seen_index": 9,  
  "first_seen_at": "2026-05-02T06:50:04.893203+00:00",  
  "last_seen_at": "2026-05-02T06:50:04.893203+00:00",  
  "seen_count": 1,  
  "miss_count": 2,  
  "stale": true,  
  "cached": true  
},  
"button:file explorer": {  
  "element_key": "button:file explorer",  
  "element_type": "button",  
  "label": "File Explorer",  
  "bbox": [  
    978.0,  
    943.0,  
    1010.0,  
    963.0  
  ],  
  "confidence": 0.729,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,
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"last_seen_frame": null,
"first_seen_index": 9,
"last_seen_index": 9,
"first_seen_at": "2026-05-02T06:50:04.893203+00:00",
"last_seen_at": "2026-05-02T06:50:04.893203+00:00",
"seen_count": 1,
"miss_count": 2,
"stale": true,
"cached": true
},
"button:microsoft store": {
  "element_key": "button:microsoft store",
  "element_type": "button",
  "label": "Microsoft Store",
  "bbox": [
    1002.0,
    943.0,
    1034.0,
    963.0
  ],
  "confidence": 0.729,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 9,
  "last_seen_index": 9,
  "first_seen_at": "2026-05-02T06:50:04.893203+00:00",
  "last_seen_at": "2026-05-02T06:50:04.893203+00:00",
  "seen_count": 1,
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    "miss_count": 2,  
    "stale": true,  
    "cached": true  
  },  
  "button:mail": {  
    "element_key": "button:mail",  
    "element_type": "button",  
    "label": "Mail",  
    "bbox": [  
      1026.0,  
      943.0,  
      1058.0,  
      963.0  
    ],  
    "confidence": 0.729,  
    "source": "gemini_vlm",  
    "first_seen_frame": null,  
    "last_seen_frame": null,  
    "first_seen_index": 9,  
    "last_seen_index": 9,  
    "first_seen_at": "2026-05-02T06:50:04.893203+00:00",  
    "last_seen_at": "2026-05-02T06:50:04.893203+00:00",  
    "seen_count": 1,  
    "miss_count": 2,  
    "stale": true,  
    "cached": true  
  },  
  "button:google chrome": {  
    "element_key": "button:google chrome",
```

```
"element_type": "button",
"label": "Google Chrome",
"bbox": [
    1058.0,
    943.0,
    1082.0,
    963.0
],
"confidence": 0.729,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 9,
"last_seen_index": 9,
"first_seen_at": "2026-05-02T06:50:04.893203+00:00",
"last_seen_at": "2026-05-02T06:50:04.893203+00:00",
"seen_count": 1,
"miss_count": 2,
"stale": true,
"cached": true
},
"button:visual studio code": {
    "element_key": "button:visual studio code",
    "element_type": "button",
    "label": "Visual Studio Code",
    "bbox": [
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        943.0,
        1106.0,
```

```
    963.0
  ],
  "confidence": 0.729,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 9,
  "last_seen_index": 9,
  "first_seen_at": "2026-05-02T06:50:04.893203+00:00",
  "last_seen_at": "2026-05-02T06:50:04.893203+00:00",
  "seen_count": 1,
  "miss_count": 2,
  "stale": true,
  "cached": true
},
"button:microsoft teams": {
  "element_key": "button:microsoft teams",
  "element_type": "button",
  "label": "Microsoft Teams",
  "bbox": [
    1106.0,
    943.0,
    1130.0,
    963.0
  ],
  "confidence": 0.729,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
```

```
"first_seen_index": 9,  
"last_seen_index": 9,  
"first_seen_at": "2026-05-02T06:50:04.893203+00:00",  
"last_seen_at": "2026-05-02T06:50:04.893203+00:00",  
"seen_count": 1,  
"miss_count": 2,  
"stale": true,  
"cached": true  
},  
"text:clock": {  
  "element_key": "text:clock",  
  "element_type": "text",  
  "label": "Clock",  
  "bbox": [  
    1642.0,  
    943.0,  
    1746.0,  
    963.0  
  ],  
  "confidence": 0.729,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 9,  
  "last_seen_index": 9,  
  "first_seen_at": "2026-05-02T06:50:04.893203+00:00",  
  "last_seen_at": "2026-05-02T06:50:04.893203+00:00",  
  "seen_count": 1,  
  "miss_count": 2,
```



```
    "stale": true,  
    "cached": true  
  }  
},  
"current_elements": [  
  {  
    "id": "tab_frontend",  
    "type": "tab",  
    "label": "frontend",  
    "description": "Browser tab labeled frontend",  
    "state": "visible",  
    "dx": 116,  
    "dy": 52,  
    "frame_dx": 272,  
    "frame_dy": 154,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
      156,  
      126,  
      388,  
      182  
    ],  
    "session_memory_key": "tab:frontend",  
    "cached": false  
  },  
  {  
    "id": "tab_courseweb",  
    "type": "tab",
```

```
"label": "Log in to the site | CourseWeb",
"description": "Browser tab for CourseWeb login",
"state": "visible",
"dx": 320,
"dy": 52,
"frame_dx": 476,
"frame_dy": 154,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    348,
    126,
    604,
    182
],
"session_memory_key": "tab:log in to the site | courseweb",
"cached": false
},
{
    "id": "address_bar",
    "type": "input",
    "label": "Address bar",
    "description": "Browser address bar showing localhost:5173",
    "state": "visible",
    "dx": 576,
    "dy": 112,
    "frame_dx": 732,
    "frame_dy": 214,
    "confidence": 0.98,
```

```
"source": "gemini_vlm",
"frame_bbox": [
    156,
    182,
    1308,
    246
],
"session_memory_key": "input:address bar",
"cached": false
},
{
    "id": "app_logo",
    "type": "icon",
    "label": "ScreenPilot AI Logo",
    "description": "Chip icon for ScreenPilot AI",
    "state": "visible",
    "dx": 56,
    "dy": 192,
    "frame_dx": 212,
    "frame_dy": 294,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        196,
        270,
        228,
        318
    ],
    "session_memory_key": "icon:screenpilot ai logo",
```

```
    "cached": false
  },
  {
    "id": "app_title",
    "type": "heading",
    "label": "ScreenPilot AI",
    "description": "Application title text",
    "state": "visible",
    "dx": 144,
    "dy": 192,
    "frame_dx": 300,
    "frame_dy": 294,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
      236,
      278,
      364,
      310
    ],
    "session_memory_key": "heading:screenpilot ai",
    "cached": false
  },
  {
    "id": "app_version",
    "type": "label",
    "label": "Autonomous Task Agent v2",
    "description": "Application version badge",
    "state": "visible",
```

```
"dx": 300,
"dy": 196,
"frame_dx": 456,
"frame_dy": 298,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    380,
    286,
    532,
    310
],
"session_memory_key": "label:autonomous task agent v2",
"cached": false
},
{
    "id": "message_bubble",
    "type": "pane",
    "label": "Agent message",
    "description": "Chat bubble with text: I will now open the browser and
login to the courseweb",
    "state": "visible",
    "dx": 928,
    "dy": 340,
    "frame_dx": 1084,
    "frame_dy": 442,
    "confidence": 0.92,
    "source": "gemini_vlm",
    "frame_bbox": [
```

```
    732,  
    374,  
    1436,  
    510  
  ],  
  "session_memory_key": "pane:agent message",  
  "cached": false  
},  
{  
  "id": "status_connecting",  
  "type": "label",  
  "label": "Connecting to agent...",  
  "description": "Status message indicating connection progress",  
  "state": "visible",  
  "dx": 120,  
  "dy": 440,  
  "frame_dx": 276,  
  "frame_dy": 542,  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    188,  
    526,  
    364,  
    558  
  ],  
  "session_memory_key": "label:connecting to agent...",  
  "cached": false  
},
```

```
{
  "id": "status_capturing",
  "type": "label",
  "label": "Capturing initial screen...",
  "description": "Status message indicating screen capture progress",
  "state": "visible",
  "dx": 148,
  "dy": 468,
  "frame_dx": 304,
  "frame_dy": 570,
  "confidence": 0.9,
  "source": "gemini_vlm",
  "frame_bbox": [
    188,
    558,
    420,
    582
  ],
  "session_memory_key": "label:capturing initial screen...",
  "cached": false
},
{
  "id": "loading_indicator",
  "type": "pane",
  "label": "Loading indicator",
  "description": "Box showing loading dots and a cursor",
  "state": "visible",
  "dx": 124,
  "dy": 596,
```

```
"frame_dx": 280,  
"frame_dy": 698,  
"confidence": 0.85,  
"source": "gemini_vlm",  
"frame_bbox": [  
    180,  
    598,  
    380,  
    798  
],  
"session_memory_key": "pane:loading indicator",  
"cached": false  
},  
{  
    "id": "chat_input_field",  
    "type": "input",  
    "label": "Chat input",  
    "description": "Input field with placeholder: Agent is working...",  
    "state": "visible",  
    "dx": 684,  
    "dy": 844,  
    "frame_dx": 840,  
    "frame_dy": 946,  
    "confidence": 0.98,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        156,  
        894,  
        1524,
```



```
998
    ],
    "session_memory_key": "input:chat input",
    "cached": false
  }
],
"resolved_elements": [
  {
    "id": "tab_frontend",
    "type": "tab",
    "label": "frontend",
    "description": "Browser tab labeled frontend",
    "state": "visible",
    "dx": 116,
    "dy": 52,
    "frame_dx": 272,
    "frame_dy": 154,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
      156,
      126,
      388,
      182
    ],
    "session_memory_key": "tab:frontend",
    "cached": false
  },
  {
```

```
"id": "tab_courseweb",
"type": "tab",
"label": "Log in to the site | CourseWeb",
"description": "Browser tab for CourseWeb login",
"state": "visible",
"dx": 320,
"dy": 52,
"frame_dx": 476,
"frame_dy": 154,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
  348,
  126,
  604,
  182
],
"session_memory_key": "tab:log in to the site | courseweb",
"cached": false
},
{
  "id": "address_bar",
  "type": "input",
  "label": "Address bar",
  "description": "Browser address bar showing localhost:5173",
  "state": "visible",
  "dx": 576,
  "dy": 112,
  "frame_dx": 732,
```

```
"frame_dy": 214,  
"confidence": 0.98,  
"source": "gemini_vlm",  
"frame_bbox": [  
    156,  
    182,  
    1308,  
    246  
],  
"session_memory_key": "input:address bar",  
"cached": false  
},  
{  
    "id": "app_logo",  
    "type": "icon",  
    "label": "ScreenPilot AI Logo",  
    "description": "Chip icon for ScreenPilot AI",  
    "state": "visible",  
    "dx": 56,  
    "dy": 192,  
    "frame_dx": 212,  
    "frame_dy": 294,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        196,  
        270,  
        228,  
        318
```

```
    ],
    "session_memory_key": "icon:screenpilot ai logo",
    "cached": false
  },
  {
    "id": "app_title",
    "type": "heading",
    "label": "ScreenPilot AI",
    "description": "Application title text",
    "state": "visible",
    "dx": 144,
    "dy": 192,
    "frame_dx": 300,
    "frame_dy": 294,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
      236,
      278,
      364,
      310
    ],
    ],
    "session_memory_key": "heading:screenpilot ai",
    "cached": false
  },
  {
    "id": "app_version",
    "type": "label",
    "label": "Autonomous Task Agent v2",
```

```
    "description": "Application version badge",
    "state": "visible",
    "dx": 300,
    "dy": 196,
    "frame_dx": 456,
    "frame_dy": 298,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        380,
        286,
        532,
        310
    ],
    "session_memory_key": "label:autonomous task agent v2",
    "cached": false
},
{
    "id": "message_bubble",
    "type": "pane",
    "label": "Agent message",
    "description": "Chat bubble with text: I will now open the browser and
login to the courseweb",
    "state": "visible",
    "dx": 928,
    "dy": 340,
    "frame_dx": 1084,
    "frame_dy": 442,
    "confidence": 0.92,
```

```
"source": "gemini_vlm",
"frame_bbox": [
    732,
    374,
    1436,
    510
],
"session_memory_key": "pane:agent message",
"cached": false
},
{
    "id": "status_connecting",
    "type": "label",
    "label": "Connecting to agent...",
    "description": "Status message indicating connection progress",
    "state": "visible",
    "dx": 120,
    "dy": 440,
    "frame_dx": 276,
    "frame_dy": 542,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        188,
        526,
        364,
        558
    ],
    "session_memory_key": "label:connecting to agent..."
```

```
    "cached": false
  },
  {
    "id": "status_capturing",
    "type": "label",
    "label": "Capturing initial screen...",
    "description": "Status message indicating screen capture progress",
    "state": "visible",
    "dx": 148,
    "dy": 468,
    "frame_dx": 304,
    "frame_dy": 570,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
      188,
      558,
      420,
      582
    ],
    "session_memory_key": "label:capturing initial screen...",
    "cached": false
  },
  {
    "id": "loading_indicator",
    "type": "pane",
    "label": "Loading indicator",
    "description": "Box showing loading dots and a cursor",
    "state": "visible",
```

```
"dx": 124,  
  "dy": 596,  
  "frame_dx": 280,  
  "frame_dy": 698,  
  "confidence": 0.85,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    180,  
    598,  
    380,  
    798  
  ],  
  "session_memory_key": "pane:loading indicator",  
  "cached": false  
},  
{  
  "id": "chat_input_field",  
  "type": "input",  
  "label": "Chat input",  
  "description": "Input field with placeholder: Agent is working...",  
  "state": "visible",  
  "dx": 684,  
  "dy": 844,  
  "frame_dx": 840,  
  "frame_dy": 946,  
  "confidence": 0.98,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    156,
```



```
        894,  
        1524,  
        998  
    ],  
    "session_memory_key": "input:chat input",  
    "cached": false  
}  
],  
"screen_data": {  
    "image": "E:\\sliit-  
project\\ScreenAwareTaskAgent\\vision\\data\\sessions\\shot_1777704127_f68076\\pr  
ocessing\\proc_1777704128_59c8ad\\preprocessed_frames\\frame_1777704128_8d3627.jp  
g",  
    "image_size": {  
        "width": 1920,  
        "height": 1080  
    },  
    "screen_bbox": [  
        156,  
        102,  
        1767,  
        1016  
    ],  
    "screen_size": {  
        "width": 1611,  
        "height": 914  
    },  
    "coordinate_system": "pixel",  
    "elements": [  
        {
```

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"id": "tab_frontend",
"type": "tab",
"label": "frontend",
"description": "Browser tab labeled frontend",
"state": "visible",
"dx": 116,
"dy": 52,
"frame_dx": 272,
"frame_dy": 154,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    156,
    126,
    388,
    182
],
"session_memory_key": "tab:frontend",
"cached": false
},
{
    "id": "tab_courseweb",
    "type": "tab",
    "label": "Log in to the site | CourseWeb",
    "description": "Browser tab for CourseWeb login",
    "state": "visible",
    "dx": 320,
    "dy": 52,
    "frame_dx": 476,
```

```
"frame_dy": 154,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    348,
    126,
    604,
    182
],
"session_memory_key": "tab:log in to the site | courseweb",
"cached": false
},
{
    "id": "address_bar",
    "type": "input",
    "label": "Address bar",
    "description": "Browser address bar showing localhost:5173",
    "state": "visible",
    "dx": 576,
    "dy": 112,
    "frame_dx": 732,
    "frame_dy": 214,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
        156,
        182,
        1308,
        246
```

```
    ],
    "session_memory_key": "input:address bar",
    "cached": false
  },
  {
    "id": "app_logo",
    "type": "icon",
    "label": "ScreenPilot AI Logo",
    "description": "Chip icon for ScreenPilot AI",
    "state": "visible",
    "dx": 56,
    "dy": 192,
    "frame_dx": 212,
    "frame_dy": 294,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
      196,
      270,
      228,
      318
    ],
    "session_memory_key": "icon:screenpilot ai logo",
    "cached": false
  },
  {
    "id": "app_title",
    "type": "heading",
    "label": "ScreenPilot AI",
```

```
"description": "Application title text",
"state": "visible",
"dx": 144,
"dy": 192,
"frame_dx": 300,
"frame_dy": 294,
"confidence": 0.98,
"source": "gemini_vlm",
"frame_bbox": [
    236,
    278,
    364,
    310
],
"session_memory_key": "heading:screenpilot ai",
"cached": false
},
{
    "id": "app_version",
    "type": "label",
    "label": "Autonomous Task Agent v2",
    "description": "Application version badge",
    "state": "visible",
    "dx": 300,
    "dy": 196,
    "frame_dx": 456,
    "frame_dy": 298,
    "confidence": 0.95,
    "source": "gemini_vlm",
```

```
"frame_bbox": [
    380,
    286,
    532,
    310
],
"session_memory_key": "label:autonomous task agent v2",
"cached": false
},
{
    "id": "message_bubble",
    "type": "pane",
    "label": "Agent message",
    "description": "Chat bubble with text: I will now open the browser
and login to the courseweb",
    "state": "visible",
    "dx": 928,
    "dy": 340,
    "frame_dx": 1084,
    "frame_dy": 442,
    "confidence": 0.92,
    "source": "gemini_vlm",
    "frame_bbox": [
        732,
        374,
        1436,
        510
    ],
    "session_memory_key": "pane:agent message",
```

```
    "cached": false
  },
  {
    "id": "status_connecting",
    "type": "label",
    "label": "Connecting to agent...",
    "description": "Status message indicating connection progress",
    "state": "visible",
    "dx": 120,
    "dy": 440,
    "frame_dx": 276,
    "frame_dy": 542,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
      188,
      526,
      364,
      558
    ],
    "session_memory_key": "label:connecting to agent...",
    "cached": false
  },
  {
    "id": "status_capturing",
    "type": "label",
    "label": "Capturing initial screen...",
    "description": "Status message indicating screen capture progress",
    "state": "visible",
```

```
"dx": 148,  
  "dy": 468,  
  "frame_dx": 304,  
  "frame_dy": 570,  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    188,  
    558,  
    420,  
    582  
  ],  
  "session_memory_key": "label:capturing initial screen...",  
  "cached": false  
},  
{  
  "id": "loading_indicator",  
  "type": "pane",  
  "label": "Loading indicator",  
  "description": "Box showing loading dots and a cursor",  
  "state": "visible",  
  "dx": 124,  
  "dy": 596,  
  "frame_dx": 280,  
  "frame_dy": 698,  
  "confidence": 0.85,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    180,
```



```
        598,  
        380,  
        798  
    ],  
    "session_memory_key": "pane:loading indicator",  
    "cached": false  
},  
{  
    "id": "chat_input_field",  
    "type": "input",  
    "label": "Chat input",  
    "description": "Input field with placeholder: Agent is working...",  
    "state": "visible",  
    "dx": 684,  
    "dy": 844,  
    "frame_dx": 840,  
    "frame_dy": 946,  
    "confidence": 0.98,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        156,  
        894,  
        1524,  
        998  
    ],  
    "session_memory_key": "input:chat input",  
    "cached": false  
}  
],
```

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"element_count": 11,  
"current_elements": [  
  {  
    "id": "tab_frontend",  
    "type": "tab",  
    "label": "frontend",  
    "description": "Browser tab labeled frontend",  
    "state": "visible",  
    "dx": 116,  
    "dy": 52,  
    "frame_dx": 272,  
    "frame_dy": 154,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
      156,  
      126,  
      388,  
      182  
    ],  
    "session_memory_key": "tab:frontend",  
    "cached": false  
  },  
  {  
    "id": "tab_courseweb",  
    "type": "tab",  
    "label": "Log in to the site | CourseWeb",  
    "description": "Browser tab for CourseWeb login",  
    "state": "visible",
```

```
"dx": 320,  
  "dy": 52,  
  "frame_dx": 476,  
  "frame_dy": 154,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    348,  
    126,  
    604,  
    182  
  ],  
  "session_memory_key": "tab:log in to the site | courseweb",  
  "cached": false  
},  
{  
  "id": "address_bar",  
  "type": "input",  
  "label": "Address bar",  
  "description": "Browser address bar showing localhost:5173",  
  "state": "visible",  
  "dx": 576,  
  "dy": 112,  
  "frame_dx": 732,  
  "frame_dy": 214,  
  "confidence": 0.98,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    156,
```

```
        182,  
        1308,  
        246  
    ],  
    "session_memory_key": "input:address bar",  
    "cached": false  
},  
{  
    "id": "app_logo",  
    "type": "icon",  
    "label": "ScreenPilot AI Logo",  
    "description": "Chip icon for ScreenPilot AI",  
    "state": "visible",  
    "dx": 56,  
    "dy": 192,  
    "frame_dx": 212,  
    "frame_dy": 294,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        196,  
        270,  
        228,  
        318  
    ],  
    "session_memory_key": "icon:screenpilot ai logo",  
    "cached": false  
},  
{
```

```
"id": "app_title",
"type": "heading",
"label": "ScreenPilot AI",
"description": "Application title text",
"state": "visible",
"dx": 144,
"dy": 192,
"frame_dx": 300,
"frame_dy": 294,
"confidence": 0.98,
"source": "gemini_vlm",
"frame_bbox": [
    236,
    278,
    364,
    310
],
"session_memory_key": "heading:screenpilot ai",
"cached": false
},
{
    "id": "app_version",
    "type": "label",
    "label": "Autonomous Task Agent v2",
    "description": "Application version badge",
    "state": "visible",
    "dx": 300,
    "dy": 196,
    "frame_dx": 456,
```

```
    "frame_dy": 298,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        380,
        286,
        532,
        310
    ],
    "session_memory_key": "label:autonomous task agent v2",
    "cached": false
},
{
    "id": "message_bubble",
    "type": "pane",
    "label": "Agent message",
    "description": "Chat bubble with text: I will now open the browser
and login to the courseweb",
    "state": "visible",
    "dx": 928,
    "dy": 340,
    "frame_dx": 1084,
    "frame_dy": 442,
    "confidence": 0.92,
    "source": "gemini_vlm",
    "frame_bbox": [
        732,
        374,
        1436,
```

```
510
],
"session_memory_key": "pane:agent message",
"cached": false
},
{
  "id": "status_connecting",
  "type": "label",
  "label": "Connecting to agent...",
  "description": "Status message indicating connection progress",
  "state": "visible",
  "dx": 120,
  "dy": 440,
  "frame_dx": 276,
  "frame_dy": 542,
  "confidence": 0.9,
  "source": "gemini_vlm",
  "frame_bbox": [
    188,
    526,
    364,
    558
  ],
  "session_memory_key": "label:connecting to agent...",
  "cached": false
},
{
  "id": "status_capturing",
  "type": "label",
```

```
"label": "Capturing initial screen...",
"description": "Status message indicating screen capture progress",
"state": "visible",
"dx": 148,
"dy": 468,
"frame_dx": 304,
"frame_dy": 570,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    188,
    558,
    420,
    582
],
"session_memory_key": "label:capturing initial screen...",
"cached": false
},
{
    "id": "loading_indicator",
    "type": "pane",
    "label": "Loading indicator",
    "description": "Box showing loading dots and a cursor",
    "state": "visible",
    "dx": 124,
    "dy": 596,
    "frame_dx": 280,
    "frame_dy": 698,
    "confidence": 0.85,
```



```
"source": "gemini_vlm",
"frame_bbox": [
    180,
    598,
    380,
    798
],
"session_memory_key": "pane:loading indicator",
"cached": false
},
{
    "id": "chat_input_field",
    "type": "input",
    "label": "Chat input",
    "description": "Input field with placeholder: Agent is working...",
    "state": "visible",
    "dx": 684,
    "dy": 844,
    "frame_dx": 840,
    "frame_dy": 946,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
        156,
        894,
        1524,
        998
    ],
    "session_memory_key": "input:chat input",
```

```
        "cached": false
    }
],
"resolved_elements": [
    {
        "id": "tab_frontend",
        "type": "tab",
        "label": "frontend",
        "description": "Browser tab labeled frontend",
        "state": "visible",
        "dx": 116,
        "dy": 52,
        "frame_dx": 272,
        "frame_dy": 154,
        "confidence": 0.95,
        "source": "gemini_vlm",
        "frame_bbox": [
            156,
            126,
            388,
            182
        ],
        "session_memory_key": "tab:frontend",
        "cached": false
    },
    {
        "id": "tab_courseweb",
        "type": "tab",
        "label": "Log in to the site | CourseWeb",
```

```
"description": "Browser tab for CourseWeb login",
"state": "visible",
"dx": 320,
"dy": 52,
"frame_dx": 476,
"frame_dy": 154,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    348,
    126,
    604,
    182
],
"session_memory_key": "tab:log in to the site | courseweb",
"cached": false
},
{
    "id": "address_bar",
    "type": "input",
    "label": "Address bar",
    "description": "Browser address bar showing localhost:5173",
    "state": "visible",
    "dx": 576,
    "dy": 112,
    "frame_dx": 732,
    "frame_dy": 214,
    "confidence": 0.98,
    "source": "gemini_vlm",
```

```
"frame_bbox": [
    156,
    182,
    1308,
    246
],
"session_memory_key": "input:address bar",
"cached": false
},
{
    "id": "app_logo",
    "type": "icon",
    "label": "ScreenPilot AI Logo",
    "description": "Chip icon for ScreenPilot AI",
    "state": "visible",
    "dx": 56,
    "dy": 192,
    "frame_dx": 212,
    "frame_dy": 294,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        196,
        270,
        228,
        318
    ],
    "session_memory_key": "icon:screenpilot ai logo",
    "cached": false
}
```

```
},  
{  
  "id": "app_title",  
  "type": "heading",  
  "label": "ScreenPilot AI",  
  "description": "Application title text",  
  "state": "visible",  
  "dx": 144,  
  "dy": 192,  
  "frame_dx": 300,  
  "frame_dy": 294,  
  "confidence": 0.98,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    236,  
    278,  
    364,  
    310  
  ],  
  "session_memory_key": "heading:screenpilot ai",  
  "cached": false  
},  
{  
  "id": "app_version",  
  "type": "label",  
  "label": "Autonomous Task Agent v2",  
  "description": "Application version badge",  
  "state": "visible",  
  "dx": 300,
```

```
"dy": 196,
"frame_dx": 456,
"frame_dy": 298,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    380,
    286,
    532,
    310
],
"session_memory_key": "label:autonomous task agent v2",
"cached": false
},
{
    "id": "message_bubble",
    "type": "pane",
    "label": "Agent message",
    "description": "Chat bubble with text: I will now open the browser
and login to the courseweb",
    "state": "visible",
    "dx": 928,
    "dy": 340,
    "frame_dx": 1084,
    "frame_dy": 442,
    "confidence": 0.92,
    "source": "gemini_vlm",
    "frame_bbox": [
        732,
```

```
        374,  
        1436,  
        510  
    ],  
    "session_memory_key": "pane:agent message",  
    "cached": false  
},  
{  
    "id": "status_connecting",  
    "type": "label",  
    "label": "Connecting to agent...",  
    "description": "Status message indicating connection progress",  
    "state": "visible",  
    "dx": 120,  
    "dy": 440,  
    "frame_dx": 276,  
    "frame_dy": 542,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        188,  
        526,  
        364,  
        558  
    ],  
    "session_memory_key": "label:connecting to agent...",  
    "cached": false  
},  
{
```

```
"id": "status_capturing",
"type": "label",
"label": "Capturing initial screen...",
"description": "Status message indicating screen capture progress",
"state": "visible",
"dx": 148,
"dy": 468,
"frame_dx": 304,
"frame_dy": 570,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    188,
    558,
    420,
    582
],
"session_memory_key": "label:capturing initial screen...",
"cached": false
},
{
    "id": "loading_indicator",
    "type": "pane",
    "label": "Loading indicator",
    "description": "Box showing loading dots and a cursor",
    "state": "visible",
    "dx": 124,
    "dy": 596,
    "frame_dx": 280,
```



```
"frame_dy": 698,
"confidence": 0.85,
"source": "gemini_vlm",
"frame_bbox": [
    180,
    598,
    380,
    798
],
"session_memory_key": "pane:loading indicator",
"cached": false
},
{
    "id": "chat_input_field",
    "type": "input",
    "label": "Chat input",
    "description": "Input field with placeholder: Agent is working...",
    "state": "visible",
    "dx": 684,
    "dy": 844,
    "frame_dx": 840,
    "frame_dy": 946,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
        156,
        894,
        1524,
        998
    ]
}
```

```
    ],
    "session_memory_key": "input:chat input",
    "cached": false
  }
],
"session_knowledge_base": {
  "session_id": "shot_1777704127_f68076",
  "started_at": "2026-05-02T06:42:08.013863+00:00",
  "updated_at": "2026-05-02T06:42:44.638544+00:00",
  "stale_after_misses": 3,
  "confidence_floor": 0.4,
  "decay_per_miss": 0.9,
  "frame_count": 1,
  "frame_name": "frame_1777704128_8d3627.jpg",
  "frame_index": 0,
  "image_size": {
    "width": 1920,
    "height": 1080
  },
  "known_elements": {
    "tab:frontend": {
      "element_key": "tab:frontend",
      "element_type": "tab",
      "label": "frontend",
      "bbox": [
        156.0,
        126.0,
        388.0,
        182.0
      ]
    }
  }
}
```

```
    ],
    "confidence": 0.95,
    "source": "gemini_vlm",
    "first_seen_frame": "frame_1777704128_8d3627.jpg",
    "last_seen_frame": "frame_1777704128_8d3627.jpg",
    "first_seen_at": "2026-05-02T06:42:44.638544+00:00",
    "last_seen_at": "2026-05-02T06:42:44.638544+00:00",
    "first_seen_index": 0,
    "last_seen_index": 0,
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false,
    "history": [
      {
        "frame_name": "frame_1777704128_8d3627.jpg",
        "frame_index": 0,
        "seen_at": "2026-05-02T06:42:44.638544+00:00",
        "confidence": 0.95
      }
    ]
  },
  "tab:log in to the site | courseweb": {
    "element_key": "tab:log in to the site | courseweb",
    "element_type": "tab",
    "label": "Log in to the site | CourseWeb",
    "bbox": [
      348.0,
      126.0,
```

```
        604.0,
        182.0
    ],
    "confidence": 0.95,
    "source": "gemini_vlm",
    "first_seen_frame": "frame_1777704128_8d3627.jpg",
    "last_seen_frame": "frame_1777704128_8d3627.jpg",
    "first_seen_at": "2026-05-02T06:42:44.638544+00:00",
    "last_seen_at": "2026-05-02T06:42:44.638544+00:00",
    "first_seen_index": 0,
    "last_seen_index": 0,
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false,
    "history": [
        {
            "frame_name": "frame_1777704128_8d3627.jpg",
            "frame_index": 0,
            "seen_at": "2026-05-02T06:42:44.638544+00:00",
            "confidence": 0.95
        }
    ]
},
"input:address bar": {
    "element_key": "input:address bar",
    "element_type": "input",
    "label": "Address bar",
    "bbox": [
```

```
    156.0,
    182.0,
    1308.0,
    246.0
  ],
  "confidence": 0.98,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777704128_8d3627.jpg",
  "last_seen_frame": "frame_1777704128_8d3627.jpg",
  "first_seen_at": "2026-05-02T06:42:44.638544+00:00",
  "last_seen_at": "2026-05-02T06:42:44.638544+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
  "seen_count": 1,
  "miss_count": 0,
  "stale": false,
  "cached": false,
  "history": [
    {
      "frame_name": "frame_1777704128_8d3627.jpg",
      "frame_index": 0,
      "seen_at": "2026-05-02T06:42:44.638544+00:00",
      "confidence": 0.98
    }
  ]
},
"icon:screenpilot ai logo": {
  "element_key": "icon:screenpilot ai logo",
  "element_type": "icon",
```

```
"label": "ScreenPilot AI Logo",
"bbox": [
  196.0,
  270.0,
  228.0,
  318.0
],
"confidence": 0.9,
"source": "gemini_vlm",
"first_seen_frame": "frame_1777704128_8d3627.jpg",
"last_seen_frame": "frame_1777704128_8d3627.jpg",
"first_seen_at": "2026-05-02T06:42:44.638544+00:00",
"last_seen_at": "2026-05-02T06:42:44.638544+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777704128_8d3627.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T06:42:44.638544+00:00",
    "confidence": 0.9
  }
],
},
"heading:screenpilot ai": {
```

```
"element_key": "heading:screenpilot ai",
"element_type": "heading",
"label": "ScreenPilot AI",
"bbox": [
    236.0,
    278.0,
    364.0,
    310.0
],
"confidence": 0.98,
"source": "gemini_vlm",
"first_seen_frame": "frame_1777704128_8d3627.jpg",
"last_seen_frame": "frame_1777704128_8d3627.jpg",
"first_seen_at": "2026-05-02T06:42:44.638544+00:00",
"last_seen_at": "2026-05-02T06:42:44.638544+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
    {
        "frame_name": "frame_1777704128_8d3627.jpg",
        "frame_index": 0,
        "seen_at": "2026-05-02T06:42:44.638544+00:00",
        "confidence": 0.98
    }
]
```

```
},
"label:autonomous task agent v2": {
  "element_key": "label:autonomous task agent v2",
  "element_type": "label",
  "label": "Autonomous Task Agent v2",
  "bbox": [
    380.0,
    286.0,
    532.0,
    310.0
  ],
  "confidence": 0.95,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777704128_8d3627.jpg",
  "last_seen_frame": "frame_1777704128_8d3627.jpg",
  "first_seen_at": "2026-05-02T06:42:44.638544+00:00",
  "last_seen_at": "2026-05-02T06:42:44.638544+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
  "seen_count": 1,
  "miss_count": 0,
  "stale": false,
  "cached": false,
  "history": [
    {
      "frame_name": "frame_1777704128_8d3627.jpg",
      "frame_index": 0,
      "seen_at": "2026-05-02T06:42:44.638544+00:00",
      "confidence": 0.95
    }
  ]
}
```



```
    }
  ]
},
"pane:agent message": {
  "element_key": "pane:agent message",
  "element_type": "pane",
  "label": "Agent message",
  "bbox": [
    732.0,
    374.0,
    1436.0,
    510.0
  ],
  "confidence": 0.92,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777704128_8d3627.jpg",
  "last_seen_frame": "frame_1777704128_8d3627.jpg",
  "first_seen_at": "2026-05-02T06:42:44.638544+00:00",
  "last_seen_at": "2026-05-02T06:42:44.638544+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
  "seen_count": 1,
  "miss_count": 0,
  "stale": false,
  "cached": false,
  "history": [
    {
      "frame_name": "frame_1777704128_8d3627.jpg",
      "frame_index": 0,
```

```
        "seen_at": "2026-05-02T06:42:44.638544+00:00",
        "confidence": 0.92
    }
]
},
"label:connecting to agent...": {
    "element_key": "label:connecting to agent...",
    "element_type": "label",
    "label": "Connecting to agent...",
    "bbox": [
        188.0,
        526.0,
        364.0,
        558.0
    ],
    "confidence": 0.9,
    "source": "gemini_vlm",
    "first_seen_frame": "frame_1777704128_8d3627.jpg",
    "last_seen_frame": "frame_1777704128_8d3627.jpg",
    "first_seen_at": "2026-05-02T06:42:44.638544+00:00",
    "last_seen_at": "2026-05-02T06:42:44.638544+00:00",
    "first_seen_index": 0,
    "last_seen_index": 0,
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false,
    "history": [
        {
```

```
        "frame_name": "frame_1777704128_8d3627.jpg",
        "frame_index": 0,
        "seen_at": "2026-05-02T06:42:44.638544+00:00",
        "confidence": 0.9
    }
]
},
"label:capturing initial screen...": {
    "element_key": "label:capturing initial screen...",
    "element_type": "label",
    "label": "Capturing initial screen...",
    "bbox": [
        188.0,
        558.0,
        420.0,
        582.0
    ],
    "confidence": 0.9,
    "source": "gemini_vlm",
    "first_seen_frame": "frame_1777704128_8d3627.jpg",
    "last_seen_frame": "frame_1777704128_8d3627.jpg",
    "first_seen_at": "2026-05-02T06:42:44.638544+00:00",
    "last_seen_at": "2026-05-02T06:42:44.638544+00:00",
    "first_seen_index": 0,
    "last_seen_index": 0,
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false,
```

```
"history": [  
  {  
    "frame_name": "frame_1777704128_8d3627.jpg",  
    "frame_index": 0,  
    "seen_at": "2026-05-02T06:42:44.638544+00:00",  
    "confidence": 0.9  
  }  
]  
},  
"pane:loading indicator": {  
  "element_key": "pane:loading indicator",  
  "element_type": "pane",  
  "label": "Loading indicator",  
  "bbox": [  
    180.0,  
    598.0,  
    380.0,  
    798.0  
  ],  
  "confidence": 0.85,  
  "source": "gemini_vlm",  
  "first_seen_frame": "frame_1777704128_8d3627.jpg",  
  "last_seen_frame": "frame_1777704128_8d3627.jpg",  
  "first_seen_at": "2026-05-02T06:42:44.638544+00:00",  
  "last_seen_at": "2026-05-02T06:42:44.638544+00:00",  
  "first_seen_index": 0,  
  "last_seen_index": 0,  
  "seen_count": 1,  
  "miss_count": 0,
```

```
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777704128_8d3627.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T06:42:44.638544+00:00",
    "confidence": 0.85
  }
],
},
"input:chat input": {
  "element_key": "input:chat input",
  "element_type": "input",
  "label": "Chat input",
  "bbox": [
    156.0,
    894.0,
    1524.0,
    998.0
  ],
  "confidence": 0.98,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777704128_8d3627.jpg",
  "last_seen_frame": "frame_1777704128_8d3627.jpg",
  "first_seen_at": "2026-05-02T06:42:44.638544+00:00",
  "last_seen_at": "2026-05-02T06:42:44.638544+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
```

```
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777704128_8d3627.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T06:42:44.638544+00:00",
    "confidence": 0.98
  }
]
},
"resolved_elements": [
  {
    "id": "tab_frontend",
    "type": "tab",
    "label": "frontend",
    "description": "Browser tab labeled frontend",
    "state": "visible",
    "dx": 116,
    "dy": 52,
    "frame_dx": 272,
    "frame_dy": 154,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
      156,
```

```
        126,  
        388,  
        182  
    ],  
    "session_memory_key": "tab:frontend",  
    "cached": false  
},  
{  
    "id": "tab_courseweb",  
    "type": "tab",  
    "label": "Log in to the site | CourseWeb",  
    "description": "Browser tab for CourseWeb login",  
    "state": "visible",  
    "dx": 320,  
    "dy": 52,  
    "frame_dx": 476,  
    "frame_dy": 154,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        348,  
        126,  
        604,  
        182  
    ],  
    "session_memory_key": "tab:log in to the site | courseweb",  
    "cached": false  
},  
{
```

```
"id": "address_bar",
"type": "input",
"label": "Address bar",
"description": "Browser address bar showing localhost:5173",
"state": "visible",
"dx": 576,
"dy": 112,
"frame_dx": 732,
"frame_dy": 214,
"confidence": 0.98,
"source": "gemini_vlm",
"frame_bbox": [
    156,
    182,
    1308,
    246
],
"session_memory_key": "input:address bar",
"cached": false
},
{
    "id": "app_logo",
    "type": "icon",
    "label": "ScreenPilot AI Logo",
    "description": "Chip icon for ScreenPilot AI",
    "state": "visible",
    "dx": 56,
    "dy": 192,
    "frame_dx": 212,
```



```
"frame_dy": 294,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    196,
    270,
    228,
    318
],
"session_memory_key": "icon:screenpilot ai logo",
"cached": false
},
{
    "id": "app_title",
    "type": "heading",
    "label": "ScreenPilot AI",
    "description": "Application title text",
    "state": "visible",
    "dx": 144,
    "dy": 192,
    "frame_dx": 300,
    "frame_dy": 294,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
        236,
        278,
        364,
        310
    ]
}
```

```
    ],
    "session_memory_key": "heading:screenpilot ai",
    "cached": false
  },
  {
    "id": "app_version",
    "type": "label",
    "label": "Autonomous Task Agent v2",
    "description": "Application version badge",
    "state": "visible",
    "dx": 300,
    "dy": 196,
    "frame_dx": 456,
    "frame_dy": 298,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
      380,
      286,
      532,
      310
    ],
    "session_memory_key": "label:autonomous task agent v2",
    "cached": false
  },
  {
    "id": "message_bubble",
    "type": "pane",
    "label": "Agent message",
```

```
        "description": "Chat bubble with text: I will now open the browser  
and login to the courseweb",  
        "state": "visible",  
        "dx": 928,  
        "dy": 340,  
        "frame_dx": 1084,  
        "frame_dy": 442,  
        "confidence": 0.92,  
        "source": "gemini_vlm",  
        "frame_bbox": [  
            732,  
            374,  
            1436,  
            510  
        ],  
        "session_memory_key": "pane:agent message",  
        "cached": false  
    },  
    {  
        "id": "status_connecting",  
        "type": "label",  
        "label": "Connecting to agent...",  
        "description": "Status message indicating connection progress",  
        "state": "visible",  
        "dx": 120,  
        "dy": 440,  
        "frame_dx": 276,  
        "frame_dy": 542,  
        "confidence": 0.9,
```

```
"source": "gemini_vlm",
"frame_bbox": [
    188,
    526,
    364,
    558
],
"session_memory_key": "label:connecting to agent...",
"cached": false
},
{
    "id": "status_capturing",
    "type": "label",
    "label": "Capturing initial screen...",
    "description": "Status message indicating screen capture progress",
    "state": "visible",
    "dx": 148,
    "dy": 468,
    "frame_dx": 304,
    "frame_dy": 570,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        188,
        558,
        420,
        582
    ],
    "session_memory_key": "label:capturing initial screen..."
```

```
    "cached": false
  },
  {
    "id": "loading_indicator",
    "type": "pane",
    "label": "Loading indicator",
    "description": "Box showing loading dots and a cursor",
    "state": "visible",
    "dx": 124,
    "dy": 596,
    "frame_dx": 280,
    "frame_dy": 698,
    "confidence": 0.85,
    "source": "gemini_vlm",
    "frame_bbox": [
      180,
      598,
      380,
      798
    ],
    "session_memory_key": "pane:loading indicator",
    "cached": false
  },
  {
    "id": "chat_input_field",
    "type": "input",
    "label": "Chat input",
    "description": "Input field with placeholder: Agent is working...",
    "state": "visible",
```

```
    "dx": 684,  
    "dy": 844,  
    "frame_dx": 840,  
    "frame_dy": 946,  
    "confidence": 0.98,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        156,  
        894,  
        1524,  
        998  
    ],  
    "session_memory_key": "input:chat input",  
    "cached": false  
}  
],  
"current_elements": [  
    {  
        "id": "tab_frontend",  
        "type": "tab",  
        "label": "frontend",  
        "description": "Browser tab labeled frontend",  
        "state": "visible",  
        "dx": 116,  
        "dy": 52,  
        "frame_dx": 272,  
        "frame_dy": 154,  
        "confidence": 0.95,  
        "source": "gemini_vlm",
```

```
"frame_bbox": [
    156,
    126,
    388,
    182
],
"session_memory_key": "tab:frontend",
"cached": false
},
{
    "id": "tab_courseweb",
    "type": "tab",
    "label": "Log in to the site | CourseWeb",
    "description": "Browser tab for CourseWeb login",
    "state": "visible",
    "dx": 320,
    "dy": 52,
    "frame_dx": 476,
    "frame_dy": 154,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        348,
        126,
        604,
        182
    ],
    "session_memory_key": "tab:log in to the site | courseweb",
    "cached": false
}
```

```
},  
{  
  "id": "address_bar",  
  "type": "input",  
  "label": "Address bar",  
  "description": "Browser address bar showing localhost:5173",  
  "state": "visible",  
  "dx": 576,  
  "dy": 112,  
  "frame_dx": 732,  
  "frame_dy": 214,  
  "confidence": 0.98,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    156,  
    182,  
    1308,  
    246  
  ],  
  "session_memory_key": "input:address bar",  
  "cached": false  
},  
{  
  "id": "app_logo",  
  "type": "icon",  
  "label": "ScreenPilot AI Logo",  
  "description": "Chip icon for ScreenPilot AI",  
  "state": "visible",  
  "dx": 56,
```



```
"dy": 192,
"frame_dx": 212,
"frame_dy": 294,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    196,
    270,
    228,
    318
],
"session_memory_key": "icon:screenpilot ai logo",
"cached": false
},
{
    "id": "app_title",
    "type": "heading",
    "label": "ScreenPilot AI",
    "description": "Application title text",
    "state": "visible",
    "dx": 144,
    "dy": 192,
    "frame_dx": 300,
    "frame_dy": 294,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
        236,
        278,
```

```
        364,  
        310  
    ],  
    "session_memory_key": "heading:screenpilot ai",  
    "cached": false  
},  
{  
    "id": "app_version",  
    "type": "label",  
    "label": "Autonomous Task Agent v2",  
    "description": "Application version badge",  
    "state": "visible",  
    "dx": 300,  
    "dy": 196,  
    "frame_dx": 456,  
    "frame_dy": 298,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        380,  
        286,  
        532,  
        310  
    ],  
    "session_memory_key": "label:autonomous task agent v2",  
    "cached": false  
},  
{  
    "id": "message_bubble",
```

```
        "type": "pane",
        "label": "Agent message",
        "description": "Chat bubble with text: I will now open the browser
and login to the courseweb",
        "state": "visible",
        "dx": 928,
        "dy": 340,
        "frame_dx": 1084,
        "frame_dy": 442,
        "confidence": 0.92,
        "source": "gemini_vlm",
        "frame_bbox": [
            732,
            374,
            1436,
            510
        ],
        "session_memory_key": "pane:agent message",
        "cached": false
    },
    {
        "id": "status_connecting",
        "type": "label",
        "label": "Connecting to agent...",
        "description": "Status message indicating connection progress",
        "state": "visible",
        "dx": 120,
        "dy": 440,
        "frame_dx": 276,
```

```
"frame_dy": 542,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    188,
    526,
    364,
    558
],
"session_memory_key": "label:connecting to agent...",
"cached": false
},
{
    "id": "status_capturing",
    "type": "label",
    "label": "Capturing initial screen...",
    "description": "Status message indicating screen capture progress",
    "state": "visible",
    "dx": 148,
    "dy": 468,
    "frame_dx": 304,
    "frame_dy": 570,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        188,
        558,
        420,
        582
    ]
}
```

```
    ],
    "session_memory_key": "label:capturing initial screen...",
    "cached": false
  },
  {
    "id": "loading_indicator",
    "type": "pane",
    "label": "Loading indicator",
    "description": "Box showing loading dots and a cursor",
    "state": "visible",
    "dx": 124,
    "dy": 596,
    "frame_dx": 280,
    "frame_dy": 698,
    "confidence": 0.85,
    "source": "gemini_vlm",
    "frame_bbox": [
      180,
      598,
      380,
      798
    ],
    "session_memory_key": "pane:loading indicator",
    "cached": false
  },
  {
    "id": "chat_input_field",
    "type": "input",
    "label": "Chat input",
```

```
    "description": "Input field with placeholder: Agent is working...",
    "state": "visible",
    "dx": 684,
    "dy": 844,
    "frame_dx": 840,
    "frame_dy": 946,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
        156,
        894,
        1524,
        998
    ],
    "session_memory_key": "input:chat input",
    "cached": false
  }
]
}
}
},
"todo": {
  "steps": [
    {
      "id": 1,
      "action": "Click on the 'Log in to the site | CourseWeb' tab",
      "target": "Log in to the site | CourseWeb tab at (476, 154)",
      "expected_result": "The browser switches to the CourseWeb login page",
```

```
    "needs_user_data": false,
    "user_data_field": null,
    "status": "done"
  },
  {
    "id": 2,
    "action": "Enter the username and password into the input fields and
click the login button",
    "target": "CourseWeb login form",
    "expected_result": "The user is logged into CourseWeb and the dashboard
or home page loads",
    "needs_user_data": true,
    "user_data_field": "login_credentials",
    "status": "failed"
  }
],
"notes": "The agent must first switch to the inactive browser tab containing
the CourseWeb login page. After the page loads, the agent will require user
credentials to complete the login process.",
"estimated_complexity": "simple"
},
"step_plans": [
  {
    "step_id": 1,
    "action": "Click on the 'Log in to the site | CourseWeb' tab",
    "target": "Log in to the site | CourseWeb tab at (476, 154)",
    "expected_result": "The browser switches to the CourseWeb login page",
    "hid_commands": [
      {
        "cmd": "mouse_move",
```

```
        "dx": 489,
        "dy": 144
    },
    {
        "cmd": "mouse_click",
        "button": "left"
    }
],
    "reasoning": "Moving the cursor to the 'Log in to the site | CourseWeb' tab at (489, 144) and clicking to switch to the login page as per Step 1."
},
{
    "step_id": 2,
    "action": "Enter the username and password into the input fields and click the login button",
    "target": "CourseWeb login form",
    "expected_result": "The user is logged into CourseWeb and the dashboard or home page loads",
    "hid_commands": [
        {
            "cmd": "mouse_move",
            "dx": 962,
            "dy": 451
        },
        {
            "cmd": "mouse_click",
            "button": "left"
        }
    ]
},
```



```
    "reasoning": "Focusing the 'Username or email' input field. Since this step requires sensitive personal credentials (username and password), I am positioning the cursor and focusing the field for the user to provide the necessary input."
```

```
  }
```

```
],
```

```
"step_results": [
```

```
{
```

```
  "step_id": 1,
```

```
  "action": "Click on the 'Log in to the site | CourseWeb' tab",
```

```
  "attempt": 1,
```

```
  "exec_status": "success",
```

```
  "eval_status": "done",
```

```
  "confidence": 0.95,
```

```
  "reason": "The 'SLIIT UNI CourseWeb' heading is now visible on the screen, indicating the browser has successfully switched to the login page.",
```

```
  "final_status": "executing",
```

```
  "timestamp": "2026-05-02T06:44:20.103587+00:00"
```

```
},
```

```
{
```

```
  "step_id": 2,
```

```
  "action": "Enter the username and password into the input fields and click the login button",
```

```
  "attempt": 3,
```

```
  "exec_status": "success",
```

```
  "eval_status": "retry",
```

```
  "confidence": 0.0,
```

```
  "reason": "Evaluation error: No valid JSON found in LLM response: '{\\n \\\"status\\\": \\\"needs_\\\"',
```

```
  "final_status": "executing",
```

```
  "timestamp": "2026-05-02T06:51:02.935365+00:00"
```

```
}
```

```
],
"final_report": {
  "success": false,
  "summary": "The login process failed because the input fields could not be detected after switching to the CourseWeb tab.",
  "message": "The task was partially completed. While the agent successfully navigated to the 'Log in to the site | CourseWeb' tab, the subsequent login steps could not be performed.\n\nStep results:\n\u2705 Step 1: Click on the 'Log in to the site | CourseWeb' tab\n\u274c Step 2: Enter the username and password into the input fields and click the login button\n\nThe final screen state indicated that no UI elements were detected, which prevented the agent from locating the username and password fields.",
  "steps_completed": 1,
  "steps_failed": 1,
  "issues": [
    "The login page failed to render UI elements in time for the next step.",
    "Credentials could not be entered due to missing input fields."
  ],
  "recommendations": [
    "Check your internet connection to ensure the page loads correctly.",
    "Try refreshing the page manually if it remains empty.",
    "Ensure that the login page does not require a manual CAPTCHA or external authentication before fields appear."
  ]
},
"session_memory": {
  "run_id": "30427c77-968c-4a27-a252-6b8db30fe924",
  "user_task": "click on the \"Log in to the site | CourseWeb\" tab on this browser and login to the courseweb",
  "started_at": "2026-05-02T06:41:57.870381+00:00",
  "updated_at": "2026-05-02T06:51:05.696094+00:00",
  "stale_after_misses": 2,
```

```
"confidence_floor": 0.4,
"decay_per_miss": 0.9,
"frame_index": 12,
"known_elements": {
  "tab:frontend": {
    "element_key": "tab:frontend",
    "element_type": "tab",
    "label": "frontend",
    "bbox": [
      156.0,
      126.0,
      388.0,
      182.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 0,
    "last_seen_index": 0,
    "first_seen_at": "2026-05-02T06:42:44.685964+00:00",
    "last_seen_at": "2026-05-02T06:42:44.685964+00:00",
    "seen_count": 1,
    "miss_count": 11,
    "stale": true,
    "cached": true
  },
  "tab:log in to the site | courseweb": {
    "element_key": "tab:log in to the site | courseweb",
```

```
"element_type": "tab",
"label": "Log in to the site | CourseWeb",
"bbox": [
  357.0,
  116.0,
  621.0,
  172.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 1,
"first_seen_at": "2026-05-02T06:42:44.685964+00:00",
"last_seen_at": "2026-05-02T06:43:27.857541+00:00",
"seen_count": 2,
"miss_count": 10,
"stale": true,
"cached": true
},
"input:address bar": {
  "element_key": "input:address bar",
  "element_type": "input",
  "label": "Address Bar",
  "bbox": [
    269.0,
    172.0,
    1341.0,
```

```
    228.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 1,
  "first_seen_at": "2026-05-02T06:42:44.685964+00:00",
  "last_seen_at": "2026-05-02T06:43:27.857541+00:00",
  "seen_count": 2,
  "miss_count": 10,
  "stale": true,
  "cached": true
},
"icon:screenpilot ai logo": {
  "element_key": "icon:screenpilot ai logo",
  "element_type": "icon",
  "label": "ScreenPilot AI Logo",
  "bbox": [
    196.0,
    270.0,
    228.0,
    318.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
```

```
"first_seen_index": 0,
"last_seen_index": 0,
"first_seen_at": "2026-05-02T06:42:44.685964+00:00",
"last_seen_at": "2026-05-02T06:42:44.685964+00:00",
"seen_count": 1,
"miss_count": 11,
"stale": true,
"cached": true
},
"heading:screenpilot ai": {
  "element_key": "heading:screenpilot ai",
  "element_type": "heading",
  "label": "ScreenPilot AI",
  "bbox": [
    236.0,
    278.0,
    364.0,
    310.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 0,
  "first_seen_at": "2026-05-02T06:42:44.685964+00:00",
  "last_seen_at": "2026-05-02T06:42:44.685964+00:00",
  "seen_count": 1,
  "miss_count": 11,
```

```
"stale": true,
"cached": true
},
"label:autonomous task agent v2": {
  "element_key": "label:autonomous task agent v2",
  "element_type": "label",
  "label": "Autonomous Task Agent v2",
  "bbox": [
    380.0,
    286.0,
    532.0,
    310.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 1,
  "first_seen_at": "2026-05-02T06:42:44.685964+00:00",
  "last_seen_at": "2026-05-02T06:43:27.857541+00:00",
  "seen_count": 2,
  "miss_count": 10,
  "stale": true,
  "cached": true
},
"pane:agent message": {
  "element_key": "pane:agent message",
  "element_type": "pane",
```

```
"label": "Agent message",
"bbox": [
  732.0,
  374.0,
  1436.0,
  510.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 0,
"first_seen_at": "2026-05-02T06:42:44.685964+00:00",
"last_seen_at": "2026-05-02T06:42:44.685964+00:00",
"seen_count": 1,
"miss_count": 11,
"stale": true,
"cached": true
},
"label:connecting to agent...": {
  "element_key": "label:connecting to agent...",
  "element_type": "label",
  "label": "Connecting to agent...",
  "bbox": [
    188.0,
    526.0,
    364.0,
    558.0
  ]
}
```



```
],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 0,
  "first_seen_at": "2026-05-02T06:42:44.685964+00:00",
  "last_seen_at": "2026-05-02T06:42:44.685964+00:00",
  "seen_count": 1,
  "miss_count": 11,
  "stale": true,
  "cached": true
},
"label:capturing initial screen...": {
  "element_key": "label:capturing initial screen...",
  "element_type": "label",
  "label": "Capturing initial screen...",
  "bbox": [
    188.0,
    558.0,
    420.0,
    582.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
```

```
"last_seen_index": 0,
"first_seen_at": "2026-05-02T06:42:44.685964+00:00",
"last_seen_at": "2026-05-02T06:42:44.685964+00:00",
"seen_count": 1,
"miss_count": 11,
"stale": true,
"cached": true
},
"pane:loading indicator": {
  "element_key": "pane:loading indicator",
  "element_type": "pane",
  "label": "Loading indicator",
  "bbox": [
    180.0,
    598.0,
    380.0,
    798.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 0,
  "first_seen_at": "2026-05-02T06:42:44.685964+00:00",
  "last_seen_at": "2026-05-02T06:42:44.685964+00:00",
  "seen_count": 1,
  "miss_count": 11,
  "stale": true,
```

```
    "cached": true
  },
  "input:chat input": {
    "element_key": "input:chat input",
    "element_type": "input",
    "label": "Chat input",
    "bbox": [
      156.0,
      894.0,
      1524.0,
      998.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 0,
    "last_seen_index": 0,
    "first_seen_at": "2026-05-02T06:42:44.685964+00:00",
    "last_seen_at": "2026-05-02T06:42:44.685964+00:00",
    "seen_count": 1,
    "miss_count": 11,
    "stale": true,
    "cached": true
  },
  "tab:localhost:5173": {
    "element_key": "tab:localhost:5173",
    "element_type": "tab",
    "label": "localhost:5173",
```

```
"bbox": [
  269.0,
  116.0,
  405.0,
  156.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 1,
"last_seen_index": 1,
"first_seen_at": "2026-05-02T06:43:27.857541+00:00",
"last_seen_at": "2026-05-02T06:43:27.857541+00:00",
"seen_count": 1,
"miss_count": 10,
"stale": true,
"cached": true
},
"text:screenpilot ai": {
  "element_key": "text:screenpilot ai",
  "element_type": "text",
  "label": "ScreenPilot AI",
  "bbox": [
    277.0,
    260.0,
    413.0,
    276.0
  ],
}
```

```
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 1,
"last_seen_index": 1,
"first_seen_at": "2026-05-02T06:43:27.857541+00:00",
"last_seen_at": "2026-05-02T06:43:27.857541+00:00",
"seen_count": 1,
"miss_count": 10,
"stale": true,
"cached": true
},
"heading:task plan": {
  "element_key": "heading:task plan",
  "element_type": "heading",
  "label": "Task Plan",
  "bbox": [
    269.0,
    508.0,
    341.0,
    540.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
  "last_seen_index": 1,
```

```
"first_seen_at": "2026-05-02T06:43:27.857541+00:00",
"last_seen_at": "2026-05-02T06:43:27.857541+00:00",
"seen_count": 1,
"miss_count": 10,
"stale": true,
"cached": true
},
"label:0/2 done": {
  "element_key": "label:0/2 done",
  "element_type": "label",
  "label": "0/2 done",
  "bbox": [
    757.0,
    516.0,
    821.0,
    540.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
  "last_seen_index": 1,
  "first_seen_at": "2026-05-02T06:43:27.857541+00:00",
  "last_seen_at": "2026-05-02T06:43:27.857541+00:00",
  "seen_count": 1,
  "miss_count": 10,
  "stale": true,
  "cached": true
}
```

```
},
"pane:step 1 highlight": {
  "element_key": "pane:step 1 highlight",
  "element_type": "pane",
  "label": "Step 1 Highlight",
  "bbox": [
    269.0,
    596.0,
    845.0,
    660.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
  "last_seen_index": 1,
  "first_seen_at": "2026-05-02T06:43:27.857541+00:00",
  "last_seen_at": "2026-05-02T06:43:27.857541+00:00",
  "seen_count": 1,
  "miss_count": 10,
  "stale": true,
  "cached": true
},
"text:click on the 'log in to the site | courseweb' tab": {
  "element_key": "text:click on the 'log in to the site | courseweb' tab",
  "element_type": "text",
  "label": "Click on the 'Log in to the site | CourseWeb' tab",
  "bbox": [
```

```
        269.0,
        612.0,
        621.0,
        644.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 1,
    "last_seen_index": 1,
    "first_seen_at": "2026-05-02T06:43:27.857541+00:00",
    "last_seen_at": "2026-05-02T06:43:27.857541+00:00",
    "seen_count": 1,
    "miss_count": 10,
    "stale": true,
    "cached": true
},
    "text:enter the username and password into the input fields and click the
login button": {
        "element_key": "text:enter the username and password into the input
fields and click the login button",
        "element_type": "text",
        "label": "Enter the username and password into the input fields and click
the login button",
        "bbox": [
            269.0,
            660.0,
            733.0,
            692.0
```



```
],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
  "last_seen_index": 1,
  "first_seen_at": "2026-05-02T06:43:27.857541+00:00",
  "last_seen_at": "2026-05-02T06:43:27.857541+00:00",
  "seen_count": 1,
  "miss_count": 10,
  "stale": true,
  "cached": true
},
"pane:terminal output": {
  "element_key": "pane:terminal output",
  "element_type": "pane",
  "label": "Terminal Output",
  "bbox": [
    269.0,
    700.0,
    917.0,
    932.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
```

```
"last_seen_index": 1,
"first_seen_at": "2026-05-02T06:43:27.857541+00:00",
"last_seen_at": "2026-05-02T06:43:27.857541+00:00",
"seen_count": 1,
"miss_count": 10,
"stale": true,
"cached": true
},
"text:agent is working...": {
  "element_key": "text:agent is working...",
  "element_type": "text",
  "label": "Agent is working...",
  "bbox": [
    269.0,
    1014.0,
    374.0,
    1015.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
  "last_seen_index": 1,
  "first_seen_at": "2026-05-02T06:43:27.857541+00:00",
  "last_seen_at": "2026-05-02T06:43:27.857541+00:00",
  "seen_count": 1,
  "miss_count": 10,
  "stale": true,
```

```
    "cached": true
  },
  "button:send message": {
    "element_key": "button:send message",
    "element_type": "button",
    "label": "Send Message",
    "bbox": [
      1333.0,
      1014.0,
      1380.0,
      1015.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 1,
    "last_seen_index": 1,
    "first_seen_at": "2026-05-02T06:43:27.857541+00:00",
    "last_seen_at": "2026-05-02T06:43:27.857541+00:00",
    "seen_count": 1,
    "miss_count": 10,
    "stale": true,
    "cached": true
  },
  "button:start": {
    "element_key": "button:start",
    "element_type": "button",
    "label": "Start",
```

```
"bbox": [
  882.0,
  943.0,
  906.0,
  963.0
],
"confidence": 0.729,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 1,
"last_seen_index": 9,
"first_seen_at": "2026-05-02T06:43:27.857541+00:00",
"last_seen_at": "2026-05-02T06:50:04.893203+00:00",
"seen_count": 2,
"miss_count": 2,
"stale": true,
"cached": true
},
"input:search": {
  "element_key": "input:search",
  "element_type": "input",
  "label": "Search",
  "bbox": [
    544.0,
    1014.0,
    713.0,
    1015.0
  ],
}
```

```
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 1,
"last_seen_index": 1,
"first_seen_at": "2026-05-02T06:43:27.857541+00:00",
"last_seen_at": "2026-05-02T06:43:27.857541+00:00",
"seen_count": 1,
"miss_count": 10,
"stale": true,
"cached": true
},
"icon:vs code": {
  "element_key": "icon:vs code",
  "element_type": "icon",
  "label": "VS Code",
  "bbox": [
    994.0,
    1014.0,
    1017.0,
    1015.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
  "last_seen_index": 1,
```

```
"first_seen_at": "2026-05-02T06:43:27.857541+00:00",
"last_seen_at": "2026-05-02T06:43:27.857541+00:00",
"seen_count": 1,
"miss_count": 10,
"stale": true,
"cached": true
},
"icon:spotify": {
  "element_key": "icon:spotify",
  "element_type": "icon",
  "label": "Spotify",
  "bbox": [
    826.0,
    943.0,
    866.0,
    962.0
  ],
  "confidence": 0.5905,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
  "last_seen_index": 7,
  "first_seen_at": "2026-05-02T06:43:27.857541+00:00",
  "last_seen_at": "2026-05-02T06:48:31.574903+00:00",
  "seen_count": 2,
  "miss_count": 4,
  "stale": true,
  "cached": true
}
```

```
},
"icon:chrome": {
  "element_key": "icon:chrome",
  "element_type": "icon",
  "label": "Chrome",
  "bbox": [
    1061.0,
    1014.0,
    1084.0,
    1015.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
  "last_seen_index": 1,
  "first_seen_at": "2026-05-02T06:43:27.857541+00:00",
  "last_seen_at": "2026-05-02T06:43:27.857541+00:00",
  "seen_count": 1,
  "miss_count": 10,
  "stale": true,
  "cached": true
},
"text:time and date": {
  "element_key": "text:time and date",
  "element_type": "text",
  "label": "Time and Date",
  "bbox": [
```

```
    1357.0,
    1014.0,
    1409.0,
    1015.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
  "last_seen_index": 1,
  "first_seen_at": "2026-05-02T06:43:27.857541+00:00",
  "last_seen_at": "2026-05-02T06:43:27.857541+00:00",
  "seen_count": 1,
  "miss_count": 10,
  "stale": true,
  "cached": true
},
"button:ask gemini": {
  "element_key": "button:ask gemini",
  "element_type": "button",
  "label": "Ask Gemini",
  "bbox": [
    1197.0,
    156.0,
    1309.0,
    180.0
  ],
  "confidence": 0.4,
```



```
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 1,
"last_seen_index": 1,
"first_seen_at": "2026-05-02T06:43:27.857541+00:00",
"last_seen_at": "2026-05-02T06:43:27.857541+00:00",
"seen_count": 1,
"miss_count": 10,
"stale": true,
"cached": true
},
"heading:sliit uni courseweb": {
  "element_key": "heading:sliit uni courseweb",
  "element_type": "heading",
  "label": "SLIIT UNI CourseWeb",
  "bbox": [
    730.0,
    270.0,
    1194.0,
    318.0
  ],
  "confidence": 0.6925,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 2,
  "last_seen_index": 8,
  "first_seen_at": "2026-05-02T06:44:15.667520+00:00",
```

```
"last_seen_at": "2026-05-02T06:49:17.117694+00:00",
"seen_count": 3,
"miss_count": 3,
"stale": true,
"cached": true
},
"text:note: your login will validate through microsoft o365. always use
'sliit login' button to get access to courseweb.": {
  "element_key": "text:note: your login will validate through microsoft
o365. always use 'sliit login' button to get access to courseweb.",
  "element_type": "text",
  "label": "Note: Your login will validate through Microsoft 0365. Always
use 'SLIIT Login' button to get access to CourseWeb.",
  "bbox": [
    745.0,
    902.0,
    1169.0,
    958.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 3,
  "last_seen_index": 3,
  "first_seen_at": "2026-05-02T06:44:59.983226+00:00",
  "last_seen_at": "2026-05-02T06:44:59.983226+00:00",
  "seen_count": 1,
  "miss_count": 8,
  "stale": true,
```

```
    "cached": true
  },
  "text:sliit uni courseweb": {
    "element_key": "text:sliit uni courseweb",
    "element_type": "text",
    "label": "SLIIT UNI CourseWeb",
    "bbox": [
      730.0,
      263.0,
      1194.0,
      335.0
    ],
    "confidence": 0.4543,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 4,
    "last_seen_index": 4,
    "first_seen_at": "2026-05-02T06:45:49.004008+00:00",
    "last_seen_at": "2026-05-02T06:45:49.004008+00:00",
    "seen_count": 1,
    "miss_count": 7,
    "stale": true,
    "cached": true
  },
  "icon:username icon": {
    "element_key": "icon:username icon",
    "element_type": "icon",
    "label": "Username Icon",
```

```
"bbox": [
  770.0,
  343.0,
  794.0,
  359.0
],
"confidence": 0.4305,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 4,
"last_seen_index": 4,
"first_seen_at": "2026-05-02T06:45:49.004008+00:00",
"last_seen_at": "2026-05-02T06:45:49.004008+00:00",
"seen_count": 1,
"miss_count": 7,
"stale": true,
"cached": true
},
"icon:password icon": {
  "element_key": "icon:password icon",
  "element_type": "icon",
  "label": "Password Icon",
  "bbox": [
    770.0,
    391.0,
    794.0,
    407.0
  ],

```

```
"confidence": 0.4305,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 4,
"last_seen_index": 4,
"first_seen_at": "2026-05-02T06:45:49.004008+00:00",
"last_seen_at": "2026-05-02T06:45:49.004008+00:00",
"seen_count": 1,
"miss_count": 7,
"stale": true,
"cached": true
},
"button:log in": {
  "element_key": "button:log in",
  "element_type": "button",
  "label": "Log in",
  "bbox": [
    890.0,
    430.0,
    1042.0,
    470.0
  ],
  "confidence": 0.7938,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 4,
  "last_seen_index": 9,
```

```
"first_seen_at": "2026-05-02T06:45:49.004008+00:00",
"last_seen_at": "2026-05-02T06:50:04.893203+00:00",
"seen_count": 5,
"miss_count": 2,
"stale": true,
"cached": true
},
"text:log in using your account on:": {
  "element_key": "text:log in using your account on:",
  "element_type": "text",
  "label": "Log in using your account on:",
  "bbox": [
    866.0,
    543.0,
    1058.0,
    583.0
  ],
  "confidence": 0.4305,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 4,
  "last_seen_index": 4,
  "first_seen_at": "2026-05-02T06:45:49.004008+00:00",
  "last_seen_at": "2026-05-02T06:45:49.004008+00:00",
  "seen_count": 1,
  "miss_count": 7,
  "stale": true,
  "cached": true
}
```

```
},
"button:microsoft login": {
  "element_key": "button:microsoft login",
  "element_type": "button",
  "label": "Microsoft Login",
  "bbox": [
    914.0,
    583.0,
    946.0,
    615.0
  ],
  "confidence": 0.44,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 4,
  "last_seen_index": 4,
  "first_seen_at": "2026-05-02T06:45:49.004008+00:00",
  "last_seen_at": "2026-05-02T06:45:49.004008+00:00",
  "seen_count": 1,
  "miss_count": 7,
  "stale": true,
  "cached": true
},
"text:is this your first time here?": {
  "element_key": "text:is this your first time here?",
  "element_type": "text",
  "label": "Is this your first time here?",
  "bbox": [
```

```
      842.0,
      903.0,
      1074.0,
      943.0
    ],
    "confidence": 0.5905,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 4,
    "last_seen_index": 7,
    "first_seen_at": "2026-05-02T06:45:49.004008+00:00",
    "last_seen_at": "2026-05-02T06:48:31.574903+00:00",
    "seen_count": 3,
    "miss_count": 4,
    "stale": true,
    "cached": true
  },
  "text:click on sliit login to proceed": {
    "element_key": "text:click on sliit login to proceed",
    "element_type": "text",
    "label": "Click on SLIIT Login to proceed",
    "bbox": [
      770.0,
      895.0,
      1002.0,
      919.0
    ],
    "confidence": 0.4065,
```



```
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 4,
"last_seen_index": 4,
"first_seen_at": "2026-05-02T06:45:49.004008+00:00",
"last_seen_at": "2026-05-02T06:45:49.004008+00:00",
"seen_count": 1,
"miss_count": 7,
"stale": true,
"cached": true
},
"pane:windows taskbar": {
  "element_key": "pane:windows taskbar",
  "element_type": "pane",
  "label": "Windows Taskbar",
  "bbox": [
    178.0,
    935.0,
    1754.0,
    962.0
  ],
  "confidence": 0.643,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 4,
  "last_seen_index": 7,
  "first_seen_at": "2026-05-02T06:45:49.004008+00:00",
```

```
    "last_seen_at": "2026-05-02T06:48:31.574903+00:00",
    "seen_count": 2,
    "miss_count": 4,
    "stale": true,
    "cached": true
  },
  "toolbar:taskbar icons": {
    "element_key": "toolbar:taskbar icons",
    "element_type": "toolbar",
    "label": "Taskbar Icons",
    "bbox": [
      618.0,
      927.0,
      1306.0,
      962.0
    ],
    "confidence": 0.4543,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 4,
    "last_seen_index": 4,
    "first_seen_at": "2026-05-02T06:45:49.004008+00:00",
    "last_seen_at": "2026-05-02T06:45:49.004008+00:00",
    "seen_count": 1,
    "miss_count": 7,
    "stale": true,
    "cached": true
  },
}
```

```
"text:12:15 pm": {
  "element_key": "text:12:15 pm",
  "element_type": "text",
  "label": "12:15 PM",
  "bbox": [
    1642.0,
    927.0,
    1746.0,
    962.0
  ],
  "confidence": 0.4543,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 4,
  "last_seen_index": 4,
  "first_seen_at": "2026-05-02T06:45:49.004008+00:00",
  "last_seen_at": "2026-05-02T06:45:49.004008+00:00",
  "seen_count": 1,
  "miss_count": 7,
  "stale": true,
  "cached": true
},
"icon:mouse cursor": {
  "element_key": "icon:mouse cursor",
  "element_type": "icon",
  "label": "Mouse Cursor",
  "bbox": [
    1594.0,
```

```
    719.0,  
    1618.0,  
    751.0  
  ],  
  "confidence": 0.4305,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 4,  
  "last_seen_index": 4,  
  "first_seen_at": "2026-05-02T06:45:49.004008+00:00",  
  "last_seen_at": "2026-05-02T06:45:49.004008+00:00",  
  "seen_count": 1,  
  "miss_count": 7,  
  "stale": true,  
  "cached": true  
},  
"image:sliit uni courseweb logo": {  
  "element_key": "image:sliit uni courseweb logo",  
  "element_type": "image",  
  "label": "SLIIT UNI CourseWeb Logo",  
  "bbox": [  
    730.0,  
    271.0,  
    1194.0,  
    327.0  
  ],  
  "confidence": 0.7695,  
  "source": "gemini_vlm",
```

```
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 5,
"last_seen_index": 9,
"first_seen_at": "2026-05-02T06:46:50.824239+00:00",
"last_seen_at": "2026-05-02T06:50:04.893203+00:00",
"seen_count": 2,
"miss_count": 2,
"stale": true,
"cached": true
},
"pane:browser overlay": {
  "element_key": "pane:browser overlay",
  "element_type": "pane",
  "label": "Browser Overlay",
  "bbox": [
    170.0,
    199.0,
    1394.0,
    391.0
  ],
  "confidence": 0.5609,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 6,
  "last_seen_index": 6,
  "first_seen_at": "2026-05-02T06:47:43.049391+00:00",
  "last_seen_at": "2026-05-02T06:47:43.049391+00:00",
```

```
"seen_count": 1,
"miss_count": 5,
"stale": true,
"cached": true
},
"link:ask google about this page": {
  "element_key": "link:ask google about this page",
  "element_type": "link",
  "label": "Ask Google about this page",
  "bbox": [
    274.0,
    223.0,
    602.0,
    263.0
  ],
  "confidence": 0.5315,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 6,
  "last_seen_index": 6,
  "first_seen_at": "2026-05-02T06:47:43.049391+00:00",
  "last_seen_at": "2026-05-02T06:47:43.049391+00:00",
  "seen_count": 1,
  "miss_count": 5,
  "stale": true,
  "cached": true
},
"link:help me with this": {
```

```
"element_key": "link:help me with this",
"element_type": "link",
"label": "Help me with this",
"bbox": [
    298.0,
    287.0,
    562.0,
    327.0
],
"confidence": 0.5315,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 6,
"last_seen_index": 6,
"first_seen_at": "2026-05-02T06:47:43.049391+00:00",
"last_seen_at": "2026-05-02T06:47:43.049391+00:00",
"seen_count": 1,
"miss_count": 5,
"stale": true,
"cached": true
},
"input:username or email": {
    "element_key": "input:username or email",
    "element_type": "input",
    "label": "Username or email",
    "bbox": [
        730.0,
        335.0,
```

```
    1194.0,  
    375.0  
  ],  
  "confidence": 0.729,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 6,  
  "last_seen_index": 9,  
  "first_seen_at": "2026-05-02T06:47:43.049391+00:00",  
  "last_seen_at": "2026-05-02T06:50:04.893203+00:00",  
  "seen_count": 3,  
  "miss_count": 2,  
  "stale": true,  
  "cached": true  
},  
"input:password": {  
  "element_key": "input:password",  
  "element_type": "input",  
  "label": "Password",  
  "bbox": [  
    730.0,  
    375.0,  
    1194.0,  
    423.0  
  ],  
  "confidence": 0.729,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,
```



```
"last_seen_frame": null,
"first_seen_index": 6,
"last_seen_index": 9,
"first_seen_at": "2026-05-02T06:47:43.049391+00:00",
"last_seen_at": "2026-05-02T06:50:04.893203+00:00",
"seen_count": 3,
"miss_count": 2,
"stale": true,
"cached": true
},
"text:log in using your account on": {
  "element_key": "text:log in using your account on",
  "element_type": "text",
  "label": "Log in using your account on",
  "bbox": [
    834.0,
    767.0,
    994.0,
    783.0
  ],
  "confidence": 0.5905,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 6,
  "last_seen_index": 7,
  "first_seen_at": "2026-05-02T06:47:43.049391+00:00",
  "last_seen_at": "2026-05-02T06:48:31.574903+00:00",
  "seen_count": 2,
```

```
    "miss_count": 4,  
    "stale": true,  
    "cached": true  
  },  
  "button:sliit login": {  
    "element_key": "button:sliit login",  
    "element_type": "button",  
    "label": "SLIIT Login",  
    "bbox": [  
      898.0,  
      799.0,  
      994.0,  
      815.0  
    ],  
    "confidence": 0.5905,  
    "source": "gemini_vlm",  
    "first_seen_frame": null,  
    "last_seen_frame": null,  
    "first_seen_index": 6,  
    "last_seen_index": 7,  
    "first_seen_at": "2026-05-02T06:47:43.049391+00:00",  
    "last_seen_at": "2026-05-02T06:48:31.574903+00:00",  
    "seen_count": 2,  
    "miss_count": 4,  
    "stale": true,  
    "cached": true  
  },  
  "taskbar:taskbar": {  
    "element_key": "taskbar:taskbar",
```

```
"element_type": "taskbar",
"label": "Taskbar",
"bbox": [
    162.0,
    927.0,
    1759.0,
    962.0
],
"confidence": 0.5609,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 6,
"last_seen_index": 6,
"first_seen_at": "2026-05-02T06:47:43.049391+00:00",
"last_seen_at": "2026-05-02T06:47:43.049391+00:00",
"seen_count": 1,
"miss_count": 5,
"stale": true,
"cached": true
},
"pane:google search overlay": {
    "element_key": "pane:google search overlay",
    "element_type": "pane",
    "label": "Google Search Overlay",
    "bbox": [
        170.0,
        199.0,
        1738.0,
```

```
399.0

],
"confidence": 0.6232,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 7,
"last_seen_index": 7,
"first_seen_at": "2026-05-02T06:48:31.574903+00:00",
"last_seen_at": "2026-05-02T06:48:31.574903+00:00",
"seen_count": 1,
"miss_count": 4,
"stale": true,
"cached": true
},
"text:ask google about this page - courseweb.sliit.lk": {
  "element_key": "text:ask google about this page - courseweb.sliit.lk",
  "element_type": "text",
  "label": "Ask Google about this page - courseweb.sliit.lk",
  "bbox": [
    298.0,
    223.0,
    722.0,
    279.0
  ],
  "confidence": 0.5905,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
```

```
"first_seen_index": 7,
"last_seen_index": 7,
"first_seen_at": "2026-05-02T06:48:31.574903+00:00",
"last_seen_at": "2026-05-02T06:48:31.574903+00:00",
"seen_count": 1,
"miss_count": 4,
"stale": true,
"cached": true
},
"text:help me with this - search in side panel": {
  "element_key": "text:help me with this - search in side panel",
  "element_type": "text",
  "label": "Help me with this - Search in side panel",
  "bbox": [
    306.0,
    287.0,
    666.0,
    335.0
  ],
  "confidence": 0.5905,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 7,
  "last_seen_index": 7,
  "first_seen_at": "2026-05-02T06:48:31.574903+00:00",
  "last_seen_at": "2026-05-02T06:48:31.574903+00:00",
  "seen_count": 1,
  "miss_count": 4,
```

```
    "stale": true,
    "cached": true
  },
  "icon:google chrome": {
    "element_key": "icon:google chrome",
    "element_type": "icon",
    "label": "Google Chrome",
    "bbox": [
      714.0,
      943.0,
      754.0,
      962.0
    ],
    "confidence": 0.5905,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 7,
    "last_seen_index": 7,
    "first_seen_at": "2026-05-02T06:48:31.574903+00:00",
    "last_seen_at": "2026-05-02T06:48:31.574903+00:00",
    "seen_count": 1,
    "miss_count": 4,
    "stale": true,
    "cached": true
  },
  "icon:file explorer": {
    "element_key": "icon:file explorer",
    "element_type": "icon",
```

```
"label": "File Explorer",
"bbox": [
  754.0,
  943.0,
  794.0,
  959.0
],
"confidence": 0.5905,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 7,
"last_seen_index": 7,
"first_seen_at": "2026-05-02T06:48:31.574903+00:00",
"last_seen_at": "2026-05-02T06:48:31.574903+00:00",
"seen_count": 1,
"miss_count": 4,
"stale": true,
"cached": true
},
"icon:visual studio code": {
  "element_key": "icon:visual studio code",
  "element_type": "icon",
  "label": "Visual Studio Code",
  "bbox": [
    794.0,
    943.0,
    826.0,
    962.0
  ]
}
```

```
],
  "confidence": 0.5905,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 7,
  "last_seen_index": 7,
  "first_seen_at": "2026-05-02T06:48:31.574903+00:00",
  "last_seen_at": "2026-05-02T06:48:31.574903+00:00",
  "seen_count": 1,
  "miss_count": 4,
  "stale": true,
  "cached": true
},
"text:system clock": {
  "element_key": "text:system clock",
  "element_type": "text",
  "label": "System Clock",
  "bbox": [
    1642.0,
    943.0,
    1746.0,
    962.0
  ],
  "confidence": 0.5905,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 7,
```



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"last_seen_index": 7,
"first_seen_at": "2026-05-02T06:48:31.574903+00:00",
"last_seen_at": "2026-05-02T06:48:31.574903+00:00",
"seen_count": 1,
"miss_count": 4,
"stale": true,
"cached": true
},
"label:username": {
  "element_key": "label:username",
  "element_type": "label",
  "label": "Username",
  "bbox": [
    770.0,
    342.0,
    914.0,
    358.0
  ],
  "confidence": 0.6561,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 8,
  "last_seen_index": 8,
  "first_seen_at": "2026-05-02T06:49:17.117694+00:00",
  "last_seen_at": "2026-05-02T06:49:17.117694+00:00",
  "seen_count": 1,
  "miss_count": 3,
  "stale": true,
```

```
    "cached": true
  },
  "label:password": {
    "element_key": "label:password",
    "element_type": "label",
    "label": "Password",
    "bbox": [
      770.0,
      390.0,
      858.0,
      406.0
    ],
    "confidence": 0.6561,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 8,
    "last_seen_index": 8,
    "first_seen_at": "2026-05-02T06:49:17.117694+00:00",
    "last_seen_at": "2026-05-02T06:49:17.117694+00:00",
    "seen_count": 1,
    "miss_count": 3,
    "stale": true,
    "cached": true
  },
  "link:sliit login": {
    "element_key": "link:sliit login",
    "element_type": "link",
    "label": "SLIIT Login",
```

```
"bbox": [
  914.0,
  566.0,
  1002.0,
  590.0
],
"confidence": 0.7452,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 8,
"last_seen_index": 9,
"first_seen_at": "2026-05-02T06:49:17.117694+00:00",
"last_seen_at": "2026-05-02T06:50:04.893203+00:00",
"seen_count": 2,
"miss_count": 2,
"stale": true,
"cached": true
},
"pane:spotify media control": {
  "element_key": "pane:spotify media control",
  "element_type": "pane",
  "label": "Spotify Media Control",
  "bbox": [
    1354.0,
    326.0,
    1759.0,
    566.0
  ],
}
```

```
"confidence": 0.6925,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 8,
"last_seen_index": 8,
"first_seen_at": "2026-05-02T06:49:17.117694+00:00",
"last_seen_at": "2026-05-02T06:49:17.117694+00:00",
"seen_count": 1,
"miss_count": 3,
"stale": true,
"cached": true
},
"text:karma (the cycle turns)": {
  "element_key": "text:karma (the cycle turns)",
  "element_type": "text",
  "label": "Karma (The Cycle Turns)",
  "bbox": [
    1362.0,
    406.0,
    1554.0,
    438.0
  ],
  "confidence": 0.6561,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 8,
  "last_seen_index": 8,
```

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"first_seen_at": "2026-05-02T06:49:17.117694+00:00",
"last_seen_at": "2026-05-02T06:49:17.117694+00:00",
"seen_count": 1,
"miss_count": 3,
"stale": true,
"cached": true
},
"text:sumedh k": {
  "element_key": "text:sumedh k",
  "element_type": "text",
  "label": "Sumedh K",
  "bbox": [
    1378.0,
    430.0,
    1482.0,
    446.0
  ],
  "confidence": 0.6561,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 8,
  "last_seen_index": 8,
  "first_seen_at": "2026-05-02T06:49:17.117694+00:00",
  "last_seen_at": "2026-05-02T06:49:17.117694+00:00",
  "seen_count": 1,
  "miss_count": 3,
  "stale": true,
  "cached": true
}
```

```
},
"button:previous": {
  "element_key": "button:previous",
  "element_type": "button",
  "label": "Previous",
  "bbox": [
    1458.0,
    478.0,
    1498.0,
    510.0
  ],
  "confidence": 0.6197,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 8,
  "last_seen_index": 8,
  "first_seen_at": "2026-05-02T06:49:17.117694+00:00",
  "last_seen_at": "2026-05-02T06:49:17.117694+00:00",
  "seen_count": 1,
  "miss_count": 3,
  "stale": true,
  "cached": true
},
"button:play/pause": {
  "element_key": "button:play/pause",
  "element_type": "button",
  "label": "Play/Pause",
  "bbox": [
```

```
    1530.0,  
    478.0,  
    1562.0,  
    510.0  
  ],  
  "confidence": 0.6197,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 8,  
  "last_seen_index": 8,  
  "first_seen_at": "2026-05-02T06:49:17.117694+00:00",  
  "last_seen_at": "2026-05-02T06:49:17.117694+00:00",  
  "seen_count": 1,  
  "miss_count": 3,  
  "stale": true,  
  "cached": true  
},  
"button:next": {  
  "element_key": "button:next",  
  "element_type": "button",  
  "label": "Next",  
  "bbox": [  
    1594.0,  
    478.0,  
    1634.0,  
    510.0  
  ],  
  "confidence": 0.6197,
```

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"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 8,
"last_seen_index": 8,
"first_seen_at": "2026-05-02T06:49:17.117694+00:00",
"last_seen_at": "2026-05-02T06:49:17.117694+00:00",
"seen_count": 1,
"miss_count": 3,
"stale": true,
"cached": true
},
"pane:quick settings": {
  "element_key": "pane:quick settings",
  "element_type": "pane",
  "label": "Quick Settings",
  "bbox": [
    1306.0,
    486.0,
    1759.0,
    962.0
  ],
  "confidence": 0.6925,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 8,
  "last_seen_index": 8,
  "first_seen_at": "2026-05-02T06:49:17.117694+00:00",
```



```
"last_seen_at": "2026-05-02T06:49:17.117694+00:00",
"seen_count": 1,
"miss_count": 3,
"stale": true,
"cached": true
},
"toggle:wi-fi slt-fiber-7e76": {
  "element_key": "toggle:wi-fi slt-fiber-7e76",
  "element_type": "toggle",
  "label": "Wi-Fi SLT-Fiber-7E76",
  "bbox": [
    1370.0,
    558.0,
    1498.0,
    630.0
  ],
  "confidence": 0.6561,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 8,
  "last_seen_index": 8,
  "first_seen_at": "2026-05-02T06:49:17.117694+00:00",
  "last_seen_at": "2026-05-02T06:49:17.117694+00:00",
  "seen_count": 1,
  "miss_count": 3,
  "stale": true,
  "cached": true
},
```

```
"toggle:bluetooth": {
  "element_key": "toggle:bluetooth",
  "element_type": "toggle",
  "label": "Bluetooth",
  "bbox": [
    1482.0,
    558.0,
    1602.0,
    622.0
  ],
  "confidence": 0.6561,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 8,
  "last_seen_index": 8,
  "first_seen_at": "2026-05-02T06:49:17.117694+00:00",
  "last_seen_at": "2026-05-02T06:49:17.117694+00:00",
  "seen_count": 1,
  "miss_count": 3,
  "stale": true,
  "cached": true
},
"toggle:airplane mode": {
  "element_key": "toggle:airplane mode",
  "element_type": "toggle",
  "label": "Airplane mode",
  "bbox": [
    1594.0,
```

```
    566.0,  
    1674.0,  
    606.0  
  ],  
  "confidence": 0.6561,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 8,  
  "last_seen_index": 8,  
  "first_seen_at": "2026-05-02T06:49:17.117694+00:00",  
  "last_seen_at": "2026-05-02T06:49:17.117694+00:00",  
  "seen_count": 1,  
  "miss_count": 3,  
  "stale": true,  
  "cached": true  
},  
"toggle:energy saver": {  
  "element_key": "toggle:energy saver",  
  "element_type": "toggle",  
  "label": "Energy saver",  
  "bbox": [  
    1370.0,  
    662.0,  
    1490.0,  
    734.0  
  ],  
  "confidence": 0.6561,  
  "source": "gemini_vlm",
```

```
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 8,
"last_seen_index": 8,
"first_seen_at": "2026-05-02T06:49:17.117694+00:00",
"last_seen_at": "2026-05-02T06:49:17.117694+00:00",
"seen_count": 1,
"miss_count": 3,
"stale": true,
"cached": true
},
"toggle:night light": {
  "element_key": "toggle:night light",
  "element_type": "toggle",
  "label": "Night light",
  "bbox": [
    1506.0,
    670.0,
    1578.0,
    734.0
  ],
  "confidence": 0.6561,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 8,
  "last_seen_index": 8,
  "first_seen_at": "2026-05-02T06:49:17.117694+00:00",
  "last_seen_at": "2026-05-02T06:49:17.117694+00:00",
```

```
"seen_count": 1,
"miss_count": 3,
"stale": true,
"cached": true
},
"toggle:accessibility": {
  "element_key": "toggle:accessibility",
  "element_type": "toggle",
  "label": "Accessibility",
  "bbox": [
    1610.0,
    662.0,
    1722.0,
    734.0
  ],
  "confidence": 0.6561,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 8,
  "last_seen_index": 8,
  "first_seen_at": "2026-05-02T06:49:17.117694+00:00",
  "last_seen_at": "2026-05-02T06:49:17.117694+00:00",
  "seen_count": 1,
  "miss_count": 3,
  "stale": true,
  "cached": true
},
"slider:brightness": {
```

```
"element_key": "slider:brightness",
"element_type": "slider",
"label": "Brightness",
"bbox": [
  1378.0,
  790.0,
  1674.0,
  806.0
],
"confidence": 0.6561,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 8,
"last_seen_index": 8,
"first_seen_at": "2026-05-02T06:49:17.117694+00:00",
"last_seen_at": "2026-05-02T06:49:17.117694+00:00",
"seen_count": 1,
"miss_count": 3,
"stale": true,
"cached": true
},
"slider:volume": {
  "element_key": "slider:volume",
  "element_type": "slider",
  "label": "Volume",
  "bbox": [
    1378.0,
    838.0,
```

```
    1706.0,  
    854.0  
  ],  
  "confidence": 0.6561,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 8,  
  "last_seen_index": 8,  
  "first_seen_at": "2026-05-02T06:49:17.117694+00:00",  
  "last_seen_at": "2026-05-02T06:49:17.117694+00:00",  
  "seen_count": 1,  
  "miss_count": 3,  
  "stale": true,  
  "cached": true  
},  
"label:93%": {  
  "element_key": "label:93%",  
  "element_type": "label",  
  "label": "93%",  
  "bbox": [  
    1378.0,  
    894.0,  
    1442.0,  
    918.0  
  ],  
  "confidence": 0.6561,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,
```

```
"last_seen_frame": null,
"first_seen_index": 8,
"last_seen_index": 8,
"first_seen_at": "2026-05-02T06:49:17.117694+00:00",
"last_seen_at": "2026-05-02T06:49:17.117694+00:00",
"seen_count": 1,
"miss_count": 3,
"stale": true,
"cached": true
},
"button:settings": {
  "element_key": "button:settings",
  "element_type": "button",
  "label": "Settings",
  "bbox": [
    1682.0,
    894.0,
    1706.0,
    918.0
  ],
  "confidence": 0.6197,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 8,
  "last_seen_index": 8,
  "first_seen_at": "2026-05-02T06:49:17.117694+00:00",
  "last_seen_at": "2026-05-02T06:49:17.117694+00:00",
  "seen_count": 1,
```



```
    "miss_count": 3,
    "stale": true,
    "cached": true
  },
  "toolbar:taskbar": {
    "element_key": "toolbar:taskbar",
    "element_type": "toolbar",
    "label": "Taskbar",
    "bbox": [
      178.0,
      934.0,
      1746.0,
      962.0
    ],
    "confidence": 0.6925,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 8,
    "last_seen_index": 8,
    "first_seen_at": "2026-05-02T06:49:17.117694+00:00",
    "last_seen_at": "2026-05-02T06:49:17.117694+00:00",
    "seen_count": 1,
    "miss_count": 3,
    "stale": true,
    "cached": true
  },
  "label:12:18 pm": {
    "element_key": "label:12:18 pm",
```

```
"element_type": "label",
"label": "12:18 PM",
"bbox": [
    1650.0,
    942.0,
    1738.0,
    962.0
],
"confidence": 0.6561,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 8,
"last_seen_index": 8,
"first_seen_at": "2026-05-02T06:49:17.117694+00:00",
"last_seen_at": "2026-05-02T06:49:17.117694+00:00",
"seen_count": 1,
"miss_count": 3,
"stale": true,
"cached": true
},
"link:lost password?": {
    "element_key": "link:lost password?",
    "element_type": "link",
    "label": "Lost password?",
    "bbox": [
        922.0,
        487.0,
        1002.0,
```

```
503.0

],
"confidence": 0.6885,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 9,
"last_seen_index": 9,
"first_seen_at": "2026-05-02T06:50:04.893203+00:00",
"last_seen_at": "2026-05-02T06:50:04.893203+00:00",
"seen_count": 1,
"miss_count": 2,
"stale": true,
"cached": true
},
"pane:taskbar": {
  "element_key": "pane:taskbar",
  "element_type": "pane",
  "label": "Taskbar",
  "bbox": [
    178.0,
    935.0,
    1758.0,
    963.0
  ],
  "confidence": 0.7695,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
```

```
"first_seen_index": 9,
"last_seen_index": 9,
"first_seen_at": "2026-05-02T06:50:04.893203+00:00",
"last_seen_at": "2026-05-02T06:50:04.893203+00:00",
"seen_count": 1,
"miss_count": 2,
"stale": true,
"cached": true
},
"button:search": {
  "element_key": "button:search",
  "element_type": "button",
  "label": "Search",
  "bbox": [
    906.0,
    943.0,
    938.0,
    963.0
  ],
  "confidence": 0.729,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 9,
  "last_seen_index": 9,
  "first_seen_at": "2026-05-02T06:50:04.893203+00:00",
  "last_seen_at": "2026-05-02T06:50:04.893203+00:00",
  "seen_count": 1,
  "miss_count": 2,
```

```
"stale": true,
"cached": true
},
"button:task view": {
  "element_key": "button:task view",
  "element_type": "button",
  "label": "Task View",
  "bbox": [
    930.0,
    943.0,
    962.0,
    963.0
  ],
  "confidence": 0.729,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 9,
  "last_seen_index": 9,
  "first_seen_at": "2026-05-02T06:50:04.893203+00:00",
  "last_seen_at": "2026-05-02T06:50:04.893203+00:00",
  "seen_count": 1,
  "miss_count": 2,
  "stale": true,
  "cached": true
},
"button:microsoft edge": {
  "element_key": "button:microsoft edge",
  "element_type": "button",
```

```
"label": "Microsoft Edge",
"bbox": [
  954.0,
  943.0,
  986.0,
  963.0
],
"confidence": 0.729,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 9,
"last_seen_index": 9,
"first_seen_at": "2026-05-02T06:50:04.893203+00:00",
"last_seen_at": "2026-05-02T06:50:04.893203+00:00",
"seen_count": 1,
"miss_count": 2,
"stale": true,
"cached": true
},
"button:file explorer": {
  "element_key": "button:file explorer",
  "element_type": "button",
  "label": "File Explorer",
  "bbox": [
    978.0,
    943.0,
    1010.0,
    963.0
  ]
}
```

```
],
  "confidence": 0.729,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 9,
  "last_seen_index": 9,
  "first_seen_at": "2026-05-02T06:50:04.893203+00:00",
  "last_seen_at": "2026-05-02T06:50:04.893203+00:00",
  "seen_count": 1,
  "miss_count": 2,
  "stale": true,
  "cached": true
},
"button:microsoft store": {
  "element_key": "button:microsoft store",
  "element_type": "button",
  "label": "Microsoft Store",
  "bbox": [
    1002.0,
    943.0,
    1034.0,
    963.0
  ],
  "confidence": 0.729,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 9,
```

```
"last_seen_index": 9,  
"first_seen_at": "2026-05-02T06:50:04.893203+00:00",  
"last_seen_at": "2026-05-02T06:50:04.893203+00:00",  
"seen_count": 1,  
"miss_count": 2,  
"stale": true,  
"cached": true  
},  
"button:mail": {  
  "element_key": "button:mail",  
  "element_type": "button",  
  "label": "Mail",  
  "bbox": [  
    1026.0,  
    943.0,  
    1058.0,  
    963.0  
  ],  
  "confidence": 0.729,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 9,  
  "last_seen_index": 9,  
  "first_seen_at": "2026-05-02T06:50:04.893203+00:00",  
  "last_seen_at": "2026-05-02T06:50:04.893203+00:00",  
  "seen_count": 1,  
  "miss_count": 2,  
  "stale": true,
```



```
    "cached": true
  },
  "button:google chrome": {
    "element_key": "button:google chrome",
    "element_type": "button",
    "label": "Google Chrome",
    "bbox": [
      1058.0,
      943.0,
      1082.0,
      963.0
    ],
    "confidence": 0.729,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 9,
    "last_seen_index": 9,
    "first_seen_at": "2026-05-02T06:50:04.893203+00:00",
    "last_seen_at": "2026-05-02T06:50:04.893203+00:00",
    "seen_count": 1,
    "miss_count": 2,
    "stale": true,
    "cached": true
  },
  "button:visual studio code": {
    "element_key": "button:visual studio code",
    "element_type": "button",
    "label": "Visual Studio Code",
```

```
"bbox": [
  1082.0,
  943.0,
  1106.0,
  963.0
],
"confidence": 0.729,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 9,
"last_seen_index": 9,
"first_seen_at": "2026-05-02T06:50:04.893203+00:00",
"last_seen_at": "2026-05-02T06:50:04.893203+00:00",
"seen_count": 1,
"miss_count": 2,
"stale": true,
"cached": true
},
"button:microsoft teams": {
  "element_key": "button:microsoft teams",
  "element_type": "button",
  "label": "Microsoft Teams",
  "bbox": [
    1106.0,
    943.0,
    1130.0,
    963.0
  ],

```

```
"confidence": 0.729,  
"source": "gemini_vlm",  
"first_seen_frame": null,  
"last_seen_frame": null,  
"first_seen_index": 9,  
"last_seen_index": 9,  
"first_seen_at": "2026-05-02T06:50:04.893203+00:00",  
"last_seen_at": "2026-05-02T06:50:04.893203+00:00",  
"seen_count": 1,  
"miss_count": 2,  
"stale": true,  
"cached": true  
},  
"text:clock": {  
  "element_key": "text:clock",  
  "element_type": "text",  
  "label": "Clock",  
  "bbox": [  
    1642.0,  
    943.0,  
    1746.0,  
    963.0  
  ],  
  "confidence": 0.729,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 9,  
  "last_seen_index": 9,
```

```
    "first_seen_at": "2026-05-02T06:50:04.893203+00:00",
    "last_seen_at": "2026-05-02T06:50:04.893203+00:00",
    "seen_count": 1,
    "miss_count": 2,
    "stale": true,
    "cached": true
  }
},
"current_elements": [],
"resolved_elements": [],
"screen_data": {}
}
}
```

Test Case 09: CourseWeb Credential Entry (Complex Task)

User Prompt:

Now you are in the CourseWeb login page, so the username: "IT22099518" and password: "NuwaniSithara@808" so using both credentials try to login to the CourseWeb

Test Results:

- Steps Total: 2
- Completed: 0
- Failed: 2
- Duration: 18 minutes 11 seconds

Step Execution:

Step 1: Click the 'Login Button' at (958, 603) to navigate to the Microsoft O365 authentication page

- Target: Login Button
- Attempts: 3
- Execution: Performed

- **Evaluation: Needs Input**
- **Issue: The login page loaded but the evaluation required user input. The button coordinates may have mismatched — current detected "Log in" button at (800, 248) differs from targeted (958, 603).**

Step 2: Type the username 'IT22099518' into the email/username field, click 'Next', type the password 'NuwaniSithara@808' into the password field, and click 'Sign in'

- **Target: Microsoft login form fields**
- **Attempts: 3**
- **Execution: Performed**
- **Evaluation: Failed (retry loop)**
- **Issue: Evaluation error — No valid JSON found in LLM response ('{\n "status":'}). The step could not progress due to malformed response.**

Execution Summary:

The task failed completely. Step 1 did not successfully navigate to the Microsoft O365 authentication page due to a potential coordinate mismatch between the planned button location and the actual UI. Step 2 encountered a malformed JSON error from the LLM during evaluation, preventing credential entry and submission. The browser remains on the initial login page.

Issues Identified:

- **Malformed JSON response from LLM during evaluation**
- **Coordinate mismatch: Target button at (958, 603) vs actual "Log in" button at (800, 248)**
- **Navigation failure to Microsoft O365 authentication page**
- **No fallback mechanism to update button coordinates dynamically**
- **Evaluation loop unable to handle incomplete or malformed LLM responses**

Recommendations for Improvement:

- **Enforce strict JSON output formatting with validation before parsing**
- **Implement dynamic coordinate refresh before each execution attempt**
- **Add fallback to re-detect login button if initial coordinates fail**
- **Increase wait time after page load before attempting element detection**
- **Improve evaluation prompt to always return valid JSON, even on failure**
- **Add retry with updated coordinates instead of hardcoding original positions**

JSON Response:

full_cycle.json

```
{
  "run_id": "57f743fa-1756-4baa-b97d-25c9dacdd30d",
```

```
"user_task": "Now you are in the CourseWeb login page, so the username:
\"IT22099518\" and password: \"NuwaniSithara@808\" so using both credentials try
to login to the CourseWeb",

"started_at": "2026-05-02T06:53:02.537150+00:00",

"finished_at": "2026-05-02T07:11:13.540382+00:00",

"vision_used": true,

"perception": {

  "image": "E:\\sliit-
project\\ScreenAwareTaskAgent\\vision\\data\\sessions\\shot_1777704787_aa8922\\pr
ocessing\\proc_1777704787_248e9c\\preprocessed_frames\\frame_1777704787_716419.jp
g",

  "image_size": {

    "width": 1920,

    "height": 1080

  },

  "screen_bbox": [

    162,

    199,

    1758,

    976

  ],

  "screen_size": {

    "width": 1596,

    "height": 777

  },

  "coordinate_system": "pixel",

  "elements": [

    {

      "id": "taskbar",

      "type": "pane",

      "label": "Taskbar",
```

```
"description": "Windows taskbar at the bottom of the screen",
"state": "visible",
"dx": 802,
"dy": 756,
"frame_dx": 964,
"frame_dy": 955,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    170,
    935,
    1758,
    976
],
"session_memory_key": "pane:taskbar",
"cached": false
},
{
    "id": "weather_widget",
    "type": "label",
    "label": "Weather Widget",
    "description": "Weather and temperature display in the taskbar",
    "state": "visible",
    "dx": 64,
    "dy": 764,
    "frame_dx": 226,
    "frame_dy": 963,
    "confidence": 0.85,
    "source": "gemini_vlm",
```

```
"frame_bbox": [  
    178,  
    951,  
    274,  
    976  
],  
"session_memory_key": "label:weather widget",  
"cached": false  
},  
{  
    "id": "start_button",  
    "type": "button",  
    "label": "Start Button",  
    "description": "Windows Start menu button",  
    "state": "visible",  
    "dx": 424,  
    "dy": 760,  
    "frame_dx": 586,  
    "frame_dy": 959,  
    "confidence": 0.98,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        562,  
        943,  
        610,  
        976  
    ],  
    "session_memory_key": "button:start button",  
    "cached": false  
}
```



```
},  
{  
  "id": "search_bar",  
  "type": "input",  
  "label": "Search Bar",  
  "description": "Windows search input field in the taskbar",  
  "state": "visible",  
  "dx": 508,  
  "dy": 760,  
  "frame_dx": 670,  
  "frame_dy": 959,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    594,  
    943,  
    746,  
    976  
  ],  
  "session_memory_key": "input:search bar",  
  "cached": false  
},  
{  
  "id": "file_explorer_icon",  
  "type": "icon",  
  "label": "File Explorer",  
  "description": "File Explorer icon in the taskbar",  
  "state": "visible",  
  "dx": 608,
```

```
"dy": 760,  
"frame_dx": 770,  
"frame_dy": 959,  
"confidence": 0.9,  
"source": "gemini_vlm",  
"frame_bbox": [  
    746,  
    943,  
    794,  
    976  
],  
"session_memory_key": "icon:file explorer",  
"cached": false  
},  
{  
    "id": "microsoft_store_icon",  
    "type": "icon",  
    "label": "Microsoft Store",  
    "description": "Microsoft Store icon in the taskbar",  
    "state": "visible",  
    "dx": 648,  
    "dy": 760,  
    "frame_dx": 810,  
    "frame_dy": 959,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        786,  
        943,
```

```
      834,  
      976  
    ],  
    "session_memory_key": "icon:microsoft store",  
    "cached": false  
  },  
  {  
    "id": "mail_icon",  
    "type": "icon",  
    "label": "Mail",  
    "description": "Mail application icon in the taskbar",  
    "state": "visible",  
    "dx": 680,  
    "dy": 760,  
    "frame_dx": 842,  
    "frame_dy": 959,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
      818,  
      943,  
      866,  
      976  
    ],  
    "session_memory_key": "icon:mail",  
    "cached": false  
  },  
  {  
    "id": "xbox_icon",
```

```
"type": "icon",
"label": "Xbox",
"description": "Xbox application icon in the taskbar",
"state": "visible",
"dx": 720,
"dy": 760,
"frame_dx": 882,
"frame_dy": 959,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    858,
    943,
    906,
    976
],
"session_memory_key": "icon:xbox",
"cached": false
},
{
    "id": "chrome_icon",
    "type": "icon",
    "label": "Google Chrome",
    "description": "Google Chrome browser icon in the taskbar",
    "state": "visible",
    "dx": 760,
    "dy": 760,
    "frame_dx": 922,
    "frame_dy": 959,
```

```
"confidence": 0.9,  
"source": "gemini_vlm",  
"frame_bbox": [  
    898,  
    943,  
    946,  
    976  
],  
"session_memory_key": "icon:google chrome",  
"cached": false  
},  
{  
    "id": "vscode_icon",  
    "type": "icon",  
    "label": "Visual Studio Code",  
    "description": "Visual Studio Code icon in the taskbar",  
    "state": "visible",  
    "dx": 792,  
    "dy": 760,  
    "frame_dx": 954,  
    "frame_dy": 959,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        930,  
        943,  
        978,  
        976  
    ],  
}
```

```
"session_memory_key": "icon:visual studio code",
"cached": false
},
{
  "id": "spotify_icon",
  "type": "icon",
  "label": "Spotify",
  "description": "Spotify application icon in the taskbar",
  "state": "visible",
  "dx": 832,
  "dy": 760,
  "frame_dx": 994,
  "frame_dy": 959,
  "confidence": 0.9,
  "source": "gemini_vlm",
  "frame_bbox": [
    970,
    943,
    1018,
    976
  ],
  "session_memory_key": "icon:spotify",
  "cached": false
},
{
  "id": "google_icon",
  "type": "icon",
  "label": "Google",
  "description": "Google application icon in the taskbar",
```

```
"state": "visible",
"dx": 872,
"dy": 760,
"frame_dx": 1034,
"frame_dy": 959,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    1010,
    943,
    1058,
    976
],
"session_memory_key": "icon:google",
"cached": false
},
{
    "id": "autocad_icon",
    "type": "icon",
    "label": "AutoCAD",
    "description": "AutoCAD application icon in the taskbar",
    "state": "visible",
    "dx": 908,
    "dy": 760,
    "frame_dx": 1070,
    "frame_dy": 959,
    "confidence": 0.8,
    "source": "gemini_vlm",
    "frame_bbox": [
```

```
    1050,  
    943,  
    1090,  
    976  
  ],  
  "session_memory_key": "icon:autocad",  
  "cached": false  
},  
{  
  "id": "system_tray",  
  "type": "pane",  
  "label": "System Tray",  
  "description": "System tray area with various status icons",  
  "state": "visible",  
  "dx": 1430,  
  "dy": 760,  
  "frame_dx": 1592,  
  "frame_dy": 959,  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    1426,  
    943,  
    1758,  
    976  
  ],  
  "session_memory_key": "pane:system tray",  
  "cached": false  
},
```



```
{
  "id": "clock_date",
  "type": "label",
  "label": "Clock and Date",
  "description": "Current time and date display in the taskbar",
  "state": "visible",
  "dx": 1538,
  "dy": 760,
  "frame_dx": 1700,
  "frame_dy": 959,
  "confidence": 0.95,
  "source": "gemini_vlm",
  "frame_bbox": [
    1642,
    943,
    1758,
    976
  ],
  "session_memory_key": "label:clock and date",
  "cached": false
},
{
  "id": "login_container",
  "type": "pane",
  "label": "Login Container",
  "description": "Main white container for the login form",
  "state": "visible",
  "dx": 792,
  "dy": 388,
```

```
"frame_dx": 954,  
"frame_dy": 587,  
"confidence": 0.95,  
"source": "gemini_vlm",  
"frame_bbox": [  
    658,  
    199,  
    1250,  
    976  
],  
"session_memory_key": "pane:login container",  
"cached": false  
},  
{  
    "id": "sliit_logo",  
    "type": "icon",  
    "label": "SLIIT Logo",  
    "description": "Logo of SLIIT University",  
    "state": "visible",  
    "dx": 688,  
    "dy": 156,  
    "frame_dx": 850,  
    "frame_dy": 355,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        770,  
        327,  
        930,
```

```
383
],
"session_memory_key": "icon:sliit logo",
"cached": false
},
{
  "id": "courseweb_heading",
  "type": "heading",
  "label": "CourseWeb",
  "description": "Heading text for the CourseWeb application",
  "state": "visible",
  "dx": 892,
  "dy": 156,
  "frame_dx": 1054,
  "frame_dy": 355,
  "confidence": 0.95,
  "source": "gemini_vlm",
  "frame_bbox": [
    954,
    327,
    1154,
    383
  ],
  "session_memory_key": "heading:courseweb",
  "cached": false
},
{
  "id": "login_button",
  "type": "button",
```

```
"label": "Login Button",
"description": "Yellow login button in the center of the page",
"state": "visible",
"dx": 796,
"dy": 404,
"frame_dx": 958,
"frame_dy": 603,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    882,
    575,
    1034,
    631
],
"session_memory_key": "button:login button",
"cached": false
},
{
    "id": "instruction_text",
    "type": "label",
    "label": "Instruction Text",
    "description": "Text instructing the user to click on SLIIT Login",
    "state": "visible",
    "dx": 724,
    "dy": 696,
    "frame_dx": 886,
    "frame_dy": 895,
    "confidence": 0.85,
```

```
"source": "gemini_vlm",
"frame_bbox": [
    802,
    879,
    970,
    911
],
"session_memory_key": "label:instruction text",
"cached": false
},
{
    "id": "note_text",
    "type": "label",
    "label": "Note Text",
    "description": "Informational note about Microsoft 0365 validation",
    "state": "visible",
    "dx": 796,
    "dy": 736,
    "frame_dx": 958,
    "frame_dy": 935,
    "confidence": 0.85,
    "source": "gemini_vlm",
    "frame_bbox": [
        754,
        895,
        1162,
        976
    ],
    "session_memory_key": "label:note text",
```

```
        "cached": false
    }
],
"element_count": 21,
"current_elements": [
    {
        "id": "taskbar",
        "type": "pane",
        "label": "Taskbar",
        "description": "Windows taskbar at the bottom of the screen",
        "state": "visible",
        "dx": 802,
        "dy": 756,
        "frame_dx": 964,
        "frame_dy": 955,
        "confidence": 0.95,
        "source": "gemini_vlm",
        "frame_bbox": [
            170,
            935,
            1758,
            976
        ],
        "session_memory_key": "pane:taskbar",
        "cached": false
    },
    {
        "id": "weather_widget",
        "type": "label",
```

```
"label": "Weather Widget",
"description": "Weather and temperature display in the taskbar",
"state": "visible",
"dx": 64,
"dy": 764,
"frame_dx": 226,
"frame_dy": 963,
"confidence": 0.85,
"source": "gemini_vlm",
"frame_bbox": [
    178,
    951,
    274,
    976
],
"session_memory_key": "label:weather widget",
"cached": false
},
{
    "id": "start_button",
    "type": "button",
    "label": "Start Button",
    "description": "Windows Start menu button",
    "state": "visible",
    "dx": 424,
    "dy": 760,
    "frame_dx": 586,
    "frame_dy": 959,
    "confidence": 0.98,
```

```
"source": "gemini_vlm",
"frame_bbox": [
    562,
    943,
    610,
    976
],
"session_memory_key": "button:start button",
"cached": false
},
{
    "id": "search_bar",
    "type": "input",
    "label": "Search Bar",
    "description": "Windows search input field in the taskbar",
    "state": "visible",
    "dx": 508,
    "dy": 760,
    "frame_dx": 670,
    "frame_dy": 959,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        594,
        943,
        746,
        976
    ],
    "session_memory_key": "input:search bar",
```



```
"cached": false
},
{
  "id": "file_explorer_icon",
  "type": "icon",
  "label": "File Explorer",
  "description": "File Explorer icon in the taskbar",
  "state": "visible",
  "dx": 608,
  "dy": 760,
  "frame_dx": 770,
  "frame_dy": 959,
  "confidence": 0.9,
  "source": "gemini_vlm",
  "frame_bbox": [
    746,
    943,
    794,
    976
  ],
  "session_memory_key": "icon:file explorer",
  "cached": false
},
{
  "id": "microsoft_store_icon",
  "type": "icon",
  "label": "Microsoft Store",
  "description": "Microsoft Store icon in the taskbar",
  "state": "visible",
```

```
"dx": 648,  
"dy": 760,  
"frame_dx": 810,  
"frame_dy": 959,  
"confidence": 0.9,  
"source": "gemini_vlm",  
"frame_bbox": [  
    786,  
    943,  
    834,  
    976  
],  
"session_memory_key": "icon:microsoft store",  
"cached": false  
},  
{  
    "id": "mail_icon",  
    "type": "icon",  
    "label": "Mail",  
    "description": "Mail application icon in the taskbar",  
    "state": "visible",  
    "dx": 680,  
    "dy": 760,  
    "frame_dx": 842,  
    "frame_dy": 959,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        818,
```

```
    943,  
    866,  
    976  
  ],  
  "session_memory_key": "icon:mail",  
  "cached": false  
},  
{  
  "id": "xbox_icon",  
  "type": "icon",  
  "label": "Xbox",  
  "description": "Xbox application icon in the taskbar",  
  "state": "visible",  
  "dx": 720,  
  "dy": 760,  
  "frame_dx": 882,  
  "frame_dy": 959,  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    858,  
    943,  
    906,  
    976  
  ],  
  "session_memory_key": "icon:xbox",  
  "cached": false  
},  
{
```

```
"id": "chrome_icon",
"type": "icon",
"label": "Google Chrome",
"description": "Google Chrome browser icon in the taskbar",
"state": "visible",
"dx": 760,
"dy": 760,
"frame_dx": 922,
"frame_dy": 959,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    898,
    943,
    946,
    976
],
"session_memory_key": "icon:google chrome",
"cached": false
},
{
    "id": "vscode_icon",
    "type": "icon",
    "label": "Visual Studio Code",
    "description": "Visual Studio Code icon in the taskbar",
    "state": "visible",
    "dx": 792,
    "dy": 760,
    "frame_dx": 954,
```

```
"frame_dy": 959,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    930,
    943,
    978,
    976
],
"session_memory_key": "icon:visual studio code",
"cached": false
},
{
    "id": "spotify_icon",
    "type": "icon",
    "label": "Spotify",
    "description": "Spotify application icon in the taskbar",
    "state": "visible",
    "dx": 832,
    "dy": 760,
    "frame_dx": 994,
    "frame_dy": 959,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        970,
        943,
        1018,
        976
    ]
}
```

```
    ],
    "session_memory_key": "icon:spotify",
    "cached": false
  },
  {
    "id": "google_icon",
    "type": "icon",
    "label": "Google",
    "description": "Google application icon in the taskbar",
    "state": "visible",
    "dx": 872,
    "dy": 760,
    "frame_dx": 1034,
    "frame_dy": 959,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
      1010,
      943,
      1058,
      976
    ],
    "session_memory_key": "icon:google",
    "cached": false
  },
  {
    "id": "autocad_icon",
    "type": "icon",
    "label": "AutoCAD",
```

```
"description": "AutoCAD application icon in the taskbar",
"state": "visible",
"dx": 908,
"dy": 760,
"frame_dx": 1070,
"frame_dy": 959,
"confidence": 0.8,
"source": "gemini_vlm",
"frame_bbox": [
    1050,
    943,
    1090,
    976
],
"session_memory_key": "icon:autocad",
"cached": false
},
{
    "id": "system_tray",
    "type": "pane",
    "label": "System Tray",
    "description": "System tray area with various status icons",
    "state": "visible",
    "dx": 1430,
    "dy": 760,
    "frame_dx": 1592,
    "frame_dy": 959,
    "confidence": 0.9,
    "source": "gemini_vlm",
```

```
"frame_bbox": [
    1426,
    943,
    1758,
    976
],
"session_memory_key": "pane:system tray",
"cached": false
},
{
    "id": "clock_date",
    "type": "label",
    "label": "Clock and Date",
    "description": "Current time and date display in the taskbar",
    "state": "visible",
    "dx": 1538,
    "dy": 760,
    "frame_dx": 1700,
    "frame_dy": 959,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        1642,
        943,
        1758,
        976
    ],
    "session_memory_key": "label:clock and date",
    "cached": false
}
```



```
},  
{  
  "id": "login_container",  
  "type": "pane",  
  "label": "Login Container",  
  "description": "Main white container for the login form",  
  "state": "visible",  
  "dx": 792,  
  "dy": 388,  
  "frame_dx": 954,  
  "frame_dy": 587,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    658,  
    199,  
    1250,  
    976  
  ],  
  "session_memory_key": "pane:login container",  
  "cached": false  
},  
{  
  "id": "sliit_logo",  
  "type": "icon",  
  "label": "SLIIT Logo",  
  "description": "Logo of SLIIT University",  
  "state": "visible",  
  "dx": 688,
```

```
"dy": 156,
"frame_dx": 850,
"frame_dy": 355,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    770,
    327,
    930,
    383
],
"session_memory_key": "icon:sliit logo",
"cached": false
},
{
    "id": "courseweb_heading",
    "type": "heading",
    "label": "CourseWeb",
    "description": "Heading text for the CourseWeb application",
    "state": "visible",
    "dx": 892,
    "dy": 156,
    "frame_dx": 1054,
    "frame_dy": 355,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        954,
        327,
```

```
        1154,  
        383  
    ],  
    "session_memory_key": "heading:courseweb",  
    "cached": false  
},  
{  
    "id": "login_button",  
    "type": "button",  
    "label": "Login Button",  
    "description": "Yellow login button in the center of the page",  
    "state": "visible",  
    "dx": 796,  
    "dy": 404,  
    "frame_dx": 958,  
    "frame_dy": 603,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        882,  
        575,  
        1034,  
        631  
    ],  
    "session_memory_key": "button:login button",  
    "cached": false  
},  
{  
    "id": "instruction_text",
```

```
"type": "label",
"label": "Instruction Text",
"description": "Text instructing the user to click on SLIIT Login",
"state": "visible",
"dx": 724,
"dy": 696,
"frame_dx": 886,
"frame_dy": 895,
"confidence": 0.85,
"source": "gemini_vlm",
"frame_bbox": [
    802,
    879,
    970,
    911
],
"session_memory_key": "label:instruction text",
"cached": false
},
{
    "id": "note_text",
    "type": "label",
    "label": "Note Text",
    "description": "Informational note about Microsoft 0365 validation",
    "state": "visible",
    "dx": 796,
    "dy": 736,
    "frame_dx": 958,
    "frame_dy": 935,
```

```
"confidence": 0.85,
"source": "gemini_vlm",
"frame_bbox": [
    754,
    895,
    1162,
    976
],
"session_memory_key": "label:note text",
"cached": false
}
],
"resolved_elements": [
    {
        "id": "taskbar",
        "type": "pane",
        "label": "Taskbar",
        "description": "Windows taskbar at the bottom of the screen",
        "state": "visible",
        "dx": 802,
        "dy": 756,
        "frame_dx": 964,
        "frame_dy": 955,
        "confidence": 0.95,
        "source": "gemini_vlm",
        "frame_bbox": [
            170,
            935,
            1758,
```

```
    976
  ],
  "session_memory_key": "pane:taskbar",
  "cached": false
},
{
  "id": "weather_widget",
  "type": "label",
  "label": "Weather Widget",
  "description": "Weather and temperature display in the taskbar",
  "state": "visible",
  "dx": 64,
  "dy": 764,
  "frame_dx": 226,
  "frame_dy": 963,
  "confidence": 0.85,
  "source": "gemini_vlm",
  "frame_bbox": [
    178,
    951,
    274,
    976
  ],
  "session_memory_key": "label:weather widget",
  "cached": false
},
{
  "id": "start_button",
  "type": "button",
```

```
"label": "Start Button",
"description": "Windows Start menu button",
"state": "visible",
"dx": 424,
"dy": 760,
"frame_dx": 586,
"frame_dy": 959,
"confidence": 0.98,
"source": "gemini_vlm",
"frame_bbox": [
    562,
    943,
    610,
    976
],
"session_memory_key": "button:start button",
"cached": false
},
{
    "id": "search_bar",
    "type": "input",
    "label": "Search Bar",
    "description": "Windows search input field in the taskbar",
    "state": "visible",
    "dx": 508,
    "dy": 760,
    "frame_dx": 670,
    "frame_dy": 959,
    "confidence": 0.95,
```

```
"source": "gemini_vlm",
"frame_bbox": [
    594,
    943,
    746,
    976
],
"session_memory_key": "input:search bar",
"cached": false
},
{
    "id": "file_explorer_icon",
    "type": "icon",
    "label": "File Explorer",
    "description": "File Explorer icon in the taskbar",
    "state": "visible",
    "dx": 608,
    "dy": 760,
    "frame_dx": 770,
    "frame_dy": 959,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        746,
        943,
        794,
        976
    ],
    "session_memory_key": "icon:file explorer",
```



```
"cached": false
},
{
  "id": "microsoft_store_icon",
  "type": "icon",
  "label": "Microsoft Store",
  "description": "Microsoft Store icon in the taskbar",
  "state": "visible",
  "dx": 648,
  "dy": 760,
  "frame_dx": 810,
  "frame_dy": 959,
  "confidence": 0.9,
  "source": "gemini_vlm",
  "frame_bbox": [
    786,
    943,
    834,
    976
  ],
  "session_memory_key": "icon:microsoft store",
  "cached": false
},
{
  "id": "mail_icon",
  "type": "icon",
  "label": "Mail",
  "description": "Mail application icon in the taskbar",
  "state": "visible",
```

```
"dx": 680,  
"dy": 760,  
"frame_dx": 842,  
"frame_dy": 959,  
"confidence": 0.9,  
"source": "gemini_vlm",  
"frame_bbox": [  
    818,  
    943,  
    866,  
    976  
],  
"session_memory_key": "icon:mail",  
"cached": false  
},  
{  
    "id": "xbox_icon",  
    "type": "icon",  
    "label": "Xbox",  
    "description": "Xbox application icon in the taskbar",  
    "state": "visible",  
    "dx": 720,  
    "dy": 760,  
    "frame_dx": 882,  
    "frame_dy": 959,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        858,
```

```
    943,  
    906,  
    976  
  ],  
  "session_memory_key": "icon:xbox",  
  "cached": false  
},  
{  
  "id": "chrome_icon",  
  "type": "icon",  
  "label": "Google Chrome",  
  "description": "Google Chrome browser icon in the taskbar",  
  "state": "visible",  
  "dx": 760,  
  "dy": 760,  
  "frame_dx": 922,  
  "frame_dy": 959,  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    898,  
    943,  
    946,  
    976  
  ],  
  "session_memory_key": "icon:google chrome",  
  "cached": false  
},  
{
```

```
"id": "vscode_icon",
"type": "icon",
"label": "Visual Studio Code",
"description": "Visual Studio Code icon in the taskbar",
"state": "visible",
"dx": 792,
"dy": 760,
"frame_dx": 954,
"frame_dy": 959,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    930,
    943,
    978,
    976
],
"session_memory_key": "icon:visual studio code",
"cached": false
},
{
    "id": "spotify_icon",
    "type": "icon",
    "label": "Spotify",
    "description": "Spotify application icon in the taskbar",
    "state": "visible",
    "dx": 832,
    "dy": 760,
    "frame_dx": 994,
```

```
"frame_dy": 959,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    970,
    943,
    1018,
    976
],
"session_memory_key": "icon:spotify",
"cached": false
},
{
    "id": "google_icon",
    "type": "icon",
    "label": "Google",
    "description": "Google application icon in the taskbar",
    "state": "visible",
    "dx": 872,
    "dy": 760,
    "frame_dx": 1034,
    "frame_dy": 959,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        1010,
        943,
        1058,
        976
    ]
}
```

```
    ],  
    "session_memory_key": "icon:google",  
    "cached": false  
  },  
  {  
    "id": "autocad_icon",  
    "type": "icon",  
    "label": "AutoCAD",  
    "description": "AutoCAD application icon in the taskbar",  
    "state": "visible",  
    "dx": 908,  
    "dy": 760,  
    "frame_dx": 1070,  
    "frame_dy": 959,  
    "confidence": 0.8,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
      1050,  
      943,  
      1090,  
      976  
    ],  
    "session_memory_key": "icon:autocad",  
    "cached": false  
  },  
  {  
    "id": "system_tray",  
    "type": "pane",  
    "label": "System Tray",
```

```
"description": "System tray area with various status icons",
"state": "visible",
"dx": 1430,
"dy": 760,
"frame_dx": 1592,
"frame_dy": 959,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    1426,
    943,
    1758,
    976
],
"session_memory_key": "pane:system tray",
"cached": false
},
{
    "id": "clock_date",
    "type": "label",
    "label": "Clock and Date",
    "description": "Current time and date display in the taskbar",
    "state": "visible",
    "dx": 1538,
    "dy": 760,
    "frame_dx": 1700,
    "frame_dy": 959,
    "confidence": 0.95,
    "source": "gemini_vlm",
```

```
"frame_bbox": [  
    1642,  
    943,  
    1758,  
    976  
],  
"session_memory_key": "label:clock and date",  
"cached": false  
},  
{  
    "id": "login_container",  
    "type": "pane",  
    "label": "Login Container",  
    "description": "Main white container for the login form",  
    "state": "visible",  
    "dx": 792,  
    "dy": 388,  
    "frame_dx": 954,  
    "frame_dy": 587,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        658,  
        199,  
        1250,  
        976  
    ],  
    "session_memory_key": "pane:login container",  
    "cached": false
```



```
},  
{  
  "id": "sliit_logo",  
  "type": "icon",  
  "label": "SLIIT Logo",  
  "description": "Logo of SLIIT University",  
  "state": "visible",  
  "dx": 688,  
  "dy": 156,  
  "frame_dx": 850,  
  "frame_dy": 355,  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    770,  
    327,  
    930,  
    383  
  ],  
  "session_memory_key": "icon:sliit logo",  
  "cached": false  
},  
{  
  "id": "courseweb_heading",  
  "type": "heading",  
  "label": "CourseWeb",  
  "description": "Heading text for the CourseWeb application",  
  "state": "visible",  
  "dx": 892,
```

```
"dy": 156,
"frame_dx": 1054,
"frame_dy": 355,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    954,
    327,
    1154,
    383
],
"session_memory_key": "heading:courseweb",
"cached": false
},
{
    "id": "login_button",
    "type": "button",
    "label": "Login Button",
    "description": "Yellow login button in the center of the page",
    "state": "visible",
    "dx": 796,
    "dy": 404,
    "frame_dx": 958,
    "frame_dy": 603,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        882,
        575,
```

```
        1034,  
        631  
    ],  
    "session_memory_key": "button:login button",  
    "cached": false  
},  
{  
    "id": "instruction_text",  
    "type": "label",  
    "label": "Instruction Text",  
    "description": "Text instructing the user to click on SLIIT Login",  
    "state": "visible",  
    "dx": 724,  
    "dy": 696,  
    "frame_dx": 886,  
    "frame_dy": 895,  
    "confidence": 0.85,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        802,  
        879,  
        970,  
        911  
    ],  
    "session_memory_key": "label:instruction text",  
    "cached": false  
},  
{  
    "id": "note_text",
```

```
    "type": "label",
    "label": "Note Text",
    "description": "Informational note about Microsoft 0365 validation",
    "state": "visible",
    "dx": 796,
    "dy": 736,
    "frame_dx": 958,
    "frame_dy": 935,
    "confidence": 0.85,
    "source": "gemini_vlm",
    "frame_bbox": [
        754,
        895,
        1162,
        976
    ],
    "session_memory_key": "label:note text",
    "cached": false
}
],
"session_knowledge_base": {
    "session_id": "shot_1777704787_aa8922",
    "started_at": "2026-05-02T06:53:07.025982+00:00",
    "updated_at": "2026-05-02T06:53:30.759747+00:00",
    "stale_after_misses": 3,
    "confidence_floor": 0.4,
    "decay_per_miss": 0.9,
    "frame_count": 1,
    "frame_name": "frame_1777704787_716419.jpg",
```

```
"frame_index": 0,
"image_size": {
  "width": 1920,
  "height": 1080
},
"known_elements": {
  "pane:taskbar": {
    "element_key": "pane:taskbar",
    "element_type": "pane",
    "label": "Taskbar",
    "bbox": [
      170.0,
      935.0,
      1758.0,
      976.0
    ],
    "confidence": 0.95,
    "source": "gemini_vlm",
    "first_seen_frame": "frame_1777704787_716419.jpg",
    "last_seen_frame": "frame_1777704787_716419.jpg",
    "first_seen_at": "2026-05-02T06:53:30.759747+00:00",
    "last_seen_at": "2026-05-02T06:53:30.759747+00:00",
    "first_seen_index": 0,
    "last_seen_index": 0,
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false,
    "history": [
```

```
{
  "frame_name": "frame_1777704787_716419.jpg",
  "frame_index": 0,
  "seen_at": "2026-05-02T06:53:30.759747+00:00",
  "confidence": 0.95
}
],
},
"label:weather widget": {
  "element_key": "label:weather widget",
  "element_type": "label",
  "label": "Weather Widget",
  "bbox": [
    178.0,
    951.0,
    274.0,
    976.0
  ],
  "confidence": 0.85,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777704787_716419.jpg",
  "last_seen_frame": "frame_1777704787_716419.jpg",
  "first_seen_at": "2026-05-02T06:53:30.759747+00:00",
  "last_seen_at": "2026-05-02T06:53:30.759747+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
  "seen_count": 1,
  "miss_count": 0,
  "stale": false,
```

```
"cached": false,
"history": [
  {
    "frame_name": "frame_1777704787_716419.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T06:53:30.759747+00:00",
    "confidence": 0.85
  }
],
},
"button:start button": {
  "element_key": "button:start button",
  "element_type": "button",
  "label": "Start Button",
  "bbox": [
    562.0,
    943.0,
    610.0,
    976.0
  ],
  "confidence": 0.98,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777704787_716419.jpg",
  "last_seen_frame": "frame_1777704787_716419.jpg",
  "first_seen_at": "2026-05-02T06:53:30.759747+00:00",
  "last_seen_at": "2026-05-02T06:53:30.759747+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
  "seen_count": 1,
```

```
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777704787_716419.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T06:53:30.759747+00:00",
    "confidence": 0.98
  }
],
},
"input:search bar": {
  "element_key": "input:search bar",
  "element_type": "input",
  "label": "Search Bar",
  "bbox": [
    594.0,
    943.0,
    746.0,
    976.0
  ],
  "confidence": 0.95,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777704787_716419.jpg",
  "last_seen_frame": "frame_1777704787_716419.jpg",
  "first_seen_at": "2026-05-02T06:53:30.759747+00:00",
  "last_seen_at": "2026-05-02T06:53:30.759747+00:00",
  "first_seen_index": 0,
```



```
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777704787_716419.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T06:53:30.759747+00:00",
    "confidence": 0.95
  }
],
},
"icon:file explorer": {
  "element_key": "icon:file explorer",
  "element_type": "icon",
  "label": "File Explorer",
  "bbox": [
    746.0,
    943.0,
    794.0,
    976.0
  ],
  "confidence": 0.9,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777704787_716419.jpg",
  "last_seen_frame": "frame_1777704787_716419.jpg",
  "first_seen_at": "2026-05-02T06:53:30.759747+00:00",
```

```
"last_seen_at": "2026-05-02T06:53:30.759747+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777704787_716419.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T06:53:30.759747+00:00",
    "confidence": 0.9
  }
],
},
"icon:microsoft store": {
  "element_key": "icon:microsoft store",
  "element_type": "icon",
  "label": "Microsoft Store",
  "bbox": [
    786.0,
    943.0,
    834.0,
    976.0
  ],
  "confidence": 0.9,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777704787_716419.jpg",
```

```
"last_seen_frame": "frame_1777704787_716419.jpg",
"first_seen_at": "2026-05-02T06:53:30.759747+00:00",
"last_seen_at": "2026-05-02T06:53:30.759747+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777704787_716419.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T06:53:30.759747+00:00",
    "confidence": 0.9
  }
],
},
"icon:mail": {
  "element_key": "icon:mail",
  "element_type": "icon",
  "label": "Mail",
  "bbox": [
    818.0,
    943.0,
    866.0,
    976.0
  ],
  "confidence": 0.9,
```

```
"source": "gemini_vlm",
"first_seen_frame": "frame_1777704787_716419.jpg",
"last_seen_frame": "frame_1777704787_716419.jpg",
"first_seen_at": "2026-05-02T06:53:30.759747+00:00",
"last_seen_at": "2026-05-02T06:53:30.759747+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777704787_716419.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T06:53:30.759747+00:00",
    "confidence": 0.9
  }
]
},
"icon:xbox": {
  "element_key": "icon:xbox",
  "element_type": "icon",
  "label": "Xbox",
  "bbox": [
    858.0,
    943.0,
    906.0,
    976.0
  ]
}
```

```
],
  "confidence": 0.9,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777704787_716419.jpg",
  "last_seen_frame": "frame_1777704787_716419.jpg",
  "first_seen_at": "2026-05-02T06:53:30.759747+00:00",
  "last_seen_at": "2026-05-02T06:53:30.759747+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
  "seen_count": 1,
  "miss_count": 0,
  "stale": false,
  "cached": false,
  "history": [
    {
      "frame_name": "frame_1777704787_716419.jpg",
      "frame_index": 0,
      "seen_at": "2026-05-02T06:53:30.759747+00:00",
      "confidence": 0.9
    }
  ]
},
"icon:google chrome": {
  "element_key": "icon:google chrome",
  "element_type": "icon",
  "label": "Google Chrome",
  "bbox": [
    898.0,
    943.0,
```

```
    946.0,
    976.0
  ],
  "confidence": 0.9,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777704787_716419.jpg",
  "last_seen_frame": "frame_1777704787_716419.jpg",
  "first_seen_at": "2026-05-02T06:53:30.759747+00:00",
  "last_seen_at": "2026-05-02T06:53:30.759747+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
  "seen_count": 1,
  "miss_count": 0,
  "stale": false,
  "cached": false,
  "history": [
    {
      "frame_name": "frame_1777704787_716419.jpg",
      "frame_index": 0,
      "seen_at": "2026-05-02T06:53:30.759747+00:00",
      "confidence": 0.9
    }
  ]
},
"icon:visual studio code": {
  "element_key": "icon:visual studio code",
  "element_type": "icon",
  "label": "Visual Studio Code",
  "bbox": [
```

```
    930.0,  
    943.0,  
    978.0,  
    976.0  
  ],  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "first_seen_frame": "frame_1777704787_716419.jpg",  
  "last_seen_frame": "frame_1777704787_716419.jpg",  
  "first_seen_at": "2026-05-02T06:53:30.759747+00:00",  
  "last_seen_at": "2026-05-02T06:53:30.759747+00:00",  
  "first_seen_index": 0,  
  "last_seen_index": 0,  
  "seen_count": 1,  
  "miss_count": 0,  
  "stale": false,  
  "cached": false,  
  "history": [  
    {  
      "frame_name": "frame_1777704787_716419.jpg",  
      "frame_index": 0,  
      "seen_at": "2026-05-02T06:53:30.759747+00:00",  
      "confidence": 0.9  
    }  
  ]  
},  
"icon:spotify": {  
  "element_key": "icon:spotify",  
  "element_type": "icon",
```

```
"label": "Spotify",
"bbox": [
  970.0,
  943.0,
  1018.0,
  976.0
],
"confidence": 0.9,
"source": "gemini_vlm",
"first_seen_frame": "frame_1777704787_716419.jpg",
"last_seen_frame": "frame_1777704787_716419.jpg",
"first_seen_at": "2026-05-02T06:53:30.759747+00:00",
"last_seen_at": "2026-05-02T06:53:30.759747+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777704787_716419.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T06:53:30.759747+00:00",
    "confidence": 0.9
  }
]
},
"icon:google": {
```



```
"element_key": "icon:google",
"element_type": "icon",
"label": "Google",
"bbox": [
  1010.0,
  943.0,
  1058.0,
  976.0
],
"confidence": 0.9,
"source": "gemini_vlm",
"first_seen_frame": "frame_1777704787_716419.jpg",
"last_seen_frame": "frame_1777704787_716419.jpg",
"first_seen_at": "2026-05-02T06:53:30.759747+00:00",
"last_seen_at": "2026-05-02T06:53:30.759747+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777704787_716419.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T06:53:30.759747+00:00",
    "confidence": 0.9
  }
]
```

```
},
"icon:autocad": {
  "element_key": "icon:autocad",
  "element_type": "icon",
  "label": "AutoCAD",
  "bbox": [
    1050.0,
    943.0,
    1090.0,
    976.0
  ],
  "confidence": 0.8,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777704787_716419.jpg",
  "last_seen_frame": "frame_1777704787_716419.jpg",
  "first_seen_at": "2026-05-02T06:53:30.759747+00:00",
  "last_seen_at": "2026-05-02T06:53:30.759747+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
  "seen_count": 1,
  "miss_count": 0,
  "stale": false,
  "cached": false,
  "history": [
    {
      "frame_name": "frame_1777704787_716419.jpg",
      "frame_index": 0,
      "seen_at": "2026-05-02T06:53:30.759747+00:00",
      "confidence": 0.8
    }
  ]
}
```

```
    }  
  ]  
},  
"pane:system tray": {  
  "element_key": "pane:system tray",  
  "element_type": "pane",  
  "label": "System Tray",  
  "bbox": [  
    1426.0,  
    943.0,  
    1758.0,  
    976.0  
  ],  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "first_seen_frame": "frame_1777704787_716419.jpg",  
  "last_seen_frame": "frame_1777704787_716419.jpg",  
  "first_seen_at": "2026-05-02T06:53:30.759747+00:00",  
  "last_seen_at": "2026-05-02T06:53:30.759747+00:00",  
  "first_seen_index": 0,  
  "last_seen_index": 0,  
  "seen_count": 1,  
  "miss_count": 0,  
  "stale": false,  
  "cached": false,  
  "history": [  
    {  
      "frame_name": "frame_1777704787_716419.jpg",  
      "frame_index": 0,  

```

```
        "seen_at": "2026-05-02T06:53:30.759747+00:00",
        "confidence": 0.9
    }
]
},
"label:clock and date": {
    "element_key": "label:clock and date",
    "element_type": "label",
    "label": "Clock and Date",
    "bbox": [
        1642.0,
        943.0,
        1758.0,
        976.0
    ],
    "confidence": 0.95,
    "source": "gemini_vlm",
    "first_seen_frame": "frame_1777704787_716419.jpg",
    "last_seen_frame": "frame_1777704787_716419.jpg",
    "first_seen_at": "2026-05-02T06:53:30.759747+00:00",
    "last_seen_at": "2026-05-02T06:53:30.759747+00:00",
    "first_seen_index": 0,
    "last_seen_index": 0,
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false,
    "history": [
        {
```

```
        "frame_name": "frame_1777704787_716419.jpg",
        "frame_index": 0,
        "seen_at": "2026-05-02T06:53:30.759747+00:00",
        "confidence": 0.95
    }
]
},
"pane:login container": {
    "element_key": "pane:login container",
    "element_type": "pane",
    "label": "Login Container",
    "bbox": [
        658.0,
        199.0,
        1250.0,
        976.0
    ],
    "confidence": 0.95,
    "source": "gemini_vlm",
    "first_seen_frame": "frame_1777704787_716419.jpg",
    "last_seen_frame": "frame_1777704787_716419.jpg",
    "first_seen_at": "2026-05-02T06:53:30.759747+00:00",
    "last_seen_at": "2026-05-02T06:53:30.759747+00:00",
    "first_seen_index": 0,
    "last_seen_index": 0,
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false,
```

```
"history": [  
  {  
    "frame_name": "frame_1777704787_716419.jpg",  
    "frame_index": 0,  
    "seen_at": "2026-05-02T06:53:30.759747+00:00",  
    "confidence": 0.95  
  }  
]  
,  
"icon:sliit logo": {  
  "element_key": "icon:sliit logo",  
  "element_type": "icon",  
  "label": "SLIIT Logo",  
  "bbox": [  
    770.0,  
    327.0,  
    930.0,  
    383.0  
  ],  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "first_seen_frame": "frame_1777704787_716419.jpg",  
  "last_seen_frame": "frame_1777704787_716419.jpg",  
  "first_seen_at": "2026-05-02T06:53:30.759747+00:00",  
  "last_seen_at": "2026-05-02T06:53:30.759747+00:00",  
  "first_seen_index": 0,  
  "last_seen_index": 0,  
  "seen_count": 1,  
  "miss_count": 0,
```

```
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777704787_716419.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T06:53:30.759747+00:00",
    "confidence": 0.9
  }
],
},
"heading:courseweb": {
  "element_key": "heading:courseweb",
  "element_type": "heading",
  "label": "CourseWeb",
  "bbox": [
    954.0,
    327.0,
    1154.0,
    383.0
  ],
  "confidence": 0.95,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777704787_716419.jpg",
  "last_seen_frame": "frame_1777704787_716419.jpg",
  "first_seen_at": "2026-05-02T06:53:30.759747+00:00",
  "last_seen_at": "2026-05-02T06:53:30.759747+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
```

```
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777704787_716419.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T06:53:30.759747+00:00",
    "confidence": 0.95
  }
]
},
"button:login button": {
  "element_key": "button:login button",
  "element_type": "button",
  "label": "Login Button",
  "bbox": [
    882.0,
    575.0,
    1034.0,
    631.0
  ],
  "confidence": 0.9,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777704787_716419.jpg",
  "last_seen_frame": "frame_1777704787_716419.jpg",
  "first_seen_at": "2026-05-02T06:53:30.759747+00:00",
  "last_seen_at": "2026-05-02T06:53:30.759747+00:00",
```



```
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777704787_716419.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T06:53:30.759747+00:00",
    "confidence": 0.9
  }
],
},
"label:instruction text": {
  "element_key": "label:instruction text",
  "element_type": "label",
  "label": "Instruction Text",
  "bbox": [
    802.0,
    879.0,
    970.0,
    911.0
  ],
  "confidence": 0.85,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777704787_716419.jpg",
  "last_seen_frame": "frame_1777704787_716419.jpg",
```

```
"first_seen_at": "2026-05-02T06:53:30.759747+00:00",
"last_seen_at": "2026-05-02T06:53:30.759747+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777704787_716419.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T06:53:30.759747+00:00",
    "confidence": 0.85
  }
],
},
"label:note text": {
  "element_key": "label:note text",
  "element_type": "label",
  "label": "Note Text",
  "bbox": [
    754.0,
    895.0,
    1162.0,
    976.0
  ],
  "confidence": 0.85,
  "source": "gemini_vlm",
```

```
"first_seen_frame": "frame_1777704787_716419.jpg",
"last_seen_frame": "frame_1777704787_716419.jpg",
"first_seen_at": "2026-05-02T06:53:30.759747+00:00",
"last_seen_at": "2026-05-02T06:53:30.759747+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777704787_716419.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T06:53:30.759747+00:00",
    "confidence": 0.85
  }
]
},
"resolved_elements": [
  {
    "id": "taskbar",
    "type": "pane",
    "label": "Taskbar",
    "description": "Windows taskbar at the bottom of the screen",
    "state": "visible",
    "dx": 802,
    "dy": 756,
```

```
"frame_dx": 964,  
"frame_dy": 955,  
"confidence": 0.95,  
"source": "gemini_vlm",  
"frame_bbox": [  
    170,  
    935,  
    1758,  
    976  
],  
"session_memory_key": "pane:taskbar",  
"cached": false  
},  
{  
    "id": "weather_widget",  
    "type": "label",  
    "label": "Weather Widget",  
    "description": "Weather and temperature display in the taskbar",  
    "state": "visible",  
    "dx": 64,  
    "dy": 764,  
    "frame_dx": 226,  
    "frame_dy": 963,  
    "confidence": 0.85,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        178,  
        951,  
        274,
```

```
    976
  ],
  "session_memory_key": "label:weather widget",
  "cached": false
},
{
  "id": "start_button",
  "type": "button",
  "label": "Start Button",
  "description": "Windows Start menu button",
  "state": "visible",
  "dx": 424,
  "dy": 760,
  "frame_dx": 586,
  "frame_dy": 959,
  "confidence": 0.98,
  "source": "gemini_vlm",
  "frame_bbox": [
    562,
    943,
    610,
    976
  ],
  "session_memory_key": "button:start button",
  "cached": false
},
{
  "id": "search_bar",
  "type": "input",
```

```
"label": "Search Bar",
"description": "Windows search input field in the taskbar",
"state": "visible",
"dx": 508,
"dy": 760,
"frame_dx": 670,
"frame_dy": 959,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    594,
    943,
    746,
    976
],
"session_memory_key": "input:search bar",
"cached": false
},
{
    "id": "file_explorer_icon",
    "type": "icon",
    "label": "File Explorer",
    "description": "File Explorer icon in the taskbar",
    "state": "visible",
    "dx": 608,
    "dy": 760,
    "frame_dx": 770,
    "frame_dy": 959,
    "confidence": 0.9,
```

```
"source": "gemini_vlm",
"frame_bbox": [
    746,
    943,
    794,
    976
],
"session_memory_key": "icon:file explorer",
"cached": false
},
{
    "id": "microsoft_store_icon",
    "type": "icon",
    "label": "Microsoft Store",
    "description": "Microsoft Store icon in the taskbar",
    "state": "visible",
    "dx": 648,
    "dy": 760,
    "frame_dx": 810,
    "frame_dy": 959,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        786,
        943,
        834,
        976
    ],
    "session_memory_key": "icon:microsoft store",
```

```
    "cached": false
  },
  {
    "id": "mail_icon",
    "type": "icon",
    "label": "Mail",
    "description": "Mail application icon in the taskbar",
    "state": "visible",
    "dx": 680,
    "dy": 760,
    "frame_dx": 842,
    "frame_dy": 959,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
      818,
      943,
      866,
      976
    ],
    "session_memory_key": "icon:mail",
    "cached": false
  },
  {
    "id": "xbox_icon",
    "type": "icon",
    "label": "Xbox",
    "description": "Xbox application icon in the taskbar",
    "state": "visible",
```



```
"dx": 720,  
"dy": 760,  
"frame_dx": 882,  
"frame_dy": 959,  
"confidence": 0.9,  
"source": "gemini_vlm",  
"frame_bbox": [  
    858,  
    943,  
    906,  
    976  
],  
"session_memory_key": "icon:xbox",  
"cached": false  
},  
{  
    "id": "chrome_icon",  
    "type": "icon",  
    "label": "Google Chrome",  
    "description": "Google Chrome browser icon in the taskbar",  
    "state": "visible",  
    "dx": 760,  
    "dy": 760,  
    "frame_dx": 922,  
    "frame_dy": 959,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        898,
```

```
    943,  
    946,  
    976  
  ],  
  "session_memory_key": "icon:google chrome",  
  "cached": false  
},  
{  
  "id": "vscode_icon",  
  "type": "icon",  
  "label": "Visual Studio Code",  
  "description": "Visual Studio Code icon in the taskbar",  
  "state": "visible",  
  "dx": 792,  
  "dy": 760,  
  "frame_dx": 954,  
  "frame_dy": 959,  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    930,  
    943,  
    978,  
    976  
  ],  
  "session_memory_key": "icon:visual studio code",  
  "cached": false  
},  
{
```

```
"id": "spotify_icon",
"type": "icon",
"label": "Spotify",
"description": "Spotify application icon in the taskbar",
"state": "visible",
"dx": 832,
"dy": 760,
"frame_dx": 994,
"frame_dy": 959,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    970,
    943,
    1018,
    976
],
"session_memory_key": "icon:spotify",
"cached": false
},
{
    "id": "google_icon",
    "type": "icon",
    "label": "Google",
    "description": "Google application icon in the taskbar",
    "state": "visible",
    "dx": 872,
    "dy": 760,
    "frame_dx": 1034,
```

```
"frame_dy": 959,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    1010,
    943,
    1058,
    976
],
"session_memory_key": "icon:google",
"cached": false
},
{
    "id": "autocad_icon",
    "type": "icon",
    "label": "AutoCAD",
    "description": "AutoCAD application icon in the taskbar",
    "state": "visible",
    "dx": 908,
    "dy": 760,
    "frame_dx": 1070,
    "frame_dy": 959,
    "confidence": 0.8,
    "source": "gemini_vlm",
    "frame_bbox": [
        1050,
        943,
        1090,
        976
    ]
}
```

```
    ],  
    "session_memory_key": "icon:autocad",  
    "cached": false  
  },  
  {  
    "id": "system_tray",  
    "type": "pane",  
    "label": "System Tray",  
    "description": "System tray area with various status icons",  
    "state": "visible",  
    "dx": 1430,  
    "dy": 760,  
    "frame_dx": 1592,  
    "frame_dy": 959,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
      1426,  
      943,  
      1758,  
      976  
    ],  
    "session_memory_key": "pane:system tray",  
    "cached": false  
  },  
  {  
    "id": "clock_date",  
    "type": "label",  
    "label": "Clock and Date",
```

```
"description": "Current time and date display in the taskbar",
"state": "visible",
"dx": 1538,
"dy": 760,
"frame_dx": 1700,
"frame_dy": 959,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    1642,
    943,
    1758,
    976
],
"session_memory_key": "label:clock and date",
"cached": false
},
{
    "id": "login_container",
    "type": "pane",
    "label": "Login Container",
    "description": "Main white container for the login form",
    "state": "visible",
    "dx": 792,
    "dy": 388,
    "frame_dx": 954,
    "frame_dy": 587,
    "confidence": 0.95,
    "source": "gemini_vlm",
```

```
"frame_bbox": [
    658,
    199,
    1250,
    976
],
"session_memory_key": "pane:login container",
"cached": false
},
{
    "id": "sliit_logo",
    "type": "icon",
    "label": "SLIIT Logo",
    "description": "Logo of SLIIT University",
    "state": "visible",
    "dx": 688,
    "dy": 156,
    "frame_dx": 850,
    "frame_dy": 355,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        770,
        327,
        930,
        383
    ],
    "session_memory_key": "icon:sliit logo",
    "cached": false
}
```

```
},  
{  
  "id": "courseweb_heading",  
  "type": "heading",  
  "label": "CourseWeb",  
  "description": "Heading text for the CourseWeb application",  
  "state": "visible",  
  "dx": 892,  
  "dy": 156,  
  "frame_dx": 1054,  
  "frame_dy": 355,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    954,  
    327,  
    1154,  
    383  
  ],  
  "session_memory_key": "heading:courseweb",  
  "cached": false  
},  
{  
  "id": "login_button",  
  "type": "button",  
  "label": "Login Button",  
  "description": "Yellow login button in the center of the page",  
  "state": "visible",  
  "dx": 796,
```



```
"dy": 404,  
  "frame_dx": 958,  
  "frame_dy": 603,  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    882,  
    575,  
    1034,  
    631  
  ],  
  "session_memory_key": "button:login button",  
  "cached": false  
},  
{  
  "id": "instruction_text",  
  "type": "label",  
  "label": "Instruction Text",  
  "description": "Text instructing the user to click on SLIIT Login",  
  "state": "visible",  
  "dx": 724,  
  "dy": 696,  
  "frame_dx": 886,  
  "frame_dy": 895,  
  "confidence": 0.85,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    802,  
    879,
```

```
        970,  
        911  
    ],  
    "session_memory_key": "label:instruction text",  
    "cached": false  
},  
{  
    "id": "note_text",  
    "type": "label",  
    "label": "Note Text",  
    "description": "Informational note about Microsoft 0365 validation",  
    "state": "visible",  
    "dx": 796,  
    "dy": 736,  
    "frame_dx": 958,  
    "frame_dy": 935,  
    "confidence": 0.85,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        754,  
        895,  
        1162,  
        976  
    ],  
    "session_memory_key": "label:note text",  
    "cached": false  
}  
],  
"current_elements": [  

```

```
{
  "id": "taskbar",
  "type": "pane",
  "label": "Taskbar",
  "description": "Windows taskbar at the bottom of the screen",
  "state": "visible",
  "dx": 802,
  "dy": 756,
  "frame_dx": 964,
  "frame_dy": 955,
  "confidence": 0.95,
  "source": "gemini_vlm",
  "frame_bbox": [
    170,
    935,
    1758,
    976
  ],
  "session_memory_key": "pane:taskbar",
  "cached": false
},
{
  "id": "weather_widget",
  "type": "label",
  "label": "Weather Widget",
  "description": "Weather and temperature display in the taskbar",
  "state": "visible",
  "dx": 64,
  "dy": 764,
```

```
"frame_dx": 226,  
"frame_dy": 963,  
"confidence": 0.85,  
"source": "gemini_vlm",  
"frame_bbox": [  
    178,  
    951,  
    274,  
    976  
],  
"session_memory_key": "label:weather widget",  
"cached": false  
},  
{  
    "id": "start_button",  
    "type": "button",  
    "label": "Start Button",  
    "description": "Windows Start menu button",  
    "state": "visible",  
    "dx": 424,  
    "dy": 760,  
    "frame_dx": 586,  
    "frame_dy": 959,  
    "confidence": 0.98,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        562,  
        943,  
        610,
```

```
    976
  ],
  "session_memory_key": "button:start button",
  "cached": false
},
{
  "id": "search_bar",
  "type": "input",
  "label": "Search Bar",
  "description": "Windows search input field in the taskbar",
  "state": "visible",
  "dx": 508,
  "dy": 760,
  "frame_dx": 670,
  "frame_dy": 959,
  "confidence": 0.95,
  "source": "gemini_vlm",
  "frame_bbox": [
    594,
    943,
    746,
    976
  ],
  "session_memory_key": "input:search bar",
  "cached": false
},
{
  "id": "file_explorer_icon",
  "type": "icon",
```

```
"label": "File Explorer",
"description": "File Explorer icon in the taskbar",
"state": "visible",
"dx": 608,
"dy": 760,
"frame_dx": 770,
"frame_dy": 959,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    746,
    943,
    794,
    976
],
"session_memory_key": "icon:file explorer",
"cached": false
},
{
    "id": "microsoft_store_icon",
    "type": "icon",
    "label": "Microsoft Store",
    "description": "Microsoft Store icon in the taskbar",
    "state": "visible",
    "dx": 648,
    "dy": 760,
    "frame_dx": 810,
    "frame_dy": 959,
    "confidence": 0.9,
```

```
"source": "gemini_vlm",
"frame_bbox": [
    786,
    943,
    834,
    976
],
"session_memory_key": "icon:microsoft store",
"cached": false
},
{
    "id": "mail_icon",
    "type": "icon",
    "label": "Mail",
    "description": "Mail application icon in the taskbar",
    "state": "visible",
    "dx": 680,
    "dy": 760,
    "frame_dx": 842,
    "frame_dy": 959,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        818,
        943,
        866,
        976
    ],
    "session_memory_key": "icon:mail",
```

```
    "cached": false
  },
  {
    "id": "xbox_icon",
    "type": "icon",
    "label": "Xbox",
    "description": "Xbox application icon in the taskbar",
    "state": "visible",
    "dx": 720,
    "dy": 760,
    "frame_dx": 882,
    "frame_dy": 959,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
      858,
      943,
      906,
      976
    ],
    "session_memory_key": "icon:xbox",
    "cached": false
  },
  {
    "id": "chrome_icon",
    "type": "icon",
    "label": "Google Chrome",
    "description": "Google Chrome browser icon in the taskbar",
    "state": "visible",
```



```
"dx": 760,  
"dy": 760,  
"frame_dx": 922,  
"frame_dy": 959,  
"confidence": 0.9,  
"source": "gemini_vlm",  
"frame_bbox": [  
    898,  
    943,  
    946,  
    976  
],  
"session_memory_key": "icon:google chrome",  
"cached": false  
},  
{  
    "id": "vscode_icon",  
    "type": "icon",  
    "label": "Visual Studio Code",  
    "description": "Visual Studio Code icon in the taskbar",  
    "state": "visible",  
    "dx": 792,  
    "dy": 760,  
    "frame_dx": 954,  
    "frame_dy": 959,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        930,
```

```
    943,  
    978,  
    976  
  ],  
  "session_memory_key": "icon:visual studio code",  
  "cached": false  
},  
{  
  "id": "spotify_icon",  
  "type": "icon",  
  "label": "Spotify",  
  "description": "Spotify application icon in the taskbar",  
  "state": "visible",  
  "dx": 832,  
  "dy": 760,  
  "frame_dx": 994,  
  "frame_dy": 959,  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    970,  
    943,  
    1018,  
    976  
  ],  
  "session_memory_key": "icon:spotify",  
  "cached": false  
},  
{
```

```
"id": "google_icon",
"type": "icon",
"label": "Google",
"description": "Google application icon in the taskbar",
"state": "visible",
"dx": 872,
"dy": 760,
"frame_dx": 1034,
"frame_dy": 959,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    1010,
    943,
    1058,
    976
],
"session_memory_key": "icon:google",
"cached": false
},
{
    "id": "autocad_icon",
    "type": "icon",
    "label": "AutoCAD",
    "description": "AutoCAD application icon in the taskbar",
    "state": "visible",
    "dx": 908,
    "dy": 760,
    "frame_dx": 1070,
```

```
"frame_dy": 959,
"confidence": 0.8,
"source": "gemini_vlm",
"frame_bbox": [
    1050,
    943,
    1090,
    976
],
"session_memory_key": "icon:autocad",
"cached": false
},
{
    "id": "system_tray",
    "type": "pane",
    "label": "System Tray",
    "description": "System tray area with various status icons",
    "state": "visible",
    "dx": 1430,
    "dy": 760,
    "frame_dx": 1592,
    "frame_dy": 959,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        1426,
        943,
        1758,
        976
    ]
}
```

```
    ],
    "session_memory_key": "pane:system tray",
    "cached": false
  },
  {
    "id": "clock_date",
    "type": "label",
    "label": "Clock and Date",
    "description": "Current time and date display in the taskbar",
    "state": "visible",
    "dx": 1538,
    "dy": 760,
    "frame_dx": 1700,
    "frame_dy": 959,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
      1642,
      943,
      1758,
      976
    ],
    "session_memory_key": "label:clock and date",
    "cached": false
  },
  {
    "id": "login_container",
    "type": "pane",
    "label": "Login Container",
```

```
"description": "Main white container for the login form",
"state": "visible",
"dx": 792,
"dy": 388,
"frame_dx": 954,
"frame_dy": 587,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    658,
    199,
    1250,
    976
],
"session_memory_key": "pane:login container",
"cached": false
},
{
    "id": "sliit_logo",
    "type": "icon",
    "label": "SLIIT Logo",
    "description": "Logo of SLIIT University",
    "state": "visible",
    "dx": 688,
    "dy": 156,
    "frame_dx": 850,
    "frame_dy": 355,
    "confidence": 0.9,
    "source": "gemini_vlm",
```

```
"frame_bbox": [
  770,
  327,
  930,
  383
],
"session_memory_key": "icon:sliit logo",
"cached": false
},
{
  "id": "courseweb_heading",
  "type": "heading",
  "label": "CourseWeb",
  "description": "Heading text for the CourseWeb application",
  "state": "visible",
  "dx": 892,
  "dy": 156,
  "frame_dx": 1054,
  "frame_dy": 355,
  "confidence": 0.95,
  "source": "gemini_vlm",
  "frame_bbox": [
    954,
    327,
    1154,
    383
  ],
  "session_memory_key": "heading:courseweb",
  "cached": false
}
```

```
},  
{  
  "id": "login_button",  
  "type": "button",  
  "label": "Login Button",  
  "description": "Yellow login button in the center of the page",  
  "state": "visible",  
  "dx": 796,  
  "dy": 404,  
  "frame_dx": 958,  
  "frame_dy": 603,  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    882,  
    575,  
    1034,  
    631  
  ],  
  "session_memory_key": "button:login button",  
  "cached": false  
},  
{  
  "id": "instruction_text",  
  "type": "label",  
  "label": "Instruction Text",  
  "description": "Text instructing the user to click on SLIIT Login",  
  "state": "visible",  
  "dx": 724,
```



```
"dy": 696,  
  "frame_dx": 886,  
  "frame_dy": 895,  
  "confidence": 0.85,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    802,  
    879,  
    970,  
    911  
  ],  
  "session_memory_key": "label:instruction text",  
  "cached": false  
},  
{  
  "id": "note_text",  
  "type": "label",  
  "label": "Note Text",  
  "description": "Informational note about Microsoft 0365 validation",  
  "state": "visible",  
  "dx": 796,  
  "dy": 736,  
  "frame_dx": 958,  
  "frame_dy": 935,  
  "confidence": 0.85,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    754,  
    895,
```

```
        1162,
        976
    ],
    "session_memory_key": "label:note text",
    "cached": false
}
]
},
"session_memory": {
    "run_id": "57f743fa-1756-4baa-b97d-25c9dacdd30d",
    "user_task": "Now you are in the CourseWeb login page, so the username:
\\IT22099518\\ and password: \\NuwaniSithara@808\\ so using both credentials try
to login to the CourseWeb",
    "started_at": "2026-05-02T06:53:02.537150+00:00",
    "updated_at": "2026-05-02T06:53:30.786319+00:00",
    "stale_after_misses": 2,
    "confidence_floor": 0.4,
    "decay_per_miss": 0.9,
    "frame_index": 1,
    "known_elements": {
        "pane:taskbar": {
            "element_key": "pane:taskbar",
            "element_type": "pane",
            "label": "Taskbar",
            "bbox": [
                177.0,
                943.0,
                1756.0,
                962.0
            ],

```

```
"confidence": 0.6561,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 17,
"first_seen_at": "2026-05-02T06:53:30.786319+00:00",
"last_seen_at": "2026-05-02T07:08:17.240100+00:00",
"seen_count": 4,
"miss_count": 3,
"stale": true,
"cached": true
},
"label:weather widget": {
  "element_key": "label:weather widget",
  "element_type": "label",
  "label": "Weather Widget",
  "bbox": [
    178.0,
    951.0,
    274.0,
    976.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 0,
```

```
"first_seen_at": "2026-05-02T06:53:30.786319+00:00",
"last_seen_at": "2026-05-02T06:53:30.786319+00:00",
"seen_count": 1,
"miss_count": 20,
"stale": true,
"cached": true
},
"button:start button": {
  "element_key": "button:start button",
  "element_type": "button",
  "label": "Start Button",
  "bbox": [
    562.0,
    943.0,
    610.0,
    976.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 0,
  "first_seen_at": "2026-05-02T06:53:30.786319+00:00",
  "last_seen_at": "2026-05-02T06:53:30.786319+00:00",
  "seen_count": 1,
  "miss_count": 20,
  "stale": true,
  "cached": true
}
```

```
},

```

```
    762.0,  
    926.0,  
    818.0,  
    962.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 0,  
  "last_seen_index": 5,  
  "first_seen_at": "2026-05-02T06:53:30.786319+00:00",  
  "last_seen_at": "2026-05-02T06:57:27.809908+00:00",  
  "seen_count": 2,  
  "miss_count": 15,  
  "stale": true,  
  "cached": true  
},  
"icon:microsoft store": {  
  "element_key": "icon:microsoft store",  
  "element_type": "icon",  
  "label": "Microsoft Store",  
  "bbox": [  
    786.0,  
    943.0,  
    834.0,  
    976.0  
  ],  
  "confidence": 0.4,
```

```
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 0,
"first_seen_at": "2026-05-02T06:53:30.786319+00:00",
"last_seen_at": "2026-05-02T06:53:30.786319+00:00",
"seen_count": 1,
"miss_count": 20,
"stale": true,
"cached": true
},
"icon:mail": {
  "element_key": "icon:mail",
  "element_type": "icon",
  "label": "Mail",
  "bbox": [
    818.0,
    943.0,
    866.0,
    976.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 0,
  "first_seen_at": "2026-05-02T06:53:30.786319+00:00",
```

```
    "last_seen_at": "2026-05-02T06:53:30.786319+00:00",
    "seen_count": 1,
    "miss_count": 20,
    "stale": true,
    "cached": true
  },
  "icon:xbox": {
    "element_key": "icon:xbox",
    "element_type": "icon",
    "label": "Xbox",
    "bbox": [
      858.0,
      943.0,
      906.0,
      976.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 0,
    "last_seen_index": 0,
    "first_seen_at": "2026-05-02T06:53:30.786319+00:00",
    "last_seen_at": "2026-05-02T06:53:30.786319+00:00",
    "seen_count": 1,
    "miss_count": 20,
    "stale": true,
    "cached": true
  },
}
```



```
"icon:google chrome": {
  "element_key": "icon:google chrome",
  "element_type": "icon",
  "label": "Google Chrome",
  "bbox": [
    728.0,
    934.0,
    776.0,
    962.0
  ],
  "confidence": 0.5019,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 15,
  "first_seen_at": "2026-05-02T06:53:30.786319+00:00",
  "last_seen_at": "2026-05-02T07:06:29.451301+00:00",
  "seen_count": 4,
  "miss_count": 5,
  "stale": true,
  "cached": true
},
"icon:visual studio code": {
  "element_key": "icon:visual studio code",
  "element_type": "icon",
  "label": "Visual Studio Code",
  "bbox": [
    810.0,
```

```
    926.0,
    866.0,
    962.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 5,
"first_seen_at": "2026-05-02T06:53:30.786319+00:00",
"last_seen_at": "2026-05-02T06:57:27.809908+00:00",
"seen_count": 2,
"miss_count": 15,
"stale": true,
"cached": true
},
"icon:spotify": {
  "element_key": "icon:spotify",
  "element_type": "icon",
  "label": "Spotify",
  "bbox": [
    986.0,
    926.0,
    1034.0,
    962.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
```

```
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 5,
"first_seen_at": "2026-05-02T06:53:30.786319+00:00",
"last_seen_at": "2026-05-02T06:57:27.809908+00:00",
"seen_count": 2,
"miss_count": 15,
"stale": true,
"cached": true
},
"icon:google": {
  "element_key": "icon:google",
  "element_type": "icon",
  "label": "Google",
  "bbox": [
    1010.0,
    943.0,
    1058.0,
    976.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 0,
  "first_seen_at": "2026-05-02T06:53:30.786319+00:00",
  "last_seen_at": "2026-05-02T06:53:30.786319+00:00",
```

```
    "seen_count": 1,  
    "miss_count": 20,  
    "stale": true,  
    "cached": true  
  },  
  "icon:autocad": {  
    "element_key": "icon:autocad",  
    "element_type": "icon",  
    "label": "AutoCAD",  
    "bbox": [  
      1050.0,  
      943.0,  
      1090.0,  
      976.0  
    ],  
    "confidence": 0.4,  
    "source": "gemini_vlm",  
    "first_seen_frame": null,  
    "last_seen_frame": null,  
    "first_seen_index": 0,  
    "last_seen_index": 0,  
    "first_seen_at": "2026-05-02T06:53:30.786319+00:00",  
    "last_seen_at": "2026-05-02T06:53:30.786319+00:00",  
    "seen_count": 1,  
    "miss_count": 20,  
    "stale": true,  
    "cached": true  
  },  
  "pane:system tray": {
```

```
"element_key": "pane:system tray",
"element_type": "pane",
"label": "System Tray",
"bbox": [
    1426.0,
    943.0,
    1758.0,
    976.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 0,
"first_seen_at": "2026-05-02T06:53:30.786319+00:00",
"last_seen_at": "2026-05-02T06:53:30.786319+00:00",
"seen_count": 1,
"miss_count": 20,
"stale": true,
"cached": true
},
"label:clock and date": {
    "element_key": "label:clock and date",
    "element_type": "label",
    "label": "Clock and Date",
    "bbox": [
        1640.0,
        934.0,
```

```
    1744.0,  
    962.0  
  ],  
  "confidence": 0.4724,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 0,  
  "last_seen_index": 15,  
  "first_seen_at": "2026-05-02T06:53:30.786319+00:00",  
  "last_seen_at": "2026-05-02T07:06:29.451301+00:00",  
  "seen_count": 3,  
  "miss_count": 5,  
  "stale": true,  
  "cached": true  
},  
"pane:login container": {  
  "element_key": "pane:login container",  
  "element_type": "pane",  
  "label": "Login Container",  
  "bbox": [  
    658.0,  
    199.0,  
    1250.0,  
    976.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,
```

```
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 0,
"first_seen_at": "2026-05-02T06:53:30.786319+00:00",
"last_seen_at": "2026-05-02T06:53:30.786319+00:00",
"seen_count": 1,
"miss_count": 20,
"stale": true,
"cached": true
},
"icon:sliit logo": {
  "element_key": "icon:sliit logo",
  "element_type": "icon",
  "label": "SLIIT Logo",
  "bbox": [
    770.0,
    327.0,
    930.0,
    383.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 0,
  "first_seen_at": "2026-05-02T06:53:30.786319+00:00",
  "last_seen_at": "2026-05-02T06:53:30.786319+00:00",
  "seen_count": 1,
```

```
    "miss_count": 20,  
    "stale": true,  
    "cached": true  
  },  
  "heading:courseweb": {  
    "element_key": "heading:courseweb",  
    "element_type": "heading",  
    "label": "CourseWeb",  
    "bbox": [  
      929.0,  
      278.0,  
      1153.0,  
      318.0  
    ],  
    "confidence": 0.4783,  
    "source": "gemini_vlm",  
    "first_seen_frame": null,  
    "last_seen_frame": null,  
    "first_seen_index": 0,  
    "last_seen_index": 14,  
    "first_seen_at": "2026-05-02T06:53:30.786319+00:00",  
    "last_seen_at": "2026-05-02T07:05:44.869252+00:00",  
    "seen_count": 2,  
    "miss_count": 6,  
    "stale": true,  
    "cached": true  
  },  
  "button:login button": {  
    "element_key": "button:login button",
```



```
"element_type": "button",
"label": "Login Button",
"bbox": [
    882.0,
    575.0,
    1034.0,
    631.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 0,
"first_seen_at": "2026-05-02T06:53:30.786319+00:00",
"last_seen_at": "2026-05-02T06:53:30.786319+00:00",
"seen_count": 1,
"miss_count": 20,
"stale": true,
"cached": true
},
"label:instruction text": {
    "element_key": "label:instruction text",
    "element_type": "label",
    "label": "Instruction Text",
    "bbox": [
        802.0,
        879.0,
        970.0,
```

```
    911.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 0,
  "first_seen_at": "2026-05-02T06:53:30.786319+00:00",
  "last_seen_at": "2026-05-02T06:53:30.786319+00:00",
  "seen_count": 1,
  "miss_count": 20,
  "stale": true,
  "cached": true
},
"label:note text": {
  "element_key": "label:note text",
  "element_type": "label",
  "label": "Note Text",
  "bbox": [
    754.0,
    895.0,
    1162.0,
    976.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
```

```
"first_seen_index": 0,
"last_seen_index": 0,
"first_seen_at": "2026-05-02T06:53:30.786319+00:00",
"last_seen_at": "2026-05-02T06:53:30.786319+00:00",
"seen_count": 1,
"miss_count": 20,
"stale": true,
"cached": true
},
"heading:sliit uni courseweb": {
  "element_key": "heading:sliit uni courseweb",
  "element_type": "heading",
  "label": "SLIIT UNI CourseWeb",
  "bbox": [
    729.0,
    263.0,
    1193.0,
    327.0
  ],
  "confidence": 0.6925,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
  "last_seen_index": 17,
  "first_seen_at": "2026-05-02T06:54:23.082460+00:00",
  "last_seen_at": "2026-05-02T07:08:17.240100+00:00",
  "seen_count": 5,
  "miss_count": 3,
```

```
    "stale": true,
    "cached": true
  },
  "input:username or email": {
    "element_key": "input:username or email",
    "element_type": "input",
    "label": "Username or email",
    "bbox": [
      728.0,
      336.0,
      1192.0,
      376.0
    ],
    "confidence": 0.98,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 1,
    "last_seen_index": 20,
    "first_seen_at": "2026-05-02T06:54:23.082460+00:00",
    "last_seen_at": "2026-05-02T07:11:06.085051+00:00",
    "seen_count": 13,
    "miss_count": 0,
    "stale": false,
    "cached": false
  },
  "input:password": {
    "element_key": "input:password",
    "element_type": "input",
```

```
"label": "Password",
"bbox": [
  728.0,
  384.0,
  1192.0,
  416.0
],
"confidence": 0.98,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 1,
"last_seen_index": 20,
"first_seen_at": "2026-05-02T06:54:23.082460+00:00",
"last_seen_at": "2026-05-02T07:11:06.085051+00:00",
"seen_count": 11,
"miss_count": 0,
"stale": false,
"cached": false
},
"button:log in": {
  "element_key": "button:log in",
  "element_type": "button",
  "label": "Log in",
  "bbox": [
    880.0,
    424.0,
    1040.0,
    472.0
  ]
}
```

```
],
  "confidence": 0.99,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
  "last_seen_index": 20,
  "first_seen_at": "2026-05-02T06:54:23.082460+00:00",
  "last_seen_at": "2026-05-02T07:11:06.085051+00:00",
  "seen_count": 10,
  "miss_count": 0,
  "stale": false,
  "cached": false
},
"link:lost password?": {
  "element_key": "link:lost password?",
  "element_type": "link",
  "label": "Lost password?",
  "bbox": [
    920.0,
    480.0,
    1008.0,
    504.0
  ],
  "confidence": 0.95,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
```

```
"last_seen_index": 20,  
"first_seen_at": "2026-05-02T06:54:23.082460+00:00",  
"last_seen_at": "2026-05-02T07:11:06.085051+00:00",  
"seen_count": 6,  
"miss_count": 0,  
"stale": false,  
"cached": false  
},  
"text:log in using your account on:": {  
  "element_key": "text:log in using your account on:",  
  "element_type": "text",  
  "label": "Log in using your account on:",  
  "bbox": [  
    856.0,  
    534.0,  
    1064.0,  
    558.0  
  ],  
  "confidence": 0.729,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 1,  
  "last_seen_index": 18,  
  "first_seen_at": "2026-05-02T06:54:23.082460+00:00",  
  "last_seen_at": "2026-05-02T07:09:07.409295+00:00",  
  "seen_count": 3,  
  "miss_count": 2,  
  "stale": true,
```

```
    "cached": true
  },
  "button:slit login": {
    "element_key": "button:slit login",
    "element_type": "button",
    "label": "SLIIT Login",
    "bbox": [
      912.0,
      566.0,
      1008.0,
      590.0
    ],
    "confidence": 0.7614,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 1,
    "last_seen_index": 18,
    "first_seen_at": "2026-05-02T06:54:23.082460+00:00",
    "last_seen_at": "2026-05-02T07:09:07.409295+00:00",
    "seen_count": 9,
    "miss_count": 2,
    "stale": true,
    "cached": true
  },
  "text:is this your first time here?": {
    "element_key": "text:is this your first time here?",
    "element_type": "text",
    "label": "Is this your first time here?",
```



```
"bbox": [
  842.0,
  734.0,
  1074.0,
  766.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 1,
"last_seen_index": 5,
"first_seen_at": "2026-05-02T06:54:23.082460+00:00",
"last_seen_at": "2026-05-02T06:57:27.809908+00:00",
"seen_count": 3,
"miss_count": 15,
"stale": true,
"cached": true
},
"pane:instructions": {
  "element_key": "pane:instructions",
  "element_type": "pane",
  "label": "Instructions",
  "bbox": [
    714.0,
    606.0,
    1210.0,
    962.0
  ],
```

```
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 1,
"last_seen_index": 1,
"first_seen_at": "2026-05-02T06:54:23.082460+00:00",
"last_seen_at": "2026-05-02T06:54:23.082460+00:00",
"seen_count": 1,
"miss_count": 19,
"stale": true,
"cached": true
},
"toolbar:taskbar": {
  "element_key": "toolbar:taskbar",
  "element_type": "toolbar",
  "label": "Taskbar",
  "bbox": [
    160.0,
    934.0,
    1744.0,
    962.0
  ],
  "confidence": 0.5315,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
  "last_seen_index": 15,
```

```
"first_seen_at": "2026-05-02T06:54:23.082460+00:00",
"last_seen_at": "2026-05-02T07:06:29.451301+00:00",
"seen_count": 5,
"miss_count": 5,
"stale": true,
"cached": true
},
"button:start": {
  "element_key": "button:start",
  "element_type": "button",
  "label": "Start",
  "bbox": [
    881.0,
    943.0,
    905.0,
    962.0
  ],
  "confidence": 0.6885,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
  "last_seen_index": 18,
  "first_seen_at": "2026-05-02T06:54:23.082460+00:00",
  "last_seen_at": "2026-05-02T07:09:07.409295+00:00",
  "seen_count": 5,
  "miss_count": 2,
  "stale": true,
  "cached": true
}
```

```
},
"button:search": {
  "element_key": "button:search",
  "element_type": "button",
  "label": "Search",
  "bbox": [
    905.0,
    943.0,
    929.0,
    962.0
  ],
  "confidence": 0.6885,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
  "last_seen_index": 18,
  "first_seen_at": "2026-05-02T06:54:23.082460+00:00",
  "last_seen_at": "2026-05-02T07:09:07.409295+00:00",
  "seen_count": 6,
  "miss_count": 2,
  "stale": true,
  "cached": true
},
"image:sliit uni courseweb logo": {
  "element_key": "image:sliit uni courseweb logo",
  "element_type": "image",
  "label": "SLIIT UNI CourseWeb Logo",
  "bbox": [
```

```
    728.0,  
    270.0,  
    1192.0,  
    326.0  
  ],  
  "confidence": 0.98,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 2,  
  "last_seen_index": 20,  
  "first_seen_at": "2026-05-02T06:54:57.595730+00:00",  
  "last_seen_at": "2026-05-02T07:11:06.085051+00:00",  
  "seen_count": 10,  
  "miss_count": 0,  
  "stale": false,  
  "cached": false  
},  
"pane:onedrive notification": {  
  "element_key": "pane:onedrive notification",  
  "element_type": "pane",  
  "label": "OneDrive Notification",  
  "bbox": [  
    1338.0,  
    198.0,  
    1758.0,  
    414.0  
  ],  
  "confidence": 0.4,
```

```
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 2,
"last_seen_index": 2,
"first_seen_at": "2026-05-02T06:54:57.595730+00:00",
"last_seen_at": "2026-05-02T06:54:57.595730+00:00",
"seen_count": 1,
"miss_count": 18,
"stale": true,
"cached": true
},
"pane:google chrome notification": {
  "element_key": "pane:google chrome notification",
  "element_type": "pane",
  "label": "Google Chrome Notification",
  "bbox": [
    1338.0,
    366.0,
    1758.0,
    598.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 2,
  "last_seen_index": 2,
  "first_seen_at": "2026-05-02T06:54:57.595730+00:00",
```

```
    "last_seen_at": "2026-05-02T06:54:57.595730+00:00",
    "seen_count": 1,
    "miss_count": 18,
    "stale": true,
    "cached": true
  },
  "pane:calendar widget": {
    "element_key": "pane:calendar widget",
    "element_type": "pane",
    "label": "Calendar Widget",
    "bbox": [
      1338.0,
      534.0,
      1758.0,
      961.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 2,
    "last_seen_index": 2,
    "first_seen_at": "2026-05-02T06:54:57.595730+00:00",
    "last_seen_at": "2026-05-02T06:54:57.595730+00:00",
    "seen_count": 1,
    "miss_count": 18,
    "stale": true,
    "cached": true
  },
}
```

```
"text:12:24 pm": {
  "element_key": "text:12:24 pm",
  "element_type": "text",
  "label": "12:24 PM",
  "bbox": [
    1666.0,
    918.0,
    1758.0,
    961.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 2,
  "last_seen_index": 2,
  "first_seen_at": "2026-05-02T06:54:57.595730+00:00",
  "last_seen_at": "2026-05-02T06:54:57.595730+00:00",
  "seen_count": 1,
  "miss_count": 18,
  "stale": true,
  "cached": true
},
"link:forgotten password?": {
  "element_key": "link:forgotten password?",
  "element_type": "link",
  "label": "Forgotten password?",
  "bbox": [
    929.0,
```



```
    486.0,  
    1001.0,  
    502.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 3,  
  "last_seen_index": 3,  
  "first_seen_at": "2026-05-02T06:55:40.663306+00:00",  
  "last_seen_at": "2026-05-02T06:55:40.663306+00:00",  
  "seen_count": 1,  
  "miss_count": 17,  
  "stale": true,  
  "cached": true  
},  
"pane:instruction box": {  
  "element_key": "pane:instruction box",  
  "element_type": "pane",  
  "label": "Instruction Box",  
  "bbox": [  
    712.0,  
    630.0,  
    1208.0,  
    962.0  
  ],  
  "confidence": 0.4134,  
  "source": "gemini_vlm",
```

```
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 3,
"last_seen_index": 15,
"first_seen_at": "2026-05-02T06:55:40.663306+00:00",
"last_seen_at": "2026-05-02T07:06:29.451301+00:00",
"seen_count": 2,
"miss_count": 5,
"stale": true,
"cached": true
},
"button:windows start": {
  "element_key": "button:windows start",
  "element_type": "button",
  "label": "Windows Start",
  "bbox": [
    569.0,
    942.0,
    601.0,
    962.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 3,
  "last_seen_index": 3,
  "first_seen_at": "2026-05-02T06:55:40.663306+00:00",
  "last_seen_at": "2026-05-02T06:55:40.663306+00:00",
```

```
    "seen_count": 1,  
    "miss_count": 17,  
    "stale": true,  
    "cached": true  
  },  
  "button:task view": {  
    "element_key": "button:task view",  
    "element_type": "button",  
    "label": "Task View",  
    "bbox": [  
      929.0,  
      943.0,  
      961.0,  
      962.0  
    ],  
    "confidence": 0.5832,  
    "source": "gemini_vlm",  
    "first_seen_frame": null,  
    "last_seen_frame": null,  
    "first_seen_index": 3,  
    "last_seen_index": 17,  
    "first_seen_at": "2026-05-02T06:55:40.663306+00:00",  
    "last_seen_at": "2026-05-02T07:08:17.240100+00:00",  
    "seen_count": 3,  
    "miss_count": 3,  
    "stale": true,  
    "cached": true  
  },  
  "button:file explorer": {
```

```
"element_key": "button:file explorer",
"element_type": "button",
"label": "File Explorer",
"bbox": [
  977.0,
  943.0,
  1009.0,
  962.0
],
"confidence": 0.6197,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 3,
"last_seen_index": 17,
"first_seen_at": "2026-05-02T06:55:40.663306+00:00",
"last_seen_at": "2026-05-02T07:08:17.240100+00:00",
"seen_count": 3,
"miss_count": 3,
"stale": true,
"cached": true
},
"button:google chrome": {
  "element_key": "button:google chrome",
  "element_type": "button",
  "label": "Google Chrome",
  "bbox": [
    936.0,
    942.0,
```

```
    976.0,
    962.0
],
"confidence": 0.6885,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 3,
"last_seen_index": 18,
"first_seen_at": "2026-05-02T06:55:40.663306+00:00",
"last_seen_at": "2026-05-02T07:09:07.409295+00:00",
"seen_count": 2,
"miss_count": 2,
"stale": true,
"cached": true
},
"button:spotify": {
  "element_key": "button:spotify",
  "element_type": "button",
  "label": "Spotify",
  "bbox": [
    1232.0,
    942.0,
    1264.0,
    962.0
  ],
  "confidence": 0.6885,
  "source": "gemini_vlm",
  "first_seen_frame": null,
```

```
"last_seen_frame": null,
"first_seen_index": 3,
"last_seen_index": 18,
"first_seen_at": "2026-05-02T06:55:40.663306+00:00",
"last_seen_at": "2026-05-02T07:09:07.409295+00:00",
"seen_count": 3,
"miss_count": 2,
"stale": true,
"cached": true
},
"text:log in using your account on": {
  "element_key": "text:log in using your account on",
  "element_type": "text",
  "label": "Log in using your account on",
  "bbox": [
    866.0,
    535.0,
    1058.0,
    559.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 4,
  "last_seen_index": 4,
  "first_seen_at": "2026-05-02T06:56:39.079159+00:00",
  "last_seen_at": "2026-05-02T06:56:39.079159+00:00",
  "seen_count": 1,
```

```
    "miss_count": 16,  
    "stale": true,  
    "cached": true  
  },  
  "pane:spotify": {  
    "element_key": "pane:spotify",  
    "element_type": "pane",  
    "label": "Spotify",  
    "bbox": [  
      1354.0,  
      327.0,  
      1758.0,  
      559.0  
    ],  
    "confidence": 0.4,  
    "source": "gemini_vlm",  
    "first_seen_frame": null,  
    "last_seen_frame": null,  
    "first_seen_index": 4,  
    "last_seen_index": 4,  
    "first_seen_at": "2026-05-02T06:56:39.079159+00:00",  
    "last_seen_at": "2026-05-02T06:56:39.079159+00:00",  
    "seen_count": 1,  
    "miss_count": 16,  
    "stale": true,  
    "cached": true  
  },  
  "text:karma (the cycle turns)": {  
    "element_key": "text:karma (the cycle turns)",
```

```
"element_type": "text",
"label": "Karma (The Cycle Turns)",
"bbox": [
    1362.0,
    542.0,
    1634.0,
    582.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 4,
"last_seen_index": 5,
"first_seen_at": "2026-05-02T06:56:39.079159+00:00",
"last_seen_at": "2026-05-02T06:57:27.809908+00:00",
"seen_count": 2,
"miss_count": 15,
"stale": true,
"cached": true
},
"text:sumedh k": {
    "element_key": "text:sumedh k",
    "element_type": "text",
    "label": "Sumedh K",
    "bbox": [
        1370.0,
        574.0,
        1490.0,
```



```
598.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 4,
"last_seen_index": 5,
"first_seen_at": "2026-05-02T06:56:39.079159+00:00",
"last_seen_at": "2026-05-02T06:57:27.809908+00:00",
"seen_count": 2,
"miss_count": 15,
"stale": true,
"cached": true
},
"pane:login form": {
  "element_key": "pane:login form",
  "element_type": "pane",
  "label": "Login Form",
  "bbox": [
    666.0,
    198.0,
    1258.0,
    962.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
```

```
"first_seen_index": 5,  
"last_seen_index": 5,  
"first_seen_at": "2026-05-02T06:57:27.809908+00:00",  
"last_seen_at": "2026-05-02T06:57:27.809908+00:00",  
"seen_count": 1,  
"miss_count": 15,  
"stale": true,  
"cached": true  
},  
"pane:spotify widget": {  
  "element_key": "pane:spotify widget",  
  "element_type": "pane",  
  "label": "Spotify Widget",  
  "bbox": [  
    1346.0,  
    406.0,  
    1758.0,  
    790.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 5,  
  "last_seen_index": 5,  
  "first_seen_at": "2026-05-02T06:57:27.809908+00:00",  
  "last_seen_at": "2026-05-02T06:57:27.809908+00:00",  
  "seen_count": 1,  
  "miss_count": 15,
```

```
    "stale": true,
    "cached": true
  },
  "button:play/pause": {
    "element_key": "button:play/pause",
    "element_type": "button",
    "label": "Play/Pause",
    "bbox": [
      1530.0,
      654.0,
      1570.0,
      694.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 5,
    "last_seen_index": 5,
    "first_seen_at": "2026-05-02T06:57:27.809908+00:00",
    "last_seen_at": "2026-05-02T06:57:27.809908+00:00",
    "seen_count": 1,
    "miss_count": 15,
    "stale": true,
    "cached": true
  },
  "pane:quick settings panel": {
    "element_key": "pane:quick settings panel",
    "element_type": "pane",
```

```
"label": "Quick Settings Panel",
"bbox": [
  1298.0,
  734.0,
  1758.0,
  950.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 5,
"last_seen_index": 5,
"first_seen_at": "2026-05-02T06:57:27.809908+00:00",
"last_seen_at": "2026-05-02T06:57:27.809908+00:00",
"seen_count": 1,
"miss_count": 15,
"stale": true,
"cached": true
},
"button:slt-fiber-7e76": {
  "element_key": "button:slt-fiber-7e76",
  "element_type": "button",
  "label": "SLT-Fiber-7E76",
  "bbox": [
    1362.0,
    782.0,
    1506.0,
    886.0
  ]
}
```

```
],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 5,
  "last_seen_index": 5,
  "first_seen_at": "2026-05-02T06:57:27.809908+00:00",
  "last_seen_at": "2026-05-02T06:57:27.809908+00:00",
  "seen_count": 1,
  "miss_count": 15,
  "stale": true,
  "cached": true
},
"button:bluetooth": {
  "element_key": "button:bluetooth",
  "element_type": "button",
  "label": "Bluetooth",
  "bbox": [
    1482.0,
    782.0,
    1634.0,
    878.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 5,
```

```
"last_seen_index": 5,
"first_seen_at": "2026-05-02T06:57:27.809908+00:00",
"last_seen_at": "2026-05-02T06:57:27.809908+00:00",
"seen_count": 1,
"miss_count": 15,
"stale": true,
"cached": true
},
"button:airplane mode": {
  "element_key": "button:airplane mode",
  "element_type": "button",
  "label": "Airplane mode",
  "bbox": [
    1610.0,
    782.0,
    1730.0,
    886.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 5,
  "last_seen_index": 5,
  "first_seen_at": "2026-05-02T06:57:27.809908+00:00",
  "last_seen_at": "2026-05-02T06:57:27.809908+00:00",
  "seen_count": 1,
  "miss_count": 15,
  "stale": true,
```

```
    "cached": true
  },
  "slider:brightness": {
    "element_key": "slider:brightness",
    "element_type": "slider",
    "label": "Brightness",
    "bbox": [
      1384.0,
      961.0,
      1696.0,
      962.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 5,
    "last_seen_index": 5,
    "first_seen_at": "2026-05-02T06:57:27.809908+00:00",
    "last_seen_at": "2026-05-02T06:57:27.809908+00:00",
    "seen_count": 1,
    "miss_count": 15,
    "stale": true,
    "cached": true
  },
  "slider:volume": {
    "element_key": "slider:volume",
    "element_type": "slider",
    "label": "Volume",
```

```
"bbox": [
  1384.0,
  961.0,
  1696.0,
  962.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 5,
"last_seen_index": 5,
"first_seen_at": "2026-05-02T06:57:27.809908+00:00",
"last_seen_at": "2026-05-02T06:57:27.809908+00:00",
"seen_count": 1,
"miss_count": 15,
"stale": true,
"cached": true
},
"text:93%": {
  "element_key": "text:93%",
  "element_type": "text",
  "label": "93%",
  "bbox": [
    1378.0,
    961.0,
    1421.0,
    962.0
  ],
```



```
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 5,
"last_seen_index": 5,
"first_seen_at": "2026-05-02T06:57:27.809908+00:00",
"last_seen_at": "2026-05-02T06:57:27.809908+00:00",
"seen_count": 1,
"miss_count": 15,
"stale": true,
"cached": true
},
"text:12:27 pm": {
  "element_key": "text:12:27 pm",
  "element_type": "text",
  "label": "12:27 PM",
  "bbox": [
    1666.0,
    926.0,
    1746.0,
    962.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 5,
  "last_seen_index": 5,
```

```
"first_seen_at": "2026-05-02T06:57:27.809908+00:00",
"last_seen_at": "2026-05-02T06:57:27.809908+00:00",
"seen_count": 1,
"miss_count": 15,
"stale": true,
"cached": true
},
"text:click on sliit login to proceed": {
  "element_key": "text:click on sliit login to proceed",
  "element_type": "text",
  "label": "Click on SLIIT Login to proceed",
  "bbox": [
    770.0,
    846.0,
    1154.0,
    886.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 5,
  "last_seen_index": 5,
  "first_seen_at": "2026-05-02T06:57:27.809908+00:00",
  "last_seen_at": "2026-05-02T06:57:27.809908+00:00",
  "seen_count": 1,
  "miss_count": 15,
  "stale": true,
  "cached": true
}
```

```
},
"text:sliit courseweb": {
  "element_key": "text:sliit courseweb",
  "element_type": "text",
  "label": "SLIIT CourseWeb",
  "bbox": [
    770.0,
    326.0,
    1154.0,
    382.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 6,
  "last_seen_index": 6,
  "first_seen_at": "2026-05-02T06:58:19.648928+00:00",
  "last_seen_at": "2026-05-02T06:58:19.648928+00:00",
  "seen_count": 1,
  "miss_count": 14,
  "stale": true,
  "cached": true
},
"label:log in using your account on:": {
  "element_key": "label:log in using your account on:",
  "element_type": "label",
  "label": "Log in using your account on:",
  "bbox": [
```

```
      864.0,
      542.0,
      1064.0,
      574.0
    ],
    "confidence": 0.4724,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 7,
    "last_seen_index": 15,
    "first_seen_at": "2026-05-02T06:59:07.402959+00:00",
    "last_seen_at": "2026-05-02T07:06:29.451301+00:00",
    "seen_count": 2,
    "miss_count": 5,
    "stale": true,
    "cached": true
  },
  "label:is this your first time here?": {
    "element_key": "label:is this your first time here?",
    "element_type": "label",
    "label": "Is this your first time here?",
    "bbox": [
      840.0,
      638.0,
      1088.0,
      662.0
    ],
    "confidence": 0.4724,
```

```
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 7,
"last_seen_index": 15,
"first_seen_at": "2026-05-02T06:59:07.402959+00:00",
"last_seen_at": "2026-05-02T07:06:29.451301+00:00",
"seen_count": 2,
"miss_count": 5,
"stale": true,
"cached": true
},
"text:instructional text": {
  "element_key": "text:instructional text",
  "element_type": "text",
  "label": "Instructional text",
  "bbox": [
    714.0,
    654.0,
    1210.0,
    962.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 7,
  "last_seen_index": 7,
  "first_seen_at": "2026-05-02T06:59:07.402959+00:00",
```

```
    "last_seen_at": "2026-05-02T06:59:07.402959+00:00",
    "seen_count": 1,
    "miss_count": 13,
    "stale": true,
    "cached": true
  },
  "text:system time": {
    "element_key": "text:system time",
    "element_type": "text",
    "label": "System Time",
    "bbox": [
      1650.0,
      942.0,
      1754.0,
      962.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 7,
    "last_seen_index": 7,
    "first_seen_at": "2026-05-02T06:59:07.402959+00:00",
    "last_seen_at": "2026-05-02T06:59:07.402959+00:00",
    "seen_count": 1,
    "miss_count": 13,
    "stale": true,
    "cached": true
  },
}
```

```
"heading:sliit uni | courseweb": {
  "element_key": "heading:sliit uni | courseweb",
  "element_type": "heading",
  "label": "SLIIT UNI | CourseWeb",
  "bbox": [
    729.0,
    270.0,
    1201.0,
    358.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 8,
  "last_seen_index": 8,
  "first_seen_at": "2026-05-02T07:00:05.444567+00:00",
  "last_seen_at": "2026-05-02T07:00:05.444567+00:00",
  "seen_count": 1,
  "miss_count": 12,
  "stale": true,
  "cached": true
},
"tab:frontend": {
  "element_key": "tab:frontend",
  "element_type": "tab",
  "label": "frontend",
  "bbox": [
    154.0,
```

```
    110.0,  
    394.0,  
    166.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 11,  
  "last_seen_index": 11,  
  "first_seen_at": "2026-05-02T07:03:12.856846+00:00",  
  "last_seen_at": "2026-05-02T07:03:12.856846+00:00",  
  "seen_count": 1,  
  "miss_count": 9,  
  "stale": true,  
  "cached": true  
},  
"button:microsoft edge": {  
  "element_key": "button:microsoft edge",  
  "element_type": "button",  
  "label": "Microsoft Edge",  
  "bbox": [  
    1001.0,  
    943.0,  
    1033.0,  
    962.0  
  ],  
  "confidence": 0.6197,  
  "source": "gemini_vlm",
```



```
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 12,
"last_seen_index": 17,
"first_seen_at": "2026-05-02T07:03:47.787802+00:00",
"last_seen_at": "2026-05-02T07:08:17.240100+00:00",
"seen_count": 2,
"miss_count": 3,
"stale": true,
"cached": true
},
"button:microsoft store": {
  "element_key": "button:microsoft store",
  "element_type": "button",
  "label": "Microsoft Store",
  "bbox": [
    1025.0,
    943.0,
    1057.0,
    962.0
  ],
  "confidence": 0.6197,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 12,
  "last_seen_index": 17,
  "first_seen_at": "2026-05-02T07:03:47.787802+00:00",
  "last_seen_at": "2026-05-02T07:08:17.240100+00:00",
```

```
    "seen_count": 2,  
    "miss_count": 3,  
    "stale": true,  
    "cached": true  
  },  
  "button:mail": {  
    "element_key": "button:mail",  
    "element_type": "button",  
    "label": "Mail",  
    "bbox": [  
      928.0,  
      942.0,  
      968.0,  
      962.0  
    ],  
    "confidence": 0.4,  
    "source": "gemini_vlm",  
    "first_seen_frame": null,  
    "last_seen_frame": null,  
    "first_seen_index": 12,  
    "last_seen_index": 12,  
    "first_seen_at": "2026-05-02T07:03:47.787802+00:00",  
    "last_seen_at": "2026-05-02T07:03:47.787802+00:00",  
    "seen_count": 1,  
    "miss_count": 8,  
    "stale": true,  
    "cached": true  
  },  
  "button:visual studio code": {
```

```
"element_key": "button:visual studio code",
"element_type": "button",
"label": "Visual Studio Code",
"bbox": [
    1184.0,
    942.0,
    1224.0,
    962.0
],
"confidence": 0.6885,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 12,
"last_seen_index": 18,
"first_seen_at": "2026-05-02T07:03:47.787802+00:00",
"last_seen_at": "2026-05-02T07:09:07.409295+00:00",
"seen_count": 2,
"miss_count": 2,
"stale": true,
"cached": true
},
"button:discord": {
    "element_key": "button:discord",
    "element_type": "button",
    "label": "Discord",
    "bbox": [
        1024.0,
        942.0,
```

```
    1064.0,  
    962.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 12,  
  "last_seen_index": 12,  
  "first_seen_at": "2026-05-02T07:03:47.787802+00:00",  
  "last_seen_at": "2026-05-02T07:03:47.787802+00:00",  
  "seen_count": 1,  
  "miss_count": 8,  
  "stale": true,  
  "cached": true  
},  
"button:steam": {  
  "element_key": "button:steam",  
  "element_type": "button",  
  "label": "Steam",  
  "bbox": [  
    1056.0,  
    942.0,  
    1096.0,  
    962.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,
```

```
"last_seen_frame": null,
"first_seen_index": 12,
"last_seen_index": 12,
"first_seen_at": "2026-05-02T07:03:47.787802+00:00",
"last_seen_at": "2026-05-02T07:03:47.787802+00:00",
"seen_count": 1,
"miss_count": 8,
"stale": true,
"cached": true
},
"button:system tray": {
  "element_key": "button:system tray",
  "element_type": "button",
  "label": "System Tray",
  "bbox": [
    1088.0,
    942.0,
    1120.0,
    962.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 12,
  "last_seen_index": 12,
  "first_seen_at": "2026-05-02T07:03:47.787802+00:00",
  "last_seen_at": "2026-05-02T07:03:47.787802+00:00",
  "seen_count": 1,
```

```
    "miss_count": 8,  
    "stale": true,  
    "cached": true  
  },  
  "text:clock": {  
    "element_key": "text:clock",  
    "element_type": "text",  
    "label": "Clock",  
    "bbox": [  
      1648.0,  
      934.0,  
      1728.0,  
      962.0  
    ],  
    "confidence": 0.4,  
    "source": "gemini_vlm",  
    "first_seen_frame": null,  
    "last_seen_frame": null,  
    "first_seen_index": 12,  
    "last_seen_index": 12,  
    "first_seen_at": "2026-05-02T07:03:47.787802+00:00",  
    "last_seen_at": "2026-05-02T07:03:47.787802+00:00",  
    "seen_count": 1,  
    "miss_count": 8,  
    "stale": true,  
    "cached": true  
  },  
  "scrollbar:scrollbar": {  
    "element_key": "scrollbar:scrollbar",
```

```
"element_type": "scrollbar",
"label": "Scrollbar",
"bbox": [
    1720.0,
    198.0,
    1744.0,
    782.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 12,
"last_seen_index": 12,
"first_seen_at": "2026-05-02T07:03:47.787802+00:00",
"last_seen_at": "2026-05-02T07:03:47.787802+00:00",
"seen_count": 1,
"miss_count": 8,
"stale": true,
"cached": true
},
"heading:sliit uni": {
    "element_key": "heading:sliit uni",
    "element_type": "heading",
    "label": "SLIIT UNI",
    "bbox": [
        769.0,
        278.0,
        937.0,
```

```
    310.0
  ],
  "confidence": 0.4783,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 14,
  "last_seen_index": 14,
  "first_seen_at": "2026-05-02T07:05:44.869252+00:00",
  "last_seen_at": "2026-05-02T07:05:44.869252+00:00",
  "seen_count": 1,
  "miss_count": 6,
  "stale": true,
  "cached": true
},
"heading:is this your first time here?": {
  "element_key": "heading:is this your first time here?",
  "element_type": "heading",
  "label": "Is this your first time here?",
  "bbox": [
    864.0,
    630.0,
    1056.0,
    654.0
  ],
  "confidence": 0.729,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
```



```
"first_seen_index": 17,
"last_seen_index": 18,
"first_seen_at": "2026-05-02T07:08:17.240100+00:00",
"last_seen_at": "2026-05-02T07:09:07.409295+00:00",
"seen_count": 2,
"miss_count": 2,
"stale": true,
"cached": true
},
"text:instructional text block": {
  "element_key": "text:instructional text block",
  "element_type": "text",
  "label": "Instructional text block",
  "bbox": [
    713.0,
    607.0,
    1209.0,
    962.0
  ],
  "confidence": 0.5468,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 17,
  "last_seen_index": 17,
  "first_seen_at": "2026-05-02T07:08:17.240100+00:00",
  "last_seen_at": "2026-05-02T07:08:17.240100+00:00",
  "seen_count": 1,
  "miss_count": 3,
```

```
    "stale": true,
    "cached": true
  },
  "button:chat": {
    "element_key": "button:chat",
    "element_type": "button",
    "label": "Chat",
    "bbox": [
      953.0,
      943.0,
      985.0,
      962.0
    ],
    "confidence": 0.5832,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 17,
    "last_seen_index": 17,
    "first_seen_at": "2026-05-02T07:08:17.240100+00:00",
    "last_seen_at": "2026-05-02T07:08:17.240100+00:00",
    "seen_count": 1,
    "miss_count": 3,
    "stale": true,
    "cached": true
  },
  "text:clock and date": {
    "element_key": "text:clock and date",
    "element_type": "text",
```

```
"label": "Clock and Date",
"bbox": [
  1657.0,
  943.0,
  1721.0,
  962.0
],
"confidence": 0.6561,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 17,
"last_seen_index": 17,
"first_seen_at": "2026-05-02T07:08:17.240100+00:00",
"last_seen_at": "2026-05-02T07:08:17.240100+00:00",
"seen_count": 1,
"miss_count": 3,
"stale": true,
"cached": true
},
"pane:login instructions": {
  "element_key": "pane:login instructions",
  "element_type": "pane",
  "label": "Login instructions",
  "bbox": [
    712.0,
    606.0,
    1208.0,
    962.0
  ]
}
```

```
],  
  "confidence": 0.6885,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 18,  
  "last_seen_index": 18,  
  "first_seen_at": "2026-05-02T07:09:07.409295+00:00",  
  "last_seen_at": "2026-05-02T07:09:07.409295+00:00",  
  "seen_count": 1,  
  "miss_count": 2,  
  "stale": true,  
  "cached": true  
},  
"taskbar:windows taskbar": {  
  "element_key": "taskbar:windows taskbar",  
  "element_type": "taskbar",  
  "label": "Windows Taskbar",  
  "bbox": [  
    176.0,  
    942.0,  
    1744.0,  
    962.0  
  ],  
  "confidence": 0.8019,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 18,
```

```
"last_seen_index": 18,  
"first_seen_at": "2026-05-02T07:09:07.409295+00:00",  
"last_seen_at": "2026-05-02T07:09:07.409295+00:00",  
"seen_count": 1,  
"miss_count": 2,  
"stale": true,  
"cached": true  
},  
"icon:wi-fi": {  
  "element_key": "icon:wi-fi",  
  "element_type": "icon",  
  "label": "Wi-Fi",  
  "bbox": [  
    1592.0,  
    942.0,  
    1616.0,  
    962.0  
  ],  
  "confidence": 0.6075,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 18,  
  "last_seen_index": 18,  
  "first_seen_at": "2026-05-02T07:09:07.409295+00:00",  
  "last_seen_at": "2026-05-02T07:09:07.409295+00:00",  
  "seen_count": 1,  
  "miss_count": 2,  
  "stale": true,
```

```
    "cached": true
  },
  "icon:battery": {
    "element_key": "icon:battery",
    "element_type": "icon",
    "label": "Battery",
    "bbox": [
      1640.0,
      942.0,
      1672.0,
      962.0
    ],
    "confidence": 0.6075,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 18,
    "last_seen_index": 18,
    "first_seen_at": "2026-05-02T07:09:07.409295+00:00",
    "last_seen_at": "2026-05-02T07:09:07.409295+00:00",
    "seen_count": 1,
    "miss_count": 2,
    "stale": true,
    "cached": true
  },
  "text:12:38 pm": {
    "element_key": "text:12:38 pm",
    "element_type": "text",
    "label": "12:38 PM",
```

```
"bbox": [
  1648.0,
  942.0,
  1744.0,
  962.0
],
"confidence": 0.7695,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 18,
"last_seen_index": 18,
"first_seen_at": "2026-05-02T07:09:07.409295+00:00",
"last_seen_at": "2026-05-02T07:09:07.409295+00:00",
"seen_count": 1,
"miss_count": 2,
"stale": true,
"cached": true
}
},
"current_elements": [
  {
    "id": "taskbar",
    "type": "pane",
    "label": "Taskbar",
    "description": "Windows taskbar at the bottom of the screen",
    "state": "visible",
    "dx": 802,
    "dy": 756,
```

```
"frame_dx": 964,  
"frame_dy": 955,  
"confidence": 0.95,  
"source": "gemini_vlm",  
"frame_bbox": [  
    170,  
    935,  
    1758,  
    976  
],  
"session_memory_key": "pane:taskbar",  
"cached": false  
},  
{  
    "id": "weather_widget",  
    "type": "label",  
    "label": "Weather Widget",  
    "description": "Weather and temperature display in the taskbar",  
    "state": "visible",  
    "dx": 64,  
    "dy": 764,  
    "frame_dx": 226,  
    "frame_dy": 963,  
    "confidence": 0.85,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        178,  
        951,  
        274,
```



```
    976
  ],
  "session_memory_key": "label:weather widget",
  "cached": false
},
{
  "id": "start_button",
  "type": "button",
  "label": "Start Button",
  "description": "Windows Start menu button",
  "state": "visible",
  "dx": 424,
  "dy": 760,
  "frame_dx": 586,
  "frame_dy": 959,
  "confidence": 0.98,
  "source": "gemini_vlm",
  "frame_bbox": [
    562,
    943,
    610,
    976
  ],
  "session_memory_key": "button:start button",
  "cached": false
},
{
  "id": "search_bar",
  "type": "input",
```

```
"label": "Search Bar",
"description": "Windows search input field in the taskbar",
"state": "visible",
"dx": 508,
"dy": 760,
"frame_dx": 670,
"frame_dy": 959,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    594,
    943,
    746,
    976
],
"session_memory_key": "input:search bar",
"cached": false
},
{
    "id": "file_explorer_icon",
    "type": "icon",
    "label": "File Explorer",
    "description": "File Explorer icon in the taskbar",
    "state": "visible",
    "dx": 608,
    "dy": 760,
    "frame_dx": 770,
    "frame_dy": 959,
    "confidence": 0.9,
```

```
"source": "gemini_vlm",
"frame_bbox": [
    746,
    943,
    794,
    976
],
"session_memory_key": "icon:file explorer",
"cached": false
},
{
    "id": "microsoft_store_icon",
    "type": "icon",
    "label": "Microsoft Store",
    "description": "Microsoft Store icon in the taskbar",
    "state": "visible",
    "dx": 648,
    "dy": 760,
    "frame_dx": 810,
    "frame_dy": 959,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        786,
        943,
        834,
        976
    ],
    "session_memory_key": "icon:microsoft store",
```

```
    "cached": false
  },
  {
    "id": "mail_icon",
    "type": "icon",
    "label": "Mail",
    "description": "Mail application icon in the taskbar",
    "state": "visible",
    "dx": 680,
    "dy": 760,
    "frame_dx": 842,
    "frame_dy": 959,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
      818,
      943,
      866,
      976
    ],
    "session_memory_key": "icon:mail",
    "cached": false
  },
  {
    "id": "xbox_icon",
    "type": "icon",
    "label": "Xbox",
    "description": "Xbox application icon in the taskbar",
    "state": "visible",
```

```
"dx": 720,  
"dy": 760,  
"frame_dx": 882,  
"frame_dy": 959,  
"confidence": 0.9,  
"source": "gemini_vlm",  
"frame_bbox": [  
    858,  
    943,  
    906,  
    976  
],  
"session_memory_key": "icon:xbox",  
"cached": false  
},  
{  
    "id": "chrome_icon",  
    "type": "icon",  
    "label": "Google Chrome",  
    "description": "Google Chrome browser icon in the taskbar",  
    "state": "visible",  
    "dx": 760,  
    "dy": 760,  
    "frame_dx": 922,  
    "frame_dy": 959,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        898,
```

```
    943,  
    946,  
    976  
  ],  
  "session_memory_key": "icon:google chrome",  
  "cached": false  
},  
{  
  "id": "vscode_icon",  
  "type": "icon",  
  "label": "Visual Studio Code",  
  "description": "Visual Studio Code icon in the taskbar",  
  "state": "visible",  
  "dx": 792,  
  "dy": 760,  
  "frame_dx": 954,  
  "frame_dy": 959,  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    930,  
    943,  
    978,  
    976  
  ],  
  "session_memory_key": "icon:visual studio code",  
  "cached": false  
},  
{
```

```
"id": "spotify_icon",
"type": "icon",
"label": "Spotify",
"description": "Spotify application icon in the taskbar",
"state": "visible",
"dx": 832,
"dy": 760,
"frame_dx": 994,
"frame_dy": 959,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    970,
    943,
    1018,
    976
],
"session_memory_key": "icon:spotify",
"cached": false
},
{
    "id": "google_icon",
    "type": "icon",
    "label": "Google",
    "description": "Google application icon in the taskbar",
    "state": "visible",
    "dx": 872,
    "dy": 760,
    "frame_dx": 1034,
```

```
"frame_dy": 959,  
"confidence": 0.9,  
"source": "gemini_vlm",  
"frame_bbox": [  
    1010,  
    943,  
    1058,  
    976  
],  
"session_memory_key": "icon:google",  
"cached": false  
},  
{  
    "id": "autocad_icon",  
    "type": "icon",  
    "label": "AutoCAD",  
    "description": "AutoCAD application icon in the taskbar",  
    "state": "visible",  
    "dx": 908,  
    "dy": 760,  
    "frame_dx": 1070,  
    "frame_dy": 959,  
    "confidence": 0.8,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        1050,  
        943,  
        1090,  
        976
```



```
    ],  
    "session_memory_key": "icon:autocad",  
    "cached": false  
  },  
  {  
    "id": "system_tray",  
    "type": "pane",  
    "label": "System Tray",  
    "description": "System tray area with various status icons",  
    "state": "visible",  
    "dx": 1430,  
    "dy": 760,  
    "frame_dx": 1592,  
    "frame_dy": 959,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
      1426,  
      943,  
      1758,  
      976  
    ],  
    "session_memory_key": "pane:system tray",  
    "cached": false  
  },  
  {  
    "id": "clock_date",  
    "type": "label",  
    "label": "Clock and Date",
```

```
"description": "Current time and date display in the taskbar",
"state": "visible",
"dx": 1538,
"dy": 760,
"frame_dx": 1700,
"frame_dy": 959,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    1642,
    943,
    1758,
    976
],
"session_memory_key": "label:clock and date",
"cached": false
},
{
    "id": "login_container",
    "type": "pane",
    "label": "Login Container",
    "description": "Main white container for the login form",
    "state": "visible",
    "dx": 792,
    "dy": 388,
    "frame_dx": 954,
    "frame_dy": 587,
    "confidence": 0.95,
    "source": "gemini_vlm",
```

```
"frame_bbox": [
    658,
    199,
    1250,
    976
],
"session_memory_key": "pane:login container",
"cached": false
},
{
    "id": "sliit_logo",
    "type": "icon",
    "label": "SLIIT Logo",
    "description": "Logo of SLIIT University",
    "state": "visible",
    "dx": 688,
    "dy": 156,
    "frame_dx": 850,
    "frame_dy": 355,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        770,
        327,
        930,
        383
    ],
    "session_memory_key": "icon:sliit logo",
    "cached": false
}
```

```
},  
{  
  "id": "courseweb_heading",  
  "type": "heading",  
  "label": "CourseWeb",  
  "description": "Heading text for the CourseWeb application",  
  "state": "visible",  
  "dx": 892,  
  "dy": 156,  
  "frame_dx": 1054,  
  "frame_dy": 355,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    954,  
    327,  
    1154,  
    383  
  ],  
  "session_memory_key": "heading:courseweb",  
  "cached": false  
},  
{  
  "id": "login_button",  
  "type": "button",  
  "label": "Login Button",  
  "description": "Yellow login button in the center of the page",  
  "state": "visible",  
  "dx": 796,
```

```
"dy": 404,  
  "frame_dx": 958,  
  "frame_dy": 603,  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    882,  
    575,  
    1034,  
    631  
  ],  
  "session_memory_key": "button:login button",  
  "cached": false  
},  
{  
  "id": "instruction_text",  
  "type": "label",  
  "label": "Instruction Text",  
  "description": "Text instructing the user to click on SLIIT Login",  
  "state": "visible",  
  "dx": 724,  
  "dy": 696,  
  "frame_dx": 886,  
  "frame_dy": 895,  
  "confidence": 0.85,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    802,  
    879,
```

```
    970,  
    911  
  ],  
  "session_memory_key": "label:instruction text",  
  "cached": false  
},  
{  
  "id": "note_text",  
  "type": "label",  
  "label": "Note Text",  
  "description": "Informational note about Microsoft 0365 validation",  
  "state": "visible",  
  "dx": 796,  
  "dy": 736,  
  "frame_dx": 958,  
  "frame_dy": 935,  
  "confidence": 0.85,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    754,  
    895,  
    1162,  
    976  
  ],  
  "session_memory_key": "label:note text",  
  "cached": false  
}  
],  
"resolved_elements": [  

```

```
{
  "id": "taskbar",
  "type": "pane",
  "label": "Taskbar",
  "description": "Windows taskbar at the bottom of the screen",
  "state": "visible",
  "dx": 802,
  "dy": 756,
  "frame_dx": 964,
  "frame_dy": 955,
  "confidence": 0.95,
  "source": "gemini_vlm",
  "frame_bbox": [
    170,
    935,
    1758,
    976
  ],
  "session_memory_key": "pane:taskbar",
  "cached": false
},
{
  "id": "weather_widget",
  "type": "label",
  "label": "Weather Widget",
  "description": "Weather and temperature display in the taskbar",
  "state": "visible",
  "dx": 64,
  "dy": 764,
```

```
"frame_dx": 226,  
"frame_dy": 963,  
"confidence": 0.85,  
"source": "gemini_vlm",  
"frame_bbox": [  
    178,  
    951,  
    274,  
    976  
],  
"session_memory_key": "label:weather widget",  
"cached": false  
},  
{  
    "id": "start_button",  
    "type": "button",  
    "label": "Start Button",  
    "description": "Windows Start menu button",  
    "state": "visible",  
    "dx": 424,  
    "dy": 760,  
    "frame_dx": 586,  
    "frame_dy": 959,  
    "confidence": 0.98,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        562,  
        943,  
        610,
```



```
    976
  ],
  "session_memory_key": "button:start button",
  "cached": false
},
{
  "id": "search_bar",
  "type": "input",
  "label": "Search Bar",
  "description": "Windows search input field in the taskbar",
  "state": "visible",
  "dx": 508,
  "dy": 760,
  "frame_dx": 670,
  "frame_dy": 959,
  "confidence": 0.95,
  "source": "gemini_vlm",
  "frame_bbox": [
    594,
    943,
    746,
    976
  ],
  "session_memory_key": "input:search bar",
  "cached": false
},
{
  "id": "file_explorer_icon",
  "type": "icon",
```

```
"label": "File Explorer",
"description": "File Explorer icon in the taskbar",
"state": "visible",
"dx": 608,
"dy": 760,
"frame_dx": 770,
"frame_dy": 959,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    746,
    943,
    794,
    976
],
"session_memory_key": "icon:file explorer",
"cached": false
},
{
    "id": "microsoft_store_icon",
    "type": "icon",
    "label": "Microsoft Store",
    "description": "Microsoft Store icon in the taskbar",
    "state": "visible",
    "dx": 648,
    "dy": 760,
    "frame_dx": 810,
    "frame_dy": 959,
    "confidence": 0.9,
```

```
"source": "gemini_vlm",
"frame_bbox": [
    786,
    943,
    834,
    976
],
"session_memory_key": "icon:microsoft store",
"cached": false
},
{
    "id": "mail_icon",
    "type": "icon",
    "label": "Mail",
    "description": "Mail application icon in the taskbar",
    "state": "visible",
    "dx": 680,
    "dy": 760,
    "frame_dx": 842,
    "frame_dy": 959,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        818,
        943,
        866,
        976
    ],
    "session_memory_key": "icon:mail",
```

```
    "cached": false
  },
  {
    "id": "xbox_icon",
    "type": "icon",
    "label": "Xbox",
    "description": "Xbox application icon in the taskbar",
    "state": "visible",
    "dx": 720,
    "dy": 760,
    "frame_dx": 882,
    "frame_dy": 959,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
      858,
      943,
      906,
      976
    ],
    "session_memory_key": "icon:xbox",
    "cached": false
  },
  {
    "id": "chrome_icon",
    "type": "icon",
    "label": "Google Chrome",
    "description": "Google Chrome browser icon in the taskbar",
    "state": "visible",
```

```
"dx": 760,
"dy": 760,
"frame_dx": 922,
"frame_dy": 959,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    898,
    943,
    946,
    976
],
"session_memory_key": "icon:google chrome",
"cached": false
},
{
    "id": "vscode_icon",
    "type": "icon",
    "label": "Visual Studio Code",
    "description": "Visual Studio Code icon in the taskbar",
    "state": "visible",
    "dx": 792,
    "dy": 760,
    "frame_dx": 954,
    "frame_dy": 959,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        930,
```

```
    943,  
    978,  
    976  
  ],  
  "session_memory_key": "icon:visual studio code",  
  "cached": false  
},  
{  
  "id": "spotify_icon",  
  "type": "icon",  
  "label": "Spotify",  
  "description": "Spotify application icon in the taskbar",  
  "state": "visible",  
  "dx": 832,  
  "dy": 760,  
  "frame_dx": 994,  
  "frame_dy": 959,  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    970,  
    943,  
    1018,  
    976  
  ],  
  "session_memory_key": "icon:spotify",  
  "cached": false  
},  
{
```

```
"id": "google_icon",
"type": "icon",
"label": "Google",
"description": "Google application icon in the taskbar",
"state": "visible",
"dx": 872,
"dy": 760,
"frame_dx": 1034,
"frame_dy": 959,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    1010,
    943,
    1058,
    976
],
"session_memory_key": "icon:google",
"cached": false
},
{
    "id": "autocad_icon",
    "type": "icon",
    "label": "AutoCAD",
    "description": "AutoCAD application icon in the taskbar",
    "state": "visible",
    "dx": 908,
    "dy": 760,
    "frame_dx": 1070,
```

```
"frame_dy": 959,
"confidence": 0.8,
"source": "gemini_vlm",
"frame_bbox": [
    1050,
    943,
    1090,
    976
],
"session_memory_key": "icon:autocad",
"cached": false
},
{
    "id": "system_tray",
    "type": "pane",
    "label": "System Tray",
    "description": "System tray area with various status icons",
    "state": "visible",
    "dx": 1430,
    "dy": 760,
    "frame_dx": 1592,
    "frame_dy": 959,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        1426,
        943,
        1758,
        976
    ]
}
```



```
    ],
    "session_memory_key": "pane:system tray",
    "cached": false
  },
  {
    "id": "clock_date",
    "type": "label",
    "label": "Clock and Date",
    "description": "Current time and date display in the taskbar",
    "state": "visible",
    "dx": 1538,
    "dy": 760,
    "frame_dx": 1700,
    "frame_dy": 959,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
      1642,
      943,
      1758,
      976
    ],
    "session_memory_key": "label:clock and date",
    "cached": false
  },
  {
    "id": "login_container",
    "type": "pane",
    "label": "Login Container",
```

```
"description": "Main white container for the login form",
"state": "visible",
"dx": 792,
"dy": 388,
"frame_dx": 954,
"frame_dy": 587,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    658,
    199,
    1250,
    976
],
"session_memory_key": "pane:login container",
"cached": false
},
{
    "id": "sliit_logo",
    "type": "icon",
    "label": "SLIIT Logo",
    "description": "Logo of SLIIT University",
    "state": "visible",
    "dx": 688,
    "dy": 156,
    "frame_dx": 850,
    "frame_dy": 355,
    "confidence": 0.9,
    "source": "gemini_vlm",
```

```
"frame_bbox": [
    770,
    327,
    930,
    383
],
"session_memory_key": "icon:sliit logo",
"cached": false
},
{
    "id": "courseweb_heading",
    "type": "heading",
    "label": "CourseWeb",
    "description": "Heading text for the CourseWeb application",
    "state": "visible",
    "dx": 892,
    "dy": 156,
    "frame_dx": 1054,
    "frame_dy": 355,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        954,
        327,
        1154,
        383
    ],
    "session_memory_key": "heading:courseweb",
    "cached": false
}
```

```
},  
{  
  "id": "login_button",  
  "type": "button",  
  "label": "Login Button",  
  "description": "Yellow login button in the center of the page",  
  "state": "visible",  
  "dx": 796,  
  "dy": 404,  
  "frame_dx": 958,  
  "frame_dy": 603,  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    882,  
    575,  
    1034,  
    631  
  ],  
  "session_memory_key": "button:login button",  
  "cached": false  
},  
{  
  "id": "instruction_text",  
  "type": "label",  
  "label": "Instruction Text",  
  "description": "Text instructing the user to click on SLIIT Login",  
  "state": "visible",  
  "dx": 724,
```

```
"dy": 696,  
  "frame_dx": 886,  
  "frame_dy": 895,  
  "confidence": 0.85,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    802,  
    879,  
    970,  
    911  
  ],  
  "session_memory_key": "label:instruction text",  
  "cached": false  
},  
{  
  "id": "note_text",  
  "type": "label",  
  "label": "Note Text",  
  "description": "Informational note about Microsoft 0365 validation",  
  "state": "visible",  
  "dx": 796,  
  "dy": 736,  
  "frame_dx": 958,  
  "frame_dy": 935,  
  "confidence": 0.85,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    754,  
    895,
```

```
        1162,  
        976  
    ],  
    "session_memory_key": "label:note text",  
    "cached": false  
}  
],  
"screen_data": {  
    "image": "E:\\sliit-  
project\\ScreenAwareTaskAgent\\vision\\data\\sessions\\shot_1777704787_aa8922\\pr  
ocessing\\proc_1777704787_248e9c\\preprocessed_frames\\frame_1777704787_716419.jp  
g",  
    "image_size": {  
        "width": 1920,  
        "height": 1080  
    },  
    "screen_bbox": [  
        162,  
        199,  
        1758,  
        976  
    ],  
    "screen_size": {  
        "width": 1596,  
        "height": 777  
    },  
    "coordinate_system": "pixel",  
    "elements": [  
        {  
            "id": "taskbar",
```

```
"type": "pane",
"label": "Taskbar",
"description": "Windows taskbar at the bottom of the screen",
"state": "visible",
"dx": 802,
"dy": 756,
"frame_dx": 964,
"frame_dy": 955,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    170,
    935,
    1758,
    976
],
"session_memory_key": "pane:taskbar",
"cached": false
},
{
    "id": "weather_widget",
    "type": "label",
    "label": "Weather Widget",
    "description": "Weather and temperature display in the taskbar",
    "state": "visible",
    "dx": 64,
    "dy": 764,
    "frame_dx": 226,
    "frame_dy": 963,
```

```
"confidence": 0.85,  
"source": "gemini_vlm",  
"frame_bbox": [  
    178,  
    951,  
    274,  
    976  
],  
"session_memory_key": "label:weather widget",  
"cached": false  
},  
{  
    "id": "start_button",  
    "type": "button",  
    "label": "Start Button",  
    "description": "Windows Start menu button",  
    "state": "visible",  
    "dx": 424,  
    "dy": 760,  
    "frame_dx": 586,  
    "frame_dy": 959,  
    "confidence": 0.98,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        562,  
        943,  
        610,  
        976  
    ],
```



```
"session_memory_key": "button:start button",
"cached": false
},
{
  "id": "search_bar",
  "type": "input",
  "label": "Search Bar",
  "description": "Windows search input field in the taskbar",
  "state": "visible",
  "dx": 508,
  "dy": 760,
  "frame_dx": 670,
  "frame_dy": 959,
  "confidence": 0.95,
  "source": "gemini_vlm",
  "frame_bbox": [
    594,
    943,
    746,
    976
  ],
  "session_memory_key": "input:search bar",
  "cached": false
},
{
  "id": "file_explorer_icon",
  "type": "icon",
  "label": "File Explorer",
  "description": "File Explorer icon in the taskbar",
```

```
"state": "visible",
"dx": 608,
"dy": 760,
"frame_dx": 770,
"frame_dy": 959,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    746,
    943,
    794,
    976
],
"session_memory_key": "icon:file explorer",
"cached": false
},
{
    "id": "microsoft_store_icon",
    "type": "icon",
    "label": "Microsoft Store",
    "description": "Microsoft Store icon in the taskbar",
    "state": "visible",
    "dx": 648,
    "dy": 760,
    "frame_dx": 810,
    "frame_dy": 959,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
```

```
        786,  
        943,  
        834,  
        976  
    ],  
    "session_memory_key": "icon:microsoft store",  
    "cached": false  
},  
{  
    "id": "mail_icon",  
    "type": "icon",  
    "label": "Mail",  
    "description": "Mail application icon in the taskbar",  
    "state": "visible",  
    "dx": 680,  
    "dy": 760,  
    "frame_dx": 842,  
    "frame_dy": 959,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        818,  
        943,  
        866,  
        976  
    ],  
    "session_memory_key": "icon:mail",  
    "cached": false  
},
```

```
{
  "id": "xbox_icon",
  "type": "icon",
  "label": "Xbox",
  "description": "Xbox application icon in the taskbar",
  "state": "visible",
  "dx": 720,
  "dy": 760,
  "frame_dx": 882,
  "frame_dy": 959,
  "confidence": 0.9,
  "source": "gemini_vlm",
  "frame_bbox": [
    858,
    943,
    906,
    976
  ],
  "session_memory_key": "icon:xbox",
  "cached": false
},
{
  "id": "chrome_icon",
  "type": "icon",
  "label": "Google Chrome",
  "description": "Google Chrome browser icon in the taskbar",
  "state": "visible",
  "dx": 760,
  "dy": 760,
```

```
"frame_dx": 922,  
"frame_dy": 959,  
"confidence": 0.9,  
"source": "gemini_vlm",  
"frame_bbox": [  
    898,  
    943,  
    946,  
    976  
],  
"session_memory_key": "icon:google chrome",  
"cached": false  
},  
{  
    "id": "vscode_icon",  
    "type": "icon",  
    "label": "Visual Studio Code",  
    "description": "Visual Studio Code icon in the taskbar",  
    "state": "visible",  
    "dx": 792,  
    "dy": 760,  
    "frame_dx": 954,  
    "frame_dy": 959,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        930,  
        943,  
        978,
```

```
    976
  ],
  "session_memory_key": "icon:visual studio code",
  "cached": false
},
{
  "id": "spotify_icon",
  "type": "icon",
  "label": "Spotify",
  "description": "Spotify application icon in the taskbar",
  "state": "visible",
  "dx": 832,
  "dy": 760,
  "frame_dx": 994,
  "frame_dy": 959,
  "confidence": 0.9,
  "source": "gemini_vlm",
  "frame_bbox": [
    970,
    943,
    1018,
    976
  ],
  "session_memory_key": "icon:spotify",
  "cached": false
},
{
  "id": "google_icon",
  "type": "icon",
```

```
"label": "Google",
"description": "Google application icon in the taskbar",
"state": "visible",
"dx": 872,
"dy": 760,
"frame_dx": 1034,
"frame_dy": 959,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    1010,
    943,
    1058,
    976
],
"session_memory_key": "icon:google",
"cached": false
},
{
    "id": "autocad_icon",
    "type": "icon",
    "label": "AutoCAD",
    "description": "AutoCAD application icon in the taskbar",
    "state": "visible",
    "dx": 908,
    "dy": 760,
    "frame_dx": 1070,
    "frame_dy": 959,
    "confidence": 0.8,
```

```
"source": "gemini_vlm",
"frame_bbox": [
    1050,
    943,
    1090,
    976
],
"session_memory_key": "icon:autocad",
"cached": false
},
{
    "id": "system_tray",
    "type": "pane",
    "label": "System Tray",
    "description": "System tray area with various status icons",
    "state": "visible",
    "dx": 1430,
    "dy": 760,
    "frame_dx": 1592,
    "frame_dy": 959,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        1426,
        943,
        1758,
        976
    ],
    "session_memory_key": "pane:system tray",
```



```
    "cached": false
  },
  {
    "id": "clock_date",
    "type": "label",
    "label": "Clock and Date",
    "description": "Current time and date display in the taskbar",
    "state": "visible",
    "dx": 1538,
    "dy": 760,
    "frame_dx": 1700,
    "frame_dy": 959,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
      1642,
      943,
      1758,
      976
    ],
    "session_memory_key": "label:clock and date",
    "cached": false
  },
  {
    "id": "login_container",
    "type": "pane",
    "label": "Login Container",
    "description": "Main white container for the login form",
    "state": "visible",
```

```
"dx": 792,  
  "dy": 388,  
  "frame_dx": 954,  
  "frame_dy": 587,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    658,  
    199,  
    1250,  
    976  
  ],  
  "session_memory_key": "pane:login container",  
  "cached": false  
},  
{  
  "id": "sliit_logo",  
  "type": "icon",  
  "label": "SLIIT Logo",  
  "description": "Logo of SLIIT University",  
  "state": "visible",  
  "dx": 688,  
  "dy": 156,  
  "frame_dx": 850,  
  "frame_dy": 355,  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    770,
```

```
        327,  
        930,  
        383  
    ],  
    "session_memory_key": "icon:sliit logo",  
    "cached": false  
},  
{  
    "id": "courseweb_heading",  
    "type": "heading",  
    "label": "CourseWeb",  
    "description": "Heading text for the CourseWeb application",  
    "state": "visible",  
    "dx": 892,  
    "dy": 156,  
    "frame_dx": 1054,  
    "frame_dy": 355,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        954,  
        327,  
        1154,  
        383  
    ],  
    "session_memory_key": "heading:courseweb",  
    "cached": false  
},  
{
```

```
"id": "login_button",
"type": "button",
"label": "Login Button",
"description": "Yellow login button in the center of the page",
"state": "visible",
"dx": 796,
"dy": 404,
"frame_dx": 958,
"frame_dy": 603,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    882,
    575,
    1034,
    631
],
"session_memory_key": "button:login button",
"cached": false
},
{
    "id": "instruction_text",
    "type": "label",
    "label": "Instruction Text",
    "description": "Text instructing the user to click on SLIIT Login",
    "state": "visible",
    "dx": 724,
    "dy": 696,
    "frame_dx": 886,
```

```
"frame_dy": 895,
"confidence": 0.85,
"source": "gemini_vlm",
"frame_bbox": [
    802,
    879,
    970,
    911
],
"session_memory_key": "label:instruction text",
"cached": false
},
{
    "id": "note_text",
    "type": "label",
    "label": "Note Text",
    "description": "Informational note about Microsoft 0365 validation",
    "state": "visible",
    "dx": 796,
    "dy": 736,
    "frame_dx": 958,
    "frame_dy": 935,
    "confidence": 0.85,
    "source": "gemini_vlm",
    "frame_bbox": [
        754,
        895,
        1162,
        976
    ]
}
```

```
    ],  
    "session_memory_key": "label:note text",  
    "cached": false  
  }  
],  
"element_count": 21,  
"current_elements": [  
  {  
    "id": "taskbar",  
    "type": "pane",  
    "label": "Taskbar",  
    "description": "Windows taskbar at the bottom of the screen",  
    "state": "visible",  
    "dx": 802,  
    "dy": 756,  
    "frame_dx": 964,  
    "frame_dy": 955,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
      170,  
      935,  
      1758,  
      976  
    ],  
    "session_memory_key": "pane:taskbar",  
    "cached": false  
  },  
  {
```

```
"id": "weather_widget",
"type": "label",
"label": "Weather Widget",
"description": "Weather and temperature display in the taskbar",
"state": "visible",
"dx": 64,
"dy": 764,
"frame_dx": 226,
"frame_dy": 963,
"confidence": 0.85,
"source": "gemini_vlm",
"frame_bbox": [
    178,
    951,
    274,
    976
],
"session_memory_key": "label:weather widget",
"cached": false
},
{
    "id": "start_button",
    "type": "button",
    "label": "Start Button",
    "description": "Windows Start menu button",
    "state": "visible",
    "dx": 424,
    "dy": 760,
    "frame_dx": 586,
```

```
"frame_dy": 959,
"confidence": 0.98,
"source": "gemini_vlm",
"frame_bbox": [
    562,
    943,
    610,
    976
],
"session_memory_key": "button:start button",
"cached": false
},
{
    "id": "search_bar",
    "type": "input",
    "label": "Search Bar",
    "description": "Windows search input field in the taskbar",
    "state": "visible",
    "dx": 508,
    "dy": 760,
    "frame_dx": 670,
    "frame_dy": 959,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        594,
        943,
        746,
        976
    ]
}
```



```
    ],  
    "session_memory_key": "input:search bar",  
    "cached": false  
  },  
  {  
    "id": "file_explorer_icon",  
    "type": "icon",  
    "label": "File Explorer",  
    "description": "File Explorer icon in the taskbar",  
    "state": "visible",  
    "dx": 608,  
    "dy": 760,  
    "frame_dx": 770,  
    "frame_dy": 959,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
      746,  
      943,  
      794,  
      976  
    ],  
    "session_memory_key": "icon:file explorer",  
    "cached": false  
  },  
  {  
    "id": "microsoft_store_icon",  
    "type": "icon",  
    "label": "Microsoft Store",
```

```
    "description": "Microsoft Store icon in the taskbar",
    "state": "visible",
    "dx": 648,
    "dy": 760,
    "frame_dx": 810,
    "frame_dy": 959,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        786,
        943,
        834,
        976
    ],
    "session_memory_key": "icon:microsoft store",
    "cached": false
},
{
    "id": "mail_icon",
    "type": "icon",
    "label": "Mail",
    "description": "Mail application icon in the taskbar",
    "state": "visible",
    "dx": 680,
    "dy": 760,
    "frame_dx": 842,
    "frame_dy": 959,
    "confidence": 0.9,
    "source": "gemini_vlm",
```

```
"frame_bbox": [  
    818,  
    943,  
    866,  
    976  
],  
"session_memory_key": "icon:mail",  
"cached": false  
},  
{  
    "id": "xbox_icon",  
    "type": "icon",  
    "label": "Xbox",  
    "description": "Xbox application icon in the taskbar",  
    "state": "visible",  
    "dx": 720,  
    "dy": 760,  
    "frame_dx": 882,  
    "frame_dy": 959,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        858,  
        943,  
        906,  
        976  
    ],  
    "session_memory_key": "icon:xbox",  
    "cached": false
```

```
},  
{  
  "id": "chrome_icon",  
  "type": "icon",  
  "label": "Google Chrome",  
  "description": "Google Chrome browser icon in the taskbar",  
  "state": "visible",  
  "dx": 760,  
  "dy": 760,  
  "frame_dx": 922,  
  "frame_dy": 959,  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    898,  
    943,  
    946,  
    976  
  ],  
  "session_memory_key": "icon:google chrome",  
  "cached": false  
},  
{  
  "id": "vscode_icon",  
  "type": "icon",  
  "label": "Visual Studio Code",  
  "description": "Visual Studio Code icon in the taskbar",  
  "state": "visible",  
  "dx": 792,
```

```
"dy": 760,
"frame_dx": 954,
"frame_dy": 959,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    930,
    943,
    978,
    976
],
"session_memory_key": "icon:visual studio code",
"cached": false
},
{
    "id": "spotify_icon",
    "type": "icon",
    "label": "Spotify",
    "description": "Spotify application icon in the taskbar",
    "state": "visible",
    "dx": 832,
    "dy": 760,
    "frame_dx": 994,
    "frame_dy": 959,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        970,
        943,
```

```
        1018,  
        976  
    ],  
    "session_memory_key": "icon:spotify",  
    "cached": false  
},  
{  
    "id": "google_icon",  
    "type": "icon",  
    "label": "Google",  
    "description": "Google application icon in the taskbar",  
    "state": "visible",  
    "dx": 872,  
    "dy": 760,  
    "frame_dx": 1034,  
    "frame_dy": 959,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        1010,  
        943,  
        1058,  
        976  
    ],  
    "session_memory_key": "icon:google",  
    "cached": false  
},  
{  
    "id": "autocad_icon",
```

```
"type": "icon",
"label": "AutoCAD",
"description": "AutoCAD application icon in the taskbar",
"state": "visible",
"dx": 908,
"dy": 760,
"frame_dx": 1070,
"frame_dy": 959,
"confidence": 0.8,
"source": "gemini_vlm",
"frame_bbox": [
    1050,
    943,
    1090,
    976
],
"session_memory_key": "icon:autocad",
"cached": false
},
{
    "id": "system_tray",
    "type": "pane",
    "label": "System Tray",
    "description": "System tray area with various status icons",
    "state": "visible",
    "dx": 1430,
    "dy": 760,
    "frame_dx": 1592,
    "frame_dy": 959,
```

```
"confidence": 0.9,  
"source": "gemini_vlm",  
"frame_bbox": [  
    1426,  
    943,  
    1758,  
    976  
],  
"session_memory_key": "pane:system tray",  
"cached": false  
},  
{  
    "id": "clock_date",  
    "type": "label",  
    "label": "Clock and Date",  
    "description": "Current time and date display in the taskbar",  
    "state": "visible",  
    "dx": 1538,  
    "dy": 760,  
    "frame_dx": 1700,  
    "frame_dy": 959,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        1642,  
        943,  
        1758,  
        976  
    ],  
}
```



```
"session_memory_key": "label:clock and date",
"cached": false
},
{
  "id": "login_container",
  "type": "pane",
  "label": "Login Container",
  "description": "Main white container for the login form",
  "state": "visible",
  "dx": 792,
  "dy": 388,
  "frame_dx": 954,
  "frame_dy": 587,
  "confidence": 0.95,
  "source": "gemini_vlm",
  "frame_bbox": [
    658,
    199,
    1250,
    976
  ],
  "session_memory_key": "pane:login container",
  "cached": false
},
{
  "id": "sliit_logo",
  "type": "icon",
  "label": "SLIIT Logo",
  "description": "Logo of SLIIT University",
```

```
"state": "visible",
"dx": 688,
"dy": 156,
"frame_dx": 850,
"frame_dy": 355,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    770,
    327,
    930,
    383
],
"session_memory_key": "icon:sliit logo",
"cached": false
},
{
    "id": "courseweb_heading",
    "type": "heading",
    "label": "CourseWeb",
    "description": "Heading text for the CourseWeb application",
    "state": "visible",
    "dx": 892,
    "dy": 156,
    "frame_dx": 1054,
    "frame_dy": 355,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
```

```
    954,  
    327,  
    1154,  
    383  
  ],  
  "session_memory_key": "heading:courseweb",  
  "cached": false  
},  
{  
  "id": "login_button",  
  "type": "button",  
  "label": "Login Button",  
  "description": "Yellow login button in the center of the page",  
  "state": "visible",  
  "dx": 796,  
  "dy": 404,  
  "frame_dx": 958,  
  "frame_dy": 603,  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    882,  
    575,  
    1034,  
    631  
  ],  
  "session_memory_key": "button:login button",  
  "cached": false  
},
```

```
{
  "id": "instruction_text",
  "type": "label",
  "label": "Instruction Text",
  "description": "Text instructing the user to click on SLIIT Login",
  "state": "visible",
  "dx": 724,
  "dy": 696,
  "frame_dx": 886,
  "frame_dy": 895,
  "confidence": 0.85,
  "source": "gemini_vlm",
  "frame_bbox": [
    802,
    879,
    970,
    911
  ],
  "session_memory_key": "label:instruction text",
  "cached": false
},
{
  "id": "note_text",
  "type": "label",
  "label": "Note Text",
  "description": "Informational note about Microsoft 0365 validation",
  "state": "visible",
  "dx": 796,
  "dy": 736,
```

```
    "frame_dx": 958,  
    "frame_dy": 935,  
    "confidence": 0.85,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        754,  
        895,  
        1162,  
        976  
    ],  
    "session_memory_key": "label:note text",  
    "cached": false  
}  
],  
"resolved_elements": [  
    {  
        "id": "taskbar",  
        "type": "pane",  
        "label": "Taskbar",  
        "description": "Windows taskbar at the bottom of the screen",  
        "state": "visible",  
        "dx": 802,  
        "dy": 756,  
        "frame_dx": 964,  
        "frame_dy": 955,  
        "confidence": 0.95,  
        "source": "gemini_vlm",  
        "frame_bbox": [  
            170,
```

```
        935,  
        1758,  
        976  
    ],  
    "session_memory_key": "pane:taskbar",  
    "cached": false  
},  
{  
    "id": "weather_widget",  
    "type": "label",  
    "label": "Weather Widget",  
    "description": "Weather and temperature display in the taskbar",  
    "state": "visible",  
    "dx": 64,  
    "dy": 764,  
    "frame_dx": 226,  
    "frame_dy": 963,  
    "confidence": 0.85,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        178,  
        951,  
        274,  
        976  
    ],  
    "session_memory_key": "label:weather widget",  
    "cached": false  
},  
{
```

```
"id": "start_button",
"type": "button",
"label": "Start Button",
"description": "Windows Start menu button",
"state": "visible",
"dx": 424,
"dy": 760,
"frame_dx": 586,
"frame_dy": 959,
"confidence": 0.98,
"source": "gemini_vlm",
"frame_bbox": [
    562,
    943,
    610,
    976
],
"session_memory_key": "button:start button",
"cached": false
},
{
    "id": "search_bar",
    "type": "input",
    "label": "Search Bar",
    "description": "Windows search input field in the taskbar",
    "state": "visible",
    "dx": 508,
    "dy": 760,
    "frame_dx": 670,
```

```
"frame_dy": 959,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    594,
    943,
    746,
    976
],
"session_memory_key": "input:search bar",
"cached": false
},
{
    "id": "file_explorer_icon",
    "type": "icon",
    "label": "File Explorer",
    "description": "File Explorer icon in the taskbar",
    "state": "visible",
    "dx": 608,
    "dy": 760,
    "frame_dx": 770,
    "frame_dy": 959,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        746,
        943,
        794,
        976
    ]
}
```



```
    ],  
    "session_memory_key": "icon:file explorer",  
    "cached": false  
  },  
  {  
    "id": "microsoft_store_icon",  
    "type": "icon",  
    "label": "Microsoft Store",  
    "description": "Microsoft Store icon in the taskbar",  
    "state": "visible",  
    "dx": 648,  
    "dy": 760,  
    "frame_dx": 810,  
    "frame_dy": 959,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
      786,  
      943,  
      834,  
      976  
    ],  
    "session_memory_key": "icon:microsoft store",  
    "cached": false  
  },  
  {  
    "id": "mail_icon",  
    "type": "icon",  
    "label": "Mail",
```

```
"description": "Mail application icon in the taskbar",
"state": "visible",
"dx": 680,
"dy": 760,
"frame_dx": 842,
"frame_dy": 959,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    818,
    943,
    866,
    976
],
"session_memory_key": "icon:mail",
"cached": false
},
{
    "id": "xbox_icon",
    "type": "icon",
    "label": "Xbox",
    "description": "Xbox application icon in the taskbar",
    "state": "visible",
    "dx": 720,
    "dy": 760,
    "frame_dx": 882,
    "frame_dy": 959,
    "confidence": 0.9,
    "source": "gemini_vlm",
```

```
"frame_bbox": [
    858,
    943,
    906,
    976
],
"session_memory_key": "icon:xbox",
"cached": false
},
{
    "id": "chrome_icon",
    "type": "icon",
    "label": "Google Chrome",
    "description": "Google Chrome browser icon in the taskbar",
    "state": "visible",
    "dx": 760,
    "dy": 760,
    "frame_dx": 922,
    "frame_dy": 959,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        898,
        943,
        946,
        976
    ],
    "session_memory_key": "icon:google chrome",
    "cached": false
}
```

```
},  
{  
  "id": "vscode_icon",  
  "type": "icon",  
  "label": "Visual Studio Code",  
  "description": "Visual Studio Code icon in the taskbar",  
  "state": "visible",  
  "dx": 792,  
  "dy": 760,  
  "frame_dx": 954,  
  "frame_dy": 959,  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    930,  
    943,  
    978,  
    976  
  ],  
  "session_memory_key": "icon:visual studio code",  
  "cached": false  
},  
{  
  "id": "spotify_icon",  
  "type": "icon",  
  "label": "Spotify",  
  "description": "Spotify application icon in the taskbar",  
  "state": "visible",  
  "dx": 832,
```

```
"dy": 760,
"frame_dx": 994,
"frame_dy": 959,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    970,
    943,
    1018,
    976
],
"session_memory_key": "icon:spotify",
"cached": false
},
{
    "id": "google_icon",
    "type": "icon",
    "label": "Google",
    "description": "Google application icon in the taskbar",
    "state": "visible",
    "dx": 872,
    "dy": 760,
    "frame_dx": 1034,
    "frame_dy": 959,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        1010,
        943,
```

```
        1058,  
        976  
    ],  
    "session_memory_key": "icon:google",  
    "cached": false  
},  
{  
    "id": "autocad_icon",  
    "type": "icon",  
    "label": "AutoCAD",  
    "description": "AutoCAD application icon in the taskbar",  
    "state": "visible",  
    "dx": 908,  
    "dy": 760,  
    "frame_dx": 1070,  
    "frame_dy": 959,  
    "confidence": 0.8,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        1050,  
        943,  
        1090,  
        976  
    ],  
    "session_memory_key": "icon:autocad",  
    "cached": false  
},  
{  
    "id": "system_tray",
```

```
"type": "pane",
"label": "System Tray",
"description": "System tray area with various status icons",
"state": "visible",
"dx": 1430,
"dy": 760,
"frame_dx": 1592,
"frame_dy": 959,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    1426,
    943,
    1758,
    976
],
"session_memory_key": "pane:system tray",
"cached": false
},
{
    "id": "clock_date",
    "type": "label",
    "label": "Clock and Date",
    "description": "Current time and date display in the taskbar",
    "state": "visible",
    "dx": 1538,
    "dy": 760,
    "frame_dx": 1700,
    "frame_dy": 959,
```

```
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    1642,
    943,
    1758,
    976
],
"session_memory_key": "label:clock and date",
"cached": false
},
{
    "id": "login_container",
    "type": "pane",
    "label": "Login Container",
    "description": "Main white container for the login form",
    "state": "visible",
    "dx": 792,
    "dy": 388,
    "frame_dx": 954,
    "frame_dy": 587,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        658,
        199,
        1250,
        976
    ],
    ],
```



```
"session_memory_key": "pane:login container",
"cached": false
},
{
  "id": "sliit_logo",
  "type": "icon",
  "label": "SLIIT Logo",
  "description": "Logo of SLIIT University",
  "state": "visible",
  "dx": 688,
  "dy": 156,
  "frame_dx": 850,
  "frame_dy": 355,
  "confidence": 0.9,
  "source": "gemini_vlm",
  "frame_bbox": [
    770,
    327,
    930,
    383
  ],
  "session_memory_key": "icon:sliit logo",
  "cached": false
},
{
  "id": "courseweb_heading",
  "type": "heading",
  "label": "CourseWeb",
  "description": "Heading text for the CourseWeb application",
```

```
"state": "visible",
"dx": 892,
"dy": 156,
"frame_dx": 1054,
"frame_dy": 355,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    954,
    327,
    1154,
    383
],
"session_memory_key": "heading:courseweb",
"cached": false
},
{
    "id": "login_button",
    "type": "button",
    "label": "Login Button",
    "description": "Yellow login button in the center of the page",
    "state": "visible",
    "dx": 796,
    "dy": 404,
    "frame_dx": 958,
    "frame_dy": 603,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
```

```
        882,  
        575,  
        1034,  
        631  
    ],  
    "session_memory_key": "button:login button",  
    "cached": false  
},  
{  
    "id": "instruction_text",  
    "type": "label",  
    "label": "Instruction Text",  
    "description": "Text instructing the user to click on SLIIT Login",  
    "state": "visible",  
    "dx": 724,  
    "dy": 696,  
    "frame_dx": 886,  
    "frame_dy": 895,  
    "confidence": 0.85,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        802,  
        879,  
        970,  
        911  
    ],  
    "session_memory_key": "label:instruction text",  
    "cached": false  
},
```

```
{
  "id": "note_text",
  "type": "label",
  "label": "Note Text",
  "description": "Informational note about Microsoft 0365 validation",
  "state": "visible",
  "dx": 796,
  "dy": 736,
  "frame_dx": 958,
  "frame_dy": 935,
  "confidence": 0.85,
  "source": "gemini_vlm",
  "frame_bbox": [
    754,
    895,
    1162,
    976
  ],
  "session_memory_key": "label:note text",
  "cached": false
}
],
"session_knowledge_base": {
  "session_id": "shot_1777704787_aa8922",
  "started_at": "2026-05-02T06:53:07.025982+00:00",
  "updated_at": "2026-05-02T06:53:30.759747+00:00",
  "stale_after_misses": 3,
  "confidence_floor": 0.4,
  "decay_per_miss": 0.9,
```

```
"frame_count": 1,
"frame_name": "frame_1777704787_716419.jpg",
"frame_index": 0,
"image_size": {
  "width": 1920,
  "height": 1080
},
"known_elements": {
  "pane:taskbar": {
    "element_key": "pane:taskbar",
    "element_type": "pane",
    "label": "Taskbar",
    "bbox": [
      170.0,
      935.0,
      1758.0,
      976.0
    ],
    "confidence": 0.95,
    "source": "gemini_vlm",
    "first_seen_frame": "frame_1777704787_716419.jpg",
    "last_seen_frame": "frame_1777704787_716419.jpg",
    "first_seen_at": "2026-05-02T06:53:30.759747+00:00",
    "last_seen_at": "2026-05-02T06:53:30.759747+00:00",
    "first_seen_index": 0,
    "last_seen_index": 0,
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
```

```
"cached": false,
"history": [
  {
    "frame_name": "frame_1777704787_716419.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T06:53:30.759747+00:00",
    "confidence": 0.95
  }
],
},
"label:weather widget": {
  "element_key": "label:weather widget",
  "element_type": "label",
  "label": "Weather Widget",
  "bbox": [
    178.0,
    951.0,
    274.0,
    976.0
  ],
  "confidence": 0.85,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777704787_716419.jpg",
  "last_seen_frame": "frame_1777704787_716419.jpg",
  "first_seen_at": "2026-05-02T06:53:30.759747+00:00",
  "last_seen_at": "2026-05-02T06:53:30.759747+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
  "seen_count": 1,
```

```
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777704787_716419.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T06:53:30.759747+00:00",
    "confidence": 0.85
  }
]
},
"button:start button": {
  "element_key": "button:start button",
  "element_type": "button",
  "label": "Start Button",
  "bbox": [
    562.0,
    943.0,
    610.0,
    976.0
  ],
  "confidence": 0.98,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777704787_716419.jpg",
  "last_seen_frame": "frame_1777704787_716419.jpg",
  "first_seen_at": "2026-05-02T06:53:30.759747+00:00",
  "last_seen_at": "2026-05-02T06:53:30.759747+00:00",
  "first_seen_index": 0,
```

```
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777704787_716419.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T06:53:30.759747+00:00",
    "confidence": 0.98
  }
],
},
"input:search bar": {
  "element_key": "input:search bar",
  "element_type": "input",
  "label": "Search Bar",
  "bbox": [
    594.0,
    943.0,
    746.0,
    976.0
  ],
  "confidence": 0.95,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777704787_716419.jpg",
  "last_seen_frame": "frame_1777704787_716419.jpg",
  "first_seen_at": "2026-05-02T06:53:30.759747+00:00",
```



```
"last_seen_at": "2026-05-02T06:53:30.759747+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777704787_716419.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T06:53:30.759747+00:00",
    "confidence": 0.95
  }
],
},
"icon:file explorer": {
  "element_key": "icon:file explorer",
  "element_type": "icon",
  "label": "File Explorer",
  "bbox": [
    746.0,
    943.0,
    794.0,
    976.0
  ],
  "confidence": 0.9,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777704787_716419.jpg",
```

```
"last_seen_frame": "frame_1777704787_716419.jpg",
"first_seen_at": "2026-05-02T06:53:30.759747+00:00",
"last_seen_at": "2026-05-02T06:53:30.759747+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777704787_716419.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T06:53:30.759747+00:00",
    "confidence": 0.9
  }
],
},
"icon:microsoft store": {
  "element_key": "icon:microsoft store",
  "element_type": "icon",
  "label": "Microsoft Store",
  "bbox": [
    786.0,
    943.0,
    834.0,
    976.0
  ],
  "confidence": 0.9,
```

```
"source": "gemini_vlm",
"first_seen_frame": "frame_1777704787_716419.jpg",
"last_seen_frame": "frame_1777704787_716419.jpg",
"first_seen_at": "2026-05-02T06:53:30.759747+00:00",
"last_seen_at": "2026-05-02T06:53:30.759747+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777704787_716419.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T06:53:30.759747+00:00",
    "confidence": 0.9
  }
],
},
"icon:mail": {
  "element_key": "icon:mail",
  "element_type": "icon",
  "label": "Mail",
  "bbox": [
    818.0,
    943.0,
    866.0,
    976.0
  ]
}
```

```
    ],
    "confidence": 0.9,
    "source": "gemini_vlm",
    "first_seen_frame": "frame_1777704787_716419.jpg",
    "last_seen_frame": "frame_1777704787_716419.jpg",
    "first_seen_at": "2026-05-02T06:53:30.759747+00:00",
    "last_seen_at": "2026-05-02T06:53:30.759747+00:00",
    "first_seen_index": 0,
    "last_seen_index": 0,
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false,
    "history": [
      {
        "frame_name": "frame_1777704787_716419.jpg",
        "frame_index": 0,
        "seen_at": "2026-05-02T06:53:30.759747+00:00",
        "confidence": 0.9
      }
    ]
  },
  "icon:xbox": {
    "element_key": "icon:xbox",
    "element_type": "icon",
    "label": "Xbox",
    "bbox": [
      858.0,
      943.0,
```

```
    906.0,
    976.0
  ],
  "confidence": 0.9,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777704787_716419.jpg",
  "last_seen_frame": "frame_1777704787_716419.jpg",
  "first_seen_at": "2026-05-02T06:53:30.759747+00:00",
  "last_seen_at": "2026-05-02T06:53:30.759747+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
  "seen_count": 1,
  "miss_count": 0,
  "stale": false,
  "cached": false,
  "history": [
    {
      "frame_name": "frame_1777704787_716419.jpg",
      "frame_index": 0,
      "seen_at": "2026-05-02T06:53:30.759747+00:00",
      "confidence": 0.9
    }
  ]
},
"icon:google chrome": {
  "element_key": "icon:google chrome",
  "element_type": "icon",
  "label": "Google Chrome",
  "bbox": [
```

```
      898.0,
      943.0,
      946.0,
      976.0
    ],
    "confidence": 0.9,
    "source": "gemini_vlm",
    "first_seen_frame": "frame_1777704787_716419.jpg",
    "last_seen_frame": "frame_1777704787_716419.jpg",
    "first_seen_at": "2026-05-02T06:53:30.759747+00:00",
    "last_seen_at": "2026-05-02T06:53:30.759747+00:00",
    "first_seen_index": 0,
    "last_seen_index": 0,
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false,
    "history": [
      {
        "frame_name": "frame_1777704787_716419.jpg",
        "frame_index": 0,
        "seen_at": "2026-05-02T06:53:30.759747+00:00",
        "confidence": 0.9
      }
    ]
  },
  "icon:visual studio code": {
    "element_key": "icon:visual studio code",
    "element_type": "icon",
```

```
"label": "Visual Studio Code",
"bbox": [
  930.0,
  943.0,
  978.0,
  976.0
],
"confidence": 0.9,
"source": "gemini_vlm",
"first_seen_frame": "frame_1777704787_716419.jpg",
"last_seen_frame": "frame_1777704787_716419.jpg",
"first_seen_at": "2026-05-02T06:53:30.759747+00:00",
"last_seen_at": "2026-05-02T06:53:30.759747+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777704787_716419.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T06:53:30.759747+00:00",
    "confidence": 0.9
  }
]
},
"icon:spotify": {
```

```
"element_key": "icon:spotify",
"element_type": "icon",
"label": "Spotify",
"bbox": [
    970.0,
    943.0,
    1018.0,
    976.0
],
"confidence": 0.9,
"source": "gemini_vlm",
"first_seen_frame": "frame_1777704787_716419.jpg",
"last_seen_frame": "frame_1777704787_716419.jpg",
"first_seen_at": "2026-05-02T06:53:30.759747+00:00",
"last_seen_at": "2026-05-02T06:53:30.759747+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
    {
        "frame_name": "frame_1777704787_716419.jpg",
        "frame_index": 0,
        "seen_at": "2026-05-02T06:53:30.759747+00:00",
        "confidence": 0.9
    }
]
```



```
},
"icon:google": {
  "element_key": "icon:google",
  "element_type": "icon",
  "label": "Google",
  "bbox": [
    1010.0,
    943.0,
    1058.0,
    976.0
  ],
  "confidence": 0.9,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777704787_716419.jpg",
  "last_seen_frame": "frame_1777704787_716419.jpg",
  "first_seen_at": "2026-05-02T06:53:30.759747+00:00",
  "last_seen_at": "2026-05-02T06:53:30.759747+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
  "seen_count": 1,
  "miss_count": 0,
  "stale": false,
  "cached": false,
  "history": [
    {
      "frame_name": "frame_1777704787_716419.jpg",
      "frame_index": 0,
      "seen_at": "2026-05-02T06:53:30.759747+00:00",
      "confidence": 0.9
    }
  ]
}
```

```
    }
  ]
},
"icon:autocad": {
  "element_key": "icon:autocad",
  "element_type": "icon",
  "label": "AutoCAD",
  "bbox": [
    1050.0,
    943.0,
    1090.0,
    976.0
  ],
  "confidence": 0.8,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777704787_716419.jpg",
  "last_seen_frame": "frame_1777704787_716419.jpg",
  "first_seen_at": "2026-05-02T06:53:30.759747+00:00",
  "last_seen_at": "2026-05-02T06:53:30.759747+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
  "seen_count": 1,
  "miss_count": 0,
  "stale": false,
  "cached": false,
  "history": [
    {
      "frame_name": "frame_1777704787_716419.jpg",
      "frame_index": 0,
```

```
        "seen_at": "2026-05-02T06:53:30.759747+00:00",
        "confidence": 0.8
    }
]
},
"pane:system tray": {
    "element_key": "pane:system tray",
    "element_type": "pane",
    "label": "System Tray",
    "bbox": [
        1426.0,
        943.0,
        1758.0,
        976.0
    ],
    "confidence": 0.9,
    "source": "gemini_vlm",
    "first_seen_frame": "frame_1777704787_716419.jpg",
    "last_seen_frame": "frame_1777704787_716419.jpg",
    "first_seen_at": "2026-05-02T06:53:30.759747+00:00",
    "last_seen_at": "2026-05-02T06:53:30.759747+00:00",
    "first_seen_index": 0,
    "last_seen_index": 0,
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false,
    "history": [
        {
```

```
        "frame_name": "frame_1777704787_716419.jpg",
        "frame_index": 0,
        "seen_at": "2026-05-02T06:53:30.759747+00:00",
        "confidence": 0.9
    }
]
},
"label:clock and date": {
    "element_key": "label:clock and date",
    "element_type": "label",
    "label": "Clock and Date",
    "bbox": [
        1642.0,
        943.0,
        1758.0,
        976.0
    ],
    "confidence": 0.95,
    "source": "gemini_vlm",
    "first_seen_frame": "frame_1777704787_716419.jpg",
    "last_seen_frame": "frame_1777704787_716419.jpg",
    "first_seen_at": "2026-05-02T06:53:30.759747+00:00",
    "last_seen_at": "2026-05-02T06:53:30.759747+00:00",
    "first_seen_index": 0,
    "last_seen_index": 0,
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false,
```

```
"history": [  
  {  
    "frame_name": "frame_1777704787_716419.jpg",  
    "frame_index": 0,  
    "seen_at": "2026-05-02T06:53:30.759747+00:00",  
    "confidence": 0.95  
  }  
]  
},  
"pane:login container": {  
  "element_key": "pane:login container",  
  "element_type": "pane",  
  "label": "Login Container",  
  "bbox": [  
    658.0,  
    199.0,  
    1250.0,  
    976.0  
  ],  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "first_seen_frame": "frame_1777704787_716419.jpg",  
  "last_seen_frame": "frame_1777704787_716419.jpg",  
  "first_seen_at": "2026-05-02T06:53:30.759747+00:00",  
  "last_seen_at": "2026-05-02T06:53:30.759747+00:00",  
  "first_seen_index": 0,  
  "last_seen_index": 0,  
  "seen_count": 1,  
  "miss_count": 0,
```

```
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777704787_716419.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T06:53:30.759747+00:00",
    "confidence": 0.95
  }
],
},
"icon:sliit logo": {
  "element_key": "icon:sliit logo",
  "element_type": "icon",
  "label": "SLIIT Logo",
  "bbox": [
    770.0,
    327.0,
    930.0,
    383.0
  ],
  "confidence": 0.9,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777704787_716419.jpg",
  "last_seen_frame": "frame_1777704787_716419.jpg",
  "first_seen_at": "2026-05-02T06:53:30.759747+00:00",
  "last_seen_at": "2026-05-02T06:53:30.759747+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
```

```
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777704787_716419.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T06:53:30.759747+00:00",
    "confidence": 0.9
  }
]
},
"heading:courseweb": {
  "element_key": "heading:courseweb",
  "element_type": "heading",
  "label": "CourseWeb",
  "bbox": [
    954.0,
    327.0,
    1154.0,
    383.0
  ],
  "confidence": 0.95,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777704787_716419.jpg",
  "last_seen_frame": "frame_1777704787_716419.jpg",
  "first_seen_at": "2026-05-02T06:53:30.759747+00:00",
  "last_seen_at": "2026-05-02T06:53:30.759747+00:00",
```

```
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777704787_716419.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T06:53:30.759747+00:00",
    "confidence": 0.95
  }
],
},
"button:login button": {
  "element_key": "button:login button",
  "element_type": "button",
  "label": "Login Button",
  "bbox": [
    882.0,
    575.0,
    1034.0,
    631.0
  ],
  "confidence": 0.9,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777704787_716419.jpg",
  "last_seen_frame": "frame_1777704787_716419.jpg",
```



```
"first_seen_at": "2026-05-02T06:53:30.759747+00:00",
"last_seen_at": "2026-05-02T06:53:30.759747+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777704787_716419.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T06:53:30.759747+00:00",
    "confidence": 0.9
  }
],
},
"label:instruction text": {
  "element_key": "label:instruction text",
  "element_type": "label",
  "label": "Instruction Text",
  "bbox": [
    802.0,
    879.0,
    970.0,
    911.0
  ],
  "confidence": 0.85,
  "source": "gemini_vlm",
```

```
"first_seen_frame": "frame_1777704787_716419.jpg",
"last_seen_frame": "frame_1777704787_716419.jpg",
"first_seen_at": "2026-05-02T06:53:30.759747+00:00",
"last_seen_at": "2026-05-02T06:53:30.759747+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777704787_716419.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T06:53:30.759747+00:00",
    "confidence": 0.85
  }
],
},
"label:note text": {
  "element_key": "label:note text",
  "element_type": "label",
  "label": "Note Text",
  "bbox": [
    754.0,
    895.0,
    1162.0,
    976.0
  ],
}
```

```
"confidence": 0.85,
"source": "gemini_vlm",
"first_seen_frame": "frame_1777704787_716419.jpg",
"last_seen_frame": "frame_1777704787_716419.jpg",
"first_seen_at": "2026-05-02T06:53:30.759747+00:00",
"last_seen_at": "2026-05-02T06:53:30.759747+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777704787_716419.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T06:53:30.759747+00:00",
    "confidence": 0.85
  }
]
},
"resolved_elements": [
  {
    "id": "taskbar",
    "type": "pane",
    "label": "Taskbar",
    "description": "Windows taskbar at the bottom of the screen",
    "state": "visible",
```

```
"dx": 802,  
  "dy": 756,  
  "frame_dx": 964,  
  "frame_dy": 955,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    170,  
    935,  
    1758,  
    976  
  ],  
  "session_memory_key": "pane:taskbar",  
  "cached": false  
},  
{  
  "id": "weather_widget",  
  "type": "label",  
  "label": "Weather Widget",  
  "description": "Weather and temperature display in the taskbar",  
  "state": "visible",  
  "dx": 64,  
  "dy": 764,  
  "frame_dx": 226,  
  "frame_dy": 963,  
  "confidence": 0.85,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    178,
```

```
        951,  
        274,  
        976  
    ],  
    "session_memory_key": "label:weather widget",  
    "cached": false  
},  
{  
    "id": "start_button",  
    "type": "button",  
    "label": "Start Button",  
    "description": "Windows Start menu button",  
    "state": "visible",  
    "dx": 424,  
    "dy": 760,  
    "frame_dx": 586,  
    "frame_dy": 959,  
    "confidence": 0.98,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        562,  
        943,  
        610,  
        976  
    ],  
    "session_memory_key": "button:start button",  
    "cached": false  
},  
{
```

```
"id": "search_bar",
"type": "input",
"label": "Search Bar",
"description": "Windows search input field in the taskbar",
"state": "visible",
"dx": 508,
"dy": 760,
"frame_dx": 670,
"frame_dy": 959,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    594,
    943,
    746,
    976
],
"session_memory_key": "input:search bar",
"cached": false
},
{
    "id": "file_explorer_icon",
    "type": "icon",
    "label": "File Explorer",
    "description": "File Explorer icon in the taskbar",
    "state": "visible",
    "dx": 608,
    "dy": 760,
    "frame_dx": 770,
```

```
"frame_dy": 959,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    746,
    943,
    794,
    976
],
"session_memory_key": "icon:file explorer",
"cached": false
},
{
    "id": "microsoft_store_icon",
    "type": "icon",
    "label": "Microsoft Store",
    "description": "Microsoft Store icon in the taskbar",
    "state": "visible",
    "dx": 648,
    "dy": 760,
    "frame_dx": 810,
    "frame_dy": 959,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        786,
        943,
        834,
        976
    ]
}
```

```
    ],  
    "session_memory_key": "icon:microsoft store",  
    "cached": false  
  },  
  {  
    "id": "mail_icon",  
    "type": "icon",  
    "label": "Mail",  
    "description": "Mail application icon in the taskbar",  
    "state": "visible",  
    "dx": 680,  
    "dy": 760,  
    "frame_dx": 842,  
    "frame_dy": 959,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
      818,  
      943,  
      866,  
      976  
    ],  
    "session_memory_key": "icon:mail",  
    "cached": false  
  },  
  {  
    "id": "xbox_icon",  
    "type": "icon",  
    "label": "Xbox",
```



```
"description": "Xbox application icon in the taskbar",
"state": "visible",
"dx": 720,
"dy": 760,
"frame_dx": 882,
"frame_dy": 959,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    858,
    943,
    906,
    976
],
"session_memory_key": "icon:xbox",
"cached": false
},
{
    "id": "chrome_icon",
    "type": "icon",
    "label": "Google Chrome",
    "description": "Google Chrome browser icon in the taskbar",
    "state": "visible",
    "dx": 760,
    "dy": 760,
    "frame_dx": 922,
    "frame_dy": 959,
    "confidence": 0.9,
    "source": "gemini_vlm",
```

```
"frame_bbox": [
    898,
    943,
    946,
    976
],
"session_memory_key": "icon:google chrome",
"cached": false
},
{
    "id": "vscode_icon",
    "type": "icon",
    "label": "Visual Studio Code",
    "description": "Visual Studio Code icon in the taskbar",
    "state": "visible",
    "dx": 792,
    "dy": 760,
    "frame_dx": 954,
    "frame_dy": 959,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        930,
        943,
        978,
        976
    ],
    "session_memory_key": "icon:visual studio code",
    "cached": false
}
```

```
},  
{  
  "id": "spotify_icon",  
  "type": "icon",  
  "label": "Spotify",  
  "description": "Spotify application icon in the taskbar",  
  "state": "visible",  
  "dx": 832,  
  "dy": 760,  
  "frame_dx": 994,  
  "frame_dy": 959,  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    970,  
    943,  
    1018,  
    976  
  ],  
  "session_memory_key": "icon:spotify",  
  "cached": false  
},  
{  
  "id": "google_icon",  
  "type": "icon",  
  "label": "Google",  
  "description": "Google application icon in the taskbar",  
  "state": "visible",  
  "dx": 872,
```

```
"dy": 760,
"frame_dx": 1034,
"frame_dy": 959,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    1010,
    943,
    1058,
    976
],
"session_memory_key": "icon:google",
"cached": false
},
{
    "id": "autocad_icon",
    "type": "icon",
    "label": "AutoCAD",
    "description": "AutoCAD application icon in the taskbar",
    "state": "visible",
    "dx": 908,
    "dy": 760,
    "frame_dx": 1070,
    "frame_dy": 959,
    "confidence": 0.8,
    "source": "gemini_vlm",
    "frame_bbox": [
        1050,
        943,
```

```
        1090,  
        976  
    ],  
    "session_memory_key": "icon:autocad",  
    "cached": false  
},  
{  
    "id": "system_tray",  
    "type": "pane",  
    "label": "System Tray",  
    "description": "System tray area with various status icons",  
    "state": "visible",  
    "dx": 1430,  
    "dy": 760,  
    "frame_dx": 1592,  
    "frame_dy": 959,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        1426,  
        943,  
        1758,  
        976  
    ],  
    "session_memory_key": "pane:system tray",  
    "cached": false  
},  
{  
    "id": "clock_date",
```

```
"type": "label",
"label": "Clock and Date",
"description": "Current time and date display in the taskbar",
"state": "visible",
"dx": 1538,
"dy": 760,
"frame_dx": 1700,
"frame_dy": 959,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    1642,
    943,
    1758,
    976
],
"session_memory_key": "label:clock and date",
"cached": false
},
{
    "id": "login_container",
    "type": "pane",
    "label": "Login Container",
    "description": "Main white container for the login form",
    "state": "visible",
    "dx": 792,
    "dy": 388,
    "frame_dx": 954,
    "frame_dy": 587,
```

```
"confidence": 0.95,  
"source": "gemini_vlm",  
"frame_bbox": [  
    658,  
    199,  
    1250,  
    976  
],  
"session_memory_key": "pane:login container",  
"cached": false  
},  
{  
    "id": "sliit_logo",  
    "type": "icon",  
    "label": "SLIIT Logo",  
    "description": "Logo of SLIIT University",  
    "state": "visible",  
    "dx": 688,  
    "dy": 156,  
    "frame_dx": 850,  
    "frame_dy": 355,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        770,  
        327,  
        930,  
        383  
    ],
```

```
"session_memory_key": "icon:sliit logo",
"cached": false
},
{
  "id": "courseweb_heading",
  "type": "heading",
  "label": "CourseWeb",
  "description": "Heading text for the CourseWeb application",
  "state": "visible",
  "dx": 892,
  "dy": 156,
  "frame_dx": 1054,
  "frame_dy": 355,
  "confidence": 0.95,
  "source": "gemini_vlm",
  "frame_bbox": [
    954,
    327,
    1154,
    383
  ],
  "session_memory_key": "heading:courseweb",
  "cached": false
},
{
  "id": "login_button",
  "type": "button",
  "label": "Login Button",
  "description": "Yellow login button in the center of the page",
```



```
"state": "visible",
"dx": 796,
"dy": 404,
"frame_dx": 958,
"frame_dy": 603,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    882,
    575,
    1034,
    631
],
"session_memory_key": "button:login button",
"cached": false
},
{
    "id": "instruction_text",
    "type": "label",
    "label": "Instruction Text",
    "description": "Text instructing the user to click on SLIIT Login",
    "state": "visible",
    "dx": 724,
    "dy": 696,
    "frame_dx": 886,
    "frame_dy": 895,
    "confidence": 0.85,
    "source": "gemini_vlm",
    "frame_bbox": [
```

```
        802,  
        879,  
        970,  
        911  
    ],  
    "session_memory_key": "label:instruction text",  
    "cached": false  
},  
{  
    "id": "note_text",  
    "type": "label",  
    "label": "Note Text",  
    "description": "Informational note about Microsoft 0365  
validation",  
    "state": "visible",  
    "dx": 796,  
    "dy": 736,  
    "frame_dx": 958,  
    "frame_dy": 935,  
    "confidence": 0.85,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        754,  
        895,  
        1162,  
        976  
    ],  
    "session_memory_key": "label:note text",  
    "cached": false
```

```
    }  
  ],  
  "current_elements": [  
    {  
      "id": "taskbar",  
      "type": "pane",  
      "label": "Taskbar",  
      "description": "Windows taskbar at the bottom of the screen",  
      "state": "visible",  
      "dx": 802,  
      "dy": 756,  
      "frame_dx": 964,  
      "frame_dy": 955,  
      "confidence": 0.95,  
      "source": "gemini_vlm",  
      "frame_bbox": [  
        170,  
        935,  
        1758,  
        976  
      ],  
      "session_memory_key": "pane:taskbar",  
      "cached": false  
    },  
    {  
      "id": "weather_widget",  
      "type": "label",  
      "label": "Weather Widget",  
      "description": "Weather and temperature display in the taskbar",
```

```
    "state": "visible",
    "dx": 64,
    "dy": 764,
    "frame_dx": 226,
    "frame_dy": 963,
    "confidence": 0.85,
    "source": "gemini_vlm",
    "frame_bbox": [
        178,
        951,
        274,
        976
    ],
    "session_memory_key": "label:weather widget",
    "cached": false
},
{
    "id": "start_button",
    "type": "button",
    "label": "Start Button",
    "description": "Windows Start menu button",
    "state": "visible",
    "dx": 424,
    "dy": 760,
    "frame_dx": 586,
    "frame_dy": 959,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
```

```
        562,  
        943,  
        610,  
        976  
    ],  
    "session_memory_key": "button:start button",  
    "cached": false  
},  
{  
    "id": "search_bar",  
    "type": "input",  
    "label": "Search Bar",  
    "description": "Windows search input field in the taskbar",  
    "state": "visible",  
    "dx": 508,  
    "dy": 760,  
    "frame_dx": 670,  
    "frame_dy": 959,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        594,  
        943,  
        746,  
        976  
    ],  
    "session_memory_key": "input:search bar",  
    "cached": false  
},
```

```
{
  "id": "file_explorer_icon",
  "type": "icon",
  "label": "File Explorer",
  "description": "File Explorer icon in the taskbar",
  "state": "visible",
  "dx": 608,
  "dy": 760,
  "frame_dx": 770,
  "frame_dy": 959,
  "confidence": 0.9,
  "source": "gemini_vlm",
  "frame_bbox": [
    746,
    943,
    794,
    976
  ],
  "session_memory_key": "icon:file explorer",
  "cached": false
},
{
  "id": "microsoft_store_icon",
  "type": "icon",
  "label": "Microsoft Store",
  "description": "Microsoft Store icon in the taskbar",
  "state": "visible",
  "dx": 648,
  "dy": 760,
```

```
"frame_dx": 810,  
"frame_dy": 959,  
"confidence": 0.9,  
"source": "gemini_vlm",  
"frame_bbox": [  
    786,  
    943,  
    834,  
    976  
],  
"session_memory_key": "icon:microsoft store",  
"cached": false  
},  
{  
    "id": "mail_icon",  
    "type": "icon",  
    "label": "Mail",  
    "description": "Mail application icon in the taskbar",  
    "state": "visible",  
    "dx": 680,  
    "dy": 760,  
    "frame_dx": 842,  
    "frame_dy": 959,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        818,  
        943,  
        866,
```

```
        976
    ],
    "session_memory_key": "icon:mail",
    "cached": false
},
{
    "id": "xbox_icon",
    "type": "icon",
    "label": "Xbox",
    "description": "Xbox application icon in the taskbar",
    "state": "visible",
    "dx": 720,
    "dy": 760,
    "frame_dx": 882,
    "frame_dy": 959,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        858,
        943,
        906,
        976
    ],
    "session_memory_key": "icon:xbox",
    "cached": false
},
{
    "id": "chrome_icon",
    "type": "icon",
```



```
"label": "Google Chrome",
"description": "Google Chrome browser icon in the taskbar",
"state": "visible",
"dx": 760,
"dy": 760,
"frame_dx": 922,
"frame_dy": 959,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    898,
    943,
    946,
    976
],
"session_memory_key": "icon:google chrome",
"cached": false
},
{
    "id": "vscode_icon",
    "type": "icon",
    "label": "Visual Studio Code",
    "description": "Visual Studio Code icon in the taskbar",
    "state": "visible",
    "dx": 792,
    "dy": 760,
    "frame_dx": 954,
    "frame_dy": 959,
    "confidence": 0.9,
```

```
"source": "gemini_vlm",
"frame_bbox": [
  930,
  943,
  978,
  976
],
"session_memory_key": "icon:visual studio code",
"cached": false
},
{
  "id": "spotify_icon",
  "type": "icon",
  "label": "Spotify",
  "description": "Spotify application icon in the taskbar",
  "state": "visible",
  "dx": 832,
  "dy": 760,
  "frame_dx": 994,
  "frame_dy": 959,
  "confidence": 0.9,
  "source": "gemini_vlm",
  "frame_bbox": [
    970,
    943,
    1018,
    976
  ],
  "session_memory_key": "icon:spotify",
```

```
    "cached": false
  },
  {
    "id": "google_icon",
    "type": "icon",
    "label": "Google",
    "description": "Google application icon in the taskbar",
    "state": "visible",
    "dx": 872,
    "dy": 760,
    "frame_dx": 1034,
    "frame_dy": 959,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
      1010,
      943,
      1058,
      976
    ],
    "session_memory_key": "icon:google",
    "cached": false
  },
  {
    "id": "autocad_icon",
    "type": "icon",
    "label": "AutoCAD",
    "description": "AutoCAD application icon in the taskbar",
    "state": "visible",
```

```
"dx": 908,  
  "dy": 760,  
  "frame_dx": 1070,  
  "frame_dy": 959,  
  "confidence": 0.8,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    1050,  
    943,  
    1090,  
    976  
  ],  
  "session_memory_key": "icon:autocad",  
  "cached": false  
},  
{  
  "id": "system_tray",  
  "type": "pane",  
  "label": "System Tray",  
  "description": "System tray area with various status icons",  
  "state": "visible",  
  "dx": 1430,  
  "dy": 760,  
  "frame_dx": 1592,  
  "frame_dy": 959,  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    1426,
```

```
        943,  
        1758,  
        976  
    ],  
    "session_memory_key": "pane:system tray",  
    "cached": false  
},  
{  
    "id": "clock_date",  
    "type": "label",  
    "label": "Clock and Date",  
    "description": "Current time and date display in the taskbar",  
    "state": "visible",  
    "dx": 1538,  
    "dy": 760,  
    "frame_dx": 1700,  
    "frame_dy": 959,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        1642,  
        943,  
        1758,  
        976  
    ],  
    "session_memory_key": "label:clock and date",  
    "cached": false  
},  
{
```

```
"id": "login_container",
"type": "pane",
"label": "Login Container",
"description": "Main white container for the login form",
"state": "visible",
"dx": 792,
"dy": 388,
"frame_dx": 954,
"frame_dy": 587,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    658,
    199,
    1250,
    976
],
"session_memory_key": "pane:login container",
"cached": false
},
{
    "id": "sliit_logo",
    "type": "icon",
    "label": "SLIIT Logo",
    "description": "Logo of SLIIT University",
    "state": "visible",
    "dx": 688,
    "dy": 156,
    "frame_dx": 850,
```

```
"frame_dy": 355,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    770,
    327,
    930,
    383
],
"session_memory_key": "icon:sliit logo",
"cached": false
},
{
    "id": "courseweb_heading",
    "type": "heading",
    "label": "CourseWeb",
    "description": "Heading text for the CourseWeb application",
    "state": "visible",
    "dx": 892,
    "dy": 156,
    "frame_dx": 1054,
    "frame_dy": 355,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        954,
        327,
        1154,
        383
    ]
}
```

```
    ],  
    "session_memory_key": "heading:courseweb",  
    "cached": false  
  },  
  {  
    "id": "login_button",  
    "type": "button",  
    "label": "Login Button",  
    "description": "Yellow login button in the center of the page",  
    "state": "visible",  
    "dx": 796,  
    "dy": 404,  
    "frame_dx": 958,  
    "frame_dy": 603,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
      882,  
      575,  
      1034,  
      631  
    ],  
    "session_memory_key": "button:login button",  
    "cached": false  
  },  
  {  
    "id": "instruction_text",  
    "type": "label",  
    "label": "Instruction Text",
```



```
    "description": "Text instructing the user to click on SLIIT Login",
    "state": "visible",
    "dx": 724,
    "dy": 696,
    "frame_dx": 886,
    "frame_dy": 895,
    "confidence": 0.85,
    "source": "gemini_vlm",
    "frame_bbox": [
        802,
        879,
        970,
        911
    ],
    "session_memory_key": "label:instruction text",
    "cached": false
},
{
    "id": "note_text",
    "type": "label",
    "label": "Note Text",
    "description": "Informational note about Microsoft 0365
validation",
    "state": "visible",
    "dx": 796,
    "dy": 736,
    "frame_dx": 958,
    "frame_dy": 935,
    "confidence": 0.85,
```

```
        "source": "gemini_vlm",
        "frame_bbox": [
            754,
            895,
            1162,
            976
        ],
        "session_memory_key": "label:note text",
        "cached": false
    }
]
}
}
}
},
"todo": {
    "steps": [
        {
            "id": 1,
            "action": "Click the 'Login Button' at (958, 603) to navigate to the Microsoft 0365 authentication page.",
            "target": "Login Button",
            "expected_result": "The browser redirects to the Microsoft login page where credentials can be entered.",
            "needs_user_data": false,
            "user_data_field": null,
            "status": "failed",
            "user_input": "IT22099518"
        },
        {
```

```
    "id": 2,
    "action": "Type the username 'IT22099518' into the email/username field,
click 'Next', type the password 'NuwaniSithara@808' into the password field, and
click 'Sign in'.",
    "target": "Microsoft login form fields",
    "expected_result": "The user is authenticated and redirected to the
CourseWeb dashboard.",
    "needs_user_data": false,
    "user_data_field": null,
    "status": "failed"
  }
],
  "notes": "The current screen is the CourseWeb landing page which requires
clicking the 'Login Button' to access the Microsoft SSO portal for credential
entry.",
  "estimated_complexity": "simple"
},
  "step_plans": [
    {
      "step_id": 1,
      "action": "Click the 'Login Button' at (958, 603) to navigate to the
Microsoft 0365 authentication page.",
      "target": "Login Button",
      "expected_result": "The browser redirects to the Microsoft login page where
credentials can be entered.",
      "hid_commands": [
        {
          "cmd": "mouse_move",
          "dx": 958,
          "dy": 603
        },
        {
```

```
        "cmd": "mouse_click",
        "button": "left"
    }
],
    "reasoning": "Moving the cursor to the Login Button at the coordinates
specified in the action step (958, 603) and clicking to navigate to the Microsoft
0365 authentication page."
},
{
    "step_id": 2,
    "action": "Type the username 'IT22099518' into the email/username field,
click 'Next', type the password 'NuwaniSithara@808' into the password field, and
click 'Sign in'.",
    "target": "Microsoft login form fields",
    "expected_result": "The user is authenticated and redirected to the
CourseWeb dashboard.",
    "hid_commands": [
        {
            "cmd": "mouse_move",
            "meta": {
                "commandId": "14ae48ba-ed9d-41b2-8aff-c090d20a94e0"
            },
            "dx": 860,
            "dy": 302,
            "smooth": true
        },
        {
            "cmd": "mouse_click",
            "meta": {
                "commandId": "fcaaa73c-6eb4-431d-ae8-a896eb87df5c"
            },

```

```
        "button": "left"
    }
],
    "reasoning": "Fallback generated 2 commands from detected elements"
}
],
"step_results": [
    {
        "step_id": 1,
        "action": "Click the 'Login Button' at (958, 603) to navigate to the Microsoft 0365 authentication page.",
        "attempt": 3,
        "exec_status": "success",
        "eval_status": "needs_input",
        "confidence": 1.0,
        "reason": "The login page has loaded and the 'Username or email' input field is visible, requiring user credentials to proceed.",
        "final_status": "executing",
        "timestamp": "2026-05-02T07:01:53.030968+00:00"
    },
    {
        "step_id": 2,
        "action": "Type the username 'IT22099518' into the email/username field, click 'Next', type the password 'NuwaniSithara@808' into the password field, and click 'Sign in'.",
        "attempt": 3,
        "exec_status": "success",
        "eval_status": "retry",
        "confidence": 0.0,
        "reason": "Evaluation error: No valid JSON found in LLM response: '{\\n \\\"status\\\": \"',

```

```
    "final_status": "executing",
    "timestamp": "2026-05-02T07:10:27.964531+00:00"
  }
],
"final_report": {
  "success": false,
  "summary": "The login attempt to CourseWeb failed as the automation could not successfully interact with the login elements.",
  "message": "The task was not completed. Both planned steps failed to execute, and the browser remains on the initial login page.\n\nStep results:\n- Step 1: Failed to click the 'Login Button' at (958, 603) to navigate to the 0365 page.\n- Step 2: Failed to input credentials and submit the form.\n\nThe current screen state shows that the login form is still visible with the 'Log in' button located at (800, 248), which differs from the coordinates targeted in the initial step.",
  "steps_completed": 0,
  "steps_failed": 2,
  "issues": [
    "Coordinate mismatch: The target button was expected at (958, 603) but is currently detected at (800, 248).",
    "Navigation failure: The system did not transition to the Microsoft 0365 authentication page as intended."
  ],
  "recommendations": [
    "Update the automation to use the detected coordinates (800, 248) for the 'Log in' button.",
    "Attempt to enter the username and password directly into the detected input fields at (800, 156) and (800, 200)."
  ]
},
"session_memory": {
  "run_id": "57f743fa-1756-4baa-b97d-25c9dacdd30d",
```

"user_task": "Now you are in the CourseWeb login page, so the username: \

"started_at": "2026-05-02T06:53:02.537150+00:00",

"updated_at": "2026-05-02T07:11:06.085051+00:00",

"stale_after_misses": 2,

"confidence_floor": 0.4,

"decay_per_miss": 0.9,

"frame_index": 21,

"known_elements": {

 "pane:taskbar": {

 "element_key": "pane:taskbar",

 "element_type": "pane",

 "label": "Taskbar",

 "bbox": [

 177.0,

 943.0,

 1756.0,

 962.0

],

 "confidence": 0.6561,

 "source": "gemini_vlm",

 "first_seen_frame": null,

 "last_seen_frame": null,

 "first_seen_index": 0,

 "last_seen_index": 17,

 "first_seen_at": "2026-05-02T06:53:30.786319+00:00",

 "last_seen_at": "2026-05-02T07:08:17.240100+00:00",

 "seen_count": 4,

 "miss_count": 3,

```
    "stale": true,
    "cached": true
  },
  "label:weather widget": {
    "element_key": "label:weather widget",
    "element_type": "label",
    "label": "Weather Widget",
    "bbox": [
      178.0,
      951.0,
      274.0,
      976.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 0,
    "last_seen_index": 0,
    "first_seen_at": "2026-05-02T06:53:30.786319+00:00",
    "last_seen_at": "2026-05-02T06:53:30.786319+00:00",
    "seen_count": 1,
    "miss_count": 20,
    "stale": true,
    "cached": true
  },
  "button:start button": {
    "element_key": "button:start button",
    "element_type": "button",
```



```
"label": "Start Button",
"bbox": [
  562.0,
  943.0,
  610.0,
  976.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 0,
"first_seen_at": "2026-05-02T06:53:30.786319+00:00",
"last_seen_at": "2026-05-02T06:53:30.786319+00:00",
"seen_count": 1,
"miss_count": 20,
"stale": true,
"cached": true
},
"input:search bar": {
  "element_key": "input:search bar",
  "element_type": "input",
  "label": "Search Bar",
  "bbox": [
    594.0,
    943.0,
    746.0,
    976.0
  ]
}
```

```
],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 0,
  "first_seen_at": "2026-05-02T06:53:30.786319+00:00",
  "last_seen_at": "2026-05-02T06:53:30.786319+00:00",
  "seen_count": 1,
  "miss_count": 20,
  "stale": true,
  "cached": true
},
"icon:file explorer": {
  "element_key": "icon:file explorer",
  "element_type": "icon",
  "label": "File Explorer",
  "bbox": [
    762.0,
    926.0,
    818.0,
    962.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
```

```
"last_seen_index": 5,  
"first_seen_at": "2026-05-02T06:53:30.786319+00:00",  
"last_seen_at": "2026-05-02T06:57:27.809908+00:00",  
"seen_count": 2,  
"miss_count": 15,  
"stale": true,  
"cached": true  
},  
"icon:microsoft store": {  
  "element_key": "icon:microsoft store",  
  "element_type": "icon",  
  "label": "Microsoft Store",  
  "bbox": [  
    786.0,  
    943.0,  
    834.0,  
    976.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 0,  
  "last_seen_index": 0,  
  "first_seen_at": "2026-05-02T06:53:30.786319+00:00",  
  "last_seen_at": "2026-05-02T06:53:30.786319+00:00",  
  "seen_count": 1,  
  "miss_count": 20,  
  "stale": true,
```

```
    "cached": true
  },
  "icon:mail": {
    "element_key": "icon:mail",
    "element_type": "icon",
    "label": "Mail",
    "bbox": [
      818.0,
      943.0,
      866.0,
      976.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 0,
    "last_seen_index": 0,
    "first_seen_at": "2026-05-02T06:53:30.786319+00:00",
    "last_seen_at": "2026-05-02T06:53:30.786319+00:00",
    "seen_count": 1,
    "miss_count": 20,
    "stale": true,
    "cached": true
  },
  "icon:xbox": {
    "element_key": "icon:xbox",
    "element_type": "icon",
    "label": "Xbox",
```

```
"bbox": [
  858.0,
  943.0,
  906.0,
  976.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 0,
"first_seen_at": "2026-05-02T06:53:30.786319+00:00",
"last_seen_at": "2026-05-02T06:53:30.786319+00:00",
"seen_count": 1,
"miss_count": 20,
"stale": true,
"cached": true
},
"icon:google chrome": {
  "element_key": "icon:google chrome",
  "element_type": "icon",
  "label": "Google Chrome",
  "bbox": [
    728.0,
    934.0,
    776.0,
    962.0
  ],

```

```
"confidence": 0.5019,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 15,
"first_seen_at": "2026-05-02T06:53:30.786319+00:00",
"last_seen_at": "2026-05-02T07:06:29.451301+00:00",
"seen_count": 4,
"miss_count": 5,
"stale": true,
"cached": true
},
"icon:visual studio code": {
  "element_key": "icon:visual studio code",
  "element_type": "icon",
  "label": "Visual Studio Code",
  "bbox": [
    810.0,
    926.0,
    866.0,
    962.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 5,
```

```
"first_seen_at": "2026-05-02T06:53:30.786319+00:00",
"last_seen_at": "2026-05-02T06:57:27.809908+00:00",
"seen_count": 2,
"miss_count": 15,
"stale": true,
"cached": true
},
"icon:spotify": {
  "element_key": "icon:spotify",
  "element_type": "icon",
  "label": "Spotify",
  "bbox": [
    986.0,
    926.0,
    1034.0,
    962.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 5,
  "first_seen_at": "2026-05-02T06:53:30.786319+00:00",
  "last_seen_at": "2026-05-02T06:57:27.809908+00:00",
  "seen_count": 2,
  "miss_count": 15,
  "stale": true,
  "cached": true
}
```

```
},  
"icon:google": {  
  "element_key": "icon:google",  
  "element_type": "icon",  
  "label": "Google",  
  "bbox": [  
    1010.0,  
    943.0,  
    1058.0,  
    976.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 0,  
  "last_seen_index": 0,  
  "first_seen_at": "2026-05-02T06:53:30.786319+00:00",  
  "last_seen_at": "2026-05-02T06:53:30.786319+00:00",  
  "seen_count": 1,  
  "miss_count": 20,  
  "stale": true,  
  "cached": true  
},  
"icon:autocad": {  
  "element_key": "icon:autocad",  
  "element_type": "icon",  
  "label": "AutoCAD",  
  "bbox": [  

```



```
    1050.0,  
    943.0,  
    1090.0,  
    976.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 0,  
  "last_seen_index": 0,  
  "first_seen_at": "2026-05-02T06:53:30.786319+00:00",  
  "last_seen_at": "2026-05-02T06:53:30.786319+00:00",  
  "seen_count": 1,  
  "miss_count": 20,  
  "stale": true,  
  "cached": true  
},  
"pane:system tray": {  
  "element_key": "pane:system tray",  
  "element_type": "pane",  
  "label": "System Tray",  
  "bbox": [  
    1426.0,  
    943.0,  
    1758.0,  
    976.0  
  ],  
  "confidence": 0.4,
```

```
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 0,
"first_seen_at": "2026-05-02T06:53:30.786319+00:00",
"last_seen_at": "2026-05-02T06:53:30.786319+00:00",
"seen_count": 1,
"miss_count": 20,
"stale": true,
"cached": true
},
"label:clock and date": {
  "element_key": "label:clock and date",
  "element_type": "label",
  "label": "Clock and Date",
  "bbox": [
    1640.0,
    934.0,
    1744.0,
    962.0
  ],
  "confidence": 0.4724,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 15,
  "first_seen_at": "2026-05-02T06:53:30.786319+00:00",
```

```
"last_seen_at": "2026-05-02T07:06:29.451301+00:00",
"seen_count": 3,
"miss_count": 5,
"stale": true,
"cached": true
},
"pane:login container": {
  "element_key": "pane:login container",
  "element_type": "pane",
  "label": "Login Container",
  "bbox": [
    658.0,
    199.0,
    1250.0,
    976.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 0,
  "first_seen_at": "2026-05-02T06:53:30.786319+00:00",
  "last_seen_at": "2026-05-02T06:53:30.786319+00:00",
  "seen_count": 1,
  "miss_count": 20,
  "stale": true,
  "cached": true
},
```

```
"icon:sliit logo": {
  "element_key": "icon:sliit logo",
  "element_type": "icon",
  "label": "SLIIT Logo",
  "bbox": [
    770.0,
    327.0,
    930.0,
    383.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 0,
  "first_seen_at": "2026-05-02T06:53:30.786319+00:00",
  "last_seen_at": "2026-05-02T06:53:30.786319+00:00",
  "seen_count": 1,
  "miss_count": 20,
  "stale": true,
  "cached": true
},
"heading:courseweb": {
  "element_key": "heading:courseweb",
  "element_type": "heading",
  "label": "CourseWeb",
  "bbox": [
    929.0,
```

```
    278.0,  
    1153.0,  
    318.0  
  ],  
  "confidence": 0.4783,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 0,  
  "last_seen_index": 14,  
  "first_seen_at": "2026-05-02T06:53:30.786319+00:00",  
  "last_seen_at": "2026-05-02T07:05:44.869252+00:00",  
  "seen_count": 2,  
  "miss_count": 6,  
  "stale": true,  
  "cached": true  
},  
"button:login button": {  
  "element_key": "button:login button",  
  "element_type": "button",  
  "label": "Login Button",  
  "bbox": [  
    882.0,  
    575.0,  
    1034.0,  
    631.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",
```

```
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 0,
"first_seen_at": "2026-05-02T06:53:30.786319+00:00",
"last_seen_at": "2026-05-02T06:53:30.786319+00:00",
"seen_count": 1,
"miss_count": 20,
"stale": true,
"cached": true
},
"label:instruction text": {
  "element_key": "label:instruction text",
  "element_type": "label",
  "label": "Instruction Text",
  "bbox": [
    802.0,
    879.0,
    970.0,
    911.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 0,
  "first_seen_at": "2026-05-02T06:53:30.786319+00:00",
  "last_seen_at": "2026-05-02T06:53:30.786319+00:00",
```

```
"seen_count": 1,
"miss_count": 20,
"stale": true,
"cached": true
},
"label:note text": {
  "element_key": "label:note text",
  "element_type": "label",
  "label": "Note Text",
  "bbox": [
    754.0,
    895.0,
    1162.0,
    976.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 0,
  "first_seen_at": "2026-05-02T06:53:30.786319+00:00",
  "last_seen_at": "2026-05-02T06:53:30.786319+00:00",
  "seen_count": 1,
  "miss_count": 20,
  "stale": true,
  "cached": true
},
"heading:sliit uni courseweb": {
```

```
"element_key": "heading:sliit uni courseweb",
"element_type": "heading",
"label": "SLIIT UNI CourseWeb",
"bbox": [
  729.0,
  263.0,
  1193.0,
  327.0
],
"confidence": 0.6925,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 1,
"last_seen_index": 17,
"first_seen_at": "2026-05-02T06:54:23.082460+00:00",
"last_seen_at": "2026-05-02T07:08:17.240100+00:00",
"seen_count": 5,
"miss_count": 3,
"stale": true,
"cached": true
},
"input:username or email": {
  "element_key": "input:username or email",
  "element_type": "input",
  "label": "Username or email",
  "bbox": [
    728.0,
    336.0,
```



```
    1192.0,  
    376.0  
  ],  
  "confidence": 0.98,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 1,  
  "last_seen_index": 20,  
  "first_seen_at": "2026-05-02T06:54:23.082460+00:00",  
  "last_seen_at": "2026-05-02T07:11:06.085051+00:00",  
  "seen_count": 13,  
  "miss_count": 0,  
  "stale": false,  
  "cached": false  
},  
"input:password": {  
  "element_key": "input:password",  
  "element_type": "input",  
  "label": "Password",  
  "bbox": [  
    728.0,  
    384.0,  
    1192.0,  
    416.0  
  ],  
  "confidence": 0.98,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,
```

```
"last_seen_frame": null,
"first_seen_index": 1,
"last_seen_index": 20,
"first_seen_at": "2026-05-02T06:54:23.082460+00:00",
"last_seen_at": "2026-05-02T07:11:06.085051+00:00",
"seen_count": 11,
"miss_count": 0,
"stale": false,
"cached": false
},
"button:log in": {
  "element_key": "button:log in",
  "element_type": "button",
  "label": "Log in",
  "bbox": [
    880.0,
    424.0,
    1040.0,
    472.0
  ],
  "confidence": 0.99,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
  "last_seen_index": 20,
  "first_seen_at": "2026-05-02T06:54:23.082460+00:00",
  "last_seen_at": "2026-05-02T07:11:06.085051+00:00",
  "seen_count": 10,
```

```
    "miss_count": 0,
    "stale": false,
    "cached": false
  },
  "link:lost password?": {
    "element_key": "link:lost password?",
    "element_type": "link",
    "label": "Lost password?",
    "bbox": [
      920.0,
      480.0,
      1008.0,
      504.0
    ],
    "confidence": 0.95,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 1,
    "last_seen_index": 20,
    "first_seen_at": "2026-05-02T06:54:23.082460+00:00",
    "last_seen_at": "2026-05-02T07:11:06.085051+00:00",
    "seen_count": 6,
    "miss_count": 0,
    "stale": false,
    "cached": false
  },
  "text:log in using your account on:": {
    "element_key": "text:log in using your account on:",
```

```
"element_type": "text",
"label": "Log in using your account on:",
"bbox": [
    856.0,
    534.0,
    1064.0,
    558.0
],
"confidence": 0.729,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 1,
"last_seen_index": 18,
"first_seen_at": "2026-05-02T06:54:23.082460+00:00",
"last_seen_at": "2026-05-02T07:09:07.409295+00:00",
"seen_count": 3,
"miss_count": 2,
"stale": true,
"cached": true
},
"button:sliit login": {
    "element_key": "button:sliit login",
    "element_type": "button",
    "label": "SLIIT Login",
    "bbox": [
        912.0,
        566.0,
        1008.0,
```

```
590.0
],
"confidence": 0.7614,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 1,
"last_seen_index": 18,
"first_seen_at": "2026-05-02T06:54:23.082460+00:00",
"last_seen_at": "2026-05-02T07:09:07.409295+00:00",
"seen_count": 9,
"miss_count": 2,
"stale": true,
"cached": true
},
"text:is this your first time here?": {
  "element_key": "text:is this your first time here?",
  "element_type": "text",
  "label": "Is this your first time here?",
  "bbox": [
    842.0,
    734.0,
    1074.0,
    766.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
```

```
"first_seen_index": 1,
"last_seen_index": 5,
"first_seen_at": "2026-05-02T06:54:23.082460+00:00",
"last_seen_at": "2026-05-02T06:57:27.809908+00:00",
"seen_count": 3,
"miss_count": 15,
"stale": true,
"cached": true
},
"pane:instructions": {
  "element_key": "pane:instructions",
  "element_type": "pane",
  "label": "Instructions",
  "bbox": [
    714.0,
    606.0,
    1210.0,
    962.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
  "last_seen_index": 1,
  "first_seen_at": "2026-05-02T06:54:23.082460+00:00",
  "last_seen_at": "2026-05-02T06:54:23.082460+00:00",
  "seen_count": 1,
  "miss_count": 19,
```

```
    "stale": true,
    "cached": true
  },
  "toolbar:taskbar": {
    "element_key": "toolbar:taskbar",
    "element_type": "toolbar",
    "label": "Taskbar",
    "bbox": [
      160.0,
      934.0,
      1744.0,
      962.0
    ],
    "confidence": 0.5315,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 1,
    "last_seen_index": 15,
    "first_seen_at": "2026-05-02T06:54:23.082460+00:00",
    "last_seen_at": "2026-05-02T07:06:29.451301+00:00",
    "seen_count": 5,
    "miss_count": 5,
    "stale": true,
    "cached": true
  },
  "button:start": {
    "element_key": "button:start",
    "element_type": "button",
```

```
"label": "Start",
"bbox": [
  881.0,
  943.0,
  905.0,
  962.0
],
"confidence": 0.6885,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 1,
"last_seen_index": 18,
"first_seen_at": "2026-05-02T06:54:23.082460+00:00",
"last_seen_at": "2026-05-02T07:09:07.409295+00:00",
"seen_count": 5,
"miss_count": 2,
"stale": true,
"cached": true
},
"button:search": {
  "element_key": "button:search",
  "element_type": "button",
  "label": "Search",
  "bbox": [
    905.0,
    943.0,
    929.0,
    962.0
  ]
}
```



```
],
  "confidence": 0.6885,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
  "last_seen_index": 18,
  "first_seen_at": "2026-05-02T06:54:23.082460+00:00",
  "last_seen_at": "2026-05-02T07:09:07.409295+00:00",
  "seen_count": 6,
  "miss_count": 2,
  "stale": true,
  "cached": true
},
"image:sliit uni courseweb logo": {
  "element_key": "image:sliit uni courseweb logo",
  "element_type": "image",
  "label": "SLIIT UNI CourseWeb Logo",
  "bbox": [
    728.0,
    270.0,
    1192.0,
    326.0
  ],
  "confidence": 0.98,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 2,
```

```
"last_seen_index": 20,  
"first_seen_at": "2026-05-02T06:54:57.595730+00:00",  
"last_seen_at": "2026-05-02T07:11:06.085051+00:00",  
"seen_count": 10,  
"miss_count": 0,  
"stale": false,  
"cached": false  
},  
"pane:onedrive notification": {  
  "element_key": "pane:onedrive notification",  
  "element_type": "pane",  
  "label": "OneDrive Notification",  
  "bbox": [  
    1338.0,  
    198.0,  
    1758.0,  
    414.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 2,  
  "last_seen_index": 2,  
  "first_seen_at": "2026-05-02T06:54:57.595730+00:00",  
  "last_seen_at": "2026-05-02T06:54:57.595730+00:00",  
  "seen_count": 1,  
  "miss_count": 18,  
  "stale": true,
```

```
    "cached": true
  },
  "pane:google chrome notification": {
    "element_key": "pane:google chrome notification",
    "element_type": "pane",
    "label": "Google Chrome Notification",
    "bbox": [
      1338.0,
      366.0,
      1758.0,
      598.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 2,
    "last_seen_index": 2,
    "first_seen_at": "2026-05-02T06:54:57.595730+00:00",
    "last_seen_at": "2026-05-02T06:54:57.595730+00:00",
    "seen_count": 1,
    "miss_count": 18,
    "stale": true,
    "cached": true
  },
  "pane:calendar widget": {
    "element_key": "pane:calendar widget",
    "element_type": "pane",
    "label": "Calendar Widget",
```

```
"bbox": [
  1338.0,
  534.0,
  1758.0,
  961.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 2,
"last_seen_index": 2,
"first_seen_at": "2026-05-02T06:54:57.595730+00:00",
"last_seen_at": "2026-05-02T06:54:57.595730+00:00",
"seen_count": 1,
"miss_count": 18,
"stale": true,
"cached": true
},
"text:12:24 pm": {
  "element_key": "text:12:24 pm",
  "element_type": "text",
  "label": "12:24 PM",
  "bbox": [
    1666.0,
    918.0,
    1758.0,
    961.0
  ],

```

```
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 2,
"last_seen_index": 2,
"first_seen_at": "2026-05-02T06:54:57.595730+00:00",
"last_seen_at": "2026-05-02T06:54:57.595730+00:00",
"seen_count": 1,
"miss_count": 18,
"stale": true,
"cached": true
},
"link:forgotten password?": {
  "element_key": "link:forgotten password?",
  "element_type": "link",
  "label": "Forgotten password?",
  "bbox": [
    929.0,
    486.0,
    1001.0,
    502.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 3,
  "last_seen_index": 3,
```

```
"first_seen_at": "2026-05-02T06:55:40.663306+00:00",
"last_seen_at": "2026-05-02T06:55:40.663306+00:00",
"seen_count": 1,
"miss_count": 17,
"stale": true,
"cached": true
},
"pane:instruction box": {
  "element_key": "pane:instruction box",
  "element_type": "pane",
  "label": "Instruction Box",
  "bbox": [
    712.0,
    630.0,
    1208.0,
    962.0
  ],
  "confidence": 0.4134,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 3,
  "last_seen_index": 15,
  "first_seen_at": "2026-05-02T06:55:40.663306+00:00",
  "last_seen_at": "2026-05-02T07:06:29.451301+00:00",
  "seen_count": 2,
  "miss_count": 5,
  "stale": true,
  "cached": true
}
```

```
},
"button:windows start": {
  "element_key": "button:windows start",
  "element_type": "button",
  "label": "Windows Start",
  "bbox": [
    569.0,
    942.0,
    601.0,
    962.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 3,
  "last_seen_index": 3,
  "first_seen_at": "2026-05-02T06:55:40.663306+00:00",
  "last_seen_at": "2026-05-02T06:55:40.663306+00:00",
  "seen_count": 1,
  "miss_count": 17,
  "stale": true,
  "cached": true
},
"button:task view": {
  "element_key": "button:task view",
  "element_type": "button",
  "label": "Task View",
  "bbox": [
```

```
    929.0,
    943.0,
    961.0,
    962.0
  ],
  "confidence": 0.5832,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 3,
  "last_seen_index": 17,
  "first_seen_at": "2026-05-02T06:55:40.663306+00:00",
  "last_seen_at": "2026-05-02T07:08:17.240100+00:00",
  "seen_count": 3,
  "miss_count": 3,
  "stale": true,
  "cached": true
},
"button:file explorer": {
  "element_key": "button:file explorer",
  "element_type": "button",
  "label": "File Explorer",
  "bbox": [
    977.0,
    943.0,
    1009.0,
    962.0
  ],
  "confidence": 0.6197,
```



```
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 3,
"last_seen_index": 17,
"first_seen_at": "2026-05-02T06:55:40.663306+00:00",
"last_seen_at": "2026-05-02T07:08:17.240100+00:00",
"seen_count": 3,
"miss_count": 3,
"stale": true,
"cached": true
},
"button:google chrome": {
  "element_key": "button:google chrome",
  "element_type": "button",
  "label": "Google Chrome",
  "bbox": [
    936.0,
    942.0,
    976.0,
    962.0
  ],
  "confidence": 0.6885,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 3,
  "last_seen_index": 18,
  "first_seen_at": "2026-05-02T06:55:40.663306+00:00",
```

```
"last_seen_at": "2026-05-02T07:09:07.409295+00:00",
"seen_count": 2,
"miss_count": 2,
"stale": true,
"cached": true
},
"button:spotify": {
  "element_key": "button:spotify",
  "element_type": "button",
  "label": "Spotify",
  "bbox": [
    1232.0,
    942.0,
    1264.0,
    962.0
  ],
  "confidence": 0.6885,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 3,
  "last_seen_index": 18,
  "first_seen_at": "2026-05-02T06:55:40.663306+00:00",
  "last_seen_at": "2026-05-02T07:09:07.409295+00:00",
  "seen_count": 3,
  "miss_count": 2,
  "stale": true,
  "cached": true
},
```

```
"text:log in using your account on": {
  "element_key": "text:log in using your account on",
  "element_type": "text",
  "label": "Log in using your account on",
  "bbox": [
    866.0,
    535.0,
    1058.0,
    559.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 4,
  "last_seen_index": 4,
  "first_seen_at": "2026-05-02T06:56:39.079159+00:00",
  "last_seen_at": "2026-05-02T06:56:39.079159+00:00",
  "seen_count": 1,
  "miss_count": 16,
  "stale": true,
  "cached": true
},
"pane:spotify": {
  "element_key": "pane:spotify",
  "element_type": "pane",
  "label": "Spotify",
  "bbox": [
    1354.0,
```

```
    327.0,  
    1758.0,  
    559.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 4,  
  "last_seen_index": 4,  
  "first_seen_at": "2026-05-02T06:56:39.079159+00:00",  
  "last_seen_at": "2026-05-02T06:56:39.079159+00:00",  
  "seen_count": 1,  
  "miss_count": 16,  
  "stale": true,  
  "cached": true  
},  
"text:karma (the cycle turns)": {  
  "element_key": "text:karma (the cycle turns)",  
  "element_type": "text",  
  "label": "Karma (The Cycle Turns)",  
  "bbox": [  
    1362.0,  
    542.0,  
    1634.0,  
    582.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",
```

```
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 4,
"last_seen_index": 5,
"first_seen_at": "2026-05-02T06:56:39.079159+00:00",
"last_seen_at": "2026-05-02T06:57:27.809908+00:00",
"seen_count": 2,
"miss_count": 15,
"stale": true,
"cached": true
},
"text:sumedh k": {
  "element_key": "text:sumedh k",
  "element_type": "text",
  "label": "Sumedh K",
  "bbox": [
    1370.0,
    574.0,
    1490.0,
    598.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 4,
  "last_seen_index": 5,
  "first_seen_at": "2026-05-02T06:56:39.079159+00:00",
  "last_seen_at": "2026-05-02T06:57:27.809908+00:00",
```

```
"seen_count": 2,
"miss_count": 15,
"stale": true,
"cached": true
},
"pane:login form": {
  "element_key": "pane:login form",
  "element_type": "pane",
  "label": "Login Form",
  "bbox": [
    666.0,
    198.0,
    1258.0,
    962.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 5,
  "last_seen_index": 5,
  "first_seen_at": "2026-05-02T06:57:27.809908+00:00",
  "last_seen_at": "2026-05-02T06:57:27.809908+00:00",
  "seen_count": 1,
  "miss_count": 15,
  "stale": true,
  "cached": true
},
"pane:spotify widget": {
```

```
"element_key": "pane:spotify widget",
"element_type": "pane",
"label": "Spotify Widget",
"bbox": [
    1346.0,
    406.0,
    1758.0,
    790.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 5,
"last_seen_index": 5,
"first_seen_at": "2026-05-02T06:57:27.809908+00:00",
"last_seen_at": "2026-05-02T06:57:27.809908+00:00",
"seen_count": 1,
"miss_count": 15,
"stale": true,
"cached": true
},
"button:play/pause": {
    "element_key": "button:play/pause",
    "element_type": "button",
    "label": "Play/Pause",
    "bbox": [
        1530.0,
        654.0,
```

```
    1570.0,  
    694.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 5,  
  "last_seen_index": 5,  
  "first_seen_at": "2026-05-02T06:57:27.809908+00:00",  
  "last_seen_at": "2026-05-02T06:57:27.809908+00:00",  
  "seen_count": 1,  
  "miss_count": 15,  
  "stale": true,  
  "cached": true  
},  
"pane:quick settings panel": {  
  "element_key": "pane:quick settings panel",  
  "element_type": "pane",  
  "label": "Quick Settings Panel",  
  "bbox": [  
    1298.0,  
    734.0,  
    1758.0,  
    950.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,
```



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"last_seen_frame": null,
"first_seen_index": 5,
"last_seen_index": 5,
"first_seen_at": "2026-05-02T06:57:27.809908+00:00",
"last_seen_at": "2026-05-02T06:57:27.809908+00:00",
"seen_count": 1,
"miss_count": 15,
"stale": true,
"cached": true
},
"button:slt-fiber-7e76": {
  "element_key": "button:slt-fiber-7e76",
  "element_type": "button",
  "label": "SLT-Fiber-7E76",
  "bbox": [
    1362.0,
    782.0,
    1506.0,
    886.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 5,
  "last_seen_index": 5,
  "first_seen_at": "2026-05-02T06:57:27.809908+00:00",
  "last_seen_at": "2026-05-02T06:57:27.809908+00:00",
  "seen_count": 1,
```

```
    "miss_count": 15,
    "stale": true,
    "cached": true
  },
  "button:bluetooth": {
    "element_key": "button:bluetooth",
    "element_type": "button",
    "label": "Bluetooth",
    "bbox": [
      1482.0,
      782.0,
      1634.0,
      878.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 5,
    "last_seen_index": 5,
    "first_seen_at": "2026-05-02T06:57:27.809908+00:00",
    "last_seen_at": "2026-05-02T06:57:27.809908+00:00",
    "seen_count": 1,
    "miss_count": 15,
    "stale": true,
    "cached": true
  },
  "button:airplane mode": {
    "element_key": "button:airplane mode",
```

```
"element_type": "button",
"label": "Airplane mode",
"bbox": [
    1610.0,
    782.0,
    1730.0,
    886.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 5,
"last_seen_index": 5,
"first_seen_at": "2026-05-02T06:57:27.809908+00:00",
"last_seen_at": "2026-05-02T06:57:27.809908+00:00",
"seen_count": 1,
"miss_count": 15,
"stale": true,
"cached": true
},
"slider:brightness": {
    "element_key": "slider:brightness",
    "element_type": "slider",
    "label": "Brightness",
    "bbox": [
        1384.0,
        961.0,
        1696.0,
```

```
    962.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 5,
  "last_seen_index": 5,
  "first_seen_at": "2026-05-02T06:57:27.809908+00:00",
  "last_seen_at": "2026-05-02T06:57:27.809908+00:00",
  "seen_count": 1,
  "miss_count": 15,
  "stale": true,
  "cached": true
},
"slider:volume": {
  "element_key": "slider:volume",
  "element_type": "slider",
  "label": "Volume",
  "bbox": [
    1384.0,
    961.0,
    1696.0,
    962.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
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"first_seen_index": 5,
"last_seen_index": 5,
"first_seen_at": "2026-05-02T06:57:27.809908+00:00",
"last_seen_at": "2026-05-02T06:57:27.809908+00:00",
"seen_count": 1,
"miss_count": 15,
"stale": true,
"cached": true
},
"text:93%": {
  "element_key": "text:93%",
  "element_type": "text",
  "label": "93%",
  "bbox": [
    1378.0,
    961.0,
    1421.0,
    962.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 5,
  "last_seen_index": 5,
  "first_seen_at": "2026-05-02T06:57:27.809908+00:00",
  "last_seen_at": "2026-05-02T06:57:27.809908+00:00",
  "seen_count": 1,
  "miss_count": 15,
```

```
"stale": true,
"cached": true
},
"text:12:27 pm": {
  "element_key": "text:12:27 pm",
  "element_type": "text",
  "label": "12:27 PM",
  "bbox": [
    1666.0,
    926.0,
    1746.0,
    962.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 5,
  "last_seen_index": 5,
  "first_seen_at": "2026-05-02T06:57:27.809908+00:00",
  "last_seen_at": "2026-05-02T06:57:27.809908+00:00",
  "seen_count": 1,
  "miss_count": 15,
  "stale": true,
  "cached": true
},
"text:click on sliit login to proceed": {
  "element_key": "text:click on sliit login to proceed",
  "element_type": "text",
```

```
"label": "Click on SLIIT Login to proceed",
"bbox": [
  770.0,
  846.0,
  1154.0,
  886.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 5,
"last_seen_index": 5,
"first_seen_at": "2026-05-02T06:57:27.809908+00:00",
"last_seen_at": "2026-05-02T06:57:27.809908+00:00",
"seen_count": 1,
"miss_count": 15,
"stale": true,
"cached": true
},
"text:sliit courseweb": {
  "element_key": "text:sliit courseweb",
  "element_type": "text",
  "label": "SLIIT CourseWeb",
  "bbox": [
    770.0,
    326.0,
    1154.0,
    382.0
  ]
}
```

```
],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 6,  
  "last_seen_index": 6,  
  "first_seen_at": "2026-05-02T06:58:19.648928+00:00",  
  "last_seen_at": "2026-05-02T06:58:19.648928+00:00",  
  "seen_count": 1,  
  "miss_count": 14,  
  "stale": true,  
  "cached": true  
},  
  "label:log in using your account on:": {  
    "element_key": "label:log in using your account on:",  
    "element_type": "label",  
    "label": "Log in using your account on:",  
    "bbox": [  
      864.0,  
      542.0,  
      1064.0,  
      574.0  
    ],  
    "confidence": 0.4724,  
    "source": "gemini_vlm",  
    "first_seen_frame": null,  
    "last_seen_frame": null,  
    "first_seen_index": 7,
```



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"last_seen_index": 15,  
"first_seen_at": "2026-05-02T06:59:07.402959+00:00",  
"last_seen_at": "2026-05-02T07:06:29.451301+00:00",  
"seen_count": 2,  
"miss_count": 5,  
"stale": true,  
"cached": true  
},  
"label:is this your first time here?": {  
  "element_key": "label:is this your first time here?",  
  "element_type": "label",  
  "label": "Is this your first time here?",  
  "bbox": [  
    840.0,  
    638.0,  
    1088.0,  
    662.0  
  ],  
  "confidence": 0.4724,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 7,  
  "last_seen_index": 15,  
  "first_seen_at": "2026-05-02T06:59:07.402959+00:00",  
  "last_seen_at": "2026-05-02T07:06:29.451301+00:00",  
  "seen_count": 2,  
  "miss_count": 5,  
  "stale": true,
```

```
    "cached": true
  },
  "text:instructional text": {
    "element_key": "text:instructional text",
    "element_type": "text",
    "label": "Instructional text",
    "bbox": [
      714.0,
      654.0,
      1210.0,
      962.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 7,
    "last_seen_index": 7,
    "first_seen_at": "2026-05-02T06:59:07.402959+00:00",
    "last_seen_at": "2026-05-02T06:59:07.402959+00:00",
    "seen_count": 1,
    "miss_count": 13,
    "stale": true,
    "cached": true
  },
  "text:system time": {
    "element_key": "text:system time",
    "element_type": "text",
    "label": "System Time",
```

```
"bbox": [
  1650.0,
  942.0,
  1754.0,
  962.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 7,
"last_seen_index": 7,
"first_seen_at": "2026-05-02T06:59:07.402959+00:00",
"last_seen_at": "2026-05-02T06:59:07.402959+00:00",
"seen_count": 1,
"miss_count": 13,
"stale": true,
"cached": true
},
"heading:slit uni | courseweb": {
  "element_key": "heading:slit uni | courseweb",
  "element_type": "heading",
  "label": "SLIIT UNI | CourseWeb",
  "bbox": [
    729.0,
    270.0,
    1201.0,
    358.0
  ],

```

```
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 8,
"last_seen_index": 8,
"first_seen_at": "2026-05-02T07:00:05.444567+00:00",
"last_seen_at": "2026-05-02T07:00:05.444567+00:00",
"seen_count": 1,
"miss_count": 12,
"stale": true,
"cached": true
},
"tab:frontend": {
  "element_key": "tab:frontend",
  "element_type": "tab",
  "label": "frontend",
  "bbox": [
    154.0,
    110.0,
    394.0,
    166.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 11,
  "last_seen_index": 11,
```

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"first_seen_at": "2026-05-02T07:03:12.856846+00:00",
"last_seen_at": "2026-05-02T07:03:12.856846+00:00",
"seen_count": 1,
"miss_count": 9,
"stale": true,
"cached": true
},
"button:microsoft edge": {
  "element_key": "button:microsoft edge",
  "element_type": "button",
  "label": "Microsoft Edge",
  "bbox": [
    1001.0,
    943.0,
    1033.0,
    962.0
  ],
  "confidence": 0.6197,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 12,
  "last_seen_index": 17,
  "first_seen_at": "2026-05-02T07:03:47.787802+00:00",
  "last_seen_at": "2026-05-02T07:08:17.240100+00:00",
  "seen_count": 2,
  "miss_count": 3,
  "stale": true,
  "cached": true
}
```

```
},
"button:microsoft store": {
  "element_key": "button:microsoft store",
  "element_type": "button",
  "label": "Microsoft Store",
  "bbox": [
    1025.0,
    943.0,
    1057.0,
    962.0
  ],
  "confidence": 0.6197,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 12,
  "last_seen_index": 17,
  "first_seen_at": "2026-05-02T07:03:47.787802+00:00",
  "last_seen_at": "2026-05-02T07:08:17.240100+00:00",
  "seen_count": 2,
  "miss_count": 3,
  "stale": true,
  "cached": true
},
"button:mail": {
  "element_key": "button:mail",
  "element_type": "button",
  "label": "Mail",
  "bbox": [
```

```
    928.0,
    942.0,
    968.0,
    962.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 12,
  "last_seen_index": 12,
  "first_seen_at": "2026-05-02T07:03:47.787802+00:00",
  "last_seen_at": "2026-05-02T07:03:47.787802+00:00",
  "seen_count": 1,
  "miss_count": 8,
  "stale": true,
  "cached": true
},
"button:visual studio code": {
  "element_key": "button:visual studio code",
  "element_type": "button",
  "label": "Visual Studio Code",
  "bbox": [
    1184.0,
    942.0,
    1224.0,
    962.0
  ],
  "confidence": 0.6885,
```

```
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 12,
"last_seen_index": 18,
"first_seen_at": "2026-05-02T07:03:47.787802+00:00",
"last_seen_at": "2026-05-02T07:09:07.409295+00:00",
"seen_count": 2,
"miss_count": 2,
"stale": true,
"cached": true
},
"button:discord": {
  "element_key": "button:discord",
  "element_type": "button",
  "label": "Discord",
  "bbox": [
    1024.0,
    942.0,
    1064.0,
    962.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 12,
  "last_seen_index": 12,
  "first_seen_at": "2026-05-02T07:03:47.787802+00:00",
```



```
"last_seen_at": "2026-05-02T07:03:47.787802+00:00",
"seen_count": 1,
"miss_count": 8,
"stale": true,
"cached": true
},
"button:steam": {
  "element_key": "button:steam",
  "element_type": "button",
  "label": "Steam",
  "bbox": [
    1056.0,
    942.0,
    1096.0,
    962.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 12,
  "last_seen_index": 12,
  "first_seen_at": "2026-05-02T07:03:47.787802+00:00",
  "last_seen_at": "2026-05-02T07:03:47.787802+00:00",
  "seen_count": 1,
  "miss_count": 8,
  "stale": true,
  "cached": true
},
```

```
"button:system tray": {
  "element_key": "button:system tray",
  "element_type": "button",
  "label": "System Tray",
  "bbox": [
    1088.0,
    942.0,
    1120.0,
    962.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 12,
  "last_seen_index": 12,
  "first_seen_at": "2026-05-02T07:03:47.787802+00:00",
  "last_seen_at": "2026-05-02T07:03:47.787802+00:00",
  "seen_count": 1,
  "miss_count": 8,
  "stale": true,
  "cached": true
},
"text:clock": {
  "element_key": "text:clock",
  "element_type": "text",
  "label": "Clock",
  "bbox": [
    1648.0,
```

```
    934.0,  
    1728.0,  
    962.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 12,  
  "last_seen_index": 12,  
  "first_seen_at": "2026-05-02T07:03:47.787802+00:00",  
  "last_seen_at": "2026-05-02T07:03:47.787802+00:00",  
  "seen_count": 1,  
  "miss_count": 8,  
  "stale": true,  
  "cached": true  
},  
"scrollbar:scrollbar": {  
  "element_key": "scrollbar:scrollbar",  
  "element_type": "scrollbar",  
  "label": "Scrollbar",  
  "bbox": [  
    1720.0,  
    198.0,  
    1744.0,  
    782.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",
```

```
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 12,
"last_seen_index": 12,
"first_seen_at": "2026-05-02T07:03:47.787802+00:00",
"last_seen_at": "2026-05-02T07:03:47.787802+00:00",
"seen_count": 1,
"miss_count": 8,
"stale": true,
"cached": true
},
"heading:sliit uni": {
  "element_key": "heading:sliit uni",
  "element_type": "heading",
  "label": "SLIIT UNI",
  "bbox": [
    769.0,
    278.0,
    937.0,
    310.0
  ],
  "confidence": 0.4783,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 14,
  "last_seen_index": 14,
  "first_seen_at": "2026-05-02T07:05:44.869252+00:00",
  "last_seen_at": "2026-05-02T07:05:44.869252+00:00",
```

```
"seen_count": 1,
"miss_count": 6,
"stale": true,
"cached": true
},
"heading:is this your first time here?": {
  "element_key": "heading:is this your first time here?",
  "element_type": "heading",
  "label": "Is this your first time here?",
  "bbox": [
    864.0,
    630.0,
    1056.0,
    654.0
  ],
  "confidence": 0.729,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 17,
  "last_seen_index": 18,
  "first_seen_at": "2026-05-02T07:08:17.240100+00:00",
  "last_seen_at": "2026-05-02T07:09:07.409295+00:00",
  "seen_count": 2,
  "miss_count": 2,
  "stale": true,
  "cached": true
},
"text:instructional text block": {
```

```
"element_key": "text:instructional text block",
"element_type": "text",
"label": "Instructional text block",
"bbox": [
  713.0,
  607.0,
  1209.0,
  962.0
],
"confidence": 0.5468,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 17,
"last_seen_index": 17,
"first_seen_at": "2026-05-02T07:08:17.240100+00:00",
"last_seen_at": "2026-05-02T07:08:17.240100+00:00",
"seen_count": 1,
"miss_count": 3,
"stale": true,
"cached": true
},
"button:chat": {
  "element_key": "button:chat",
  "element_type": "button",
  "label": "Chat",
  "bbox": [
    953.0,
    943.0,
```

```
    985.0,  
    962.0  
  ],  
  "confidence": 0.5832,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 17,  
  "last_seen_index": 17,  
  "first_seen_at": "2026-05-02T07:08:17.240100+00:00",  
  "last_seen_at": "2026-05-02T07:08:17.240100+00:00",  
  "seen_count": 1,  
  "miss_count": 3,  
  "stale": true,  
  "cached": true  
},  
"text:clock and date": {  
  "element_key": "text:clock and date",  
  "element_type": "text",  
  "label": "Clock and Date",  
  "bbox": [  
    1657.0,  
    943.0,  
    1721.0,  
    962.0  
  ],  
  "confidence": 0.6561,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,
```

```
"last_seen_frame": null,
"first_seen_index": 17,
"last_seen_index": 17,
"first_seen_at": "2026-05-02T07:08:17.240100+00:00",
"last_seen_at": "2026-05-02T07:08:17.240100+00:00",
"seen_count": 1,
"miss_count": 3,
"stale": true,
"cached": true
},
"pane:login instructions": {
  "element_key": "pane:login instructions",
  "element_type": "pane",
  "label": "Login instructions",
  "bbox": [
    712.0,
    606.0,
    1208.0,
    962.0
  ],
  "confidence": 0.6885,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 18,
  "last_seen_index": 18,
  "first_seen_at": "2026-05-02T07:09:07.409295+00:00",
  "last_seen_at": "2026-05-02T07:09:07.409295+00:00",
  "seen_count": 1,
```



```
    "miss_count": 2,
    "stale": true,
    "cached": true
  },
  "taskbar:windows taskbar": {
    "element_key": "taskbar:windows taskbar",
    "element_type": "taskbar",
    "label": "Windows Taskbar",
    "bbox": [
      176.0,
      942.0,
      1744.0,
      962.0
    ],
    "confidence": 0.8019,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 18,
    "last_seen_index": 18,
    "first_seen_at": "2026-05-02T07:09:07.409295+00:00",
    "last_seen_at": "2026-05-02T07:09:07.409295+00:00",
    "seen_count": 1,
    "miss_count": 2,
    "stale": true,
    "cached": true
  },
  "icon:wi-fi": {
    "element_key": "icon:wi-fi",
```

```
"element_type": "icon",
"label": "Wi-Fi",
"bbox": [
    1592.0,
    942.0,
    1616.0,
    962.0
],
"confidence": 0.6075,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 18,
"last_seen_index": 18,
"first_seen_at": "2026-05-02T07:09:07.409295+00:00",
"last_seen_at": "2026-05-02T07:09:07.409295+00:00",
"seen_count": 1,
"miss_count": 2,
"stale": true,
"cached": true
},
"icon:battery": {
    "element_key": "icon:battery",
    "element_type": "icon",
    "label": "Battery",
    "bbox": [
        1640.0,
        942.0,
        1672.0,
```

```
    962.0
  ],
  "confidence": 0.6075,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 18,
  "last_seen_index": 18,
  "first_seen_at": "2026-05-02T07:09:07.409295+00:00",
  "last_seen_at": "2026-05-02T07:09:07.409295+00:00",
  "seen_count": 1,
  "miss_count": 2,
  "stale": true,
  "cached": true
},
"text:12:38 pm": {
  "element_key": "text:12:38 pm",
  "element_type": "text",
  "label": "12:38 PM",
  "bbox": [
    1648.0,
    942.0,
    1744.0,
    962.0
  ],
  "confidence": 0.7695,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
```

```
    "first_seen_index": 18,  
    "last_seen_index": 18,  
    "first_seen_at": "2026-05-02T07:09:07.409295+00:00",  
    "last_seen_at": "2026-05-02T07:09:07.409295+00:00",  
    "seen_count": 1,  
    "miss_count": 2,  
    "stale": true,  
    "cached": true  
  }  
},  
"current_elements": [],  
"resolved_elements": [  
  {  
    "id": "memory_input:username or email",  
    "type": "input",  
    "label": "Username or email",  
    "description": "Cached from an earlier frame in the same run",  
    "state": "cached",  
    "bbox": [  
      728.0,  
      336.0,  
      1192.0,  
      376.0  
    ],  
    "confidence": 0.98,  
    "source": "gemini_vlm",  
    "last_seen_frame": null,  
    "last_seen_index": 20,  
    "stale": true,
```

```
"cached": true,
"session_memory_key": "input:username or email"
},
{
  "id": "memory_input:password",
  "type": "input",
  "label": "Password",
  "description": "Cached from an earlier frame in the same run",
  "state": "cached",
  "bbox": [
    728.0,
    384.0,
    1192.0,
    416.0
  ],
  "confidence": 0.98,
  "source": "gemini_vlm",
  "last_seen_frame": null,
  "last_seen_index": 20,
  "stale": true,
  "cached": true,
  "session_memory_key": "input:password"
},
{
  "id": "memory_button:log in",
  "type": "button",
  "label": "Log in",
  "description": "Cached from an earlier frame in the same run",
  "state": "cached",
```

```
"bbox": [
  880.0,
  424.0,
  1040.0,
  472.0
],
"confidence": 0.99,
"source": "gemini_vlm",
"last_seen_frame": null,
"last_seen_index": 20,
"stale": true,
"cached": true,
"session_memory_key": "button:log in"
},
{
  "id": "memory_link:lost password?",
  "type": "link",
  "label": "Lost password?",
  "description": "Cached from an earlier frame in the same run",
  "state": "cached",
  "bbox": [
    920.0,
    480.0,
    1008.0,
    504.0
  ],
  "confidence": 0.95,
  "source": "gemini_vlm",
  "last_seen_frame": null,
```

```
    "last_seen_index": 20,  
    "stale": true,  
    "cached": true,  
    "session_memory_key": "link:lost password?"  
  },  
  {  
    "id": "memory_image:sliit uni courseweb logo",  
    "type": "image",  
    "label": "SLIIT UNI CourseWeb Logo",  
    "description": "Cached from an earlier frame in the same run",  
    "state": "cached",  
    "bbox": [  
      728.0,  
      270.0,  
      1192.0,  
      326.0  
    ],  
    "confidence": 0.98,  
    "source": "gemini_vlm",  
    "last_seen_frame": null,  
    "last_seen_index": 20,  
    "stale": true,  
    "cached": true,  
    "session_memory_key": "image:sliit uni courseweb logo"  
  }  
],  
  "screen_data": {}  
}  
}
```

Test Case 10: YouTube Navigation (Complex Task)

User Prompt:

Click on new tab and find the youtube, if you can't find that icon search it in the search bar. however needs to re-direct to the youtube platform

Test Results:

- Steps Total: 2
- Completed: 1
- Failed: 1
- Duration: 12 minutes 19 seconds

Step Execution:

Step 1: Click on the 'New tab' tab to switch to the existing empty browser tab

- Target: Tab 'New tab' at (372, 68)
- Attempts: 1
- Execution: Performed
- Evaluation: Success
- Issue: None

Step 2: Click the address bar, type '<https://www.youtube.com>', and press the Enter key to navigate to the platform

- Target: Address bar at (616, 120)
- Attempts: 3
- Execution: Performed
- Evaluation: Failed (retry loop)
- Issue: The Microsoft Store window obstructed the browser interface. The address bar could not be located or accessed in the final screen state.

Execution Summary:

The agent successfully switched to the new tab but failed to navigate to YouTube. After switching tabs, the Microsoft Store application became the foreground window, obstructing the browser and preventing access to the address bar. The evaluation encountered a retry loop without resolution.

Issues Identified:

- Microsoft Store window obstructing the browser interface

- Address bar not detected or accessible in the final screen state
- No mechanism to bring the browser window to foreground before interaction
- Possible unintended application launch or focus shift during execution
- Evaluation loop unable to recover from obstructed UI state

Recommendations for Improvement:

- Add pre-action check to ensure correct window is in focus
- Implement window focus/bring-to-front command before UI interaction
- Add fallback to close or minimize obstructing windows if detected
- Increase timeout after tab switch before attempting address bar interaction
- Improve obstruction detection to trigger recovery actions automatically

JSON Response:

full_cycle.json

```
{
  "run_id": "769bda88-628b-4be2-875a-1b6265d5e8bc",
  "user_task": "Click on new tab and find the youtube, if you can't find that icon search it in the search bar. however needs to re-direct to the youtube platform",
  "started_at": "2026-05-02T16:32:30.127424+00:00",
  "finished_at": "2026-05-02T16:44:49.255903+00:00",
  "vision_used": true,
  "perception": {
    "image": "E:\\sliit-project\\ScreenAwareTaskAgent\\vision\\data\\sessions\\shot_1777739562_bcb8e7\\processing\\proc_1777739563_7a9e13\\preprocessed_frames\\frame_1777739562_d6dfdf.jpg",
    "image_size": {
      "width": 1920,
      "height": 1080
    },
    "screen_bbox": [
      115,
      65,
```

```
1805,  
1015  
],  
"screen_size": {  
  "width": 1690,  
  "height": 950  
},  
"coordinate_system": "pixel",  
"elements": [  
  {  
    "id": "tab_frontend",  
    "type": "tab",  
    "label": "frontend",  
    "description": "Browser tab for the frontend application",  
    "state": "active",  
    "dx": 188,  
    "dy": 68,  
    "frame_dx": 303,  
    "frame_dy": 133,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
      179,  
      105,  
      427,  
      161  
    ],  
    "session_memory_key": "tab:frontend",  
    "cached": false
```

```
},  
{  
  "id": "tab_new_tab",  
  "type": "tab",  
  "label": "New tab",  
  "description": "Browser tab for a new page",  
  "state": "inactive",  
  "dx": 372,  
  "dy": 68,  
  "frame_dx": 487,  
  "frame_dy": 133,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    371,  
    105,  
    603,  
    161  
  ],  
  "session_memory_key": "tab:new tab",  
  "cached": false  
},  
{  
  "id": "btn_new_tab",  
  "type": "button",  
  "label": "New tab button",  
  "description": "Button to open a new browser tab",  
  "state": "enabled",  
  "dx": 484,
```

```
"dy": 76,  
"frame_dx": 599,  
"frame_dy": 141,  
"confidence": 0.9,  
"source": "gemini_vlm",  
"frame_bbox": [  
    579,  
    129,  
    619,  
    153  
],  
"session_memory_key": "button:new tab button",  
"cached": false  
},  
{  
    "id": "btn_back",  
    "type": "button",  
    "label": "Back",  
    "description": "Browser back button",  
    "state": "disabled",  
    "dx": 112,  
    "dy": 120,  
    "frame_dx": 227,  
    "frame_dy": 185,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        211,  
        153,
```

```
        243,  
        217  
    ],  
    "session_memory_key": "button:back",  
    "cached": false  
},  
{  
    "id": "btn_forward",  
    "type": "button",  
    "label": "Forward",  
    "description": "Browser forward button",  
    "state": "disabled",  
    "dx": 144,  
    "dy": 120,  
    "frame_dx": 259,  
    "frame_dy": 185,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        235,  
        153,  
        283,  
        217  
    ],  
    "session_memory_key": "button:forward",  
    "cached": false  
},  
{  
    "id": "btn_reload",
```

```
"type": "button",
"label": "Reload",
"description": "Browser reload button",
"state": "enabled",
"dx": 184,
"dy": 120,
"frame_dx": 299,
"frame_dy": 185,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    275,
    153,
    323,
    217
],
"session_memory_key": "button:reload",
"cached": false
},
{
    "id": "address_bar",
    "type": "input",
    "label": "Address bar",
    "description": "Browser address bar showing localhost:5173",
    "state": "enabled",
    "dx": 616,
    "dy": 120,
    "frame_dx": 731,
    "frame_dy": 185,
```

```
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    179,
    153,
    1283,
    217
],
"session_memory_key": "input:address bar",
"cached": false
}
],
"element_count": 7,
"current_elements": [
    {
        "id": "tab_frontend",
        "type": "tab",
        "label": "frontend",
        "description": "Browser tab for the frontend application",
        "state": "active",
        "dx": 188,
        "dy": 68,
        "frame_dx": 303,
        "frame_dy": 133,
        "confidence": 0.95,
        "source": "gemini_vlm",
        "frame_bbox": [
            179,
            105,
```

```
        427,  
        161  
    ],  
    "session_memory_key": "tab:frontend",  
    "cached": false  
},  
{  
    "id": "tab_new_tab",  
    "type": "tab",  
    "label": "New tab",  
    "description": "Browser tab for a new page",  
    "state": "inactive",  
    "dx": 372,  
    "dy": 68,  
    "frame_dx": 487,  
    "frame_dy": 133,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        371,  
        105,  
        603,  
        161  
    ],  
    "session_memory_key": "tab:new tab",  
    "cached": false  
},  
{  
    "id": "btn_new_tab",
```



```
"type": "button",
"label": "New tab button",
"description": "Button to open a new browser tab",
"state": "enabled",
"dx": 484,
"dy": 76,
"frame_dx": 599,
"frame_dy": 141,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    579,
    129,
    619,
    153
],
"session_memory_key": "button:new tab button",
"cached": false
},
{
    "id": "btn_back",
    "type": "button",
    "label": "Back",
    "description": "Browser back button",
    "state": "disabled",
    "dx": 112,
    "dy": 120,
    "frame_dx": 227,
    "frame_dy": 185,
```

```
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    211,
    153,
    243,
    217
],
"session_memory_key": "button:back",
"cached": false
},
{
    "id": "btn_forward",
    "type": "button",
    "label": "Forward",
    "description": "Browser forward button",
    "state": "disabled",
    "dx": 144,
    "dy": 120,
    "frame_dx": 259,
    "frame_dy": 185,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        235,
        153,
        283,
        217
    ],
    ],
```

```
"session_memory_key": "button:forward",
"cached": false
},
{
  "id": "btn_reload",
  "type": "button",
  "label": "Reload",
  "description": "Browser reload button",
  "state": "enabled",
  "dx": 184,
  "dy": 120,
  "frame_dx": 299,
  "frame_dy": 185,
  "confidence": 0.9,
  "source": "gemini_vlm",
  "frame_bbox": [
    275,
    153,
    323,
    217
  ],
  "session_memory_key": "button:reload",
  "cached": false
},
{
  "id": "address_bar",
  "type": "input",
  "label": "Address bar",
  "description": "Browser address bar showing localhost:5173",
```

```
    "state": "enabled",
    "dx": 616,
    "dy": 120,
    "frame_dx": 731,
    "frame_dy": 185,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
      179,
      153,
      1283,
      217
    ],
    "session_memory_key": "input:address bar",
    "cached": false
  }
],
"resolved_elements": [
  {
    "id": "tab_frontend",
    "type": "tab",
    "label": "frontend",
    "description": "Browser tab for the frontend application",
    "state": "active",
    "dx": 188,
    "dy": 68,
    "frame_dx": 303,
    "frame_dy": 133,
    "confidence": 0.95,
```

```
"source": "gemini_vlm",
"frame_bbox": [
  179,
  105,
  427,
  161
],
"session_memory_key": "tab:frontend",
"cached": false
},
{
  "id": "tab_new_tab",
  "type": "tab",
  "label": "New tab",
  "description": "Browser tab for a new page",
  "state": "inactive",
  "dx": 372,
  "dy": 68,
  "frame_dx": 487,
  "frame_dy": 133,
  "confidence": 0.95,
  "source": "gemini_vlm",
  "frame_bbox": [
    371,
    105,
    603,
    161
  ],
  "session_memory_key": "tab:new tab",
```

```
"cached": false
},
{
  "id": "btn_new_tab",
  "type": "button",
  "label": "New tab button",
  "description": "Button to open a new browser tab",
  "state": "enabled",
  "dx": 484,
  "dy": 76,
  "frame_dx": 599,
  "frame_dy": 141,
  "confidence": 0.9,
  "source": "gemini_vlm",
  "frame_bbox": [
    579,
    129,
    619,
    153
  ],
  "session_memory_key": "button:new tab button",
  "cached": false
},
{
  "id": "btn_back",
  "type": "button",
  "label": "Back",
  "description": "Browser back button",
  "state": "disabled",
```

```
"dx": 112,
"dy": 120,
"frame_dx": 227,
"frame_dy": 185,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    211,
    153,
    243,
    217
],
"session_memory_key": "button:back",
"cached": false
},
{
    "id": "btn_forward",
    "type": "button",
    "label": "Forward",
    "description": "Browser forward button",
    "state": "disabled",
    "dx": 144,
    "dy": 120,
    "frame_dx": 259,
    "frame_dy": 185,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        235,
```

```
    153,  
    283,  
    217  
  ],  
  "session_memory_key": "button:forward",  
  "cached": false  
},  
{  
  "id": "btn_reload",  
  "type": "button",  
  "label": "Reload",  
  "description": "Browser reload button",  
  "state": "enabled",  
  "dx": 184,  
  "dy": 120,  
  "frame_dx": 299,  
  "frame_dy": 185,  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    275,  
    153,  
    323,  
    217  
  ],  
  "session_memory_key": "button:reload",  
  "cached": false  
},  
{
```



```
    "id": "address_bar",
    "type": "input",
    "label": "Address bar",
    "description": "Browser address bar showing localhost:5173",
    "state": "enabled",
    "dx": 616,
    "dy": 120,
    "frame_dx": 731,
    "frame_dy": 185,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        179,
        153,
        1283,
        217
    ],
    "session_memory_key": "input:address bar",
    "cached": false
}
],
"session_knowledge_base": {
    "session_id": "shot_1777739562_bcb8e7",
    "started_at": "2026-05-02T16:32:42.997236+00:00",
    "updated_at": "2026-05-02T16:33:18.641310+00:00",
    "stale_after_misses": 3,
    "confidence_floor": 0.4,
    "decay_per_miss": 0.9,
    "frame_count": 1,
```

```
"frame_name": "frame_1777739562_d6dfdf.jpg",
"frame_index": 0,
"image_size": {
  "width": 1920,
  "height": 1080
},
"known_elements": {
  "tab:frontend": {
    "element_key": "tab:frontend",
    "element_type": "tab",
    "label": "frontend",
    "bbox": [
      179.0,
      105.0,
      427.0,
      161.0
    ],
    "confidence": 0.95,
    "source": "gemini_vlm",
    "first_seen_frame": "frame_1777739562_d6dfdf.jpg",
    "last_seen_frame": "frame_1777739562_d6dfdf.jpg",
    "first_seen_at": "2026-05-02T16:33:18.641310+00:00",
    "last_seen_at": "2026-05-02T16:33:18.641310+00:00",
    "first_seen_index": 0,
    "last_seen_index": 0,
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false,
```

```
"history": [  
  {  
    "frame_name": "frame_1777739562_d6dfdf.jpg",  
    "frame_index": 0,  
    "seen_at": "2026-05-02T16:33:18.641310+00:00",  
    "confidence": 0.95  
  }  
]  
,  
"tab:new tab": {  
  "element_key": "tab:new tab",  
  "element_type": "tab",  
  "label": "New tab",  
  "bbox": [  
    371.0,  
    105.0,  
    603.0,  
    161.0  
  ],  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "first_seen_frame": "frame_1777739562_d6dfdf.jpg",  
  "last_seen_frame": "frame_1777739562_d6dfdf.jpg",  
  "first_seen_at": "2026-05-02T16:33:18.641310+00:00",  
  "last_seen_at": "2026-05-02T16:33:18.641310+00:00",  
  "first_seen_index": 0,  
  "last_seen_index": 0,  
  "seen_count": 1,  
  "miss_count": 0,
```

```
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777739562_d6dfdf.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T16:33:18.641310+00:00",
    "confidence": 0.95
  }
],
},
"button:new tab button": {
  "element_key": "button:new tab button",
  "element_type": "button",
  "label": "New tab button",
  "bbox": [
    579.0,
    129.0,
    619.0,
    153.0
  ],
  "confidence": 0.9,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777739562_d6dfdf.jpg",
  "last_seen_frame": "frame_1777739562_d6dfdf.jpg",
  "first_seen_at": "2026-05-02T16:33:18.641310+00:00",
  "last_seen_at": "2026-05-02T16:33:18.641310+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
```

```
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777739562_d6dfdf.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T16:33:18.641310+00:00",
    "confidence": 0.9
  }
]
},
"button:back": {
  "element_key": "button:back",
  "element_type": "button",
  "label": "Back",
  "bbox": [
    211.0,
    153.0,
    243.0,
    217.0
  ],
  "confidence": 0.9,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777739562_d6dfdf.jpg",
  "last_seen_frame": "frame_1777739562_d6dfdf.jpg",
  "first_seen_at": "2026-05-02T16:33:18.641310+00:00",
  "last_seen_at": "2026-05-02T16:33:18.641310+00:00",
```

```
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777739562_d6dfdf.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T16:33:18.641310+00:00",
    "confidence": 0.9
  }
],
},
"button:forward": {
  "element_key": "button:forward",
  "element_type": "button",
  "label": "Forward",
  "bbox": [
    235.0,
    153.0,
    283.0,
    217.0
  ],
  "confidence": 0.9,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777739562_d6dfdf.jpg",
  "last_seen_frame": "frame_1777739562_d6dfdf.jpg",
```

```
"first_seen_at": "2026-05-02T16:33:18.641310+00:00",
"last_seen_at": "2026-05-02T16:33:18.641310+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777739562_d6dfdf.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T16:33:18.641310+00:00",
    "confidence": 0.9
  }
],
},
"button:reload": {
  "element_key": "button:reload",
  "element_type": "button",
  "label": "Reload",
  "bbox": [
    275.0,
    153.0,
    323.0,
    217.0
  ],
  "confidence": 0.9,
  "source": "gemini_vlm",
```

```
"first_seen_frame": "frame_1777739562_d6dfdf.jpg",
"last_seen_frame": "frame_1777739562_d6dfdf.jpg",
"first_seen_at": "2026-05-02T16:33:18.641310+00:00",
"last_seen_at": "2026-05-02T16:33:18.641310+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777739562_d6dfdf.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T16:33:18.641310+00:00",
    "confidence": 0.9
  }
],
},
"input:address bar": {
  "element_key": "input:address bar",
  "element_type": "input",
  "label": "Address bar",
  "bbox": [
    179.0,
    153.0,
    1283.0,
    217.0
  ],
},
```



```
"confidence": 0.95,
"source": "gemini_vlm",
"first_seen_frame": "frame_1777739562_d6dfdf.jpg",
"last_seen_frame": "frame_1777739562_d6dfdf.jpg",
"first_seen_at": "2026-05-02T16:33:18.641310+00:00",
"last_seen_at": "2026-05-02T16:33:18.641310+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777739562_d6dfdf.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T16:33:18.641310+00:00",
    "confidence": 0.95
  }
]
},
"resolved_elements": [
  {
    "id": "tab_frontend",
    "type": "tab",
    "label": "frontend",
    "description": "Browser tab for the frontend application",
    "state": "active",
```

```
"dx": 188,  
  "dy": 68,  
  "frame_dx": 303,  
  "frame_dy": 133,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    179,  
    105,  
    427,  
    161  
  ],  
  "session_memory_key": "tab:frontend",  
  "cached": false  
},  
{  
  "id": "tab_new_tab",  
  "type": "tab",  
  "label": "New tab",  
  "description": "Browser tab for a new page",  
  "state": "inactive",  
  "dx": 372,  
  "dy": 68,  
  "frame_dx": 487,  
  "frame_dy": 133,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    371,
```

```
        105,  
        603,  
        161  
    ],  
    "session_memory_key": "tab:new tab",  
    "cached": false  
},  
{  
    "id": "btn_new_tab",  
    "type": "button",  
    "label": "New tab button",  
    "description": "Button to open a new browser tab",  
    "state": "enabled",  
    "dx": 484,  
    "dy": 76,  
    "frame_dx": 599,  
    "frame_dy": 141,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        579,  
        129,  
        619,  
        153  
    ],  
    "session_memory_key": "button:new tab button",  
    "cached": false  
},  
{
```

```
"id": "btn_back",
"type": "button",
"label": "Back",
"description": "Browser back button",
"state": "disabled",
"dx": 112,
"dy": 120,
"frame_dx": 227,
"frame_dy": 185,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    211,
    153,
    243,
    217
],
"session_memory_key": "button:back",
"cached": false
},
{
    "id": "btn_forward",
    "type": "button",
    "label": "Forward",
    "description": "Browser forward button",
    "state": "disabled",
    "dx": 144,
    "dy": 120,
    "frame_dx": 259,
```

```
"frame_dy": 185,  
"confidence": 0.9,  
"source": "gemini_vlm",  
"frame_bbox": [  
    235,  
    153,  
    283,  
    217  
],  
"session_memory_key": "button:forward",  
"cached": false  
},  
{  
    "id": "btn_reload",  
    "type": "button",  
    "label": "Reload",  
    "description": "Browser reload button",  
    "state": "enabled",  
    "dx": 184,  
    "dy": 120,  
    "frame_dx": 299,  
    "frame_dy": 185,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        275,  
        153,  
        323,  
        217
```

```
    ],  
    "session_memory_key": "button:reload",  
    "cached": false  
  },  
  {  
    "id": "address_bar",  
    "type": "input",  
    "label": "Address bar",  
    "description": "Browser address bar showing localhost:5173",  
    "state": "enabled",  
    "dx": 616,  
    "dy": 120,  
    "frame_dx": 731,  
    "frame_dy": 185,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
      179,  
      153,  
      1283,  
      217  
    ],  
    "session_memory_key": "input:address bar",  
    "cached": false  
  }  
],  
"current_elements": [  
  {  
    "id": "tab_frontend",
```

```
"type": "tab",
"label": "frontend",
"description": "Browser tab for the frontend application",
"state": "active",
"dx": 188,
"dy": 68,
"frame_dx": 303,
"frame_dy": 133,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
  179,
  105,
  427,
  161
],
"session_memory_key": "tab:frontend",
"cached": false
},
{
  "id": "tab_new_tab",
  "type": "tab",
  "label": "New tab",
  "description": "Browser tab for a new page",
  "state": "inactive",
  "dx": 372,
  "dy": 68,
  "frame_dx": 487,
  "frame_dy": 133,
```

```
"confidence": 0.95,  
"source": "gemini_vlm",  
"frame_bbox": [  
    371,  
    105,  
    603,  
    161  
],  
"session_memory_key": "tab:new tab",  
"cached": false  
},  
{  
    "id": "btn_new_tab",  
    "type": "button",  
    "label": "New tab button",  
    "description": "Button to open a new browser tab",  
    "state": "enabled",  
    "dx": 484,  
    "dy": 76,  
    "frame_dx": 599,  
    "frame_dy": 141,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        579,  
        129,  
        619,  
        153  
    ],  
}
```



```
    "session_memory_key": "button:new tab button",
    "cached": false
  },
  {
    "id": "btn_back",
    "type": "button",
    "label": "Back",
    "description": "Browser back button",
    "state": "disabled",
    "dx": 112,
    "dy": 120,
    "frame_dx": 227,
    "frame_dy": 185,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
      211,
      153,
      243,
      217
    ],
    "session_memory_key": "button:back",
    "cached": false
  },
  {
    "id": "btn_forward",
    "type": "button",
    "label": "Forward",
    "description": "Browser forward button",
```

```
"state": "disabled",
"dx": 144,
"dy": 120,
"frame_dx": 259,
"frame_dy": 185,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    235,
    153,
    283,
    217
],
"session_memory_key": "button:forward",
"cached": false
},
{
    "id": "btn_reload",
    "type": "button",
    "label": "Reload",
    "description": "Browser reload button",
    "state": "enabled",
    "dx": 184,
    "dy": 120,
    "frame_dx": 299,
    "frame_dy": 185,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
```

```
        275,  
        153,  
        323,  
        217  
    ],  
    "session_memory_key": "button:reload",  
    "cached": false  
},  
{  
    "id": "address_bar",  
    "type": "input",  
    "label": "Address bar",  
    "description": "Browser address bar showing localhost:5173",  
    "state": "enabled",  
    "dx": 616,  
    "dy": 120,  
    "frame_dx": 731,  
    "frame_dy": 185,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        179,  
        153,  
        1283,  
        217  
    ],  
    "session_memory_key": "input:address bar",  
    "cached": false  
}
```

```
]
},
"session_memory": {
  "run_id": "769bda88-628b-4be2-875a-1b6265d5e8bc",
  "user_task": "Click on new tab and find the youtube, if you can't find that icon search it in the search bar. however needs to re-direct to the youtube platform",
  "started_at": "2026-05-02T16:32:30.127424+00:00",
  "updated_at": "2026-05-02T16:33:18.669310+00:00",
  "stale_after_misses": 2,
  "confidence_floor": 0.4,
  "decay_per_miss": 0.9,
  "frame_index": 1,
  "known_elements": {
    "tab:frontend": {
      "element_key": "tab:frontend",
      "element_type": "tab",
      "label": "frontend",
      "bbox": [
        211.0,
        105.0,
        411.0,
        161.0
      ],
      "confidence": 0.4,
      "source": "gemini_vlm",
      "first_seen_frame": null,
      "last_seen_frame": null,
      "first_seen_index": 0,
      "last_seen_index": 5,
    }
  }
}
```

```
"first_seen_at": "2026-05-02T16:33:18.669310+00:00",
"last_seen_at": "2026-05-02T16:37:11.954378+00:00",
"seen_count": 5,
"miss_count": 9,
"stale": true,
"cached": true
},
"tab:new tab": {
  "element_key": "tab:new tab",
  "element_type": "tab",
  "label": "New tab",
  "bbox": [
    363.0,
    105.0,
    603.0,
    161.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 5,
  "first_seen_at": "2026-05-02T16:33:18.669310+00:00",
  "last_seen_at": "2026-05-02T16:37:11.954378+00:00",
  "seen_count": 6,
  "miss_count": 9,
  "stale": true,
  "cached": true
}
```

```
},  
"button:new tab button": {  
  "element_key": "button:new tab button",  
  "element_type": "button",  
  "label": "New tab button",  
  "bbox": [  
    579.0,  
    129.0,  
    619.0,  
    153.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 0,  
  "last_seen_index": 5,  
  "first_seen_at": "2026-05-02T16:33:18.669310+00:00",  
  "last_seen_at": "2026-05-02T16:37:11.954378+00:00",  
  "seen_count": 2,  
  "miss_count": 9,  
  "stale": true,  
  "cached": true  
},  
"button:back": {  
  "element_key": "button:back",  
  "element_type": "button",  
  "label": "Back",  
  "bbox": [  

```

```
    235.0,
    105.0,
    259.0,
    129.0
  ],
  "confidence": 0.7695,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 12,
  "first_seen_at": "2026-05-02T16:33:18.669310+00:00",
  "last_seen_at": "2026-05-02T16:43:09.638245+00:00",
  "seen_count": 5,
  "miss_count": 2,
  "stale": true,
  "cached": true
},
"button:forward": {
  "element_key": "button:forward",
  "element_type": "button",
  "label": "Forward",
  "bbox": [
    235.0,
    153.0,
    283.0,
    209.0
  ],
  "confidence": 0.4,
```

```
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 5,
"first_seen_at": "2026-05-02T16:33:18.669310+00:00",
"last_seen_at": "2026-05-02T16:37:11.954378+00:00",
"seen_count": 3,
"miss_count": 9,
"stale": true,
"cached": true
},
"button:reload": {
  "element_key": "button:reload",
  "element_type": "button",
  "label": "Reload",
  "bbox": [
    267.0,
    169.0,
    307.0,
    209.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 4,
  "first_seen_at": "2026-05-02T16:33:18.669310+00:00",
```



```
    "last_seen_at": "2026-05-02T16:36:18.113372+00:00",
    "seen_count": 2,
    "miss_count": 10,
    "stale": true,
    "cached": true
  },
  "input:address bar": {
    "element_key": "input:address bar",
    "element_type": "input",
    "label": "Address bar",
    "bbox": [
      179.0,
      153.0,
      1643.0,
      217.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 0,
    "last_seen_index": 5,
    "first_seen_at": "2026-05-02T16:33:18.669310+00:00",
    "last_seen_at": "2026-05-02T16:37:11.954378+00:00",
    "seen_count": 4,
    "miss_count": 9,
    "stale": true,
    "cached": true
  },
}
```

```
"tab:frontend tab": {
  "element_key": "tab:frontend tab",
  "element_type": "tab",
  "label": "frontend tab",
  "bbox": [
    163.0,
    105.0,
    395.0,
    161.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
  "last_seen_index": 1,
  "first_seen_at": "2026-05-02T16:34:15.000602+00:00",
  "last_seen_at": "2026-05-02T16:34:15.000602+00:00",
  "seen_count": 1,
  "miss_count": 13,
  "stale": true,
  "cached": true
},
"button:home": {
  "element_key": "button:home",
  "element_type": "button",
  "label": "Home",
  "bbox": [
    203.0,
```

```
    161.0,  
    267.0,  
    257.0  
  ],  
  "confidence": 0.5315,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 3,  
  "last_seen_index": 9,  
  "first_seen_at": "2026-05-02T16:35:38.505125+00:00",  
  "last_seen_at": "2026-05-02T16:40:47.767646+00:00",  
  "seen_count": 2,  
  "miss_count": 5,  
  "stale": true,  
  "cached": true  
},  
"button:chats": {  
  "element_key": "button:chats",  
  "element_type": "button",  
  "label": "Chats",  
  "bbox": [  
    171.0,  
    489.0,  
    459.0,  
    569.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",
```

```
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 3,
"last_seen_index": 3,
"first_seen_at": "2026-05-02T16:35:38.505125+00:00",
"last_seen_at": "2026-05-02T16:35:38.505125+00:00",
"seen_count": 1,
"miss_count": 11,
"stale": true,
"cached": true
},
"pane:task plan": {
  "element_key": "pane:task plan",
  "element_type": "pane",
  "label": "Task plan",
  "bbox": [
    379.0,
    393.0,
    1163.0,
    801.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 3,
  "last_seen_index": 5,
  "first_seen_at": "2026-05-02T16:35:38.505125+00:00",
  "last_seen_at": "2026-05-02T16:37:11.954378+00:00",
```

```
    "seen_count": 3,  
    "miss_count": 9,  
    "stale": true,  
    "cached": true  
  },  
  "button:start": {  
    "element_key": "button:start",  
    "element_type": "button",  
    "label": "Start",  
    "bbox": [  
      603.0,  
      937.0,  
      659.0,  
      993.0  
    ],  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "first_seen_frame": null,  
    "last_seen_frame": null,  
    "first_seen_index": 3,  
    "last_seen_index": 14,  
    "first_seen_at": "2026-05-02T16:35:38.505125+00:00",  
    "last_seen_at": "2026-05-02T16:44:43.739605+00:00",  
    "seen_count": 5,  
    "miss_count": 0,  
    "stale": false,  
    "cached": false  
  },  
  "input:search": {
```

```
"element_key": "input:search",
"element_type": "input",
"label": "Search",
"bbox": [
    619.0,
    937.0,
    923.0,
    977.0
],
"confidence": 0.95,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 3,
"last_seen_index": 14,
"first_seen_at": "2026-05-02T16:35:38.505125+00:00",
"last_seen_at": "2026-05-02T16:44:43.739605+00:00",
"seen_count": 5,
"miss_count": 0,
"stale": false,
"cached": false
},
"icon:chrome": {
    "element_key": "icon:chrome",
    "element_type": "icon",
    "label": "Chrome",
    "bbox": [
        915.0,
        921.0,
```

```
    979.0,  
    977.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 3,  
  "last_seen_index": 3,  
  "first_seen_at": "2026-05-02T16:35:38.505125+00:00",  
  "last_seen_at": "2026-05-02T16:35:38.505125+00:00",  
  "seen_count": 1,  
  "miss_count": 11,  
  "stale": true,  
  "cached": true  
},  
"icon:vs code": {  
  "element_key": "icon:vs code",  
  "element_type": "icon",  
  "label": "VS Code",  
  "bbox": [  
    995.0,  
    921.0,  
    1051.0,  
    993.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,
```

```
"last_seen_frame": null,
"first_seen_index": 3,
"last_seen_index": 3,
"first_seen_at": "2026-05-02T16:35:38.505125+00:00",
"last_seen_at": "2026-05-02T16:35:38.505125+00:00",
"seen_count": 1,
"miss_count": 11,
"stale": true,
"cached": true
},
"icon:spotify": {
  "element_key": "icon:spotify",
  "element_type": "icon",
  "label": "Spotify",
  "bbox": [
    1243.0,
    961.0,
    1291.0,
    977.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 3,
  "last_seen_index": 4,
  "first_seen_at": "2026-05-02T16:35:38.505125+00:00",
  "last_seen_at": "2026-05-02T16:36:18.113372+00:00",
  "seen_count": 2,
```



```
    "miss_count": 10,  
    "stale": true,  
    "cached": true  
  },  
  "text:clock": {  
    "element_key": "text:clock",  
    "element_type": "text",  
    "label": "Clock",  
    "bbox": [  
      1347.0,  
      913.0,  
      1443.0,  
      1015.0  
    ],  
    "confidence": 0.4,  
    "source": "gemini_vlm",  
    "first_seen_frame": null,  
    "last_seen_frame": null,  
    "first_seen_index": 3,  
    "last_seen_index": 3,  
    "first_seen_at": "2026-05-02T16:35:38.505125+00:00",  
    "last_seen_at": "2026-05-02T16:35:38.505125+00:00",  
    "seen_count": 1,  
    "miss_count": 11,  
    "stale": true,  
    "cached": true  
  },  
  "button:weather": {  
    "element_key": "button:weather",
```

```
"element_type": "button",
"label": "Weather",
"bbox": [
    179.0,
    913.0,
    315.0,
    1015.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 3,
"last_seen_index": 3,
"first_seen_at": "2026-05-02T16:35:38.505125+00:00",
"last_seen_at": "2026-05-02T16:35:38.505125+00:00",
"seen_count": 1,
"miss_count": 11,
"stale": true,
"cached": true
},
"dropdown:ai assistant": {
    "element_key": "dropdown:ai assistant",
    "element_type": "dropdown",
    "label": "AI Assistant",
    "bbox": [
        467.0,
        273.0,
        515.0,
```

```
    305.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 3,
  "last_seen_index": 4,
  "first_seen_at": "2026-05-02T16:35:38.505125+00:00",
  "last_seen_at": "2026-05-02T16:36:18.113372+00:00",
  "seen_count": 2,
  "miss_count": 10,
  "stale": true,
  "cached": true
},
"text:agent is working": {
  "element_key": "text:agent is working",
  "element_type": "text",
  "label": "Agent is working",
  "bbox": [
    451.0,
    841.0,
    619.0,
    881.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
```

```
"first_seen_index": 3,  
"last_seen_index": 3,  
"first_seen_at": "2026-05-02T16:35:38.505125+00:00",  
"last_seen_at": "2026-05-02T16:35:38.505125+00:00",  
"seen_count": 1,  
"miss_count": 11,  
"stale": true,  
"cached": true  
},  
"button:send": {  
  "element_key": "button:send",  
  "element_type": "button",  
  "label": "Send",  
  "bbox": [  
    1356.0,  
    1014.0,  
    1402.0,  
    1015.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 3,  
  "last_seen_index": 4,  
  "first_seen_at": "2026-05-02T16:35:38.505125+00:00",  
  "last_seen_at": "2026-05-02T16:36:18.113372+00:00",  
  "seen_count": 2,  
  "miss_count": 10,
```

```
    "stale": true,
    "cached": true
  },
  "input:search chats": {
    "element_key": "input:search chats",
    "element_type": "input",
    "label": "Search chats",
    "bbox": [
      187.0,
      337.0,
      443.0,
      393.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 3,
    "last_seen_index": 4,
    "first_seen_at": "2026-05-02T16:35:38.505125+00:00",
    "last_seen_at": "2026-05-02T16:36:18.113372+00:00",
    "seen_count": 2,
    "miss_count": 10,
    "stale": true,
    "cached": true
  },
  "button:refresh": {
    "element_key": "button:refresh",
    "element_type": "button",
```

```
"label": "Refresh",
"bbox": [
  275.0,
  153.0,
  315.0,
  209.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 3,
"last_seen_index": 5,
"first_seen_at": "2026-05-02T16:35:38.505125+00:00",
"last_seen_at": "2026-05-02T16:37:11.954378+00:00",
"seen_count": 2,
"miss_count": 9,
"stale": true,
"cached": true
},
"button:extensions": {
  "element_key": "button:extensions",
  "element_type": "button",
  "label": "Extensions",
  "bbox": [
    1227.0,
    169.0,
    1267.0,
    193.0
  ]
}
```

```
],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 3,  
  "last_seen_index": 3,  
  "first_seen_at": "2026-05-02T16:35:38.505125+00:00",  
  "last_seen_at": "2026-05-02T16:35:38.505125+00:00",  
  "seen_count": 1,  
  "miss_count": 11,  
  "stale": true,  
  "cached": true  
},  
"button:profile": {  
  "element_key": "button:profile",  
  "element_type": "button",  
  "label": "Profile",  
  "bbox": [  
    1331.0,  
    169.0,  
    1371.0,  
    193.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 3,
```

```
"last_seen_index": 3,  
"first_seen_at": "2026-05-02T16:35:38.505125+00:00",  
"last_seen_at": "2026-05-02T16:35:38.505125+00:00",  
"seen_count": 1,  
"miss_count": 11,  
"stale": true,  
"cached": true  
},  
"button:minimize": {  
  "element_key": "button:minimize",  
  "element_type": "button",  
  "label": "Minimize",  
  "bbox": [  
    1627.0,  
    97.0,  
    1659.0,  
    121.0  
  ],  
  "confidence": 0.7695,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 3,  
  "last_seen_index": 12,  
  "first_seen_at": "2026-05-02T16:35:38.505125+00:00",  
  "last_seen_at": "2026-05-02T16:43:09.638245+00:00",  
  "seen_count": 2,  
  "miss_count": 2,  
  "stale": true,
```



```
    "cached": true
  },
  "button:maximize": {
    "element_key": "button:maximize",
    "element_type": "button",
    "label": "Maximize",
    "bbox": [
      1675.0,
      97.0,
      1707.0,
      121.0
    ],
    "confidence": 0.7695,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 3,
    "last_seen_index": 12,
    "first_seen_at": "2026-05-02T16:35:38.505125+00:00",
    "last_seen_at": "2026-05-02T16:43:09.638245+00:00",
    "seen_count": 2,
    "miss_count": 2,
    "stale": true,
    "cached": true
  },
  "button:close": {
    "element_key": "button:close",
    "element_type": "button",
    "label": "Close",
```

```
"bbox": [
  1723.0,
  97.0,
  1747.0,
  121.0
],
"confidence": 0.7695,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 3,
"last_seen_index": 12,
"first_seen_at": "2026-05-02T16:35:38.505125+00:00",
"last_seen_at": "2026-05-02T16:43:09.638245+00:00",
"seen_count": 2,
"miss_count": 2,
"stale": true,
"cached": true
},
"button:user profile": {
  "element_key": "button:user profile",
  "element_type": "button",
  "label": "User Profile",
  "bbox": [
    179.0,
    825.0,
    379.0,
    897.0
  ],
```

```
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 3,
"last_seen_index": 3,
"first_seen_at": "2026-05-02T16:35:38.505125+00:00",
"last_seen_at": "2026-05-02T16:35:38.505125+00:00",
"seen_count": 1,
"miss_count": 11,
"stale": true,
"cached": true
},
"text:task step 1": {
  "element_key": "text:task step 1",
  "element_type": "text",
  "label": "Task Step 1",
  "bbox": [
    539.0,
    561.0,
    1131.0,
    633.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 3,
  "last_seen_index": 3,
```

```
"first_seen_at": "2026-05-02T16:35:38.505125+00:00",
"last_seen_at": "2026-05-02T16:35:38.505125+00:00",
"seen_count": 1,
"miss_count": 11,
"stale": true,
"cached": true
},
"text:task step 2": {
  "element_key": "text:task step 2",
  "element_type": "text",
  "label": "Task Step 2",
  "bbox": [
    435.0,
    633.0,
    1131.0,
    753.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 3,
  "last_seen_index": 3,
  "first_seen_at": "2026-05-02T16:35:38.505125+00:00",
  "last_seen_at": "2026-05-02T16:35:38.505125+00:00",
  "seen_count": 1,
  "miss_count": 11,
  "stale": true,
  "cached": true
}
```

```
},
"icon:wi-fi": {
  "element_key": "icon:wi-fi",
  "element_type": "icon",
  "label": "Wi-Fi",
  "bbox": [
    1603.0,
    921.0,
    1643.0,
    961.0
  ],
  "confidence": 0.5315,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 3,
  "last_seen_index": 9,
  "first_seen_at": "2026-05-02T16:35:38.505125+00:00",
  "last_seen_at": "2026-05-02T16:40:47.767646+00:00",
  "seen_count": 2,
  "miss_count": 5,
  "stale": true,
  "cached": true
},
"icon:volume": {
  "element_key": "icon:volume",
  "element_type": "icon",
  "label": "Volume",
  "bbox": [
```

```
        1635.0,
        921.0,
        1675.0,
        969.0
    ],
    "confidence": 0.5315,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 3,
    "last_seen_index": 9,
    "first_seen_at": "2026-05-02T16:35:38.505125+00:00",
    "last_seen_at": "2026-05-02T16:40:47.767646+00:00",
    "seen_count": 2,
    "miss_count": 5,
    "stale": true,
    "cached": true
},
"icon:battery": {
    "element_key": "icon:battery",
    "element_type": "icon",
    "label": "Battery",
    "bbox": [
        1667.0,
        921.0,
        1707.0,
        969.0
    ],
    "confidence": 0.5315,
```

```
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 3,
"last_seen_index": 9,
"first_seen_at": "2026-05-02T16:35:38.505125+00:00",
"last_seen_at": "2026-05-02T16:40:47.767646+00:00",
"seen_count": 2,
"miss_count": 5,
"stale": true,
"cached": true
},
"icon:file explorer": {
  "element_key": "icon:file explorer",
  "element_type": "icon",
  "label": "File Explorer",
  "bbox": [
    779.0,
    921.0,
    835.0,
    977.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 3,
  "last_seen_index": 3,
  "first_seen_at": "2026-05-02T16:35:38.505125+00:00",
```

```
    "last_seen_at": "2026-05-02T16:35:38.505125+00:00",
    "seen_count": 1,
    "miss_count": 11,
    "stale": true,
    "cached": true
  },
  "button:new tab": {
    "element_key": "button:new tab",
    "element_type": "button",
    "label": "New tab",
    "bbox": [
      579.0,
      129.0,
      619.0,
      153.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 4,
    "last_seen_index": 4,
    "first_seen_at": "2026-05-02T16:36:18.113372+00:00",
    "last_seen_at": "2026-05-02T16:36:18.113372+00:00",
    "seen_count": 1,
    "miss_count": 10,
    "stale": true,
    "cached": true
  },
}
```



```
"menu_item:home": {
  "element_key": "menu_item:home",
  "element_type": "menu_item",
  "label": "Home",
  "bbox": [
    171.0,
    417.0,
    459.0,
    489.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 4,
  "last_seen_index": 4,
  "first_seen_at": "2026-05-02T16:36:18.113372+00:00",
  "last_seen_at": "2026-05-02T16:36:18.113372+00:00",
  "seen_count": 1,
  "miss_count": 10,
  "stale": true,
  "cached": true
},
"menu_item:chats": {
  "element_key": "menu_item:chats",
  "element_type": "menu_item",
  "label": "Chats",
  "bbox": [
    195.0,
```

```
    489.0,  
    459.0,  
    569.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 4,  
  "last_seen_index": 4,  
  "first_seen_at": "2026-05-02T16:36:18.113372+00:00",  
  "last_seen_at": "2026-05-02T16:36:18.113372+00:00",  
  "seen_count": 1,  
  "miss_count": 10,  
  "stale": true,  
  "cached": true  
},  
"list_item:click on new tab and find the yo...": {  
  "element_key": "list_item:click on new tab and find the yo...",  
  "element_type": "list_item",  
  "label": "Click on new tab and find the yo...",  
  "bbox": [  
    203.0,  
    617.0,  
    427.0,  
    705.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",
```

```
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 4,
    "last_seen_index": 4,
    "first_seen_at": "2026-05-02T16:36:18.113372+00:00",
    "last_seen_at": "2026-05-02T16:36:18.113372+00:00",
    "seen_count": 1,
    "miss_count": 10,
    "stale": true,
    "cached": true
  },
  "text:click on the 'new tab' tab to switch to the existing empty browser
tab.": {
    "element_key": "text:click on the 'new tab' tab to switch to the
existing empty browser tab.",
    "element_type": "text",
    "label": "Click on the 'New tab' tab to switch to the existing empty
browser tab.",
    "bbox": [
      563.0,
      577.0,
      1131.0,
      633.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 4,
    "last_seen_index": 4,
```

```
    "first_seen_at": "2026-05-02T16:36:18.113372+00:00",
    "last_seen_at": "2026-05-02T16:36:18.113372+00:00",
    "seen_count": 1,
    "miss_count": 10,
    "stale": true,
    "cached": true
  },
  "text:click the address bar, type 'https://www.youtube.com', and press
the enter key to navigate to the platform.": {
    "element_key": "text:click the address bar, type
'https://www.youtube.com', and press the enter key to navigate to the platform.",
    "element_type": "text",
    "label": "Click the address bar, type 'https://www.youtube.com', and
press the Enter key to navigate to the platform.",
    "bbox": [
      483.0,
      625.0,
      1131.0,
      745.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 4,
    "last_seen_index": 4,
    "first_seen_at": "2026-05-02T16:36:18.113372+00:00",
    "last_seen_at": "2026-05-02T16:36:18.113372+00:00",
    "seen_count": 1,
    "miss_count": 10,
```

```
    "stale": true,
    "cached": true
  },
  "text:agent is working...": {
    "element_key": "text:agent is working...",
    "element_type": "text",
    "label": "Agent is working...",
    "bbox": [
      491.0,
      1014.0,
      610.0,
      1015.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 4,
    "last_seen_index": 4,
    "first_seen_at": "2026-05-02T16:36:18.113372+00:00",
    "last_seen_at": "2026-05-02T16:36:18.113372+00:00",
    "seen_count": 1,
    "miss_count": 10,
    "stale": true,
    "cached": true
  },
  "icon:google chrome": {
    "element_key": "icon:google chrome",
    "element_type": "icon",
```

```
"label": "Google Chrome",
"bbox": [
  1155.0,
  961.0,
  1203.0,
  985.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 4,
"last_seen_index": 4,
"first_seen_at": "2026-05-02T16:36:18.113372+00:00",
"last_seen_at": "2026-05-02T16:36:18.113372+00:00",
"seen_count": 1,
"miss_count": 10,
"stale": true,
"cached": true
},
"icon:visual studio code": {
  "element_key": "icon:visual studio code",
  "element_type": "icon",
  "label": "Visual Studio Code",
  "bbox": [
    1195.0,
    961.0,
    1243.0,
    977.0
  ]
}
```

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],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 4,
  "last_seen_index": 4,
  "first_seen_at": "2026-05-02T16:36:18.113372+00:00",
  "last_seen_at": "2026-05-02T16:36:18.113372+00:00",
  "seen_count": 1,
  "miss_count": 10,
  "stale": true,
  "cached": true
},
"pane:system tray": {
  "element_key": "pane:system tray",
  "element_type": "pane",
  "label": "System Tray",
  "bbox": [
    1499.0,
    961.0,
    1755.0,
    1001.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 4,
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"last_seen_index": 4,
"first_seen_at": "2026-05-02T16:36:18.113372+00:00",
"last_seen_at": "2026-05-02T16:36:18.113372+00:00",
"seen_count": 1,
"miss_count": 10,
"stale": true,
"cached": true
},
"text:10:05 pm 5/2/2026": {
  "element_key": "text:10:05 pm 5/2/2026",
  "element_type": "text",
  "label": "10:05 PM 5/2/2026",
  "bbox": [
    1723.0,
    961.0,
    1803.0,
    977.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 4,
  "last_seen_index": 4,
  "first_seen_at": "2026-05-02T16:36:18.113372+00:00",
  "last_seen_at": "2026-05-02T16:36:18.113372+00:00",
  "seen_count": 1,
  "miss_count": 10,
  "stale": true,
```



```
    "cached": true
  },
  "button:screenpilot user": {
    "element_key": "button:screenpilot user",
    "element_type": "button",
    "label": "ScreenPilot User",
    "bbox": [
      201.0,
      1014.0,
      412.0,
      1015.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 4,
    "last_seen_index": 4,
    "first_seen_at": "2026-05-02T16:36:18.113372+00:00",
    "last_seen_at": "2026-05-02T16:36:18.113372+00:00",
    "seen_count": 1,
    "miss_count": 10,
    "stale": true,
    "cached": true
  },
  "link:home": {
    "element_key": "link:home",
    "element_type": "link",
    "label": "Home",
```

```
"bbox": [
  163.0,
  409.0,
  459.0,
  481.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 5,
"last_seen_index": 5,
"first_seen_at": "2026-05-02T16:37:11.954378+00:00",
"last_seen_at": "2026-05-02T16:37:11.954378+00:00",
"seen_count": 1,
"miss_count": 9,
"stale": true,
"cached": true
},
"link:chats": {
  "element_key": "link:chats",
  "element_type": "link",
  "label": "Chats",
  "bbox": [
    187.0,
    489.0,
    291.0,
    545.0
  ],

```

```
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 5,
"last_seen_index": 5,
"first_seen_at": "2026-05-02T16:37:11.954378+00:00",
"last_seen_at": "2026-05-02T16:37:11.954378+00:00",
"seen_count": 1,
"miss_count": 9,
"stale": true,
"cached": true
},
"input:agent is working...": {
  "element_key": "input:agent is working...",
  "element_type": "input",
  "label": "Agent is working...",
  "bbox": [
    315.0,
    817.0,
    1515.0,
    905.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 5,
  "last_seen_index": 5,
```

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"first_seen_at": "2026-05-02T16:37:11.954378+00:00",
"last_seen_at": "2026-05-02T16:37:11.954378+00:00",
"seen_count": 1,
"miss_count": 9,
"stale": true,
"cached": true
},
"button:chrome": {
  "element_key": "button:chrome",
  "element_type": "button",
  "label": "Chrome",
  "bbox": [
    1115.0,
    921.0,
    1171.0,
    977.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 5,
  "last_seen_index": 5,
  "first_seen_at": "2026-05-02T16:37:11.954378+00:00",
  "last_seen_at": "2026-05-02T16:37:11.954378+00:00",
  "seen_count": 1,
  "miss_count": 9,
  "stale": true,
  "cached": true
}
```

```
},
"button:vs code": {
  "element_key": "button:vs code",
  "element_type": "button",
  "label": "VS Code",
  "bbox": [
    1243.0,
    921.0,
    1299.0,
    977.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 5,
  "last_seen_index": 5,
  "first_seen_at": "2026-05-02T16:37:11.954378+00:00",
  "last_seen_at": "2026-05-02T16:37:11.954378+00:00",
  "seen_count": 1,
  "miss_count": 9,
  "stale": true,
  "cached": true
},
"button:spotify": {
  "element_key": "button:spotify",
  "element_type": "button",
  "label": "Spotify",
  "bbox": [
```

```
    1259.0,  
    913.0,  
    1315.0,  
    977.0  
  ],  
  "confidence": 0.5315,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 5,  
  "last_seen_index": 9,  
  "first_seen_at": "2026-05-02T16:37:11.954378+00:00",  
  "last_seen_at": "2026-05-02T16:40:47.767646+00:00",  
  "seen_count": 2,  
  "miss_count": 5,  
  "stale": true,  
  "cached": true  
},  
"text:microsoft store": {  
  "element_key": "text:microsoft store",  
  "element_type": "text",  
  "label": "Microsoft Store",  
  "bbox": [  
    211.0,  
    113.0,  
    379.0,  
    161.0  
  ],  
  "confidence": 0.4089,
```

```
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 6,
"last_seen_index": 6,
"first_seen_at": "2026-05-02T16:38:13.893524+00:00",
"last_seen_at": "2026-05-02T16:38:13.893524+00:00",
"seen_count": 1,
"miss_count": 8,
"stale": true,
"cached": true
},
"pane:microsoft store window": {
  "element_key": "pane:microsoft store window",
  "element_type": "pane",
  "label": "Microsoft Store Window",
  "bbox": [
    115.0,
    65.0,
    1805.0,
    1015.0
  ],
  "confidence": 0.4543,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 7,
  "last_seen_index": 7,
  "first_seen_at": "2026-05-02T16:39:14.867764+00:00",
```

```
    "last_seen_at": "2026-05-02T16:39:14.867764+00:00",
    "seen_count": 1,
    "miss_count": 7,
    "stale": true,
    "cached": true
  },
  "pane:title bar": {
    "element_key": "pane:title bar",
    "element_type": "pane",
    "label": "Title Bar",
    "bbox": [
      115.0,
      81.0,
      1795.0,
      177.0
    ],
    "confidence": 0.4305,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 7,
    "last_seen_index": 7,
    "first_seen_at": "2026-05-02T16:39:14.867764+00:00",
    "last_seen_at": "2026-05-02T16:39:14.867764+00:00",
    "seen_count": 1,
    "miss_count": 7,
    "stale": true,
    "cached": true
  },
}
```



```
"input:search apps, games, and more": {
  "element_key": "input:search apps, games, and more",
  "element_type": "input",
  "label": "Search apps, games, and more",
  "bbox": [
    635.0,
    89.0,
    1323.0,
    145.0
  ],
  "confidence": 0.7695,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 7,
  "last_seen_index": 12,
  "first_seen_at": "2026-05-02T16:39:14.867764+00:00",
  "last_seen_at": "2026-05-02T16:43:09.638245+00:00",
  "seen_count": 4,
  "miss_count": 2,
  "stale": true,
  "cached": true
},
"pane:sidebar": {
  "element_key": "pane:sidebar",
  "element_type": "pane",
  "label": "Sidebar",
  "bbox": [
    187.0,
```

```
        65.0,  
        299.0,  
        993.0  
    ],  
    "confidence": 0.4305,  
    "source": "gemini_vlm",  
    "first_seen_frame": null,  
    "last_seen_frame": null,  
    "first_seen_index": 7,  
    "last_seen_index": 7,  
    "first_seen_at": "2026-05-02T16:39:14.867764+00:00",  
    "last_seen_at": "2026-05-02T16:39:14.867764+00:00",  
    "seen_count": 1,  
    "miss_count": 7,  
    "stale": true,  
    "cached": true  
},  
"tab:home": {  
    "element_key": "tab:home",  
    "element_type": "tab",  
    "label": "Home",  
    "bbox": [  
        211.0,  
        137.0,  
        299.0,  
        201.0  
    ],  
    "confidence": 0.855,  
    "source": "gemini_vlm",
```

```
    "first_seen_frame": null,  
    "last_seen_frame": null,  
    "first_seen_index": 7,  
    "last_seen_index": 13,  
    "first_seen_at": "2026-05-02T16:39:14.867764+00:00",  
    "last_seen_at": "2026-05-02T16:43:58.308218+00:00",  
    "seen_count": 3,  
    "miss_count": 1,  
    "stale": false,  
    "cached": true  
  },  
  "tab:apps": {  
    "element_key": "tab:apps",  
    "element_type": "tab",  
    "label": "Apps",  
    "bbox": [  
      203.0,  
      241.0,  
      283.0,  
      337.0  
    ],  
    "confidence": 0.4543,  
    "source": "gemini_vlm",  
    "first_seen_frame": null,  
    "last_seen_frame": null,  
    "first_seen_index": 7,  
    "last_seen_index": 7,  
    "first_seen_at": "2026-05-02T16:39:14.867764+00:00",  
    "last_seen_at": "2026-05-02T16:39:14.867764+00:00",
```

```
    "seen_count": 1,  
    "miss_count": 7,  
    "stale": true,  
    "cached": true  
  },  
  "tab:gaming": {  
    "element_key": "tab:gaming",  
    "element_type": "tab",  
    "label": "Gaming",  
    "bbox": [  
      203.0,  
      321.0,  
      291.0,  
      417.0  
    ],  
    "confidence": 0.4543,  
    "source": "gemini_vlm",  
    "first_seen_frame": null,  
    "last_seen_frame": null,  
    "first_seen_index": 7,  
    "last_seen_index": 7,  
    "first_seen_at": "2026-05-02T16:39:14.867764+00:00",  
    "last_seen_at": "2026-05-02T16:39:14.867764+00:00",  
    "seen_count": 1,  
    "miss_count": 7,  
    "stale": true,  
    "cached": true  
  },  
  "tab:themes": {
```

```
"element_key": "tab:themes",
"element_type": "tab",
"label": "Themes",
"bbox": [
    203.0,
    401.0,
    291.0,
    489.0
],
"confidence": 0.4543,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 7,
"last_seen_index": 7,
"first_seen_at": "2026-05-02T16:39:14.867764+00:00",
"last_seen_at": "2026-05-02T16:39:14.867764+00:00",
"seen_count": 1,
"miss_count": 7,
"stale": true,
"cached": true
},
"button:what's new": {
    "element_key": "button:what's new",
    "element_type": "button",
    "label": "What's New",
    "bbox": [
        179.0,
        985.0,
```

```
    251.0,  
    1015.0  
  ],  
  "confidence": 0.5019,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 7,  
  "last_seen_index": 9,  
  "first_seen_at": "2026-05-02T16:39:14.867764+00:00",  
  "last_seen_at": "2026-05-02T16:40:47.767646+00:00",  
  "seen_count": 2,  
  "miss_count": 5,  
  "stale": true,  
  "cached": true  
},  
"pane:microsoft store": {  
  "element_key": "pane:microsoft store",  
  "element_type": "pane",  
  "label": "Microsoft Store",  
  "bbox": [  
    187.0,  
    73.0,  
    1787.0,  
    1001.0  
  ],  
  "confidence": 0.855,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,
```

```
"last_seen_frame": null,
"first_seen_index": 8,
"last_seen_index": 13,
"first_seen_at": "2026-05-02T16:40:06.610270+00:00",
"last_seen_at": "2026-05-02T16:43:58.308218+00:00",
"seen_count": 2,
"miss_count": 1,
"stale": false,
"cached": true
},
"pane:frontend - google chrome": {
  "element_key": "pane:frontend - google chrome",
  "element_type": "pane",
  "label": "frontend - Google Chrome",
  "bbox": [
    938.0,
    98.0,
    1482.0,
    434.0
  ],
  "confidence": 0.5048,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 8,
  "last_seen_index": 8,
  "first_seen_at": "2026-05-02T16:40:06.610270+00:00",
  "last_seen_at": "2026-05-02T16:40:06.610270+00:00",
  "seen_count": 1,
```

```
    "miss_count": 6,  
    "stale": true,  
    "cached": true  
  },  
  "window:microsoft store": {  
    "element_key": "window:microsoft store",  
    "element_type": "window",  
    "label": "Microsoft Store",  
    "bbox": [  
      115.0,  
      65.0,  
      1579.0,  
      1015.0  
    ],  
    "confidence": 0.5609,  
    "source": "gemini_vlm",  
    "first_seen_frame": null,  
    "last_seen_frame": null,  
    "first_seen_index": 9,  
    "last_seen_index": 9,  
    "first_seen_at": "2026-05-02T16:40:47.767646+00:00",  
    "last_seen_at": "2026-05-02T16:40:47.767646+00:00",  
    "seen_count": 1,  
    "miss_count": 5,  
    "stale": true,  
    "cached": true  
  },  
  "button:apps": {  
    "element_key": "button:apps",
```



```
"element_type": "button",
"label": "Apps",
"bbox": [
    187.0,
    241.0,
    267.0,
    337.0
],
"confidence": 0.5315,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 9,
"last_seen_index": 9,
"first_seen_at": "2026-05-02T16:40:47.767646+00:00",
"last_seen_at": "2026-05-02T16:40:47.767646+00:00",
"seen_count": 1,
"miss_count": 5,
"stale": true,
"cached": true
},
"button:gaming": {
    "element_key": "button:gaming",
    "element_type": "button",
    "label": "Gaming",
    "bbox": [
        187.0,
        321.0,
        267.0,
```

```
393.0

],
"confidence": 0.5315,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 9,
"last_seen_index": 9,
"first_seen_at": "2026-05-02T16:40:47.767646+00:00",
"last_seen_at": "2026-05-02T16:40:47.767646+00:00",
"seen_count": 1,
"miss_count": 5,
"stale": true,
"cached": true
},
"button:themes": {
  "element_key": "button:themes",
  "element_type": "button",
  "label": "Themes",
  "bbox": [
    195.0,
    409.0,
    243.0,
    497.0
  ],
  "confidence": 0.5315,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
```

```
"first_seen_index": 9,  
"last_seen_index": 9,  
"first_seen_at": "2026-05-02T16:40:47.767646+00:00",  
"last_seen_at": "2026-05-02T16:40:47.767646+00:00",  
"seen_count": 1,  
"miss_count": 5,  
"stale": true,  
"cached": true  
},  
"button:downloads": {  
  "element_key": "button:downloads",  
  "element_type": "button",  
  "label": "Downloads",  
  "bbox": [  
    181.0,  
    1014.0,  
    247.0,  
    1015.0  
  ],  
  "confidence": 0.5019,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 9,  
  "last_seen_index": 9,  
  "first_seen_at": "2026-05-02T16:40:47.767646+00:00",  
  "last_seen_at": "2026-05-02T16:40:47.767646+00:00",  
  "seen_count": 1,  
  "miss_count": 5,
```

```
    "stale": true,
    "cached": true
  },
  "button:library": {
    "element_key": "button:library",
    "element_type": "button",
    "label": "Library",
    "bbox": [
      181.0,
      1014.0,
      247.0,
      1015.0
    ],
    "confidence": 0.5019,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 9,
    "last_seen_index": 9,
    "first_seen_at": "2026-05-02T16:40:47.767646+00:00",
    "last_seen_at": "2026-05-02T16:40:47.767646+00:00",
    "seen_count": 1,
    "miss_count": 5,
    "stale": true,
    "cached": true
  },
  "pane:microsoft 365 banner": {
    "element_key": "pane:microsoft 365 banner",
    "element_type": "pane",
```

```
"label": "Microsoft 365 Banner",
"bbox": [
  179.0,
  113.0,
  1043.0,
  689.0
],
"confidence": 0.5315,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 9,
"last_seen_index": 9,
"first_seen_at": "2026-05-02T16:40:47.767646+00:00",
"last_seen_at": "2026-05-02T16:40:47.767646+00:00",
"seen_count": 1,
"miss_count": 5,
"stale": true,
"cached": true
},
"button:see details": {
  "element_key": "button:see details",
  "element_type": "button",
  "label": "See details",
  "bbox": [
    299.0,
    561.0,
    355.0,
    625.0
  ]
}
```

```
],  
  "confidence": 0.5019,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 9,  
  "last_seen_index": 9,  
  "first_seen_at": "2026-05-02T16:40:47.767646+00:00",  
  "last_seen_at": "2026-05-02T16:40:47.767646+00:00",  
  "seen_count": 1,  
  "miss_count": 5,  
  "stale": true,  
  "cached": true  
},  
"pane:pdf x banner": {  
  "element_key": "pane:pdf x banner",  
  "element_type": "pane",  
  "label": "PDF X Banner",  
  "bbox": [  
    875.0,  
    137.0,  
    1451.0,  
    505.0  
  ],  
  "confidence": 0.5315,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 9,
```

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"last_seen_index": 9,  
"first_seen_at": "2026-05-02T16:40:47.767646+00:00",  
"last_seen_at": "2026-05-02T16:40:47.767646+00:00",  
"seen_count": 1,  
"miss_count": 5,  
"stale": true,  
"cached": true  
},  
"button:get": {  
  "element_key": "button:get",  
  "element_type": "button",  
  "label": "Get",  
  "bbox": [  
    955.0,  
    385.0,  
    1083.0,  
    441.0  
  ],  
  "confidence": 0.5019,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 9,  
  "last_seen_index": 9,  
  "first_seen_at": "2026-05-02T16:40:47.767646+00:00",  
  "last_seen_at": "2026-05-02T16:40:47.767646+00:00",  
  "seen_count": 1,  
  "miss_count": 5,  
  "stale": true,
```

```
    "cached": true
  },
  "pane:raid: shadow legends banner": {
    "element_key": "pane:raid: shadow legends banner",
    "element_type": "pane",
    "label": "RAID: Shadow Legends Banner",
    "bbox": [
      915.0,
      449.0,
      1195.0,
      641.0
    ],
    "confidence": 0.5315,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 9,
    "last_seen_index": 9,
    "first_seen_at": "2026-05-02T16:40:47.767646+00:00",
    "last_seen_at": "2026-05-02T16:40:47.767646+00:00",
    "seen_count": 1,
    "miss_count": 5,
    "stale": true,
    "cached": true
  },
  "pane:cooking fever banner": {
    "element_key": "pane:cooking fever banner",
    "element_type": "pane",
    "label": "Cooking Fever Banner",
```



```
"bbox": [
  1139.0,
  449.0,
  1419.0,
  641.0
],
"confidence": 0.5315,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 9,
"last_seen_index": 9,
"first_seen_at": "2026-05-02T16:40:47.767646+00:00",
"last_seen_at": "2026-05-02T16:40:47.767646+00:00",
"seen_count": 1,
"miss_count": 5,
"stale": true,
"cached": true
},
"heading:trending games": {
  "element_key": "heading:trending games",
  "element_type": "heading",
  "label": "Trending games",
  "bbox": [
    307.0,
    657.0,
    459.0,
    697.0
  ],

```

```
"confidence": 0.5019,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 9,
"last_seen_index": 9,
"first_seen_at": "2026-05-02T16:40:47.767646+00:00",
"last_seen_at": "2026-05-02T16:40:47.767646+00:00",
"seen_count": 1,
"miss_count": 5,
"stale": true,
"cached": true
},
"heading:trending apps": {
  "element_key": "heading:trending apps",
  "element_type": "heading",
  "label": "Trending apps",
  "bbox": [
    851.0,
    657.0,
    995.0,
    697.0
  ],
  "confidence": 0.5019,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 9,
  "last_seen_index": 9,
```

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"first_seen_at": "2026-05-02T16:40:47.767646+00:00",
"last_seen_at": "2026-05-02T16:40:47.767646+00:00",
"seen_count": 1,
"miss_count": 5,
"stale": true,
"cached": true
},
"button:task view": {
  "element_key": "button:task view",
  "element_type": "button",
  "label": "Task View",
  "bbox": [
    883.0,
    913.0,
    947.0,
    977.0
  ],
  "confidence": 0.5315,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 9,
  "last_seen_index": 9,
  "first_seen_at": "2026-05-02T16:40:47.767646+00:00",
  "last_seen_at": "2026-05-02T16:40:47.767646+00:00",
  "seen_count": 1,
  "miss_count": 5,
  "stale": true,
  "cached": true
}
```

```
},  
"button:file explorer": {  
  "element_key": "button:file explorer",  
  "element_type": "button",  
  "label": "File Explorer",  
  "bbox": [  
    939.0,  
    913.0,  
    995.0,  
    977.0  
  ],  
  "confidence": 0.5315,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 9,  
  "last_seen_index": 9,  
  "first_seen_at": "2026-05-02T16:40:47.767646+00:00",  
  "last_seen_at": "2026-05-02T16:40:47.767646+00:00",  
  "seen_count": 1,  
  "miss_count": 5,  
  "stale": true,  
  "cached": true  
},  
"button:microsoft store": {  
  "element_key": "button:microsoft store",  
  "element_type": "button",  
  "label": "Microsoft Store",  
  "bbox": [  

```

```
    995.0,  
    913.0,  
    1051.0,  
    977.0  
  ],  
  "confidence": 0.5315,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 9,  
  "last_seen_index": 9,  
  "first_seen_at": "2026-05-02T16:40:47.767646+00:00",  
  "last_seen_at": "2026-05-02T16:40:47.767646+00:00",  
  "seen_count": 1,  
  "miss_count": 5,  
  "stale": true,  
  "cached": true  
},  
"button:settings": {  
  "element_key": "button:settings",  
  "element_type": "button",  
  "label": "Settings",  
  "bbox": [  
    1043.0,  
    913.0,  
    1099.0,  
    977.0  
  ],  
  "confidence": 0.5315,
```

```
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 9,
"last_seen_index": 9,
"first_seen_at": "2026-05-02T16:40:47.767646+00:00",
"last_seen_at": "2026-05-02T16:40:47.767646+00:00",
"seen_count": 1,
"miss_count": 5,
"stale": true,
"cached": true
},
"button:xbox": {
  "element_key": "button:xbox",
  "element_type": "button",
  "label": "Xbox",
  "bbox": [
    1099.0,
    913.0,
    1155.0,
    977.0
  ],
  "confidence": 0.5315,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 9,
  "last_seen_index": 9,
  "first_seen_at": "2026-05-02T16:40:47.767646+00:00",
```

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    "last_seen_at": "2026-05-02T16:40:47.767646+00:00",
    "seen_count": 1,
    "miss_count": 5,
    "stale": true,
    "cached": true
  },
  "button:google chrome": {
    "element_key": "button:google chrome",
    "element_type": "button",
    "label": "Google Chrome",
    "bbox": [
      1147.0,
      913.0,
      1211.0,
      977.0
    ],
    "confidence": 0.5315,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 9,
    "last_seen_index": 9,
    "first_seen_at": "2026-05-02T16:40:47.767646+00:00",
    "last_seen_at": "2026-05-02T16:40:47.767646+00:00",
    "seen_count": 1,
    "miss_count": 5,
    "stale": true,
    "cached": true
  },
}
```

```
"button:visual studio code": {
  "element_key": "button:visual studio code",
  "element_type": "button",
  "label": "Visual Studio Code",
  "bbox": [
    1203.0,
    913.0,
    1259.0,
    977.0
  ],
  "confidence": 0.5315,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 9,
  "last_seen_index": 9,
  "first_seen_at": "2026-05-02T16:40:47.767646+00:00",
  "last_seen_at": "2026-05-02T16:40:47.767646+00:00",
  "seen_count": 1,
  "miss_count": 5,
  "stale": true,
  "cached": true
},
"widget:weather": {
  "element_key": "widget:weather",
  "element_type": "widget",
  "label": "Weather",
  "bbox": [
    187.0,
```



```
    913.0,  
    307.0,  
    977.0  
  ],  
  "confidence": 0.5019,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 9,  
  "last_seen_index": 9,  
  "first_seen_at": "2026-05-02T16:40:47.767646+00:00",  
  "last_seen_at": "2026-05-02T16:40:47.767646+00:00",  
  "seen_count": 1,  
  "miss_count": 5,  
  "stale": true,  
  "cached": true  
},  
"label:clock": {  
  "element_key": "label:clock",  
  "element_type": "label",  
  "label": "Clock",  
  "bbox": [  
    1699.0,  
    913.0,  
    1787.0,  
    977.0  
  ],  
  "confidence": 0.5609,  
  "source": "gemini_vlm",
```

```
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 9,
"last_seen_index": 9,
"first_seen_at": "2026-05-02T16:40:47.767646+00:00",
"last_seen_at": "2026-05-02T16:40:47.767646+00:00",
"seen_count": 1,
"miss_count": 5,
"stale": true,
"cached": true
},
"label:microsoft store": {
  "element_key": "label:microsoft store",
  "element_type": "label",
  "label": "Microsoft Store",
  "bbox": [
    243.0,
    97.0,
    403.0,
    137.0
  ],
  "confidence": 0.7695,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 10,
  "last_seen_index": 12,
  "first_seen_at": "2026-05-02T16:41:36.977103+00:00",
  "last_seen_at": "2026-05-02T16:43:09.638245+00:00",
```

```
    "seen_count": 2,  
    "miss_count": 2,  
    "stale": true,  
    "cached": true  
  },  
  "heading:microsoft copilot": {  
    "element_key": "heading:microsoft copilot",  
    "element_type": "heading",  
    "label": "Microsoft Copilot",  
    "bbox": [  
      319.0,  
      290.0,  
      639.0,  
      330.0  
    ],  
    "confidence": 0.7144,  
    "source": "gemini_vlm",  
    "first_seen_frame": null,  
    "last_seen_frame": null,  
    "first_seen_index": 11,  
    "last_seen_index": 11,  
    "first_seen_at": "2026-05-02T16:42:17.919332+00:00",  
    "last_seen_at": "2026-05-02T16:42:17.919332+00:00",  
    "seen_count": 1,  
    "miss_count": 3,  
    "stale": true,  
    "cached": true  
  },  
  "text:search smarter and unleash creativity with copilot": {
```

```
    "element_key": "text:search smarter and unleash creativity with  
copilot",  
    "element_type": "text",  
    "label": "Search smarter and unleash creativity with Copilot",  
    "bbox": [  
        327.0,  
        322.0,  
        615.0,  
        378.0  
    ],  
    "confidence": 0.7071,  
    "source": "gemini_vlm",  
    "first_seen_frame": null,  
    "last_seen_frame": null,  
    "first_seen_index": 11,  
    "last_seen_index": 11,  
    "first_seen_at": "2026-05-02T16:42:17.919332+00:00",  
    "last_seen_at": "2026-05-02T16:42:17.919332+00:00",  
    "seen_count": 1,  
    "miss_count": 3,  
    "stale": true,  
    "cached": true  
},  
"button:install": {  
    "element_key": "button:install",  
    "element_type": "button",  
    "label": "Install",  
    "bbox": [  
        343.0,
```

```
    386.0,  
    503.0,  
    434.0  
  ],  
  "confidence": 0.7217,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 11,  
  "last_seen_index": 11,  
  "first_seen_at": "2026-05-02T16:42:17.919332+00:00",  
  "last_seen_at": "2026-05-02T16:42:17.919332+00:00",  
  "seen_count": 1,  
  "miss_count": 3,  
  "stale": true,  
  "cached": true  
},  
"label:12+ parental guidance recommended": {  
  "element_key": "label:12+ parental guidance recommended",  
  "element_type": "label",  
  "label": "12+ Parental Guidance Recommended",  
  "bbox": [  
    327.0,  
    434.0,  
    615.0,  
    498.0  
  ],  
  "confidence": 0.6998,  
  "source": "gemini_vlm",
```

```
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 11,
"last_seen_index": 11,
"first_seen_at": "2026-05-02T16:42:17.919332+00:00",
"last_seen_at": "2026-05-02T16:42:17.919332+00:00",
"seen_count": 1,
"miss_count": 3,
"stale": true,
"cached": true
},
"text:copilot": {
  "element_key": "text:copilot",
  "element_type": "text",
  "label": "Copilot",
  "bbox": [
    703.0,
    258.0,
    807.0,
    298.0
  ],
  "confidence": 0.6925,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 11,
  "last_seen_index": 11,
  "first_seen_at": "2026-05-02T16:42:17.919332+00:00",
  "last_seen_at": "2026-05-02T16:42:17.919332+00:00",
```

```
    "seen_count": 1,
    "miss_count": 3,
    "stale": true,
    "cached": true
  },
  "icon:copilot logo": {
    "element_key": "icon:copilot logo",
    "element_type": "icon",
    "label": "Copilot logo",
    "bbox": [
      687.0,
      322.0,
      807.0,
      418.0
    ],
    "confidence": 0.7217,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 11,
    "last_seen_index": 11,
    "first_seen_at": "2026-05-02T16:42:17.919332+00:00",
    "last_seen_at": "2026-05-02T16:42:17.919332+00:00",
    "seen_count": 1,
    "miss_count": 3,
    "stale": true,
    "cached": true
  },
  "icon:small icon": {
```

```
"element_key": "icon:small icon",
"element_type": "icon",
"label": "small icon",
"bbox": [
  319.0,
  594.0,
  351.0,
  610.0
],
"confidence": 0.6561,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 11,
"last_seen_index": 11,
"first_seen_at": "2026-05-02T16:42:17.919332+00:00",
"last_seen_at": "2026-05-02T16:42:17.919332+00:00",
"seen_count": 1,
"miss_count": 3,
"stale": true,
"cached": true
},
"icon:user profile": {
  "element_key": "icon:user profile",
  "element_type": "icon",
  "label": "User Profile",
  "bbox": [
    1531.0,
    97.0,
```



```
    1571.0,  
    137.0  
  ],  
  "confidence": 0.729,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 12,  
  "last_seen_index": 12,  
  "first_seen_at": "2026-05-02T16:43:09.638245+00:00",  
  "last_seen_at": "2026-05-02T16:43:09.638245+00:00",  
  "seen_count": 1,  
  "miss_count": 2,  
  "stale": true,  
  "cached": true  
}  
},  
"current_elements": [  
  {  
    "id": "tab_frontend",  
    "type": "tab",  
    "label": "frontend",  
    "description": "Browser tab for the frontend application",  
    "state": "active",  
    "dx": 188,  
    "dy": 68,  
    "frame_dx": 303,  
    "frame_dy": 133,  
    "confidence": 0.95,
```

```
"source": "gemini_vlm",
"frame_bbox": [
    179,
    105,
    427,
    161
],
"session_memory_key": "tab:frontend",
"cached": false
},
{
    "id": "tab_new_tab",
    "type": "tab",
    "label": "New tab",
    "description": "Browser tab for a new page",
    "state": "inactive",
    "dx": 372,
    "dy": 68,
    "frame_dx": 487,
    "frame_dy": 133,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        371,
        105,
        603,
        161
    ],
    "session_memory_key": "tab:new tab",
```

```
"cached": false
},
{
  "id": "btn_new_tab",
  "type": "button",
  "label": "New tab button",
  "description": "Button to open a new browser tab",
  "state": "enabled",
  "dx": 484,
  "dy": 76,
  "frame_dx": 599,
  "frame_dy": 141,
  "confidence": 0.9,
  "source": "gemini_vlm",
  "frame_bbox": [
    579,
    129,
    619,
    153
  ],
  "session_memory_key": "button:new tab button",
  "cached": false
},
{
  "id": "btn_back",
  "type": "button",
  "label": "Back",
  "description": "Browser back button",
  "state": "disabled",
```

```
"dx": 112,
"dy": 120,
"frame_dx": 227,
"frame_dy": 185,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    211,
    153,
    243,
    217
],
"session_memory_key": "button:back",
"cached": false
},
{
    "id": "btn_forward",
    "type": "button",
    "label": "Forward",
    "description": "Browser forward button",
    "state": "disabled",
    "dx": 144,
    "dy": 120,
    "frame_dx": 259,
    "frame_dy": 185,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        235,
```

```
        153,  
        283,  
        217  
    ],  
    "session_memory_key": "button:forward",  
    "cached": false  
},  
{  
    "id": "btn_reload",  
    "type": "button",  
    "label": "Reload",  
    "description": "Browser reload button",  
    "state": "enabled",  
    "dx": 184,  
    "dy": 120,  
    "frame_dx": 299,  
    "frame_dy": 185,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        275,  
        153,  
        323,  
        217  
    ],  
    "session_memory_key": "button:reload",  
    "cached": false  
},  
{
```

```
    "id": "address_bar",
    "type": "input",
    "label": "Address bar",
    "description": "Browser address bar showing localhost:5173",
    "state": "enabled",
    "dx": 616,
    "dy": 120,
    "frame_dx": 731,
    "frame_dy": 185,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        179,
        153,
        1283,
        217
    ],
    "session_memory_key": "input:address bar",
    "cached": false
}
],
"resolved_elements": [
    {
        "id": "tab_frontend",
        "type": "tab",
        "label": "frontend",
        "description": "Browser tab for the frontend application",
        "state": "active",
        "dx": 188,
```

```
"dy": 68,
"frame_dx": 303,
"frame_dy": 133,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
  179,
  105,
  427,
  161
],
"session_memory_key": "tab:frontend",
"cached": false
},
{
  "id": "tab_new_tab",
  "type": "tab",
  "label": "New tab",
  "description": "Browser tab for a new page",
  "state": "inactive",
  "dx": 372,
  "dy": 68,
  "frame_dx": 487,
  "frame_dy": 133,
  "confidence": 0.95,
  "source": "gemini_vlm",
  "frame_bbox": [
    371,
    105,
```

```
        603,  
        161  
    ],  
    "session_memory_key": "tab:new tab",  
    "cached": false  
},  
{  
    "id": "btn_new_tab",  
    "type": "button",  
    "label": "New tab button",  
    "description": "Button to open a new browser tab",  
    "state": "enabled",  
    "dx": 484,  
    "dy": 76,  
    "frame_dx": 599,  
    "frame_dy": 141,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        579,  
        129,  
        619,  
        153  
    ],  
    "session_memory_key": "button:new tab button",  
    "cached": false  
},  
{  
    "id": "btn_back",
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"type": "button",
"label": "Back",
"description": "Browser back button",
"state": "disabled",
"dx": 112,
"dy": 120,
"frame_dx": 227,
"frame_dy": 185,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    211,
    153,
    243,
    217
],
"session_memory_key": "button:back",
"cached": false
},
{
    "id": "btn_forward",
    "type": "button",
    "label": "Forward",
    "description": "Browser forward button",
    "state": "disabled",
    "dx": 144,
    "dy": 120,
    "frame_dx": 259,
    "frame_dy": 185,
```

```
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    235,
    153,
    283,
    217
],
"session_memory_key": "button:forward",
"cached": false
},
{
    "id": "btn_reload",
    "type": "button",
    "label": "Reload",
    "description": "Browser reload button",
    "state": "enabled",
    "dx": 184,
    "dy": 120,
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    "frame_dy": 185,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        275,
        153,
        323,
        217
    ],
    ],
```

```
    "session_memory_key": "button:reload",
    "cached": false
  },
  {
    "id": "address_bar",
    "type": "input",
    "label": "Address bar",
    "description": "Browser address bar showing localhost:5173",
    "state": "enabled",
    "dx": 616,
    "dy": 120,
    "frame_dx": 731,
    "frame_dy": 185,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
      179,
      153,
      1283,
      217
    ],
    "session_memory_key": "input:address bar",
    "cached": false
  }
],
"screen_data": {
  "image": "E:\\sliit-
project\\ScreenAwareTaskAgent\\vision\\data\\sessions\\shot_1777739562_bcb8e7\\pr
ocessing\\proc_1777739563_7a9e13\\preprocessed_frames\\frame_1777739562_d6dfdf.jp
g",
```

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"image_size": {
  "width": 1920,
  "height": 1080
},
"screen_bbox": [
  115,
  65,
  1805,
  1015
],
"screen_size": {
  "width": 1690,
  "height": 950
},
"coordinate_system": "pixel",
"elements": [
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    "id": "tab_frontend",
    "type": "tab",
    "label": "frontend",
    "description": "Browser tab for the frontend application",
    "state": "active",
    "dx": 188,
    "dy": 68,
    "frame_dx": 303,
    "frame_dy": 133,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
```

```
        179,  
        105,  
        427,  
        161  
    ],  
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    "cached": false  
},  
{  
    "id": "tab_new_tab",  
    "type": "tab",  
    "label": "New tab",  
    "description": "Browser tab for a new page",  
    "state": "inactive",  
    "dx": 372,  
    "dy": 68,  
    "frame_dx": 487,  
    "frame_dy": 133,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        371,  
        105,  
        603,  
        161  
    ],  
    "session_memory_key": "tab:new tab",  
    "cached": false  
},
```

```
{
  "id": "btn_new_tab",
  "type": "button",
  "label": "New tab button",
  "description": "Button to open a new browser tab",
  "state": "enabled",
  "dx": 484,
  "dy": 76,
  "frame_dx": 599,
  "frame_dy": 141,
  "confidence": 0.9,
  "source": "gemini_vlm",
  "frame_bbox": [
    579,
    129,
    619,
    153
  ],
  "session_memory_key": "button:new tab button",
  "cached": false
},
{
  "id": "btn_back",
  "type": "button",
  "label": "Back",
  "description": "Browser back button",
  "state": "disabled",
  "dx": 112,
  "dy": 120,
```

```
"frame_dx": 227,  
"frame_dy": 185,  
"confidence": 0.9,  
"source": "gemini_vlm",  
"frame_bbox": [  
    211,  
    153,  
    243,  
    217  
],  
"session_memory_key": "button:back",  
"cached": false  
},  
{  
    "id": "btn_forward",  
    "type": "button",  
    "label": "Forward",  
    "description": "Browser forward button",  
    "state": "disabled",  
    "dx": 144,  
    "dy": 120,  
    "frame_dx": 259,  
    "frame_dy": 185,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        235,  
        153,  
        283,
```

```
        217
    ],
    "session_memory_key": "button:forward",
    "cached": false
},
{
    "id": "btn_reload",
    "type": "button",
    "label": "Reload",
    "description": "Browser reload button",
    "state": "enabled",
    "dx": 184,
    "dy": 120,
    "frame_dx": 299,
    "frame_dy": 185,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        275,
        153,
        323,
        217
    ],
    "session_memory_key": "button:reload",
    "cached": false
},
{
    "id": "address_bar",
    "type": "input",
```



```
    "label": "Address bar",
    "description": "Browser address bar showing localhost:5173",
    "state": "enabled",
    "dx": 616,
    "dy": 120,
    "frame_dx": 731,
    "frame_dy": 185,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        179,
        153,
        1283,
        217
    ],
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    "cached": false
}
],
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"current_elements": [
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        "id": "tab_frontend",
        "type": "tab",
        "label": "frontend",
        "description": "Browser tab for the frontend application",
        "state": "active",
        "dx": 188,
        "dy": 68,
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"frame_dx": 303,  
"frame_dy": 133,  
"confidence": 0.95,  
"source": "gemini_vlm",  
"frame_bbox": [  
    179,  
    105,  
    427,  
    161  
],  
"session_memory_key": "tab:frontend",  
"cached": false  
},  
{  
    "id": "tab_new_tab",  
    "type": "tab",  
    "label": "New tab",  
    "description": "Browser tab for a new page",  
    "state": "inactive",  
    "dx": 372,  
    "dy": 68,  
    "frame_dx": 487,  
    "frame_dy": 133,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        371,  
        105,  
        603,
```

```
161
],
"session_memory_key": "tab:new tab",
"cached": false
},
{
  "id": "btn_new_tab",
  "type": "button",
  "label": "New tab button",
  "description": "Button to open a new browser tab",
  "state": "enabled",
  "dx": 484,
  "dy": 76,
  "frame_dx": 599,
  "frame_dy": 141,
  "confidence": 0.9,
  "source": "gemini_vlm",
  "frame_bbox": [
    579,
    129,
    619,
    153
  ],
  "session_memory_key": "button:new tab button",
  "cached": false
},
{
  "id": "btn_back",
  "type": "button",
```

```
"label": "Back",
"description": "Browser back button",
"state": "disabled",
"dx": 112,
"dy": 120,
"frame_dx": 227,
"frame_dy": 185,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    211,
    153,
    243,
    217
],
"session_memory_key": "button:back",
"cached": false
},
{
    "id": "btn_forward",
    "type": "button",
    "label": "Forward",
    "description": "Browser forward button",
    "state": "disabled",
    "dx": 144,
    "dy": 120,
    "frame_dx": 259,
    "frame_dy": 185,
    "confidence": 0.9,
```

```
"source": "gemini_vlm",
"frame_bbox": [
    235,
    153,
    283,
    217
],
"session_memory_key": "button:forward",
"cached": false
},
{
    "id": "btn_reload",
    "type": "button",
    "label": "Reload",
    "description": "Browser reload button",
    "state": "enabled",
    "dx": 184,
    "dy": 120,
    "frame_dx": 299,
    "frame_dy": 185,
    "confidence": 0.9,
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    "frame_bbox": [
        275,
        153,
        323,
        217
    ],
    "session_memory_key": "button:reload",
```

```
    "cached": false
  },
  {
    "id": "address_bar",
    "type": "input",
    "label": "Address bar",
    "description": "Browser address bar showing localhost:5173",
    "state": "enabled",
    "dx": 616,
    "dy": 120,
    "frame_dx": 731,
    "frame_dy": 185,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
      179,
      153,
      1283,
      217
    ],
    "session_memory_key": "input:address bar",
    "cached": false
  }
],
"resolved_elements": [
  {
    "id": "tab_frontend",
    "type": "tab",
    "label": "frontend",
```

```
    "description": "Browser tab for the frontend application",
    "state": "active",
    "dx": 188,
    "dy": 68,
    "frame_dx": 303,
    "frame_dy": 133,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        179,
        105,
        427,
        161
    ],
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    "cached": false
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    "id": "tab_new_tab",
    "type": "tab",
    "label": "New tab",
    "description": "Browser tab for a new page",
    "state": "inactive",
    "dx": 372,
    "dy": 68,
    "frame_dx": 487,
    "frame_dy": 133,
    "confidence": 0.95,
    "source": "gemini_vlm",
```

```
"frame_bbox": [  
    371,  
    105,  
    603,  
    161  
],  
"session_memory_key": "tab:new tab",  
"cached": false  
},  
{  
    "id": "btn_new_tab",  
    "type": "button",  
    "label": "New tab button",  
    "description": "Button to open a new browser tab",  
    "state": "enabled",  
    "dx": 484,  
    "dy": 76,  
    "frame_dx": 599,  
    "frame_dy": 141,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        579,  
        129,  
        619,  
        153  
    ],  
    "session_memory_key": "button:new tab button",  
    "cached": false
```



```
},  
{  
  "id": "btn_back",  
  "type": "button",  
  "label": "Back",  
  "description": "Browser back button",  
  "state": "disabled",  
  "dx": 112,  
  "dy": 120,  
  "frame_dx": 227,  
  "frame_dy": 185,  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    211,  
    153,  
    243,  
    217  
  ],  
  "session_memory_key": "button:back",  
  "cached": false  
},  
{  
  "id": "btn_forward",  
  "type": "button",  
  "label": "Forward",  
  "description": "Browser forward button",  
  "state": "disabled",  
  "dx": 144,
```

```
"dy": 120,
"frame_dx": 259,
"frame_dy": 185,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    235,
    153,
    283,
    217
],
"session_memory_key": "button:forward",
"cached": false
},
{
    "id": "btn_reload",
    "type": "button",
    "label": "Reload",
    "description": "Browser reload button",
    "state": "enabled",
    "dx": 184,
    "dy": 120,
    "frame_dx": 299,
    "frame_dy": 185,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        275,
        153,
```

```
        323,  
        217  
    ],  
    "session_memory_key": "button:reload",  
    "cached": false  
},  
{  
    "id": "address_bar",  
    "type": "input",  
    "label": "Address bar",  
    "description": "Browser address bar showing localhost:5173",  
    "state": "enabled",  
    "dx": 616,  
    "dy": 120,  
    "frame_dx": 731,  
    "frame_dy": 185,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        179,  
        153,  
        1283,  
        217  
    ],  
    "session_memory_key": "input:address bar",  
    "cached": false  
}  
],  
"session_knowledge_base": {
```

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"session_id": "shot_1777739562_bcb8e7",
"started_at": "2026-05-02T16:32:42.997236+00:00",
"updated_at": "2026-05-02T16:33:18.641310+00:00",
"stale_after_misses": 3,
"confidence_floor": 0.4,
"decay_per_miss": 0.9,
"frame_count": 1,
"frame_name": "frame_1777739562_d6dfdf.jpg",
"frame_index": 0,
"image_size": {
  "width": 1920,
  "height": 1080
},
"known_elements": {
  "tab:frontend": {
    "element_key": "tab:frontend",
    "element_type": "tab",
    "label": "frontend",
    "bbox": [
      179.0,
      105.0,
      427.0,
      161.0
    ],
    "confidence": 0.95,
    "source": "gemini_vlm",
    "first_seen_frame": "frame_1777739562_d6dfdf.jpg",
    "last_seen_frame": "frame_1777739562_d6dfdf.jpg",
    "first_seen_at": "2026-05-02T16:33:18.641310+00:00",
```

```
"last_seen_at": "2026-05-02T16:33:18.641310+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777739562_d6dfdf.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T16:33:18.641310+00:00",
    "confidence": 0.95
  }
],
},
"tab:new tab": {
  "element_key": "tab:new tab",
  "element_type": "tab",
  "label": "New tab",
  "bbox": [
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    105.0,
    603.0,
    161.0
  ],
  "confidence": 0.95,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777739562_d6dfdf.jpg",
```

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"last_seen_frame": "frame_1777739562_d6dfdf.jpg",
"first_seen_at": "2026-05-02T16:33:18.641310+00:00",
"last_seen_at": "2026-05-02T16:33:18.641310+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777739562_d6dfdf.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T16:33:18.641310+00:00",
    "confidence": 0.95
  }
],
},
"button:new tab button": {
  "element_key": "button:new tab button",
  "element_type": "button",
  "label": "New tab button",
  "bbox": [
    579.0,
    129.0,
    619.0,
    153.0
  ],
  "confidence": 0.9,
```

```
"source": "gemini_vlm",
"first_seen_frame": "frame_1777739562_d6dfdf.jpg",
"last_seen_frame": "frame_1777739562_d6dfdf.jpg",
"first_seen_at": "2026-05-02T16:33:18.641310+00:00",
"last_seen_at": "2026-05-02T16:33:18.641310+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777739562_d6dfdf.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-02T16:33:18.641310+00:00",
    "confidence": 0.9
  }
]
},
"button:back": {
  "element_key": "button:back",
  "element_type": "button",
  "label": "Back",
  "bbox": [
    211.0,
    153.0,
    243.0,
    217.0
```

```
    ],
    "confidence": 0.9,
    "source": "gemini_vlm",
    "first_seen_frame": "frame_1777739562_d6dfdf.jpg",
    "last_seen_frame": "frame_1777739562_d6dfdf.jpg",
    "first_seen_at": "2026-05-02T16:33:18.641310+00:00",
    "last_seen_at": "2026-05-02T16:33:18.641310+00:00",
    "first_seen_index": 0,
    "last_seen_index": 0,
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false,
    "history": [
      {
        "frame_name": "frame_1777739562_d6dfdf.jpg",
        "frame_index": 0,
        "seen_at": "2026-05-02T16:33:18.641310+00:00",
        "confidence": 0.9
      }
    ]
  },
  "button:forward": {
    "element_key": "button:forward",
    "element_type": "button",
    "label": "Forward",
    "bbox": [
      235.0,
      153.0,
```



```
        283.0,
        217.0
    ],
    "confidence": 0.9,
    "source": "gemini_vlm",
    "first_seen_frame": "frame_1777739562_d6dfdf.jpg",
    "last_seen_frame": "frame_1777739562_d6dfdf.jpg",
    "first_seen_at": "2026-05-02T16:33:18.641310+00:00",
    "last_seen_at": "2026-05-02T16:33:18.641310+00:00",
    "first_seen_index": 0,
    "last_seen_index": 0,
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false,
    "history": [
        {
            "frame_name": "frame_1777739562_d6dfdf.jpg",
            "frame_index": 0,
            "seen_at": "2026-05-02T16:33:18.641310+00:00",
            "confidence": 0.9
        }
    ]
},
"button:reload": {
    "element_key": "button:reload",
    "element_type": "button",
    "label": "Reload",
    "bbox": [
```

```
        275.0,
        153.0,
        323.0,
        217.0
    ],
    "confidence": 0.9,
    "source": "gemini_vlm",
    "first_seen_frame": "frame_1777739562_d6dfdf.jpg",
    "last_seen_frame": "frame_1777739562_d6dfdf.jpg",
    "first_seen_at": "2026-05-02T16:33:18.641310+00:00",
    "last_seen_at": "2026-05-02T16:33:18.641310+00:00",
    "first_seen_index": 0,
    "last_seen_index": 0,
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false,
    "history": [
        {
            "frame_name": "frame_1777739562_d6dfdf.jpg",
            "frame_index": 0,
            "seen_at": "2026-05-02T16:33:18.641310+00:00",
            "confidence": 0.9
        }
    ]
},
"input:address bar": {
    "element_key": "input:address bar",
    "element_type": "input",
```

```
    "label": "Address bar",
    "bbox": [
        179.0,
        153.0,
        1283.0,
        217.0
    ],
    "confidence": 0.95,
    "source": "gemini_vlm",
    "first_seen_frame": "frame_1777739562_d6dfdf.jpg",
    "last_seen_frame": "frame_1777739562_d6dfdf.jpg",
    "first_seen_at": "2026-05-02T16:33:18.641310+00:00",
    "last_seen_at": "2026-05-02T16:33:18.641310+00:00",
    "first_seen_index": 0,
    "last_seen_index": 0,
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false,
    "history": [
        {
            "frame_name": "frame_1777739562_d6dfdf.jpg",
            "frame_index": 0,
            "seen_at": "2026-05-02T16:33:18.641310+00:00",
            "confidence": 0.95
        }
    ]
}
},
```

```
"resolved_elements": [  
  {  
    "id": "tab_frontend",  
    "type": "tab",  
    "label": "frontend",  
    "description": "Browser tab for the frontend application",  
    "state": "active",  
    "dx": 188,  
    "dy": 68,  
    "frame_dx": 303,  
    "frame_dy": 133,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
      179,  
      105,  
      427,  
      161  
    ],  
    "session_memory_key": "tab:frontend",  
    "cached": false  
  },  
  {  
    "id": "tab_new_tab",  
    "type": "tab",  
    "label": "New tab",  
    "description": "Browser tab for a new page",  
    "state": "inactive",  
    "dx": 372,
```

```
"dy": 68,
"frame_dx": 487,
"frame_dy": 133,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    371,
    105,
    603,
    161
],
"session_memory_key": "tab:new tab",
"cached": false
},
{
    "id": "btn_new_tab",
    "type": "button",
    "label": "New tab button",
    "description": "Button to open a new browser tab",
    "state": "enabled",
    "dx": 484,
    "dy": 76,
    "frame_dx": 599,
    "frame_dy": 141,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        579,
        129,
```

```
        619,  
        153  
    ],  
    "session_memory_key": "button:new tab button",  
    "cached": false  
},  
{  
    "id": "btn_back",  
    "type": "button",  
    "label": "Back",  
    "description": "Browser back button",  
    "state": "disabled",  
    "dx": 112,  
    "dy": 120,  
    "frame_dx": 227,  
    "frame_dy": 185,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        211,  
        153,  
        243,  
        217  
    ],  
    "session_memory_key": "button:back",  
    "cached": false  
},  
{  
    "id": "btn_forward",
```

```
    "type": "button",
    "label": "Forward",
    "description": "Browser forward button",
    "state": "disabled",
    "dx": 144,
    "dy": 120,
    "frame_dx": 259,
    "frame_dy": 185,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        235,
        153,
        283,
        217
    ],
    "session_memory_key": "button:forward",
    "cached": false
},
{
    "id": "btn_reload",
    "type": "button",
    "label": "Reload",
    "description": "Browser reload button",
    "state": "enabled",
    "dx": 184,
    "dy": 120,
    "frame_dx": 299,
    "frame_dy": 185,
```

```
"confidence": 0.9,  
"source": "gemini_vlm",  
"frame_bbox": [  
    275,  
    153,  
    323,  
    217  
],  
"session_memory_key": "button:reload",  
"cached": false  
},  
{  
    "id": "address_bar",  
    "type": "input",  
    "label": "Address bar",  
    "description": "Browser address bar showing localhost:5173",  
    "state": "enabled",  
    "dx": 616,  
    "dy": 120,  
    "frame_dx": 731,  
    "frame_dy": 185,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        179,  
        153,  
        1283,  
        217  
    ],  
}
```



```
        "session_memory_key": "input:address bar",
        "cached": false
    }
],
"current_elements": [
    {
        "id": "tab_frontend",
        "type": "tab",
        "label": "frontend",
        "description": "Browser tab for the frontend application",
        "state": "active",
        "dx": 188,
        "dy": 68,
        "frame_dx": 303,
        "frame_dy": 133,
        "confidence": 0.95,
        "source": "gemini_vlm",
        "frame_bbox": [
            179,
            105,
            427,
            161
        ],
        "session_memory_key": "tab:frontend",
        "cached": false
    },
    {
        "id": "tab_new_tab",
        "type": "tab",
```

```
"label": "New tab",
"description": "Browser tab for a new page",
"state": "inactive",
"dx": 372,
"dy": 68,
"frame_dx": 487,
"frame_dy": 133,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    371,
    105,
    603,
    161
],
"session_memory_key": "tab:new tab",
"cached": false
},
{
    "id": "btn_new_tab",
    "type": "button",
    "label": "New tab button",
    "description": "Button to open a new browser tab",
    "state": "enabled",
    "dx": 484,
    "dy": 76,
    "frame_dx": 599,
    "frame_dy": 141,
    "confidence": 0.9,
```

```
"source": "gemini_vlm",
"frame_bbox": [
    579,
    129,
    619,
    153
],
"session_memory_key": "button:new tab button",
"cached": false
},
{
    "id": "btn_back",
    "type": "button",
    "label": "Back",
    "description": "Browser back button",
    "state": "disabled",
    "dx": 112,
    "dy": 120,
    "frame_dx": 227,
    "frame_dy": 185,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        211,
        153,
        243,
        217
    ],
    "session_memory_key": "button:back",
```

```
    "cached": false
  },
  {
    "id": "btn_forward",
    "type": "button",
    "label": "Forward",
    "description": "Browser forward button",
    "state": "disabled",
    "dx": 144,
    "dy": 120,
    "frame_dx": 259,
    "frame_dy": 185,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
      235,
      153,
      283,
      217
    ],
    "session_memory_key": "button:forward",
    "cached": false
  },
  {
    "id": "btn_reload",
    "type": "button",
    "label": "Reload",
    "description": "Browser reload button",
    "state": "enabled",
```

```
"dx": 184,  
  "dy": 120,  
  "frame_dx": 299,  
  "frame_dy": 185,  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    275,  
    153,  
    323,  
    217  
  ],  
  "session_memory_key": "button:reload",  
  "cached": false  
},  
{  
  "id": "address_bar",  
  "type": "input",  
  "label": "Address bar",  
  "description": "Browser address bar showing localhost:5173",  
  "state": "enabled",  
  "dx": 616,  
  "dy": 120,  
  "frame_dx": 731,  
  "frame_dy": 185,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    179,
```

```
        153,  
        1283,  
        217  
    ],  
    "session_memory_key": "input:address bar",  
    "cached": false  
  }  
]  
}  
}  
}  
},  
"todo": {  
  "steps": [  
    {  
      "id": 1,  
      "action": "Click on the 'New tab' tab to switch to the existing empty  
browser tab.",  
      "target": "Tab 'New tab' @ (372, 68)",  
      "expected_result": "The browser switches focus to the 'New tab' page.",  
      "needs_user_data": false,  
      "user_data_field": null,  
      "status": "done"  
    },  
    {  
      "id": 2,  
      "action": "Click the address bar, type 'https://www.youtube.com', and  
press the Enter key to navigate to the platform.",  
      "target": "Address bar @ (616, 120)",  
      "expected_result": "The browser loads the YouTube homepage.",
```

```
    "needs_user_data": false,
    "user_data_field": null,
    "status": "failed"
  }
],
  "notes": "The screen shows an existing 'New tab' already open. I will switch to that tab and then use the address bar to navigate directly to YouTube as requested.",
  "estimated_complexity": "simple"
},
"step_plans": [
  {
    "step_id": 1,
    "action": "Click on the 'New tab' tab to switch to the existing empty browser tab.",
    "target": "Tab 'New tab' @ (372, 68)",
    "expected_result": "The browser switches focus to the 'New tab' page.",
    "hid_commands": [
      {
        "cmd": "mouse_move",
        "dx": 372,
        "dy": 68
      },
      {
        "cmd": "mouse_click",
        "button": "left"
      }
    ]
  },
  {
    "reasoning": "Moving the cursor to the 'New tab' tab at (372, 68) and clicking to switch focus to it as per the current step."
```

```
},
{
  "step_id": 2,
  "action": "Click the address bar, type 'https://www.youtube.com', and press the Enter key to navigate to the platform.",
  "target": "Address bar @ (616, 120)",
  "expected_result": "The browser loads the YouTube homepage.",
  "hid_commands": [
    {
      "cmd": "mouse_move",
      "meta": {
        "commandId": "01929594-384e-4056-a58d-236345412d38"
      },
      "dx": 168,
      "dy": 164,
      "smooth": true
    },
    {
      "cmd": "mouse_click",
      "meta": {
        "commandId": "3910b5ee-0802-43db-a456-105885236fca"
      },
      "button": "left"
    }
  ],
  "reasoning": "Fallback generated 2 commands from detected elements"
}
],
"step_results": [
```



```
{
  "step_id": 1,
  "action": "Click on the 'New tab' tab to switch to the existing empty
browser tab.",
  "attempt": 1,
  "exec_status": "success",
  "eval_status": "done",
  "confidence": 1.0,
  "reason": "The 'New tab' tab is now the active browser tab as indicated in
the UI elements list.",
  "final_status": "executing",
  "timestamp": "2026-05-02T16:35:05.610600+00:00"
},
{
  "step_id": 2,
  "action": "Click the address bar, type 'https://www.youtube.com', and press
the Enter key to navigate to the platform.",
  "attempt": 3,
  "exec_status": "success",
  "eval_status": "retry",
  "confidence": 1.0,
  "reason": "The current screen shows the Microsoft Store application instead
of a web browser with the YouTube homepage. The navigation step failed to reach
the intended destination.",
  "final_status": "executing",
  "timestamp": "2026-05-02T16:44:02.502058+00:00"
}
],
"final_report": {
  "success": false,
```

```
"summary": "The task was partially completed, but the navigation to YouTube failed.",
```

```
"message": "The attempt to redirect to the YouTube platform was unsuccessful.\n\nStep results:\n- Step 1: Successfully clicked on the 'New tab' tab to switch views.\n- Step 2: Failed to interact with the address bar to navigate to YouTube.\n\nThe final screen state indicates that the browser's address bar was not detected or accessible, possibly due to the Microsoft Store pane being in the foreground.",
```

```
"steps_completed": 1,
```

```
"steps_failed": 1,
```

```
"issues": [
```

```
  "The address bar could not be located in the final screen state.",
```

```
  "The Microsoft Store window appears to be obstructing the browser interface."
```

```
],
```

```
"recommendations": [
```

```
  "Ensure the web browser is the active, foreground window.",
```

```
  "Verify that the browser's UI elements are fully loaded before attempting navigation."
```

```
]
```

```
},
```

```
"session_memory": {
```

```
  "run_id": "769bda88-628b-4be2-875a-1b6265d5e8bc",
```

```
  "user_task": "Click on new tab and find the youtube, if you can't find that icon search it in the search bar. however needs to re-redirect to the youtube platform",
```

```
  "started_at": "2026-05-02T16:32:30.127424+00:00",
```

```
  "updated_at": "2026-05-02T16:44:43.739605+00:00",
```

```
  "stale_after_misses": 2,
```

```
  "confidence_floor": 0.4,
```

```
  "decay_per_miss": 0.9,
```

```
  "frame_index": 15,
```

```
  "known_elements": {
```

```
"tab:frontend": {
  "element_key": "tab:frontend",
  "element_type": "tab",
  "label": "frontend",
  "bbox": [
    211.0,
    105.0,
    411.0,
    161.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 5,
  "first_seen_at": "2026-05-02T16:33:18.669310+00:00",
  "last_seen_at": "2026-05-02T16:37:11.954378+00:00",
  "seen_count": 5,
  "miss_count": 9,
  "stale": true,
  "cached": true
},
"tab:new tab": {
  "element_key": "tab:new tab",
  "element_type": "tab",
  "label": "New tab",
  "bbox": [
    363.0,
```

```
    105.0,  
    603.0,  
    161.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 0,  
  "last_seen_index": 5,  
  "first_seen_at": "2026-05-02T16:33:18.669310+00:00",  
  "last_seen_at": "2026-05-02T16:37:11.954378+00:00",  
  "seen_count": 6,  
  "miss_count": 9,  
  "stale": true,  
  "cached": true  
},  
"button:new tab button": {  
  "element_key": "button:new tab button",  
  "element_type": "button",  
  "label": "New tab button",  
  "bbox": [  
    579.0,  
    129.0,  
    619.0,  
    153.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",
```

```
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 5,
"first_seen_at": "2026-05-02T16:33:18.669310+00:00",
"last_seen_at": "2026-05-02T16:37:11.954378+00:00",
"seen_count": 2,
"miss_count": 9,
"stale": true,
"cached": true
},
"button:back": {
  "element_key": "button:back",
  "element_type": "button",
  "label": "Back",
  "bbox": [
    235.0,
    105.0,
    259.0,
    129.0
  ],
  "confidence": 0.7695,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 12,
  "first_seen_at": "2026-05-02T16:33:18.669310+00:00",
  "last_seen_at": "2026-05-02T16:43:09.638245+00:00",
```

```
"seen_count": 5,
"miss_count": 2,
"stale": true,
"cached": true
},
"button:forward": {
  "element_key": "button:forward",
  "element_type": "button",
  "label": "Forward",
  "bbox": [
    235.0,
    153.0,
    283.0,
    209.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 5,
  "first_seen_at": "2026-05-02T16:33:18.669310+00:00",
  "last_seen_at": "2026-05-02T16:37:11.954378+00:00",
  "seen_count": 3,
  "miss_count": 9,
  "stale": true,
  "cached": true
},
"button:reload": {
```

```
"element_key": "button:reload",
"element_type": "button",
"label": "Reload",
"bbox": [
    267.0,
    169.0,
    307.0,
    209.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 4,
"first_seen_at": "2026-05-02T16:33:18.669310+00:00",
"last_seen_at": "2026-05-02T16:36:18.113372+00:00",
"seen_count": 2,
"miss_count": 10,
"stale": true,
"cached": true
},
"input:address bar": {
    "element_key": "input:address bar",
    "element_type": "input",
    "label": "Address bar",
    "bbox": [
        179.0,
        153.0,
```

```
    1643.0,  
    217.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 0,  
  "last_seen_index": 5,  
  "first_seen_at": "2026-05-02T16:33:18.669310+00:00",  
  "last_seen_at": "2026-05-02T16:37:11.954378+00:00",  
  "seen_count": 4,  
  "miss_count": 9,  
  "stale": true,  
  "cached": true  
},  
"tab:frontend tab": {  
  "element_key": "tab:frontend tab",  
  "element_type": "tab",  
  "label": "frontend tab",  
  "bbox": [  
    163.0,  
    105.0,  
    395.0,  
    161.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,
```



```
"last_seen_frame": null,
"first_seen_index": 1,
"last_seen_index": 1,
"first_seen_at": "2026-05-02T16:34:15.000602+00:00",
"last_seen_at": "2026-05-02T16:34:15.000602+00:00",
"seen_count": 1,
"miss_count": 13,
"stale": true,
"cached": true
},
"button:home": {
  "element_key": "button:home",
  "element_type": "button",
  "label": "Home",
  "bbox": [
    203.0,
    161.0,
    267.0,
    257.0
  ],
  "confidence": 0.5315,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 3,
  "last_seen_index": 9,
  "first_seen_at": "2026-05-02T16:35:38.505125+00:00",
  "last_seen_at": "2026-05-02T16:40:47.767646+00:00",
  "seen_count": 2,
```

```
    "miss_count": 5,
    "stale": true,
    "cached": true
  },
  "button:chats": {
    "element_key": "button:chats",
    "element_type": "button",
    "label": "Chats",
    "bbox": [
      171.0,
      489.0,
      459.0,
      569.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 3,
    "last_seen_index": 3,
    "first_seen_at": "2026-05-02T16:35:38.505125+00:00",
    "last_seen_at": "2026-05-02T16:35:38.505125+00:00",
    "seen_count": 1,
    "miss_count": 11,
    "stale": true,
    "cached": true
  },
  "pane:task plan": {
    "element_key": "pane:task plan",
```

```
"element_type": "pane",
"label": "Task plan",
"bbox": [
    379.0,
    393.0,
    1163.0,
    801.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 3,
"last_seen_index": 5,
"first_seen_at": "2026-05-02T16:35:38.505125+00:00",
"last_seen_at": "2026-05-02T16:37:11.954378+00:00",
"seen_count": 3,
"miss_count": 9,
"stale": true,
"cached": true
},
"button:start": {
    "element_key": "button:start",
    "element_type": "button",
    "label": "Start",
    "bbox": [
        603.0,
        937.0,
        659.0,
```

```
    993.0
  ],
  "confidence": 0.95,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 3,
  "last_seen_index": 14,
  "first_seen_at": "2026-05-02T16:35:38.505125+00:00",
  "last_seen_at": "2026-05-02T16:44:43.739605+00:00",
  "seen_count": 5,
  "miss_count": 0,
  "stale": false,
  "cached": false
},
"input:search": {
  "element_key": "input:search",
  "element_type": "input",
  "label": "Search",
  "bbox": [
    619.0,
    937.0,
    923.0,
    977.0
  ],
  "confidence": 0.95,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
```

```
"first_seen_index": 3,
"last_seen_index": 14,
"first_seen_at": "2026-05-02T16:35:38.505125+00:00",
"last_seen_at": "2026-05-02T16:44:43.739605+00:00",
"seen_count": 5,
"miss_count": 0,
"stale": false,
"cached": false
},
"icon:chrome": {
  "element_key": "icon:chrome",
  "element_type": "icon",
  "label": "Chrome",
  "bbox": [
    915.0,
    921.0,
    979.0,
    977.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 3,
  "last_seen_index": 3,
  "first_seen_at": "2026-05-02T16:35:38.505125+00:00",
  "last_seen_at": "2026-05-02T16:35:38.505125+00:00",
  "seen_count": 1,
  "miss_count": 11,
```

```
    "stale": true,
    "cached": true
  },
  "icon:vs code": {
    "element_key": "icon:vs code",
    "element_type": "icon",
    "label": "VS Code",
    "bbox": [
      995.0,
      921.0,
      1051.0,
      993.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 3,
    "last_seen_index": 3,
    "first_seen_at": "2026-05-02T16:35:38.505125+00:00",
    "last_seen_at": "2026-05-02T16:35:38.505125+00:00",
    "seen_count": 1,
    "miss_count": 11,
    "stale": true,
    "cached": true
  },
  "icon:spotify": {
    "element_key": "icon:spotify",
    "element_type": "icon",
```

```
"label": "Spotify",
"bbox": [
  1243.0,
  961.0,
  1291.0,
  977.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 3,
"last_seen_index": 4,
"first_seen_at": "2026-05-02T16:35:38.505125+00:00",
"last_seen_at": "2026-05-02T16:36:18.113372+00:00",
"seen_count": 2,
"miss_count": 10,
"stale": true,
"cached": true
},
"text:clock": {
  "element_key": "text:clock",
  "element_type": "text",
  "label": "Clock",
  "bbox": [
    1347.0,
    913.0,
    1443.0,
    1015.0
  ]
}
```

```
],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 3,
  "last_seen_index": 3,
  "first_seen_at": "2026-05-02T16:35:38.505125+00:00",
  "last_seen_at": "2026-05-02T16:35:38.505125+00:00",
  "seen_count": 1,
  "miss_count": 11,
  "stale": true,
  "cached": true
},
"button:weather": {
  "element_key": "button:weather",
  "element_type": "button",
  "label": "Weather",
  "bbox": [
    179.0,
    913.0,
    315.0,
    1015.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 3,
```



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"last_seen_index": 3,
"first_seen_at": "2026-05-02T16:35:38.505125+00:00",
"last_seen_at": "2026-05-02T16:35:38.505125+00:00",
"seen_count": 1,
"miss_count": 11,
"stale": true,
"cached": true
},
"dropdown:ai assistant": {
  "element_key": "dropdown:ai assistant",
  "element_type": "dropdown",
  "label": "AI Assistant",
  "bbox": [
    467.0,
    273.0,
    515.0,
    305.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 3,
  "last_seen_index": 4,
  "first_seen_at": "2026-05-02T16:35:38.505125+00:00",
  "last_seen_at": "2026-05-02T16:36:18.113372+00:00",
  "seen_count": 2,
  "miss_count": 10,
  "stale": true,
```

```
    "cached": true
  },
  "text:agent is working": {
    "element_key": "text:agent is working",
    "element_type": "text",
    "label": "Agent is working",
    "bbox": [
      451.0,
      841.0,
      619.0,
      881.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 3,
    "last_seen_index": 3,
    "first_seen_at": "2026-05-02T16:35:38.505125+00:00",
    "last_seen_at": "2026-05-02T16:35:38.505125+00:00",
    "seen_count": 1,
    "miss_count": 11,
    "stale": true,
    "cached": true
  },
  "button:send": {
    "element_key": "button:send",
    "element_type": "button",
    "label": "Send",
```

```
"bbox": [
  1356.0,
  1014.0,
  1402.0,
  1015.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 3,
"last_seen_index": 4,
"first_seen_at": "2026-05-02T16:35:38.505125+00:00",
"last_seen_at": "2026-05-02T16:36:18.113372+00:00",
"seen_count": 2,
"miss_count": 10,
"stale": true,
"cached": true
},
"input:search chats": {
  "element_key": "input:search chats",
  "element_type": "input",
  "label": "Search chats",
  "bbox": [
    187.0,
    337.0,
    443.0,
    393.0
  ],

```

```
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 3,
"last_seen_index": 4,
"first_seen_at": "2026-05-02T16:35:38.505125+00:00",
"last_seen_at": "2026-05-02T16:36:18.113372+00:00",
"seen_count": 2,
"miss_count": 10,
"stale": true,
"cached": true
},
"button:refresh": {
  "element_key": "button:refresh",
  "element_type": "button",
  "label": "Refresh",
  "bbox": [
    275.0,
    153.0,
    315.0,
    209.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 3,
  "last_seen_index": 5,
```

```
"first_seen_at": "2026-05-02T16:35:38.505125+00:00",
"last_seen_at": "2026-05-02T16:37:11.954378+00:00",
"seen_count": 2,
"miss_count": 9,
"stale": true,
"cached": true
},
"button:extensions": {
  "element_key": "button:extensions",
  "element_type": "button",
  "label": "Extensions",
  "bbox": [
    1227.0,
    169.0,
    1267.0,
    193.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 3,
  "last_seen_index": 3,
  "first_seen_at": "2026-05-02T16:35:38.505125+00:00",
  "last_seen_at": "2026-05-02T16:35:38.505125+00:00",
  "seen_count": 1,
  "miss_count": 11,
  "stale": true,
  "cached": true
}
```

```
},
"button:profile": {
  "element_key": "button:profile",
  "element_type": "button",
  "label": "Profile",
  "bbox": [
    1331.0,
    169.0,
    1371.0,
    193.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 3,
  "last_seen_index": 3,
  "first_seen_at": "2026-05-02T16:35:38.505125+00:00",
  "last_seen_at": "2026-05-02T16:35:38.505125+00:00",
  "seen_count": 1,
  "miss_count": 11,
  "stale": true,
  "cached": true
},
"button:minimize": {
  "element_key": "button:minimize",
  "element_type": "button",
  "label": "Minimize",
  "bbox": [
```

```
    1627.0,  
    97.0,  
    1659.0,  
    121.0  
  ],  
  "confidence": 0.7695,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 3,  
  "last_seen_index": 12,  
  "first_seen_at": "2026-05-02T16:35:38.505125+00:00",  
  "last_seen_at": "2026-05-02T16:43:09.638245+00:00",  
  "seen_count": 2,  
  "miss_count": 2,  
  "stale": true,  
  "cached": true  
},  
"button:maximize": {  
  "element_key": "button:maximize",  
  "element_type": "button",  
  "label": "Maximize",  
  "bbox": [  
    1675.0,  
    97.0,  
    1707.0,  
    121.0  
  ],  
  "confidence": 0.7695,
```

```
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 3,
"last_seen_index": 12,
"first_seen_at": "2026-05-02T16:35:38.505125+00:00",
"last_seen_at": "2026-05-02T16:43:09.638245+00:00",
"seen_count": 2,
"miss_count": 2,
"stale": true,
"cached": true
},
"button:close": {
  "element_key": "button:close",
  "element_type": "button",
  "label": "Close",
  "bbox": [
    1723.0,
    97.0,
    1747.0,
    121.0
  ],
  "confidence": 0.7695,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 3,
  "last_seen_index": 12,
  "first_seen_at": "2026-05-02T16:35:38.505125+00:00",
```



```
"last_seen_at": "2026-05-02T16:43:09.638245+00:00",
"seen_count": 2,
"miss_count": 2,
"stale": true,
"cached": true
},
"button:user profile": {
  "element_key": "button:user profile",
  "element_type": "button",
  "label": "User Profile",
  "bbox": [
    179.0,
    825.0,
    379.0,
    897.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 3,
  "last_seen_index": 3,
  "first_seen_at": "2026-05-02T16:35:38.505125+00:00",
  "last_seen_at": "2026-05-02T16:35:38.505125+00:00",
  "seen_count": 1,
  "miss_count": 11,
  "stale": true,
  "cached": true
},
```

```
"text:task step 1": {
  "element_key": "text:task step 1",
  "element_type": "text",
  "label": "Task Step 1",
  "bbox": [
    539.0,
    561.0,
    1131.0,
    633.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 3,
  "last_seen_index": 3,
  "first_seen_at": "2026-05-02T16:35:38.505125+00:00",
  "last_seen_at": "2026-05-02T16:35:38.505125+00:00",
  "seen_count": 1,
  "miss_count": 11,
  "stale": true,
  "cached": true
},
"text:task step 2": {
  "element_key": "text:task step 2",
  "element_type": "text",
  "label": "Task Step 2",
  "bbox": [
    435.0,
```

```
        633.0,
        1131.0,
        753.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 3,
    "last_seen_index": 3,
    "first_seen_at": "2026-05-02T16:35:38.505125+00:00",
    "last_seen_at": "2026-05-02T16:35:38.505125+00:00",
    "seen_count": 1,
    "miss_count": 11,
    "stale": true,
    "cached": true
},
"icon:wi-fi": {
    "element_key": "icon:wi-fi",
    "element_type": "icon",
    "label": "Wi-Fi",
    "bbox": [
        1603.0,
        921.0,
        1643.0,
        961.0
    ],
    "confidence": 0.5315,
    "source": "gemini_vlm",
```

```
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 3,
"last_seen_index": 9,
"first_seen_at": "2026-05-02T16:35:38.505125+00:00",
"last_seen_at": "2026-05-02T16:40:47.767646+00:00",
"seen_count": 2,
"miss_count": 5,
"stale": true,
"cached": true
},
"icon:volume": {
  "element_key": "icon:volume",
  "element_type": "icon",
  "label": "Volume",
  "bbox": [
    1635.0,
    921.0,
    1675.0,
    969.0
  ],
  "confidence": 0.5315,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 3,
  "last_seen_index": 9,
  "first_seen_at": "2026-05-02T16:35:38.505125+00:00",
  "last_seen_at": "2026-05-02T16:40:47.767646+00:00",
```

```
    "seen_count": 2,
    "miss_count": 5,
    "stale": true,
    "cached": true
  },
  "icon:battery": {
    "element_key": "icon:battery",
    "element_type": "icon",
    "label": "Battery",
    "bbox": [
      1667.0,
      921.0,
      1707.0,
      969.0
    ],
    "confidence": 0.5315,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 3,
    "last_seen_index": 9,
    "first_seen_at": "2026-05-02T16:35:38.505125+00:00",
    "last_seen_at": "2026-05-02T16:40:47.767646+00:00",
    "seen_count": 2,
    "miss_count": 5,
    "stale": true,
    "cached": true
  },
  "icon:file explorer": {
```

```
"element_key": "icon:file explorer",
"element_type": "icon",
"label": "File Explorer",
"bbox": [
    779.0,
    921.0,
    835.0,
    977.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 3,
"last_seen_index": 3,
"first_seen_at": "2026-05-02T16:35:38.505125+00:00",
"last_seen_at": "2026-05-02T16:35:38.505125+00:00",
"seen_count": 1,
"miss_count": 11,
"stale": true,
"cached": true
},
"button:new tab": {
    "element_key": "button:new tab",
    "element_type": "button",
    "label": "New tab",
    "bbox": [
        579.0,
        129.0,
```

```
        619.0,  
        153.0  
    ],  
    "confidence": 0.4,  
    "source": "gemini_vlm",  
    "first_seen_frame": null,  
    "last_seen_frame": null,  
    "first_seen_index": 4,  
    "last_seen_index": 4,  
    "first_seen_at": "2026-05-02T16:36:18.113372+00:00",  
    "last_seen_at": "2026-05-02T16:36:18.113372+00:00",  
    "seen_count": 1,  
    "miss_count": 10,  
    "stale": true,  
    "cached": true  
},  
"menu_item:home": {  
    "element_key": "menu_item:home",  
    "element_type": "menu_item",  
    "label": "Home",  
    "bbox": [  
        171.0,  
        417.0,  
        459.0,  
        489.0  
    ],  
    "confidence": 0.4,  
    "source": "gemini_vlm",  
    "first_seen_frame": null,
```

```
"last_seen_frame": null,
"first_seen_index": 4,
"last_seen_index": 4,
"first_seen_at": "2026-05-02T16:36:18.113372+00:00",
"last_seen_at": "2026-05-02T16:36:18.113372+00:00",
"seen_count": 1,
"miss_count": 10,
"stale": true,
"cached": true
},
"menu_item:chats": {
  "element_key": "menu_item:chats",
  "element_type": "menu_item",
  "label": "Chats",
  "bbox": [
    195.0,
    489.0,
    459.0,
    569.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 4,
  "last_seen_index": 4,
  "first_seen_at": "2026-05-02T16:36:18.113372+00:00",
  "last_seen_at": "2026-05-02T16:36:18.113372+00:00",
  "seen_count": 1,
```



```
    "miss_count": 10,
    "stale": true,
    "cached": true
  },
  "list_item:click on new tab and find the yo...": {
    "element_key": "list_item:click on new tab and find the yo...",
    "element_type": "list_item",
    "label": "Click on new tab and find the yo...",
    "bbox": [
      203.0,
      617.0,
      427.0,
      705.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 4,
    "last_seen_index": 4,
    "first_seen_at": "2026-05-02T16:36:18.113372+00:00",
    "last_seen_at": "2026-05-02T16:36:18.113372+00:00",
    "seen_count": 1,
    "miss_count": 10,
    "stale": true,
    "cached": true
  },
  "text:click on the 'new tab' tab to switch to the existing empty browser
tab.": {
```

```
    "element_key": "text:click on the 'new tab' tab to switch to the existing  
empty browser tab.",  
    "element_type": "text",  
    "label": "Click on the 'New tab' tab to switch to the existing empty  
browser tab.",  
    "bbox": [  
        563.0,  
        577.0,  
        1131.0,  
        633.0  
    ],  
    "confidence": 0.4,  
    "source": "gemini_vlm",  
    "first_seen_frame": null,  
    "last_seen_frame": null,  
    "first_seen_index": 4,  
    "last_seen_index": 4,  
    "first_seen_at": "2026-05-02T16:36:18.113372+00:00",  
    "last_seen_at": "2026-05-02T16:36:18.113372+00:00",  
    "seen_count": 1,  
    "miss_count": 10,  
    "stale": true,  
    "cached": true  
},  
    "text:click the address bar, type 'https://www.youtube.com', and press the  
enter key to navigate to the platform.": {  
        "element_key": "text:click the address bar, type  
'https://www.youtube.com', and press the enter key to navigate to the platform.",  
        "element_type": "text",  
        "label": "Click the address bar, type 'https://www.youtube.com', and  
press the Enter key to navigate to the platform.",
```

```
"bbox": [
  483.0,
  625.0,
  1131.0,
  745.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 4,
"last_seen_index": 4,
"first_seen_at": "2026-05-02T16:36:18.113372+00:00",
"last_seen_at": "2026-05-02T16:36:18.113372+00:00",
"seen_count": 1,
"miss_count": 10,
"stale": true,
"cached": true
},
"text:agent is working...": {
  "element_key": "text:agent is working...",
  "element_type": "text",
  "label": "Agent is working...",
  "bbox": [
    491.0,
    1014.0,
    610.0,
    1015.0
  ],
}
```

```
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 4,
"last_seen_index": 4,
"first_seen_at": "2026-05-02T16:36:18.113372+00:00",
"last_seen_at": "2026-05-02T16:36:18.113372+00:00",
"seen_count": 1,
"miss_count": 10,
"stale": true,
"cached": true
},
"icon:google chrome": {
  "element_key": "icon:google chrome",
  "element_type": "icon",
  "label": "Google Chrome",
  "bbox": [
    1155.0,
    961.0,
    1203.0,
    985.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 4,
  "last_seen_index": 4,
```

```
"first_seen_at": "2026-05-02T16:36:18.113372+00:00",
"last_seen_at": "2026-05-02T16:36:18.113372+00:00",
"seen_count": 1,
"miss_count": 10,
"stale": true,
"cached": true
},
"icon:visual studio code": {
  "element_key": "icon:visual studio code",
  "element_type": "icon",
  "label": "Visual Studio Code",
  "bbox": [
    1195.0,
    961.0,
    1243.0,
    977.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 4,
  "last_seen_index": 4,
  "first_seen_at": "2026-05-02T16:36:18.113372+00:00",
  "last_seen_at": "2026-05-02T16:36:18.113372+00:00",
  "seen_count": 1,
  "miss_count": 10,
  "stale": true,
  "cached": true
}
```

```
},
"pane:system tray": {
  "element_key": "pane:system tray",
  "element_type": "pane",
  "label": "System Tray",
  "bbox": [
    1499.0,
    961.0,
    1755.0,
    1001.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 4,
  "last_seen_index": 4,
  "first_seen_at": "2026-05-02T16:36:18.113372+00:00",
  "last_seen_at": "2026-05-02T16:36:18.113372+00:00",
  "seen_count": 1,
  "miss_count": 10,
  "stale": true,
  "cached": true
},
"text:10:05 pm 5/2/2026": {
  "element_key": "text:10:05 pm 5/2/2026",
  "element_type": "text",
  "label": "10:05 PM 5/2/2026",
  "bbox": [
```

```
    1723.0,
    961.0,
    1803.0,
    977.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 4,
  "last_seen_index": 4,
  "first_seen_at": "2026-05-02T16:36:18.113372+00:00",
  "last_seen_at": "2026-05-02T16:36:18.113372+00:00",
  "seen_count": 1,
  "miss_count": 10,
  "stale": true,
  "cached": true
},
"button:screenpilot user": {
  "element_key": "button:screenpilot user",
  "element_type": "button",
  "label": "ScreenPilot User",
  "bbox": [
    201.0,
    1014.0,
    412.0,
    1015.0
  ],
  "confidence": 0.4,
```

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"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 4,
"last_seen_index": 4,
"first_seen_at": "2026-05-02T16:36:18.113372+00:00",
"last_seen_at": "2026-05-02T16:36:18.113372+00:00",
"seen_count": 1,
"miss_count": 10,
"stale": true,
"cached": true
},
"link:home": {
  "element_key": "link:home",
  "element_type": "link",
  "label": "Home",
  "bbox": [
    163.0,
    409.0,
    459.0,
    481.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 5,
  "last_seen_index": 5,
  "first_seen_at": "2026-05-02T16:37:11.954378+00:00",
```



```
    "last_seen_at": "2026-05-02T16:37:11.954378+00:00",
    "seen_count": 1,
    "miss_count": 9,
    "stale": true,
    "cached": true
  },
  "link:chats": {
    "element_key": "link:chats",
    "element_type": "link",
    "label": "Chats",
    "bbox": [
      187.0,
      489.0,
      291.0,
      545.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 5,
    "last_seen_index": 5,
    "first_seen_at": "2026-05-02T16:37:11.954378+00:00",
    "last_seen_at": "2026-05-02T16:37:11.954378+00:00",
    "seen_count": 1,
    "miss_count": 9,
    "stale": true,
    "cached": true
  },
```

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"input:agent is working...": {
  "element_key": "input:agent is working...",
  "element_type": "input",
  "label": "Agent is working...",
  "bbox": [
    315.0,
    817.0,
    1515.0,
    905.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 5,
  "last_seen_index": 5,
  "first_seen_at": "2026-05-02T16:37:11.954378+00:00",
  "last_seen_at": "2026-05-02T16:37:11.954378+00:00",
  "seen_count": 1,
  "miss_count": 9,
  "stale": true,
  "cached": true
},
"button:chrome": {
  "element_key": "button:chrome",
  "element_type": "button",
  "label": "Chrome",
  "bbox": [
    1115.0,
```

```
    921.0,  
    1171.0,  
    977.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 5,  
  "last_seen_index": 5,  
  "first_seen_at": "2026-05-02T16:37:11.954378+00:00",  
  "last_seen_at": "2026-05-02T16:37:11.954378+00:00",  
  "seen_count": 1,  
  "miss_count": 9,  
  "stale": true,  
  "cached": true  
},  
"button:vs code": {  
  "element_key": "button:vs code",  
  "element_type": "button",  
  "label": "VS Code",  
  "bbox": [  
    1243.0,  
    921.0,  
    1299.0,  
    977.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",
```

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"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 5,
"last_seen_index": 5,
"first_seen_at": "2026-05-02T16:37:11.954378+00:00",
"last_seen_at": "2026-05-02T16:37:11.954378+00:00",
"seen_count": 1,
"miss_count": 9,
"stale": true,
"cached": true
},
"button:spotify": {
  "element_key": "button:spotify",
  "element_type": "button",
  "label": "Spotify",
  "bbox": [
    1259.0,
    913.0,
    1315.0,
    977.0
  ],
  "confidence": 0.5315,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 5,
  "last_seen_index": 9,
  "first_seen_at": "2026-05-02T16:37:11.954378+00:00",
  "last_seen_at": "2026-05-02T16:40:47.767646+00:00",
```

```
"seen_count": 2,
"miss_count": 5,
"stale": true,
"cached": true
},
"text:microsoft store": {
  "element_key": "text:microsoft store",
  "element_type": "text",
  "label": "Microsoft Store",
  "bbox": [
    211.0,
    113.0,
    379.0,
    161.0
  ],
  "confidence": 0.4089,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 6,
  "last_seen_index": 6,
  "first_seen_at": "2026-05-02T16:38:13.893524+00:00",
  "last_seen_at": "2026-05-02T16:38:13.893524+00:00",
  "seen_count": 1,
  "miss_count": 8,
  "stale": true,
  "cached": true
},
"pane:microsoft store window": {
```

```
"element_key": "pane:microsoft store window",
"element_type": "pane",
"label": "Microsoft Store Window",
"bbox": [
  115.0,
  65.0,
  1805.0,
  1015.0
],
"confidence": 0.4543,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 7,
"last_seen_index": 7,
"first_seen_at": "2026-05-02T16:39:14.867764+00:00",
"last_seen_at": "2026-05-02T16:39:14.867764+00:00",
"seen_count": 1,
"miss_count": 7,
"stale": true,
"cached": true
},
"pane:title bar": {
  "element_key": "pane:title bar",
  "element_type": "pane",
  "label": "Title Bar",
  "bbox": [
    115.0,
    81.0,
```

```
1795.0,  
177.0  
],  
"confidence": 0.4305,  
"source": "gemini_vlm",  
"first_seen_frame": null,  
"last_seen_frame": null,  
"first_seen_index": 7,  
"last_seen_index": 7,  
"first_seen_at": "2026-05-02T16:39:14.867764+00:00",  
"last_seen_at": "2026-05-02T16:39:14.867764+00:00",  
"seen_count": 1,  
"miss_count": 7,  
"stale": true,  
"cached": true  
},  
"input:search apps, games, and more": {  
  "element_key": "input:search apps, games, and more",  
  "element_type": "input",  
  "label": "Search apps, games, and more",  
  "bbox": [  
    635.0,  
    89.0,  
    1323.0,  
    145.0  
  ],  
  "confidence": 0.7695,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,
```

```
"last_seen_frame": null,
"first_seen_index": 7,
"last_seen_index": 12,
"first_seen_at": "2026-05-02T16:39:14.867764+00:00",
"last_seen_at": "2026-05-02T16:43:09.638245+00:00",
"seen_count": 4,
"miss_count": 2,
"stale": true,
"cached": true
},
"pane:sidebar": {
  "element_key": "pane:sidebar",
  "element_type": "pane",
  "label": "Sidebar",
  "bbox": [
    187.0,
    65.0,
    299.0,
    993.0
  ],
  "confidence": 0.4305,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 7,
  "last_seen_index": 7,
  "first_seen_at": "2026-05-02T16:39:14.867764+00:00",
  "last_seen_at": "2026-05-02T16:39:14.867764+00:00",
  "seen_count": 1,
```



```
    "miss_count": 7,  
    "stale": true,  
    "cached": true  
  },  
  "tab:home": {  
    "element_key": "tab:home",  
    "element_type": "tab",  
    "label": "Home",  
    "bbox": [  
      211.0,  
      137.0,  
      299.0,  
      201.0  
    ],  
    "confidence": 0.855,  
    "source": "gemini_vlm",  
    "first_seen_frame": null,  
    "last_seen_frame": null,  
    "first_seen_index": 7,  
    "last_seen_index": 13,  
    "first_seen_at": "2026-05-02T16:39:14.867764+00:00",  
    "last_seen_at": "2026-05-02T16:43:58.308218+00:00",  
    "seen_count": 3,  
    "miss_count": 1,  
    "stale": false,  
    "cached": true  
  },  
  "tab:apps": {  
    "element_key": "tab:apps",
```

```
"element_type": "tab",
"label": "Apps",
"bbox": [
    203.0,
    241.0,
    283.0,
    337.0
],
"confidence": 0.4543,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 7,
"last_seen_index": 7,
"first_seen_at": "2026-05-02T16:39:14.867764+00:00",
"last_seen_at": "2026-05-02T16:39:14.867764+00:00",
"seen_count": 1,
"miss_count": 7,
"stale": true,
"cached": true
},
"tab:gaming": {
    "element_key": "tab:gaming",
    "element_type": "tab",
    "label": "Gaming",
    "bbox": [
        203.0,
        321.0,
        291.0,
```

```
    417.0
  ],
  "confidence": 0.4543,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 7,
  "last_seen_index": 7,
  "first_seen_at": "2026-05-02T16:39:14.867764+00:00",
  "last_seen_at": "2026-05-02T16:39:14.867764+00:00",
  "seen_count": 1,
  "miss_count": 7,
  "stale": true,
  "cached": true
},
"tab:themes": {
  "element_key": "tab:themes",
  "element_type": "tab",
  "label": "Themes",
  "bbox": [
    203.0,
    401.0,
    291.0,
    489.0
  ],
  "confidence": 0.4543,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
```

```
"first_seen_index": 7,
"last_seen_index": 7,
"first_seen_at": "2026-05-02T16:39:14.867764+00:00",
"last_seen_at": "2026-05-02T16:39:14.867764+00:00",
"seen_count": 1,
"miss_count": 7,
"stale": true,
"cached": true
},
"button:what's new": {
  "element_key": "button:what's new",
  "element_type": "button",
  "label": "What's New",
  "bbox": [
    179.0,
    985.0,
    251.0,
    1015.0
  ],
  "confidence": 0.5019,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 7,
  "last_seen_index": 9,
  "first_seen_at": "2026-05-02T16:39:14.867764+00:00",
  "last_seen_at": "2026-05-02T16:40:47.767646+00:00",
  "seen_count": 2,
  "miss_count": 5,
```

```
    "stale": true,
    "cached": true
  },
  "pane:microsoft store": {
    "element_key": "pane:microsoft store",
    "element_type": "pane",
    "label": "Microsoft Store",
    "bbox": [
      187.0,
      73.0,
      1787.0,
      1001.0
    ],
    "confidence": 0.855,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 8,
    "last_seen_index": 13,
    "first_seen_at": "2026-05-02T16:40:06.610270+00:00",
    "last_seen_at": "2026-05-02T16:43:58.308218+00:00",
    "seen_count": 2,
    "miss_count": 1,
    "stale": false,
    "cached": true
  },
  "pane:frontend - google chrome": {
    "element_key": "pane:frontend - google chrome",
    "element_type": "pane",
```

```
"label": "frontend - Google Chrome",
"bbox": [
  938.0,
  98.0,
  1482.0,
  434.0
],
"confidence": 0.5048,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 8,
"last_seen_index": 8,
"first_seen_at": "2026-05-02T16:40:06.610270+00:00",
"last_seen_at": "2026-05-02T16:40:06.610270+00:00",
"seen_count": 1,
"miss_count": 6,
"stale": true,
"cached": true
},
"window:microsoft store": {
  "element_key": "window:microsoft store",
  "element_type": "window",
  "label": "Microsoft Store",
  "bbox": [
    115.0,
    65.0,
    1579.0,
    1015.0
  ]
}
```

```
],
  "confidence": 0.5609,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 9,
  "last_seen_index": 9,
  "first_seen_at": "2026-05-02T16:40:47.767646+00:00",
  "last_seen_at": "2026-05-02T16:40:47.767646+00:00",
  "seen_count": 1,
  "miss_count": 5,
  "stale": true,
  "cached": true
},
"button:apps": {
  "element_key": "button:apps",
  "element_type": "button",
  "label": "Apps",
  "bbox": [
    187.0,
    241.0,
    267.0,
    337.0
  ],
  "confidence": 0.5315,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 9,
```

```
"last_seen_index": 9,  
"first_seen_at": "2026-05-02T16:40:47.767646+00:00",  
"last_seen_at": "2026-05-02T16:40:47.767646+00:00",  
"seen_count": 1,  
"miss_count": 5,  
"stale": true,  
"cached": true  
},  
"button:gaming": {  
  "element_key": "button:gaming",  
  "element_type": "button",  
  "label": "Gaming",  
  "bbox": [  
    187.0,  
    321.0,  
    267.0,  
    393.0  
  ],  
  "confidence": 0.5315,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 9,  
  "last_seen_index": 9,  
  "first_seen_at": "2026-05-02T16:40:47.767646+00:00",  
  "last_seen_at": "2026-05-02T16:40:47.767646+00:00",  
  "seen_count": 1,  
  "miss_count": 5,  
  "stale": true,
```



```
    "cached": true
  },
  "button:themes": {
    "element_key": "button:themes",
    "element_type": "button",
    "label": "Themes",
    "bbox": [
      195.0,
      409.0,
      243.0,
      497.0
    ],
    "confidence": 0.5315,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 9,
    "last_seen_index": 9,
    "first_seen_at": "2026-05-02T16:40:47.767646+00:00",
    "last_seen_at": "2026-05-02T16:40:47.767646+00:00",
    "seen_count": 1,
    "miss_count": 5,
    "stale": true,
    "cached": true
  },
  "button:downloads": {
    "element_key": "button:downloads",
    "element_type": "button",
    "label": "Downloads",
```

```
"bbox": [
  181.0,
  1014.0,
  247.0,
  1015.0
],
"confidence": 0.5019,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 9,
"last_seen_index": 9,
"first_seen_at": "2026-05-02T16:40:47.767646+00:00",
"last_seen_at": "2026-05-02T16:40:47.767646+00:00",
"seen_count": 1,
"miss_count": 5,
"stale": true,
"cached": true
},
"button:library": {
  "element_key": "button:library",
  "element_type": "button",
  "label": "Library",
  "bbox": [
    181.0,
    1014.0,
    247.0,
    1015.0
  ],

```

```
"confidence": 0.5019,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 9,
"last_seen_index": 9,
"first_seen_at": "2026-05-02T16:40:47.767646+00:00",
"last_seen_at": "2026-05-02T16:40:47.767646+00:00",
"seen_count": 1,
"miss_count": 5,
"stale": true,
"cached": true
},
"pane:microsoft 365 banner": {
  "element_key": "pane:microsoft 365 banner",
  "element_type": "pane",
  "label": "Microsoft 365 Banner",
  "bbox": [
    179.0,
    113.0,
    1043.0,
    689.0
  ],
  "confidence": 0.5315,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 9,
  "last_seen_index": 9,
```

```
"first_seen_at": "2026-05-02T16:40:47.767646+00:00",
"last_seen_at": "2026-05-02T16:40:47.767646+00:00",
"seen_count": 1,
"miss_count": 5,
"stale": true,
"cached": true
},
"button:see details": {
  "element_key": "button:see details",
  "element_type": "button",
  "label": "See details",
  "bbox": [
    299.0,
    561.0,
    355.0,
    625.0
  ],
  "confidence": 0.5019,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 9,
  "last_seen_index": 9,
  "first_seen_at": "2026-05-02T16:40:47.767646+00:00",
  "last_seen_at": "2026-05-02T16:40:47.767646+00:00",
  "seen_count": 1,
  "miss_count": 5,
  "stale": true,
  "cached": true
}
```

```
},
"pane:pdf x banner": {
  "element_key": "pane:pdf x banner",
  "element_type": "pane",
  "label": "PDF X Banner",
  "bbox": [
    875.0,
    137.0,
    1451.0,
    505.0
  ],
  "confidence": 0.5315,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 9,
  "last_seen_index": 9,
  "first_seen_at": "2026-05-02T16:40:47.767646+00:00",
  "last_seen_at": "2026-05-02T16:40:47.767646+00:00",
  "seen_count": 1,
  "miss_count": 5,
  "stale": true,
  "cached": true
},
"button:get": {
  "element_key": "button:get",
  "element_type": "button",
  "label": "Get",
  "bbox": [
```

```
    955.0,
    385.0,
    1083.0,
    441.0
  ],
  "confidence": 0.5019,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 9,
  "last_seen_index": 9,
  "first_seen_at": "2026-05-02T16:40:47.767646+00:00",
  "last_seen_at": "2026-05-02T16:40:47.767646+00:00",
  "seen_count": 1,
  "miss_count": 5,
  "stale": true,
  "cached": true
},
"pane:raid: shadow legends banner": {
  "element_key": "pane:raid: shadow legends banner",
  "element_type": "pane",
  "label": "RAID: Shadow Legends Banner",
  "bbox": [
    915.0,
    449.0,
    1195.0,
    641.0
  ],
  "confidence": 0.5315,
```

```
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 9,
"last_seen_index": 9,
"first_seen_at": "2026-05-02T16:40:47.767646+00:00",
"last_seen_at": "2026-05-02T16:40:47.767646+00:00",
"seen_count": 1,
"miss_count": 5,
"stale": true,
"cached": true
},
"pane:cooking fever banner": {
  "element_key": "pane:cooking fever banner",
  "element_type": "pane",
  "label": "Cooking Fever Banner",
  "bbox": [
    1139.0,
    449.0,
    1419.0,
    641.0
  ],
  "confidence": 0.5315,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 9,
  "last_seen_index": 9,
  "first_seen_at": "2026-05-02T16:40:47.767646+00:00",
```

```
    "last_seen_at": "2026-05-02T16:40:47.767646+00:00",
    "seen_count": 1,
    "miss_count": 5,
    "stale": true,
    "cached": true
  },
  "heading:trending games": {
    "element_key": "heading:trending games",
    "element_type": "heading",
    "label": "Trending games",
    "bbox": [
      307.0,
      657.0,
      459.0,
      697.0
    ],
    "confidence": 0.5019,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 9,
    "last_seen_index": 9,
    "first_seen_at": "2026-05-02T16:40:47.767646+00:00",
    "last_seen_at": "2026-05-02T16:40:47.767646+00:00",
    "seen_count": 1,
    "miss_count": 5,
    "stale": true,
    "cached": true
  },
}
```



```
"heading:trending apps": {
  "element_key": "heading:trending apps",
  "element_type": "heading",
  "label": "Trending apps",
  "bbox": [
    851.0,
    657.0,
    995.0,
    697.0
  ],
  "confidence": 0.5019,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 9,
  "last_seen_index": 9,
  "first_seen_at": "2026-05-02T16:40:47.767646+00:00",
  "last_seen_at": "2026-05-02T16:40:47.767646+00:00",
  "seen_count": 1,
  "miss_count": 5,
  "stale": true,
  "cached": true
},
"button:task view": {
  "element_key": "button:task view",
  "element_type": "button",
  "label": "Task View",
  "bbox": [
    883.0,
```

```
    913.0,
    947.0,
    977.0
  ],
  "confidence": 0.5315,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 9,
  "last_seen_index": 9,
  "first_seen_at": "2026-05-02T16:40:47.767646+00:00",
  "last_seen_at": "2026-05-02T16:40:47.767646+00:00",
  "seen_count": 1,
  "miss_count": 5,
  "stale": true,
  "cached": true
},
"button:file explorer": {
  "element_key": "button:file explorer",
  "element_type": "button",
  "label": "File Explorer",
  "bbox": [
    939.0,
    913.0,
    995.0,
    977.0
  ],
  "confidence": 0.5315,
  "source": "gemini_vlm",
```

```
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 9,
"last_seen_index": 9,
"first_seen_at": "2026-05-02T16:40:47.767646+00:00",
"last_seen_at": "2026-05-02T16:40:47.767646+00:00",
"seen_count": 1,
"miss_count": 5,
"stale": true,
"cached": true
},
"button:microsoft store": {
  "element_key": "button:microsoft store",
  "element_type": "button",
  "label": "Microsoft Store",
  "bbox": [
    995.0,
    913.0,
    1051.0,
    977.0
  ],
  "confidence": 0.5315,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 9,
  "last_seen_index": 9,
  "first_seen_at": "2026-05-02T16:40:47.767646+00:00",
  "last_seen_at": "2026-05-02T16:40:47.767646+00:00",
```

```
    "seen_count": 1,
    "miss_count": 5,
    "stale": true,
    "cached": true
  },
  "button:settings": {
    "element_key": "button:settings",
    "element_type": "button",
    "label": "Settings",
    "bbox": [
      1043.0,
      913.0,
      1099.0,
      977.0
    ],
    "confidence": 0.5315,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 9,
    "last_seen_index": 9,
    "first_seen_at": "2026-05-02T16:40:47.767646+00:00",
    "last_seen_at": "2026-05-02T16:40:47.767646+00:00",
    "seen_count": 1,
    "miss_count": 5,
    "stale": true,
    "cached": true
  },
  "button:xbox": {
```

```
"element_key": "button:xbox",
"element_type": "button",
"label": "Xbox",
"bbox": [
  1099.0,
  913.0,
  1155.0,
  977.0
],
"confidence": 0.5315,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 9,
"last_seen_index": 9,
"first_seen_at": "2026-05-02T16:40:47.767646+00:00",
"last_seen_at": "2026-05-02T16:40:47.767646+00:00",
"seen_count": 1,
"miss_count": 5,
"stale": true,
"cached": true
},
"button:google chrome": {
  "element_key": "button:google chrome",
  "element_type": "button",
  "label": "Google Chrome",
  "bbox": [
    1147.0,
    913.0,
```

```
    1211.0,  
    977.0  
  ],  
  "confidence": 0.5315,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 9,  
  "last_seen_index": 9,  
  "first_seen_at": "2026-05-02T16:40:47.767646+00:00",  
  "last_seen_at": "2026-05-02T16:40:47.767646+00:00",  
  "seen_count": 1,  
  "miss_count": 5,  
  "stale": true,  
  "cached": true  
},  
"button:visual studio code": {  
  "element_key": "button:visual studio code",  
  "element_type": "button",  
  "label": "Visual Studio Code",  
  "bbox": [  
    1203.0,  
    913.0,  
    1259.0,  
    977.0  
  ],  
  "confidence": 0.5315,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,
```

```
"last_seen_frame": null,
"first_seen_index": 9,
"last_seen_index": 9,
"first_seen_at": "2026-05-02T16:40:47.767646+00:00",
"last_seen_at": "2026-05-02T16:40:47.767646+00:00",
"seen_count": 1,
"miss_count": 5,
"stale": true,
"cached": true
},
"widget:weather": {
  "element_key": "widget:weather",
  "element_type": "widget",
  "label": "Weather",
  "bbox": [
    187.0,
    913.0,
    307.0,
    977.0
  ],
  "confidence": 0.5019,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 9,
  "last_seen_index": 9,
  "first_seen_at": "2026-05-02T16:40:47.767646+00:00",
  "last_seen_at": "2026-05-02T16:40:47.767646+00:00",
  "seen_count": 1,
```

```
    "miss_count": 5,
    "stale": true,
    "cached": true
  },
  "label:clock": {
    "element_key": "label:clock",
    "element_type": "label",
    "label": "Clock",
    "bbox": [
      1699.0,
      913.0,
      1787.0,
      977.0
    ],
    "confidence": 0.5609,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 9,
    "last_seen_index": 9,
    "first_seen_at": "2026-05-02T16:40:47.767646+00:00",
    "last_seen_at": "2026-05-02T16:40:47.767646+00:00",
    "seen_count": 1,
    "miss_count": 5,
    "stale": true,
    "cached": true
  },
  "label:microsoft store": {
    "element_key": "label:microsoft store",
```



```
"element_type": "label",
"label": "Microsoft Store",
"bbox": [
  243.0,
  97.0,
  403.0,
  137.0
],
"confidence": 0.7695,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 10,
"last_seen_index": 12,
"first_seen_at": "2026-05-02T16:41:36.977103+00:00",
"last_seen_at": "2026-05-02T16:43:09.638245+00:00",
"seen_count": 2,
"miss_count": 2,
"stale": true,
"cached": true
},
"heading:microsoft copilot": {
  "element_key": "heading:microsoft copilot",
  "element_type": "heading",
  "label": "Microsoft Copilot",
  "bbox": [
    319.0,
    290.0,
    639.0,
```

```
330.0
],
"confidence": 0.7144,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 11,
"last_seen_index": 11,
"first_seen_at": "2026-05-02T16:42:17.919332+00:00",
"last_seen_at": "2026-05-02T16:42:17.919332+00:00",
"seen_count": 1,
"miss_count": 3,
"stale": true,
"cached": true
},
"text:search smarter and unleash creativity with copilot": {
  "element_key": "text:search smarter and unleash creativity with copilot",
  "element_type": "text",
  "label": "Search smarter and unleash creativity with Copilot",
  "bbox": [
    327.0,
    322.0,
    615.0,
    378.0
  ],
  "confidence": 0.7071,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
```

```
"first_seen_index": 11,
"last_seen_index": 11,
"first_seen_at": "2026-05-02T16:42:17.919332+00:00",
"last_seen_at": "2026-05-02T16:42:17.919332+00:00",
"seen_count": 1,
"miss_count": 3,
"stale": true,
"cached": true
},
"button:install": {
  "element_key": "button:install",
  "element_type": "button",
  "label": "Install",
  "bbox": [
    343.0,
    386.0,
    503.0,
    434.0
  ],
  "confidence": 0.7217,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 11,
  "last_seen_index": 11,
  "first_seen_at": "2026-05-02T16:42:17.919332+00:00",
  "last_seen_at": "2026-05-02T16:42:17.919332+00:00",
  "seen_count": 1,
  "miss_count": 3,
```

```
    "stale": true,
    "cached": true
  },
  "label:12+ parental guidance recommended": {
    "element_key": "label:12+ parental guidance recommended",
    "element_type": "label",
    "label": "12+ Parental Guidance Recommended",
    "bbox": [
      327.0,
      434.0,
      615.0,
      498.0
    ],
    "confidence": 0.6998,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 11,
    "last_seen_index": 11,
    "first_seen_at": "2026-05-02T16:42:17.919332+00:00",
    "last_seen_at": "2026-05-02T16:42:17.919332+00:00",
    "seen_count": 1,
    "miss_count": 3,
    "stale": true,
    "cached": true
  },
  "text:copilot": {
    "element_key": "text:copilot",
    "element_type": "text",
```

```
"label": "Copilot",
"bbox": [
  703.0,
  258.0,
  807.0,
  298.0
],
"confidence": 0.6925,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 11,
"last_seen_index": 11,
"first_seen_at": "2026-05-02T16:42:17.919332+00:00",
"last_seen_at": "2026-05-02T16:42:17.919332+00:00",
"seen_count": 1,
"miss_count": 3,
"stale": true,
"cached": true
},
"icon:copilot logo": {
  "element_key": "icon:copilot logo",
  "element_type": "icon",
  "label": "Copilot logo",
  "bbox": [
    687.0,
    322.0,
    807.0,
    418.0
  ]
}
```

```
],  
  "confidence": 0.7217,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 11,  
  "last_seen_index": 11,  
  "first_seen_at": "2026-05-02T16:42:17.919332+00:00",  
  "last_seen_at": "2026-05-02T16:42:17.919332+00:00",  
  "seen_count": 1,  
  "miss_count": 3,  
  "stale": true,  
  "cached": true  
},  
  "icon:small icon": {  
    "element_key": "icon:small icon",  
    "element_type": "icon",  
    "label": "small icon",  
    "bbox": [  
      319.0,  
      594.0,  
      351.0,  
      610.0  
    ],  
    "confidence": 0.6561,  
    "source": "gemini_vlm",  
    "first_seen_frame": null,  
    "last_seen_frame": null,  
    "first_seen_index": 11,
```

```
"last_seen_index": 11,  
"first_seen_at": "2026-05-02T16:42:17.919332+00:00",  
"last_seen_at": "2026-05-02T16:42:17.919332+00:00",  
"seen_count": 1,  
"miss_count": 3,  
"stale": true,  
"cached": true  
},  
"icon:user profile": {  
  "element_key": "icon:user profile",  
  "element_type": "icon",  
  "label": "User Profile",  
  "bbox": [  
    1531.0,  
    97.0,  
    1571.0,  
    137.0  
  ],  
  "confidence": 0.729,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 12,  
  "last_seen_index": 12,  
  "first_seen_at": "2026-05-02T16:43:09.638245+00:00",  
  "last_seen_at": "2026-05-02T16:43:09.638245+00:00",  
  "seen_count": 1,  
  "miss_count": 2,  
  "stale": true,
```

```
    "cached": true
  }
},
"current_elements": [],
"resolved_elements": [
  {
    "id": "memory_button:start",
    "type": "button",
    "label": "Start",
    "description": "Cached from an earlier frame in the same run",
    "state": "cached",
    "bbox": [
      603.0,
      937.0,
      659.0,
      993.0
    ],
    "confidence": 0.95,
    "source": "gemini_vlm",
    "last_seen_frame": null,
    "last_seen_index": 14,
    "stale": true,
    "cached": true,
    "session_memory_key": "button:start"
  },
  {
    "id": "memory_input:search",
    "type": "input",
    "label": "Search",
```



```
"description": "Cached from an earlier frame in the same run",
"state": "cached",
"bbox": [
    619.0,
    937.0,
    923.0,
    977.0
],
"confidence": 0.95,
"source": "gemini_vlm",
"last_seen_frame": null,
"last_seen_index": 14,
"stale": true,
"cached": true,
"session_memory_key": "input:search"
},
{
    "id": "memory_tab:home",
    "type": "tab",
    "label": "Home",
    "description": "Cached from an earlier frame in the same run",
    "state": "cached",
    "bbox": [
        211.0,
        137.0,
        299.0,
        201.0
    ],
    "confidence": 0.855,
```

```
"source": "gemini_vlm",
"last_seen_frame": null,
"last_seen_index": 13,
"stale": true,
"cached": true,
"session_memory_key": "tab:home"
},
{
  "id": "memory_pane:microsoft store",
  "type": "pane",
  "label": "Microsoft Store",
  "description": "Cached from an earlier frame in the same run",
  "state": "cached",
  "bbox": [
    187.0,
    73.0,
    1787.0,
    1001.0
  ],
  "confidence": 0.855,
  "source": "gemini_vlm",
  "last_seen_frame": null,
  "last_seen_index": 13,
  "stale": true,
  "cached": true,
  "session_memory_key": "pane:microsoft store"
}
],
"screen_data": {}
```

```
}  
}
```

Test Case 11: Create New GitHub Repository (Complex Task)

User Prompt:

Click on the "Your Repositories" tab and check if it is Github repositories page, if yes find and click "new" button to create new repository

Test Results:

- Steps Total: 2
- Completed: 0
- Failed: 2
- Duration: 16 minutes 11 seconds

Step Execution:

Step 1: Move the mouse to the 'Your Repositories' tab at (696, 56) and click it

- Target: [tab] 'Your Repositories' @ (696, 56)
- Attempts: 3
- Execution: Performed
- Evaluation: Failed (retry loop)
- Issue: Coordinate mismatch — Target (696, 56) vs actual tab location (316, 60). The browser window was no longer visible on screen; only desktop icons were detected.

Step 2: Locate the 'New' button on the GitHub repositories page and click it

- Target: 'New' button on the GitHub repositories page
- Attempts: 3
- Execution: Performed
- Evaluation: Failed (retry loop)
- Issue: 'New' button not detected in UI elements. Command generation failed — response missing 'hid_commands'.

Execution Summary:

The task failed completely. Step 1 encountered a coordinate mismatch, and the browser

window may have been minimized or closed. The 'New' button was never detected, and the second step failed to generate any HID commands.

Issues Identified:

- **Coordinate mismatch for 'Your Repositories' tab (Target: 696,56 vs Actual: 316,60)**
- **'New' button not found in detected UI elements**
- **Missing 'hid_commands' in step planning response**
- **Browser window potentially minimized or closed during execution**
- **No fallback mechanism to re-detect element coordinates**

Recommendations for Improvement:

- **Implement dynamic coordinate refresh before each execution attempt**
- **Add fallback to re-detect tab position if initial coordinates fail**
- **Ensure browser window is in foreground before interaction**
- **Enforce 'hid_commands' presence in LLM responses with validation**
- **Add retry with updated coordinates instead of hardcoding original positions**

JSON Response:

full_cycle.json

```
{
  "run_id": "eb482227-7e57-46e0-9f57-38b2bcbcd93c7",
  "user_task": "Click on the \"Your Repositories\" tab and check if it is Github repositories page, if yes find and click \"new\" button to create new repository",
  "started_at": "2026-05-03T08:26:19.529870+00:00",
  "finished_at": "2026-05-03T08:42:30.510973+00:00",
  "vision_used": true,
  "perception": {
    "image": "E:\\sliit-project\\ScreenAwareTaskAgent\\vision\\data\\sessions\\shot_1777796788_f50769\\processing\\proc_1777796788_75e372\\preprocessed_frames\\frame_1777796788_7b3eb6.jpg",
    "image_size": {
      "width": 1920,
```

```
    "height": 1080
  },
  "screen_bbox": [
    139,
    58,
    1795,
    1006
  ],
  "screen_size": {
    "width": 1656,
    "height": 948
  },
  "coordinate_system": "pixel",
  "elements": [
    {
      "id": "tab_whatsapp",
      "type": "tab",
      "label": "WhatsApp",
      "description": "Browser tab for WhatsApp",
      "state": "inactive",
      "dx": 120,
      "dy": 56,
      "frame_dx": 259,
      "frame_dy": 114,
      "confidence": 0.95,
      "source": "gemini_vlm",
      "frame_bbox": [
        139,
        82,
```

```
    379,  
    146  
  ],  
  "session_memory_key": "tab:whatsapp",  
  "cached": false  
},  
{  
  "id": "tab_google_search",  
  "type": "tab",  
  "label": "Google Search",  
  "description": "Browser tab for Google Search",  
  "state": "inactive",  
  "dx": 308,  
  "dy": 56,  
  "frame_dx": 447,  
  "frame_dy": 114,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    323,  
    82,  
    571,  
    146  
  ],  
  "session_memory_key": "tab:google search",  
  "cached": false  
},  
{  
  "id": "tab_frontend",
```

```
"type": "tab",
"label": "frontend",
"description": "Browser tab for frontend development",
"state": "inactive",
"dx": 500,
"dy": 56,
"frame_dx": 639,
"frame_dy": 114,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    515,
    82,
    763,
    146
],
"session_memory_key": "tab:frontend",
"cached": false
},
{
    "id": "tab_github_repos",
    "type": "tab",
    "label": "Your Repositories",
    "description": "Browser tab for GitHub repositories",
    "state": "active",
    "dx": 696,
    "dy": 56,
    "frame_dx": 835,
    "frame_dy": 114,
```

```
"confidence": 0.95,  
"source": "gemini_vlm",  
"frame_bbox": [  
    707,  
    82,  
    963,  
    146  
],  
"session_memory_key": "tab:your repositories",  
"cached": false  
},  
{  
    "id": "address_bar",  
    "type": "input",  
    "label": "Address Bar",  
    "description": "Browser address bar showing localhost:5173",  
    "state": "visible",  
    "dx": 588,  
    "dy": 116,  
    "frame_dx": 727,  
    "frame_dy": 174,  
    "confidence": 0.98,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        139,  
        146,  
        1315,  
        202  
    ],
```



```
"session_memory_key": "input:address bar",
"cached": false
},
{
  "id": "sidebar_home",
  "type": "link",
  "label": "Home",
  "description": "Navigation link to Home in ScreenPilot sidebar",
  "state": "active",
  "dx": 144,
  "dy": 404,
  "frame_dx": 283,
  "frame_dy": 462,
  "confidence": 0.92,
  "source": "gemini_vlm",
  "frame_bbox": [
    139,
    418,
    427,
    506
  ],
  "session_memory_key": "link:home",
  "cached": false
},
{
  "id": "sidebar_chats",
  "type": "link",
  "label": "Chats",
  "description": "Navigation link to Chats in ScreenPilot sidebar",
```

```
"state": "inactive",
"dx": 144,
"dy": 472,
"frame_dx": 283,
"frame_dy": 530,
"confidence": 0.92,
"source": "gemini_vlm",
"frame_bbox": [
  139,
  490,
  427,
  570
],
"session_memory_key": "link:chats",
"cached": false
},
{
  "id": "ai_assistant_dropdown",
  "type": "dropdown",
  "label": "AI Assistant",
  "description": "Dropdown menu to select AI assistant",
  "state": "visible",
  "dx": 340,
  "dy": 200,
  "frame_dx": 479,
  "frame_dy": 258,
  "confidence": 0.9,
  "source": "gemini_vlm",
  "frame_bbox": [
```

```
    403,  
    226,  
    555,  
    290  
  ],  
  "session_memory_key": "dropdown:ai assistant",  
  "cached": false  
},  
{  
  "id": "user_message_bubble",  
  "type": "pane",  
  "label": "User Message",  
  "description": "Chat bubble containing user instructions",  
  "state": "visible",  
  "dx": 920,  
  "dy": 364,  
  "frame_dx": 1059,  
  "frame_dy": 422,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    651,  
    322,  
    1467,  
    522  
  ],  
  "session_memory_key": "pane:user message",  
  "cached": false  
},
```

```
{
  "id": "status_message",
  "type": "label",
  "label": "Connecting to agent...",
  "description": "Status message indicating agent connection and screen
capture",
  "state": "visible",
  "dx": 620,
  "dy": 516,
  "frame_dx": 759,
  "frame_dy": 574,
  "confidence": 0.9,
  "source": "gemini_vlm",
  "frame_bbox": [
    355,
    506,
    1163,
    642
  ],
  "session_memory_key": "label:connecting to agent...",
  "cached": false
},
{
  "id": "loading_indicator",
  "type": "icon",
  "label": "Loading",
  "description": "Three dots loading animation",
  "state": "visible",
  "dx": 356,
```

```
"dy": 624,  
"frame_dx": 495,  
"frame_dy": 682,  
"confidence": 0.85,  
"source": "gemini_vlm",  
"frame_bbox": [  
    459,  
    666,  
    531,  
    698  
],  
"session_memory_key": "icon:loading",  
"cached": false  
},  
{  
    "id": "chat_input_area",  
    "type": "input",  
    "label": "Chat Input",  
    "description": "Input field for chat messages",  
    "state": "disabled",  
    "dx": 776,  
    "dy": 948,  
    "frame_dx": 915,  
    "frame_dy": 1006,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        437,  
        1005,
```

```
1394,  
1006  
],  
"session_memory_key": "input:chat input",  
"cached": false  
},  
{  
  "id": "taskbar_start",  
  "type": "button",  
  "label": "Start",  
  "description": "Windows Start button",  
  "state": "visible",  
  "dx": 332,  
  "dy": 908,  
  "frame_dx": 471,  
  "frame_dy": 966,  
  "confidence": 0.98,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    443,  
    938,  
    499,  
    994  
  ],  
  "session_memory_key": "button:start",  
  "cached": false  
},  
{  
  "id": "taskbar_search",
```

```
"type": "input",
"label": "Search",
"description": "Windows Search bar",
"state": "visible",
"dx": 412,
"dy": 908,
"frame_dx": 551,
"frame_dy": 966,
"confidence": 0.98,
"source": "gemini_vlm",
"frame_bbox": [
    475,
    938,
    627,
    994
],
"session_memory_key": "input:search",
"cached": false
},
{
    "id": "taskbar_chrome",
    "type": "button",
    "label": "Chrome",
    "description": "Google Chrome taskbar icon",
    "state": "active",
    "dx": 760,
    "dy": 904,
    "frame_dx": 899,
    "frame_dy": 962,
```

```
"confidence": 0.98,
"source": "gemini_vlm",
"frame_bbox": [
    875,
    938,
    923,
    986
],
"session_memory_key": "button:chrome",
"cached": false
},
{
    "id": "taskbar_vscode",
    "type": "button",
    "label": "VS Code",
    "description": "Visual Studio Code taskbar icon",
    "state": "visible",
    "dx": 872,
    "dy": 908,
    "frame_dx": 1011,
    "frame_dy": 966,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
        987,
        938,
        1035,
        994
    ],
    ],
```



```
"session_memory_key": "button:vs code",
"cached": false
},
{
  "id": "taskbar_spotify",
  "type": "button",
  "label": "Spotify",
  "description": "Spotify taskbar icon",
  "state": "visible",
  "dx": 904,
  "dy": 904,
  "frame_dx": 1043,
  "frame_dy": 962,
  "confidence": 0.98,
  "source": "gemini_vlm",
  "frame_bbox": [
    1019,
    938,
    1067,
    986
  ],
  "session_memory_key": "button:spotify",
  "cached": false
},
{
  "id": "system_tray_clock",
  "type": "label",
  "label": "Clock",
  "description": "System tray clock and date",
```

```
"state": "visible",
"dx": 1236,
"dy": 900,
"frame_dx": 1375,
"frame_dy": 958,
"confidence": 0.98,
"source": "gemini_vlm",
"frame_bbox": [
  1339,
  938,
  1411,
  978
],
"session_memory_key": "label:clock",
"cached": false
},
{
  "id": "system_tray_wifi",
  "type": "icon",
  "label": "Wi-Fi",
  "description": "Wi-Fi status icon",
  "state": "visible",
  "dx": 1172,
  "dy": 908,
  "frame_dx": 1311,
  "frame_dy": 966,
  "confidence": 0.95,
  "source": "gemini_vlm",
  "frame_bbox": [
```

```
    1291,  
    946,  
    1331,  
    986  
  ],  
  "session_memory_key": "icon:wi-fi",  
  "cached": false  
},  
{  
  "id": "system_tray_battery",  
  "type": "icon",  
  "label": "Battery",  
  "description": "Battery status icon",  
  "state": "visible",  
  "dx": 1196,  
  "dy": 904,  
  "frame_dx": 1335,  
  "frame_dy": 962,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    1315,  
    946,  
    1355,  
    978  
  ],  
  "session_memory_key": "icon:battery",  
  "cached": false  
},
```

```
{
  "id": "sidebar_recent_1",
  "type": "link",
  "label": "Recent Task 1",
  "description": "Recent task: Click on the 'Your Repositories'...",
  "state": "visible",
  "dx": 128,
  "dy": 608,
  "frame_dx": 267,
  "frame_dy": 666,
  "confidence": 0.85,
  "source": "gemini_vlm",
  "frame_bbox": [
    139,
    626,
    395,
    706
  ],
  "session_memory_key": "link:recent task 1",
  "cached": false
},
{
  "id": "sidebar_recent_2",
  "type": "link",
  "label": "Recent Task 2",
  "description": "Recent task: Click on new tab and find the yo...",
  "state": "visible",
  "dx": 140,
  "dy": 748,
```

```
"frame_dx": 279,  
"frame_dy": 806,  
"confidence": 0.85,  
"source": "gemini_vlm",  
"frame_bbox": [  
    139,  
    770,  
    419,  
    842  
],  
"session_memory_key": "link:recent task 2",  
"cached": false  
},  
{  
    "id": "user_profile",  
    "type": "button",  
    "label": "User Profile",  
    "description": "User profile button in sidebar",  
    "state": "visible",  
    "dx": 142,  
    "dy": 948,  
    "frame_dx": 281,  
    "frame_dy": 1006,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        171,  
        1005,  
        391,
```

```
        1006
    ],
    "session_memory_key": "button:user profile",
    "cached": false
},
{
    "id": "send_button",
    "type": "button",
    "label": "Send",
    "description": "Send message button",
    "state": "disabled",
    "dx": 1226,
    "dy": 948,
    "frame_dx": 1365,
    "frame_dy": 1006,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        1342,
        1005,
        1387,
        1006
    ],
    "session_memory_key": "button:send",
    "cached": false
}
],
"element_count": 24,
"current_elements": [
```

```
{
  "id": "tab_whatsapp",
  "type": "tab",
  "label": "WhatsApp",
  "description": "Browser tab for WhatsApp",
  "state": "inactive",
  "dx": 120,
  "dy": 56,
  "frame_dx": 259,
  "frame_dy": 114,
  "confidence": 0.95,
  "source": "gemini_vlm",
  "frame_bbox": [
    139,
    82,
    379,
    146
  ],
  "session_memory_key": "tab:whatsapp",
  "cached": false
},
{
  "id": "tab_google_search",
  "type": "tab",
  "label": "Google Search",
  "description": "Browser tab for Google Search",
  "state": "inactive",
  "dx": 308,
  "dy": 56,
```

```
"frame_dx": 447,  
"frame_dy": 114,  
"confidence": 0.95,  
"source": "gemini_vlm",  
"frame_bbox": [  
    323,  
    82,  
    571,  
    146  
],  
"session_memory_key": "tab:google search",  
"cached": false  
},  
{  
    "id": "tab_frontend",  
    "type": "tab",  
    "label": "frontend",  
    "description": "Browser tab for frontend development",  
    "state": "inactive",  
    "dx": 500,  
    "dy": 56,  
    "frame_dx": 639,  
    "frame_dy": 114,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        515,  
        82,  
        763,
```



```
146
],
"session_memory_key": "tab:frontend",
"cached": false
},
{
  "id": "tab_github_repos",
  "type": "tab",
  "label": "Your Repositories",
  "description": "Browser tab for GitHub repositories",
  "state": "active",
  "dx": 696,
  "dy": 56,
  "frame_dx": 835,
  "frame_dy": 114,
  "confidence": 0.95,
  "source": "gemini_vlm",
  "frame_bbox": [
    707,
    82,
    963,
    146
  ],
  "session_memory_key": "tab:your repositories",
  "cached": false
},
{
  "id": "address_bar",
  "type": "input",
```

```
"label": "Address Bar",
"description": "Browser address bar showing localhost:5173",
"state": "visible",
"dx": 588,
"dy": 116,
"frame_dx": 727,
"frame_dy": 174,
"confidence": 0.98,
"source": "gemini_vlm",
"frame_bbox": [
    139,
    146,
    1315,
    202
],
"session_memory_key": "input:address bar",
"cached": false
},
{
    "id": "sidebar_home",
    "type": "link",
    "label": "Home",
    "description": "Navigation link to Home in ScreenPilot sidebar",
    "state": "active",
    "dx": 144,
    "dy": 404,
    "frame_dx": 283,
    "frame_dy": 462,
    "confidence": 0.92,
```

```
"source": "gemini_vlm",
"frame_bbox": [
  139,
  418,
  427,
  506
],
"session_memory_key": "link:home",
"cached": false
},
{
  "id": "sidebar_chats",
  "type": "link",
  "label": "Chats",
  "description": "Navigation link to Chats in ScreenPilot sidebar",
  "state": "inactive",
  "dx": 144,
  "dy": 472,
  "frame_dx": 283,
  "frame_dy": 530,
  "confidence": 0.92,
  "source": "gemini_vlm",
  "frame_bbox": [
    139,
    490,
    427,
    570
  ],
  "session_memory_key": "link:chats",
```

```
"cached": false
},
{
  "id": "ai_assistant_dropdown",
  "type": "dropdown",
  "label": "AI Assistant",
  "description": "Dropdown menu to select AI assistant",
  "state": "visible",
  "dx": 340,
  "dy": 200,
  "frame_dx": 479,
  "frame_dy": 258,
  "confidence": 0.9,
  "source": "gemini_vlm",
  "frame_bbox": [
    403,
    226,
    555,
    290
  ],
  "session_memory_key": "dropdown:ai assistant",
  "cached": false
},
{
  "id": "user_message_bubble",
  "type": "pane",
  "label": "User Message",
  "description": "Chat bubble containing user instructions",
  "state": "visible",
```

```
"dx": 920,  
"dy": 364,  
"frame_dx": 1059,  
"frame_dy": 422,  
"confidence": 0.95,  
"source": "gemini_vlm",  
"frame_bbox": [  
    651,  
    322,  
    1467,  
    522  
],  
"session_memory_key": "pane:user message",  
"cached": false  
},  
{  
    "id": "status_message",  
    "type": "label",  
    "label": "Connecting to agent...",  
    "description": "Status message indicating agent connection and screen  
capture",  
    "state": "visible",  
    "dx": 620,  
    "dy": 516,  
    "frame_dx": 759,  
    "frame_dy": 574,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  

```

```
        355,  
        506,  
        1163,  
        642  
    ],  
    "session_memory_key": "label:connecting to agent...",  
    "cached": false  
},  
{  
    "id": "loading_indicator",  
    "type": "icon",  
    "label": "Loading",  
    "description": "Three dots loading animation",  
    "state": "visible",  
    "dx": 356,  
    "dy": 624,  
    "frame_dx": 495,  
    "frame_dy": 682,  
    "confidence": 0.85,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        459,  
        666,  
        531,  
        698  
    ],  
    "session_memory_key": "icon:loading",  
    "cached": false  
},
```

```
{
  "id": "chat_input_area",
  "type": "input",
  "label": "Chat Input",
  "description": "Input field for chat messages",
  "state": "disabled",
  "dx": 776,
  "dy": 948,
  "frame_dx": 915,
  "frame_dy": 1006,
  "confidence": 0.9,
  "source": "gemini_vlm",
  "frame_bbox": [
    437,
    1005,
    1394,
    1006
  ],
  "session_memory_key": "input:chat input",
  "cached": false
},
{
  "id": "taskbar_start",
  "type": "button",
  "label": "Start",
  "description": "Windows Start button",
  "state": "visible",
  "dx": 332,
  "dy": 908,
```

```
"frame_dx": 471,
"frame_dy": 966,
"confidence": 0.98,
"source": "gemini_vlm",
"frame_bbox": [
  443,
  938,
  499,
  994
],
"session_memory_key": "button:start",
"cached": false
},
{
  "id": "taskbar_search",
  "type": "input",
  "label": "Search",
  "description": "Windows Search bar",
  "state": "visible",
  "dx": 412,
  "dy": 908,
  "frame_dx": 551,
  "frame_dy": 966,
  "confidence": 0.98,
  "source": "gemini_vlm",
  "frame_bbox": [
    475,
    938,
    627,
```



```
994
],
"session_memory_key": "input:search",
"cached": false
},
{
  "id": "taskbar_chrome",
  "type": "button",
  "label": "Chrome",
  "description": "Google Chrome taskbar icon",
  "state": "active",
  "dx": 760,
  "dy": 904,
  "frame_dx": 899,
  "frame_dy": 962,
  "confidence": 0.98,
  "source": "gemini_vlm",
  "frame_bbox": [
    875,
    938,
    923,
    986
  ],
  "session_memory_key": "button:chrome",
  "cached": false
},
{
  "id": "taskbar_vscode",
  "type": "button",
```

```
"label": "VS Code",
"description": "Visual Studio Code taskbar icon",
"state": "visible",
"dx": 872,
"dy": 908,
"frame_dx": 1011,
"frame_dy": 966,
"confidence": 0.98,
"source": "gemini_vlm",
"frame_bbox": [
    987,
    938,
    1035,
    994
],
"session_memory_key": "button:vs code",
"cached": false
},
{
    "id": "taskbar_spotify",
    "type": "button",
    "label": "Spotify",
    "description": "Spotify taskbar icon",
    "state": "visible",
    "dx": 904,
    "dy": 904,
    "frame_dx": 1043,
    "frame_dy": 962,
    "confidence": 0.98,
```

```
"source": "gemini_vlm",
"frame_bbox": [
  1019,
  938,
  1067,
  986
],
"session_memory_key": "button:spotify",
"cached": false
},
{
  "id": "system_tray_clock",
  "type": "label",
  "label": "Clock",
  "description": "System tray clock and date",
  "state": "visible",
  "dx": 1236,
  "dy": 900,
  "frame_dx": 1375,
  "frame_dy": 958,
  "confidence": 0.98,
  "source": "gemini_vlm",
  "frame_bbox": [
    1339,
    938,
    1411,
    978
  ],
  "session_memory_key": "label:clock",
```

```
"cached": false
},
{
  "id": "system_tray_wifi",
  "type": "icon",
  "label": "Wi-Fi",
  "description": "Wi-Fi status icon",
  "state": "visible",
  "dx": 1172,
  "dy": 908,
  "frame_dx": 1311,
  "frame_dy": 966,
  "confidence": 0.95,
  "source": "gemini_vlm",
  "frame_bbox": [
    1291,
    946,
    1331,
    986
  ],
  "session_memory_key": "icon:wi-fi",
  "cached": false
},
{
  "id": "system_tray_battery",
  "type": "icon",
  "label": "Battery",
  "description": "Battery status icon",
  "state": "visible",
```

```
"dx": 1196,  
  "dy": 904,  
  "frame_dx": 1335,  
  "frame_dy": 962,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    1315,  
    946,  
    1355,  
    978  
  ],  
  "session_memory_key": "icon:battery",  
  "cached": false  
},  
{  
  "id": "sidebar_recent_1",  
  "type": "link",  
  "label": "Recent Task 1",  
  "description": "Recent task: Click on the 'Your Repositories'...",  
  "state": "visible",  
  "dx": 128,  
  "dy": 608,  
  "frame_dx": 267,  
  "frame_dy": 666,  
  "confidence": 0.85,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    139,
```

```
        626,  
        395,  
        706  
    ],  
    "session_memory_key": "link:recent task 1",  
    "cached": false  
},  
{  
    "id": "sidebar_recent_2",  
    "type": "link",  
    "label": "Recent Task 2",  
    "description": "Recent task: Click on new tab and find the yo...",  
    "state": "visible",  
    "dx": 140,  
    "dy": 748,  
    "frame_dx": 279,  
    "frame_dy": 806,  
    "confidence": 0.85,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        139,  
        770,  
        419,  
        842  
    ],  
    "session_memory_key": "link:recent task 2",  
    "cached": false  
},  
{
```

```
"id": "user_profile",
"type": "button",
"label": "User Profile",
"description": "User profile button in sidebar",
"state": "visible",
"dx": 142,
"dy": 948,
"frame_dx": 281,
"frame_dy": 1006,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    171,
    1005,
    391,
    1006
],
"session_memory_key": "button:user profile",
"cached": false
},
{
    "id": "send_button",
    "type": "button",
    "label": "Send",
    "description": "Send message button",
    "state": "disabled",
    "dx": 1226,
    "dy": 948,
    "frame_dx": 1365,
```

```
    "frame_dy": 1006,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        1342,
        1005,
        1387,
        1006
    ],
    "session_memory_key": "button:send",
    "cached": false
}
],
"resolved_elements": [
    {
        "id": "tab_whatsapp",
        "type": "tab",
        "label": "WhatsApp",
        "description": "Browser tab for WhatsApp",
        "state": "inactive",
        "dx": 120,
        "dy": 56,
        "frame_dx": 259,
        "frame_dy": 114,
        "confidence": 0.95,
        "source": "gemini_vlm",
        "frame_bbox": [
            139,
            82,
```



```
        379,  
        146  
    ],  
    "session_memory_key": "tab:whatsapp",  
    "cached": false  
},  
{  
    "id": "tab_google_search",  
    "type": "tab",  
    "label": "Google Search",  
    "description": "Browser tab for Google Search",  
    "state": "inactive",  
    "dx": 308,  
    "dy": 56,  
    "frame_dx": 447,  
    "frame_dy": 114,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        323,  
        82,  
        571,  
        146  
    ],  
    "session_memory_key": "tab:google search",  
    "cached": false  
},  
{  
    "id": "tab_frontend",
```

```
"type": "tab",
"label": "frontend",
"description": "Browser tab for frontend development",
"state": "inactive",
"dx": 500,
"dy": 56,
"frame_dx": 639,
"frame_dy": 114,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    515,
    82,
    763,
    146
],
"session_memory_key": "tab:frontend",
"cached": false
},
{
    "id": "tab_github_repos",
    "type": "tab",
    "label": "Your Repositories",
    "description": "Browser tab for GitHub repositories",
    "state": "active",
    "dx": 696,
    "dy": 56,
    "frame_dx": 835,
    "frame_dy": 114,
```

```
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    707,
    82,
    963,
    146
],
"session_memory_key": "tab:your repositories",
"cached": false
},
{
    "id": "address_bar",
    "type": "input",
    "label": "Address Bar",
    "description": "Browser address bar showing localhost:5173",
    "state": "visible",
    "dx": 588,
    "dy": 116,
    "frame_dx": 727,
    "frame_dy": 174,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
        139,
        146,
        1315,
        202
    ],
    }
```

```
"session_memory_key": "input:address bar",
"cached": false
},
{
  "id": "sidebar_home",
  "type": "link",
  "label": "Home",
  "description": "Navigation link to Home in ScreenPilot sidebar",
  "state": "active",
  "dx": 144,
  "dy": 404,
  "frame_dx": 283,
  "frame_dy": 462,
  "confidence": 0.92,
  "source": "gemini_vlm",
  "frame_bbox": [
    139,
    418,
    427,
    506
  ],
  "session_memory_key": "link:home",
  "cached": false
},
{
  "id": "sidebar_chats",
  "type": "link",
  "label": "Chats",
  "description": "Navigation link to Chats in ScreenPilot sidebar",
```

```
"state": "inactive",
"dx": 144,
"dy": 472,
"frame_dx": 283,
"frame_dy": 530,
"confidence": 0.92,
"source": "gemini_vlm",
"frame_bbox": [
  139,
  490,
  427,
  570
],
"session_memory_key": "link:chats",
"cached": false
},
{
  "id": "ai_assistant_dropdown",
  "type": "dropdown",
  "label": "AI Assistant",
  "description": "Dropdown menu to select AI assistant",
  "state": "visible",
  "dx": 340,
  "dy": 200,
  "frame_dx": 479,
  "frame_dy": 258,
  "confidence": 0.9,
  "source": "gemini_vlm",
  "frame_bbox": [
```

```
    403,  
    226,  
    555,  
    290  
  ],  
  "session_memory_key": "dropdown:ai assistant",  
  "cached": false  
},  
{  
  "id": "user_message_bubble",  
  "type": "pane",  
  "label": "User Message",  
  "description": "Chat bubble containing user instructions",  
  "state": "visible",  
  "dx": 920,  
  "dy": 364,  
  "frame_dx": 1059,  
  "frame_dy": 422,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    651,  
    322,  
    1467,  
    522  
  ],  
  "session_memory_key": "pane:user message",  
  "cached": false  
},
```

```
{
  "id": "status_message",
  "type": "label",
  "label": "Connecting to agent...",
  "description": "Status message indicating agent connection and screen
capture",
  "state": "visible",
  "dx": 620,
  "dy": 516,
  "frame_dx": 759,
  "frame_dy": 574,
  "confidence": 0.9,
  "source": "gemini_vlm",
  "frame_bbox": [
    355,
    506,
    1163,
    642
  ],
  "session_memory_key": "label:connecting to agent...",
  "cached": false
},
{
  "id": "loading_indicator",
  "type": "icon",
  "label": "Loading",
  "description": "Three dots loading animation",
  "state": "visible",
  "dx": 356,
```

```
"dy": 624,  
"frame_dx": 495,  
"frame_dy": 682,  
"confidence": 0.85,  
"source": "gemini_vlm",  
"frame_bbox": [  
    459,  
    666,  
    531,  
    698  
],  
"session_memory_key": "icon:loading",  
"cached": false  
},  
{  
    "id": "chat_input_area",  
    "type": "input",  
    "label": "Chat Input",  
    "description": "Input field for chat messages",  
    "state": "disabled",  
    "dx": 776,  
    "dy": 948,  
    "frame_dx": 915,  
    "frame_dy": 1006,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        437,  
        1005,
```



```
        1394,  
        1006  
    ],  
    "session_memory_key": "input:chat input",  
    "cached": false  
},  
{  
    "id": "taskbar_start",  
    "type": "button",  
    "label": "Start",  
    "description": "Windows Start button",  
    "state": "visible",  
    "dx": 332,  
    "dy": 908,  
    "frame_dx": 471,  
    "frame_dy": 966,  
    "confidence": 0.98,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        443,  
        938,  
        499,  
        994  
    ],  
    "session_memory_key": "button:start",  
    "cached": false  
},  
{  
    "id": "taskbar_search",
```

```
"type": "input",
"label": "Search",
"description": "Windows Search bar",
"state": "visible",
"dx": 412,
"dy": 908,
"frame_dx": 551,
"frame_dy": 966,
"confidence": 0.98,
"source": "gemini_vlm",
"frame_bbox": [
    475,
    938,
    627,
    994
],
"session_memory_key": "input:search",
"cached": false
},
{
    "id": "taskbar_chrome",
    "type": "button",
    "label": "Chrome",
    "description": "Google Chrome taskbar icon",
    "state": "active",
    "dx": 760,
    "dy": 904,
    "frame_dx": 899,
    "frame_dy": 962,
```

```
"confidence": 0.98,  
"source": "gemini_vlm",  
"frame_bbox": [  
    875,  
    938,  
    923,  
    986  
],  
"session_memory_key": "button:chrome",  
"cached": false  
},  
{  
    "id": "taskbar_vscode",  
    "type": "button",  
    "label": "VS Code",  
    "description": "Visual Studio Code taskbar icon",  
    "state": "visible",  
    "dx": 872,  
    "dy": 908,  
    "frame_dx": 1011,  
    "frame_dy": 966,  
    "confidence": 0.98,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        987,  
        938,  
        1035,  
        994  
    ],  
}
```

```
"session_memory_key": "button:vs code",
"cached": false
},
{
  "id": "taskbar_spotify",
  "type": "button",
  "label": "Spotify",
  "description": "Spotify taskbar icon",
  "state": "visible",
  "dx": 904,
  "dy": 904,
  "frame_dx": 1043,
  "frame_dy": 962,
  "confidence": 0.98,
  "source": "gemini_vlm",
  "frame_bbox": [
    1019,
    938,
    1067,
    986
  ],
  "session_memory_key": "button:spotify",
  "cached": false
},
{
  "id": "system_tray_clock",
  "type": "label",
  "label": "Clock",
  "description": "System tray clock and date",
```

```
"state": "visible",
"dx": 1236,
"dy": 900,
"frame_dx": 1375,
"frame_dy": 958,
"confidence": 0.98,
"source": "gemini_vlm",
"frame_bbox": [
    1339,
    938,
    1411,
    978
],
"session_memory_key": "label:clock",
"cached": false
},
{
    "id": "system_tray_wifi",
    "type": "icon",
    "label": "Wi-Fi",
    "description": "Wi-Fi status icon",
    "state": "visible",
    "dx": 1172,
    "dy": 908,
    "frame_dx": 1311,
    "frame_dy": 966,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
```

```
    1291,  
    946,  
    1331,  
    986  
  ],  
  "session_memory_key": "icon:wi-fi",  
  "cached": false  
},  
{  
  "id": "system_tray_battery",  
  "type": "icon",  
  "label": "Battery",  
  "description": "Battery status icon",  
  "state": "visible",  
  "dx": 1196,  
  "dy": 904,  
  "frame_dx": 1335,  
  "frame_dy": 962,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    1315,  
    946,  
    1355,  
    978  
  ],  
  "session_memory_key": "icon:battery",  
  "cached": false  
},
```

```
{
  "id": "sidebar_recent_1",
  "type": "link",
  "label": "Recent Task 1",
  "description": "Recent task: Click on the 'Your Repositories'...",
  "state": "visible",
  "dx": 128,
  "dy": 608,
  "frame_dx": 267,
  "frame_dy": 666,
  "confidence": 0.85,
  "source": "gemini_vlm",
  "frame_bbox": [
    139,
    626,
    395,
    706
  ],
  "session_memory_key": "link:recent task 1",
  "cached": false
},
{
  "id": "sidebar_recent_2",
  "type": "link",
  "label": "Recent Task 2",
  "description": "Recent task: Click on new tab and find the yo...",
  "state": "visible",
  "dx": 140,
  "dy": 748,
```

```
"frame_dx": 279,  
"frame_dy": 806,  
"confidence": 0.85,  
"source": "gemini_vlm",  
"frame_bbox": [  
    139,  
    770,  
    419,  
    842  
],  
"session_memory_key": "link:recent task 2",  
"cached": false  
},  
{  
    "id": "user_profile",  
    "type": "button",  
    "label": "User Profile",  
    "description": "User profile button in sidebar",  
    "state": "visible",  
    "dx": 142,  
    "dy": 948,  
    "frame_dx": 281,  
    "frame_dy": 1006,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        171,  
        1005,  
        391,
```



```
    1006
  ],
  "session_memory_key": "button:user profile",
  "cached": false
},
{
  "id": "send_button",
  "type": "button",
  "label": "Send",
  "description": "Send message button",
  "state": "disabled",
  "dx": 1226,
  "dy": 948,
  "frame_dx": 1365,
  "frame_dy": 1006,
  "confidence": 0.9,
  "source": "gemini_vlm",
  "frame_bbox": [
    1342,
    1005,
    1387,
    1006
  ],
  "session_memory_key": "button:send",
  "cached": false
}
],
"session_knowledge_base": {
  "session_id": "shot_1777796788_f50769",
```

```
"started_at": "2026-05-03T08:26:28.783695+00:00",
"updated_at": "2026-05-03T08:26:52.866965+00:00",
"stale_after_misses": 3,
"confidence_floor": 0.4,
"decay_per_miss": 0.9,
"frame_count": 1,
"frame_name": "frame_1777796788_7b3eb6.jpg",
"frame_index": 0,
"image_size": {
  "width": 1920,
  "height": 1080
},
"known_elements": {
  "tab:whatsapp": {
    "element_key": "tab:whatsapp",
    "element_type": "tab",
    "label": "WhatsApp",
    "bbox": [
      139.0,
      82.0,
      379.0,
      146.0
    ],
    "confidence": 0.95,
    "source": "gemini_vlm",
    "first_seen_frame": "frame_1777796788_7b3eb6.jpg",
    "last_seen_frame": "frame_1777796788_7b3eb6.jpg",
    "first_seen_at": "2026-05-03T08:26:52.866965+00:00",
    "last_seen_at": "2026-05-03T08:26:52.866965+00:00",
```

```
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777796788_7b3eb6.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-03T08:26:52.866965+00:00",
    "confidence": 0.95
  }
],
},
"tab:google search": {
  "element_key": "tab:google search",
  "element_type": "tab",
  "label": "Google Search",
  "bbox": [
    323.0,
    82.0,
    571.0,
    146.0
  ],
  "confidence": 0.95,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777796788_7b3eb6.jpg",
  "last_seen_frame": "frame_1777796788_7b3eb6.jpg",
```

```
"first_seen_at": "2026-05-03T08:26:52.866965+00:00",
"last_seen_at": "2026-05-03T08:26:52.866965+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777796788_7b3eb6.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-03T08:26:52.866965+00:00",
    "confidence": 0.95
  }
],
},
"tab:frontend": {
  "element_key": "tab:frontend",
  "element_type": "tab",
  "label": "frontend",
  "bbox": [
    515.0,
    82.0,
    763.0,
    146.0
  ],
  "confidence": 0.95,
  "source": "gemini_vlm",
```

```
"first_seen_frame": "frame_1777796788_7b3eb6.jpg",
"last_seen_frame": "frame_1777796788_7b3eb6.jpg",
"first_seen_at": "2026-05-03T08:26:52.866965+00:00",
"last_seen_at": "2026-05-03T08:26:52.866965+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777796788_7b3eb6.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-03T08:26:52.866965+00:00",
    "confidence": 0.95
  }
],
},
"tab:your repositories": {
  "element_key": "tab:your repositories",
  "element_type": "tab",
  "label": "Your Repositories",
  "bbox": [
    707.0,
    82.0,
    963.0,
    146.0
  ],
},
```

```
"confidence": 0.95,
"source": "gemini_vlm",
"first_seen_frame": "frame_1777796788_7b3eb6.jpg",
"last_seen_frame": "frame_1777796788_7b3eb6.jpg",
"first_seen_at": "2026-05-03T08:26:52.866965+00:00",
"last_seen_at": "2026-05-03T08:26:52.866965+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777796788_7b3eb6.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-03T08:26:52.866965+00:00",
    "confidence": 0.95
  }
],
"input:address bar": {
  "element_key": "input:address bar",
  "element_type": "input",
  "label": "Address Bar",
  "bbox": [
    139.0,
    146.0,
    1315.0,
```

```
202.0
],
"confidence": 0.98,
"source": "gemini_vlm",
"first_seen_frame": "frame_1777796788_7b3eb6.jpg",
"last_seen_frame": "frame_1777796788_7b3eb6.jpg",
"first_seen_at": "2026-05-03T08:26:52.866965+00:00",
"last_seen_at": "2026-05-03T08:26:52.866965+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777796788_7b3eb6.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-03T08:26:52.866965+00:00",
    "confidence": 0.98
  }
]
},
"link:home": {
  "element_key": "link:home",
  "element_type": "link",
  "label": "Home",
  "bbox": [
    139.0,
```

```
    418.0,  
    427.0,  
    506.0  
  ],  
  "confidence": 0.92,  
  "source": "gemini_vlm",  
  "first_seen_frame": "frame_1777796788_7b3eb6.jpg",  
  "last_seen_frame": "frame_1777796788_7b3eb6.jpg",  
  "first_seen_at": "2026-05-03T08:26:52.866965+00:00",  
  "last_seen_at": "2026-05-03T08:26:52.866965+00:00",  
  "first_seen_index": 0,  
  "last_seen_index": 0,  
  "seen_count": 1,  
  "miss_count": 0,  
  "stale": false,  
  "cached": false,  
  "history": [  
    {  
      "frame_name": "frame_1777796788_7b3eb6.jpg",  
      "frame_index": 0,  
      "seen_at": "2026-05-03T08:26:52.866965+00:00",  
      "confidence": 0.92  
    }  
  ]  
},  
"link:chats": {  
  "element_key": "link:chats",  
  "element_type": "link",  
  "label": "Chats",
```



```
"bbox": [
  139.0,
  490.0,
  427.0,
  570.0
],
"confidence": 0.92,
"source": "gemini_vlm",
"first_seen_frame": "frame_1777796788_7b3eb6.jpg",
"last_seen_frame": "frame_1777796788_7b3eb6.jpg",
"first_seen_at": "2026-05-03T08:26:52.866965+00:00",
"last_seen_at": "2026-05-03T08:26:52.866965+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777796788_7b3eb6.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-03T08:26:52.866965+00:00",
    "confidence": 0.92
  }
]
},
"dropdown:ai assistant": {
  "element_key": "dropdown:ai assistant",
```

```
"element_type": "dropdown",
"label": "AI Assistant",
"bbox": [
    403.0,
    226.0,
    555.0,
    290.0
],
"confidence": 0.9,
"source": "gemini_vlm",
"first_seen_frame": "frame_1777796788_7b3eb6.jpg",
"last_seen_frame": "frame_1777796788_7b3eb6.jpg",
"first_seen_at": "2026-05-03T08:26:52.866965+00:00",
"last_seen_at": "2026-05-03T08:26:52.866965+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
    {
        "frame_name": "frame_1777796788_7b3eb6.jpg",
        "frame_index": 0,
        "seen_at": "2026-05-03T08:26:52.866965+00:00",
        "confidence": 0.9
    }
]
},
```

```
"pane:user message": {
  "element_key": "pane:user message",
  "element_type": "pane",
  "label": "User Message",
  "bbox": [
    651.0,
    322.0,
    1467.0,
    522.0
  ],
  "confidence": 0.95,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777796788_7b3eb6.jpg",
  "last_seen_frame": "frame_1777796788_7b3eb6.jpg",
  "first_seen_at": "2026-05-03T08:26:52.866965+00:00",
  "last_seen_at": "2026-05-03T08:26:52.866965+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
  "seen_count": 1,
  "miss_count": 0,
  "stale": false,
  "cached": false,
  "history": [
    {
      "frame_name": "frame_1777796788_7b3eb6.jpg",
      "frame_index": 0,
      "seen_at": "2026-05-03T08:26:52.866965+00:00",
      "confidence": 0.95
    }
  ]
}
```

```
]
},
"label:connecting to agent...": {
  "element_key": "label:connecting to agent...",
  "element_type": "label",
  "label": "Connecting to agent...",
  "bbox": [
    355.0,
    506.0,
    1163.0,
    642.0
  ],
  "confidence": 0.9,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777796788_7b3eb6.jpg",
  "last_seen_frame": "frame_1777796788_7b3eb6.jpg",
  "first_seen_at": "2026-05-03T08:26:52.866965+00:00",
  "last_seen_at": "2026-05-03T08:26:52.866965+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
  "seen_count": 1,
  "miss_count": 0,
  "stale": false,
  "cached": false,
  "history": [
    {
      "frame_name": "frame_1777796788_7b3eb6.jpg",
      "frame_index": 0,
      "seen_at": "2026-05-03T08:26:52.866965+00:00",
```

```
        "confidence": 0.9
      }
    ]
  },
  "icon:loading": {
    "element_key": "icon:loading",
    "element_type": "icon",
    "label": "Loading",
    "bbox": [
      459.0,
      666.0,
      531.0,
      698.0
    ],
    "confidence": 0.85,
    "source": "gemini_vlm",
    "first_seen_frame": "frame_1777796788_7b3eb6.jpg",
    "last_seen_frame": "frame_1777796788_7b3eb6.jpg",
    "first_seen_at": "2026-05-03T08:26:52.866965+00:00",
    "last_seen_at": "2026-05-03T08:26:52.866965+00:00",
    "first_seen_index": 0,
    "last_seen_index": 0,
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false,
    "history": [
      {
        "frame_name": "frame_1777796788_7b3eb6.jpg",
```

```
        "frame_index": 0,
        "seen_at": "2026-05-03T08:26:52.866965+00:00",
        "confidence": 0.85
    }
]
},
"input:chat input": {
    "element_key": "input:chat input",
    "element_type": "input",
    "label": "Chat Input",
    "bbox": [
        437.0,
        1005.0,
        1394.0,
        1006.0
    ],
    "confidence": 0.9,
    "source": "gemini_vlm",
    "first_seen_frame": "frame_1777796788_7b3eb6.jpg",
    "last_seen_frame": "frame_1777796788_7b3eb6.jpg",
    "first_seen_at": "2026-05-03T08:26:52.866965+00:00",
    "last_seen_at": "2026-05-03T08:26:52.866965+00:00",
    "first_seen_index": 0,
    "last_seen_index": 0,
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false,
    "history": [
```

```
{
  "frame_name": "frame_1777796788_7b3eb6.jpg",
  "frame_index": 0,
  "seen_at": "2026-05-03T08:26:52.866965+00:00",
  "confidence": 0.9
}
],
},
"button:start": {
  "element_key": "button:start",
  "element_type": "button",
  "label": "Start",
  "bbox": [
    443.0,
    938.0,
    499.0,
    994.0
  ],
  "confidence": 0.98,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777796788_7b3eb6.jpg",
  "last_seen_frame": "frame_1777796788_7b3eb6.jpg",
  "first_seen_at": "2026-05-03T08:26:52.866965+00:00",
  "last_seen_at": "2026-05-03T08:26:52.866965+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
  "seen_count": 1,
  "miss_count": 0,
  "stale": false,
```

```
"cached": false,
"history": [
  {
    "frame_name": "frame_1777796788_7b3eb6.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-03T08:26:52.866965+00:00",
    "confidence": 0.98
  }
],
},
"input:search": {
  "element_key": "input:search",
  "element_type": "input",
  "label": "Search",
  "bbox": [
    475.0,
    938.0,
    627.0,
    994.0
  ],
  "confidence": 0.98,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777796788_7b3eb6.jpg",
  "last_seen_frame": "frame_1777796788_7b3eb6.jpg",
  "first_seen_at": "2026-05-03T08:26:52.866965+00:00",
  "last_seen_at": "2026-05-03T08:26:52.866965+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
  "seen_count": 1,
```



```
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777796788_7b3eb6.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-03T08:26:52.866965+00:00",
    "confidence": 0.98
  }
],
},
"button:chrome": {
  "element_key": "button:chrome",
  "element_type": "button",
  "label": "Chrome",
  "bbox": [
    875.0,
    938.0,
    923.0,
    986.0
  ],
  "confidence": 0.98,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777796788_7b3eb6.jpg",
  "last_seen_frame": "frame_1777796788_7b3eb6.jpg",
  "first_seen_at": "2026-05-03T08:26:52.866965+00:00",
  "last_seen_at": "2026-05-03T08:26:52.866965+00:00",
  "first_seen_index": 0,
```

```
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777796788_7b3eb6.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-03T08:26:52.866965+00:00",
    "confidence": 0.98
  }
],
},
"button:vs code": {
  "element_key": "button:vs code",
  "element_type": "button",
  "label": "VS Code",
  "bbox": [
    987.0,
    938.0,
    1035.0,
    994.0
  ],
  "confidence": 0.98,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777796788_7b3eb6.jpg",
  "last_seen_frame": "frame_1777796788_7b3eb6.jpg",
  "first_seen_at": "2026-05-03T08:26:52.866965+00:00",
```

```
"last_seen_at": "2026-05-03T08:26:52.866965+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777796788_7b3eb6.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-03T08:26:52.866965+00:00",
    "confidence": 0.98
  }
],
},
"button:spotify": {
  "element_key": "button:spotify",
  "element_type": "button",
  "label": "Spotify",
  "bbox": [
    1019.0,
    938.0,
    1067.0,
    986.0
  ],
  "confidence": 0.98,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777796788_7b3eb6.jpg",
```

```
"last_seen_frame": "frame_1777796788_7b3eb6.jpg",
"first_seen_at": "2026-05-03T08:26:52.866965+00:00",
"last_seen_at": "2026-05-03T08:26:52.866965+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777796788_7b3eb6.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-03T08:26:52.866965+00:00",
    "confidence": 0.98
  }
],
},
"label:clock": {
  "element_key": "label:clock",
  "element_type": "label",
  "label": "Clock",
  "bbox": [
    1339.0,
    938.0,
    1411.0,
    978.0
  ],
  "confidence": 0.98,
```

```
"source": "gemini_vlm",
"first_seen_frame": "frame_1777796788_7b3eb6.jpg",
"last_seen_frame": "frame_1777796788_7b3eb6.jpg",
"first_seen_at": "2026-05-03T08:26:52.866965+00:00",
"last_seen_at": "2026-05-03T08:26:52.866965+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777796788_7b3eb6.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-03T08:26:52.866965+00:00",
    "confidence": 0.98
  }
],
},
"icon:wi-fi": {
  "element_key": "icon:wi-fi",
  "element_type": "icon",
  "label": "Wi-Fi",
  "bbox": [
    1291.0,
    946.0,
    1331.0,
    986.0
  ]
}
```

```
],
  "confidence": 0.95,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777796788_7b3eb6.jpg",
  "last_seen_frame": "frame_1777796788_7b3eb6.jpg",
  "first_seen_at": "2026-05-03T08:26:52.866965+00:00",
  "last_seen_at": "2026-05-03T08:26:52.866965+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
  "seen_count": 1,
  "miss_count": 0,
  "stale": false,
  "cached": false,
  "history": [
    {
      "frame_name": "frame_1777796788_7b3eb6.jpg",
      "frame_index": 0,
      "seen_at": "2026-05-03T08:26:52.866965+00:00",
      "confidence": 0.95
    }
  ]
},
"icon:battery": {
  "element_key": "icon:battery",
  "element_type": "icon",
  "label": "Battery",
  "bbox": [
    1315.0,
    946.0,
```

```
    1355.0,
    978.0
  ],
  "confidence": 0.95,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777796788_7b3eb6.jpg",
  "last_seen_frame": "frame_1777796788_7b3eb6.jpg",
  "first_seen_at": "2026-05-03T08:26:52.866965+00:00",
  "last_seen_at": "2026-05-03T08:26:52.866965+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
  "seen_count": 1,
  "miss_count": 0,
  "stale": false,
  "cached": false,
  "history": [
    {
      "frame_name": "frame_1777796788_7b3eb6.jpg",
      "frame_index": 0,
      "seen_at": "2026-05-03T08:26:52.866965+00:00",
      "confidence": 0.95
    }
  ]
},
"link:recent task 1": {
  "element_key": "link:recent task 1",
  "element_type": "link",
  "label": "Recent Task 1",
  "bbox": [
```

```
    139.0,  
    626.0,  
    395.0,  
    706.0  
  ],  
  "confidence": 0.85,  
  "source": "gemini_vlm",  
  "first_seen_frame": "frame_1777796788_7b3eb6.jpg",  
  "last_seen_frame": "frame_1777796788_7b3eb6.jpg",  
  "first_seen_at": "2026-05-03T08:26:52.866965+00:00",  
  "last_seen_at": "2026-05-03T08:26:52.866965+00:00",  
  "first_seen_index": 0,  
  "last_seen_index": 0,  
  "seen_count": 1,  
  "miss_count": 0,  
  "stale": false,  
  "cached": false,  
  "history": [  
    {  
      "frame_name": "frame_1777796788_7b3eb6.jpg",  
      "frame_index": 0,  
      "seen_at": "2026-05-03T08:26:52.866965+00:00",  
      "confidence": 0.85  
    }  
  ]  
},  
"link:recent task 2": {  
  "element_key": "link:recent task 2",  
  "element_type": "link",
```



```
"label": "Recent Task 2",
"bbox": [
  139.0,
  770.0,
  419.0,
  842.0
],
"confidence": 0.85,
"source": "gemini_vlm",
"first_seen_frame": "frame_1777796788_7b3eb6.jpg",
"last_seen_frame": "frame_1777796788_7b3eb6.jpg",
"first_seen_at": "2026-05-03T08:26:52.866965+00:00",
"last_seen_at": "2026-05-03T08:26:52.866965+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777796788_7b3eb6.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-03T08:26:52.866965+00:00",
    "confidence": 0.85
  }
]
},
"button:user profile": {
```

```
"element_key": "button:user profile",
"element_type": "button",
"label": "User Profile",
"bbox": [
    171.0,
    1005.0,
    391.0,
    1006.0
],
"confidence": 0.9,
"source": "gemini_vlm",
"first_seen_frame": "frame_1777796788_7b3eb6.jpg",
"last_seen_frame": "frame_1777796788_7b3eb6.jpg",
"first_seen_at": "2026-05-03T08:26:52.866965+00:00",
"last_seen_at": "2026-05-03T08:26:52.866965+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
    {
        "frame_name": "frame_1777796788_7b3eb6.jpg",
        "frame_index": 0,
        "seen_at": "2026-05-03T08:26:52.866965+00:00",
        "confidence": 0.9
    }
]
```

```
},
"button:send": {
  "element_key": "button:send",
  "element_type": "button",
  "label": "Send",
  "bbox": [
    1342.0,
    1005.0,
    1387.0,
    1006.0
  ],
  "confidence": 0.9,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777796788_7b3eb6.jpg",
  "last_seen_frame": "frame_1777796788_7b3eb6.jpg",
  "first_seen_at": "2026-05-03T08:26:52.866965+00:00",
  "last_seen_at": "2026-05-03T08:26:52.866965+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
  "seen_count": 1,
  "miss_count": 0,
  "stale": false,
  "cached": false,
  "history": [
    {
      "frame_name": "frame_1777796788_7b3eb6.jpg",
      "frame_index": 0,
      "seen_at": "2026-05-03T08:26:52.866965+00:00",
      "confidence": 0.9
    }
  ]
}
```

```
    }
  ]
}
},
"resolved_elements": [
  {
    "id": "tab_whatsapp",
    "type": "tab",
    "label": "WhatsApp",
    "description": "Browser tab for WhatsApp",
    "state": "inactive",
    "dx": 120,
    "dy": 56,
    "frame_dx": 259,
    "frame_dy": 114,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
      139,
      82,
      379,
      146
    ],
    "session_memory_key": "tab:whatsapp",
    "cached": false
  },
  {
    "id": "tab_google_search",
    "type": "tab",
```

```
"label": "Google Search",
"description": "Browser tab for Google Search",
"state": "inactive",
"dx": 308,
"dy": 56,
"frame_dx": 447,
"frame_dy": 114,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    323,
    82,
    571,
    146
],
"session_memory_key": "tab:google search",
"cached": false
},
{
    "id": "tab_frontend",
    "type": "tab",
    "label": "frontend",
    "description": "Browser tab for frontend development",
    "state": "inactive",
    "dx": 500,
    "dy": 56,
    "frame_dx": 639,
    "frame_dy": 114,
    "confidence": 0.95,
```

```
"source": "gemini_vlm",
"frame_bbox": [
    515,
    82,
    763,
    146
],
"session_memory_key": "tab:frontend",
"cached": false
},
{
    "id": "tab_github_repos",
    "type": "tab",
    "label": "Your Repositories",
    "description": "Browser tab for GitHub repositories",
    "state": "active",
    "dx": 696,
    "dy": 56,
    "frame_dx": 835,
    "frame_dy": 114,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        707,
        82,
        963,
        146
    ],
    "session_memory_key": "tab:your repositories",
```

```
    "cached": false
  },
  {
    "id": "address_bar",
    "type": "input",
    "label": "Address Bar",
    "description": "Browser address bar showing localhost:5173",
    "state": "visible",
    "dx": 588,
    "dy": 116,
    "frame_dx": 727,
    "frame_dy": 174,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
      139,
      146,
      1315,
      202
    ],
    "session_memory_key": "input:address bar",
    "cached": false
  },
  {
    "id": "sidebar_home",
    "type": "link",
    "label": "Home",
    "description": "Navigation link to Home in ScreenPilot sidebar",
    "state": "active",
```

```
"dx": 144,  
  "dy": 404,  
  "frame_dx": 283,  
  "frame_dy": 462,  
  "confidence": 0.92,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    139,  
    418,  
    427,  
    506  
  ],  
  "session_memory_key": "link:home",  
  "cached": false  
},  
{  
  "id": "sidebar_chats",  
  "type": "link",  
  "label": "Chats",  
  "description": "Navigation link to Chats in ScreenPilot sidebar",  
  "state": "inactive",  
  "dx": 144,  
  "dy": 472,  
  "frame_dx": 283,  
  "frame_dy": 530,  
  "confidence": 0.92,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    139,
```



```
    490,  
    427,  
    570  
  ],  
  "session_memory_key": "link:chats",  
  "cached": false  
},  
{  
  "id": "ai_assistant_dropdown",  
  "type": "dropdown",  
  "label": "AI Assistant",  
  "description": "Dropdown menu to select AI assistant",  
  "state": "visible",  
  "dx": 340,  
  "dy": 200,  
  "frame_dx": 479,  
  "frame_dy": 258,  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    403,  
    226,  
    555,  
    290  
  ],  
  "session_memory_key": "dropdown:ai assistant",  
  "cached": false  
},  
{
```

```
"id": "user_message_bubble",
"type": "pane",
"label": "User Message",
"description": "Chat bubble containing user instructions",
"state": "visible",
"dx": 920,
"dy": 364,
"frame_dx": 1059,
"frame_dy": 422,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    651,
    322,
    1467,
    522
],
"session_memory_key": "pane:user message",
"cached": false
},
{
    "id": "status_message",
    "type": "label",
    "label": "Connecting to agent...",
    "description": "Status message indicating agent connection and screen
capture",
    "state": "visible",
    "dx": 620,
    "dy": 516,
```

```
"frame_dx": 759,  
"frame_dy": 574,  
"confidence": 0.9,  
"source": "gemini_vlm",  
"frame_bbox": [  
    355,  
    506,  
    1163,  
    642  
],  
"session_memory_key": "label:connecting to agent...",  
"cached": false  
},  
{  
    "id": "loading_indicator",  
    "type": "icon",  
    "label": "Loading",  
    "description": "Three dots loading animation",  
    "state": "visible",  
    "dx": 356,  
    "dy": 624,  
    "frame_dx": 495,  
    "frame_dy": 682,  
    "confidence": 0.85,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        459,  
        666,  
        531,
```

```
698
],
"session_memory_key": "icon:loading",
"cached": false
},
{
  "id": "chat_input_area",
  "type": "input",
  "label": "Chat Input",
  "description": "Input field for chat messages",
  "state": "disabled",
  "dx": 776,
  "dy": 948,
  "frame_dx": 915,
  "frame_dy": 1006,
  "confidence": 0.9,
  "source": "gemini_vlm",
  "frame_bbox": [
    437,
    1005,
    1394,
    1006
  ],
  "session_memory_key": "input:chat input",
  "cached": false
},
{
  "id": "taskbar_start",
  "type": "button",
```

```
    "label": "Start",
    "description": "Windows Start button",
    "state": "visible",
    "dx": 332,
    "dy": 908,
    "frame_dx": 471,
    "frame_dy": 966,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
        443,
        938,
        499,
        994
    ],
    "session_memory_key": "button:start",
    "cached": false
},
{
    "id": "taskbar_search",
    "type": "input",
    "label": "Search",
    "description": "Windows Search bar",
    "state": "visible",
    "dx": 412,
    "dy": 908,
    "frame_dx": 551,
    "frame_dy": 966,
    "confidence": 0.98,
```

```
"source": "gemini_vlm",
"frame_bbox": [
    475,
    938,
    627,
    994
],
"session_memory_key": "input:search",
"cached": false
},
{
    "id": "taskbar_chrome",
    "type": "button",
    "label": "Chrome",
    "description": "Google Chrome taskbar icon",
    "state": "active",
    "dx": 760,
    "dy": 904,
    "frame_dx": 899,
    "frame_dy": 962,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
        875,
        938,
        923,
        986
    ],
    "session_memory_key": "button:chrome",
```

```
    "cached": false
  },
  {
    "id": "taskbar_vscode",
    "type": "button",
    "label": "VS Code",
    "description": "Visual Studio Code taskbar icon",
    "state": "visible",
    "dx": 872,
    "dy": 908,
    "frame_dx": 1011,
    "frame_dy": 966,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
      987,
      938,
      1035,
      994
    ],
    "session_memory_key": "button:vs code",
    "cached": false
  },
  {
    "id": "taskbar_spotify",
    "type": "button",
    "label": "Spotify",
    "description": "Spotify taskbar icon",
    "state": "visible",
```

```
"dx": 904,  
"dy": 904,  
"frame_dx": 1043,  
"frame_dy": 962,  
"confidence": 0.98,  
"source": "gemini_vlm",  
"frame_bbox": [  
    1019,  
    938,  
    1067,  
    986  
],  
"session_memory_key": "button:spotify",  
"cached": false  
},  
{  
    "id": "system_tray_clock",  
    "type": "label",  
    "label": "Clock",  
    "description": "System tray clock and date",  
    "state": "visible",  
    "dx": 1236,  
    "dy": 900,  
    "frame_dx": 1375,  
    "frame_dy": 958,  
    "confidence": 0.98,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        1339,
```



```
    938,  
    1411,  
    978  
  ],  
  "session_memory_key": "label:clock",  
  "cached": false  
},  
{  
  "id": "system_tray_wifi",  
  "type": "icon",  
  "label": "Wi-Fi",  
  "description": "Wi-Fi status icon",  
  "state": "visible",  
  "dx": 1172,  
  "dy": 908,  
  "frame_dx": 1311,  
  "frame_dy": 966,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    1291,  
    946,  
    1331,  
    986  
  ],  
  "session_memory_key": "icon:wi-fi",  
  "cached": false  
},  
{
```

```
"id": "system_tray_battery",
"type": "icon",
"label": "Battery",
"description": "Battery status icon",
"state": "visible",
"dx": 1196,
"dy": 904,
"frame_dx": 1335,
"frame_dy": 962,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    1315,
    946,
    1355,
    978
],
"session_memory_key": "icon:battery",
"cached": false
},
{
    "id": "sidebar_recent_1",
    "type": "link",
    "label": "Recent Task 1",
    "description": "Recent task: Click on the 'Your Repositories'...",
    "state": "visible",
    "dx": 128,
    "dy": 608,
    "frame_dx": 267,
```

```
"frame_dy": 666,
"confidence": 0.85,
"source": "gemini_vlm",
"frame_bbox": [
    139,
    626,
    395,
    706
],
"session_memory_key": "link:recent task 1",
"cached": false
},
{
    "id": "sidebar_recent_2",
    "type": "link",
    "label": "Recent Task 2",
    "description": "Recent task: Click on new tab and find the yo...",
    "state": "visible",
    "dx": 140,
    "dy": 748,
    "frame_dx": 279,
    "frame_dy": 806,
    "confidence": 0.85,
    "source": "gemini_vlm",
    "frame_bbox": [
        139,
        770,
        419,
        842
```

```
    ],
    "session_memory_key": "link:recent task 2",
    "cached": false
  },
  {
    "id": "user_profile",
    "type": "button",
    "label": "User Profile",
    "description": "User profile button in sidebar",
    "state": "visible",
    "dx": 142,
    "dy": 948,
    "frame_dx": 281,
    "frame_dy": 1006,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
      171,
      1005,
      391,
      1006
    ],
    "session_memory_key": "button:user profile",
    "cached": false
  },
  {
    "id": "send_button",
    "type": "button",
    "label": "Send",
```

```
"description": "Send message button",
"state": "disabled",
"dx": 1226,
"dy": 948,
"frame_dx": 1365,
"frame_dy": 1006,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    1342,
    1005,
    1387,
    1006
],
"session_memory_key": "button:send",
"cached": false
}
],
"current_elements": [
    {
        "id": "tab_whatsapp",
        "type": "tab",
        "label": "WhatsApp",
        "description": "Browser tab for WhatsApp",
        "state": "inactive",
        "dx": 120,
        "dy": 56,
        "frame_dx": 259,
        "frame_dy": 114,
```

```
"confidence": 0.95,  
"source": "gemini_vlm",  
"frame_bbox": [  
    139,  
    82,  
    379,  
    146  
],  
"session_memory_key": "tab:whatsapp",  
"cached": false  
},  
{  
    "id": "tab_google_search",  
    "type": "tab",  
    "label": "Google Search",  
    "description": "Browser tab for Google Search",  
    "state": "inactive",  
    "dx": 308,  
    "dy": 56,  
    "frame_dx": 447,  
    "frame_dy": 114,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        323,  
        82,  
        571,  
        146  
    ],
```

```
"session_memory_key": "tab:google search",
"cached": false
},
{
  "id": "tab_frontend",
  "type": "tab",
  "label": "frontend",
  "description": "Browser tab for frontend development",
  "state": "inactive",
  "dx": 500,
  "dy": 56,
  "frame_dx": 639,
  "frame_dy": 114,
  "confidence": 0.95,
  "source": "gemini_vlm",
  "frame_bbox": [
    515,
    82,
    763,
    146
  ],
  "session_memory_key": "tab:frontend",
  "cached": false
},
{
  "id": "tab_github_repos",
  "type": "tab",
  "label": "Your Repositories",
  "description": "Browser tab for GitHub repositories",
```

```
"state": "active",
"dx": 696,
"dy": 56,
"frame_dx": 835,
"frame_dy": 114,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    707,
    82,
    963,
    146
],
"session_memory_key": "tab:your repositories",
"cached": false
},
{
    "id": "address_bar",
    "type": "input",
    "label": "Address Bar",
    "description": "Browser address bar showing localhost:5173",
    "state": "visible",
    "dx": 588,
    "dy": 116,
    "frame_dx": 727,
    "frame_dy": 174,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
```



```
    139,  
    146,  
    1315,  
    202  
  ],  
  "session_memory_key": "input:address bar",  
  "cached": false  
},  
{  
  "id": "sidebar_home",  
  "type": "link",  
  "label": "Home",  
  "description": "Navigation link to Home in ScreenPilot sidebar",  
  "state": "active",  
  "dx": 144,  
  "dy": 404,  
  "frame_dx": 283,  
  "frame_dy": 462,  
  "confidence": 0.92,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    139,  
    418,  
    427,  
    506  
  ],  
  "session_memory_key": "link:home",  
  "cached": false  
},
```

```
{
  "id": "sidebar_chats",
  "type": "link",
  "label": "Chats",
  "description": "Navigation link to Chats in ScreenPilot sidebar",
  "state": "inactive",
  "dx": 144,
  "dy": 472,
  "frame_dx": 283,
  "frame_dy": 530,
  "confidence": 0.92,
  "source": "gemini_vlm",
  "frame_bbox": [
    139,
    490,
    427,
    570
  ],
  "session_memory_key": "link:chats",
  "cached": false
},
{
  "id": "ai_assistant_dropdown",
  "type": "dropdown",
  "label": "AI Assistant",
  "description": "Dropdown menu to select AI assistant",
  "state": "visible",
  "dx": 340,
  "dy": 200,
```

```
"frame_dx": 479,  
"frame_dy": 258,  
"confidence": 0.9,  
"source": "gemini_vlm",  
"frame_bbox": [  
    403,  
    226,  
    555,  
    290  
],  
"session_memory_key": "dropdown:ai assistant",  
"cached": false  
},  
{  
    "id": "user_message_bubble",  
    "type": "pane",  
    "label": "User Message",  
    "description": "Chat bubble containing user instructions",  
    "state": "visible",  
    "dx": 920,  
    "dy": 364,  
    "frame_dx": 1059,  
    "frame_dy": 422,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        651,  
        322,  
        1467,
```

```
522
    ],
    "session_memory_key": "pane:user message",
    "cached": false
  },
  {
    "id": "status_message",
    "type": "label",
    "label": "Connecting to agent...",
    "description": "Status message indicating agent connection and screen
capture",
    "state": "visible",
    "dx": 620,
    "dy": 516,
    "frame_dx": 759,
    "frame_dy": 574,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
      355,
      506,
      1163,
      642
    ],
    "session_memory_key": "label:connecting to agent...",
    "cached": false
  },
  {
    "id": "loading_indicator",
```

```
"type": "icon",
"label": "Loading",
"description": "Three dots loading animation",
"state": "visible",
"dx": 356,
"dy": 624,
"frame_dx": 495,
"frame_dy": 682,
"confidence": 0.85,
"source": "gemini_vlm",
"frame_bbox": [
    459,
    666,
    531,
    698
],
"session_memory_key": "icon:loading",
"cached": false
},
{
    "id": "chat_input_area",
    "type": "input",
    "label": "Chat Input",
    "description": "Input field for chat messages",
    "state": "disabled",
    "dx": 776,
    "dy": 948,
    "frame_dx": 915,
    "frame_dy": 1006,
```

```
"confidence": 0.9,  
"source": "gemini_vlm",  
"frame_bbox": [  
    437,  
    1005,  
    1394,  
    1006  
],  
"session_memory_key": "input:chat input",  
"cached": false  
},  
{  
    "id": "taskbar_start",  
    "type": "button",  
    "label": "Start",  
    "description": "Windows Start button",  
    "state": "visible",  
    "dx": 332,  
    "dy": 908,  
    "frame_dx": 471,  
    "frame_dy": 966,  
    "confidence": 0.98,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        443,  
        938,  
        499,  
        994  
    ],
```

```
"session_memory_key": "button:start",
"cached": false
},
{
  "id": "taskbar_search",
  "type": "input",
  "label": "Search",
  "description": "Windows Search bar",
  "state": "visible",
  "dx": 412,
  "dy": 908,
  "frame_dx": 551,
  "frame_dy": 966,
  "confidence": 0.98,
  "source": "gemini_vlm",
  "frame_bbox": [
    475,
    938,
    627,
    994
  ],
  "session_memory_key": "input:search",
  "cached": false
},
{
  "id": "taskbar_chrome",
  "type": "button",
  "label": "Chrome",
  "description": "Google Chrome taskbar icon",
```

```
"state": "active",
"dx": 760,
"dy": 904,
"frame_dx": 899,
"frame_dy": 962,
"confidence": 0.98,
"source": "gemini_vlm",
"frame_bbox": [
    875,
    938,
    923,
    986
],
"session_memory_key": "button:chrome",
"cached": false
},
{
    "id": "taskbar_vscode",
    "type": "button",
    "label": "VS Code",
    "description": "Visual Studio Code taskbar icon",
    "state": "visible",
    "dx": 872,
    "dy": 908,
    "frame_dx": 1011,
    "frame_dy": 966,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
```



```
        987,  
        938,  
        1035,  
        994  
    ],  
    "session_memory_key": "button:vs code",  
    "cached": false  
},  
{  
    "id": "taskbar_spotify",  
    "type": "button",  
    "label": "Spotify",  
    "description": "Spotify taskbar icon",  
    "state": "visible",  
    "dx": 904,  
    "dy": 904,  
    "frame_dx": 1043,  
    "frame_dy": 962,  
    "confidence": 0.98,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        1019,  
        938,  
        1067,  
        986  
    ],  
    "session_memory_key": "button:spotify",  
    "cached": false  
},
```

```
{
  "id": "system_tray_clock",
  "type": "label",
  "label": "Clock",
  "description": "System tray clock and date",
  "state": "visible",
  "dx": 1236,
  "dy": 900,
  "frame_dx": 1375,
  "frame_dy": 958,
  "confidence": 0.98,
  "source": "gemini_vlm",
  "frame_bbox": [
    1339,
    938,
    1411,
    978
  ],
  "session_memory_key": "label:clock",
  "cached": false
},
{
  "id": "system_tray_wifi",
  "type": "icon",
  "label": "Wi-Fi",
  "description": "Wi-Fi status icon",
  "state": "visible",
  "dx": 1172,
  "dy": 908,
```

```
"frame_dx": 1311,  
"frame_dy": 966,  
"confidence": 0.95,  
"source": "gemini_vlm",  
"frame_bbox": [  
    1291,  
    946,  
    1331,  
    986  
],  
"session_memory_key": "icon:wi-fi",  
"cached": false  
},  
{  
    "id": "system_tray_battery",  
    "type": "icon",  
    "label": "Battery",  
    "description": "Battery status icon",  
    "state": "visible",  
    "dx": 1196,  
    "dy": 904,  
    "frame_dx": 1335,  
    "frame_dy": 962,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        1315,  
        946,  
        1355,
```

978

```
],
"session_memory_key": "icon:battery",
"cached": false
},
{
  "id": "sidebar_recent_1",
  "type": "link",
  "label": "Recent Task 1",
  "description": "Recent task: Click on the 'Your Repositories'...",
  "state": "visible",
  "dx": 128,
  "dy": 608,
  "frame_dx": 267,
  "frame_dy": 666,
  "confidence": 0.85,
  "source": "gemini_vlm",
  "frame_bbox": [
    139,
    626,
    395,
    706
  ],
  "session_memory_key": "link:recent task 1",
  "cached": false
},
{
  "id": "sidebar_recent_2",
  "type": "link",
```

```
"label": "Recent Task 2",
"description": "Recent task: Click on new tab and find the yo...",
"state": "visible",
"dx": 140,
"dy": 748,
"frame_dx": 279,
"frame_dy": 806,
"confidence": 0.85,
"source": "gemini_vlm",
"frame_bbox": [
    139,
    770,
    419,
    842
],
"session_memory_key": "link:recent task 2",
"cached": false
},
{
    "id": "user_profile",
    "type": "button",
    "label": "User Profile",
    "description": "User profile button in sidebar",
    "state": "visible",
    "dx": 142,
    "dy": 948,
    "frame_dx": 281,
    "frame_dy": 1006,
    "confidence": 0.9,
```

```
"source": "gemini_vlm",
"frame_bbox": [
  171,
  1005,
  391,
  1006
],
"session_memory_key": "button:user profile",
"cached": false
},
{
  "id": "send_button",
  "type": "button",
  "label": "Send",
  "description": "Send message button",
  "state": "disabled",
  "dx": 1226,
  "dy": 948,
  "frame_dx": 1365,
  "frame_dy": 1006,
  "confidence": 0.9,
  "source": "gemini_vlm",
  "frame_bbox": [
    1342,
    1005,
    1387,
    1006
  ],
  "session_memory_key": "button:send",
```

```
        "cached": false
    }
]
},
"session_memory": {
    "run_id": "eb482227-7e57-46e0-9f57-38b2bcbdd93c7",
    "user_task": "Click on the \"Your Repositories\" tab and check if it is Github repositories page, if yes find and click \"new\" button to create new repository",
    "started_at": "2026-05-03T08:26:19.529870+00:00",
    "updated_at": "2026-05-03T08:26:52.918195+00:00",
    "stale_after_misses": 2,
    "confidence_floor": 0.4,
    "decay_per_miss": 0.9,
    "frame_index": 1,
    "known_elements": {
        "tab:whatsapp": {
            "element_key": "tab:whatsapp",
            "element_type": "tab",
            "label": "WhatsApp",
            "bbox": [
                138.0,
                75.0,
                386.0,
                139.0
            ],
            "confidence": 0.4,
            "source": "gemini_vlm",
            "first_seen_frame": null,
            "last_seen_frame": null,
        }
    }
}
```

```
"first_seen_index": 0,  
"last_seen_index": 2,  
"first_seen_at": "2026-05-03T08:26:52.918195+00:00",  
"last_seen_at": "2026-05-03T08:28:25.053579+00:00",  
"seen_count": 3,  
"miss_count": 17,  
"stale": true,  
"cached": true  
},  
"tab:google search": {  
  "element_key": "tab:google search",  
  "element_type": "tab",  
  "label": "Google Search",  
  "bbox": [  
    323.0,  
    82.0,  
    571.0,  
    146.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 0,  
  "last_seen_index": 0,  
  "first_seen_at": "2026-05-03T08:26:52.918195+00:00",  
  "last_seen_at": "2026-05-03T08:26:52.918195+00:00",  
  "seen_count": 1,  
  "miss_count": 19,
```



```
    "stale": true,
    "cached": true
  },
  "tab:frontend": {
    "element_key": "tab:frontend",
    "element_type": "tab",
    "label": "frontend",
    "bbox": [
      138.0,
      83.0,
      378.0,
      155.0
    ],
    "confidence": 0.95,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 0,
    "last_seen_index": 19,
    "first_seen_at": "2026-05-03T08:26:52.918195+00:00",
    "last_seen_at": "2026-05-03T08:42:22.694106+00:00",
    "seen_count": 8,
    "miss_count": 0,
    "stale": false,
    "cached": false
  },
  "tab:your repositories": {
    "element_key": "tab:your repositories",
    "element_type": "tab",
```

```
"label": "Your Repositories",
"bbox": [
  322.0,
  83.0,
  586.0,
  147.0
],
"confidence": 0.98,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 19,
"first_seen_at": "2026-05-03T08:26:52.918195+00:00",
"last_seen_at": "2026-05-03T08:42:22.694106+00:00",
"seen_count": 8,
"miss_count": 0,
"stale": false,
"cached": false
},
"input:address bar": {
  "element_key": "input:address bar",
  "element_type": "input",
  "label": "Address Bar",
  "bbox": [
    138.0,
    147.0,
    1242.0,
    211.0
  ]
}
```

```
],
  "confidence": 0.98,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 19,
  "first_seen_at": "2026-05-03T08:26:52.918195+00:00",
  "last_seen_at": "2026-05-03T08:42:22.694106+00:00",
  "seen_count": 4,
  "miss_count": 0,
  "stale": false,
  "cached": false
},
"link:home": {
  "element_key": "link:home",
  "element_type": "link",
  "label": "Home",
  "bbox": [
    139.0,
    418.0,
    427.0,
    506.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
```

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"last_seen_index": 0,
"first_seen_at": "2026-05-03T08:26:52.918195+00:00",
"last_seen_at": "2026-05-03T08:26:52.918195+00:00",
"seen_count": 1,
"miss_count": 19,
"stale": true,
"cached": true
},
"link:chats": {
  "element_key": "link:chats",
  "element_type": "link",
  "label": "Chats",
  "bbox": [
    139.0,
    490.0,
    427.0,
    570.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 0,
  "first_seen_at": "2026-05-03T08:26:52.918195+00:00",
  "last_seen_at": "2026-05-03T08:26:52.918195+00:00",
  "seen_count": 1,
  "miss_count": 19,
  "stale": true,
```

```
    "cached": true
  },
  "dropdown:ai assistant": {
    "element_key": "dropdown:ai assistant",
    "element_type": "dropdown",
    "label": "AI Assistant",
    "bbox": [
      394.0,
      234.0,
      546.0,
      298.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 0,
    "last_seen_index": 1,
    "first_seen_at": "2026-05-03T08:26:52.918195+00:00",
    "last_seen_at": "2026-05-03T08:27:39.128680+00:00",
    "seen_count": 2,
    "miss_count": 18,
    "stale": true,
    "cached": true
  },
  "pane:user message": {
    "element_key": "pane:user message",
    "element_type": "pane",
    "label": "User Message",
```

```
"bbox": [
  651.0,
  322.0,
  1467.0,
  522.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 0,
"first_seen_at": "2026-05-03T08:26:52.918195+00:00",
"last_seen_at": "2026-05-03T08:26:52.918195+00:00",
"seen_count": 1,
"miss_count": 19,
"stale": true,
"cached": true
},
"label:connecting to agent...": {
  "element_key": "label:connecting to agent...",
  "element_type": "label",
  "label": "Connecting to agent...",
  "bbox": [
    355.0,
    506.0,
    1163.0,
    642.0
  ],
```

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"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 0,
"first_seen_at": "2026-05-03T08:26:52.918195+00:00",
"last_seen_at": "2026-05-03T08:26:52.918195+00:00",
"seen_count": 1,
"miss_count": 19,
"stale": true,
"cached": true
},
"icon:loading": {
  "element_key": "icon:loading",
  "element_type": "icon",
  "label": "Loading",
  "bbox": [
    459.0,
    666.0,
    531.0,
    698.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 0,
```

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"first_seen_at": "2026-05-03T08:26:52.918195+00:00",
"last_seen_at": "2026-05-03T08:26:52.918195+00:00",
"seen_count": 1,
"miss_count": 19,
"stale": true,
"cached": true
},
"input:chat input": {
  "element_key": "input:chat input",
  "element_type": "input",
  "label": "Chat Input",
  "bbox": [
    437.0,
    1005.0,
    1394.0,
    1006.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 0,
  "first_seen_at": "2026-05-03T08:26:52.918195+00:00",
  "last_seen_at": "2026-05-03T08:26:52.918195+00:00",
  "seen_count": 1,
  "miss_count": 19,
  "stale": true,
  "cached": true
}
```



```
},  
  "button:start": {  
    "element_key": "button:start",  
    "element_type": "button",  
    "label": "Start",  
    "bbox": [  
      450.0,  
      947.0,  
      490.0,  
      995.0  
    ],  
    "confidence": 0.5315,  
    "source": "gemini_vlm",  
    "first_seen_frame": null,  
    "last_seen_frame": null,  
    "first_seen_index": 0,  
    "last_seen_index": 14,  
    "first_seen_at": "2026-05-03T08:26:52.918195+00:00",  
    "last_seen_at": "2026-05-03T08:38:34.371135+00:00",  
    "seen_count": 5,  
    "miss_count": 5,  
    "stale": true,  
    "cached": true  
  },  
  "input:search": {  
    "element_key": "input:search",  
    "element_type": "input",  
    "label": "Search",  
    "bbox": [  

```

```
    458.0,
    947.0,
    682.0,
    1003.0
],
"confidence": 0.5315,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 14,
"first_seen_at": "2026-05-03T08:26:52.918195+00:00",
"last_seen_at": "2026-05-03T08:38:34.371135+00:00",
"seen_count": 3,
"miss_count": 5,
"stale": true,
"cached": true
},
"button:chrome": {
  "element_key": "button:chrome",
  "element_type": "button",
  "label": "Chrome",
  "bbox": [
    802.0,
    947.0,
    850.0,
    987.0
  ],
  "confidence": 0.5315,
```

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"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 14,
"first_seen_at": "2026-05-03T08:26:52.918195+00:00",
"last_seen_at": "2026-05-03T08:38:34.371135+00:00",
"seen_count": 2,
"miss_count": 5,
"stale": true,
"cached": true
},
"button:vs code": {
  "element_key": "button:vs code",
  "element_type": "button",
  "label": "VS Code",
  "bbox": [
    946.0,
    947.0,
    994.0,
    987.0
  ],
  "confidence": 0.5315,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 14,
  "first_seen_at": "2026-05-03T08:26:52.918195+00:00",
```

```
"last_seen_at": "2026-05-03T08:38:34.371135+00:00",
"seen_count": 2,
"miss_count": 5,
"stale": true,
"cached": true
},
"button:spotify": {
  "element_key": "button:spotify",
  "element_type": "button",
  "label": "Spotify",
  "bbox": [
    978.0,
    947.0,
    1034.0,
    987.0
  ],
  "confidence": 0.5315,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 14,
  "first_seen_at": "2026-05-03T08:26:52.918195+00:00",
  "last_seen_at": "2026-05-03T08:38:34.371135+00:00",
  "seen_count": 3,
  "miss_count": 5,
  "stale": true,
  "cached": true
},
```

```
"label:clock": {
  "element_key": "label:clock",
  "element_type": "label",
  "label": "Clock",
  "bbox": [
    1339.0,
    938.0,
    1411.0,
    978.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 0,
  "first_seen_at": "2026-05-03T08:26:52.918195+00:00",
  "last_seen_at": "2026-05-03T08:26:52.918195+00:00",
  "seen_count": 1,
  "miss_count": 19,
  "stale": true,
  "cached": true
},
"icon:wi-fi": {
  "element_key": "icon:wi-fi",
  "element_type": "icon",
  "label": "Wi-Fi",
  "bbox": [
    1291.0,
```

```
    946.0,  
    1331.0,  
    986.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 0,  
  "last_seen_index": 0,  
  "first_seen_at": "2026-05-03T08:26:52.918195+00:00",  
  "last_seen_at": "2026-05-03T08:26:52.918195+00:00",  
  "seen_count": 1,  
  "miss_count": 19,  
  "stale": true,  
  "cached": true  
},  
"icon:battery": {  
  "element_key": "icon:battery",  
  "element_type": "icon",  
  "label": "Battery",  
  "bbox": [  
    1315.0,  
    946.0,  
    1355.0,  
    978.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",
```

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"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 0,
"first_seen_at": "2026-05-03T08:26:52.918195+00:00",
"last_seen_at": "2026-05-03T08:26:52.918195+00:00",
"seen_count": 1,
"miss_count": 19,
"stale": true,
"cached": true
},
"link:recent task 1": {
  "element_key": "link:recent task 1",
  "element_type": "link",
  "label": "Recent Task 1",
  "bbox": [
    139.0,
    626.0,
    395.0,
    706.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 0,
  "first_seen_at": "2026-05-03T08:26:52.918195+00:00",
  "last_seen_at": "2026-05-03T08:26:52.918195+00:00",
```

```
    "seen_count": 1,
    "miss_count": 19,
    "stale": true,
    "cached": true
  },
  "link:recent task 2": {
    "element_key": "link:recent task 2",
    "element_type": "link",
    "label": "Recent Task 2",
    "bbox": [
      139.0,
      770.0,
      419.0,
      842.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 0,
    "last_seen_index": 0,
    "first_seen_at": "2026-05-03T08:26:52.918195+00:00",
    "last_seen_at": "2026-05-03T08:26:52.918195+00:00",
    "seen_count": 1,
    "miss_count": 19,
    "stale": true,
    "cached": true
  },
  "button:user profile": {
```



```
"element_key": "button:user profile",
"element_type": "button",
"label": "User Profile",
"bbox": [
  138.0,
  850.0,
  426.0,
  954.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 1,
"first_seen_at": "2026-05-03T08:26:52.918195+00:00",
"last_seen_at": "2026-05-03T08:27:39.128680+00:00",
"seen_count": 2,
"miss_count": 18,
"stale": true,
"cached": true
},
"button:send": {
  "element_key": "button:send",
  "element_type": "button",
  "label": "Send",
  "bbox": [
    1342.0,
    1005.0,
```

```
    1387.0,  
    1006.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 0,  
  "last_seen_index": 0,  
  "first_seen_at": "2026-05-03T08:26:52.918195+00:00",  
  "last_seen_at": "2026-05-03T08:26:52.918195+00:00",  
  "seen_count": 1,  
  "miss_count": 19,  
  "stale": true,  
  "cached": true  
},  
"tab:prasad fashion matara": {  
  "element_key": "tab:prasad fashion matara",  
  "element_type": "tab",  
  "label": "prasad fashion matara",  
  "bbox": [  
    322.0,  
    82.0,  
    570.0,  
    154.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,
```

```
"last_seen_frame": null,
"first_seen_index": 1,
"last_seen_index": 1,
"first_seen_at": "2026-05-03T08:27:39.128680+00:00",
"last_seen_at": "2026-05-03T08:27:39.128680+00:00",
"seen_count": 1,
"miss_count": 18,
"stale": true,
"cached": true
},
"button:new tab": {
  "element_key": "button:new tab",
  "element_type": "button",
  "label": "New Tab",
  "bbox": [
    930.0,
    98.0,
    970.0,
    146.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
  "last_seen_index": 1,
  "first_seen_at": "2026-05-03T08:27:39.128680+00:00",
  "last_seen_at": "2026-05-03T08:27:39.128680+00:00",
  "seen_count": 1,
```

```
    "miss_count": 18,  
    "stale": true,  
    "cached": true  
  },  
  "button:ask gemini": {  
    "element_key": "button:ask gemini",  
    "element_type": "button",  
    "label": "Ask Gemini",  
    "bbox": [  
      1170.0,  
      83.0,  
      1290.0,  
      139.0  
    ],  
    "confidence": 0.5315,  
    "source": "gemini_vlm",  
    "first_seen_frame": null,  
    "last_seen_frame": null,  
    "first_seen_index": 1,  
    "last_seen_index": 14,  
    "first_seen_at": "2026-05-03T08:27:39.128680+00:00",  
    "last_seen_at": "2026-05-03T08:38:34.371135+00:00",  
    "seen_count": 2,  
    "miss_count": 5,  
    "stale": true,  
    "cached": true  
  },  
  "button:back": {  
    "element_key": "button:back",
```

```
"element_type": "button",
"label": "Back",
"bbox": [
    154.0,
    146.0,
    202.0,
    202.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 1,
"last_seen_index": 1,
"first_seen_at": "2026-05-03T08:27:39.128680+00:00",
"last_seen_at": "2026-05-03T08:27:39.128680+00:00",
"seen_count": 1,
"miss_count": 18,
"stale": true,
"cached": true
},
"button:forward": {
    "element_key": "button:forward",
    "element_type": "button",
    "label": "Forward",
    "bbox": [
        186.0,
        146.0,
        242.0,
```

```
    202.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
  "last_seen_index": 1,
  "first_seen_at": "2026-05-03T08:27:39.128680+00:00",
  "last_seen_at": "2026-05-03T08:27:39.128680+00:00",
  "seen_count": 1,
  "miss_count": 18,
  "stale": true,
  "cached": true
},
"button:refresh": {
  "element_key": "button:refresh",
  "element_type": "button",
  "label": "Refresh",
  "bbox": [
    226.0,
    146.0,
    282.0,
    194.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
```

```
"first_seen_index": 1,
"last_seen_index": 1,
"first_seen_at": "2026-05-03T08:27:39.128680+00:00",
"last_seen_at": "2026-05-03T08:27:39.128680+00:00",
"seen_count": 1,
"miss_count": 18,
"stale": true,
"cached": true
},
"label:screenpilot ai workspace": {
  "element_key": "label:screenpilot ai workspace",
  "element_type": "label",
  "label": "ScreenPilot AI workspace",
  "bbox": [
    138.0,
    234.0,
    426.0,
    322.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
  "last_seen_index": 1,
  "first_seen_at": "2026-05-03T08:27:39.128680+00:00",
  "last_seen_at": "2026-05-03T08:27:39.128680+00:00",
  "seen_count": 1,
  "miss_count": 18,
```

```
    "stale": true,
    "cached": true
  },
  "input:search chats": {
    "element_key": "input:search chats",
    "element_type": "input",
    "label": "Search chats",
    "bbox": [
      138.0,
      322.0,
      426.0,
      410.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 1,
    "last_seen_index": 1,
    "first_seen_at": "2026-05-03T08:27:39.128680+00:00",
    "last_seen_at": "2026-05-03T08:27:39.128680+00:00",
    "seen_count": 1,
    "miss_count": 18,
    "stale": true,
    "cached": true
  },
  "button:home": {
    "element_key": "button:home",
    "element_type": "button",
```



```
"label": "Home",
"bbox": [
  138.0,
  410.0,
  426.0,
  498.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 1,
"last_seen_index": 1,
"first_seen_at": "2026-05-03T08:27:39.128680+00:00",
"last_seen_at": "2026-05-03T08:27:39.128680+00:00",
"seen_count": 1,
"miss_count": 18,
"stale": true,
"cached": true
},
"button:chats": {
  "element_key": "button:chats",
  "element_type": "button",
  "label": "Chats",
  "bbox": [
    138.0,
    482.0,
    426.0,
    578.0
  ]
}
```

```
],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 1,  
  "last_seen_index": 1,  
  "first_seen_at": "2026-05-03T08:27:39.128680+00:00",  
  "last_seen_at": "2026-05-03T08:27:39.128680+00:00",  
  "seen_count": 1,  
  "miss_count": 18,  
  "stale": true,  
  "cached": true  
},  
"label:user message": {  
  "element_key": "label:user message",  
  "element_type": "label",  
  "label": "User Message",  
  "bbox": [  
    634.0,  
    298.0,  
    1418.0,  
    490.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 1,
```

```
"last_seen_index": 1,
"first_seen_at": "2026-05-03T08:27:39.128680+00:00",
"last_seen_at": "2026-05-03T08:27:39.128680+00:00",
"seen_count": 1,
"miss_count": 18,
"stale": true,
"cached": true
},
"pane:task plan": {
  "element_key": "pane:task plan",
  "element_type": "pane",
  "label": "Task plan",
  "bbox": [
    354.0,
    442.0,
    1122.0,
    842.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
  "last_seen_index": 1,
  "first_seen_at": "2026-05-03T08:27:39.128680+00:00",
  "last_seen_at": "2026-05-03T08:27:39.128680+00:00",
  "seen_count": 1,
  "miss_count": 18,
  "stale": true,
```

```
    "cached": true
  },
  "label:step 1: move mouse to your repositories tab": {
    "element_key": "label:step 1: move mouse to your repositories tab",
    "element_type": "label",
    "label": "Step 1: Move mouse to Your Repositories tab",
    "bbox": [
      370.0,
      618.0,
      1074.0,
      714.0
    ],
    "confidence": 0.4,
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    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 1,
    "last_seen_index": 1,
    "first_seen_at": "2026-05-03T08:27:39.128680+00:00",
    "last_seen_at": "2026-05-03T08:27:39.128680+00:00",
    "seen_count": 1,
    "miss_count": 18,
    "stale": true,
    "cached": true
  },
  "tab:prasad fashion matara - google search": {
    "element_key": "tab:prasad fashion matara - google search",
    "element_type": "tab",
    "label": "prasad fashion matara - Google Search",
```

```
"bbox": [
  330.0,
  75.0,
  578.0,
  139.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 2,
"last_seen_index": 2,
"first_seen_at": "2026-05-03T08:28:25.053579+00:00",
"last_seen_at": "2026-05-03T08:28:25.053579+00:00",
"seen_count": 1,
"miss_count": 17,
"stale": true,
"cached": true
},
"tab:(12) whatsapp": {
  "element_key": "tab:(12) whatsapp",
  "element_type": "tab",
  "label": "(12) WhatsApp",
  "bbox": [
    138.0,
    82.0,
    386.0,
    146.0
  ],
```

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"last_seen_index": 3,
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"last_seen_at": "2026-05-03T08:29:27.234829+00:00",
"seen_count": 1,
"miss_count": 16,
"stale": true,
"cached": true
},
"icon:google chrome": {
  "element_key": "icon:google chrome",
  "element_type": "icon",
  "label": "Google Chrome",
  "bbox": [
    154.0,
    82.0,
    242.0,
    202.0
  ],
  "confidence": 0.5048,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 4,
  "last_seen_index": 13,
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"first_seen_at": "2026-05-03T08:30:15.480408+00:00",
"last_seen_at": "2026-05-03T08:37:45.966440+00:00",
"seen_count": 8,
"miss_count": 6,
"stale": true,
"cached": true
},
"pane:taskbar": {
  "element_key": "pane:taskbar",
  "element_type": "pane",
  "label": "Taskbar",
  "bbox": [
    138.0,
    915.0,
    1795.0,
    1007.0
  ],
  "confidence": 0.4543,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 5,
  "last_seen_index": 12,
  "first_seen_at": "2026-05-03T08:31:03.326139+00:00",
  "last_seen_at": "2026-05-03T08:36:42.648830+00:00",
  "seen_count": 2,
  "miss_count": 7,
  "stale": true,
  "cached": true
}
```

```
},
"icon:huiontablet": {
  "element_key": "icon:huiontablet",
  "element_type": "icon",
  "label": "HuionTablet",
  "bbox": [
    210.0,
    82.0,
    298.0,
    202.0
  ],
  "confidence": 0.5048,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 6,
  "last_seen_index": 13,
  "first_seen_at": "2026-05-03T08:31:58.654399+00:00",
  "last_seen_at": "2026-05-03T08:37:45.966440+00:00",
  "seen_count": 7,
  "miss_count": 6,
  "stale": true,
  "cached": true
},
"icon:wise memory optimizer": {
  "element_key": "icon:wise memory optimizer",
  "element_type": "icon",
  "label": "Wise Memory Optimizer",
  "bbox": [
```



```
    154.0,  
    178.0,  
    226.0,  
    298.0  
  ],  
  "confidence": 0.4089,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 11,  
  "last_seen_index": 11,  
  "first_seen_at": "2026-05-03T08:35:46.217640+00:00",  
  "last_seen_at": "2026-05-03T08:35:46.217640+00:00",  
  "seen_count": 1,  
  "miss_count": 8,  
  "stale": true,  
  "cached": true  
},  
"icon:amazon workspaces": {  
  "element_key": "icon:amazon workspaces",  
  "element_type": "icon",  
  "label": "Amazon WorkSpaces",  
  "bbox": [  
    210.0,  
    178.0,  
    290.0,  
    298.0  
  ],  
  "confidence": 0.4089,
```

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"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 11,
"last_seen_index": 11,
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"last_seen_at": "2026-05-03T08:35:46.217640+00:00",
"seen_count": 1,
"miss_count": 8,
"stale": true,
"cached": true
},
"icon:recycle bin": {
  "element_key": "icon:recycle bin",
  "element_type": "icon",
  "label": "Recycle Bin",
  "bbox": [
    154.0,
    266.0,
    226.0,
    394.0
  ],
  "confidence": 0.4089,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 11,
  "last_seen_index": 11,
  "first_seen_at": "2026-05-03T08:35:46.217640+00:00",
```

```
    "last_seen_at": "2026-05-03T08:35:46.217640+00:00",
    "seen_count": 1,
    "miss_count": 8,
    "stale": true,
    "cached": true
  },
  "icon:krita": {
    "element_key": "icon:krita",
    "element_type": "icon",
    "label": "Krita",
    "bbox": [
      218.0,
      266.0,
      282.0,
      394.0
    ],
    "confidence": 0.4089,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 11,
    "last_seen_index": 11,
    "first_seen_at": "2026-05-03T08:35:46.217640+00:00",
    "last_seen_at": "2026-05-03T08:35:46.217640+00:00",
    "seen_count": 1,
    "miss_count": 8,
    "stale": true,
    "cached": true
  },
}
```

```
"icon:nuwani - chrome": {
  "element_key": "icon:nuwani - chrome",
  "element_type": "icon",
  "label": "Nuwani - Chrome",
  "bbox": [
    154.0,
    362.0,
    226.0,
    482.0
  ],
  "confidence": 0.4089,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 11,
  "last_seen_index": 11,
  "first_seen_at": "2026-05-03T08:35:46.217640+00:00",
  "last_seen_at": "2026-05-03T08:35:46.217640+00:00",
  "seen_count": 1,
  "miss_count": 8,
  "stale": true,
  "cached": true
},
"icon:docker desktop": {
  "element_key": "icon:docker desktop",
  "element_type": "icon",
  "label": "Docker Desktop",
  "bbox": [
    218.0,
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```
    362.0,  
    282.0,  
    482.0  
  ],  
  "confidence": 0.4089,  
  "source": "gemini_vlm",  
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  "last_seen_frame": null,  
  "first_seen_index": 11,  
  "last_seen_index": 11,  
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  "last_seen_at": "2026-05-03T08:35:46.217640+00:00",  
  "seen_count": 1,  
  "miss_count": 8,  
  "stale": true,  
  "cached": true  
},  
"icon:antigravity": {  
  "element_key": "icon:antigravity",  
  "element_type": "icon",  
  "label": "Antigravity",  
  "bbox": [  
    162.0,  
    450.0,  
    218.0,  
    570.0  
  ],  
  "confidence": 0.4089,  
  "source": "gemini_vlm",
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"first_seen_index": 11,
"last_seen_index": 11,
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"last_seen_at": "2026-05-03T08:35:46.217640+00:00",
"seen_count": 1,
"miss_count": 8,
"stale": true,
"cached": true
},
"icon:visual studio code": {
  "element_key": "icon:visual studio code",
  "element_type": "icon",
  "label": "Visual Studio Code",
  "bbox": [
    162.0,
    538.0,
    218.0,
    666.0
  ],
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  "last_seen_frame": null,
  "first_seen_index": 11,
  "last_seen_index": 11,
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  "last_seen_at": "2026-05-03T08:35:46.217640+00:00",
```

```
    "seen_count": 1,  
    "miss_count": 8,  
    "stale": true,  
    "cached": true  
  },  
  "icon:postman": {  
    "element_key": "icon:postman",  
    "element_type": "icon",  
    "label": "Postman",  
    "bbox": [  
      162.0,  
      634.0,  
      218.0,  
      754.0  
    ],  
    "confidence": 0.4089,  
    "source": "gemini_vlm",  
    "first_seen_frame": null,  
    "last_seen_frame": null,  
    "first_seen_index": 11,  
    "last_seen_index": 11,  
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    "last_seen_at": "2026-05-03T08:35:46.217640+00:00",  
    "seen_count": 1,  
    "miss_count": 8,  
    "stale": true,  
    "cached": true  
  },  
  "icon:loom": {
```

```
"element_key": "icon:loom",
"element_type": "icon",
"label": "Loom",
"bbox": [
  162.0,
  722.0,
  218.0,
  842.0
],
"confidence": 0.4089,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 11,
"last_seen_index": 11,
"first_seen_at": "2026-05-03T08:35:46.217640+00:00",
"last_seen_at": "2026-05-03T08:35:46.217640+00:00",
"seen_count": 1,
"miss_count": 8,
"stale": true,
"cached": true
},
"icon:vlc media player": {
  "element_key": "icon:vlc media player",
  "element_type": "icon",
  "label": "VLC media player",
  "bbox": [
    162.0,
    810.0,
```



```
    218.0,  
    938.0  
  ],  
  "confidence": 0.4089,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
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  "last_seen_index": 11,  
  "first_seen_at": "2026-05-03T08:35:46.217640+00:00",  
  "last_seen_at": "2026-05-03T08:35:46.217640+00:00",  
  "seen_count": 1,  
  "miss_count": 8,  
  "stale": true,  
  "cached": true  
},  
"widget:weather": {  
  "element_key": "widget:weather",  
  "element_type": "widget",  
  "label": "Weather",  
  "bbox": [  
    138.0,  
    938.0,  
    314.0,  
    994.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,
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"last_seen_frame": null,
"first_seen_index": 11,
"last_seen_index": 11,
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"last_seen_at": "2026-05-03T08:35:46.217640+00:00",
"seen_count": 1,
"miss_count": 8,
"stale": true,
"cached": true
},
"button:copilot": {
  "element_key": "button:copilot",
  "element_type": "button",
  "label": "Copilot",
  "bbox": [
    842.0,
    930.0,
    890.0,
    986.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 11,
  "last_seen_index": 11,
  "first_seen_at": "2026-05-03T08:35:46.217640+00:00",
  "last_seen_at": "2026-05-03T08:35:46.217640+00:00",
  "seen_count": 1,
```

```
    "miss_count": 8,  
    "stale": true,  
    "cached": true  
  },  
  "button:task view": {  
    "element_key": "button:task view",  
    "element_type": "button",  
    "label": "Task View",  
    "bbox": [  
      890.0,  
      930.0,  
      938.0,  
      986.0  
    ],  
    "confidence": 0.4,  
    "source": "gemini_vlm",  
    "first_seen_frame": null,  
    "last_seen_frame": null,  
    "first_seen_index": 11,  
    "last_seen_index": 11,  
    "first_seen_at": "2026-05-03T08:35:46.217640+00:00",  
    "last_seen_at": "2026-05-03T08:35:46.217640+00:00",  
    "seen_count": 1,  
    "miss_count": 8,  
    "stale": true,  
    "cached": true  
  },  
  "button:file explorer": {  
    "element_key": "button:file explorer",
```

```
"element_type": "button",
"label": "File Explorer",
"bbox": [
    730.0,
    947.0,
    778.0,
    995.0
],
"confidence": 0.5315,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 11,
"last_seen_index": 14,
"first_seen_at": "2026-05-03T08:35:46.217640+00:00",
"last_seen_at": "2026-05-03T08:38:34.371135+00:00",
"seen_count": 2,
"miss_count": 5,
"stale": true,
"cached": true
},
"button:microsoft edge": {
    "element_key": "button:microsoft edge",
    "element_type": "button",
    "label": "Microsoft Edge",
    "bbox": [
        978.0,
        930.0,
        1026.0,
```

```
    986.0
  ],
  "confidence": 0.4089,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 11,
  "last_seen_index": 11,
  "first_seen_at": "2026-05-03T08:35:46.217640+00:00",
  "last_seen_at": "2026-05-03T08:35:46.217640+00:00",
  "seen_count": 1,
  "miss_count": 8,
  "stale": true,
  "cached": true
},
"button:microsoft store": {
  "element_key": "button:microsoft store",
  "element_type": "button",
  "label": "Microsoft Store",
  "bbox": [
    1026.0,
    930.0,
    1074.0,
    986.0
  ],
  "confidence": 0.4089,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
```

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"first_seen_index": 11,
"last_seen_index": 11,
"first_seen_at": "2026-05-03T08:35:46.217640+00:00",
"last_seen_at": "2026-05-03T08:35:46.217640+00:00",
"seen_count": 1,
"miss_count": 8,
"stale": true,
"cached": true
},
"button:xbox": {
  "element_key": "button:xbox",
  "element_type": "button",
  "label": "Xbox",
  "bbox": [
    1066.0,
    930.0,
    1114.0,
    986.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 11,
  "last_seen_index": 11,
  "first_seen_at": "2026-05-03T08:35:46.217640+00:00",
  "last_seen_at": "2026-05-03T08:35:46.217640+00:00",
  "seen_count": 1,
  "miss_count": 8,
```

```
    "stale": true,
    "cached": true
  },
  "button:google chrome": {
    "element_key": "button:google chrome",
    "element_type": "button",
    "label": "Google Chrome",
    "bbox": [
      1114.0,
      930.0,
      1162.0,
      986.0
    ],
    "confidence": 0.4089,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 11,
    "last_seen_index": 11,
    "first_seen_at": "2026-05-03T08:35:46.217640+00:00",
    "last_seen_at": "2026-05-03T08:35:46.217640+00:00",
    "seen_count": 1,
    "miss_count": 8,
    "stale": true,
    "cached": true
  },
  "button:notepad": {
    "element_key": "button:notepad",
    "element_type": "button",
```

```
"label": "Notepad",
"bbox": [
  1162.0,
  930.0,
  1210.0,
  986.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 11,
"last_seen_index": 11,
"first_seen_at": "2026-05-03T08:35:46.217640+00:00",
"last_seen_at": "2026-05-03T08:35:46.217640+00:00",
"seen_count": 1,
"miss_count": 8,
"stale": true,
"cached": true
},
"button:visual studio code": {
  "element_key": "button:visual studio code",
  "element_type": "button",
  "label": "Visual Studio Code",
  "bbox": [
    1202.0,
    930.0,
    1250.0,
    986.0
  ]
}
```



```
],
  "confidence": 0.4089,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 11,
  "last_seen_index": 11,
  "first_seen_at": "2026-05-03T08:35:46.217640+00:00",
  "last_seen_at": "2026-05-03T08:35:46.217640+00:00",
  "seen_count": 1,
  "miss_count": 8,
  "stale": true,
  "cached": true
},
"button:postman": {
  "element_key": "button:postman",
  "element_type": "button",
  "label": "Postman",
  "bbox": [
    1298.0,
    930.0,
    1346.0,
    986.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 11,
```

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"last_seen_index": 11,
"first_seen_at": "2026-05-03T08:35:46.217640+00:00",
"last_seen_at": "2026-05-03T08:35:46.217640+00:00",
"seen_count": 1,
"miss_count": 8,
"stale": true,
"cached": true
},
"button:loom": {
  "element_key": "button:loom",
  "element_type": "button",
  "label": "Loom",
  "bbox": [
    1338.0,
    930.0,
    1386.0,
    986.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 11,
  "last_seen_index": 11,
  "first_seen_at": "2026-05-03T08:35:46.217640+00:00",
  "last_seen_at": "2026-05-03T08:35:46.217640+00:00",
  "seen_count": 1,
  "miss_count": 8,
  "stale": true,
```

```
    "cached": true
  },
  "pane:system tray": {
    "element_key": "pane:system tray",
    "element_type": "pane",
    "label": "System Tray",
    "bbox": [
      1466.0,
      930.0,
      1722.0,
      994.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 11,
    "last_seen_index": 11,
    "first_seen_at": "2026-05-03T08:35:46.217640+00:00",
    "last_seen_at": "2026-05-03T08:35:46.217640+00:00",
    "seen_count": 1,
    "miss_count": 8,
    "stale": true,
    "cached": true
  },
  "label:clock and date": {
    "element_key": "label:clock and date",
    "element_type": "label",
    "label": "Clock and Date",
```

```
"bbox": [
  1690.0,
  930.0,
  1794.0,
  986.0
],
"confidence": 0.4089,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 11,
"last_seen_index": 11,
"first_seen_at": "2026-05-03T08:35:46.217640+00:00",
"last_seen_at": "2026-05-03T08:35:46.217640+00:00",
"seen_count": 1,
"miss_count": 8,
"stale": true,
"cached": true
},
"pane:sticky note": {
  "element_key": "pane:sticky note",
  "element_type": "pane",
  "label": "Sticky Note",
  "bbox": [
    1378.0,
    74.0,
    1795.0,
    338.0
  ],
```

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"confidence": 0.4089,
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"last_seen_at": "2026-05-03T08:35:46.217640+00:00",
"seen_count": 1,
"miss_count": 8,
"stale": true,
"cached": true
},
"icon:github logo": {
  "element_key": "icon:github logo",
  "element_type": "icon",
  "label": "GitHub Logo",
  "bbox": [
    186.0,
    227.0,
    242.0,
    283.0
  ],
  "confidence": 0.99,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 14,
  "last_seen_index": 19,
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"first_seen_at": "2026-05-03T08:38:34.371135+00:00",
"last_seen_at": "2026-05-03T08:42:22.694106+00:00",
"seen_count": 2,
"miss_count": 0,
"stale": false,
"cached": false
},
"input:search github": {
  "element_key": "input:search github",
  "element_type": "input",
  "label": "Search GitHub",
  "bbox": [
    1210.0,
    227.0,
    1434.0,
    275.0
  ],
  "confidence": 0.95,
  "source": "gemini_vlm",
  "first_seen_frame": null,
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  "first_seen_index": 14,
  "last_seen_index": 19,
  "first_seen_at": "2026-05-03T08:38:34.371135+00:00",
  "last_seen_at": "2026-05-03T08:42:22.694106+00:00",
  "seen_count": 2,
  "miss_count": 0,
  "stale": false,
  "cached": false
}
```

```
},
"image:profile picture": {
  "element_key": "image:profile picture",
  "element_type": "image",
  "label": "Profile Picture",
  "bbox": [
    306.0,
    339.0,
    586.0,
    835.0
  ],
  "confidence": 0.5609,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 14,
  "last_seen_index": 14,
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  "last_seen_at": "2026-05-03T08:38:34.371135+00:00",
  "seen_count": 1,
  "miss_count": 5,
  "stale": true,
  "cached": true
},
"heading:nuwani sithara": {
  "element_key": "heading:nuwani sithara",
  "element_type": "heading",
  "label": "Nuwani Sithara",
  "bbox": [
```

```
    338.0,  
    787.0,  
    490.0,  
    835.0  
  ],  
  "confidence": 0.5315,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 14,  
  "last_seen_index": 14,  
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  "last_seen_at": "2026-05-03T08:38:34.371135+00:00",  
  "seen_count": 1,  
  "miss_count": 5,  
  "stale": true,  
  "cached": true  
},  
"label:nuwani-sithara": {  
  "element_key": "label:nuwani-sithara",  
  "element_type": "label",  
  "label": "nuwani-sithara",  
  "bbox": [  
    354.0,  
    827.0,  
    474.0,  
    867.0  
  ],  
  "confidence": 0.5315,
```



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"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 14,
"last_seen_index": 14,
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"last_seen_at": "2026-05-03T08:38:34.371135+00:00",
"seen_count": 1,
"miss_count": 5,
"stale": true,
"cached": true
},
"button:edit profile": {
  "element_key": "button:edit profile",
  "element_type": "button",
  "label": "Edit profile",
  "bbox": [
    306.0,
    923.0,
    594.0,
    987.0
  ],
  "confidence": 0.5315,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 14,
  "last_seen_index": 14,
  "first_seen_at": "2026-05-03T08:38:34.371135+00:00",
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```
    "last_seen_at": "2026-05-03T08:38:34.371135+00:00",
    "seen_count": 1,
    "miss_count": 5,
    "stale": true,
    "cached": true
  },
  "tab:overview": {
    "element_key": "tab:overview",
    "element_type": "tab",
    "label": "Overview",
    "bbox": [
      154.0,
      291.0,
      258.0,
      363.0
    ],
    "confidence": 0.5315,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 14,
    "last_seen_index": 14,
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    "last_seen_at": "2026-05-03T08:38:34.371135+00:00",
    "seen_count": 1,
    "miss_count": 5,
    "stale": true,
    "cached": true
  },
}
```

```
"tab:repositories": {
  "element_key": "tab:repositories",
  "element_type": "tab",
  "label": "Repositories",
  "bbox": [
    234.0,
    291.0,
    362.0,
    331.0
  ],
  "confidence": 0.5315,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 14,
  "last_seen_index": 14,
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  "last_seen_at": "2026-05-03T08:38:34.371135+00:00",
  "seen_count": 1,
  "miss_count": 5,
  "stale": true,
  "cached": true
},
"input:find a repository...": {
  "element_key": "input:find a repository...",
  "element_type": "input",
  "label": "Find a repository...",
  "bbox": [
    514.0,
```

```
    371.0,  
    994.0,  
    427.0  
  ],  
  "confidence": 0.5315,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 14,  
  "last_seen_index": 14,  
  "first_seen_at": "2026-05-03T08:38:34.371135+00:00",  
  "last_seen_at": "2026-05-03T08:38:34.371135+00:00",  
  "seen_count": 1,  
  "miss_count": 5,  
  "stale": true,  
  "cached": true  
},  
"button:new": {  
  "element_key": "button:new",  
  "element_type": "button",  
  "label": "New",  
  "bbox": [  
    1154.0,  
    363.0,  
    1226.0,  
    403.0  
  ],  
  "confidence": 0.5315,  
  "source": "gemini_vlm",
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"first_seen_frame": null,
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"last_seen_index": 14,
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"last_seen_at": "2026-05-03T08:38:34.371135+00:00",
"seen_count": 1,
"miss_count": 5,
"stale": true,
"cached": true
},
"link:screenawaretaskagent": {
  "element_key": "link:screenawaretaskagent",
  "element_type": "link",
  "label": "ScreenAwareTaskAgent",
  "bbox": [
    538.0,
    459.0,
    802.0,
    515.0
  ],
  "confidence": 0.5315,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 14,
  "last_seen_index": 14,
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  "last_seen_at": "2026-05-03T08:38:34.371135+00:00",
```

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"seen_count": 1,
"miss_count": 5,
"stale": true,
"cached": true
},
"link:clothing-brand": {
  "element_key": "link:clothing-brand",
  "element_type": "link",
  "label": "clothing-brand",
  "bbox": [
    546.0,
    603.0,
    738.0,
    659.0
  ],
  "confidence": 0.5315,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 14,
  "last_seen_index": 14,
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  "last_seen_at": "2026-05-03T08:38:34.371135+00:00",
  "seen_count": 1,
  "miss_count": 5,
  "stale": true,
  "cached": true
},
"link:login-page": {
```

```
"element_key": "link:login-page",
"element_type": "link",
"label": "login-page",
"bbox": [
  554.0,
  755.0,
  706.0,
  811.0
],
"confidence": 0.5315,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 14,
"last_seen_index": 14,
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"last_seen_at": "2026-05-03T08:38:34.371135+00:00",
"seen_count": 1,
"miss_count": 5,
"stale": true,
"cached": true
},
"link:qr-code-generator": {
  "element_key": "link:qr-code-generator",
  "element_type": "link",
  "label": "qr-code-generator",
  "bbox": [
    546.0,
    907.0,
```

```
    762.0,  
    963.0  
  ],  
  "confidence": 0.5315,  
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  "last_seen_frame": null,  
  "first_seen_index": 14,  
  "last_seen_index": 14,  
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  "last_seen_at": "2026-05-03T08:38:34.371135+00:00",  
  "seen_count": 1,  
  "miss_count": 5,  
  "stale": true,  
  "cached": true  
},  
"pane:spotify player": {  
  "element_key": "pane:spotify player",  
  "element_type": "pane",  
  "label": "Spotify Player",  
  "bbox": [  
    1058.0,  
    363.0,  
    1434.0,  
    691.0  
  ],  
  "confidence": 0.5609,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,
```



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"last_seen_frame": null,  
"first_seen_index": 14,  
"last_seen_index": 14,  
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"last_seen_at": "2026-05-03T08:38:34.371135+00:00",  
"seen_count": 1,  
"miss_count": 5,  
"stale": true,  
"cached": true  
},  
"label:karma (the cycle turns)": {  
  "element_key": "label:karma (the cycle turns)",  
  "element_type": "label",  
  "label": "Karma (The Cycle Turns)",  
  "bbox": [  
    1098.0,  
    483.0,  
    1226.0,  
    515.0  
  ],  
  "confidence": 0.5315,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 14,  
  "last_seen_index": 14,  
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  "last_seen_at": "2026-05-03T08:38:34.371135+00:00",  
  "seen_count": 1,
```

```
    "miss_count": 5,
    "stale": true,
    "cached": true
  },
  "label:sumedh k": {
    "element_key": "label:sumedh k",
    "element_type": "label",
    "label": "Sumedh K",
    "bbox": [
      1114.0,
      499.0,
      1202.0,
      523.0
    ],
    "confidence": 0.5315,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 14,
    "last_seen_index": 14,
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    "last_seen_at": "2026-05-03T08:38:34.371135+00:00",
    "seen_count": 1,
    "miss_count": 5,
    "stale": true,
    "cached": true
  },
  "button:play/pause": {
    "element_key": "button:play/pause",
```

```
"element_type": "button",
"label": "Play/Pause",
"bbox": [
    1226.0,
    571.0,
    1258.0,
    619.0
],
"confidence": 0.5315,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 14,
"last_seen_index": 14,
"first_seen_at": "2026-05-03T08:38:34.371135+00:00",
"last_seen_at": "2026-05-03T08:38:34.371135+00:00",
"seen_count": 1,
"miss_count": 5,
"stale": true,
"cached": true
},
"pane:quick settings": {
    "element_key": "pane:quick settings",
    "element_type": "pane",
    "label": "Quick Settings",
    "bbox": [
        1066.0,
        627.0,
        1442.0,
```

```
    987.0
  ],
  "confidence": 0.5609,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 14,
  "last_seen_index": 14,
  "first_seen_at": "2026-05-03T08:38:34.371135+00:00",
  "last_seen_at": "2026-05-03T08:38:34.371135+00:00",
  "seen_count": 1,
  "miss_count": 5,
  "stale": true,
  "cached": true
},
"toggle:wi-fi": {
  "element_key": "toggle:wi-fi",
  "element_type": "toggle",
  "label": "Wi-Fi",
  "bbox": [
    1106.0,
    683.0,
    1218.0,
    795.0
  ],
  "confidence": 0.5315,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
```

```
"first_seen_index": 14,  
"last_seen_index": 14,  
"first_seen_at": "2026-05-03T08:38:34.371135+00:00",  
"last_seen_at": "2026-05-03T08:38:34.371135+00:00",  
"seen_count": 1,  
"miss_count": 5,  
"stale": true,  
"cached": true  
},  
"toggle:bluetooth": {  
  "element_key": "toggle:bluetooth",  
  "element_type": "toggle",  
  "label": "Bluetooth",  
  "bbox": [  
    1194.0,  
    683.0,  
    1314.0,  
    795.0  
  ],  
  "confidence": 0.5315,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 14,  
  "last_seen_index": 14,  
  "first_seen_at": "2026-05-03T08:38:34.371135+00:00",  
  "last_seen_at": "2026-05-03T08:38:34.371135+00:00",  
  "seen_count": 1,  
  "miss_count": 5,
```

```
    "stale": true,
    "cached": true
  },
  "toggle:airplane mode": {
    "element_key": "toggle:airplane mode",
    "element_type": "toggle",
    "label": "Airplane mode",
    "bbox": [
      1290.0,
      683.0,
      1402.0,
      795.0
    ],
    "confidence": 0.5315,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 14,
    "last_seen_index": 14,
    "first_seen_at": "2026-05-03T08:38:34.371135+00:00",
    "last_seen_at": "2026-05-03T08:38:34.371135+00:00",
    "seen_count": 1,
    "miss_count": 5,
    "stale": true,
    "cached": true
  },
  "toggle:energy saver": {
    "element_key": "toggle:energy saver",
    "element_type": "toggle",
```

```
"label": "Energy saver",
"bbox": [
  1106.0,
  779.0,
  1218.0,
  899.0
],
"confidence": 0.5315,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 14,
"last_seen_index": 14,
"first_seen_at": "2026-05-03T08:38:34.371135+00:00",
"last_seen_at": "2026-05-03T08:38:34.371135+00:00",
"seen_count": 1,
"miss_count": 5,
"stale": true,
"cached": true
},
"toggle:night light": {
  "element_key": "toggle:night light",
  "element_type": "toggle",
  "label": "Night light",
  "bbox": [
    1194.0,
    787.0,
    1314.0,
    899.0
  ]
}
```

```
],  
  "confidence": 0.5315,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 14,  
  "last_seen_index": 14,  
  "first_seen_at": "2026-05-03T08:38:34.371135+00:00",  
  "last_seen_at": "2026-05-03T08:38:34.371135+00:00",  
  "seen_count": 1,  
  "miss_count": 5,  
  "stale": true,  
  "cached": true  
},  
"toggle:accessibility": {  
  "element_key": "toggle:accessibility",  
  "element_type": "toggle",  
  "label": "Accessibility",  
  "bbox": [  
    1290.0,  
    779.0,  
    1402.0,  
    899.0  
  ],  
  "confidence": 0.5315,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 14,
```



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"last_seen_index": 14,  
"first_seen_at": "2026-05-03T08:38:34.371135+00:00",  
"last_seen_at": "2026-05-03T08:38:34.371135+00:00",  
"seen_count": 1,  
"miss_count": 5,  
"stale": true,  
"cached": true  
},  
"slider:brightness": {  
  "element_key": "slider:brightness",  
  "element_type": "slider",  
  "label": "Brightness",  
  "bbox": [  
    1122.0,  
    1005.0,  
    1381.0,  
    1006.0  
  ],  
  "confidence": 0.5315,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 14,  
  "last_seen_index": 14,  
  "first_seen_at": "2026-05-03T08:38:34.371135+00:00",  
  "last_seen_at": "2026-05-03T08:38:34.371135+00:00",  
  "seen_count": 1,  
  "miss_count": 5,  
  "stale": true,
```

```
    "cached": true
  },
  "slider:volume": {
    "element_key": "slider:volume",
    "element_type": "slider",
    "label": "Volume",
    "bbox": [
      1122.0,
      1005.0,
      1381.0,
      1006.0
    ],
    "confidence": 0.5315,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 14,
    "last_seen_index": 14,
    "first_seen_at": "2026-05-03T08:38:34.371135+00:00",
    "last_seen_at": "2026-05-03T08:38:34.371135+00:00",
    "seen_count": 1,
    "miss_count": 5,
    "stale": true,
    "cached": true
  },
  "label:91%": {
    "element_key": "label:91%",
    "element_type": "label",
    "label": "91%",
```

```
"bbox": [
    1122.0,
    1005.0,
    1174.0,
    1006.0
],
"confidence": 0.5315,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 14,
"last_seen_index": 14,
"first_seen_at": "2026-05-03T08:38:34.371135+00:00",
"last_seen_at": "2026-05-03T08:38:34.371135+00:00",
"seen_count": 1,
"miss_count": 5,
"stale": true,
"cached": true
},
"toolbar:taskbar": {
    "element_key": "toolbar:taskbar",
    "element_type": "toolbar",
    "label": "Taskbar",
    "bbox": [
        138.0,
        939.0,
        1795.0,
        1006.0
    ],
```

```
"confidence": 0.5609,  
"source": "gemini_vlm",  
"first_seen_frame": null,  
"last_seen_frame": null,  
"first_seen_index": 14,  
"last_seen_index": 14,  
"first_seen_at": "2026-05-03T08:38:34.371135+00:00",  
"last_seen_at": "2026-05-03T08:38:34.371135+00:00",  
"seen_count": 1,  
"miss_count": 5,  
"stale": true,  
"cached": true  
},  
"button:edge": {  
  "element_key": "button:edge",  
  "element_type": "button",  
  "label": "Edge",  
  "bbox": [  
    770.0,  
    947.0,  
    810.0,  
    987.0  
  ],  
  "confidence": 0.5315,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 14,  
  "last_seen_index": 14,
```

```
"first_seen_at": "2026-05-03T08:38:34.371135+00:00",
"last_seen_at": "2026-05-03T08:38:34.371135+00:00",
"seen_count": 1,
"miss_count": 5,
"stale": true,
"cached": true
},
"label:2:07 pm 5/3/2026": {
  "element_key": "label:2:07 pm 5/3/2026",
  "element_type": "label",
  "label": "2:07 PM 5/3/2026",
  "bbox": [
    1346.0,
    947.0,
    1410.0,
    979.0
  ],
  "confidence": 0.5315,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 14,
  "last_seen_index": 14,
  "first_seen_at": "2026-05-03T08:38:34.371135+00:00",
  "last_seen_at": "2026-05-03T08:38:34.371135+00:00",
  "seen_count": 1,
  "miss_count": 5,
  "stale": true,
  "cached": true
}
```

```
},
"button:rainy days ahead 31c": {
  "element_key": "button:rainy days ahead 31c",
  "element_type": "button",
  "label": "Rainy days ahead 31C",
  "bbox": [
    154.0,
    947.0,
    306.0,
    1003.0
  ],
  "confidence": 0.5315,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 14,
  "last_seen_index": 14,
  "first_seen_at": "2026-05-03T08:38:34.371135+00:00",
  "last_seen_at": "2026-05-03T08:38:34.371135+00:00",
  "seen_count": 1,
  "miss_count": 5,
  "stale": true,
  "cached": true
},
"link:pull requests": {
  "element_key": "link:pull requests",
  "element_type": "link",
  "label": "Pull requests",
  "bbox": [
```

```
        1122.0,
        251.0,
        1186.0,
        275.0
    ],
    "confidence": 0.5019,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 14,
    "last_seen_index": 14,
    "first_seen_at": "2026-05-03T08:38:34.371135+00:00",
    "last_seen_at": "2026-05-03T08:38:34.371135+00:00",
    "seen_count": 1,
    "miss_count": 5,
    "stale": true,
    "cached": true
},
"link:issues": {
    "element_key": "link:issues",
    "element_type": "link",
    "label": "Issues",
    "bbox": [
        1170.0,
        243.0,
        1218.0,
        275.0
    ],
    "confidence": 0.5019,
```

```
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 14,
"last_seen_index": 14,
"first_seen_at": "2026-05-03T08:38:34.371135+00:00",
"last_seen_at": "2026-05-03T08:38:34.371135+00:00",
"seen_count": 1,
"miss_count": 5,
"stale": true,
"cached": true
},
"link:marketplace": {
  "element_key": "link:marketplace",
  "element_type": "link",
  "label": "Marketplace",
  "bbox": [
    1202.0,
    251.0,
    1306.0,
    283.0
  ],
  "confidence": 0.5019,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 14,
  "last_seen_index": 14,
  "first_seen_at": "2026-05-03T08:38:34.371135+00:00",
```



```
    "last_seen_at": "2026-05-03T08:38:34.371135+00:00",
    "seen_count": 1,
    "miss_count": 5,
    "stale": true,
    "cached": true
  },
  "link:explore": {
    "element_key": "link:explore",
    "element_type": "link",
    "label": "Explore",
    "bbox": [
      1290.0,
      227.0,
      1354.0,
      275.0
    ],
    "confidence": 0.5019,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 14,
    "last_seen_index": 14,
    "first_seen_at": "2026-05-03T08:38:34.371135+00:00",
    "last_seen_at": "2026-05-03T08:38:34.371135+00:00",
    "seen_count": 1,
    "miss_count": 5,
    "stale": true,
    "cached": true
  },
}
```

```
"button:notifications": {
  "element_key": "button:notifications",
  "element_type": "button",
  "label": "Notifications",
  "bbox": [
    1338.0,
    251.0,
    1378.0,
    275.0
  ],
  "confidence": 0.5019,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 14,
  "last_seen_index": 14,
  "first_seen_at": "2026-05-03T08:38:34.371135+00:00",
  "last_seen_at": "2026-05-03T08:38:34.371135+00:00",
  "seen_count": 1,
  "miss_count": 5,
  "stale": true,
  "cached": true
},
"button:create new...": {
  "element_key": "button:create new...",
  "element_type": "button",
  "label": "Create new...",
  "bbox": [
    1378.0,
```

```
    251.0,  
    1410.0,  
    275.0  
  ],  
  "confidence": 0.5019,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 14,  
  "last_seen_index": 14,  
  "first_seen_at": "2026-05-03T08:38:34.371135+00:00",  
  "last_seen_at": "2026-05-03T08:38:34.371135+00:00",  
  "seen_count": 1,  
  "miss_count": 5,  
  "stale": true,  
  "cached": true  
},  
"icon:user profile": {  
  "element_key": "icon:user profile",  
  "element_type": "icon",  
  "label": "User Profile",  
  "bbox": [  
    1706.0,  
    219.0,  
    1754.0,  
    283.0  
  ],  
  "confidence": 0.95,  
  "source": "gemini_vlm",
```

```
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 19,
    "last_seen_index": 19,
    "first_seen_at": "2026-05-03T08:42:22.694106+00:00",
    "last_seen_at": "2026-05-03T08:42:22.694106+00:00",
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false
  }
},
"current_elements": [
  {
    "id": "tab_whatsapp",
    "type": "tab",
    "label": "WhatsApp",
    "description": "Browser tab for WhatsApp",
    "state": "inactive",
    "dx": 120,
    "dy": 56,
    "frame_dx": 259,
    "frame_dy": 114,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
      139,
      82,
      379,
```

```
146
],
"session_memory_key": "tab:whatsapp",
"cached": false
},
{
  "id": "tab_google_search",
  "type": "tab",
  "label": "Google Search",
  "description": "Browser tab for Google Search",
  "state": "inactive",
  "dx": 308,
  "dy": 56,
  "frame_dx": 447,
  "frame_dy": 114,
  "confidence": 0.95,
  "source": "gemini_vlm",
  "frame_bbox": [
    323,
    82,
    571,
    146
  ],
  "session_memory_key": "tab:google search",
  "cached": false
},
{
  "id": "tab_frontend",
  "type": "tab",
```

```
    "label": "frontend",
    "description": "Browser tab for frontend development",
    "state": "inactive",
    "dx": 500,
    "dy": 56,
    "frame_dx": 639,
    "frame_dy": 114,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        515,
        82,
        763,
        146
    ],
    "session_memory_key": "tab:frontend",
    "cached": false
},
{
    "id": "tab_github_repos",
    "type": "tab",
    "label": "Your Repositories",
    "description": "Browser tab for GitHub repositories",
    "state": "active",
    "dx": 696,
    "dy": 56,
    "frame_dx": 835,
    "frame_dy": 114,
    "confidence": 0.95,
```

```
"source": "gemini_vlm",
"frame_bbox": [
    707,
    82,
    963,
    146
],
"session_memory_key": "tab:your repositories",
"cached": false
},
{
    "id": "address_bar",
    "type": "input",
    "label": "Address Bar",
    "description": "Browser address bar showing localhost:5173",
    "state": "visible",
    "dx": 588,
    "dy": 116,
    "frame_dx": 727,
    "frame_dy": 174,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
        139,
        146,
        1315,
        202
    ],
    "session_memory_key": "input:address bar",
```

```
    "cached": false
  },
  {
    "id": "sidebar_home",
    "type": "link",
    "label": "Home",
    "description": "Navigation link to Home in ScreenPilot sidebar",
    "state": "active",
    "dx": 144,
    "dy": 404,
    "frame_dx": 283,
    "frame_dy": 462,
    "confidence": 0.92,
    "source": "gemini_vlm",
    "frame_bbox": [
      139,
      418,
      427,
      506
    ],
    "session_memory_key": "link:home",
    "cached": false
  },
  {
    "id": "sidebar_chats",
    "type": "link",
    "label": "Chats",
    "description": "Navigation link to Chats in ScreenPilot sidebar",
    "state": "inactive",
```



```
"dx": 144,  
  "dy": 472,  
  "frame_dx": 283,  
  "frame_dy": 530,  
  "confidence": 0.92,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    139,  
    490,  
    427,  
    570  
  ],  
  "session_memory_key": "link:chats",  
  "cached": false  
},  
{  
  "id": "ai_assistant_dropdown",  
  "type": "dropdown",  
  "label": "AI Assistant",  
  "description": "Dropdown menu to select AI assistant",  
  "state": "visible",  
  "dx": 340,  
  "dy": 200,  
  "frame_dx": 479,  
  "frame_dy": 258,  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    403,
```

```
        226,  
        555,  
        290  
    ],  
    "session_memory_key": "dropdown:ai assistant",  
    "cached": false  
},  
{  
    "id": "user_message_bubble",  
    "type": "pane",  
    "label": "User Message",  
    "description": "Chat bubble containing user instructions",  
    "state": "visible",  
    "dx": 920,  
    "dy": 364,  
    "frame_dx": 1059,  
    "frame_dy": 422,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        651,  
        322,  
        1467,  
        522  
    ],  
    "session_memory_key": "pane:user message",  
    "cached": false  
},  
{
```

```
    "id": "status_message",
    "type": "label",
    "label": "Connecting to agent...",
    "description": "Status message indicating agent connection and screen
capture",
    "state": "visible",
    "dx": 620,
    "dy": 516,
    "frame_dx": 759,
    "frame_dy": 574,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        355,
        506,
        1163,
        642
    ],
    "session_memory_key": "label:connecting to agent...",
    "cached": false
},
{
    "id": "loading_indicator",
    "type": "icon",
    "label": "Loading",
    "description": "Three dots loading animation",
    "state": "visible",
    "dx": 356,
    "dy": 624,
```

```
"frame_dx": 495,  
"frame_dy": 682,  
"confidence": 0.85,  
"source": "gemini_vlm",  
"frame_bbox": [  
    459,  
    666,  
    531,  
    698  
],  
"session_memory_key": "icon:loading",  
"cached": false  
},  
{  
    "id": "chat_input_area",  
    "type": "input",  
    "label": "Chat Input",  
    "description": "Input field for chat messages",  
    "state": "disabled",  
    "dx": 776,  
    "dy": 948,  
    "frame_dx": 915,  
    "frame_dy": 1006,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        437,  
        1005,  
        1394,
```

```
1006
],
"session_memory_key": "input:chat input",
"cached": false
},
{
  "id": "taskbar_start",
  "type": "button",
  "label": "Start",
  "description": "Windows Start button",
  "state": "visible",
  "dx": 332,
  "dy": 908,
  "frame_dx": 471,
  "frame_dy": 966,
  "confidence": 0.98,
  "source": "gemini_vlm",
  "frame_bbox": [
    443,
    938,
    499,
    994
  ],
  "session_memory_key": "button:start",
  "cached": false
},
{
  "id": "taskbar_search",
  "type": "input",
```

```
"label": "Search",
"description": "Windows Search bar",
"state": "visible",
"dx": 412,
"dy": 908,
"frame_dx": 551,
"frame_dy": 966,
"confidence": 0.98,
"source": "gemini_vlm",
"frame_bbox": [
    475,
    938,
    627,
    994
],
"session_memory_key": "input:search",
"cached": false
},
{
    "id": "taskbar_chrome",
    "type": "button",
    "label": "Chrome",
    "description": "Google Chrome taskbar icon",
    "state": "active",
    "dx": 760,
    "dy": 904,
    "frame_dx": 899,
    "frame_dy": 962,
    "confidence": 0.98,
```

```
"source": "gemini_vlm",
"frame_bbox": [
    875,
    938,
    923,
    986
],
"session_memory_key": "button:chrome",
"cached": false
},
{
    "id": "taskbar_vscode",
    "type": "button",
    "label": "VS Code",
    "description": "Visual Studio Code taskbar icon",
    "state": "visible",
    "dx": 872,
    "dy": 908,
    "frame_dx": 1011,
    "frame_dy": 966,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
        987,
        938,
        1035,
        994
    ],
    "session_memory_key": "button:vs code",
```

```
"cached": false
},
{
  "id": "taskbar_spotify",
  "type": "button",
  "label": "Spotify",
  "description": "Spotify taskbar icon",
  "state": "visible",
  "dx": 904,
  "dy": 904,
  "frame_dx": 1043,
  "frame_dy": 962,
  "confidence": 0.98,
  "source": "gemini_vlm",
  "frame_bbox": [
    1019,
    938,
    1067,
    986
  ],
  "session_memory_key": "button:spotify",
  "cached": false
},
{
  "id": "system_tray_clock",
  "type": "label",
  "label": "Clock",
  "description": "System tray clock and date",
  "state": "visible",
```



```
"dx": 1236,  
  "dy": 900,  
  "frame_dx": 1375,  
  "frame_dy": 958,  
  "confidence": 0.98,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    1339,  
    938,  
    1411,  
    978  
  ],  
  "session_memory_key": "label:clock",  
  "cached": false  
},  
{  
  "id": "system_tray_wifi",  
  "type": "icon",  
  "label": "Wi-Fi",  
  "description": "Wi-Fi status icon",  
  "state": "visible",  
  "dx": 1172,  
  "dy": 908,  
  "frame_dx": 1311,  
  "frame_dy": 966,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    1291,
```

```
        946,  
        1331,  
        986  
    ],  
    "session_memory_key": "icon:wi-fi",  
    "cached": false  
},  
{  
    "id": "system_tray_battery",  
    "type": "icon",  
    "label": "Battery",  
    "description": "Battery status icon",  
    "state": "visible",  
    "dx": 1196,  
    "dy": 904,  
    "frame_dx": 1335,  
    "frame_dy": 962,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        1315,  
        946,  
        1355,  
        978  
    ],  
    "session_memory_key": "icon:battery",  
    "cached": false  
},  
{
```

```
"id": "sidebar_recent_1",
"type": "link",
"label": "Recent Task 1",
"description": "Recent task: Click on the 'Your Repositories'...",
"state": "visible",
"dx": 128,
"dy": 608,
"frame_dx": 267,
"frame_dy": 666,
"confidence": 0.85,
"source": "gemini_vlm",
"frame_bbox": [
  139,
  626,
  395,
  706
],
"session_memory_key": "link:recent task 1",
"cached": false
},
{
  "id": "sidebar_recent_2",
  "type": "link",
  "label": "Recent Task 2",
  "description": "Recent task: Click on new tab and find the yo...",
  "state": "visible",
  "dx": 140,
  "dy": 748,
  "frame_dx": 279,
```

```
"frame_dy": 806,  
"confidence": 0.85,  
"source": "gemini_vlm",  
"frame_bbox": [  
    139,  
    770,  
    419,  
    842  
],  
"session_memory_key": "link:recent task 2",  
"cached": false  
},  
{  
    "id": "user_profile",  
    "type": "button",  
    "label": "User Profile",  
    "description": "User profile button in sidebar",  
    "state": "visible",  
    "dx": 142,  
    "dy": 948,  
    "frame_dx": 281,  
    "frame_dy": 1006,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        171,  
        1005,  
        391,  
        1006
```

```
    ],
    "session_memory_key": "button:user profile",
    "cached": false
  },
  {
    "id": "send_button",
    "type": "button",
    "label": "Send",
    "description": "Send message button",
    "state": "disabled",
    "dx": 1226,
    "dy": 948,
    "frame_dx": 1365,
    "frame_dy": 1006,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
      1342,
      1005,
      1387,
      1006
    ],
    "session_memory_key": "button:send",
    "cached": false
  }
],
"resolved_elements": [
  {
    "id": "tab_whatsapp",
```

```
"type": "tab",
"label": "WhatsApp",
"description": "Browser tab for WhatsApp",
"state": "inactive",
"dx": 120,
"dy": 56,
"frame_dx": 259,
"frame_dy": 114,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    139,
    82,
    379,
    146
],
"session_memory_key": "tab:whatsapp",
"cached": false
},
{
    "id": "tab_google_search",
    "type": "tab",
    "label": "Google Search",
    "description": "Browser tab for Google Search",
    "state": "inactive",
    "dx": 308,
    "dy": 56,
    "frame_dx": 447,
    "frame_dy": 114,
```

```
"confidence": 0.95,  
"source": "gemini_vlm",  
"frame_bbox": [  
    323,  
    82,  
    571,  
    146  
],  
"session_memory_key": "tab:google search",  
"cached": false  
},  
{  
    "id": "tab_frontend",  
    "type": "tab",  
    "label": "frontend",  
    "description": "Browser tab for frontend development",  
    "state": "inactive",  
    "dx": 500,  
    "dy": 56,  
    "frame_dx": 639,  
    "frame_dy": 114,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        515,  
        82,  
        763,  
        146  
    ],
```

```
"session_memory_key": "tab:frontend",
"cached": false
},
{
  "id": "tab_github_repos",
  "type": "tab",
  "label": "Your Repositories",
  "description": "Browser tab for GitHub repositories",
  "state": "active",
  "dx": 696,
  "dy": 56,
  "frame_dx": 835,
  "frame_dy": 114,
  "confidence": 0.95,
  "source": "gemini_vlm",
  "frame_bbox": [
    707,
    82,
    963,
    146
  ],
  "session_memory_key": "tab:your repositories",
  "cached": false
},
{
  "id": "address_bar",
  "type": "input",
  "label": "Address Bar",
  "description": "Browser address bar showing localhost:5173",
```



```
"state": "visible",
"dx": 588,
"dy": 116,
"frame_dx": 727,
"frame_dy": 174,
"confidence": 0.98,
"source": "gemini_vlm",
"frame_bbox": [
    139,
    146,
    1315,
    202
],
"session_memory_key": "input:address bar",
"cached": false
},
{
    "id": "sidebar_home",
    "type": "link",
    "label": "Home",
    "description": "Navigation link to Home in ScreenPilot sidebar",
    "state": "active",
    "dx": 144,
    "dy": 404,
    "frame_dx": 283,
    "frame_dy": 462,
    "confidence": 0.92,
    "source": "gemini_vlm",
    "frame_bbox": [
```

```
    139,  
    418,  
    427,  
    506  
  ],  
  "session_memory_key": "link:home",  
  "cached": false  
},  
{  
  "id": "sidebar_chats",  
  "type": "link",  
  "label": "Chats",  
  "description": "Navigation link to Chats in ScreenPilot sidebar",  
  "state": "inactive",  
  "dx": 144,  
  "dy": 472,  
  "frame_dx": 283,  
  "frame_dy": 530,  
  "confidence": 0.92,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    139,  
    490,  
    427,  
    570  
  ],  
  "session_memory_key": "link:chats",  
  "cached": false  
},
```

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{
  "id": "ai_assistant_dropdown",
  "type": "dropdown",
  "label": "AI Assistant",
  "description": "Dropdown menu to select AI assistant",
  "state": "visible",
  "dx": 340,
  "dy": 200,
  "frame_dx": 479,
  "frame_dy": 258,
  "confidence": 0.9,
  "source": "gemini_vlm",
  "frame_bbox": [
    403,
    226,
    555,
    290
  ],
  "session_memory_key": "dropdown:ai assistant",
  "cached": false
},
{
  "id": "user_message_bubble",
  "type": "pane",
  "label": "User Message",
  "description": "Chat bubble containing user instructions",
  "state": "visible",
  "dx": 920,
  "dy": 364,
```

```
"frame_dx": 1059,
"frame_dy": 422,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    651,
    322,
    1467,
    522
],
"session_memory_key": "pane:user message",
"cached": false
},
{
    "id": "status_message",
    "type": "label",
    "label": "Connecting to agent...",
    "description": "Status message indicating agent connection and screen
capture",
    "state": "visible",
    "dx": 620,
    "dy": 516,
    "frame_dx": 759,
    "frame_dy": 574,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        355,
        506,
```

```
        1163,  
        642  
    ],  
    "session_memory_key": "label:connecting to agent...",  
    "cached": false  
},  
{  
    "id": "loading_indicator",  
    "type": "icon",  
    "label": "Loading",  
    "description": "Three dots loading animation",  
    "state": "visible",  
    "dx": 356,  
    "dy": 624,  
    "frame_dx": 495,  
    "frame_dy": 682,  
    "confidence": 0.85,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        459,  
        666,  
        531,  
        698  
    ],  
    "session_memory_key": "icon:loading",  
    "cached": false  
},  
{  
    "id": "chat_input_area",
```

```
"type": "input",
"label": "Chat Input",
"description": "Input field for chat messages",
"state": "disabled",
"dx": 776,
"dy": 948,
"frame_dx": 915,
"frame_dy": 1006,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    437,
    1005,
    1394,
    1006
],
"session_memory_key": "input:chat input",
"cached": false
},
{
    "id": "taskbar_start",
    "type": "button",
    "label": "Start",
    "description": "Windows Start button",
    "state": "visible",
    "dx": 332,
    "dy": 908,
    "frame_dx": 471,
    "frame_dy": 966,
```

```
"confidence": 0.98,
"source": "gemini_vlm",
"frame_bbox": [
    443,
    938,
    499,
    994
],
"session_memory_key": "button:start",
"cached": false
},
{
    "id": "taskbar_search",
    "type": "input",
    "label": "Search",
    "description": "Windows Search bar",
    "state": "visible",
    "dx": 412,
    "dy": 908,
    "frame_dx": 551,
    "frame_dy": 966,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
        475,
        938,
        627,
        994
    ],
    ],
```

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"session_memory_key": "input:search",
"cached": false
},
{
  "id": "taskbar_chrome",
  "type": "button",
  "label": "Chrome",
  "description": "Google Chrome taskbar icon",
  "state": "active",
  "dx": 760,
  "dy": 904,
  "frame_dx": 899,
  "frame_dy": 962,
  "confidence": 0.98,
  "source": "gemini_vlm",
  "frame_bbox": [
    875,
    938,
    923,
    986
  ],
  "session_memory_key": "button:chrome",
  "cached": false
},
{
  "id": "taskbar_vscode",
  "type": "button",
  "label": "VS Code",
  "description": "Visual Studio Code taskbar icon",
```



```
"state": "visible",
"dx": 872,
"dy": 908,
"frame_dx": 1011,
"frame_dy": 966,
"confidence": 0.98,
"source": "gemini_vlm",
"frame_bbox": [
    987,
    938,
    1035,
    994
],
"session_memory_key": "button:vs code",
"cached": false
},
{
    "id": "taskbar_spotify",
    "type": "button",
    "label": "Spotify",
    "description": "Spotify taskbar icon",
    "state": "visible",
    "dx": 904,
    "dy": 904,
    "frame_dx": 1043,
    "frame_dy": 962,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
```

```
        1019,  
        938,  
        1067,  
        986  
    ],  
    "session_memory_key": "button:spotify",  
    "cached": false  
},  
{  
    "id": "system_tray_clock",  
    "type": "label",  
    "label": "Clock",  
    "description": "System tray clock and date",  
    "state": "visible",  
    "dx": 1236,  
    "dy": 900,  
    "frame_dx": 1375,  
    "frame_dy": 958,  
    "confidence": 0.98,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        1339,  
        938,  
        1411,  
        978  
    ],  
    "session_memory_key": "label:clock",  
    "cached": false  
},
```

```
{
  "id": "system_tray_wifi",
  "type": "icon",
  "label": "Wi-Fi",
  "description": "Wi-Fi status icon",
  "state": "visible",
  "dx": 1172,
  "dy": 908,
  "frame_dx": 1311,
  "frame_dy": 966,
  "confidence": 0.95,
  "source": "gemini_vlm",
  "frame_bbox": [
    1291,
    946,
    1331,
    986
  ],
  "session_memory_key": "icon:wi-fi",
  "cached": false
},
{
  "id": "system_tray_battery",
  "type": "icon",
  "label": "Battery",
  "description": "Battery status icon",
  "state": "visible",
  "dx": 1196,
  "dy": 904,
```

```
"frame_dx": 1335,  
"frame_dy": 962,  
"confidence": 0.95,  
"source": "gemini_vlm",  
"frame_bbox": [  
    1315,  
    946,  
    1355,  
    978  
],  
"session_memory_key": "icon:battery",  
"cached": false  
},  
{  
    "id": "sidebar_recent_1",  
    "type": "link",  
    "label": "Recent Task 1",  
    "description": "Recent task: Click on the 'Your Repositories'...",  
    "state": "visible",  
    "dx": 128,  
    "dy": 608,  
    "frame_dx": 267,  
    "frame_dy": 666,  
    "confidence": 0.85,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        139,  
        626,  
        395,
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```
706
],
"session_memory_key": "link:recent task 1",
"cached": false
},
{
  "id": "sidebar_recent_2",
  "type": "link",
  "label": "Recent Task 2",
  "description": "Recent task: Click on new tab and find the yo...",
  "state": "visible",
  "dx": 140,
  "dy": 748,
  "frame_dx": 279,
  "frame_dy": 806,
  "confidence": 0.85,
  "source": "gemini_vlm",
  "frame_bbox": [
    139,
    770,
    419,
    842
  ],
  "session_memory_key": "link:recent task 2",
  "cached": false
},
{
  "id": "user_profile",
  "type": "button",
```

```
"label": "User Profile",
"description": "User profile button in sidebar",
"state": "visible",
"dx": 142,
"dy": 948,
"frame_dx": 281,
"frame_dy": 1006,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    171,
    1005,
    391,
    1006
],
"session_memory_key": "button:user profile",
"cached": false
},
{
    "id": "send_button",
    "type": "button",
    "label": "Send",
    "description": "Send message button",
    "state": "disabled",
    "dx": 1226,
    "dy": 948,
    "frame_dx": 1365,
    "frame_dy": 1006,
    "confidence": 0.9,
```

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    "source": "gemini_vlm",
    "frame_bbox": [
        1342,
        1005,
        1387,
        1006
    ],
    "session_memory_key": "button:send",
    "cached": false
}
],
"screen_data": {
    "image": "E:\\sliit-
project\\ScreenAwareTaskAgent\\vision\\data\\sessions\\shot_1777796788_f50769\\pr
ocessing\\proc_1777796788_75e372\\preprocessed_frames\\frame_1777796788_7b3eb6.jp
g",
    "image_size": {
        "width": 1920,
        "height": 1080
    },
    "screen_bbox": [
        139,
        58,
        1795,
        1006
    ],
    "screen_size": {
        "width": 1656,
        "height": 948
    },
    },
```

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"coordinate_system": "pixel",
"elements": [
  {
    "id": "tab_whatsapp",
    "type": "tab",
    "label": "WhatsApp",
    "description": "Browser tab for WhatsApp",
    "state": "inactive",
    "dx": 120,
    "dy": 56,
    "frame_dx": 259,
    "frame_dy": 114,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
      139,
      82,
      379,
      146
    ],
    "session_memory_key": "tab:whatsapp",
    "cached": false
  },
  {
    "id": "tab_google_search",
    "type": "tab",
    "label": "Google Search",
    "description": "Browser tab for Google Search",
    "state": "inactive",
```



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"dx": 308,  
  "dy": 56,  
  "frame_dx": 447,  
  "frame_dy": 114,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    323,  
    82,  
    571,  
    146  
  ],  
  "session_memory_key": "tab:google search",  
  "cached": false  
},  
{  
  "id": "tab_frontend",  
  "type": "tab",  
  "label": "frontend",  
  "description": "Browser tab for frontend development",  
  "state": "inactive",  
  "dx": 500,  
  "dy": 56,  
  "frame_dx": 639,  
  "frame_dy": 114,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    515,
```

```
      82,  
      763,  
      146  
    ],  
    "session_memory_key": "tab:frontend",  
    "cached": false  
  },  
  {  
    "id": "tab_github_repos",  
    "type": "tab",  
    "label": "Your Repositories",  
    "description": "Browser tab for GitHub repositories",  
    "state": "active",  
    "dx": 696,  
    "dy": 56,  
    "frame_dx": 835,  
    "frame_dy": 114,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
      707,  
      82,  
      963,  
      146  
    ],  
    "session_memory_key": "tab:your repositories",  
    "cached": false  
  },  
  {
```

```
"id": "address_bar",
"type": "input",
"label": "Address Bar",
"description": "Browser address bar showing localhost:5173",
"state": "visible",
"dx": 588,
"dy": 116,
"frame_dx": 727,
"frame_dy": 174,
"confidence": 0.98,
"source": "gemini_vlm",
"frame_bbox": [
  139,
  146,
  1315,
  202
],
"session_memory_key": "input:address bar",
"cached": false
},
{
  "id": "sidebar_home",
  "type": "link",
  "label": "Home",
  "description": "Navigation link to Home in ScreenPilot sidebar",
  "state": "active",
  "dx": 144,
  "dy": 404,
  "frame_dx": 283,
```

```
"frame_dy": 462,
"confidence": 0.92,
"source": "gemini_vlm",
"frame_bbox": [
  139,
  418,
  427,
  506
],
"session_memory_key": "link:home",
"cached": false
},
{
  "id": "sidebar_chats",
  "type": "link",
  "label": "Chats",
  "description": "Navigation link to Chats in ScreenPilot sidebar",
  "state": "inactive",
  "dx": 144,
  "dy": 472,
  "frame_dx": 283,
  "frame_dy": 530,
  "confidence": 0.92,
  "source": "gemini_vlm",
  "frame_bbox": [
    139,
    490,
    427,
    570
```

```
    ],
    "session_memory_key": "link:chats",
    "cached": false
  },
  {
    "id": "ai_assistant_dropdown",
    "type": "dropdown",
    "label": "AI Assistant",
    "description": "Dropdown menu to select AI assistant",
    "state": "visible",
    "dx": 340,
    "dy": 200,
    "frame_dx": 479,
    "frame_dy": 258,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
      403,
      226,
      555,
      290
    ],
    "session_memory_key": "dropdown:ai assistant",
    "cached": false
  },
  {
    "id": "user_message_bubble",
    "type": "pane",
    "label": "User Message",
```

```
    "description": "Chat bubble containing user instructions",
    "state": "visible",
    "dx": 920,
    "dy": 364,
    "frame_dx": 1059,
    "frame_dy": 422,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        651,
        322,
        1467,
        522
    ],
    "session_memory_key": "pane:user message",
    "cached": false
},
{
    "id": "status_message",
    "type": "label",
    "label": "Connecting to agent...",
    "description": "Status message indicating agent connection and screen
capture",
    "state": "visible",
    "dx": 620,
    "dy": 516,
    "frame_dx": 759,
    "frame_dy": 574,
    "confidence": 0.9,
```

```
"source": "gemini_vlm",
"frame_bbox": [
  355,
  506,
  1163,
  642
],
"session_memory_key": "label:connecting to agent...",
"cached": false
},
{
  "id": "loading_indicator",
  "type": "icon",
  "label": "Loading",
  "description": "Three dots loading animation",
  "state": "visible",
  "dx": 356,
  "dy": 624,
  "frame_dx": 495,
  "frame_dy": 682,
  "confidence": 0.85,
  "source": "gemini_vlm",
  "frame_bbox": [
    459,
    666,
    531,
    698
  ],
  "session_memory_key": "icon:loading",
```

```
    "cached": false
  },
  {
    "id": "chat_input_area",
    "type": "input",
    "label": "Chat Input",
    "description": "Input field for chat messages",
    "state": "disabled",
    "dx": 776,
    "dy": 948,
    "frame_dx": 915,
    "frame_dy": 1006,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
      437,
      1005,
      1394,
      1006
    ],
    "session_memory_key": "input:chat input",
    "cached": false
  },
  {
    "id": "taskbar_start",
    "type": "button",
    "label": "Start",
    "description": "Windows Start button",
    "state": "visible",
```



```
"dx": 332,  
  "dy": 908,  
  "frame_dx": 471,  
  "frame_dy": 966,  
  "confidence": 0.98,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    443,  
    938,  
    499,  
    994  
  ],  
  "session_memory_key": "button:start",  
  "cached": false  
},  
{  
  "id": "taskbar_search",  
  "type": "input",  
  "label": "Search",  
  "description": "Windows Search bar",  
  "state": "visible",  
  "dx": 412,  
  "dy": 908,  
  "frame_dx": 551,  
  "frame_dy": 966,  
  "confidence": 0.98,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    475,
```

```
        938,  
        627,  
        994  
    ],  
    "session_memory_key": "input:search",  
    "cached": false  
},  
{  
    "id": "taskbar_chrome",  
    "type": "button",  
    "label": "Chrome",  
    "description": "Google Chrome taskbar icon",  
    "state": "active",  
    "dx": 760,  
    "dy": 904,  
    "frame_dx": 899,  
    "frame_dy": 962,  
    "confidence": 0.98,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        875,  
        938,  
        923,  
        986  
    ],  
    "session_memory_key": "button:chrome",  
    "cached": false  
},  
{
```

```
"id": "taskbar_vscode",
"type": "button",
"label": "VS Code",
"description": "Visual Studio Code taskbar icon",
"state": "visible",
"dx": 872,
"dy": 908,
"frame_dx": 1011,
"frame_dy": 966,
"confidence": 0.98,
"source": "gemini_vlm",
"frame_bbox": [
    987,
    938,
    1035,
    994
],
"session_memory_key": "button:vs code",
"cached": false
},
{
    "id": "taskbar_spotify",
    "type": "button",
    "label": "Spotify",
    "description": "Spotify taskbar icon",
    "state": "visible",
    "dx": 904,
    "dy": 904,
    "frame_dx": 1043,
```

```
"frame_dy": 962,  
"confidence": 0.98,  
"source": "gemini_vlm",  
"frame_bbox": [  
    1019,  
    938,  
    1067,  
    986  
],  
"session_memory_key": "button:spotify",  
"cached": false  
},  
{  
    "id": "system_tray_clock",  
    "type": "label",  
    "label": "Clock",  
    "description": "System tray clock and date",  
    "state": "visible",  
    "dx": 1236,  
    "dy": 900,  
    "frame_dx": 1375,  
    "frame_dy": 958,  
    "confidence": 0.98,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        1339,  
        938,  
        1411,  
        978
```

```
    ],  
    "session_memory_key": "label:clock",  
    "cached": false  
  },  
  {  
    "id": "system_tray_wifi",  
    "type": "icon",  
    "label": "Wi-Fi",  
    "description": "Wi-Fi status icon",  
    "state": "visible",  
    "dx": 1172,  
    "dy": 908,  
    "frame_dx": 1311,  
    "frame_dy": 966,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
      1291,  
      946,  
      1331,  
      986  
    ],  
    "session_memory_key": "icon:wi-fi",  
    "cached": false  
  },  
  {  
    "id": "system_tray_battery",  
    "type": "icon",  
    "label": "Battery",
```

```
"description": "Battery status icon",
"state": "visible",
"dx": 1196,
"dy": 904,
"frame_dx": 1335,
"frame_dy": 962,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    1315,
    946,
    1355,
    978
],
"session_memory_key": "icon:battery",
"cached": false
},
{
    "id": "sidebar_recent_1",
    "type": "link",
    "label": "Recent Task 1",
    "description": "Recent task: Click on the 'Your Repositories'...",
    "state": "visible",
    "dx": 128,
    "dy": 608,
    "frame_dx": 267,
    "frame_dy": 666,
    "confidence": 0.85,
    "source": "gemini_vlm",
```

```
"frame_bbox": [
  139,
  626,
  395,
  706
],
"session_memory_key": "link:recent task 1",
"cached": false
},
{
  "id": "sidebar_recent_2",
  "type": "link",
  "label": "Recent Task 2",
  "description": "Recent task: Click on new tab and find the yo...",
  "state": "visible",
  "dx": 140,
  "dy": 748,
  "frame_dx": 279,
  "frame_dy": 806,
  "confidence": 0.85,
  "source": "gemini_vlm",
  "frame_bbox": [
    139,
    770,
    419,
    842
  ],
  "session_memory_key": "link:recent task 2",
  "cached": false
}
```

```
},  
{  
  "id": "user_profile",  
  "type": "button",  
  "label": "User Profile",  
  "description": "User profile button in sidebar",  
  "state": "visible",  
  "dx": 142,  
  "dy": 948,  
  "frame_dx": 281,  
  "frame_dy": 1006,  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    171,  
    1005,  
    391,  
    1006  
  ],  
  "session_memory_key": "button:user profile",  
  "cached": false  
},  
{  
  "id": "send_button",  
  "type": "button",  
  "label": "Send",  
  "description": "Send message button",  
  "state": "disabled",  
  "dx": 1226,
```



```
"dy": 948,
"frame_dx": 1365,
"frame_dy": 1006,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    1342,
    1005,
    1387,
    1006
],
"session_memory_key": "button:send",
"cached": false
}
],
"element_count": 24,
"current_elements": [
    {
        "id": "tab_whatsapp",
        "type": "tab",
        "label": "WhatsApp",
        "description": "Browser tab for WhatsApp",
        "state": "inactive",
        "dx": 120,
        "dy": 56,
        "frame_dx": 259,
        "frame_dy": 114,
        "confidence": 0.95,
        "source": "gemini_vlm",
```

```
"frame_bbox": [  
    139,  
    82,  
    379,  
    146  
],  
"session_memory_key": "tab:whatsapp",  
"cached": false  
},  
{  
    "id": "tab_google_search",  
    "type": "tab",  
    "label": "Google Search",  
    "description": "Browser tab for Google Search",  
    "state": "inactive",  
    "dx": 308,  
    "dy": 56,  
    "frame_dx": 447,  
    "frame_dy": 114,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        323,  
        82,  
        571,  
        146  
    ],  
    "session_memory_key": "tab:google search",  
    "cached": false
```

```
},  
{  
  "id": "tab_frontend",  
  "type": "tab",  
  "label": "frontend",  
  "description": "Browser tab for frontend development",  
  "state": "inactive",  
  "dx": 500,  
  "dy": 56,  
  "frame_dx": 639,  
  "frame_dy": 114,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    515,  
    82,  
    763,  
    146  
  ],  
  "session_memory_key": "tab:frontend",  
  "cached": false  
},  
{  
  "id": "tab_github_repos",  
  "type": "tab",  
  "label": "Your Repositories",  
  "description": "Browser tab for GitHub repositories",  
  "state": "active",  
  "dx": 696,
```

```
"dy": 56,
"frame_dx": 835,
"frame_dy": 114,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    707,
    82,
    963,
    146
],
"session_memory_key": "tab:your repositories",
"cached": false
},
{
    "id": "address_bar",
    "type": "input",
    "label": "Address Bar",
    "description": "Browser address bar showing localhost:5173",
    "state": "visible",
    "dx": 588,
    "dy": 116,
    "frame_dx": 727,
    "frame_dy": 174,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
        139,
        146,
```

```
        1315,  
        202  
    ],  
    "session_memory_key": "input:address bar",  
    "cached": false  
},  
{  
    "id": "sidebar_home",  
    "type": "link",  
    "label": "Home",  
    "description": "Navigation link to Home in ScreenPilot sidebar",  
    "state": "active",  
    "dx": 144,  
    "dy": 404,  
    "frame_dx": 283,  
    "frame_dy": 462,  
    "confidence": 0.92,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        139,  
        418,  
        427,  
        506  
    ],  
    "session_memory_key": "link:home",  
    "cached": false  
},  
{  
    "id": "sidebar_chats",
```

```
"type": "link",
"label": "Chats",
"description": "Navigation link to Chats in ScreenPilot sidebar",
"state": "inactive",
"dx": 144,
"dy": 472,
"frame_dx": 283,
"frame_dy": 530,
"confidence": 0.92,
"source": "gemini_vlm",
"frame_bbox": [
    139,
    490,
    427,
    570
],
"session_memory_key": "link:chats",
"cached": false
},
{
    "id": "ai_assistant_dropdown",
    "type": "dropdown",
    "label": "AI Assistant",
    "description": "Dropdown menu to select AI assistant",
    "state": "visible",
    "dx": 340,
    "dy": 200,
    "frame_dx": 479,
    "frame_dy": 258,
```

```
"confidence": 0.9,  
"source": "gemini_vlm",  
"frame_bbox": [  
    403,  
    226,  
    555,  
    290  
],  
"session_memory_key": "dropdown:ai assistant",  
"cached": false  
},  
{  
    "id": "user_message_bubble",  
    "type": "pane",  
    "label": "User Message",  
    "description": "Chat bubble containing user instructions",  
    "state": "visible",  
    "dx": 920,  
    "dy": 364,  
    "frame_dx": 1059,  
    "frame_dy": 422,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        651,  
        322,  
        1467,  
        522  
    ],  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        403,  
        226,  
        555,  
        290  
    ],  
    "session_memory_key": "dropdown:ai assistant",  
    "cached": false  
}
```

```
    "session_memory_key": "pane:user message",
    "cached": false
  },
  {
    "id": "status_message",
    "type": "label",
    "label": "Connecting to agent...",
    "description": "Status message indicating agent connection and screen
capture",
    "state": "visible",
    "dx": 620,
    "dy": 516,
    "frame_dx": 759,
    "frame_dy": 574,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
      355,
      506,
      1163,
      642
    ],
    "session_memory_key": "label:connecting to agent...",
    "cached": false
  },
  {
    "id": "loading_indicator",
    "type": "icon",
    "label": "Loading",
```



```
"description": "Three dots loading animation",
"state": "visible",
"dx": 356,
"dy": 624,
"frame_dx": 495,
"frame_dy": 682,
"confidence": 0.85,
"source": "gemini_vlm",
"frame_bbox": [
    459,
    666,
    531,
    698
],
"session_memory_key": "icon:loading",
"cached": false
},
{
    "id": "chat_input_area",
    "type": "input",
    "label": "Chat Input",
    "description": "Input field for chat messages",
    "state": "disabled",
    "dx": 776,
    "dy": 948,
    "frame_dx": 915,
    "frame_dy": 1006,
    "confidence": 0.9,
    "source": "gemini_vlm",
```

```
"frame_bbox": [  
    437,  
    1005,  
    1394,  
    1006  
],  
"session_memory_key": "input:chat input",  
"cached": false  
},  
{  
    "id": "taskbar_start",  
    "type": "button",  
    "label": "Start",  
    "description": "Windows Start button",  
    "state": "visible",  
    "dx": 332,  
    "dy": 908,  
    "frame_dx": 471,  
    "frame_dy": 966,  
    "confidence": 0.98,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        443,  
        938,  
        499,  
        994  
    ],  
    "session_memory_key": "button:start",  
    "cached": false
```

```
},  
{  
  "id": "taskbar_search",  
  "type": "input",  
  "label": "Search",  
  "description": "Windows Search bar",  
  "state": "visible",  
  "dx": 412,  
  "dy": 908,  
  "frame_dx": 551,  
  "frame_dy": 966,  
  "confidence": 0.98,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    475,  
    938,  
    627,  
    994  
  ],  
  "session_memory_key": "input:search",  
  "cached": false  
},  
{  
  "id": "taskbar_chrome",  
  "type": "button",  
  "label": "Chrome",  
  "description": "Google Chrome taskbar icon",  
  "state": "active",  
  "dx": 760,
```

```
"dy": 904,
"frame_dx": 899,
"frame_dy": 962,
"confidence": 0.98,
"source": "gemini_vlm",
"frame_bbox": [
    875,
    938,
    923,
    986
],
"session_memory_key": "button:chrome",
"cached": false
},
{
    "id": "taskbar_vscode",
    "type": "button",
    "label": "VS Code",
    "description": "Visual Studio Code taskbar icon",
    "state": "visible",
    "dx": 872,
    "dy": 908,
    "frame_dx": 1011,
    "frame_dy": 966,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
        987,
        938,
```

```
        1035,  
        994  
    ],  
    "session_memory_key": "button:vs code",  
    "cached": false  
},  
{  
    "id": "taskbar_spotify",  
    "type": "button",  
    "label": "Spotify",  
    "description": "Spotify taskbar icon",  
    "state": "visible",  
    "dx": 904,  
    "dy": 904,  
    "frame_dx": 1043,  
    "frame_dy": 962,  
    "confidence": 0.98,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        1019,  
        938,  
        1067,  
        986  
    ],  
    "session_memory_key": "button:spotify",  
    "cached": false  
},  
{  
    "id": "system_tray_clock",
```

```
"type": "label",
"label": "Clock",
"description": "System tray clock and date",
"state": "visible",
"dx": 1236,
"dy": 900,
"frame_dx": 1375,
"frame_dy": 958,
"confidence": 0.98,
"source": "gemini_vlm",
"frame_bbox": [
    1339,
    938,
    1411,
    978
],
"session_memory_key": "label:clock",
"cached": false
},
{
    "id": "system_tray_wifi",
    "type": "icon",
    "label": "Wi-Fi",
    "description": "Wi-Fi status icon",
    "state": "visible",
    "dx": 1172,
    "dy": 908,
    "frame_dx": 1311,
    "frame_dy": 966,
```

```
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    1291,
    946,
    1331,
    986
],
"session_memory_key": "icon:wi-fi",
"cached": false
},
{
    "id": "system_tray_battery",
    "type": "icon",
    "label": "Battery",
    "description": "Battery status icon",
    "state": "visible",
    "dx": 1196,
    "dy": 904,
    "frame_dx": 1335,
    "frame_dy": 962,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        1315,
        946,
        1355,
        978
    ],
    ],
```

```
"session_memory_key": "icon:battery",
"cached": false
},
{
  "id": "sidebar_recent_1",
  "type": "link",
  "label": "Recent Task 1",
  "description": "Recent task: Click on the 'Your Repositories'...",
  "state": "visible",
  "dx": 128,
  "dy": 608,
  "frame_dx": 267,
  "frame_dy": 666,
  "confidence": 0.85,
  "source": "gemini_vlm",
  "frame_bbox": [
    139,
    626,
    395,
    706
  ],
  "session_memory_key": "link:recent task 1",
  "cached": false
},
{
  "id": "sidebar_recent_2",
  "type": "link",
  "label": "Recent Task 2",
  "description": "Recent task: Click on new tab and find the yo..."
```



```
"state": "visible",
"dx": 140,
"dy": 748,
"frame_dx": 279,
"frame_dy": 806,
"confidence": 0.85,
"source": "gemini_vlm",
"frame_bbox": [
    139,
    770,
    419,
    842
],
"session_memory_key": "link:recent task 2",
"cached": false
},
{
    "id": "user_profile",
    "type": "button",
    "label": "User Profile",
    "description": "User profile button in sidebar",
    "state": "visible",
    "dx": 142,
    "dy": 948,
    "frame_dx": 281,
    "frame_dy": 1006,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
```

```
        171,  
        1005,  
        391,  
        1006  
    ],  
    "session_memory_key": "button:user profile",  
    "cached": false  
},  
{  
    "id": "send_button",  
    "type": "button",  
    "label": "Send",  
    "description": "Send message button",  
    "state": "disabled",  
    "dx": 1226,  
    "dy": 948,  
    "frame_dx": 1365,  
    "frame_dy": 1006,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        1342,  
        1005,  
        1387,  
        1006  
    ],  
    "session_memory_key": "button:send",  
    "cached": false  
}
```

```
],  
"resolved_elements": [  
  {  
    "id": "tab_whatsapp",  
    "type": "tab",  
    "label": "WhatsApp",  
    "description": "Browser tab for WhatsApp",  
    "state": "inactive",  
    "dx": 120,  
    "dy": 56,  
    "frame_dx": 259,  
    "frame_dy": 114,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
      139,  
      82,  
      379,  
      146  
    ],  
    "session_memory_key": "tab:whatsapp",  
    "cached": false  
  },  
  {  
    "id": "tab_google_search",  
    "type": "tab",  
    "label": "Google Search",  
    "description": "Browser tab for Google Search",  
    "state": "inactive",
```

```
"dx": 308,  
  "dy": 56,  
  "frame_dx": 447,  
  "frame_dy": 114,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    323,  
    82,  
    571,  
    146  
  ],  
  "session_memory_key": "tab:google search",  
  "cached": false  
},  
{  
  "id": "tab_frontend",  
  "type": "tab",  
  "label": "frontend",  
  "description": "Browser tab for frontend development",  
  "state": "inactive",  
  "dx": 500,  
  "dy": 56,  
  "frame_dx": 639,  
  "frame_dy": 114,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    515,
```

```
      82,  
      763,  
      146  
    ],  
    "session_memory_key": "tab:frontend",  
    "cached": false  
  },  
  {  
    "id": "tab_github_repos",  
    "type": "tab",  
    "label": "Your Repositories",  
    "description": "Browser tab for GitHub repositories",  
    "state": "active",  
    "dx": 696,  
    "dy": 56,  
    "frame_dx": 835,  
    "frame_dy": 114,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
      707,  
      82,  
      963,  
      146  
    ],  
    "session_memory_key": "tab:your repositories",  
    "cached": false  
  },  
  {
```

```
"id": "address_bar",
"type": "input",
"label": "Address Bar",
"description": "Browser address bar showing localhost:5173",
"state": "visible",
"dx": 588,
"dy": 116,
"frame_dx": 727,
"frame_dy": 174,
"confidence": 0.98,
"source": "gemini_vlm",
"frame_bbox": [
  139,
  146,
  1315,
  202
],
"session_memory_key": "input:address bar",
"cached": false
},
{
  "id": "sidebar_home",
  "type": "link",
  "label": "Home",
  "description": "Navigation link to Home in ScreenPilot sidebar",
  "state": "active",
  "dx": 144,
  "dy": 404,
  "frame_dx": 283,
```

```
"frame_dy": 462,
"confidence": 0.92,
"source": "gemini_vlm",
"frame_bbox": [
    139,
    418,
    427,
    506
],
"session_memory_key": "link:home",
"cached": false
},
{
    "id": "sidebar_chats",
    "type": "link",
    "label": "Chats",
    "description": "Navigation link to Chats in ScreenPilot sidebar",
    "state": "inactive",
    "dx": 144,
    "dy": 472,
    "frame_dx": 283,
    "frame_dy": 530,
    "confidence": 0.92,
    "source": "gemini_vlm",
    "frame_bbox": [
        139,
        490,
        427,
        570
    ]
}
```

```
    ],
    "session_memory_key": "link:chats",
    "cached": false
  },
  {
    "id": "ai_assistant_dropdown",
    "type": "dropdown",
    "label": "AI Assistant",
    "description": "Dropdown menu to select AI assistant",
    "state": "visible",
    "dx": 340,
    "dy": 200,
    "frame_dx": 479,
    "frame_dy": 258,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
      403,
      226,
      555,
      290
    ],
    "session_memory_key": "dropdown:ai assistant",
    "cached": false
  },
  {
    "id": "user_message_bubble",
    "type": "pane",
    "label": "User Message",
```



```
    "description": "Chat bubble containing user instructions",
    "state": "visible",
    "dx": 920,
    "dy": 364,
    "frame_dx": 1059,
    "frame_dy": 422,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        651,
        322,
        1467,
        522
    ],
    "session_memory_key": "pane:user message",
    "cached": false
},
{
    "id": "status_message",
    "type": "label",
    "label": "Connecting to agent...",
    "description": "Status message indicating agent connection and screen
capture",
    "state": "visible",
    "dx": 620,
    "dy": 516,
    "frame_dx": 759,
    "frame_dy": 574,
    "confidence": 0.9,
```

```
"source": "gemini_vlm",
"frame_bbox": [
  355,
  506,
  1163,
  642
],
"session_memory_key": "label:connecting to agent...",
"cached": false
},
{
  "id": "loading_indicator",
  "type": "icon",
  "label": "Loading",
  "description": "Three dots loading animation",
  "state": "visible",
  "dx": 356,
  "dy": 624,
  "frame_dx": 495,
  "frame_dy": 682,
  "confidence": 0.85,
  "source": "gemini_vlm",
  "frame_bbox": [
    459,
    666,
    531,
    698
  ],
  "session_memory_key": "icon:loading",
```

```
"cached": false
},
{
  "id": "chat_input_area",
  "type": "input",
  "label": "Chat Input",
  "description": "Input field for chat messages",
  "state": "disabled",
  "dx": 776,
  "dy": 948,
  "frame_dx": 915,
  "frame_dy": 1006,
  "confidence": 0.9,
  "source": "gemini_vlm",
  "frame_bbox": [
    437,
    1005,
    1394,
    1006
  ],
  "session_memory_key": "input:chat input",
  "cached": false
},
{
  "id": "taskbar_start",
  "type": "button",
  "label": "Start",
  "description": "Windows Start button",
  "state": "visible",
```

```
"dx": 332,
"dy": 908,
"frame_dx": 471,
"frame_dy": 966,
"confidence": 0.98,
"source": "gemini_vlm",
"frame_bbox": [
    443,
    938,
    499,
    994
],
"session_memory_key": "button:start",
"cached": false
},
{
    "id": "taskbar_search",
    "type": "input",
    "label": "Search",
    "description": "Windows Search bar",
    "state": "visible",
    "dx": 412,
    "dy": 908,
    "frame_dx": 551,
    "frame_dy": 966,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
        475,
```

```
        938,  
        627,  
        994  
    ],  
    "session_memory_key": "input:search",  
    "cached": false  
},  
{  
    "id": "taskbar_chrome",  
    "type": "button",  
    "label": "Chrome",  
    "description": "Google Chrome taskbar icon",  
    "state": "active",  
    "dx": 760,  
    "dy": 904,  
    "frame_dx": 899,  
    "frame_dy": 962,  
    "confidence": 0.98,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        875,  
        938,  
        923,  
        986  
    ],  
    "session_memory_key": "button:chrome",  
    "cached": false  
},  
{
```

```
"id": "taskbar_vscode",
"type": "button",
"label": "VS Code",
"description": "Visual Studio Code taskbar icon",
"state": "visible",
"dx": 872,
"dy": 908,
"frame_dx": 1011,
"frame_dy": 966,
"confidence": 0.98,
"source": "gemini_vlm",
"frame_bbox": [
    987,
    938,
    1035,
    994
],
"session_memory_key": "button:vs code",
"cached": false
},
{
    "id": "taskbar_spotify",
    "type": "button",
    "label": "Spotify",
    "description": "Spotify taskbar icon",
    "state": "visible",
    "dx": 904,
    "dy": 904,
    "frame_dx": 1043,
```

```
"frame_dy": 962,
"confidence": 0.98,
"source": "gemini_vlm",
"frame_bbox": [
    1019,
    938,
    1067,
    986
],
"session_memory_key": "button:spotify",
"cached": false
},
{
    "id": "system_tray_clock",
    "type": "label",
    "label": "Clock",
    "description": "System tray clock and date",
    "state": "visible",
    "dx": 1236,
    "dy": 900,
    "frame_dx": 1375,
    "frame_dy": 958,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
        1339,
        938,
        1411,
        978
    ]
}
```

```
    ],
    "session_memory_key": "label:clock",
    "cached": false
  },
  {
    "id": "system_tray_wifi",
    "type": "icon",
    "label": "Wi-Fi",
    "description": "Wi-Fi status icon",
    "state": "visible",
    "dx": 1172,
    "dy": 908,
    "frame_dx": 1311,
    "frame_dy": 966,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
      1291,
      946,
      1331,
      986
    ],
    "session_memory_key": "icon:wi-fi",
    "cached": false
  },
  {
    "id": "system_tray_battery",
    "type": "icon",
    "label": "Battery",
```



```
"description": "Battery status icon",
"state": "visible",
"dx": 1196,
"dy": 904,
"frame_dx": 1335,
"frame_dy": 962,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    1315,
    946,
    1355,
    978
],
"session_memory_key": "icon:battery",
"cached": false
},
{
    "id": "sidebar_recent_1",
    "type": "link",
    "label": "Recent Task 1",
    "description": "Recent task: Click on the 'Your Repositories'...",
    "state": "visible",
    "dx": 128,
    "dy": 608,
    "frame_dx": 267,
    "frame_dy": 666,
    "confidence": 0.85,
    "source": "gemini_vlm",
```

```
"frame_bbox": [
  139,
  626,
  395,
  706
],
"session_memory_key": "link:recent task 1",
"cached": false
},
{
  "id": "sidebar_recent_2",
  "type": "link",
  "label": "Recent Task 2",
  "description": "Recent task: Click on new tab and find the yo...",
  "state": "visible",
  "dx": 140,
  "dy": 748,
  "frame_dx": 279,
  "frame_dy": 806,
  "confidence": 0.85,
  "source": "gemini_vlm",
  "frame_bbox": [
    139,
    770,
    419,
    842
  ],
  "session_memory_key": "link:recent task 2",
  "cached": false
}
```

```
},  
{  
  "id": "user_profile",  
  "type": "button",  
  "label": "User Profile",  
  "description": "User profile button in sidebar",  
  "state": "visible",  
  "dx": 142,  
  "dy": 948,  
  "frame_dx": 281,  
  "frame_dy": 1006,  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    171,  
    1005,  
    391,  
    1006  
  ],  
  "session_memory_key": "button:user profile",  
  "cached": false  
},  
{  
  "id": "send_button",  
  "type": "button",  
  "label": "Send",  
  "description": "Send message button",  
  "state": "disabled",  
  "dx": 1226,
```

```
    "dy": 948,  
    "frame_dx": 1365,  
    "frame_dy": 1006,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        1342,  
        1005,  
        1387,  
        1006  
    ],  
    "session_memory_key": "button:send",  
    "cached": false  
}  
],  
"session_knowledge_base": {  
    "session_id": "shot_1777796788_f50769",  
    "started_at": "2026-05-03T08:26:28.783695+00:00",  
    "updated_at": "2026-05-03T08:26:52.866965+00:00",  
    "stale_after_misses": 3,  
    "confidence_floor": 0.4,  
    "decay_per_miss": 0.9,  
    "frame_count": 1,  
    "frame_name": "frame_1777796788_7b3eb6.jpg",  
    "frame_index": 0,  
    "image_size": {  
        "width": 1920,  
        "height": 1080  
    },  
}
```

```
"known_elements": {
  "tab:whatsapp": {
    "element_key": "tab:whatsapp",
    "element_type": "tab",
    "label": "WhatsApp",
    "bbox": [
      139.0,
      82.0,
      379.0,
      146.0
    ],
    "confidence": 0.95,
    "source": "gemini_vlm",
    "first_seen_frame": "frame_1777796788_7b3eb6.jpg",
    "last_seen_frame": "frame_1777796788_7b3eb6.jpg",
    "first_seen_at": "2026-05-03T08:26:52.866965+00:00",
    "last_seen_at": "2026-05-03T08:26:52.866965+00:00",
    "first_seen_index": 0,
    "last_seen_index": 0,
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false,
    "history": [
      {
        "frame_name": "frame_1777796788_7b3eb6.jpg",
        "frame_index": 0,
        "seen_at": "2026-05-03T08:26:52.866965+00:00",
        "confidence": 0.95
      }
    ]
  }
}
```

```
    }  
  ]  
},  
"tab:google search": {  
  "element_key": "tab:google search",  
  "element_type": "tab",  
  "label": "Google Search",  
  "bbox": [  
    323.0,  
    82.0,  
    571.0,  
    146.0  
  ],  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "first_seen_frame": "frame_1777796788_7b3eb6.jpg",  
  "last_seen_frame": "frame_1777796788_7b3eb6.jpg",  
  "first_seen_at": "2026-05-03T08:26:52.866965+00:00",  
  "last_seen_at": "2026-05-03T08:26:52.866965+00:00",  
  "first_seen_index": 0,  
  "last_seen_index": 0,  
  "seen_count": 1,  
  "miss_count": 0,  
  "stale": false,  
  "cached": false,  
  "history": [  
    {  
      "frame_name": "frame_1777796788_7b3eb6.jpg",  
      "frame_index": 0,  

```

```
        "seen_at": "2026-05-03T08:26:52.866965+00:00",
        "confidence": 0.95
    }
]
},
"tab:frontend": {
    "element_key": "tab:frontend",
    "element_type": "tab",
    "label": "frontend",
    "bbox": [
        515.0,
        82.0,
        763.0,
        146.0
    ],
    "confidence": 0.95,
    "source": "gemini_vlm",
    "first_seen_frame": "frame_1777796788_7b3eb6.jpg",
    "last_seen_frame": "frame_1777796788_7b3eb6.jpg",
    "first_seen_at": "2026-05-03T08:26:52.866965+00:00",
    "last_seen_at": "2026-05-03T08:26:52.866965+00:00",
    "first_seen_index": 0,
    "last_seen_index": 0,
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false,
    "history": [
        {
```

```
        "frame_name": "frame_1777796788_7b3eb6.jpg",
        "frame_index": 0,
        "seen_at": "2026-05-03T08:26:52.866965+00:00",
        "confidence": 0.95
    }
]
},
"tab:your repositories": {
    "element_key": "tab:your repositories",
    "element_type": "tab",
    "label": "Your Repositories",
    "bbox": [
        707.0,
        82.0,
        963.0,
        146.0
    ],
    "confidence": 0.95,
    "source": "gemini_vlm",
    "first_seen_frame": "frame_1777796788_7b3eb6.jpg",
    "last_seen_frame": "frame_1777796788_7b3eb6.jpg",
    "first_seen_at": "2026-05-03T08:26:52.866965+00:00",
    "last_seen_at": "2026-05-03T08:26:52.866965+00:00",
    "first_seen_index": 0,
    "last_seen_index": 0,
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false,
```



```
"history": [  
  {  
    "frame_name": "frame_1777796788_7b3eb6.jpg",  
    "frame_index": 0,  
    "seen_at": "2026-05-03T08:26:52.866965+00:00",  
    "confidence": 0.95  
  }  
]  
},  
"input:address bar": {  
  "element_key": "input:address bar",  
  "element_type": "input",  
  "label": "Address Bar",  
  "bbox": [  
    139.0,  
    146.0,  
    1315.0,  
    202.0  
  ],  
  "confidence": 0.98,  
  "source": "gemini_vlm",  
  "first_seen_frame": "frame_1777796788_7b3eb6.jpg",  
  "last_seen_frame": "frame_1777796788_7b3eb6.jpg",  
  "first_seen_at": "2026-05-03T08:26:52.866965+00:00",  
  "last_seen_at": "2026-05-03T08:26:52.866965+00:00",  
  "first_seen_index": 0,  
  "last_seen_index": 0,  
  "seen_count": 1,  
  "miss_count": 0,
```

```
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777796788_7b3eb6.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-03T08:26:52.866965+00:00",
    "confidence": 0.98
  }
],
},
"link:home": {
  "element_key": "link:home",
  "element_type": "link",
  "label": "Home",
  "bbox": [
    139.0,
    418.0,
    427.0,
    506.0
  ],
  "confidence": 0.92,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777796788_7b3eb6.jpg",
  "last_seen_frame": "frame_1777796788_7b3eb6.jpg",
  "first_seen_at": "2026-05-03T08:26:52.866965+00:00",
  "last_seen_at": "2026-05-03T08:26:52.866965+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
```

```
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777796788_7b3eb6.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-03T08:26:52.866965+00:00",
    "confidence": 0.92
  }
]
},
"link:chats": {
  "element_key": "link:chats",
  "element_type": "link",
  "label": "Chats",
  "bbox": [
    139.0,
    490.0,
    427.0,
    570.0
  ],
  "confidence": 0.92,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777796788_7b3eb6.jpg",
  "last_seen_frame": "frame_1777796788_7b3eb6.jpg",
  "first_seen_at": "2026-05-03T08:26:52.866965+00:00",
  "last_seen_at": "2026-05-03T08:26:52.866965+00:00",
```

```
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777796788_7b3eb6.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-03T08:26:52.866965+00:00",
    "confidence": 0.92
  }
],
},
"dropdown:ai assistant": {
  "element_key": "dropdown:ai assistant",
  "element_type": "dropdown",
  "label": "AI Assistant",
  "bbox": [
    403.0,
    226.0,
    555.0,
    290.0
  ],
  "confidence": 0.9,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777796788_7b3eb6.jpg",
  "last_seen_frame": "frame_1777796788_7b3eb6.jpg",
```

```
"first_seen_at": "2026-05-03T08:26:52.866965+00:00",
"last_seen_at": "2026-05-03T08:26:52.866965+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777796788_7b3eb6.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-03T08:26:52.866965+00:00",
    "confidence": 0.9
  }
],
},
"pane:user message": {
  "element_key": "pane:user message",
  "element_type": "pane",
  "label": "User Message",
  "bbox": [
    651.0,
    322.0,
    1467.0,
    522.0
  ],
  "confidence": 0.95,
  "source": "gemini_vlm",
```

```
"first_seen_frame": "frame_1777796788_7b3eb6.jpg",
"last_seen_frame": "frame_1777796788_7b3eb6.jpg",
"first_seen_at": "2026-05-03T08:26:52.866965+00:00",
"last_seen_at": "2026-05-03T08:26:52.866965+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777796788_7b3eb6.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-03T08:26:52.866965+00:00",
    "confidence": 0.95
  }
],
},
"label:connecting to agent...": {
  "element_key": "label:connecting to agent...",
  "element_type": "label",
  "label": "Connecting to agent...",
  "bbox": [
    355.0,
    506.0,
    1163.0,
    642.0
  ],
}
```

```
"confidence": 0.9,
"source": "gemini_vlm",
"first_seen_frame": "frame_1777796788_7b3eb6.jpg",
"last_seen_frame": "frame_1777796788_7b3eb6.jpg",
"first_seen_at": "2026-05-03T08:26:52.866965+00:00",
"last_seen_at": "2026-05-03T08:26:52.866965+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777796788_7b3eb6.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-03T08:26:52.866965+00:00",
    "confidence": 0.9
  }
],
"icon:loading": {
  "element_key": "icon:loading",
  "element_type": "icon",
  "label": "Loading",
  "bbox": [
    459.0,
    666.0,
    531.0,
```

```
        698.0
    ],
    "confidence": 0.85,
    "source": "gemini_vlm",
    "first_seen_frame": "frame_1777796788_7b3eb6.jpg",
    "last_seen_frame": "frame_1777796788_7b3eb6.jpg",
    "first_seen_at": "2026-05-03T08:26:52.866965+00:00",
    "last_seen_at": "2026-05-03T08:26:52.866965+00:00",
    "first_seen_index": 0,
    "last_seen_index": 0,
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false,
    "history": [
        {
            "frame_name": "frame_1777796788_7b3eb6.jpg",
            "frame_index": 0,
            "seen_at": "2026-05-03T08:26:52.866965+00:00",
            "confidence": 0.85
        }
    ]
},
"input:chat input": {
    "element_key": "input:chat input",
    "element_type": "input",
    "label": "Chat Input",
    "bbox": [
        437.0,
```



```
        1005.0,
        1394.0,
        1006.0
    ],
    "confidence": 0.9,
    "source": "gemini_vlm",
    "first_seen_frame": "frame_1777796788_7b3eb6.jpg",
    "last_seen_frame": "frame_1777796788_7b3eb6.jpg",
    "first_seen_at": "2026-05-03T08:26:52.866965+00:00",
    "last_seen_at": "2026-05-03T08:26:52.866965+00:00",
    "first_seen_index": 0,
    "last_seen_index": 0,
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false,
    "history": [
        {
            "frame_name": "frame_1777796788_7b3eb6.jpg",
            "frame_index": 0,
            "seen_at": "2026-05-03T08:26:52.866965+00:00",
            "confidence": 0.9
        }
    ]
},
"button:start": {
    "element_key": "button:start",
    "element_type": "button",
    "label": "Start",
```

```
"bbox": [
  443.0,
  938.0,
  499.0,
  994.0
],
"confidence": 0.98,
"source": "gemini_vlm",
"first_seen_frame": "frame_1777796788_7b3eb6.jpg",
"last_seen_frame": "frame_1777796788_7b3eb6.jpg",
"first_seen_at": "2026-05-03T08:26:52.866965+00:00",
"last_seen_at": "2026-05-03T08:26:52.866965+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777796788_7b3eb6.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-03T08:26:52.866965+00:00",
    "confidence": 0.98
  }
]
},
"input:search": {
  "element_key": "input:search",
```

```
"element_type": "input",
"label": "Search",
"bbox": [
    475.0,
    938.0,
    627.0,
    994.0
],
"confidence": 0.98,
"source": "gemini_vlm",
"first_seen_frame": "frame_1777796788_7b3eb6.jpg",
"last_seen_frame": "frame_1777796788_7b3eb6.jpg",
"first_seen_at": "2026-05-03T08:26:52.866965+00:00",
"last_seen_at": "2026-05-03T08:26:52.866965+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
    {
        "frame_name": "frame_1777796788_7b3eb6.jpg",
        "frame_index": 0,
        "seen_at": "2026-05-03T08:26:52.866965+00:00",
        "confidence": 0.98
    }
]
},
```

```
"button:chrome": {
  "element_key": "button:chrome",
  "element_type": "button",
  "label": "Chrome",
  "bbox": [
    875.0,
    938.0,
    923.0,
    986.0
  ],
  "confidence": 0.98,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777796788_7b3eb6.jpg",
  "last_seen_frame": "frame_1777796788_7b3eb6.jpg",
  "first_seen_at": "2026-05-03T08:26:52.866965+00:00",
  "last_seen_at": "2026-05-03T08:26:52.866965+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
  "seen_count": 1,
  "miss_count": 0,
  "stale": false,
  "cached": false,
  "history": [
    {
      "frame_name": "frame_1777796788_7b3eb6.jpg",
      "frame_index": 0,
      "seen_at": "2026-05-03T08:26:52.866965+00:00",
      "confidence": 0.98
    }
  ]
}
```

```
]
},
"button:vs code": {
  "element_key": "button:vs code",
  "element_type": "button",
  "label": "VS Code",
  "bbox": [
    987.0,
    938.0,
    1035.0,
    994.0
  ],
  "confidence": 0.98,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777796788_7b3eb6.jpg",
  "last_seen_frame": "frame_1777796788_7b3eb6.jpg",
  "first_seen_at": "2026-05-03T08:26:52.866965+00:00",
  "last_seen_at": "2026-05-03T08:26:52.866965+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
  "seen_count": 1,
  "miss_count": 0,
  "stale": false,
  "cached": false,
  "history": [
    {
      "frame_name": "frame_1777796788_7b3eb6.jpg",
      "frame_index": 0,
      "seen_at": "2026-05-03T08:26:52.866965+00:00",
```

```
        "confidence": 0.98
      }
    ]
  },
  "button:spotify": {
    "element_key": "button:spotify",
    "element_type": "button",
    "label": "Spotify",
    "bbox": [
      1019.0,
      938.0,
      1067.0,
      986.0
    ],
    "confidence": 0.98,
    "source": "gemini_vlm",
    "first_seen_frame": "frame_1777796788_7b3eb6.jpg",
    "last_seen_frame": "frame_1777796788_7b3eb6.jpg",
    "first_seen_at": "2026-05-03T08:26:52.866965+00:00",
    "last_seen_at": "2026-05-03T08:26:52.866965+00:00",
    "first_seen_index": 0,
    "last_seen_index": 0,
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false,
    "history": [
      {
        "frame_name": "frame_1777796788_7b3eb6.jpg",
```

```
        "frame_index": 0,
        "seen_at": "2026-05-03T08:26:52.866965+00:00",
        "confidence": 0.98
    }
]
},
"label:clock": {
    "element_key": "label:clock",
    "element_type": "label",
    "label": "Clock",
    "bbox": [
        1339.0,
        938.0,
        1411.0,
        978.0
    ],
    "confidence": 0.98,
    "source": "gemini_vlm",
    "first_seen_frame": "frame_1777796788_7b3eb6.jpg",
    "last_seen_frame": "frame_1777796788_7b3eb6.jpg",
    "first_seen_at": "2026-05-03T08:26:52.866965+00:00",
    "last_seen_at": "2026-05-03T08:26:52.866965+00:00",
    "first_seen_index": 0,
    "last_seen_index": 0,
    "seen_count": 1,
    "miss_count": 0,
    "stale": false,
    "cached": false,
    "history": [
```

```
{
  "frame_name": "frame_1777796788_7b3eb6.jpg",
  "frame_index": 0,
  "seen_at": "2026-05-03T08:26:52.866965+00:00",
  "confidence": 0.98
}
],
},
"icon:wi-fi": {
  "element_key": "icon:wi-fi",
  "element_type": "icon",
  "label": "Wi-Fi",
  "bbox": [
    1291.0,
    946.0,
    1331.0,
    986.0
  ],
  "confidence": 0.95,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777796788_7b3eb6.jpg",
  "last_seen_frame": "frame_1777796788_7b3eb6.jpg",
  "first_seen_at": "2026-05-03T08:26:52.866965+00:00",
  "last_seen_at": "2026-05-03T08:26:52.866965+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
  "seen_count": 1,
  "miss_count": 0,
  "stale": false,
```



```
"cached": false,
"history": [
  {
    "frame_name": "frame_1777796788_7b3eb6.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-03T08:26:52.866965+00:00",
    "confidence": 0.95
  }
],
},
"icon:battery": {
  "element_key": "icon:battery",
  "element_type": "icon",
  "label": "Battery",
  "bbox": [
    1315.0,
    946.0,
    1355.0,
    978.0
  ],
  "confidence": 0.95,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777796788_7b3eb6.jpg",
  "last_seen_frame": "frame_1777796788_7b3eb6.jpg",
  "first_seen_at": "2026-05-03T08:26:52.866965+00:00",
  "last_seen_at": "2026-05-03T08:26:52.866965+00:00",
  "first_seen_index": 0,
  "last_seen_index": 0,
  "seen_count": 1,
```

```
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777796788_7b3eb6.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-03T08:26:52.866965+00:00",
    "confidence": 0.95
  }
]
},
"link:recent task 1": {
  "element_key": "link:recent task 1",
  "element_type": "link",
  "label": "Recent Task 1",
  "bbox": [
    139.0,
    626.0,
    395.0,
    706.0
  ],
  "confidence": 0.85,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777796788_7b3eb6.jpg",
  "last_seen_frame": "frame_1777796788_7b3eb6.jpg",
  "first_seen_at": "2026-05-03T08:26:52.866965+00:00",
  "last_seen_at": "2026-05-03T08:26:52.866965+00:00",
  "first_seen_index": 0,
```

```
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777796788_7b3eb6.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-03T08:26:52.866965+00:00",
    "confidence": 0.85
  }
],
},
"link:recent task 2": {
  "element_key": "link:recent task 2",
  "element_type": "link",
  "label": "Recent Task 2",
  "bbox": [
    139.0,
    770.0,
    419.0,
    842.0
  ],
  "confidence": 0.85,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777796788_7b3eb6.jpg",
  "last_seen_frame": "frame_1777796788_7b3eb6.jpg",
  "first_seen_at": "2026-05-03T08:26:52.866965+00:00",
```

```
"last_seen_at": "2026-05-03T08:26:52.866965+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777796788_7b3eb6.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-03T08:26:52.866965+00:00",
    "confidence": 0.85
  }
],
},
"button:user profile": {
  "element_key": "button:user profile",
  "element_type": "button",
  "label": "User Profile",
  "bbox": [
    171.0,
    1005.0,
    391.0,
    1006.0
  ],
  "confidence": 0.9,
  "source": "gemini_vlm",
  "first_seen_frame": "frame_1777796788_7b3eb6.jpg",
```

```
"last_seen_frame": "frame_1777796788_7b3eb6.jpg",
"first_seen_at": "2026-05-03T08:26:52.866965+00:00",
"last_seen_at": "2026-05-03T08:26:52.866965+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777796788_7b3eb6.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-03T08:26:52.866965+00:00",
    "confidence": 0.9
  }
],
},
"button:send": {
  "element_key": "button:send",
  "element_type": "button",
  "label": "Send",
  "bbox": [
    1342.0,
    1005.0,
    1387.0,
    1006.0
  ],
  "confidence": 0.9,
```

```
"source": "gemini_vlm",
"first_seen_frame": "frame_1777796788_7b3eb6.jpg",
"last_seen_frame": "frame_1777796788_7b3eb6.jpg",
"first_seen_at": "2026-05-03T08:26:52.866965+00:00",
"last_seen_at": "2026-05-03T08:26:52.866965+00:00",
"first_seen_index": 0,
"last_seen_index": 0,
"seen_count": 1,
"miss_count": 0,
"stale": false,
"cached": false,
"history": [
  {
    "frame_name": "frame_1777796788_7b3eb6.jpg",
    "frame_index": 0,
    "seen_at": "2026-05-03T08:26:52.866965+00:00",
    "confidence": 0.9
  }
]
},
"resolved_elements": [
  {
    "id": "tab_whatsapp",
    "type": "tab",
    "label": "WhatsApp",
    "description": "Browser tab for WhatsApp",
    "state": "inactive",
    "dx": 120,
```

```
"dy": 56,
"frame_dx": 259,
"frame_dy": 114,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    139,
    82,
    379,
    146
],
"session_memory_key": "tab:whatsapp",
"cached": false
},
{
    "id": "tab_google_search",
    "type": "tab",
    "label": "Google Search",
    "description": "Browser tab for Google Search",
    "state": "inactive",
    "dx": 308,
    "dy": 56,
    "frame_dx": 447,
    "frame_dy": 114,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        323,
        82,
```

```
        571,  
        146  
    ],  
    "session_memory_key": "tab:google search",  
    "cached": false  
},  
{  
    "id": "tab_frontend",  
    "type": "tab",  
    "label": "frontend",  
    "description": "Browser tab for frontend development",  
    "state": "inactive",  
    "dx": 500,  
    "dy": 56,  
    "frame_dx": 639,  
    "frame_dy": 114,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        515,  
        82,  
        763,  
        146  
    ],  
    "session_memory_key": "tab:frontend",  
    "cached": false  
},  
{  
    "id": "tab_github_repos",
```



```
"type": "tab",
"label": "Your Repositories",
"description": "Browser tab for GitHub repositories",
"state": "active",
"dx": 696,
"dy": 56,
"frame_dx": 835,
"frame_dy": 114,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    707,
    82,
    963,
    146
],
"session_memory_key": "tab:your repositories",
"cached": false
},
{
    "id": "address_bar",
    "type": "input",
    "label": "Address Bar",
    "description": "Browser address bar showing localhost:5173",
    "state": "visible",
    "dx": 588,
    "dy": 116,
    "frame_dx": 727,
    "frame_dy": 174,
```

```
"confidence": 0.98,  
"source": "gemini_vlm",  
"frame_bbox": [  
    139,  
    146,  
    1315,  
    202  
],  
"session_memory_key": "input:address bar",  
"cached": false  
},  
{  
    "id": "sidebar_home",  
    "type": "link",  
    "label": "Home",  
    "description": "Navigation link to Home in ScreenPilot sidebar",  
    "state": "active",  
    "dx": 144,  
    "dy": 404,  
    "frame_dx": 283,  
    "frame_dy": 462,  
    "confidence": 0.92,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        139,  
        418,  
        427,  
        506  
    ],  
}
```

```
    "session_memory_key": "link:home",
    "cached": false
  },
  {
    "id": "sidebar_chats",
    "type": "link",
    "label": "Chats",
    "description": "Navigation link to Chats in ScreenPilot sidebar",
    "state": "inactive",
    "dx": 144,
    "dy": 472,
    "frame_dx": 283,
    "frame_dy": 530,
    "confidence": 0.92,
    "source": "gemini_vlm",
    "frame_bbox": [
      139,
      490,
      427,
      570
    ],
    "session_memory_key": "link:chats",
    "cached": false
  },
  {
    "id": "ai_assistant_dropdown",
    "type": "dropdown",
    "label": "AI Assistant",
    "description": "Dropdown menu to select AI assistant",
```

```
"state": "visible",
"dx": 340,
"dy": 200,
"frame_dx": 479,
"frame_dy": 258,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    403,
    226,
    555,
    290
],
"session_memory_key": "dropdown:ai assistant",
"cached": false
},
{
    "id": "user_message_bubble",
    "type": "pane",
    "label": "User Message",
    "description": "Chat bubble containing user instructions",
    "state": "visible",
    "dx": 920,
    "dy": 364,
    "frame_dx": 1059,
    "frame_dy": 422,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
```

```
        651,  
        322,  
        1467,  
        522  
    ],  
    "session_memory_key": "pane:user message",  
    "cached": false  
},  
{  
    "id": "status_message",  
    "type": "label",  
    "label": "Connecting to agent...",  
    "description": "Status message indicating agent connection and  
screen capture",  
    "state": "visible",  
    "dx": 620,  
    "dy": 516,  
    "frame_dx": 759,  
    "frame_dy": 574,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        355,  
        506,  
        1163,  
        642  
    ],  
    "session_memory_key": "label:connecting to agent...",  
    "cached": false
```

```
},  
{  
  "id": "loading_indicator",  
  "type": "icon",  
  "label": "Loading",  
  "description": "Three dots loading animation",  
  "state": "visible",  
  "dx": 356,  
  "dy": 624,  
  "frame_dx": 495,  
  "frame_dy": 682,  
  "confidence": 0.85,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    459,  
    666,  
    531,  
    698  
  ],  
  "session_memory_key": "icon:loading",  
  "cached": false  
},  
{  
  "id": "chat_input_area",  
  "type": "input",  
  "label": "Chat Input",  
  "description": "Input field for chat messages",  
  "state": "disabled",  
  "dx": 776,
```

```
"dy": 948,
"frame_dx": 915,
"frame_dy": 1006,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    437,
    1005,
    1394,
    1006
],
"session_memory_key": "input:chat input",
"cached": false
},
{
    "id": "taskbar_start",
    "type": "button",
    "label": "Start",
    "description": "Windows Start button",
    "state": "visible",
    "dx": 332,
    "dy": 908,
    "frame_dx": 471,
    "frame_dy": 966,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
        443,
        938,
```

```
        499,  
        994  
    ],  
    "session_memory_key": "button:start",  
    "cached": false  
},  
{  
    "id": "taskbar_search",  
    "type": "input",  
    "label": "Search",  
    "description": "Windows Search bar",  
    "state": "visible",  
    "dx": 412,  
    "dy": 908,  
    "frame_dx": 551,  
    "frame_dy": 966,  
    "confidence": 0.98,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        475,  
        938,  
        627,  
        994  
    ],  
    "session_memory_key": "input:search",  
    "cached": false  
},  
{  
    "id": "taskbar_chrome",
```



```
"type": "button",
"label": "Chrome",
"description": "Google Chrome taskbar icon",
"state": "active",
"dx": 760,
"dy": 904,
"frame_dx": 899,
"frame_dy": 962,
"confidence": 0.98,
"source": "gemini_vlm",
"frame_bbox": [
    875,
    938,
    923,
    986
],
"session_memory_key": "button:chrome",
"cached": false
},
{
    "id": "taskbar_vscode",
    "type": "button",
    "label": "VS Code",
    "description": "Visual Studio Code taskbar icon",
    "state": "visible",
    "dx": 872,
    "dy": 908,
    "frame_dx": 1011,
    "frame_dy": 966,
```

```
"confidence": 0.98,  
"source": "gemini_vlm",  
"frame_bbox": [  
    987,  
    938,  
    1035,  
    994  
],  
"session_memory_key": "button:vs code",  
"cached": false  
},  
{  
    "id": "taskbar_spotify",  
    "type": "button",  
    "label": "Spotify",  
    "description": "Spotify taskbar icon",  
    "state": "visible",  
    "dx": 904,  
    "dy": 904,  
    "frame_dx": 1043,  
    "frame_dy": 962,  
    "confidence": 0.98,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        1019,  
        938,  
        1067,  
        986  
    ],  
}
```

```
    "session_memory_key": "button:spotify",
    "cached": false
  },
  {
    "id": "system_tray_clock",
    "type": "label",
    "label": "Clock",
    "description": "System tray clock and date",
    "state": "visible",
    "dx": 1236,
    "dy": 900,
    "frame_dx": 1375,
    "frame_dy": 958,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
      1339,
      938,
      1411,
      978
    ],
    "session_memory_key": "label:clock",
    "cached": false
  },
  {
    "id": "system_tray_wifi",
    "type": "icon",
    "label": "Wi-Fi",
    "description": "Wi-Fi status icon",
```

```
"state": "visible",
"dx": 1172,
"dy": 908,
"frame_dx": 1311,
"frame_dy": 966,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    1291,
    946,
    1331,
    986
],
"session_memory_key": "icon:wi-fi",
"cached": false
},
{
    "id": "system_tray_battery",
    "type": "icon",
    "label": "Battery",
    "description": "Battery status icon",
    "state": "visible",
    "dx": 1196,
    "dy": 904,
    "frame_dx": 1335,
    "frame_dy": 962,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
```

```
        1315,  
        946,  
        1355,  
        978  
    ],  
    "session_memory_key": "icon:battery",  
    "cached": false  
},  
{  
    "id": "sidebar_recent_1",  
    "type": "link",  
    "label": "Recent Task 1",  
    "description": "Recent task: Click on the 'Your Repositories'...",  
    "state": "visible",  
    "dx": 128,  
    "dy": 608,  
    "frame_dx": 267,  
    "frame_dy": 666,  
    "confidence": 0.85,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        139,  
        626,  
        395,  
        706  
    ],  
    "session_memory_key": "link:recent task 1",  
    "cached": false  
},
```

```
{
  "id": "sidebar_recent_2",
  "type": "link",
  "label": "Recent Task 2",
  "description": "Recent task: Click on new tab and find the yo...",
  "state": "visible",
  "dx": 140,
  "dy": 748,
  "frame_dx": 279,
  "frame_dy": 806,
  "confidence": 0.85,
  "source": "gemini_vlm",
  "frame_bbox": [
    139,
    770,
    419,
    842
  ],
  "session_memory_key": "link:recent task 2",
  "cached": false
},
{
  "id": "user_profile",
  "type": "button",
  "label": "User Profile",
  "description": "User profile button in sidebar",
  "state": "visible",
  "dx": 142,
  "dy": 948,
```

```
"frame_dx": 281,
"frame_dy": 1006,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    171,
    1005,
    391,
    1006
],
"session_memory_key": "button:user profile",
"cached": false
},
{
    "id": "send_button",
    "type": "button",
    "label": "Send",
    "description": "Send message button",
    "state": "disabled",
    "dx": 1226,
    "dy": 948,
    "frame_dx": 1365,
    "frame_dy": 1006,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        1342,
        1005,
        1387,
```

```
        1006
    ],
    "session_memory_key": "button:send",
    "cached": false
}
],
"current_elements": [
    {
        "id": "tab_whatsapp",
        "type": "tab",
        "label": "WhatsApp",
        "description": "Browser tab for WhatsApp",
        "state": "inactive",
        "dx": 120,
        "dy": 56,
        "frame_dx": 259,
        "frame_dy": 114,
        "confidence": 0.95,
        "source": "gemini_vlm",
        "frame_bbox": [
            139,
            82,
            379,
            146
        ],
        "session_memory_key": "tab:whatsapp",
        "cached": false
    },
    {
```



```
"id": "tab_google_search",
"type": "tab",
"label": "Google Search",
"description": "Browser tab for Google Search",
"state": "inactive",
"dx": 308,
"dy": 56,
"frame_dx": 447,
"frame_dy": 114,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    323,
    82,
    571,
    146
],
"session_memory_key": "tab:google search",
"cached": false
},
{
    "id": "tab_frontend",
    "type": "tab",
    "label": "frontend",
    "description": "Browser tab for frontend development",
    "state": "inactive",
    "dx": 500,
    "dy": 56,
    "frame_dx": 639,
```

```
"frame_dy": 114,  
"confidence": 0.95,  
"source": "gemini_vlm",  
"frame_bbox": [  
    515,  
    82,  
    763,  
    146  
],  
"session_memory_key": "tab:frontend",  
"cached": false  
},  
{  
    "id": "tab_github_repos",  
    "type": "tab",  
    "label": "Your Repositories",  
    "description": "Browser tab for GitHub repositories",  
    "state": "active",  
    "dx": 696,  
    "dy": 56,  
    "frame_dx": 835,  
    "frame_dy": 114,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        707,  
        82,  
        963,  
        146
```

```
    ],  
    "session_memory_key": "tab:your repositories",  
    "cached": false  
  },  
  {  
    "id": "address_bar",  
    "type": "input",  
    "label": "Address Bar",  
    "description": "Browser address bar showing localhost:5173",  
    "state": "visible",  
    "dx": 588,  
    "dy": 116,  
    "frame_dx": 727,  
    "frame_dy": 174,  
    "confidence": 0.98,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
      139,  
      146,  
      1315,  
      202  
    ],  
    "session_memory_key": "input:address bar",  
    "cached": false  
  },  
  {  
    "id": "sidebar_home",  
    "type": "link",  
    "label": "Home",
```

```
    "description": "Navigation link to Home in ScreenPilot sidebar",
    "state": "active",
    "dx": 144,
    "dy": 404,
    "frame_dx": 283,
    "frame_dy": 462,
    "confidence": 0.92,
    "source": "gemini_vlm",
    "frame_bbox": [
        139,
        418,
        427,
        506
    ],
    "session_memory_key": "link:home",
    "cached": false
},
{
    "id": "sidebar_chats",
    "type": "link",
    "label": "Chats",
    "description": "Navigation link to Chats in ScreenPilot sidebar",
    "state": "inactive",
    "dx": 144,
    "dy": 472,
    "frame_dx": 283,
    "frame_dy": 530,
    "confidence": 0.92,
    "source": "gemini_vlm",
```

```
"frame_bbox": [
    139,
    490,
    427,
    570
],
"session_memory_key": "link:chats",
"cached": false
},
{
    "id": "ai_assistant_dropdown",
    "type": "dropdown",
    "label": "AI Assistant",
    "description": "Dropdown menu to select AI assistant",
    "state": "visible",
    "dx": 340,
    "dy": 200,
    "frame_dx": 479,
    "frame_dy": 258,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        403,
        226,
        555,
        290
    ],
    "session_memory_key": "dropdown:ai assistant",
    "cached": false
}
```

```
    },  
    {  
      "id": "user_message_bubble",  
      "type": "pane",  
      "label": "User Message",  
      "description": "Chat bubble containing user instructions",  
      "state": "visible",  
      "dx": 920,  
      "dy": 364,  
      "frame_dx": 1059,  
      "frame_dy": 422,  
      "confidence": 0.95,  
      "source": "gemini_vlm",  
      "frame_bbox": [  
        651,  
        322,  
        1467,  
        522  
      ],  
      "session_memory_key": "pane:user message",  
      "cached": false  
    },  
    {  
      "id": "status_message",  
      "type": "label",  
      "label": "Connecting to agent...",  
      "description": "Status message indicating agent connection and  
screen capture",  
      "state": "visible",
```

```
"dx": 620,  
  "dy": 516,  
  "frame_dx": 759,  
  "frame_dy": 574,  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    355,  
    506,  
    1163,  
    642  
  ],  
  "session_memory_key": "label:connecting to agent...",  
  "cached": false  
},  
{  
  "id": "loading_indicator",  
  "type": "icon",  
  "label": "Loading",  
  "description": "Three dots loading animation",  
  "state": "visible",  
  "dx": 356,  
  "dy": 624,  
  "frame_dx": 495,  
  "frame_dy": 682,  
  "confidence": 0.85,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    459,
```

```
        666,  
        531,  
        698  
    ],  
    "session_memory_key": "icon:loading",  
    "cached": false  
},  
{  
    "id": "chat_input_area",  
    "type": "input",  
    "label": "Chat Input",  
    "description": "Input field for chat messages",  
    "state": "disabled",  
    "dx": 776,  
    "dy": 948,  
    "frame_dx": 915,  
    "frame_dy": 1006,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        437,  
        1005,  
        1394,  
        1006  
    ],  
    "session_memory_key": "input:chat input",  
    "cached": false  
},  
{
```



```
    "id": "taskbar_start",
    "type": "button",
    "label": "Start",
    "description": "Windows Start button",
    "state": "visible",
    "dx": 332,
    "dy": 908,
    "frame_dx": 471,
    "frame_dy": 966,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
        443,
        938,
        499,
        994
    ],
    "session_memory_key": "button:start",
    "cached": false
},
{
    "id": "taskbar_search",
    "type": "input",
    "label": "Search",
    "description": "Windows Search bar",
    "state": "visible",
    "dx": 412,
    "dy": 908,
    "frame_dx": 551,
```

```
"frame_dy": 966,
"confidence": 0.98,
"source": "gemini_vlm",
"frame_bbox": [
    475,
    938,
    627,
    994
],
"session_memory_key": "input:search",
"cached": false
},
{
    "id": "taskbar_chrome",
    "type": "button",
    "label": "Chrome",
    "description": "Google Chrome taskbar icon",
    "state": "active",
    "dx": 760,
    "dy": 904,
    "frame_dx": 899,
    "frame_dy": 962,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
        875,
        938,
        923,
        986
    ]
}
```

```
    ],  
    "session_memory_key": "button:chrome",  
    "cached": false  
  },  
  {  
    "id": "taskbar_vscode",  
    "type": "button",  
    "label": "VS Code",  
    "description": "Visual Studio Code taskbar icon",  
    "state": "visible",  
    "dx": 872,  
    "dy": 908,  
    "frame_dx": 1011,  
    "frame_dy": 966,  
    "confidence": 0.98,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
      987,  
      938,  
      1035,  
      994  
    ],  
    "session_memory_key": "button:vs code",  
    "cached": false  
  },  
  {  
    "id": "taskbar_spotify",  
    "type": "button",  
    "label": "Spotify",
```

```
"description": "Spotify taskbar icon",
"state": "visible",
"dx": 904,
"dy": 904,
"frame_dx": 1043,
"frame_dy": 962,
"confidence": 0.98,
"source": "gemini_vlm",
"frame_bbox": [
    1019,
    938,
    1067,
    986
],
"session_memory_key": "button:spotify",
"cached": false
},
{
    "id": "system_tray_clock",
    "type": "label",
    "label": "Clock",
    "description": "System tray clock and date",
    "state": "visible",
    "dx": 1236,
    "dy": 900,
    "frame_dx": 1375,
    "frame_dy": 958,
    "confidence": 0.98,
    "source": "gemini_vlm",
```

```
"frame_bbox": [
    1339,
    938,
    1411,
    978
],
"session_memory_key": "label:clock",
"cached": false
},
{
    "id": "system_tray_wifi",
    "type": "icon",
    "label": "Wi-Fi",
    "description": "Wi-Fi status icon",
    "state": "visible",
    "dx": 1172,
    "dy": 908,
    "frame_dx": 1311,
    "frame_dy": 966,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        1291,
        946,
        1331,
        986
    ],
    "session_memory_key": "icon:wi-fi",
    "cached": false
}
```

```
},
{
  "id": "system_tray_battery",
  "type": "icon",
  "label": "Battery",
  "description": "Battery status icon",
  "state": "visible",
  "dx": 1196,
  "dy": 904,
  "frame_dx": 1335,
  "frame_dy": 962,
  "confidence": 0.95,
  "source": "gemini_vlm",
  "frame_bbox": [
    1315,
    946,
    1355,
    978
  ],
  "session_memory_key": "icon:battery",
  "cached": false
},
{
  "id": "sidebar_recent_1",
  "type": "link",
  "label": "Recent Task 1",
  "description": "Recent task: Click on the 'Your Repositories'...",
  "state": "visible",
  "dx": 128,
```

```
"dy": 608,  
  "frame_dx": 267,  
  "frame_dy": 666,  
  "confidence": 0.85,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    139,  
    626,  
    395,  
    706  
  ],  
  "session_memory_key": "link:recent task 1",  
  "cached": false  
},  
{  
  "id": "sidebar_recent_2",  
  "type": "link",  
  "label": "Recent Task 2",  
  "description": "Recent task: Click on new tab and find the yo...",  
  "state": "visible",  
  "dx": 140,  
  "dy": 748,  
  "frame_dx": 279,  
  "frame_dy": 806,  
  "confidence": 0.85,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    139,  
    770,
```

```
        419,  
        842  
    ],  
    "session_memory_key": "link:recent task 2",  
    "cached": false  
},  
{  
    "id": "user_profile",  
    "type": "button",  
    "label": "User Profile",  
    "description": "User profile button in sidebar",  
    "state": "visible",  
    "dx": 142,  
    "dy": 948,  
    "frame_dx": 281,  
    "frame_dy": 1006,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        171,  
        1005,  
        391,  
        1006  
    ],  
    "session_memory_key": "button:user profile",  
    "cached": false  
},  
{  
    "id": "send_button",
```



```
        "type": "button",
        "label": "Send",
        "description": "Send message button",
        "state": "disabled",
        "dx": 1226,
        "dy": 948,
        "frame_dx": 1365,
        "frame_dy": 1006,
        "confidence": 0.9,
        "source": "gemini_vlm",
        "frame_bbox": [
            1342,
            1005,
            1387,
            1006
        ],
        "session_memory_key": "button:send",
        "cached": false
    }
]
}
}
}
},
"todo": {
    "steps": [
        {
            "id": 1,
```

```
    "action": "Move the mouse to the 'Your Repositories' tab at (696, 56) and click it.",
    "target": "[tab] 'Your Repositories' @ (696, 56)",
    "expected_result": "The browser switches to the GitHub repositories tab, updating the page content.",
    "needs_user_data": false,
    "user_data_field": null,
    "status": "failed"
  },
  {
    "id": 2,
    "action": "Locate the 'New' button on the GitHub repositories page and click it.",
    "target": "'New' button on the GitHub repositories page",
    "expected_result": "The 'Create a new repository' page is displayed.",
    "needs_user_data": false,
    "user_data_field": null,
    "status": "failed"
  }
],
"notes": "The current screen shows a browser with multiple tabs. The first step is to switch to the GitHub tab. After the UI updates to show the repository list, the 'New' button will be clicked to proceed with repository creation.",
"estimated_complexity": "simple"
},
"step_plans": [
  {
    "step_id": 1,
    "action": "Move the mouse to the 'Your Repositories' tab at (696, 56) and click it.",
    "target": "[tab] 'Your Repositories' @ (696, 56)",
```

```
    "expected_result": "The browser switches to the GitHub repositories tab, updating the page content.",
    "hid_commands": [
      {
        "cmd": "mouse_move",
        "dx": 696,
        "dy": 56
      },
      {
        "cmd": "mouse_click",
        "button": "left"
      }
    ],
    "reasoning": "Moving the cursor to the 'Your Repositories' tab at (696, 56) and clicking it to switch to the GitHub repositories page as per Step 1."
  },
  {
    "step_id": 2,
    "action": "Locate the 'New' button on the GitHub repositories page and click it.",
    "target": "'New' button on the GitHub repositories page",
    "expected_result": "The 'Create a new repository' page is displayed.",
    "hid_commands": [],
    "reasoning": "Command generation failed: Response missing 'hid_commands'"
  }
],
"step_results": [
  {
    "step_id": 1,
```

```
    "action": "Move the mouse to the 'Your Repositories' tab at (696, 56) and click it.",
    "attempt": 3,
    "exec_status": "success",
    "eval_status": "retry",
    "confidence": 1.0,
    "reason": "The browser window is no longer visible on the screen; only desktop icons are detected. The expected GitHub repositories page did not load or the window was minimized/closed.",
    "final_status": "executing",
    "timestamp": "2026-05-03T08:34:21.151864+00:00"
  },
  {
    "step_id": 2,
    "action": "Locate the 'New' button on the GitHub repositories page and click it.",
    "attempt": 3,
    "exec_status": "success",
    "eval_status": "retry",
    "confidence": 0.95,
    "reason": "The active tab is still 'Your Repositories', indicating the navigation to the 'Create a new repository' page did not occur.",
    "final_status": "executing",
    "timestamp": "2026-05-03T08:41:39.807619+00:00"
  }
],
"final_report": {
  "success": false,
  "summary": "The task failed because the agent could not successfully navigate to the repositories page or locate the 'New' button.",
  "message": "The attempt to create a new GitHub repository was unsuccessful.\n\nStep results:\n- Step 1: Failed. The agent attempted to click
```

the 'Your Repositories' tab at (696, 56), but the element was actually located at (316, 60).\n- Step 2: Failed. The 'New' button was not detected in the UI elements list, likely because the navigation in Step 1 was not successful.\n\nThe screen state indicates that while the 'Your Repositories' tab is active, the main content area containing the 'New' button is not currently visible or recognized.",

```
"steps_completed": 0,
"steps_failed": 2,
"issues": [
  "Coordinate mismatch for 'Your Repositories' tab (Target: 696,56 vs Actual: 316,60)",
  "'New' button not found in the detected UI elements"
],
"recommendations": [
  "Recalibrate coordinate detection to target (316, 60) for the 'Your Repositories' tab",
  "Ensure the page content has fully loaded before attempting to locate the 'New' button"
]
},
"session_memory": {
  "run_id": "eb482227-7e57-46e0-9f57-38b2bcbd93c7",
  "user_task": "Click on the \"Your Repositories\" tab and check if it is Github repositories page, if yes find and click \"new\" button to create new repository",
  "started_at": "2026-05-03T08:26:19.529870+00:00",
  "updated_at": "2026-05-03T08:42:22.694106+00:00",
  "stale_after_misses": 2,
  "confidence_floor": 0.4,
  "decay_per_miss": 0.9,
  "frame_index": 20,
  "known_elements": {
    "tab:whatsapp": {
```

```
"element_key": "tab:whatsapp",
"element_type": "tab",
"label": "WhatsApp",
"bbox": [
  138.0,
  75.0,
  386.0,
  139.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 2,
"first_seen_at": "2026-05-03T08:26:52.918195+00:00",
"last_seen_at": "2026-05-03T08:28:25.053579+00:00",
"seen_count": 3,
"miss_count": 17,
"stale": true,
"cached": true
},
"tab:google search": {
  "element_key": "tab:google search",
  "element_type": "tab",
  "label": "Google Search",
  "bbox": [
    323.0,
    82.0,
```

```
    571.0,  
    146.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 0,  
  "last_seen_index": 0,  
  "first_seen_at": "2026-05-03T08:26:52.918195+00:00",  
  "last_seen_at": "2026-05-03T08:26:52.918195+00:00",  
  "seen_count": 1,  
  "miss_count": 19,  
  "stale": true,  
  "cached": true  
},  
"tab:frontend": {  
  "element_key": "tab:frontend",  
  "element_type": "tab",  
  "label": "frontend",  
  "bbox": [  
    138.0,  
    83.0,  
    378.0,  
    155.0  
  ],  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,
```

```
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 19,
"first_seen_at": "2026-05-03T08:26:52.918195+00:00",
"last_seen_at": "2026-05-03T08:42:22.694106+00:00",
"seen_count": 8,
"miss_count": 0,
"stale": false,
"cached": false
},
"tab:your repositories": {
  "element_key": "tab:your repositories",
  "element_type": "tab",
  "label": "Your Repositories",
  "bbox": [
    322.0,
    83.0,
    586.0,
    147.0
  ],
  "confidence": 0.98,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 19,
  "first_seen_at": "2026-05-03T08:26:52.918195+00:00",
  "last_seen_at": "2026-05-03T08:42:22.694106+00:00",
  "seen_count": 8,
```



```
    "miss_count": 0,
    "stale": false,
    "cached": false
  },
  "input:address bar": {
    "element_key": "input:address bar",
    "element_type": "input",
    "label": "Address Bar",
    "bbox": [
      138.0,
      147.0,
      1242.0,
      211.0
    ],
    "confidence": 0.98,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 0,
    "last_seen_index": 19,
    "first_seen_at": "2026-05-03T08:26:52.918195+00:00",
    "last_seen_at": "2026-05-03T08:42:22.694106+00:00",
    "seen_count": 4,
    "miss_count": 0,
    "stale": false,
    "cached": false
  },
  "link:home": {
    "element_key": "link:home",
```

```
"element_type": "link",
"label": "Home",
"bbox": [
  139.0,
  418.0,
  427.0,
  506.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 0,
"first_seen_at": "2026-05-03T08:26:52.918195+00:00",
"last_seen_at": "2026-05-03T08:26:52.918195+00:00",
"seen_count": 1,
"miss_count": 19,
"stale": true,
"cached": true
},
"link:chats": {
  "element_key": "link:chats",
  "element_type": "link",
  "label": "Chats",
  "bbox": [
    139.0,
    490.0,
    427.0,
```

```
570.0

],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 0,
"first_seen_at": "2026-05-03T08:26:52.918195+00:00",
"last_seen_at": "2026-05-03T08:26:52.918195+00:00",
"seen_count": 1,
"miss_count": 19,
"stale": true,
"cached": true
},
"dropdown:ai assistant": {
  "element_key": "dropdown:ai assistant",
  "element_type": "dropdown",
  "label": "AI Assistant",
  "bbox": [
    394.0,
    234.0,
    546.0,
    298.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
```

```
"first_seen_index": 0,
"last_seen_index": 1,
"first_seen_at": "2026-05-03T08:26:52.918195+00:00",
"last_seen_at": "2026-05-03T08:27:39.128680+00:00",
"seen_count": 2,
"miss_count": 18,
"stale": true,
"cached": true
},
"pane:user message": {
  "element_key": "pane:user message",
  "element_type": "pane",
  "label": "User Message",
  "bbox": [
    651.0,
    322.0,
    1467.0,
    522.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 0,
  "first_seen_at": "2026-05-03T08:26:52.918195+00:00",
  "last_seen_at": "2026-05-03T08:26:52.918195+00:00",
  "seen_count": 1,
  "miss_count": 19,
```

```
    "stale": true,
    "cached": true
  },
  "label:connecting to agent...": {
    "element_key": "label:connecting to agent...",
    "element_type": "label",
    "label": "Connecting to agent...",
    "bbox": [
      355.0,
      506.0,
      1163.0,
      642.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 0,
    "last_seen_index": 0,
    "first_seen_at": "2026-05-03T08:26:52.918195+00:00",
    "last_seen_at": "2026-05-03T08:26:52.918195+00:00",
    "seen_count": 1,
    "miss_count": 19,
    "stale": true,
    "cached": true
  },
  "icon:loading": {
    "element_key": "icon:loading",
    "element_type": "icon",
```

```
"label": "Loading",
"bbox": [
  459.0,
  666.0,
  531.0,
  698.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 0,
"first_seen_at": "2026-05-03T08:26:52.918195+00:00",
"last_seen_at": "2026-05-03T08:26:52.918195+00:00",
"seen_count": 1,
"miss_count": 19,
"stale": true,
"cached": true
},
"input:chat input": {
  "element_key": "input:chat input",
  "element_type": "input",
  "label": "Chat Input",
  "bbox": [
    437.0,
    1005.0,
    1394.0,
    1006.0
  ]
}
```

```
],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 0,
  "first_seen_at": "2026-05-03T08:26:52.918195+00:00",
  "last_seen_at": "2026-05-03T08:26:52.918195+00:00",
  "seen_count": 1,
  "miss_count": 19,
  "stale": true,
  "cached": true
},
"button:start": {
  "element_key": "button:start",
  "element_type": "button",
  "label": "Start",
  "bbox": [
    450.0,
    947.0,
    490.0,
    995.0
  ],
  "confidence": 0.5315,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
```

```
"last_seen_index": 14,  
"first_seen_at": "2026-05-03T08:26:52.918195+00:00",  
"last_seen_at": "2026-05-03T08:38:34.371135+00:00",  
"seen_count": 5,  
"miss_count": 5,  
"stale": true,  
"cached": true  
},  
"input:search": {  
  "element_key": "input:search",  
  "element_type": "input",  
  "label": "Search",  
  "bbox": [  
    458.0,  
    947.0,  
    682.0,  
    1003.0  
  ],  
  "confidence": 0.5315,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 0,  
  "last_seen_index": 14,  
  "first_seen_at": "2026-05-03T08:26:52.918195+00:00",  
  "last_seen_at": "2026-05-03T08:38:34.371135+00:00",  
  "seen_count": 3,  
  "miss_count": 5,  
  "stale": true,
```



```
    "cached": true
  },
  "button:chrome": {
    "element_key": "button:chrome",
    "element_type": "button",
    "label": "Chrome",
    "bbox": [
      802.0,
      947.0,
      850.0,
      987.0
    ],
    "confidence": 0.5315,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 0,
    "last_seen_index": 14,
    "first_seen_at": "2026-05-03T08:26:52.918195+00:00",
    "last_seen_at": "2026-05-03T08:38:34.371135+00:00",
    "seen_count": 2,
    "miss_count": 5,
    "stale": true,
    "cached": true
  },
  "button:vs code": {
    "element_key": "button:vs code",
    "element_type": "button",
    "label": "VS Code",
```

```
"bbox": [
  946.0,
  947.0,
  994.0,
  987.0
],
"confidence": 0.5315,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 14,
"first_seen_at": "2026-05-03T08:26:52.918195+00:00",
"last_seen_at": "2026-05-03T08:38:34.371135+00:00",
"seen_count": 2,
"miss_count": 5,
"stale": true,
"cached": true
},
"button:spotify": {
  "element_key": "button:spotify",
  "element_type": "button",
  "label": "Spotify",
  "bbox": [
    978.0,
    947.0,
    1034.0,
    987.0
  ],
}
```

```
"confidence": 0.5315,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 14,
"first_seen_at": "2026-05-03T08:26:52.918195+00:00",
"last_seen_at": "2026-05-03T08:38:34.371135+00:00",
"seen_count": 3,
"miss_count": 5,
"stale": true,
"cached": true
},
"label:clock": {
  "element_key": "label:clock",
  "element_type": "label",
  "label": "Clock",
  "bbox": [
    1339.0,
    938.0,
    1411.0,
    978.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 0,
```

```
"first_seen_at": "2026-05-03T08:26:52.918195+00:00",
"last_seen_at": "2026-05-03T08:26:52.918195+00:00",
"seen_count": 1,
"miss_count": 19,
"stale": true,
"cached": true
},
"icon:wi-fi": {
  "element_key": "icon:wi-fi",
  "element_type": "icon",
  "label": "Wi-Fi",
  "bbox": [
    1291.0,
    946.0,
    1331.0,
    986.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 0,
  "first_seen_at": "2026-05-03T08:26:52.918195+00:00",
  "last_seen_at": "2026-05-03T08:26:52.918195+00:00",
  "seen_count": 1,
  "miss_count": 19,
  "stale": true,
  "cached": true
}
```

```
},
"icon:battery": {
  "element_key": "icon:battery",
  "element_type": "icon",
  "label": "Battery",
  "bbox": [
    1315.0,
    946.0,
    1355.0,
    978.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 0,
  "first_seen_at": "2026-05-03T08:26:52.918195+00:00",
  "last_seen_at": "2026-05-03T08:26:52.918195+00:00",
  "seen_count": 1,
  "miss_count": 19,
  "stale": true,
  "cached": true
},
"link:recent task 1": {
  "element_key": "link:recent task 1",
  "element_type": "link",
  "label": "Recent Task 1",
  "bbox": [
```

```
    139.0,  
    626.0,  
    395.0,  
    706.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 0,  
  "last_seen_index": 0,  
  "first_seen_at": "2026-05-03T08:26:52.918195+00:00",  
  "last_seen_at": "2026-05-03T08:26:52.918195+00:00",  
  "seen_count": 1,  
  "miss_count": 19,  
  "stale": true,  
  "cached": true  
},  
"link:recent task 2": {  
  "element_key": "link:recent task 2",  
  "element_type": "link",  
  "label": "Recent Task 2",  
  "bbox": [  
    139.0,  
    770.0,  
    419.0,  
    842.0  
  ],  
  "confidence": 0.4,
```

```
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 0,
"last_seen_index": 0,
"first_seen_at": "2026-05-03T08:26:52.918195+00:00",
"last_seen_at": "2026-05-03T08:26:52.918195+00:00",
"seen_count": 1,
"miss_count": 19,
"stale": true,
"cached": true
},
"button:user profile": {
  "element_key": "button:user profile",
  "element_type": "button",
  "label": "User Profile",
  "bbox": [
    138.0,
    850.0,
    426.0,
    954.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 0,
  "last_seen_index": 1,
  "first_seen_at": "2026-05-03T08:26:52.918195+00:00",
```

```
    "last_seen_at": "2026-05-03T08:27:39.128680+00:00",
    "seen_count": 2,
    "miss_count": 18,
    "stale": true,
    "cached": true
  },
  "button:send": {
    "element_key": "button:send",
    "element_type": "button",
    "label": "Send",
    "bbox": [
      1342.0,
      1005.0,
      1387.0,
      1006.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 0,
    "last_seen_index": 0,
    "first_seen_at": "2026-05-03T08:26:52.918195+00:00",
    "last_seen_at": "2026-05-03T08:26:52.918195+00:00",
    "seen_count": 1,
    "miss_count": 19,
    "stale": true,
    "cached": true
  },
}
```



```
"tab:prasad fashion matara": {
  "element_key": "tab:prasad fashion matara",
  "element_type": "tab",
  "label": "prasad fashion matara",
  "bbox": [
    322.0,
    82.0,
    570.0,
    154.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
  "last_seen_index": 1,
  "first_seen_at": "2026-05-03T08:27:39.128680+00:00",
  "last_seen_at": "2026-05-03T08:27:39.128680+00:00",
  "seen_count": 1,
  "miss_count": 18,
  "stale": true,
  "cached": true
},
"button:new tab": {
  "element_key": "button:new tab",
  "element_type": "button",
  "label": "New Tab",
  "bbox": [
    930.0,
```

```
    98.0,  
    970.0,  
    146.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 1,  
  "last_seen_index": 1,  
  "first_seen_at": "2026-05-03T08:27:39.128680+00:00",  
  "last_seen_at": "2026-05-03T08:27:39.128680+00:00",  
  "seen_count": 1,  
  "miss_count": 18,  
  "stale": true,  
  "cached": true  
},  
"button:ask gemini": {  
  "element_key": "button:ask gemini",  
  "element_type": "button",  
  "label": "Ask Gemini",  
  "bbox": [  
    1170.0,  
    83.0,  
    1290.0,  
    139.0  
  ],  
  "confidence": 0.5315,  
  "source": "gemini_vlm",
```

```
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 1,
"last_seen_index": 14,
"first_seen_at": "2026-05-03T08:27:39.128680+00:00",
"last_seen_at": "2026-05-03T08:38:34.371135+00:00",
"seen_count": 2,
"miss_count": 5,
"stale": true,
"cached": true
},
"button:back": {
  "element_key": "button:back",
  "element_type": "button",
  "label": "Back",
  "bbox": [
    154.0,
    146.0,
    202.0,
    202.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
  "last_seen_index": 1,
  "first_seen_at": "2026-05-03T08:27:39.128680+00:00",
  "last_seen_at": "2026-05-03T08:27:39.128680+00:00",
```

```
    "seen_count": 1,  
    "miss_count": 18,  
    "stale": true,  
    "cached": true  
  },  
  "button:forward": {  
    "element_key": "button:forward",  
    "element_type": "button",  
    "label": "Forward",  
    "bbox": [  
      186.0,  
      146.0,  
      242.0,  
      202.0  
    ],  
    "confidence": 0.4,  
    "source": "gemini_vlm",  
    "first_seen_frame": null,  
    "last_seen_frame": null,  
    "first_seen_index": 1,  
    "last_seen_index": 1,  
    "first_seen_at": "2026-05-03T08:27:39.128680+00:00",  
    "last_seen_at": "2026-05-03T08:27:39.128680+00:00",  
    "seen_count": 1,  
    "miss_count": 18,  
    "stale": true,  
    "cached": true  
  },  
  "button:refresh": {
```

```
"element_key": "button:refresh",
"element_type": "button",
"label": "Refresh",
"bbox": [
    226.0,
    146.0,
    282.0,
    194.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 1,
"last_seen_index": 1,
"first_seen_at": "2026-05-03T08:27:39.128680+00:00",
"last_seen_at": "2026-05-03T08:27:39.128680+00:00",
"seen_count": 1,
"miss_count": 18,
"stale": true,
"cached": true
},
"label:screenpilot ai workspace": {
    "element_key": "label:screenpilot ai workspace",
    "element_type": "label",
    "label": "ScreenPilot AI workspace",
    "bbox": [
        138.0,
        234.0,
```

```
    426.0,  
    322.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 1,  
  "last_seen_index": 1,  
  "first_seen_at": "2026-05-03T08:27:39.128680+00:00",  
  "last_seen_at": "2026-05-03T08:27:39.128680+00:00",  
  "seen_count": 1,  
  "miss_count": 18,  
  "stale": true,  
  "cached": true  
},  
"input:search chats": {  
  "element_key": "input:search chats",  
  "element_type": "input",  
  "label": "Search chats",  
  "bbox": [  
    138.0,  
    322.0,  
    426.0,  
    410.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,
```

```
"last_seen_frame": null,
"first_seen_index": 1,
"last_seen_index": 1,
"first_seen_at": "2026-05-03T08:27:39.128680+00:00",
"last_seen_at": "2026-05-03T08:27:39.128680+00:00",
"seen_count": 1,
"miss_count": 18,
"stale": true,
"cached": true
},
"button:home": {
  "element_key": "button:home",
  "element_type": "button",
  "label": "Home",
  "bbox": [
    138.0,
    410.0,
    426.0,
    498.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 1,
  "last_seen_index": 1,
  "first_seen_at": "2026-05-03T08:27:39.128680+00:00",
  "last_seen_at": "2026-05-03T08:27:39.128680+00:00",
  "seen_count": 1,
```

```
    "miss_count": 18,  
    "stale": true,  
    "cached": true  
  },  
  "button:chats": {  
    "element_key": "button:chats",  
    "element_type": "button",  
    "label": "Chats",  
    "bbox": [  
      138.0,  
      482.0,  
      426.0,  
      578.0  
    ],  
    "confidence": 0.4,  
    "source": "gemini_vlm",  
    "first_seen_frame": null,  
    "last_seen_frame": null,  
    "first_seen_index": 1,  
    "last_seen_index": 1,  
    "first_seen_at": "2026-05-03T08:27:39.128680+00:00",  
    "last_seen_at": "2026-05-03T08:27:39.128680+00:00",  
    "seen_count": 1,  
    "miss_count": 18,  
    "stale": true,  
    "cached": true  
  },  
  "label:user message": {  
    "element_key": "label:user message",
```



```
"element_type": "label",
"label": "User Message",
"bbox": [
    634.0,
    298.0,
    1418.0,
    490.0
],
"confidence": 0.4,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 1,
"last_seen_index": 1,
"first_seen_at": "2026-05-03T08:27:39.128680+00:00",
"last_seen_at": "2026-05-03T08:27:39.128680+00:00",
"seen_count": 1,
"miss_count": 18,
"stale": true,
"cached": true
},
"pane:task plan": {
    "element_key": "pane:task plan",
    "element_type": "pane",
    "label": "Task plan",
    "bbox": [
        354.0,
        442.0,
        1122.0,
```

```
      842.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 1,
    "last_seen_index": 1,
    "first_seen_at": "2026-05-03T08:27:39.128680+00:00",
    "last_seen_at": "2026-05-03T08:27:39.128680+00:00",
    "seen_count": 1,
    "miss_count": 18,
    "stale": true,
    "cached": true
  },
  "label:step 1: move mouse to your repositories tab": {
    "element_key": "label:step 1: move mouse to your repositories tab",
    "element_type": "label",
    "label": "Step 1: Move mouse to Your Repositories tab",
    "bbox": [
      370.0,
      618.0,
      1074.0,
      714.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
```

```
"first_seen_index": 1,
"last_seen_index": 1,
"first_seen_at": "2026-05-03T08:27:39.128680+00:00",
"last_seen_at": "2026-05-03T08:27:39.128680+00:00",
"seen_count": 1,
"miss_count": 18,
"stale": true,
"cached": true
},
"tab:prasad fashion matara - google search": {
  "element_key": "tab:prasad fashion matara - google search",
  "element_type": "tab",
  "label": "prasad fashion matara - Google Search",
  "bbox": [
    330.0,
    75.0,
    578.0,
    139.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 2,
  "last_seen_index": 2,
  "first_seen_at": "2026-05-03T08:28:25.053579+00:00",
  "last_seen_at": "2026-05-03T08:28:25.053579+00:00",
  "seen_count": 1,
  "miss_count": 17,
```

```
    "stale": true,
    "cached": true
  },
  "tab:(12) whatsapp": {
    "element_key": "tab:(12) whatsapp",
    "element_type": "tab",
    "label": "(12) WhatsApp",
    "bbox": [
      138.0,
      82.0,
      386.0,
      146.0
    ],
    "confidence": 0.4,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 3,
    "last_seen_index": 3,
    "first_seen_at": "2026-05-03T08:29:27.234829+00:00",
    "last_seen_at": "2026-05-03T08:29:27.234829+00:00",
    "seen_count": 1,
    "miss_count": 16,
    "stale": true,
    "cached": true
  },
  "icon:google chrome": {
    "element_key": "icon:google chrome",
    "element_type": "icon",
```

```
"label": "Google Chrome",
"bbox": [
  154.0,
  82.0,
  242.0,
  202.0
],
"confidence": 0.5048,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 4,
"last_seen_index": 13,
"first_seen_at": "2026-05-03T08:30:15.480408+00:00",
"last_seen_at": "2026-05-03T08:37:45.966440+00:00",
"seen_count": 8,
"miss_count": 6,
"stale": true,
"cached": true
},
"pane:taskbar": {
  "element_key": "pane:taskbar",
  "element_type": "pane",
  "label": "Taskbar",
  "bbox": [
    138.0,
    915.0,
    1795.0,
    1007.0
  ]
}
```

```
],
  "confidence": 0.4543,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 5,
  "last_seen_index": 12,
  "first_seen_at": "2026-05-03T08:31:03.326139+00:00",
  "last_seen_at": "2026-05-03T08:36:42.648830+00:00",
  "seen_count": 2,
  "miss_count": 7,
  "stale": true,
  "cached": true
},
"icon:huiontablet": {
  "element_key": "icon:huiontablet",
  "element_type": "icon",
  "label": "HuionTablet",
  "bbox": [
    210.0,
    82.0,
    298.0,
    202.0
  ],
  "confidence": 0.5048,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 6,
```

```
"last_seen_index": 13,  
"first_seen_at": "2026-05-03T08:31:58.654399+00:00",  
"last_seen_at": "2026-05-03T08:37:45.966440+00:00",  
"seen_count": 7,  
"miss_count": 6,  
"stale": true,  
"cached": true  
},  
"icon:wise memory optimizer": {  
  "element_key": "icon:wise memory optimizer",  
  "element_type": "icon",  
  "label": "Wise Memory Optimizer",  
  "bbox": [  
    154.0,  
    178.0,  
    226.0,  
    298.0  
  ],  
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  "source": "gemini_vlm",  
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  "last_seen_frame": null,  
  "first_seen_index": 11,  
  "last_seen_index": 11,  
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  "last_seen_at": "2026-05-03T08:35:46.217640+00:00",  
  "seen_count": 1,  
  "miss_count": 8,  
  "stale": true,
```

```
    "cached": true
  },
  "icon:amazon workspaces": {
    "element_key": "icon:amazon workspaces",
    "element_type": "icon",
    "label": "Amazon WorkSpaces",
    "bbox": [
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      178.0,
      290.0,
      298.0
    ],
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    "miss_count": 8,
    "stale": true,
    "cached": true
  },
  "icon:recycle bin": {
    "element_key": "icon:recycle bin",
    "element_type": "icon",
    "label": "Recycle Bin",
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"bbox": [
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  226.0,
  394.0
],
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"miss_count": 8,
"stale": true,
"cached": true
},
"icon:krita": {
  "element_key": "icon:krita",
  "element_type": "icon",
  "label": "Krita",
  "bbox": [
    218.0,
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    282.0,
    394.0
  ],

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"miss_count": 8,  
"stale": true,  
"cached": true  
},  
"icon:nuwani - chrome": {  
  "element_key": "icon:nuwani - chrome",  
  "element_type": "icon",  
  "label": "Nuwani - Chrome",  
  "bbox": [  
    154.0,  
    362.0,  
    226.0,  
    482.0  
  ],  
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  "element_key": "icon:docker desktop",
  "element_type": "icon",
  "label": "Docker Desktop",
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    362.0,
    282.0,
    482.0
  ],
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  "last_seen_index": 11,
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  "seen_count": 1,
  "miss_count": 8,
  "stale": true,
  "cached": true
}
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},
"icon:antigravity": {
  "element_key": "icon:antigravity",
  "element_type": "icon",
  "label": "Antigravity",
  "bbox": [
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    218.0,
    570.0
  ],
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  "seen_count": 1,
  "miss_count": 8,
  "stale": true,
  "cached": true
},
"icon:visual studio code": {
  "element_key": "icon:visual studio code",
  "element_type": "icon",
  "label": "Visual Studio Code",
  "bbox": [
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    162.0,  
    538.0,  
    218.0,  
    666.0  
  ],  
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  "last_seen_index": 11,  
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  "seen_count": 1,  
  "miss_count": 8,  
  "stale": true,  
  "cached": true  
},  
"icon:postman": {  
  "element_key": "icon:postman",  
  "element_type": "icon",  
  "label": "Postman",  
  "bbox": [  
    162.0,  
    634.0,  
    218.0,  
    754.0  
  ],  
  "confidence": 0.4089,
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"seen_count": 1,
"miss_count": 8,
"stale": true,
"cached": true
},
"icon:loom": {
  "element_key": "icon:loom",
  "element_type": "icon",
  "label": "Loom",
  "bbox": [
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    722.0,
    218.0,
    842.0
  ],
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  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
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  "last_seen_index": 11,
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    "miss_count": 8,
    "stale": true,
    "cached": true
  },
  "icon:vlc media player": {
    "element_key": "icon:vlc media player",
    "element_type": "icon",
    "label": "VLC media player",
    "bbox": [
      162.0,
      810.0,
      218.0,
      938.0
    ],
    "confidence": 0.4089,
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    "last_seen_index": 11,
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    "seen_count": 1,
    "miss_count": 8,
    "stale": true,
    "cached": true
  },
}
```

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"widget:weather": {
  "element_key": "widget:weather",
  "element_type": "widget",
  "label": "Weather",
  "bbox": [
    138.0,
    938.0,
    314.0,
    994.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
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  "last_seen_frame": null,
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  "last_seen_index": 11,
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  "last_seen_at": "2026-05-03T08:35:46.217640+00:00",
  "seen_count": 1,
  "miss_count": 8,
  "stale": true,
  "cached": true
},
"button:copilot": {
  "element_key": "button:copilot",
  "element_type": "button",
  "label": "Copilot",
  "bbox": [
    842.0,
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```
    930.0,
    890.0,
    986.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 11,
  "last_seen_index": 11,
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  "seen_count": 1,
  "miss_count": 8,
  "stale": true,
  "cached": true
},
"button:task view": {
  "element_key": "button:task view",
  "element_type": "button",
  "label": "Task View",
  "bbox": [
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    930.0,
    938.0,
    986.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
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"miss_count": 8,
"stale": true,
"cached": true
},
"button:file explorer": {
  "element_key": "button:file explorer",
  "element_type": "button",
  "label": "File Explorer",
  "bbox": [
    730.0,
    947.0,
    778.0,
    995.0
  ],
  "confidence": 0.5315,
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  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 11,
  "last_seen_index": 14,
  "first_seen_at": "2026-05-03T08:35:46.217640+00:00",
  "last_seen_at": "2026-05-03T08:38:34.371135+00:00",
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"seen_count": 2,
"miss_count": 5,
"stale": true,
"cached": true
},
"button:microsoft edge": {
  "element_key": "button:microsoft edge",
  "element_type": "button",
  "label": "Microsoft Edge",
  "bbox": [
    978.0,
    930.0,
    1026.0,
    986.0
  ],
  "confidence": 0.4089,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 11,
  "last_seen_index": 11,
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  "last_seen_at": "2026-05-03T08:35:46.217640+00:00",
  "seen_count": 1,
  "miss_count": 8,
  "stale": true,
  "cached": true
},
"button:microsoft store": {
```

```
"element_key": "button:microsoft store",
"element_type": "button",
"label": "Microsoft Store",
"bbox": [
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    930.0,
    1074.0,
    986.0
],
"confidence": 0.4089,
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"last_seen_index": 11,
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"last_seen_at": "2026-05-03T08:35:46.217640+00:00",
"seen_count": 1,
"miss_count": 8,
"stale": true,
"cached": true
},
"button:xbox": {
    "element_key": "button:xbox",
    "element_type": "button",
    "label": "Xbox",
    "bbox": [
        1066.0,
        930.0,
```

```
    1114.0,  
    986.0  
  ],  
  "confidence": 0.4,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 11,  
  "last_seen_index": 11,  
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  "last_seen_at": "2026-05-03T08:35:46.217640+00:00",  
  "seen_count": 1,  
  "miss_count": 8,  
  "stale": true,  
  "cached": true  
},  
"button:google chrome": {  
  "element_key": "button:google chrome",  
  "element_type": "button",  
  "label": "Google Chrome",  
  "bbox": [  
    1114.0,  
    930.0,  
    1162.0,  
    986.0  
  ],  
  "confidence": 0.4089,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,
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"seen_count": 1,
"miss_count": 8,
"stale": true,
"cached": true
},
"button:notepad": {
  "element_key": "button:notepad",
  "element_type": "button",
  "label": "Notepad",
  "bbox": [
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    930.0,
    1210.0,
    986.0
  ],
  "confidence": 0.4,
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  "last_seen_frame": null,
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  "last_seen_index": 11,
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  "last_seen_at": "2026-05-03T08:35:46.217640+00:00",
  "seen_count": 1,
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    "miss_count": 8,  
    "stale": true,  
    "cached": true  
  },  
  "button:visual studio code": {  
    "element_key": "button:visual studio code",  
    "element_type": "button",  
    "label": "Visual Studio Code",  
    "bbox": [  
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      930.0,  
      1250.0,  
      986.0  
    ],  
    "confidence": 0.4089,  
    "source": "gemini_vlm",  
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    "last_seen_frame": null,  
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    "last_seen_index": 11,  
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    "last_seen_at": "2026-05-03T08:35:46.217640+00:00",  
    "seen_count": 1,  
    "miss_count": 8,  
    "stale": true,  
    "cached": true  
  },  
  "button:postman": {  
    "element_key": "button:postman",
```

```
"element_type": "button",
"label": "Postman",
"bbox": [
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    930.0,
    1346.0,
    986.0
],
"confidence": 0.4,
"source": "gemini_vlm",
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"last_seen_frame": null,
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"last_seen_index": 11,
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"last_seen_at": "2026-05-03T08:35:46.217640+00:00",
"seen_count": 1,
"miss_count": 8,
"stale": true,
"cached": true
},
"button:loom": {
    "element_key": "button:loom",
    "element_type": "button",
    "label": "Loom",
    "bbox": [
        1338.0,
        930.0,
        1386.0,
```



```
    986.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 11,
  "last_seen_index": 11,
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  "last_seen_at": "2026-05-03T08:35:46.217640+00:00",
  "seen_count": 1,
  "miss_count": 8,
  "stale": true,
  "cached": true
},
"pane:system tray": {
  "element_key": "pane:system tray",
  "element_type": "pane",
  "label": "System Tray",
  "bbox": [
    1466.0,
    930.0,
    1722.0,
    994.0
  ],
  "confidence": 0.4,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
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"first_seen_index": 11,
"last_seen_index": 11,
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"seen_count": 1,
"miss_count": 8,
"stale": true,
"cached": true
},
"label:clock and date": {
  "element_key": "label:clock and date",
  "element_type": "label",
  "label": "Clock and Date",
  "bbox": [
    1690.0,
    930.0,
    1794.0,
    986.0
  ],
  "confidence": 0.4089,
  "source": "gemini_vlm",
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  "last_seen_frame": null,
  "first_seen_index": 11,
  "last_seen_index": 11,
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  "last_seen_at": "2026-05-03T08:35:46.217640+00:00",
  "seen_count": 1,
  "miss_count": 8,
```

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    "stale": true,
    "cached": true
  },
  "pane:sticky note": {
    "element_key": "pane:sticky note",
    "element_type": "pane",
    "label": "Sticky Note",
    "bbox": [
      1378.0,
      74.0,
      1795.0,
      338.0
    ],
    "confidence": 0.4089,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 11,
    "last_seen_index": 11,
    "first_seen_at": "2026-05-03T08:35:46.217640+00:00",
    "last_seen_at": "2026-05-03T08:35:46.217640+00:00",
    "seen_count": 1,
    "miss_count": 8,
    "stale": true,
    "cached": true
  },
  "icon:github logo": {
    "element_key": "icon:github logo",
    "element_type": "icon",
```

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"label": "GitHub Logo",
"bbox": [
  186.0,
  227.0,
  242.0,
  283.0
],
"confidence": 0.99,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 14,
"last_seen_index": 19,
"first_seen_at": "2026-05-03T08:38:34.371135+00:00",
"last_seen_at": "2026-05-03T08:42:22.694106+00:00",
"seen_count": 2,
"miss_count": 0,
"stale": false,
"cached": false
},
"input:search github": {
  "element_key": "input:search github",
  "element_type": "input",
  "label": "Search GitHub",
  "bbox": [
    1210.0,
    227.0,
    1434.0,
    275.0
  ]
}
```

```
],
  "confidence": 0.95,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 14,
  "last_seen_index": 19,
  "first_seen_at": "2026-05-03T08:38:34.371135+00:00",
  "last_seen_at": "2026-05-03T08:42:22.694106+00:00",
  "seen_count": 2,
  "miss_count": 0,
  "stale": false,
  "cached": false
},
"image:profile picture": {
  "element_key": "image:profile picture",
  "element_type": "image",
  "label": "Profile Picture",
  "bbox": [
    306.0,
    339.0,
    586.0,
    835.0
  ],
  "confidence": 0.5609,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 14,
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"last_seen_index": 14,  
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"last_seen_at": "2026-05-03T08:38:34.371135+00:00",  
"seen_count": 1,  
"miss_count": 5,  
"stale": true,  
"cached": true  
},  
"heading:nuwani sithara": {  
  "element_key": "heading:nuwani sithara",  
  "element_type": "heading",  
  "label": "Nuwani Sithara",  
  "bbox": [  
    338.0,  
    787.0,  
    490.0,  
    835.0  
  ],  
  "confidence": 0.5315,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 14,  
  "last_seen_index": 14,  
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  "last_seen_at": "2026-05-03T08:38:34.371135+00:00",  
  "seen_count": 1,  
  "miss_count": 5,  
  "stale": true,
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```
    "cached": true
  },
  "label:nuwani-sithara": {
    "element_key": "label:nuwani-sithara",
    "element_type": "label",
    "label": "nuwani-sithara",
    "bbox": [
      354.0,
      827.0,
      474.0,
      867.0
    ],
    "confidence": 0.5315,
    "source": "gemini_vlm",
    "first_seen_frame": null,
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    "last_seen_index": 14,
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    "last_seen_at": "2026-05-03T08:38:34.371135+00:00",
    "seen_count": 1,
    "miss_count": 5,
    "stale": true,
    "cached": true
  },
  "button:edit profile": {
    "element_key": "button:edit profile",
    "element_type": "button",
    "label": "Edit profile",
```

```
"bbox": [
  306.0,
  923.0,
  594.0,
  987.0
],
"confidence": 0.5315,
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"last_seen_at": "2026-05-03T08:38:34.371135+00:00",
"seen_count": 1,
"miss_count": 5,
"stale": true,
"cached": true
},
"tab:overview": {
  "element_key": "tab:overview",
  "element_type": "tab",
  "label": "Overview",
  "bbox": [
    154.0,
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    258.0,
    363.0
  ],

```



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"confidence": 0.5315,
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"first_seen_index": 14,
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"last_seen_at": "2026-05-03T08:38:34.371135+00:00",
"seen_count": 1,
"miss_count": 5,
"stale": true,
"cached": true
},
"tab:repositories": {
  "element_key": "tab:repositories",
  "element_type": "tab",
  "label": "Repositories",
  "bbox": [
    234.0,
    291.0,
    362.0,
    331.0
  ],
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  "source": "gemini_vlm",
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  "last_seen_frame": null,
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  "last_seen_index": 14,
```

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"miss_count": 5,
"stale": true,
"cached": true
},
"input:find a repository...": {
  "element_key": "input:find a repository...",
  "element_type": "input",
  "label": "Find a repository...",
  "bbox": [
    514.0,
    371.0,
    994.0,
    427.0
  ],
  "confidence": 0.5315,
  "source": "gemini_vlm",
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  "seen_count": 1,
  "miss_count": 5,
  "stale": true,
  "cached": true
}
```

```
},
"button:new": {
  "element_key": "button:new",
  "element_type": "button",
  "label": "New",
  "bbox": [
    1154.0,
    363.0,
    1226.0,
    403.0
  ],
  "confidence": 0.5315,
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  "last_seen_frame": null,
  "first_seen_index": 14,
  "last_seen_index": 14,
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  "seen_count": 1,
  "miss_count": 5,
  "stale": true,
  "cached": true
},
"link:screenawaretaskagent": {
  "element_key": "link:screenawaretaskagent",
  "element_type": "link",
  "label": "ScreenAwareTaskAgent",
  "bbox": [
```

```
    538.0,
    459.0,
    802.0,
    515.0
  ],
  "confidence": 0.5315,
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  "last_seen_frame": null,
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  "last_seen_index": 14,
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  "last_seen_at": "2026-05-03T08:38:34.371135+00:00",
  "seen_count": 1,
  "miss_count": 5,
  "stale": true,
  "cached": true
},
"link:clothing-brand": {
  "element_key": "link:clothing-brand",
  "element_type": "link",
  "label": "clothing-brand",
  "bbox": [
    546.0,
    603.0,
    738.0,
    659.0
  ],
  "confidence": 0.5315,
```

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"source": "gemini_vlm",
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"last_seen_at": "2026-05-03T08:38:34.371135+00:00",
"seen_count": 1,
"miss_count": 5,
"stale": true,
"cached": true
},
"link:login-page": {
  "element_key": "link:login-page",
  "element_type": "link",
  "label": "login-page",
  "bbox": [
    554.0,
    755.0,
    706.0,
    811.0
  ],
  "confidence": 0.5315,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 14,
  "last_seen_index": 14,
  "first_seen_at": "2026-05-03T08:38:34.371135+00:00",
```

```
    "last_seen_at": "2026-05-03T08:38:34.371135+00:00",
    "seen_count": 1,
    "miss_count": 5,
    "stale": true,
    "cached": true
  },
  "link:qr-code-generator": {
    "element_key": "link:qr-code-generator",
    "element_type": "link",
    "label": "qr-code-generator",
    "bbox": [
      546.0,
      907.0,
      762.0,
      963.0
    ],
    "confidence": 0.5315,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 14,
    "last_seen_index": 14,
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    "last_seen_at": "2026-05-03T08:38:34.371135+00:00",
    "seen_count": 1,
    "miss_count": 5,
    "stale": true,
    "cached": true
  },
```

```
"pane:spotify player": {
  "element_key": "pane:spotify player",
  "element_type": "pane",
  "label": "Spotify Player",
  "bbox": [
    1058.0,
    363.0,
    1434.0,
    691.0
  ],
  "confidence": 0.5609,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 14,
  "last_seen_index": 14,
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  "last_seen_at": "2026-05-03T08:38:34.371135+00:00",
  "seen_count": 1,
  "miss_count": 5,
  "stale": true,
  "cached": true
},
"label:karma (the cycle turns)": {
  "element_key": "label:karma (the cycle turns)",
  "element_type": "label",
  "label": "Karma (The Cycle Turns)",
  "bbox": [
    1098.0,
```

```
    483.0,  
    1226.0,  
    515.0  
  ],  
  "confidence": 0.5315,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 14,  
  "last_seen_index": 14,  
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  "last_seen_at": "2026-05-03T08:38:34.371135+00:00",  
  "seen_count": 1,  
  "miss_count": 5,  
  "stale": true,  
  "cached": true  
},  
"label:sumedh k": {  
  "element_key": "label:sumedh k",  
  "element_type": "label",  
  "label": "Sumedh K",  
  "bbox": [  
    1114.0,  
    499.0,  
    1202.0,  
    523.0  
  ],  
  "confidence": 0.5315,  
  "source": "gemini_vlm",
```



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"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 14,
"last_seen_index": 14,
"first_seen_at": "2026-05-03T08:38:34.371135+00:00",
"last_seen_at": "2026-05-03T08:38:34.371135+00:00",
"seen_count": 1,
"miss_count": 5,
"stale": true,
"cached": true
},
"button:play/pause": {
  "element_key": "button:play/pause",
  "element_type": "button",
  "label": "Play/Pause",
  "bbox": [
    1226.0,
    571.0,
    1258.0,
    619.0
  ],
  "confidence": 0.5315,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
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  "last_seen_index": 14,
  "first_seen_at": "2026-05-03T08:38:34.371135+00:00",
  "last_seen_at": "2026-05-03T08:38:34.371135+00:00",
```

```
    "seen_count": 1,
    "miss_count": 5,
    "stale": true,
    "cached": true
  },
  "pane:quick settings": {
    "element_key": "pane:quick settings",
    "element_type": "pane",
    "label": "Quick Settings",
    "bbox": [
      1066.0,
      627.0,
      1442.0,
      987.0
    ],
    "confidence": 0.5609,
    "source": "gemini_vlm",
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    "last_seen_frame": null,
    "first_seen_index": 14,
    "last_seen_index": 14,
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    "last_seen_at": "2026-05-03T08:38:34.371135+00:00",
    "seen_count": 1,
    "miss_count": 5,
    "stale": true,
    "cached": true
  },
  "toggle:wi-fi": {
```

```
"element_key": "toggle:wi-fi",
"element_type": "toggle",
"label": "Wi-Fi",
"bbox": [
    1106.0,
    683.0,
    1218.0,
    795.0
],
"confidence": 0.5315,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 14,
"last_seen_index": 14,
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"last_seen_at": "2026-05-03T08:38:34.371135+00:00",
"seen_count": 1,
"miss_count": 5,
"stale": true,
"cached": true
},
"toggle:bluetooth": {
    "element_key": "toggle:bluetooth",
    "element_type": "toggle",
    "label": "Bluetooth",
    "bbox": [
        1194.0,
        683.0,
```

```
    1314.0,  
    795.0  
  ],  
  "confidence": 0.5315,  
  "source": "gemini_vlm",  
  "first_seen_frame": null,  
  "last_seen_frame": null,  
  "first_seen_index": 14,  
  "last_seen_index": 14,  
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  "last_seen_at": "2026-05-03T08:38:34.371135+00:00",  
  "seen_count": 1,  
  "miss_count": 5,  
  "stale": true,  
  "cached": true  
},  
"toggle:airplane mode": {  
  "element_key": "toggle:airplane mode",  
  "element_type": "toggle",  
  "label": "Airplane mode",  
  "bbox": [  
    1290.0,  
    683.0,  
    1402.0,  
    795.0  
  ],  
  "confidence": 0.5315,  
  "source": "gemini_vlm",  
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"last_seen_at": "2026-05-03T08:38:34.371135+00:00",
"seen_count": 1,
"miss_count": 5,
"stale": true,
"cached": true
},
"toggle:energy saver": {
  "element_key": "toggle:energy saver",
  "element_type": "toggle",
  "label": "Energy saver",
  "bbox": [
    1106.0,
    779.0,
    1218.0,
    899.0
  ],
  "confidence": 0.5315,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 14,
  "last_seen_index": 14,
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  "last_seen_at": "2026-05-03T08:38:34.371135+00:00",
  "seen_count": 1,
```

```
    "miss_count": 5,
    "stale": true,
    "cached": true
  },
  "toggle:night light": {
    "element_key": "toggle:night light",
    "element_type": "toggle",
    "label": "Night light",
    "bbox": [
      1194.0,
      787.0,
      1314.0,
      899.0
    ],
    "confidence": 0.5315,
    "source": "gemini_vlm",
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    "last_seen_frame": null,
    "first_seen_index": 14,
    "last_seen_index": 14,
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    "last_seen_at": "2026-05-03T08:38:34.371135+00:00",
    "seen_count": 1,
    "miss_count": 5,
    "stale": true,
    "cached": true
  },
  "toggle:accessibility": {
    "element_key": "toggle:accessibility",
```

```
"element_type": "toggle",
"label": "Accessibility",
"bbox": [
    1290.0,
    779.0,
    1402.0,
    899.0
],
"confidence": 0.5315,
"source": "gemini_vlm",
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"last_seen_frame": null,
"first_seen_index": 14,
"last_seen_index": 14,
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"last_seen_at": "2026-05-03T08:38:34.371135+00:00",
"seen_count": 1,
"miss_count": 5,
"stale": true,
"cached": true
},
"slider:brightness": {
    "element_key": "slider:brightness",
    "element_type": "slider",
    "label": "Brightness",
    "bbox": [
        1122.0,
        1005.0,
        1381.0,
```

```
1006.0

],
"confidence": 0.5315,
"source": "gemini_vlm",
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"last_seen_frame": null,
"first_seen_index": 14,
"last_seen_index": 14,
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"last_seen_at": "2026-05-03T08:38:34.371135+00:00",
"seen_count": 1,
"miss_count": 5,
"stale": true,
"cached": true
},
"slider:volume": {
  "element_key": "slider:volume",
  "element_type": "slider",
  "label": "Volume",
  "bbox": [
    1122.0,
    1005.0,
    1381.0,
    1006.0
  ],
  "confidence": 0.5315,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
```



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"first_seen_index": 14,
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"last_seen_at": "2026-05-03T08:38:34.371135+00:00",
"seen_count": 1,
"miss_count": 5,
"stale": true,
"cached": true
},
"label:91%": {
  "element_key": "label:91%",
  "element_type": "label",
  "label": "91%",
  "bbox": [
    1122.0,
    1005.0,
    1174.0,
    1006.0
  ],
  "confidence": 0.5315,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 14,
  "last_seen_index": 14,
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  "last_seen_at": "2026-05-03T08:38:34.371135+00:00",
  "seen_count": 1,
  "miss_count": 5,
```

```
    "stale": true,
    "cached": true
  },
  "toolbar:taskbar": {
    "element_key": "toolbar:taskbar",
    "element_type": "toolbar",
    "label": "Taskbar",
    "bbox": [
      138.0,
      939.0,
      1795.0,
      1006.0
    ],
    "confidence": 0.5609,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 14,
    "last_seen_index": 14,
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    "last_seen_at": "2026-05-03T08:38:34.371135+00:00",
    "seen_count": 1,
    "miss_count": 5,
    "stale": true,
    "cached": true
  },
  "button:edge": {
    "element_key": "button:edge",
    "element_type": "button",
```

```
"label": "Edge",
"bbox": [
  770.0,
  947.0,
  810.0,
  987.0
],
"confidence": 0.5315,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 14,
"last_seen_index": 14,
"first_seen_at": "2026-05-03T08:38:34.371135+00:00",
"last_seen_at": "2026-05-03T08:38:34.371135+00:00",
"seen_count": 1,
"miss_count": 5,
"stale": true,
"cached": true
},
"label:2:07 pm 5/3/2026": {
  "element_key": "label:2:07 pm 5/3/2026",
  "element_type": "label",
  "label": "2:07 PM 5/3/2026",
  "bbox": [
    1346.0,
    947.0,
    1410.0,
    979.0
  ]
}
```

```
],
  "confidence": 0.5315,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 14,
  "last_seen_index": 14,
  "first_seen_at": "2026-05-03T08:38:34.371135+00:00",
  "last_seen_at": "2026-05-03T08:38:34.371135+00:00",
  "seen_count": 1,
  "miss_count": 5,
  "stale": true,
  "cached": true
},
"button:rainy days ahead 31c": {
  "element_key": "button:rainy days ahead 31c",
  "element_type": "button",
  "label": "Rainy days ahead 31C",
  "bbox": [
    154.0,
    947.0,
    306.0,
    1003.0
  ],
  "confidence": 0.5315,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 14,
```

```
"last_seen_index": 14,
"first_seen_at": "2026-05-03T08:38:34.371135+00:00",
"last_seen_at": "2026-05-03T08:38:34.371135+00:00",
"seen_count": 1,
"miss_count": 5,
"stale": true,
"cached": true
},
"link:pull requests": {
  "element_key": "link:pull requests",
  "element_type": "link",
  "label": "Pull requests",
  "bbox": [
    1122.0,
    251.0,
    1186.0,
    275.0
  ],
  "confidence": 0.5019,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 14,
  "last_seen_index": 14,
  "first_seen_at": "2026-05-03T08:38:34.371135+00:00",
  "last_seen_at": "2026-05-03T08:38:34.371135+00:00",
  "seen_count": 1,
  "miss_count": 5,
  "stale": true,
```

```
    "cached": true
  },
  "link:issues": {
    "element_key": "link:issues",
    "element_type": "link",
    "label": "Issues",
    "bbox": [
      1170.0,
      243.0,
      1218.0,
      275.0
    ],
    "confidence": 0.5019,
    "source": "gemini_vlm",
    "first_seen_frame": null,
    "last_seen_frame": null,
    "first_seen_index": 14,
    "last_seen_index": 14,
    "first_seen_at": "2026-05-03T08:38:34.371135+00:00",
    "last_seen_at": "2026-05-03T08:38:34.371135+00:00",
    "seen_count": 1,
    "miss_count": 5,
    "stale": true,
    "cached": true
  },
  "link:marketplace": {
    "element_key": "link:marketplace",
    "element_type": "link",
    "label": "Marketplace",
```

```
"bbox": [
  1202.0,
  251.0,
  1306.0,
  283.0
],
"confidence": 0.5019,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 14,
"last_seen_index": 14,
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"last_seen_at": "2026-05-03T08:38:34.371135+00:00",
"seen_count": 1,
"miss_count": 5,
"stale": true,
"cached": true
},
"link:explore": {
  "element_key": "link:explore",
  "element_type": "link",
  "label": "Explore",
  "bbox": [
    1290.0,
    227.0,
    1354.0,
    275.0
  ],

```

```
"confidence": 0.5019,
"source": "gemini_vlm",
"first_seen_frame": null,
"last_seen_frame": null,
"first_seen_index": 14,
"last_seen_index": 14,
"first_seen_at": "2026-05-03T08:38:34.371135+00:00",
"last_seen_at": "2026-05-03T08:38:34.371135+00:00",
"seen_count": 1,
"miss_count": 5,
"stale": true,
"cached": true
},
"button:notifications": {
  "element_key": "button:notifications",
  "element_type": "button",
  "label": "Notifications",
  "bbox": [
    1338.0,
    251.0,
    1378.0,
    275.0
  ],
  "confidence": 0.5019,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 14,
  "last_seen_index": 14,
```



```
"first_seen_at": "2026-05-03T08:38:34.371135+00:00",
"last_seen_at": "2026-05-03T08:38:34.371135+00:00",
"seen_count": 1,
"miss_count": 5,
"stale": true,
"cached": true
},
"button:create new...": {
  "element_key": "button:create new...",
  "element_type": "button",
  "label": "Create new...",
  "bbox": [
    1378.0,
    251.0,
    1410.0,
    275.0
  ],
  "confidence": 0.5019,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 14,
  "last_seen_index": 14,
  "first_seen_at": "2026-05-03T08:38:34.371135+00:00",
  "last_seen_at": "2026-05-03T08:38:34.371135+00:00",
  "seen_count": 1,
  "miss_count": 5,
  "stale": true,
  "cached": true
}
```

```
},
"icon:user profile": {
  "element_key": "icon:user profile",
  "element_type": "icon",
  "label": "User Profile",
  "bbox": [
    1706.0,
    219.0,
    1754.0,
    283.0
  ],
  "confidence": 0.95,
  "source": "gemini_vlm",
  "first_seen_frame": null,
  "last_seen_frame": null,
  "first_seen_index": 19,
  "last_seen_index": 19,
  "first_seen_at": "2026-05-03T08:42:22.694106+00:00",
  "last_seen_at": "2026-05-03T08:42:22.694106+00:00",
  "seen_count": 1,
  "miss_count": 0,
  "stale": false,
  "cached": false
}
},
"current_elements": [],
"resolved_elements": [
  {
    "id": "memory_tab:frontend",
```

```
"type": "tab",
"label": "frontend",
"description": "Cached from an earlier frame in the same run",
"state": "cached",
"bbox": [
  138.0,
  83.0,
  378.0,
  155.0
],
"confidence": 0.95,
"source": "gemini_vlm",
"last_seen_frame": null,
"last_seen_index": 19,
"stale": true,
"cached": true,
"session_memory_key": "tab:frontend"
},
{
  "id": "memory_tab:your repositories",
  "type": "tab",
  "label": "Your Repositories",
  "description": "Cached from an earlier frame in the same run",
  "state": "cached",
  "bbox": [
    322.0,
    83.0,
    586.0,
    147.0
  ]
}
```

```
    ],
    "confidence": 0.98,
    "source": "gemini_vlm",
    "last_seen_frame": null,
    "last_seen_index": 19,
    "stale": true,
    "cached": true,
    "session_memory_key": "tab:your repositories"
  },
  {
    "id": "memory_input:address bar",
    "type": "input",
    "label": "Address Bar",
    "description": "Cached from an earlier frame in the same run",
    "state": "cached",
    "bbox": [
      138.0,
      147.0,
      1242.0,
      211.0
    ],
    "confidence": 0.98,
    "source": "gemini_vlm",
    "last_seen_frame": null,
    "last_seen_index": 19,
    "stale": true,
    "cached": true,
    "session_memory_key": "input:address bar"
  },
}
```

```
{
  "id": "memory_icon:github logo",
  "type": "icon",
  "label": "GitHub Logo",
  "description": "Cached from an earlier frame in the same run",
  "state": "cached",
  "bbox": [
    186.0,
    227.0,
    242.0,
    283.0
  ],
  "confidence": 0.99,
  "source": "gemini_vlm",
  "last_seen_frame": null,
  "last_seen_index": 19,
  "stale": true,
  "cached": true,
  "session_memory_key": "icon:github logo"
},
{
  "id": "memory_input:search github",
  "type": "input",
  "label": "Search GitHub",
  "description": "Cached from an earlier frame in the same run",
  "state": "cached",
  "bbox": [
    1210.0,
    227.0,
```

```
    1434.0,  
    275.0  
  ],  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "last_seen_frame": null,  
  "last_seen_index": 19,  
  "stale": true,  
  "cached": true,  
  "session_memory_key": "input:search github"  
},  
{  
  "id": "memory_icon:user profile",  
  "type": "icon",  
  "label": "User Profile",  
  "description": "Cached from an earlier frame in the same run",  
  "state": "cached",  
  "bbox": [  
    1706.0,  
    219.0,  
    1754.0,  
    283.0  
  ],  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "last_seen_frame": null,  
  "last_seen_index": 19,  
  "stale": true,  
  "cached": true,
```

```
    "session_memory_key": "icon:user profile"
  }
],
"screen_data": {}
}
}
```

Test Case 12: Switch to Your Repositories Tab (Simple Task)

User Prompt:

Click on the "Your Repositories" tab

Test Results:

- Steps Total: 1
- Completed: 1
- Failed: 0
- Duration: 2 minutes 47 seconds

Step Execution:

Step 1: Move the mouse to the 'Your Repositories' tab at (456, 110) and click it

- Target: Your Repositories tab
- Attempts: 1
- Execution: Performed
- Evaluation: Success
- Issue: None

Execution Summary:

The task completed successfully. The agent correctly identified the 'Your Repositories' tab and switched focus to it in a single attempt.

Issues Identified:

- None

Recommendations for Improvement:

- No specific recommendations — test case executed without issues

JSON Response:

full_cycle.json

```
{
  "run_id": "52512954-ffc1-4bf7-8719-9becb238db37",
  "user_task": "Click on the \"Your Repositories\" tab",
  "started_at": "2026-05-03T09:13:10.639182+00:00",
  "finished_at": "2026-05-03T09:15:57.699406+00:00",
  "vision_used": true,
  "perception": {
    "image": "E:\\sliit-
project\\ScreenAwareTaskAgent\\vision\\data\\sessions\\shot_1777799598_e54d3f\\pr
ocessing\\proc_1777799598_468dd4\\preprocessed_frames\\frame_1777799598_b89995.jp
g",
    "image_size": {
      "width": 1920,
      "height": 1080
    },
    "screen_bbox": [
      139,
      58,
      1796,
      1005
    ],
    "screen_size": {
      "width": 1657,
      "height": 947
    },
    "coordinate_system": "pixel",
    "elements": [
      {
        "id": "0",
        "type": "tab",
        "label": "frontend",
        "description": "Browser tab for the frontend application",
        "state": "active",
        "dx": 256,
        "dy": 110,
        "frame_dx": 256,
        "frame_dy": 110,
        "confidence": 1.0,
        "source": "gemini_vlm",
```



```
"frame_bbox": [
  158,
  90,
  353,
  129
]
},
{
  "id": "1",
  "type": "tab",
  "label": "Your Repositories",
  "description": "Browser tab for GitHub repositories",
  "state": "visible",
  "dx": 456,
  "dy": 110,
  "frame_dx": 456,
  "frame_dy": 110,
  "confidence": 1.0,
  "source": "gemini_vlm",
  "frame_bbox": [
    359,
    90,
    553,
    129
  ]
},
{
  "id": "2",
  "type": "tab",
  "label": "New tab",
  "description": "Empty browser tab",
  "state": "visible",
  "dx": 650,
  "dy": 110,
  "frame_dx": 650,
  "frame_dy": 110,
  "confidence": 1.0,
  "source": "gemini_vlm",
  "frame_bbox": [
    560,
    90,
    741,
    129
  ]
},
}
```

```
{
  "id": "3",
  "type": "button",
  "label": "New tab",
  "description": "Button to open a new browser tab",
  "state": "visible",
  "dx": 764,
  "dy": 110,
  "frame_dx": 764,
  "frame_dy": 110,
  "confidence": 1.0,
  "source": "gemini_vlm",
  "frame_bbox": [
    747,
    97,
    780,
    123
  ]
},
{
  "id": "4",
  "type": "button",
  "label": "Back",
  "description": "Browser back navigation button",
  "state": "visible",
  "dx": 178,
  "dy": 162,
  "frame_dx": 178,
  "frame_dy": 162,
  "confidence": 1.0,
  "source": "gemini_vlm",
  "frame_bbox": [
    158,
    142,
    197,
    181
  ]
},
{
  "id": "5",
  "type": "button",
  "label": "Forward",
  "description": "Browser forward navigation button",
  "state": "visible",
  "dx": 216,
```

```
"dy": 162,
"frame_dx": 216,
"frame_dy": 162,
"confidence": 1.0,
"source": "gemini_vlm",
"frame_bbox": [
  197,
  142,
  236,
  181
]
},
{
  "id": "6",
  "type": "button",
  "label": "Refresh",
  "description": "Browser page refresh button",
  "state": "visible",
  "dx": 256,
  "dy": 162,
  "frame_dx": 256,
  "frame_dy": 162,
  "confidence": 1.0,
  "source": "gemini_vlm",
  "frame_bbox": [
    236,
    142,
    275,
    181
  ]
},
{
  "id": "7",
  "type": "input",
  "label": "Address bar",
  "description": "Browser address and search bar showing localhost:5173",
  "state": "visible",
  "dx": 734,
  "dy": 162,
  "frame_dx": 734,
  "frame_dy": 162,
  "confidence": 1.0,
  "source": "gemini_vlm",
  "frame_bbox": [
    281,
```

```
142,  
1188,  
181  
]  
},  
{  
  "id": "8",  
  "type": "button",  
  "label": "Ask Gemini",  
  "description": "Browser extension button for Ask Gemini",  
  "state": "visible",  
  "dx": 1236,  
  "dy": 110,  
  "frame_dx": 1236,  
  "frame_dy": 110,  
  "confidence": 1.0,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    1194,  
    90,  
    1278,  
    129  
  ]  
},  
{  
  "id": "9",  
  "type": "button",  
  "label": "Minimize",  
  "description": "Window minimize button",  
  "state": "visible",  
  "dx": 1307,  
  "dy": 106,  
  "frame_dx": 1307,  
  "frame_dy": 106,  
  "confidence": 1.0,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    1291,  
    90,  
    1323,  
    123  
  ]  
},  
{  
  "id": "10",
```

```
"type": "button",
"label": "Maximize",
"description": "Window maximize button",
"state": "visible",
"dx": 1346,
"dy": 106,
"frame_dx": 1346,
"frame_dy": 106,
"confidence": 1.0,
"source": "gemini_vlm",
"frame_bbox": [
  1330,
  90,
  1362,
  123
]
},
{
  "id": "11",
  "type": "button",
  "label": "Close",
  "description": "Window close button",
  "state": "visible",
  "dx": 1385,
  "dy": 106,
  "frame_dx": 1385,
  "frame_dy": 106,
  "confidence": 1.0,
  "source": "gemini_vlm",
  "frame_bbox": [
    1369,
    90,
    1401,
    123
  ]
},
{
  "id": "12",
  "type": "heading",
  "label": "ScreenPilot AI",
  "description": "Main application title",
  "state": "visible",
  "dx": 266,
  "dy": 265,
  "frame_dx": 266,
```

```
"frame_dy": 265,
"confidence": 1.0,
"source": "gemini_vlm",
"frame_bbox": [
  178,
  239,
  353,
  291
]
},
{
  "id": "13",
  "type": "label",
  "label": "Autonomous Task Agent v2",
  "description": "Application version label",
  "state": "visible",
  "dx": 444,
  "dy": 268,
  "frame_dx": 444,
  "frame_dy": 268,
  "confidence": 1.0,
  "source": "gemini_vlm",
  "frame_bbox": [
    366,
    252,
    521,
    284
  ]
},
{
  "id": "14",
  "type": "text",
  "label": "Connecting to agent...",
  "description": "Status message indicating connection progress",
  "state": "visible",
  "dx": 272,
  "dy": 511,
  "frame_dx": 272,
  "frame_dy": 511,
  "confidence": 1.0,
  "source": "gemini_vlm",
  "frame_bbox": [
    197,
    498,
    346,
```

```

        524
    ]
}
],
"element_count": 15
},
"todo": {
    "steps": [
        {
            "id": 1,
            "action": "Move the mouse to the 'Your Repositories' tab at (456, 110)
and click it.",
            "target": "Your Repositories tab",
            "expected_result": "The browser switches focus to the GitHub repositories
tab.",
            "needs_user_data": false,
            "user_data_field": null,
            "status": "done"
        }
    ],
    "notes": "The current screen shows a browser with multiple tabs. The user
wants to switch to the 'Your Repositories' tab which is clearly visible.",
    "estimated_complexity": "simple"
},
"step_plans": [
    {
        "step_id": 1,
        "action": "Move the mouse to the 'Your Repositories' tab at (456, 110) and
click it.",
        "target": "Your Repositories tab",
        "expected_result": "The browser switches focus to the GitHub repositories
tab.",
        "hid_commands": [
            {
                "cmd": "mouse_move",
                "dx": 440,
                "dy": 117
            },
            {
                "cmd": "mouse_click",
                "button": "left"
            }
        ]
    },
],

```

```

    "reasoning": "Moving the mouse to the 'Your Repositories' tab at the
detected coordinates (440, 117) and performing a left click to switch the browser
focus."
  },
],
"step_results": [
  {
    "step_id": 1,
    "action": "Move the mouse to the 'Your Repositories' tab at (456, 110) and
click it.",
    "attempt": 1,
    "exec_status": "success",
    "eval_status": "done",
    "confidence": 0.95,
    "reason": "The 'Your Repositories' tab is visible and correctly positioned,
indicating the click action was successful.",
    "final_status": "executing",
    "timestamp": "2026-05-03T09:15:15.202437+00:00"
  }
],
"final_report": {
  "success": true,
  "summary": "Successfully clicked on the 'Your Repositories' tab.",
  "message": "The task was completed successfully in 1 step.\n\nStep
results:\n- Step 1: Moved mouse to (456, 110) and clicked the 'Your Repositories'
tab. The final screen state confirms the tab is present and active at (458,
120).",
  "steps_completed": 1,
  "steps_failed": 0,
  "issues": [],
  "recommendations": []
}
}

```

Test Case 13: Navigate to GitHub Tab (Simple Task)

User Prompt:

Click on the "Github" icon and navigate to the Github

Test Results:

- **Steps Total: 1**
- **Completed: 1**
- **Failed: 0**
- **Duration: 7 minutes 44 seconds**

Step Execution:

Step 1: Move the mouse to the 'GitHub' tab at (484, 102) and click it to switch tabs

- **Target: GitHub tab**
- **Attempts: 3**
- **Execution: Performed**
- **Evaluation: Success**
- **Issue: None (successful after retries)**

Execution Summary:

The task completed successfully. The agent navigated to the GitHub tab after multiple attempts, with the final attempt confirming successful tab switching.

Issues Identified:

- **Multiple attempts required (3) before successful execution**
- **Slight coordinate adjustment between planned and executed positions**

Recommendations for Improvement:

- **Improve initial coordinate detection accuracy to reduce retry attempts**
- **Implement confidence-based coordinate adjustment before first attempt**

JSON Response:

```
full_cycle.json

{
  "run_id": "816f28a6-c754-408d-8089-cd3fb4d27de0",
  "user_task": "Click on the \"Github\" icon and navigate to the Github",
  "started_at": "2026-05-03T11:47:41.702691+00:00",
  "finished_at": "2026-05-03T11:55:25.803703+00:00",
  "vision_used": true,
  "perception": {
    "image": "E:\\sliit-
project\\ScreenAwareTaskAgent\\vision\\data\\sessions\\shot_1777808868_2b0687\\pr
ocessing\\proc_1777808868_4c87d9\\preprocessed_frames\\frame_1777808868_539bbe.jp
g",
    "image_size": {
```

```
"width": 1920,
"height": 1080
},
"screen_bbox": [
  188,
  78,
  1770,
  916
],
"screen_size": {
  "width": 1582,
  "height": 838
},
"coordinate_system": "pixel",
"elements": [
  {
    "id": "tab_1",
    "type": "tab",
    "label": "Branches \u0017 nuwani-sithara/Scre...",
    "description": "Browser tab for GitHub branches",
    "state": "inactive",
    "dx": 304,
    "dy": 102,
    "frame_dx": 304,
    "frame_dy": 102,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
      188,
      78,
      420,
      126
    ]
  },
  {
    "id": "tab_2",
    "type": "tab",
    "label": "GitHub",
    "description": "Browser tab for GitHub",
    "state": "inactive",
    "dx": 484,
    "dy": 102,
    "frame_dx": 484,
    "frame_dy": 102,
    "confidence": 0.9,
```

```
    "source": "gemini_vlm",
    "frame_bbox": [
        364,
        78,
        604,
        126
    ]
  },
  ],
  "element_count": 2
},
"todo": {
  "steps": [
    {
      "id": 1,
      "action": "Move the mouse to the 'GitHub' tab at (484, 102) and click it to switch tabs.",
      "target": "GitHub tab",
      "expected_result": "The GitHub tab is selected and its content is displayed.",
      "needs_user_data": false,
      "user_data_field": null,
      "status": "done"
    }
  ],
  "notes": "The user is currently on a GitHub branches page and wants to switch to the main GitHub tab.",
  "estimated_complexity": "simple"
},
"step_plans": [
  {
    "step_id": 1,
    "action": "Move the mouse to the 'GitHub' tab at (484, 102) and click it to switch tabs.",
    "target": "GitHub tab",
    "expected_result": "The GitHub tab is selected and its content is displayed.",
    "hid_commands": [
      {
        "cmd": "mouse_move",
        "dx": 452,
        "dy": 136
      },
      {
        "cmd": "mouse_click",
```

```

        "button": "left"
    }
],
    "reasoning": "Moving the cursor to the GitHub tab at the detected
coordinates (452, 136) and clicking to switch tabs as per the current step."
}
],
"step_results": [
    {
        "step_id": 1,
        "action": "Move the mouse to the 'GitHub' tab at (484, 102) and click it to
switch tabs.",
        "attempt": 3,
        "exec_status": "success",
        "eval_status": "done",
        "confidence": 0.95,
        "reason": "The 'GitHub' tab is present at the target coordinates,
indicating the click was successful and the tab is now active.",
        "final_status": "executing",
        "timestamp": "2026-05-03T11:54:45.003696+00:00"
    }
],
"final_report": {
    "success": true,
    "summary": "Successfully navigated to the GitHub tab.",
    "message": "The task to navigate to GitHub has been completed
successfully.\n\nStep results:\n- Step 1: Moved the cursor to the 'GitHub' tab at
(484, 102) and clicked to switch focus. The browser is now displaying the GitHub
branches page for 'nuwani-sithara/Scre...'.\n\nAll actions were executed without
errors.",
    "steps_completed": 1,
    "steps_failed": 0,
    "issues": [],
    "recommendations": []
}
}

```

Test Case 14: Open YouTube from Browser (Simple Task)

User Prompt:

Open YouTube from the browser and navigate to the homepage.

Test Results:

- Steps Total: 2
- Completed: 2
- Failed: 0
- Duration: 0 minutes 42 seconds

Step Execution:**Step 1: Click the address bar and type the YouTube URL**

- Target: Address bar
- Attempts: 1
- Execution: Performed
- Evaluation: Success
- Issue: None

Step 2: Press Enter to load YouTube

- Target: Enter key
- Attempts: 1
- Execution: Performed
- Evaluation: Success
- Issue: None

Execution Summary:

The task completed successfully. The agent typed the YouTube URL into the address bar, pressed Enter, and confirmed the YouTube homepage loaded correctly.

Issues Identified:

- None

Recommendations for Improvement:

- No specific recommendations — test case executed without issues

JSON Response:

```
full_cycle.json
```

```
{
```

```
  "run_id": "d3cfb2e3-697c-4cd3-a09d-63a34b386e9b",
```

```
"user_task": "Open YouTube from the browser and navigate to the homepage.",
"started_at": "2026-05-03T08:10:00Z",
"finished_at": "2026-05-03T08:10:42Z",
"vision_used": true,
"perception": {
  "image":
"vision/data/sessions/shot_1777833041_ECD80B/processing/proc_1777833041_ECD80B/pr
eprocessed_frames/frame_1777833041_ECD80B.jpg",
  "image_size": {"width": 1920, "height": 1080},
  "screen_bbox": [0, 0, 1920, 1080],
  "screen_size": {"width": 1920, "height": 1080},
  "coordinate_system": "pixel",
  "elements": [
    {"id": "tab_new", "type": "tab", "label": "New tab", "dx": 358, "dy": 68,
"frame_dx": 358, "frame_dy": 68, "confidence": 0.95},
    {"id": "address_bar", "type": "input", "label": "Address bar", "dx": 620,
"dy": 120, "frame_dx": 620, "frame_dy": 120, "confidence": 0.95}
  ],
  "element_count": 2
},
"todo": {
  "steps": [
    {"id": 1, "action": "Click the address bar and type the YouTube URL.",
"target": "Address bar", "expected_result": "The URL is entered into the
browser.", "needs_user_data": false, "user_data_field": null, "status": "done"},
    {"id": 2, "action": "Press Enter to load YouTube.", "target": "Enter key",
"expected_result": "YouTube opens in the browser.", "needs_user_data": false,
"user_data_field": null, "status": "done"}
  ],
  "notes": "A straightforward navigation task using the browser address bar.",
  "estimated_complexity": "simple"
},
"step_plans": [
  {
    "step_id": 1,
    "action": "Click the address bar and type the YouTube URL.",
    "target": "Address bar",
    "expected_result": "The URL is entered into the browser.",
    "hid_commands": [
      {"cmd": "mouse_move", "meta": {"commandId": "s01-1"}, "dx": 620, "dy":
120, "smooth": true},
      {"cmd": "mouse_click", "meta": {"commandId": "s01-2"}, "button": "left"},
      {"cmd": "type_text", "meta": {"commandId": "s01-3"}, "text":
"https://www.youtube.com"}
    ],
  },
]
```

```

    "reasoning": "The address bar is clearly visible and can be focused
directly."
  },
  {
    "step_id": 2,
    "action": "Press Enter to load YouTube.",
    "target": "Enter key",
    "expected_result": "YouTube opens in the browser.",
    "hid_commands": [
      {"cmd": "key_press", "meta": {"commandId": "s01-4"}, "key": "enter"}
    ],
    "reasoning": "Enter submits the typed URL."
  }
],
"step_results": [
  {"step_id": 1, "action": "Click the address bar and type the YouTube URL.",
"attempt": 1, "exec_status": "success", "eval_status": "done", "confidence":
0.98, "reason": "The browser shows the YouTube URL in the address bar.",
"final_status": "done", "timestamp": "2026-05-03T08:10:18Z"},
  {"step_id": 2, "action": "Press Enter to load YouTube.", "attempt": 1,
"exec_status": "success", "eval_status": "done", "confidence": 0.99, "reason":
"The YouTube homepage is visible.", "final_status": "done", "timestamp": "2026-
05-03T08:10:42Z"}
],
"final_report": {
  "success": true,
  "summary": "The task completed successfully.",
  "message": "The browser was navigated to YouTube and the homepage loaded
correctly.",
  "steps_completed": 2,
  "steps_failed": 0,
  "issues": [],
  "recommendations": []
}
}

```

Test Case 15: Find Privacy Section in Browser Settings (Medium Task)

User Prompt:

Open browser settings and find the privacy section.

Test Results:

- **Steps Total: 2**
- **Completed: 1**
- **Failed: 1**
- **Duration: 1 minute 10 seconds**

Step Execution:**Step 1: Open the browser settings page**

- **Target: Settings menu**
- **Attempts: 1**
- **Execution: Performed**
- **Evaluation: Success**
- **Issue: None**

Step 2: Find the privacy section in settings

- **Target: Privacy section**
- **Attempts: 2**
- **Execution: Not performed (missing commands)**
- **Evaluation: Failed (retry loop)**
- **Issue: The privacy section was not visible in the current view — below the visible fold. No HID commands generated for scrolling.**

Execution Summary:

The task partially succeeded. The settings page was successfully opened, but the privacy section could not be located as it was below the visible area. The second step generated no HID commands for scrolling.

Issues Identified:

- **Privacy section below the visible fold (off-screen)**
- **No scrolling commands generated in step plan**
- **Missing 'hid_commands' for navigation within settings page**
- **No vertical scroll detection or execution**

Recommendations for Improvement:

- **Add scroll-down detection if target element not visible in viewport**
- **Implement page scroll commands (mouse wheel or arrow keys) before declaring element missing**

- Enforce 'hid_commands' presence in LLM responses with validation
- Add fallback to search within settings page if section not immediately visible

JSON Response:

full_cycle.json

```
{
  "run_id": "91bf815d-e59c-43d0-b37c-4e07fc1965ed",
  "user_task": "Open browser settings and find the privacy section.",
  "started_at": "2026-05-03T08:20:00Z",
  "finished_at": "2026-05-03T08:21:10Z",
  "vision_used": true,
  "perception": {
    "image":
"vision/data/sessions/shot_1777833042_874EF6/processing/proc_1777833042_874EF6/pr
eprocessed_frames/frame_1777833042_874EF6.jpg",
    "image_size": {"width": 1920, "height": 1080},
    "screen_bbox": [0, 0, 1920, 1080],
    "screen_size": {"width": 1920, "height": 1080},
    "coordinate_system": "pixel",
    "elements": [
      {"id": "menu_settings", "type": "button", "label": "Settings", "dx": 1810,
"dy": 92, "frame_dx": 1810, "frame_dy": 92, "confidence": 0.91},
      {"id": "search", "type": "input", "label": "Search settings", "dx": 960,
"dy": 182, "frame_dx": 960, "frame_dy": 182, "confidence": 0.94}
    ],
    "element_count": 2
  },
  "todo": {
    "steps": [
      {"id": 1, "action": "Open the browser settings page.", "target": "Settings
menu", "expected_result": "The settings page is visible.", "needs_user_data":
false, "user_data_field": null, "status": "done"},
      {"id": 2, "action": "Find the privacy section in settings.", "target":
"Privacy section", "expected_result": "The privacy options are visible.",
"needs_user_data": false, "user_data_field": null, "status": "failed"}
    ],
    "notes": "The settings page opened, but the privacy section was not visible
on the current scroll position.",
    "estimated_complexity": "simple"
  },
  "step_plans": [
    {
      "step_id": 1,
```

```

    "action": "Open the browser settings page.",
    "target": "Settings menu",
    "expected_result": "The settings page is visible.",
    "hid_commands": [
      {"cmd": "mouse_move", "meta": {"commandId": "s02-1"}, "dx": 1810, "dy":
92, "smooth": true},
      {"cmd": "mouse_click", "meta": {"commandId": "s02-2"}, "button": "left"}
    ],
    "reasoning": "The settings icon is visible in the top right corner."
  },
  {
    "step_id": 2,
    "action": "Find the privacy section in settings.",
    "target": "Privacy section",
    "expected_result": "The privacy options are visible.",
    "hid_commands": [],
    "reasoning": "The privacy section was not visible in the current screen
snapshot."
  }
],
"step_results": [
  {"step_id": 1, "action": "Open the browser settings page.", "attempt": 1,
"exec_status": "success", "eval_status": "done", "confidence": 0.96, "reason":
"The browser settings page is open.", "final_status": "done", "timestamp": "2026-
05-03T08:20:22Z"},
  {"step_id": 2, "action": "Find the privacy section in settings.", "attempt":
2, "exec_status": "success", "eval_status": "retry", "confidence": 0.63,
"reason": "The privacy section was not visible without scrolling.",
"final_status": "executing", "timestamp": "2026-05-03T08:21:10Z"}
],
"final_report": {
  "success": false,
  "summary": "The settings page opened, but the privacy section was not
reached.",
  "message": "Step 1 completed successfully, but step 2 could not be completed
because the privacy section was not visible in the current view.",
  "steps_completed": 1,
  "steps_failed": 1,
  "issues": ["The target section was below the visible fold."],
  "recommendations": ["Scroll further down the settings page before retrying
the privacy search."]
}
}

```

Test Case 16: Open Repository and Click Pull Requests (Medium Task)

User Prompt:

Open the ScreenAwareTaskAutomation repository and click the Pull requests tab.

Test Results:

- **Steps Total: 2**
- **Completed: 1**
- **Failed: 1**
- **Duration: 3 minutes 28 seconds**

Step Execution:

Step 1: Click the ScreenAwareTaskAutomation repository link

- **Target: ScreenAwareTaskAutomation repository link**
- **Attempts: 1**
- **Execution: Performed**
- **Evaluation: Success**
- **Issue: None**

Step 2: Click the Pull requests tab

- **Target: Pull requests tab**
- **Attempts: 2**
- **Execution: Performed**
- **Evaluation: Failed (retry loop)**
- **Issue: Browser still on repository overview page instead of pull requests page. Navigation mismatch after repository page loaded — possibly affected by stale tab or timing issue.**

Execution Summary:

The task partially succeeded. The repository was successfully opened, but the pull requests tab click did not result in the expected page load. The second step failed due to a navigation mismatch, likely caused by a stale tab reference or timing issue after the repository page loaded.

Issues Identified:

- Navigation to pull requests tab did not change the page content
- Potential stale tab or timing issue after repository page load
- No validation that pull requests page loaded before considering step complete
- Element coordinates may have shifted after page navigation

Recommendations for Improvement:

- Re-capture the repository page after Step 1 before clicking the Pull requests tab
- Add explicit page load wait after navigation before next interaction
- Implement element re-detection after each page change
- Verify tab activation state before proceeding to next step

JSON Response:

full_cycle.json

```
{
  "run_id": "1e17e84a-8e72-4098-a566-d9a5bb9b7c31",
  "user_task": "Open the ScreenAwareTaskAutomation repository and click the Pull requests tab.",
  "started_at": "2026-05-03T08:30:00Z",
  "finished_at": "2026-05-03T08:33:28Z",
  "vision_used": true,
  "perception": {
    "image":
"vision/data/sessions/shot_1777833043_F70B14/processing/proc_1777833043_F70B14/processed_frames/frame_1777833043_F70B14.jpg",
    "image_size": {"width": 1920, "height": 1080},
    "screen_bbox": [0, 0, 1920, 1080],
    "screen_size": {"width": 1920, "height": 1080},
    "coordinate_system": "pixel",
    "elements": [
      {"id": "repo_link", "type": "link", "label": "ScreenAwareTaskAutomation",
"dx": 312, "dy": 438, "frame_dx": 312, "frame_dy": 438, "confidence": 0.93},
      {"id": "tab_pull", "type": "tab", "label": "Pull requests", "dx": 736,
"dy": 202, "frame_dx": 736, "frame_dy": 202, "confidence": 0.91},
      {"id": "tab_code", "type": "tab", "label": "Code", "dx": 580, "dy": 202,
"frame_dx": 580, "frame_dy": 202, "confidence": 0.9}
    ],
    "element_count": 3
  },
  "todo": {
    "steps": [
      {"id": 1, "action": "Click the ScreenAwareTaskAutomation repository link.",
"target": "ScreenAwareTaskAutomation repository link", "expected_result": "The
```

```
repository main page opens.", "needs_user_data": false, "user_data_field": null,
"status": "done"},
  {"id": 2, "action": "Click the Pull requests tab.", "target": "Pull
requests tab", "expected_result": "The pull requests list opens.",
"needs_user_data": false, "user_data_field": null, "status": "failed"}
],
  "notes": "The repository page was opened, but the active tab navigation
changed during the run and the final click landed on an incorrect page.",
  "estimated_complexity": "medium"
},
"step_plans": [
  {
    "step_id": 1,
    "action": "Click the ScreenAwareTaskAutomation repository link.",
    "target": "ScreenAwareTaskAutomation repository link",
    "expected_result": "The repository main page opens.",
    "hid_commands": [
      {"cmd": "mouse_move", "meta": {"commandId": "s03-1"}, "dx": 312, "dy":
438, "smooth": true},
      {"cmd": "mouse_click", "meta": {"commandId": "s03-2"}, "button": "left"}
    ],
    "reasoning": "The repository link is visible in the list and can be clicked
directly."
  },
  {
    "step_id": 2,
    "action": "Click the Pull requests tab.",
    "target": "Pull requests tab",
    "expected_result": "The pull requests list opens.",
    "hid_commands": [
      {"cmd": "mouse_move", "meta": {"commandId": "s03-3"}, "dx": 736, "dy":
202, "smooth": true},
      {"cmd": "mouse_click", "meta": {"commandId": "s03-4"}, "button": "left"}
    ],
    "reasoning": "The repository navigation is visible at the top of the page."
  }
],
"step_results": [
  {"step_id": 1, "action": "Click the ScreenAwareTaskAutomation repository
link.", "attempt": 1, "exec_status": "success", "eval_status": "done",
"confidence": 0.97, "reason": "The repository main page loaded successfully.",
"final_status": "done", "timestamp": "2026-05-03T08:31:05Z"},
  {"step_id": 2, "action": "Click the Pull requests tab.", "attempt": 2,
"exec_status": "success", "eval_status": "retry", "confidence": 0.72, "reason":
"The browser is still on the repository overview page instead of the pull
```

```
requests page.", "final_status": "executing", "timestamp": "2026-05-03T08:33:28Z"}
],
"final_report": {
  "success": false,
  "summary": "The repository opened, but the pull requests tab was not reached reliably.",
  "message": "Step 1 succeeded, but step 2 failed due to a navigation mismatch after the repository page loaded.",
  "steps_completed": 1,
  "steps_failed": 1,
  "issues": ["The second navigation step was likely affected by a stale tab or timing issue."],
  "recommendations": ["Re-capture the repository page after step 1 before clicking the Pull requests tab."]
}
```

Test Case 17: Search Gmail for Acme Invoices (Medium Task)

User Prompt:

Search Gmail for invoices from Acme and open the latest one.

Test Results:

- Steps Total: 2
- Completed: 1
- Failed: 1
- Duration: 3 minutes 41 seconds

Step Execution:

Step 1: Click the Gmail search field and search for Acme invoices

- Target: Search mail
- Attempts: 1
- Execution: Performed
- Evaluation: Success
- Issue: None

Step 2: Open the latest invoice email

- **Target:** Acme invoice email
- **Attempts:** 3
- **Execution:** Failed
- **Evaluation:** Failed (retry loop)
- **Issue:** The latest matching email was not clearly visible in the current results list. Message ordering in the inbox was ambiguous after search.

Execution Summary:

The search was successful, but the target message could not be opened with confidence. The latest invoice email was not at the top of the visible list, and the agent could not reliably identify the correct message to click.

Issues Identified:

- Latest matching email not visible in current viewport
- Message ordering ambiguous after search
- No scroll or sort functionality applied to bring latest email into view
- Low confidence in target identification

Recommendations for Improvement:

- Scroll through filtered results to locate latest message
- Apply "Newest first" sort before attempting to locate target
- Implement confidence threshold for email selection before clicking
- Add fallback to select first result if latest cannot be determined

JSON Response:

full_cycle.json

```
{
  "run_id": "3fc60555-e84a-450c-98f9-ef48050ee4d0",
  "user_task": "Search Gmail for invoices from Acme and open the latest one.",
  "started_at": "2026-05-03T09:00:00Z",
  "finished_at": "2026-05-03T09:03:41Z",
  "vision_used": true,
  "perception": {
    "image":
"vision/data/sessions/shot_1777833044_B0A31D/processing/proc_1777833044_B0A31D/pr
eprocessed_frames/frame_1777833044_B0A31D.jpg",
    "image_size": {"width": 1920, "height": 1080},
    "screen_bbox": [0, 0, 1920, 1080],
    "screen_size": {"width": 1920, "height": 1080},
```

```

    "coordinate_system": "pixel",
    "elements": [
      {"id": "search", "type": "input", "label": "Search mail", "dx": 943, "dy":
218, "frame_dx": 943, "frame_dy": 218, "confidence": 0.94},
      {"id": "message_1", "type": "email", "label": "Acme invoice", "dx": 612,
"dy": 424, "frame_dx": 612, "frame_dy": 424, "confidence": 0.88}
    ],
    "element_count": 2
  },
  "todo": {
    "steps": [
      {"id": 1, "action": "Click the Gmail search field and search for Acme
invoices.", "target": "Search mail", "expected_result": "Search results are
shown.", "needs_user_data": false, "user_data_field": null, "status": "done"},
      {"id": 2, "action": "Open the latest invoice email.", "target": "Acme
invoice email", "expected_result": "The invoice email opens.", "needs_user_data":
false, "user_data_field": null, "status": "failed"}
    ],
    "notes": "The search field was used successfully, but the latest matching
message was not at the top of the visible list.",
    "estimated_complexity": "medium"
  },
  "step_plans": [
    {
      "step_id": 1,
      "action": "Click the Gmail search field and search for Acme invoices.",
      "target": "Search mail",
      "expected_result": "Search results are shown.",
      "hid_commands": [
        {"cmd": "mouse_move", "meta": {"commandId": "s04-1"}, "dx": 943, "dy":
218, "smooth": true},
        {"cmd": "mouse_click", "meta": {"commandId": "s04-2"}, "button": "left"},
        {"cmd": "type_text", "meta": {"commandId": "s04-3"}, "text": "from:acme
invoice"}
      ],
      "reasoning": "The search field is visible and supports direct text entry."
    },
    {
      "step_id": 2,
      "action": "Open the latest invoice email.",
      "target": "Acme invoice email",
      "expected_result": "The invoice email opens.",
      "hid_commands": [],
      "reasoning": "The latest matching email was not clearly visible in the
current results list."
    }
  ]
}

```



```

    }
  ],
  "step_results": [
    {
      "step_id": 1,
      "action": "Click the Gmail search field and search for Acme invoices.",
      "attempt": 1,
      "exec_status": "success",
      "eval_status": "done",
      "confidence": 0.95,
      "reason": "The search query was entered and results appeared.",
      "final_status": "done",
      "timestamp": "2026-05-03T09:01:15Z"
    },
    {
      "step_id": 2,
      "action": "Open the latest invoice email.",
      "attempt": 3,
      "exec_status": "failed",
      "eval_status": "retry",
      "confidence": 0.55,
      "reason": "The visible email list did not clearly show the latest invoice message.",
      "final_status": "executing",
      "timestamp": "2026-05-03T09:03:41Z"
    }
  ],
  "final_report": {
    "success": false,
    "summary": "Search succeeded, but the target message was not confidently identified.",
    "message": "The search field was used successfully, but the latest invoice email could not be opened because it was not clearly visible in the current results.",
    "steps_completed": 1,
    "steps_failed": 1,
    "issues": ["Message ordering in the inbox was ambiguous after search."],
    "recommendations": ["Scroll through the filtered results or sort by newest first before retrying."]
  }
}

```

Test Case 18: Login to Test Portal (Simple Task)

User Prompt:

Login to the test portal using the on-screen form and submit the credentials.

Test Results:

- Steps Total: 3
- Completed: 3
- Failed: 0
- Duration: 1 minute 28 seconds

Step Execution:

Step 1: Click the username field and enter the username

- **Target:** Username field
- **Attempts:** 1
- **Execution:** Performed
- **Evaluation:** Success
- **Issue:** None

Step 2: Click the password field and enter the password

- **Target:** Password field
- **Attempts:** 1
- **Execution:** Performed
- **Evaluation:** Success
- **Issue:** None

Step 3: Click Sign in to submit the form

- **Target:** Sign in button
- **Attempts:** 1
- **Execution:** Performed
- **Evaluation:** Success
- **Issue:** None

Execution Summary:

The task completed successfully. The username and password were entered correctly, and the form was submitted. The portal home screen loaded after login confirmation.

Issues Identified:

- None

Recommendations for Improvement:

- No specific recommendations — test case executed without issues

JSON Response:

```
full_cycle.json

{
  "run_id": "f33335d8-7f20-4ea0-b286-30967107c630",
  "user_task": "Login to the test portal using the on-screen form and submit the credentials.",
  "started_at": "2026-05-03T09:20:00Z",
  "finished_at": "2026-05-03T09:21:28Z",
  "vision_used": true,
```

```
"perception": {
  "image":
"vision/data/sessions/shot_1777833045_CB43DA/processing/proc_1777833045_CB43DA/pr
eprocessed_frames/frame_1777833045_CB43DA.jpg",
  "image_size": {"width": 1920, "height": 1080},
  "screen_bbox": [0, 0, 1920, 1080],
  "screen_size": {"width": 1920, "height": 1080},
  "coordinate_system": "pixel",
  "elements": [
    {"id": "username", "type": "input", "label": "Username", "dx": 796, "dy":
392, "frame_dx": 796, "frame_dy": 392, "confidence": 0.95},
    {"id": "password", "type": "input", "label": "Password", "dx": 796, "dy":
472, "frame_dx": 796, "frame_dy": 472, "confidence": 0.95},
    {"id": "login", "type": "button", "label": "Sign in", "dx": 796, "dy": 560,
"frame_dx": 796, "frame_dy": 560, "confidence": 0.94}
  ],
  "element_count": 3
},
"todo": {
  "steps": [
    {"id": 1, "action": "Click the username field and enter the username.",
"target": "Username field", "expected_result": "Username is entered.",
"needs_user_data": true, "user_data_field": "username", "status": "done"},
    {"id": 2, "action": "Click the password field and enter the password.",
"target": "Password field", "expected_result": "Password is entered.",
"needs_user_data": true, "user_data_field": "password", "status": "done"},
    {"id": 3, "action": "Click Sign in to submit the form.", "target": "Sign in
button", "expected_result": "The portal opens after login.", "needs_user_data":
false, "user_data_field": null, "status": "done"}
  ],
  "notes": "A standard form-login flow with secure user input handled
separately.",
  "estimated_complexity": "medium"
},
"step_plans": [
  {
    "step_id": 1,
    "action": "Click the username field and enter the username.",
    "target": "Username field",
    "expected_result": "Username is entered.",
    "hid_commands": [
      {"cmd": "mouse_move", "meta": {"commandId": "s05-1"}, "dx": 796, "dy":
392, "smooth": true},
      {"cmd": "mouse_click", "meta": {"commandId": "s05-2"}, "button": "left"},
```

```
        {"cmd": "type_text", "meta": {"commandId": "s05-3"}, "text":  
"<SECURE_INPUT_PROVIDED>"}  
    ],  
    "reasoning": "The username field is visible and focused directly."  
},  
{  
    "step_id": 2,  
    "action": "Click the password field and enter the password.",  
    "target": "Password field",  
    "expected_result": "Password is entered.",  
    "hid_commands": [  
        {"cmd": "mouse_move", "meta": {"commandId": "s05-4"}, "dx": 796, "dy":  
472, "smooth": true},  
        {"cmd": "mouse_click", "meta": {"commandId": "s05-5"}, "button": "left"},  
        {"cmd": "type_text", "meta": {"commandId": "s05-6"}, "text":  
"<SECURE_INPUT_PROVIDED>"}  
    ],  
    "reasoning": "The password field is visible and can be used securely."  
},  
{  
    "step_id": 3,  
    "action": "Click Sign in to submit the form.",  
    "target": "Sign in button",  
    "expected_result": "The portal opens after login.",  
    "hid_commands": [  
        {"cmd": "mouse_move", "meta": {"commandId": "s05-7"}, "dx": 796, "dy":  
560, "smooth": true},  
        {"cmd": "mouse_click", "meta": {"commandId": "s05-8"}, "button": "left"}  
    ],  
    "reasoning": "The sign-in button is visible below the fields."  
}  
],  
"step_results": [  
    {"step_id": 1, "action": "Click the username field and enter the username.",  
"attempt": 1, "exec_status": "success", "eval_status": "done", "confidence":  
0.98, "reason": "Username field populated correctly.", "final_status": "done",  
"timestamp": "2026-05-03T09:20:22Z"},  
    {"step_id": 2, "action": "Click the password field and enter the password.",  
"attempt": 1, "exec_status": "success", "eval_status": "done", "confidence":  
0.97, "reason": "Password field populated correctly.", "final_status": "done",  
"timestamp": "2026-05-03T09:20:48Z"},  
    {"step_id": 3, "action": "Click Sign in to submit the form.", "attempt": 1,  
"exec_status": "success", "eval_status": "done", "confidence": 0.96, "reason":  
"The portal home screen loaded after submission.", "final_status": "done",  
"timestamp": "2026-05-03T09:21:28Z"}  
]
```

```
],  
  "final_report": {  
    "success": true,  
    "summary": "The login form was completed successfully.",  
    "message": "The username and password were entered securely and the form was  
submitted successfully.",  
    "steps_completed": 3,  
    "steps_failed": 0,  
    "issues": [],  
    "recommendations": []  
  }  
}
```

Test Case 19: Download Latest Installer (Medium Task)

User Prompt:

Go to the downloads page and start the latest installer download.

Test Results:

- Steps Total: 2
- Completed: 1
- Failed: 1
- Duration: 2 minutes 16 seconds

Step Execution:

Step 1: Open the downloads page

- Target: Downloads
- Attempts: 1
- Execution: Performed
- Evaluation: Success
- Issue: None

Step 2: Click the latest installer download button

- Target: Latest installer
- Attempts: 2
- Execution: Performed

- **Evaluation: Failed (retry loop)**
- **Issue: The visible button no longer matched the latest installer after page refresh. Dynamic content changed the button coordinates or target.**

Execution Summary:

The task partially succeeded. The downloads page was successfully opened, but the installer button target changed after the page refreshed. The originally detected button was no longer the correct target for the latest installer.

Issues Identified:

- **Dynamic content changed button after page load/refresh**
- **Original coordinates (621, 486) no longer valid for target**
- **No re-detection of element after page stabilization**
- **Page likely contains version-specific installer buttons that shift with updates**

Recommendations for Improvement:

- **Re-capture the page immediately before clicking the installer button**
- **Implement element re-detection after page load before each click**
- **Use label-based (e.g., "Latest installer") rather than coordinate-based targeting**
- **Add wait for page stabilization before detecting download buttons**

JSON Response:

full_cycle.json

```
{
  "run_id": "0679ee52-75fa-4821-86f7-da2fea450b27",
  "user_task": "Go to the downloads page and start the latest installer
download.",
  "started_at": "2026-05-03T09:40:00Z",
  "finished_at": "2026-05-03T09:42:16Z",
  "vision_used": true,
  "perception": {
    "image":
"vision/data/sessions/shot_1777833046_8E2F43/processing/proc_1777833046_8E2F43/pr
eprocessed_frames/frame_1777833046_8E2F43.jpg",
    "image_size": {"width": 1920, "height": 1080},
    "screen_bbox": [0, 0, 1920, 1080],
    "screen_size": {"width": 1920, "height": 1080},
    "coordinate_system": "pixel",
    "elements": [
      {"id": "downloads", "type": "button", "label": "Downloads", "dx": 1540,
"dy": 92, "frame_dx": 1540, "frame_dy": 92, "confidence": 0.93},
```

```
    {"id": "installer", "type": "button", "label": "Latest installer", "dx":  
621, "dy": 486, "frame_dx": 621, "frame_dy": 486, "confidence": 0.89}  
  ],  
  "element_count": 2  
},  
"todo": {  
  "steps": [  
    {"id": 1, "action": "Open the downloads page.", "target": "Downloads",  
"expected_result": "The downloads page is visible.", "needs_user_data": false,  
"user_data_field": null, "status": "done"},  
    {"id": 2, "action": "Click the latest installer download button.",  
"target": "Latest installer", "expected_result": "The download begins.",  
"needs_user_data": false, "user_data_field": null, "status": "failed"}  
  ],  
  "notes": "The downloads page loaded, but the installer button changed after  
the page refreshed.",  
  "estimated_complexity": "medium"  
},  
"step_plans": [  
  {  
    "step_id": 1,  
    "action": "Open the downloads page.",  
    "target": "Downloads",  
    "expected_result": "The downloads page is visible.",  
    "hid_commands": [  
      {"cmd": "mouse_move", "meta": {"commandId": "s06-1"}, "dx": 1540, "dy":  
92, "smooth": true},  
      {"cmd": "mouse_click", "meta": {"commandId": "s06-2"}, "button": "left"}  
    ],  
    "reasoning": "The downloads control is visible in the top navigation."  
  },  
  {  
    "step_id": 2,  
    "action": "Click the latest installer download button.",  
    "target": "Latest installer",  
    "expected_result": "The download begins.",  
    "hid_commands": [  
      {"cmd": "mouse_move", "meta": {"commandId": "s06-3"}, "dx": 621, "dy":  
486, "smooth": true},  
      {"cmd": "mouse_click", "meta": {"commandId": "s06-4"}, "button": "left"}  
    ],  
    "reasoning": "The installer button was visible in the initial capture."  
  }  
],  
"step_results": [  

```

```
{
  "step_id": 1, "action": "Open the downloads page.", "attempt": 1,
  "exec_status": "success", "eval_status": "done", "confidence": 0.96, "reason":
  "The downloads page is visible.", "final_status": "done", "timestamp": "2026-05-
  03T09:40:20Z"},
  {
    "step_id": 2, "action": "Click the latest installer download button.",
    "attempt": 2, "exec_status": "success", "eval_status": "retry", "confidence":
    0.61, "reason": "The visible button no longer matched the latest installer after
    refresh.", "final_status": "executing", "timestamp": "2026-05-03T09:42:16Z"}
  ],
  "final_report": {
    "success": false,
    "summary": "The downloads page was reached, but the installer target changed
    before execution completed.",
    "message": "Step 1 succeeded. Step 2 did not complete because the page
    refreshed and the originally detected installer button was no longer the same
    target.",
    "steps_completed": 1,
    "steps_failed": 1,
    "issues": ["Dynamic content changed the button coordinates."],
    "recommendations": ["Re-capture the page immediately before clicking the
    installer button."]
  }
}
```

Test Case 20: GitHub Issues Filter by Bug Label (Complex Task)

User Prompt:

Switch to the GitHub window, open the Issues tab, and filter issues by the bug label.

Test Results:

- Steps Total: 3
- Completed: 2
- Failed: 1
- Duration: 5 minutes 55 seconds

Step Execution:

Step 1: Switch to the GitHub window

- **Target:** GitHub tab
- **Attempts:** 1
- **Execution:** Performed
- **Evaluation:** Success
- **Issue:** None

Step 2: Open the Issues tab

- **Target:** Issues tab
- **Attempts:** 1
- **Execution:** Performed
- **Evaluation:** Success
- **Issue:** None

Step 3: Filter issues by the bug label

- **Target:** Filter issues field
- **Attempts:** 2
- **Execution:** Performed
- **Evaluation:** Failed (retry loop)
- **Issue:** The filter field could not be stably matched after the page changed layout. Dynamic page layout shifted after navigation.

Execution Summary:

The task partially succeeded. The GitHub window and Issues tab were successfully opened, but the final filter step failed because the issues page layout shifted and the filter field could not be targeted with confidence.

Issues Identified:

- Dynamic page layout changed after navigation to Issues tab
- Filter field not stable in captured screen
- No re-detection of filter field after page load

Recommendations for Improvement:

- Capture a fresh screen and retry filter field detection immediately after opening Issues tab
- Implement page load wait before detecting filter element
- Add fallback to locate filter by label text rather than fixed coordinates

JSON Response:

full_cycle.json

```
{
```

```
"run_id": "3fb6213d-137d-4b4d-9ffa-ff963a482e51",
"user_task": "Switch to the GitHub window, open the Issues tab, and filter
issues by the bug label.",
"started_at": "2026-05-03T10:00:00Z",
"finished_at": "2026-05-03T10:05:55Z",
"vision_used": true,
"perception": {
  "image":
"vision/data/sessions/shot_1777833047_15374B/processing/proc_1777833047_15374B/pr
eprocessed_frames/frame_1777833047_15374B.jpg",
  "image_size": {"width": 1920, "height": 1080},
  "screen_bbox": [0, 0, 1920, 1080],
  "screen_size": {"width": 1920, "height": 1080},
  "coordinate_system": "pixel",
  "elements": [
    {"id": "tab_git", "type": "tab", "label": "GitHub", "dx": 402, "dy": 66,
"frame_dx": 402, "frame_dy": 66, "confidence": 0.92},
    {"id": "tab_issues", "type": "tab", "label": "Issues", "dx": 796, "dy":
204, "frame_dx": 796, "frame_dy": 204, "confidence": 0.9},
    {"id": "filter", "type": "input", "label": "Filter issues", "dx": 880,
"dy": 268, "frame_dx": 880, "frame_dy": 268, "confidence": 0.9}
  ],
  "element_count": 3
},
"todo": {
  "steps": [
    {"id": 1, "action": "Switch to the GitHub window.", "target": "GitHub tab",
"expected_result": "The GitHub window is active.", "needs_user_data": false,
"user_data_field": null, "status": "done"},
    {"id": 2, "action": "Open the Issues tab.", "target": "Issues tab",
"expected_result": "The issues list is visible.", "needs_user_data": false,
"user_data_field": null, "status": "done"},
    {"id": 3, "action": "Filter issues by the bug label.", "target": "Filter
issues field", "expected_result": "Only bug-labeled issues remain.",
"needs_user_data": false, "user_data_field": null, "status": "failed"}
  ],
  "notes": "A multi-window navigation task where the final filter step became
unstable after the issues page loaded.",
  "estimated_complexity": "complex"
},
"step_plans": [
  {
    "step_id": 1,
    "action": "Switch to the GitHub window.",
    "target": "GitHub tab",
```

```

    "expected_result": "The GitHub window is active.",
    "hid_commands": [
      {"cmd": "mouse_move", "meta": {"commandId": "s07-1"}, "dx": 402, "dy":
66, "smooth": true},
      {"cmd": "mouse_click", "meta": {"commandId": "s07-2"}, "button": "left"}
    ],
    "reasoning": "The GitHub browser tab is visible and can be activated."
  },
  {
    "step_id": 2,
    "action": "Open the Issues tab.",
    "target": "Issues tab",
    "expected_result": "The issues list is visible.",
    "hid_commands": [
      {"cmd": "mouse_move", "meta": {"commandId": "s07-3"}, "dx": 796, "dy":
204, "smooth": true},
      {"cmd": "mouse_click", "meta": {"commandId": "s07-4"}, "button": "left"}
    ],
    "reasoning": "The Issues tab is visible in the repository navigation."
  },
  {
    "step_id": 3,
    "action": "Filter issues by the bug label.",
    "target": "Filter issues field",
    "expected_result": "Only bug-labeled issues remain.",
    "hid_commands": [],
    "reasoning": "The filter field was not stable in the captured screen and
could not be targeted confidently."
  }
],
"step_results": [
  {"step_id": 1, "action": "Switch to the GitHub window.", "attempt": 1,
"exec_status": "success", "eval_status": "done", "confidence": 0.96, "reason":
"The GitHub window became active.", "final_status": "done", "timestamp": "2026-
05-03T10:00:34Z"},
  {"step_id": 2, "action": "Open the Issues tab.", "attempt": 1, "exec_status":
"success", "eval_status": "done", "confidence": 0.95, "reason": "The Issues page
is visible.", "final_status": "done", "timestamp": "2026-05-03T10:01:18Z"},
  {"step_id": 3, "action": "Filter issues by the bug label.", "attempt": 2,
"exec_status": "success", "eval_status": "retry", "confidence": 0.58, "reason":
"The filter field could not be stably matched after the page changed layout.",
"final_status": "executing", "timestamp": "2026-05-03T10:05:55Z"}
],
"final_report": {
  "success": false,

```

```
    "summary": "The GitHub window and Issues tab were opened, but filtering was not completed reliably.",
    "message": "The task progressed through the first two steps successfully. The final filter step failed because the issues page layout shifted and the filter field could not be matched with confidence.",
    "steps_completed": 2,
    "steps_failed": 1,
    "issues": ["Dynamic page layout changed after navigation."],
    "recommendations": ["Capture a fresh screen and retry the bug label filter immediately after opening Issues."]
  }
}
```

Test Case 21: Spreadsheet Copy Quarterly Total to Summary Cell (Complex Task)

User Prompt:

Open the spreadsheet, go to the Sales sheet, and copy the quarterly total into the summary cell.

Test Results:

- Steps Total: 3
- Completed: 2
- Failed: 1
- Duration: 7 minutes 4 seconds

Step Execution:

Step 1: Open the spreadsheet and select the Sales sheet

- Target: Sales sheet tab
- Attempts: 1
- Execution: Performed
- Evaluation: Success
- Issue: None

Step 2: Copy the quarterly total value

- Target: Quarterly total cell
- Attempts: 1

- **Execution: Performed**
- **Evaluation: Success**
- **Issue: None**

Step 3: Paste the value into the summary cell

- **Target: Summary cell**
- **Attempts: 2**
- **Execution: Performed**
- **Evaluation: Failed (retry loop)**
- **Issue: Summary cell was not confirmed after sheet redraw. Cell focus was lost after the redraw.**

Execution Summary:

The task partially succeeded. The spreadsheet navigation and copy step completed successfully. The final paste step could not be validated because the cell view changed after the sheet redraw, and focus was lost.

Issues Identified:

- **Summary cell focus lost after sheet redraw**
- **No confirmation of paste action before proceeding**
- **Potential timing issue between copy and paste operations**

Recommendations for Improvement:

- **Click the summary cell again and paste immediately after sheet stabilizes**
- **Add explicit wait between copy and paste operations**
- **Implement cell selection verification before pasting**

JSON Response:

```
full_cycle.json

{
  "run_id": "47615049-8d34-4f98-bc94-aaa5723aaeb6",
  "user_task": "Open the spreadsheet, go to the Sales sheet, and copy the quarterly total into the summary cell.",
  "started_at": "2026-05-03T10:20:00Z",
  "finished_at": "2026-05-03T10:27:04Z",
  "vision_used": true,
  "perception": {
    "image":
"vision/data/sessions/shot_1777833048_F9A1C8/processing/proc_1777833048_F9A1C8/processed_frames/frame_1777833048_F9A1C8.jpg",
  }
}
```

```
"image_size": {"width": 1920, "height": 1080},
"screen_bbox": [0, 0, 1920, 1080],
"screen_size": {"width": 1920, "height": 1080},
"coordinate_system": "pixel",
"elements": [
  {"id": "sheet_sales", "type": "tab", "label": "Sales", "dx": 512, "dy":
136, "frame_dx": 512, "frame_dy": 136, "confidence": 0.92},
  {"id": "total_cell", "type": "cell", "label": "Quarterly total", "dx":
1084, "dy": 604, "frame_dx": 1084, "frame_dy": 604, "confidence": 0.9},
  {"id": "summary_cell", "type": "cell", "label": "Summary cell", "dx": 1380,
"dy": 604, "frame_dx": 1380, "frame_dy": 604, "confidence": 0.9}
],
"element_count": 3
},
"todo": {
  "steps": [
    {"id": 1, "action": "Open the spreadsheet and select the Sales sheet.",
"target": "Sales sheet tab", "expected_result": "The Sales sheet is visible.",
"needs_user_data": false, "user_data_field": null, "status": "done"},
    {"id": 2, "action": "Copy the quarterly total value.", "target": "Quarterly
total cell", "expected_result": "The value is copied to the clipboard.",
"needs_user_data": false, "user_data_field": null, "status": "done"},
    {"id": 3, "action": "Paste the value into the summary cell.", "target":
"Summary cell", "expected_result": "The summary cell contains the copied total.",
"needs_user_data": false, "user_data_field": null, "status": "failed"}
  ],
  "notes": "A spreadsheet workflow that mostly succeeded, but the paste target
was not retained after a sheet repaint.",
  "estimated_complexity": "complex"
},
"step_plans": [
  {
    "step_id": 1,
    "action": "Open the spreadsheet and select the Sales sheet.",
    "target": "Sales sheet tab",
    "expected_result": "The Sales sheet is visible.",
    "hid_commands": [
      {"cmd": "mouse_move", "meta": {"commandId": "s08-1"}, "dx": 512, "dy":
136, "smooth": true},
      {"cmd": "mouse_click", "meta": {"commandId": "s08-2"}, "button": "left"}
    ],
    "reasoning": "The Sales sheet tab is visible in the workbook navigation."
  },
  {
    "step_id": 2,
```

```
        "action": "Copy the quarterly total value.",
        "target": "Quarterly total cell",
        "expected_result": "The value is copied to the clipboard.",
        "hid_commands": [
            {"cmd": "mouse_move", "meta": {"commandId": "s08-3"}, "dx": 1084, "dy":
604, "smooth": true},
            {"cmd": "mouse_click", "meta": {"commandId": "s08-4"}, "button": "left"},
            {"cmd": "key_press", "meta": {"commandId": "s08-5"}, "key": "ctrl+c"}
        ],
        "reasoning": "The total cell is visible and can be selected before
copying."
    },
    {
        "step_id": 3,
        "action": "Paste the value into the summary cell.",
        "target": "Summary cell",
        "expected_result": "The summary cell contains the copied total.",
        "hid_commands": [
            {"cmd": "mouse_move", "meta": {"commandId": "s08-6"}, "dx": 1380, "dy":
604, "smooth": true},
            {"cmd": "mouse_click", "meta": {"commandId": "s08-7"}, "button": "left"},
            {"cmd": "key_press", "meta": {"commandId": "s08-8"}, "key": "ctrl+v"}
        ],
        "reasoning": "The summary cell is visible near the total cell."
    }
],
"step_results": [
    {"step_id": 1, "action": "Open the spreadsheet and select the Sales sheet.",
"attempt": 1, "exec_status": "success", "eval_status": "done", "confidence":
0.96, "reason": "The Sales sheet is visible.", "final_status": "done",
"timestamp": "2026-05-03T10:20:40Z"},
    {"step_id": 2, "action": "Copy the quarterly total value.", "attempt": 1,
"exec_status": "success", "eval_status": "done", "confidence": 0.94, "reason":
"The highlighted cell value was copied.", "final_status": "done", "timestamp":
"2026-05-03T10:22:11Z"},
    {"step_id": 3, "action": "Paste the value into the summary cell.", "attempt":
2, "exec_status": "success", "eval_status": "retry", "confidence": 0.67,
"reason": "The summary cell was not confirmed after the sheet redraw.",
"final_status": "executing", "timestamp": "2026-05-03T10:27:04Z"}
],
"final_report": {
    "success": false,
    "summary": "The spreadsheet data was copied, but the paste outcome was not
confirmed.",
}
```

```
    "message": "The workbook navigation and copy step completed successfully. The final paste step could not be validated because the cell view changed after the sheet redraw.",
    "steps_completed": 2,
    "steps_failed": 1,
    "issues": ["Cell focus was lost after the redraw."],
    "recommendations": ["Click the summary cell again and paste immediately after the sheet stabilizes."]
  }
}
```

Test Case 22: Chat App - Switch Channel and Upload Screenshot (Complex Task)

User Prompt:

Open the chat app, switch to the project channel, and upload the latest screenshot.

Test Results:

- Steps Total: 3
- Completed: 1
- Failed: 2
- Duration: 4 minutes 33 seconds

Step Execution:

Step 1: Open the chat application

- Target: Chat app
- Attempts: 1
- Execution: Performed
- Evaluation: Success
- Issue: None

Step 2: Switch to the project channel

- Target: Project channel
- Attempts: 3
- Execution: Failed
- Evaluation: Failed (retry loop)
- Issue: The project channel was not visible in the sidebar. The sidebar did not expose the project channel in the current view.

Step 3: Upload the latest screenshot

- **Target:** Upload file button
- **Attempts:** 1
- **Execution:** Failed
- **Evaluation:** Failed (retry loop)
- **Issue:** Prerequisite channel switch failed.

Execution Summary:

The task partially succeeded in opening the chat app but failed completely on the remaining steps. The project channel was not visible in the sidebar, preventing channel switching and subsequent screenshot upload.

Issues Identified:

- Project channel not visible in sidebar
- No scroll or search functionality applied to locate channel
- Upload step blocked by prerequisite failure
- Low confidence in channel detection

Recommendations for Improvement:

- Scroll the channel list to reveal hidden channels
- Use search functionality to find channel by name
- Implement fallback to show all channels if not visible
- Reorder steps to ensure channel availability before upload

JSON Response:

```
full_cycle.json

{
  "run_id": "874bde9e-8388-4864-a14d-f0ddfbccfcac",
  "user_task": "Open the chat app, switch to the project channel, and upload the latest screenshot.",
  "started_at": "2026-05-03T10:40:00Z",
  "finished_at": "2026-05-03T10:44:33Z",
  "vision_used": true,
  "perception": {
    "image":
"vision/data/sessions/shot_1777833049_E8B2F1/processing/proc_1777833049_E8B2F1/preprocessed_frames/frame_1777833049_E8B2F1.jpg",
    "image_size": {"width": 1920, "height": 1080},
    "screen_bbox": [0, 0, 1920, 1080],
    "screen_size": {"width": 1920, "height": 1080},
```

```
"coordinate_system": "pixel",
"elements": [
  {"id": "channel_general", "type": "channel", "label": "general", "dx": 248,
"dy": 238, "frame_dx": 248, "frame_dy": 238, "confidence": 0.9},
  {"id": "upload", "type": "button", "label": "Upload file", "dx": 1672,
"dy": 1000, "frame_dx": 1672, "frame_dy": 1000, "confidence": 0.9}
],
"element_count": 2
},
"todo": {
  "steps": [
    {"id": 1, "action": "Open the chat application.", "target": "Chat app",
"expected_result": "The chat app is visible.", "needs_user_data": false,
"user_data_field": null, "status": "done"},
    {"id": 2, "action": "Switch to the project channel.", "target": "Project
channel", "expected_result": "The project channel is active.", "needs_user_data":
false, "user_data_field": null, "status": "failed"},
    {"id": 3, "action": "Upload the latest screenshot.", "target": "Upload file
button", "expected_result": "The screenshot is attached.", "needs_user_data":
false, "user_data_field": null, "status": "failed"}
  ],
  "notes": "The app opened, but the desired channel was not visible in the
current sidebar state.",
  "estimated_complexity": "complex"
},
"step_plans": [
  {
    "step_id": 1,
    "action": "Open the chat application.",
    "target": "Chat app",
    "expected_result": "The chat app is visible.",
    "hid_commands": [
      {"cmd": "mouse_move", "meta": {"commandId": "s09-1"}, "dx": 248, "dy":
238, "smooth": true},
      {"cmd": "mouse_click", "meta": {"commandId": "s09-2"}, "button": "left"}
    ],
    "reasoning": "The chat application is already visible on screen."
  },
  {
    "step_id": 2,
    "action": "Switch to the project channel.",
    "target": "Project channel",
    "expected_result": "The project channel is active.",
    "hid_commands": [],

```

```
    "reasoning": "The project channel was not detected in the visible channel list."
  },
  {
    "step_id": 3,
    "action": "Upload the latest screenshot.",
    "target": "Upload file button",
    "expected_result": "The screenshot is attached.",
    "hid_commands": [],
    "reasoning": "The upload button was present, but the task could not continue because the correct channel was not found."
  }
],
"step_results": [
  {"step_id": 1, "action": "Open the chat application.", "attempt": 1, "exec_status": "success", "eval_status": "done", "confidence": 0.95, "reason": "The chat app is open.", "final_status": "done", "timestamp": "2026-05-03T10:40:20Z"},
  {"step_id": 2, "action": "Switch to the project channel.", "attempt": 3, "exec_status": "failed", "eval_status": "retry", "confidence": 0.52, "reason": "The project channel was not visible in the sidebar.", "final_status": "failed", "timestamp": "2026-05-03T10:43:05Z"},
  {"step_id": 3, "action": "Upload the latest screenshot.", "attempt": 1, "exec_status": "failed", "eval_status": "retry", "confidence": 0.49, "reason": "Prerequisite channel switch failed.", "final_status": "failed", "timestamp": "2026-05-03T10:44:33Z"}
],
"final_report": {
  "success": false,
  "summary": "The chat app opened, but the target channel could not be selected.",
  "message": "The task failed because the project channel was not visible in the current sidebar state, so the screenshot upload step could not proceed.",
  "steps_completed": 1,
  "steps_failed": 2,
  "issues": ["The sidebar did not expose the project channel in the current view."],
  "recommendations": ["Scroll the channel list or use search to locate the project channel before uploading."]
}
```

Test Case 23: File Explorer - Rename Report File (Simple Task)

User Prompt:

Open File Explorer, go to the Documents folder, and rename the report file to final_report.txt.

Test Results:

- **Steps Total: 3**
- **Completed: 3**
- **Failed: 0**
- **Duration: 4 minutes 49 seconds**

Step Execution:

Step 1: Open File Explorer and navigate to Documents

- **Target: Documents folder**
- **Attempts: 1**
- **Execution: Performed**
- **Evaluation: Success**
- **Issue: None**

Step 2: Select the report file

- **Target: report.docx**
- **Attempts: 1**
- **Execution: Performed**
- **Evaluation: Success**
- **Issue: None**

Step 3: Rename the file to final_report.txt

- **Target: Rename action**
- **Attempts: 1**
- **Execution: Performed**
- **Evaluation: Success**
- **Issue: None**

Execution Summary:

The task completed successfully. File Explorer was opened, the Documents folder was reached, and the report file was renamed to final_report.txt without issues.

Issues Identified:

- None

Recommendations for Improvement:

- No specific recommendations — test case executed without issues

JSON Response:

```
full_cycle.json

{
  "run_id": "333378f9-a8ed-425f-9e14-08fdbaeefe393",
  "user_task": "Open File Explorer, go to the Documents folder, and rename the
report file to final_report.txt.",
  "started_at": "2026-05-03T11:00:00Z",
  "finished_at": "2026-05-03T11:04:49Z",
  "vision_used": true,
  "perception": {
    "image":
"vision/data/sessions/shot_1777833050_EC72F5/processing/proc_1777833050_EC72F5/pr
eprocessed_frames/frame_1777833050_EC72F5.jpg",
    "image_size": {"width": 1920, "height": 1080},
    "screen_bbox": [0, 0, 1920, 1080],
    "screen_size": {"width": 1920, "height": 1080},
    "coordinate_system": "pixel",
    "elements": [
      {"id": "documents", "type": "folder", "label": "Documents", "dx": 468,
"dy": 312, "frame_dx": 468, "frame_dy": 312, "confidence": 0.93},
      {"id": "report_file", "type": "file", "label": "report.docx", "dx": 728,
"dy": 426, "frame_dx": 728, "frame_dy": 426, "confidence": 0.91},
      {"id": "rename", "type": "menu_item", "label": "Rename", "dx": 1112, "dy":
288, "frame_dx": 1112, "frame_dy": 288, "confidence": 0.9}
    ],
    "element_count": 3
  },
  "todo": {
    "steps": [
      {"id": 1, "action": "Open File Explorer and navigate to Documents.",
"target": "Documents folder", "expected_result": "The Documents folder contents
are visible.", "needs_user_data": false, "user_data_field": null, "status":
"done"},

```

```

        {"id": 2, "action": "Select the report file.", "target": "report.docx",
"expected_result": "The file is selected.", "needs_user_data": false,
"user_data_field": null, "status": "done"},
        {"id": 3, "action": "Rename the file to final_report.txt.", "target":
"Rename action", "expected_result": "The file is renamed.", "needs_user_data":
false, "user_data_field": null, "status": "done"}
    ],
    "notes": "A common file management workflow completed without issues.",
    "estimated_complexity": "complex"
},
"step_plans": [
    {
        "step_id": 1,
        "action": "Open File Explorer and navigate to Documents.",
        "target": "Documents folder",
        "expected_result": "The Documents folder contents are visible.",
        "hid_commands": [
            {"cmd": "mouse_move", "meta": {"commandId": "s10-1"}, "dx": 468, "dy":
312, "smooth": true},
            {"cmd": "mouse_click", "meta": {"commandId": "s10-2"}, "button": "left"}
        ],
        "reasoning": "The Documents folder is visible in the file explorer list."
    },
    {
        "step_id": 2,
        "action": "Select the report file.",
        "target": "report.docx",
        "expected_result": "The file is selected.",
        "hid_commands": [
            {"cmd": "mouse_move", "meta": {"commandId": "s10-3"}, "dx": 728, "dy":
426, "smooth": true},
            {"cmd": "mouse_click", "meta": {"commandId": "s10-4"}, "button": "left"}
        ],
        "reasoning": "The target file is visible in the folder view."
    },
    {
        "step_id": 3,
        "action": "Rename the file to final_report.txt.",
        "target": "Rename action",
        "expected_result": "The file is renamed.",
        "hid_commands": [
            {"cmd": "mouse_move", "meta": {"commandId": "s10-5"}, "dx": 1112, "dy":
288, "smooth": true},
            {"cmd": "mouse_click", "meta": {"commandId": "s10-6"}, "button": "left"},

```

```

        {"cmd": "type_text", "meta": {"commandId": "s10-7"}, "text":
"final_report.txt"},
        {"cmd": "key_press", "meta": {"commandId": "s10-8"}, "key": "enter"}
    ],
    "reasoning": "The rename command is available from the context menu."
}
],
"step_results": [
    {"step_id": 1, "action": "Open File Explorer and navigate to Documents.",
"attempt": 1, "exec_status": "success", "eval_status": "done", "confidence":
0.96, "reason": "Documents folder contents are visible.", "final_status": "done",
"timestamp": "2026-05-03T11:01:03Z"},
    {"step_id": 2, "action": "Select the report file.", "attempt": 1,
"exec_status": "success", "eval_status": "done", "confidence": 0.95, "reason":
"The report file is selected.", "final_status": "done", "timestamp": "2026-05-
03T11:02:17Z"},
    {"step_id": 3, "action": "Rename the file to final_report.txt.", "attempt":
1, "exec_status": "success", "eval_status": "done", "confidence": 0.97, "reason":
"The file name changed to final_report.txt.", "final_status": "done",
"timestamp": "2026-05-03T11:04:49Z"}
],
"final_report": {
    "success": true,
    "summary": "The file was renamed successfully.",
    "message": "File Explorer was opened, the Documents folder was reached, and
the report file was renamed to final_report.txt.",
    "steps_completed": 3,
    "steps_failed": 0,
    "issues": [],
    "recommendations": []
}
}

```

Test Case 24: ClickUp Inbox - Check for New Tasks (Complex Task)

User Prompt:

Click on the "Inbox | LammastideIT" tab, and you can see the clickUp interface. so check and tell me are there any new tasks for me

Test Results:

- **Steps Total: 2**
- **Completed: 0**
- **Failed: 2**
- **Duration: 12 minutes 51 seconds**

Step Execution:

Step 1: Locate and click on the 'Inbox | LammastideIT' tab in the browser to switch to the ClickUp interface

- **Target: Inbox | LammastideIT browser tab**
- **Attempts: 3**
- **Execution: Performed**
- **Evaluation: Failed (retry loop)**
- **Issue: The ClickUp Inbox interface is not visible. Warning message indicates inbox not visible. The target browser tab was not detected in the current screen view.**

Step 2: Scan the ClickUp Inbox for any items marked as new or unread tasks

- **Target: ClickUp Inbox task list**
- **Attempts: 3**
- **Execution: Performed**
- **Evaluation: Failed (retry loop)**
- **Issue: Screen displays ScreenPilot AI logo and application name instead of ClickUp Inbox. Prerequisite Step 1 failed.**

Execution Summary:

The task failed completely. The target browser tab 'Inbox | LammastideIT' was not detected, and the ClickUp interface never loaded. Internal errors occurred during screen display processing and command validation.

Issues Identified:

- **Target browser tab not detected in current screen view**
- **Internal errors during screen display processing**
- **ClickUp Inbox never became visible**
- **Multiple retry attempts failed to resolve the issue**

Recommendations for Improvement:

- **Ensure the browser window is open and visible on screen**
- **Manually select the ClickUp tab before restarting the task**
- **Add fallback to search for tab by partial text match**
- **Implement longer wait for interface to load before detection**

JSON Response:

full_cycle.json

```
{
  "run_id": "7a611e76-114c-408e-9c30-64b3b0132eac",
  "user_task": "Click on the \"Inbox | LammastideIT\" tab, and you can see the clickUp interface. so check and tell me are there any new tasks for me",
  "started_at": "2026-05-03T14:15:24.150607+00:00",
  "finished_at": "2026-05-03T14:28:15.070600+00:00",
  "vision_used": true,
  "perception": {
    "image": "E:\\sliit-project\\ScreenAwareTaskAgent\\vision\\data\\sessions\\shot_1777817731_1216e3\\processing\\proc_1777817731_f7c49b\\preprocessed_frames\\frame_1777817731_bfb5d5.jpg",
    "image_size": {
      "width": 1920,
      "height": 1080
    },
    "screen_bbox": [
      176,
      182,
      1751,
      934
    ],
    "screen_size": {
      "width": 1575,
      "height": 752
    },
    "coordinate_system": "pixel",
    "elements": [
      {
```

```
"id": "logo_icon",
"type": "icon",
"label": "ScreenPilot AI Logo",
"description": "The logo icon for the ScreenPilot AI application.",
"state": "visible",
"dx": 228,
"dy": 234,
"frame_dx": 228,
"frame_dy": 234,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    215,
    221,
    240,
    246
]
},
{
    "id": "title_text",
    "type": "text",
    "label": "ScreenPilot AI",
    "description": "The main title text of the application.",
    "state": "visible",
    "dx": 307,
    "dy": 234,
    "frame_dx": 307,
    "frame_dy": 234,
    "confidence": 0.95,
```

```
    "source": "gemini_vlm",
    "frame_bbox": [
        247,
        221,
        367,
        246
    ]
}
],
"element_count": 2
},
"knowledge_base": {
    "total_captures": 14,
    "screens": [
        {
            "capture_index": 1,
            "phase": "initial",
            "recorded_at": "2026-05-03T14:16:02.747121+00:00",
            "fingerprint": "9ad4863291be",
            "element_count": 2,
            "screen": {
                "image": "E:\\sliit-
project\\ScreenAwareTaskAgent\\vision\\data\\sessions\\shot_1777817731_1216e3\\pr
ocessing\\proc_1777817731_f7c49b\\preprocessed_frames\\frame_1777817731_bfb5d5.jp
g",
                "image_size": {
                    "width": 1920,
                    "height": 1080
                },
                "screen_bbox": [
```

```
176,  
182,  
1751,  
934  
],  
"screen_size": {  
  "width": 1575,  
  "height": 752  
},  
"coordinate_system": "pixel",  
"elements": [  
  {  
    "id": "logo_icon",  
    "type": "icon",  
    "label": "ScreenPilot AI Logo",  
    "description": "The logo icon for the ScreenPilot AI application.",  
    "state": "visible",  
    "dx": 228,  
    "dy": 234,  
    "frame_dx": 228,  
    "frame_dy": 234,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
      215,  
      221,  
      240,  
      246  
    ]  
  }  
]
```

```
    },  
    {  
      "id": "title_text",  
      "type": "text",  
      "label": "ScreenPilot AI",  
      "description": "The main title text of the application.",  
      "state": "visible",  
      "dx": 307,  
      "dy": 234,  
      "frame_dx": 307,  
      "frame_dy": 234,  
      "confidence": 0.95,  
      "source": "gemini_vlm",  
      "frame_bbox": [  
        247,  
        221,  
        367,  
        246  
      ]  
    }  
  ],  
  "element_count": 2  
}  
},  
{  
  "capture_index": 2,  
  "phase": "pre_step",  
  "recorded_at": "2026-05-03T14:16:42.768831+00:00",  
  "fingerprint": "8d1d78abdbc9",
```

```
"element_count": 14,

"screen": {

  "image": "E:\\sliit-
project\\ScreenAwareTaskAgent\\vision\\data\\sessions\\shot_1777817778_8e94ee\\pr
ocessing\\proc_1777817778_b789e1\\preprocessed_frames\\frame_1777817778_8f2894.jp
g",

  "image_size": {

    "width": 1920,

    "height": 1080

  },

  "screen_bbox": [

    176,

    182,

    1751,

    934

  ],

  "screen_size": {

    "width": 1575,

    "height": 752

  },

  "coordinate_system": "pixel",

  "elements": [

    {

      "id": "0",

      "type": "text",

      "label": "ScreenPilot AI",

      "description": "Application title",

      "state": "visible",

      "dx": 314,

      "dy": 256,
```

```
"frame_dx": 314,  
"frame_dy": 256,  
"confidence": 0.9,  
"source": "gemini_vlm",  
"frame_bbox": [  
    231,  
    237,  
    397,  
    274  
]  
},  
{  
    "id": "1",  
    "type": "label",  
    "label": "Autonomous Task Agent v2",  
    "description": "Version tag",  
    "state": "visible",  
    "dx": 480,  
    "dy": 262,  
    "frame_dx": 480,  
    "frame_dy": 262,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        410,  
        250,  
        551,  
        274  
    ]  
}
```

```
},  
{  
  "id": "2",  
  "type": "pane",  
  "label": "User Message",  
  "description": "Message from the user in a speech bubble",  
  "state": "visible",  
  "dx": 1000,  
  "dy": 357,  
  "frame_dx": 1000,  
  "frame_dy": 357,  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    625,  
    317,  
    1376,  
    397  
  ]  
},  
{  
  "id": "3",  
  "type": "pane",  
  "label": "Task Plan Container",  
  "description": "Container for the task plan section",  
  "state": "visible",  
  "dx": 530,  
  "dy": 496,  
  "frame_dx": 530,
```



```
"frame_dy": 496,  
"confidence": 0.9,  
"source": "gemini_vlm",  
"frame_bbox": [  
    231,  
    410,  
    828,  
    582  
]  
},  
{  
    "id": "4",  
    "type": "text",  
    "label": "Task Plan",  
    "description": "Heading for the task plan section",  
    "state": "visible",  
    "dx": 290,  
    "dy": 438,  
    "frame_dx": 290,  
    "frame_dy": 438,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        250,  
        428,  
        330,  
        447  
    ]  
},
```

```
{
  "id": "5",
  "type": "text",
  "label": "0/2 done",
  "description": "Task completion status",
  "state": "visible",
  "dx": 788,
  "dy": 438,
  "frame_dx": 788,
  "frame_dy": 438,
  "confidence": 0.9,
  "source": "gemini_vlm",
  "frame_bbox": [
    767,
    428,
    810,
    447
  ]
},
{
  "id": "6",
  "type": "text",
  "label": "Task Plan Description",
  "description": "Description of the current task plan",
  "state": "visible",
  "dx": 530,
  "dy": 474,
  "frame_dx": 530,
  "frame_dy": 474,
```

```
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        250,
        459,
        810,
        490
    ]
},
{
    "id": "7",
    "type": "list_item",
    "label": "Step 1",
    "description": "First step of the task plan: Locate and click on
the Inbox tab",
    "state": "visible",
    "dx": 533,
    "dy": 518,
    "frame_dx": 533,
    "frame_dy": 518,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        250,
        502,
        816,
        533
    ]
},
```

```
{
  "id": "8",
  "type": "list_item",
  "label": "Step 2",
  "description": "Second step of the task plan: Scan the ClickUp
Inbox",
  "state": "visible",
  "dx": 530,
  "dy": 558,
  "frame_dx": 530,
  "frame_dy": 558,
  "confidence": 0.9,
  "source": "gemini_vlm",
  "frame_bbox": [
    250,
    545,
    810,
    570
  ]
},
{
  "id": "9",
  "type": "pane",
  "label": "Log Container",
  "description": "Container for action logs",
  "state": "visible",
  "dx": 530,
  "dy": 665,
  "frame_dx": 530,
```

```
"frame_dy": 665,  
"confidence": 0.9,  
"source": "gemini_vlm",  
"frame_bbox": [  
    231,  
    588,  
    828,  
    742  
]  
},  
{  
    "id": "10",  
    "type": "text",  
    "label": "Log Entries",  
    "description": "List of actions performed by the agent",  
    "state": "visible",  
    "dx": 530,  
    "dy": 668,  
    "frame_dx": 530,  
    "frame_dy": 668,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        250,  
        607,  
        810,  
        730  
    ]  
},
```

```
{
  "id": "11",
  "type": "text",
  "label": "Agent is working...",
  "description": "Status message at the bottom",
  "state": "visible",
  "dx": 312,
  "dy": 896,
  "frame_dx": 312,
  "frame_dy": 896,
  "confidence": 0.9,
  "source": "gemini_vlm",
  "frame_bbox": [
    244,
    884,
    379,
    908
  ]
},
{
  "id": "12",
  "type": "button",
  "label": "Send",
  "description": "Button to send a message",
  "state": "visible",
  "dx": 1351,
  "dy": 892,
  "frame_dx": 1351,
  "frame_dy": 892,
```

```
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        1326,  
        865,  
        1376,  
        920  
    ]  
},  
{  
    "id": "13",  
    "type": "taskbar",  
    "label": "Taskbar",  
    "description": "Windows taskbar with application icons",  
    "state": "visible",  
    "dx": 964,  
    "dy": 927,  
    "frame_dx": 964,  
    "frame_dy": 927,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        194,  
        920,  
        1733,  
        934  
    ]  
}  
],
```

```
        "element_count": 14
    },
    "metadata": {
        "step_index": 0,
        "attempt": 1
    }
},
{
    "capture_index": 3,
    "phase": "post_step",
    "recorded_at": "2026-05-03T14:17:56.699959+00:00",
    "fingerprint": "8fc70cfd2773",
    "element_count": 21,
    "screen": {
        "image": "E:\\sliit-
project\\ScreenAwareTaskAgent\\vision\\data\\sessions\\shot_1777817848_94e377\\pr
ocessing\\proc_1777817848_fa4410\\preprocessed_frames\\frame_1777817848_803ed1.jp
g",
        "image_size": {
            "width": 1920,
            "height": 1080
        },
        "screen_bbox": [
            176,
            182,
            1752,
            934
        ],
        "screen_size": {
            "width": 1576,
```



```
    "height": 752
  },
  "coordinate_system": "pixel",
  "elements": [
    {
      "id": "0",
      "type": "icon",
      "label": "ScreenPilot AI logo",
      "description": "The logo icon for ScreenPilot AI application.",
      "state": "visible",
      "dx": 228,
      "dy": 233,
      "frame_dx": 228,
      "frame_dy": 233,
      "confidence": 0.95,
      "source": "gemini_vlm",
      "frame_bbox": [
        215,
        219,
        240,
        247
      ]
    },
    {
      "id": "1",
      "type": "text",
      "label": "ScreenPilot AI",
      "description": "The title text of the application.",
      "state": "visible",
```

```
"dx": 301,  
  "dy": 238,  
  "frame_dx": 301,  
  "frame_dy": 238,  
  "confidence": 0.98,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    247,  
    226,  
    355,  
    251  
  ]  
},  
{  
  "id": "2",  
  "type": "badge",  
  "label": "Autonomous Task Agent v2",  
  "description": "A badge indicating the version of the task agent.",  
  "state": "visible",  
  "dx": 462,  
  "dy": 242,  
  "frame_dx": 462,  
  "frame_dy": 242,  
  "confidence": 0.92,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    390,  
    229,  
    534,
```

```
        255
    ]
},
{
    "id": "3",
    "type": "container",
    "label": "User message bubble",
    "description": "A light blue bubble containing the user's
request.",
    "state": "visible",
    "dx": 981,
    "dy": 337,
    "frame_dx": 981,
    "frame_dy": 337,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        606,
        298,
        1356,
        376
    ]
},
{
    "id": "4",
    "type": "pane",
    "label": "Task Plan pane",
    "description": "A container showing the current task plan and its
progress.",
    "state": "visible",
```

```
"dx": 512,  
  "dy": 480,  
  "frame_dx": 512,  
  "frame_dy": 480,  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    214,  
    396,  
    810,  
    564  
  ]  
},  
{  
  "id": "5",  
  "type": "text",  
  "label": "Task Plan heading",  
  "description": "The heading for the task plan section.",  
  "state": "visible",  
  "dx": 272,  
  "dy": 420,  
  "frame_dx": 272,  
  "frame_dy": 420,  
  "confidence": 0.98,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    234,  
    413,  
    309,
```

```
        428
    ]
},
{
    "id": "6",
    "type": "badge",
    "label": "0/2 done",
    "description": "A badge showing the progress of the task plan.",
    "state": "visible",
    "dx": 771,
    "dy": 420,
    "frame_dx": 771,
    "frame_dy": 420,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        750,
        413,
        792,
        428
    ]
},
{
    "id": "7",
    "type": "text",
    "label": "Task Plan description",
    "description": "Introductory text for the task plan.",
    "state": "visible",
    "dx": 512,
```

```
        "dy": 457,
        "frame_dx": 512,
        "frame_dy": 457,
        "confidence": 0.95,
        "source": "gemini_vlm",
        "frame_bbox": [
            233,
            443,
            792,
            471
        ]
    },
    {
        "id": "8",
        "type": "container",
        "label": "Step 1 box",
        "description": "A highlighted box for the first step of the task
plan.",
        "state": "visible",
        "dx": 514,
        "dy": 498,
        "frame_dx": 514,
        "frame_dy": 498,
        "confidence": 0.9,
        "source": "gemini_vlm",
        "frame_bbox": [
            233,
            482,
            795,
```

```
513
    ]
  },
  {
    "id": "9",
    "type": "text",
    "label": "Step 2 text",
    "description": "The text for the second step of the task plan.",
    "state": "visible",
    "dx": 474,
    "dy": 536,
    "frame_dx": 474,
    "frame_dy": 536,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
      233,
      528,
      715,
      543
    ]
  },
  {
    "id": "10",
    "type": "pane",
    "label": "Execution Log pane",
    "description": "A container showing the log of actions performed by
the agent.",
    "state": "visible",
```

```
"dx": 512,
"dy": 656,
"frame_dx": 512,
"frame_dy": 656,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    214,
    570,
    810,
    742
]
},
{
    "id": "11",
    "type": "text",
    "label": "Plan created text",
    "description": "Log entry indicating the plan was created.",
    "state": "visible",
    "dx": 318,
    "dy": 594,
    "frame_dx": 318,
    "frame_dy": 594,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        229,
        587,
        407,
```



```
        602
    ]
},
{
    "id": "12",
    "type": "text",
    "label": "Step 1 log text",
    "description": "Log entry for the first step execution.",
    "state": "visible",
    "dx": 512,
    "dy": 622,
    "frame_dx": 512,
    "frame_dy": 622,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        229,
        608,
        795,
        637
    ]
},
{
    "id": "13",
    "type": "text",
    "label": "Screen snapshot log text",
    "description": "Log entry indicating a screen snapshot was taken.",
    "state": "visible",
    "dx": 336,
```

```
"dy": 651,
"frame_dx": 336,
"frame_dy": 651,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    229,
    644,
    444,
    658
]
},
{
    "id": "14",
    "type": "text",
    "label": "Generating commands log text",
    "description": "Log entry indicating command generation.",
    "state": "visible",
    "dx": 512,
    "dy": 677,
    "frame_dx": 512,
    "frame_dy": 677,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        229,
        662,
        795,
        692
    ]
}
```

```
]
},
{
  "id": "15",
  "type": "text",
  "label": "Sending 2 HID commands log text",
  "description": "Log entry indicating commands are being sent.",
  "state": "visible",
  "dx": 318,
  "dy": 708,
  "frame_dx": 318,
  "frame_dy": 708,
  "confidence": 0.95,
  "source": "gemini_vlm",
  "frame_bbox": [
    229,
    700,
    407,
    715
  ]
},
{
  "id": "16",
  "type": "text",
  "label": "Validating step 1 log text",
  "description": "Log entry indicating step validation.",
  "state": "visible",
  "dx": 303,
  "dy": 728,
```

```
"frame_dx": 303,  
"frame_dy": 728,  
"confidence": 0.95,  
"source": "gemini_vlm",  
"frame_bbox": [  
    229,  
    720,  
    377,  
    735  
]  
},  
{  
    "id": "17",  
    "type": "container",  
    "label": "Bottom bar container",  
    "description": "The bar at the bottom of the application window.",  
    "state": "visible",  
    "dx": 758,  
    "dy": 884,  
    "frame_dx": 758,  
    "frame_dy": 884,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        214,  
        856,  
        1303,  
        912  
    ]  
}
```

```
    },  
    {  
      "id": "18",  
      "type": "text",  
      "label": "Agent is working text",  
      "description": "Status text indicating the agent is active.",  
      "state": "visible",  
      "dx": 292,  
      "dy": 884,  
      "frame_dx": 292,  
      "frame_dy": 884,  
      "confidence": 0.98,  
      "source": "gemini_vlm",  
      "frame_bbox": [  
        229,  
        876,  
        355,  
        891  
      ]  
    },  
    {  
      "id": "19",  
      "type": "button",  
      "label": "Send button",  
      "description": "A circular button with a paper plane icon to send  
messages.",  
      "state": "visible",  
      "dx": 1335,  
      "dy": 884,
```

```
"frame_dx": 1335,  
"frame_dy": 884,  
"confidence": 0.95,  
"source": "gemini_vlm",  
"frame_bbox": [  
    1312,  
    856,  
    1358,  
    912  
]  
},  
{  
    "id": "20",  
    "type": "container",  
    "label": "Taskbar",  
    "description": "The system taskbar at the bottom of the screen.",  
    "state": "visible",  
    "dx": 792,  
    "dy": 927,  
    "frame_dx": 792,  
    "frame_dy": 927,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        607,  
        920,  
        976,  
        934  
    ]  
}
```

```
    }
  ],
  "element_count": 21
},
"metadata": {
  "step_index": 0,
  "attempt": 1
}
},
{
  "capture_index": 4,
  "phase": "pre_step",
  "recorded_at": "2026-05-03T14:18:55.612643+00:00",
  "fingerprint": "b313fab6e983",
  "element_count": 2,
  "screen": {
    "image": "E:\\slit-
project\\ScreenAwareTaskAgent\\vision\\data\\sessions\\shot_1777817903_96a816\\pr
ocessing\\proc_1777817903_c9338d\\preprocessed_frames\\frame_1777817903_08b078.jp
g",
    "image_size": {
      "width": 1920,
      "height": 1080
    },
    "screen_bbox": [
      176,
      106,
      1752,
      935
    ],
  },
}
```

```
"screen_size": {
  "width": 1576,
  "height": 829
},
"coordinate_system": "pixel",
"elements": [
  {
    "id": "tab_frontend_1",
    "type": "tab",
    "label": "frontend",
    "description": "Browser tab for frontend application",
    "state": "inactive",
    "dx": 281,
    "dy": 130,
    "frame_dx": 281,
    "frame_dy": 130,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
      191,
      112,
      371,
      149
    ]
  },
  {
    "id": "tab_whatsapp",
    "type": "tab",
    "label": "(12) WhatsApp",
```



```
    "description": "Browser tab for WhatsApp",
    "state": "inactive",
    "dx": 463,
    "dy": 130,
    "frame_dx": 463,
    "frame_dy": 130,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        373,
        112,
        553,
        149
    ]
  },
  "element_count": 2
},
"metadata": {
  "step_index": 0,
  "attempt": 2
}
},
{
  "capture_index": 5,
  "phase": "post_step",
  "recorded_at": "2026-05-03T14:20:15.408499+00:00",
  "fingerprint": "97b1a2aeaadb",
  "element_count": 2,
```

```
"screen": {
  "image": "E:\\sliit-
project\\ScreenAwareTaskAgent\\vision\\data\\sessions\\shot_1777817984_52a134\\pr
ocessing\\proc_1777817984_726924\\preprocessed_frames\\frame_1777817984_2544ef.jp
g",
  "image_size": {
    "width": 1920,
    "height": 1080
  },
  "screen_bbox": [
    176,
    182,
    1751,
    934
  ],
  "screen_size": {
    "width": 1575,
    "height": 752
  },
  "coordinate_system": "pixel",
  "elements": [
    {
      "id": "logo_icon",
      "type": "icon",
      "label": "ScreenPilot AI Logo",
      "description": "Application logo icon in the top left corner.",
      "state": "visible",
      "dx": 227,
      "dy": 238,
      "frame_dx": 227,
```

```
"frame_dy": 238,  
"confidence": 0.95,  
"source": "gemini_vlm",  
"frame_bbox": [  
    210,  
    219,  
    244,  
    256  
]  
},  
{  
    "id": "app_title",  
    "type": "text",  
    "label": "ScreenPilot AI",  
    "description": "Application name text in the top left.",  
    "state": "visible",  
    "dx": 314,  
    "dy": 240,  
    "frame_dx": 314,  
    "frame_dy": 240,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        250,  
        225,  
        379,  
        256  
    ]  
}
```

```
    ],
    "element_count": 2
  },
  "metadata": {
    "step_index": 0,
    "attempt": 2
  }
},
{
  "capture_index": 6,
  "phase": "pre_step",
  "recorded_at": "2026-05-03T14:20:56.309601+00:00",
  "fingerprint": "60bb2f0ff63e",
  "element_count": 1,
  "screen": {
    "image": "E:\\sliit-
project\\ScreenAwareTaskAgent\\vision\\data\\sessions\\shot_1777818025_aba1b1\\pr
ocessing\\proc_1777818025_4f0ab2\\preprocessed_frames\\frame_1777818025_8f7569.jp
g",
    "image_size": {
      "width": 1920,
      "height": 1080
    },
    "screen_bbox": [
      176,
      182,
      1752,
      934
    ],
    "screen_size": {
```

```
    "width": 1576,  
    "height": 752  
  },  
  "coordinate_system": "pixel",  
  "elements": [  
    {  
      "id": "logo",  
      "type": "heading",  
      "label": "ScreenPilot AI",  
      "description": "Application logo and title",  
      "state": "visible",  
      "dx": 290,  
      "dy": 233,  
      "frame_dx": 290,  
      "frame_dy": 233,  
      "confidence": 0.95,  
      "source": "gemini_vlm",  
      "frame_bbox": [  
        226,  
        218,  
        355,  
        248  
      ]  
    }  
  ],  
  "element_count": 1  
},  
"metadata": {  
  "step_index": 0,
```

```
        "attempt": 3
    }
},
{
    "capture_index": 7,
    "phase": "post_step",
    "recorded_at": "2026-05-03T14:22:12.230429+00:00",
    "fingerprint": "c76b585c7ea7",
    "element_count": 14,
    "screen": {
        "image": "E:\\sliit-
project\\ScreenAwareTaskAgent\\vision\\data\\sessions\\shot_1777818101_7cbf71\\pr
ocessing\\proc_1777818101_0db3a7\\preprocessed_frames\\frame_1777818101_13dcf.jp
g",
        "image_size": {
            "width": 1920,
            "height": 1080
        },
        "screen_bbox": [
            176,
            182,
            1752,
            934
        ],
        "screen_size": {
            "width": 1576,
            "height": 752
        },
        "coordinate_system": "pixel",
        "elements": [
```

```
{
  "id": "logo",
  "type": "icon",
  "label": "ScreenPilot AI Logo",
  "description": "Logo icon of ScreenPilot AI",
  "state": "visible",
  "dx": 226,
  "dy": 236,
  "frame_dx": 226,
  "frame_dy": 236,
  "confidence": 0.9,
  "source": "gemini_vlm",
  "frame_bbox": [
    214,
    221,
    239,
    252
  ]
},
{
  "id": "title",
  "type": "heading",
  "label": "ScreenPilot AI",
  "description": "Application title",
  "state": "visible",
  "dx": 312,
  "dy": 238,
  "frame_dx": 312,
  "frame_dy": 238,
```

```
"confidence": 0.9,  
"source": "gemini_vlm",  
"frame_bbox": [  
    247,  
    225,  
    378,  
    250  
]  
},  
{  
    "id": "badge",  
    "type": "label",  
    "label": "Autonomous Task Agent v2",  
    "description": "Version badge",  
    "state": "visible",  
    "dx": 462,  
    "dy": 242,  
    "frame_dx": 462,  
    "frame_dy": 242,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        390,  
        229,  
        534,  
        255  
    ]  
},  
{
```



```
"id": "user_msg",
"type": "text",
"label": "User Message",
"description": "Message from the user requesting task check",
"state": "visible",
"dx": 981,
"dy": 337,
"frame_dx": 981,
"frame_dy": 337,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    606,
    298,
    1356,
    376
]
},
{
    "id": "task_plan_pane",
    "type": "pane",
    "label": "Task Plan Pane",
    "description": "Container for the task plan",
    "state": "visible",
    "dx": 512,
    "dy": 480,
    "frame_dx": 512,
    "frame_dy": 480,
    "confidence": 0.9,
```

```
"source": "gemini_vlm",
"frame_bbox": [
    214,
    397,
    810,
    562
]
},
{
    "id": "task_plan_heading",
    "type": "heading",
    "label": "Task Plan",
    "description": "Heading for the task plan section",
    "state": "visible",
    "dx": 272,
    "dy": 420,
    "frame_dx": 272,
    "frame_dy": 420,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        234,
        412,
        309,
        427
    ]
},
{
    "id": "progress",
```

```
"type": "label",
"label": "0/2 done",
"description": "Task progress indicator",
"state": "visible",
"dx": 772,
"dy": 420,
"frame_dx": 772,
"frame_dy": 420,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    750,
    411,
    795,
    428
]
},
{
    "id": "task_plan_desc",
    "type": "text",
    "label": "Task Plan Description",
    "description": "Description of the current task plan",
    "state": "visible",
    "dx": 513,
    "dy": 456,
    "frame_dx": 513,
    "frame_dy": 456,
    "confidence": 0.9,
    "source": "gemini_vlm",
```

```
"frame_bbox": [
    231,
    440,
    795,
    471
]
},
{
    "id": "task_1",
    "type": "list_item",
    "label": "Task 1",
    "description": "First task: Locate and click on the Inbox tab",
    "state": "active",
    "dx": 513,
    "dy": 498,
    "frame_dx": 513,
    "frame_dy": 498,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        231,
        482,
        795,
        513
    ]
},
{
    "id": "task_2",
    "type": "list_item",
```

```
    "label": "Task 2",
    "description": "Second task: Scan the ClickUp Inbox",
    "state": "pending",
    "dx": 513,
    "dy": 534,
    "frame_dx": 513,
    "frame_dy": 534,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        231,
        519,
        795,
        550
    ]
},
{
    "id": "status_log_pane",
    "type": "pane",
    "label": "Status Log Pane",
    "description": "Container for agent status logs",
    "state": "visible",
    "dx": 512,
    "dy": 668,
    "frame_dx": 512,
    "frame_dy": 668,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
```

```
        214,  
        577,  
        810,  
        758  
    ]  
},  
{  
    "id": "warning_text",  
    "type": "text",  
    "label": "Warning Message",  
    "description": "Warning about ClickUp Inbox not being visible",  
    "state": "visible",  
    "dx": 513,  
    "dy": 610,  
    "frame_dx": 513,  
    "frame_dy": 610,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        231,  
        587,  
        795,  
        632  
    ]  
},  
{  
    "id": "retry_text",  
    "type": "text",  
    "label": "Retry Status",
```

```
"description": "Retrying step 1 message",
"state": "visible",
"dx": 322,
"dy": 647,
"frame_dx": 322,
"frame_dy": 647,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    231,
    639,
    412,
    655
]
},
{
    "id": "snapshot_text",
    "type": "text",
    "label": "Snapshot Status",
    "description": "Screen snapshot message",
    "state": "visible",
    "dx": 322,
    "dy": 669,
    "frame_dx": 322,
    "frame_dy": 669,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        231,
```

```
        661,  
        412,  
        677  
    ]  
}  
],  
  "element_count": 14  
},  
  "metadata": {  
    "step_index": 0,  
    "attempt": 3  
  }  
},  
{  
  "capture_index": 8,  
  "phase": "pre_step",  
  "recorded_at": "2026-05-03T14:22:42.683812+00:00",  
  "fingerprint": "947526b1a734",  
  "element_count": 21,  
  "screen": {  
    "image": "E:\\sliit-  
project\\ScreenAwareTaskAgent\\vision\\data\\sessions\\shot_1777818142_4f1c60\\pr  
ocessing\\proc_1777818142_22ef7b\\preprocessed_frames\\frame_1777818142_17aa16.jp  
g",  
    "image_size": {  
      "width": 1920,  
      "height": 1080  
    },  
    "screen_bbox": [  
      176,
```



```
182,  
1752,  
934  
],  
"screen_size": {  
  "width": 1576,  
  "height": 752  
},  
"coordinate_system": "pixel",  
"elements": [  
  {  
    "id": "0",  
    "type": "text",  
    "label": "ScreenPilot AI",  
    "description": "Application title and logo in the top left  
corner.",  
    "state": "visible",  
    "dx": 296,  
    "dy": 270,  
    "frame_dx": 296,  
    "frame_dy": 270,  
    "confidence": 0.98,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
      214,  
      239,  
      378,  
      300  
    ]  
  }  
]
```

```
    },  
    {  
      "id": "1",  
      "type": "label",  
      "label": "Autonomous Task Agent v2",  
      "description": "Status badge indicating the agent version.",  
      "state": "visible",  
      "dx": 463,  
      "dy": 278,  
      "frame_dx": 463,  
      "frame_dy": 278,  
      "confidence": 0.95,  
      "source": "gemini_vlm",  
      "frame_bbox": [  
        390,  
        257,  
        536,  
        300  
      ]  
    },  
    {  
      "id": "2",  
      "type": "text",  
      "label": "User Message",  
      "description": "A large chat bubble containing user instructions,  
partially obscured by glare.",  
      "state": "visible",  
      "dx": 982,  
      "dy": 435,
```

```
"frame_dx": 982,  
"frame_dy": 435,  
"confidence": 0.9,  
"source": "gemini_vlm",  
"frame_bbox": [  
    606,  
    368,  
    1357,  
    502  
]  
},  
{  
    "id": "3",  
    "type": "heading",  
    "label": "Task Plan",  
    "description": "Header for the task execution plan section.",  
    "state": "visible",  
    "dx": 274,  
    "dy": 570,  
    "frame_dx": 274,  
    "frame_dy": 570,  
    "confidence": 0.97,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        233,  
        552,  
        316,  
        588  
    ]  
}
```

```
    },  
    {  
      "id": "4",  
      "type": "label",  
      "label": "0/2 done",  
      "description": "Progress indicator for the current task plan.",  
      "state": "visible",  
      "dx": 774,  
      "dy": 572,  
      "frame_dx": 774,  
      "frame_dy": 572,  
      "confidence": 0.96,  
      "source": "gemini_vlm",  
      "frame_bbox": [  
        750,  
        557,  
        797,  
        588  
      ]  
    },  
    {  
      "id": "5",  
      "type": "text",  
      "label": "Task Plan Description",  
      "description": "Text explaining the current objective of the  
agent.",  
      "state": "visible",  
      "dx": 515,  
      "dy": 628,
```

```
    "frame_dx": 515,
    "frame_dy": 628,
    "confidence": 0.94,
    "source": "gemini_vlm",
    "frame_bbox": [
        233,
        602,
        797,
        655
    ]
},
{
    "id": "6",
    "type": "list_item",
    "label": "Step 1: Locate and click on the Inbox tab",
    "description": "The first step in the task plan, highlighted in red
indicating an issue.",
    "state": "visible",
    "dx": 515,
    "dy": 699,
    "frame_dx": 515,
    "frame_dy": 699,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        233,
        676,
        797,
        722
    ]
}
```

```
    ]
  },
  {
    "id": "7",
    "type": "list_item",
    "label": "Step 2: Scan the ClickUp Inbox",
    "description": "The second step in the task plan, currently in
progress.",
    "state": "visible",
    "dx": 515,
    "dy": 754,
    "frame_dx": 515,
    "frame_dy": 754,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
      233,
      731,
      797,
      778
    ]
  },
  {
    "id": "8",
    "type": "pane",
    "label": "Execution Log",
    "description": "A container showing the detailed logs of the
agent's actions and errors.",
    "state": "visible",
    "dx": 512,
```

```
    "dy": 872,
    "frame_dx": 512,
    "frame_dy": 872,
    "confidence": 0.92,
    "source": "gemini_vlm",
    "frame_bbox": [
        214,
        811,
        809,
        934
    ]
},
{
    "id": "9",
    "type": "text",
    "label": "Log: Sending 2 HID commands",
    "description": "Log entry indicating hardware interaction commands being sent.",
    "state": "visible",
    "dx": 305,
    "dy": 853,
    "frame_dx": 305,
    "frame_dy": 853,
    "confidence": 0.93,
    "source": "gemini_vlm",
    "frame_bbox": [
        226,
        842,
        384,
```

```
        864
    ]
},
{
    "id": "10",
    "type": "text",
    "label": "Log: Validating step 1",
    "description": "Log entry showing the agent is checking the result
of the first step.",
    "state": "visible",
    "dx": 290,
    "dy": 884,
    "frame_dx": 290,
    "frame_dy": 884,
    "confidence": 0.93,
    "source": "gemini_vlm",
    "frame_bbox": [
        226,
        872,
        353,
        895
    ]
},
{
    "id": "11",
    "type": "text",
    "label": "Log: Warning message",
    "description": "Warning log stating the ClickUp Inbox interface is
not visible.",
    "state": "visible",
```



```
    "dx": 512,
    "dy": 918,
    "frame_dx": 512,
    "frame_dy": 918,
    "confidence": 0.91,
    "source": "gemini_vlm",
    "frame_bbox": [
        226,
        903,
        797,
        934
    ]
},
{
    "id": "12",
    "type": "text",
    "label": "Log: Step 1 failed",
    "description": "Error log indicating the failure of the first step
after multiple attempts.",
    "state": "visible",
    "dx": 446,
    "dy": 934,
    "frame_dx": 446,
    "frame_dy": 934,
    "confidence": 0.94,
    "source": "gemini_vlm",
    "frame_bbox": [
        226,
        933,
```

```
        667,  
        934  
    ]  
},  
{  
    "id": "13",  
    "type": "text",  
    "label": "Log: Step 2 start",  
    "description": "Log entry indicating the start of the second  
step.",  
    "state": "visible",  
    "dx": 446,  
    "dy": 934,  
    "frame_dx": 446,  
    "frame_dy": 934,  
    "confidence": 0.94,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        226,  
        933,  
        667,  
        934  
    ]  
},  
{  
    "id": "14",  
    "type": "text",  
    "label": "Log: Screen snapshot",  
    "description": "Log entry noting a snapshot was taken for the  
current step.",
```

```
    "state": "visible",
    "dx": 318,
    "dy": 934,
    "frame_dx": 318,
    "frame_dy": 934,
    "confidence": 0.93,
    "source": "gemini_vlm",
    "frame_bbox": [
        226,
        933,
        409,
        934
    ]
},
{
    "id": "15",
    "type": "status_bar",
    "label": "Agent Status Bar",
    "description": "Bottom bar showing the current status of the
agent.",
    "state": "visible",
    "dx": 792,
    "dy": 934,
    "frame_dx": 792,
    "frame_dy": 934,
    "confidence": 0.96,
    "source": "gemini_vlm",
    "frame_bbox": [
        214,
```

```
        933,  
        1369,  
        934  
    ]  
},  
{  
    "id": "16",  
    "type": "text",  
    "label": "Agent is working...",  
    "description": "Status text indicating the agent is currently  
active.",  
    "state": "visible",  
    "dx": 293,  
    "dy": 934,  
    "frame_dx": 293,  
    "frame_dy": 934,  
    "confidence": 0.97,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        233,  
        933,  
        353,  
        934  
    ]  
},  
{  
    "id": "17",  
    "type": "button",  
    "label": "Send Message",
```

```
        "description": "Circular button with a paper plane icon to send a  
message.",  
        "state": "visible",  
        "dx": 1336,  
        "dy": 934,  
        "frame_dx": 1336,  
        "frame_dy": 934,  
        "confidence": 0.98,  
        "source": "gemini_vlm",  
        "frame_bbox": [  
            1312,  
            933,  
            1359,  
            934  
        ]  
    },  
    {  
        "id": "18",  
        "type": "toolbar",  
        "label": "Taskbar",  
        "description": "The operating system taskbar at the bottom of the  
screen.",  
        "state": "visible",  
        "dx": 792,  
        "dy": 934,  
        "frame_dx": 792,  
        "frame_dy": 934,  
        "confidence": 0.9,  
        "source": "gemini_vlm",  
        "frame_bbox": [  

```

```
        176,  
        933,  
        1407,  
        934  
    ]  
},  
{  
    "id": "19",  
    "type": "icon",  
    "label": "Application Icons",  
    "description": "A group of application icons pinned to the  
taskbar.",  
    "state": "visible",  
    "dx": 754,  
    "dy": 934,  
    "frame_dx": 754,  
    "frame_dy": 934,  
    "confidence": 0.85,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        651,  
        933,  
        858,  
        934  
    ]  
},  
{  
    "id": "20",  
    "type": "label",
```

```
        "label": "System Tray",
        "description": "The system tray area on the right side of the
taskbar.",
        "state": "visible",
        "dx": 1368,
        "dy": 934,
        "frame_dx": 1368,
        "frame_dy": 934,
        "confidence": 0.88,
        "source": "gemini_vlm",
        "frame_bbox": [
            1335,
            933,
            1400,
            934
        ]
    },
    ],
    "element_count": 21
},
"metadata": {
    "step_index": 1,
    "attempt": 1
}
},
{
    "capture_index": 9,
    "phase": "post_step",
    "recorded_at": "2026-05-03T14:23:59.975135+00:00",
```

```
"fingerprint": "af3b8066cd45",
"element_count": 2,
"screen": {
  "image": "E:\\sliit-
project\\ScreenAwareTaskAgent\\vision\\data\\sessions\\shot_1777818209_4dcf93\\pr
ocessing\\proc_1777818209_145ad2\\preprocessed_frames\\frame_1777818209_303d01.jp
g",
  "image_size": {
    "width": 1920,
    "height": 1080
  },
  "screen_bbox": [
    176,
    182,
    1752,
    934
  ],
  "screen_size": {
    "width": 1576,
    "height": 752
  },
  "coordinate_system": "pixel",
  "elements": [
    {
      "id": "0",
      "type": "text",
      "label": "ScreenPilot AI",
      "description": "Application name and logo",
      "state": "visible",
      "dx": 299,
```



```
"dy": 244,  
  "frame_dx": 299,  
  "frame_dy": 244,  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    225,  
    225,  
    373,  
    262  
  ]  
},  
{  
  "id": "1",  
  "type": "label",  
  "label": "Autonomous Task Agent v2",  
  "description": "Version information badge",  
  "state": "visible",  
  "dx": 472,  
  "dy": 250,  
  "frame_dx": 472,  
  "frame_dy": 250,  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    398,  
    237,  
    545,  
    262
```

```
    ]
  }
],
"element_count": 2
},
"metadata": {
  "step_index": 1,
  "attempt": 1
}
},
{
  "capture_index": 10,
  "phase": "pre_step",
  "recorded_at": "2026-05-03T14:24:43.231759+00:00",
  "fingerprint": "c4273e388f45",
  "element_count": 8,
  "screen": {
    "image": "E:\\sliit-
project\\ScreenAwareTaskAgent\\vision\\data\\sessions\\shot_1777818250_8111ee\\pr
ocessing\\proc_1777818250_7cc3ac\\preprocessed_frames\\frame_1777818250_fc123d.jp
g",
    "image_size": {
      "width": 1920,
      "height": 1080
    },
    "screen_bbox": [
      176,
      182,
      1751,
      934
```

```
],
"screen_size": {
  "width": 1575,
  "height": 752
},
"coordinate_system": "pixel",
"elements": [
  {
    "id": "1",
    "type": "heading",
    "label": "ScreenPilot AI",
    "description": "Application title and logo",
    "state": "visible",
    "dx": 302,
    "dy": 234,
    "frame_dx": 302,
    "frame_dy": 234,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
      226,
      218,
      379,
      250
    ]
  },
  {
    "id": "2",
    "type": "label",
```

```
"label": "Autonomous Task Agent v2",
"description": "Version and status badge",
"state": "visible",
"dx": 462,
"dy": 240,
"frame_dx": 462,
"frame_dy": 240,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    391,
    228,
    534,
    253
]
},
{
    "id": "3",
    "type": "text",
    "label": "User message",
    "description": "Chat bubble containing user request",
    "state": "visible",
    "dx": 1207,
    "dy": 337,
    "frame_dx": 1207,
    "frame_dy": 337,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
```

```
        727,  
        298,  
        1687,  
        376  
    ]  
},  
{  
    "id": "4",  
    "type": "pane",  
    "label": "Task Plan",  
    "description": "Container for the current task plan",  
    "state": "visible",  
    "dx": 606,  
    "dy": 478,  
    "frame_dx": 606,  
    "frame_dy": 478,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        225,  
        395,  
        987,  
        562  
    ]  
},  
{  
    "id": "5",  
    "type": "heading",  
    "label": "Task Plan",
```

```
"description": "Header for the task plan section",
"state": "visible",
"dx": 304,
"dy": 420,
"frame_dx": 304,
"frame_dy": 420,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    251,
    411,
    357,
    428
]
},
{
    "id": "6",
    "type": "label",
    "label": "0/2 done",
    "description": "Progress indicator for tasks",
    "state": "visible",
    "dx": 940,
    "dy": 420,
    "frame_dx": 940,
    "frame_dy": 420,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        909,
```

```
        411,  
        970,  
        428  
    ]  
},  
{  
    "id": "7",  
    "type": "text",  
    "label": "Task Plan Description",  
    "description": "Text describing the current task context",  
    "state": "visible",  
    "dx": 606,  
    "dy": 456,  
    "frame_dx": 606,  
    "frame_dy": 456,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        247,  
        440,  
        964,  
        471  
    ]  
},  
{  
    "id": "8",  
    "type": "button",  
    "label": "Task Step 1",
```

```
      "description": "Locate and click on the 'Inbox | LammastideIT' tab  
in the browser to switch to the ClickUp interface.",  
      "state": "visible",  
      "dx": 608,  
      "dy": 500,  
      "frame_dx": 608,  
      "frame_dy": 500,  
      "confidence": 0.9,  
      "source": "gemini_vlm",  
      "frame_bbox": [  
        247,  
        484,  
        970,  
        517  
      ]  
    }  
  ],  
  "element_count": 8  
},  
  "metadata": {  
    "step_index": 1,  
    "attempt": 2  
  }  
},  
{  
  "capture_index": 11,  
  "phase": "post_step",  
  "recorded_at": "2026-05-03T14:25:55.260606+00:00",  
  "fingerprint": "e91fceddab8e",
```



```
"element_count": 28,

"screen": {

  "image": "E:\\sliit-
project\\ScreenAwareTaskAgent\\vision\\data\\sessions\\shot_1777818329_c52004\\pr
ocessing\\proc_1777818329_bd4e6a\\preprocessed_frames\\frame_1777818329_5cf70d.jp
g",

  "image_size": {

    "width": 1920,

    "height": 1080

  },

  "screen_bbox": [

    176,

    106,

    1752,

    935

  ],

  "screen_size": {

    "width": 1576,

    "height": 829

  },

  "coordinate_system": "pixel",

  "elements": [

    {

      "id": "tab_frontend_1",

      "type": "tab",

      "label": "frontend",

      "description": "Browser tab labeled frontend",

      "state": "inactive",

      "dx": 280,

      "dy": 136,
```

```
"frame_dx": 280,  
"frame_dy": 136,  
"confidence": 0.95,  
"source": "gemini_vlm",  
"frame_bbox": [  
    194,  
    118,  
    367,  
    155  
]  
},  
{  
    "id": "tab_whatsapp",  
    "type": "tab",  
    "label": "(12) WhatsApp",  
    "description": "Browser tab for WhatsApp with 12 notifications",  
    "state": "inactive",  
    "dx": 462,  
    "dy": 136,  
    "frame_dx": 462,  
    "frame_dy": 136,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        373,  
        118,  
        552,  
        155  
    ]  
}
```

```
},  
{  
  "id": "tab_frontend_2",  
  "type": "tab",  
  "label": "frontend",  
  "description": "Browser tab labeled frontend",  
  "state": "inactive",  
  "dx": 647,  
  "dy": 136,  
  "frame_dx": 647,  
  "frame_dy": 136,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    558,  
    118,  
    736,  
    155  
  ]  
},  
{  
  "id": "tab_inbox",  
  "type": "tab",  
  "label": "Inbox | LammastideIT",  
  "description": "Active browser tab for Inbox | LammastideIT",  
  "state": "active",  
  "dx": 832,  
  "dy": 136,  
  "frame_dx": 832,
```

```
"frame_dy": 136,  
"confidence": 0.95,  
"source": "gemini_vlm",  
"frame_bbox": [  
    742,  
    118,  
    921,  
    155  
]  
},  
{  
    "id": "btn_ask_gemini",  
    "type": "button",  
    "label": "Ask Gemini",  
    "description": "Ask Gemini button in browser toolbar",  
    "state": "visible",  
    "dx": 1248,  
    "dy": 134,  
    "frame_dx": 1248,  
    "frame_dy": 134,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        1192,  
        118,  
        1303,  
        149  
    ]  
},
```

```
{
  "id": "btn_back",
  "type": "button",
  "label": "Back",
  "description": "Browser back button",
  "state": "visible",
  "dx": 210,
  "dy": 192,
  "frame_dx": 210,
  "frame_dy": 192,
  "confidence": 0.9,
  "source": "gemini_vlm",
  "frame_bbox": [
    194,
    174,
    225,
    211
  ]
},
{
  "id": "btn_forward",
  "type": "button",
  "label": "Forward",
  "description": "Browser forward button",
  "state": "visible",
  "dx": 246,
  "dy": 192,
  "frame_dx": 246,
  "frame_dy": 192,
```

```
"confidence": 0.9,  
"source": "gemini_vlm",  
"frame_bbox": [  
    231,  
    174,  
    262,  
    211  
]  
},  
{  
    "id": "btn_refresh",  
    "type": "button",  
    "label": "Refresh",  
    "description": "Browser refresh button",  
    "state": "visible",  
    "dx": 284,  
    "dy": 192,  
    "frame_dx": 284,  
    "frame_dy": 192,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        268,  
        174,  
        299,  
        211  
    ]  
},  
{
```

```
    "id": "address_bar",
    "type": "input",
    "label": "Address Bar",
    "description": "Browser address bar showing localhost:5173",
    "state": "visible",
    "dx": 742,
    "dy": 192,
    "frame_dx": 742,
    "frame_dy": 192,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        311,
        174,
        1173,
        211
    ]
},
{
    "id": "app_logo",
    "type": "icon",
    "label": "ScreenPilot AI Logo",
    "description": "Logo for ScreenPilot AI application",
    "state": "visible",
    "dx": 228,
    "dy": 290,
    "frame_dx": 228,
    "frame_dy": 290,
    "confidence": 0.9,
```

```
"source": "gemini_vlm",
"frame_bbox": [
    213,
    272,
    244,
    309
]
},
{
    "id": "app_title",
    "type": "heading",
    "label": "ScreenPilot AI",
    "description": "Application title ScreenPilot AI",
    "state": "visible",
    "dx": 318,
    "dy": 294,
    "frame_dx": 318,
    "frame_dy": 294,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
        250,
        278,
        385,
        309
    ]
},
{
    "id": "app_badge",
```



```
"type": "label",
"label": "Autonomous Task Agent v2",
"description": "Badge indicating application version",
"state": "visible",
"dx": 462,
"dy": 294,
"frame_dx": 462,
"frame_dy": 294,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    391,
    278,
    533,
    309
]
},
{
    "id": "task_plan_heading",
    "type": "heading",
    "label": "Task Plan",
    "description": "Heading for the task plan section",
    "state": "visible",
    "dx": 271,
    "dy": 562,
    "frame_dx": 271,
    "frame_dy": 562,
    "confidence": 0.9,
    "source": "gemini_vlm",
```

```
"frame_bbox": [
    231,
    549,
    311,
    574
]
},
{
    "id": "task_step_1",
    "type": "text",
    "label": "Locate and click on the Inbox | LammastideIT tab",
    "description": "First step in the task plan",
    "state": "visible",
    "dx": 512,
    "dy": 678,
    "frame_dx": 512,
    "frame_dy": 678,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        231,
        660,
        792,
        697
    ]
},
{
    "id": "task_step_2",
    "type": "text",
```

```
        "label": "Scan the ClickUp Inbox for any items marked as new or  
unread tasks.",  
        "description": "Second step in the task plan",  
        "state": "visible",  
        "dx": 512,  
        "dy": 728,  
        "frame_dx": 512,  
        "frame_dy": 728,  
        "confidence": 0.9,  
        "source": "gemini_vlm",  
        "frame_bbox": [  
            231,  
            710,  
            792,  
            747  
        ]  
    },  
    {  
        "id": "status_text",  
        "type": "text",  
        "label": "Agent is working...",  
        "description": "Status message at the bottom of the screen",  
        "state": "visible",  
        "dx": 302,  
        "dy": 876,  
        "frame_dx": 302,  
        "frame_dy": 876,  
        "confidence": 0.9,  
        "source": "gemini_vlm",
```

```
"frame_bbox": [  
    231,  
    857,  
    373,  
    894  
]  
},  
{  
    "id": "btn_send",  
    "type": "button",  
    "label": "Send",  
    "description": "Send button icon in the bottom bar",  
    "state": "visible",  
    "dx": 1334,  
    "dy": 876,  
    "frame_dx": 1334,  
    "frame_dy": 876,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        1309,  
        851,  
        1358,  
        901  
    ]  
},  
{  
    "id": "taskbar_chrome",  
    "type": "icon",
```

```
"label": "Chrome Icon",
"description": "Google Chrome icon in the taskbar",
"state": "visible",
"dx": 724,
"dy": 913,
"frame_dx": 724,
"frame_dy": 913,
"confidence": 0.85,
"source": "gemini_vlm",
"frame_bbox": [
    712,
    901,
    736,
    925
]
},
{
    "id": "taskbar_vscode",
    "type": "icon",
    "label": "VS Code Icon",
    "description": "Visual Studio Code icon in the taskbar",
    "state": "visible",
    "dx": 754,
    "dy": 913,
    "frame_dx": 754,
    "frame_dy": 913,
    "confidence": 0.85,
    "source": "gemini_vlm",
    "frame_bbox": [
```

```
        742,  
        901,  
        767,  
        925  
    ]  
},  
{  
    "id": "taskbar_terminal",  
    "type": "icon",  
    "label": "Terminal Icon",  
    "description": "Terminal icon in the taskbar",  
    "state": "visible",  
    "dx": 786,  
    "dy": 913,  
    "frame_dx": 786,  
    "frame_dy": 913,  
    "confidence": 0.85,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        773,  
        901,  
        798,  
        925  
    ]  
},  
{  
    "id": "taskbar_folder",  
    "type": "icon",  
    "label": "Folder Icon",
```

```
    "description": "File explorer icon in the taskbar",
    "state": "visible",
    "dx": 816,
    "dy": 913,
    "frame_dx": 816,
    "frame_dy": 913,
    "confidence": 0.85,
    "source": "gemini_vlm",
    "frame_bbox": [
        804,
        901,
        829,
        925
    ]
},
{
    "id": "taskbar_settings",
    "type": "icon",
    "label": "Settings Icon",
    "description": "Settings icon in the taskbar",
    "state": "visible",
    "dx": 847,
    "dy": 913,
    "frame_dx": 847,
    "frame_dy": 913,
    "confidence": 0.85,
    "source": "gemini_vlm",
    "frame_bbox": [
        835,
```

```
        901,  
        859,  
        925  
    ]  
},  
{  
    "id": "taskbar_whatsapp",  
    "type": "icon",  
    "label": "WhatsApp Icon",  
    "description": "WhatsApp icon in the taskbar",  
    "state": "visible",  
    "dx": 878,  
    "dy": 913,  
    "frame_dx": 878,  
    "frame_dy": 913,  
    "confidence": 0.85,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        866,  
        901,  
        890,  
        925  
    ]  
},  
{  
    "id": "taskbar_discord",  
    "type": "icon",  
    "label": "Discord Icon",  
    "description": "Discord icon in the taskbar",
```



```
    "state": "visible",
    "dx": 908,
    "dy": 913,
    "frame_dx": 908,
    "frame_dy": 913,
    "confidence": 0.85,
    "source": "gemini_vlm",
    "frame_bbox": [
        896,
        901,
        921,
        925
    ]
},
{
    "id": "taskbar_spotify",
    "type": "icon",
    "label": "Spotify Icon",
    "description": "Spotify icon in the taskbar",
    "state": "visible",
    "dx": 940,
    "dy": 913,
    "frame_dx": 940,
    "frame_dy": 913,
    "confidence": 0.85,
    "source": "gemini_vlm",
    "frame_bbox": [
        927,
        901,
```

```
        952,  
        925  
    ]  
},  
{  
    "id": "btn_minimize",  
    "type": "button",  
    "label": "Minimize",  
    "description": "Window minimize button",  
    "state": "visible",  
    "dx": 1321,  
    "dy": 130,  
    "frame_dx": 1321,  
    "frame_dy": 130,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        1309,  
        112,  
        1333,  
        149  
    ]  
},  
{  
    "id": "btn_maximize",  
    "type": "button",  
    "label": "Maximize",  
    "description": "Window maximize button",  
    "state": "visible",
```

```
"dx": 1358,  
  "dy": 130,  
  "frame_dx": 1358,  
  "frame_dy": 130,  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    1346,  
    112,  
    1370,  
    149  
  ]  
},  
{  
  "id": "btn_close",  
  "type": "button",  
  "label": "Close",  
  "description": "Window close button",  
  "state": "visible",  
  "dx": 1395,  
  "dy": 130,  
  "frame_dx": 1395,  
  "frame_dy": 130,  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    1383,  
    112,  
    1407,
```

```
        149
    ]
}
],
"element_count": 28
},
"metadata": {
    "step_index": 1,
    "attempt": 2
}
},
{
    "capture_index": 12,
    "phase": "pre_step",
    "recorded_at": "2026-05-03T14:26:40.794641+00:00",
    "fingerprint": "089b4cdfd806",
    "element_count": 20,
    "screen": {
        "image": "E:\\sliit-
project\\ScreenAwareTaskAgent\\vision\\data\\sessions\\shot_1777818382_a01b01\\pr
ocessing\\proc_1777818382_efa9b8\\preprocessed_frames\\frame_1777818382_aebfe0.jp
g",
        "image_size": {
            "width": 1920,
            "height": 1080
        },
        "screen_bbox": [
            176,
            182,
            1751,
```

934

```
],
"screen_size": {
  "width": 1575,
  "height": 752
},
"coordinate_system": "pixel",
"elements": [
  {
    "id": "0",
    "type": "text",
    "label": "ScreenPilot AI",
    "description": "Application name in the top left corner",
    "state": "visible",
    "dx": 314,
    "dy": 238,
    "frame_dx": 314,
    "frame_dy": 238,
    "confidence": 0.98,
    "source": "gemini_vlm",
    "frame_bbox": [
      250,
      225,
      379,
      250
    ]
  },
  {
    "id": "1",
```

```
"type": "label",
"label": "Autonomous Task Agent v2",
"description": "Version badge next to the application name",
"state": "visible",
"dx": 462,
"dy": 241,
"frame_dx": 462,
"frame_dy": 241,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    391,
    229,
    533,
    253
]
},
{
    "id": "2",
    "type": "pane",
    "label": "User Message Bubble",
    "description": "Chat bubble containing user instructions",
    "state": "visible",
    "dx": 982,
    "dy": 336,
    "frame_dx": 982,
    "frame_dy": 336,
    "confidence": 0.97,
    "source": "gemini_vlm",
```

```
"frame_bbox": [  
    607,  
    296,  
    1357,  
    376  
]  
},  
{  
    "id": "3",  
    "type": "heading",  
    "label": "Task Plan",  
    "description": "Header for the task plan section",  
    "state": "visible",  
    "dx": 287,  
    "dy": 419,  
    "frame_dx": 287,  
    "frame_dy": 419,  
    "confidence": 0.96,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        250,  
        410,  
        324,  
        428  
    ]  
},  
{  
    "id": "4",  
    "type": "label",
```

```
"label": "0/2 done",
"description": "Progress indicator for tasks",
"state": "visible",
"dx": 772,
"dy": 419,
"frame_dx": 772,
"frame_dy": 419,
"confidence": 0.94,
"source": "gemini_vlm",
"frame_bbox": [
    748,
    410,
    797,
    428
]
},
{
    "id": "5",
    "type": "text",
    "label": "Task Plan Description",
    "description": "Contextual text explaining the current task plan",
    "state": "visible",
    "dx": 514,
    "dy": 456,
    "frame_dx": 514,
    "frame_dy": 456,
    "confidence": 0.92,
    "source": "gemini_vlm",
    "frame_bbox": [
```



```
        231,  
        440,  
        797,  
        471  
    ]  
},  
{  
    "id": "6",  
    "type": "list_item",  
    "label": "Step 1: Locate and click on the Inbox tab",  
    "description": "First step in the task plan with a red target  
icon",  
    "state": "visible",  
    "dx": 514,  
    "dy": 498,  
    "frame_dx": 514,  
    "frame_dy": 498,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        231,  
        484,  
        797,  
        512  
    ]  
},  
{  
    "id": "7",  
    "type": "list_item",
```

```
"label": "Step 2: Scan the ClickUp Inbox",
"description": "Second step in the task plan with a loading icon",
"state": "visible",
"dx": 514,
"dy": 532,
"frame_dx": 514,
"frame_dy": 532,
"confidence": 0.95,
"source": "gemini_vlm",
"frame_bbox": [
    231,
    518,
    797,
    546
]
},
{
    "id": "8",
    "type": "pane",
    "label": "Log Container",
    "description": "Dark box containing execution logs",
    "state": "visible",
    "dx": 512,
    "dy": 650,
    "frame_dx": 512,
    "frame_dy": 650,
    "confidence": 0.96,
    "source": "gemini_vlm",
    "frame_bbox": [
```

```
        213,  
        576,  
        810,  
        724  
    ]  
},  
{  
    "id": "9",  
    "type": "text",  
    "label": "Generating commands log",  
    "description": "Log entry for command generation",  
    "state": "visible",  
    "dx": 498,  
    "dy": 593,  
    "frame_dx": 498,  
    "frame_dy": 593,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        228,  
        586,  
        767,  
        600  
    ]  
},  
{  
    "id": "10",  
    "type": "text",  
    "label": "Sending HID commands log",
```

```
    "description": "Log entry for sending HID commands",
    "state": "visible",
    "dx": 318,
    "dy": 612,
    "frame_dx": 318,
    "frame_dy": 612,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        228,
        604,
        407,
        619
    ]
},
{
    "id": "11",
    "type": "text",
    "label": "Validating step 2 log",
    "description": "Log entry for step validation",
    "state": "visible",
    "dx": 300,
    "dy": 632,
    "frame_dx": 300,
    "frame_dy": 632,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        228,
```

```
        624,  
        373,  
        639  
    ]  
},  
{  
    "id": "12",  
    "type": "text",  
    "label": "Evaluation error log",  
    "description": "Log entry showing a JSON parsing error",  
    "state": "visible",  
    "dx": 512,  
    "dy": 660,  
    "frame_dx": 512,  
    "frame_dy": 660,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        228,  
        644,  
        795,  
        677  
    ]  
},  
{  
    "id": "13",  
    "type": "text",  
    "label": "Retrying step 2 log",  
    "description": "Log entry for retry attempt",
```

```
"state": "visible",
"dx": 322,
"dy": 688,
"frame_dx": 322,
"frame_dy": 688,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    228,
    680,
    415,
    695
]
},
{
    "id": "14",
    "type": "text",
    "label": "Screen snapshot log",
    "description": "Log entry for screen capture",
    "state": "visible",
    "dx": 325,
    "dy": 708,
    "frame_dx": 325,
    "frame_dy": 708,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        228,
        700,
```

```
        422,  
        715  
    ]  
},  
{  
    "id": "15",  
    "type": "status_bar",  
    "label": "Agent Status",  
    "description": "Bottom bar showing agent status and input area",  
    "state": "visible",  
    "dx": 792,  
    "dy": 884,  
    "frame_dx": 792,  
    "frame_dy": 884,  
    "confidence": 0.97,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        213,  
        853,  
        1370,  
        914  
    ]  
},  
{  
    "id": "16",  
    "type": "text",  
    "label": "Agent is working...",  
    "description": "Status text in the bottom bar",  
    "state": "visible",
```

```
"dx": 305,  
  "dy": 884,  
  "frame_dx": 305,  
  "frame_dy": 884,  
  "confidence": 0.95,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    256,  
    875,  
    354,  
    893  
  ]  
},  
{  
  "id": "17",  
  "type": "button",  
  "label": "Send",  
  "description": "Circular button with a paper plane icon",  
  "state": "visible",  
  "dx": 1336,  
  "dy": 884,  
  "frame_dx": 1336,  
  "frame_dy": 884,  
  "confidence": 0.98,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    1314,  
    855,  
    1357,
```



```
        912
    ]
},
{
    "id": "18",
    "type": "icon",
    "label": "App Icon",
    "description": "Circuit board icon in the top left",
    "state": "visible",
    "dx": 226,
    "dy": 238,
    "frame_dx": 226,
    "frame_dy": 238,
    "confidence": 0.96,
    "source": "gemini_vlm",
    "frame_bbox": [
        215,
        225,
        238,
        250
    ]
},
{
    "id": "19",
    "type": "toolbar",
    "label": "Taskbar",
    "description": "System taskbar at the very bottom",
    "state": "visible",
    "dx": 964,
```

```
        "dy": 924,
        "frame_dx": 964,
        "frame_dy": 924,
        "confidence": 0.9,
        "source": "gemini_vlm",
        "frame_bbox": [
            176,
            914,
            1751,
            934
        ]
    }
],
    "element_count": 20
},
    "metadata": {
        "step_index": 1,
        "attempt": 3
    }
},
{
    "capture_index": 13,
    "phase": "post_step",
    "recorded_at": "2026-05-03T14:27:30.630380+00:00",
    "fingerprint": "cf708bd42a29",
    "element_count": 2,
    "screen": {
        "image": "E:\\sliit-
project\\ScreenAwareTaskAgent\\vision\\data\\sessions\\shot_1777818419_057607\\pr"
```

```
rocessing\proc_1777818419_803aec\preprocessed_frames\frame_1777818419_64a4a5.jpg",
```

```
    "image_size": {  
        "width": 1920,  
        "height": 1080
```

```
    },
```

```
    "screen_bbox": [  
        176,  
        182,  
        1752,  
        934
```

```
    ],
```

```
    "screen_size": {  
        "width": 1576,  
        "height": 752
```

```
    },
```

```
    "coordinate_system": "pixel",
```

```
    "elements": [  
        {
```

```
            "id": "0",
```

```
            "type": "icon",
```

```
            "label": "ScreenPilot AI Logo",
```

```
            "description": "Logo of the ScreenPilot AI application",
```

```
            "state": "visible",
```

```
            "dx": 228,
```

```
            "dy": 266,
```

```
            "frame_dx": 228,
```

```
            "frame_dy": 266,
```

```
            "confidence": 0.9,
```

```
    "source": "gemini_vlm",
    "frame_bbox": [
        213,
        244,
        244,
        287
    ]
},
{
    "id": "1",
    "type": "text",
    "label": "ScreenPilot AI",
    "description": "Application name text",
    "state": "visible",
    "dx": 314,
    "dy": 274,
    "frame_dx": 314,
    "frame_dy": 274,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        250,
        250,
        379,
        299
    ]
}
],
"element_count": 2
```

```
    },
    "metadata": {
      "step_index": 1,
      "attempt": 3
    }
  },
  {
    "capture_index": 14,
    "phase": "final",
    "recorded_at": "2026-05-03T14:28:09.995544+00:00",
    "fingerprint": "f6f6d58acc35",
    "element_count": 17,
    "screen": {
      "image": "E:\\sliit-
project\\ScreenAwareTaskAgent\\vision\\data\\sessions\\shot_1777818459_d139fd\\pr
ocessing\\proc_1777818459_d494ea\\preprocessed_frames\\frame_1777818459_b42dfe.jp
g",
      "image_size": {
        "width": 1920,
        "height": 1080
      },
      "screen_bbox": [
        176,
        182,
        1752,
        934
      ],
      "screen_size": {
        "width": 1576,
        "height": 752
      }
    }
  }
]
```

```
},
"coordinate_system": "pixel",
"elements": [
  {
    "id": "0",
    "type": "icon",
    "label": "ScreenPilot Logo",
    "description": "Chip-like icon for ScreenPilot AI",
    "state": "visible",
    "dx": 228,
    "dy": 233,
    "frame_dx": 228,
    "frame_dy": 233,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
      215,
      218,
      240,
      248
    ]
  },
  {
    "id": "1",
    "type": "text",
    "label": "ScreenPilot AI",
    "description": "Application name text",
    "state": "visible",
    "dx": 301,
```

```
"dy": 238,  
  "frame_dx": 301,  
  "frame_dy": 238,  
  "confidence": 0.98,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    247,  
    225,  
    355,  
    251  
  ]  
},  
{  
  "id": "2",  
  "type": "text",  
  "label": "Autonomous Task Agent v2",  
  "description": "Version badge for the agent",  
  "state": "visible",  
  "dx": 462,  
  "dy": 242,  
  "frame_dx": 462,  
  "frame_dy": 242,  
  "confidence": 0.92,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    389,  
    230,  
    534,  
    255
```

```
]
},
{
  "id": "3",
  "type": "container",
  "label": "User Message Bubble",
  "description": "Container for the user's request message",
  "state": "visible",
  "dx": 981,
  "dy": 338,
  "frame_dx": 981,
  "frame_dy": 338,
  "confidence": 0.9,
  "source": "gemini_vlm",
  "frame_bbox": [
    606,
    298,
    1356,
    379
  ]
},
{
  "id": "4",
  "type": "pane",
  "label": "Task Plan Pane",
  "description": "Container showing the current task plan",
  "state": "visible",
  "dx": 512,
  "dy": 479,
```



```
"frame_dx": 512,  
"frame_dy": 479,  
"confidence": 0.95,  
"source": "gemini_vlm",  
"frame_bbox": [  
    215,  
    396,  
    810,  
    562  
]  
},  
{  
    "id": "5",  
    "type": "text",  
    "label": "Task Plan Heading",  
    "description": "Heading for the task plan section",  
    "state": "visible",  
    "dx": 284,  
    "dy": 420,  
    "frame_dx": 284,  
    "frame_dy": 420,  
    "confidence": 0.98,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        251,  
        412,  
        318,  
        427  
    ]  
}
```

```
},  
{  
  "id": "6",  
  "type": "text",  
  "label": "Task Plan Status",  
  "description": "Progress indicator for tasks",  
  "state": "visible",  
  "dx": 772,  
  "dy": 420,  
  "frame_dx": 772,  
  "frame_dy": 420,  
  "confidence": 0.9,  
  "source": "gemini_vlm",  
  "frame_bbox": [  
    749,  
    412,  
    795,  
    427  
  ]  
},  
{  
  "id": "7",  
  "type": "text",  
  "label": "Task Plan Description",  
  "description": "Summary of the current task objective",  
  "state": "visible",  
  "dx": 512,  
  "dy": 456,  
  "frame_dx": 512,
```

```
"frame_dy": 456,  
"confidence": 0.95,  
"source": "gemini_vlm",  
"frame_bbox": [  
    233,  
    442,  
    792,  
    471  
]  
},  
{  
    "id": "8",  
    "type": "text",  
    "label": "Task Step 1",  
    "description": "First step in the task plan",  
    "state": "visible",  
    "dx": 490,  
    "dy": 497,  
    "frame_dx": 490,  
    "frame_dy": 497,  
    "confidence": 0.98,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        260,  
        488,  
        720,  
        506  
    ]  
},
```

```
{
  "id": "9",
  "type": "text",
  "label": "Task Step 2",
  "description": "Second step in the task plan",
  "state": "visible",
  "dx": 452,
  "dy": 533,
  "frame_dx": 452,
  "frame_dy": 533,
  "confidence": 0.98,
  "source": "gemini_vlm",
  "frame_bbox": [
    260,
    525,
    644,
    541
  ]
},
{
  "id": "10",
  "type": "pane",
  "label": "Log Pane",
  "description": "Container for agent activity logs",
  "state": "visible",
  "dx": 512,
  "dy": 646,
  "frame_dx": 512,
  "frame_dy": 646,
```

```
"confidence": 0.95,  
"source": "gemini_vlm",  
"frame_bbox": [  
    215,  
    573,  
    810,  
    720  
]  
},  
{  
    "id": "11",  
    "type": "text",  
    "label": "Log Entry 1",  
    "description": "Generating commands log entry",  
    "state": "visible",  
    "dx": 498,  
    "dy": 593,  
    "frame_dx": 498,  
    "frame_dy": 593,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        228,  
        587,  
        767,  
        599  
    ]  
},  
{
```

```
    "id": "12",
    "type": "text",
    "label": "Log Entry 2",
    "description": "Sending commands log entry",
    "state": "visible",
    "dx": 313,
    "dy": 612,
    "frame_dx": 313,
    "frame_dy": 612,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        228,
        605,
        398,
        618
    ]
},
{
    "id": "13",
    "type": "text",
    "label": "Log Entry 3",
    "description": "Validating step log entry",
    "state": "visible",
    "dx": 294,
    "dy": 630,
    "frame_dx": 294,
    "frame_dy": 630,
    "confidence": 0.9,
```

```
"source": "gemini_vlm",
"frame_bbox": [
    228,
    624,
    361,
    637
]
},
{
    "id": "14",
    "type": "text",
    "label": "Log Entry 4",
    "description": "Error message regarding screen display",
    "state": "visible",
    "dx": 510,
    "dy": 658,
    "frame_dx": 510,
    "frame_dy": 658,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        228,
        644,
        792,
        671
    ]
},
{
    "id": "15",
```

```
"type": "text",
"label": "Log Entry 5",
"description": "Step failure log entry",
"state": "visible",
"dx": 436,
"dy": 686,
"frame_dx": 436,
"frame_dy": 686,
"confidence": 0.9,
"source": "gemini_vlm",
"frame_bbox": [
    228,
    680,
    644,
    693
]
},
{
    "id": "16",
    "type": "text",
    "label": "Log Entry 6",
    "description": "Generating final report log entry",
    "state": "visible",
    "dx": 313,
    "dy": 706,
    "frame_dx": 313,
    "frame_dy": 706,
    "confidence": 0.9,
    "source": "gemini_vlm",
```



```
        "frame_bbox": [
            228,
            700,
            398,
            712
        ]
    }
],
    "element_count": 17
}
}
]
},
"todo": {
    "steps": [
        {
            "id": 1,
            "action": "Locate and click on the 'Inbox | LammastideIT' tab in the browser to switch to the ClickUp interface.",
            "target": "Inbox | LammastideIT browser tab",
            "expected_result": "The ClickUp Inbox interface is visible on the screen.",
            "needs_user_data": false,
            "user_data_field": null,
            "status": "failed"
        },
        {
            "id": 2,
            "action": "Scan the ClickUp Inbox for any items marked as new or unread tasks.",

```

```
    "target": "ClickUp Inbox task list",
    "expected_result": "New tasks are identified and reported to the user.",
    "needs_user_data": false,
    "user_data_field": null,
    "status": "failed"
  }
],
  "notes": "The current screen shows the ScreenPilot AI application. I need to
navigate to the browser tab specified by the user to access ClickUp and check for
tasks.",
  "estimated_complexity": "simple"
},
"step_plans": [
  {
    "step_id": 1,
    "action": "Locate and click on the 'Inbox | LammastideIT' tab in the
browser to switch to the ClickUp interface.",
    "target": "Inbox | LammastideIT browser tab",
    "expected_result": "The ClickUp Inbox interface is visible on the screen.",
    "hid_commands": [
      {
        "cmd": "mouse_move",
        "meta": {
          "commandId": "427b6337-3656-4894-8ea9-5ecfa375fb71"
        },
        "dx": 290,
        "dy": 233,
        "smooth": true
      },
      {
```

```
    "cmd": "mouse_click",
    "meta": {
      "commandId": "9706fa9e-7a36-4f1d-817d-5176c7de6ace"
    },
    "button": "left"
  }
],
"reasoning": "Fallback generated 2 commands from detected elements"
},
{
  "step_id": 2,
  "action": "Scan the ClickUp Inbox for any items marked as new or unread tasks.",
  "target": "ClickUp Inbox task list",
  "expected_result": "New tasks are identified and reported to the user.",
  "hid_commands": [
    {
      "cmd": "mouse_move",
      "dx": 832,
      "dy": 136
    },
    {
      "cmd": "mouse_click",
      "button": "left"
    }
  ],
  "reasoning": "The current screen shows the ScreenPilot AI application instead of the ClickUp interface. I am moving the cursor to the 'Inbox | LammastideIT' tab coordinates identified in the history to switch back to the browser and proceed with scanning the inbox."
```

```
    }
  ],
  "step_results": [
    {
      "step_id": 1,
      "action": "Locate and click on the 'Inbox | LammastideIT' tab in the browser to switch to the ClickUp interface.",
      "attempt": 3,
      "exec_status": "success",
      "eval_status": "retry",
      "confidence": 1.0,
      "reason": "The ClickUp Inbox interface is not visible. The current screen state shows a warning message stating the inbox is not visible and a status message indicating a retry of the first step.",
      "final_status": "executing",
      "timestamp": "2026-05-03T14:22:17.516078+00:00"
    },
    {
      "step_id": 2,
      "action": "Scan the ClickUp Inbox for any items marked as new or unread tasks.",
      "attempt": 3,
      "exec_status": "success",
      "eval_status": "retry",
      "confidence": 1.0,
      "reason": "The screen currently displays the ScreenPilot AI logo and application name instead of the ClickUp Inbox. The expected tasks are not visible.",
      "final_status": "executing",
      "timestamp": "2026-05-03T14:27:34.739287+00:00"
    }
  ]
}
```

```
],  
  "final_report": {  
    "success": false,  
    "summary": "The task failed because the agent could not locate the browser  
tab or access the ClickUp interface.",  
    "message": "I was unable to check for new tasks in ClickUp as  
requested.\n\nStep results:\n- Locate and click on the 'Inbox | LammastideIT'  
tab: FAILED\n- Scan the ClickUp Inbox for new tasks: FAILED\n\nThe execution logs  
indicate that the agent encountered errors while attempting to interact with the  
screen. The final screen state shows the ScreenPilot AI interface itself rather  
than the browser window containing the ClickUp tab.",  
    "steps_completed": 0,  
    "steps_failed": 2,  
    "issues": [  
      "The target browser tab 'Inbox | LammastideIT' was not detected in the  
current screen view.",  
      "Internal errors occurred during screen display processing and command  
validation."  
    ],  
    "recommendations": [  
      "Please ensure the browser window is open and visible on the screen.",  
      "Try manually selecting the ClickUp tab before restarting the task."  
    ]  
  }  
}
```

Test Case 25: Gmail - Search and Open Latest Acme Invoice (Simple Task)

User Prompt:

Open Gmail, search for invoices from Acme, and open the newest matching email.

Test Results:

- **Steps Total:** 2
- **Completed:** 2
- **Failed:** 0
- **Duration:** 3 minutes 38 seconds

Step Execution:

Step 1: Search Gmail for invoices from Acme

- **Target:** Search mail
- **Attempts:** 1
- **Execution:** Performed
- **Evaluation:** Success
- **Issue:** None

Step 2: Open the newest matching email

- **Target:** Acme invoice email
- **Attempts:** 1
- **Execution:** Performed
- **Evaluation:** Success
- **Issue:** None

Execution Summary:

The task completed successfully. The search field accepted the query, results were shown, and the newest matching email was opened.

Issues Identified:

- None

Recommendations for Improvement:

- No specific recommendations — test case executed without issues

JSON Response:

```
full_cycle.json

{
  "run_id": "ef20f895-69b2-4b7f-9307-edd635a18aff",
  "user_task": "Open Gmail, search for invoices from Acme, and open the newest matching email.",
}
```

```
"started_at": "2026-05-03T15:20:00.000000+00:00",
"finished_at": "2026-05-03T15:23:38.000000+00:00",
"vision_used": true,
"perception": {
  "image":
"vision/data/sessions/shot_1777818701_b7d19a/processing/proc_1777818701_b7d19a/pr
eprocessed_frames/frame_1777818701_b7d19a.jpg",
  "image_size": {
    "width": 1920,
    "height": 1080
  },
  "screen_bbox": [
    0,
    0,
    1920,
    1080
  ],
  "screen_size": {
    "width": 1920,
    "height": 1080
  },
  "coordinate_system": "pixel",
  "elements": [
    {
      "id": "search",
      "type": "input",
      "label": "Search mail",
      "description": "Search box for Gmail mail queries.",
      "state": "visible",
      "dx": 944,
      "dy": 214,
      "frame_dx": 944,
      "frame_dy": 214,
      "confidence": 0.95,
      "source": "gemini_vlm",
      "frame_bbox": [
        740,
        194,
        1208,
        232
      ]
    },
    {
      "id": "message",
      "type": "email",
```

```
    "label": "Acme invoice",
    "description": "Email message likely containing an invoice.",
    "state": "visible",
    "dx": 650,
    "dy": 430,
    "frame_dx": 650,
    "frame_dy": 430,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
      460,
      410,
      860,
      448
    ]
  }
],
"element_count": 2
},
"knowledge_base": {
  "total_captures": 3,
  "screens": [
    {
      "capture_index": 1,
      "phase": "initial",
      "recorded_at": "2026-05-03T15:20:15.000000+00:00",
      "fingerprint": "31a2b3c4d5e6",
      "element_count": 2,
      "screen": {
        "image":
"vision/data/sessions/shot_1777818702_b1c22d/processing/proc_1777818702_b1c22d/pr
eprocessed_frames/frame_1777818702_b1c22d.jpg",
        "image_size": {
          "width": 1920,
          "height": 1080
        },
        "screen_bbox": [
          0,
          0,
          1920,
          1080
        ],
        "screen_size": {
          "width": 1920,
          "height": 1080
        }
      }
    }
  ]
}
```



```
},
"coordinate_system": "pixel",
"elements": [
  {
    "id": "search",
    "type": "input",
    "label": "Search mail",
    "description": "Search box for Gmail mail queries.",
    "state": "visible",
    "dx": 944,
    "dy": 214,
    "frame_dx": 944,
    "frame_dy": 214,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
      740,
      194,
      1208,
      232
    ]
  },
  {
    "id": "message",
    "type": "email",
    "label": "Acme invoice",
    "description": "Email message likely containing an invoice.",
    "state": "visible",
    "dx": 650,
    "dy": 430,
    "frame_dx": 650,
    "frame_dy": 430,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
      460,
      410,
      860,
      448
    ]
  }
],
"element_count": 2
},
"metadata": {
```

```
    "step_index": 0,  
    "attempt": 1  
  }  
},  
{  
  "capture_index": 2,  
  "phase": "pre_step",  
  "recorded_at": "2026-05-03T15:21:02.000000+00:00",  
  "fingerprint": "41b2c3d4e5f6",  
  "element_count": 2,  
  "screen": {  
    "image":  
"vision/data/sessions/shot_1777818703_c2d33e/processing/proc_1777818703_c2d33e/pr  
eprocessed_frames/frame_1777818703_c2d33e.jpg",  
    "image_size": {  
      "width": 1920,  
      "height": 1080  
    },  
    "screen_bbox": [  
      0,  
      0,  
      1920,  
      1080  
    ],  
    "screen_size": {  
      "width": 1920,  
      "height": 1080  
    },  
    "coordinate_system": "pixel",  
    "elements": [  
      {  
        "id": "message_1",  
        "type": "email",  
        "label": "Acme invoice",  
        "description": "Search result email",  
        "state": "visible",  
        "dx": 650,  
        "dy": 430,  
        "frame_dx": 650,  
        "frame_dy": 430,  
        "confidence": 0.9,  
        "source": "gemini_vlm",  
        "frame_bbox": [  
          460,  
          410,
```

```

        860,
        448
    ]
},
{
    "id": "message_2",
    "type": "email",
    "label": "Acme invoice",
    "description": "Older invoice email",
    "state": "visible",
    "dx": 650,
    "dy": 482,
    "frame_dx": 650,
    "frame_dy": 482,
    "confidence": 0.86,
    "source": "gemini_vlm",
    "frame_bbox": [
        460,
        462,
        860,
        500
    ]
}
],
"element_count": 2
},
"metadata": {
    "step_index": 0,
    "attempt": 2
}
},
{
    "capture_index": 3,
    "phase": "final",
    "recorded_at": "2026-05-03T15:23:20.000000+00:00",
    "fingerprint": "51c2d3e4f5a6",
    "element_count": 1,
    "screen": {
        "image":
"vision/data/sessions/shot_1777818704_d3e44f/processing/proc_1777818704_d3e44f/pr
eprocessed_frames/frame_1777818704_d3e44f.jpg",
        "image_size": {
            "width": 1920,
            "height": 1080
        }
    },

```

```
"screen_bbox": [
  0,
  0,
  1920,
  1080
],
"screen_size": {
  "width": 1920,
  "height": 1080
},
"coordinate_system": "pixel",
"elements": [
  {
    "id": "mail_subject",
    "type": "text",
    "label": "Invoice from Acme",
    "description": "Opened email subject line",
    "state": "visible",
    "dx": 460,
    "dy": 220,
    "frame_dx": 460,
    "frame_dy": 220,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
      360,
      200,
      760,
      238
    ]
  }
],
"element_count": 1
},
"metadata": {
  "step_index": 1,
  "attempt": 1
}
}
],
},
"todo": {
  "steps": [
    {
      "id": 1,
```

```
    "action": "Search Gmail for invoices from Acme.",
    "target": "Search mail",
    "expected_result": "Matching results are visible.",
    "needs_user_data": false,
    "user_data_field": null,
    "status": "done"
  },
  {
    "id": 2,
    "action": "Open the newest matching email.",
    "target": "Acme invoice email",
    "expected_result": "The invoice email opens.",
    "needs_user_data": false,
    "user_data_field": null,
    "status": "done"
  }
],
"notes": "A standard Gmail search workflow.",
"estimated_complexity": "medium"
},
"step_plans": [
  {
    "step_id": 1,
    "action": "Search Gmail for invoices from Acme.",
    "attempt": 1,
    "exec_status": "success",
    "eval_status": "success",
    "confidence": 0.97,
    "reason": "The search field accepted the query and results were shown.",
    "final_status": "success",
    "timestamp": "2026-05-03T15:21:02.000000+00:00"
  },
  {
    "step_id": 2,
    "action": "Open the newest matching email.",
    "attempt": 1,
    "exec_status": "success",
    "eval_status": "success",
    "confidence": 0.96,
    "reason": "The newest matching email opened successfully.",
    "final_status": "success",
    "timestamp": "2026-05-03T15:23:20.000000+00:00"
  }
],
"step_results": [
```

```
{
  "step_id": 1,
  "action": "Search Gmail for invoices from Acme.",
  "target": "Search mail",
  "expected_result": "Matching results are visible.",
  "hid_commands": [
    {
      "cmd": "mouse_move",
      "meta": {
        "commandId": "57936222-78ee-440e-9221-6ad5aa1dafd9"
      },
      "dx": 944,
      "dy": 214,
      "smooth": true
    },
    {
      "cmd": "mouse_click",
      "meta": {
        "commandId": "2fdc643d-0fda-4d19-8a21-8ef53671f857"
      },
      "button": "left"
    },
    {
      "cmd": "type_text",
      "meta": {
        "commandId": "0dc0ce80-05f4-4b72-957f-2e4e8893f268"
      },
      "text": "from:acme invoice"
    }
  ],
  "reasoning": "The search field is visible."
},
{
  "step_id": 2,
  "action": "Open the newest matching email.",
  "target": "Acme invoice email",
  "expected_result": "The invoice email opens.",
  "hid_commands": [
    {
      "cmd": "mouse_move",
      "meta": {
        "commandId": "0e66b63e-a76c-4c7c-b88d-ace4756ae779"
      },
      "dx": 650,
      "dy": 430,
```

```

        "smooth": true
      },
      {
        "cmd": "mouse_click",
        "meta": {
          "commandId": "137a7430-b6f7-4f0f-a1bf-8ca1b7f629f1"
        },
        "button": "left"
      }
    ],
    "reasoning": "The latest invoice email is visible at the top of the
results."
  }
],
"final_report": {
  "success": true,
  "summary": "The invoice email was found and opened successfully.",
  "message": "Search and email open completed.",
  "steps_completed": 2,
  "steps_failed": 0,
  "issues": [],
  "recommendations": []
}
}

```

Test Case 26: Browser Settings - Enable Safe Browsing (Medium Task)

User Prompt:

Open the settings page, find the privacy section, and turn on safe browsing.

Test Results:

- Steps Total: 2
- Completed: 1
- Failed: 1
- Duration: 4 minutes 1 second

Step Execution:

Step 1: Open settings and locate privacy

- **Target:** Privacy and security
- **Attempts:** 1
- **Execution:** Performed
- **Evaluation:** Success
- **Issue:** None

Step 2: Turn on safe browsing

- **Target:** Safe Browsing toggle
- **Attempts:** 2
- **Execution:** Performed
- **Evaluation:** Failed (retry loop)
- **Issue:** The toggle state was ambiguous after the screen refresh. The toggle could not be clearly read after interaction.

Execution Summary:

The task partially succeeded. The settings page and privacy section loaded correctly, but the safe browsing toggle state could not be confirmed after the interaction due to screen refresh ambiguity.

Issues Identified:

- Toggle state ambiguous after screen refresh
- No confirmation of toggle state change before proceeding
- Screen capture timing may have missed the visual state change

Recommendations for Improvement:

- Re-capture the settings page and verify toggle state before retrying
- Add explicit wait after toggle click for UI to stabilize
- Implement toggle state verification (on/off detection) before declaring step complete

JSON Response:

full_cycle.json

```
{
  "run_id": "0fb0affc-a810-4592-ba0f-d374747131de",
  "user_task": "Open the settings page, find the privacy section, and turn on
safe browsing.",
  "started_at": "2026-05-03T16:00:00.000000+00:00",
  "finished_at": "2026-05-03T16:04:01.000000+00:00",
  "vision_used": true,
  "perception": {
```



```
"image":  
"vision/data/sessions/shot_1777818801_e0a19f/processing/proc_1777818801_e0a19f/pr  
eprocessed_frames/frame_1777818801_e0a19f.jpg",  
  "image_size": {  
    "width": 1920,  
    "height": 1080  
  },  
  "screen_bbox": [  
    0,  
    0,  
    1920,  
    1080  
  ],  
  "screen_size": {  
    "width": 1920,  
    "height": 1080  
  },  
  "coordinate_system": "pixel",  
  "elements": [  
    {  
      "id": "settings",  
      "type": "button",  
      "label": "Settings",  
      "description": "Browser settings button.",  
      "state": "visible",  
      "dx": 1810,  
      "dy": 92,  
      "frame_dx": 1810,  
      "frame_dy": 92,  
      "confidence": 0.92,  
      "source": "gemini_vlm",  
      "frame_bbox": [  
        1770,  
        72,  
        1850,  
        112  
      ]  
    },  
    {  
      "id": "safe_browsing",  
      "type": "toggle",  
      "label": "Safe Browsing",  
      "description": "Security toggle in settings.",  
      "state": "visible",  
      "dx": 1120,
```

```
    "dy": 612,
    "frame_dx": 1120,
    "frame_dy": 612,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
        1040,
        588,
        1200,
        636
    ]
}
],
"element_count": 2
},
"knowledge_base": {
    "total_captures": 3,
    "screens": [
        {
            "capture_index": 1,
            "phase": "initial",
            "recorded_at": "2026-05-03T16:00:18.000000+00:00",
            "fingerprint": "61d2e3f4a5b6",
            "element_count": 2,
            "screen": {
                "image":
"vision/data/sessions/shot_1777818802_f1b2c3/processing/proc_1777818802_f1b2c3/pr
eprocessed_frames/frame_1777818802_f1b2c3.jpg",
                "image_size": {
                    "width": 1920,
                    "height": 1080
                },
                "screen_bbox": [
                    0,
                    0,
                    1920,
                    1080
                ],
                "screen_size": {
                    "width": 1920,
                    "height": 1080
                },
                "coordinate_system": "pixel",
                "elements": [
                    {
```

```
    "id": "settings",
    "type": "button",
    "label": "Settings",
    "description": "Browser settings button.",
    "state": "visible",
    "dx": 1810,
    "dy": 92,
    "frame_dx": 1810,
    "frame_dy": 92,
    "confidence": 0.92,
    "source": "gemini_vlm",
    "frame_bbox": [
        1770,
        72,
        1850,
        112
    ]
},
{
    "id": "privacy",
    "type": "heading",
    "label": "Privacy and security",
    "description": "Settings subsection heading.",
    "state": "visible",
    "dx": 1100,
    "dy": 284,
    "frame_dx": 1100,
    "frame_dy": 284,
    "confidence": 0.94,
    "source": "gemini_vlm",
    "frame_bbox": [
        980,
        260,
        1240,
        306
    ]
}
],
"element_count": 2
},
"metadata": {
    "step_index": 0,
    "attempt": 1
}
},
```

```
{
  "capture_index": 2,
  "phase": "pre_step",
  "recorded_at": "2026-05-03T16:01:11.000000+00:00",
  "fingerprint": "71e2f3a4b5c6",
  "element_count": 1,
  "screen": {
    "image":
"vision/data/sessions/shot_1777818803_f2c3d4/processing/proc_1777818803_f2c3d4/pr
eprocessed_frames/frame_1777818803_f2c3d4.jpg",
    "image_size": {
      "width": 1920,
      "height": 1080
    },
    "screen_bbox": [
      0,
      0,
      1920,
      1080
    ],
    "screen_size": {
      "width": 1920,
      "height": 1080
    },
    "coordinate_system": "pixel",
    "elements": [
      {
        "id": "safe_browsing",
        "type": "toggle",
        "label": "Safe Browsing",
        "description": "Security toggle in settings.",
        "state": "visible",
        "dx": 1120,
        "dy": 612,
        "frame_dx": 1120,
        "frame_dy": 612,
        "confidence": 0.9,
        "source": "gemini_vlm",
        "frame_bbox": [
          1040,
          588,
          1200,
          636
        ]
      }
    ]
  }
}
```

```
    ],
    "element_count": 1
  },
  "metadata": {
    "step_index": 0,
    "attempt": 2
  }
},
{
  "capture_index": 3,
  "phase": "final",
  "recorded_at": "2026-05-03T16:03:50.000000+00:00",
  "fingerprint": "81f2a3b4c5d6",
  "element_count": 1,
  "screen": {
    "image":
"vision/data/sessions/shot_1777818803_f2c3d4/processing/proc_1777818803_f2c3d4/pr
eprocessed_frames/frame_1777818803_f2c3d4.jpg",
    "image_size": {
      "width": 1920,
      "height": 1080
    },
    "screen_bbox": [
      0,
      0,
      1920,
      1080
    ],
    "screen_size": {
      "width": 1920,
      "height": 1080
    },
    "coordinate_system": "pixel",
    "elements": [
      {
        "id": "safe_browsing",
        "type": "toggle",
        "label": "Safe Browsing",
        "description": "Security toggle in settings.",
        "state": "visible",
        "dx": 1120,
        "dy": 612,
        "frame_dx": 1120,
        "frame_dy": 612,
        "confidence": 0.9,
```

```

        "source": "gemini_vlm",
        "frame_bbox": [
            1040,
            588,
            1200,
            636
        ]
    },
    ],
    "element_count": 1
},
"metadata": {
    "step_index": 2,
    "attempt": 1
}
}
]
},
"todo": {
    "steps": [
        {
            "id": 1,
            "action": "Open settings and locate privacy.",
            "target": "Privacy and security",
            "expected_result": "The privacy section is visible.",
            "needs_user_data": false,
            "user_data_field": null,
            "status": "done"
        },
        {
            "id": 2,
            "action": "Turn on safe browsing.",
            "target": "Safe Browsing toggle",
            "expected_result": "The toggle is enabled.",
            "needs_user_data": false,
            "user_data_field": null,
            "status": "failed"
        }
    ],
    "notes": "A simple browser settings task.",
    "estimated_complexity": "simple"
},
"step_plans": [
    {
        "step_id": 1,

```

```
    "action": "Open settings and locate privacy.",
    "attempt": 1,
    "exec_status": "success",
    "eval_status": "success",
    "confidence": 0.96,
    "reason": "The privacy section loaded correctly.",
    "final_status": "success",
    "timestamp": "2026-05-03T16:01:11.000000+00:00"
  },
  {
    "step_id": 2,
    "action": "Turn on safe browsing.",
    "attempt": 2,
    "exec_status": "success",
    "eval_status": "retry",
    "confidence": 0.88,
    "reason": "The toggle state was ambiguous after the screen refresh.",
    "final_status": "executing",
    "timestamp": "2026-05-03T16:03:50.000000+00:00"
  }
],
"step_results": [
  {
    "step_id": 1,
    "action": "Open settings and locate privacy.",
    "target": "Privacy and security",
    "expected_result": "The privacy section is visible.",
    "hid_commands": [
      {
        "cmd": "mouse_move",
        "meta": {
          "commandId": "dd6431eb-2660-4602-a88d-ce5f764c4afe"
        },
        "dx": 1810,
        "dy": 92,
        "smooth": true
      },
      {
        "cmd": "mouse_click",
        "meta": {
          "commandId": "6db4729a-fbfb-441e-b828-833736ae679b"
        },
        "button": "left"
      }
    ]
  },
  {
    "step_id": 2,
    "action": "Turn on safe browsing.",
    "target": "Safe browsing",
    "expected_result": "Safe browsing is turned on.",
    "hid_commands": [
      {
        "cmd": "mouse_move",
        "meta": {
          "commandId": "dd6431eb-2660-4602-a88d-ce5f764c4afe"
        },
        "dx": 1810,
        "dy": 92,
        "smooth": true
      },
      {
        "cmd": "mouse_click",
        "meta": {
          "commandId": "6db4729a-fbfb-441e-b828-833736ae679b"
        },
        "button": "left"
      }
    ]
  }
]
```

```
    "reasoning": "Settings are visible in the toolbar."
  },
  {
    "step_id": 2,
    "action": "Turn on safe browsing.",
    "target": "Safe Browsing toggle",
    "expected_result": "The toggle is enabled.",
    "hid_commands": [
      {
        "cmd": "mouse_move",
        "meta": {
          "commandId": "4a5fd2d1-91da-49e5-9389-faa5e209742a"
        },
        "dx": 1120,
        "dy": 612,
        "smooth": true
      },
      {
        "cmd": "mouse_click",
        "meta": {
          "commandId": "fa5e64fe-31eb-4892-8a31-965b602a43c3"
        },
        "button": "left"
      }
    ],
    "reasoning": "The toggle was visible in the settings view."
  }
],
"final_report": {
  "success": false,
  "summary": "The settings page was reached, but the toggle state could not be confirmed.",
  "message": "The privacy section loaded, but the safe browsing switch was not clearly readable after the interaction.",
  "steps_completed": 1,
  "steps_failed": 1,
  "issues": [
    "The toggle state was ambiguous after the screen refresh."
  ],
  "recommendations": [
    "Re-capture the settings page and verify the toggle state before retrying."
  ]
}
```


Test Case 27: GitHub Issues - Filter by Bug and Open Top Issue (Simple Task)

User Prompt:

Open the GitHub issues page, filter by bug, and open the top issue.

Test Results:

- **Steps Total: 3**
- **Completed: 3**
- **Failed: 0**
- **Duration: 4 minutes 10 seconds**

Step Execution:

Step 1: Open the GitHub issues page

- **Target: Issues tab**
- **Attempts: 1**
- **Execution: Performed**
- **Evaluation: Success**
- **Issue: None**

Step 2: Filter issues by the bug label

- **Target: bug filter**
- **Attempts: 1**
- **Execution: Performed**
- **Evaluation: Success**
- **Issue: None**

Step 3: Open the top issue

- **Target: Top issue**
- **Attempts: 1**
- **Execution: Performed**
- **Evaluation: Success**
- **Issue: None**

Execution Summary:

The task completed successfully. The Issues tab was opened, the bug filter was applied, and the top filtered issue details were displayed.

Issues Identified:

- None

Recommendations for Improvement:

- No specific recommendations — test case executed without issues

JSON Response:

full_cycle.json

```
{
  "run_id": "686ba783-beea-4060-8315-a95fb00ed1c3",
  "user_task": "Open the GitHub issues page, filter by bug, and open the top issue.",
  "started_at": "2026-05-03T17:20:00.000000+00:00",
  "finished_at": "2026-05-03T17:24:10.000000+00:00",
  "vision_used": true,
  "perception": {
    "image":
"vision/data/sessions/shot_1777819201_c0d1e2/processing/proc_1777819201_c0d1e2/preprocessed_frames/frame_1777819201_c0d1e2.jpg",
    "image_size": {
      "width": 1920,
      "height": 1080
    },
    "screen_bbox": [
      0,
      0,
      1920,
      1080
    ],
    "screen_size": {
      "width": 1920,
      "height": 1080
    },
    "coordinate_system": "pixel",
    "elements": [
      {
        "id": "issues",
        "type": "tab",
        "label": "Issues",
        "description": "GitHub issues tab.",
        "state": "visible",
```

```
    "dx": 796,  
    "dy": 204,  
    "frame_dx": 796,  
    "frame_dy": 204,  
    "confidence": 0.91,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        736,  
        184,  
        860,  
        226  
    ]  
},  
{  
    "id": "filter",  
    "type": "input",  
    "label": "Filter issues",  
    "description": "Search field for GitHub issue filters.",  
    "state": "visible",  
    "dx": 880,  
    "dy": 268,  
    "frame_dx": 880,  
    "frame_dy": 268,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
        760,  
        248,  
        1080,  
        290  
    ]  
}  
],  
    "element_count": 2  
},  
"knowledge_base": {  
    "total_captures": 3,  
    "screens": [  
        {  
            "capture_index": 1,  
            "phase": "initial",  
            "recorded_at": "2026-05-03T17:20:14.000000+00:00",  
            "fingerprint": "a0b1c2d3e4f5",  
            "element_count": 2,  
            "screen": {
```

```
    "image":  
    "vision/data/sessions/shot_1777819202_d1e2f3/processing/proc_1777819202_d1e2f3/pr  
eprocessed_frames/frame_1777819202_d1e2f3.jpg",  
    "image_size": {  
        "width": 1920,  
        "height": 1080  
    },  
    "screen_bbox": [  
        0,  
        0,  
        1920,  
        1080  
    ],  
    "screen_size": {  
        "width": 1920,  
        "height": 1080  
    },  
    "coordinate_system": "pixel",  
    "elements": [  
        {  
            "id": "issues",  
            "type": "tab",  
            "label": "Issues",  
            "description": "GitHub issues tab.",  
            "state": "visible",  
            "dx": 796,  
            "dy": 204,  
            "frame_dx": 796,  
            "frame_dy": 204,  
            "confidence": 0.91,  
            "source": "gemini_vlm",  
            "frame_bbox": [  
                736,  
                184,  
                860,  
                226  
            ]  
        },  
        {  
            "id": "bug_filter",  
            "type": "text",  
            "label": "label:bug",  
            "description": "Applied bug filter",  
            "state": "visible",  
            "dx": 796,
```

```
        "dy": 204,
        "frame_dx": 796,
        "frame_dy": 204,
        "confidence": 0.9,
        "source": "gemini_vlm",
        "frame_bbox": [
            736,
            184,
            860,
            226
        ]
    }
],
    "element_count": 2
},
    "metadata": {
        "step_index": 0,
        "attempt": 1
    }
},
{
    "capture_index": 2,
    "phase": "pre_step",
    "recorded_at": "2026-05-03T17:21:12.000000+00:00",
    "fingerprint": "b1c2d3e4f5a6",
    "element_count": 1,
    "screen": {
        "image":
"vision/data/sessions/shot_1777819203_e2f304/processing/proc_1777819203_e2f304/pr
eprocessed_frames/frame_1777819203_e2f304.jpg",
        "image_size": {
            "width": 1920,
            "height": 1080
        },
        "screen_bbox": [
            0,
            0,
            1920,
            1080
        ],
        "screen_size": {
            "width": 1920,
            "height": 1080
        },
        "coordinate_system": "pixel",
```

```

    "elements": [
      {
        "id": "issue_top",
        "type": "link",
        "label": "Crash on startup",
        "description": "Top bug issue in the filtered list.",
        "state": "visible",
        "dx": 612,
        "dy": 420,
        "frame_dx": 612,
        "frame_dy": 420,
        "confidence": 0.92,
        "source": "gemini_vlm",
        "frame_bbox": [
          500,
          398,
          860,
          446
        ]
      }
    ],
    "element_count": 1
  },
  "metadata": {
    "step_index": 1,
    "attempt": 1
  }
},
{
  "capture_index": 3,
  "phase": "final",
  "recorded_at": "2026-05-03T17:24:00.000000+00:00",
  "fingerprint": "c2d3e4f5a6b7",
  "element_count": 1,
  "screen": {
    "image":
"vision/data/sessions/shot_1777819203_e2f304/processing/proc_1777819203_e2f304/pr
eprocessed_frames/frame_1777819203_e2f304.jpg",
    "image_size": {
      "width": 1920,
      "height": 1080
    },
    "screen_bbox": [
      0,
      0,

```

```
        1920,
        1080
    ],
    "screen_size": {
        "width": 1920,
        "height": 1080
    },
    "coordinate_system": "pixel",
    "elements": [
        {
            "id": "issue_top",
            "type": "link",
            "label": "Crash on startup",
            "description": "Top bug issue in the filtered list.",
            "state": "visible",
            "dx": 612,
            "dy": 420,
            "frame_dx": 612,
            "frame_dy": 420,
            "confidence": 0.92,
            "source": "gemini_vlm",
            "frame_bbox": [
                500,
                398,
                860,
                446
            ]
        }
    ],
    "element_count": 1
},
"metadata": {
    "step_index": 2,
    "attempt": 1
}
}
]
},
"todo": {
    "steps": [
        {
            "id": 1,
            "action": "Open the GitHub issues page.",
            "target": "Issues tab",
            "expected_result": "The issues list is visible.",
```

```
    "needs_user_data": false,
    "user_data_field": null,
    "status": "done"
  },
  {
    "id": 2,
    "action": "Filter issues by the bug label.",
    "target": "bug filter",
    "expected_result": "Only bug issues are shown.",
    "needs_user_data": false,
    "user_data_field": null,
    "status": "done"
  },
  {
    "id": 3,
    "action": "Open the top issue.",
    "target": "Top issue",
    "expected_result": "The issue details open.",
    "needs_user_data": false,
    "user_data_field": null,
    "status": "done"
  }
],
"notes": "A complex GitHub navigation task.",
"estimated_complexity": "complex"
},
"step_plans": [
  {
    "step_id": 1,
    "action": "Open the GitHub issues page.",
    "target": "Issues tab",
    "expected_result": "The issues list is visible.",
    "hid_commands": [
      {
        "cmd": "mouse_move",
        "meta": {
          "commandId": "7bbe0323-2e7e-44ea-a1ab-51d84cd717e6"
        },
        "dx": 796,
        "dy": 204,
        "smooth": true
      },
      {
        "cmd": "mouse_click",
        "meta": {
```



```
        "commandId": "92273538-af8d-4ce9-b827-39bd94c49944"
    },
    "button": "left"
  }
],
"reasoning": "The Issues tab is visible in the repository navigation."
},
{
  "step_id": 2,
  "action": "Filter issues by the bug label.",
  "target": "bug filter",
  "expected_result": "Only bug issues are shown.",
  "hid_commands": [
    {
      "cmd": "mouse_move",
      "meta": {
        "commandId": "02ec36e1-786a-4b31-ad19-2db48fa489fb"
      },
      "dx": 880,
      "dy": 268,
      "smooth": true
    },
    {
      "cmd": "mouse_click",
      "meta": {
        "commandId": "c6441bac-df8d-4e98-b095-4a48b6ff69c9"
      },
      "button": "left"
    },
    {
      "cmd": "type_text",
      "meta": {
        "commandId": "bba3f468-8e62-4d7a-aecc-312ca6d33ee2"
      },
      "text": "label:bug"
    }
  ],
  "reasoning": "The filter field is visible at the top of the issues list."
},
{
  "step_id": 3,
  "action": "Open the top issue.",
  "target": "Top issue",
  "expected_result": "The issue details open.",
  "hid_commands": [
```

```
{
  "cmd": "mouse_move",
  "meta": {
    "commandId": "abc6fec8-f3c7-4caa-9882-e2b96862a196"
  },
  "dx": 612,
  "dy": 420,
  "smooth": true
},
{
  "cmd": "mouse_click",
  "meta": {
    "commandId": "2dbae063-be88-4154-85cf-c299f9335ffa"
  },
  "button": "left"
}
],
"reasoning": "The top issue is visible in the filtered results."
}
],
"step_results": [
  {
    "step_id": 1,
    "action": "Open the GitHub issues page.",
    "attempt": 1,
    "exec_status": "success",
    "eval_status": "success",
    "confidence": 0.96,
    "reason": "The issues list was opened.",
    "final_status": "success",
    "timestamp": "2026-05-03T17:21:12.000000+00:00"
  },
  {
    "step_id": 2,
    "action": "Filter issues by the bug label.",
    "attempt": 1,
    "exec_status": "success",
    "eval_status": "success",
    "confidence": 0.95,
    "reason": "The bug filter was applied.",
    "final_status": "success",
    "timestamp": "2026-05-03T17:22:20.000000+00:00"
  },
  {
    "step_id": 3,
```

```
    "action": "Open the top issue.",
    "attempt": 1,
    "exec_status": "success",
    "eval_status": "success",
    "confidence": 0.96,
    "reason": "The top filtered issue opened successfully.",
    "final_status": "success",
    "timestamp": "2026-05-03T17:24:00.000000+00:00"
  }
],
"final_report": {
  "success": true,
  "summary": "The bug-filtered issue was opened successfully.",
  "message": "The Issues tab was opened, the bug filter was applied, and the top issue details were displayed.",
  "steps_completed": 3,
  "steps_failed": 0,
  "issues": [],
  "recommendations": []
}
```

Test Case 28: Check 10 AM Meeting on Next Monday (Simple Task)

User Prompt:

Open the calendar app, switch to next Monday, and check whether there is a 10 AM meeting.

Test Results:

- Steps Total: 2
- Completed: 2
- Failed: 0
- Duration: 3 minutes 22 seconds

Step Execution:

Step 1: Move to next Monday in the calendar

- Target: Next button

- **Attempts:** 1
- **Execution:** Performed
- **Evaluation:** Success
- **Issue:** None

Step 2: Check whether there is a 10 AM meeting

- **Target:** 10 AM meeting slot
- **Attempts:** 1
- **Execution:** Performed
- **Evaluation:** Success
- **Issue:** None

Execution Summary:

The task completed successfully. The calendar advanced to next Monday, and a 10 AM "Team sync" meeting was identified and confirmed.

Issues Identified:

- None

Recommendations for Improvement:

- No specific recommendations — test case executed without issues

JSON Response:

full_cycle.json

```
{
  "run_id": "d747b369-c1da-4b4d-9187-be096197d2a5",
  "user_task": "Open the calendar app, switch to next Monday, and check whether there is a 10 AM meeting.",
  "started_at": "2026-05-03T17:40:00.000000+00:00",
  "finished_at": "2026-05-03T17:43:22.000000+00:00",
  "vision_used": true,
  "perception": {
    "image":
"vision/data/sessions/shot_1777819301_a1b2c3/processing/proc_1777819301_a1b2c3/processed_frames/frame_1777819301_a1b2c3.jpg",
    "image_size": {
      "width": 1920,
      "height": 1080
    },
    "screen_bbox": [
```

```
0,
0,
1920,
1080
],
"screen_size": {
  "width": 1920,
  "height": 1080
},
"coordinate_system": "pixel",
"elements": [
  {
    "id": "calendar_title",
    "type": "heading",
    "label": "Calendar",
    "description": "Calendar application title.",
    "state": "visible",
    "dx": 286,
    "dy": 168,
    "frame_dx": 286,
    "frame_dy": 168,
    "confidence": 0.96,
    "source": "gemini_vlm",
    "frame_bbox": [
      210,
      146,
      364,
      188
    ]
  },
  {
    "id": "next_button",
    "type": "button",
    "label": "Next",
    "description": "Button to move to the next date range.",
    "state": "visible",
    "dx": 1362,
    "dy": 170,
    "frame_dx": 1362,
    "frame_dy": 170,
    "confidence": 0.94,
    "source": "gemini_vlm",
    "frame_bbox": [
      1332,
      146,
```

```

        1390,
        194
    ]
}
],
"element_count": 2
},
"knowledge_base": {
    "total_captures": 3,
    "screens": [
        {
            "capture_index": 1,
            "phase": "initial",
            "recorded_at": "2026-05-03T17:40:12.000000+00:00",
            "fingerprint": "d9e8f7a6b5c4",
            "element_count": 2,
            "screen": {
                "image":
"vision/data/sessions/shot_1777819302_b2c3d4/processing/proc_1777819302_b2c3d4/pr
eprocessed_frames/frame_1777819302_b2c3d4.jpg",
                "image_size": {
                    "width": 1920,
                    "height": 1080
                },
                "screen_bbox": [
                    0,
                    0,
                    1920,
                    1080
                ],
                "screen_size": {
                    "width": 1920,
                    "height": 1080
                },
                "coordinate_system": "pixel",
                "elements": [
                    {
                        "id": "week_label",
                        "type": "text",
                        "label": "Mon May 4",
                        "description": "Calendar week label after navigation.",
                        "state": "visible",
                        "dx": 332,
                        "dy": 220,
                        "frame_dx": 332,

```

```
        "frame_dy": 220,  
        "confidence": 0.94,  
        "source": "gemini_vlm",  
        "frame_bbox": [  
            260,  
            200,  
            412,  
            240  
        ]  
    },  
    {  
        "id": "meeting_slot",  
        "type": "event",  
        "label": "Team sync",  
        "description": "10 AM meeting slot.",  
        "state": "visible",  
        "dx": 1022,  
        "dy": 456,  
        "frame_dx": 1022,  
        "frame_dy": 456,  
        "confidence": 0.9,  
        "source": "gemini_vlm",  
        "frame_bbox": [  
            954,  
            430,  
            1122,  
            484  
        ]  
    }  
],  
    "element_count": 2  
},  
    "metadata": {  
        "step_index": 0,  
        "attempt": 1  
    }  
},  
    {  
        "capture_index": 2,  
        "phase": "pre_step",  
        "recorded_at": "2026-05-03T17:41:04.000000+00:00",  
        "fingerprint": "e8f7a6b5c4d3",  
        "element_count": 2,  
        "screen": {
```

```
    "image":  
    "vision/data/sessions/shot_1777819303_c3d4e5/processing/proc_1777819303_c3d4e5/pr  
eprocessed_frames/frame_1777819303_c3d4e5.jpg",  
    "image_size": {  
        "width": 1920,  
        "height": 1080  
    },  
    "screen_bbox": [  
        0,  
        0,  
        1920,  
        1080  
    ],  
    "screen_size": {  
        "width": 1920,  
        "height": 1080  
    },  
    "coordinate_system": "pixel",  
    "elements": [  
        {  
            "id": "meeting_slot",  
            "type": "event",  
            "label": "Team sync",  
            "description": "10 AM meeting slot.",  
            "state": "visible",  
            "dx": 1022,  
            "dy": 456,  
            "frame_dx": 1022,  
            "frame_dy": 456,  
            "confidence": 0.9,  
            "source": "gemini_vlm",  
            "frame_bbox": [  
                954,  
                430,  
                1122,  
                484  
            ]  
        },  
        {  
            "id": "no_meeting",  
            "type": "text",  
            "label": "No events after 10:30",  
            "description": "Status message for the selected day.",  
            "state": "visible",  
            "dx": 330,
```



```
        "dy": 716,
        "frame_dx": 330,
        "frame_dy": 716,
        "confidence": 0.9,
        "source": "gemini_vlm",
        "frame_bbox": [
            250,
            694,
            480,
            742
        ]
    }
],
    "element_count": 2
},
    "metadata": {
        "step_index": 1,
        "attempt": 1
    }
},
{
    "capture_index": 3,
    "phase": "final",
    "recorded_at": "2026-05-03T17:43:10.000000+00:00",
    "fingerprint": "f7a6b5c4d3e2",
    "element_count": 2,
    "screen": {
        "image":
"vision/data/sessions/shot_1777819303_c3d4e5/processing/proc_1777819303_c3d4e5/pr
eprocessed_frames/frame_1777819303_c3d4e5.jpg",
        "image_size": {
            "width": 1920,
            "height": 1080
        },
        "screen_bbox": [
            0,
            0,
            1920,
            1080
        ],
        "screen_size": {
            "width": 1920,
            "height": 1080
        },
        "coordinate_system": "pixel",
```

```
"elements": [  
  {  
    "id": "meeting_slot",  
    "type": "event",  
    "label": "Team sync",  
    "description": "10 AM meeting slot.",  
    "state": "visible",  
    "dx": 1022,  
    "dy": 456,  
    "frame_dx": 1022,  
    "frame_dy": 456,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
      954,  
      430,  
      1122,  
      484  
    ]  
  },  
  {  
    "id": "no_meeting",  
    "type": "text",  
    "label": "No events after 10:30",  
    "description": "Status message for the selected day.",  
    "state": "visible",  
    "dx": 330,  
    "dy": 716,  
    "frame_dx": 330,  
    "frame_dy": 716,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
      250,  
      694,  
      480,  
      742  
    ]  
  }  
],  
"element_count": 2  
},  
"metadata": {  
  "step_index": 1,  
  "attempt": 1
```

```

    }
  }
]
},
"todo": {
  "steps": [
    {
      "id": 1,
      "action": "Move to next Monday in the calendar.",
      "target": "Next button",
      "expected_result": "The calendar shows next Monday.",
      "needs_user_data": false,
      "user_data_field": null,
      "status": "done"
    },
    {
      "id": 2,
      "action": "Check whether there is a 10 AM meeting.",
      "target": "10 AM meeting slot",
      "expected_result": "The meeting is identified or confirmed absent.",
      "needs_user_data": false,
      "user_data_field": null,
      "status": "done"
    }
  ],
  "notes": "A simple calendar check.",
  "estimated_complexity": "simple"
},
"step_plans": [
  {
    "step_id": 1,
    "action": "Move to next Monday in the calendar.",
    "attempt": 1,
    "exec_status": "success",
    "eval_status": "success",
    "confidence": 0.96,
    "reason": "The calendar advanced to the next week.",
    "final_status": "success",
    "timestamp": "2026-05-03T17:41:04.000000+00:00"
  },
  {
    "step_id": 2,
    "action": "Check whether there is a 10 AM meeting.",
    "attempt": 1,
    "exec_status": "success",

```

```
    "eval_status": "success",
    "confidence": 0.92,
    "reason": "A 10 AM team sync event was visible.",
    "final_status": "success",
    "timestamp": "2026-05-03T17:43:10.000000+00:00"
  }
],
"step_results": [
  {
    "step_id": 1,
    "action": "Move to next Monday in the calendar.",
    "target": "Next button",
    "expected_result": "The calendar shows next Monday.",
    "hid_commands": [
      {
        "cmd": "mouse_move",
        "meta": {
          "commandId": "d13ebee9-e955-483b-be11-f5fb627fe1cd"
        },
        "dx": 1362,
        "dy": 170,
        "smooth": true
      },
      {
        "cmd": "mouse_click",
        "meta": {
          "commandId": "6e0fc6f3-a93e-40a2-a407-82893f0289ac"
        },
        "button": "left"
      }
    ],
    "reasoning": "The Next button is visible in the calendar header."
  },
  {
    "step_id": 2,
    "action": "Check whether there is a 10 AM meeting.",
    "target": "10 AM meeting slot",
    "expected_result": "The meeting is identified or confirmed absent.",
    "hid_commands": [
      {
        "cmd": "mouse_move",
        "meta": {
          "commandId": "49e331be-18ad-459c-999d-a9a8d7a6eaec"
        },
        "dx": 1022,
```

```
        "dy": 456,  
        "smooth": true  
      },  
      {  
        "cmd": "mouse_click",  
        "meta": {  
          "commandId": "9c50d741-8400-434f-a31a-96bd16dc1c5b"  
        },  
        "button": "left"  
      }  
    ],  
    "reasoning": "The 10 AM meeting slot is visible after navigation."  
  }  
],  
"final_report": {  
  "success": true,  
  "summary": "The calendar was advanced and the 10 AM meeting was found.",  
  "message": "Next Monday was opened and a team sync at 10 AM was visible.",  
  "steps_completed": 2,  
  "steps_failed": 0,  
  "issues": [],  
  "recommendations": []  
}  
}
```

Test Case 29: File Explorer - Move Latest PDF to Reports Folder (Medium Task)

User Prompt:

Open the downloads folder, find the most recent PDF, and move it into the Reports folder.

Test Results:

- Steps Total: 2
- Completed: 1
- Failed: 1
- Duration: 5 minutes 14 seconds

Step Execution:

Step 1: Open Downloads and find the latest PDF

- **Target:** latest PDF
- **Attempts:** 1
- **Execution:** Performed
- **Evaluation:** Success
- **Issue:** None

Step 2: Move the PDF into Reports

- **Target:** Reports folder
- **Attempts:** 2
- **Execution:** Performed
- **Evaluation:** Failed (retry loop)
- **Issue:** The move confirmation dialog appeared, but the action was not finalized. The move confirmation dialog remained open.

Execution Summary:

The task partially succeeded. The latest PDF was found in Downloads, but the move action into Reports did not complete. The confirmation dialog appeared but was not finalized before the run ended.

Issues Identified:

- Move confirmation dialog remained open
- File transfer not confirmed before task completion
- Missing click on confirmation button to finalize move

Recommendations for Improvement:

- Finalize the move by clicking confirmation button in the dialog before ending task
- Add explicit detection and handling of confirmation dialogs
- Implement dialog state verification before proceeding

JSON Response:

full_cycle.json

```
{
  "run_id": "d5bddbb9-4e4b-496e-8f65-3f8517aabf92",
  "user_task": "Open the downloads folder, find the most recent PDF, and move it into the Reports folder.",
  "started_at": "2026-05-03T18:00:00.000000+00:00",
  "finished_at": "2026-05-03T18:05:14.000000+00:00",
  "vision_used": true,
```

```
"perception": {
  "image":
"vision/data/sessions/shot_1777819401_b4c5d6/processing/proc_1777819401_b4c5d6/pr
eprocessed_frames/frame_1777819401_b4c5d6.jpg",
  "image_size": {
    "width": 1920,
    "height": 1080
  },
  "screen_bbox": [
    0,
    0,
    1920,
    1080
  ],
  "screen_size": {
    "width": 1920,
    "height": 1080
  },
  "coordinate_system": "pixel",
  "elements": [
    {
      "id": "downloads",
      "type": "folder",
      "label": "Downloads",
      "description": "Downloads folder in file explorer.",
      "state": "visible",
      "dx": 452,
      "dy": 284,
      "frame_dx": 452,
      "frame_dy": 284,
      "confidence": 0.94,
      "source": "gemini_vlm",
      "frame_bbox": [
        402,
        264,
        540,
        312
      ]
    },
    {
      "id": "reports",
      "type": "folder",
      "label": "Reports",
      "description": "Destination folder for report files.",
      "state": "visible",
```

```
    "dx": 690,
    "dy": 284,
    "frame_dx": 690,
    "frame_dy": 284,
    "confidence": 0.93,
    "source": "gemini_vlm",
    "frame_bbox": [
        638,
        264,
        744,
        312
    ]
  }
],
"element_count": 2
},
"knowledge_base": {
  "total_captures": 3,
  "screens": [
    {
      "capture_index": 1,
      "phase": "initial",
      "recorded_at": "2026-05-03T18:00:16.000000+00:00",
      "fingerprint": "aabbccddeeff",
      "element_count": 2,
      "screen": {
        "image":
"vision/data/sessions/shot_1777819402_c5d6e7/processing/proc_1777819402_c5d6e7/pr
eprocessed_frames/frame_1777819402_c5d6e7.jpg",
        "image_size": {
          "width": 1920,
          "height": 1080
        },
        "screen_bbox": [
          0,
          0,
          1920,
          1080
        ],
        "screen_size": {
          "width": 1920,
          "height": 1080
        },
        "coordinate_system": "pixel",
        "elements": [
```



```
{
  "id": "pdf_1",
  "type": "file",
  "label": "specs.pdf",
  "description": "Older PDF file in Downloads.",
  "state": "visible",
  "dx": 672,
  "dy": 430,
  "frame_dx": 672,
  "frame_dy": 430,
  "confidence": 0.9,
  "source": "gemini_vlm",
  "frame_bbox": [
    610,
    408,
    780,
    452
  ]
},
{
  "id": "pdf_2",
  "type": "file",
  "label": "release_notes.pdf",
  "description": "Most recent PDF in Downloads.",
  "state": "visible",
  "dx": 676,
  "dy": 486,
  "frame_dx": 676,
  "frame_dy": 486,
  "confidence": 0.92,
  "source": "gemini_vlm",
  "frame_bbox": [
    580,
    464,
    806,
    508
  ]
}
],
"element_count": 2
},
"metadata": {
  "step_index": 0,
  "attempt": 1
}
```

```
},
{
  "capture_index": 2,
  "phase": "pre_step",
  "recorded_at": "2026-05-03T18:01:10.000000+00:00",
  "fingerprint": "bbccddee1122",
  "element_count": 2,
  "screen": {
    "image":
"vision/data/sessions/shot_1777819403_d6e7f8/processing/proc_1777819403_d6e7f8/pr
eprocessed_frames/frame_1777819403_d6e7f8.jpg",
    "image_size": {
      "width": 1920,
      "height": 1080
    },
    "screen_bbox": [
      0,
      0,
      1920,
      1080
    ],
    "screen_size": {
      "width": 1920,
      "height": 1080
    },
    "coordinate_system": "pixel",
    "elements": [
      {
        "id": "reports_folder",
        "type": "folder",
        "label": "Reports",
        "description": "Destination folder in file explorer.",
        "state": "visible",
        "dx": 690,
        "dy": 284,
        "frame_dx": 690,
        "frame_dy": 284,
        "confidence": 0.93,
        "source": "gemini_vlm",
        "frame_bbox": [
          638,
          264,
          744,
          312
        ]
      }
    ]
  }
}
```

```

    },
    {
      "id": "move_confirm",
      "type": "dialog",
      "label": "Move item",
      "description": "Confirmation dialog for moving the selected file.",
      "state": "visible",
      "dx": 960,
      "dy": 590,
      "frame_dx": 960,
      "frame_dy": 590,
      "confidence": 0.89,
      "source": "gemini_vlm",
      "frame_bbox": [
        860,
        550,
        1060,
        632
      ]
    }
  ],
  "element_count": 2
},
"metadata": {
  "step_index": 1,
  "attempt": 1
}
},
{
  "capture_index": 3,
  "phase": "final",
  "recorded_at": "2026-05-03T18:05:00.000000+00:00",
  "fingerprint": "ccddee112233",
  "element_count": 2,
  "screen": {
    "image":
"vision/data/sessions/shot_1777819403_d6e7f8/processing/proc_1777819403_d6e7f8/pr
eprocessed_frames/frame_1777819403_d6e7f8.jpg",
    "image_size": {
      "width": 1920,
      "height": 1080
    },
    "screen_bbox": [
      0,
      0,

```

```
1920,
1080
],
"screen_size": {
  "width": 1920,
  "height": 1080
},
"coordinate_system": "pixel",
"elements": [
  {
    "id": "reports_folder",
    "type": "folder",
    "label": "Reports",
    "description": "Destination folder in file explorer.",
    "state": "visible",
    "dx": 690,
    "dy": 284,
    "frame_dx": 690,
    "frame_dy": 284,
    "confidence": 0.93,
    "source": "gemini_vlm",
    "frame_bbox": [
      638,
      264,
      744,
      312
    ]
  },
  {
    "id": "move_confirm",
    "type": "dialog",
    "label": "Move item",
    "description": "Confirmation dialog for moving the selected file.",
    "state": "visible",
    "dx": 960,
    "dy": 590,
    "frame_dx": 960,
    "frame_dy": 590,
    "confidence": 0.89,
    "source": "gemini_vlm",
    "frame_bbox": [
      860,
      550,
      1060,
      632
    ]
  }
]
```

```

        ]
      }
    ],
    "element_count": 2
  },
  "metadata": {
    "step_index": 2,
    "attempt": 1
  }
}
]
},
"todo": {
  "steps": [
    {
      "id": 1,
      "action": "Open Downloads and find the latest PDF.",
      "target": "latest PDF",
      "expected_result": "The most recent PDF is visible.",
      "needs_user_data": false,
      "user_data_field": null,
      "status": "done"
    },
    {
      "id": 2,
      "action": "Move the PDF into Reports.",
      "target": "Reports folder",
      "expected_result": "The PDF is moved into Reports.",
      "needs_user_data": false,
      "user_data_field": null,
      "status": "failed"
    }
  ],
  "notes": "A medium file-management workflow.",
  "estimated_complexity": "medium"
},
"step_plans": [
  {
    "step_id": 1,
    "action": "Open Downloads and find the latest PDF.",
    "attempt": 1,
    "exec_status": "success",
    "eval_status": "success",
    "confidence": 0.95,
    "reason": "The newest PDF was visible in the list.",
  }
]

```

```

    "final_status": "success",
    "timestamp": "2026-05-03T18:01:10.000000+00:00"
  },
  {
    "step_id": 2,
    "action": "Move the PDF into Reports.",
    "attempt": 2,
    "exec_status": "success",
    "eval_status": "retry",
    "confidence": 0.88,
    "reason": "The move confirmation dialog appeared, but the action was not
finalized.",
    "final_status": "executing",
    "timestamp": "2026-05-03T18:05:00.000000+00:00"
  }
],
"step_results": [
  {
    "step_id": 1,
    "action": "Open Downloads and find the latest PDF.",
    "target": "latest PDF",
    "expected_result": "The most recent PDF is visible.",
    "hid_commands": [
      {
        "cmd": "mouse_move",
        "meta": {
          "commandId": "24813e5a-862d-428a-972c-cd82ccc3168b"
        },
        "dx": 676,
        "dy": 486,
        "smooth": true
      },
      {
        "cmd": "mouse_click",
        "meta": {
          "commandId": "f75056c2-799e-4cc6-b237-80bed7f3411d"
        },
        "button": "left"
      }
    ],
    "reasoning": "The latest PDF is visible in Downloads."
  },
  {
    "step_id": 2,
    "action": "Move the PDF into Reports.",

```

```
"target": "Reports folder",
"expected_result": "The PDF is moved into Reports.",
"hid_commands": [
  {
    "cmd": "mouse_move",
    "meta": {
      "commandId": "e4c78f8b-dbbf-405b-8f1a-8c3659997d77"
    },
    "dx": 690,
    "dy": 284,
    "smooth": true
  },
  {
    "cmd": "mouse_click",
    "meta": {
      "commandId": "5701cab7-a13c-4781-b919-4561a88af60c"
    },
    "button": "left"
  }
],
"reasoning": "The Reports folder is visible as the drop target."
},
],
"final_report": {
  "success": false,
  "summary": "The latest PDF was found, but moving it into Reports did not complete.",
  "message": "The move action started, but the file transfer was not confirmed before the run ended.",
  "steps_completed": 1,
  "steps_failed": 1,
  "issues": [
    "The move confirmation dialog remained open."
  ],
  "recommendations": [
    "Finalize the move in the dialog before ending the task."
  ]
}
```

Test Case 30: Slack - Check for Pinned Deployment Note (Simple Task)

User Prompt:

Open Slack, jump to the team channel, and check whether there is a pinned deployment note.

Test Results:

- **Steps Total: 2**
- **Completed: 2**
- **Failed: 0**
- **Duration: 3 minutes 40 seconds**

Step Execution:

Step 1: Jump to the team channel

- **Target: team channel**
- **Attempts: 1**
- **Execution: Performed**
- **Evaluation: Success**
- **Issue: None**

Step 2: Check for a pinned deployment note

- **Target: Pinned button**
- **Attempts: 1**
- **Execution: Performed**
- **Evaluation: Success**
- **Issue: None**

Execution Summary:

The task completed successfully. The team channel was activated, and a pinned deployment note ("Deploy tonight at 8 PM") was confirmed.

Issues Identified:

- **None**

Recommendations for Improvement:

- **No specific recommendations — test case executed without issues**

JSON Response:

full_cycle.json

```
{
  "run_id": "e77b8944-584d-4ac4-aa99-bd8e6096d6fd",
  "user_task": "Open Slack, jump to the team channel, and check whether there is
a pinned deployment note.",
  "started_at": "2026-05-03T18:20:00.000000+00:00",
  "finished_at": "2026-05-03T18:23:40.000000+00:00",
  "vision_used": true,
  "perception": {
    "image":
"vision/data/sessions/shot_1777819501_d7e8f9/processing/proc_1777819501_d7e8f9/pr
eprocessed_frames/frame_1777819501_d7e8f9.jpg",
    "image_size": {
      "width": 1920,
      "height": 1080
    },
    "screen_bbox": [
      0,
      0,
      1920,
      1080
    ],
    "screen_size": {
      "width": 1920,
      "height": 1080
    },
    "coordinate_system": "pixel",
    "elements": [
      {
        "id": "channel_team",
        "type": "channel",
        "label": "team",
        "description": "Team channel in Slack.",
        "state": "visible",
        "dx": 250,
        "dy": 320,
        "frame_dx": 250,
        "frame_dy": 320,
        "confidence": 0.94,
        "source": "gemini_vlm",
        "frame_bbox": [
          194,
          300,
```

```
        314,  
        342  
    ]  
  },  
  {  
    "id": "pinned",  
    "type": "button",  
    "label": "Pinned",  
    "description": "Pinned items button in the channel header.",  
    "state": "visible",  
    "dx": 1542,  
    "dy": 186,  
    "frame_dx": 1542,  
    "frame_dy": 186,  
    "confidence": 0.93,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
      1496,  
      164,  
      1586,  
      208  
    ]  
  }  
],  
"element_count": 2  
},  
"knowledge_base": {  
  "total_captures": 3,  
  "screens": [  
    {  
      "capture_index": 1,  
      "phase": "initial",  
      "recorded_at": "2026-05-03T18:20:20.000000+00:00",  
      "fingerprint": "112233445566",  
      "element_count": 2,  
      "screen": {  
        "image":  
"vision/data/sessions/shot_1777819502_e8f9a0/processing/proc_1777819502_e8f9a0/pr  
eprocessed_frames/frame_1777819502_e8f9a0.jpg",  
        "image_size": {  
          "width": 1920,  
          "height": 1080  
        },  
        "screen_bbox": [  
          0,
```

```
0,
1920,
1080
],
"screen_size": {
  "width": 1920,
  "height": 1080
},
"coordinate_system": "pixel",
"elements": [
  {
    "id": "channel_team",
    "type": "channel",
    "label": "team",
    "description": "Team channel in Slack.",
    "state": "active",
    "dx": 250,
    "dy": 320,
    "frame_dx": 250,
    "frame_dy": 320,
    "confidence": 0.94,
    "source": "gemini_vlm",
    "frame_bbox": [
      194,
      300,
      314,
      342
    ]
  },
  {
    "id": "note",
    "type": "message",
    "label": "Deploy tonight at 8 PM",
    "description": "Pinned deployment note.",
    "state": "visible",
    "dx": 816,
    "dy": 404,
    "frame_dx": 816,
    "frame_dy": 404,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
      672,
      384,
      972,
```

```

        430
    ]
}
],
"element_count": 2
},
"metadata": {
    "step_index": 0,
    "attempt": 1
}
},
{
    "capture_index": 2,
    "phase": "pre_step",
    "recorded_at": "2026-05-03T18:21:04.000000+00:00",
    "fingerprint": "223344556677",
    "element_count": 2,
    "screen": {
        "image":
"vision/data/sessions/shot_1777819503_f9a0b1/processing/proc_1777819503_f9a0b1/pr
eprocessed_frames/frame_1777819503_f9a0b1.jpg",
        "image_size": {
            "width": 1920,
            "height": 1080
        },
        "screen_bbox": [
            0,
            0,
            1920,
            1080
        ],
        "screen_size": {
            "width": 1920,
            "height": 1080
        },
        "coordinate_system": "pixel",
        "elements": [
            {
                "id": "note",
                "type": "message",
                "label": "Deploy tonight at 8 PM",
                "description": "Pinned deployment note.",
                "state": "visible",
                "dx": 816,
                "dy": 404,

```

```
        "frame_dx": 816,
        "frame_dy": 404,
        "confidence": 0.9,
        "source": "gemini_vlm",
        "frame_bbox": [
            672,
            384,
            972,
            430
        ]
    },
    {
        "id": "reply_box",
        "type": "input",
        "label": "Message team",
        "description": "Message composer at the bottom of Slack.",
        "state": "visible",
        "dx": 990,
        "dy": 920,
        "frame_dx": 990,
        "frame_dy": 920,
        "confidence": 0.9,
        "source": "gemini_vlm",
        "frame_bbox": [
            666,
            884,
            1314,
            950
        ]
    }
],
"element_count": 2
},
"metadata": {
    "step_index": 1,
    "attempt": 1
}
},
{
    "capture_index": 3,
    "phase": "final",
    "recorded_at": "2026-05-03T18:23:28.000000+00:00",
    "fingerprint": "334455667788",
    "element_count": 2,
    "screen": {
```

```
    "image":  
    "vision/data/sessions/shot_1777819503_f9a0b1/processing/proc_1777819503_f9a0b1/pr  
eprocessed_frames/frame_1777819503_f9a0b1.jpg",  
    "image_size": {  
        "width": 1920,  
        "height": 1080  
    },  
    "screen_bbox": [  
        0,  
        0,  
        1920,  
        1080  
    ],  
    "screen_size": {  
        "width": 1920,  
        "height": 1080  
    },  
    "coordinate_system": "pixel",  
    "elements": [  
        {  
            "id": "note",  
            "type": "message",  
            "label": "Deploy tonight at 8 PM",  
            "description": "Pinned deployment note.",  
            "state": "visible",  
            "dx": 816,  
            "dy": 404,  
            "frame_dx": 816,  
            "frame_dy": 404,  
            "confidence": 0.9,  
            "source": "gemini_vlm",  
            "frame_bbox": [  
                672,  
                384,  
                972,  
                430  
            ]  
        },  
        {  
            "id": "reply_box",  
            "type": "input",  
            "label": "Message team",  
            "description": "Message composer at the bottom of Slack.",  
            "state": "visible",  
            "dx": 990,
```

```

        "dy": 920,
        "frame_dx": 990,
        "frame_dy": 920,
        "confidence": 0.9,
        "source": "gemini_vlm",
        "frame_bbox": [
            666,
            884,
            1314,
            950
        ]
    }
],
    "element_count": 2
},
    "metadata": {
        "step_index": 1,
        "attempt": 1
    }
}
]
},
"todo": {
    "steps": [
        {
            "id": 1,
            "action": "Jump to the team channel.",
            "target": "team channel",
            "expected_result": "The team channel becomes active.",
            "needs_user_data": false,
            "user_data_field": null,
            "status": "done"
        },
        {
            "id": 2,
            "action": "Check for a pinned deployment note.",
            "target": "Pinned button",
            "expected_result": "A pinned deployment note is identified.",
            "needs_user_data": false,
            "user_data_field": null,
            "status": "done"
        }
    ],
    "notes": "A simple collaboration-app check.",
    "estimated_complexity": "simple"
}

```

```
},
"step_plans": [
  {
    "step_id": 1,
    "action": "Jump to the team channel.",
    "attempt": 1,
    "exec_status": "success",
    "eval_status": "success",
    "confidence": 0.96,
    "reason": "The team channel became active.",
    "final_status": "success",
    "timestamp": "2026-05-03T18:21:04.000000+00:00"
  },
  {
    "step_id": 2,
    "action": "Check for a pinned deployment note.",
    "attempt": 1,
    "exec_status": "success",
    "eval_status": "success",
    "confidence": 0.94,
    "reason": "A pinned deployment note was visible in the channel.",
    "final_status": "success",
    "timestamp": "2026-05-03T18:23:28.000000+00:00"
  }
],
"step_results": [
  {
    "step_id": 1,
    "action": "Jump to the team channel.",
    "target": "team channel",
    "expected_result": "The team channel becomes active.",
    "hid_commands": [
      {
        "cmd": "mouse_move",
        "meta": {
          "commandId": "6414e9e2-8106-4172-a842-893b4257d898"
        },
        "dx": 250,
        "dy": 320,
        "smooth": true
      },
      {
        "cmd": "mouse_click",
        "meta": {
          "commandId": "5855ebfd-8fa0-498e-ae98-5b6303b804d1"
        }
      }
    ]
  }
]
```



```

        },
        "button": "left"
    }
],
"reasoning": "The team channel is visible in the channel list."
},
{
    "step_id": 2,
    "action": "Check for a pinned deployment note.",
    "target": "Pinned button",
    "expected_result": "A pinned deployment note is identified.",
    "hid_commands": [
        {
            "cmd": "mouse_move",
            "meta": {
                "commandId": "0f2f8bdc-dfa0-4dc5-8661-a8d5f60a11fe"
            },
            "dx": 1542,
            "dy": 186,
            "smooth": true
        },
        {
            "cmd": "mouse_click",
            "meta": {
                "commandId": "91699cf6-96c6-40fb-ad14-b1338956d3a0"
            },
            "button": "left"
        }
    ],
    "reasoning": "The Pinned button is visible in the channel header."
}
],
"final_report": {
    "success": true,
    "summary": "The deployment note was found in the pinned items.",
    "message": "The Slack team channel was opened and the pinned deployment note
was confirmed.",
    "steps_completed": 2,
    "steps_failed": 0,
    "issues": [],
    "recommendations": []
}
}

```

Test Case 31: Ticketing System - Check Latest Open Ticket Assignment (Medium Task)

User Prompt:

Open the browser tab with the ticketing system, search for the latest open ticket, and note whether it is assigned.

Test Results:

- **Steps Total: 2**
- **Completed: 1**
- **Failed: 1**
- **Duration: 4 minutes 58 seconds**

Step Execution:

Step 1: Search for the latest open ticket

- **Target: Search tickets**
- **Attempts: 1**
- **Execution: Performed**
- **Evaluation: Success**
- **Issue: None**

Step 2: Check whether the ticket is assigned

- **Target: Assignment badge**
- **Attempts: 1**
- **Execution: Performed**
- **Evaluation: Failed (retry loop)**
- **Issue: The opened ticket showed "Unassigned", but the final confirmation was not captured. The assignment badge changed after the ticket opened.**

Execution Summary:

The task partially succeeded. The ticketing system was reached and the latest open ticket was identified, but the assignment status check was not reliably confirmed due to the badge changing after the ticket opened.

Issues Identified:

- **Assignment badge changed after ticket opened**
- **No re-capture of ticket details after navigation**

- Final confirmation not captured before task completion

Recommendations for Improvement:

- Re-capture the ticket details page before checking assignment status
- Add explicit wait for page to stabilize after opening ticket
- Implement assignment verification after page fully loads

JSON Response:

full_cycle.json

```
{
  "run_id": "31c925d3-ccb0-455f-b8c0-6e3fb415cab7",
  "user_task": "Open the browser tab with the ticketing system, search for the latest open ticket, and note whether it is assigned.",
  "started_at": "2026-05-03T18:40:00.000000+00:00",
  "finished_at": "2026-05-03T18:44:58.000000+00:00",
  "vision_used": true,
  "perception": {
    "image":
"vision/data/sessions/shot_1777819601_g0h1i2/processing/proc_1777819601_g0h1i2/preprocessed_frames/frame_1777819601_g0h1i2.jpg",
    "image_size": {
      "width": 1920,
      "height": 1080
    },
    "screen_bbox": [
      0,
      0,
      1920,
      1080
    ],
    "screen_size": {
      "width": 1920,
      "height": 1080
    },
    "coordinate_system": "pixel",
    "elements": [
      {
        "id": "tab_ticketing",
        "type": "tab",
        "label": "Tickets - LammastideIT",
        "description": "Browser tab for the ticketing system.",
        "state": "active",
```

```
    "dx": 832,
    "dy": 136,
    "frame_dx": 832,
    "frame_dy": 136,
    "confidence": 0.95,
    "source": "gemini_vlm",
    "frame_bbox": [
      742,
      118,
      952,
      155
    ]
  },
  {
    "id": "search",
    "type": "input",
    "label": "Search tickets",
    "description": "Search field within the ticketing system.",
    "state": "visible",
    "dx": 1012,
    "dy": 212,
    "frame_dx": 1012,
    "frame_dy": 212,
    "confidence": 0.93,
    "source": "gemini_vlm",
    "frame_bbox": [
      820,
      190,
      1230,
      232
    ]
  }
],
"element_count": 2
},
"knowledge_base": {
  "total_captures": 3,
  "screens": [
    {
      "capture_index": 1,
      "phase": "initial",
      "recorded_at": "2026-05-03T18:40:11.000000+00:00",
      "fingerprint": "445566778899",
      "element_count": 2,
      "screen": {
```

```
    "image":  
    "vision/data/sessions/shot_1777819602_h1i2j3/processing/proc_1777819602_h1i2j3/pr  
    eprocessed_frames/frame_1777819602_h1i2j3.jpg",  
    "image_size": {  
        "width": 1920,  
        "height": 1080  
    },  
    "screen_bbox": [  
        0,  
        0,  
        1920,  
        1080  
    ],  
    "screen_size": {  
        "width": 1920,  
        "height": 1080  
    },  
    "coordinate_system": "pixel",  
    "elements": [  
        {  
            "id": "ticket_top",  
            "type": "link",  
            "label": "Cannot sync account",  
            "description": "Latest open ticket in the list.",  
            "state": "visible",  
            "dx": 644,  
            "dy": 424,  
            "frame_dx": 644,  
            "frame_dy": 424,  
            "confidence": 0.92,  
            "source": "gemini_vlm",  
            "frame_bbox": [  
                520,  
                402,  
                880,  
                450  
            ]  
        },  
        {  
            "id": "assigned_badge",  
            "type": "label",  
            "label": "Unassigned",  
            "description": "Assignment badge for the top ticket.",  
            "state": "visible",  
            "dx": 1010,
```

```
        "dy": 424,
        "frame_dx": 1010,
        "frame_dy": 424,
        "confidence": 0.9,
        "source": "gemini_vlm",
        "frame_bbox": [
            944,
            402,
            1092,
            450
        ]
    }
],
"element_count": 2
},
"metadata": {
    "step_index": 0,
    "attempt": 1
}
},
{
    "capture_index": 2,
    "phase": "pre_step",
    "recorded_at": "2026-05-03T18:41:08.000000+00:00",
    "fingerprint": "5566778899aa",
    "element_count": 2,
    "screen": {
        "image":
"vision/data/sessions/shot_1777819603_i2j3k4/processing/proc_1777819603_i2j3k4/pr
eprocessed_frames/frame_1777819603_i2j3k4.jpg",
        "image_size": {
            "width": 1920,
            "height": 1080
        },
        "screen_bbox": [
            0,
            0,
            1920,
            1080
        ],
        "screen_size": {
            "width": 1920,
            "height": 1080
        },
        "coordinate_system": "pixel",
```

```
"elements": [  
  {  
    "id": "ticket_detail",  
    "type": "heading",  
    "label": "Cannot sync account",  
    "description": "Opened ticket detail page.",  
    "state": "visible",  
    "dx": 614,  
    "dy": 186,  
    "frame_dx": 614,  
    "frame_dy": 186,  
    "confidence": 0.95,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
      480,  
      160,  
      760,  
      214  
    ]  
  },  
  {  
    "id": "assignment",  
    "type": "label",  
    "label": "Unassigned",  
    "description": "Ticket assignment status.",  
    "state": "visible",  
    "dx": 1024,  
    "dy": 240,  
    "frame_dx": 1024,  
    "frame_dy": 240,  
    "confidence": 0.9,  
    "source": "gemini_vlm",  
    "frame_bbox": [  
      944,  
      220,  
      1104,  
      264  
    ]  
  }  
],  
"element_count": 2  
},  
"metadata": {  
  "step_index": 1,  
  "attempt": 1
```

```
}
},
{
  "capture_index": 3,
  "phase": "final",
  "recorded_at": "2026-05-03T18:44:44.000000+00:00",
  "fingerprint": "66778899aabb",
  "element_count": 2,
  "screen": {
    "image":
"vision/data/sessions/shot_1777819603_i2j3k4/processing/proc_1777819603_i2j3k4/pr
eprocessed_frames/frame_1777819603_i2j3k4.jpg",
    "image_size": {
      "width": 1920,
      "height": 1080
    },
    "screen_bbox": [
      0,
      0,
      1920,
      1080
    ],
    "screen_size": {
      "width": 1920,
      "height": 1080
    },
    "coordinate_system": "pixel",
    "elements": [
      {
        "id": "ticket_detail",
        "type": "heading",
        "label": "Cannot sync account",
        "description": "Opened ticket detail page.",
        "state": "visible",
        "dx": 614,
        "dy": 186,
        "frame_dx": 614,
        "frame_dy": 186,
        "confidence": 0.95,
        "source": "gemini_vlm",
        "frame_bbox": [
          480,
          160,
          760,
          214
```



```

    ]
  },
  {
    "id": "assignment",
    "type": "label",
    "label": "Unassigned",
    "description": "Ticket assignment status.",
    "state": "visible",
    "dx": 1024,
    "dy": 240,
    "frame_dx": 1024,
    "frame_dy": 240,
    "confidence": 0.9,
    "source": "gemini_vlm",
    "frame_bbox": [
      944,
      220,
      1104,
      264
    ]
  }
],
"element_count": 2
},
"metadata": {
  "step_index": 2,
  "attempt": 1
}
}
]
},
"todo": {
  "steps": [
    {
      "id": 1,
      "action": "Search for the latest open ticket.",
      "target": "Search tickets",
      "expected_result": "The latest ticket is visible.",
      "needs_user_data": false,
      "user_data_field": null,
      "status": "done"
    },
    {
      "id": 2,
      "action": "Check whether the ticket is assigned.",

```

```
    "target": "Assignment badge",
    "expected_result": "The assignment state is identified.",
    "needs_user_data": false,
    "user_data_field": null,
    "status": "failed"
  }
],
"notes": "A slightly more complex support-ticket workflow.",
"estimated_complexity": "medium"
},
"step_plans": [
  {
    "step_id": 1,
    "action": "Search for the latest open ticket.",
    "attempt": 1,
    "exec_status": "success",
    "eval_status": "success",
    "confidence": 0.96,
    "reason": "The latest ticket appeared in the results list.",
    "final_status": "success",
    "timestamp": "2026-05-03T18:41:08.000000+00:00"
  },
  {
    "step_id": 2,
    "action": "Check whether the ticket is assigned.",
    "attempt": 1,
    "exec_status": "success",
    "eval_status": "retry",
    "confidence": 0.89,
    "reason": "The opened ticket showed Unassigned, but the final confirmation
was not captured.",
    "final_status": "executing",
    "timestamp": "2026-05-03T18:44:44.000000+00:00"
  }
],
"step_results": [
  {
    "step_id": 1,
    "action": "Search for the latest open ticket.",
    "target": "Search tickets",
    "expected_result": "The latest ticket is visible.",
    "hid_commands": [
      {
        "cmd": "mouse_move",
        "meta": {
```

```
        "commandId": "19837a2f-e3ba-4ec6-ae5c-a13c4b4ee2c1"
    },
    "dx": 1012,
    "dy": 212,
    "smooth": true
},
{
    "cmd": "mouse_click",
    "meta": {
        "commandId": "693acaf5-4ffb-42b7-b86f-f649d057446c"
    },
    "button": "left"
},
{
    "cmd": "type_text",
    "meta": {
        "commandId": "1e9a232a-e8df-481c-b8d6-cc6506fca6c9"
    },
    "text": "status:open sort:latest"
}
],
"reasoning": "The search field is visible in the ticketing system."
},
{
    "step_id": 2,
    "action": "Check whether the ticket is assigned.",
    "target": "Assignment badge",
    "expected_result": "The assignment state is identified.",
    "hid_commands": [
        {
            "cmd": "mouse_move",
            "meta": {
                "commandId": "d71d647c-2d7c-4689-9b29-e24c96204266"
            },
            "dx": 1010,
            "dy": 424,
            "smooth": true
        },
        {
            "cmd": "mouse_click",
            "meta": {
                "commandId": "42319b0c-1851-47b2-acb8-356dabb54ef1"
            },
            "button": "left"
        }
    ]
}
```

```
    ],
    "reasoning": "The assignment badge is visible on the latest ticket."
  }
],
"final_report": {
  "success": false,
  "summary": "The latest ticket was opened, but the assignment status was not reliably confirmed.",
  "message": "The ticketing system was reached and the latest open ticket was identified, but the assignment check stayed ambiguous.",
  "steps_completed": 1,
  "steps_failed": 1,
  "issues": [
    "The assignment badge changed after the ticket opened."
  ],
  "recommendations": [
    "Re-capture the ticket details page before checking assignment status."
  ]
}
```